On the Distribution of Prime Squares and Its Applications

Full Unedited Blueprint

Historical development archive

Prepared for Hamed Mousavi
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This volume brings together every one of the 1,303 original files preserved in the saved blueprint archive. The Markdown notes contain 493,795 whitespace-separated words. Their contents have not been mathematically rewritten or condensed.

The archive includes 54 topic and historical planning notes, 1,224 individual checkpoint notes (42 through 1265), seven batch summaries, an earlier README, four LaTeX source files, three CSV inventories, nine JSON records, and the original initial-blueprint PDF.

Historical status. Earlier notes contain intermediate errors, superseded replacements, earlier titles, and progress statements that were true only at the recorded checkpoint. They are preserved here as development history. The current compact blueprint and current assumption catalogue remain the guides to the final conditional scope.

Formatting and preservation. Markdown and mathematics are typeset, long source lines are wrapped, and the files are placed in a new reading order. Original source files and their checksum manifest are supplied in the companion source bundle; a byte-identical copy of that bundle is embedded in this PDF. The original initial-blueprint PDF is appended at the end.

Verification scope. This compilation changes no Lean proofs and performs no new Lean build or mathematical audit. The verification claims inside individual notes are reproduced historical reports, not new claims made by this compilation.

Contents and reading order

Use the PDF outline to jump to any individual source file. Checkpoint notes are in numerical order. Relative links in the archived Markdown are printed as text; their original destinations remain in the attached source files.

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Topic reference notes

Annular_Boundary_and_Remainder_Bounds.md

Bounding the actual integration-by-parts terms

For $W_\beta(t) = t^{\beta-2}$ and $E_\theta(t) = e^{-2\pi i\theta t^2}$, the previous identity reduces decay to the boundary $[W_\beta(t)E_\theta(t)]_{N/2}^{N}$ and remainder $\int_{N/2}^{N} W'_\beta(t)E_\theta(t)\,dt$.

For $N \geq 2,\ 0 \leq \beta \leq 1$, every t in the annulus is at least one. Monotonicity in the exponent and in the positive base gives

$$|W_\beta(t)| \leq 2/N, \qquad |W'_\beta(t)| \leq 8/N^2.$$

The first estimate only requires beta<=1. The second also uses beta>=0 to bound $|\beta - 2|$ by two. The oscillation has norm one.

The triangle inequality bounds the two boundary terms together by $4/N$. The annulus has length $N/2$, so bounding the remainder integrand by $8/N^2$ also gives $4/N$. These estimates concern the actual integrals, with no hypothesis asserting their decay.

Module	New named theorems
AnnularBoundaryBounds.lean	7

Scope

The proof is unconditional. The next unit will combine these bounds with exact integration by parts and the normalization 1/N. This is a supporting step toward the source model estimates; it does not complete the full minor-arc lemma or the high-low decomposition.

<!-- checkpoint-verification -->

Verification

All 99 project modules and the root were freshly rebuilt: 3614 jobs pass without warnings/errors. All 789 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 101 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. No new deep analytic input or own-paper lemma is assumed.

Annular_Gauss_Model_Tails.md

Uniform tails for the actual Gauss models

The model is

$$G_{N,q,\chi,\kappa,\beta,a}(\theta) = \mathfrak{s}(a,q)M_N^1(\theta) - \mathfrak{s}_\chi(a,q)\kappa M_N^\beta(\theta).$$

It uses the project's genuine totient-normalized sums and the genuine annular integrals, with the source sign and phase.

The already proved unconditional bound for all complex Dirichlet characters, applied at epsilon=1/12, gives one constant C such that both Gauss coefficients are at most $Cq^{-5/12}$ for unit a. The principal coefficient is the character-one specialization of the same proved theorem.

For $N \geq 2,\ 0 \leq \beta \leq 1,\ |\kappa| \leq 1$, the half bound for each integral gives

$$|G(\theta)| \leq Cq^{-5/12}.$$

If $q \leq R$ and $|\theta| \geq R/(qN^2)$, each integral is at most $2q/(\pi R)$, so

$$|G(\theta)| \leq (4C/\pi)R^{-5/12}.$$

The required power comparison $q^{-5/12}q/R \le R^{-5/12}$ is proved with positive bases. Multiplication by the actual cutoff $\eta_s \in [0,1]$ does not increase the norm.

Module	New named theorems
AnnularGaussModelBounds.lean	5

Scope

All inputs used here are proved in the pinned project or Mathlib. No new external analytic estimate is assumed. The coefficient kappa remains a parameter with norm at most one; the exceptional-conductor indicator will be an allowed specialization. This is a model-tail result, not a minor-arc bound for the actual prime sum, and it does not complete the global approximation.

<!-- checkpoint-verification -->

Verification

All 103 project modules and the root were freshly rebuilt: 3622 jobs pass without warnings/errors. All 818 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 105 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. No new deep analytic input or own-paper lemma is assumed.

`Annular_Model_Frequency_Decay.md`

Frequency decay for the genuine annular models

The exact integration-by-parts identity and the two bounds $4/N$ give, for $N \ge 2,\ 0 \le \beta \le 1$,

$$4\pi|\theta|\,|M_N^\beta(\theta)| \le 8/N^2.$$

Thus, at nonzero frequency,

$$|M_N^\beta(\theta)| \le \frac{2}{\pi|\theta|N^2}.$$

Both statements are proved for the actual real integral. Its interval avoids the stationary point zero; no external oscillatory-integral theorem is assumed.

Combining the decay with the already proved half bound yields a global profile, valid also at zero:

$$|M_N^\beta(\theta)| \le \frac{1/2 + 2/\pi}{1 + N^2|\theta|}.$$

Finally, for positive real q and R, outside the rational arc

$$|\theta| \ge R/(qN^2) \quad \Longrightarrow \quad |M_N^\beta(\theta)| \le 2q/(\pi R).$$

This last estimate is intended for the model-tail argument with q<=R. It retains prime-square spatial scale N^2.

Module	New named theorems
AnnularModelDecay.lean	5

Scope

No deep analytic input is used. These uniform-beta estimates supply the decay needed for model tails. They do not claim the extra N^(beta-1) factor in the manuscript's displayed estimate. The independent minor-arc gap is unchanged, and the full approximation and decomposition remain open.

<!-- checkpoint-verification -->

Verification

All 100 project modules and the root were freshly rebuilt: 3615 jobs pass without warnings/errors. All 794 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 102 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. No new deep analytic input or own-paper lemma is assumed.

`Annular_Model_Integration_by_Parts.md`

Exact integration by parts on the quadratic annulus

Write $E_\theta(t) = e^{-2\pi i\theta t^2}$, $W_\beta(t) = t^{\beta-2}$, and $K_\theta = -4\pi i\theta$. For $N \geq 1$, all powers in this proof are evaluated on $[N/2, N] \subset (0, \infty)$.

The new module proves the derivatives and integrability needed for the exact complex identity

$$K_\theta \int_{N/2}^{N} t^{\beta-1} E_\theta(t)\, dt = [t^{\beta-2} E_\theta(t)]_{N/2}^{N} - \int_{N/2}^{N} (\beta-2) t^{\beta-3} E_\theta(t)\, dt.$$

It holds for every real beta and theta. The normalization $1/N$ is not yet applied in this identity. Multiplying by K_θ instead of dividing by it keeps zero frequency within the statement.

The power identity $t^{\beta-2}t = t^{\beta-1}$ is justified for positive real t. No periodicity is assumed for a noninteger integration variable, and the genuine phase derivative from QuadraticOscillation.lean is used.

Module	New named theorems
AnnularModelIntegrationByParts.lean	6

Scope

These are unconditional calculus proofs. Bounds for the two boundary terms and the remainder integral are the next step. The full minor-arc deduction and the analytic decomposition remain open. This unit requests no manuscript change.

<!-- checkpoint-verification -->

Verification

All 98 project modules and the root were freshly rebuilt: 3613 jobs pass without warnings/errors. All 782 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 100 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. No new deep analytic input or own-paper lemma is assumed.

`Annular_Model_Size_and_Mass.md`

Uniform bounds and mass for the actual annular models

This unit concerns the integral already used in the genuine major-arc formula:

$$M_N^\beta(\theta) = \frac{1}{N} \int_{N/2}^{N} t^{\beta-1} e^{-2\pi i\theta t^2}\, dt.$$

The endpoints are real, including when N is odd. The integral is not replaced by a discrete sum or by an abstract bounded function.

For $N \geq 1$ and $\beta \leq 1$, the power density decreases on this positive annulus. Since the oscillation has norm one,

$$|M_N^\beta(\theta)| \leq \tfrac{1}{2}(N/2)^{\beta-1}.$$

Consequently $N \geq 2,\ \beta \leq 1$ gives the uniform bound $1/2$, including the ordinary case $\beta = 1$. Positivity of beta is not needed for these size bounds because the interval is bounded away from zero.

For $\beta > 0$, the derivative of t^β/β gives the exact identity

$$M_N^\beta(0) = \frac{N^\beta - (N/2)^\beta}{N\beta}.$$

The Lean proof uses the previously proved derivative, proves interval integrability on the positive annulus, and applies the complex-valued fundamental theorem of calculus.

Module	New named theorems
AnnularModelSize.lean	5

Scope

No external analytic input is used. This proves the uniform-size part needed in the approximation argument; frequency decay is the next task. The full source minor-arc deduction remains unresolved as documented in Ren_Input_and_Annular_Bound.md. No manuscript modification is requested by this unit.

<!-- checkpoint-verification -->

Verification

All 97 project modules and the root were freshly rebuilt: 3612 jobs pass without warnings/errors. All 776 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 99 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. No new deep analytic input or own-paper lemma is assumed.

`Cohen_Identity.md`

Quadratic Cohen identity: completed formalization

Source: `sources/current/Draft5.tex`, Lemma 3.1, label `l:cohenquadratic`, on page 17 of the supplied 99-page PDF. The complete identity and its uniform bound are proved in Lean. This is the first complete numbered paper lemma in the project; the other paper results remain open.

Statement and definitions

For a positive integer q, write A_q for the units of Z/qZ and define

$$c_q(m) = \sum_{a \in A_q} \exp(2\pi i a m/q), \qquad C_q(n) = \sum_{r \in A_q} c_q(n - r^2).$$

Let $a = v_2(q)$ and $d = q/2^a$. The checked global statement is

$$C_q(n) = |\mu(d)| T_a(n) \prod_{\substack{p \mid q \\ p > 2}} \begin{cases} p+1, & \left(\frac{n}{p}\right) = 1, \\ 1-p, & \left(\frac{n}{p}\right) \neq 1. \end{cases}$$

It holds for every positive q and every integer n, including negative n. The modulus-one product is empty and has value 1. The Legendre symbol has value zero when p divides n, and that case is included in the second branch.

The 2-adic factor is represented explicitly in the code:

Exponent a	Nonzero values of T_a(n)	All other residues
0	1 for every n	—
1	+1 at n=1 modulo 2; -1 at n=0 modulo 2	—
2	+4 at n=1 modulo 4; -4 at n=3 modulo 4	0
3	+16 at n=1 modulo 8; -16 at n=5 modulo 8	0
a>=4	None	0

This is the paper's T_a, with the signs spelled out. The equivalence is also checked: `twoAdicCohenFactor_one` proves $T_1(n) = (-1)^{n+1}$, using integer powers, and `twoAdicCohenFactor_small` proves $T_a(n) = \phi(2^a)c_{2^a}(n-1)$ for a=2,3. The explicit residue cases give the zero and signed-square assertions in the paper.

The formalization also proves

$$|C_q(n)| \le 16\,|\mu(q/2^a)| \prod_{p|q}(p+1).$$

The absolute constant 16 suffices for the paper's implied-constant bound. The sharper constant 16/3 is not asserted as a universal Lean theorem here.

1. Genuine exponential sums

`PrimeSquares/QuadraticCohen.lean` uses Mathlib's `ZMod.stdAddChar`, whose value on an integer residue is the displayed complex exponential. It defines c_q and C_q by finite sums over a complete residue system filtered by `IsUnit`. `sum_unit_filter_eq_sum_units` proves the equivalence with summing directly over the unit group. Modulus zero is excluded from these definitions by `NeZero q`.

The assembly file introduces `cohenCorrelation q n` as an integer-input interface. `cohenCorrelation_eq` proves that it is exactly C_q(n) whenever q>0. Its arbitrary zero extension at q=0 is used only to apply the standard multiplicative-function factorization theorem; q=0 is excluded from the main identity and bound.

2. Odd prime factor

Additive-character orthogonality gives

$$\sum_{a \bmod p} e(am/p) = \begin{cases} p, & m = 0 \pmod p, \\ 0, & m \neq 0 \pmod p. \end{cases}$$

For prime p, units are precisely nonzero residues, so subtracting the zero term proves $c_p(m) = p-1$ when m=0 and -1 otherwise.

Let R be the number of all roots of $r^2 = n$ modulo p. Summing over all r and subtracting the r=0 summand gives

$$C_p(n) = pR - p - c_p(n).$$

Mathlib's `legendreSym.card_sqrts` proves $R = (n/p) + 1$ for odd p. If p divides n, R=1 and the formula gives 1-p. Otherwise the Legendre symbol is +1 or -1, giving p+1 or 1-p. This proves `quadraticRamanujanSum_odd_prime` with no assumption that n is a unit.

3. Higher prime powers by a unit permutation

The code uses one algebraic argument for both higher-power vanishings, in place of the paper's separate Hensel-lifting count and 2-adic permutation.

Let R be a commutative ring, $t^2 = 0$, and r a unit. Then

$$F_t(r) = r + t/r$$

is a unit: its multiplicative inverse is $r^{-1} - tr^{-3}$. Moreover F_{-t} is the inverse permutation to F_t, and

$$F_t(r)^2 = r^2 + 2t.$$

All three assertions are proved in `UnitSquareShift.lean`, including bijectivity and the explicit inverse. Define the unit quadratic Gauss sum

$$S_q(b) = \sum_{r \in A_q} e(br^2/q).$$

For unit b, reindexing by F_t gives

$$S_q(b) = e(2bt/q) S_q(b).$$

When $2t \neq 0$ modulo q, primitivity of the standard character and invertibility of b make the multiplier different from 1. Thus S_q(b)=0. Expanding c_q and interchanging the finite sums then gives

$$C_q(n) = \sum_{b \in A_q} e(bn/q) S_q(-b) = 0.$$

For $q = p^k$, p odd and k>=2, take $t = p^{k-1}$. The divisibility $q \mid t^2$ follows from 2k-2>=k, while 2t is nonzero since p does not divide 2. For $q = 2^a$, a>=4, take $t = 2^{a-2}$. Again t squared is zero modulo q, and $2t = 2^{a-1} \neq 0$. The inequalities, divisibility, and nonvanishing are proved for all exponents, not checked up to a bound.

4. Small powers of two

For q=2,4,8 every unit square is 1. Kernel-reduced `decide` checks these finite unit assertions and their counts, giving

$$C_2(n) = c_2(n-1), \quad C_4(n) = 2c_4(n-1), \quad C_8(n) = 4c_8(n-1).$$

To evaluate c_4 and c_8, subtract the even residues from the complete additive-character sum. `stdAddChar_scale` identifies the even-residue sum with the complete character sum at half the modulus. Orthogonality gives

$$c_4(m) = \begin{cases} 2, & m = 0 \pmod 4, \\ -2, & m = 2 \pmod 4, \\ 0, & \text{otherwise}, \end{cases}$$

and c_8 has values +4 at 0, -4 at 4 modulo 8, and zero elsewhere. These are symbolic proofs of the complex-valued sums. Finite unit sets and projection kernels are the only computational residue checks. The modulus-one case uses the trivial ring's unit and character conventions and gives C_1=1.

5. CRT multiplicativity with the normalization proved

For coprime positive u,v, CRT identifies the units modulo uv with the product of the two unit groups. For the coefficients inside c_q use

$$(b, c) \longmapsto vb + uc \pmod{uv}.$$

Its CRT coordinates are (vb,uc). Multiplication by v modulo u and by u modulo v are unit permutations, so this is a bijection of the coefficient sets. The character-scaling identities

$$e(vj/(uv)) = e(j/u), \qquad e(uj/(uv)) = e(j/v)$$

are proved from the actual exponential formula. Reindexing the coefficient sum therefore proves c_(uv)(m)=c_u(m)c_v(m). Reindexing the square-shift variable by ordinary CRT then proves

$$C_{uv}(n) = C_u(n) C_v(n).$$

This avoids assuming an incorrect unscaled factorization of the standard character itself. Both multiplicativity theorems include u=1 or v=1.

6. Prime-factor and Mobius assembly

`cohenCorrelation_factorization` applies multiplicativity to the canonical prime-power factorization. At an odd prime, the local contribution is its Legendre factor if the exponent is one, and zero for every higher exponent. Consequently the product of exponent-one indicators is 1 exactly when the odd modulus is squarefree. `primeFactor_indicator_eq_abs_moebius` identifies this product with $|\mu(d)|$.

Mathlib's prime-power projection gives

$$q = 2^a d, \qquad d > 0, \qquad 2 \nmid d,$$

and the prime factors of d are exactly the prime factors of q greater than 2. Combining the odd product with the complete T_a factor proves the global identity. This includes q=1 and all nonsquarefree odd parts.

7. Uniform bound

The explicit local cases prove $|T_a(n)| \leq 16$ for every a,n. Each odd Legendre factor has absolute value at most p+1. The Mobius absolute value is nonnegative, so taking norms of the product gives the displayed bound with constant 16, after enlarging the odd-prime product to all prime factors. This is `quadratic_cohen_bound`.

Checked declarations and source correspondence

Paper component	Main Lean declaration
Definition of c_q and C_q	`ramanujanSum`, `quadraticRamanujanSum`
Positive-modulus integer interface	`cohenCorrelation_eq`
Odd-prime formula, e:Cp	`quadraticRamanujanSum_odd_prime`
Odd higher powers vanish	`quadraticRamanujanSum_odd_prime_pow`
Higher powers of two vanish	`quadraticRamanujanSum_two_pow_ge_four`
Complete factors at 4 and 8	`quadraticRamanujanSum_four_explicit`, `quadraticRamanujanSum_eight_explicit`
Parity and totient expressions, e:Ta	`twoAdicCohenFactor_one`, `twoAdicCohenFactor_small`
Multiplicativity, e:Cmultiplicative	`quadraticRamanujanSum_coprime_mul`
Global product, e:cohenquadratic	`quadratic_cohen_identity`
Uniform bound, e:CohenQuadratic	`quadratic_cohen_bound`

Verification and scope

The root project imports all nine modules. The complete build and transitive axiom audit cover all 57 named project theorem declarations, including the initial support lemmas. Only `propext`, `Classical.choice`, and `Quot.sound` occur. No deep analytic input, extra mathematical axiom, proof placeholder, or `native_decide` is used in this development.

The exact Python experiment over every n modulo every 1<=q<=512 passed 131,328 cases. It uses the classical Ramanujan divisor formula and is separate from the universal Lean proof. Its sharper finite bound check with constant 16/3 is not needed for the formal bound with constant 16.

Lemma 3.1 is complete. The next finite-arithmetic targets are the ordinary Cohen identity, `l:progressioncohen`, and `l:sumoverramanujan`. The rest of the paper, including its headline theorems, remains open.

`Conductor_Refinement.md`

Lemma 3.27: the complete conductor-sensitive refinement

Checkpoint 12, 22 October 2026. The full statement is proved in Lean by an alternative elementary argument, including all three estimates, the logarithmic consequence and the support condition. No extra mathematical hypothesis is used. The proof actually permits any real character at the original level; primitivity is unnecessary for this choice of majorant.

The main declaration is `PrimeSquares.conductor_sensitive_refinement` in `PrimeSquares/ConductorRefinement.lean`. This checkpoint is a proof of this lemma, not of the paper's headline theorems.

Source identification and exact scope

The archived `sources/current/Draft5.tex` and its reference map identify:

Source item	Stable label	Typeset number	PDF page
Previous uniform twisted Gauss bound	`l:Gcharbound`	3.24	22
Current lemma	`l:exceptionalarithmetic`	3.27	23
Three estimates	`e:exceptionalarithmetic`	3.28	23
Support condition	`e:sqfreeconds`	3.29	23

Earlier checkpoint 11 prose incorrectly used 3.22 for `l:Gcharbound` and 3.23 for `l:exceptionalarithmetic`. The current reports, inventory and provenance use the reference-map numbers. The Lean statements were already tied to the correct stable source labels. No manuscript correction is needed.

Fix a positive integer q_0 and a real Dirichlet character χ_0 modulo q_0. For each positive integer d, let χ be its actual induced character modulo $q_0 d$, using Mathlib's `DirichletCharacter.changeLevel`. Write

$$s_\chi(a,q)=\frac{1}{\phi(q)}\sum_{r\in(\mathbb{Z}/q\mathbb{Z})^\times}\chi(r)e(-ar^2/q),\qquad F_\chi(x,q)=\sum_{a\in(\mathbb{Z}/q\mathbb{Z})^\times}s_\chi(a,q)e(ax/q).$$

These are the actual definitions `twistedSquareSum` and `twistedSquareCorrelation`, with the paper's minus sign and totient. All residue classes x are included; the second estimate requires a unit coefficient a, exactly as in the paper. Integer inputs are their residue classes. Modulus one and imprimitive characters are included.

`conductorSigma` is the maximum of the norms over $j=0,\dots,5$ and the units modulo $q_0 2^j$. Since multiplication is commutative, these are exactly the paper's moduli $2^j q_0$. `dyadicTwistNorm` forms the inner finite maximum; `conductorSigma` forms the outer one. Finite suprema are taken in the nonnegative reals and then coerced to the reals. The checked lower-bound and upper-bound lemmas ensure this is the actual maximum, rather than an assumed upper bound. `conductorRho` and `conductorCorrelationProduct` are exactly

$$\rho(d)=\prod_{p\mid d}(1-p^{-1/2})^{-1},\qquad G(d)=\prod_{p\mid d}\frac{p+1}{p-1}.$$

The explicit admissible choice in this proof is

$$\mathcal{B}_0=64\,8^{\omega(q_0)}\sqrt{q_0}.$$

The combined Lean theorem proves, with an absolute positive constant C, uniform in q_0, χ_0, d, x and a,

$$\mathcal{B}_0\ge 1,\qquad |F_{\chi_{q_0 d}}(x,q_0 d)|\le \mathcal{B}_0 G(d),$$
$$|s_{\chi_{q_0 d}}(a,q_0 d)|\le 32\sigma_0 d^{-1/2}\rho(d),\qquad \sigma_0\sqrt{\mathcal{B}_0}\le C q_0^{-1/25},$$
$$(1+\log q_0)\sigma_0\sqrt{\mathcal{B}_0}\le C.$$

If the sum is nonzero for any unit coefficient, it also proves

$$d=2^\nu d_1,\quad \nu\le 5,\quad d_1>0,\quad d_1 \text{ squarefree},\quad (d_1,2q_0)=1.$$

The natural-number convention makes $\nu\ge 0$ automatic. For positive d_1, squarefreeness is the paper's condition $\mu^2(d_1)=1$.

The paper's proof proposes the sharper choice $\mathcal{B}_0\asymp\prod_{p\mid q_0,\,p>2}2p/(p-1)$. That additional comparison is not part of the lemma's statement and is not claimed here. The larger explicit choice above satisfies every stated conclusion and avoids real primitive conductor classification. The archived manuscript has not been edited.

1. Real characters and multiplication by squares

For a character on a finite unit group, complex conjugation equals inversion. Consequently χ has real values if and only if $\chi^2=1$, equivalently χ takes only the values 0, 1 and -1. Lean proves this equivalence in `dirichletCharacter_quadratic_iff_real`. It also proves that `changeLevel` preserves the quadratic predicate.

At every unit v, a quadratic character satisfies $\chi(v^2)=1$. Thus multiplication of a unit r by a unit square leaves $\chi(r)$ unchanged. This fact supplies the character invariance needed for the cancellation argument; it is not an unstated assumption about induced characters.

2. A square-preserving shift forces exact cancellation

Let t,s be residues modulo q with

$$t^2 = s^2 = 0, \qquad 2s = t, \qquad 2t \neq 0.$$

The existing `squareShiftEquiv` is the unit permutation r↦r+t/r. Its inverse and the identity

$$(r + t/r)^2 = r^2 + 2t$$

were already checked in `UnitSquareShift.lean`. The new proof shows that

$$r + t/r = r(1 + s/r^2)^2.$$

Here $1+s/r^2$ is a unit, with inverse $1-s/r^2$. Thus the permutation also preserves χ(r) for every quadratic χ. Reindexing the genuine twisted Gauss sum U gives

$$U_\chi(a, q) = e(2at/q)U_\chi(a, q).$$

For a unit a, the additive character is injective and 2at≠0. The phase is therefore different from one, forcing Uχ(a,q)=0. This is an exact identity, with no asymptotic input. The normalized sum vanishes as well.

There is also a separate checked parity cancellation: if χ(-1)≠1, then reindexing by r↦-r gives a zero twisted sum for every coefficient, including nonunits. The main support proof does not need to classify which local characters are odd.

3. Vanishing at the full modulus

The shift can be applied directly at any modulus q, without first decomposing χ into local characters.

If an odd prime square p^2 divides q, write $q=p^2m$ and take

$$t = pm, \qquad s = \frac{p+1}{2}t \pmod q.$$

Then pt=0 modulo q, $t^2=s^2=0$, and 2s=t. Moreover $0<2pm<p^2m$, so 2t≠0.

If 64 divides q, write q=64m and take t=16m, s=8m. Again $t^2=s^2=0$, 2s=t, and 2t=32m is nonzero modulo q. Lean checks all divisibility, positivity, unit and phase conditions, including even moduli in the odd-prime argument.

It follows that a nonzero unit-coefficient sum has

$$v_2(q) \leq 5, \qquad v_p(q) = 1 \text{ for every odd } p \mid q.$$

For $q=q_0d$, remove the power of two from d to obtain d_1. Its odd prime exponents are at most one. An odd prime dividing both d_1 and q_0 would have its square dividing q, a contradiction. This proves the entire support condition. It holds even for a quadratic character at level q_0d that was not specified as induced.

4. Exact CRT for the induced sums

Let (m,n)=1 and let χ modulo mn be induced from a character χm modulo m. The unit CRT equivalence sends r to (rm,rn), and the induced character has value χm(rm). This follows from the actual definition of `changeLevel`.

For a residue a modulo mn, put

$$a_m = (a \bmod m)n^{-1} \pmod m, \qquad a_n = (a \bmod n)m^{-1} \pmod n.$$

Both are units when a is a unit. The standard additive character satisfies the checked CRT identity with these inverse-modulus factors. Finite reindexing and φ(mn)=φ(m)φ(n) then give

$$s_\chi(a, mn) = s_{\chi_m}(a_m, m)s_1(a_n, n).$$

The inverse coefficients are retained explicitly in `induced_twistedSquareSum_coprime_mul`; unscaled multiplicativity at a fixed coefficient is not assumed.

Opening the weighted Fourier sum in the other order gives

$$F_\chi(x,q)=\frac{1}{\phi(q)}\sum_{r\in(\mathbb{Z}/q\mathbb{Z})^\times}\chi(r)c_q(x-r^2).$$

Ramanujan-sum CRT and character induction now give

$$F_\chi(x,mn)=F_{\chi_m}(x \bmod m,m)F_1(x \bmod n,n).$$

There are no inverse factors in x in this latter identity. The Fourier identity and correlation factorization are proved separately from the quadratic Gauss factorization.

5. The correlation estimate

The earlier unconditional Lemma 3.24 gives the finite bound

$$|s_\chi(a,m)|\le 8^{\omega(m)}/\sqrt{m}.$$

Summing norms over the φ(m) unit coefficients and using φ(m)≤m yields

$$|F_\chi(x,m)|\le 8^{\omega(m)}\sqrt{m}.$$

If m=q₀2^ν with ν≤5, then ω(m)≤ω(q₀)+1 and √m≤8√q₀. Hence this is at most the chosen $\mathcal{B}_0$.

If all unit-coefficient sums at q vanish, the correlation is zero. Otherwise the support theorem supplies d=2^νd₁. Set m=q₀2^ν, n=d₁; they are coprime. For the principal correlation at the odd squarefree n, the earlier exact Cohen factors and φ(n)=∏p|n(p-1) give

$$|F_1(x,n)|\le G(n).$$

Every factor in G is at least one. Since n divides d, G(n)≤G(d). The checked CRT identity proves the first estimate with constant one.

6. The individual-sum estimate

Use the same support decomposition. The norm of the m-part of the CRT product is bounded by σ₀ by its exact definition. The previous unconditional principal product bound, with constant four, gives

$$|s_1(a_n,n)|\le\frac{4}{\sqrt{n}}\prod_{\substack{p\mid n\\ p>2}}(1-p^{-1/2})^{-1}.$$

Each factor of ρ is positive and at least one, so this product is at most ρ(n)≤ρ(d). Also √d≤8√n. Consequently

$$|s_\chi(a,q)|\le\sigma_0\frac{4}{\sqrt{n}}\rho(n)\le 32\sigma_0 d^{-1/2}\rho(d).$$

Zero sums are handled explicitly. The proof never divides by σ₀ and thus includes the σ₀=0 case.

7. Decay and the logarithmic consequence

Apply the earlier uniform bound with ε=1/8. Since q₀2^j≥q₀ and the exponent -3/8 is negative, the finite maximum defining σ₀ satisfies

$$\sigma_0\le Sq_0^{-3/8}$$

with an absolute S>0, for every complex χ₀. Put D=8^ω(q₀)≥1. The majorant satisfies

$$\sqrt{\mathcal{B}_0}=8\sqrt{D}\,q_0^{1/4}\le 8Dq_0^{1/4}.$$

The already proved elementary prime-factor absorption, used with ε=1/16, gives D≤Aq₀^(1/16), with an absolute A>0. Multiplication gives

$$\sigma_0\sqrt{\mathcal{B}_0} \le 8SAq_0^{-1/16} \le 8SAq_0^{-1/25}.$$

The comparison uses $q_0 \ge 1$. No restriction $p \equiv 1 \pmod 4$, prime-distribution estimate, Weil bound or real-character classification is required for this alternative choice of majorant.

Finally, the checked elementary logarithm bound gives

$$\log q_0 \le 25q_0^{1/25}, \qquad 1 + \log q_0 \le 26q_0^{1/25}.$$

Multiplying the decay estimate proves the logarithmic conclusion. The combined theorem takes one positive constant large enough for both conclusions, before quantifying q_0 and χ_0. Uniformity is therefore part of the formal statement.

Files and proof boundary

Module	Checked role
`QuadraticTwistCancellation.lean`	Real/quadratic equivalence, induction, square shift, parity and full-modulus vanishing
`TwistedSupport.lean`	Prime exponents and the exact relative support decomposition
`InducedTwistCRT.lean`	Actual `changeLevel`, numerator CRT and normalized CRT
`TwistedCorrelation.lean`	Actual Fourier sum, Ramanujan identity, induced CRT and finite bound
`ConductorFactors.lean`	Exact σ_0 maximum, ρ, G, B_0, positivity and divisor monotonicity
`ConductorEstimates.lean`	The first two estimates with constants 1 and 32
`ConductorDecay.lean`	Uniform $q_0^{-1/25}$ decay and logarithmic conclusion
`ConductorRefinement.lean`	Full combined source statement, strengthened to all real characters

All are under `PrimeSquares/`. No proof placeholders or new mathematical axioms are used. The ordinary Lean foundations `propext`, `Classical.choice` and `Quot.sound` remain the audit whitelist. Neither Siegel--Walfisz nor PNT is an input to this lemma. The separate Lemma 3.21 still has its two explicit PNT-limit hypotheses; their external bridge remains unintegrated.

Verification and independent experiments

The fresh project build checks all 40 project modules. The transitive axiom audit covers all 321 named project theorems, including the 49 added here. The source scan covers 42 Lean files. These are infrastructure counts, not counts of numbered paper results. Full logs are in `verification/`.

`scripts/check_conductor_refinement.py` uses exact signed integer polynomials, cyclotomic reduction, integer Ramanujan sums and rational comparisons. The recorded experiment has zero failures:

Check	Range	Cases
All real characters and all unit coefficients	$q=1,\dots,128$	452 characters; 16,754 coefficients
Higher-power vanishing	Same range	2,968
Odd-character vanishing, including nonunit coefficients	Same range	13,937
Explicit shift, square ratio, character invariance and unit permutation	Same range	2,968 unit cases
Relative conductor support	Same range	723
Fourier identity with source sign	$q \le 32$	1,561
Induced Gauss CRT with inverse coefficients	$mn \le 96$	20,331
Induced correlation CRT	$mn \le 96$	41,144
First estimate with the explicit B_0, checked by squaring	$q \le 128$	111,520

The induced-level checks include 1,048 pairs of a real base character and a strictly larger level. The σ_0 estimate and its unbounded uniform decay are established by Lean, not by a finite numerical claim. These experiments are independent finite error checks and do not replace the formal proof.

The next dependency stage is the abstract high--low argument in Theorem 4.1 (`t:improvinggeneral`), followed by its application to the prime-square operators. That work is not part of this checkpoint.

`Current_Roadmap.md`

Dependency-driven roadmap for Draft5

Source: the archived 99-page paper in `sources/current/`, with the accepted Lemma 3.14 fragment in `sources/updates/` and status in `docs/SOURCE_UPDATE_08.md`. Progress is organized by mathematical dependencies; the statement inventory preserves the section and label correspondence. This avoids redoing definitions when a proof in a later section supplies an input used earlier.

Stage	Main work	Paper location	Checkpoint status
0	Lock sources, reproduce references, state the trust boundary	Whole paper	Current files locked; 58 statements inventoried
1A	Quadratic Cohen identity: prime and prime-power local factors	Section 3, `l:cohenquadratic`	Complete Lemma 3.1; see Cohen_Identity.md
1B	CRT and the global product; progression-restricted correlations	Remaining first part of Section 3	Complete 3.11, 3.12, accepted 3.14/3.16 and original 3.20
1C	Gauss-sum bounds and exceptional-character arithmetic	Remaining Section 3	Full 3.21 with explicit PNT limits; full 3.24 for all complex characters and full 3.27 for all real characters, both unconditional
2	Abstract high--low optimization and improving inequalities	Section 4, Theorem 4.1 and Corollary 4.15	Complete: both abstract endpoints, full strong interior interpolation and p=2, actual complex lp operators, exact N scaling and uniform constants
3	Establish the prime-square high--low inputs and connect the first main theorem	Section 6, Proposition 4.19 and Theorem 1.1	The actual current user upload is matched to the completed formalization in the agreed conditional scope. No internal numbered source claim remains open.
4	Stopping intervals, local estimates, sparse recursion	Section 5, `t:sparsebound`	Both source Theorem1.3 clauses complete conditional at checkpoint828; actual sparse major subsets and all-cutoff maximum are proved. General auxiliary source lemmas remain partial.
5	Quantitative maximal and Orlicz estimates	Section 7, `t:orlicz-maximal`	Partial at checkpoint846: gauge, scalar growth/moment, finite layers and general modular deductions proved; actual IW low/high, restricted bound, strong lp and ergodic conclusions unfinished.
6	Local density correlations and the badness criterion	Section 8, first part of `t:pet`	Open
7	Sieve reductions, local densities, circle method, final error assembly	Sections 9--11, `t:repmotthm`	Open; earlier finite support lemmas available

Within each stage:

1. Expand the selected statement into definitions and a dependency graph, with all parameter restrictions and exceptional cases visible.
2. Find the relevant Mathlib declarations in the pinned source before writing replacement infrastructure.
3. Prove a coherent mathematical unit; include the q=1, small-prime, zero, support, and endpoint cases relevant to that unit.
4. Build all project imports and audit every named theorem's transitive axioms.
5. Use exact finite experiments when they can expose an error. Record their range and arithmetic; never count them as an unbounded proof.
6. Update the status, proof specification, and continuation note; save the checkpoint before proceeding to the next substantial unit.

The current branch does not contain unfinished Lean proof stubs. Open targets live in the blueprint. No main theorem is counted as proved until its precise statement and every input have been connected,

with any deep external hypotheses displayed in the final theorem.

Euler_Product.md

Corrected progression Euler product: checked proof specification

Checkpoint 07 update: the user has supplied the corrected Lemma 3.14. It matches this checked statement; the earlier statement issues are resolved for that revision. The full archived source remains unchanged. See `docs/SOURCE_UPDATE_07.md`. The proof below is the completed checkpoint 06 proof.

Checkpoint 06, 22 October 2026. Source: the locked `sources/current/Draft5.tex`, Lemma 3.14, label `l:motamakemajorprimesquareprog`, equation `e:corupss`. Its SHA256 remains `d2c31b96cb15b00feac09ed4fd9a2b0482726c719c4b772320c0711875d04660`.

The complete Euler-product identity is proved with **Q>2 and positive k**. The last vanishing assertion is proved with the necessary restriction **p<Q** on the obstructing prime. The original statement needs these qualifications; it is not counted as proved verbatim. Both missing qualifications have kernel-checked counterexamples. See `docs/SOURCE_FINDINGS_06.md` and the replacement statement in `corrections/Lemma_3_14_Replacement.tex`.

Statement being proved

For q,k positive and b,x arbitrary integers, use exactly the previously formalized exponential sums from Lemma 3.12 and put

$$B(q;k,b,x)=\sum_{a\in A_q}\Upsilon(a,q,b,k)\mathfrak{s}(a,q)e(ax/q).$$

For Q>2, k>=1 and gcd(b,k)=1, the smooth-modulus sum equals

$$\sum_{P^+(q)<Q}B(q;k,b,x)=\mathcal{T}_2\prod_{2<p<Q}E_p(k,b,x),$$

where h=v_2(k),

$$\mathcal{T}_2=\begin{cases}1+(-1)^x, & h=0,\\ 2^{\min(h,3)}\mathbf{1}_{x-b\equiv 1 \pmod{2^{\min(h,3)}}}, & h>0,\end{cases}$$

and the four cases of the odd-prime factor are

$$E_p=\begin{cases}1+(p+1)/(p-1), & p\mid k,\ (\frac{x-b}{p})=1,\\ 0, & p\mid k,\ (\frac{x-b}{p})\neq 1,\\ 1-(p+1)/(p-1)^2, & p\nmid k,\ (\frac{x}{p})=1,\\ 1+1/(p-1), & p\nmid k,\ (\frac{x}{p})\neq 1.\end{cases}$$

A single product of this case-defined factor is the same mathematical formula as the four products in the source: the four conditions partition the odd primes below Q. No condition has been omitted at Legendre-symbol value zero. The formal local definition uses a harmless unused branch for nonprime p.

The vanishing consequence is: if an odd prime **p<Q** divides k and ((x-b)/p) is not 1, the smooth-modulus sum is zero.

Source component	Checked declaration
Actual coefficient B, not an assumed local formula	`progressionCoefficient`, `progressionCoefficient_eq`
Its exact normalization and C_q representation	`progressionCoefficient_as_quadratic`
Multiplicativity in q	`progressionCoefficient_mul`
Uniform support at all moduli	`progressionCoefficient_odd_prime_sq`, `progressionCoefficient_sixteen_dvd`
Genuine reduction of the series to a finite sum	`smoothProgressionSeries_eq_divisor_sum`
Euler product of actual local coefficients	`smoothProgressionSeries_euler_product`
Four explicit odd-prime factors	`progressionCoefficient_odd_euler_factor`
Complete T_2 factor	`progressionCoefficient_two_euler_factor`
Corrected identity, integer cutoff Q>=3	`progression_euler_product`
Corrected identity, every real cutoff Q>2	`progression_euler_product_real`

Source component	Checked declaration
Corrected obstruction consequence	`progression_euler_product_eq_zero_of_obstruction`
Counterexamples to the omitted restrictions	`cutoff_two_counterexample`, `obstructing_prime_above_cutoff_counterexample`

All names are in namespace `PrimeSquares`. The files added are `EulerCoefficient.lean`, `SmoothEuler.lean` and `LocalEuler.lean`.

1. Open the sums in the useful order

Write g=gcd(q,k), and retain the genuine quadratic Cohen correlation C_q(n)=sum_(u in A_q) c_q(n-u^2), completed in Lemma 3.1. Opening the two exponential sums, then summing over a and next over u, gives the exact identity

$$B(q;k,b,x)=\frac{\phi(k)}{\phi(q)\phi([k,q])}\sum_{\substack{r\in A_q\\ r\equiv b \pmod g}} C_q(x-r).$$

The existing normalized expansion from Lemma 3.12 proves this finite reordering. The new proof explicitly checks x-u^2-r=x-r-u^2. For every multiplicative arithmetic function f, f(lcm(q,k)) f(gcd(q,k))=f(q)f(k). Applying Mathlib's proved identity to f=phi gives phi(lcm(q,k)) phi(g)=phi(q) phi(k), hence

$$B(q;k,b,x)=\frac{\phi(g)}{\phi(q)^2}\sum_{\substack{r\in A_q\\ r\equiv b \pmod g}} C_q(x-r). \qquad (1)$$

All cancelled totients are first proved nonzero. The coefficient is extended by zero at q=0 solely to use natural-number arithmetic-function interfaces; agreement with the actual exponential sum at every positive q is proved.

Formula (1) needs neither coprimality of b and k nor a distribution theorem. This stronger formulation simplifies the support and multiplicativity proofs.

2. Multiplicativity includes the restriction and weight

Let gcd(m,n)=1. Then

$$\gcd(mn,k)=\gcd(m,k)\gcd(n,k),\qquad \gcd(\gcd(m,k),\gcd(n,k))=1.$$

The ring CRT and its checked unit equivalence identify A_(mn) with A_m×A_n. The progression restriction modulo gcd(mn,k) is exactly the conjunction of the two coordinate restrictions. This equivalence is proved through the canonical reduction homomorphisms, rather than postulated.

The already checked identity C_(mn)(z)=C_m(z)C_n(z), with explicit reductions, separates the summand. The two finite unit sums consequently factor. Totient multiplicativity separates phi(gcd(mn,k))/phi(mn)^2 into the corresponding weights at m and n. Thus

$$B(mn;k,b,x)=B(m;k,b,x)B(n;k,b,x).$$

At modulus one the unit group has one element, its character is one and B(1;k,b,x)=1. The proofs include this case and the zero extension.

3. Uniform vanishing gives finite support

For an odd prime p, if p^2 divides q then C_q(n)=0 for every n. If 16 divides q then C_q(n)=0 for every n. These follow from the completed global prime-power factorization of C: one local factor in its product vanishes. The divisibility-to-valuation bridge is formally checked.

Formula (1) therefore proves, for arbitrary b and x,

$$p^2\mid q,\ p>2 \implies B(q;k,b,x)=0,\qquad 16\mid q \implies B(q;k,b,x)=0.$$

This is a statement about all moduli, not just isolated prime powers. Only squarefree odd parts and a power of two of exponent at most three can contribute.

For an integer cutoff Q>=3, choose the convenient bound

$$N_Q = ((Q-1)!)^3.$$

It is not intended as an optimal bound. If every prime divisor of q is below Q and B(q;k,b,x) is nonzero, then q divides N_Q. Indeed, every prime of q divides (Q-1)!, so its exponent in N_Q is at least three, while its exponent in a nonzero coefficient is at most three. Comparing all prime valuations proves the divisibility.

Conversely, every divisor of N_Q has all prime factors below Q. Consequently

$$\sum_{P^+(q)<Q} B(q;k,b,x) = \sum_{q \mid N_Q} B(q;k,b,x). \qquad (2)$$

The formal proof uses `tsum_eq_sum`, with a proof that every summand outside the divisor set is zero. Thus the complex-valued series has finite support; no conditional rearrangement, unproved convergence or analytic input occurs. Smoothness is modeled by the equivalent prime-divisor condition. For q=1, this agrees with the source convention P^+(1)=1 because Q>2. The extension at q=0 contributes zero.

4. The finite divisor Euler product

For a multiplicative function f with f(0)=0 and f(1)=1, the divisor sum is the Dirichlet convolution (f*zeta)(N). Mathlib proves multiplicativity of this convolution. Applying the natural-number prime-power factorization to it, then identifying the divisors of p^a with p^j, 0<=j<=a, yields

$$\sum_{d \mid N} f(d) = \prod_{p \mid N} \left(\sum_{j=0}^{v_p(N)} f(p^j) \right).$$

This argument is formalized in `sum_divisors_multiplicative` and is applied to the actual B. The primes of N_Q are exactly the primes below Q, each with valuation at least three. The previously proved vanishings truncate the odd-prime exponent sums after one, and the exponent sum at two after three. Equation (2) becomes

$$(1 + B(2) + B(4) + B(8)) \prod_{2<p<Q} (1 + B(p)). \qquad (3)$$

This proves the Euler-product step before any explicit local evaluation.

5. Local evaluations

If q divides k and b is a unit modulo q, the restriction in (1) fixes r=b. Thus B(q)=C_q(x-b)/phi(q). If gcd(q,k)=1 the restriction disappears; opening C and applying the ordinary linear Cohen identity gives

$$\sum_{r \in A_q} C_q(x-r) = \mu(q) C_q(x), \qquad B(q) = \mu(q) C_q(x)/\phi(q)^2.$$

For p^a with a>=2 which does not divide k, write gcd(p^a,k)=p^j. If j=0, mu(p^a)=0. If 0<j<a, the gcd and complementary modulus share p, so the completed progression Cohen identity kills the coefficient. This also proves the partial-overlap cases for powers of two. Coprimality of b and k supplies every unit-class hypothesis via divisibility.

For odd primes, insert the completed evaluation C_p(n)=p+1 for Legendre symbol 1, and 1-p otherwise, and use mu(p)=-1 and phi(p)=p-1. Adding 1 to the coefficient gives precisely the four factors E_p displayed above. Nonzero denominators are proved from primality.

6. The 2-adic sum

If v_2(k)=0, only B(2) can contribute and B(2)=(-1)^x. Otherwise the allowed terms are C_(2^a)(x-b)/phi(2^a), for 1<=a<=min(v_2(k),3). The complete signed factors from Lemma 3.1 give

$$\begin{aligned} 1 + C_2(n) &= 2\mathbf{1}_{n \equiv 1 \pmod 2}, \\ 1 + C_2(n) + C_4(n)/2 &= 4\mathbf{1}_{n \equiv 1 \pmod 4}, \\ 1 + C_2(n) + C_4(n)/2 + C_8(n)/4 &= 8\mathbf{1}_{n \equiv 1 \pmod 8}. \end{aligned}$$

The small congruence predicates are checked by Lean's kernel using `decide`; the complex arithmetic and the relation of the residues at 2,4,8 are proved explicitly. This is not `native_decide` or an external numeric oracle. Splitting the valuation of k into 0,1,2,>=3 proves exactly T_2 in the source, including arbitrary negative integer x and b.

7. Real cutoffs and the corrections

For every real Q>2 and integer prime p, p<ceil(Q) if and only if p<Q. Moreover ceil(Q)>=3. Mathlib's proved `Nat.lt_ceil` transports the integer cutoff theorem to every real cutoff Q>2 without changing the included primes.

At Q=2 the literal smooth series is 1. With k=1, b=x=0 the displayed 2-adic factor is 2, so the unqualified identity fails. For the vanishing sentence, Q=3, k=5, b=1, x=4 gives a series equal to 2, while x-b=3 is a nonsquare modulo five. Both assertions are proved in Lean. The corrected obstruction theorem explicitly requires p<Q; the prime then supplies an actual zero factor in (3).

The replacement LaTeX fragment changes only the statement's cutoff, modulus range and final obstruction restriction. It preserves the displayed identity, T_2 formula and labels. The locked authoritative source is unchanged.

Verification and remaining work

All fifteen project modules build. The source scan covers seventeen Lean files and rejects proof placeholders, extra axioms, native_decide and unsafe code. All 120 named project theorems pass the transitive axiom audit, allowing only `propext`, `Classical.choice` and `Quot.sound`.

The independent exact-integer/rational experiment passes 94,208 fixed-modulus cases with 1<=q<=128, 1<=k<=12, all unit b modulo k and 0<=x<=15. It also passes 3,680 finite Euler-product comparisons for integer cutoffs 3 through 7. These experiments use the classical Ramanujan divisor formula; they do not supply the universal support proof or the formal counterexamples.

Three earlier numbered paper lemmas are complete as stated within positive modulus conventions; this checkpoint adds the corrected version of Lemma 3.14. No headline theorem or general-coefficient Corollary 3.16 is complete. No deep analytic theorem has been assumed. The next target is Corollary 3.16; the new multiplicativity, support and finite Euler-product infrastructure should be reused, with the progression coefficient multiplied by its general integer ell.

`Exceptional_Density_Partial_Summation.md`

The source densities, oscillatory integrals and conditional major-arc deduction

Checkpoint 26. This unit specializes the proved Abel identity and quantitative error transport to the manuscript's actual power model. All derivatives, continuity and integrability requirements are discharged. A bound on the genuine unweighted progression prefix remains an explicit hypothesis.

Exact source objects

For an integer N>=1, a real frequency theta and beta>0, write

$$g_\theta(t) = \exp(-2\pi i\theta t^2), \qquad C_{q,r}(x) = \sum_{0\le n\le \lfloor x\rfloor} \mathbf{1}_{n\equiv r \pmod q}\Lambda(n).$$

The Lean prefix includes n=0, whose von Mangoldt coefficient is zero. It includes every prime power. The lower endpoint in the integral is the real number N/2. The corresponding finite sum uses the natural floor N/2, with that correspondence explicitly proved in the Abel calculation. The statements therefore also cover odd N; dyadic N is not required here.

Define the complex progression model and density by

$$H_{q,z,\beta}(t) = \frac{1}{\phi(q)}\left(t - z\frac{t^\beta}{\beta}\right), \qquad D_{q,z,\beta}(t) = \frac{1}{\phi(q)}(1 - zt^{\beta-1}).$$

The new Lean definitions are `progressionPowerModel` and `progressionPowerDensity`. The primitive is `exceptionalPrimitive`. All real quantities are explicitly embedded into the complex numbers where needed. These are actual functions, rather than unspecified model parameters supplied with regularity assumptions.

The normalized oscillatory integral is

$$M_{N,\beta}(\theta) = \frac{1}{N}\int_{N/2}^{N} t^{\beta-1} g_\theta(t)\,dt.$$

`annularOscillatoryModel` initially uses the Bochner integral over (N/2,N]. `annularOscillatoryModel_interval` identifies it with the interval integral in the manuscript's equation e:M. That identity also holds at N=0 with Lean's inverse convention, although subsequent estimates require N>=1. The integral parameter theta is real: the phase is not assumed periodic in theta when t is nonintegral.

Two calibration results are proved. At beta=1 the real power disappears, and at beta=1, theta=0 the model has value exactly 1/2 for N>=1. This checks the annular interval length and the factor 1/N together.

Derivative and integrability

For t>0 and beta>0, Mathlib's derivative of the real power gives

$$\frac{d}{dt}\frac{t^\beta}{\beta} = \frac{\beta t^{\beta-1}}{\beta} = t^{\beta-1}.$$

Positivity of beta justifies cancellation; positivity of t avoids the singularity at zero. Applying the complex embedding, subtraction and constant-multiple derivative rules proves H'=D. Every point in [N/2,N] is positive when N>=1, so this proves all the regularity needed by the previously established integration-by-parts theorem.

Continuity of the real power on this positive interval gives continuity and interval integrability of D. Continuity of both the power and the quadratic phase proves integrability of their product. Integrability is proved before distributing the Bochner integral over subtraction. The calculus only requires beta>0; the manuscript's additional beta<=1 can be imposed when applying it to an exceptional zero.

Distributing the actual density integral then proves

$$\frac{1}{N}\int_{N/2}^{N} g_\theta(t) D_{q,z,\beta}(t)\,dt = \frac{1}{\phi(q)}\left(M_{N,1}(\theta) - zM_{N,\beta}(\theta)\right).$$

This is `progressionPowerDensity_integral`.

Exact residue error and its quantitative bound

Let

$$W_{q,r} = \sum_{\substack{N/2<n\le N\\ n\equiv r \pmod q}} \Lambda(n) g_\theta(n), \qquad E(t) = C_{q,r}(t) - H_{q,z,\beta}(t).$$

The exact identity `annularProgressionPowerModel_error_identity` is

$$\frac{W_{q,r}}{N} - \frac{M_{N,1}(\theta) - zM_{N,\beta}(\theta)}{\phi(q)} = \frac{1}{N}\left(g_\theta(N)E(N) - g_\theta(N/2)E(N/2) - \int_{N/2}^{N} g_\theta'(t)E(t)\,dt\right).$$

There is no assumed residue estimate in this identity. Both endpoint terms and the exact derivative integral are present. All model regularity conditions have been proved for the displayed H and D.

Suppose delta>=0 and, throughout the actual real interval [N/2,N],

$$|C_{q,r}(t) - H_{q,z,\beta}(t)| \le N\delta.$$

The previously proved derivative bound and exact normalization imply

$$\left|\frac{W_{q,r}}{N} - \frac{M_{N,1}(\theta) - zM_{N,\beta}(\theta)}{\phi(q)}\right| \le \delta(2 + 4\pi|\theta|N^2).$$

This is `annularProgressionPowerModel_error_bound`. The term 2 records both endpoints. The N in the raw prefix error cancels against the normalizing 1/N. No derivative or integrability hypothesis remains for the model in this bound.

Gauss sums and the actual annular symbol

Let q>=1, let chi be a complex Dirichlet character modulo q, and let kappa be a complex coefficient. Assume the displayed raw prefix bound holds for every reduced residue r with z=kappa*chi(r), using the same beta, chi, kappa and delta throughout this one interval. Define

$$U(N,q)=\frac{\log q+2\sqrt{N}\log N}{N}.$$

The earlier exact finite residue grouping, proved nonunit contribution bound and Gauss-sum identity now give

$$\begin{aligned}\left|\widehat{\widetilde{A}}_{N,\Lambda}(a/q+\theta)-\left(\mathfrak{s}(a,q)M_{N,1}(\theta)-\mathfrak{s}_{\chi}(a,q)\kappa M_{N,\beta}(\theta)\right)\right|\\ \le U(N,q)+\phi(q)\delta(2+4\pi|\theta|N^2).\end{aligned}$$

This is `annularMangoldt_majorArc_from_prefix_bound`. Its symbol is the actual finite von Mangoldt annular Fourier sum, not an abstract surrogate. The Gauss sums use the manuscript's negative phase and division by phi(q). The proof sums the per-residue errors over exactly phi(q) units. The nonunit error U(N,q) is already proved for the actual coefficients.

Taking kappa=0 omits the exceptional term; taking kappa=1 includes it. `annularMangoldt_ordinary_from_prefix_bound` separately gives the ordinary case directly from a prefix bound around t/phi(q), with no exceptional parameter in the statement. The deduction does not require primitivity of chi or coprimality of a and q; a published analytic input may impose those additional hypotheses. Exceptional data may change with N. The theorem makes no claim that the same data work at different scales.

Source correspondence and remaining analytic work

This verifies the power-model differentiation, actual partial summation and Gauss-sum substitution in the proof of Lemma 6.5, after equation e:M. It does not prove or silently assume the manuscript's raw progression distribution estimate. To use a published theorem as an explicit deep input, its primary-source statement, range, exceptional data and uniformity must first be matched to the displayed prefix hypothesis.

The final conversion to O(R^-5), with the stated N,Q,R range and choice of constants, remains open. The independent prime-square minor-arc estimate also remains open. Thus full Lemma 6.5, Proposition 4.19 and the headline applications are not marked complete. No new PNT bridge, Siegel-Walfisz theorem or Landau-Page theorem is asserted here.

Files

Module	New named theorems	Result
ExceptionalProgressionDensity.lean	12	Actual model derivative, continuity and integrability, source interval identity, normalization and density integral
PrimeSquarePartialSummation.lean	4	Exact source-model error, quantitative residue bound and conditional ordinary/exceptional major-arc deductions

No numerical experiment is used to justify these analytic statements. They are checked Lean proofs with the raw prefix-error bound displayed as a hypothesis where it is needed.

<!-- checkpoint-verification -->

Verification

All 90 project modules and the root were freshly rebuilt: 3605 jobs pass without warnings/errors. All 739 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 92 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. No new deep analytic input or own-paper lemma is assumed.

`Finite_Major_Arc_Model_Split.md`

Finite major-arc symbols and the exact high-low split

For a finite rational center set Gamma, the module uses the actual reduced numerator and denominator of each center r. It defines

$$P_r(\xi) = \mathfrak{s}(r_{\mathrm{num}}, r_{\mathrm{den}}) M_N^1(\xi - r), \qquad X_r(\xi) = -\mathfrak{s}_{\chi_r}(r_{\mathrm{num}}, r_{\mathrm{den}}) \kappa_r M_N^\beta(\xi - r).$$

One beta and one family of character/coefficient data are shared over Gamma. The broad model term is $(P_r + X_r)\eta_{\log_2(r_{\mathrm{den}})}(\xi - r)$, and V is their finite sum. Its equality to the previously defined localized genuine Gauss model is proved.

The ordinary low term uses the cutoff $\eta((qN^2/Q)(\xi - r))$ when q<=Q. Its high complement uses broad-minus-narrow in that case and the full broad term otherwise. The exceptional terms use $\eta((B_0(q/q_0)N^2/Q)(\xi - r))$, with selection $q/q_0 \le Q/B_0$. For the source application, the coefficient kappa vanishes when q0 does not divide q, and q/q0 is the source d. The algebra is valid for the broader parameter family without asserting the existence of source exceptional data.

Splitting the ordinary and exceptional cutoff cases gives, exactly,

$$\mathrm{Low} + \mathrm{High} = V, \qquad \widehat{\widetilde{A}}_{N,\Lambda} = \mathrm{Low} + \mathrm{High} + (\widehat{\widetilde{A}}_{N,\Lambda} - V).$$

The error here is the difference from the actual annular von Mangoldt symbol, including prime powers.

Module	New named theorems
FiniteMajorArcModels.lean	5

Scope

These are real-frequency finite symbols and exact identities. The selected finite center set remains an argument. No global periodization or analytic norm estimate is claimed by defining the error as a difference. The full approximation, low/high operator estimates, minor-arc bridge and Theorem 1.1 remain open.

<!-- checkpoint-verification -->

Verification

All 104 project modules and the root were freshly rebuilt: 3623 jobs pass without warnings/errors. All 823 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 106 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. No new deep analytic input or own-paper lemma is assumed.

`Finite_Overlap_and_Major_Arc_Error.md`

Finite overlap and the genuine major-arc model error

This checkpoint combines actual finite rational-center symbols, genuine Gauss sums, the normalized annular integrals, and the constructed smooth cutoff. It proves a major-arc error estimate while preserving the unresolved minor-arc and global-model obligations.

Counting all active scales

For a canonical center $r \in [0,1)$, a reduced denominator below four forces

$$r \in \{0, \tfrac{1}{2}, \tfrac{1}{3}, \tfrac{2}{3}\}.$$

This is an exact rational proof, not a numerical probe. The first two dyadic scales therefore contribute at most four active centers. For larger scales, the previously proved support separation makes the map $r \mapsto \lfloor \log_2 \operatorname{den}(r) \rfloor$ injective on active centers.

If every denominator in a finite canonical center set Gamma is at most M, the total active count is at most $\lfloor \log_2 M \rfloor + 5$. With denominators at most a real $R \geq 1$, a proved floor/log comparison gives

$$\#\{\text{active centers}\} \leq 5 + \frac{\log R}{\log 2}.$$

The finite-norm theorem first removes all inactive terms exactly and only then applies this count.

Geometry of the selected major arc

For $N > 0, R > 0$, rational arcs of denominators at most R and radii $R/(qN^2)$ are disjoint under the explicit sufficient condition

$$2R^3 < N^2.$$

Hence every other center lies outside its own major arc. The actual dyadic cutoff at the chosen center equals one under

$$2R^4 \leq N^2.$$

The real rational center plus its real offset is proved to equal the target circle frequency after coercion, retaining the true negative-phase annular symbol.

Bounding the actual error

Let $C_G > 0$ be one unconditional constant selected from the already proved uniform Gauss-model theorem. It is uniform in N, modulus, character, unit coefficient and allowed exceptional coefficient. Assume $N \geq 2,\ 0 < \beta \leq 1,\ |\kappa_r| \leq 1$, one shared character/coefficient family, and Gamma contained in [0,1) with denominators at most R.

At a selected $r_0 \in \Gamma$, put $q_* = \operatorname{den}(r_0)$, $\theta = \xi - r_0$, and assume

$$|\theta| \leq R/(q_* N^2).$$

The raw progression-prefix estimate is the same explicit hypothesis used in the earlier major-arc proof:

$$\left|\Psi_{q_*,u}(t) - H_{q_*,\kappa_{r_0}\chi_{r_0}(u),\beta}(t)\right| \leq NCe^{-c\sqrt{\log N}}, \qquad N/2 \leq t \leq N.$$

The theorem also retains the earlier explicit parameter budgets $\log R \leq \log N$, $R^5 \log N \leq \sqrt{N}$, and $7 \log R \leq c\sqrt{\log N}$, together with the geometry conditions above.

Removing the chosen center exactly, applying its cutoff core identity, and bounding all remaining model terms gives

$$\left|\widehat{\widetilde{A}}_{N,\Lambda}(\xi) - V_\Gamma(\xi)\right| \leq \frac{4 + (2+4\pi)C}{R^5} + \left(5 + \frac{\log R}{\log 2}\right)\frac{4C_G}{\pi} R^{-5/12}.$$

This is a proved bound for the actual von Mangoldt symbol, including prime powers. Its analytic caller consumes the raw prefix estimate; the paper's own minor-arc lemma is not an assumption.

For every fixed C>=0, the complete displayed right-hand side is also proved eventually at most $R^{-1/6}$. The prior checkpoint provides the source-scale conversion of that rate to $Q^{-1/2}(1+\log N)^{-A}$.

Module	New named theorems
RationalArcOverlap.lean	6
MajorArcGeometry.lean	4
MajorArcModelError.lean	7

Scope and next obligations

The raw progression estimate remains an explicit deep input. No existence theorem for one uniform choice of exceptional data is registered. Gamma is an explicit finite center set; this checkpoint does not construct the global source center set or identify a periodized model on the circle. In particular, the real integral with noninteger integration variable is not silently treated as periodic.

The full-range minor-arc deduction, global approximation 6.9, low 6.14, multifrequency 6.16, high/maximal 6.21, Proposition 4.19 and Theorem 1.1 remain open. The frozen manuscript and accepted revisions are unchanged. These open proof obligations do not establish a counterexample or prescribe a manuscript correction.

<!-- checkpoint-verification -->

Verification

All 108 project modules and the root were freshly rebuilt: 3627 jobs pass without warnings/errors. All 845 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 110 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`General_Euler_Product.md`

Corollary 3.16: checked proof and source correspondence

Checkpoint 08 update: the user has confirmed applying the supplied statement replacement. Its explicit ranges are accepted; no further edit to that supplied statement is requested. The proof below was completed in checkpoint 07. See `docs/SOURCE_UPDATE_08.md`.

Checkpoint 07, 22 October 2026. Source label: `c:generalmajorprimesquareprog`; equations `e:generalcorupss`, `e:generalT2` and `e:fixedqcoefficient`. The archived source is `sources/current/Draft5.tex`, SHA256 `d2c31b96cb15b00feac09ed4fd9a2b0482726c719c4b772320c0711875d04660`.

The identity is proved for every real Q>2, every positive integer k, and all integers ell,b,n satisfying gcd(b,k)=1. There is **no coprimality or nonzero condition on ell**. The actual exponential sums, local counts, normalization, finite support and complete Euler product are connected in Lean.

The corollary should state the same cutoff and modulus conventions now explicit in the user's revised Lemma 3.14. A statement fragment with this opening is in `corrections/Corollary_3_16_Replacement.tex`. Its displayed formulas are unchanged. See `docs/SOURCE_UPDATE_07.md` for the accepted Lemma 3.14 revision and the distinction between the supplied update and the archived full draft.

Definitions and target

For a positive modulus q let A_q be its unit residue classes and set

$$B_\ell(q;k,b,n)=\sum_{a\in A_q}\Upsilon(a\ell,q,b,k)\mathfrak{s}(a,q)e(an/q).$$

The definition `generalCoefficient` uses exactly the existing `upsilon`, `principalSquareSum` and standard complex additive character. It is extended by zero at q=0, solely to use natural-number arithmetic functions. The paper sums over positive q. The extension contributes zero to the formal series.

For each odd prime p define the genuine finite cardinality

$$N_p=\#\{(u,v)\in A_p^2 : u^2+\ell v\equiv n \pmod p,\quad v\equiv b \pmod{(p,k)}\}.$$

The formal definition `primeSolutionCount` is the cardinality of this filtered product of unit groups. Congruence modulo gcd(p,k) is implemented by the canonical reduction homomorphism. Equality in ZMod p implements the first congruence. Its irrelevant value at p=0 is defined as zero.

The target is

$$\sum_{P^+(q)<Q} B_\ell(q;k,b,n) = \mathcal{T}_2(\ell,k,n;b)\prod_{2<p<Q}\frac{p\phi((p,k))}{(p-1)^2}N_p,$$

with exactly the finite 2-adic factor given in the source. Products over an empty prime set have value one. At q=1, the coefficient is one; this endpoint is proved directly, without a prime-factor calculation at a zero modulus.

1. Expand the actual coefficient

Put g=gcd(q,k) and write psi_q for the standard additive character. Opening Upsilon(a*ell) contributes psi_q(-a*ell*v), the square sum contributes psi_q(-a*u^2), and the remaining phase contributes psi_q(a*n). Their product is psi_q(a*(n-u^2-ell*v)). Summing over unit a gives the Ramanujan sum c_q(n-u^2-ell*v). All sums in this step are finite, so their order may be interchanged.

Thus, with

$$R_\ell(q) = \sum_{\substack{u,v\in A_q\\ v\equiv b \pmod g}} c_q(n-u^2-\ell v),$$

the exact source normalization is

$$B_\ell(q) = \frac{\phi(k)}{\phi(q)\phi([k,q])}R_\ell(q).$$

`general_correlation_expansion` checks the phases and signs. `generalCoefficient_fixed_modulus` proves equation `e:fixedqcoefficient`. The scalar identity phi(lcm(q,k))*phi(gcd(q,k))=phi(q)*phi(k), with positive totients when cancelled, gives the equivalent normalization

$$B_\ell(q) = \frac{\phi(g)}{\phi(q)^2}\sum_{\substack{v\in A_q\\ v\equiv b \pmod g}} C_q(n-\ell v), \qquad C_q(t)=\sum_{u\in A_q} c_q(t-u^2).$$

This is `generalCoefficient_as_quadratic`. It also proves that ell=0 causes no difficulty: no inverse of ell and no substitution dividing by ell is used.

2. Multiplicativity

For coprime m and r, the unit CRT identifies A_(mr) with A_m times A_r. The previously proved identity for C_(mr) separates its two coordinates. Reduction maps preserve subtraction, multiplication and integer casts, so n-ell*v becomes the corresponding n-ell*v in each coordinate.

The progression restrictions separate because gcd(m*r,k)=gcd(m,k)*gcd(r,k), and those two gcds are coprime. The totient normalization separates by totient multiplicativity. Consequently

$$B_\ell(mr) = B_\ell(m)B_\ell(r), \qquad B_\ell(1)=1.$$

These statements concern the actual coefficient, established by `scaled_quadratic_filtered_crt`, `generalCoefficient_mul` and `generalCoefficient_one`. Neither b being a unit nor any restriction on ell is needed for this part when k is positive.

3. Prove finite support before forming an Euler product

The completed quadratic Cohen identity implies pointwise C_q(t)=0 for every integer t if an odd prime square divides q or if 16 divides q. Substitution t=n-ell*v transfers this vanishing to B_ell(q), uniformly in every parameter. Thus a nonzero coefficient can only have odd-prime exponents zero or one and exponent at two at most three.

For an integer cutoff Q>=3, put M=((Q-1)!)^3. Every nonzero smooth coefficient has q dividing M. Every divisor of M has all its prime divisors below Q. The formal infinite sum therefore equals the finite divisor sum

$$\sum_{P^+(q)<Q} B_\ell(q) = \sum_{q\mid M} B_\ell(q).$$

This is `smoothGeneralSeries_eq_divisor_sum`. The finite divisor Euler identity `sum_divisors_multiplicative`, already proved in checkpoint 06, then applies. The vanishing rules shorten the exponent sums and yield

$$\sum_{P^+(q)<Q} B_\ell(q) = (1 + B_\ell(2) + B_\ell(4) + B_\ell(8)) \prod_{2<p<Q} (1 + B_\ell(p)).$$

This is `smoothGeneralSeries_euler_product`. No analytic convergence theorem or rearrangement assumption is used.

4. Evaluate each prime factor by an actual pair count

At a prime p, the exponential definition of the Ramanujan sum has already been proved to give c_p(t)=p*1_(t=0)-1. Let M_p be the number of unit pairs satisfying only the progression restriction on v. Opening this formula gives

$$R_\ell(p) = pN_p - M_p.$$

To count M_p there are two cases. If p does not divide k, the gcd is one, so every v is allowed and M_p=(p-1)^2. If p divides k, coprimality of b and k ensures b is a unit modulo p; the condition fixes exactly v=b and leaves p-1 choices for u. In both cases

$$\phi((p,k))M_p = (p-1)^2.$$

The proofs `prime_progression_unit_mass` and `prime_progression_pair_mass` establish this count and use the actual unit-group cardinality phi(p). With w_p=phi((p,k))/(p-1)^2, one obtains

$$1 + B_\ell(p) = 1 + w_p(pN_p - M_p) = pw_pN_p.$$

This is `generalCoefficient_prime_euler_factor`. In particular, zero solution counts, primes dividing ell, ell=0, and n=0 need no separate exclusions.

5. Keep the exact source factor at two

For nu=1,2,3 use the proved fixed-q identity at q=2^nu. The sum 1+B_ell(2)+B_ell(4)+B_ell(8) is exactly

$$1 + \sum_{\nu=1}^{3} \frac{\phi(k)}{\phi(2^\nu)\phi([k,2^\nu])} \sum_{\substack{u,v \in A_{2^\nu} \\ v \equiv b \pmod{(2^\nu,k)}}} c_{2^\nu}(n - u^2 - \ell v).$$

This is the definition `generalTwoEulerFactor`, with the source's exponent range represented by j=0,1,2 and nu=j+1. Its identification with the local coefficient sum is `generalCoefficient_two_euler_factor`. Higher powers have already been proved to vanish. No further assumption on a 2-adic factor is made.

6. Assemble the identity and handle real cutoffs

`general_progression_euler_product` combines the finite product and the local evaluations for integer Q>=3. For any real Q>2, an integer prime satisfies p<Q exactly when p<ceil(Q). Using this equivalence proves `general_progression_euler_product_real`, with precisely the strict prime cutoff in the paper. This includes nonintegral Q and exact prime endpoints.

Two consistency results connect the new definitions to the earlier lemma: `generalCoefficient_one_multiplier` identifies B_1 with the old coefficient at every natural q, and `generalTwoEulerFactor_one_multiplier` identifies its 2-adic factor with the previous closed valuation formula.

Verification and limits

All 18 project modules build with pinned Lean 4.34.1 and Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612`. The 150 named project theorems, including 30 new declarations, pass the transitive axiom audit: only propext, Classical.choice and Quot.sound occur. There are no proof placeholders or additional mathematical axioms. The source audit scans 20 Lean files.

The separate script `check_general_euler_product.py` passes 109,824 exact fixed-modulus comparisons and 13,728 finite Euler comparisons. The ranges, signed representatives and zero/negative multipliers are recorded in `verification/general-euler-numerical.json`. All arithmetic uses integers and rational Fractions. These are finite experiments, not inputs to Lean.

This finishes the corollary with the stated ranges. It does not complete a headline theorem of the paper or the general character-twisted square sum. No deep analytic result has yet been assumed.

`High_Low_B_Endpoint.md`

Theorem 4.1: the complete B_N endpoint deduction

Checkpoint 13, 22 October 2026.

The B_N conclusion, equation `e:endimprovbn` (4.9), is proved in Lean from the theorem's local estimate and actual operator norm assumptions. This includes the exact logarithmic exponent, dyadic selection, the truncated parameter range, empty admissible ranges and empty input/test sets.

Theorem 4.1 is still partially complete. Its A_N conclusion, the sharp coefficient in `e:cstar`, and the subsequent strong-type interpolation corollary are not asserted by this checkpoint. The prime-square operator estimates supplying the abstract hypotheses are also later obligations.

Source and theorem correspondence

The frozen source is `sources/current/Draft5.tex`. Its saved reference map is `verification/Draft5.aux`.

Item	Stable label	Typeset number	PDF page
Full abstract high--low theorem	`t:improvinggeneral`	4.1	25
Local crude input	`e:bnv`	4.3	26
Low operator norm input	`e:lobn1`	4.4	26
High operator norm input	`e:hibn1`	4.5	26
B_N endpoint conclusion	`e:endimprovbn`	4.9	26
Pairing/balancing estimate	`e:Bbalance`	4.12	26

The preferred scale is the expression labeled `e:Jchoice`. The source's logarithmic conventions are retained exactly:

$$\operatorname{Log} x = 2 + |\log x|, \qquad \operatorname{Log}\operatorname{Log} x = 2 + \log(2 + |\log x|),$$
$$\Phi_{c,\kappa}(x) = \exp\left(c\frac{(\operatorname{Log} x)^{\kappa}}{\operatorname{Log}\operatorname{Log} x}\right).$$

These are `paperLog`, `paperLogLog` and `paperPhi`. The last definition is recorded for the next stage; defining it is not a proof of the A_N estimate. The definitions have harmless total extensions at x=0 in Lean. Empty-set cases are proved separately, with endpoint value zero.

Exact formal statement for the B_N clause

Fix s,r>0, β≥0, γ≥β and C>0. In the paper, s≥1-v and v<1 imply s>0. Let N>0, let K be a complex-valued convolution kernel on the integers, and let I_x be any finite local window. Write

$$(K * \mathbf{1}_F)(x) = \sum_{y\in F} K(x-y), \qquad \langle K * \mathbf{1}_F, \mathbf{1}_G\rangle = \sum_{x\in G}\sum_{y\in F} K(x-y).$$

The formal definitions are `indicatorConvolution` and `restrictedKernelPairing`. Lean also proves that these finite sums equal the corresponding infinite sums with the actual indicator functions and complex conjugation in the second factor.

Assume the local inequality

$$|(K * \mathbf{1}_F)(x)| \le CN^{-v}\|\mathbf{1}_F \mathbf{1}_{I_x}\|_1.$$

For each admissible dyadic J=2^k with

$$2 \le J \le N^{(1-v)/s}/10,$$

assume operators Lo:$\ell^1\to\ell\infty$ and Hi:$\ell^2\to\ell^2$ satisfying the decomposition on indicators and the genuine norm bounds

$$\|Lof\|_\infty \le CN^{-1}J^s(\log J)^\gamma\|f\|_1, \qquad \|Hif\|_2 \le CJ^{-r}(\log J)^\beta\|f\|_2.$$

These are the actual normed spaces `lp` in Mathlib. The proof does not need linearity of the pieces, so the formal input permits a wider class of maps. The decomposition is required only on indicators, since that is all the restricted endpoint uses. The source's full operator decomposition implies these hypotheses directly.

Then a single D>0, depending only on the fixed parameters and C, works for every such N, kernel, local-window family and finite F,G with |F||G|≤N²:

$$\frac{|\langle K * \mathbf{1}_F, \mathbf{1}_G\rangle|}{N} \le D\,\delta^{1-s/(2(s+r))}(\operatorname{Log}\delta)^{\gamma+s(\beta-\gamma)/(s+r)}, \qquad \delta = \frac{|F||G|}{N^2}.$$

This is `highLow_B_endpoint`. The constant is quantified before N, K, the window family and F,G, so the required uniformity is formal. Common interval support of F,G and positivity of K are not needed for this B_N deduction; the paper itself notes the extension to arbitrary finite sets with the stated cardinality product. Those assumptions may have other uses in the remaining theorem and its applications.

The local windows in the source are integer intervals about x. The proof uses only their finite-set interpretation. The exact local indicator is `finiteIndicatorLp (F ∩ I_x) 1`; its norm is proved to be |F∩I_x|.

1. Indicators, actual norms and Cauchy--Schwarz

`finiteIndicatorLp` constructs the indicator as an element of `lp` at any exponent, using its finite support. For positive finite p, Lean proves

$$\|\mathbf{1}_F\|_p = |F|^{1/p}.$$

In particular, the input norms are exactly |F| and √|F| at exponents one and two. Restricting an indicator to I is proved to give the indicator of F∩I. These identities are not left as informal conversions between counts and norms.

For the high output h in genuine ℓ^2, Mathlib's finite-energy bound gives

$$\sum_{x\in G} |h(x)|^2 \le \|h\|_2^2.$$

The checked finite Cauchy--Schwarz inequality then proves

$$\left|\sum_{x\in G} h(x)\right| \le \sqrt{|G|}\,\|h\|_2.$$

The low output is bounded using its actual $\ell\infty$ norm and summing over G. Triangle inequality and the decomposition therefore yield

$$\frac{|\langle K * \mathbf{1}_F, \mathbf{1}_G\rangle|}{N} \le C\left(\delta J^s(\log J)^\gamma + \delta^{1/2}J^{-r}(\log J)^\beta\right).$$

Lean checks the factors of N through

$$\sqrt{\delta} = \frac{\sqrt{|F|}\sqrt{|G|}}{N}, \qquad N > 0.$$

The scalar balancing inequality is derived from these norm inputs; it is not an extra hypothesis in the final operator theorem.

2. The crude estimate, including empty parameter ranges

Apply the local assumption to F={0}. Since |F∩I_x|≤1, this gives

$$|K(z)| \le CN^{-v}.$$

Two finite sums and triangle inequality imply

$$\frac{|\langle K * \mathbf{1}_F, \mathbf{1}_G\rangle|}{N} \le C\delta N^{1-v}.$$

The real-power identity used to normalize this expression is checked for every N>0. In particular, the proof does not request a decomposition at J=2 when the admissible range is empty.

3. Exact balance identities

Put

$$u = s + r, \quad L = \operatorname{Log} \delta, \quad \theta = 1 - \frac{s}{2u}, \quad \tau = \gamma + \frac{s(\beta - \gamma)}{u} = \frac{r\gamma + s\beta}{u},$$
$$T = \delta^{-1/(2u)} L^{(\beta-\gamma)/u}, \qquad E = \delta^{\theta} L^{\tau}.$$

For 0<δ≤1, all bases of real powers are positive, L≥2, θ>0 and τ≥0. Lean proves the exact identities

$$\delta T^s L^\gamma = E, \qquad \delta^{1/2} T^{-r} L^\beta = E.$$

It also proves δ≤E and δT^s≤E. The first follows from θ≤1 and L^τ≥1; the second follows from the first balance identity and L^γ≥1. None of these steps invokes an unproved asymptotic or optimization result.

4. Dyadic rounding that preserves the cutoff

When T≥2, choose a dyadic J rounded down:

$$2 \le J \le T < 2J.$$

Its existence is proved using the Archimedean power lemma. If T is within the admissible range, J remains within that range. An upward rounding can fail this requirement near the cutoff.

Set

$$A = 1 + \frac{1}{2(s + r)}.$$

Because β≤γ and log L≥0,

$$\log T = -\frac{\log \delta}{2u} + \frac{\beta - \gamma}{u} \log L \le \frac{L}{2u} \le AL.$$

Thus log J≤AL, J^s≤T^s and J^(-r)≤2^r T^(-r). As β,γ≥0, the two exact balance identities give

$$\delta J^s (\log J)^\gamma + \delta^{1/2} J^{-r} (\log J)^\beta \le (A^\gamma + 2^r A^\beta) E.$$

This is the entire interior case, with an explicit parameter-dependent constant and the printed logarithmic exponent.

5. Every boundary and fallback case

Write M=N^((1-v)/s)/10 for the upper cutoff.

Situation	Checked choice or estimate	Resulting coefficient multiplying C E
T≥2 and T≤M	Dyadic J≤T<2J	A^γ+2^r A^β
T≥2 and M<T	Crude estimate	10^s
T<2 and 2≤M	J=2	2^s+1
T<2 and M<2	Crude estimate; no admissible scale required	20^s

For the exceeded-cutoff case, raising N^((1-v)/s)<10T to the positive power s gives

$$N^{1-v} \le 10^s T^s.$$

Then the crude bound is at most C10^s δT^s≤C10^s E. The exponent cancellation ((1-v)/s)s=1-v is explicitly checked with s≠0.

For T<2 and admissible J=2, use log 2≤1 for the low term, δ≤E, and 2^(-r)≤T^(-r), (log 2)^β≤L^β for the high term. This yields C(2^s+1)E. This argument also covers preferred scales T<1.

For an empty admissible range M<2, the same cutoff calculation gives N^(1-v)≤20^s. The crude bound is at most C20^s δ≤C20^s E. All positive scales N are covered; there is no unproved sufficiently-large-N exception.

One explicit final choice is therefore

$$D = C\left(A^{\gamma} + 2^{r}A^{\beta} + 10^{s} + 2^{s} + 1 + 20^{s}\right) > 0.$$

No optimal constant is claimed. If F or G is empty, the pairing and density are zero. Since θ>0, the endpoint is zero and the conclusion follows directly, before any use of a logarithm of a positive density.

Lean files and trust boundary

Module	Checked content
`HighLowNotation.lean`	Exact Log/LogLog/Phi definitions, positivity, dyadic rounding down
`HighLowBalance.lean`	Density/log exponents, exact preferred-scale identities, logarithm bound
`HighLowOptimization.lean`	Full scalar optimization with all four range cases and an explicit constant
`HighLowPairing.lean`	Actual finite convolution, finite complex pairings, genuine lp-two energy and Cauchy--Schwarz
`HighLowEndpoint.lean`	Normalizations, local-kernel crude bound, endpoint from indicator-output inputs
`HighLowOperators.lean`	Exact indicator norms, standard infinite-sum identities, operator norm bridge and `highLow_B_endpoint`

The operator norm estimates are the hypotheses of the abstract theorem. This checkpoint proves their implication, rather than asserting them for the prime-square kernels. No own-paper result is replaced by an axiom. There are no extra mathematical axioms or proof placeholders.

The previous eight completed numbered results remain unchanged. Lemma 3.21 still has its two explicit PNT-limit hypotheses. No PNT or Siegel--Walfisz input enters the new B_N argument. This checkpoint does not increase the count of completely proved numbered results, because Theorem 4.1 also has the unproved A_N conclusion.

Verification and exact finite checks

The fresh build checks all 46 project modules. The source audit scans 48 Lean files and the transitive axiom audit covers 372 named project theorems, including the 51 added here. The only permitted foundational axioms are `propext`, `Classical.choice` and `Quot.sound`.

`scripts/check_high_low.py` records zero failures using integer arithmetic and exact fractions:

Check	Cases
Complete scalar optimization, rational subfamily s=r=1, β=γ=0, δ=T^(-4)	56,164
Admissible dyadic scales examined	122,193
Empty admissible ranges	7,979
Preferred scales below two	2,220
Interior-balance cases	14,667
Exceeded-cutoff cases	39,277
Examples where upward rounding exceeds the cutoff	4,071
Actual complex finite-kernel pairings	6,144
Empty input/test-set pairings	378

The kernel tests use six finite kernels with Gaussian-integer coefficients and every pair of subsets of {-2,-1,0,1,2}. They check the decomposition, global high-output square energy, low-output supremum bound, combined pairing bound and crude kernel estimate. Noninterval and empty sets are included. Irrational norm inequalities are tested by squaring with the required sign case, without converting to floating point.

The scalar experiment is explicitly restricted to its rational subfamily; it does not numerically certify general logarithmic exponents. Lean proves the full real-parameter result and its unbounded quantifiers. These finite checks supplement that proof and do not replace it.

Remaining part of Theorem 4.1

The next target is the A_N conclusion with every fixed coefficient larger than

$$\frac{c_1 s}{2^{\kappa}(s+r)^{\kappa+1}}.$$

It requires the actual low/high/error pairing bound, the two density/scale ranges, dyadic Q selection, domination of logarithmic losses by the exact Phi function, and a uniform epsilon formulation of the sharp coefficient. The displayed first-order Q choice should be handled with explicit epsilon margins; uniform comparability of its first two contributions must not be silently assumed. The subsequent interpolation corollary also remains open.

`High_Low_Theorem.md`

Theorem 4.1: complete restricted endpoint deduction

Current checkpoint 20: the full abstract Corollary 4.15 is complete, including strong interior interpolation and p=2. This document records an earlier stage; see Strong_Improving_Complete.md for the completed proof and current scope.

Checkpoint 15 completes both restricted endpoint conclusions of Theorem 4.1, `t:improvinggeneral`, from its abstract operator norm hypotheses. The main Lean declaration is `PrimeSquares.highLow_AB_endpoint` in `PrimeSquares/HighLowTheorem.lean`.

The source conclusions are equation 4.9 (`e:endimprovbn`), equation 4.10 (`e:endimprovan`) and the coefficient in equation 4.11 (`e:cstar`). The epsilon interpretation of the coefficient is included. Theorem 4.1 is now counted as a complete numbered result. The prime-square estimates that supply its abstract hypotheses and the following interpolation corollary remain open.

Exact conclusion

Let delta=|F||G|/N^2. One DB>0 and one DA>0, chosen before N, both kernels, local windows and F,G, satisfy

$$\frac{|\langle B_N 1_F, 1_G\rangle|}{N} \le D_B \delta^{1-\frac{s}{2(s+r)}} (\operatorname{Log}\delta)^{\gamma+\frac{s(\beta-\gamma)}{s+r}},$$

and

$$\frac{|\langle A_N 1_F, 1_G\rangle|}{N} \le D_A \delta^{1-\frac{s}{2(s+r)}} \Phi_{cstar,\kappa}(\delta)$$

for every fixed

$$cstar > \frac{c_1 s}{2^\kappa (s+r)^{\kappa+1}}.$$

The proof includes all positive real N and all finite F,G in the integers with |F||G|<=N^2, including empty sets and sets outside a common interval. It therefore includes the paper's positive integer scales and its concluding arbitrary-finite-set extension. The exact definitions `Log x=2+abs(log x)` and `LogLog x=2+log(Log x)` are retained.

The finite convolution sum is defined with the source orientation x-y, and the pairing is formally identified with the ordinary infinite sum against the conjugate indicator in `restrictedKernelPairing_as_tsum`. Finite sums are not used as a substitute for a different pairing.

Source hypotheses and Lean correspondence

`HighLowEndpointInputs` collects only the indicator restrictions of the source assumptions. Its operator norm fields use genuine Mathlib lp spaces.

Source	Lean field or declaration	Exact restriction used
Positive B_N convolution kernel	`nonnegative`, `NonnegativeKernel`	Every kernel value has imaginary part zero and nonnegative real part
Equation 4.2, `e:anbnsupbound`	`domination`	Norm of A_N applied to any finite indicator is at most `(log N)^a` times the real part of B_N applied to that indicator
Equation 4.3, `e:bnv`	`local_bound`	The stated local l1 norm on a finite window, for all finite indicators
`e:lobn1`, `e:hibn1`	`B_decomposition`, `OperatorHighLowBound`	Genuine l1-to-linfinity and l2-to-l2 maps at every admissible dyadic J
`e:decompan--e:errorgeneral`	`A_decomposition`, `OperatorOrliczBound`	Genuine l1-to-linfinity and two l2-to-l2 maps at every admissible dyadic Q

The final theorem assumes v<1, 1-v<=s, r>0, 0<=beta<=gamma, a>=0, c0,c1,c>0, 0<kappa<1, and the strict coefficient inequality. These include the printed parameter ranges. The printed requirements v>=0 and alpha>0 can be omitted for this deduction. Only the positivity of B_N is used; the A_N kernel may be complex. Linearity and convolution formulas for the individual decomposition pieces are unnecessary once their actual norm bounds and indicator decomposition identities are given. These are relaxations of source hypotheses, with no additional hypothesis introduced.

The local windows are arbitrary finite sets in the formal theorem. The source intervals give a particular choice. No claim is made that the prime-square kernels satisfy these hypotheses merely because the abstract implication has been proved.

Separate three-piece constants and error input

The A_N predicate retains three separate real constants CLo, CHi, CErr:

$$\begin{aligned}\|\mathrm{Low} f\|_\infty &\le C_{Lo} N^{-1} Q^s (\log Q)^\alpha \|f\|_1,\\ \|\mathrm{High} f\|_2 &\le C_{Hi} Q^{-r} \exp\left(c_1 \frac{(\log Q)^\kappa}{\log\log Q}\right)\|f\|_2,\\ \|Ef\|_2 &\le C_{Err} Q^{-r} (\log Q)^M (1+\log N)^{-1}\|f\|_2.\end{aligned}$$

The exponent M and the norm constant CErr are distinct fixed parameters. The source provides every A>0; the proof uses its permitted value A=1, with the associated power M=C_1. Thus it requires a consequence of the source error hypothesis. The bound at every positive A is not assumed as an unproved analytic statement within this abstract theorem; it is a given operator hypothesis of the source, restricted to one value.

To use the scalar bound, the proof forms

$$C_A = 1 + |C_{Lo}| + |C_{Hi}| + |C_{Err}|.$$

It proves that this number is positive and majorizes the three constants. Every multiplier multiplying a constant is checked to be nonnegative before it is enlarged. All constants and the power M are fixed before N and Q. The notation C_A here denotes this auxiliary common constant; it is not an identification of the source's error exponent with a norm constant.

Actual operator-to-pairing bridge

`finiteIndicatorLp` is a genuine element of lp, with checked identities

$$\|1_F\|_1 = |F|, \qquad \|1_F\|_2 = \sqrt{|F|}.$$

The three output functions for an indicator are a bounded low function and two actual l2 functions. Triangle inequality and the proved finite l2 Cauchy--Schwarz estimate give

$$|\langle A_N 1_F, 1_G\rangle| \le |G|\|\mathrm{Low} 1_F\|_\infty + (\|\mathrm{High} 1_F\|_2 + \|E1_F\|_2)\sqrt{|G|}.$$

The input norm identities and exact normalization by N then yield

$$\begin{aligned}\frac{|\langle A_N 1_F, 1_G\rangle|}{N} \le C_A\Bigg[&\delta Q^s (\log Q)^\alpha + \delta^{1/2} Q^{-r} \exp\left(c_1 \frac{(\log Q)^\kappa}{\log\log Q}\right)\\ &+ \delta^{1/2} Q^{-r} (\log Q)^M (1+\log N)^{-1}\Bigg].\end{aligned}$$

This is `restrictedKernelPairing_orlicz_bound`, deduced from actual output norms. The final theorem does not take this paper-specific scalar pairing bound as a hypothesis.

Positive-kernel pairing domination

For a nonnegative B_N kernel, every B_N indicator output is real and nonnegative. Consequently its finite pairing equals the sum of those nonnegative real outputs, and its norm equals that same sum. This identity is proved in `restrictedKernelPairing_norm_positive`.

Summing the source pointwise domination and using triangle inequality gives

$$|\langle A_N 1_F, 1_G\rangle| \le (\log N)^a |\langle B_N 1_F, 1_G\rangle|.$$

The normalized identity is `restrictedKernelPairing_domination_normalized`. It accounts explicitly for the positivity needed to pass from pointwise domination to a norm of the B_N pairing. The completed B_N endpoint from checkpoint 13 is then applicable.

Uniform scalar optimization: every density range

Set u=s+r, k=1/(2u), L=Log(delta) and W(L)=L^kappa/(2+log L). Checkpoint 14 proved the actual dyadic balancing bound with every coefficient above the printed leading one. The full scalar implication, including the remaining ranges, is now `orlicz_scalar_endpoint`.

The proof chooses

$$d = \frac{c_0}{k + 1 + \log 4} > 0.$$

It proves d<=c0/(k+1). It also proves the following useful implication: if L>=2 and L<=d*t, then log4<=c0*t. This is an elementary multiplication by a positive denominator, checked in `orlicz_small_cutoff`.

There are three cases:

1. **Complementary range:** d*(log N)^c<=L. Positive domination and the B_N endpoint bound the A_N pairing by a fixed constant times `(log N)^a*delta^theta*L^tau`. The already proved uniform logarithmic absorption `log_density_domination_le_orlicz` bounds this by `D*delta^theta*Phi(cstar,kappa,delta)` over the entire density interval.
2. **Large L in the other range:** L>T, or L>=T at the chosen threshold, and L<=d*(log N)^c. The earlier constructed dyadic scale is admissible since d<=c0/(k+1), and its three scalar contributions have the required bound. The actual error decay factor is included.
3. **Bounded L in the other range:** L<T and L<=d*(log N)^c. The special choice of d proves that the fixed dyadic Q=4 is admissible. For theta=1-s/(2u)<=1 the proof checks `delta<=delta^theta` and `sqrt(delta)<=exp(T)*delta^theta`. These follow from 0<delta<=1 and L<=T. Substitution at Q=4 gives a fixed multiple of delta^theta; Phi>=1 completes the estimate.

An empty admissible Q range is covered by the first case: either remaining case entails an admissible Q, as proved. No decomposition at an inadmissible scale is invoked. The low/high/error terms include their separate constants through C_A, and their common endpoint constant is selected before N,delta.

One valid unoptimized scalar constant is

$$D_A = D_B D_C + 3C_A + C_A C_{\mathrm{compact}},$$

where D_C is the uniform complementary-range absorption constant and C_compact is the explicitly defined fixed-Q coefficient in `orliczCompactConstant`. The density threshold T depends only on the fixed parameters, alpha and M. Its dependence is permitted and never involves N, F or G.

The sharp asymptotic step uses the strict-margin scale `log Q approximately kL+bW(L)`, with `c1*k^kappa/(s+r)<b<cstar/s`. It proves the literal log(log Q) scaling and rounding limits. The particular first-order expansion in `e:Qchoice` is not formalized; the epsilon-margin construction proves the same endpoint conclusion directly. See `blueprint/Orlicz_A_Scalar_Balance.md` for that step.

Small scales, zero density and uniformity

The operator bridge's growth and error-decay inequalities use N>=1. For nonempty F and G, their cardinalities are positive integers and `1<=|F||G|<=N^2`. If N>0, this implies N>=1. This deduction is proved in the final theorem. Thus every smaller positive N already has an empty F or G.

If either set is empty, the actual pairing vanishes. Since `theta=1-s/(2(s+r))>0`, the density factor at zero vanishes too. These cases are handled before applying any positive-density scalar result. In particular, no logarithm at a zero density is used in a limit argument. The total Lean extensions of Log and Phi cause no ambiguity in the zero endpoint.

Verification and numerical scope

Checkpoint 15 adds four modules and 18 named theorems. The whole project has 54 modules and 424 named theorem declarations. Every module and the root receive a fresh build with the artifact cache disabled, passing 3451 jobs. The complete source scan covers 56 Lean files. Every named theorem receives a transitive axiom audit with exactly the whitelist propext, Classical.choice and Quot.sound. See the current verification logs.

The new independent exact finite experiment checks 6144 actual complex convolution pairings, including 378 empty pairs, the three-piece norm bound, positive-kernel identities, 18432 domination cases, 110592 normalized domination cases, and 13104 normalization identities with separate constants. Square-root sums are certified by integer isolating intervals with exact equality detection; no floating-point arithmetic is used. The report and reproducible script are `verification/three-piece-pairing-numerical.json` and `scripts/check_three_piece_pairing.py`.

These numerical checks concern finite pairing components. They do not test the unbounded analytic optimization, all-scale operator hypotheses or the sharp coefficient; those implications are established by Lean. No external analytic input, PNT or new mathematical axiom is used in Theorem 4.1.

Status and next dependency

The nine unconditional completed numbered results are 3.1, 3.11, 3.12, accepted revised 3.14, accepted revised Corollary 3.16, and original 3.20, 3.24, 3.27 and Theorem 4.1. Lemma 3.21 keeps its explicit two PNT-limit hypotheses. The full archived sources and accepted revision fragments are unchanged. No new manuscript statement correction is required by this proof.

Next is Corollary 4.15 (`c:improvinggeneral`): deduce the strong lp-improving bound from the restricted endpoints and uniform l2 bounds. That interpolation argument remains an open obligation; no strong lp or prime-square main theorem is claimed here.

`Integer_Fourier_and_Multipliers.md`

Integer Fourier transform and the actual square multipliers

Checkpoint 22. Source: Draft5, equations `e:basicaverages` and `e:tildedef2`, and the Fourier framework preceding Lemma 6.5. This unit connects the completed actual averaging operators to Mathlib's genuine Plancherel theory. It supplies prerequisites for Proposition 4.19; it does not prove its major-arc, low, high, error or maximal estimates.

Exact conventions

Let T = R/Z with Haar measure of total mass one. Put

$$c_n(\theta) = \exp(-2\pi i n\theta), \qquad n \in \mathbb{Z}.$$

Mathlib's `fourier n` uses the positive phase. We explicitly define `negativeFourierCharacter n = fourier (-n)` at period one. The theorem `negativeFourierCharacter_real` proves the displayed exponential formula for every real lift theta. The character product law and unit magnitude are proved without restricting theta to rational frequencies.

The measure is `AddCircle.haarAddCircle`, not an unspecified volume normalization. Elements of circle L2 are equivalence classes modulo almost-everywhere equality. All formulas involving their chosen representatives are correctly stated almost everywhere.

The actual unitary transform

Mathlib proves that its Fourier monomials form an orthonormal Hilbert basis. Negation is a bijection on the integers, so the family c_n has the same orthonormality and the same dense span. We prove these facts and construct `negativeFourierBasis` using `HilbertBasis.mk`.

Inverting the basis representation defines the complex-linear isometric equivalence

$$\mathcal{F} : \ell^2(\mathbb{Z};\mathbb{C}) \simeq L^2(T;\mathbb{C}).$$

Consequently, for every actual l2 sequence f, rather than only finite inputs,

$$\|\mathcal{F} f\|_2 = \|f\|_2.$$

This is `integerFourier_norm`. The inverse is part of the constructed isometric equivalence; `integerFourier_inverse` records one inverse identity. No Plancherel assumption, postulate or finite-grid approximation is used. The pinned Mathlib supplies the underlying density and Hilbert space arguments.

The inverse coefficients have the required positive inverse phase:

$$(\mathcal{F}^{-1}g)(n) = \text{fourierCoeff}(g)(-n) = \int_T e(n\theta)g(\theta)\,d\theta.$$

`integerFourier_coefficient` proves the coefficient identity, using the actual Mathlib coefficient definition and the reflected Hilbert basis.

For the finite sequence supported on F with coefficients f(n), we prove

$$\mathcal{F}\left(\sum_{n\in F} f(n)\delta_n\right) = \left[\theta \mapsto \sum_{n\in F} f(n)e(-n\theta)\right]_{L^2}.$$

This is `integerFourier_finiteInput`. The continuous polynomial's exact real-lift formula is `finiteFourierPolynomial_real`; its squared L2 norm is exactly the finite sum of |f(n)|^2, by `finiteFourierPolynomial_parseval`. Empty support and zero coefficients are included.

Continuous symbols and all-input operator bounds

For a continuous symbol m on T, the product m g is in circle L2 for every g in circle L2, since

$$|m(\theta)g(\theta)| \le \|m\|_\infty |g(\theta)|.$$

We prove membership, representative compatibility, complex linearity and continuity, constructing `circleMultiply m`. Define

$$U_m = \mathcal{F}^{-1}\ \text{Mul}_m\ \mathcal{F}.$$

`integerMultiplier m` is this actual continuous complex-linear map on integer l2. We prove

$$\|U_m\| \le \|m\|_\infty, \qquad \mathcal{F}(U_m f) = m\mathcal{F}f \quad \text{a.e.}$$

The operator bound holds for every l2 input. Addition, scalar multiplication, zero, negation and subtraction of symbols have the exact corresponding operator identities. In particular, if B >= 0 and |m(theta)-v(theta)| <= B for every theta, then

$$\|U_m - U_v\|_{\ell^2\to\ell^2} \le B.$$

`integerMultiplier_error_bound` proves this implication. The uniform symbol inequality is an explicit hypothesis of that transport lemma. It is not an assertion that the paper's proposed approximation already satisfies it. This unit treats continuous symbols; it makes no claim to have implemented every bounded measurable multiplier or a maximal multiplier theorem.

Translations and weighted square sums

Let tau_d f(x) = f(x-d). First we prove exactly

$$\tau_d\delta_n = \delta_{n+d}.$$

Multiplying c_n by c_d gives c_{n+d}. The transform identity therefore holds on every single-coordinate input. Equality of continuous linear maps follows from density of finite-support l2 sequences, via `lp.ext_continuousLinearMap`. Thus

$$\tau_d = U_{c_d}, \qquad \mathcal{F}(\tau_d f)(\theta) = e(-d\theta)\mathcal{F}f(\theta) \quad \text{a.e.}$$

The equality is `integerTranslate_l2_multiplier`, and the representative formula is `integerFourier_translate_ae`. This step checks the sign against the actual translation definition; it does not change that definition to fit a desired multiplier.

For any finite set S of natural numbers, real scalar c and real weights w, put

$$T_{S,c,w}f(x) = c\sum_{n\in S} w(n)f(x-n^2), \qquad m_{S,c,w}(\theta) = c\sum_{n\in S} w(n)e(-n^2\theta).$$

`finiteSquareSymbol_real` proves this symbol's real-lift formula, and `finiteSquareOperator_l2_multiplier` proves

$$T_{S,c,w} = U_{m_{S,c,w}}.$$

Finite linear combinations of the proved translation identities give this equality. It holds on all of integer l2, for arbitrary signs of c and w, including empty S. No distribution-of-primes input is involved.

The full and annular symbols specialize to

$$\widehat{A}_{N,w}(\theta) = \frac{1}{N}\sum_{0<n\le N} w(n)e(-n^2\theta), \qquad \widehat{\widetilde{A}}_{N,w}(\theta) = \frac{1}{N}\sum_{\lfloor N/2\rfloor<n\le N} w(n)e(-n^2\theta).$$

The exact operator equalities are `weightedSquareAverage_l2_multiplier` and `annularWeightedSquareAverage_l2_multiplier`. The two von Mangoldt real-lift formulas explicitly use `ArithmeticFunction.vonMangoldt`. Every prime power remains in the sum. The formal definitions also cover N=0, using Lean's total inverse 0^-1=0; the source application uses positive dyadic N.

The theorem `annularWeightedSquareAverage_fourier_error_bound` now turns any future proved uniform error estimate for the actual annular symbol into the same l2 operator bound. The hard estimate itself remains open.

Inverse kernels and convolution agreement

For a continuous m, define the genuine l2 sequence

$$K_m = \mathcal{F}^{-1}[m]_{L^2}.$$

`fourierKernel_coefficient` identifies K_m(x) with the actual coefficient `fourierCoeff m (-x)`. Using the already proved translation identity, commutativity of symbol multiplication gives

$$U_m(z\delta_i)(x) = K_m(x-i)z.$$

Finite summation then proves

$$U_m\left(\sum_{i\in F} f(i)\delta_i\right)(x) = \sum_{i\in F} K_m(x-i)f(i)$$

for every finite F, every complex coefficient function f and every integer x. These are actual pointwise integer-sequence equalities, not just circle a.e. statements. They supply `KernelL2Bound K_m ||m||` and its version with any proved uniform bound B. For the finite square symbol, the new inverse kernel is proved equal at every integer to the original `finiteSquareKernel`. Thus both constructions agree exactly.

A general smooth-cutoff inverse kernel need not have the original averaging kernel's compact support. No positivity, compact support or absolute summability is inferred for general K_m by this unit.

Rational frequencies and the earlier Gauss sums

For every positive modulus q and all integer a,n, we prove

$$c_n(a/q) = \mathrm{stdAddChar}_q(-an).$$

The squared version gives

$$e(-an^2/q) = \mathrm{stdAddChar}_q(-an^2).$$

These are `negativeFourierCharacter_rational` and `negativeSquareCharacter_rational`. The proof uses the actual complex exponential definitions, without a coprimality restriction on a.

The earlier principal and twisted quadratic sums therefore equal, respectively,

$$\frac{1}{\varphi(q)}\sum_{u\in(\mathbb{Z}/q\mathbb{Z})^\times} c_{\widetilde{u^2}}(a/q), \qquad \frac{1}{\varphi(q)}\sum_{u\in(\mathbb{Z}/q\mathbb{Z})^\times} \chi(u)c_{\widetilde{u^2}}(a/q),$$

where tilde u is the standard natural representative. Both identities are formally proved, for arbitrary complex Dirichlet characters in the twisted case, and include q=1. They connect the arithmetic modules' minus sign and totient normalization to the analytic symbols.

Files and verification scope

Module	New named theorems	Result
CircleFourier.lean	15	Reflected Hilbert basis, actual unitary transform, finite formulas and Parseval
CircleMultipliers.lean	21	Actual all-input multiplier maps, norm bounds and linear identities
SquareFourier.lean	16	Exact translations, full/annular square symbols and error-bound transport
FourierKernels.lean	9	Actual inverse kernels, finite convolution and genuine kernel l2 bounds
FourierGaussCompatibility.lean	4	Rational phase, principal and twisted normalization compatibility

The five modules add 65 named theorems, for 678 in the whole project. The total includes supporting lemmas; it is not a count of completed numbered paper statements. Every module is root-imported and every named theorem is included exactly once in the transitive axiom audit.

The full fresh artifact-cache-disabled build passes 3595 jobs, including every one of the 80 project modules and the root, with no warnings/errors. All 678 named project theorems pass the complete transitive whitelist: propext, Classical.choice and Quot.sound. The source scan checks 82 Lean files and finds no placeholders or extra mathematical axioms. Lean 4.34.1 and Mathlib d13f23b723b8a846827a245b89c10fc7d3f11612 remain pinned.

Independent finite probes

`scripts/check_fourier_conventions.py` uses exact Gaussian rationals and formal independent log(prime) symbols. It checks 532 translations, 2856 actual finite square multiplier identities, 189 dyadic symbol reconstructions, 7 Gauss sign/normalization cases and 27 finite cyclic Parseval cases. The rational frequency denominators are 1,2,4. No floating point or sampling tolerance is used.

At theta=1/4 the explicit diagnostics include

$$\mathcal{F}\delta_1 = -i, \qquad \widehat{A}_{4,1} = (1-i)/2, \qquad \widehat{A}_{4,\Lambda} = (2\log 2 - i\log 3)/4.$$

The annular von Mangoldt value is (log2-i log3)/4. Reversing the sign changes -i to +i, and omitting the prime power 4 loses log2/4 from the full symbol. Both mutations are detected. These finite probes do not prove general Plancherel or any asymptotic estimate. The genuine infinite-dimensional bridge is the separate Lean construction above.

What remains before Proposition 4.19

Dependency	Status
Actual averages, positivity, elementary local and uniform lp/l2 bounds	Complete at checkpoint 21
Actual negative-phase Fourier/Plancherel and exact square multipliers	Complete in this unit
Published deep analytic input boundaries	Must be verified and stated before use
Lemma 6.5: actual major-arc prime-square input and paper-specific bookkeeping	Open
Lemma 6.9: actual approximation by V_N	Open
Proposition 6.14: low contribution	Open
Lemma 6.16 and Orlicz prefix estimate	Open
Proposition 6.21: high and maximal high estimates	Open
Proposition 4.19 assembly and Theorem 1.1 application	Open

The next step is Lemma 6.5: exact residue grouping of the actual finite Lambda sum, non-unit and prime-power bookkeeping, partial summation, and consumption of precisely quantified published analytic inputs. Own-paper deductions remain proof obligations. Exceptional conductor, character and beta may vary with N and must not be fixed across scales in later maximal arguments.

The ten completed numbered results remain unchanged. This unit introduces no new deep hypothesis. Lemma 3.21 still has its earlier two explicit PNT limits; their external bridge is not integrated and is not used here. The frozen manuscript and accepted revisions are unchanged.

`Interpolation_Geometry.md`

Corollary 4.15: uniform geometric weights and actual complex block bounds

Current checkpoint 20: the full abstract Corollary 4.15 is complete, including strong interior interpolation and p=2. This document records an earlier stage; see Strong_Improving_Complete.md for the completed proof and current scope.

Checkpoint 19 update: actual rank blocks, coefficient mass and the strong finite bilinear estimate are now proved. See `Rank_Blocks_and_Finite_Bilinear.md`. The remaining-obligation list below records checkpoint 18; output norm duality and final strong operator assembly still remain open.

Checkpoint 18 proves the actual geometric summation estimate needed for strong interpolation, its exact N scaling, and the estimate for any specified finite family of complex convolution blocks. Two modules add 22 named theorems. **The full strong interior estimate in Corollary 4.15 remains open:** the rank partition and its coefficient mass bound, finite duality, and final assembly are still required.

Source and checked scope

Source: Corollary 4.15 (`c:improvinggeneral`), near line 2431 of `sources/current/Draft5.tex`. Equation 4.17, the global restricted estimate, and the strong p=2 case are already proved. The target equation 4.16 is

$$\|A_N f\|_{p'} + \|B_N f\|_{p'} \le C_p N^{1/p'-1/p} \|f\|_p, \qquad p_0 < p \le 2, \qquad p_0 = \frac{2(s+r)}{2r+s}.$$

Fix p0<q<p<2. For one kernel K, the available global restricted bound has constant A, and its genuine continuous l2 realization has norm <=B. Put

$$a = \frac{1}{q} - \frac{1}{p} > 0, \qquad b = \frac{1}{p} - \frac{1}{2} > 0.$$

`interpolation_gaps_pos` proves these strict inequalities. The geometric module works for arbitrary a,b>0 and A,B>=0. No prime distribution input is used by this deduction.

Module	Named theorems	Checked content
`InterpolationGeometry.lean`	15	Actual weights, summability, uniform total sum, actual Schur estimates
`InterpolationBlockBounds.lean`	7	Exact scaling, complex block bound, finite family estimate

The actual summable weight

For X>0 and n>=0, define

$$h_X(n) = \min\left\{ A\left(\frac{2^n}{X}\right)^a, B\left(\frac{2^n}{X}\right)^{-b} \right\}.$$

This is the Lean definition `interpolationWeight`. The proved constant is

$$C(A,B,a,b) = \frac{A}{1-2^{-a}} + \frac{2^b B}{1-2^{-b}}.$$

`interpolationWeightConstant_nonneg` proves C>=0. Both geometric ratios are strictly between zero and one by `dyadic_decay_ratio`.

First, summability itself follows for every X>0 from

$$0 \le h_X(n) \le B X^b (2^{-b})^n.$$

The factorization and comparison are proved, then the checked Mathlib geometric-series theorem gives `interpolationWeight_summable`. This initial majorant depends on X; the next argument removes that dependence.

For X>=1, construct k>=0 with 2^k<=X<2^(k+1), using the checked Archimedean power-selection theorem. For n<k,

$$h_X(n) \le A(2^{-a})^{k-n}.$$

The map n->k-n is injective on {0,...,k-1}. Therefore this finite sum is bounded by the full nonnegative geometric series A/(1-2^(-a)). This argument is checked in `finite_reversed_geometric_sum_le`; truncation of natural subtraction is handled using the explicit range condition.

For the tail, indexed by n+k with n>=0, X<=2*2^k gives

$$h_X(n+k) \le 2^b B(2^{-b})^n.$$

The negative power reverses the ratio comparison, with positivity proved before applying it. Summing the tail yields 2^b*B/(1-2^(-b)). The original convergent series is exactly its finite head plus this shifted tail.

For 0<X<1, take k=0. The head is empty, and X<=2*2^0 supplies the same tail estimate. No assertion 2^k<=X is used in this case.

Thus `interpolationWeight_tsum_le` proves, for every real X>0,

$$\sum_{n\ge 0} h_X(n) \le C(A,B,a,b).$$

The constant contains no X or N. `interpolationWeight_N_squared` gives the actual application X=N^2 for every real N>0, including N<1.

Schur with the actual weights

For any finite index sets S,T, the matrix w(i,j)=h_(N^2)(i+j) has finite row and column sums <=C. The earlier injective-shift lemma bounds each row by the now-proved total series. The finite Schur theorem therefore applies without an additional summability or geometric-bound hypothesis.

`interpolationWeight_finite_schur` proves the coefficient l2 estimate. `interpolationWeight_finite_schur_lp` gives, for 1<=P<=2,

$$\sum_{i\in S, j\in T} h_{N^2}(i+j)c_i d_j \le C\left(\sum_{i\in S}|c_i|^P\right)^{1/P}\left(\sum_{j\in T}|d_j|^P\right)^{1/P}.$$

The formal exponent is ENNReal P, with ordinary value P.toReal. Empty sets, zero weights/constants and signed real coefficients are included.

Exact scale algebra

`interpolation_scale_balance` proves, for N,f,g>0 and arbitrary real x,y,

$$N^{1-2x}f^x g^x = N^{1-2y}f^y g^y\left(\frac{fg}{N^2}\right)^{x-y}.$$

The proof expands the quotient and product powers, then combines powers of positive bases. Specializing f=2^i, g=2^j gives the dyadic identity.

For nonnegative amplitude bounds u,v, put

$$c_i = 2^{i/p}u, \qquad d_j = 2^{j/p}v.$$

The Lean definition `interpolationBlockCoefficient` uses (2^i)^(1/p), and `dyadic_pow_rpow` justifies interchange of the natural and real powers. The restricted and l2 block scalars satisfy the exact identities

$$uvAN^{1-2/q}2^{i/q}2^{j/q} = N^{1-2/p}c_i d_j A\left(\frac{2^{i+j}}{N^2}\right)^a,$$

$$uvB2^{i/2}2^{j/2} = N^{1-2/p}c_i d_j B\left(\frac{2^{i+j}}{N^2}\right)^{-b}.$$

`interpolation_block_min_identity` factors their minimum as N^(1-2/p)*c_i*d_j*h_(N^2)(i+j). Its common factor is nonnegative, including zero amplitudes. The previously proved `holderConjugate_scaling` identifies 1-2/p with the paper's 1/p'-1/p when p>1.

Actual complex convolution blocks and finite families

Suppose |F|<=2^i, |G|<=2^j, |f|<=u on F and |g|<=v on G, with u,v>=0. `finiteKernelPairing_dyadic_block_bound` proves for the actual finite sesquilinear convolution pairing

$$|B_{F,G}(f,g)| \le N^{1-2/p}c_i d_j h_{N^2}(i+j).$$

Its input is the already available restricted indicator bound at q, positivity of K, and a genuine continuous l2 realization with norm <=B. The l2 pairing estimate is derived from that actual operator. Monotonicity of positive powers gives the dyadic cardinality bounds. The checked positive-kernel extension handles all complex phases. The minimum identity above then gives the literal scale factor, without dropping powers of N.

`finite_complex_block_family_bound` sums these actual complex pairings over any specified finite families F_i,G_j with their support-cardinality and amplitude bounds. Triangle inequality and the actual-weight Schur estimate give

$$\left|\sum_{i\in S, j\in T} B_{F_i,G_j}(f_i,g_j)\right| \le N^{1-2/p}C\left(\sum_{i\in S}|c_i|^p\right)^{1/p}\left(\sum_{j\in T}|d_j|^p\right)^{1/p}.$$

The constants depend on A,B,p,q and not on N or the number of blocks. The generic theorem takes the restricted bound and genuine l2 bound as explicit inputs, along with the stated finite block data. The restricted bound is already proved from the paper's abstract high-low inputs in checkpoint 16. No partition or desired coefficient mass estimate is silently supplied.

What remains before closing the corollary

The finite family estimate does not yet compare its coefficient norms with those of arbitrary original inputs. The next steps are:

1. Construct the decreasing-magnitude dyadic rank blocks for every finite complex input, prove coverage/disjointness, and prove sum c_i^p<=3 sum |f|^p, including ties, zero input, and first/last blocks.
2. Identify the original finite bilinear pairing with the sum over these actual blocks and apply `finite_complex_block_family_bound`.
3. Prove finite-output lp norm duality with the correct complex phase, converting the finite bilinear estimate to the finite-input norm bound.
4. Apply the already checked lp realization and exact-constant density transfer, retain the two operators' l2 constants, and assemble the strong interior range with p=2.

Corollary 4.15 remains partial until these connections are complete. Its desired conclusion is not an axiom or external analytic hypothesis.

Verification and finite experiments

All 64 project modules and the root receive a fresh build with the artifact cache disabled: 3461 jobs pass without warnings or errors. All 511 named theorems pass the transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source audit scans 66 Lean files. No new mathematical axiom or deep analytic input is used.

`scripts/check_interpolation_geometry.py` independently checks 2592 finite 96-term geometric sums, enclosing 248832 terms. The six positive exponents include 1/16, and eight N values range from 1/16 to 2^20. Zero constants, N=1 and both sides of that scale are included. Fractional powers have certified rational enclosures of width at most 2^(-40): each root enclosure is checked by exact integer/rational powers. The finite sum is bounded from above and its proposed constant from below before comparison. No floating-point arithmetic or tolerance is used.

The script also checks 19440 block-minimum identities exactly with rational powers of two, including 8505 zero-amplitude cases. All pass. These finite experiments do not establish the infinite series the-

orem; Lean proves that theorem for all allowed parameters. They do not verify the remaining rank construction, duality, full strong corollary or prime-square operator inputs.

The nine completely proved unconditional numbered paper results remain unchanged. Lemma 3.21 retains its earlier explicit PNT limits independently. The archived manuscript and accepted revision fragments are unchanged; no new manuscript correction has been established.

`Inverse_Fifth_Major_Arc_Error.md`

Explicit absorption into the inverse fifth power of R

Checkpoint 28. This unit proves the R^-5 major-arc error from the previous explicit error budget, under three displayed numerical conditions. The next unit establishes these conditions uniformly for the manuscript's choice of R. The raw progression estimate remains a hypothesis.

Assume N>=1, R>=1, C>=0 and an actual progression-prefix error bounded throughout [N/2,N] by

$$NC\exp(-c\sqrt{\log N}).$$

The same beta>0 and character data must be used over that interval. The major-arc range is q<=R^(11/10), abs(theta)<=R/(q*N^2).

Exponential absorption

For R>0, the condition 7*log(R)<=t implies

$$R^2e^{-t}=\exp(2\log R-t)\le\exp(-5\log R)=R^{-5}.$$

`exponential_majorArc_absorption` proves this for every real t. In the application t=c*sqrt(log N). There is no unspecified loss in its constant.

The nonunit contribution

Suppose additionally

$$\log R\le\log N,\qquad R^5\log N\le\sqrt{N}.$$

Since sqrt(N)<=N for N>=1, these hypotheses give R^5*log(R)<=N and R^5*sqrt(N)*log(N)<=N. Multiplying the nonunit budget by the positive R^5 and N therefore proves

$$\frac{2\log R+2\sqrt{N}\log N}{N}\le\frac{4}{R^5}.$$

The exact square-root identity, positivity of N, and every denominator cancellation are checked in `nonunit_majorArc_absorption`.

Bound for the actual symbol

Combining the two calculations with checkpoint 27 proves

$$\begin{aligned}&\left|\widehat{\widetilde{A}}_{N,\Lambda}(a/q+\theta)-(\mathfrak{s}(a,q)M_{N,1}(\theta)-\mathfrak{s}_\chi(a,q)\kappa M_{N,\beta}(\theta))\right|\\&\le\frac{4+(2+4\pi)C}{R^5},\end{aligned}$$

provided the three displayed conditions hold, including 7*log(R)<=c*sqrt(log N). This is `annularMangoldt_majorArc_inverse_fifth`. It is a theorem about the genuine annular Fourier sum, including all von Mangoldt prime powers. The constant depends only on the raw prefix-error constant C.

This proves the conditional inverse-fifth deduction with finite, checkable parameter hypotheses. It does not yet remove those numerical hypotheses for the source's N,Q range, prove existence of the analytic prefix estimate, or establish the independent minor-arc bound. Full Lemma 6.5, Proposition 4.19 and Theorem 1.1 remain open.

Module	New named theorems	Result
ExponentialMajorArcBudget.lean	4	Exponential absorption, nonunit absorption, combined error and the actual conditional inverse-fifth major-arc theorem

No new deep theorem is assumed in this unit, and no numerical experiment is used as a substitute for these Lean proofs.

<!-- checkpoint-verification -->

Verification

All 92 project modules and the root were freshly rebuilt: 3607 jobs pass without warnings/errors. All 749 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 94 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. No new deep analytic input or own-paper lemma is assumed.

`Low_Ramanujan_Bound.md`

Lemma 3.20: a complete checked logarithmic bound

Checkpoint 08, 22 October 2026. Source: `Draft5.tex`, label `l:lowramanujan`, near line 1885; typeset number 3.20. The archived full-source SHA256 remains `d2c31b96cb15b00feac09ed4fd9a2b0482726c719c4b772320c0711875d04660`.

The lemma is proved as stated, for every real Q>=3 and every integer n. The implied constant is absolute, independent of both variables. The manuscript proof cites Mertens' theorem. The formal proof supplies an elementary sufficient prime-product bound, so **Mertens' theorem is not an assumed input**. No change to the statement of this lemma is needed.

The statement and the actual sums

Keep the checked definition

$$C_q(n) = \sum_{u \in A_q} c_q(n-u^2), \qquad w_n(q) = \frac{|C_q(n)|}{q\phi(q)}.$$

The Ramanujan sums are the genuine sums of the standard complex additive character over units. `cohenWeight_eq` explicitly identifies the formal `cohenWeight` with this source expression at every positive q. The harmless natural-number extension has weight zero at q=0 and weight one at q=1.

The proved bound is

$$\sum_{P^+(q)<Q} w_n(q) \le C \log Q, \qquad C = 5e^2\left(12 + \frac{2}{\log 3}\right).$$

The constant is explicit and deliberately unoptimized. The theorem `low_ramanujan_uniform_bound` states the uniform quantifiers as an existence of one positive C valid simultaneously for all real Q>=3 and all integer n.

Source/proof component	Lean declaration
Exact source absolute sum at a positive q	`cohenWeight_eq`
Nonnegative multiplicative weights	`cohenWeight_nonneg`, `cohenWeight_mul`
Finite support of the smooth series	`cohenWeight_nonzero_dvd_bound`, `smoothCohenSeries_eq_divisor_sum`
Euler product of the actual weights	`smoothCohenSeries_euler_product`
Uniform local factors	`cohenWeight_two_factor_le`, `cohenWeight_odd_prime_le`
Reduction to the source prime-product majorant	`smoothCohenSeries_le_prime_product`

Source/proof component	Lean declaration
Elementary weighted prime sum	`sum_log_div_prime_le`
Finite subset mass and logarithmic moment	`primeSubsetMass_expansion`, `primeSubsetMass_log_moment`
Half-mass concentration and logarithmic product bound	`primeSubsetMass_le_affine_log`, `primeSubsetMass_le_log`
Control of the extra local correction	`sum_inv_mul_pred_le_one`, `cohen_prime_product_le_log`
Full lemma for integer cutoffs	`low_ramanujan_bound_integer`
Full lemma for real cutoffs	`low_ramanujan_bound`, `low_ramanujan_uniform_bound`

1. Establish finite support and the exact Euler product

The checked Cohen factorization gives C_q(n)=0 whenever an odd prime square divides q or 16 divides q. This is pointwise in every integer n, so the same vanishing holds for the nonnegative weight w_n(q).

CRT multiplicativity of C_q and multiplicativity of the totient imply w_n(mr)=w_n(m)w_n(r) for coprime m,r. Complex norm multiplicativity handles the absolute values. No sign restriction on C_q is assumed.

For an integer cutoff N>=3, every possible nonzero smooth coefficient divides M=((N-1)!)^3. Conversely every divisor of M has all prime factors below N. The formal infinite sum is therefore exactly its finite divisor sum over M. The previously checked divisor Euler identity has been generalized from complex values to any commutative semiring; this proof now also applies to real weights.

After the proved vanishing removes higher exponents, the exact product is

$$\sum_{P^+(q)<N} w_n(q) = (1 + w_n(2) + w_n(4) + w_n(8)) \prod_{2<p<N} (1 + w_n(p)).$$

This step uses proved finite support rather than an unproved interchange of an infinite series and a product.

2. Bound every local factor uniformly in n

The explicit Cohen factors at 2,4,8 have absolute values at most 1,4,16. Their denominators q*phi(q) are respectively 2,8,32. Hence

$$1 + w_n(2) + w_n(4) + w_n(8) \le \frac{5}{2}.$$

At an odd prime p the two Cohen values are p+1 and 1-p, so

$$w_n(p) \le \frac{p+1}{p(p-1)}.$$

Consequently the entire series is at most

$$\frac{5}{2} \prod_{2<p<N} \left(1 + \frac{p+1}{p(p-1)}\right).$$

The estimates include n=0, negative n, primes dividing n, and every residue case; no restriction on n enters the following prime-product argument.

3. An elementary logarithmically weighted prime sum

Let S be any finite set of primes p<=N. The first term of Legendre's formula for the factorial gives floor(N/p)<=v_p(N!). Multiplication by log(p)>=0 and summation therefore give

$$\sum_{p \in S} \left\lfloor \frac{N}{p} \right\rfloor \log p \le \log(N!).$$

The elementary bound N!<=N^N implies log(N!)<=N log(N). Also

$$\sum_{p \in S} \log p \le \log\left(\prod_{p \le N} p\right) \le N \log 4,$$

using the primorial bound already proved in pinned Mathlib. Since N/p<=floor(N/p)+1, the two estimates imply

$$\sum_{p\in S}\frac{\log p}{p}\le \log N+\log 4.$$

The actual factorial, prime factorization and primorial inequalities are used in Lean. This is not an assumed weighted-prime estimate.

4. The subset expansion and its first logarithmic moment

For t a subset of S put d(t)=product_(p in t) p, including d(empty)=1. Define

$$P(S)=\prod_{p\in S}\left(1+\frac{1}{p}\right)=\sum_{t\subseteq S}\frac{1}{d(t)}.$$

Distinct subsets have distinct products, by unique prime factorization. Expanding the finite product and its first logarithmic moment gives

$$\sum_{t\subseteq S}\frac{\log d(t)}{d(t)}=P(S)\sum_{p\in S}\frac{\log p}{p+1}.$$

`weighted_powerset_moment` proves the underlying product identity by induction on S. It imposes no probability-theoretic assumption. The logarithm of the product is then identified with the sum of the logarithms.

For N>=3, log4<=2 log(N). The preceding weighted-prime sum therefore yields

$$\sum_{t\subseteq S}\frac{\log d(t)}{d(t)}\le 3P(S)\log N.$$

5. Concentrate the mass below N^6

Let A be the reciprocal mass of subsets with d(t)<=N^6, and B the remaining mass. Then P=A+B. Every product in B has log d(t)>=6 log N, so the moment bound gives

$$6B\log N\le 3P\log N,\qquad B\le P/2.$$

The products counted in A are distinct positive integers at most N^6. Thus A<=H_(N^6), where H is the harmonic sum. The checked bound H_m<=1+log(m) gives

$$P\le 2A\le 2+12\log N\le\left(12+\frac{2}{\log 3}\right)\log N.$$

This proves the sufficient Mertens-type upper bound by finite sums and an elementary harmonic estimate. It does not assert Mertens' asymptotic formula.

6. Control the summable correction in the source product

For p>=2,

$$1+\frac{p+1}{p(p-1)}=1+\frac{1}{p}+\frac{2}{p(p-1)}\le\left(1+\frac{1}{p}\right)\left(1+\frac{2}{p(p-1)}\right).$$

The elementary telescoping identity

$$\sum_{j=0}^{m-1}\frac{1}{(j+1)(j+2)}=1-\frac{1}{m+1}$$

implies sum_(p in S) 1/(p(p-1))<=1. Applying 1+u<=exp(u) to the nonnegative correction factors shows their product is at most e^2. Hence

$$\prod_{p\in S}\left(1+\frac{p+1}{p(p-1)}\right)\le e^2\left(12+\frac{2}{\log 3}\right)\log N.$$

Combining with the factor 5/2 proves the integer-cutoff estimate with constant C/2. Every sum and product in this argument is finite.

7. Preserve the strict cutoff for all real Q>=3

Put N=ceil(Q). For an integer prime p, p<N is equivalent to p<Q, including exact prime endpoints. Thus the real-cutoff series is exactly the already bounded integer-cutoff series. Since Q>=3, N<=Q^2 and log N<=2 log Q. The additional factor two gives the stated constant C for all real Q>=3.

Verification and scope

All 21 modules were freshly compiled locally with pinned Lean 4.34.1 and Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612`. All 186 named project theorems pass the transitive axiom audit: only propext, Classical.choice and Quot.sound occur. The source scan passes for 23 Lean files. There are no proof placeholders, native proof shortcuts, or added mathematical axioms.

The exact rational experiment checks 4,640 cases, covering every residue class in the full period for each of twelve integer/noninteger cutoffs. The actual supported-modulus sum agrees with its Euler product and satisfies the rational local-factor majorant in every case. See `verification/low-ramanujan-numerical.json`. The logarithmic estimate is proved in Lean; these finite experiments supply supplementary evidence.

The user has confirmed applying both the Lemma 3.14 and Corollary 3.16 statement replacements. No further correction is requested for those supplied updates. The archived full source and PDF retain their original hashes. No headline theorem is yet complete and no deep analytic input has been assumed.

`Major_Arc_Error_Budget.md`

A uniform explicit major-arc error budget

Checkpoint 27. This unit reduces the proved error for the actual annular von Mangoldt symbol to a single R-dependent bound. It uses the raw progression-prefix hypothesis from checkpoint 26; no new distribution theorem or own-paper conclusion is assumed.

For N>=1, q>=1, R>=1 and delta>=0, suppose

$$q \le R^{11/10}, \qquad |\theta| \le \frac{R}{qN^2}, \qquad |C_{q,u}(t) - H_{q,\kappa\chi(u),\beta}(t)| \le N\delta \quad (N/2 \le t \le N)$$

for every reduced residue u. The actual C,H,chi and beta>0 are the objects defined in the previous unit. The same exceptional data are used throughout this interval; no consistency across different N is asserted.

The denominator and frequency calculation

The width condition implies q*abs(theta)*N^2<=R, by multiplying by the positive q*N^2. Since R>=1, real-power monotonicity gives R^(11/10)<=R^2, and also R<=R^2. Totient(q)<=q therefore implies

$$\phi(q)\delta(2 + 4\pi|\theta|N^2) \le \delta(2q + 4\pi R) \le (2 + 4\pi)R^2\delta.$$

This explicitly removes the dependence on q and theta from the residue error in the whole source major-arc range. Positivity conditions are displayed in Lean rather than used implicitly in denominator cancellation.

Monotonicity of the logarithm and log(R^2)=2*log(R) give log(q)<=2*log(R). Substitution into the previously proved nonunit bound produces the following result for the genuine finite Fourier symbol:

$$\begin{aligned}\left|\widetilde{A}_{N,\Lambda}(a/q+\theta) - (\mathfrak{s}(a,q)M_{N,1}(\theta) - \mathfrak{s}_\chi(a,q)\kappa M_{N,\beta}(\theta))\right| \\ \le \frac{2\log R + 2\sqrt{N}\log N}{N} + (2+4\pi)R^2\delta.\end{aligned}$$

`annularMangoldt_majorArc_source_R_budget` is the paper-facing theorem. An earlier theorem proves the same bound under the weaker denominator restriction q<=R^2. This is a quantitative implication, with an explicit constant, all prime powers retained and the exact annular normalization.

Scope and continuation

The next step absorbs these two positive errors into a constant times R^-5 and proves the required parameter inequalities for the source's R=max(exp((log N)^(1/10)),Q^8). The raw progression estimate

remains an explicit analytic input. The independent minor-arc statement and full Lemma 6.5 remain open. Theorem 1.1 is an explicit later target, after Proposition 4.19 and the application of the proved abstract high-low result.

Module	New named theorems	Result
MajorArcErrorBudget.lean	6	Positive-denominator phase budget, real-power and logarithmic bounds, totient error control and actual major-arc error reduction

There is no new numerical experiment or new deep assumption in this unit.

<!-- checkpoint-verification -->

Verification

All 91 project modules and the root were freshly rebuilt: 3606 jobs pass without warnings/errors. All 745 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 93 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. No new deep analytic input or own-paper lemma is assumed.

`Orlicz_A_Scalar_Balance.md`

Theorem 4.1: Orlicz growth and sharp A_N scalar balance

Checkpoint 14 continues the B_N endpoint proved in checkpoint 13. It proves four new modules, containing 34 named theorems, for the A_N proof. Theorem 4.1 remains partial: its uniform A_N conclusion from actual operator norm hypotheses has not yet been assembled.

Source and exact scope

The source is `sources/current/Draft5.tex`, Theorem 4.1, `t:improvinggeneral`, with `e:Abalance`, `e:endimprovan`, `e:cstar`, `e:Lognotation` and `e:defPhi`. The archived TeX/PDF are unchanged.

The new modules are:

Module	Checked content
`OrliczGrowth.lean`	Exact Phi identities, three limits, bounded-error control, uniform logarithmic absorption and uniform sub-power bounds
`OrliczBalance.lean`	Literal high-weight scaling, sharp leading coefficient, strict balancing margins, actual dyadic choice and all three exponent limits
`OrliczScalar.lean`	Exact normalization of low/high terms and a bound for the actual three-term scalar sum at that dyadic choice
`OrliczCutoff.lean`	Complementary density-range absorption, admissibility of the selected Q and the actual error decay factor

No endpoint estimate for A_N, scalar optimization theorem, analytic asymptotic or prime distribution theorem is assumed. Each limiting and inequality statement is proved from the pinned Mathlib. The functions q used in intermediate scaling lemmas are general real functions; the final scalar lemmas construct an actual dyadic Q. The final `exists_admissible_dyadic_orlicz_bound` does not take a desired asymptotic of Q as a hypothesis.

Exact logarithmic conventions

Write

$$L = \mathrm{Log}\,\delta = 2 + |\log \delta|, \qquad W_\kappa(L) = \frac{L^\kappa}{2 + \log L}.$$

The definition `orliczWeight` is exactly W. The checked identity is

$$\Phi_{c,\kappa}(\delta) = \exp(cW_\kappa(\operatorname{Log}\delta)).$$

For 0<delta<=1, L=2-log(delta), and delta=exp(2-L). The identity `paperLog_exp_two_sub` is proved for every L>=2. The additive twos in the paper's definitions are retained throughout.

For kappa>0 the module proves

$$\frac{\log L}{W_\kappa(L)} \to 0, \qquad \frac{1}{W_\kappa(L)} \to 0.$$

For kappa<1 it proves W_kappa(L)/L -> 0. For the first limit, the exact quotient is `(2 log L + (log L)^2)/L^kappa`. Both terms tend to zero by Mathlib's proved logarithm-versus-power limits. The inverse-weight limit uses `(2+log L)/L^kappa`. The sublinear-weight limit factors as `L^(kappa-1)/(2+log L)`.

These limits yield actual uniform estimates, including their bounded range:

- For every kappa>0, c>0 and real m, there is D>0 such that `(Log x)^m <= D*Phi(c,kappa,x)` for every x. In particular this holds for every positive density.
- For 0<=kappa<1, c>=0 and epsilon>0, there is D>0 such that `Phi(c,kappa,delta) <= D*delta^(-epsilon)` simultaneously for 0<delta<=1.
- Consequently `delta^theta*Phi(c,kappa,delta)` is bounded by `D*delta^(theta-epsilon)` on that same interval, for arbitrary real theta.

Constants are quantified before delta. The bounded-range argument selects a fixed threshold U>=2 and uses elementary real-power inequalities below U. Thus these are not only statements about an unspecified sufficiently small density. The last estimate is useful for the later interpolation argument; that corollary itself is still open.

Literal high denominator

The high term in `e:highgeneral` uses `(log Q)^kappa/log(log Q)`. Put q=log Q. For any function q(L) with q(L)/L -> k>0, the proof establishes

$$\frac{\log q(L)}{\log L} \to 1, \qquad \frac{q(L)^\kappa/\log q(L)}{W_\kappa(L)} \to k^\kappa.$$

The logarithmic ratio follows by taking the logarithm of q(L)/L, whose limit is log(k), and dividing by log L -> infinity. For the second ratio, the exact algebra is

$$\frac{q(L)^\kappa/\log q(L)}{L^\kappa/(2+\log L)} = \left(\frac{q(L)}{L}\right)^\kappa \frac{1+2/\log L}{\log q(L)/\log L}.$$

All needed positivity and nonvanishing conditions hold eventually and are checked in Lean. The literal denominator and the paper's LogLog denominator are connected by this proved limit, with no informal replacement of one by the other.

Sharp coefficient and actual dyadic scale

Let u=s+r, k=1/(2u), where s,r>0. Define

$$c_{\text{lead}} = \frac{c_1 s}{2^\kappa u^{\kappa+1}} = s\frac{c_1 k^\kappa}{u}.$$

`orliczLeadingCoefficient_eq` proves this equality with real powers. Given cstar>c_lead and c1>0, the proof chooses b satisfying

$$\frac{c_1 k^\kappa}{u} < b < \frac{cstar}{s}.$$

In particular b>0 and

$$sb < cstar, \qquad c_1 k^\kappa - rb < sb < cstar, \qquad -rb < cstar.$$

The candidate logarithm is y(L)=kL+bW_kappa(L). Since W/L -> 0, y/L -> k, so y -> infinity. An actual dyadic Q=2^n is chosen by rounding exp(y) down:

$$Q \le e^{y(L)} < 2Q.$$

For all L the chosen Q is at least two. The two-sided rounding bounds hold for sufficiently large L. They give `abs(log Q-y(L)) <= log 2`. The inverse-weight limit proves

$$\frac{\log Q - kL}{W_\kappa(L)} \to b.$$

The index function n(L) is fixed before the low logarithmic power alpha, the error power M, the error decay exponent A and N. Its existence is proved using the earlier dyadic rounding lemma; it is not an extra hypothesis.

The three actual scalar terms

Set delta=exp(2-L), q=log Q and theta=1-s/(2u). After division by delta^theta, the low, high and error terms are exponentials with respective exponents

$$\begin{aligned} E_{\text{low}} &= s(q-kL) + \alpha \log q + 2sk, \\ E_{\text{high}} &= -r(q-kL) + c_1 q^\kappa / \log q - 2rk, \\ E_{\text{err}} &= -r(q-kL) + M \log q - 2rk. \end{aligned}$$

The equalities for this normalization are proved in `orlicz_low_term_exact` and `orlicz_high_term_exact`; the constants from the additive two in L are included. The three exponent ratios tend to

$$E_{\text{low}}/W \to sb, \quad E_{\text{high}}/W \to c_1 k^\kappa - rb, \quad E_{\text{err}}/W \to -rb.$$

The polynomial logarithms vanish in these ratios by the proved limits. The strict margins therefore give, for all sufficiently large L,

$$\begin{aligned} \delta Q^s (\log Q)^\alpha + \delta^{1/2} Q^{-r} \exp\left(c_1 \frac{(\log Q)^\kappa}{\log\log Q}\right) + \delta^{1/2} Q^{-r} (\log Q)^M \\ \le 3\delta^\theta \Phi_{cstar,\kappa}(\delta). \end{aligned}$$

This is `exists_dyadic_orlicz_scalar_bound`. It is an inequality for the actual terms, beyond a limit calculation of their logarithms. The density threshold may depend on alpha and M, as allowed for fixed theorem parameters. The proof uses strict margins for the epsilon formulation of the conclusion. It does not need bounded-factor comparability of the low and high terms at the paper's first-order displayed choice.

Cutoff and error decay

The dyadic choice satisfies log Q<=y(L), while y(L)/L -> k. Consequently log Q <= (k+1)L eventually. If

$$L \le \frac{c_0}{k+1} (\log N)^c,$$

the source cutoff log Q <= c0*(log N)^c follows. This is checked for the same Q as in the scalar inequality.

For N>=1 and A>=0, `error_decay_factor_le_one` proves `(1+log N)^(-A) <= 1`. All other factors of the error term are nonnegative, so multiplying by this decay factor preserves the scalar upper bound. `exists_admissible_dyadic_orlicz_bound` combines the cutoff, the actual dyadic choice and the three-term bound with this decay factor. It works uniformly in N and A once the sufficiently-large-L condition holds. The source's separate constants implicit in the operator bounds and C_A must still be carried through the operator bridge; they are not certified as equal to one.

Complementary density range

For any d>0, a>=0, c>0 and x>=0, the elementary cutoff implication is proved:

$$dx^c \le L \quad \Longrightarrow \quad x^a \le d^{-a/c} L^{a/c}.$$

It follows that for any real theta,tau and cstar,kappa>0, one D>0 works for all N>=1 and 0<delta<=1 satisfying `d*(log N)^c <= Log delta`:

$$(\log N)^a \delta^\theta (\mathrm{Log}\,\delta)^\tau \le D\delta^\theta \Phi_{cstar,\kappa}(\delta).$$

This is `log_density_domination_le_orlicz`. Applied to the already proved B_N exponent tau, it is the scalar absorption required in the complementary range of the source proof. The positive-operator domination and finite pairing bridge needed for that application remain to be connected.

What remains before marking Theorem 4.1 complete

1. Deduce the actual three-piece A_N pairing inequality from its genuine l1-to-linfinity and l2-to-l2 operator hypotheses, retaining the fixed operator constants and the chosen C_A. Use the existing indicator/lp infrastructure from the B_N proof.
2. Connect positive B_N kernels and the stated A_N domination to a pairing domination; use the B_N theorem and the complementary-range scalar bound.
3. Handle the bounded-density range uniformly, including bounded N and an empty admissible Q range. A fixed Q=2 or 4 may only be used after proving it is admissible. State the discrete N domain explicitly; the new cutoff and error-decay results assume N>=1.
4. Include empty input/test sets and combine all ranges with one constant quantified before N,F,G. Only then mark `e:endimprovan` and the full theorem complete. The following interpolation corollary and the prime-square operator norm estimates are later obligations.

No new source correction has been established in this checkpoint. No unproved own-paper lemma has been classified as a deep external input. The existing explicit PNT hypotheses in Lemma 3.21 remain separate and are not used anywhere in these four modules.

Verification

The checkpoint contains 50 project modules and 406 named project theorems. All 50 modules receive a fresh local build with the artifact cache disabled; all 406 named theorem declarations receive the complete transitive axiom audit. The whitelist is exactly propext, Classical.choice and Quot.sound. The source scan covers 52 Lean files. See the accompanying current logs.

No new numerical experiment is claimed for these unbounded real-parameter analytic estimates. Their verification is the Lean proof. Earlier finite experiments are preserved with their original scopes, including the B_N rational scalar and Gaussian-integer convolution checks; those experiments do not validate the new A_N asymptotics. The ZIP manifest is checked against every archived entry, and the frozen paper sources keep their recorded hashes.

`Prefix_Error_Transport.md`

Quantitative transport of the actual progression-prefix error

Checkpoint 25. This unit turns a bound on the actual progression prefix into a bound on the quadratically weighted residue sum. It proves the integration-by-parts deduction; the progression distribution estimate itself remains an explicit input boundary.

Prefix-error integral

Let g(t)=exp(-2*pi*i*theta*t^2), as constructed previously. Suppose 0<=a<=b<=N and R is a complex function satisfying |R(t)|<=B on [a,b]. The exact derivative bound gives

$$\left| \int_{(a,b]} g'(t) R(t)\, dt \right| \le 4\pi |\theta| N B (b-a).$$

Both endpoint terms have norm at most B, because |g|=1. The triangle inequality therefore proves

$$|g(b)R(b) - g(a)R(a) - \int_{(a,b]} g'(t)R(t)\, dt| \le B(2 + 4\pi|\theta|N(b-a)) \le B(2 + 4\pi|\theta|N^2).$$

The last inequality uses b-a<=N and B>=0. The explicit 2 is the sum of two endpoint bounds. The integral estimate uses Mathlib's actual Bochner integral norm bound. No differentiability assumption on the prefix-error function is needed.

Subtracting a differentiable model

Let C(t) be the actual complex progression von Mangoldt prefix from `progressionMangoldtPrefix`. Let H,H' be complex functions such that H has derivative H' at every point of [a,b], and H' is interval-integrable. All these regularity conditions will be discharged for the source model in the next unit.

We first prove integrability of g'(t)C(t), using Mathlib's theorem for a smooth factor times a floor-indexed finite prefix. Continuity of H and integrability of g' give integrability of g'H. Thus subtracting their integrals is justified, rather than used as a formal rule without its hypotheses.

Integration by parts for gH gives

$$\int_{(a,b]} g(t)H'(t)\,dt = g(b)H(b) - g(a)H(a) - \int_{(a,b]} g'(t)H(t)\,dt.$$

Subtract this from the actual Abel identity proved in checkpoint 24:

$$W_r - \int_{(a,b]} g(t)H'(t)\,dt = g(b)(C(b)-H(b)) - g(a)(C(a)-H(a)) - \int_{(a,b]} g'(t)(C(t)-H(t))\,dt.$$

`squareResidueSum_model_error_identity` proves this exact identity. W_r is the actual unnormalized residue sum with the floor cutoffs at a,b. It includes all von Mangoldt prime powers and the paper's negative phase.

Applying the prefix-error bound with R=C-H proves

$$|W_r - \int gH'| \le B(2+4\pi|\theta|N^2).$$

The hypothesis is a bound on the genuine unweighted prefix C-H; no weighted-residue estimate or own-paper lemma is assumed.

Annular normalization

Take a=N/2, b=N with N>=1. If the actual prefix difference is bounded by N*delta throughout [N/2,N], then the normalized residue error is

$$\left|\frac{W_r}{N} - \frac{1}{N}\int_{N/2}^{N} g(t)H'(t)\,dt\right| \le \delta(2+4\pi|\theta|N^2).$$

`annularProgressionModel_error_bound` proves this statement. The real lower endpoint and natural floor cutoff are matched explicitly. Positivity of N justifies cancellation of 1/N against the prefix error N*delta. No additional N factor is lost. Delta>=0 is displayed.

Files

Module	New named theorems	Result
PrefixErrorBounds.lean	2	Endpoint plus derivative error bounds, including the N^2 version
ProgressionModelError.lean	4	Actual prefix-product integrability, exact model subtraction, quantitative and normalized annular bounds

The next unit proves regularity and the derivative of the source model x/phi(q)-z*x^beta/(beta*phi(q)), for beta>0 on a positive interval, and identifies its integral with the actual ordinary and exceptional oscillatory integrals. The raw progression-prefix estimate remains an explicit analytic boundary. The final R^-5 and minor-arc statements of Lemma 6.5 are not concluded here.

<!-- checkpoint-verification -->

Verification

All 88 project modules and the root were freshly rebuilt: 3603 jobs pass without warnings/errors. All 723 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 90 Lean files. Exact unique import/audit coverage and unchanged

frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. No new deep analytic input or own-paper lemma is assumed.

`Prime_Square_Operators.md`

Actual prime-square operators and annular reconstruction

Checkpoint 21 constructs the source's actual von Mangoldt and ordinary square averages, proves their elementary operator inputs, and proves the exact annular identities and dyadic reconstruction. Five modules add 48 named theorems. All 75 modules and the root have a fresh clean build (3482 jobs); all 613 named theorems pass the transitive axiom audit, and 77 Lean files pass the source scan.

Proposition 4.19 is still open. Its Fourier low, high, error and maximal estimates are not proved or assumed in these new results. No headline prime-square application is counted as complete. The abstract Theorem 4.1 and full abstract Corollary 4.15 remain complete.

Source definitions and domains

The source averages are equation `e:basicaverages`, near line 993 of `sources/current/Draft5.tex`:

$$A_{N,w}f(x) = \frac{1}{N}\sum_{0<n\le N} w(n)f(x-n^2).$$

The two weights are w(n)=1 and Mathlib's actual ArithmeticFunction.vonMangoldt. In particular, Lambda(p^j)=log(p) for every j>=1. Prime powers are retained. Replacing Lambda by a sum over primes alone would change the source operator.

N is the arithmetic cutoff; the spatial scale is N^2. The formal definitions take N:Nat, which includes all dyadic scales in Proposition 4.19. The elementary uniform norm theorems apply to every N>0. The definitions also have a total N=0 extension, giving zero operators; the source's positive-scale statements do not rely on that extension.

For every finite sequence-space exponent P with 1<=P and p=P.toReal>0, weightedSquareAverage hp N w is a genuine continuous complex-linear map lp(P)->lp(P). Its underlying coordinate expression is precisely the displayed sum. The annular definition is

$$\widetilde{A}_{N,w}f(x) = \frac{1}{N}\sum_{N/2<n\le N} w(n)f(x-n^2),$$

using natural integer division for N/2. This is the source's `e:tildedef2` at dyadic scales. No finite-support restriction on f is imposed on either continuous operator.

Constructing the actual operators and kernels

integerTranslate hp d sends f(x) to f(x-d). Its membership in lp follows by reindexing the summable norm powers through the integer translation bijection. Reindexing the norm sum proves exact norm preservation. The map is complex-linear and continuous with operator norm at most one.

For a finite S:Finset Nat and a real normalizer c, define

$$T_{S,c,w} = c\sum_{n\in S} w(n)\tau_{n^2}, \qquad K_{S,c,w}(z) = c\sum_{n\in S}\mathbf{1}_{z=n^2}w(n).$$

finiteSquareOperator and finiteSquareKernel are these actual objects, with real coefficients embedded in the complex scalars. The norm bound

$$\|T_{S,c,w}\| \le |c|\sum_{n\in S}|w(n)|$$

follows from the triangle inequality and the translation norm bound. For c>=0 and nonnegative weights this is at most c*sum w(n).

The proof natSquare_int_injective checks injectivity of n->n^2 from nonnegative natural indices into the integers. Consequently the kernel at m^2 is c*w(m) when m belongs to S and is zero otherwise. It is zero outside the finite image of S. Positive coefficients give a nonnegative real-valued kernel, and coefficientwise domination gives kernel domination.

finiteSquareKernel_finiteInput proves the exact identity with the existing finiteConvolution for every finite set and arbitrary complex coefficients. It interchanges the two finite sums and uses x-y=n^2 iff y=x-n^2. Thus the continuous maps agree with actual convolution; they are not unrelated maps with only a scalar norm model.

Module	New named theorems	Checked role
FiniteSquareOperators.lean	8	Genuine lp translations, coordinates and finite operator norm
SquareAverageKernels.lean	12	Actual kernel, finite convolution, positivity, support and local bound
WeightedSquareAverages.lean	10	Full/annular definitions, normalization, mass and spatial scale
PrimeSquareAverages.lean	10	Actual Lambda, elementary Chebyshev bound and source elementary inputs
DyadicSquareReconstruction.lean	8	Exact half subtraction, telescoping, reindexing and both bottom terms

Uniform norms without a PNT assumption

Mathlib's pinned Chebyshev theorem psi_le_const_mul_self proves

$$\sum_{0<n\le N}\Lambda(n)=\psi(N)\le(\log 4+4)N.$$

The equality of the sums, including their endpoint conventions, is checked in mangoldt_square_mass_bound. It uses the natural-floor identity at N. This is an elementary upper bound already proved in Mathlib, not an unformalized PNT input. Put C_Lambda=log4+4>0.

The actual full and annular operator norms are at most C_Lambda for Lambda and at most one for ordinary squares, uniformly over N>0 and every finite p>=1. The annular mass is bounded by the full mass because its index set is a subset and the weights are nonnegative. Normalizing by 1/N gives the bounds. In particular, the actual kernels satisfy the existing genuine KernelL2Bound definition with constants C_Lambda and 1.

These are bounds on each operator, uniform in N. They do **not** assert an l2 bound for the supremum over N. Proposition 4.19's maximal high estimate remains a separate open theorem.

Positivity, domination and the local spatial bound

For S contained in (0,N], the kernel vanishes outside [1,N^2] in the integers. The support theorem proves this from the actual square indices; it is not a support assumption.

Use the finite window W_N(x)=[x-N^2,x] intersected with the integers. The kernel support gives the exact localization

$$(K*\mathbf{1}_F)(x)=(K*\mathbf{1}_{F\cap W_N(x)})(x).$$

Injectivity of the square indices and the ordinary coefficient 1/N then give

$$|\widetilde{A}_{N,\mathbb{N}^2}\mathbf{1}_F(x)|\le N^{-1}\|\mathbf{1}_{F\cap W_N(x)}\|_1=(N^2)^{-1/2}\|\mathbf{1}_{F\cap W_N(x)}\|_1.$$

The equality of scale factors is proved in square_inverse_spatial_scale. The one-sided window is smaller than the source's symmetric window; the estimate supplies the indicator-local bound used by HighLowEndpointInputs.

Mathlib proves 0<=Lambda(n)<=log(n). Since 0<n<=N, logarithm monotonicity gives Lambda(n)<=log(N). Both actual kernels are positive, so summing coefficientwise domination proves

$$|\widetilde{A}_{N,\Lambda}\mathbf{1}_F(x)|\le\log N\,\widetilde{A}_{N,\mathbb{N}^2}\mathbf{1}_F(x)\le\log(N^2)\,\widetilde{A}_{N,\mathbb{N}^2}\mathbf{1}_F(x).$$

The last comparison is used at spatial scale N^2 with source a=1. These are the indicator versions consumed by the already checked abstract theorem. The structure AnnularSquareElementaryInputs packages both positive kernels, domination, localization bound and two genuine l2 bounds. annular_square_elementary_inputs constructs it for every N>0; none of its fields is an unproved assumption.

Exact annular difference and dyadic reconstruction

For every natural M and arbitrary real weight w, the actual continuous maps obey

$$\widetilde{A}_{2M,w} = A_{2M,w} - \tfrac{1}{2}A_{M,w}.$$

The proof splits (0,2M] into (0,M] and (M,2M], then checks the normalizers (1/2)*(1/M)=1/(2M), including M=0 in the total extension. There is no unjustified replacement of floor(N/2) by N/2 for odd N.

The base scale is A_(1,w)=w(1)*tau_1. Induction proves the weighted telescoping identity for N=2^k. Normalizing and reflecting the finite index range gives the source's descending order:

$$A_{2^k,w} = \frac{w(1)}{2^k}\tau_1 + \sum_{j=0}^{k-1} 2^{-j}\widetilde{A}_{2^{k-j},w}.$$

The sum is empty at k=0. The complex scalar coefficients are exactly the real dyadic coefficients embedded in C. The reflection and coefficient cancellation are checked, using nonvanishing of powers of two.

For Lambda, Mathlib proves Lambda(1)=0, so the bottom term vanishes. For ordinary squares, w(1)=1, so it is N^-1*f(x-1). The final declarations mangoldtSquareAverage_dyadic_reconstruction and ordinarySquareAverage_dyadic_reconstruction match the source formulas `e:annularreconstruction` and the following square-average identity. These are equalities of actual continuous maps, not only identities for indicators or finite coefficient arrays.

Remaining Proposition 4.19 dependencies

The source correspondence below distinguishes completed prerequisites from unproved conclusions. No listed open own-paper lemma is treated as a deep published theorem or assumed in the new Lean results.

Source	Dependency and remaining work
e:basicaverages, e:tildedef2, e:annularaverages, e:annularreconstruction	Actual definitions and operator identities now checked
Elementary hypotheses in the proof of Theorem 1.1	Positivity, indicator domination/localization and genuine uniform l2 bounds now checked at N^2
Lemma 6.5, l:prime-square-input	Open: exact annular Fourier sums, non-unit/prime-power bookkeeping, residue grouping and partial summation; published Landau--Page and minor-arc inputs need precise boundaries before use
Lemma 6.9, l:aaplemma	Open: actual major-arc model, separation, complementary range, approximation error and its Fourier-to-l2 bridge
Proposition 6.14, p:lowcontribution	Open: actual smooth low multiplier, inverse-transform decay, ordinary/exceptional arithmetic assembly and local maximal average
Lemma 6.16, l:notBourgain, and c:orlicz-prefix	Open: own-paper multifrequency and prefix estimates; no blanket transference or maximal assumption introduced
Proposition 6.21, p:highupperbound	Open: high and maximal high estimates, actual Gauss/conductor bounds, and exceptional characters varying with scale
Proposition 4.19, p:decompositionofan	Open: construct and join the low/high/error operators and every stated bound; elementary averages alone do not prove it
Theorem 1.1, t:improvingfixed	Open: apply the checked abstract theorems after the actual decompositions, then connect the annular endpoint and the general cutoff

The existing arithmetic results remain available. Lemma 3.21's sharp asymptotic still has its two earlier explicit PNT-limit hypotheses; any future consumption must retain that boundary. It is not used in this checkpoint's Chebyshev norm proof.

The next coherent unit is the genuine integer Fourier/Plancherel bridge and the exact finite multiplier identities. The pinned Mathlib source Mathlib.Analysis.Fourier.AddCircle already contains a Fourier Hilbert basis and Parseval infrastructure, but it has not yet been imported or connected here. Its basis

uses the positive phase, while the paper's transform uses e(-n*theta); the index/sign must be explicitly matched.

For later maximal bounds, q0, beta and the exceptional character are scale-dependent in Lemma 6.5. They must not be silently frozen across N. The low term's exceptional cutoff d<=Q/B0 and the complementary high terms must retain both the conductor and current-scale restrictions. The actual low and high kernels need not have the compact support of the original averaging kernel.

Verification and independent exact experiments

All 75 project modules and the root were freshly compiled with the artifact cache disabled: 3482 jobs pass without warnings or errors. Every one of the 613 named project theorems passes the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 77 Lean files with no placeholders or added mathematical axioms. Exact import/module and unique theorem/audit coverage were checked, as were all frozen source hashes. No Lean source changed after these checks.

scripts/check_square_averages.py uses rational coefficients of formal log(prime) symbols and exact Gaussian rational arithmetic; no floating point is used. It checks 1542 support cases, 1028 positivity cases, 387 annular difference identities, 39 ascending and 39 descending dyadic reconstructions through arithmetic cutoff 4096, 11256 complex finite convolution identities, 14280 local indicator bounds and 184 translation norm cases at exponents two and four. All pass.

Three diagnostics detect changed conventions: omitting prime powers loses log(2)/4 at shift 16 for N=4; omitting the ordinary-square bottom term loses 1/4 at shift one for N=4; using the half subtraction at odd N=3 incorrectly leaves -1/6 at shift one. The last diagnostic is outside the source's dyadic hypothesis and is not a manuscript error.

The experiment checks finite identities and elementary probes, not Chebyshev's analytic inequality, the full operator norm, a Fourier estimate or Proposition 4.19. Those current general elementary norm statements are Lean proofs. Exact ranges, digests and limitations are in verification/square-averages-numerical.json. The source manuscript and accepted revisions are unchanged; no new source edit is pending.

`Principal_Gauss_Bounds.md`

Lemma 3.21: the finite Gauss-sum argument

Historical checkpoint 09 exposition. Checkpoint 10 completes the remaining paper-specific asymptotic deduction with explicit PNT hypotheses; read `Sharp_Gauss_Asymptotic.md` for the current complete conditional statement.

Checkpoint 09, 22 October 2026. Source: `Draft5.tex`, label `l:inftyboundsaq`, especially `e:primepowerestimate` and `e:sproductbound`. The archived source SHA256 remains `d2c31b96cb15b00feac09ed4fd9a2b0482726c719c4b772320c0711875d04660`.

The finite arithmetic part is complete. The full sharp asymptotic statement of Lemma 3.21 is not yet complete. All prime estimates, prime-power vanishings, CRT coefficients, global support and the actual finite product bound below are proved in Lean. The remaining prime-distribution/asymptotic step is identified at the end. No new mathematical axiom or deep analytic input has been assumed.

Exact proved statement

For every positive integer q and every integer a coprime to q,

$$|\mathfrak{s}(a,q)| \le \frac{4}{\sqrt{q}} \prod_{\substack{p\mid q\\ p>2}} (1-p^{-1/2})^{-1}.$$

The sum vanishes if 16 divides q or an odd prime square divides q. More explicitly, a nonzero sum implies

$$q = 2^j d, \quad 0 \le j \le 3, \quad d \ge 1, \quad 2 \nmid d, \quad d \text{ squarefree}.$$

These are unbounded results for the actual exponential sums, not numerical observations or hypotheses about a proposed local model. The constant four is explicit and unoptimized. Modulus one is included; modulus zero is excluded by the positive-modulus convention.

Component	Checked declaration
Actual complete quadratic exponential sum	`completeQuadraticGaussSum`
Complete prime sum times its conjugate equals p	`completeQuadraticGaussSum_mul_conj`
Exact prime norm sqrt(p)	`norm_completeQuadraticGaussSum_prime`
Unit prime sum is complete sum minus one	`unitQuadraticGaussSum_prime`
Unit prime sum bound sqrt(p)+1	`norm_unitQuadraticGaussSum_prime_le`
Actual source normalization and sign	`principalSquareSum_as_unitGauss`, `norm_principalSquareSum`
Uniform normalized bound at every modulus	`norm_principalSquareSum_le_one`
Local odd-power and 2-power cancellation	`principalSquareSum_odd_prime_pow_eq_zero`, `principalSquareSum_two_pow_eq_zero`
Standard character with correct CRT scaling	`stdAddChar_crt`
Exact CRT for actual sums	`unitQuadraticGaussSum_coprime_mul`, `principalSquareSum_coprime_mul`
Actual sum bounded by proved prime-power majorants	`scaled_norm_principalSquareSum_le_product`
Global vanishings	`principalSquareSum_odd_prime_sq`, `principalSquareSum_sixteen_dvd`
Support as a squarefree odd-part decomposition	`principalSquareSum_support_decomposition`
Explicit finite product bound	`principalSquareSum_finite_product_bound`, its `_int` version

1. Retain the actual definitions

Write

$$U_q(a) = \sum_{u \in A_q} e(au^2/q), \qquad G_q(a) = \sum_{r \bmod q} e(ar^2/q).$$

`unitQuadraticGaussSum` was already defined in the Cohen development. The new `completeQuadraticGaussSum` is exactly the second sum. The source's principal normalized square sum is

$$\mathfrak{s}(a,q) = \phi(q)^{-1} U_q(-a).$$

`principalSquareSum_as_unitGauss` checks this sign and normalization explicitly. The norm formula divides |U_q(-a)| by the real positive totient phi(q). All character values have norm one, so the triangle inequality and the actual unit-group cardinality give |s(a,q)|<=1, even without a being a unit.

2. Evaluate the complete sum at an odd prime

Let p be odd prime and a nonzero modulo p. Multiplying the complete sum by its complex conjugate and combining the characters gives

$$G_p(a)\overline{G_p(a)} = \sum_{u,v \bmod p} e(a(u^2 - v^2)/p).$$

The change of variables s=u-v, t=u+v is a bijection since two is invertible modulo p. Its inverse is u=(s+t)/2, v=(t-s)/2. The formal equivalence proves both inverse identities. Consequently the double sum is

$$\sum_{s,t \bmod p} e(ast/p) = p.$$

For s=0, every term in the inner sum is one. For s nonzero, additive-character orthogonality makes that inner sum zero. Taking the real norm square proves |G_p(a)|=sqrt(p).

The nonunits modulo a prime consist exactly of zero, whose quadratic phase is one. Thus U_p(a)=G_p(a)-1 and

$$|U_p(a)| \le \sqrt{p} + 1, \qquad |\mathfrak{s}(a,p)| \le \frac{\sqrt{p}+1}{p-1} = \frac{1}{\sqrt{p}-1}.$$

This proof uses a finite bijection and the already checked standard character orthogonality. No Gauss-sum evaluation theorem is assumed as a new input.

3. Reuse the checked cancellation at prime powers

The earlier Cohen work constructed a permutation of the units

$$u \longmapsto u + t/u$$

when t^2=0. It changes u^2 by 2t. If 2t is nonzero and a is a unit, the quadratic sum is multiplied by a character value different from one and therefore vanishes.

For q=p^j with p odd and j>=2, take t=p^(j-1). For q=2^j with j>=4, take t=2^(j-2). The required square-zero and nonzero-phase conditions are proved from exact divisibility inequalities. The new local theorems apply the previous generic cancellation directly to U_q(a), then to s(a,q).

This establishes the local support without relying on an unproved recurrence for the complete Gauss sum. At the three remaining powers of two the simple bound |s|<=1 gives sqrt(2^j)*|s|<=sqrt(8)<=4.

4. CRT with the correct coefficients

For coprime positive m,n, define the local coefficients

$$a_m = (a \bmod m)n^{-1} \pmod m, \qquad a_n = (a \bmod n)m^{-1} \pmod n.$$

The exact standard-character identity has these inverse-modulus factors. Reindexing the unit sum by the checked CRT equivalence proves

$$U_{mn}(a) = U_m(a_m)U_n(a_n), \qquad \mathfrak{s}(a, mn) = \mathfrak{s}(a_m, m)\mathfrak{s}(a_n, n).$$

The coefficient transforms are retained in the formal statements. This is the CRT factorization needed in the source's multiplicativity argument. Reduction preserves units, and each inverse-modulus factor is a unit, so unit a produces unit a_m,a_n. The local norm bounds are uniform in these coefficients. Multiplicativity of the totient and sqrt(mn)=sqrt(m)*sqrt(n) give the corresponding identity for sqrt(q)*|s(a,q)|.

5. Assemble a proved global majorant

For a prime power p^j define a real local majorant L(p,j). Its value at j=0 is one. For j>0 it is

$$L(p,j) = \begin{cases} 4, & p = 2,\ 1 \le j \le 3, \\ 0, & p = 2,\ j \ge 4, \\ (1 - p^{-1/2})^{-1}, & p > 2,\ j = 1, \\ 0, & p > 2,\ j \ge 2. \end{cases}$$

Each local majorant is proved to bound the actual scaled norm at that prime power. Define E(q) as the product of these local majorants over the canonical prime-power factorization of q. Disjoint prime supports prove multiplicativity of E on coprime inputs.

Induction on positive prime powers and coprime products, uniform in every unit coefficient, then proves

$$\sqrt{q}\,|\mathfrak{s}(a,q)| \le E(q).$$

Zero local factors force the global actual sum to vanish, since sqrt(q)>0. This proves the global support criterion. Extracting the full power of two from q and checking the remaining factorization gives the explicit odd squarefree decomposition above, including the case d=1.

The product E has at most one factor at two, bounded by four. Every other factor is bounded by (1-p^(-1/2))^(-1). Dividing the resulting inequality by sqrt(q) proves the stated finite product bound for every positive q and every unit coefficient. The integer wrapper uses the checked equivalence between a unit integer residue modulo q and coprimality of a,q.

6. The remaining analytic step: precise proposed route

This subsection is a continuation specification, **not a completed Lean proof and not an assumed input of the current project**. The full source lemma requires the asymptotic coefficient 2+o(1) and its sufficiently-large-q consequence with coefficient three.

Let L=log(q), let S be the prime divisors of q, and suppose L>=3. The squarefree prime product divides q, so sum_(p in S) log(p)<=L. Put

$$\theta(y) = \sum_{p \le y} \log p, \qquad W(y) = \sum_{p \le y} p^{-1/2}.$$

A useful finite threshold inequality is

$$\sum_{p\in S} p^{-1/2} \le W(y) + \frac{L-\theta(y)}{\sqrt{y}\log y}, \qquad y > 2.$$

To prove it, subtract log(p)/(sqrt(y)*log(y)) from each term. The difference is nonnegative below y and nonpositive above y, because sqrt(p)*log(p) is increasing for p>=2. Extend the nonnegative portion to all primes below y and use the logarithmic budget for S. This avoids needing an exact discrete maximizer for the source's smallest-primes argument.

The standard PNT consequences theta(y)=y+o(y) and W(y)=(2+o(1))*sqrt(y)/log(y), applied with y=L, give the leading bound needed for the sum over S. Any use of these results must expose their exact quantified statements and published source; the user has authorized leaving deep published theorems unformalized. No own-paper obligation should be silently replaced by an axiom.

The remaining correction admits an elementary bound:

$$\sum_{p\in S} \frac{1}{p} \le 1 + \log L + \frac{1}{\log 2}.$$

Below L, compare with the harmonic sum; above L, use the logarithmic budget to bound the number of primes. Also

$$\log(1-p^{-1/2})^{-1} \le p^{-1/2} + 4/p \quad (p \ge 2).$$

This follows from log(x)<=x-1 and (1-p^(-1/2))^(-1)<=4. The correction O(log L) and the fixed log(4) are o(sqrt(L)/log(L)). Formalizing these steps, the threshold argument, and the PNT input boundary is the next target. Only after those are connected should the full Lemma 3.21 be marked complete.

Verification and evidence

All 24 modules were freshly compiled with pinned Lean 4.34.1 and Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612`. All 227 named project theorems pass the transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan passes for 26 Lean files. No proof placeholders, native_decide, unsafe declarations or added mathematical axioms occur.

The independent script `check_principal_gauss.py` evaluates the actual Gauss sums as integer polynomials modulo the cyclotomic polynomial Phi_q. These are exact values at the primitive q-th root of unity; no floating point arithmetic is used. For q=1,...,128 and every unit a, it checks:

- 5,022 unit coefficients in total;
- 1,688 complete odd-prime norm-square identities;
- 1,689 prime zero-term removal identities;
- 1,088 bad-modulus vanishings;
- 8,630 exact CRT comparisons, including transformed coefficients and totients;
- the modulus-one value.

These categories overlap and are not a count of distinct paper statements. The data are in `verification/principal-gauss-numerical.json`. They are finite supplementary checks, not inputs to Lean and not evidence for the uncompleted sharp asymptotic. The earlier six completed paper results remain complete; no headline theorem is yet formally verified.

`Progression_Cohen.md`

Progression Cohen identities: checked proof specification

Checkpoint 05, 22 October 2026. Authoritative source: `sources/current/Draft5.tex`, SHA256 `d2c31b96cb15b00feac09ed4fd9a2b0482726c719c4b772320c0711875d04660`. The targets are Lemmas **3.11** (`l:progressioncohen`) and **3.12** (`l:sumoverramanujan`). Both are proved, including the latter's vanishing and prime-power factorization clauses. This document explains the actual checked proofs; it does not certify other statements in the paper.

Statements and source correspondence

Write A_q = (Z/qZ)^×, ψ_q(t)=exp(2πit/q), and

$$c_q(n) = \sum_{a \in A_q} \psi_q(na), \qquad C_q(n) = \sum_{u \in A_q} c_q(n - u^2).$$

These are the existing genuine complex exponential definitions in `QuadraticCohen.lean`. Complete-residue filters and unit-group sums were already proved equivalent. There is no replacement of these functions by an assumed closed formula.

All moduli q are positive. The progression modulus g divides q, so g and h=q/g are positive as well. Lemma 3.12 uses positive y. Inputs b,x,n can be arbitrary integers, including negative ones.

The congruence r≡b mod g is expressed by the canonical ring homomorphism ρ_g:ZMod q→ZMod g. In Lean, its divisibility proof is an argument of `ZMod.castHom`. The paper's gcd(b,g)=1 becomes `IsUnit (b : ZMod g)` or `IsCoprime (g : ℤ) b`; Mathlib's `ZMod.coe_int_isUnit_iff_isCoprime` supplies the formal equivalence. Using an actual unit b in the CRT proof adds no mathematical restriction.

Define

$$R_{q,g}(b, n) = \sum_{\substack{r \in A_q \\ \rho_g(r) = b}} c_q(n - r).$$

Lemma 3.11 is

$$R_{q,g}(b, n) = \begin{cases} \mu(h) c_h(n) c_g(n - b), & \gcd(g, h) = 1, \\ 0, & \gcd(g, h) > 1. \end{cases}$$

The product-modulus theorem takes q=gh with arbitrary positive g,h; this covers every positive q and divisor g. The divisor/quotient wrappers state the same result in the paper's notation. Their `NeZero (q/g)` instance only expresses the positivity already implied by g|q and q>0. The gcd and quotient instances used for Lemma 3.12 are constructed explicitly in its module.

Source component	Checked declarations
Ordinary Cohen identity used in the proof	`linear_cohen_identity`
Genuine Ramanujan sum at a unit equals μ(q)	`ramanujanAtOne_eq_moebius`, `ramanujanSum_unit_eq_moebius`
Lemma 3.11, product modulus	`progression_cohen_identity_mul`
Lemma 3.11, divisor notation and integer input	`progression_cohen_identity`, `progression_cohen_identity_int`
Lemma 3.11, arbitrary modular input	`progression_cohen_identity_residue`
Definitions e:mathfraks and e:upsilondef	`principalSquareSum`, `upsilon`
Opening the actual normalized sums	`normalized_upsilon_correlation_expansion`
Lemma 3.12, equation e:eqjustforLHS	`sum_upsilon_square_identity`
Its squarefree vanishing clause	`sum_upsilon_square_eq_zero_of_not_squarefree`
Its prime-power factorization clause	`sum_upsilon_square_prime_factorization`

All declarations are in namespace `PrimeSquares`. The general character-twisted sum s_χ from e:mathfraksgeneral is not defined or formalized in this checkpoint. Only the principal-character sum needed for these two lemmas is used.

1. Translation of the unit group

The proofs use an elementary fact valid in every commutative ring. If t²=0 and u is a unit, then u+t is a unit with inverse

$$u^{-1} - tu^{-2}.$$

Multiplication verifies this formula using t²=0 and uu⁻¹=1. Translation by −t is the inverse permutation. Lean constructs the units and both inverse identities explicitly in `nilpotentAddUnit` and `nilpotentAddEquiv`.

This is an additive translation of units. The earlier quadratic identity used the different permutation u↦u+t/u. Both are proved from ring identities; neither presupposes root lifting or any analytic theorem.

2. The ordinary linear Cohen identity

First prove c_q(1)=μ(q). At a prime p the existing exact evaluation gives c_p(1)=−1. At p^(k+2), take t=p^(k+1). Then t²=0 in the residue ring, but t≠0. Reindexing the unit sum by u↦u+t gives

$$c_{p^{k+2}}(1) = \psi_{p^{k+2}}(t) c_{p^{k+2}}(1).$$

The standard character is injective, so ψ(t)≠1. Therefore the sum is zero. CRT multiplicativity, together with the proved prime-power evaluations of the Möbius function, identifies c_q(1) with μ(q) at every positive modulus. The auxiliary function `ramanujanAtOne` extends this actual sum by zero at q=0 solely for a natural-number induction interface. Its agreement with c_q(1) at every positive q is proved.

Multiplication by any unit permutes the unit group, so c_q(a)=μ(q) for unit a. Now open the two finite sums:

$$\begin{aligned}\sum_{r \in A_q} c_q(n-r) &= \sum_{a \in A_q} \psi_q(na) \sum_{r \in A_q} \psi_q(-ar) \\ &= \sum_{a \in A_q} \psi_q(na) c_q(-a) = \mu(q) c_q(n).\end{aligned}$$

The coefficient −a is again a unit, including for even q. The Lean proof checks the reordering and phase signs before extracting the common factor. Modulus one is handled explicitly: it has one unit, its character is identically one, and μ(1)=1.

3. Lemma 3.11: the coprime case

Suppose gcd(g,h)=1 and q=gh. The existing checked CRT equivalence identifies A_q with A_g×A_h. The progression condition fixes the first coordinate to b; the second coordinate ranges over all units modulo h. The previously proved Ramanujan multiplicativity then gives

$$R_{q,g}(b,n) = c_g(n-b) \sum_{v \in A_h} c_h(n-v) = \mu(h) c_h(n) c_g(n-b).$$

Lean transports the unit sum by `crtUnitsEquiv`, proves that the canonical reductions are the two coordinates, evaluates the fixed-coordinate filter, and applies the ordinary Cohen identity. There is no assumption that the unscaled standard additive character factors under CRT: the earlier character scaling proofs establish Ramanujan multiplicativity with the correct phases.

4. Lemma 3.11: the shared-prime case

Suppose gcd(g,h)>1. Choose a prime p dividing both, and write g=pℓ, h=pk. Set t=gk=q/p. Then

$$t^2 = q\ell k, \qquad g \mid t, \qquad 0 < t < q.$$

Consequently t²=0 in ZMod q, ρ_g(t)=0, and t≠0. Translation by t therefore permutes precisely the units in the chosen progression class.

For a fixed unit coefficient a in the opened Ramanujan sum, reindexing this progression-restricted sum multiplies it by ψ_q(−ta). The phase is not one: injectivity of the character would give ta=0, and a is invertible, forcing t=0. Thus every such coefficient sum is zero, and so is R_(q,g)(b,n).

`progressionRamanujanSum_eq_zero_of_square_zero` proves the general translation argument. `progressionRamanujanSum_non_coprime` constructs t with all the required properties from the shared prime. This branch in fact holds for any b; coprimality of b is needed only for the nonzero coprime branch.

The argument covers all overlaps, not merely prime-power q. Degenerate cases g=1, g=q and q=1 enter the coprime branch and require no omitted exception.

5. Lemma 3.12: expansion and normalization

Use exactly the definitions in the paper:

$$\mathfrak{s}(a,q) = \frac{1}{\phi(q)} \sum_{u \in A_q} \psi_q(-au^2), \quad \Upsilon(a,q,b,y) = \frac{\phi(y)}{\phi([y,q])} \sum_{\substack{r \in A_q \\ r \equiv b \pmod g}} \psi_q(-ar), \quad g = \gcd(q,y).$$

All three totients φ(q), φ(y), φ([y,q]) are strictly positive. The formal proof establishes their nonvanishing in ℂ before cancelling the factors. The product of the three phases is

$$\psi_q(-ar)\psi_q(-au^2)\psi_q(ax) = \psi_q(a(x-u^2-r)).$$

Summing first over a gives the genuine Ramanujan sum, hence

$$\frac{\phi(q)\phi([y,q])}{\phi(y)}\sum_{a\in A_q}\Upsilon(a,q,b,y)\mathfrak{s}(a,q)\psi_q(ax)=\sum_{u\in A_q}R_{q,g}(b,x-u^2).$$

Applying the residue-input form of Lemma 3.11 term by term gives exactly

$$\mu(h)\mathbf{1}_{\gcd(g,h)=1}\sum_{u\in A_q}c_h(x-u^2)c_g(x-b-u^2),\qquad h=q/g.$$

The two reductions are explicit ring homomorphisms in Lean. In particular, the shift in the second factor is x−b−u², with the same sign as the source.

If h is not squarefree, Mathlib's proved Möbius evaluation gives μ(h)=0. This proves the stated vanishing, without assuming a squarefree-support rule.

6. Complete prime-power factorization

If gcd(g,h)=1, CRT also separates the two coordinates of u in the last sum:

$$\sum_{u\in A_q}c_h(x-u^2)c_g(x-b-u^2)=C_g(x-b)C_h(x).$$

`progression_quadratic_correlation_coprime` proves this by the actual CRT unit equivalence. The earlier prime-power factorization of C and the proved multiplicativity of μ now give the fully local expression for the normalized correlation:

$$\left(\prod_{p|g}C_{p^{v_p(g)}}(x-b)\right)\left(\prod_{p|h}\mu(p^{v_p(h)})C_{p^{v_p(h)}}(x)\right).$$

Because q=gh and gcd(g,h)=1, the two prime supports are disjoint and together cover the prime powers dividing q. Empty products have value one. The local C factors already have checked evaluations from Lemma 3.1. This completes the factorization assertion of Lemma 3.12; evaluating and summing the later Euler products of Lemma 3.14 remains a separate target.

Verification and limits

The three new modules are `LinearCohen.lean`, `ProgressionCohen.lean` and `ProgressionCorrelation.lean`. All twelve project modules build. The source scan and the transitive axiom audit of all 78 named project theorems pass, allowing only `propext`, `Classical.choice` and `Quot.sound`. No deep analytic input or additional mathematical axiom is used.

A separate exact-integer experiment checks Lemma 3.11 for every 1≤q≤128, every divisor g, every unit class b modulo g and every n modulo q: 707,264 cases, including 89,364 cases in the shared-prime branch. There are no failures. The script uses the classical integer divisor formula for Ramanujan sums and records an output digest. This is finite experimental evidence; it is not used to prove any Lean theorem.

The paper's main analytic and application theorems remain open. The formal results above certify these specified statements, not the complete paper or all of its exposition. Source-to-formal-statement correspondence must continue to be reviewed at each milestone.

`Quadratic_Oscillation_and_Abel.md`

Quadratic oscillation and actual progression Abel summation

Checkpoint 24. This is the exact partial-summation identity used in the proof of Lemma 6.5. It does not yet estimate the resulting error terms or assert a progression prime-number theorem.

The actual phase and its derivative

For real theta and real t define

$$g_\theta(t)=\exp(-2\pi i\theta t^2).$$

`quadraticOscillation` uses the real coefficient -2*pi*theta*t^2, coerced to C and multiplied by i before exponentiation. It has exactly the paper's negative phase. At every natural n it is proved equal to `negativeFourierCharacter ((n:Z)^2) (theta:PrimeCircle)`.

The offset theta is a real number here. No periodicity in theta is claimed for arbitrary noninteger t; the oscillatory integral uses the chosen real major-arc offset. The integer-indexed Fourier character agreement is proved separately.

Ordinary real differentiation, then complex coercion and the complex exponential chain rule, prove

$$g'_\theta(t) = -4\pi i\theta t\, g_\theta(t).$$

This is an actual `HasDerivAt` statement for every real theta,t. `quadraticOscillation_deriv` identifies the bundled derivative function. Both the phase and its derivative are continuous, so they are integrable on every compact real interval. No differentiability or integrability hypothesis is passed to the final Abel theorem.

The actual complex exponential has unit norm. Consequently

$$|g_\theta(t)| = 1, \qquad |g'_\theta(t)| = 4\pi|\theta||t|.$$

For 0<=t<=N this yields the explicit uniform derivative bound 4*pi*abs(theta)*N. These are unbounded Lean proofs, not numerical samples.

Actual progression coefficients and prefixes

For a natural modulus q, a residue r in ZMod q and n natural, define

$$c_{q,r}(n) = \begin{cases} \Lambda(n), & n \equiv r \pmod q, \\ 0, & \text{otherwise}, \end{cases} \qquad C_{q,r}(x) = \sum_{0\le n\le \lfloor x\rfloor} c_{q,r}(n).$$

Both are complex-valued in Lean. They are actual sums of Mathlib's von Mangoldt function; all prime powers remain. Since Lambda(0)=0, the coefficient at zero and the prefix at zero are proved to vanish.

`squareResidueSum_as_coeff` proves the exact identification

$$W_r(\theta) = \sum_{n\in S} g_\theta(n) c_{q,r}(n),$$

where W_r is the earlier unnormalized residue sum. This connects the actual residue filter to the coefficients used by Mathlib's Abel formula, including the phase and the order of complex multiplication.

The actual Abel formula

Let 0<=a<=b be reals. Applying the pinned Mathlib theorem `sum_mul_eq_sub_sub_integral_mul` to these actual coefficients and the proved differentiable phase gives

$$\sum_{\lfloor a\rfloor < n\le \lfloor b\rfloor} g_\theta(n) c_{q,r}(n) = g_\theta(b) C_{q,r}(b) - g_\theta(a) C_{q,r}(a) - \int_{(a,b]} g'_\theta(t) C_{q,r}(t)\, dt.$$

This is `progression_quadratic_abel`; `squareResidueSum_abel` rewrites its left side as the actual square residue sum. Every differentiability and derivative-integrability obligation is discharged using the proved phase lemmas. The step-function prefix is the genuine floor-indexed prefix from the Abel theorem.

Taking a=N/2 and b=N, with N natural, yields

$$\sum_{\substack{N/2<n\le N \\ n\equiv r \pmod q}} \Lambda(n) e(-n^2\theta) = g_\theta(N) C_{q,r}(N) - g_\theta(N/2) C_{q,r}(N/2) - \int_{(N/2,N]} g'_\theta(t) C_{q,r}(t)\, dt.$$

`annularMangoldtResidue_abel` proves this identity with the real lower endpoint N/2. The natural cutoff agrees because floor((N:R)/2)=N/2 in natural division; that equality is explicitly used. Thus odd N and N=0 are covered, without substituting a half-subtraction identity that requires an even cutoff. The residue sum is unnormalized; the 1/N factor is not inserted or canceled at this stage.

Files and remaining scope

Module	New named theorems	Result
QuadraticOscillation.lean	10	Actual phase, derivative, norm, continuity and integrability
ProgressionAbel.lean	6	Actual Lambda progression prefixes, residue identification and exact real-endpoint Abel formula

The next unit subtracts a differentiable model and bounds the resulting weighted prefix error, retaining both endpoints and the derivative contribution. The source model x/phi(q)-chi(r)x^beta/(beta*phi(q)) and its actual oscillatory integrals will then be specialized explicitly. The analytic prefix bound itself remains a separate published input boundary. No such theorem or own-paper deduction is assumed here.

<!-- checkpoint-verification -->

Verification

All 86 project modules and the root were freshly rebuilt: 3601 jobs pass without warnings/errors. All 717 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 88 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. No new deep analytic input or own-paper lemma is assumed.

Rank_Blocks_and_Finite_Bilinear.md

Corollary 4.15: constructed rank blocks and the finite bilinear estimate

Current checkpoint 20: the full abstract Corollary 4.15 is complete, including strong interior interpolation and p=2. This document records an earlier stage; see Strong_Improving_Complete.md for the completed proof and current scope.

Checkpoint 19 constructs the dyadic rank decomposition for every finite complex input and proves its coefficient mass estimate. It then proves the strong finite bilinear estimate with the exact N scaling. Three modules add 40 named theorems. **The full strong operator corollary remains partial:** finite-output norm duality and final assembly are still required.

Source correspondence and inputs

The source is Corollary 4.15 (`c:improvinggeneral`), near line 2431 of `sources/current/Draft5.tex`. Fix p0<q<p<2, where p0=2(s+r)/(2r+s). The previously proved global equation 4.17 supplies

$$|\langle T\mathbf{1}_F, \mathbf{1}_G\rangle| \le AN^{1-2/q}|F|^{1/q}|G|^{1/q}.$$

Here A is the restricted-estimate constant; B below is the norm bound of the genuine continuous l2 realization of this one kernel. They are not names of the paper's two operators. Kernel positivity is used for the complex block estimate, exactly as in checkpoint 18.

Module	Named theorems	Checked content
`DyadicRankMass.lean`	12	Rank intervals, coverage/disjointness, decreasing-sequence mass bound
`FiniteMagnitudeBlocks.lean`	25	Actual sorting, finite blocks, amplitudes, partition and coefficient mass
`FiniteBilinearImproving.lean`	3	Exact complex regrouping, coefficient norm bound, finite bilinear improving

Sorting the actual indices, including ties

Given a finite F and arbitrary f:Z->C, `decreasingMagnitudeList F f` merge-sorts the actual indices of F by decreasing |f(x)|. Its permutation, length, membership, absence of duplicate indices, and ordering properties are proved. Equal magnitudes do not require distinct values or a strict ordering. Every original index is retained, including those with f(x)=0.

Write the resulting list as x_0,...,x_(L-1), with L=|F|, and define

$$m_n = \begin{cases} |f(x_n)|, & n < L, \\ 0, & n \geq L. \end{cases}$$

`rankMagnitude_nonneg`, `rankMagnitude_antitone`, and `rankMagnitude_tail` prove that this is a nonnegative decreasing sequence with the stated zero extension. In particular, no nonempty-support assumption is introduced.

The finite rank partition

Put s_j=2^j-1 and I_j=[s_j,s_(j+1)) in the natural ranks. Then |I_j|=2^j. The intervals are pairwise disjoint, and each rank n belongs to I_(floor(log2(n+1))). For n<L, this block index is <=L.

The actual block `magnitudeBlock F f j` is the image under n->x_n of the ranks in I_j with n<L. Only j=0,...,L are needed. The construction uses a finite subset of Fin L, so no out-of-range list lookup occurs.

The proved consequences are:

- `magnitudeBlock_card_le`: |F_j|<=2^j;
- `magnitudeBlock_disjoint`: the actual finite sets are pairwise disjoint;
- `magnitudeBlock_cover`: their finite union is exactly F;
- `magnitudeBlock_sum`: summing any function over the blocks equals its sum over F, for every additive commutative monoid;
- `magnitudeBlock_norm_le`: |f(x)|<=m_(s_j) on F_j.

The amplitude is `magnitudeAmplitude F f j = m_(s_j)`. It is zero for an empty block beyond the input length. A block containing zero-valued indices also has the correct amplitude; the construction does not remove such indices. The last nonempty block may have fewer than 2^j elements.

The coefficient mass estimate

For every real p>0, set c_j=(2^j)^(1/p)*m_(s_j). The exact power identity c_j^p=2^j*m_(s_j)^p is proved, including zero amplitudes.

For the next block, all ranks n in I_j precede s_(j+1). Monotonicity gives m_(s_(j+1))^p<=m_n^p, and the full rank interval has 2^j elements. Thus

$$2^{j+1} m_{s_{j+1}}^p \leq 2 \sum_{n \in I_j} m_n^p.$$

This is `dyadic_rank_next_mass`. It holds even if part or all of the rank interval is past L: the zero extension makes those terms zero.

Induction, using the exact finite interval-sum split, proves

$$\sum_{j=0}^{K} 2^j m_{s_j}^p \leq m_0^p + 2 \sum_{n < s_K} m_n^p.$$

Every finite prefix sum of the zero-extended sequence is bounded by its original mass. This includes the prefix of length one, so m_0^p is also bounded by the original mass when L=0. Consequently

$$\sum_{j=0}^{K} c_j^p \leq 3 \sum_{n<L} m_n^p$$

for every finite K. `dyadic_rank_coefficient_mass` proves this generic statement. `rankMagnitude_sum` identifies the ranked mass with the actual sum over F by an injective, surjective finite reindexing. Specializing K=L therefore gives the actual input theorem

$$\sum_{j=0}^{|F|} c_j^p \le 3 \sum_{x\in F} |f(x)|^p.$$

This is `magnitudeBlock_coefficient_mass`. It has no assumed partition, monotonicity or coefficient estimate: all of those have been constructed and proved for the given F and f. Its p>0 range is stronger than the p>=1 range needed for the corollary.

Using the exact norm of `finiteInputLp F f P`, the theorem `magnitudeBlock_coefficient_lp_norm` proves

$$\Big(\sum_j |c_j|^p\Big)^{1/p} \le 3^{1/p} \|f\mathbf{1}_F\|_p.$$

P is an ENNReal exponent in Lean and p=P.toReal. All norms here are Mathlib's genuine sequence-space norms. The proof covers the zero vector.

Exact regrouping with complex phases

For any complex kernel K, define the actual finite pairing

$$B_{F,G}(f,g) = \sum_{x\in G}\sum_{y\in F} K(x-y) f(y)\overline{g(x)}.$$

The rank partitions of F using f and G using g give

$$B_{F,G}(f,g) = \sum_i \sum_j B_{F_i,G_j}(f,g).$$

`finiteKernelPairing_magnitudeBlocks` proves this identity by distributing the finite convolution and applying the actual block-sum identities in both variables. Complex conjugation is retained throughout. This algebraic identity needs no positivity of K; positivity enters the bound below.

The strong finite bilinear estimate

Put a=1/q-1/p and b=1/p-1/2. The previously checked actual block-family estimate gives the right side below with the coefficient norms in place of the original input norms. The new mass bounds supply one factor 3^(1/p) for each input. Thus `restricted_l2_finite_bilinear_improving` proves, for every finite F,G and arbitrary complex f,g,

$$|B_{F,G}(f,g)| \le 3^{2/p}\left(\frac{A}{1-2^{-a}} + \frac{2^b B}{1-2^{-b}}\right) N^{1-2/p} \|f\mathbf{1}_F\|_p \|g\mathbf{1}_G\|_p.$$

The formal theorem allows 1<=p<2 and 0<q<p whenever its stated global restricted q estimate holds. The paper uses p0<q<p<2. N is any positive real number. The prefactor depends on A,B,p,q and not on N, the sets, their cardinalities, the coefficients, or a chosen partition.

The theorem's analytic inputs are exactly the global restricted estimate, source kernel positivity and a genuine continuous l2 operator norm bound. It does not assume block data, a desired coefficient inequality, norm duality or strong interpolation. The earlier high-low theorem supplies the restricted estimate in the intended application. No new deep analytic theorem or mathematical axiom is used.

Remaining operator-norm step

The finite bilinear theorem is now complete. Corollary 4.15 remains partial until it is connected to the output lp' norm and both actual operators.

One proposed finite-output duality route is as follows; it is **not yet a Lean proof**. Let p'=p/(p-1)>=2 and u be the output of the coarse actual lp-to-lp' realization. On a finite output set G, use the test

$$g(x) = |u(x)|^{p'-2}u(x).$$

Then the desired identities are

$$\sum_{x\in G} u(x)\overline{g(x)} = \sum_{x\in G} |u(x)|^{p'}, \qquad \|g\mathbf{1}_G\|_p^p = \sum_{x\in G} |u(x)|^{p'}.$$

These identities, including zero coordinates and the exponent relation (p'-1)*p=p', still need formal proofs. The nonnegative p'-2 avoids a negative power at zero; p=2 can also be handled by the existing endpoint. Apply the finite bilinear estimate, split off the zero-sum case, and derive the finite output norm bound. The existing lp finite-sum norm criterion then gives the full output norm bound. Membership of u in lp' is already supplied by `lpL2Lift`.

Finally invoke `lp_opNorm_bound_of_finiteInputs` for all lp inputs, choose q between p0 and p before N, retain both original l2 constants, translate 1-2/p to 1/p'-1/p, and assemble the interior range with p=2. No assertion that these final connections are complete is made in this checkpoint.

Verification and finite experiments

All 67 modules and the root receive a fresh build with the artifact cache disabled: 3464 jobs pass without warnings or errors. All 551 named project theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 69 Lean files.

`scripts/check_dyadic_rank_blocks.py` checks 56113 finite inputs, including all 55987 input vectors of length <=6 over a six-value Gaussian integer alphabet and 126 structured longer inputs. The largest tested length is 1024. It verifies coverage, disjointness, cardinality and magnitude bounds, then 280565 coefficient mass inequalities at p=1/2,1,3/2,2,3. The alphabet has square integer magnitudes, so all tested powers are exact integers. Ties, zero inputs, empty blocks and incomplete final blocks are included.

Another 768 exact Gaussian integer checks verify complex bilinear regrouping, allowing signed and complex kernels. All pass. These finite experiments are independent checks; Lean proves the all-input statements. They do not verify output norm duality, the full operator corollary, or the later prime-square operator hypotheses.

Workspace maintenance had cleared temporary caches. The saved checkpoint 18 was recovered and all 206 archive entries were verified. The exact Lean 4.34.1 release archive hash and pinned dependency revisions were verified before rebuilding. No version upgrade or proof-kernel change was made. The archived manuscript and accepted revision fragments are unchanged. The count of completely proved unconditional numbered results remains nine; Lemma 3.21 retains its earlier explicit PNT hypotheses independently.

`Rational_Arc_Separation.md`

Exact rational-arc separation

For integers a,b and positive q,r, a nonzero determinant $ar - bq$ gives

$$|a/q - b/r| \geq 1/(qr).$$

The proof casts the nonzero integer determinant to the reals, proves its absolute value at least one, and divides by the positive denominator product.

For rationals u,v with their reduced positive denominators, this yields

$$u \neq v \Longrightarrow |(u : \mathbb{R}) - (v : \mathbb{R})| \geq 1/(\mathrm{den}(u)\,\mathrm{den}(v)).$$

If u,v are distinct canonical representatives in [0,1), the same lower bound holds for $|u - v - k|$ for every integer k. An equality u=v+k would force -1<k<1, hence k=0 and u=v. This accounts explicitly for integer lifts when an arc crosses the boundary of a fundamental interval.

If both denominators are at most D, separation is at least $1/D^2$. Two real arcs of radius delta centered at distinct such rationals cannot contain a common point when $2\delta < 1/D^2$. The final theorem proves that uniqueness from the triangle inequality.

Module	New named theorems
RationalArcSeparation.lean	6

Scope

These are exact unconditional proofs. No numerical sampling is used. Dyadic support uniqueness will only be asserted when its strict radius inequality is proved; the smallest denominator scales will not be silently included by an inapplicable estimate. The analytic minor-arc gap remains unchanged.

<!-- checkpoint-verification -->

Verification

All 101 project modules and the root were freshly rebuilt: 3616 jobs pass without warnings/errors. All 800 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 103 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. No new deep analytic input or own-paper lemma is assumed.

`Ren_Input_and_Annular_Bound.md`

Ren's actual annular sum and a verified minor-arc consequence

Checkpoint 31. This unit connects the paper's genuine negative-phase annular Fourier sum to Ren's positive-phase sum, transports the exact three-term bound, and proves a valid decay consequence in an explicit parameter range. It leaves the manuscript's wider minor-arc claim open.

Published input and its scope

The primary text inspected is Xiumin Ren's author preprint, On exponential sum over primes and application in Waring-Goldbach problem, Theorem 1.1, corresponding to the cited 2005 paper. Its sum is already over x<m<=2x. For k=2, frequency xi=a/q+theta and reduced a/q, its displayed bound has the form

$$|S(x,\xi)| \le C\tau(q)^{3/2}(\log x)^A\left(\sqrt{x}\sqrt{T}+x^{4/5}+\frac{x}{\sqrt{T}}\right), \qquad T=q(1+|\theta|x^2).$$

Here C,A are fixed constants supplied by the analytic result. The exact journal text was not accessible; the author-hosted preprint's statement and first-page formula were checked. This result itself is not proved in Lean or added as an axiom. The consuming theorems display the actual inequality as a hypothesis. External applicability requires x in the theorem's large-parameter range and the stated rational approximation.

Actual sum, sign and endpoints

`renQuadraticSum` is the complex finite sum of Lambda(n)*e(xi*n^2) over floor(x)<n<=floor(2x). `renQuadraticSum_exp` identifies the exact positive complex exponential explicitly. Conjugation of this actual sum reverses the sign of xi and preserves its norm, because Lambda is real. These are proved identities, not assumed Fourier conventions.

Taking x=N/2 uses the real lower endpoint, with floor(N/2)=N/2 in natural division and floor(2*(N/2))=N. Thus for every natural N,

$$\widehat{\widetilde{A}}_{N,\Lambda}(\xi)=\frac{1}{N}S(N/2,-\xi), \qquad \left|\widehat{\widetilde{A}}_{N,\Lambda}(\xi)\right|=\frac{1}{N}|S(N/2,\xi)|.$$

These identities even handle odd N and the inverse convention at N=0. Applying the external analytic theorem requires N sufficiently large; the explicit source-prefactor corollary below uses N>=4. No prime powers are dropped. There is no need to reconstruct this annular sum by subtracting two initial sums.

Exact bound transported to the annulus

Put B=C*tau(q)^(3/2)*(log x)^A and

$$J(x,T)=\sqrt{x}\sqrt{T}+x^{4/5}+x/\sqrt{T}.$$

`annularMangoldt_from_Ren_bound` proves directly from the raw inequality |S(N/2,xi)|<=B*J(N/2,T) that

$$\left|\widehat{\widetilde{A}}_{N,\Lambda}(\xi)\right| \le \frac{B}{N}J(N/2,T).$$

This retains all three terms and every normalizing factor. The theorem does not assert that arbitrary independent choices of xi,theta,q satisfy the analytic input: the genuine raw inequality is displayed.

A proved decay range

For x>0, R>0, assume

$$R \le T \le x/R, \qquad R \le x^{2/5}.$$

Then each term of J(x,T) is at most x/sqrt(R). Indeed:

1. T*R<=x implies sqrt(T)*sqrt(R)<=sqrt(x), and hence sqrt(x)*sqrt(T)<=x/sqrt(R).
2. R<=x^(2/5) implies sqrt(R)<=x^(1/5), giving x^(4/5)*sqrt(R)<=x.
3. T>=R implies x/sqrt(T)<=x/sqrt(R).

Thus `ren_bracket_central_bound` proves J(x,T)<=3*x/sqrt(R). The exact N=2x normalization gives

$$\left|\widehat{\widetilde{A}}_{N,\Lambda}(\xi)\right| \le \frac{3B}{2\sqrt{R}}.$$

`annularMangoldt_Ren_central_source` substitutes the actual divisor and logarithmic prefactor and T=q*(1+abs(theta)*(N/2)^2). Its positivity is proved using N>=4 and C>=0. The divisor factor multiplies the whole bound; it has not been discarded or silently absorbed into a logarithm.

The remaining source deduction

The manuscript's Lemma 6.5 asks for a minor-arc estimate throughout R^(11/10)<q<=N^2/R with abs(theta)<=R/(q*N^2). Those conditions do not imply the central-range hypothesis T<=x/R. For example, theta=0 and q comparable to N give T comparable to x. At T=x, the first normalized term in Ren's displayed estimate is exactly

$$\frac{\sqrt{x}\sqrt{T}}{x} = 1.$$

`ren_first_term_diagonal` proves this identity for every x>0. Thus direct substitution into this particular bound does not supply a decaying estimate in the entire source range. A further argument or a complementary estimate must be supplied and checked. This is a gap in the currently verified deduction, not a counterexample to the manuscript's minor-arc assertion. No replacement of that assertion is prescribed on this evidence alone.

The conditional source-range major-arc R^-5 theorem remains intact. Full Lemma 6.5 stays open pending its independent full-range minor-arc argument and the precise external-input bridge. Proposition 4.19 and the paper's Theorem 1.1 remain explicit later targets.

Module	New named theorems	Result
RenSumCompatibility.lean	6	Positive-phase exponential, conjugation, exact floor endpoints and actual annular normalization
RenMinorArcBound.lean	5	Raw Ren bound transport, proved central-range decay with actual prefactor, and the diagonal diagnostic

All new deductions are Lean proofs. Ren's deep estimate is a displayed theorem hypothesis, with no new mathematical axiom or proof placeholder.

<!-- checkpoint-verification -->

Verification

All 96 project modules and the root were freshly rebuilt: 3611 jobs pass without warnings/errors. All 771 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 98 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Ren's displayed three-term estimate enters only as an explicit hypothesis of conditional consuming theorems; it is not added as an axiom. No own-paper lemma is assumed.

Residue_Grouping_and_Major_Arc_Assembly.md

Residue grouping and finite major-arc assembly

Checkpoint 23. Source: Draft5, proof of Lemma 6.5 (`l:prime-square-input`), immediately after `e:prime-square-minor-input`. This unit proves the finite residue decomposition, controls the discarded non-unit terms, and assembles the ordinary and twisted Gauss factors with explicit error accounting. The full major-arc and minor-arc statements of Lemma 6.5 remain open.

Actual sums and conventions

For finite S contained in {1,...,N}, real c and real weights w, the already constructed continuous Fourier symbol is

$$m_{S,c,w}(\xi) = c\sum_{n\in S} w(n)e(-n^2\xi).$$

The actual annular von Mangoldt symbol takes S = {floor(N/2)+1,...,N}, c = 1/N and w(n) = Lambda(n), including every prime power. Frequencies are in the period-one additive circle. Finite regrouping works for every positive modulus q and every integer numerator a; no coprimality or major-arc size restriction is needed for these algebraic identities.

Define the unnormalized residue sum

$$W_r(\theta) = \sum_{\substack{n\in S\\ n\equiv r \pmod q}} w(n)e(-n^2\theta).$$

This is `squareResidueSum`. It is a finite sum using the actual equality of natural-number casts in ZMod q. The normalization c is kept outside W_r, so it cannot be silently applied twice.

Non-unit support and its sharper elementary bound

If Lambda(n) is nonzero, Mathlib's actual von Mangoldt definition implies n = p^k for some prime p and k >= 1. If n is not coprime to q, then p must divide q: otherwise p is coprime to q, hence so is p^k. `mangoldt_nonunit_prime_power` proves this support statement.

Split the non-unit mass into primes and non-primes. For the prime part, only distinct primes dividing q can occur. Therefore

$$\sum_{\substack{n\in S\\ n \text{ prime}\\ (n,q)>1}} \Lambda(n) \le \sum_{p\mid q} \log p \le \log q, \qquad q>0.$$

The last inequality is the already proved `primeFactors_log_budget`: the product of the distinct primes dividing q divides q. The finite-set inclusion and the use of the actual prime value Lambda(p)=log(p) are proved in `mangoldt_prime_nonunit_mass_le_log`.

For all the non-prime terms, including those coprime to q, the pinned Mathlib identifies the full weighted mass with psi(N)-theta(N), then proves the elementary estimate

$$0 \le \psi(N) - \vartheta(N) \le 2\sqrt{N}\log N, \qquad N\ge 1.$$

The exact declarations are `Chebyshev.psi_sub_theta_eq_sum_not_prime` and `Chebyshev.psi_sub_theta_le`. By nonnegativity of Lambda, any subset of the higher prime powers has at most this mass. Our `mangoldt_nonprime_mass_bound` proves that subset implication.

Combining the two parts gives, for every S contained in {1,...,N},

$$0 \le \sum_{\substack{n\in S\\(n,q)>1}} \Lambda(n) \le \log q + 2\sqrt{N}\log N.$$

This is `mangoldtNonunitMass_le`. N >= 1 and q > 0 are its only size restrictions. For q=1 the discarded set is actually empty; `mangoldtNonunitMass_one` records exact vanishing.

The source uses the looser normalized error O(N^(-1/2)(log N)^2 + q N^(-1)(log N)^2). The proved explicit estimate above is sharper and is sufficient for this reduction step. It does not require changing the source statement. No PNT or distribution-in- progressions estimate is used; the Chebyshev bound is already proved in the same pinned Mathlib.

The exact residue identity

For every integer n and circle offsets alpha,beta, the negative Fourier character obeys c_n(alpha+beta)=c_n(alpha)c_n(beta). At a rational frequency we already know c_(n^2)(a/q)=stdAddChar_q(-a*n^2). We prove and apply the argument-addition identity, obtaining

$$e(-n^2(a/q+\theta)) = e(-an^2/q)e(-n^2\theta).$$

Within the class n congruent to r modulo q, the first factor is exactly e(-a r^2/q). Each natural n lies in one and only one residue class, so finite interchange of summation gives

$$m_{S,c,w}(a/q+\theta) = c\sum_{r\in\mathbb{Z}/q\mathbb{Z}} e(-ar^2/q)W_r(\theta).$$

`finite_sum_residue_fibers` proves the partition identity and `finiteSquareSymbol_residues` applies the actual rational phase.

A natural-number cast is a unit in ZMod q exactly when it is coprime to q. This proves that restricting S to the coprime n retains W_r precisely for unit r and makes it zero for other r. The earlier exact equivalence between the unit-filter convention and the actual unit group then gives

$$m_{S,c,w}(a/q+\theta) = c\sum_{r\in(\mathbb{Z}/q\mathbb{Z})^\times} e(-ar^2/q)W_r(\theta) + \mathcal{E}_{\text{nonunit}},$$

where the remainder is explicitly

$$\mathcal{E}_{\text{nonunit}} = c\sum_{\substack{n\in S\\(n,q)>1}} w(n)e(-n^2(a/q+\theta)).$$

This is `finiteSquareSymbol_reduced_residues`. It retains an equality with an actual finite remainder, rather than replacing the remainder by an unexplained big-O term.

For nonnegative c and weights, the triangle inequality and unit magnitude of every Fourier character give

$$|m_{S,c,w}(\xi)| \le c\sum_{n\in S} w(n).$$

Together with the mass estimate, this proves, for the actual annular Lambda symbol and N >= 1,

$$\left|\widehat{\widetilde{A}}_{N,\Lambda}(a/q+\theta) - \frac{1}{N}\sum_{r\in(\mathbb{Z}/q\mathbb{Z})^\times} e(-ar^2/q)W_r(\theta)\right| \le \frac{\log q + 2\sqrt{N}\log N}{N}.$$

`annularMangoldt_reduced_residues_bound` is this unbounded, uniform-in- frequency Lean theorem. Its q=1 remainder is exactly zero. The finite identities also cover empty S and the total N=0 definition; the logarithmic inequality is stated only for N >= 1.

Ordinary and twisted Gauss assembly

Fix q>0, an arbitrary complex Dirichlet character chi modulo q, and two complex numbers M_1 and M_beta. In a future application these will be the ordinary and exceptional oscillatory integrals. No assertion about those integrals or a progression asymptotic is assumed in this unit.

For each residue r define its actual normalized model error by

$$\varepsilon_r = cW_r(\theta) - \frac{1}{\varphi(q)}(M_1 - \chi(r)M_\beta).$$

The function `squareResidueModelError` is exactly this difference. In the Lean declaration the scalar names are `M,E`; they denote M_1,M_beta here. That scalar E is separate from both the non-unit remainder and the paper's global error operator E_(Q,N).

The Section 3 Gauss definitions have already fixed

$$\mathfrak{s}(a,q) = \frac{1}{\varphi(q)} \sum_{r\in(\mathbb{Z}/q\mathbb{Z})^\times} e(-ar^2/q), \quad \mathfrak{s}_\chi(a,q) = \frac{1}{\varphi(q)} \sum_{r\in(\mathbb{Z}/q\mathbb{Z})^\times} \chi(r)e(-ar^2/q).$$

Distributivity of these actual finite sums gives

$$\sum_r e(-ar^2/q)\frac{M_1 - \chi(r)M_\beta}{\varphi(q)} = \mathfrak{s}(a,q)M_1 - \mathfrak{s}_\chi(a,q)M_\beta.$$

This is `unitSquareGauss_model_identity`. Combining it with the exact residue decomposition and the definition of epsilon gives the identity

$$m_{S,c,w}(a/q+\theta) - (\mathfrak{s}(a,q)M_1 - \mathfrak{s}_\chi(a,q)M_\beta) = \mathcal{E}_{\text{nonunit}} + \sum_r e(-ar^2/q)\varepsilon_r.$$

`finiteSquareSymbol_majorArc_error_identity` proves this formula for all of the displayed data, without an approximation hypothesis. It preserves the exceptional minus sign and the exact totient normalization.

Taking norms gives the sum of the actual residue-error norms. For the annular Lambda average this is

$$\left|\widetilde{A}_{N,\Lambda}(a/q+\theta) - (\mathfrak{s}(a,q)M_1 - \mathfrak{s}_\chi(a,q)M_\beta)\right| \le \frac{\log q + 2\sqrt{N}\log N}{N} + \sum_r |\varepsilon_r|.$$

If a later argument proves |epsilon_r| <= delta on every unit residue, there are exactly phi(q) such residues. The final theorem therefore gives

$$\text{error} \le \frac{\log q + 2\sqrt{N}\log N}{N} + \varphi(q)\delta.$$

The per-residue bound is an explicit hypothesis of this last implication; it has not been established by naming the errors. The preceding identity and sum-of-errors inequality have no such hypothesis. Without an exceptional contribution one can put M_beta=0. All assertions are at an individual scale, and do not freeze exceptional data as N varies.

Files and proof scope

Module	New named theorems	Completed work
MangoldtNonunits.lean	6	Actual non-unit prime-power support and logarithmic/Chebyshev mass bound
SquareResidueGrouping.lean	8	Exact rational shift, full residue partition and unit-group decomposition
MangoldtResidueReduction.lean	4	Uniform explicit remainder for the actual annular Lambda symbol
SquareMajorArcAssembly.lean	5	Exact ordinary/twisted Gauss assembly and finite error accounting

These are 23 supporting theorems. They complete these finite proof steps, not either full numbered conclusion of Lemma 6.5. The ten completed numbered paper results are unchanged. No own-paper lemma or new deep analytic input has been assumed.

The full fresh artifact-cache-disabled build passes 3599 jobs, with all 84 modules and the root rebuilt and no warnings/errors. All 701 named project theorems pass the complete transitive whitelist: propext, Classical.choice and Quot.sound. The source scan checks 86 Lean files; there are no placeholders or added mathematical axioms. No dependency version was changed and no new Mathlib cache target was required.

Independent finite checks

`scripts/check_residue_grouping.py` checks 18144 finite residue identities and 18144 corresponding Gauss/error identities. It uses q=1,...,12, N=0,...,8, full and annular windows, ordinary/Lambda/signed weights, negative and positive numerators, offsets 0,1/2,-1/3,1/4, and principal or available induced real characters. Both a nonzero exceptional model and a zero exceptional model are tested.

Coefficients are exact Gaussian rationals and independent formal log(prime) symbols in a cyclic group algebra, with order the lcm of the frequency denominators. There is no transcendental evaluation or floating point. The identities hold at the formal coefficient level, hence under the usual root-of-unity evaluation. These tests complement the Lean proofs; they are not an unbounded mathematical proof.

Separate probes check 395 non-unit prime-power support instances and 576 divisibility cases for the product of prime divisors in a cutoff. The support/product ranges are q<=64 and n<=128. At N=4,q=2 the omitted annular non-unit term is log(2)/4 from n=4. At q=5 a reversed exceptional sign changes the Gauss/error identity. The q=1 and N=0,1 cases, and odd cutoffs, are explicitly included.

The tests do not numerically certify the logarithmic/Chebyshev inequality, a progression estimate, partial summation, an R^-5 error, or a maximal estimate. The general finite remainder bound is a Lean proof using Mathlib's proved Chebyshev inequality.

Next obligations

1. Prove partial summation for the actual progression-weighted Lambda sums with the quadratic oscillation. The prefix-error bounds and endpoint terms must yield the correct normalized error, including theta*N^2. Mathlib.NumberTheory.AbelSummation already supplies the general Abel formula; inspect and reuse it before rebuilding infrastructure.
2. Verify and state the exact published prime-distribution and minor-arc input boundaries, with uniform constants and parameter ranges. One scale's exceptional data must work throughout the prefix interval [N/2,N], while it may change between scales.
3. Specialize M_1,M_beta to the actual source integrals, differentiate t^beta/beta correctly, and derive the per-residue error bounds.
4. Prove the final R^-5 major-arc bound in the permitted N,Q,R ranges and the annular minor-arc statement, completing Lemma 6.5.
5. Continue approximation 6.9, low 6.14, multifrequency 6.16, high/maximal 6.21, Proposition 4.19 and the Theorem 1.1 application.

Exceptional conductor, character and beta may depend on N. Published deep inputs may be explicit hypotheses, as authorized, but the paper's own partial summation and deductions remain proof obligations. Lemma 3.21's two earlier PNT limits are unchanged and unused in this unit. The frozen manuscript and both accepted revisions are unchanged; no new source correction is pending.

`Restricted_Improving.md`

Corollary 4.15: global restricted improving and weak distribution bounds

Current checkpoint 20: the full abstract Corollary 4.15 is complete, including strong interior interpolation and p=2. This document records an earlier stage; see Strong_Improving_Complete.md for the completed proof and current scope.

Checkpoint 16 proves equation 4.17 (`e:restrictedimproving`), including its exact N scaling, from Theorem 4.1 and the stated extra l2 bounds. It also proves the actual restricted weak distribution estimate and the strong p=2 case for the unique continuous linear l2 realizations.

The strong estimate for p0<p<2 in Corollary 4.15 is still open. The interpolation step for general lp inputs is not a hypothesis or an axiom in this project. This checkpoint does not mark the whole corollary complete.

Source and exact statement

The source is `c:improvinggeneral`, Corollary 4.15, beginning near line 2431 of `sources/current/Draft5.tex`. Equation 4.17 is the restricted pairing estimate inside its proof. Equation 4.16 (`e:lpimproving`) is the full strong conclusion and remains only partially covered.

Put

$$p_0 = \frac{2(s+r)}{2r+s}, \qquad p' = \frac{p}{p-1}.$$

For every p0<p<=2, `highLow_AB_restricted_improving` proves that one D>0, quantified before N, both kernels, windows and F,G, satisfies

$$|\langle A_N 1_F, 1_G\rangle| + |\langle B_N 1_F, 1_G\rangle| \le D N^{1/p'-1/p} |F|^{1/p} |G|^{1/p}$$

for every positive real N and all finite F,G in the integers. There is no remaining cardinality-product or common-interval restriction. In particular, the sum estimate gives the source's estimate for each operator separately. Empty input and test sets are included.

The hypotheses are the completed Theorem 4.1's actual abstract operator inputs and two genuine uniform continuous linear l2 realizations. Their l2 constants may be different. This result does not establish the prime-square operator hypotheses.

Four new modules

Module	Checked content
`RestrictedPowers.lean`	Critical/conjugate exponent algebra, exact density scaling, uniform removal of losses, zero density
`RestrictedL2.lean`	Actual continuous linear l2 inputs, uniqueness, strong p=2 case, l2 pairing bound and globalization
`RestrictedImproving.lean`	Equation 4.17 for both actual kernels and all finite sets
`RestrictedWeak.lean`	Finiteness of positive superlevel sets, weak gauge/cardinality bounds and both-kernel restricted weak theorem

They add 33 named theorems. The completed Theorem 4.1 and its scalar optimization are reused rather than assumed in a new conditional statement.

Exact exponents and normalization

For s,r>0, the following identities and inequalities are proved:

$$1 < p_0 < 2, \qquad \frac{1}{p_0} = 1 - \frac{s}{2(s+r)} =: \theta, \qquad \frac{1}{p'} = 1 - \frac{1}{p}.$$

Thus p>p0 gives epsilon=theta-1/p>0. For p<=2, `1/2<=1/p`. The definitions are `improvingCriticalExponent` and `holderConjugate`. Positivity and nonvanishing needed in the divisions are checked before using these formulas.

For N>0 and nonnegative real f,g, the exact scaling identity is

$$N\left(\frac{fg}{N^2}\right)^{1/p} = N^{1-2/p} f^{1/p} g^{1/p} = N^{1/p'-1/p} f^{1/p} g^{1/p}.$$

This is proved for real powers, including f=0 or g=0. It is used to convert the normalized density estimate into the literal N power in equation 4.17. No unspecified comparison or hidden N-dependent constant is involved.

Uniform removal of the endpoint losses

The B_N endpoint has factor `(Log delta)^tau`, with the exact tau from Theorem 4.1. The A_N endpoint has Phi(cstar,kappa,delta). For any sigma<theta with sigma>0, the new modules prove, with constants uniform on 0<=delta<=1,

$$\delta^\theta (\mathrm{Log}\,\delta)^\tau \le C\delta^\sigma, \qquad \delta^\theta \Phi_{cstar,\kappa}(\delta) \le C'\delta^\sigma.$$

For positive delta these follow from the already proved uniform Phi sub-power bound, with epsilon=theta-sigma. A fixed logarithmic power is first absorbed into Phi(1,1/2,delta). At zero, positivity of theta and sigma makes both density powers vanish. This zero case is proved separately; there is no limiting substitution at log(0).

The A_N coefficient is chosen once as the printed leading coefficient plus one. Theorem 4.1 allows that choice, and the final restricted estimate is independent of density. Taking sigma=1/p gives `highLow_AB_power_small` for |F||G|<=N^2. It retains actual operator inputs through `HighLowEndpointInputs` and does not assume an endpoint deduction.

The genuine l2 hypothesis

`KernelL2Bound K C` means there exists a continuous complex-linear map

$$T : \ell^2(\mathbb{Z}) \longrightarrow \ell^2(\mathbb{Z})$$

with operator norm at most C, whose value on every finite indicator agrees pointwise with the actual convolution by K. This is the restriction of the source's uniform l2 convolution bound needed here. The source operator itself supplies the map.

`kernelL2_realization_unique` proves that two such maps agree on the whole l2 space. Agreement on singleton indicators gives agreement on their complex multiples; Mathlib's proved density of finite-support vectors and continuity then give equality. Thus the realization is determined by the kernel.

`kernelL2_pair_strong_two` supplies both realizations and proves

$$\|T_A f\|_2 + \|T_B f\|_2 \le (C_{2,A} + C_{2,B})\|f\|_2 \quad (f \in \ell^2).$$

Since p'=2 at p=2 and the N exponent is zero, this is the strong p=2 case. The interior p0<p<2 still requires interpolation.

Larger cardinality products

The actual l2 input and the proved indicator norm identities imply

$$|\langle T1_F, 1_G\rangle| \le C_2\sqrt{|F|}\sqrt{|G|}.$$

After division by N this is `C2*delta^(1/2)`. If |F||G|>N^2, then delta>1; since p<=2, real-power monotonicity gives `delta^(1/2)<=delta^(1/p)`. This proves the same normalized restricted bound in the missing range. The small-range and l2 constants are added before N,F,G are introduced.

The result is global in F,G. This provides an alternative first step for the source's interpolation/localization proof; no finite-propagation or interval partition theorem is needed for this restricted deduction. The subsequent strong argument remains to be established.

Actual superlevel sets and restricted weak type

For a nonnegative kernel K and a positive level lambda, let

$$E = \{x : \lambda < |T1_F(x)|\}.$$

`kernelL2Bound_superlevel_finite` proves that E is finite: the l2 output has summable squared norms, those squares tend to zero along the cofinite filter, and only finitely many can exceed lambda squared.

Every output is real and nonnegative. The earlier exact positive pairing identity therefore gives

$$\lambda|E| \le |\langle T1_F, 1_E\rangle| \le S|F|^{1/p}|E|^{1/p}, \qquad S = DN^{1/p'-1/p}.$$

For nonempty E the proof divides by the strictly positive power `|E|^(1/p)` and uses `1/p'+1/p=1`. The empty case is handled directly. The result is

$$\lambda|E|^{1/p'} \le S|F|^{1/p}, \qquad |E| \le \left(\frac{S|F|^{1/p}}{\lambda}\right)^{p'}.$$

The Lean result constructs an actual finite set E, proves its membership is equivalent to the displayed superlevel condition on the whole integer line, and proves the weak gauge inequality. `weak_gauge_to_`

`cardinality` proves the cardinality conversion. `highLow_AB_restricted_weak` connects this to both kernels and all the actual source norm assumptions.

The positivity of A_N is used in this distribution step, as supplied by the original theorem. The global complex pairing estimate itself did not need that additional positivity. These are indicator-input weak bounds; they are not a strong estimate for general inputs.

Verification and numerical scope

The project now has 58 modules and 457 named theorem declarations; the source scan covers 60 Lean files. All modules and the root receive a fresh build, with the artifact cache disabled. All named theorem declarations receive the complete transitive axiom audit with whitelist propext, Classical.choice and Quot.sound. See the current verification logs.

The new exact experiment checks 11250 normalization cases, including 882 zero-product cases and 9303 larger-density cases. It examines 98304 distinct finite positive-kernel pairings at three conjugate exponents (294912 pairing/exponent checks), and 36864 superlevel inequalities. Empty superlevels and strict-threshold boundary cases are included. All pass.

The finite families use six integer kernels, all input subsets of {-2,-1,0,1,2}, all output subsets of {-4,...,4}, and positive half-integer levels. The normalization exponents are p=4/3,3/2,2. Integer powers remove all radicals, so the checks use only integers and exact fractions. These finite checks do not verify the unbounded analytic operator hypotheses or the missing strong interpolation theorem. General real-parameter restricted implications are proved in Lean.

No new mathematical axiom, deep input or manuscript correction is used. The explicit PNT hypotheses in Lemma 3.21 remain independent.

Remaining obligation

The nine fully proved numbered results, including Theorem 4.1, remain unchanged. Corollary 4.15 is partial: equation 4.17, the actual restricted weak consequence and p=2 are proved; the strong interior range is open.

Checkpoint 17 connects finite complex inputs, genuine lp operator realizations, uniqueness and density, and proves the finite Schur lemmas; see `Strong_Interpolation_Foundations.md`. The strong interpolation estimate with the correct N scaling still needs to be proved. Checkpoint 18 also closes the actual geometric weight and complex block/family estimates; see `Interpolation_Geometry.md`. The rank construction, mass bound, finite-output duality and final strong assembly remain open. The local source search did not find a named Marcinkiewicz/Riesz--Thorin/restricted-weak theorem in the pinned Mathlib; that search is recorded in `verification/interpolation-availability.json`. This is a search result, not a proof that no suitable result exists under another name. A concrete proposed construction is recorded in `blueprint/Strong_Interpolation_Plan.md`; it distinguishes checked supporting steps from the remaining unproved construction, with no Lean placeholders added.

Checkpoint 19 constructs the rank blocks, proves the mass bound and derives the strong finite bilinear estimate. See `Rank_Blocks_and_Finite_Bilinear.md`. Finite-output norm duality and final strong operator assembly remain open.

`Section11_Complete.md`

Section11 is complete for the reviewed corrected statements. The joint endpoint goldbach_section11_complete uses one B,C for the actual lower and upper sieve counts, actual common main term, the lower bound k N(log N)^(-(A0/2+6)), and both original remainder families. It gives the exact lower coefficient f_(1/2)(9/8), the exact upper coefficient c_(9/8) F_1(2) W(9/8), uniform tolerances, coefficients chosen before n and a combined mean-square saving of arbitrary logarithmic order. C can exceed any prescribed minimum and C>=A0+24 is sufficient. The exponent A0/2+6 is independent of C. Strict positivity of the normalized coefficient difference is already kernel checked.

The general one-stage sieve, SW/BV, Ren/Ghosh, effective multiplicative mean and classical Mertens inputs, plus classical sieve-function continuity and normalization, remain explicit independent hypotheses. No paper endpoint is an input or an added mathematical axiom.

A complete replacement copy is saved as sources/review/Section11_Reconciled.tex. It applies the earlier presieving hypotheses and endpoint correction, both lower-bound notation fixes, the weighted integration correction, the inclusive p2=z endpoint in both occurrences, and a shorter exact numerical-comparison proof matching Lean. Original manuscript files and their accepted hashes remain unchanged. The generated copy has unique labels and balanced environment counts; it has not been typeset

as a standalone document. The reproduction script and exact replace-with manifest are included.

Crucial source distinction: the weaker original l:presieving hypothesis is not claimed proved. Its completed claim now points explicitly to the separately supplied corrected statement, whose bytes and original-statement bytes are guarded by hashes. The user manuscript still requires those edits. This review-copy completion does not assert that the user has applied them. The remaining open paper-level claim is the final quantitative Goldbach theorem; the original presieving version is retained as a superseded, unclaimed variant.

The weighted endpoint uses the checked effective upper-mean route. It does not add completion credit to every stronger asymptotic displayed in the manuscript's alternative convolution argument. The exported lemma conclusions and the joint sieve estimates are what is certified.

`Section11_Fixed_Sieve_Comparison.md`

Section 11: fixed sieve sets and progression sequences

This note explains the checked starting results through checkpoint1218. It does not claim the later local-density, presieving or final sieve propositions.

Actual sets

The lower sequence consists of integers r-1, where r is prime, r=11 modulo24 and n-r is a prime square. The original sieve uses primes p=3 modulo4 which do not divide n-1. The fixed sieve uses all primes p>3 with p=3 modulo4. Both use the strict real cutoff p<z.

The upper sequence M_n(ell) consists of integers r=ell*t+1<N² with t a prime congruent to3 modulo4 and n-r=m² for a prime m>3. The original upper sieve primes do not divide12*ell*n. The fixed upper sieve primes do not divide12*ell. The level is exactly (N²/ell)^(1/4). The actual closed multiplier set L and its subset L_n are retained.

These definitions are implemented as `goldbachFixedLowerSiftedValues`, `goldbachFixedUpperSifted` and the previously proved original sets. No formal set is replaced by a generic counting hypothesis.

Lower comparison

For every lower-sequence element, r-1=10 modulo24, so3 does not divide r-1. Every other prime in the original lower sieve belongs to the fixed prime set. Surviving the fixed sieve therefore implies surviving the original sieve. Taking cardinalities gives exactly the source lower inequality. The shift from r to r-1 is implemented on actual finite sets.

Checked declarations: `goldbach_fixed_lower_subset`, `goldbach_fixed_lower_values_subset`, `goldbach_fixed_lower_count`.

Exact upper exceptional count

Suppose a pair (ell,r), with ell in L_n, survives the original upper sieve but fails the fixed sieve. Some prime p below the actual level divides r and does not divide12*ell. If p did not divide n, it would be an original upper sieve prime, contradicting survival. Thus p divides n.

Write n-r=m² using the actual defining prime m of M_n(ell). Since p divides both n and r, p divides m². Primality gives p=m, hence r=n-p². Also r-1=ell*t>0, so ell divides the positive integer n-p²-1.

For a fixed p dividing n, the number of such multiplier pairs is therefore at most tau(n-p²-1). Summing gives the exact finite inequality

$$\sum_{\ell\in L_n} S(M_n(\ell),W_{\ell,n},z_\ell)\le\sum_{\ell\in L} S(M_n(\ell),W_\ell,z_\ell)+\sum_{p\mid n}\tau(n-p^2-1).$$

The proof counts pairs (ell,r), so multiplicities across different multipliers are retained. Enlarging L_n to L is handled by a proved finite-set inclusion. Divisor arguments equal to0 contribute0 in the finite envelope; every actual exceptional pair has a positive divisor argument.

Checked declarations: `goldbach_fixed_upper_exception`, `goldbach_fixed_sieve_exceptions_card`, `goldbach_fixed_upper_source_count_finite`.

Uniform power error

For each epsilon>0, the elementary prime-factor bound gives a constant K1 such that the number of prime divisors of n is at most K1 N^(epsilon/2), uniformly for positive n<=N². The proved divisor bound

gives a constant K2 such that tau(a)<=K2 N^(epsilon/2) for every0<=a<=N². Both constants are chosen before N and n.

Applying the second bound to each n-p²-1 gives

$$\sum_{p|n} \tau(n-p^2-1) \le K_1 K_2 N^{\varepsilon}.$$

The actual proof covers every N>=1,0<n<=N² and every real z, which contains the full retained source range. Neither PNT nor a sieve theorem is used. This closes the whole source lemma `l:fixedsievesets10`, not merely a numerical instance or an implication from the desired error bound.

Checked declarations: `goldbach_uniform_divisor_card`, `goldbach_fixed_sieve_divisor_sum`, `goldbach_fixed_sieve_sets`.

Lower progression identity

For a positive d whose prime divisors exceed3, the CRT supplies a coprime class a modulo24d satisfying a=11 modulo24 and a=1 modulo d. For every prime r, these two conditions are equivalent to r=a modulo24d and d dividing r-1.

The actual lower divisibility count is therefore exactly the number of prime pairs m,r with m²+r=n and r=a modulo24d. The finite count bijection is proved; uniqueness of the positive prime m follows from injectivity of squaring. The residue a is chosen before every n.

Using the actual definition of the already proved circle-method remainder then gives the exact G_R H_R/phi(24d) plus remainder decomposition. Normalizing this by the source X_n and omega_n still requires the local-density calculation, which remains open.

Checked declarations: `goldbach_lower_sieve_sequence_count`, `goldbach_lower_sequence_fixed_residue`, `goldbach_lower_sequence_circle_identity`.

Upper progression identity

For ell in the actual closed L and z>3, the source factorization ell=2*u*p2 gives ell=2v with v=3 modulo4 and coprime to6. This is proved from the prime support of u and the primality and lower bound of p2.

For d coprime to12*ell, the upper CRT class a modulo12d satisfies v*a=5 modulo12 and ell*a+1=0 modulo d. The congruence t=a modulo12d is proved equivalent to these two conditions with t in place of a.

For n=12 modulo24, the equation m²+ell*t=n-1 and this residue condition force m>3 and t=3 modulo4. Conversely, the actual defining conditions of M_n(ell) yield that residue condition. The proof therefore gives the exact finite identity at n-1, including the source prime exclusions and the actual r<N² cutoff. Again a is chosen before every n.

Checked declarations: `goldbach_upper_sieve_sequence_count`, `goldbach_closed_multiplier_residue_data`, `goldbach_upper_sequence_fixed_residue`, `goldbach_upper_sequence_circle_identity`.

Mean-square remainder transport

Let I be any fixed finite subset of the allowed circle-method indices. Extend its coefficients by0 outside I and extend the residue family by the coprime representative1. The full circle-method theorem applies to these extended families. A checked finite sum identity removes every added index and recovers the sum over I exactly. The constant and threshold retain their original order before all families and cutoffs.

For the upper sieve, n maps to n-1 injectively on positive integers. Thus a sum over any positive subset of n<N² is bounded by the full original mean-square sum. If the whole range0<=n<N² is used, the value at0 is counted twice after taking natural predecessors. Its square is bounded by the original full sum, giving a harmless factor2. No value of the remainder at0 is assumed to vanish.

The transport uses only the existing independent ordinary SW, ordinary BV and uniform raw Ren/Ghosh inputs of the completed circle-method theorem. No additional deep input or own-paper sieve endpoint is introduced.

Checked declarations: `goldbach_pred_square_sum_bound`, `goldbach_pred_full_square_sum_bound`, `goldbach_sieve_remainders_selected`.

Next obligations

The local-density identities and their multiplicative normalization must still be proved for the actual Euler factors. Then the two-range presieve construction and its levels, dimension bounds, coefficient estimates

and main-term analysis must be completed before either full sieve proposition is awarded source-level credit. The statement map and current paper ledger distinguish these unfinished claims from the checked reductions above.

Section11_Local_Densities.md

Section11: the two local-density lemmas

Both `l:lowerlocaldensity` and `l:upperlocaldensity` are proved for the retained source context. No new external input is used. The selected manuscript is unchanged. The clause-by-clause source review is in `verification/section11-local-density-source-review1221.json`.

The root count is the actual cardinality of the units satisfying the square equation. Consequently zero is excluded from the variable being squared, and rho is zero at the zero residue. Opening the prime Ramanujan sum proves

$$C_p(a) = p\rho_p(a) - (p-1), \qquad \rho_p(a) = \begin{cases} 2, & (a/p) = 1, \\ 0, & (a/p) \neq 1. \end{cases}$$

The normalized character coefficient agrees with the earlier original definition. Its unrestricted factor, when the multiplier is invertible, is

$$1 + G_p = \frac{p(p - 1 - \rho_p(n))}{(p-1)^2}.$$

If the progression fixes its unit variable to b, the factor becomes

$$1 + G_p = \frac{p\rho_p(n - \ell b)}{p-1}.$$

The proof also checks the multiplier-zero case, whose common factor can vanish. The actual unrestricted pair cardinality plus rho equals p-1.

The finite coefficient identity for q dividing ell*k is needed at the small primes. It is proved from reduction of units and their exact normalized fiber mass. Under the source congruence n-ell*b=1 modulo24, this gives the factor8 at2 and3 at3. In particular, the upper modulus12 with its even multiplier is treated correctly at8.

Adjoining one prime p>3 to a coprime progression modulus proves an identity between the original normalized arithmetic factors. The identity multiplies by

$$g(p) = \begin{cases} \rho_p(n-c)/(p - 1 - \rho_p(n)), & p \leq R, \\ 1/(p-1), & p > R, \end{cases}$$

when ell*b=c modulo p. The cutoff is inclusive, including a prime equal to R. Every other prime factor is proved unchanged. Induction over a finite set of distinct primes, followed by the square-free prime-factor product identity, proves the full factorization for d.

For the lower sieve, the actual CRT conditions are b=11 modulo24 and b=1 modulo d, while n=12 modulo24. They give c=1. The base arithmetic factor at modulus24 is proved nonzero, so the paper's original quotient definition is identified with this density. The prime formula is proved for every p>3; the lower sieve's additional restriction p=3 modulo4 is therefore included.

For the upper sieve, the existing source multiplier result supplies ell=2v and v coprime6. The actual conditions vb=5 modulo12 and ell*b+1=0 modulo d imply the required congruence at24 and give c=-1 at every divisor prime. Unit cancellation shows that every representative reduces to the same base class modulo12. The resulting density is

$$g_{n,\ell}(p) = \begin{cases} \rho_p(n)/(p - 1 - \rho_p(n-1)), & p \leq R, \\ 1/(p-1), & p > R. \end{cases}$$

One explicit function omega(d)=d times the product of these prime densities is constructed before all supported square-free d and all representatives of the source CRT classes. Its value at1 and its multiplicativity on coprime arguments are proved. The complete existence and factorization statement is `goldbach_upper_local_density`.

The denominator is positive for p>3. Both candidate densities are nonnegative and at most 2/(p-3), which is below1 for p>=7. At5, a kernel-checked finite calculation gives either rho_5(n)=0, or rho_5(n)=2 and rho_5(n-1)=0. The upper density is therefore at most1/2 at5. This proves its full required range [0,1).

The upper identity never divides by its base arithmetic factor. If a common obstructing local factor is zero, the multiplicative identity still proves equality with zero on both sides. A zero base is not an extra hypothesis or an omitted case.

The paper-level ledger is now19 proved claims,27 complete conditional implications,1 quoted external result and7 open claims, with4 definitions separate. Presieving and the later sieve estimates remain open. The later analytic use of these densities does not retroactively become an input to the local-density proofs.

`Section11_Presieving_Review.md`

Corrected presieving: complete conditional deduction

The corrected full presieving implication is proved in `goldbach_corrected_presieving`, conditional on the independent general one-stage Rosser–Iwaniec fundamental and coefficient inputs. Their exact interfaces are documented in `docs/EXTERNAL_SIEVE_INPUTS.md`. The literal original source lemma remains open until its weaker hypotheses are corrected. Proposed replacement passages are in `Section11_Presieving_Replacements.tex`. They have not been applied to the selected manuscript.

The primary source checked is Yuta Suzuki's The Rosser–Iwaniec sieve, dated14 March2023. Definition8.2 requires the product bound with factor 1+K/log(w), rather than an unrestricted fixed factor K0. Lemma14.7 states the corresponding coefficient estimates with D>=z>=2. Proposition11.8 controls the sieve-function tails. The original official Iwaniec1980 PDF downloads returned403; this review does not claim they were read.

The corrected statement explicitly requires X>=0 and g(1)=1; X=0 is included. Three further points need to be explicit in the manuscript. First, the refined dimension estimate is stronger than a coarse implied-constant estimate. A fixed K0 allows a concentration of local mass at a few large primes, while 1+K/log(w) approaches1 at large w. Consequently the coarse estimate alone is not the hypothesis of the cited coefficient theorem. The refined interval forms of Mertens' theorem are available for the actual outer densities in this application; they must be stated and used.

Second, the preliminary prime set includes p=R. A dimension condition restricted to p<y with y<=R omits that endpoint if R is prime. The proposed formulation bounds the truncated preliminary prime set over all relevant intervals and uses sifting limit R+1. This retains exactly the intended primes p<=R.

Third, D0<D by itself does not place the outer sieve in the range D1>=z, and log(D0)/log(R) tending to infinity by itself does not give the rate needed to absorb arbitrary outer logarithmic losses. The proposed passage states the actual scale R=(log N)^C, the bound R+1<=z<=D1<=N^A, and the intended uniformity. These conditions give preliminary relative errors bounded by a constant times exp(-c(log log N)^2), which dominate every fixed negative power of log N. Applying these scale conditions to the final lower and upper sieve parameters is a separate later obligation.

The checked finite construction does not assume the paper's two-stage conclusion. It starts from independent one-stage lower and upper weight inequalities. For an outer lower coefficient u(d), the preliminary choice is the lower weight when u(d)>=0 and the upper weight when u(d)<0. For an outer upper coefficient, the choices are reversed. Lean proves the resulting pointwise inequalities, then sums them over actual finite sequences. The sequence is represented by a finite set of indices and an arbitrary integer-valued map, so repeated values are counted correctly.

Disjoint prime supports imply coprimality of every preliminary divisor e and outer divisor d. This gives unique decomposition of m=e*d. The flattened coefficient is explicitly the sum over pairs with that product. Lean proves its absolute value is at most1 and that it vanishes for m>=D0*D1. It also proves exact reindexing into the sum over all natural m<D0*D1, using the natural ceiling of the real level. None of these coefficient definitions mentions the density function, the sequence, X or the remainders.

For the main term, let V0 be the preliminary Euler product, and suppose each preliminary coefficient sum differs from V0 by at most epsilon0*V0. The error from either signed selection is proved to be at most

$$\varepsilon_0 V_0 \sum_d |u(d)| g(d).$$

Together with the outer coefficient estimates, this proves explicit finite lower and upper inequalities with the original signed remainders. No positivity assumption on an outer signed weight is introduced. The uniform Euler-product bound and exact rate absorption are now proved and joined to the two independent classical one-stage interfaces in the complete corrected asymptotic deduction. No own-paper endpoint is assumed.

The source-level ledger remains19 proved claims,27 complete conditional implications,1 quoted external result and7 open claims. The presieving supporting theorems do not increase that count by themselves.

The signed outer mass M is proved to satisfy M/V1<=V1^(-2). The refined dimension condition at w=2 and log(z)<=A*log(N) give V1^(-1)<=K2*(log N)^kappa, with K2=(1+K/log 2)*(A/log 2)^kappa. Multiplicativity is needed only on supported square-free integers; its prime-product expression is proved internally.

For t=log(log N), D0=exp(t^3), and preliminary sifting limit exp(C*t)+1, the exact preliminary parameter satisfies s0>=t^2/(C+1) for t>=1. Lean proves this bound, eventual level admissibility and s0 tending to infinity. The explicit common relative error

$$\varepsilon(N) = K_1(\log D_1)^{-\delta} + K_0 e^{-cs_0}\left(K_2(\log N)^{\kappa}\right)^2 \longrightarrow 0$$

is proved uniformly before all admissible densities. For every sufficiently large N, the final coefficients are chosen before g, the finite sequence, X and the remainders. In the paper application they are therefore chosen before n. The main terms contain only the outer functions f_kappa and F_kappa. The natural-absolute-value bridge covers repeated and negative integer sequence values.

The original manuscript is unchanged. The corrected complete conditional variant is recorded separately from the literal-source inventory. Its independent one-stage inputs and classical function normalization remain unproved; the kernel audit verifies the consuming implication. The next mathematical milestone is the actual lower-sieve application.

Section9_Complete.md

Section9 has been reviewed against the actual source, not merely its stale planning map. Its nine numbered statements and the final quantitative exceptional-set theorem all have complete matched formal evidence. The final theorem already followed conditionally at1259; this task removes the remaining elementary arithmetic and seed hypotheses and closes additional source prerequisites.

The primitive sum-of-two-squares criterion is now proved from pinned Mathlib and project algebra. For n=2 mod4, a coprime sum-of-two-squares representation exists exactly when no3 mod4 prime divides n. The proof covers arbitrary prime powers by choosing a conjugate orientation at each split prime, and treats the single factor2 using parity. Consequently l:fromsiftedtop, l:truncationT and c:mastersievereduction are promoted from conditional to proved under their ordinary source hypotheses. The upper arithmetic reduction was already proved. Their actual finite sets, prime-factor multiplicities, closed cutoff and uniform error terms are retained.

The complete source arithmetic exports are goldbach_from_sifted_to_primes_complete, goldbach_truncation_complete, goldbach_section9_master_exact and goldbach_master_sieve_complete. No primitive criterion is a hypothesis. The source assertion that W(alpha) is finite for every alpha>1 is now proved over its full range, including the square-root endpoint singularity; nonnegativity is also proved.

A concrete compactly supported smooth function is constructed with value1 at0 and support inside(-1/8,1/8). The final export goldbach_section9_headline_complete therefore takes only GoldbachSection11Inputs and A>0, and yields the actual real-X exceptional-set estimate. It needs neither a supplied primitive criterion nor a supplied smoothing function. The almost-all statement is also exported as goldbach_section9_exceptional_density_zero.

The circle-method approximation, lower and upper sieve propositions, uniform main-term lower bound and strict numerical gap retain their complete checked conditional scopes. Remaining independent inputs are SW/BV, Ren/Ghosh, the general one-stage sieve and classical sieve normalizations/continuity, the effective multiplicative mean and classical Mertens estimates. None is a paper endpoint. The primitive criterion has been removed from every current claim's external-input list and is explicitly recorded as a proved internal arithmetic lemma.

The Section9 map is reconciled to the actual current nine source statements and their current theorem evidence, replacing the stale1085/1136 planning statuses. The old map is archived. Original manuscript and accepted source hashes remain unchanged. The user is applying the previously supplied Section11 edits; their application is not presumed. No additional Section9 LaTeX replacement was needed. The previously supplied presieving replacement remains the explicitly reviewed target.

The paper ledger now records22 proved under ordinary hypotheses,31 complete conditional implications and1 quoted external theorem among54 reviewed numbered targets, plus4 definitions. This is a statement inventory, not a percentage based on supporting theorem counts. Earlier stronger unnumbered assertions in alternative proof routes retain their separately documented scope; no new completion credit is awarded to them.

Checkpoint1260 separately saved the primitive multiplication proof, and1261 separately saved the complete criterion, general W integral and fixed smooth seed. This checkpoint saves their final Section9 consumption and source reconciliation.

`Section9_Primitive_Two_Squares.md`

The full primitive sum-of-two-squares criterion used in Section9 is now proved unconditionally: for n=2 mod4, n has a coprime sum-of-two-squares representation exactly when no prime3 mod4 divides n. The proof constructs a primitive representation of the odd cofactor using the checked prime-multiplication argument, then doubles it; parity excludes a common factor2. The converse uses Mathlib's already proved obstruction to -1 being a square modulo a3 mod4 prime. No external arithmetic criterion is assumed.

The source's unnumbered assertion that W(alpha) is finite for every alpha>1 is now proved across its full range, with interval integrability of the actual logarithmic kernel. The endpoint square-root singularity is handled by an integrable negative half power; nonnegativity is also proved for the complete range. This extends the earlier special-value formalization at alpha9/8.

A concrete smooth bump with inner radius1/32 and outer radius1/16 is converted to a Schwartz function. Lean verifies value1 at0, nonzero, compact support contained in(-1/8,1/8), and the source support cutoff. Thus the proof will no longer need an externally supplied test function at the final endpoint. No additional source replacement is required for these valid mathematical statements.

This checkpoint saves these prerequisite proofs separately. The next checked export specializes the legacy parameterized Section9 deductions with the proved criterion and fixed seed, and updates the source map. No source claim is promoted before that final statement review. The previously supplied manuscript edits remain in progress; no changed user source is assumed.

`Sharp_Gauss_Asymptotic.md`

Lemma 3.21: sharp bound with an explicit PNT boundary

Checkpoint 10, 22 October 2026. Source label `l:inftyboundsaq`, equation `e:gaussboundknown`. The complete paper-specific deduction is now proved in Lean, conditional on the two standard prime-distribution limits below. The six previously completed numbered results remain unconditional.

Exact external hypotheses

For real y define

$$\theta(y) = \sum_{p \le y} \log p, \qquad W(y) = \sum_{p \le y} p^{-1/2}, \qquad F(y) = \frac{\sqrt{y}}{\log y}.$$

In Lean the sums use `Nat.primesLE (Nat.floor y)`. The proof checks the exact equivalence of membership with p prime and p<=y for the positive cutoffs used. `PrimeAsymptoticInputs` consists of these two hypotheses, for y tending to positive infinity through the real numbers:

$$\theta(y)/y \longrightarrow 1, \qquad W(y)/F(y) \longrightarrow 2.$$

Both limits are explicit fields of a proposition supplied to the conditional theorems. This project does not construct a proof of that proposition. There are no new `axiom` declarations or proof placeholders. A clean transitive axiom audit establishes the validity of the implication; it does not establish the hypotheses.

The second limit is the usual partial-summation consequence of PNT. For clarity about scope, the bridge from PNT to these two limits is also outside the present Lean development. Thus the exact formal dependency is this pair of standard limits, not an imported PNT theorem.

PNT availability and sources

PNT has been formalized in the separate PrimeNumberTheorem+ project. The pinned Mathlib's `docs/100.yaml`, entry 5, links to it, and a targeted source inspection found no PNT asymptotic to import from our pinned Mathlib. The upstream project inspected on 22 October 2026 targets Lean/Mathlib 4.35.0; our existing verified development targets Lean 4.34.1 and Mathlib commit `d13f23b723b8a846827a245b89c10fc7d3f11612`. No upgrade, external import, or external-project axiom audit has been performed. This is an unintegrated dependency, not a claim that PNT has never been formalized. See `verification/pnt-availability.json` for the inspected file hashes.

A published reference is Iwaniec--Kowalski, *Analytic Number Theory*, AMS Colloquium Publications 53, the paper's existing bibliography key `iwaniec`. The classical PNT and its original sources are also recorded in NIST DLMF 27.2, equation 27.2.3. The following elementary bridge explains precisely why the hypotheses are standard PNT consequences; it is a mathematical explanation, not a Lean proof. Writing pi(y) for the prime counting function, partial summation gives

$$\theta(y) = \pi(y)\log y - \int_2^y \frac{\pi(t)}{t}\,dt, \qquad W(y) = \frac{\pi(y)}{\sqrt{y}} + \frac{1}{2}\int_2^y \frac{\pi(t)}{t^{3/2}}\,dt.$$

PNT says pi(y)~y/log y. The first integral is O(y/log y)=o(y), by splitting at sqrt(y) and using pi(t)=O(t/log t). In the second formula the integral is asymptotic to `2*sqrt(y)/log(y)`: replace pi(t) by t/log t above an arbitrarily large fixed lower endpoint, and compare derivatives of the integral with `2*sqrt(y)/log(y)`. Its derivative ratio tends to one. The finite initial interval contributes a bounded constant. These formulas give the two limits.

1. Logarithmic budget, proved without PNT

Let q>=1, S be its prime divisors, and L=log(q). The squarefree product of these prime divisors divides q, so

$$\sum_{p\in S} \log p \le L.$$

`primeFactors_log_budget` proves this using the actual natural prime-factor set, positive products, divisibility, and monotonicity of the real logarithm.

2. Finite threshold inequality, proved without PNT

For every real y>=2,

$$\sum_{p\in S} p^{-1/2} \le W(y) + \frac{L-\theta(y)}{\sqrt{y}\log y}.$$

Put c=1/(sqrt(y)*log(y)) and subtract c*log(p) from each summand. Since sqrt(t)*log(t) increases on t>=1, the difference is nonnegative below y and nonpositive above y. Drop the nonpositive terms, extend the small terms to all primes <=y, and apply the logarithmic budget. The general finite-set comparison and its application are both proved in `PrimeFactorEstimates.lean`. No exact extremizer or prime-factor asymptotic is assumed.

3. Reciprocal and logarithmic corrections, proved without PNT

For L>=2,

$$\sum_{p\in S} p^{-1} \le 1 + \log L + 1/\log 2.$$

For p<=L, enlarge the sum to the harmonic sum through floor(L). For p>L, use `1/p <= (1/L)*(log p/log 2)` and the same logarithmic budget. This proof also covers nonsquarefree q. It deliberately uses a sufficient O(log L) bound; the stronger O(log log L) in the source is unnecessary for 3.21.

Writing A(p)=(1-p^(-1/2))^(-1), the project proves, for every integer p>=2,

$$A(p) \le 4, \qquad \log A(p) \le p^{-1/2} + 4/p.$$

Indeed `log A <= A-1`, and the exact geometric identity is `A-1 = p^(-1/2) + A/p`. The local bound, the identity, and summation over the actual odd prime factors are all checked by Lean.

4. Connect the actual Gauss sums

Checkpoint 09 proved the genuine normalized square-sum estimate

$$|\mathfrak{s}(a,q)| \le q^{-1/2}\, 4 \prod_{p\mid q,\ p>2} A(p) \quad ((a,q)=1).$$

For y>=2 put

$$E(y) = \log 4 + 4(1 + \log y + 1/\log 2), \quad B(y) = W(y) + \frac{y-\theta(y)}{\sqrt{y}\log y} + E(y).$$

The finite proofs give, unconditionally when L>=2,

$$\log\left(4 \prod_{p\mid q,\ p>2} A(p)\right) \le B(L), \qquad |\mathfrak{s}(a,q)| \le q^{-1/2} e^{B(L)}.$$

Under the explicit analytic hypotheses,

$$\frac{B(y)}{F(y)} = \frac{W(y)}{F(y)} + 1 - \frac{\theta(y)}{y} + \frac{E(y)}{F(y)} \longrightarrow 2.$$

The identity is proved for y>1 using sqrt(y)^2=y. The final correction tends to zero by the checked Mathlib limits `(log y)^j/sqrt(y) -> 0`, j=1,2. The fixed constant four at two is therefore absorbed rather than omitted.

5. Full source statement and quantifiers

Define the explicit coefficient error

$$r(q) = B(\log q)/F(\log q) - 2.$$

`gaussAsymptoticError_tendsto` proves r(q)->0 under the two analytic inputs. `principalSquareSum_sharp_error_bound` proves the inequality with exponent `(2+r(q))*F(log q)` without those inputs. The theorem `principalSquareSum_two_add_o_one` combines these into the literal source statement, with one r independent of the integer coefficient a.

`principalSquareSum_sharp_bound_int` also states the equivalent epsilon form: for every epsilon>0 there exists an integer Q such that for every positive q>=Q and every integer a coprime to q,

$$|\mathfrak{s}(a,q)| \le q^{-1/2} \exp\left((2+\varepsilon)\frac{\sqrt{\log q}}{\log\log q}\right).$$

`principalSquareSum_three_bound_int` proves the coefficient-three consequence by choosing epsilon=1. The threshold is existential, as in the source; no effective numerical value is claimed. Modulus zero is excluded explicitly by `NeZero q`. The reciprocal square root is the source's q^(-1/2), and the integer-to-unit equivalence is the same checked one used in checkpoint 09.

There is no new manuscript correction at this checkpoint. No headline theorem or real-character-twisted Gauss bound is claimed proved.

Smooth_Cutoff_and_Dyadic_Arcs.md

An actual smooth cutoff and its rational-arc geometry

The module constructs a real smooth bump centered at zero, with inner radius 1/2 and outer radius 1. Mathlib supplies the constructed function; its properties are proved rather than postulated:

$$\eta \in C^\infty, \quad \eta(-x) = \eta(x), \quad 0 \le \eta \le 1, \quad \eta(x) = 1 \ (|x| \le 1/2), \quad \eta(x) = 0 \ (|x| \ge 1).$$

For the actual dilate $\eta_s(\theta) = \eta(16^s\theta)$, a nonzero value implies $|\theta| < 16^{-s}$. The identity $16^s = (2^s)^4$ gives the major-arc core assertion:

$$q \le R, \quad 2^s \le q, \quad 2R^4 \le N^2, \quad |\theta| \le R/(qN^2) \quad \Longrightarrow \quad \eta_s(\theta) = 1.$$

The proof keeps all constants and proves the required positivity before dividing.

For $s \ge 2$,

$$2\,16^{-s} < (2^{s+1})^{-2}.$$

The preceding rational-separation theorem therefore implies uniqueness of an active rational center whose reduced denominator is at most 2^{s+1}.

Module	New named theorems
SourceArcCutoff.lean	13

Scope and small scales

This proves uniqueness only for s>=2. The first two scales remain for a finite overlap estimate; the strict radius inequality is not silently asserted for them. These functions currently live on real frequencies. The real quadratic integral is not treated as periodic in a noninteger integration variable. The full minor-arc bridge, approximation and high-low operator conclusions remain open.

<!-- checkpoint-verification -->

Verification

All 102 project modules and the root were freshly rebuilt: 3621 jobs pass without warnings/errors. All 813 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 104 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. No new deep analytic input or own-paper lemma is assumed.

`Source_Model_Logarithmic_Rates.md`

Logarithmic rates for the actual source scale

For c,alpha>0 and A>=0, the module first proves the eventual estimate

$$e^{-cx^\alpha} \le 2^A(1+x)^{-A}.$$

It composes the established exponential-versus-power limit with x^alpha, proves the exact power substitution on positive x, and handles the change from x to 1+x. The threshold depends on the fixed parameters.

For the already defined

$$R(N,Q) = \max\{\exp((\log N)^{1/10}), Q^8\},$$

the exact exponent split gives

$$R^{-1/6} = R^{-1/16}R^{-5/48} \le Q^{-1/2}e^{-(5/48)(\log N)^{1/10}}.$$

Consequently, for each fixed A>=0, one natural N0 works for every Q>=1:

$$N \ge N_0 \Longrightarrow R(N,Q)^{-1/6} \le 2^A Q^{-1/2}(1+\log N)^{-A}.$$

No upper bound on Q is needed for this parameter implication; the separate analytic estimates retain their own ranges.

A final general estimate proves that any fixed affine logarithmic factor A+B log R is eventually at most R^epsilon for epsilon>0. In particular,

$$(A + B\log R)R^{-5/12} \le R^{-1/3}$$

eventually. This matches the loss produced by counting finitely many active dyadic model terms.

Module	New named theorems
SourceModelLogarithmicRate.lean	5

Scope

All assertions are proved in Lean from the pinned Mathlib limits; no new analytic input is assumed. These estimates convert an established error rate. They do not themselves establish that rate for the actual prime-square symbol, or complete approximation 6.9 and its remaining minor-arc/periodization obligations.

<!-- checkpoint-verification -->

Verification

All 105 project modules and the root were freshly rebuilt: 3624 jobs pass without warnings/errors. All 828 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 107 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. No new deep analytic input or own-paper lemma is assumed.

`Source_Range_Major_Arc.md`

The uniform source-range major-arc implication

Checkpoint 30. The finite Fourier reduction, actual partial summation, error absorption and source parameter inequalities are now assembled. The major-arc R^-5 implication is proved in the source N,Q range, conditional on the displayed genuine progression-prefix estimate.

Quantified statement

Fix c>0 and put c0=min(c/7,1)/8. There is an integer N0>=1 depending on c alone such that the following holds for every integer N>=N0 and every real Q>=1 satisfying

$$\log Q \le c_0\sqrt{\log N}.$$

Let R=max(exp((log N)^(1/10)),Q^8). For every positive modulus q, complex Dirichlet character chi modulo q, coefficient kappa, beta>0, integer a and real theta, suppose

$$q \le R^{11/10}, \qquad |\theta| \le R/(qN^2).$$

Suppose C>=0 and the actual progression prefix satisfies

$$\left|C_{q,u}(t) - \frac{1}{\phi(q)}\left(t - \kappa\chi(u)\frac{t^\beta}{\beta}\right)\right| \le NC\exp(-c\sqrt{\log N}) \quad (N/2 \le t \le N)$$

for every reduced residue u. Then

$$\left|\widehat{\widetilde{A}}_{N,\Lambda}(a/q+\theta)-\left(\mathfrak{s}(a,q)M_{N,1}(\theta)-\mathfrak{s}_{\chi}(a,q)\kappa M_{N,\beta}(\theta)\right)\right| \leq \frac{4+(2+4\pi)C}{R^5}.$$

`sourceMajorArc_uniform_inverse_fifth` proves exactly this implication. Its N0 is chosen before Q,q,chi,kappa,beta,a,theta and C. Thus the threshold is uniform over the displayed range; the error constant depends only on C. It applies in particular to the source's dyadic N and Q>=2.

Assembly of the proof

`sourceMajorArc_parameter_threshold` composes the proved real-variable eventual parameter conditions with the divergence of the natural numbers in the reals. It chooses a natural threshold at least one. For N>=N0, the actual R therefore satisfies R>=1, log(R)<=log(N), R^5*log(N)<=sqrt(N), and 7*log(R)<=c*sqrt(log(N)).

The fixed-parameter actual-symbol theorem from checkpoint 28 then applies directly. Every integration-by-parts, integrability, finite Gauss-sum and normalization condition has already been discharged. None is reinstated as an assumption in the final theorem.

`sourceMajorArc_uniform_ordinary` gives the separate ordinary statement from a raw prefix estimate around t/phi(q), with no exceptional parameter in its conclusion. Its proof specializes kappa=0 and uses the proved identity for the zero exceptional model. In the exceptional application kappa=1 includes the correction. Data can change between scales N; at each one scale the same data must work throughout [N/2,N].

Exact trust boundary

The raw exponential progression-prefix inequality above is still a theorem hypothesis. This unit does not construct exceptional characters or zeros, prove a Page/Siegel-Walfisz theorem, or assert that an unverified version of a published theorem supplies the hypothesis. Its applicability in the external theorem's own modulus range must be checked separately. The explicit c0 above controls the elementary error absorption.

The major-arc part of Lemma 6.5 is now a verified implication from this raw analytic input in the source range. The independent minor-arc bound has not been established over its full stated range, so full Lemma 6.5 remains open. The next unit handles the exact Ren input and its genuinely proved deduction. Proposition 4.19 and Theorem 1.1 remain later targets; Theorem 1.1 is not counted as formalized.

Module	New named theorems	Result
SourceRangeMajorArc.lean	3	Uniform integer parameter threshold and actual exceptional/ordinary source-range inverse-fifth major-arc implications

No new mathematical axiom or numerical experiment is used.

<!-- checkpoint-verification -->

Verification

All 94 project modules and the root were freshly rebuilt: 3609 jobs pass without warnings/errors. All 760 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 96 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. No new deep analytic input or own-paper lemma is assumed.

Strong_Improving_Complete.md

Corollary 4.15: complete strong improving deduction

The source is `c:improvinggeneral`, equation 4.16 (`e:lpimproving`), near line 2431 of the archived `sources/current/Draft5.tex`. All 70 modules and the root received a fresh build: 3467 jobs pass without warnings or errors. All 565 named theorems pass the transitive axiom audit; 72 Lean files pass the source scan.

The paper's abstract high-low assumptions, positivity of both kernels and two genuine uniform l2 bounds imply the full strong improving conclusion:

$$\|A_N f\|_{p'} + \|B_N f\|_{p'} \le D_p N^{1/p'-1/p}\|f\|_p, \qquad \frac{2(s+r)}{2r+s} < p \le 2.$$

The constant is chosen before N, the kernels, the windows and the input. The formal statement works for every positive real N. Both operators are actual Mathlib continuous complex-linear maps between sequence spaces, and they agree with the actual convolution on every finite complex input. The prime-square operators' high-low hypotheses are not established here.

Exact statement and new declarations

`PrimeSquares.highLow_AB_strong_improving` is the paper-facing theorem. It uses P,Q:ENNReal with P.toReal=p and Q.toReal=p/(p-1), and the ordinary Fact instances 1<=P and 1<=Q needed for Banach sequence spaces. `improving_exponents_exist` constructs these parameters for every real 1<p<=2: P=ofReal(p), Q=ofReal(p/(p-1)), with both finite and P<=2<=Q. Thus this parameterization does not restrict the source's real exponent range.

Module	New named theorems	Role
`FiniteOutputDuality.lean`	8	Conjugate algebra, complex test identities, cancellation and genuine output norm
`StrongImproving.lean`	4	Actual finite convolution agreement, strong operator norm and combining two maps
`StrongImprovingTheorem.lean`	2	Real exponent parameters and full abstract Corollary 4.15

No interpolation conclusion, duality assertion or desired strong norm is an external assumption. These are deductions from the already proved finite bilinear estimate and the source's genuine operator hypotheses.

Finite-output duality, including zero coordinates

Let 1<p<=2 and put R=p/(p-1). Then R>=2, (R-1)p=R and 1/R=1-1/p. Given a complex sequence u, define

$$g(x) = |u(x)|^{R-2}u(x).$$

The test has the same complex phase as u. The pairing conjugates the test. At a nonzero coordinate, ordinary positive-base real-power laws give

$$u(x)\overline{g(x)} = |u(x)|^R, \qquad |g(x)|^p = |u(x)|^{(R-1)p} = |u(x)|^R.$$

At a zero coordinate, g(x)=0. This includes R=2, where the real-power convention gives 0^0=1 but its product with u(x)=0 is still zero. Both identities therefore remain valid. The formal proofs split off this case before using positive-base power addition. They retain the complex conjugation and prove the pairing identity as a complex equality.

For a finite output set G let S=sum_(x in G)|u(x)|^R>=0. The identities yield

$$\sum_{x\in G} u(x)\overline{g(x)} = S, \qquad \|g\mathbf{1}_G\|_p = S^{1/p}.$$

Suppose every finite output test satisfies

$$\left|\sum_{x\in G} u(x)\overline{h(x)}\right| \le M\|h\mathbf{1}_G\|_p, \qquad M \ge 0.$$

Testing with g gives S<=M*S^(1/p). If S=0, then S<=M^R immediately. If S>0, divide by its positive 1/p power to get S^(1/R)<=M, then raise to the positive power R. Thus S<=M^R for every finite G, including the empty set. `duality_power_cancel` checks precisely this division, zero case and real-power argument.

`finite_output_power_bound` proves the finite-sum bound, and `lp_norm_le_of_finite_pairing` invokes Mathlib's finite-sum norm criterion to obtain ||u||R<=M for an actual u in lp(R). The proof does not require the untruncated g to belong to lp(p); only its finite truncation is used. No representation theorem for the Banach dual is assumed or needed.

The sharp norm on the actual operator

Fix one positive kernel K, 0<q<p<2, N>0, and constants A,B>=0. Assume

$$|\langle K * \mathbf{1}_F, \mathbf{1}_G\rangle| \le A N^{1-2/q}|F|^{1/q}|G|^{1/q}$$

for all finite sets, and an actual continuous l2 realization with norm at most B. Set

$$a = 1/q - 1/p > 0, \qquad b = 1/p - 1/2 > 0,$$
$$C(A,B,p,q) = 3^{2/p}\left(\frac{A}{1-2^{-a}} + \frac{2^b B}{1-2^{-b}}\right).$$

The already checked theorem `restricted_l2_finite_bilinear_improving` proves, for arbitrary finite F,G and complex f,h,

$$|B_{F,G}(f,h)| \le C(A,B,p,q) N^{1-2/p}\|f\mathbf{1}_F\|_p \|h\mathbf{1}_G\|_p.$$

The existing contractive inclusions lp(p)->lp(2)->lp(R), composed with the l2 realization, give a coarse actual continuous map U:lp(p)->lp(R). This supplies output membership. Its finite-input coordinate identity identifies U(f*1_F) with K*(f*1_F). Applying finite-output duality with

$$M = C(A,B,p,q) N^{1-2/p}\|f\mathbf{1}_F\|_p \ge 0$$

gives the sharper norm bound for every finite input. The previously proved `lp_opNorm_bound_of_finiteInputs` extends it by density with the same constant. This proves `restricted_l2_strong_improving`; its conclusion is a genuine `KernelLpBound`, not a scalar model or a finite-support estimate.

The generic lemmas `kernelLp_realization_single` and `kernelLp_realization_finiteInput` prove that every such realization agrees with finite convolution for arbitrary complex coefficients. Agreement on indicators is sufficient by linearity. The existing uniqueness theorem identifies any two continuous realizations with that agreement.

Both source operators and all exponent cases

Write p0=2(s+r)/(2r+s). The source parameters imply 1<p0<2. For p0<p<2 choose q=(p0+p)/2 before N. The already proved global restricted equation 4.17 supplies one constant A>0 uniform in N and both kernels. Each individual restricted pairing is bounded by their sum.

Apply the strong one-kernel theorem separately, keeping the source's two l2 constants B_A and B_B. Positivity of A_N is an explicit source hypothesis at line 2300; positivity of B_N is already included in `HighLowEndpointInputs`. It is not a new mathematical restriction.

Let C_A=C(A,B_A,p,q) and C_B=C(A,B_B,p,q). Both are nonnegative and independent of N. The checked assembly chooses D=C_A+C_B+1>0 and proves the sum of the two norms is bounded by D*N^(1-2/p)*||f||p for every f. The harmless added one ensures strict positivity even for zero operators. The checked conjugate identity rewrites the exponent as 1/p'-1/p.

For p=2 the coarse realizations already have the genuine l2 bounds; N^(1-2/p)=1. The assembly uses D=B_A+B_B+1 and requires no limiting interpolation argument. The final theorem gives the two actual maps, their finite convolution identities, and the norm estimate for every actual lp input. Uniformity is expressed by the quantifier order.

The proof establishes the global estimate directly from global equation 4.17, so the paper's last partition into intervals is unnecessary in this formal proof. This is an alternative proof of the same conclusion.

Verification and exact experiments

All 70 project modules and the root were freshly compiled with the artifact cache disabled. The 3467-job build has no warnings or errors. Every one of the 565 named project theorems passes the transitive audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 72 Lean files and finds no placeholders or additional axioms. Exact theorem/audit coverage and fresh module/import coverage are checked. No Lean source changed after these checks.

`scripts/check_finite_output_duality.py` independently checks 223988 finite identity instances: 55987 exhaustive vectors of length <=6 over a six-value Gaussian integer alphabet and 10 structured vectors through length 1024, at output exponents 2,4,6,8 and input exponents 2,4/3,6/5,8/7. All pass. Every operation is integral, including independently checked perfect roots; there is no floating point. Empty, zero and nonreal inputs are included.

The diagnostic u=(1,i) at output exponent four gives pairing 2 with the correct test and 0 with a mistakenly conjugated phase. The test suite therefore detects this particular phase error. The experiment covers sampled exponents only. Lean proves the full real exponent statements; the experiment does not independently verify the operator estimate or the prime-square high-low inputs. Exact ranges, digest and arithmetic are recorded in `verification/finite-output-duality-numerical.json`.

No new deep input is used in Corollary 4.15. The two earlier explicit PNT limits for Lemma 3.21 are unchanged and are not used here. The archived manuscript and accepted revision fragments are unchanged. The next work is to establish the prime-square operators' inputs and connect the main applications, with every genuinely external analytic assumption explicit.

`Strong_Interpolation_Foundations.md`

Corollary 4.15: checked interpolation foundations

Current checkpoint 20: the full abstract Corollary 4.15 is complete, including strong interior interpolation and p=2. This document records an earlier stage; see Strong_Improving_Complete.md for the completed proof and current scope.

Checkpoint 18 update: the actual geometric weight bound, exact N scaling, and actual complex block/family estimates are now proved; see `Interpolation_Geometry.md`. The remaining-obligation list below records checkpoint 17. Rank construction, coefficient mass, finite-output duality and final strong assembly still remain open.

Checkpoint 17 proves four supporting modules with 32 named theorems. These complete the operator-realization and density infrastructure, the finite complex-input bridge, and the finite weighted Schur estimate. **The strong interior estimate in Corollary 4.15 is still open.** In particular, the coarse bound below has no improving power of N.

Source correspondence

The source is Corollary 4.15 (`c:improvinggeneral`), near line 2431 of `sources/current/Draft5.tex`. Its equation 4.16 asks for

$$\|A_N f\|_{p'} + \|B_N f\|_{p'} \le C_p N^{1/p'-1/p} \|f\|_p, \qquad \frac{2(s+r)}{2r+s} < p \le 2.$$

Theorem 4.1, the global finite-indicator estimate 4.17, actual restricted weak distribution bounds, and the strong p=2 case were already proved. This checkpoint supplies genuine sequence-space constructions needed for a direct global proof of the remaining interpolation step. It does not invoke the interpolation theorem cited by the paper as an external input.

Module	Named theorems	Checked content
`LpInclusion.lean`	5	Contractive continuous complex-linear inclusions for finite 1<=P<=Q
`FiniteComplexInputs.lean`	11	Finite inputs and norms, actual convolution identity, complex positivity bound
`LpRealizations.lean`	10	Coarse lp realizations, uniqueness, extension of finite norm estimates by density

Module	Named theorems	Checked content
`FiniteSchur.lean`	6	Weighted Cauchy--Schwarz, Schur, Hankel sums, finite lp coefficient bounds

P and Q in the Lean declarations are ENNReal exponents, with their ordinary finite values P.toReal and Q.toReal. The spaces use counting measure on Z, represented by Mathlib's actual `lp`, not a norm formula on an arbitrary function. The Schur and inclusion lemmas also allow other index types.

Contractive inclusions

`lp_inclusion_norm` proves that the canonical inclusion from lp to lq has norm at most one whenever 1<=P<=Q<infinity. Membership comes from `Memℓp.of_exponent_ge`; the new proof establishes the norm inequality.

Writing p=P.toReal and q=Q.toReal, for every finite S it proves

$$\sum_{i\in S}|f_i|^q=\sum_{i\in S}|f_i|^{q-p}|f_i|^p\le\|f\|_p^{q-p}\sum_{i\in S}|f_i|^p\le\|f\|_p^q.$$

The coordinate bound |f_i|<=||f||p and p-summability are already checked Mathlib results. The lp norm criterion converts the finite-sum estimates into the claimed q norm bound. Real-power identities are justified with q>0, including zero coordinates, the zero vector and P=Q. The inclusion is then packaged as `lpInclusion`, an actual continuous complex-linear map. Its operator norm is proved <=1 and its coordinates are unchanged.

Finite complex inputs and positivity

`finiteInputLp F f P` is the actual finite sum

$$\sum_{y\in F}f(y)e_y.$$

It equals f on F and zero outside F. Its exact norm is proved for every positive finite real exponent. Truncating an lp sequence to F does not increase its norm. No support, phase or nonzero coefficient is assumed.

For an l2 operator T agreeing with the kernel on finite indicators, `kernelL2_realization_single` proves T(z e_y)(x)=K(x-y)z by complex linearity. `kernelL2_realization_finiteInput` then proves

$$T(\textstyle\sum_{y\in F}f(y)e_y)(x)=\sum_{y\in F}K(x-y)f(y).$$

Thus arbitrary complex coefficients refer to the same operator as the previous indicator estimates. This is an exact identity for every x.

For a nonnegative real kernel represented in C, define the actual finite sesquilinear pairing

$$B_{F,G}(f,g)=\sum_{x\in G}\sum_{y\in F}K(x-y)f(y)\overline{g(x)}.$$

`finiteKernelPairing_bound_by_indicators` proves

$$|B_{F,G}(f,g)|\le AB\,|\langle T\mathbf{1}_F,\mathbf{1}_G\rangle|$$

when |f(y)|<=A on F and |g(x)|<=B on G, with nonnegative amplitude bounds. Triangle inequality controls the phases. Positivity identifies the sum of absolute kernel coefficients with the original indicator pairing. Empty sets and zero amplitudes are included.

A genuine operator on the required sequence spaces

For 1<=P<=2<=Q<infinity, `lpL2Lift` is the composition

$$\ell^P\hookrightarrow\ell^2\xrightarrow{T}\ell^2\hookrightarrow\ell^Q.$$

Both inclusions are proved contractions, so its norm is <=||T||. It agrees with the explicit finite convolution for every finite complex input. `kernelLpBound_of_l2` packages the coarse bound with its indicator agreement. `kernelLp_realization_unique` proves that two continuous complex-linear maps agreeing on the kernel indicators coincide, using their agreement on coordinate vectors and density.

Taking P=p and Q=p' gives the spaces required by the corollary for 1<p<=2. The generic formal theorem has the explicit exponent inequalities P<=2<=Q. This only supplies a coarse bound. Since 1/p'-1/p<0 for p<2, it cannot by itself provide the required decay as N grows.

`lp_bound_of_finiteInputs` proves a separate density implication. If a continuous lp-to-lq map has norm bound D on all actual finite inputs, it has the same bound on every lp input. The proof uses the norm-convergent net of finite sums from `lp.hasSum_single`, continuity, and preservation of a nonstrict inequality under limits. `lp_opNorm_bound_of_finiteInputs` gives the operator-norm formulation for D>=0. The finite-input estimate is a hypothesis of this generic transfer lemma; its interpolation application remains to be proved, and is not assumed in a final paper theorem.

Weighted Schur, without a support-size loss

For finite S,T, nonnegative weights w(i,j), and any real coefficients c,d, `finite_weighted_schur` proves

$$\sum_{i\in S,j\in T} w_{ij}c_i d_j \le C\Big(\sum_{i\in S} c_i^2\Big)^{1/2}\Big(\sum_{j\in T} d_j^2\Big)^{1/2},$$

provided all row and column sums are <=C, with C>=0. Weighted Cauchy--Schwarz is proved by applying ordinary finite Cauchy--Schwarz to sqrt(w)c and sqrt(w)d. The row and column estimates bound the two weighted energies. Square-root identities complete the bound, including C=0.

`finite_hankel_row_sum_le` proves that a finite row of h(i+j) is bounded by sum h when h is nonnegative and summable. The indexing shift is injective, so no term is counted twice. `finite_hankel_schur` supplies the corresponding matrix estimate. These generic statements require summability and the total-sum bound; those conditions for the proposed interpolation weight are **not yet proved**.

`finite_energy_le_lp` uses the contractive counting-space inclusion to prove the finite l2 coefficient norm is <= its lp norm for 1<=P<=2. `finite_weighted_schur_lp` therefore replaces the two coefficient l2 norms by their lp norms, with the same C. The bound is independent of the number of blocks. No dyadic decomposition is assumed in these statements.

Remaining obligations

The complete strong corollary still needs these connections:

1. Construct a decreasing-magnitude partition of every finite complex input into dyadic rank blocks, proving their cardinalities and lp mass bound, including first, last, empty and tied-value cases.
2. From the two actual indicator estimates, derive the block estimate with its exact N scaling. Prove the proposed geometric weight is summable with a bound uniform in every positive N.
3. Apply the checked complex positivity and Schur estimates to that actual partition and weight. This must yield a genuine finite bilinear lp bound, without an unproved decomposition hypothesis.
4. Prove finite-output lp duality with the correct complex phase. Convert the finite bilinear estimate to the finite-input output norm bound.
5. Apply the checked realization and density lemmas, retain both l2 constants, and assemble the interior range with the existing p=2 case.

Only then can Corollary 4.15 be marked complete. A successful global route would make the paper's interval-localization step unnecessary for this deduction; no such completed global interpolation theorem is claimed yet.

Verification and exact finite probes

All 62 modules and the root receive a fresh build with the artifact cache disabled. All 489 named project theorems are checked by the transitive axiom audit; only propext, Classical.choice and Quot.sound are allowed. There are no new mathematical axioms, proof placeholders or deep analytic inputs. The declaration-discovery scanner now ignores comments and strings, after the audit caught a comment sentence being mistaken for a theorem name.

The deterministic script `scripts/check_interpolation_foundations.py` uses only integers and exact fractions. It records zero failures in 98304 finite complex positive-kernel pairing cases, 160012 vector-pair Schur checks on 259 matrices, and 3410 lp inclusion checks at integer exponents. Empty sets, zero amplitudes, zero weights, signed coefficients and equality cases are included. It also probes the proposed dyadic

weights at 63 parameter choices, including 0<N<1, with 1008 row and 504 Schur checks. Counts are experiment cases; restricted inputs may repeat across cases.

These experiments do not prove the unbounded geometric estimate, sorting, duality or full interpolation. They do not verify the later prime-square operator hypotheses. The general supporting implications above are Lean proofs. Full archived sources and accepted revisions are unchanged. The count of completely proved unconditional numbered results remains nine; Lemma 3.21 retains its earlier explicit PNT hypotheses independently.

`Strong_Interpolation_Plan.md`

Strong interpolation: completed proof route

Checkpoint 20 completes the abstract Corollary 4.15. The former proposed steps below are all proved; none is an external input.

1. Global restricted improving at q, from the actual source high-low and uniform l2 hypotheses: RestrictedImproving.lean.
2. Contractive sequence-space inclusions, actual complex finite inputs, coarse actual realizations and exact-constant density transfer: LpInclusion.lean, FiniteComplexInputs.lean, LpRealizations.lean.
3. Finite Schur and actual uniformly summable geometric weights with exact N scaling: FiniteSchur.lean, InterpolationGeometry.lean, InterpolationBlockBounds.lean.
4. Actual sorted dyadic rank partition, coefficient mass, exact complex regrouping and strong finite bilinear estimate: DyadicRankMass.lean, FiniteMagnitudeBlocks.lean, FiniteBilinearImproving.lean.
5. Complex finite-output norm duality, zero cases and genuine output norm: FiniteOutputDuality.lean.
6. Strong N-dependent norm on actual continuous operators, arbitrary finite complex convolution agreement and density extension: StrongImproving.lean.
7. Both source operators, a constant preceding N and the input, every permitted real exponent and the p=2 endpoint: StrongImprovingTheorem.lean.

The final theorem is PrimeSquares.highLow_AB_strong_improving. See Strong_Improving_Complete.md for the full proof, source correspondence, constants and verification scope. No strong interpolation work remains.

The next task is the actual prime-square operator input bridge, beginning with Proposition 4.19 and its dependencies in Section 6. Those inputs and the main applications are still open. Deep published inputs must remain explicit when not formalized; own-paper deductions are proof obligations.

`Twisted_Gauss_Bound.md`

Lemma 3.24: an unconditional twisted Gauss-sum bound

Checkpoint 11, 22 October 2026. Source label `l:Gcharbound`, equation `e:Gcharbound`, and the definition `e:mathfraksgeneral`.

The complete statement is now proved in Lean, strengthened from real to all complex Dirichlet characters. For every epsilon>0 there is C_epsilon>0 such that for every positive integer q, every Dirichlet character chi modulo q and every integer a coprime to q,

$$|\mathfrak{s}_\chi(a,q)| \le C_\varepsilon q^{-1/2+\varepsilon}.$$

The constant depends only on epsilon, not on q, chi, a, or the conductor. No primitive-character assumption is made. This proof uses no PNT input, unformalized Weil bound, or classification of real characters. The PNT hypotheses in the separate Lemma 3.21 development are unchanged.

1. Source definitions and principal specialization

For a complex Dirichlet character chi modulo q write

$$U_\chi(b) = \sum_{r\in\mathbb{A}_q} \chi(r)e(br^2/q), \qquad \mathfrak{s}_\chi(a,q) = \phi(q)^{-1}U_\chi(-a).$$

`twistedUnitQuadraticGaussSum` and `twistedSquareSum` are exactly these sums, using the actual modular unit group, the standard complex additive character, and Mathlib's complex Dirichlet character. The source

minus sign and totient normalization are retained. The principal-character specialization is proved to equal the previously formalized principal square sum.

Every character value on a unit has norm one, using the checked Mathlib lemma `DirichletCharacter.unit_norm_eq_one`. The normalized twisted sum therefore has norm at most one for every coefficient. This also handles modulus one with its one-element unit group and totient equal to one.

2. Count square fibers

Let

$$K(q) = \#\{u \in \mathbb{A}_q : u^2 = 1\}.$$

The code defines this as the cardinality of the kernel of the squaring homomorphism on the genuine modular unit group. For unit a and any residue b, the fiber of u -> a*u^2 over b has at most K(q) elements. If the fiber is nonempty, fix v in it. Sending u to u/v injects it into the square kernel: unit cancellation gives u^2=v^2, hence (u/v)^2=1. This injection is formalized, including the empty-fiber case, without assuming b is a unit in advance.

3. Orthogonality and square scaling

Reindexing r -> v*r for any unit v gives

$$\chi(v)U_\chi(av^2) = U_\chi(a), \qquad |U_\chi(av^2)| = |U_\chi(a)|.$$

The code proves the exact complex identity before taking norms. No reality condition on chi is needed.

Expanding the norm square and summing over every additive frequency gives the exact finite identity

$$\sum_{b \bmod q} |U_\chi(b)|^2 = q \sum_{\substack{r,s \in \mathbb{A}_q \\ r^2 = s^2}} \chi(r)\overline{\chi(s)}.$$

Indeed the inner frequency sum of e(b*(r^2-s^2)/q) is q if the squares agree and zero otherwise. The additive orthogonality used here was already proved for the genuine standard character in the earlier Cohen development. Taking real parts, each character product contributes at most one, and each r has at most K(q) partners s. Thus the energy E satisfies

$$E := \sum_{b \bmod q} |U_\chi(b)|^2 \le q\phi(q)K(q).$$

The square-scaling orbit has bounded multiplicity by the same fiber lemma. For a unit a, therefore,

$$\phi(q)|U_\chi(a)|^2 = \sum_{v \in \mathbb{A}_q} |U_\chi(av^2)|^2 \le K(q)E \le q\phi(q)K(q)^2.$$

The finite bounded-multiplicity summation inequality is proved separately. Since phi(q)>0, cancellation and the nonnegative square root give

$$|U_\chi(a)| \le K(q)\sqrt{q}, \qquad |\mathfrak{s}_\chi(a,q)| \le K(q)\sqrt{q}/\phi(q).$$

These inequalities are proved for every modulus, character and unit coefficient in the stated domains, not inferred from numerical tests.

4. Modular root counts, including powers of two

At an odd prime power p^j, the unit group is cyclic by the checked Mathlib theorem `ZMod.isCyclic_units_of_prime_pow`. A finite cyclic group has at most two solutions to u^2=1, giving K(p^j)<=2.

At powers of two the proof does not assume cyclicity. Exponents j<=2 are checked by kernel-reduced finite decisions. For j=k+3, the unit 5 modulo 2^(k+3) has order 2^(k+1), by `ZMod.orderOf_five`; its square has order 2^k and belongs to the range of squaring. Hence that range has at least 2^k elements. The entire unit group has 4*2^k elements. The exact finite-group kernel-times-range identity yields K(2^(k+3))<=4. The order, positivity, range membership, group cardinality and cancellation are all checked in Lean.

The unit CRT equivalence induces an explicit bijection between the square kernel at mn and the product of the square kernels at coprime positive m,n. Consequently K(mn)=K(m)K(n). Coprime induction, including q=1, gives

$$K(q) \le 4^{\omega(q)},$$

where omega(q) is the cardinality of the actual finite prime-divisor set. This loose constant is sufficient; the sharper local constant two at odd primes is also proved. The proof uses CRT for square kernels and does not postulate a character decomposition of the original exponential sum.

5. Totient and epsilon absorption

Euler's exact finite product identity and p<=2*(p-1) for every prime p give

$$q \le \phi(q)2^{\omega(q)}.$$

Combining the proved estimates yields the uniform explicit finite majorant

$$|\mathfrak{s}_\chi(a,q)| \le 8^{\omega(q)}q^{-1/2}.$$

The remaining q^epsilon estimate is elementary. More generally, fix B>=1 and epsilon>0, and take

$$N = \lceil \exp(\log B/\varepsilon) \rceil, \qquad A = \exp(N \log B) > 0.$$

There are at most N prime divisors of q not exceeding N. For p>N we have log B<=epsilon*log p. All logarithms of primes are nonnegative, so the already proved squarefree-divisor logarithmic budget gives

$$\omega(q)\log B \le N \log B + \varepsilon \sum_{p|q} \log p \le N \log B + \varepsilon \log q.$$

Exponentiating proves B^omega(q)<=A*q^epsilon, uniformly for every positive q. Applying this with B=8 establishes the full Lemma 3.24. The displayed constant is effective but intentionally very large; no optimization is claimed.

Checked declarations and scope

Component	Lean declaration
Actual twisted sum and principal specialization	`twistedSquareSum`, `twistedSquareSum_principal`
Square-scaling identity	`twistedUnitQuadraticGaussSum_mul_unit_sq`
Square-fiber bound	`unitSquareFiber_card_le`
Exact complex energy identity	`twistedGauss_energy_complex`
Energy upper bound	`twistedGauss_energy_le`
Actual numerator norm bound	`norm_twistedUnitQuadraticGaussSum_le_roots`
Odd and dyadic root counts	`unitSquareRootCount_odd_prime_pow`, `unitSquareRootCount_two_pow`
Exact CRT for square kernels	`unitSquareRootCount_coprime_mul`
Global kernel count	`unitSquareRootCount_le_pow`
Elementary totient estimate	`le_totient_mul_two_pow_primeFactors`
Fixed prime weights absorbed into q^epsilon	`primeFactor_pow_le_rpow`
Explicit 8^omega/sqrt(q) bound	`norm_twistedSquareSum_finite_bound`
Full uniform theorem and integer-coefficient form	`twistedSquareSum_uniform_bound`, `twistedSquareSum_uniform_bound_int`

All declarations are directly in namespace `PrimeSquares`. The final integer theorem explicitly states epsilon>0, one positive constant, every nonzero natural modulus q, every complex Dirichlet character and every integer a coprime to q. Nonzero natural moduli are exactly the positive ones intended in the source. The exponent is exactly -1/2+epsilon.

No manuscript correction is needed. This completes the general twisted definition and Lemma 3.24; it does not prove the conductor-sensitive refinements in Lemma 3.27 or any headline theorem.

Independent exact checks

`scripts/check_twisted_gauss.py` constructs every character for each q=1,...,32 from exact local unit-group coordinates, including non-real and imprimitive characters. It uses integer phase counters and reduction modulo cyclotomic polynomials, with no floating point. It checks multiplicativity, the exact energy identity, square scaling, fiber bounds, root-count CRT and integer root/totient inequalities. These finite checks supplement the Lean proofs; they do not establish an unbounded epsilon estimate by experimentation.

Checkpoint 12 continuation: the full conductor-sensitive Lemma 3.27 is now proved by the additional support, induced CRT and majorant arguments in `Conductor_Refinement.md`. This checkpoint 11 exposition describes the preceding uniform bound.

`Uniform_Major_Arc_Parameters.md`

Uniform parameter inequalities for the source major-arc scale

Checkpoint 29. This unit proves the elementary asymptotic parameter conditions needed for the inverse-fifth major-arc bound. It has no prime-distribution hypothesis and introduces no deep analytic input.

Define the actual source scale, for real N,Q, by

$$R(N,Q) = \max\{\exp((\log N)^{1/10}), Q^8\}.$$

The Lean definition is `sourceMajorArcScale`. It is positive everywhere, and at least one when N>=1. In applications N is a positive integer.

Uniformity in Q

For any fixed epsilon>0, real powers satisfy

$$x^{1/10} = o(x^{1/2}) \quad (x \longrightarrow \infty).$$

The supporting theorem `eventually_rpow_le_mul_rpow` proves the general comparison x^a<=epsilon*x^b eventually when a<b. Its proof uses the checked limit of x^(a-b) to zero and division by the positive x^b. Composition with log(N) tending to infinity gives the comparison needed for the first term in R.

If log(Q)<=(epsilon/8)*sqrt(log N), then Q^8<=exp(epsilon*sqrt(log N)). Both terms defining the maximum therefore satisfy the same upper bound. Taking logarithms proves, for all sufficiently large N and every Q>=1,

$$\log Q \le \frac{\epsilon}{8}\sqrt{\log N} \quad \Longrightarrow \quad \log R(N,Q) \le \epsilon\sqrt{\log N}.$$

The threshold depends on epsilon alone, not on Q. This is the quantified statement `eventually_sourceMajorArcScale_log_upper`.

The nonunit budget

For any fixed epsilon>0, eventually

$$\epsilon\sqrt{\log N} \le \frac{\log N}{20}, \qquad \log N \le \exp((\log N)/4).$$

The first follows from the same power comparison; the second follows from Mathlib's proved domination of powers by the exponential. Thus, whenever R>0 and log(R)<=epsilon*sqrt(log N),

$$\log R \le \log N, \qquad R^5 \le \exp((\log N)/4), \qquad R^5 \log N \le \exp((\log N)/2) = \sqrt{N}.$$

`eventually_majorArc_nonunit_conditions` proves both required conditions uniformly for every such R. No unspecified eventual bound is supplied as a hypothesis: the proof establishes its existence from checked limits.

An explicit admissible coefficient

Given any positive decay constant c, take

$$\epsilon = \min\{c/7, 1\}, \qquad c_0 = \epsilon/8 > 0.$$

For all sufficiently large N, simultaneously for every Q>=1 with log(Q)<=c_0*sqrt(log N), the actual source R satisfies

$$R \geq 1, \quad \log R \leq \log N, \quad R^5 \log N \leq \sqrt{N}, \quad 7 \log R \leq c\sqrt{\log N}.$$

This is `eventually_sourceMajorArc_conditions`. These are precisely the elementary inequalities used in checkpoint 28. This choice controls the error absorption; applicability of an external progression theorem in its own modulus range remains a separate condition on that external input. No existence or range of an exceptional zero is asserted here.

The next unit transfers this real-variable eventual statement to integer N and connects it to the actual ordinary and exceptional major-arc bounds. Full Lemma 6.5 and Theorem 1.1 remain open.

Module	New named theorems	Result
MajorArcParameters.lean	8	Actual source scale, positive and exponential/logarithmic bounds, proved power comparison and uniform eventual parameter conditions

All statements in this unit are unconditional elementary analysis.

<!-- checkpoint-verification -->

Verification

All 93 project modules and the root were freshly rebuilt: 3608 jobs pass without warnings/errors. All 757 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 95 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. No new deep analytic input or own-paper lemma is assumed.

`Weighted_Upper_Main_Term_Complete.md`

Weighted upper main term: complete conditional proof

This file concerns the source lemma `l:wumatomakisum`, with alpha=9/8. The checked endpoint is `goldbach_weighted_upper_main_term_complete`. For every A>0, every fixed C>=A+24, and every eta>0, it gives one threshold in N such that the source inequality holds for every n=12 modulo24 with N^2(log N)^(-A)<=n<N^2. Its coefficient is c_alpha W(alpha)+eta, where c_alpha=16 exp(-gamma/2)/sqrt(pi alpha). This is the uniform epsilon formulation of the source o(1).

The theorem uses the actual finite multiplier set, source Euler products, smoothed singular integrals, and lower main term. No weighted-prime sum, error-absorption conclusion, or Goldbach main-term comparison is an independent hypothesis.

Independent inputs

Four general interfaces remain explicit theorem parameters: `ClassicalHalfMeanValueInput`, `ClassicalPrimeLogMertens`, `ClassicalPrimeMertensRatios`, and the existing `GoldbachClassicalIntervalMertens`. The first is the nonnegative strongly multiplicative upper corollary of the corrected Tenenbaum effective mean theorem, with uniform constants and its prime-balance condition stated explicitly. The others contain ordinary fixed-progression Mertens estimates. Their exact scope and the external corollary reductions are

recorded in `docs/EXTERNAL_WEIGHTED_MEAN_INPUTS.md`. They are hypotheses, not new Lean axioms. Their truth is not asserted by this project.

Arithmetic normalization

The source multiplier ell=2up has a unique pair (u,p): the primes dividing u are 1 modulo4 and p is 3 modulo4. Lean proves the finite image identity and uniqueness, including the lower prime endpoint p=z. The geometry gives R<z_ell, u<z_ell<p, and N^2/ell>=2 for all source pairs once R<N^(2/9).

For every odd prime p>3, the joint factor U_p is positive. The associated strongly multiplicative weight h_R has nonnegative prime values at most6. The small-prime cancellation identities are proved even when an individual arithmetic factor vanishes. They yield the exact normalizations

$$X_n\Theta_n(z) = 3C_R(n)H_{1,R}(n)V_3(R,z),$$

and

$$X_n(2up)\Theta_{n,2up}(z_{2up}) = 6B_R(n)\frac{h_R(u)}{2up}H_{2up,R}(n-1)V_0(R,z_{2up}).$$

Here V_3 and V_0 are the actual finite outer Euler products. The positive factors satisfy B_R<=C_R. These are equalities in the original definitions, not asymptotic replacements.

The varying multiplicative weight

Above R the prime values of h_R equal the fixed 1 modulo4 baseline (p-1)/(p-2). The finite perturbation below R and the classical logarithmic prime sums give, uniformly in n and y>=R,

$$\left|\sum_{p\le y}(h_R(p)-1/2)\frac{\log p}{p}\right| \le 6\log R + 10K_0.$$

The application of the independent effective mean theorem is proved from this bound. The finite Euler product is exactly C_R/B_R times an ordinary 1 modulo4 product. Thus, with kappa=exp(-gamma/2)/sqrt(pi), the ordinary partial sum is at most

$$(\kappa + K\varepsilon^{\delta})(1+K_1/\log R)\frac{C_R(n)}{B_R(n)}\frac{x}{\sqrt{\log R\log x}}$$

under the explicit scale and prime-balance budgets. Constants K,delta precede R,n,x,epsilon. At R=(log N)^C and a=exp(sqrt(log N)), Lean proves every budget eventually, uniformly in n.

The small denominators do not require this sharp mean estimate. A coarser instance applies beyond R^D for a fixed D>1. Below that scale a finite Euler-product bound applies; above it a checked Abel identity applies. Together they give

$$\sum_{u\le a}\frac{h_R(u)}{u} \le \frac{C_R(n)}{B_R(n)}\left(P(\log R)^6 + A'\frac{1+\log a}{\log R}\right),$$

with constants independent of R,n,a. After multiplication by the prime reciprocal and kernel bounds this range is negligible on the required scale.

The double weighted sum and W

Two finite Abel estimates are proved with their derivatives, signs, integrability, floor conventions, and boundary terms. The denominator summation is bounded using the preceding mean. Finite variable-interval integration interchanges its order with the prime sum. The prime sum uses the 3 modulo4 logarithmic Mertens estimate. The possible prime p=z contributes at most 1/(z(log z)^2); its remaining error is at most 2(2K_0+1)/(log z)^3.

At D=2 log N, the logarithmic variables r=log(2u)/D and v=log p/D give the actual source region. The two changes of variables are checked, and

$$\int_0^{1/9}\frac{dr}{\sqrt{r}}\int_{4/9}^{(1-r)/2}\frac{dv}{v(1-r-v)^2} = 8\sqrt{2}\,\mathcal{W}(9/8).$$

The endpoint at r=0 is integrable. Replacing log u by log(2u) costs at most sqrt(1+log2/log a), which tends to one. The small-denominator, prime, and Abel boundary coefficients all tend to zero. Choosing epsilon arbitrarily small therefore proves, uniformly in n, the sharp fixed kernel bound with coefficient kappa W(9/8)+eta. A checked logarithmic-loss argument proves the same bound for every m in the entire retained interval N^2(log N)^(-A_1)<=m<=N^2.

Integration, smoothing, and relative errors

The finite sum of original Archimedean integrals is interchanged exactly with integration. On the retained region the logarithmic kernel bound is compared directly with the defining integrand of H_1(n): log(n-y^2)<=2 log N. No assertion about an unrestricted real power of log m near m=0, or a relative shift H_1(n-1)/H_1(n), is needed.

The discarded set has measure at most 2T/sqrt(n). The actual reciprocal mass over pairs is at most M(log N)^7. The already proved singular-integral smoothing bound is uniform in the multiplier, so its total error is also controlled by that mass. The lower bound for H_1(n) on the retained n range is proved internally.

For A_1=A+14 and C>=A+24, both total errors are bounded relative to the common positive quantity

$$S = C_R(n) H_1(n) \sqrt{\log R / \log N}.$$

The discarded-region contribution is at most 768 M/(log2)^3^(-3) S, and the total smoothing contribution at most 48 M K (log N)^(-2) S. The lower smoothing error is at most 8K_s/log N times H_1(n). All three coefficients tend to zero with thresholds independent of n.

Final comparison

The full Mertens ratio and log z_ell=one-quarter log(N^2/ell) give the upper coefficient 48(kappa W+o(1)) multiplying S. The exact lower normalization and half Mertens ratio give the lower coefficient 3 sqrt(9/8)+o(1). The identity 3 sqrt(alpha)c_alpha=48 kappa is checked. A positive-tolerance continuity argument concludes the original uniform source inequality.

The theorem statement needs no replacement. Two proof passages are supplied separately in `docs/WEIGHTED_UPPER_MAIN_TERM_REPLACEMENTS.tex`: the non-strict lower endpoint p>=z, and Step6 with its retained cutoff and direct H_1(n) comparison. The selected manuscript is unchanged.

Checkpoint notes 42-116

Checkpoint_42_UniformMajorArcBudget.md

Checkpoint 42: Uniform arithmetic thresholds for the major-arc model error

The finite major-arc budget is the actual sum of the prefix error and the tail from all other model terms:

$$B_C(R) = \frac{4 + (2 + 4\pi)C}{R^5} + \left(5 + \frac{\log R}{\log 2}\right) \frac{4C_G}{\pi} R^{-5/12}.$$

The new proof gives a single natural-number threshold for every real Q. First the exponential part of R=max(exp((log N)^(1/10)),Q^8) tends to infinity independently of Q. This transfers the already proved eventual R^(-1/6) budget estimate uniformly to this scale. Joining this threshold with the proved logarithmic absorption gives, for every fixed C>=0 and A>=0, one N0 such that for all N>=N0 and all Q>=1,

$$B_C(R(N, Q)) \le 2^A Q^{-1/2} (1 + \log N)^{-A}.$$

The N0 may depend on C and A; it does not depend on Q. This is a genuine quantifier strengthening needed for the source approximation. No distribution theorem, minor-arc estimate or own-paper lemma is assumed by these three results. Applying this budget to the actual finite model still requires the major-arc hypotheses and the displayed raw progression-prefix estimate from checkpoint 41.

Checked declarations

Module	Named supporting theorems
UniformMajorArcBudget.lean	3

- `sourceMajorArcScale_uniform_lower`
- `sourceMajorArcScale_uniform_model_budget`
- `sourceMajorArcScale_uniform_model_log_rate`

<!-- checkpoint-verification -->

Verification

All 109 project modules and the root were freshly rebuilt: 3628 jobs pass without warnings/errors. All 848 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 111 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

Checkpoint_43_MajorArcBudgetGeometry.md

Checkpoint 43: Source parameter budgets imply both major-arc geometric conditions

The prefix-error parameter budget also supplies the rational-arc geometry needed for the actual model comparison. For N>2, R>=1 and log N>=1, the inequality R^5 log N<=sqrt N implies both 2R^3<N^2 and 2R^4<=N^2. The proof compares R^3 and R^4 to R^5, and sqrt N to N; no asymptotic constant is hidden.

Combining this elementary deduction with the already checked source parameter threshold gives one natural-number N0, independent of Q, for both geometric conditions throughout the authorized source range log Q <= min(c/7,1)*sqrt(log N)/8. These conditions establish disjoint major arcs and force the actual smooth cutoff to equal one on the chosen major arc. The raw progression-prefix estimate and full minor-arc estimate are not established by this unit.

Checked declarations

Module	Named supporting theorems
MajorArcBudgetGeometry.lean	2

- `majorArc_geometry_of_budget`
- `sourceMajorArc_uniform_geometry`

<!-- checkpoint-verification -->

Verification

All 110 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3629-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 850 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 112 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_44_UniformFiniteMajorArcError.md`

Checkpoint 44: Uniform logarithmic major-arc approximation for the actual finite model

The actual finite model error now has the full major-arc logarithmic rate with a single threshold independent of Q, the finite set of canonical rational centers, the chosen major arc and the exceptional character data. For fixed C>=0, c>0 and A>=0, and the established source range for Q, the theorem concludes

$$\|\widehat{A}_N(\xi) - V_{N,\Gamma}(\xi)\| \le 2^A Q^{-1/2}(1+\log N)^{-A}.$$

The hypotheses are explicit: N sufficiently large; Q>=1 in its source range; canonical centers with denominator at most R(N,Q); coefficient norms at most one; beta in (0,1]; a chosen center in the finite set with the required major-arc width; and the actual von-Mangoldt progression-prefix estimate throughout the positive annulus with the same exceptional data. The proof joins the source parameter, geometric and error-budget thresholds, applies the checked prefix-to-model estimate, then absorbs the counted tail. No minor-arc estimate or existence of exceptional data is assumed implicitly. This is the major-arc branch for the finite real-frequency model; the source's global approximation is still open.

Checked declarations

Module	Named supporting theorems
UniformFiniteMajorArcError.lean	1

- `finiteMajorArcError_uniform_log_rate`

<!-- checkpoint-verification -->

Verification

All 111 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3630-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 851 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 113 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_45_AnnularFrequencyContinuity.md`

Checkpoint 45: Continuity in frequency of the genuine annular oscillatory integral

The genuine normalized annular oscillatory integral is continuous as a function of the real frequency, for every positive natural arithmetic length N and every real exponent beta. This proves regularity of the actual integral; it is not an extra hypothesis on a model function.

The proof first establishes joint continuity of the quadratic phase in frequency and integration variable. To apply compact parametric integration for arbitrary beta, the power density is continuously extended by replacing t with max(t,N/2). This extension is strictly positive and equals the original density throughout the integration annulus. The compact-interval integral is then identified with the original half-open integral, whose endpoint difference has measure zero, and the exact factor 1/N is preserved. This regularity will support continuous circle multipliers after periodization. It does not supply any prime-distribution estimate.

Checked declarations

Module	Named supporting theorems
AnnularFrequencyContinuity.lean	2

- `quadraticOscillation_joint_continuous`
- `annularOscillatoryModel_continuous`

<!-- checkpoint-verification -->

Verification

All 112 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3631-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 853 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 114 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_46_LocalizedModelContinuity.md`

Checkpoint 46: Continuity of the genuine localized Gauss models

The actual Gauss model and its localized smooth-cutoff version are continuous in real frequency. Both statements hold for arbitrary character, coefficient and exponent data at every positive arithmetic length, without any distribution assumptions. Their proofs combine the genuine annular integral regularity with the actual smooth cutoff. This discharges the continuity requirement for the forthcoming quotient construction, rather than assuming continuity of the manuscript model.

Checked declarations

Module	Named supporting theorems
LocalizedModelContinuity.lean	2

- `annularGaussModel_continuous`
- `localizedAnnularGaussModel_continuous`

<!-- checkpoint-verification -->

Verification

All 113 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3632-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 855 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 115 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_47_CompactPeriodWindow.md`

Checkpoint 47: The finite periodization window and its endpoint compatibility

For a continuous real symbol supported inside [-1,1], periodization on the centered fundamental interval uses exactly the three translates f(x-1)+f(x)+f(x+1). The new proofs establish continuity, exact equality of the values at -1/2 and 1/2, and the uniform bound 3B when the original symbol has norm at most B. The endpoint identity includes the possible overlap at the largest cutoff scale; it is not assumed that only one translate contributes. These elementary analytic facts prepare a valid continuous circle symbol. They do not assert prime-distribution estimates.

Checked declarations

Module	Named supporting theorems
CompactPeriodWindow.lean	3

- `compactPeriodWindow_continuous`
- `compactPeriodWindow_endpoints`
- `compactPeriodWindow_bound`

<!-- checkpoint-verification -->

Verification

All 114 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3633-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 858 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 116 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new

external analytic estimate or own-paper lemma is assumed.

`Checkpoint_48_CompactCircleLift.md`

Checkpoint 48: A continuous circle symbol from a compact real-frequency symbol

A real continuous symbol supported in [-1,1] now gives a genuine continuous function on the unit circle. The quotient uses the centered half-open fundamental interval and the three-translate window. Continuity follows from the proved endpoint compatibility; it is not obtained by treating a nonperiodic real function as periodic. The theorem identifies its value with the finite window on [-1/2,1/2) and proves a uniform norm bound 3B. These statements handle the largest cutoff scale, where neighboring translates can overlap.

Checked declarations

Module	Named supporting theorems
CompactCircleLift.lean	2

- `compactCircleLift_real`
- `compactCircleLift_norm_bound`

<!-- checkpoint-verification -->

Verification

All 115 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3634-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 860 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 117 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_49_CompactPeriodSum.md`

Checkpoint 49: Exact integer-sum interpretation of compact periodization

The integer-translate sum of a real symbol is proved one-periodic by an exact reindexing of the integer summation. For a symbol vanishing whenever |x|>=1, the infinite sum on the centered fundamental interval is proved equal to the three-term window. Every omitted integer translate is shown to vanish; no convergence or truncation error is assumed. This identifies the finite construction with genuine periodization, including both endpoints.

Checked declarations

Module	Named supporting theorems
CompactPeriodSum.lean	2

- `integerPeriodSum_periodic`
- `integerPeriodSum_window`

<!-- checkpoint-verification -->

Verification

All 116 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3635-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 862 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 118 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_50_CompactCircleCore.md`

Checkpoint 50: Exact local agreement for narrow periodized symbols

When a real symbol vanishes already for |x|>=1/2, the two neighboring translates in the periodization window vanish throughout the closed centered interval. The resulting continuous circle symbol therefore agrees exactly with the original real symbol on [-1/2,1/2). This proves the absence of alias terms for sufficiently narrow cutoffs; the largest cutoff remains covered by the earlier three-translate formula. No periodicity of the original oscillatory integral is asserted.

Checked declarations

Module	Named supporting theorems
CompactCircleCore.lean	2

- `compactPeriodWindow_small_support`
- `compactCircleLift_small_support_real`

<!-- checkpoint-verification -->

Verification

All 117 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3636-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 864 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 119 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_51_CompactCircleAlgebra.md`

Checkpoint 51: Linearity of compact periodization as a continuous circle map

Compact periodization is proved compatible with addition and complex scalar multiplication as an equality of continuous circle maps. The support conditions of the resulting symbols are displayed explicitly, and the proof uses the quotient representative and exact three-term window. These identities allow finite arithmetic model components to be periodized separately and then assembled without changing the circle operator.

Checked declarations

Module	Named supporting theorems
CompactCircleAlgebra.lean	2

- `compactCircleLift_add`
- `compactCircleLift_smul`

<!-- checkpoint-verification -->

Verification

All 118 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3637-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 866 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 120 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_52_CompactCirclePeriodization.md`

Checkpoint 52: Global identification with genuine integer periodization

The continuous quotient construction is identified globally with the lift of the genuine integer-translate sum. The proof chooses the unique representative in the centered half-open interval, applies the proved finite-window identity there, and uses the quotient's exact representative identity. Consequently, for every real x, the circle symbol at x modulo one equals the infinite sum of f(x+k) over all integer k. This global equality removes the restriction to one fundamental interval and proves that the construction is actual periodization, not an arbitrary endpoint-compatible extension.

Checked declarations

Module	Named supporting theorems
CompactCirclePeriodization.lean	2

- `compactCircleLift_eq_periodization`
- `compactCircleLift_all_real`

<!-- checkpoint-verification -->

Verification

All 119 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3638-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 868 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 121 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_53_CenteredRationalSymbol.md`

Checkpoint 53: The actual centered rational model and its compact support

The localized Gauss model at each rational center is packaged as a genuine continuous real-frequency symbol. Its translation back to the previously checked rationalMajorArcTerm is proved exactly, preserving the sign of the exceptional term and the 1/N annular normalization. Its support is proved to lie inside [-1,1] by the actual cutoff-radius theorem and the inequality 16^s>=1. This supplies the continuity and support required by compact periodization without an extra analytic hypothesis.

Checked declarations

Module	Named supporting theorems
CenteredRationalSymbol.lean	2

- `centeredRationalSymbol_translate`
- `centeredRationalSymbol_support`

<!-- checkpoint-verification -->

Verification

All 120 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3639-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 870 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 122 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_54_CircleRationalModel.md`

Checkpoint 54: Genuine rational Gauss terms as continuous circle symbols

Each actual rational Gauss-model term now defines a continuous unit-circle symbol, translated by its rational center. For every real frequency, this symbol equals the integer periodization of the genuine centered annular Gauss model. On the centered local interval it equals the exact three-translate window. These formulas include boundary aliases and use real-to-circle subtraction correctly. The definition consumes no distribution theorem. A finite sum of these terms will be a global continuous model, but the full source approximation and source-specific analytic estimates remain open.

Checked declarations

Module	Named supporting theorems
CircleRationalModel.lean	2

- `circleRationalModelTerm_real`
- `circleRationalModelTerm_local`

<!-- checkpoint-verification -->

Verification

All 121 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3640-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 872 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 123 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_55_CircleRationalModelNorm.md`

Checkpoint 55: Uniform Gauss decay survives actual circle periodization

The genuine centered rational model has the previously proved Gauss decay C_G q^(-5/12), uniformly in frequency, for N>=2, beta in [0,1] and exceptional coefficient of norm at most one. Its actual periodized circle term has bound 3 C_G q^(-5/12). The factor three accounts for the largest cutoff's possible aliases and is justified by the proved periodization estimate. C_G is the single unconditional constant already extracted from the formalized quadratic Gauss estimates; no prime-distribution input is added.

Checked declarations

Module	Named supporting theorems
CircleRationalModelNorm.lean	2

- `centeredRationalSymbol_norm`
- `circleRationalModelTerm_norm`

<!-- checkpoint-verification -->

Verification

All 122 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3641-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 874 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 124 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_56_FiniteCircleModel.md`

Checkpoint 56: The genuine global finite circle model and its uniform Gauss bound

A global continuous circle model is now defined for any explicit finite set of rational centers, using the genuine periodized Gauss terms. Its evaluation at every real frequency is proved to equal the finite sum of actual integer-periodized centered models. A uniform pointwise bound is proved by summing the unconditional denominator-dependent Gauss bounds, with the alias factor three included. The error is defined as the difference between the genuine von-Mangoldt annular symbol and this continuous model. This establishes a valid global finite model; identifying the manuscript's exact center set and proving its uniform approximation still require further work.

Checked declarations

Module	Named supporting theorems
FiniteCircleModel.lean	2

- `circleMajorArcModel_real`

- `circleMajorArcModel_norm`

<!-- checkpoint-verification -->

Verification

All 123 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3642-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 876 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 125 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_57_CircleModelOperators.md`

Checkpoint 57: Actual l2 model and error operators with exact annular decomposition

The global finite circle model and its exact error define genuine bounded linear operators on integer l2 through the already checked Fourier isometry. Their sum is proved to equal the actual annular von-Mangoldt averaging operator on every l2 input. A uniform error-symbol bound is transported to the error operator norm. This is an operator-construction and transport result: it does not assume or prove the manuscript's unproved global error estimate. The exact decomposition is unconditional for the defined finite model; the small norm conclusion has its displayed pointwise hypothesis.

Checked declarations

Module	Named supporting theorems
CircleModelOperators.lean	2

- `circleModel_operator_decomposition`
- `circleModel_error_operator_bound`

<!-- checkpoint-verification -->

Verification

All 124 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3643-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 878 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 126 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_58_UniformFourierKernel.md`

Checkpoint 58: Uniform inverse-kernel and finite-input convolution estimates

Every continuous circle symbol with a uniform pointwise bound B has an inverse Fourier kernel bounded by B at every integer. The proof uses the actual Fourier coefficient integral, the unit norm of the Fourier

character and the normalized Haar mass one. This gives an explicit finite-input convolution bound B times the sum of input magnitudes, valid without positivity or compact spatial support. These facts supply the basic l1-to-linfinity kernel estimate needed for low operators; the source's stronger low estimate is not claimed.

Checked declarations

Module	Named supporting theorems
UniformFourierKernel.lean	2

- `fourierKernel_uniform_bound`
- `fourierKernel_finite_input_bound`

<!-- checkpoint-verification -->

Verification

All 125 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3644-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 880 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 127 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_59_KernelL1Summation.md`

Checkpoint 59: Absolute convergence and bounds for all-input l1 convolution

Convolution by an arbitrary uniformly bounded integer kernel is defined for every l1 input by an actual infinite sum. The series is proved absolutely summable at each output point by comparison with the l1 norm. Its norm is bounded by B times the input l1 norm. This removes the finite-input restriction from the earlier estimate and justifies the infinite convolution rather than assigning a value without proving convergence.

Checked declarations

Module	Named supporting theorems
KernelL1Summation.lean	2

- `kernelL1_summable`
- `kernelL1Sum_bound`

<!-- checkpoint-verification -->

Verification

All 126 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3645-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 882 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 128 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_60_KernelL1Value.md`

Checkpoint 60: The bounded convolution sequence and its additive structure

The absolutely convergent convolution of a bounded kernel and an arbitrary l1 input is packaged as a genuine linfinity sequence. Its norm is proved at most B times the input norm, and addition of inputs is proved compatible with the sequence-valued construction using justified summation of the two absolutely convergent series. These results prepare the actual bounded linear low operator required by the abstract high-low framework.

Checked declarations

Module	Named supporting theorems
KernelL1Value.lean	2

- `kernelL1Value_norm`
- `kernelL1Value_add`

<!-- checkpoint-verification -->

Verification

All 127 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3646-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 884 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 129 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_61_KernelL1Operator.md`

Checkpoint 61: A genuine bounded linear l1-to-linfinity convolution operator

Bounded-kernel convolution is now a genuine complex continuous linear map from integer l1 to integer linfinity. Complex linearity is proved directly from the convergent series, the application formula is the actual convolution sum, and its operator norm is at most the kernel bound B. The construction does not assume positivity or finite kernel support, so it applies to smooth Fourier-cutoff inverse kernels whose spatial support need not be compact.

Checked declarations

Module	Named supporting theorems
KernelL1Operator.lean	2

- `kernelL1Operator_apply`
- `kernelL1Operator_norm`

<!-- checkpoint-verification -->

Verification

All 128 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3647-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 886 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 130 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_62_KernelL1FiniteAgreement.md`

Checkpoint 62: Exact agreement of infinite convolution with finite and indicator inputs

The newly constructed infinite l1 convolution operator is proved to agree exactly with the existing finite convolution on every finite complex input. Its indicator-input specialization agrees with indicatorConvolution. The proof removes all terms outside the finite support by an exact summation identity; it does not approximate the infinite sum. These identifications connect the new all-input low operator to the already formalized restricted high-low framework.

Checked declarations

Module	Named supporting theorems
KernelL1FiniteAgreement.lean	2

- `kernelL1Operator_finite`
- `kernelL1Operator_indicator`

<!-- checkpoint-verification -->

Verification

All 129 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3648-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 888 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 131 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_63_FourierL1Operator.md`

Checkpoint 63: Compatible l1 and l2 realizations of continuous circle multipliers

Every continuous circle multiplier now has a genuine l1-to-linfinity convolution realization using its actual inverse Fourier kernel. The operator norm is at most the symbol sup norm. On every finite complex input its values agree exactly with the independently constructed l2 Fourier multiplier, with the same negative-phase convention. This compatibility is needed when combining l1 low bounds and l2 high bounds for the same averaging operator.

Checked declarations

Module	Named supporting theorems
FourierL1Operator.lean	2

- `fourierL1Operator_norm`
- `fourierL1Operator_l2_agreement`

<!-- checkpoint-verification -->

Verification

All 130 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3649-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 890 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 132 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_64_KernelSensitiveFourierL1.md`

Checkpoint 64: Operator estimates from sharper inverse-kernel bounds

The l1-to-linfinity operator is bounded by any valid pointwise bound on its inverse kernel, even when that bound is smaller than the symbol sup norm used to construct the operator. Both the all-input norm estimate and the operator-norm estimate are proved. This distinction is useful for the source's low term, where arithmetic cancellation in the inverse kernel must provide a sharper bound than the immediate symbol estimate. No such arithmetic cancellation is assumed or claimed in this checkpoint.

Checked declarations

Module	Named supporting theorems
KernelSensitiveFourierL1.lean	2

- `fourierL1Operator_kernel_bound`
- `fourierL1Operator_opNorm_kernel_bound`

<!-- checkpoint-verification -->

Verification

All 131 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3650-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 892 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 133 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_65_FourierKernelAlgebra.md`

Checkpoint 65: Pointwise linearity of actual inverse Fourier kernels

Inverse Fourier kernels are proved compatible with addition, complex scalar multiplication and subtraction. The proof uses the actual continuous-map-to-L2 map and the inverse Fourier isometry, so the identities hold pointwise at every integer, not merely almost everywhere in frequency. These algebraic identities are required to turn a symbol decomposition into an exact spatial-kernel decomposition.

Checked declarations

Module	Named supporting theorems
FourierKernelAlgebra.lean	3

- `fourierKernel_add`
- `fourierKernel_smul`
- `fourierKernel_sub`

<!-- checkpoint-verification -->

Verification

All 132 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3651-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 895 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 134 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_66_FourierKernelSums.md`

Checkpoint 66: Exact inverse-kernel assembly over finite symbol families

The inverse kernel of the zero symbol vanishes, and the inverse kernel of any finite sum of continuous circle symbols equals the finite sum of their actual inverse kernels. The proof passes the finite sum through the genuine linear maps and then evaluates the integer l2 sequence. This supports finite rational-center assembly without introducing a formal kernel unrelated to the circle multiplier.

Checked declarations

Module	Named supporting theorems
FourierKernelSums.lean	2

- `fourierKernel_zero`
- `fourierKernel_sum`

<!-- checkpoint-verification -->

Verification

All 133 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3652-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 897 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 135 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_67_CircleModelKernels.md`

Checkpoint 67: Actual model and error kernels with exact annular decomposition

The actual global finite circle model and its error now have inverse Fourier kernels. The model kernel is proved to be the finite sum of the periodized rational-term kernels. At every integer, the genuine annular von-Mangoldt square kernel equals the model kernel plus the error kernel, using the already checked Fourier identification of the actual averaging kernel. No finite spatial support is asserted for the two new kernels.

Checked declarations

Module	Named supporting theorems
CircleModelKernels.lean	2

- `circleModelKernel_sum`
- `annularKernel_circle_decomposition`

<!-- checkpoint-verification -->

Verification

All 134 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3653-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 899 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 136 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_68_CircleModelKernelBounds.md`

Checkpoint 68: Spatial model and error bounds from the genuine circle symbols

The global finite model's actual inverse kernel satisfies the explicit denominator-weighted Gauss bound at every spatial integer. Any uniform circle error bound also bounds the actual error kernel pointwise. These are proved through the Fourier coefficient integral and the already checked circle-model estimates. The second statement is transport of a displayed hypothesis; it does not discharge the missing global minor-arc estimate.

Checked declarations

Module	Named supporting theorems
CircleModelKernelBounds.lean	2

- `circleModelKernel_bound`
- `circleModelErrorKernel_bound`

<!-- checkpoint-verification -->

Verification

All 135 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3654-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 901 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 137 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_69_CircleModelConvolution.md`

Checkpoint 69: Exact finite-input convolution formulas for model and error operators

The genuine l2 model and error operators are proved to act on every finite complex input by convolution with their actual inverse Fourier kernels. This identifies the spatial and frequency constructions for both components, preserving the input support and the sign convention. The exact annular kernel decomposition is already available; no extra positivity or spatial support of model kernels is assumed.

Checked declarations

Module	Named supporting theorems
CircleModelConvolution.lean	2

- `circleModelOperator_finite_convolution`
- `circleModelErrorOperator_finite_convolution`

<!-- checkpoint-verification -->

Verification

All 136 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3655-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 903 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 138 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_70_CircleModelOperatorBounds.md`

Checkpoint 70: All-input l2 and l1-to-linfinity bounds for the finite circle model

The explicit finite-center Gauss bound now controls both the actual l2 model operator and its compatible l1-to-linfinity realization. The first uses Plancherel and the global circle-symbol bound; the second uses the actual inverse-kernel bound. Both hold for every input in the corresponding Banach space and use the same finite center set and exceptional data. The estimate is a coarse finite-model bound, not the source-specific improved low or high estimate.

Checked declarations

Module	Named supporting theorems
CircleModelOperatorBounds.lean	2

- `circleMajorArcOperator_bound`
- `circleMajorArcL1Operator_bound`

<!-- checkpoint-verification -->

Verification

All 137 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3656-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 905 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 139 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_71_DilatedCutoffGeometry.md`

Checkpoint 71: Exact support and core widths for arithmetic narrow cutoffs

For every positive real dilation A, the actual smooth cutoff eta(A theta) has support inside |theta|<1/A and equals one when |theta|<=1/(2A). These exact width statements extend the already checked dyadic cutoff geometry to the arithmetic narrow-cutoff dilations qN^2/Q and B0(q/q0)N^2/Q used by the low model. Positivity of the dilation is explicit.

Checked declarations

Module	Named supporting theorems
DilatedCutoffGeometry.lean	2

- `dilatedSourceCutoff_radius`
- `dilatedSourceCutoff_core`

<!-- checkpoint-verification -->

Verification

All 138 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3657-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 907 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 140 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_72_NarrowCutoffContainment.md`

Checkpoint 72: The narrow cutoff lies inside the broad cutoff core

Under the proved geometric budget 2R^4<=N^2, with q<=R, Q<=R and the appropriate dyadic index, every nonzero narrow cutoff eta((qN^2/Q)theta) lies inside the region where the broad dyadic cutoff is exactly one. Hence multiplying the narrow cutoff by the broad cutoff leaves it unchanged. This is an exact support-containment identity and is needed to justify compactifying the low model without altering it in the admissible range.

Checked declarations

Module	Named supporting theorems
NarrowCutoffContainment.lean	2

- `narrowCutoff_inside_broad_core`
- `narrowCutoff_mul_broad`

<!-- checkpoint-verification -->

Verification

All 139 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3658-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 909 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 141 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_73_OrdinaryLowContinuity.md`

Checkpoint 73: Continuity of the actual ordinary low-model ingredients

The actual ordinary rational model term and its arithmetic narrow cutoff are proved continuous in real frequency, at every positive arithmetic length. The power and phase integral are the genuine annular model already checked; the cutoff has its exact qN^2/Q dilation. No restriction on Q is needed for continuity because Lean's total division and the resulting constant dilations remain continuous.

Checked declarations

Module	Named supporting theorems
OrdinaryLowContinuity.lean	2

- `rationalOrdinaryModelTerm_continuous`
- `ordinaryNarrowCutoff_continuous`

<!-- checkpoint-verification -->

Verification

All 140 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3659-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 911 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 142 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_74_ExceptionalLowContinuity.md`

Checkpoint 74: Continuity of the actual exceptional low-model ingredients

The actual exceptional rational model term is proved continuous in frequency, including its negative sign, character coefficient and arbitrary real exponent. The exceptional narrow cutoff with its exact B0(q/q0)N^2/Q dilation is also proved continuous. This establishes regularity for the low construction with its exceptional data, without assuming an analytic model-regularity lemma.

Checked declarations

Module	Named supporting theorems
ExceptionalLowContinuity.lean	2

- `rationalExceptionalModelTerm_continuous`
- `exceptionalNarrowCutoff_continuous`

<!-- checkpoint-verification -->

Verification

All 141 project modules and the root were freshly rebuilt: 3660 jobs pass without warnings/errors. All 913 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 143 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_75_LowModelContinuity.md`

Checkpoint 75: Continuity of the full arithmetic low term with both selectors

The full actual real-frequency low term, including both denominator selectors and ordinary and exceptional narrow cutoffs, is proved continuous. The selectors depend only on the fixed arithmetic data, so the proof treats their four cases and combines the already checked component regularity. This result prepares periodization of the low term; support containment and the source-specific low estimate remain separate obligations.

Checked declarations

Module	Named supporting theorems
LowModelContinuity.lean	1

- `rationalLowModelTerm_continuous`

<!-- checkpoint-verification -->

Verification

All 142 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3661-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 914 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 144 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_76_CenteredLowSymbol.md`

Checkpoint 76: A compact continuous realization of the centered low symbol

The low term is given a continuous compact real-frequency realization by multiplying its exact centered source formula by the broad dyadic cutoff. Support inside [-1,1] is proved from the actual cutoff, for all parameters. This compactification is a defined candidate, not an unconditional assertion that the extra cutoff changes nothing. The previously proved narrow-within-broad geometry, and the later ordinary/exceptional containment identities, are required to identify it with the original source low formula in the admissible range.

Checked declarations

Module	Named supporting theorems
CenteredLowSymbol.lean	2

- `sourceArcCutoffScale_zero_of_unit`
- `centeredLowSymbol_support`

<!-- checkpoint-verification -->

Verification

All 143 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3662-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 916 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 145 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_77_CenteredHighSymbol.md`

Checkpoint 77: The compact centered high symbol and exact total-model identity

The centered high symbol is constructed as the genuine localized total model minus the compact low symbol. It is continuous as a continuous-map difference, its compact support is proved, and low plus high equals the genuine centered Gauss model exactly. Source identification remains separate from this unconditional candidate decomposition: the extra low cutoff must be shown redundant under the appropriate width conditions.

Checked declarations

Module	Named supporting theorems
CenteredHighSymbol.lean	2

- `centeredHighSymbol_support`
- `centeredModel_low_add_high`

<!-- checkpoint-verification -->

Verification

All 144 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3663-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 918 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 146 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

Checkpoint_78_CircleLowTerm.md

Checkpoint 78: The actual periodized low term on the unit circle

The compact centered low term now gives a genuine continuous unit-circle symbol translated by its rational center. Its value at every real frequency is proved equal to the actual integer periodization of the centered low symbol. The construction has no implicit periodicity or convergence assumption. Identifying the centered compactification with the paper's low expression requires the explicit containment conditions recorded separately.

Checked declarations

Module	Named supporting theorems
CircleLowTerm.lean	1

- `circleLowTerm_real`

<!-- checkpoint-verification -->

Verification

All 145 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3664-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 919 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 147 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

Checkpoint_79_CircleHighTerm.md

Checkpoint 79: The periodized high term and exact low-plus-high identity

The compact high term is likewise periodized into a genuine continuous circle symbol. Its global integer-sum evaluation is proved, and the periodized low and high terms sum exactly to the previously constructed rational Gauss-model circle term. The proof uses the actual quotient representative and finite window, so it handles alias terms rather than dropping them.

Checked declarations

Module	Named supporting theorems
CircleHighTerm.lean	2

- `circleTerm_low_add_high`
- `circleHighTerm_real`

<!-- checkpoint-verification -->

Verification

All 146 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3665-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 921 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 148 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_80_FiniteCircleHighLow.md`

Checkpoint 80: Global finite low and high symbols with the actual annular error

Finite sums of the genuine low and high circle terms define continuous global finite low and high model symbols. Their sum equals the actual finite circle Gauss model. Adding the defined actual error gives exactly the annular von-Mangoldt averaging symbol, as an equality of continuous maps on the circle. This provides an unconditional candidate decomposition for the explicit finite center set; it does not supply the missing source-specific norm estimates or full minor-arc approximation.

Checked declarations

Module	Named supporting theorems
FiniteCircleHighLow.lean	2

- `circleModel_low_add_high`
- `annularSymbol_circle_low_high_error`

<!-- checkpoint-verification -->

Verification

All 147 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3666-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 923 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 149 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_81_CircleHighLowKernels.md`

Checkpoint 81: Actual inverse kernels for the finite low-high-error decomposition

The global finite low and high circle symbols now have actual inverse Fourier kernels. Their pointwise sum equals the genuine model kernel, and low plus high plus the model error kernel equals the actual annular von-Mangoldt square kernel at every integer. No compact spatial support is assumed for any inverse kernel produced by a smooth cutoff.

Checked declarations

Module	Named supporting theorems
CircleHighLowKernels.lean	2

- `circleKernel_low_add_high`
- `annularKernel_circle_low_high_error`

<!-- checkpoint-verification -->

Verification

All 148 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3667-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 925 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 150 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_82_CircleHighLowOperators.md`

Checkpoint 82: Genuine low l1 and high l2 operators with finite convolution formulas

The low circle symbol defines an actual bounded linear operator from integer l1 into linfinity; the high symbol defines an actual l2 Fourier multiplier. On every finite complex input both operators are proved to equal convolution with their respective genuine inverse kernels. This aligns the two different input norms with a common spatial decomposition and avoids treating formal symbols as already-defined convolution operators.

Checked declarations

Module	Named supporting theorems
CircleHighLowOperators.lean	2

- `circleLowOperator_finite`
- `circleHighOperator_finite`

<!-- checkpoint-verification -->

Verification

All 149 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3668-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 927 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 151 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_83_CircleFiniteDecomposition.md`

Checkpoint 83: Exact actual averaging decomposition on every finite complex input

For every finite complex input, convolution with the actual annular von-Mangoldt kernel is proved to equal the low l1 operator plus the high l2 operator plus the actual error l2 operator. This is an exact identity at every integer output. The input is represented in l1 and l2 with the same finite coefficients, and the proof uses the checked pointwise inverse-kernel decomposition. No analytic smallness hypothesis is needed for the identity.

Checked declarations

Module	Named supporting theorems
CircleFiniteDecomposition.lean	1

- `annularKernel_finite_low_high_error`

<!-- checkpoint-verification -->

Verification

All 150 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3669-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 928 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 152 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_84_CircleHighLowNormInputs.md`

Checkpoint 84: Precise kernel and symbol inputs for the actual low and high operators

The actual low operator norm is bounded by any established uniform bound on its genuine inverse kernel. The actual high l2 operator norm is bounded by any established uniform circle-symbol bound. These transport statements expose the precise analytic obligations still needed for the paper: the sharper low kernel bound and high symbol bound must be proved, not inferred merely from constructing the operators.

Checked declarations

Module	Named supporting theorems
CircleHighLowNormInputs.lean	2

- `circleLowOperator_kernel_bound`
- `circleHighOperator_symbol_bound`

<!-- checkpoint-verification -->

Verification

All 151 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3670-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 930 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 153 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_85_OrdinaryLowContainment.md`

Checkpoint 85: The extra broad cutoff is redundant for the ordinary narrow term

For a positive narrow parameter Q with Q<=R, rational denominator q<=R and 2R^4<=N^2, multiplying the exact ordinary narrow cutoff by the broad dyadic cutoff leaves it unchanged. The proof applies the real cutoff-containment theorem with the actual log2(q) index and transports the identity to complex values. This justifies redundancy of the extra broad cutoff for the ordinary part in this range.

Checked declarations

Module	Named supporting theorems
OrdinaryLowContainment.lean	1

- `ordinaryNarrowCutoff_mul_broad`

<!-- checkpoint-verification -->

Verification

All 152 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3671-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 931 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 154 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_86_ExceptionalLowContainment.md`

Checkpoint 86: The extra broad cutoff is redundant under the exceptional width condition

The same redundancy is proved for the exact exceptional narrow cutoff, with positive Q, B0 and q0 and the explicit effective-width condition Q*q0/B0<=R. Its dilation is first identified algebraically with qN^2/(Q*q0/B0), then the proved geometric containment applies. The effective-width condition is displayed rather than silently assumed to follow from conductor data; connecting the full source range to this condition remains an arithmetic parameter obligation.

Checked declarations

Module	Named supporting theorems
ExceptionalLowContainment.lean	1

- `exceptionalNarrowCutoff_mul_broad`

<!-- checkpoint-verification -->

Verification

All 153 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3672-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 932 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 155 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_87_CompactLowSourceIdentification.md`

Checkpoint 87: Exact identification of the compact low symbol with the source formula

Under both explicit width conditions Q<=R and Q*q0/B0<=R, positive Q,B0,q0, q<=R and the geometric budget 2R^4<=N^2, the compact centered low symbol is proved exactly equal to the source's original rationalLowModelTerm after translating back by its rational center. The two ordinary/exceptional selector cases are retained. Thus the extra broad cutoff is rigorously removed under stated scalar conditions; no unconditional source identification is claimed outside that range.

Checked declarations

Module	Named supporting theorems
CompactLowSourceIdentification.lean	1

- `centeredLowSymbol_source`

<!-- checkpoint-verification -->

Verification

All 154 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3673-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 933 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 156 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

Checkpoint_88_CompactHighSourceIdentification.md

Checkpoint 88: Exact identification of the compact high symbol with the source formula

Under the same explicit scalar conditions, the compact centered high symbol is proved exactly equal to the original rationalHighModelTerm after translation. The proof combines the genuine total-model identity, the just-proved low source identification and the original exact low-plus-high algebra. This completes source correspondence of both real compact components in this displayed range. Full source parameter coverage, global approximation and analytic norm estimates remain open.

Checked declarations

Module	Named supporting theorems
CompactHighSourceIdentification.lean	1

- `centeredHighSymbol_source`

<!-- checkpoint-verification -->

Verification

All 155 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3674-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 934 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 157 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

Checkpoint_89_FiniteOperatorEnvelope.md

Checkpoint 89: A genuine nonnegative l2 envelope for finite operator families

For a finite family of genuine l2 operators, the pointwise magnitudes of their outputs are assembled into an actual nonnegative real l2 envelope. Taking pointwise magnitude preserves the l2 norm exactly, and the finite envelope dominates the output magnitude of every member of the family at every integer. This prepares a rigorously defined maximal majorant; it does not claim a uniform infinite-scale maximal theorem.

Checked declarations

Module	Named supporting theorems
FiniteOperatorEnvelope.lean	3

- `magnitudeL2_norm`
- `finiteOperatorEnvelope_nonneg`
- `finiteOperatorEnvelope_dominates`

<!-- checkpoint-verification -->

Verification

All 156 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3675-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 937 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 158 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_90_FiniteMaximalEnvelopeBound.md`

Checkpoint 90: Finite-family l2 maximal control through the explicit envelope

The finite l2 envelope has norm at most the sum of the individual operator bounds times the input norm. Every nonnegative pointwise function dominated by this envelope is proved to belong to real l2 with the same bound. In particular, this supplies the finite-family maximal bound once a pointwise maximum is dominated by the envelope. The bound depends on the sum of scale bounds; proving the source's uniform infinite-family or multifrequency maximal estimate remains open.

Checked declarations

Module	Named supporting theorems
FiniteMaximalEnvelopeBound.lean	2

- `finiteOperatorEnvelope_norm`
- `finiteOperator_dominated_maximal`

<!-- checkpoint-verification -->

Verification

All 157 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3676-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 939 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 159 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_91_CircleHighLowAssembly.md`

Checkpoint 91: Actual high-low assembly and completion of the fifty-checkpoint batch

The actual finite circle construction is now connected to the previously formalized operator high-low framework. An explicit uniform bound on the genuine low inverse kernel and an explicit uniform symbol bound on the sum of high and error imply OperatorHighLowBound for the actual annular von-Mangoldt square kernel. The spatial normalization is N^2, where N is the arithmetic length. The low witness is the genuine l1-to-linfinity convolution operator; the high witness is the actual l2 Fourier multiplier of high plus error. Their agreement on indicator inputs is derived from the exact finite-input decomposition.

This is an assembly theorem with its two analytic inputs displayed. Those source-specific inputs are NOT proved by this theorem, are NOT registered as axioms, and are NOT evidence that Proposition 4.19 or Theorem 1.1 is complete. The remaining global minor-arc deduction, source center-set/parameter identification and sharp analytic low/high/maximal estimates are documented explicitly. The partial formalization has no newly added analytic axioms or own-paper lemma assumptions hidden in a completed numbered result.

This checkpoint ends the requested batch of 50 separately saved checkpoints, numbered 42 through 91. The combined Batch_42_91_Report.md gives the complete ledger, verification method and outstanding obligations. The entire project is freshly rebuilt again here, then every named theorem's complete transitive axiom set, source constraints, frozen manuscript hashes and ZIP entries are checked before saving.

Checked declarations

Module	Named supporting theorems
CircleHighLowAssembly.lean	1

- `circleModel_operatorHighLowBound`

<!-- checkpoint-verification -->

Verification

All 158 project modules and the root were freshly rebuilt: 3677 jobs pass without warnings/errors. All 940 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 160 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_92_CanonicalRationalCenters.md`

Checkpoint 92: The exact finite set of reduced canonical rational centers

A concrete finite set of canonical rational centers is defined, and its membership is proved exactly equivalent to 0<=r<1 and den(r)<=M. The construction enumerates a finite numerator-denominator box and filters by this exact predicate; it is not a sample or an incomplete denominator list. The normalized denominator is used, so fractions with different unreduced presentations do not introduce duplicate centers.

The proofs include the exact numerator and denominator ranges, monotonicity in M, the empty M=0 case, inclusion of the denominator-one center zero exactly when M>=1, and a finite cardinality bound. For a real cutoff R>=0, membership is proved equivalent to canonicality and den(r)<=R by using floor(R). This supplies the precise finite rational index set required by the model rather than leaving an arbitrary finite set as the only interface.

This identifies the reduced canonical rational index predicate. The character/coefficient data and complete summation correspondence are not silently supplied by this construction. No prime-distribution estimate, analytical approximation or deep input is used in these finite arithmetic proofs.

Checked declarations

Module	Named supporting theorems
CanonicalRationalCenters.lean	7

- `mem_canonicalRationalCenters`
- `canonicalRationalCenters_parameters`
- `canonicalRationalCenters_mono`

- `canonicalRationalCenters_zero`
- `zero_mem_canonicalRationalCenters`
- `canonicalRationalCenters_card`
- `mem_sourceRationalCenters`

<!-- checkpoint-verification -->

Verification

All 159 project modules and the root were freshly rebuilt: 3678 jobs pass without warnings/errors. All 947 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 161 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_93_WrappedRationalArcGeometry.md`

Checkpoint 93: Rational arc geometry with every integer lift

This checkpoint proves rational-arc geometry on the circle while retaining every integer lift. Two different canonical center/lift pairs with denominators at most D are separated by at least 1/D^2. The proof treats distinct reduced rationals and distinct lifts of the same rational separately. Consequently two arcs of common radius delta with 2*delta<1/D^2 cannot meet unless both their centers and integer lifts agree.

For the manuscript's major-arc widths R/(q*N^2), the condition 2*R^3<N^2 gives uniqueness of the center/lift pair. If a frequency is on the major arc of r0, every distinct center or nonzero integer shift is strictly outside its own major arc. In particular, shifted copies of r0 itself are not omitted. The same argument proves uniqueness for the actual dyadic cutoff at scales s>=2. The earlier low-scale overlap bound remains applicable at the smaller scales; no uniqueness claim is made there.

The exact quotient identity says that adding any integer to a real representative gives the same point of PrimeCircle. Together these results cover the wrap across zero and one without assuming that the real-variable oscillatory integral is periodic. The proofs use only finite rational arithmetic, inequalities and the circle quotient; no progression estimate or minor-arc conclusion is assumed.

Checked declarations

Module	Named supporting theorems
WrappedRationalArcGeometry.lean	6

- `primeCircle_coe_integer_shift`
- `canonicalLift_den_separation`
- `canonicalLift_small_arc_unique`
- `canonicalLift_major_arc_unique`
- `canonicalLift_other_outside`
- `cutoffScale_wrapped_unique`

<!-- checkpoint-verification -->

Verification

All 160 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3679-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 953 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 162 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

Checkpoint_94_PeriodizedModelTails.md

Checkpoint 94: Exact periodization and logarithmic circle-model tails

The genuine periodization is reduced exactly to the five integer shifts -2,-1,0,1,2 for every canonical rational center and every real representative xi in [-1,2]. The centered cutoff model vanishes at distance at least one; this proves that all omitted shifts are zero, including the boundary cases. Summing over the rational centers and interchanging the finite sums gives the exact same representation for the actual continuous circle model.

If a representative is outside every small major arc at each of those five shifts, the actual circle model has norm at most

```
T_circle(R) = 5\*(5+log(R)/log(2))\*(4\*C_G/pi)\*R^(-5/12),
```

where C_G is the positive constant chosen from the previously proved, unconditional ordinary/exceptional Gauss-model bounds. This is a logarithmic tail estimate, rather than the much larger sum of absolute coefficient bounds over all denominators. All cutoffs and actual integral models retain their existing definitions.

A common approximation budget is defined by

```
B(C,R) = (4+(2+4\*pi)\*C)/R^5 + C_G\*R^(-5/12) + T_circle(R).
```

Lean proves that T_circle(R)<=R^(-1/3) eventually and B(C,R)<=R^(-1/6) eventually for every fixed C>=0. Substituting the actual source scale R=max(exp((log N)^(1/10)),Q^8) gives, for every A>=0, a threshold in N independent of Q>=1 with B(C,R)<=2^A*Q^(-1/2)*(1+log N)^(-A).

This checkpoint bounds the model and its explicit error budget. It does not prove that the prime-square multiplier is close to that model at all frequencies. No distribution hypothesis is consumed by these statements. The actual approximation branches are treated in the subsequent checkpoints; the full-range minor-arc problem remains open.

Checked declarations

Module	Named supporting theorems
PeriodizedModelTails.lean	8

- `fivePeriodShifts_card`
- `integerPeriodSum_five`
- `circleRationalModelTerm_five`
- `circleMajorArcModel_five`
- `circleMajorArcModel_outside_norm`
- `eventually_circleModelTailBudget`
- `eventually_circleApproxBudget`
- `circleApproxBudget_uniform_log_rate`

<!-- checkpoint-verification -->

Verification

All 161 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3680-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 961 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 163 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

Checkpoint_95_PeriodizedMajorArcError.md

Checkpoint 95: Conditional major-arc approximation for the genuine circle model

This checkpoint transfers the progression-prefix major-arc estimate to the actual continuous circle error. It does not replace the circle model by its unperiodized real-variable counterpart. The old real-model error is combined with four additional integer shifts, obtained from the exact five-shift identity proved in checkpoint 94.

If xi is on the major arc of a canonical center r0 with denominator at most R, the separation condition 2*R^3<N^2 shows that every nonzero shift is outside every center's own major arc, including the shifted copies of r0. Their combined norm is at most four real-model tail budgets. The zero-shift error already contributes at most the inverse-fifth progression error plus one such tail. Thus the actual circle error is bounded by the common circleApproxBudget(C,R). The proof also establishes that the chosen lifted representative lies in [-1,2], so the finite-shift formula applies without an extra hypothesis.

The explicit inputs include N>=2, R>=1, 0<beta<=1, coefficient norms at most one, 2*R^3<N^2, 2*R^4<=N^2, log R<=log N, R^5*log N<=sqrt N and 7*log R<=c*sqrt(log N). The sole unproved distribution input is the displayed inequality for progressionMangoldtPrefix on the whole positive real interval [N/2,N], with error N*C*exp(-c*sqrt(log N)); the same character/coefficient/beta data must work throughout this interval. No approximation conclusion from the manuscript is assumed.

For the exact sourceRationalCenters(R) with R=sourceMajorArcScale(N,Q), the uniform theorem supplies one N threshold, independent of Q and the permitted exceptional data, giving

```
norm(circle error) <= 2^A\*Q^(-1/2)\*(1+log N)^(-A).
```

Here C>=0, c>0, A>=0, Q>=1 and log Q <= (min(c/7,1)/8)*sqrt(log N). Canonicality and denominator bounds come from the exact membership theorem of checkpoint 92. Arcs crossing the circle boundary are handled by choosing their integer-shifted real representative; checkpoint 93 proves that its circle point is unchanged.

This completes this conditional major-arc branch for the actual periodized model. It is not the global approximation Lemma 6.9: the intermediate and larger denominator branches, and the full source exceptional-data correspondence, cannot be silently inferred. No LaTeX correction is prescribed by these proofs.

Checked declarations

Module	Named supporting theorems
PeriodizedMajorArcError.lean	4

- `majorArc_representative_range`
- `realModelTailBudget_nonneg`
- `circleMajorArcError_from_prefix`
- `circleMajorArcError_uniform_log_rate`

<!-- checkpoint-verification -->

Verification

All 162 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3681-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 965 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 164 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_96_IntermediateCircleApproximation.md`

Checkpoint 96: Intermediate-denominator approximation for the genuine circle model

This checkpoint proves the intermediate-denominator part of the actual circle approximation. Suppose a reduced rational r=a/q approximates the chosen real representative with |xi-r|<=R/(q*N^2), and

```
R < q <= R^(11/10).
```

The already proved inverse-fifth progression estimate controls the difference between the actual annular von Mangoldt multiplier and its genuine Gauss model. The unconditional Gauss bounds give model size at most C_G*q^(-5/12), which is at most C_G*R^(-5/12). Lean therefore proves a bound for the actual multiplier by (4+(2+4*pi)*C)/R^5+C_G*R^(-5/12).

On the branch outside all small-denominator major arcs, including all integer lifts relevant for the representative in [-1,2], checkpoint 94 bounds the actual circle model by circleModelTailBudget(R). Subtracting it gives the same circleApproxBudget(C,R) used on the major-arc branch. The outside-arc assumption is an explicit geometric case condition, not an assumed minor-arc estimate. The approximating rational need not itself belong to the small-denominator set.

The uniform theorem specializes to the exact source rational set and source scale. For C>=0, c>0 and A>=0, one threshold in N, independent of Q and the permitted data, gives the bound 2^A*Q^(-1/2)*(1+log N)^(-A) whenever Q>=1, log Q <= (min(c/7,1)/8)*sqrt(log N), and the displayed raw progression-prefix inequality holds at the approximating denominator throughout [N/2,N]. This is the same kind of explicit deep input already used in the preceding source-range major-arc work; no new analytic axiom is introduced.

The major-arc and intermediate branches now refer to the genuine periodized circle model, with precise finite rational indexing and all integer shifts accounted for. The existence/selection of an appropriate rational approximation is not asserted by this checkpoint. The branch q>R^(11/10), including the unresolved full-range minor-arc deduction, remains open. Consequently full Lemma 6.5, global Lemma 6.9, sharp low 6.14, multifrequency 6.16, high/maximal 6.21, Proposition 4.19 and Theorem 1.1 remain unproved. The full exceptional-parameter/source correspondence also remains open. No new manuscript correction has been established.

Checked declarations

Module	Named supporting theorems
IntermediateCircleApproximation.lean	3

- `annularMangoldt_intermediate_norm`
- `circleMajorArcError_intermediate_from_prefix`
- `circleMajorArcError_intermediate_uniform_log_rate`

<!-- checkpoint-verification -->

Verification

All 163 project modules and the root were freshly rebuilt: 3682 jobs pass without warnings/errors. All 968 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 165 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The already displayed raw progression-prefix estimate is consumed as a theorem hypothesis; it remains unproved. No new external analytic estimate or own-paper lemma is assumed.

`Checkpoint_97_GhoshBalancedBound.md`

Checkpoint 97: A fixed-epsilon Ghosh bound in the balanced denominator range

This checkpoint introduces the explicit Ghosh quadratic bracket 1/q+x^(-1/2)+q/x^2 and proves that, throughout x^(1/2)<=q<=x^(3/2), it is at most 3*x^(-1/2). Raising to the fourth power root gives 3^(1/4)*x^(-1/8), with positivity proved rather than assumed.

The actual annular von Mangoldt circle multiplier is related exactly to the positive-phase Ren sum on (N/2,N], with its 1/N normalization. From the displayed raw Ghosh estimate with the fixed epsilon=1/16, Lean proves a bound (B*3^(1/4)/2)*(N/2)^(-1/16) for this actual multiplier in the balanced denominator range. The epsilon is fixed once; it is never chosen as a function of N. This gives a polynomial saving in that range.

The deep Ghosh estimate is not formalized or added as an axiom. It is an explicit inequality hypothesis of the consuming theorems. Its source is A. Ghosh, The Distribution of alpha p^2 Modulo 1, Proc. London Math. Soc. 42 (1981), Theorem 2, with coprimality and strict approximation error <1/q^2. The original article's first-page transcription was inspected at https://www.academia.edu/60387191/The_Distribution_of_p2_Modulo_1 and its publisher identity is https://doi.org/10.1112/plms/s3-42.2.252. The annular form is also explicitly displayed as equation (1.4) of Xiumin Ren's author preprint https://www.math.hku.hk/imrwww/IMRPreprintSeries/2004/IMR2004-9.pdf. The consumer assumes the displayed raw inequality itself; it does not assert it for arbitrary incompatible choices of q and xi. Exact rational selection and compatibility are subsequent tasks.

This supplies the complementary balanced-denominator branch, not yet a full minor-arc theorem. A new approximation at precision x^(-3/2) is planned so that its larger denominators lie in this range. The smaller-denominator branch will use Ren, with its divisor prefactor controlled explicitly. The source's stronger printed minor-arc estimate remains open; no LaTeX correction is established by this checkpoint.

Checked declarations

Module	Named supporting theorems
GhoshBalancedBound.lean	5

- `ghoshQuadraticBracket_nonneg`
- `ghoshQuadraticBracket_balanced`
- `ghoshBalanced_fourth_root`
- `annularMangoldt_from_Ghosh_bound`
- `annularMangoldt_Ghosh_balanced`

<!-- checkpoint-verification -->

Verification

All 164 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3683-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 973 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 166 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypothe-

ses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_98_ExponentGeometricBound.md

Checkpoint 98: Uniform prime-power exponent growth

This is the first step of an unconditional divisor-growth proof needed to retain Ren's divisor prefactor. For every fixed real ratio r>1, Lean proves a single constant A>=1 such that n+1<=A*r^n for every natural exponent n, including n=0 and the whole finite initial range. The eventual estimate is obtained from a proved Mathlib asymptotic limit; the early exponents are bounded explicitly by the selected threshold. No number-theoretic hypothesis enters.

Real powers commute with natural powers on nonnegative bases. Applying the geometric estimate at ratio 2^epsilon, and using p>=2, gives a constant A independent of the prime p and exponent n such that n+1<=A*(p^n)^epsilon. The exponent epsilon is a fixed positive real number, not a parameter chosen with the denominator or scale. This is the local prime-power inequality that will be multiplied over the factorization of q.

This checkpoint proves these uniform local growth statements only. The full divisor count and its 3/2 power are addressed subsequently. It does not assume a divisor bound, PNT, a progression estimate or an exponential-sum theorem, and it does not close the minor-arc or main theorem gaps on its own.

Checked declarations

Module	Named supporting theorems
ExponentGeometricBound.lean	3

- `exponent_add_one_le_geometric`
- `rpow_nat_power_comm`
- `primePower_exponent_bound`

<!-- checkpoint-verification -->

Verification

All 165 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3684-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 976 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 167 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_99_DivisorProductGrowth.md

Checkpoint 99: An unconditional uniform subpower bound for the divisor count

The exact divisor count is expressed as the product of exponent-plus-one factors over the distinct prime divisors of q. Using the uniform prime-power estimate of checkpoint 98, Lean bounds this product by A^omega(q)*q^epsilon. The product of prime powers is exactly q, with natural-to-real casts and real-power multiplication justified.

The existing elementary absorption of a fixed A^omega(q) into a small positive power of q then proves: for every fixed epsilon>0, there is C>0 such that tau(q)<=C*q^epsilon for every positive integer q. The constant is independent of q, includes q=1, and is selected before the denominator. The proof uses only

unique factorization, finite products and proved asymptotic limits; it assumes no divisor bound, PNT or analytic number-theory theorem.

This is a supporting arithmetic estimate needed for Ren's prefactor. It does not assert a bound for a prime exponential sum without its explicit deep-input hypothesis, or close the full minor-arc or headline theorem gaps.

Checked declarations

Module	Named supporting theorems
DivisorProductGrowth.lean	3

- `divisors_card_primePower_product`
- `divisors_card_weighted_power`
- `divisors_card_le_rpow`

<!-- checkpoint-verification -->

Verification

All 166 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3685-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 979 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 168 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_100_RenDivisorBudget.md`

Checkpoint 100: A fixed subpower budget for the exact Ren divisor prefactor

The unconditional divisor bound is raised to an arbitrary fixed positive real power gamma, retaining a constant uniform in the positive integer q. For each epsilon>0, Lean proves tau(q)^gamma<=C*q^epsilon by applying the previous bound at epsilon/gamma and justifying all real-power identities.

For the exact Ren prefactor gamma=3/2, a positive constant renDivisorConstant is selected from the proved existence theorem at the fixed exponent 1/12. Its positivity and the bound tau(q)^(3/2)<=renDivisorConstant*q^(1/12) are proved for every positive q. Neither exponent varies with q or N. No external analytic hypothesis is involved.

The aim is to insert this controlled prefactor into the subsequent Ren branch. This checkpoint does not claim that the q^(1/12) loss can be ignored or absorbed without further estimates. Full minor-arc and global approximation deductions remain open at this stage.

Checked declarations

Module	Named supporting theorems
RenDivisorBudget.lean	3

- `divisors_card_rpow_le_rpow`
- `renDivisorConstant_pos`
- `ren_divisor_prefactor_bound`

<!-- checkpoint-verification -->

Verification

All 167 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3686-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 982 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 169 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_101_BalancedEffectiveDenominator.md`

Checkpoint 101: Effective-denominator bounds for the balanced approximation

This checkpoint proves the elementary geometry for a new balanced rational approximation. If x>0, 0<q<=sqrt(x), and the real offset satisfies |theta|<=1/(q*x^(3/2)), then Ren's effective denominator T=q*(1+|theta|*x^2) lies between q and 2*sqrt(x). The upper bound is exact and uses the real-power identity x^2=x^(3/2)*x^(1/2).

A complementary lower bound is proved: T>=R/4 whenever either q>=R or |theta|>=R/(4*q*x^2), for x>0 and R>=0. With N=2*x, the latter threshold is exactly the width R/(q*N^2) of a source major arc. Thus being outside the small-denominator arcs will force T large, even when the newly selected q itself is small.

These statements are inequalities for the actual previously defined renEffectiveDenominator, not assumptions about an abstract replacement. They use no exponential-sum or distribution theorem. Selecting a compatible rational approximation and connecting the outside-arc condition are separate subsequent tasks.

Checked declarations

Module	Named supporting theorems
BalancedEffectiveDenominator.lean	3

- `renEffectiveDenominator_ge`
- `renEffectiveDenominator_balanced_upper`
- `renEffectiveDenominator_lower_from_arc`

<!-- checkpoint-verification -->

Verification

All 168 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3687-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 985 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 170 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

`Checkpoint_102_BalancedRationalApproximation.md`

Checkpoint 102: A proved rational selection at balanced precision

Mathlib's proved Dirichlet approximation theorem is applied with the integer cutoff floor(x^(3/2)). For every x>=1 and every real xi, Lean constructs a reduced rational r with den(r)<=x^(3/2), |xi-r|<=1/(den(r)*x^(3/2)), and the strict bound |xi-r|<1/den(r)^2. Floor errors, positive denominators and the cutoff's positivity are proved explicitly. This applies to rational and irrational xi alike.

The annular specialization sets x=N/2 for natural N>=2. If xi lies in [0,1], the selected rational lies in [-1,2]. No rational approximation is supplied as an unproved existence assumption. The strict reciprocal-square bound is the compatibility needed for the external Ghosh estimate; the finer x^(-3/2) precision also controls Ren's effective denominator in the smaller-denominator branch.

This is a new approximation used by the complementary argument; it need not be the manuscript's original approximant. No change to the manuscript is prescribed. The prime exponential-sum estimates remain explicit deep-input hypotheses.

Checked declarations

Module	Named supporting theorems
BalancedRationalApproximation.lean	3

- `exists_balanced_rational_approximation`
- `exists_balanced_annular_approximation`
- `balanced_approximation_representative_range`

<!-- checkpoint-verification -->

Verification

All 169 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3691-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 988 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 171 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_103_RenSmallDenominatorTerms.md`

Checkpoint 103: The first two normalized Ren terms with the divisor loss retained

The first two terms of Ren's three-term bracket are bounded while retaining the q^(1/12) loss proved necessary for its divisor prefactor. For x>0 and 0<=q<=sqrt(x), Lean proves q^(1/12)<=x^(1/24). If the effective denominator T<=2*sqrt(x), then sqrt(T)<=sqrt(2)*x^(1/4).

These yield the exact normalized inequalities

```
q^(1/12)\*sqrt(x)\*sqrt(T)/x <= sqrt(2)\*x^(-5/24),
q^(1/12)\*x^(4/5)/x <= x^(-19/120).
```

All real-power identities, positivity conditions and normalization by x are checked in Lean. The denominator q is a real parameter in these inequalities, allowing the later natural denominator to be substituted without changing the statement. No external analytic estimate is assumed. The remaining inverse-effective-denominator term is handled separately before the bracket is recombined.

Checked declarations

Module	Named supporting theorems
RenSmallDenominatorTerms.lean	5

- `rpow_product_div_self`
- `smallDenominator_twelfth`
- `effectiveDenominator_sqrt_bound`
- `ren_first_weighted_term`
- `ren_middle_weighted_term`

<!-- checkpoint-verification -->

Verification

All 170 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3692-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 993 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 172 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_104_RenInverseEffectiveTerm.md`

Checkpoint 104: The inverse effective-denominator term with divisor loss retained

The inverse-effective-denominator term is controlled with the divisor loss included. If q>=0, R>0, q<=T and R/4<=T, then

```
q^(1/12)/sqrt(T) <= 4^(5/12)\*R^(-5/12).
```

The proof first replaces q^(1/12) by T^(1/12), giving T^(-5/12), and then uses the lower bound on T. This avoids inserting an uncontrolled q^epsilon factor into an R-only estimate. The exact normalized third term q^(1/12)*(x/sqrt(T))/x satisfies the same bound for x>0; its cancellation is justified.

These are unconditional real inequalities. In the later application, the new approximation gives the upper bound on T and the outside-major-arc condition gives the lower bound. No bound on a prime sum is inferred until its explicit Ren hypothesis has been supplied.

Checked declarations

Module	Named supporting theorems
RenInverseEffectiveTerm.lean	2

- `weighted_inverse_sqrt_effective`
- `ren_third_weighted_term`

<!-- checkpoint-verification -->

Verification

All 171 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3693-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 995 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 173 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_105_RenBalancedBracket.md`

Checkpoint 105: The complete weighted Ren bracket in the balanced regime

The three previously proved weighted term bounds are combined into the explicit renBalancedBudget(x,R): sqrt(2)*x^(-5/24)+x^(-19/120)+4^(5/12)*R^(-5/12). Its nonnegativity is proved on nonnegative bases.

For positive x,R, 0<=q<=sqrt(x), q<=T<=2*sqrt(x), and T>=R/4, Lean proves q^(1/12)*renQuadraticBracket(x,T)/x<=renBalancedBudget(x,R). A second theorem substitutes the actual effective denominator associated to an approximation satisfying |theta|<=1/(q*x^(3/2)). The upper bound and q<=T then come from the checked geometry, rather than extra analytic assumptions.

This is an unconditional recombination of inequalities. The next consuming theorem inserts the raw Ren exponential-sum estimate and its genuine divisor prefactor; no prime-sum bound is assumed or asserted in the present statements.

Checked declarations

Module	Named supporting theorems
RenBalancedBracket.lean	3

- `renBalancedBudget_nonneg`
- `ren_weighted_bracket_balanced`
- `ren_weighted_bracket_from_balanced_approximation`

<!-- checkpoint-verification -->

Verification

All 172 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3694-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 998 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 174 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_106_AnnularRenBalancedBound.md`

Checkpoint 106: The actual annular multiplier under the raw Ren estimate

This checkpoint applies the explicit Ren exponential-sum hypothesis to the actual annular von Mangoldt circle multiplier, using the newly proved weighted bracket estimate and the unconditional divisor budget.

For N>=4, x=N/2, q<=sqrt(x), approximation precision |theta|<=1/(q*x^(3/2)), and effective denominator at least R/4, Lean proves

```
norm(actual annular multiplier) <= (C\*renDivisorConstant/2)\*(log x)^c\*renBalancedBudget(x,R).
```

The factor 1/2 comes from the exact N=2*x normalization. Positivity of the bracket, logarithmic base and selected divisor constant is proved. The q^(1/12) loss is consumed by the weighted estimates and has not been discarded. The raw Ren inequality still appears explicitly in the theorem hypothesis with its full tau(q)^(3/2) prefactor.

The theorem consumes a compatible raw estimate; it does not assert Ren's theorem for arbitrary unrelated xi and theta. Rational compatibility will be supplied by the selected approximation. No own-paper minor-arc conclusion or additional mathematical axiom is used.

Checked declarations

Module	Named supporting theorems
AnnularRenBalancedBound.lean	2

- `renQuadraticBracket_nonneg`
- `annularMangoldt_Ren_balanced`

<!-- checkpoint-verification -->

Verification

All 173 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3695-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1000 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 175 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_107_SourceScaleFourthPower.md`

Checkpoint 107: A uniform fourth-power budget for the source scale

The source-range major-arc parameter budget also controls the new complementary argument. For N>=4, R>=1, log N>=1 and R^5*log N<=sqrt(N), Lean proves R^4<=N/2. The proof retains the exact N/2 annulus: R^4<=R^5<=R^5*log N<=sqrt N<=N/2.

A real-power consequence is proved for every a>=0: if R>0 and R^4<=x, then x^(-a)<=R^(-4*a). This will convert the polynomial savings from Ren and Ghosh into powers of R without changing a fixed epsilon with N.

The final theorem supplies a single natural threshold, independent of Q, for the actual source scale R=sourceMajorArcScale(N,Q), over the same allowed Q range as the already proved progression budget. Positivity, log N>=1 and N>=4 are supplied by proved eventual limits and the existing parameter theorem. No deep estimate is added here.

Checked declarations

Module	Named supporting theorems
SourceScaleFourthPower.lean	4

- `sqrt_le_half_of_four_le`

- `sourceBudget_fourth_power`
- `inverse_rpow_from_fourth_power`
- `sourceMajorArcScale_uniform_fourth_power`

<!-- checkpoint-verification -->

Verification

All 174 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3696-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1004 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 176 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_108_CanonicalApproximationLift.md`

Checkpoint 108: Canonical rational centers for an arbitrary selected approximation

Any rational approximation r is normalized to its exact fractional part, a canonical rational in [0,1). Its reduced denominator is unchanged by this normalization, using Mathlib's proved rational-floor theorem. The frequency offset is exactly rewritten as an integer lift of that canonical center:

```
xi-r = xi-floor(r)-fract(r).
```

Lean proves that if den(r)<=R with R>=0, then fract(r) belongs to the exact finite sourceRationalCenters(R). The cast of the rational fractional part is explicit, avoiding an implicit change from rational to real floor operations.

This is the arithmetic bridge allowing a newly selected rational, whose numerator may fall outside the canonical interval, to be tested against all lifted source major arcs. It uses no prime-distribution or exponential-sum input.

Checked declarations

Module	Named supporting theorems
CanonicalApproximationLift.lean	3

- `ratFract_canonical`
- `rational_offset_canonical_lift`
- `ratFract_mem_sourceRationalCenters`

<!-- checkpoint-verification -->

Verification

All 175 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3697-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1007 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 177 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_109_OutsideSourceArcs.md

Checkpoint 109: Exclusion from all lifted source arcs and the effective lower bound

OutsideSourceMajorArcs is defined using the exact finite source rational set and every integer lift. It requires R/(den(r)*N^2)<=|xi+k-r| for each center and integer k. This is an explicit geometric branch condition, not a minor-arc analytic estimate.

For any rational r with den(r)<=R, the canonical-fractional-part bridge gives the same lower bound for |xi-r|. Consequently, for every rational r, the actual Ren effective denominator at x=N/2 is at least R/4: either den(r)>=R, or exclusion from its lifted small-denominator arc gives the bound. There is no restriction that the selected r already be canonical.

Finally, the existing circle-model outside-arc theorem is specialized to this exact set and all-lift condition. It bounds the genuine periodized model by circleModelTailBudget(R), including the denominator-one center and all shifted copies. The beta/coefficient bounds and representative interval [-1,2] remain explicit. No new analytic hypothesis is assumed.

Checked declarations

Module	Named supporting theorems
OutsideSourceArcs.lean	3

- `outsideSourceMajorArcs_rational`
- `outsideSourceMajorArcs_effective_lower`
- `outsideSourceMajorArcs_circleModel_norm`

<!-- checkpoint-verification -->

Verification

All 176 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3698-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1010 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 178 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_110_RenScaleAbsorption.md

Checkpoint 110: Converting the balanced Ren budget to a scale-only bound

The explicit balanced Ren budget is converted to a pure power of R under R>=1 and R^4<=x. The previously proved fourth-power comparison gives x^(-5/24)<=R^(-5/6) and x^(-19/120)<=R^(-19/30). Both are at most R^(-5/12).

Lean therefore proves renBalancedBudget(x,R)<=renGeometryConstant*R^(-5/12), with the fixed positive constant sqrt(2)+1+4^(5/12). A weaker quarter-power consequence is also recorded for later comparison with Ghosh. The fixed exponents and constants are preserved, and no uncontrolled N^epsilon loss is absorbed into a subpolynomial scale.

These are unconditional scalar deductions. They become estimates for the actual multiplier only when the preceding consuming theorem's explicit Ren inequality is supplied.

Checked declarations

Module	Named supporting theorems
RenScaleAbsorption.lean	3

- `renGeometryConstant_pos`
- `renBalancedBudget_scale_bound`
- `renBalancedBudget_quarter_bound`

<!-- checkpoint-verification -->

Verification

All 177 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3699-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1013 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 179 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_111_GhoshScaleAbsorption.md`

Checkpoint 111: Converting the balanced Ghosh saving to a scale-only bound

The balanced Ghosh saving (N/2)^(-1/16) is converted to R^(-1/4) whenever R>0 and R^4<=N/2. The fixed constant B*3^(1/4)/2 is defined and proved nonnegative for B>=0.

The resulting theorem bounds the actual annular von Mangoldt circle multiplier for (N/2)^(1/2)<=q<=(N/2)^(3/2), conditional on the displayed raw Ghosh inequality at the fixed epsilon 1/16. The fourth-power hypothesis is exactly the source-scale budget proved earlier. No additional outside-arc hypothesis is needed on this balanced-denominator branch.

Ghosh's deep estimate remains a hypothesis, not an added axiom or a placeholder. The new step is the checked scale conversion and its application to the genuine multiplier, uniform in the permitted denominator.

Checked declarations

Module	Named supporting theorems
GhoshScaleAbsorption.lean	2

- `ghoshSourceConstant_nonneg`
- `annularMangoldt_Ghosh_scale_bound`

<!-- checkpoint-verification -->

Verification

All 178 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3700-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1015 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 180 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_112_AnnularRenScaleBound.md

Checkpoint 112: The small-denominator Ren branch at the actual source scale

The raw Ren consuming theorem is combined with the proved fourth-power scale absorption. The actual annular multiplier is bounded by renSourceConstant(C)*(log(N/2))^c*R^(-5/12), provided N>=4, R>=1, R^4<=N/2, the selected denominator q<=sqrt(N/2), the balanced approximation precision holds, and its effective denominator is at least R/4.

The constant is the explicit product (C*renDivisorConstant/2)*renGeometryConstant; its nonnegativity is proved for C>=0. All divisor and logarithmic factors remain accounted for. The supplied Ren inequality involves the actual annular exponential sum at the same xi and theta as the chosen approximation.

This is the small-denominator analytic branch of the complementary argument. The deep raw estimate is explicit; the paper's own minor-arc conclusion is not assumed. The rational selection and outside-arc geometry will supply the remaining geometric hypotheses in the subsequent assembly.

Checked declarations

Module	Named supporting theorems
AnnularRenScaleBound.lean	2

- `renSourceConstant_nonneg`
- `annularMangoldt_Ren_scale_bound`

<!-- checkpoint-verification -->

Verification

All 179 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3701-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1017 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 181 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_113_ComplementaryMinorArcBound.md

Checkpoint 113: An alternative full-range minor-arc deduction under raw deep estimates

This checkpoint completes an alternative full-range minor-arc deduction for the actual annular von Mangoldt circle multiplier, conditional on displayed raw Ren and Ghosh estimates. The two input predicates state precise inequalities for the actual sum at compatible rational approximations. They are propositions in theorem hypotheses, not mathematical axioms.

Lean selects a reduced rational at precision x^(-3/2), x=N/2, using the proved existence theorem. If its denominator is at most sqrt(x), the all-lift outside-source-arc condition supplies the lower effective-denominator bound and the Ren branch applies. If its denominator exceeds sqrt(x), it lies in the proved balanced Ghosh range and satisfies the strict reciprocal-square approximation condition. There is no uncovered denominator range.

For N>=4, R>=1 and R^4<=N/2, the result is

```
norm(actual annular multiplier) <= renSourceConstant(C)\*(log(N/2))^c\*R^(-5/12)
                                 + ghoshSourceConstant(B)\*R^(-1/4).
```

The combined budget is proved nonnegative. The fixed epsilon in Ghosh is 1/16; the divisor-growth epsilon is fixed at 1/12. Neither depends on N. The estimate is valid on the full complement of the small-denominator lifted major arcs and does not rely on the manuscript's original approximant. It is weaker than the stronger printed minor-arc formula in Lemma 6.5, which remains open; it is intended to suffice for global approximation Lemma 6.9. No manuscript replacement is prescribed on this basis.

Checked declarations

Module	Named supporting theorems
ComplementaryMinorArcBound.lean	2

- `complementaryMinorBudget_nonneg`
- `annularMangoldt_complementary_minor_bound`

<!-- checkpoint-verification -->

Verification

All 180 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3702-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1019 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 182 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_114_LogPowerAbsorption.md`

Checkpoint 114: Absorbing fixed logarithmic and power budgets

Lean proves that every fixed real power of log R is eventually bounded by R^epsilon for any fixed epsilon>0. The proof consumes a proved Mathlib little-o limit and retains explicit positivity before simplifying norms. It does not use a number-theoretic asymptotic assumption.

Two constant-absorption statements are also proved. If K>=0 and a<b, then K*R^a<=R^b eventually; and if two nonnegative weighted powers both have exponent below b, their sum is eventually at most R^b. The finite constants are absorbed into thresholds, never by changing exponents with R.

These supporting inequalities will absorb the genuine logarithmic factor in the complementary minor budget. They do not by themselves assert an estimate for a prime exponential sum.

Checked declarations

Module	Named supporting theorems
LogPowerAbsorption.lean	3

- `eventually_log_rpow_le_rpow`
- `eventually_scaled_rpow_le_rpow`
- `eventually_two_rpow_budget`

<!-- checkpoint-verification -->

Verification

All 181 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3703-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1022 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 183 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_115_SourceLogPowerBudget.md`

Checkpoint 115: Uniform logarithmic absorption at the actual source scale

For the actual source scale R=max(exp((log N)^(1/10)),Q^8), Lean proves log R>=(log N)^(1/10). Hence, for N>=1 and every d>=0, (log N)^d<=(log R)^(10*d). All real-power compositions and the nonnegative logarithmic bases are justified.

Combining this with the preceding logarithmic absorption and the source scale's uniform lower growth gives one natural threshold, independent of Q, for

```
(log(N/2))^d <= R^epsilon,
```

for every fixed d>=0 and epsilon>0. The statement quantifies over all real Q; no Q-dependent threshold is hidden. This handles Ren's fixed logarithmic loss at the exact annular scale.

The proof is unconditional and uses no PNT or prime exponential-sum hypothesis. It prepares the uniform error budget needed after the complementary geometric and analytic branches have been assembled.

Checked declarations

Module	Named supporting theorems
SourceLogPowerBudget.lean	3

- `sourceMajorArcScale_log_lower`
- `source_log_power_domination`
- `source_annular_log_power_uniform`

<!-- checkpoint-verification -->

Verification

All 182 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3704-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1025 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 184 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_116_UniformComplementaryBudget.md`

Checkpoint 116: A uniform source-scale complementary minor budget

The complementary minor budget is proved uniformly small at the actual source scale. The logarithmic factor in Ren's term is bounded by R^(1/12), turning that contribution into renSourceConstant(C)*R^(-1/3). Ghosh's contribution is ghoshSourceConstant(B)*R^(-1/4). Their sum is eventually at most R^(-1/6), uniformly over Q, for fixed C,B,d>=0.

Substituting the already proved source-scale logarithmic rate gives the desired Q^(-1/2)*(1+log N)^(-A) bound with factor 2^A, for every A>=0 and Q>=1. The threshold depends on the fixed constants and A but not on Q, the selected rational or the frequency.

This checkpoint controls the explicit scalar budget. The actual minor-arc multiplier theorem still requires its displayed raw Ren and Ghosh inputs and geometric outside-arc condition; those are not hidden inside the scalar estimate. The stronger printed minor-arc formula remains open.

Checked declarations

Module	Named supporting theorems
UniformComplementaryBudget.lean	2

- `complementaryMinorBudget_uniform`
- `complementaryMinorBudget_uniform_log_rate`

<!-- checkpoint-verification -->

Verification

All 183 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3705-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1027 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 185 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint notes 117-191

Checkpoint_117_UniformCircleMinorError.md

Checkpoint 117: A uniform error bound on the full circle minor-arc complement

The alternative minor-arc estimate is transferred to the genuine periodized circle error. On the full all-lift complement of the source major arcs, the actual multiplier is bounded by complementaryMinorBudget, and the actual circle model by circleModelTailBudget. Their difference has norm at most the sum; no unperiodized model is substituted.

A separate uniform logarithmic bound is proved for circleModelTailBudget. Combining it with the complementary budget, the uniform source fourth-power condition and the displayed raw Ren/Ghosh inputs gives, for fixed nonnegative C,B,d and every A>=0, the bound

```
norm(circle error) <= 2\*2^A\*Q^(-1/2)\*(1+log N)^(-A).
```

One threshold in N works for all permitted Q, beta, character/coefficient data and frequencies on this geometric branch. The Q range is the same small-constant exp(sqrt(log N)) range as in the major-arc work. The coefficient and beta bounds, representative interval [-1,2], outside-arc condition and raw inputs are explicit. The stronger printed minor formula remains open; this alternative estimate is sufficient for the next global circle approximation assembly.

Checked declarations

Module	Named supporting theorems
UniformCircleMinorError.lean	3

- `circleModelTailBudget_uniform_log_rate`
- `circleMajorArcError_minor_from_inputs`
- `circleMajorArcError_minor_uniform`

<!-- checkpoint-verification -->

Verification

All 184 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3706-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1030 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 186 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_118_WrappedCircleMajorUniform.md

Checkpoint 118: The uniform error bound on all lifted major arcs

The actual continuous circle error is exactly unchanged when an integer is added to a real representative. Lean proves this from the quotient identity rather than asserting periodicity of the underlying real oscillatory integral.

The conditional major-arc uniform theorem is then extended to any lifted major arc, with arbitrary integer k and condition |xi+k-r0|<=R/(den(r0)*N^2). Its conclusion concerns the original circle point xi. The

same uniform N threshold, all-Q range and raw progression-prefix hypothesis are retained, and shifted copies of the denominator-one center are included.

This is the boundary-compatible major branch used by the global major/minor case split. No outside-arc or minor conclusion is assumed, and no new deep input is introduced.

Checked declarations

Module	Named supporting theorems
WrappedCircleMajorUniform.lean	2

- `circleMajorArcError_integer_shift`
- `circleMajorArcError_major_lift_uniform`

<!-- checkpoint-verification -->

Verification

All 185 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3707-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1032 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 187 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_119_GlobalCircleApproximation.md`

Checkpoint 119: A conditional global pointwise approximation on the circle

The genuine circle error is now bounded at every representative xi in [0,1], conditional on the displayed progression-prefix, Ren and Ghosh inputs. Lean makes a complete case split over the exact finite source rational set and all integer lifts. If a lifted major arc contains xi, the wrapped conditional major theorem applies. Otherwise the logical negation supplies the outside-arc condition needed for the alternative minor theorem.

For every fixed P,C,B,d>=0, c>0 and A>=0, one natural threshold works uniformly in all permitted Q and character/coefficient/beta data, giving

```
norm(actual circle error) <= 2\*2^A\*Q^(-1/2)\*(1+log N)^(-A).
```

The progression estimate must hold for every source denominator and every real endpoint in [N/2,N] with the same scale data. The Ren/Ghosh inequalities are explicit pointwise hypotheses at the compatible approximants, not hypotheses asserting the desired error conclusion. There is no gap between the two cases and no assumption of the paper's own minor-arc or approximation lemma.

This completes the conditional global pointwise analytic deduction for the genuine circle model. The next steps extend the bound to every circle point and to the actual bounded l2 operator. The stronger printed Lemma 6.5, full exceptional-data/source correspondence and sharp low/high/maximal inputs remain open; Theorem 1.1 is not proved.

Checked declarations

Module	Named supporting theorems
GlobalCircleApproximation.lean	1

- `globalCircleApproximation_uniform`

<!-- checkpoint-verification -->

Verification

All 186 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3708-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1033 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 188 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_120_CircleFundamentalBounds.md

Checkpoint 120: From a fundamental representative interval to all circle points

Every point of PrimeCircle is represented by a real xi in the half-open interval [0,1), using Mathlib's proved quotient equivalence. Consequently a bound for a continuous circle function on the closed representative interval [0,1] is a bound at every circle point. Endpoints are included consistently.

The same fundamental-interval bound controls the norm of the actual integer l2 multiplier operator through the previously proved negative-phase Fourier isometry. The function is an actual continuous circle symbol and the operator acts on all l2 inputs; no finite-input or formal-symbol substitute is used.

These quotient and operator-transfer facts introduce no analytic hypothesis or new mathematical axiom. They allow the newly proved conditional global error bound to yield an actual all-input operator estimate.

Checked declarations

Module	Named supporting theorems
CircleFundamentalBounds.lean	3

- `primeCircle_representative`
- `circleBound_from_fundamental_interval`
- `integerMultiplier_bound_fundamental_interval`

<!-- checkpoint-verification -->

Verification

All 187 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3709-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1036 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 189 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_121_CircleApproximationData.md

Checkpoint 121: A conditional global approximation for the actual l2 operator

The explicit deep analytic inputs are collected in CircleAnalyticInputs: 0<beta<=1, coefficient norms at most one, the actual raw progression-prefix inequality for every source rational denominator and every real endpoint in [N/2,N], and the compatible raw Ren/Ghosh inequalities for every representative xi in [0,1]. The structure contains no approximation, operator-norm or own-paper lemma conclusion.

From these inputs, Lean proves a bound for the actual circle error at every circle point. It then proves the norm bound for the genuine l2 error operator, with one N threshold before all permitted Q and input data:

```
norm(error operator) <= 2\*2^A\*Q^(-1/2)\*(1+log N)^(-A).
```

This is an end-to-end conditional analytic deduction for the periodized approximation: exact finite rational indexing, compatible rational selection, both major and alternative minor branches, all integer lifts, the circle representative bridge and the actual Fourier operator are included. The deep number-theory inputs remain hypotheses, as authorized, rather than added axioms or placeholders.

The stronger printed minor estimate of Lemma 6.5 has not been established, and the full source exceptional-parameter and double-sum correspondence is not silently certified by the input structure. Sharp low/high/multifrequency/maximal estimates and Theorem 1.1 remain open. This checkpoint performs a fresh full project rebuild and complete axiom/source audits before packaging.

Checked declarations

Module	Named supporting theorems
CircleApproximationData.lean	2

- `globalCircleApproximation_all_points`
- `globalCircleApproximation_l2_operator`

<!-- checkpoint-verification -->

Verification

All 188 project modules and the root were freshly rebuilt: 3710 jobs pass without warnings/errors. All 1038 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 190 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_122_SourceRangeMonotonicity.md

Checkpoint 122: Monotonicity of the allowed Q range and source cutoff

The allowed Q range is proved stable as the arithmetic scale increases. The coefficient min(c/7,1)/8 is positive for c>0, and monotonicity of log and sqrt shows that a Q allowed at N is allowed at every M>=N>=1.

The source cutoff itself is also monotone in N on N>=1. This follows from monotonicity of log, the fixed positive real power 1/10, exp and max. These facts allow one Q and one starting-scale condition to be used along an entire dyadic tail, rather than imposing unrelated conditions at each scale.

The proofs are unconditional scalar geometry. They introduce no prime-distribution or operator-norm assumptions and do not claim the remaining sharp high/low estimates.

Checked declarations

Module	Named supporting theorems
SourceRangeMonotonicity.lean	3

- `sourceQCoefficient_pos`
- `sourceQRange_upward`
- `sourceMajorArcScale_mono_N`

<!-- checkpoint-verification -->

Verification

All 189 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3711-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1041 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 191 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_123_DyadicLogarithmicBudget.md`

Checkpoint 123: Exact dyadic logarithms and inverse-power comparison

Lean records the exact arithmetic and logarithmic identities for dyadic scales 2^j. In particular log(2^j)=j*log 2 and monotonicity in the exponent is proved.

Since 0<log 2<=1, the bound log 2*(j+1)<=1+log(2^j) holds even at j=0. For every A>=0 this gives

```
(1+log(2^j))^(-A) <= (log 2)^(-A)\*(j+1)^(-A).
```

The explicit constant will convert the conditional global operator-error bound to a summable sequence in the dyadic exponent. No asymptotic substitution of j for log N is left informal.

Checked declarations

Module	Named supporting theorems
DyadicLogarithmicBudget.lean	4

- `two_pow_monotone`
- `log_two_pow_nat`
- `dyadic_log_linear_lower`
- `dyadic_log_inverse_power`

<!-- checkpoint-verification -->

Verification

All 190 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3712-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1045 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 192 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_124_DyadicCircleErrorBudget.md

Checkpoint 124: Quantitative decay of the actual dyadic error operators

An actual dyadic family of l2 error operators is defined from the genuine periodized circle model, with character, coefficient and beta data allowed to vary with the dyadic exponent. One positive constant dyadicErrorConstant(A)=2*2^A*(log 2)^(-A) is selected explicitly.

The global conditional l2 approximation, the upward stability of the allowed Q range and the dyadic logarithmic comparison prove an operator norm bound

```
norm(E_(2^j)) <= dyadicErrorConstant(A)\*Q^(-1/2)\*(j+1)^(-A)
```

for every j beyond a chosen starting exponent, under the explicit analytic inputs at those scales. A single arithmetic threshold is chosen before the starting exponent, Q and all scale-dependent data. The Q restriction is required only at the starting scale and is then proved at later scales.

These are genuine bounded operators acting on every l2 input. The external progression, Ren and Ghosh inputs remain explicit via CircleAnalyticInputs; no maximal-error or high/low conclusion is assumed.

Checked declarations

Module	Named supporting theorems
DyadicCircleErrorBudget.lean	2

- `dyadicErrorConstant_pos`
- `dyadicCircleError_operator_decay`

<!-- checkpoint-verification -->

Verification

All 191 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3713-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1047 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 193 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_125_ReciprocalSquareFiniteTail.md

Checkpoint 125: A finite reciprocal-square tail with exact telescoping

The quantitative series estimate needed for the dyadic error tail begins with a checked telescoping inequality:

```
1/(j+1)^2 <= 2\*(1/(j+1)-1/(j+2)).
```

Lean proves the exact sum of the reciprocal differences on every integer interval [J,K), including the empty interval. It follows that the sum of 1/(j+1)^2 on that interval is at most 2/(J+1). This holds for all natural endpoints, with no hidden J>0 condition.

The estimates are elementary finite real inequalities. They will provide an explicit infinite-tail bound rather than relying only on qualitative convergence of a p-series.

Checked declarations

Module	Named supporting theorems
ReciprocalSquareFiniteTail.lean	3

- `reciprocal_square_step`
- `reciprocal_step_sum`
- `reciprocal_square_finite_tail`

<!-- checkpoint-verification -->

Verification

All 192 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3717-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1050 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 194 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_126_ReciprocalSquareInfiniteTail.md`

Checkpoint 126: A quantitative infinite reciprocal-square tail

Finite sums indexed by a shifted range are identified exactly with sums on [J,J+K). The reciprocal-square finite-tail bound then bounds every partial sum of the shifted nonnegative series.

Lean proves both summability of the series sum_j 1/(J+j+1)^2 and the explicit bound 2/(J+1). The proof works for every natural J, including J=0. It uses the proved nonnegative-series criterion and the finite telescoping estimate; no numerical approximation or assumed zeta value is involved.

This prepares a quantitative infinite envelope for the actual dyadic operator errors.

Checked declarations

Module	Named supporting theorems
ReciprocalSquareInfiniteTail.lean	3

- `sum_range_shift_eq_Ico`
- `reciprocal_square_shift_summable`
- `reciprocal_square_infinite_tail`

<!-- checkpoint-verification -->

Verification

All 193 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3718-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1053 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 195 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_127_WeightedPowerFiniteTail.md

Checkpoint 127: Weighted real-power finite tails

For every fixed real s>=2 and j>=J, Lean splits (j+1)^(-s) into a nonincreasing weight and a reciprocal square. The weight is bounded by (J+1)^(2-s), giving

```
sum_(J<=j<K) (j+1)^(-s) <= 2\*(J+1)^(1-s).
```

All real-power identities and positivity conditions are proved. The bound holds for arbitrary natural endpoints, including empty intervals and the starting exponent zero. It retains the exact tail exponent needed for a maximal-error estimate with arbitrary logarithmic saving.

The statement is an unconditional scalar series bound, not an assumed operator inequality.

Checked declarations

Module	Named supporting theorems
WeightedPowerFiniteTail.lean	2

- `weighted_power_reciprocal_bound`
- `weighted_power_finite_tail`

<!-- checkpoint-verification -->

Verification

All 194 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3719-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1055 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 196 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_128_WeightedPowerInfiniteTail.md

Checkpoint 128: Summability and quantitative weighted infinite tails

The weighted finite-tail estimate is passed to the infinite nonnegative series. For every real s>=2 and every natural starting exponent J, Lean proves summability and

```
sum_j (J+j+1)^(-s) <= 2\*(J+1)^(1-s).
```

There is no dependence on an unbounded finite truncation and no interchange of conditionally convergent sums. The proof uses bounded nonnegative partial sums, so both existence and the quantitative tail are checked.

This is the scalar summability input for the countable l2 envelope constructed next.

Checked declarations

Module	Named supporting theorems
WeightedPowerInfiniteTail.lean	2

- `weighted_power_shift_summable`
- `weighted_power_infinite_tail`

<!-- checkpoint-verification -->

Verification

All 195 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3720-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1057 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 197 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_129_OperatorDecaySeries.md

Checkpoint 129: Absolute summability and quantitative tails of operator norms

An arbitrary family of actual bounded l2 operators with a fixed power decay norm(T_j)<=K*(j+1)^(-s), s>=2, has a summable norm tail. Lean proves both norm summability and convergence of the operator series in the Banach space of continuous linear maps.

The exact quantitative tail is

```
sum_j norm(T_(J+j)) <= 2\*K\*(J+1)^(1-s).
```

The estimate follows from the checked weighted scalar series and applies to every natural starting exponent. This is the summability mechanism for the actual dyadic error family, not an assumed maximal theorem. The actual decay bound from the conditional global approximation will be supplied at the later application.

Checked declarations

Module	Named supporting theorems
OperatorDecaySeries.lean	3

- `operator_norm_shift_summable`
- `operator_norm_shift_tsum_bound`
- `operator_shift_summable`

<!-- checkpoint-verification -->

Verification

All 196 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3721-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1060 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 198 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_130_CountableOperatorEnvelope.md`

Checkpoint 130: A genuine countable positive l2 operator envelope

A countable positive l2 envelope is constructed as the Banach-space sum of the existing magnitudeL2 realizations of T_j f. Under summability of the operator norms, Lean proves that both the vector magnitudes and their norms are summable for every complex l2 input f.

The resulting genuine real l2 vector has norm at most (sum_j norm(T_j))*norm(f). Completeness and absolute summability justify the vector sum; it is not treated as a formal series or a finite maximum. The magnitude construction already present in the project is reused.

The next checkpoints identify every coordinate of this vector sum and show that it dominates the true countable maximal function. No analytic number-theory or own-paper maximal hypothesis enters this functional-analytic construction.

Checked declarations

Module	Named supporting theorems
CountableOperatorEnvelope.lean	3

- `operator_magnitude_summable`
- `operator_magnitude_norm_summable`
- `countableOperatorEnvelope_norm`

<!-- checkpoint-verification -->

Verification

All 197 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3722-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1063 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 199 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_131_CountableEnvelopeCoordinates.md`

Checkpoint 131: Exact coordinate identities for the countable l2 envelope

Evaluation at each integer coordinate is a continuous linear map on real l2. Lean uses that map to commute evaluation with the proved convergent vector series, identifying the countable envelope exactly with sum_j norm(T_j f(x)) at every integer x.

The coordinate series is proved summable, and its absolute value is bounded by (sum_j norm(T_j))*norm(f). There is no almost-everywhere exception or unjustified interchange of a non-summable series with evaluation.

These are the coordinate identities needed to prove domination of each operator output and of the true countable supremum. They use only the displayed operator-norm summability assumption, which is subsequently supplied by the quantitative decay of the actual error operators.

Checked declarations

Module	Named supporting theorems
CountableEnvelopeCoordinates.lean	3

- `countableOperatorEnvelope_coordinate`
- `operator_coordinate_norm_summable`
- `countableOperatorEnvelope_coordinate_bound`

<!-- checkpoint-verification -->

Verification

All 198 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3723-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1066 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 200 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_132_CountableEnvelopeDomination.md`

Checkpoint 132: Pointwise domination by the countable envelope

The countable envelope is proved nonnegative at every integer and to dominate norm(T_j f(x)) for every index j. Every existing finite operator envelope is also bounded pointwise by the countable envelope.

The proofs use the exact coordinate series and its established summability. In particular, the finite-to-countable passage is justified for all l2 inputs rather than only for finitely supported functions. The bound contains all indices and has no dependence on the size of a finite truncation.

This is a functional-analytic consequence of norm summability, not a new deep analytic input.

Checked declarations

Module	Named supporting theorems
CountableEnvelopeDomination.lean	3

- `countableOperatorEnvelope_nonneg`
- `countableOperatorEnvelope_dominates`
- `finiteOperatorEnvelope_le_countable`

<!-- checkpoint-verification -->

Verification

All 199 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3724-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1069 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 201 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_133_CountableOperatorMaximum.md`

Checkpoint 133: A bounded true countable maximal function

The actual pointwise countable supremum sup_j norm(T_j f(x)) is defined as operatorMaximalValue. Lean proves the set of values is bounded above by the countable envelope before using conditional supremum inequalities over the real numbers.

The supremum dominates every operator output, is nonnegative, and is at most the countable envelope. This rules out interpreting an unbounded real supremum through a default total definition. All statements require and use summability of the operator norms to supply a genuine bound.

The next step proves that this actual maximal function, rather than just its majorant, belongs to l2.

Checked declarations

Module	Named supporting theorems
CountableOperatorMaximum.lean	4

- `operatorNormValues_bddAbove`
- `operatorMaximalValue_dominates`
- `operatorMaximalValue_nonneg`
- `operatorMaximalValue_le_envelope`

<!-- checkpoint-verification -->

Verification

All 200 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3725-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1073 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 202 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_134_OperatorMaximalMembership.md`

Checkpoint 134: A concrete l2 realization of the countable maximal function

The actual countable maximal value is proved to belong to real l2 by pointwise domination by the constructed l2 envelope. A concrete l2 realization operatorMaximalL2 is defined, and its value at every integer is exactly sup_j norm(T_j f(x)).

The proof applies to every complex l2 input. The function is not a finite maximum, an unspecified dominating function or an almost-everywhere representative of an unproved supremum. Boundedness of the value set and norm summability have already been established.

Checked declarations

Module	Named supporting theorems
OperatorMaximalMembership.lean	2

- `operatorMaximalValue_memLp`
- `operatorMaximalL2_apply`

<!-- checkpoint-verification -->

Verification

All 201 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3726-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1075 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 203 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_135_OperatorMaximalNorm.md`

Checkpoint 135: An all-input countable maximal l2 bound

The l2 norm of the genuine countable maximal function is bounded by the norm of the countable positive envelope. Consequently,

```
norm(sup_j |T_j f|)_2 <= (sum_j norm(T_j))\*norm(f)_2.
```

Lean proves the pointwise nonnegativity and norm comparisons required for the l2 monotonicity argument. This is a complete countable all-input estimate under operator-norm summability, not merely an inequality for finite scale collections.

Applied to the proved dyadic error decay, it will provide the maximal error term without assuming a new multifrequency theorem. It does not bound the separate high or low operators.

Checked declarations

Module	Named supporting theorems
OperatorMaximalNorm.lean	2

- `operatorMaximalL2_norm_le_envelope`
- `operatorMaximalL2_norm_bound`

<!-- checkpoint-verification -->

Verification

All 202 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3727-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1077 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 204 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_136_OperatorMaximalSublinearity.md`

Checkpoint 136: Sublinearity of the actual countable maximal function

The true countable maximal function is proved subadditive on l2 inputs and invariant under replacing f by -f. A scalar bound operatorMaximalValue(z*f)<=|z|*operatorMaximalValue(f) is also proved for every complex scalar z.

The proofs use the linearity of the actual operators, the triangle inequality and the previously established boundedness of the real supremum. They do not assume that taking a countable supremum preserves linearity. These properties will give the final Lipschitz estimate for the maximal-error map.

Checked declarations

Module	Named supporting theorems
OperatorMaximalSublinearity.lean	3

- `operatorMaximalValue_add`
- `operatorMaximalValue_neg`
- `operatorMaximalValue_smul_le`

<!-- checkpoint-verification -->

Verification

All 203 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3728-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1080 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 205 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_137_DecayingMaximalTail.md`

Checkpoint 137: A quantitative countable maximal tail from power decay

The general countable maximal theorem is combined with the quantitative power-decay tail. If norm(T_(J+j))<=K*(J+j+1)^(-s) with K>=0 and s>=2, Lean constructs the actual maximal l2 function and proves

```
norm(sup_j |T_(J+j) f|)_2 <= 2\*K\*(J+1)^(1-s)\*norm(f)_2.
```

A positive l2 majorant with the same norm bound is also constructed and proved to dominate every output on the tail. Both statements are valid for all complex l2 inputs and all natural starting exponents. No finite truncation or extra maximal assumption is present.

Checked declarations

Module	Named supporting theorems
DecayingMaximalTail.lean	2

- `decaying_operator_maximal_tail`
- `decaying_operator_maximal_tail_majorant`

<!-- checkpoint-verification -->

Verification

All 204 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3729-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1082 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 206 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_138_MaximalTailMonotonicity.md`

Checkpoint 138: Monotonicity of the true countable maximal tails

Every tail of a norm-summable operator family is norm summable. Lean proves that moving the starting exponent from J to K>=J decreases the actual pointwise countable maximal function and its l2 norm.

The proof reindexes K+j exactly as J+(K-J+j); it does not discard indices informally or identify different scale-dependent data. The nonnegative real l2 realizations are compared using the pointwise inequalities already proved for the true supremum.

This supports quantitative approximation and continuity statements for dyadic error tails.

Checked declarations

Module	Named supporting theorems
MaximalTailMonotonicity.lean	3

- `operator_tail_norm_summable`
- `operatorMaximalValue_tail_mono`
- `operatorMaximalL2_tail_mono`

<!-- checkpoint-verification -->

Verification

All 205 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3730-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1085 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 207 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_139_DyadicErrorNormSeries.md

Checkpoint 139: Absolute norm summability of the actual dyadic error family

The actual dyadic circle-error family is proved absolutely summable in operator norm on every allowed tail. For a fixed exponent s>=2, the previously established decay and upward Q-range condition give

```
sum_j norm(E_(2^(J+j))) <= 2\*dyadicErrorConstant(s)\*Q^(-1/2)\*(J+1)^(1-s).
```

Both summability and the explicit bound are proved. The arithmetic threshold is chosen before J, Q, the input and all scale-dependent character, coefficient and beta data. CircleAnalyticInputs is required at the actual scales in the tail, with uniform displayed deep-input constants.

This supplies a genuine summable family for the countable maximal construction. No maximal-error inequality is assumed in the input.

Checked declarations

Module	Named supporting theorems
DyadicErrorNormSeries.lean	1

- `dyadicCircleError_norm_series`

<!-- checkpoint-verification -->

Verification

All 206 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3731-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1086 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 208 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_140_DyadicMaximalErrorPower.md

Checkpoint 140: The genuine countable maximal error with a power tail

The genuine countable maximal function of the actual dyadic circle-error operators is now realized as a real l2 vector for every complex l2 input. Lean proves

```
norm(sup_j |E_(2^(J+j)) f|)_2
  <= 2\*dyadicErrorConstant(s)\*Q^(-1/2)\*(J+1)^(1-s)\*norm(f)_2,
```

for s>=2, under the explicit scale-wise analytic inputs and the Q condition at the starting scale. There is one uniform arithmetic threshold before all J, Q and data. The supremum ranges over the entire infinite tail, and its coordinate equality is exact at every integer.

This is the maximal error component of the approximation argument. It does not provide the separate high-operator multifrequency bound or the low-kernel bound.

Checked declarations

Module	Named supporting theorems
DyadicMaximalErrorPower.lean	1

- `dyadicCircleError_maximal_power`

<!-- checkpoint-verification -->

Verification

All 207 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3732-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1087 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 209 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_141_DyadicMaximalErrorLog.md

Checkpoint 141: The conditional logarithmic dyadic maximal error estimate

The quantitative maximal error estimate is expressed in the manuscript's arithmetic-scale logarithm. Lean proves (J+1)^(1-(A+2))<=(1+log(2^J))^(-A) for every A>=0, including J=0.

Choosing s=A+2 in the full countable maximal bound gives a fixed positive constant C_A=2*dyadicErrorConstant(A+2) and the actual all-input estimate

```
norm(sup_j |E_(2^(J+j)) f|)_2
  <= C_A\*Q^(-1/2)\*(1+log(2^J))^(-A)\*norm(f)_2.
```

The Q range is imposed only at the starting scale and propagated rigorously. Character, coefficient and beta data may vary between scales, but the raw progression-prefix estimate uses one set of data throughout each scale's real endpoint interval. All external estimates remain explicit hypotheses.

This completes the conditional global approximation error and its dyadic maximal tail for the genuine circle model. The stronger printed Lemma 6.5, complete exceptional/source correspondence and the separate sharp low/high/multifrequency estimates remain open. Theorem 1.1 is not proved.

Checked declarations

Module	Named supporting theorems
DyadicMaximalErrorLog.lean	3

- `dyadic_tail_power_log_bound`
- `maximalCircleErrorConstant_pos`
- `dyadicCircleError_maximal_log_rate`

<!-- checkpoint-verification -->

Verification

All 208 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3733-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1090 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 210 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_142_SummableOperatorLimits.md`

Checkpoint 142: Vanishing outputs of norm-summable operator families

A norm-summable family of bounded l2 operators produces an absolutely summable output series for every complex l2 input. Lean proves that the individual outputs tend to zero in l2 and at every integer coordinate.

The pointwise statement uses the continuity of coordinate evaluation on l2, so it does not rely on selecting an almost-everywhere subsequence. Applied to the norm-summable actual dyadic error tail, it shows that model approximation errors vanish along that full tail under the recorded analytic inputs.

These are consequences of the already checked summability mechanism; they are not claims of the paper's ergodic convergence theorem.

Checked declarations

Module	Named supporting theorems
SummableOperatorLimits.lean	3

- `operator_outputs_summable`
- `operator_outputs_tendsto_zero_l2`
- `operator_outputs_tendsto_zero_pointwise`

<!-- checkpoint-verification -->

Verification

All 209 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3734-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1093 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 211 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_143_MaximalTailVanishing.md`

Checkpoint 143: Vanishing of the genuine countable maximal tails

The l2 norm of the true countable maximal tail tends to zero as the starting exponent tends to infinity for every norm-summable operator family and every l2 input. The proof bounds it by the tail sum of the operator norms times norm(f), and uses the proved vanishing of absolutely convergent series tails.

Lean also proves convergence of the maximal-tail vectors themselves to zero in real l2. The positive envelope and true supremum are not conflated, and no finite-scale limit is left unproved.

This gives a rigorous continuation property for the actual error family once its norm summability is supplied. It does not assert convergence of the remaining main model or prove an ergodic endpoint theorem.

Checked declarations

Module	Named supporting theorems
MaximalTailVanishing.lean	2

- `operatorMaximalL2_tail_norm_tendsto_zero`
- `operatorMaximalL2_tail_tendsto_zero`

<!-- checkpoint-verification -->

Verification

All 210 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3735-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1095 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 212 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_144_DyadicModelDifference.md`

Checkpoint 144: Identifying the actual average-minus-model maximal error

The actual dyadic circle-model operator is defined with the same exact rational set and scale-dependent character/coefficient/beta data as the error family. Lean proves the operator identity

```
E_(2^j) = actual annular von Mangoldt average - circle model operator.
```

The identity is checked on all l2 inputs and every integer coordinate. The true countable maximal value of the error family is therefore exactly the supremum of the differences between those two actual operators on the dyadic tail.

This identifies the quantitative maximal bound with an approximation of the actual prime-power weighted square averages, rather than a bound for an abstractly named error family. It uses the existing negative-phase Fourier operator decomposition; no analytic conclusion is assumed.

Checked declarations

Module	Named supporting theorems
DyadicModelDifference.lean	3

- `dyadicCircleError_difference`
- `dyadicCircleError_difference_apply`
- `dyadicCircleError_maximal_difference`

<!-- checkpoint-verification -->

Verification

All 211 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3736-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 1098 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 213 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_145_QuantitativeErrorPairing.md`

Checkpoint 145: Finite-indicator pairing bounds for the conditional error operator

An actual l2 operator norm bound gives an explicit finite-indicator pairing bound b*sqrt(|F|)*sqrt(|G|), without kernel positivity. Lean proves this using the existing finite-indicator norm and discrete Cauchy--Schwarz estimate. Its normalized form is b*(|F||G|/L^2)^(1/2) after division by a positive spatial length L.

The conditional global circle-error norm theorem is then applied to obtain this finite-set error bound uniformly in the permitted Q and actual analytic inputs. The result is the quantitative error pairing needed in later restricted improving arguments.

This controls the error contribution only. The low and high contributions needed for Theorem 1.1 have not been supplied, so no full improving theorem is claimed.

Checked declarations

Module	Named supporting theorems
QuantitativeErrorPairing.lean	3

- `operator_finite_indicator_pairing_bound`
- `operator_finite_indicator_pairing_normalized`
- `globalCircleError_restricted_bound`

<!-- checkpoint-verification -->

Verification

All 212 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3737-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1101 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 214 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_146_MaximalErrorLipschitz.md`

Checkpoint 146: Lipschitz continuity and final verification of the maximal error milestone

The genuine countable maximal map is proved Lipschitz on all complex l2 inputs when the operator norms are summable. Lean first proves the pointwise inequality

```
|sup_j |T_j f(x)| - sup_j |T_j g(x)|| <= sup_j |T_j(f-g)(x)|.
```

It then proves the l2 difference bound by (sum_j norm(T_j))*norm(f-g) and records the resulting LipschitzWith statement. This supplies a checked continuity property for the actual error maximal map, in addition to its norm estimate and tail convergence.

Checkpoints 97-146 establish an alternative full-complement minor argument using compatible balanced rational selection and explicit Ren/Ghosh estimates, a conditional global all-point and l2 approximation for the genuine periodized circle model, and a conditional quantitative countable dyadic maximal error bound with all-input realization. The raw progression-prefix, Ren and Ghosh inequalities remain unproved deep hypotheses. They are not added as mathematical axioms.

The stronger printed minor formula of Lemma 6.5 remains open; the new complementary argument is weaker but sufficient for the checked approximation deduction. Complete source exceptional-parameter and double-sum correspondence, sharp low 6.14, multifrequency 6.16, high/maximal 6.21, Proposition 4.19 and Theorem 1.1 remain open. No new manuscript error or mandatory LaTeX replacement is asserted. Lemma 3.21's two earlier PNT limits remain explicit; the separate PNT project has not been integrated by this batch.

This final checkpoint performs a fresh full rebuild and complete source/transitive-axiom audits. All fifty checkpoints are saved separately with their own proof expositions and verification histories. The combined report records exact scope, the meaningful mathematical milestones and the next unfinished targets.

Checked declarations

Module	Named supporting theorems
MaximalErrorLipschitz.lean	3

- `operatorMaximalValue_difference_bound`
- `operatorMaximalL2_difference_bound`
- `operatorMaximalL2_lipschitz`

<!-- checkpoint-verification -->

Verification

All 213 project modules and the root were freshly rebuilt: 3738 jobs pass without warnings/errors. All 1104 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 215 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_147_TruncatedCohenBound.md`

Checkpoint 147: The truncated ordinary low arithmetic sum

The ordinary low-kernel arithmetic estimate is connected to the exact truncated denominator sum. Lean proves that the smooth absolute Cohen series is summable using its finite divisor support, dominates every finite sum whose prime factors lie below the cutoff, and gives an explicit uniform bound for the sum over 1 <= q <= Q by (cohenLogConstant/2)*log(Q+1). This is unconditional, uses the genuine correlation, and does not assume Mertens or the low-kernel conclusion. It supplies the arithmetic input for Proposition 6.14; the spatial kernel and source Fourier bridges remain to be proved.

Checked declarations

Module	Named supporting theorems
TruncatedCohenBound.lean	3

- `smoothCohen_summable`
- `finite_cohen_sum_le_smooth`
- `truncated_cohen_log_bound`

<!-- checkpoint-verification -->

Verification

All 214 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3739-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1107 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 216 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_148_DecayProfile.md`

Checkpoint 148: Polynomial spatial decay at the square scale

Introduces the exact polynomial decay profile (1+|x|/H)^(-K) using natural powers and proves positivity, the uniform bound by one, evenness and monotonicity with distance. These properties retain the spatial scale H explicitly and are used in the genuine low-kernel convolution estimates, where H=N^2. No Fourier decay or low estimate is assumed.

Checked declarations

Module	Named supporting theorems
DecayProfile.lean	4

- `decayProfile_pos`
- `decayProfile_le_one`
- `decayProfile_even`
- `decayProfile_mono_abs`

<!-- checkpoint-verification -->

Verification

All 215 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3740-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1111 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 217 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypothe-

ses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_149_ShiftedDecayProfile.md

Checkpoint 149: Decay after a bounded spatial translation

Proves the translated spatial decay inequality when the averaging measure is supported in |y| <= C H. The precise bound is profile_H,K(x-y) <= (1+C)^K profile_H,K(x), with all positivity and scale conditions present. This supplies the support comparison needed to convolve a decaying cutoff kernel with the annular square measure.

Checked declarations

Module	Named supporting theorems
ShiftedDecayProfile.lean	2

- `decayProfile_shift_base`
- `decayProfile_shift_bound`

<!-- checkpoint-verification -->

Verification

All 216 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3741-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1113 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 218 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_150_SchwartzDecayBound.md

Checkpoint 150: Quantitative polynomial decay for Schwartz kernels

Proves an explicit polynomial-decay bound for every complex Schwartz function on the real line. The constant is 2^K times the sum of its zero and K-th weighted seminorms. The proof splits at |x|=1 and retains all constants; no Fourier or kernel decay is assumed.

Checked declarations

Module	Named supporting theorems
SchwartzDecayBound.lean	2

- `schwartzDecayConstant_nonneg`
- `schwartz_polynomial_decay`

<!-- checkpoint-verification -->

Verification

All 217 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3804-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1115 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 219 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_151_SmoothCutoffSchwartz.md`

Checkpoint 151: The actual smooth cutoff as a Schwartz function

Proves compact support and infinite differentiability of the actual complex source cutoff, then constructs its SchwartzMap. This uses the genuine bump from SourceArcCutoff, without replacing the cutoff by an assumed kernel.

Checked declarations

Module	Named supporting theorems
SmoothCutoffSchwartz.lean	4

- `sourceArcCutoff_compact`
- `complexSourceCutoff_compact`
- `complexSourceCutoff_smooth`
- `sourceCutoffSchwartz_apply`

<!-- checkpoint-verification -->

Verification

All 218 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3805-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1119 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 220 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_152_CutoffInverseSchwartz.md`

Checkpoint 152: Rapid decay of the actual inverse cutoff transform

Constructs the inverse Fourier transform of the source cutoff in Schwartz space, proves that its Fourier transform is the original cutoff, and obtains a simultaneous polynomial-decay bound for every natural decay order.

Checked declarations

Module	Named supporting theorems
CutoffInverseSchwartz.lean	3

- `cutoffDecayConstant_nonneg`
- `sourceCutoffInverse_fourier`
- `sourceCutoffInverse_decay`

<!-- checkpoint-verification -->

Verification

All 219 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3826-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1122 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 221 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_153_ScaledCutoffKernel.md`

Checkpoint 153: The continuous scaled inverse-cutoff kernel

Defines the physical dilation a^(-1)*etaInverse(x/a), proves continuity and its exact scale-dependent polynomial envelope. Its identity with the inverse transform of eta(a*frequency) is a separate bridge and is not assumed here.

Checked declarations

Module	Named supporting theorems
ScaledCutoffKernel.lean	3

- `scaledCutoffKernel_continuous`
- `scaledCutoffKernel_decay`
- `scaledCutoffKernel_uniform`

<!-- checkpoint-verification -->

Verification

All 220 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3827-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1125 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 222 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_154_DecayScaleComparison.md`

Checkpoint 154: Embedding narrower cutoff decay into the square scale

Proves monotonicity of the spatial decay profile in the positive width and obtains the correct envelope at the larger square scale when a <= N^2. The normalization 1/a is retained.

Checked declarations

Module	Named supporting theorems
DecayScaleComparison.lean	2

- `decayProfile_mono_scale`
- `scaledCutoffKernel_larger_profile`

<!-- checkpoint-verification -->

Verification

All 221 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3828-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1127 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 223 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_155_SmoothedAnnularKernel.md`

Checkpoint 155: The actual annular convolution kernel

Defines the normalized integral of the scaled cutoff inverse against t^(beta-1) on the real annulus [N/2,N], with shift x-t^2. Proves square support, uniform density control and interval integrability, keeping real endpoints and the exact 1/N normalization.

Checked declarations

Module	Named supporting theorems
SmoothedAnnularKernel.lean	3

- `annular_square_support`
- `annular_density_le_one`
- `smoothedAnnularKernel_integrable`

<!-- checkpoint-verification -->

Verification

All 222 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3829-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1130 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 224 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_156_SmoothedAnnularDecay.md

Checkpoint 156: Uniform spatial decay after annular smoothing

Proves the actual smoothed annular kernel is bounded by 2^K*C_K/(2a) times (1+|x|/N^2)^(-K), for 0<a<=N^2, N>=2 and beta<=1. This is an unconditional physical-space estimate. Its Fourier identification and the arithmetic low-kernel assembly remain separate obligations.

Checked declarations

Module	Named supporting theorems
SmoothedAnnularDecay.lean	1

- `smoothedAnnularKernel_decay`

<!-- checkpoint-verification -->

Verification

All 223 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3830-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1131 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 225 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_157_FiniteDecayConvolution.md

Checkpoint 157: Finite convolution against a decaying kernel

Proves the finite complex convolution envelope with the exact polynomial spatial profile, and deduces its uniform kernel bound. The hypotheses describe an arbitrary kernel; this lemma is infrastructure for applying the proved annular kernel decay, not a claim that the source low kernel already has that bound.

Checked declarations

Module	Named supporting theorems
FiniteDecayConvolution.lean	2

- `finiteConvolution_profile_bound`
- `decay_kernel_uniform`

<!-- checkpoint-verification -->

Verification

All 224 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3831-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1133 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 226 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_158_DecayL1Convolution.md`

Checkpoint 158: All-input l1 consequences of polynomial kernel decay

Realizes a decaying kernel on every complex l1 input and proves its global l1-to-linfinity norm estimate. Also records the finite complex input bound. The exact inverse-kernel/source correspondence is still separate.

Checked declarations

Module	Named supporting theorems
DecayL1Convolution.lean	2

- `decay_kernel_L1_norm`
- `decay_kernel_finite_norm`

<!-- checkpoint-verification -->

Verification

All 225 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3832-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1135 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 227 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_159_FiniteDecayTail.md`

Checkpoint 159: Finite near-far convolution decomposition

Proves the exact finite convolution split into a spatial ball and its complement, and a quantitative far-field estimate retaining the profile at the ball radius.

Checked declarations

Module	Named supporting theorems
FiniteDecayTail.lean	2

- `finiteConvolution_outside_ball`
- `finiteConvolution_split_ball`

<!-- checkpoint-verification -->

Verification

All 226 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3833-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1137 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 228 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_160_DyadicDecayShell.md`

Checkpoint 160: Polynomial decay on dyadic spatial shells

Proves the profile bound on a shell of radius 2^j H and the exact geometric normalization when the decay order is at least one. This supplies the summable shell weights needed for the local low estimate.

Checked declarations

Module	Named supporting theorems
DyadicDecayShell.lean	2

- `decayProfile_dyadic_shell`
- `dyadic_shell_normalization`

<!-- checkpoint-verification -->

Verification

All 227 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3834-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1139 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 229 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_161_GeometricDyadicTail.md`

Checkpoint 161: The exact summable tenth-order shell budget

Proves the ninth-power dyadic geometric series has sum 512/511 and bounds every finite partial sum. This is the exact budget after the spatial-volume loss in the tenth-order low-kernel profile.

Checked declarations

Module	Named supporting theorems
GeometricDyadicTail.lean	2

- `dyadic_ninth_geometric`
- `dyadic_ninth_partial_bound`

<!-- checkpoint-verification -->

Verification

All 228 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3835-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1141 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 230 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_162_DyadicLocalMass.md`

Checkpoint 162: The bounded dyadic local-mass envelope

Defines actual normalized local masses over dyadic balls and their countable real supremum for nonnegative finite inputs. Proves nonnegativity, a uniform total-mass bound and the bounded-above condition required by the real supremum.

Checked declarations

Module	Named supporting theorems
DyadicLocalMass.lean	3

- `dyadicBallMass_nonneg`
- `dyadicBallMass_uniform`
- `dyadicLocalMass_bddAbove`

<!-- checkpoint-verification -->

Verification

All 229 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3836-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1144 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 231 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_163_DyadicLocalMassBounds.md`

Checkpoint 163: Controlling every dyadic ball by the local envelope

Proves the actual countable local envelope is nonnegative, bounded by total mass/H, and controls the mass of every dyadic ball after multiplication by its radius. These facts justify the supremum used in the shell argument.

Checked declarations

Module	Named supporting theorems
DyadicLocalMassBounds.lean	4

- `dyadicBallMass_le_local`
- `dyadicLocalMass_nonneg`
- `dyadicLocalMass_uniform`
- `dyadic_ball_sum_le_local`

<!-- checkpoint-verification -->

Verification

All 230 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3837-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1148 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 232 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_164_DyadicRadiusCover.md`

Checkpoint 164: A genuine dyadic cover of every distance

Uses the Archimedean property and the growth of powers of two to cover every real distance at positive base scale. Constructs its least dyadic level and proves the upper and strict lower minimality properties.

Checked declarations

Module	Named supporting theorems
DyadicRadiusCover.lean	3

- `dyadic_radius_cover`
- `dyadicRadiusLevel_upper`
- `dyadicRadiusLevel_minimal`

<!-- checkpoint-verification -->

Verification

All 231 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3838-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1151 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 233 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_165_DyadicRadiusShells.md`

Checkpoint 165: The core and annular cases of the least dyadic level

Identifies the zero shell exactly with the base ball, and proves both strict inner and weak outer radius bounds for each positive shell. No points are lost at shell boundaries.

Checked declarations

Module	Named supporting theorems
DyadicRadiusShells.lean	2

- `dyadicRadiusLevel_zero_iff`
- `dyadicRadiusLevel_succ_bounds`

<!-- checkpoint-verification -->

Verification

All 232 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3839-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1153 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 234 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_166_FiniteDyadicPartition.md`

Checkpoint 166: An exact finite partition into spatial shells

Defines a finite depth that covers every input point and proves the exact fiberwise partition of a finite weighted sum over its dyadic levels. This is an equality with complete boundary coverage, rather than an informal shell decomposition.

Checked declarations

Module	Named supporting theorems
FiniteDyadicPartition.lean	2

- `finiteDyadicDepth_covers`
- `finite_dyadic_partition`

<!-- checkpoint-verification -->

Verification

All 233 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3840-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1155 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 235 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_167_DyadicShellBudget.md`

Checkpoint 167: Exact volume-normalized shell summation

Proves the exact identity converting tenth-order spatial decay times a dyadic radius into 2*(1/512)^j, and the corresponding uniform finite shell budget 1024/511.

Checked declarations

Module	Named supporting theorems
DyadicShellBudget.lean	2

- `tenth_order_shell_identity`
- `tenth_order_shell_partial`

<!-- checkpoint-verification -->

Verification

All 234 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3841-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1157 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 236 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_168_DyadicShellMass.md`

Checkpoint 168: Local-mass control on each exact spatial shell

Bounds each shell mass and the core mass by the actual dyadic local envelope times the appropriate radius. The proof uses the least-level upper radius and nonnegativity, without assuming an exact-support property for the smoothed kernel.

Checked declarations

Module	Named supporting theorems
DyadicShellMass.lean	2

- `finite_dyadic_shell_mass`
- `finite_dyadic_core_mass`

<!-- checkpoint-verification -->

Verification

All 235 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3842-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1159 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 237 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_169_DyadicShellProfileMass.md`

Checkpoint 169: Summable local contributions from every shell

Combines the strict inner-radius bound, polynomial decay, and local ball-mass estimate. The core contribution is at most H times the envelope, and shell j+1 contributes at most 2H times the envelope times (1/512)^j.

Checked declarations

Module	Named supporting theorems
DyadicShellProfileMass.lean	2

- `finite_dyadic_core_profile`
- `finite_dyadic_shell_profile`

<!-- checkpoint-verification -->

Verification

All 236 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3843-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1161 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 238 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_170_PolynomialLocalMassBound.md`

Checkpoint 170: The local bound for the full polynomial profile

Proves that summing the tenth-order polynomial profile against any nonnegative finite input is at most 4H times its genuine countable dyadic local-mass envelope. The proof uses an exact finite shell partition and a uniform geometric sum.

Checked declarations

Module	Named supporting theorems
PolynomialLocalMassBound.lean	1

- `polynomial_profile_local_mass`

<!-- checkpoint-verification -->

Verification

All 237 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3844-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1162 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 239 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_171_FiniteLocalConvolution.md`

Checkpoint 171: A full finite-input local convolution estimate

For every complex kernel with norm bounded by (A/H)*(1+|z|/H)^(-10), proves the finite complex convolution is at most 4A times the dyadic local-mass envelope of input magnitudes. This fully proves the generic harmonic-analysis shell step in Proposition 6.14; identifying and assembling the source low kernel remain separate obligations. This checkpoint performs a fresh full rebuild.

Checked declarations

Module	Named supporting theorems
FiniteLocalConvolution.lean	1

- `finiteConvolution_local_mass`

<!-- checkpoint-verification -->

Verification

All 238 project modules and the root were freshly rebuilt: 3845 jobs pass without warnings/errors. All 1163 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 240 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_172_CorrelationFactorExpansion.md`

Checkpoint 172: Expanding the exceptional arithmetic factor

Proves the exact identity (p+1)/(p-1)=1+2/(p-1) at every prime and expands conductorCorrelationProduct(n) over all subsets of the prime factors of n. All summands are explicit nonnegative arithmetic quantities.

Checked declarations

Module	Named supporting theorems
CorrelationFactorExpansion.lean	3

- `correlationExcess_nonneg`
- `correlation_factor_excess`
- `conductorCorrelationProduct_subset_expansion`

<!-- checkpoint-verification -->

Verification

All 239 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3846-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1166 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 241 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_173_PrimeSubsetDivisibility.md`

Checkpoint 173: Prime subsets and exact divisibility

Proves that a product of a subset of a finite prime set has exactly those prime factors, and divides a positive integer precisely when that subset lies in its prime factors. The proof reuses unique factorization rather than assuming a squarefree Euler identity.

Checked declarations

Module	Named supporting theorems
PrimeSubsetDivisibility.lean	2

- `subsetProduct_primeFactors`
- `subsetProduct_dvd_iff`

<!-- checkpoint-verification -->

Verification

All 240 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3847-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 1168 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 242 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_174_CorrelationGlobalExpansion.md`

Checkpoint 174: A common prime-set expansion for every denominator

Expands the genuine exceptional arithmetic product over one common finite prime set using divisibility indicators. Also proves every prime factor of a positive n<=D lies in the exact primesLE D cutoff.

Checked declarations

Module	Named supporting theorems
CorrelationGlobalExpansion.lean	2

- `conductorCorrelationProduct_global_expansion`
- `primeFactors_le_cutoff`

<!-- checkpoint-verification -->

Verification

All 241 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3848-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1170 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 243 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_175_ReciprocalMultiples.md`

Checkpoint 175: A uniform harmonic bound for reciprocal multiples

Reindexes n divisible by m using n/m, proving injectivity and positivity, and obtains sum_{1<=n<=D,m|n} 1/n <= harmonic(D)/m. No asymptotic or numerical estimate is used.

Checked declarations

Module	Named supporting theorems
ReciprocalMultiples.lean	1

- `reciprocal_multiples_bound`

<!-- checkpoint-verification -->

Verification

All 242 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3849-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1171 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 244 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_176_CorrelationWeightedDoubleSum.md`

Checkpoint 176: Rearranging the actual weighted exceptional sum

Proves an exact finite double-sum identity for sum_{n<=D} conductorCorrelationProduct(n)/n. The subset factors and reciprocal-multiple sums are separated with every original divisor condition retained.

Checked declarations

Module	Named supporting theorems
CorrelationWeightedDoubleSum.lean	1

- `conductorCorrelationProduct_weighted_expansion`

<!-- checkpoint-verification -->

Verification

All 243 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3850-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1172 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 245 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_177_CorrelationExcessEuler.md`

Checkpoint 177: The summable correction Euler product

Proves the finite subset excess mass divided by its prime product is at most exp(2). The proof uses the already checked telescoping reciprocal p(p-1) sum; no Mertens theorem is needed.

Checked declarations

Module	Named supporting theorems
CorrelationExcessEuler.lean	3

- `subsetCorrelationExcess_nonneg`
- `subsetCorrelationExcess_div_product`
- `correlation_excess_euler_bound`

<!-- checkpoint-verification -->

Verification

All 244 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3851-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1175 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 246 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_178_CorrelationWeightedHarmonic.md`

Checkpoint 178: An unconditional exceptional weighted harmonic bound

Proves sum_{1<=n<=D} conductorCorrelationProduct(n)/n <= exp(2)*harmonic(D), for every natural D. This stronger sum includes all denominators, so the special support restrictions and bounded powers of two are unnecessary for this upper bound.

Checked declarations

Module	Named supporting theorems
CorrelationWeightedHarmonic.lean	1

- `conductorCorrelationProduct_weighted_harmonic`

<!-- checkpoint-verification -->

Verification

All 245 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3852-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1176 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 247 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_179_CorrelationWeightedLog.md`

Checkpoint 179: The exceptional logarithmic sum without Mertens

Derives an explicit logarithmic bound with correlationLogConstant=exp(2)*(1+1/log 2). The bound is uniform and unconditional for every positive natural cutoff.

Checked declarations

Module	Named supporting theorems
CorrelationWeightedLog.lean	3

- `correlationLogConstant_pos`
- `one_add_log_le_log_twice`
- `conductorCorrelationProduct_weighted_log`

<!-- checkpoint-verification -->

Verification

All 246 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3853-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1179 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 248 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_180_CorrelationWeightedSubsets.md`

Checkpoint 180: Logarithmic control for every contributing subset

Proves positivity of the exceptional weighted factor and the same logarithmic bound for every finite subset of the denominator cutoff. Thus q0*d<=R and nonvanishing restrictions can safely be imposed or discarded.

Checked declarations

Module	Named supporting theorems
CorrelationWeightedSubsets.lean	2

- `conductorCorrelationProduct_weight_nonneg`
- `conductorCorrelationProduct_subset_log`

<!-- checkpoint-verification -->

Verification

All 247 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3854-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1181 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 249 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

Checkpoint_181_InducedIntegerCorrelation.md

Checkpoint 181: The actual induced-character correlation at integer arguments

Defines the genuine induced twisted correlation at all natural d, with the zero-modulus case handled explicitly. Proves its exact positive-modulus formula and the previously established conductorMajorant(q0)*conductorCorrelationProduct(d) bound.

Checked declarations

Module	Named supporting theorems
InducedIntegerCorrelation.lean	2

- `inducedIntegerCorrelation_eq`
- `inducedIntegerCorrelation_bound`

<!-- checkpoint-verification -->

Verification

All 248 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3855-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1183 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 250 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_182_InducedWeightedCorrelation.md

Checkpoint 182: Weighted logarithmic control of induced correlations

Combines the actual induced-character correlation estimate with the new all-denominator logarithmic sum. The conductor majorant is kept explicitly for cancellation against the exceptional narrow spatial width.

Checked declarations

Module	Named supporting theorems
InducedWeightedCorrelation.lean	1

- `inducedIntegerCorrelation_weighted_log`

<!-- checkpoint-verification -->

Verification

All 249 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3856-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1184 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 251 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_183_NormalizedCohenCorrelation.md`

Checkpoint 183: The normalized ordinary arithmetic correlation

Defines the genuine totient-normalized Cohen correlation, proves that its norm divided by q is exactly cohenWeight(q,x), identifies it with the principal twisted correlation, and obtains the truncated logarithmic bound.

Checked declarations

Module	Named supporting theorems
NormalizedCohenCorrelation.lean	3

- `normalizedCohenCorrelation_weight`
- `normalizedCohenCorrelation_log`
- `normalizedCohenCorrelation_principal`

<!-- checkpoint-verification -->

Verification

All 250 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3857-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1187 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 252 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_184_WeightedKernelAssembly.md`

Checkpoint 184: Summing arithmetic coefficients against spatial kernels

Proves the finite complex weighted kernel inequality and a coefficient-free version. These retain each arithmetic weight and are used to assemble ordinary and exceptional physical kernels.

Checked declarations

Module	Named supporting theorems
WeightedKernelAssembly.lean	2

- `weighted_kernel_sum_bound`
- `weighted_kernel_sum_majorant`

<!-- checkpoint-verification -->

Verification

All 251 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3858-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1189 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 253 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_185_OrdinaryLowWidth.md`

Checkpoint 185: The exact ordinary low spatial width

Defines q*N^2/Q and proves its positivity, its upper bound by N^2 for q<=Q, and the exact reciprocal normalization. No rounded or substituted spatial scale is introduced.

Checked declarations

Module	Named supporting theorems
OrdinaryLowWidth.lean	3

- `ordinaryLowWidth_pos`
- `ordinaryLowWidth_le_square`
- `ordinaryLowWidth_reciprocal`

<!-- checkpoint-verification -->

Verification

All 252 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3859-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1192 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 254 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_186_OrdinaryPhysicalLow.md`

Checkpoint 186: Spatial decay of the ordinary physical low kernel

Defines the exact finite sum of normalized Cohen correlations times the actual smoothed annular cutoff kernels. Proves the Q*log(Q+1)/N^2 spatial envelope for natural Q>=2. The real/circle inverse-transform identification remains a separate source bridge.

Checked declarations

Module	Named supporting theorems
OrdinaryPhysicalLow.lean	1

- `ordinaryPhysicalLowKernel_decay`

<!-- checkpoint-verification -->

Verification

All 253 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3860-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1193 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 255 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_187_ExceptionalLowWidth.md`

Checkpoint 187: The exceptional spatial width and conductor cancellation

Defines B*d*N^2/Q, proves the scale condition under B*d<=Q, and proves the exact cancellation B/width=Q/(d*N^2). This accounts for the conductor-sensitive normalization in the low estimate.

Checked declarations

Module	Named supporting theorems
ExceptionalLowWidth.lean	3

- `exceptionalLowWidth_pos`
- `exceptionalLowWidth_le_square`
- `exceptionalLowWidth_cancel`

<!-- checkpoint-verification -->

Verification

All 254 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3861-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1196 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 256 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_188_ExceptionalPhysicalLow.md`

Checkpoint 188: Uniform spatial decay of the induced exceptional kernel

Defines the actual induced-character physical kernel and proves its Q*log(2Q)/N^2 envelope uniformly in the conductor and beta<=1. The conductorMajorant cancels exactly against the spatial width; the remaining sum uses the new unconditional harmonic argument.

Checked declarations

Module	Named supporting theorems
ExceptionalPhysicalLow.lean	1

- `exceptionalPhysicalLowKernel_decay`

<!-- checkpoint-verification -->

Verification

All 255 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3862-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1197 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 257 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_189_ExceptionalLowEmptyRange.md`

Checkpoint 189: The empty exceptional low range

Proves a contributing positive denominator set is empty when the conductor majorant exceeds Q, and checks that the corresponding exceptional physical kernel is zero. This handles the source D<1 edge case.

Checked declarations

Module	Named supporting theorems
ExceptionalLowEmptyRange.lean	2

- `exceptionalLow_set_empty`
- `exceptionalPhysicalLowKernel_empty`

<!-- checkpoint-verification -->

Verification

All 256 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3863-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1199 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 258 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_190_PhysicalLowKernel.md`

Checkpoint 190: The combined ordinary-minus-exceptional physical kernel

Combines the genuine ordinary and induced exceptional physical kernels with the correct minus sign. Proves the full tenth-order Q*log(2Q)/N^2 spatial envelope, uniformly in the conductor and beta. This is a physical-space result; full source Fourier correspondence is still open.

Checked declarations

Module	Named supporting theorems
PhysicalLowKernel.lean	2

- `lowSpatialConstant_nonneg`
- `physicalLowKernel_decay`

<!-- checkpoint-verification -->

Verification

All 257 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3864-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1201 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 259 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_191_PhysicalLowAmplitude.md`

Checkpoint 191: Uniform low amplitude across arithmetic scales

Defines the exact global low amplitude, proves nonnegativity and monotonicity as N increases. The spatial scale remains N^2 throughout.

Checked declarations

Module	Named supporting theorems
PhysicalLowAmplitude.lean	2

- `physicalLowAmplitude_nonneg`
- `physicalLowAmplitude_mono`

<!-- checkpoint-verification -->

Verification

All 258 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3865-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1203 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 260 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint notes 192-266

Checkpoint_192_PhysicalLowLocalBound.md

Checkpoint 192: The local low bound for finite complex inputs

Applies the fully proved dyadic shell argument to the combined physical low kernel. Every finite complex input is bounded by 4*lowSpatialConstant*Q*log(2Q) times its actual countable dyadic local-mass envelope.

Checked declarations

Module	Named supporting theorems
PhysicalLowLocalBound.lean	1

- `physicalLowKernel_local_finite`

<!-- checkpoint-verification -->

Verification

All 259 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3866-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1204 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 261 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_193_PhysicalLowL1Operator.md

Checkpoint 193: A genuine all-input physical low operator

Constructs a bounded complex linear map from l1 to linfinity for the actual physical low kernel, proves its norm bound, and identifies its value with the absolutely convergent convolution and every finite-indicator convolution.

Checked declarations

Module	Named supporting theorems
PhysicalLowL1Operator.lean	2

- `physicalLowKernel_uniform`
- `physicalLow_L1_operator`

<!-- checkpoint-verification -->

Verification

All 260 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3867-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1206 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 262 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_194_PhysicalLowPairing.md`

Checkpoint 194: Finite-set bounds for the physical low operator

Proves pointwise indicator and finite bilinear pairing estimates with the exact Q*log(2Q)/N^2 normalization and cardinality factors. These supply low-component estimates for the later high-low application, without asserting the unfinished high estimates.

Checked declarations

Module	Named supporting theorems
PhysicalLowPairing.lean	2

- `physicalLow_indicator_bound`
- `physicalLow_pairing_bound`

<!-- checkpoint-verification -->

Verification

All 261 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3868-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1208 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 263 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_195_PhysicalLowFarField.md`

Checkpoint 195: A quantitative far-field estimate for the physical low kernel

Proves the actual physical low convolution outside a radius R gains the tenth-order profile at R, in addition to the global l1 amplitude. No compact spatial support is assumed.

Checked declarations

Module	Named supporting theorems
PhysicalLowFarField.lean	1

- `physicalLow_far_field`

<!-- checkpoint-verification -->

Verification

All 262 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3869-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1209 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 264 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_196_UniformL1Maximum.md`

Checkpoint 196: A genuine countable l1-to-linfinity maximal construction

Constructs the actual countable pointwise supremum of a uniformly bounded family of l1-to-linfinity operators, proves boundedness and nonnegativity, realizes it in real linfinity, and proves its norm bound. A fresh full rebuild is performed.

Checked declarations

Module	Named supporting theorems
UniformL1Maximum.lean	4

- `operatorL1MaxValue_bdd`
- `operatorL1MaxValue_nonneg`
- `operatorL1MaxValue_bound`
- `operatorL1Maximum_norm`

<!-- checkpoint-verification -->

Verification

All 263 project modules and the root were freshly rebuilt: 3870 jobs pass without warnings/errors. All 1213 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 265 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_197_PhysicalLowMaximum.md`

Checkpoint 197: The all-scale physical low maximal bound

Applies the countable maximal construction to the genuine physical low kernels at arbitrary scales N>=N0. The conductor, quadratic character, beta and contributing denominator set may all vary with the scale. The result is a true all-input l1-to-linfinity maximal bound; source Fourier correspondence is still separate.

Checked declarations

Module	Named supporting theorems
PhysicalLowMaximum.lean	2

- `physicalLow_kernel_uniform_tail`
- `physicalLow_countable_L1_maximum`

<!-- checkpoint-verification -->

Verification

All 264 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3871-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1215 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 266 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_198_RealFourierPhase.md`

Checkpoint 198: The real Fourier phase and quadratic convention

The real Fourier phase and quadratic convention. The module proves the exact displayed identities for the actual source cutoff and Fourier convention. All integrals, scale normalizations and support boundaries are retained. This is a proved source-construction stage, not an assumed low-operator estimate; the denominator summation and final operator assembly are handled separately. The Lean declarations give the full quantified hypotheses.

Checked declarations

Module	Named supporting theorems
RealFourierPhase.lean	4

- `realFourierPhase_norm`
- `realFourierPhase_continuous`
- `realFourierPhase_add`
- `realFourierPhase_quadratic`

<!-- checkpoint-verification -->

Verification

All 265 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3872-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1219 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 267 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_199_RealInverseFourierIntegral.md`

Checkpoint 199: The exact real inverse Fourier integral

The exact real inverse Fourier integral. The module proves the exact displayed identities for the actual source cutoff and Fourier convention. All integrals, scale normalizations and support boundaries are retained. This is a proved source-construction stage, not an assumed low-operator estimate; the denominator summation and final operator assembly are handled separately. The Lean declarations give the full quantified hypotheses.

Checked declarations

Module	Named supporting theorems
RealInverseFourierIntegral.lean	3

- `real_inverse_fourier_integral`
- `realFourierPhase_integer_character`
- `realFourierPhase_integer`

<!-- checkpoint-verification -->

Verification

All 266 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3873-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1222 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 268 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_200_RealFourierDilation.md`

Checkpoint 200: Change of variables for the inverse Fourier transform

Change of variables for the inverse Fourier transform. The module proves the exact displayed identities for the actual source cutoff and Fourier convention. All integrals, scale normalizations and support boundaries are retained. This is a proved source-construction stage, not an assumed low-operator estimate; the denominator summation and final operator assembly are handled separately. The Lean declarations give the full quantified hypotheses.

Checked declarations

Module	Named supporting theorems
RealFourierDilation.lean	2

- `real_inverse_fourier_dilate`
- `real_inverse_fourier_const_mul`

<!-- checkpoint-verification -->

Verification

All 267 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3874-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1224 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 269 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_201_ScaledCutoffInverseIdentity.md`

Checkpoint 201: Identification of the scaled physical cutoff inverse

Identification of the scaled physical cutoff inverse. The module proves the exact displayed identities for the actual source cutoff and Fourier convention. All integrals, scale normalizations and support boundaries are retained. This is a proved source-construction stage, not an assumed low-operator estimate; the denominator summation and final operator assembly are handled separately. The Lean declarations give the full quantified hypotheses.

Checked declarations

Module	Named supporting theorems
ScaledCutoffInverseIdentity.lean	2

- `scaledCutoffKernel_inverse_transform`
- `scaledCutoffInverse_decay`

<!-- checkpoint-verification -->

Verification

All 268 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3875-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1226 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 270 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_202_NarrowFourierIntegral.md`

Checkpoint 202: Reducing narrow inverse transforms to the fundamental interval

Reducing narrow inverse transforms to the fundamental interval. The module proves the exact displayed identities for the actual source cutoff and Fourier convention. All integrals, scale normalizations and support boundaries are retained. This is a proved source-construction stage, not an assumed low-operator estimate; the denominator summation and final operator assembly are handled separately. The Lean declarations give the full quantified hypotheses.

Checked declarations

Module	Named supporting theorems
NarrowFourierIntegral.lean	2

- `narrow_inverse_fourier_interval`
- `scaled_source_cutoff_narrow`

<!-- checkpoint-verification -->

Verification

All 269 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3876-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1228 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 271 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_203_ClosedFundamentalCutoff.md`

Checkpoint 203: Exact periodization at both fundamental endpoints

Exact periodization at both fundamental endpoints. The module proves the exact displayed identities for the actual source cutoff and Fourier convention. All integrals, scale normalizations and support boundaries are retained. This is a proved source-construction stage, not an assumed low-operator estimate; the denominator summation and final operator assembly are handled separately. The Lean declarations give the full quantified hypotheses.

Checked declarations

Module	Named supporting theorems
ClosedFundamentalCutoff.lean	1

- `compactCircleLift_small_support_closed`

<!-- checkpoint-verification -->

Verification

All 270 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3877-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1229 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 272 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_204_CompactCircleInverseKernel.md`

Checkpoint 204: The real-to-circle inverse-kernel identity

The real-to-circle inverse-kernel identity. The module proves the exact displayed identities for the actual source cutoff and Fourier convention. All integrals, scale normalizations and support boundaries are

retained. This is a proved source-construction stage, not an assumed low-operator estimate; the denominator summation and final operator assembly are handled separately. The Lean declarations give the full quantified hypotheses.

Checked declarations

Module	Named supporting theorems
CompactCircleInverseKernel.lean	1

- `compactCircleLift_inverse_kernel`

<!-- checkpoint-verification -->

Verification

All 271 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3878-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1230 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 273 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_205_AnnularFourierIntegrand.md`

Checkpoint 205: Joint regularity and product integrability for the annular inverse

Joint regularity and product integrability for the annular inverse. The module proves the exact displayed identities for the actual source cutoff and Fourier convention. All integrals, scale normalizations and support boundaries are retained. This is a proved source-construction stage, not an assumed low-operator estimate; the denominator summation and final operator assembly are handled separately. The Lean declarations give the full quantified hypotheses.

Checked declarations

Module	Named supporting theorems
AnnularFourierIntegrand.lean	2

- `annularFourierIntegrand_continuousOn`
- `annularFourierIntegrand_integrable`

<!-- checkpoint-verification -->

Verification

All 272 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3879-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1232 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 274 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_206_AnnularFourierFubini.md`

Checkpoint 206: A checked Fubini interchange on the annular rectangle

A checked Fubini interchange on the annular rectangle. The module proves the exact displayed identities for the actual source cutoff and Fourier convention. All integrals, scale normalizations and support boundaries are retained. This is a proved source-construction stage, not an assumed low-operator estimate; the denominator summation and final operator assembly are handled separately. The Lean declarations give the full quantified hypotheses.

Checked declarations

Module	Named supporting theorems
AnnularFourierFubini.lean	1

- `annularFourierIntegrand_swap`

<!-- checkpoint-verification -->

Verification

All 273 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3880-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1233 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 275 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_207_AnnularCutoffSymbol.md`

Checkpoint 207: The actual narrow annular frequency symbol

The actual narrow annular frequency symbol. The module proves the exact displayed identities for the actual source cutoff and Fourier convention. All integrals, scale normalizations and support boundaries are retained. This is a proved source-construction stage, not an assumed low-operator estimate; the denominator summation and final operator assembly are handled separately. The Lean declarations give the full quantified hypotheses.

Checked declarations

Module	Named supporting theorems
AnnularCutoffSymbol.lean	2

- `annularCutoffSymbol_narrow`
- `annularCutoffSymbol_support`

<!-- checkpoint-verification -->

Verification

All 274 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3881-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1235 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 276 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_208_AnnularPhaseFactorization.md`

Checkpoint 208: Matching the annular and inverse Fourier phases

Matching the annular and inverse Fourier phases. The module proves the exact displayed identities for the actual source cutoff and Fourier convention. All integrals, scale normalizations and support boundaries are retained. This is a proved source-construction stage, not an assumed low-operator estimate; the denominator summation and final operator assembly are handled separately. The Lean declarations give the full quantified hypotheses.

Checked declarations

Module	Named supporting theorems
AnnularPhaseFactorization.lean	2

- `annular_phase_factorization`
- `annular_cutoff_phase_integral`

<!-- checkpoint-verification -->

Verification

All 275 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3882-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1237 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 277 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_209_AnnularCutoffInverseKernel.md`

Checkpoint 209: The exact annular-product inverse-transform formula

The exact annular-product inverse-transform formula. The module proves the exact displayed identities for the actual source cutoff and Fourier convention. All integrals, scale normalizations and support boundaries are retained. This is a proved source-construction stage, not an assumed low-operator estimate; the denominator summation and final operator assembly are handled separately. The Lean declarations give the full quantified hypotheses.

Checked declarations

Module	Named supporting theorems
AnnularCutoffInverseKernel.lean	1

- `annular_cutoff_inverse_transform`

<!-- checkpoint-verification -->

Verification

All 276 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3883-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1238 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 278 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_210_CircleAnnularCutoffKernel.md`

Checkpoint 210: The genuine circle annular kernel equals the physical kernel

The genuine circle annular kernel equals the physical kernel. The module proves the exact displayed identities for the actual source cutoff and Fourier convention. All integrals, scale normalizations and support boundaries are retained. This is a proved source-construction stage, not an assumed low-operator estimate; the denominator summation and final operator assembly are handled separately. The Lean declarations give the full quantified hypotheses.

Checked declarations

Module	Named supporting theorems
CircleAnnularCutoffKernel.lean	1

- `circleAnnularCutoff_kernel`

<!-- checkpoint-verification -->

Verification

All 277 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3884-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1239 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 279 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_211_CircleFrequencyShiftKernel.md`

Checkpoint 211: Rational frequency shifts and physical modulation

Rational frequency shifts and physical modulation. The module proves the exact displayed identities for the actual source cutoff and Fourier convention. All integrals, scale normalizations and support boundaries are retained. This is a proved source-construction stage, not an assumed low-operator estimate; the denominator summation and final operator assembly are handled separately. The Lean declarations give the full quantified hypotheses.

Checked declarations

Module	Named supporting theorems
CircleFrequencyShiftKernel.lean	3

- `integer_circle_character_add`
- `circleFrequencyShift_kernel`
- `circleFrequencyShift_real_kernel`

<!-- checkpoint-verification -->

Verification

All 278 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3885-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1242 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 280 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_212_IntegerRationalPhase.md`

Checkpoint 212: Exact agreement with the rational additive character

Exact agreement with the rational additive character. This module proves the stated construction or operator consequence for genuine continuous circle symbols, with the actual Gauss/twisted coefficients and the source annular smoothing. The parameter domain is natural N,Q>=2 with 2Q<=N^2, positive conductor, a quadratic character, beta<=1 and the explicit contributing-denominator width conditions. No analytic low estimate is assumed. The exact identification with the earlier rational-indexed circle approximation model, general real cutoffs, the sharp high estimates and Theorem 1.1 remain separate obligations. The Lean declarations specify every condition and normalization.

Checked declarations

Module	Named supporting theorems
IntegerRationalPhase.lean	2

- `realFourierPhase_rational`
- `realFourierPhase_unit_rational`

<!-- checkpoint-verification -->

Verification

All 279 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3886-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1244 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 281 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_213_CorrelationPhaseSums.md`

Checkpoint 213: The ordinary and twisted arithmetic phase sums

The ordinary and twisted arithmetic phase sums. This module proves the stated construction or operator consequence for genuine continuous circle symbols, with the actual Gauss/twisted coefficients and the source annular smoothing. The parameter domain is natural N,Q>=2 with 2Q<=N^2, positive conductor, a quadratic character, beta<=1 and the explicit contributing-denominator width conditions. No analytic low estimate is assumed. The exact identification with the earlier rational-indexed circle approximation model, general real cutoffs, the sharp high estimates and Theorem 1.1 remain separate obligations. The Lean declarations specify every condition and normalization.

Checked declarations

Module	Named supporting theorems
CorrelationPhaseSums.lean	2

- `normalizedCohenCorrelation_phase_sum`
- `twistedCorrelation_phase_sum`

<!-- checkpoint-verification -->

Verification

All 280 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3887-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1246 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 282 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_214_LowWidthLowerBounds.md`

Checkpoint 214: Lower spatial widths needed for exact circle lifting

Lower spatial widths needed for exact circle lifting. This module proves the stated construction or operator consequence for genuine continuous circle symbols, with the actual Gauss/twisted coefficients and the source annular smoothing. The parameter domain is natural N,Q>=2 with 2Q<=N^2, positive conductor, a quadratic character, beta<=1 and the explicit contributing-denominator width conditions. No analytic low estimate is assumed. The exact identification with the earlier rational-indexed circle approximation model, general real cutoffs, the sharp high estimates and Theorem 1.1 remain separate obligations. The Lean declarations specify every condition and normalization.

Checked declarations

Module	Named supporting theorems
LowWidthLowerBounds.lean	2

- `ordinaryLowWidth_two_le`
- `exceptionalLowWidth_two_le`

<!-- checkpoint-verification -->

Verification

All 281 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3888-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1248 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 283 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_215_OrdinaryCircleLowTerm.md`

Checkpoint 215: The genuine ordinary denominator circle term

The genuine ordinary denominator circle term. This module proves the stated construction or operator consequence for genuine continuous circle symbols, with the actual Gauss/twisted coefficients and the source annular smoothing. The parameter domain is natural N,Q>=2 with 2Q<=N^2, positive conductor, a quadratic character, beta<=1 and the explicit contributing-denominator width conditions. No analytic low estimate is assumed. The exact identification with the earlier rational-indexed circle approximation model, general real cutoffs, the sharp high estimates and Theorem 1.1 remain separate obligations. The Lean declarations specify every condition and normalization.

Checked declarations

Module	Named supporting theorems
OrdinaryCircleLowTerm.lean	2

- `ordinaryCircleLowTerm_kernel`
- `ordinaryCircleLowTerm_zero`

<!-- checkpoint-verification -->

Verification

All 282 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3889-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1250 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 284 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_216_OrdinaryCircleLowModel.md

Checkpoint 216: The exact kernel of the ordinary denominator model

The exact kernel of the ordinary denominator model. This module proves the stated construction or operator consequence for genuine continuous circle symbols, with the actual Gauss/twisted coefficients and the source annular smoothing. The parameter domain is natural N,Q>=2 with 2Q<=N^2, positive conductor, a quadratic character, beta<=1 and the explicit contributing-denominator width conditions. No analytic low estimate is assumed. The exact identification with the earlier rational-indexed circle approximation model, general real cutoffs, the sharp high estimates and Theorem 1.1 remain separate obligations. The Lean declarations specify every condition and normalization.

Checked declarations

Module	Named supporting theorems
OrdinaryCircleLowModel.lean	1

- `ordinaryCircleLowModel_kernel`

<!-- checkpoint-verification -->

Verification

All 283 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3890-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1251 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 285 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_217_ExceptionalCircleLowTerm.md

Checkpoint 217: The genuine induced exceptional circle term

The genuine induced exceptional circle term. This module proves the stated construction or operator consequence for genuine continuous circle symbols, with the actual Gauss/twisted coefficients and the source annular smoothing. The parameter domain is natural N,Q>=2 with 2Q<=N^2, positive conductor, a quadratic character, beta<=1 and the explicit contributing-denominator width conditions. No analytic low estimate is assumed. The exact identification with the earlier rational-indexed circle approximation model, general real cutoffs, the sharp high estimates and Theorem 1.1 remain separate obligations. The Lean declarations specify every condition and normalization.

Checked declarations

Module	Named supporting theorems
ExceptionalCircleLowTerm.lean	2

- `exceptionalCircleLowTerm_kernel`
- `exceptionalCircleLowTerm_zero`

<!-- checkpoint-verification -->

Verification

All 284 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3891-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1253 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 286 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_218_ExceptionalCircleLowModel.md`

Checkpoint 218: The exact induced exceptional circle kernel

The exact induced exceptional circle kernel. This module proves the stated construction or operator consequence for genuine continuous circle symbols, with the actual Gauss/twisted coefficients and the source annular smoothing. The parameter domain is natural N,Q>=2 with 2Q<=N^2, positive conductor, a quadratic character, beta<=1 and the explicit contributing-denominator width conditions. No analytic low estimate is assumed. The exact identification with the earlier rational-indexed circle approximation model, general real cutoffs, the sharp high estimates and Theorem 1.1 remain separate obligations. The Lean declarations specify every condition and normalization.

Checked declarations

Module	Named supporting theorems
ExceptionalCircleLowModel.lean	1

- `exceptionalCircleLowModel_kernel`

<!-- checkpoint-verification -->

Verification

All 285 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3892-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1254 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 287 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_219_DenominatorCircleLow.md`

Checkpoint 219: The genuine denominator-indexed low circle model

The genuine denominator-indexed low circle model. This module proves the stated construction or operator consequence for genuine continuous circle symbols, with the actual Gauss/twisted coefficients and the source annular smoothing. The parameter domain is natural N,Q>=2 with 2Q<=N^2, positive

conductor, a quadratic character, beta<=1 and the explicit contributing-denominator width conditions. No analytic low estimate is assumed. The exact identification with the earlier rational-indexed circle approximation model, general real cutoffs, the sharp high estimates and Theorem 1.1 remain separate obligations. The Lean declarations specify every condition and normalization.

Checked declarations

Module	Named supporting theorems
DenominatorCircleLow.lean	1

- `denominatorCircleLow_kernel`

<!-- checkpoint-verification -->

Verification

All 286 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3893-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1255 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 288 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_220_DenominatorLowKernelDecay.md`

Checkpoint 220: The low spatial envelope for the actual circle kernel

The low spatial envelope for the actual circle kernel. This module proves the stated construction or operator consequence for genuine continuous circle symbols, with the actual Gauss/twisted coefficients and the source annular smoothing. The parameter domain is natural N,Q>=2 with 2Q<=N^2, positive conductor, a quadratic character, beta<=1 and the explicit contributing-denominator width conditions. No analytic low estimate is assumed. The exact identification with the earlier rational-indexed circle approximation model, general real cutoffs, the sharp high estimates and Theorem 1.1 remain separate obligations. The Lean declarations specify every condition and normalization.

Checked declarations

Module	Named supporting theorems
DenominatorLowKernelDecay.lean	2

- `denominatorCircleLow_kernel_decay`
- `denominatorCircleLow_kernel_uniform`

<!-- checkpoint-verification -->

Verification

All 287 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3894-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1257 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 289 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_221_DenominatorLowLocalBound.md`

Checkpoint 221: The local low estimate for the genuine Fourier multiplier

The local low estimate for the genuine Fourier multiplier. This module proves the stated construction or operator consequence for genuine continuous circle symbols, with the actual Gauss/twisted coefficients and the source annular smoothing. The parameter domain is natural N,Q>=2 with 2Q<=N^2, positive conductor, a quadratic character, beta<=1 and the explicit contributing-denominator width conditions. No analytic low estimate is assumed. The exact identification with the earlier rational-indexed circle approximation model, general real cutoffs, the sharp high estimates and Theorem 1.1 remain separate obligations. The Lean declarations specify every condition and normalization.

Checked declarations

Module	Named supporting theorems
DenominatorLowLocalBound.lean	2

- `denominatorCircleLow_finite_convolution`
- `denominatorCircleLow_local_finite`

<!-- checkpoint-verification -->

Verification

All 288 project modules and the root were freshly rebuilt: 3895 jobs pass without warnings/errors. All 1259 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 290 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_222_DenominatorLowL1Bound.md`

Checkpoint 222: The all-input l1 bound for the genuine low multiplier

The all-input l1 bound for the genuine low multiplier. This module proves the stated construction or operator consequence for genuine continuous circle symbols, with the actual Gauss/twisted coefficients and the source annular smoothing. The parameter domain is natural N,Q>=2 with 2Q<=N^2, positive conductor, a quadratic character, beta<=1 and the explicit contributing-denominator width conditions. No analytic low estimate is assumed. The exact identification with the earlier rational-indexed circle approximation model, general real cutoffs, the sharp high estimates and Theorem 1.1 remain separate obligations. The Lean declarations specify every condition and normalization.

Checked declarations

Module	Named supporting theorems
DenominatorLowL1Bound.lean	2

- `denominatorCircleLow_L1_norm`
- `denominatorCircleLow_L1_bound`

<!-- checkpoint-verification -->

Verification

All 289 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3896-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1261 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 291 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_223_AllInputFourierConvolution.md`

Checkpoint 223: All-input Fourier convolution and l1-l2 agreement

All-input Fourier convolution and l1-l2 agreement. This module proves the stated construction or operator consequence for genuine continuous circle symbols, with the actual Gauss/twisted coefficients and the source annular smoothing. The parameter domain is natural N,Q>=2 with 2Q<=N^2, positive conductor, a quadratic character, beta<=1 and the explicit contributing-denominator width conditions. No analytic low estimate is assumed. The exact identification with the earlier rational-indexed circle approximation model, general real cutoffs, the sharp high estimates and Theorem 1.1 remain separate obligations. The Lean declarations specify every condition and normalization.

Checked declarations

Module	Named supporting theorems
AllInputFourierConvolution.lean	3

- `integerMultiplier_convolution_hasSum`
- `integerMultiplier_all_convolution`
- `fourierL1Operator_all_l2_agreement`

<!-- checkpoint-verification -->

Verification

All 290 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3897-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1264 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 292 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_224_DenominatorLowPairing.md`

Checkpoint 224: All-input and finite-set low estimates

All-input and finite-set low estimates. This module proves the stated construction or operator consequence for genuine continuous circle symbols, with the actual Gauss/twisted coefficients and the source annular smoothing. The parameter domain is natural N,Q>=2 with 2Q<=N^2, positive conductor, a quadratic character, beta<=1 and the explicit contributing-denominator width conditions. No analytic low estimate is assumed. The exact identification with the earlier rational-indexed circle approximation model, general real cutoffs, the sharp high estimates and Theorem 1.1 remain separate obligations. The Lean declarations specify every condition and normalization.

Checked declarations

Module	Named supporting theorems
DenominatorLowPairing.lean	2

- `denominatorCircleLow_all_l2_low_bound`
- `denominatorCircleLow_pairing`

<!-- checkpoint-verification -->

Verification

All 291 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3898-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1266 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 293 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_225_AdmissibleLowFamilies.md`

Checkpoint 225: Uniform countable maximal bounds for actual low families

Uniform countable maximal bounds for actual low families. This module proves the stated construction or operator consequence for genuine continuous circle symbols, with the actual Gauss/twisted coefficients and the source annular smoothing. The parameter domain is natural N,Q>=2 with 2Q<=N^2, positive conductor, a quadratic character, beta<=1 and the explicit contributing-denominator width conditions. No analytic low estimate is assumed. The exact identification with the earlier rational-indexed circle approximation model, general real cutoffs, the sharp high estimates and Theorem 1.1 remain separate obligations. The Lean declarations specify every condition and normalization.

Checked declarations

Module	Named supporting theorems
AdmissibleLowFamilies.lean	2

- `lowSymbolData_L1_norm`
- `lowSymbolFamily_countable_maximal`

<!-- checkpoint-verification -->

Verification

All 292 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3899-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 1268 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 294 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_226_DenominatorLowSourceInterface.md

Checkpoint 226: Source-range geometry and exact integer-lift formulas

Source-range geometry and exact integer-lift formulas. This module proves the stated construction or operator consequence for genuine continuous circle symbols, with the actual Gauss/twisted coefficients and the source annular smoothing. The parameter domain is natural N,Q>=2 with 2Q<=N^2, positive conductor, a quadratic character, beta<=1 and the explicit contributing-denominator width conditions. No analytic low estimate is assumed. The exact identification with the earlier rational-indexed circle approximation model, general real cutoffs, the sharp high estimates and Theorem 1.1 remain separate obligations. The Lean declarations specify every condition and normalization.

Checked declarations

Module	Named supporting theorems
DenominatorLowSourceInterface.lean	3

- `source_low_width_budget`
- `circleAnnularCutoff_real`
- `circleAnnularCutoff_shift_real`

<!-- checkpoint-verification -->

Verification

All 293 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3900-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1271 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 295 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_227_L2SquareEnergy.md

Checkpoint 227: Exact square energy of an integer l2 sequence

Exact square energy of an integer l2 sequence. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has

already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
L2SquareEnergy.lean	3

- `l2_square_hasSum`
- `l2_square_tsum`
- `l2_square_finite_bound`

<!-- checkpoint-verification -->

Verification

All 294 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3901-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1274 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 296 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_228_DoubleSquareSummability.md`

Checkpoint 228: Joint square summability over scale and space

Joint square summability over scale and space. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
DoubleSquareSummability.lean	3

- `joint_square_summable`
- `joint_square_summable_swap`
- `coordinate_square_summable`

<!-- checkpoint-verification -->

Verification

All 295 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3902-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1277 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 297 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_229_SquareEnergyFubini.md`

Checkpoint 229: A rigorous interchange of scale and spatial energy

A rigorous interchange of scale and spatial energy. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
SquareEnergyFubini.lean	2

- `coordinate_square_totals_summable`
- `coordinate_square_total`

<!-- checkpoint-verification -->

Verification

All 296 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3903-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1279 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 298 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_230_SquareFunctionValue.md`

Checkpoint 230: The actual countable pointwise square function

The actual countable pointwise square function. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
SquareFunctionValue.lean	3

- `squareFunctionValue_nonneg`
- `squareFunctionValue_sq`
- `squareFunctionValue_dominates`

<!-- checkpoint-verification -->

Verification

All 297 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3904-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1282 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 299 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_231_SquareFunctionMembership.md`

Checkpoint 231: Realizing the countable square function in l2

Realizing the countable square function in l2. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
SquareFunctionMembership.lean	1

- `squareFunctionValue_mem_l2`

<!-- checkpoint-verification -->

Verification

All 298 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3905-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1283 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 300 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypothe-

ses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_232_SquareFunctionNorm.md

Checkpoint 232: The exact l2 norm of the countable square function

The exact l2 norm of the countable square function. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
SquareFunctionNorm.lean	2

- `squareFunctionL2_norm_square`
- `squareFunctionL2_norm`

<!-- checkpoint-verification -->

Verification

All 299 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3906-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1285 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 301 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_233_SquareMaximumValue.md

Checkpoint 233: The genuine countable maximum dominated by square energy

The genuine countable maximum dominated by square energy. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
SquareMaximumValue.lean	3

- `squareMaximumValue_bddAbove`
- `squareMaximumValue_nonneg`

- `squareMaximumValue_le_squareFunction`

<!-- checkpoint-verification -->

Verification

All 300 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3907-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1288 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 302 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_234_SquareMaximumL2.md`

Checkpoint 234: The countable l2 maximal bound from square summability

The countable l2 maximal bound from square summability. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
SquareMaximumL2.lean	2

- `squareMaximumValue_mem_l2`
- `squareMaximumL2_norm`

<!-- checkpoint-verification -->

Verification

All 301 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3908-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1290 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 303 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_235_FiniteSquareEnergy.md`

Checkpoint 235: Finite energy budgets imply the full countable maximal bound

Finite energy budgets imply the full countable maximal bound. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
FiniteSquareEnergy.lean	3

- `finiteSquareEnergy_summable`
- `finiteSquareEnergy_total`
- `finiteSquareEnergy_maximal`

<!-- checkpoint-verification -->

Verification

All 302 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3909-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1293 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 304 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_236_OperatorSquareMaximal.md`

Checkpoint 236: The countable operator maximal principle from square energy

The countable operator maximal principle from square energy. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
OperatorSquareMaximal.lean	2

- `operatorSquareEnergy_finite_bound`
- `operatorSquareEnergy_countable_maximal`

<!-- checkpoint-verification -->

Verification

All 303 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3910-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1295 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 305 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_237_SummableOperatorSquares.md`

Checkpoint 237: A weaker squared-operator-norm summability criterion

A weaker squared-operator-norm summability criterion. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
SummableOperatorSquares.lean	3

- `operator_output_square_summable`
- `operator_output_square_total`
- `operator_squares_countable_maximal`

<!-- checkpoint-verification -->

Verification

All 304 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3911-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1298 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 306 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_238_SquareMaximalTailBounds.md`

Checkpoint 238: Quantitative maximal tails from square energy

Quantitative maximal tails from square energy. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit

frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
SquareMaximalTailBounds.lean	3

- `square_tail_summable`
- `squareMaximum_tail_norm`
- `squareMaximum_tail_value_mono`

<!-- checkpoint-verification -->

Verification

All 305 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3912-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1301 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 307 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_239_SquareMaximalTailVanishing.md`

Checkpoint 239: Convergence to zero of maximal square-energy tails

Convergence to zero of maximal square-energy tails. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
SquareMaximalTailVanishing.lean	2

- `squareMaximum_tail_norm_zero`
- `squareMaximum_tail_zero`

<!-- checkpoint-verification -->

Verification

All 306 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3913-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1303 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 308 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_240_SquareMaximalLipschitz.md`

Checkpoint 240: Sublinearity and Lipschitz control under square summability

Sublinearity and Lipschitz control under square summability. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
SquareMaximalLipschitz.lean	3

- `squareOperatorMax_add`
- `squareOperatorMax_difference`
- `squareOperatorMax_L2_difference`

<!-- checkpoint-verification -->

Verification

All 307 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3914-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1306 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 309 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_241_NarrowCollarOnset.md`

Checkpoint 241: The actual lower frequency at first collar onset

The actual lower frequency at first collar onset. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
NarrowCollarOnset.lean	2

- `narrow_collar_onset`
- `ordinary_collar_frequency_lower`

<!-- checkpoint-verification -->

Verification

All 308 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3915-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1308 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 310 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_242_DyadicAnnularFrequencyDecay.md`

Checkpoint 242: Geometric decay after an actual annular collar onset

Geometric decay after an actual annular collar onset. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
DyadicAnnularFrequencyDecay.lean	2

- `dyadic_square_scale`
- `dyadic_annular_frequency_decay`

<!-- checkpoint-verification -->

Verification

All 309 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3916-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1310 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 311 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypothe-

ses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_243_DyadicAnnularSquareTail.md`

Checkpoint 243: The summable squared annular frequency tail

The summable squared annular frequency tail. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
DyadicAnnularSquareTail.lean	2

- `dyadic_annular_square_summable`
- `dyadic_annular_square_total`

<!-- checkpoint-verification -->

Verification

All 310 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3917-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1312 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 312 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_244_FiniteDisjointSquareEnergy.md`

Checkpoint 244: Exact finite energy decomposition under disjoint frequency support

Exact finite energy decomposition under disjoint frequency support. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
FiniteDisjointSquareEnergy.lean	2

- `norm_sum_square_single_support`
- `finite_disjoint_square_energy`

<!-- checkpoint-verification -->

Verification

All 311 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3918-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1314 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 313 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_245_CircleSymbolSquareEnergy.md`

Checkpoint 245: A frequency square-energy criterion for actual l2 multipliers

A frequency square-energy criterion for actual l2 multipliers. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
CircleSymbolSquareEnergy.lean	5

- `circleL2_square_integrable`
- `circleL2_norm_square`
- `circleMultiply_norm_square`
- `circleMultiply_finite_square_energy`
- `circleSymbols_operator_square_energy`

<!-- checkpoint-verification -->

Verification

All 312 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3919-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1319 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 314 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_246_CircleSquareMaximal.md`

Checkpoint 246: The genuine countable maximal theorem for square-bounded symbols

The genuine countable maximal theorem for square-bounded symbols. This module proves the exact finite or infinite statement in its Lean declarations with no proof placeholder and no added mathematical axiom. The square-function construction requires summability of squared output norms, rather than the stronger sum-of-norms hypothesis used in the earlier envelope argument. The symbol criterion is a proved implication from explicit frequency square bounds; it is not a claim that the paper's complete high or multifrequency estimate has already been established. The source annular collar stages retain beta in [0,1], the exact rational-scale onset and all dyadic powers. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
CircleSquareMaximal.lean	2

- `circle_symbol_countable_maximal`
- `circle_symbol_attenuated_maximal`

<!-- checkpoint-verification -->

Verification

All 313 project modules and the root were freshly rebuilt: 3920 jobs pass without warnings/errors. All 1321 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 315 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_247_RationalNumeratorUnits.md`

Checkpoint 247: Canonical rational numerators as actual residue units

The canonical numerator of every reduced rational determines an actual unit modulo its positive reduced denominator. For a center in the finite canonical set, the least nonnegative representative of that unit is exactly the natural numerator. The proof uses the rational's proved coprimality and its canonical interval bounds; no new indexing equivalence is assumed. The denominator-one center zero is included. This is the first direction of the required rational/unit correspondence. The complete low-model correspondence and high-frequency estimates remain open at this stage.

Checked declarations

Module	Named supporting theorems
RationalNumeratorUnits.lean	2

- `rationalNumeratorUnit_coe`
- `rationalNumeratorUnit_val`

<!-- checkpoint-verification -->

Verification

All 314 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3921-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1323 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 316 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_248_UnitRationalReduction.md`

Checkpoint 248: Exact reduced numerator and denominator of a unit fraction

A unit modulo positive q defines the rational val(u)/q. Its numerator and denominator are proved to be val(u) and q, using the unit's coprimality and the exact reduced-fraction theorems. In particular no denominator can silently drop during reindexing. Modulus one remains in the domain. These statements supply the converse arithmetic direction of the rational/unit bridge, with no analytic hypothesis.

Checked declarations

Module	Named supporting theorems
UnitRationalReduction.lean	2

- `unitRational_num`
- `unitRational_den`

<!-- checkpoint-verification -->

Verification

All 315 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3922-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1325 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 317 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_249_UnitRationalCanonical.md`

Checkpoint 249: Canonical rational centers from every residue unit

Canonical rational centers from every residue unit. The definitions retain the denominator-one center and use actual least nonnegative unit representatives. All interval, denominator and recovery properties are proved, without an assumed finite-index correspondence. This supplies arithmetic infrastructure for the low-symbol identity; no high-frequency estimate or headline theorem is assumed.

Checked declarations

Module	Named supporting theorems
UnitRationalCanonical.lean	2

- `unitRational_canonical`

- `unitRational_mem_centers`

<!-- checkpoint-verification -->

Verification

All 316 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3923-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1327 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 318 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_250_RationalUnitRecovery.md`

Checkpoint 250: Recovery and injectivity of the unit-rational map

Recovery and injectivity of the unit-rational map. The definitions retain the denominator-one center and use actual least nonnegative unit representatives. All interval, denominator and recovery properties are proved, without an assumed finite-index correspondence. This supplies arithmetic infrastructure for the low-symbol identity; no high-frequency estimate or headline theorem is assumed.

Checked declarations

Module	Named supporting theorems
RationalUnitRecovery.lean	2

- `unitRational_numerator_recovery`
- `unitRational_injective`

<!-- checkpoint-verification -->

Verification

All 317 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3924-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1329 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 319 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_251_CanonicalDenominatorFibers.md`

Checkpoint 251: The exact finite denominator fibers

The exact finite denominator fibers. The definitions retain the denominator-one center and use actual least nonnegative unit representatives. All interval, denominator and recovery properties are proved,

without an assumed finite-index correspondence. This supplies arithmetic infrastructure for the low-symbol identity; no high-frequency estimate or headline theorem is assumed.

Checked declarations

Module	Named supporting theorems
CanonicalDenominatorFibers.lean	2

- `mem_canonicalDenominatorFiber`
- `unitRational_mem_fiber`

<!-- checkpoint-verification -->

Verification

All 318 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3925-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1331 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 320 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_252_UnitFiberBijection.md`

Checkpoint 252: The actual bijection between residue units and a rational fiber

For every positive q, the actual map u to val(u)/q is a bijection from residue units to the complete canonical denominator-q fiber. Surjectivity is proved by the rational numerator unit, injectivity by exact reduced numerators. No cardinality equality or missing denominator-one case is postulated. This bijection is the basis for the coming finite-sum reindexing.

Checked declarations

Module	Named supporting theorems
UnitFiberBijection.lean	2

- `unitRational_surjective_fiber`
- `unitFiberMap_bijective`

<!-- checkpoint-verification -->

Verification

All 319 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3926-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1333 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 321 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_253_UnitFiberReindexing.md`

Checkpoint 253: Exact finite-sum reindexing through the unit-fiber equivalence

Exact finite-sum reindexing through the unit-fiber equivalence. The correspondence uses the proved unit/rational bijection and retains the denominator-one center. Finite-sum and cardinality identities are proved for the full displayed sets, including empty global truncations. No analytic estimate or desired indexing identity is assumed.

Checked declarations

Module	Named supporting theorems
UnitFiberReindexing.lean	1

- `canonicalDenominatorFiber_sum`

<!-- checkpoint-verification -->

Verification

All 320 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3927-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1334 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 322 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_254_CanonicalFiberTruncation.md`

Checkpoint 254: Denominator fibers under the global rational truncation

Denominator fibers under the global rational truncation. The correspondence uses the proved unit/rational bijection and retains the denominator-one center. Finite-sum and cardinality identities are proved for the full displayed sets, including empty global truncations. No analytic estimate or desired indexing identity is assumed.

Checked declarations

Module	Named supporting theorems
CanonicalFiberTruncation.lean	2

- `canonicalDenominatorFiber_zero`
- `canonicalCenters_denominator_filter`

<!-- checkpoint-verification -->

Verification

All 321 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3928-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 1336 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 323 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_255_RationalDenominatorPartition.md`

Checkpoint 255: Exact partition of the canonical sum by denominator

Exact partition of the canonical sum by denominator. The correspondence uses the proved unit/rational bijection and retains the denominator-one center. Finite-sum and cardinality identities are proved for the full displayed sets, including empty global truncations. No analytic estimate or desired indexing identity is assumed.

Checked declarations

Module	Named supporting theorems
RationalDenominatorPartition.lean	1

- `canonicalCenters_sum_fibers`

<!-- checkpoint-verification -->

Verification

All 322 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3929-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1337 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 324 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_256_RationalUnitDoubleSum.md`

Checkpoint 256: The complete rational to denominator-unit double-sum identity

The complete rational to denominator-unit double-sum identity. The correspondence uses the proved unit/rational bijection and retains the denominator-one center. Finite-sum and cardinality identities are proved for the full displayed sets, including empty global truncations. No analytic estimate or desired indexing identity is assumed.

Checked declarations

Module	Named supporting theorems
RationalUnitDoubleSum.lean	2

- `denominatorUnitSum_pos`
- `canonicalCenters_sum_units`

<!-- checkpoint-verification -->

Verification

All 323 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3930-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1339 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 325 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_257_CanonicalUnitCardinality.md`

Checkpoint 257: Exact totient count of the canonical rational centers

Exact totient count of the canonical rational centers. The correspondence uses the proved unit/rational bijection and retains the denominator-one center. Finite-sum and cardinality identities are proved for the full displayed sets, including empty global truncations. No analytic estimate or desired indexing identity is assumed.

Checked declarations

Module	Named supporting theorems
CanonicalUnitCardinality.lean	2

- `canonicalDenominatorFiber_card`
- `canonicalCenters_card_totient`

<!-- checkpoint-verification -->

Verification

All 324 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3931-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1341 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 326 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_258_UnitRationalSourceCoordinates.md`

Checkpoint 258: Exact source frequency and numerator coordinates

Exact source frequency and numerator coordinates. These are actual rational and Dirichlet-character constructions, not substitute functions with assumed estimates. Both absence and presence of an exceptional character are represented. Existence of analytic exceptional data is a separate external input and is not asserted. The source correspondence retains every phase and normalization.

Checked declarations

Module	Named supporting theorems
UnitRationalSourceCoordinates.lean	2

- `unitRational_real`
- `unitRational_numerator_mod`

<!-- checkpoint-verification -->

Verification

All 325 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3932-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1343 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 327 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_259_SourceExceptionalSelector.md`

Checkpoint 259: The optional exceptional-conductor indicator with its actual norm

The optional exceptional-conductor indicator with its actual norm. These are actual rational and Dirichlet-character constructions, not substitute functions with assumed estimates. Both absence and presence of an exceptional character are represented. Existence of analytic exceptional data is a separate external input and is not asserted. The source correspondence retains every phase and normalization.

Checked declarations

Module	Named supporting theorems
SourceExceptionalSelector.lean	3

- `sourceExceptionalSelector_norm`
- `sourceExceptionalSelector_inactive`
- `sourceExceptionalSelector_active`

<!-- checkpoint-verification -->

Verification

All 326 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3933-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1346 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 328 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_260_SourceInducedCharacters.md`

Checkpoint 260: The induced character family at every rational denominator

The induced character family at every rational denominator. These are actual rational and Dirichlet-character constructions, not substitute functions with assumed estimates. Both absence and presence of an exceptional character are represented. Existence of analytic exceptional data is a separate external input and is not asserted. The source correspondence retains every phase and normalization.

Checked declarations

Module	Named supporting theorems
SourceInducedCharacters.lean	2

- `sourceInducedCharacter_of_dvd`
- `sourceInducedCharacter_of_not_dvd`

<!-- checkpoint-verification -->

Verification

All 327 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3934-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1348 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 329 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_261_SourceExceptionalUnitData.md`

Checkpoint 261: Exact induced-character and exceptional coefficient data on unit fractions

Exact induced-character and exceptional coefficient data on unit fractions. All cutoffs, source phases and the optional exceptional term retain their exact definitions. The broad-cutoff removal uses the already proved containment conditions Q<=R, Q*q0/B0<=R and 2R^4<=N^2. No analytic low conclusion is assumed; the remaining unit/kernel assembly and high-frequency estimates are distinct goals.

Checked declarations

Module	Named supporting theorems
SourceExceptionalUnitData.lean	3

- `sourceExceptionalSelector_unitRational`
- `sourceInducedCharacter_gauss_coordinates`
- `sourceInducedCharacter_unit_gauss`

<!-- checkpoint-verification -->

Verification

All 328 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3935-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1351 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 330 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_262_SourceCircleDoubleSum.md`

Checkpoint 262: Exact rational to unit indexing for the actual low circle model

Exact rational to unit indexing for the actual low circle model. All cutoffs, source phases and the optional exceptional term retain their exact definitions. The broad-cutoff removal uses the already proved containment conditions Q<=R, Q*q0/B0<=R and 2R^4<=N^2. No analytic low conclusion is assumed; the remaining unit/kernel assembly and high-frequency estimates are distinct goals.

Checked declarations

Module	Named supporting theorems
SourceCircleDoubleSum.lean	2

- `circleLowModel_denominator_sum`
- `sourceRationalCenters_sum_units`

<!-- checkpoint-verification -->

Verification

All 329 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3936-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1353 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 331 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_263_SourceLowCircleRealFormula.md`

Checkpoint 263: Removing the redundant broad cutoff in the actual source low symbol

Removing the redundant broad cutoff in the actual source low symbol. All cutoffs, source phases and the optional exceptional term retain their exact definitions. The broad-cutoff removal uses the already proved containment conditions Q<=R, Q*q0/B0<=R and 2R^4<=N^2. No analytic low conclusion is assumed; the remaining unit/kernel assembly and high-frequency estimates are distinct goals.

Checked declarations

Module	Named supporting theorems
SourceLowCircleRealFormula.lean	2

- `centeredLowSymbol_bare`
- `circleLowTerm_bare_real`

<!-- checkpoint-verification -->

Verification

All 330 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3937-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1355 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 332 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_264_CompactLiftSourceAlgebra.md`

Checkpoint 264: Subtraction and zero in the compact circle lift

Subtraction and zero in the compact circle lift. All cutoffs, source phases and the optional exceptional term retain their exact definitions. The broad-cutoff removal uses the already proved containment conditions Q<=R, Q*q0/B0<=R and 2R^4<=N^2. No analytic low conclusion is assumed; the remaining unit/kernel assembly and high-frequency estimates are distinct goals.

Checked declarations

Module	Named supporting theorems
CompactLiftSourceAlgebra.lean	2

- `compactCircleLift_zero`
- `compactCircleLift_sub`

<!-- checkpoint-verification -->

Verification

All 331 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3938-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1357 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 333 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

Checkpoint_265_FourierKernelOperatorEquality.md

Checkpoint 265: Equality of actual l2 and l1 operators from their inverse kernels

Equality of actual l2 and l1 operators from their inverse kernels. The smoothing width uses the real cutoff Q, retaining its precise value. The width budget is 2Q<=N^2, the scale is positive, and all positive reduced denominators are included. Every Fourier normalization and circle shift is retained. The new identities do not assert any unproved high estimate or headline conclusion.

Checked declarations

Module	Named supporting theorems
FourierKernelOperatorEquality.lean	2

- `integerMultiplier_eq_of_kernel`
- `fourierL1Operator_eq_of_kernel`

<!-- checkpoint-verification -->

Verification

All 332 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3939-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1359 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 334 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_266_CircleShiftAlgebra.md

Checkpoint 266: Exact linear algebra for shifted circle symbols

Exact linear algebra for shifted circle symbols. The smoothing width uses the real cutoff Q, retaining its precise value. The width budget is 2Q<=N^2, the scale is positive, and all positive reduced denominators are included. Every Fourier normalization and circle shift is retained. The new identities do not assert any unproved high estimate or headline conclusion.

Checked declarations

Module	Named supporting theorems
CircleShiftAlgebra.lean	3

- `circleFrequencyShift_add`
- `circleFrequencyShift_sub`
- `circleFrequencyShift_smul`

<!-- checkpoint-verification -->

Verification

All 333 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3940-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 1362 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 335 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint notes 267-341

Checkpoint_267_OrdinaryCenteredLowSource.md

Checkpoint 267: The actual centered ordinary low symbol with a real cutoff

The actual centered ordinary low symbol with a real cutoff. The smoothing width uses the real cutoff Q, retaining its precise value. The width budget is 2Q<=N^2, the scale is positive, and all positive reduced denominators are included. Every Fourier normalization and circle shift is retained. The new identities do not assert any unproved high estimate or headline conclusion.

Checked declarations

Module	Named supporting theorems
OrdinaryCenteredLowSource.lean	2

- `ordinaryCenteredLow_source`
- `ordinaryCenteredLow_support`

<!-- checkpoint-verification -->

Verification

All 334 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3941-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1364 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 336 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_268_OrdinaryRationalCircleSource.md

Checkpoint 268: The ordinary rational circle term from its actual centered source

The ordinary rational circle term from its actual centered source. The smoothing width uses the real cutoff Q, retaining its precise value. The width budget is 2Q<=N^2, the scale is positive, and all positive reduced denominators are included. Every Fourier normalization and circle shift is retained. The new identities do not assert any unproved high estimate or headline conclusion.

Checked declarations

Module	Named supporting theorems
OrdinaryRationalCircleSource.lean	1

- `ordinaryCenteredLow_circle`

<!-- checkpoint-verification -->

Verification

All 335 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3942-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1365 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 337 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_269_OrdinaryRationalUnitSymbol.md`

Checkpoint 269: Exact ordinary circle symbols under the rational-unit correspondence

Exact ordinary circle symbols under the rational-unit correspondence. The cutoff Q is real and is retained exactly in every smoothing width. The floor is used solely to enumerate integer denominators satisfying q<=Q. All sums include q=1 and the actual Gauss coefficients and rational phases. The norm estimate and source-model assembly are proved in subsequent stages, with no analytic conclusion assumed here.

Checked declarations

Module	Named supporting theorems
OrdinaryRationalUnitSymbol.lean	2

- `ordinaryRationalCircleLow_coordinates`
- `ordinaryRationalCircleLow_unit`

<!-- checkpoint-verification -->

Verification

All 336 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3943-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1367 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 338 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_270_RealOrdinaryCircleTerm.md`

Checkpoint 270: The denominator-unit ordinary term for a real cutoff

The denominator-unit ordinary term for a real cutoff. The cutoff Q is real and is retained exactly in every smoothing width. The floor is used solely to enumerate integer denominators satisfying q<=Q. All sums include q=1 and the actual Gauss coefficients and rational phases. The norm estimate and source-model assembly are proved in subsequent stages, with no analytic conclusion assumed here.

Checked declarations

Module	Named supporting theorems
RealOrdinaryCircleTerm.lean	1

- `denominatorUnitSum_ordinary`

<!-- checkpoint-verification -->

Verification

All 337 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3944-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1368 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 339 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_271_RealOrdinaryTermKernel.md`

Checkpoint 271: The inverse kernel of the real-cutoff ordinary term

The inverse kernel of the real-cutoff ordinary term. The cutoff Q is real and is retained exactly in every smoothing width. The floor is used solely to enumerate integer denominators satisfying q<=Q. All sums include q=1 and the actual Gauss coefficients and rational phases. The norm estimate and source-model assembly are proved in subsequent stages, with no analytic conclusion assumed here.

Checked declarations

Module	Named supporting theorems
RealOrdinaryTermKernel.lean	1

- `realOrdinaryCircleTerm_kernel`

<!-- checkpoint-verification -->

Verification

All 338 project modules and the root were freshly rebuilt: 3945 jobs pass without warnings/errors. All 1369 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 340 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_272_RealOrdinaryLowModel.md`

Checkpoint 272: The actual ordinary circle low model at a real cutoff

The actual ordinary circle low model at a real cutoff. The cutoff Q is real and is retained exactly in every smoothing width. The floor is used solely to enumerate integer denominators satisfying q<=Q. All sums include q=1 and the actual Gauss coefficients and rational phases. The norm estimate and source-model assembly are proved in subsequent stages, with no analytic conclusion assumed here.

Checked declarations

Module	Named supporting theorems
RealOrdinaryLowModel.lean	1

- `realOrdinaryLowModel_kernel`

<!-- checkpoint-verification -->

Verification

All 339 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3946-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1370 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 341 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_273_RealCutoffFloorBudget.md`

Checkpoint 273: Exact real-cutoff floors and logarithmic arithmetic budgets

Exact real-cutoff floors and logarithmic arithmetic budgets. The cutoff Q is real and is retained exactly in every smoothing width. The floor is used solely to enumerate integer denominators satisfying q<=Q. All sums include q=1 and the actual Gauss coefficients and rational phases. The norm estimate and source-model assembly are proved in subsequent stages, with no analytic conclusion assumed here.

Checked declarations

Module	Named supporting theorems
RealCutoffFloorBudget.lean	3

- `floor_cutoff_two`
- `floor_cutoff_log`
- `real_cutoff_denominator_bound`

<!-- checkpoint-verification -->

Verification

All 340 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3947-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1373 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 342 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

Checkpoint_274_RealOrdinaryKernelDecay.md

Checkpoint 274: The logarithmic ordinary kernel decay with the actual real cutoff

The logarithmic ordinary kernel decay with the actual real cutoff. For N>=2 and real Q>=2 the actual ordinary inverse kernel is bounded by an absolute constant times Q log(2Q)/N^2 times the tenth-order spatial decay profile. Integer denominators are enumerated by floor(Q); Q is retained unchanged in the smoothing widths. The arithmetic logarithmic estimate is already proved without PNT or Mertens. The exceptional and source assemblies are separate obligations.

Checked declarations

Module	Named supporting theorems
RealOrdinaryKernelDecay.lean	1

- `realOrdinaryPhysicalKernel_decay`

<!-- checkpoint-verification -->

Verification

All 341 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3948-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1374 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 343 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_275_RealLowAmplitude.md

Checkpoint 275: Uniform real-cutoff amplitudes and scale monotonicity

Uniform real-cutoff amplitudes and scale monotonicity. For N>=2 and real Q>=2 the actual ordinary inverse kernel is bounded by an absolute constant times Q log(2Q)/N^2 times the tenth-order spatial decay profile. Integer denominators are enumerated by floor(Q); Q is retained unchanged in the smoothing widths. The arithmetic logarithmic estimate is already proved without PNT or Mertens. The exceptional and source assemblies are separate obligations.

Checked declarations

Module	Named supporting theorems
RealLowAmplitude.lean	2

- `realLowAmplitude_nonneg`
- `realLowAmplitude_scale_mono`

<!-- checkpoint-verification -->

Verification

All 342 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3949-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 1376 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 344 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_276_RealOrdinaryAmplitudeBound.md`

Checkpoint 276: Rapid-decay and uniform ordinary low-kernel bounds at real Q

Rapid-decay and uniform ordinary low-kernel bounds at real Q. For N>=2 and real Q>=2 the actual ordinary inverse kernel is bounded by an absolute constant times Q log(2Q)/N^2 times the tenth-order spatial decay profile. Integer denominators are enumerated by floor(Q); Q is retained unchanged in the smoothing widths. The arithmetic logarithmic estimate is already proved without PNT or Mertens. The exceptional and source assemblies are separate obligations.

Checked declarations

Module	Named supporting theorems
RealOrdinaryAmplitudeBound.lean	2

- `realOrdinaryPhysicalKernel_amplitude`
- `realOrdinaryPhysicalKernel_uniform`

<!-- checkpoint-verification -->

Verification

All 343 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3950-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1378 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 345 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_277_RealOrdinaryLocalMass.md`

Checkpoint 277: The real-cutoff ordinary local convolution estimate

The real-cutoff ordinary local convolution estimate. The actual circle symbol and its proved inverse kernel are used. The finite local result holds for arbitrary finite complex inputs, and the l1-to-linfinity norm and countable maximum cover all l1 inputs. N>=2, real Q>=2 and the explicit 2Q<=N^2 width budget are retained. No desired low estimate is assumed, and no new deep theorem is used.

Checked declarations

Module	Named supporting theorems
RealOrdinaryLocalMass.lean	1

- `realOrdinaryPhysicalKernel_local`

<!-- checkpoint-verification -->

Verification

All 344 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3951-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1379 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 346 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_278_RealOrdinaryL1Bound.md`

Checkpoint 278: The real-cutoff ordinary all-input l1 operator bound

The real-cutoff ordinary all-input l1 operator bound. The actual circle symbol and its proved inverse kernel are used. The finite local result holds for arbitrary finite complex inputs, and the l1-to-linfinity norm and countable maximum cover all l1 inputs. N>=2, real Q>=2 and the explicit 2Q<=N^2 width budget are retained. No desired low estimate is assumed, and no new deep theorem is used.

Checked declarations

Module	Named supporting theorems
RealOrdinaryL1Bound.lean	3

- `realOrdinaryLowModel_kernel_uniform`
- `realOrdinaryLowModel_L1_norm`
- `realOrdinaryLowModel_L1_bound`

<!-- checkpoint-verification -->

Verification

All 345 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3952-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1382 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 347 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_279_RealOrdinaryLocalOperator.md`

Checkpoint 279: The real-cutoff ordinary local Fourier multiplier estimate

The real-cutoff ordinary local Fourier multiplier estimate. The actual circle symbol and its proved inverse kernel are used. The finite local result holds for arbitrary finite complex inputs, and the l1-to-linfinity norm and countable maximum cover all l1 inputs. N>=2, real Q>=2 and the explicit 2Q<=N^2 width budget are retained. No desired low estimate is assumed, and no new deep theorem is used.

Checked declarations

Module	Named supporting theorems
RealOrdinaryLocalOperator.lean	2

- `realOrdinaryLowModel_finite_convolution`
- `realOrdinaryLowModel_local`

<!-- checkpoint-verification -->

Verification

All 346 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3953-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1384 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 348 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_280_RealOrdinaryMaximalFamily.md`

Checkpoint 280: The genuine countable ordinary low maximum for real Q

The genuine countable ordinary low maximum for real Q. The actual circle symbol and its proved inverse kernel are used. The finite local result holds for arbitrary finite complex inputs, and the l1-to-linfinity norm and countable maximum cover all l1 inputs. N>=2, real Q>=2 and the explicit 2Q<=N^2 width budget are retained. No desired low estimate is assumed, and no new deep theorem is used.

Checked declarations

Module	Named supporting theorems
RealOrdinaryMaximalFamily.lean	1

- `realOrdinaryLow_countable_maximal`

<!-- checkpoint-verification -->

Verification

All 347 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3954-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1385 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 349 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_281_OrdinaryOnlySourceSymbol.md`

Checkpoint 281: The exact source low symbol in the absence of an exception

The exact source low symbol in the absence of an exception. The original rational-indexed circleLowModel is used, with its original broad cutoff and exact phase. Its equality with the denominator/unit model is proved when the exceptional coefficient is zero, including all integer lifts and the denominator-one term. The real cutoff is retained in the smoothing width. Under N>=2, Q>=2, Q<=R and 2R^4<=N^2 the low kernel and operator estimates are deduced without any new analytic input. The active exceptional source bridge and sharp high estimates remain separate obligations.

Checked declarations

Module	Named supporting theorems
OrdinaryOnlySourceSymbol.lean	2

- `rationalLowModelTerm_no_exception`
- `centeredLowSymbol_no_exception`

<!-- checkpoint-verification -->

Verification

All 348 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3955-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1387 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 350 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_282_CompactLiftExtensionality.md`

Checkpoint 282: Extensionality of the actual compact circle lift

Extensionality of the actual compact circle lift. The original rational-indexed circleLowModel is used, with its original broad cutoff and exact phase. Its equality with the denominator/unit model is proved when the exceptional coefficient is zero, including all integer lifts and the denominator-one term. The real cutoff is retained in the smoothing width. Under N>=2, Q>=2, Q<=R and 2R^4<=N^2 the low kernel and operator estimates are deduced without any new analytic input. The active exceptional source bridge and sharp high estimates remain separate obligations.

Checked declarations

Module	Named supporting theorems
CompactLiftExtensionality.lean	3

- `compactCircleLift_congr`
- `circleFrequencyShift_zero`

- `shiftedAnnularLift_smul`

<!-- checkpoint-verification -->

Verification

All 349 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3956-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1390 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 351 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_283_OrdinaryOnlySourceCircle.md`

Checkpoint 283: The exact ordinary-only source circle term

The exact ordinary-only source circle term. The original rational-indexed circleLowModel is used, with its original broad cutoff and exact phase. Its equality with the denominator/unit model is proved when the exceptional coefficient is zero, including all integer lifts and the denominator-one term. The real cutoff is retained in the smoothing width. Under N>=2, Q>=2, Q<=R and 2R^4<=N^2 the low kernel and operator estimates are deduced without any new analytic input. The active exceptional source bridge and sharp high estimates remain separate obligations.

Checked declarations

Module	Named supporting theorems
OrdinaryOnlySourceCircle.lean	1

- `circleLowTerm_no_exception`

<!-- checkpoint-verification -->

Verification

All 350 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3957-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1391 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 352 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_284_RealCutoffCanonicalFilter.md`

Checkpoint 284: Exact denominator filtering for real source cutoffs

Exact denominator filtering for real source cutoffs. The original rational-indexed circleLowModel is used, with its original broad cutoff and exact phase. Its equality with the denominator/unit model is proved

when the exceptional coefficient is zero, including all integer lifts and the denominator-one term. The real cutoff is retained in the smoothing width. Under N>=2, Q>=2, Q<=R and 2R^4<=N^2 the low kernel and operator estimates are deduced without any new analytic input. The active exceptional source bridge and sharp high estimates remain separate obligations.

Checked declarations

Module	Named supporting theorems
RealCutoffCanonicalFilter.lean	2

- `canonicalCenters_real_filter`
- `sourceCenters_real_filter`

<!-- checkpoint-verification -->

Verification

All 351 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3958-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1393 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 353 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_285_SourceOrdinaryLowIdentity.md`

Checkpoint 285: The full source ordinary-only low-symbol identity at real Q

The full source ordinary-only low-symbol identity at real Q. The original rational-indexed circleLowModel is used, with its original broad cutoff and exact phase. Its equality with the denominator/unit model is proved when the exceptional coefficient is zero, including all integer lifts and the denominator-one term. The real cutoff is retained in the smoothing width. Under N>=2, Q>=2, Q<=R and 2R^4<=N^2 the low kernel and operator estimates are deduced without any new analytic input. The active exceptional source bridge and sharp high estimates remain separate obligations.

Checked declarations

Module	Named supporting theorems
SourceOrdinaryLowIdentity.lean	1

- `sourceLowModel_no_exception`

<!-- checkpoint-verification -->

Verification

All 352 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3959-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1394 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 354 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_286_SourceOrdinaryLowBounds.md

Checkpoint 286: The actual source ordinary-only low norm and local estimates

The actual source ordinary-only low norm and local estimates. The original rational-indexed circleLowModel is used, with its original broad cutoff and exact phase. Its equality with the denominator/unit model is proved when the exceptional coefficient is zero, including all integer lifts and the denominator-one term. The real cutoff is retained in the smoothing width. Under N>=2, Q>=2, Q<=R and 2R^4<=N^2 the low kernel and operator estimates are deduced without any new analytic input. The active exceptional source bridge and sharp high estimates remain separate obligations.

Checked declarations

Module	Named supporting theorems
SourceOrdinaryLowBounds.lean	2

- `sourceLowModel_no_exception_L1`
- `sourceLowModel_no_exception_local`

<!-- checkpoint-verification -->

Verification

All 353 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3960-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1396 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 355 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_287_ConductorQuotientCoordinates.md

Checkpoint 287: Exact real and natural conductor quotients and cutoffs

Exact real and natural conductor quotients and cutoffs. The conductor is positive, all quotient denominators are genuine natural quotients of multiples, and the exact real cutoff is retained. The set records both B0*d<=Q and q0*d<=floor(R), including their equality boundaries. B0 is the previously proved conductor majorant. No replacement of an arithmetic or source identity by a hypothesis is made.

Checked declarations

Module	Named supporting theorems
ConductorQuotientCoordinates.lean	3

- `conductor_quotient_real`
- `conductor_quotient_positive`
- `conductor_quotient_cutoff`

<!-- checkpoint-verification -->

Verification

All 354 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3961-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1399 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 356 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_288_ExceptionalDenominatorSet.md`

Checkpoint 288: The exact finite set of contributing exceptional quotient denominators

The exact finite set of contributing exceptional quotient denominators. The conductor is positive, all quotient denominators are genuine natural quotients of multiples, and the exact real cutoff is retained. The set records both B0*d<=Q and q0*d<=floor(R), including their equality boundaries. B0 is the previously proved conductor majorant. No replacement of an arithmetic or source identity by a hypothesis is made.

Checked declarations

Module	Named supporting theorems
ExceptionalDenominatorSet.lean	2

- `mem_realExceptionalDenominators`
- `realExceptionalDenominators_subset`

<!-- checkpoint-verification -->

Verification

All 355 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3962-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1401 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 357 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_289_ExceptionalMultipleReindexing.md`

Checkpoint 289: Exact reindexing of conductor multiples and both cutoff boundaries

Exact reindexing of conductor multiples and both cutoff boundaries. The conductor is positive, all quotient denominators are genuine natural quotients of multiples, and the exact real cutoff is retained. The set records both B0*d<=Q and q0*d<=floor(R), including their equality boundaries. B0 is the previously proved conductor majorant. No replacement of an arithmetic or source identity by a hypothesis is made.

Checked declarations

Module	Named supporting theorems
ExceptionalMultipleReindexing.lean	1

- `conductor_multiple_reindex`

<!-- checkpoint-verification -->

Verification

All 356 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3963-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1402 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 358 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_290_ExceptionalCenteredLow.md`

Checkpoint 290: The actual induced exceptional centered cutoff symbol

The actual induced exceptional centered cutoff symbol. The conductor is positive, all quotient denominators are genuine natural quotients of multiples, and the exact real cutoff is retained. The set records both B0*d<=Q and q0*d<=floor(R), including their equality boundaries. B0 is the previously proved conductor majorant. No replacement of an arithmetic or source identity by a hypothesis is made.

Checked declarations

Module	Named supporting theorems
ExceptionalCenteredLow.lean	1

- `exceptionalCenteredLow_of_not_dvd`

<!-- checkpoint-verification -->

Verification

All 357 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3964-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1403 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 359 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_291_ExceptionalCenteredSource.md`

Checkpoint 291: The exact source exceptional sign and real smoothing width

The exact source exceptional sign and real smoothing width. The exceptional character is the actual character induced from the positive base level. The optional source coefficient, the negative exceptional sign and the real/natural quotient identity are retained. Compact support and the circle lift are proved with the actual cutoff and width budget 2Q<=N^2. No conductor classification, progression estimate or desired low bound is postulated.

Checked declarations

Module	Named supporting theorems
ExceptionalCenteredSource.lean	1

- `exceptionalCenteredLow_source`

<!-- checkpoint-verification -->

Verification

All 358 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3965-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1404 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 360 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_292_ExceptionalCenteredSupport.md`

Checkpoint 292: Actual compact support of the induced exceptional centered term

Actual compact support of the induced exceptional centered term. The exceptional character is the actual character induced from the positive base level. The optional source coefficient, the negative exceptional sign and the real/natural quotient identity are retained. Compact support and the circle lift are proved with the actual cutoff and width budget 2Q<=N^2. No conductor classification, progression estimate or desired low bound is postulated.

Checked declarations

Module	Named supporting theorems
ExceptionalCenteredSupport.lean	1

- `exceptionalCenteredLow_support`

<!-- checkpoint-verification -->

Verification

All 359 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3966-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1405 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 361 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_293_ExceptionalRationalCircle.md`

Checkpoint 293: The induced exceptional rational circle multiplier

The induced exceptional rational circle multiplier. The exceptional character is the actual character induced from the positive base level. The optional source coefficient, the negative exceptional sign and the real/natural quotient identity are retained. Compact support and the circle lift are proved with the actual cutoff and width budget 2Q<=N^2. No conductor classification, progression estimate or desired low bound is postulated.

Checked declarations

Module	Named supporting theorems
ExceptionalRationalCircle.lean	2

- `exceptionalRationalCircleLow_not_dvd`
- `exceptionalRationalCircleLow_dvd`

<!-- checkpoint-verification -->

Verification

All 360 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3967-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1407 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 362 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_294_ExceptionalRationalCoordinates.md`

Checkpoint 294: Exact exceptional unit and rational coordinates

Exact exceptional unit and rational coordinates. The character is induced from the fixed positive base level, the quotient d is a natural conductor quotient, and the complex Gauss sum and phase are unchanged. The cutoff Q remains real. The source truncation records q0*d<=floor(R) and B0*d<=Q exactly. Kernel identities require no quadratic character or analytic estimate; those hypotheses only enter later norm estimates.

Checked declarations

Module	Named supporting theorems
ExceptionalRationalCoordinates.lean	2

- `exceptionalRationalCircleLow_coordinates`

- `exceptionalRationalCircleLow_unit`

<!-- checkpoint-verification -->

Verification

All 361 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3968-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1409 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 363 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_295_RealExceptionalCircleTerm.md`

Checkpoint 295: The actual denominator-unit exceptional circle term

The actual denominator-unit exceptional circle term. The character is induced from the fixed positive base level, the quotient d is a natural conductor quotient, and the complex Gauss sum and phase are unchanged. The cutoff Q remains real. The source truncation records q0*d<=floor(R) and B0*d<=Q exactly. Kernel identities require no quadratic character or analytic estimate; those hypotheses only enter later norm estimates.

Checked declarations

Module	Named supporting theorems
RealExceptionalCircleTerm.lean	2

- `realExceptionalCircleTerm_dvd`
- `realExceptionalCircleTerm_not_dvd`

<!-- checkpoint-verification -->

Verification

All 362 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3969-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1411 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 364 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_296_RealExceptionalTermKernel.md`

Checkpoint 296: The exceptional inverse kernel with its exact conductor quotient

The exceptional inverse kernel with its exact conductor quotient. The character is induced from the fixed positive base level, the quotient d is a natural conductor quotient, and the complex Gauss sum and phase

are unchanged. The cutoff Q remains real. The source truncation records q0*d<=floor(R) and B0*d<=Q exactly. Kernel identities require no quadratic character or analytic estimate; those hypotheses only enter later norm estimates.

Checked declarations

Module	Named supporting theorems
RealExceptionalTermKernel.lean	1

- `realExceptionalCircleTerm_kernel`

<!-- checkpoint-verification -->

Verification

All 363 project modules and the root were freshly rebuilt: 3970 jobs pass without warnings/errors. All 1412 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 365 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_297_RealExceptionalLowModel.md

Checkpoint 297: The real-cutoff induced exceptional low model and its inverse kernel

The real-cutoff induced exceptional low model and its inverse kernel. The character is induced from the fixed positive base level, the quotient d is a natural conductor quotient, and the complex Gauss sum and phase are unchanged. The cutoff Q remains real. The source truncation records q0*d<=floor(R) and B0*d<=Q exactly. Kernel identities require no quadratic character or analytic estimate; those hypotheses only enter later norm estimates.

Checked declarations

Module	Named supporting theorems
RealExceptionalLowModel.lean	1

- `realExceptionalLowModel_kernel`

<!-- checkpoint-verification -->

Verification

All 364 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3971-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1413 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 366 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypothe-

ses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_298_RealExceptionalKernelDecay.md

Checkpoint 298: Uniform induced exceptional kernel decay for real cutoffs

Uniform induced exceptional kernel decay for real cutoffs. For a quadratic base character and beta<=1 the actual induced exceptional inverse kernel satisfies the same Q log(2Q)/N^2 rapid-decay envelope, uniformly in the global denominator truncation. The conductor majorant is cancelled by the real smoothing width. The proof uses the already checked elementary weighted correlation sum, not PNT, Mertens or an assumed analytic low estimate.

Checked declarations

Module	Named supporting theorems
RealExceptionalKernelDecay.lean	1

- `realExceptionalPhysicalKernel_decay`

<!-- checkpoint-verification -->

Verification

All 365 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3972-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1414 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 367 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_299_RealExceptionalAmplitude.md

Checkpoint 299: The real-cutoff exceptional amplitude and l1 operator norm

The real-cutoff exceptional amplitude and l1 operator norm. For a quadratic base character and beta<=1 the actual induced exceptional inverse kernel satisfies the same Q log(2Q)/N^2 rapid-decay envelope, uniformly in the global denominator truncation. The conductor majorant is cancelled by the real smoothing width. The proof uses the already checked elementary weighted correlation sum, not PNT, Mertens or an assumed analytic low estimate.

Checked declarations

Module	Named supporting theorems
RealExceptionalAmplitude.lean	3

- `realExceptionalPhysicalKernel_amplitude`
- `realExceptionalLowModel_kernel_uniform`
- `realExceptionalLowModel_L1_norm`

<!-- checkpoint-verification -->

Verification

All 366 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3973-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1417 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 368 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_300_RealExceptionalLocal.md`

Checkpoint 300: The real-cutoff exceptional local convolution estimate

The real-cutoff exceptional local convolution estimate. For a quadratic base character and beta<=1 the actual induced exceptional inverse kernel satisfies the same Q log(2Q)/N^2 rapid-decay envelope, uniformly in the global denominator truncation. The conductor majorant is cancelled by the real smoothing width. The proof uses the already checked elementary weighted correlation sum, not PNT, Mertens or an assumed analytic low estimate.

Checked declarations

Module	Named supporting theorems
RealExceptionalLocal.lean	1

- `realExceptionalPhysicalKernel_local`

<!-- checkpoint-verification -->

Verification

All 367 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3974-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1418 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 369 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_301_SourceLowCenteredIdentity.md`

Checkpoint 301: The full centered source low identity with an optional induced exception

The full centered source low identity with an optional induced exception. Both the ordinary and induced exceptional terms are identified as actual continuous symbols. The optional exceptional coefficient is zero or the exact conductor divisibility indicator. All broad/narrow cutoffs and circle shifts are proved compatible under Q<=R, Q*q0/B0<=R and 2R^4<=N^2, with the real cutoff unchanged. This is an identity, not an assumed low norm or analytic progression statement.

Checked declarations

Module	Named supporting theorems
SourceLowCenteredIdentity.lean	1

- `sourceCenteredLow_identity`

<!-- checkpoint-verification -->

Verification

All 368 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3975-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1419 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 370 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_302_SourceLowCircleIdentity.md`

Checkpoint 302: The full source circle low term and its exact exceptional sign

The full source circle low term and its exact exceptional sign. Both the ordinary and induced exceptional terms are identified as actual continuous symbols. The optional exceptional coefficient is zero or the exact conductor divisibility indicator. All broad/narrow cutoffs and circle shifts are proved compatible under Q<=R, Q*q0/B0<=R and 2R^4<=N^2, with the real cutoff unchanged. This is an identity, not an assumed low norm or analytic progression statement.

Checked declarations

Module	Named supporting theorems
SourceLowCircleIdentity.lean	2

- `shiftedCompactLift_ite`
- `sourceCircleLowTerm_identity`

<!-- checkpoint-verification -->

Verification

All 369 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3976-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1421 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 371 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

Checkpoint_303_ExceptionalUnitCutoffFilter.md

Checkpoint 303: Exact exceptional cutoff filtering in the unit sum

Exact exceptional cutoff filtering in the unit sum. The original rational-indexed source circle symbol is identified with the ordinary denominator-unit model minus the optional induced exceptional model. Every denominator and conductor quotient is reindexed through proved bijections. Real Q and R, equality cutoff boundaries, the denominator-one term, all circle lifts and the exceptional negative sign are retained. Range conditions Q<=R, Q*q0/B0<=R and 2R^4<=N^2 are explicit; existence of analytic exceptional data is not assumed as a new axiom.

Checked declarations

Module	Named supporting theorems
ExceptionalUnitCutoffFilter.lean	2

- `exceptionalUnitSum_filter`
- `exceptionalUnitSum_divisor_filter`

<!-- checkpoint-verification -->

Verification

All 370 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3977-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1423 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 372 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_304_SourceExceptionalFiniteSum.md

Checkpoint 304: The complete source exceptional finite sum and conductor reindexing

The complete source exceptional finite sum and conductor reindexing. The original rational-indexed source circle symbol is identified with the ordinary denominator-unit model minus the optional induced exceptional model. Every denominator and conductor quotient is reindexed through proved bijections. Real Q and R, equality cutoff boundaries, the denominator-one term, all circle lifts and the exceptional negative sign are retained. Range conditions Q<=R, Q*q0/B0<=R and 2R^4<=N^2 are explicit; existence of analytic exceptional data is not assumed as a new axiom.

Checked declarations

Module	Named supporting theorems
SourceExceptionalFiniteSum.lean	2

- `canonicalExceptionalSum`
- `canonicalExceptionalSum_active`

<!-- checkpoint-verification -->

Verification

All 371 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3978-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1425 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 373 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_305_SourceFullLowIdentity.md`

Checkpoint 305: The exact original source low symbol for real Q and optional exceptions

The exact original source low symbol for real Q and optional exceptions. The original rational-indexed source circle symbol is identified with the ordinary denominator-unit model minus the optional induced exceptional model. Every denominator and conductor quotient is reindexed through proved bijections. Real Q and R, equality cutoff boundaries, the denominator-one term, all circle lifts and the exceptional negative sign are retained. Range conditions Q<=R, Q*q0/B0<=R and 2R^4<=N^2 are explicit; existence of analytic exceptional data is not assumed as a new axiom.

Checked declarations

Module	Named supporting theorems
SourceFullLowIdentity.lean	3

- `canonicalOrdinarySum`
- `truncatedSourceLow_identity`
- `sourceFullLow_identity`

<!-- checkpoint-verification -->

Verification

All 372 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3979-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1428 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 374 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_306_RealCombinedLowKernel.md`

Checkpoint 306: The actual combined real-cutoff low symbol and its inverse kernel

The actual combined real-cutoff low symbol and its inverse kernel. Ordinary and optional induced exceptional kernels are combined with the exact subtraction sign. For N>=2, real Q>=2, a quadratic

base character, beta<=1 and 2Q<=N^2, the kernel has a uniform 2*C_low*Q*log(2Q)/N^2 rapid-decay bound, and the finite local estimate has constant 8*C_low*Q*log(2Q). The l1-to-linfinity operator estimate holds for all l1 inputs and is uniform in the global denominator truncation. No deep analytic input is used.

Checked declarations

Module	Named supporting theorems
RealCombinedLowKernel.lean	1

- `realCombinedLowModel_kernel`

<!-- checkpoint-verification -->

Verification

All 373 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3980-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1429 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 375 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_307_RealCombinedLowBounds.md`

Checkpoint 307: The complete real-cutoff low decay and l1 operator bounds

The complete real-cutoff low decay and l1 operator bounds. Ordinary and optional induced exceptional kernels are combined with the exact subtraction sign. For N>=2, real Q>=2, a quadratic base character, beta<=1 and 2Q<=N^2, the kernel has a uniform 2*C_low*Q*log(2Q)/N^2 rapid-decay bound, and the finite local estimate has constant 8*C_low*Q*log(2Q). The l1-to-linfinity operator estimate holds for all l1 inputs and is uniform in the global denominator truncation. No deep analytic input is used.

Checked declarations

Module	Named supporting theorems
RealCombinedLowBounds.lean	3

- `realCombinedPhysicalKernel_decay`
- `realCombinedLowModel_uniform`
- `realCombinedLowModel_L1_norm`

<!-- checkpoint-verification -->

Verification

All 374 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3981-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1432 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 376 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_308_RealCombinedLowLocal.md

Checkpoint 308: The complete real-cutoff low local Fourier estimate

The complete real-cutoff low local Fourier estimate. Ordinary and optional induced exceptional kernels are combined with the exact subtraction sign. For N>=2, real Q>=2, a quadratic base character, beta<=1 and 2Q<=N^2, the kernel has a uniform 2*C_low*Q*log(2Q)/N^2 rapid-decay bound, and the finite local estimate has constant 8*C_low*Q*log(2Q). The l1-to-linfinity operator estimate holds for all l1 inputs and is uniform in the global denominator truncation. No deep analytic input is used.

Checked declarations

Module	Named supporting theorems
RealCombinedLowLocal.lean	2

- `realCombinedPhysicalKernel_local`
- `realCombinedLowModel_local`

<!-- checkpoint-verification -->

Verification

All 375 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3982-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1434 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 377 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_309_SourceLowFamilies.md

Checkpoint 309: Actual source low families and their genuine countable l1 maximum

Actual source low families and their genuine countable l1 maximum. The source data structure contains only numerical, character and cutoff compatibility conditions, never the desired norm conclusion. Its circle symbol is the original rational-indexed low model with optional induced exceptional data. The proved all-input l1 bound gives a genuine countable maximum, allowing varying conductor, character, beta and radius at later scales. Source range conversion keeps the global denominator cap fixed while enlarging only the working geometry.

Checked declarations

Module	Named supporting theorems
SourceLowFamilies.lean	3

- `sourceLowData_symbol_eq`
- `sourceLowData_L1_norm`
- `sourceLowFamily_countable_maximal`

<!-- checkpoint-verification -->

Verification

All 376 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3983-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1437 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 378 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_310_SourceLowRangeBudget.md`

Checkpoint 310: Source scale budgets for the full conductor range

Source scale budgets for the full conductor range. The source data structure contains only numerical, character and cutoff compatibility conditions, never the desired norm conclusion. Its circle symbol is the original rational-indexed low model with optional induced exceptional data. The proved all-input l1 bound gives a genuine countable maximum, allowing varying conductor, character, beta and radius at later scales. Source range conversion keeps the global denominator cap fixed while enlarging only the working geometry.

Checked declarations

Module	Named supporting theorems
SourceLowRangeBudget.lean	4

- `radius_le_square`
- `source_squared_low_width`
- `source_exceptional_squared_radius`
- `source_fourth_budget_to_eighth`

<!-- checkpoint-verification -->

Verification

All 377 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3984-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1441 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 379 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_311_SourceLowSquaredRange.md`

Checkpoint 311: Keeping the source denominator cap while using the squared working radius

Keeping the source denominator cap while using the squared working radius. The extra Q*q0/B0<=R restriction is removed. For every positive conductor, the original source cutoff R is retained; only the

working geometric radius is enlarged to R^2. If the conductor exceeds R, its actual source coefficients and exceptional denominator set vanish. Under Q>=2, Q<=R and 2R^8<=N^2, the original low symbol has its full finite-input local estimate and all-input l1-to-linfinity norm bound. The paper range R^4<=N/2 implies the new geometric budget. Sharp high/multifrequency estimates and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceLowSquaredRange.lean	1

- `sourceFullLow_squared_level`

<!-- checkpoint-verification -->

Verification

All 378 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3985-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1442 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 380 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_312_ConductorOutsideSource.md`

Checkpoint 312: Exact disappearance of exceptional terms beyond the source cutoff

Exact disappearance of exceptional terms beyond the source cutoff. The extra Q*q0/B0<=R restriction is removed. For every positive conductor, the original source cutoff R is retained; only the working geometric radius is enlarged to R^2. If the conductor exceeds R, its actual source coefficients and exceptional denominator set vanish. Under Q>=2, Q<=R and 2R^8<=N^2, the original low symbol has its full finite-input local estimate and all-input l1-to-linfinity norm bound. The paper range R^4<=N/2 implies the new geometric budget. Sharp high/multifrequency estimates and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ConductorOutsideSource.lean	5

- `centeredLowSymbol_coefficient_zero`
- `circleLowTerm_coefficient_zero`
- `sourceLowModel_conductor_outside`
- `realExceptionalDenominators_empty`
- `realExceptionalLowModel_empty`

<!-- checkpoint-verification -->

Verification

All 379 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3986-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1447 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 381 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_313_SourceLowFullRange.md`

Checkpoint 313: The exact source low identity for every positive conductor

The exact source low identity for every positive conductor. The extra Q*q0/B0<=R restriction is removed. For every positive conductor, the original source cutoff R is retained; only the working geometric radius is enlarged to R^2. If the conductor exceeds R, its actual source coefficients and exceptional denominator set vanish. Under Q>=2, Q<=R and 2R^8<=N^2, the original low symbol has its full finite-input local estimate and all-input l1-to-linfinity norm bound. The paper range R^4<=N/2 implies the new geometric budget. Sharp high/multifrequency estimates and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceLowFullRange.lean	1

- `sourceFullLow_squared`

<!-- checkpoint-verification -->

Verification

All 380 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3987-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1448 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 382 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_314_SourceLowFullRangeBounds.md`

Checkpoint 314: The source low kernel and operator bounds throughout the full conductor range

The source low kernel and operator bounds throughout the full conductor range. The extra Q*q0/B0<=R restriction is removed. For every positive conductor, the original source cutoff R is retained; only the working geometric radius is enlarged to R^2. If the conductor exceeds R, its actual source coefficients and exceptional denominator set vanish. Under Q>=2, Q<=R and 2R^8<=N^2, the original low symbol has its full finite-input local estimate and all-input l1-to-linfinity norm bound. The paper range R^4<=N/2 implies the new geometric budget. Sharp high/multifrequency estimates and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceLowFullRangeBounds.lean	3

- `sourceFullLow_L1_norm`
- `sourceFullLow_local`
- `sourceFullLow_kernel_decay`

<!-- checkpoint-verification -->

Verification

All 381 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3988-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1451 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 383 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_315_SourceLowPairing.md`

Checkpoint 315: Actual finite-indicator pairings for the complete source low multiplier

Actual finite-indicator pairings for the complete source low multiplier. The original source rational-indexed low symbol is used with real Q, optional induced exceptional data and every positive conductor. The actual paper scale R=max(exp((log N)^(1/10)),Q^8) satisfies the required geometry uniformly for sufficiently large N in the displayed Q-log range. No Siegel-Walfisz, PNT, Ren or Ghosh estimate enters these low proofs. Finite local estimates, all-input l1 norms, pairings and genuine countable l1 maxima retain their exact domains. High/multifrequency estimates and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceLowPairing.lean	3

- `uniformKernel_indicator_bound`
- `uniformKernel_pairing_bound`
- `sourceFullLow_pairing`

<!-- checkpoint-verification -->

Verification

All 382 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3989-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1454 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 384 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_316_NaturalSourceLowFamilies.md`

Checkpoint 316: Countable actual source low families throughout the paper scale budget

Countable actual source low families throughout the paper scale budget. The original source rational-indexed low symbol is used with real Q, optional induced exceptional data and every positive conductor. The actual paper scale R=max(exp((log N)^(1/10)),Q^8) satisfies the required geometry uniformly for sufficiently large N in the displayed Q-log range. No Siegel-Walfisz, PNT, Ren or Ghosh estimate enters these low proofs. Finite local estimates, all-input l1 norms, pairings and genuine countable l1 maxima retain their exact domains. High/multifrequency estimates and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
NaturalSourceLowFamilies.lean	2

- `naturalSourceLowData_L1_norm`
- `naturalSourceLow_countable_maximal`

<!-- checkpoint-verification -->

Verification

All 383 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3990-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1456 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 385 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_317_SourceScaleLowEstimate.md`

Checkpoint 317: The complete source low estimates in the actual uniform paper parameter range

The complete source low estimates in the actual uniform paper parameter range. The original source rational-indexed low symbol is used with real Q, optional induced exceptional data and every positive conductor. The actual paper scale R=max(exp((log N)^(1/10)),Q^8) satisfies the required geometry uniformly for sufficiently large N in the displayed Q-log range. No Siegel-Walfisz, PNT, Ren or Ghosh estimate enters these low proofs. Finite local estimates, all-input l1 norms, pairings and genuine countable l1 maxima retain their exact domains. High/multifrequency estimates and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceScaleLowEstimate.lean	2

- `sourceMajorArcScale_cutoff_le`
- `sourceFullLow_uniform_source_range`

<!-- checkpoint-verification -->

Verification

All 384 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3991-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1458 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 386 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_318_GeneralGaussDecay.md`

Checkpoint 318: Uniform Gauss decay at every exponent below one half

Uniform Gauss decay at every exponent below one half. These are proved arithmetic and annular estimates for the actual source components. Gauss decay is available at every exponent delta<1/2, and the selected conductor majorant divided by its level has one absolute bound. The annular high mask is zero on the narrow core and its nonzero onset forces the displayed decay. These are steps toward a genuine single-scale high operator estimate; no full high maximal or multifrequency conclusion is assumed.

Checked declarations

Module	Named supporting theorems
GeneralGaussDecay.lean	1

- `gauss_pair_arbitrary_decay`

<!-- checkpoint-verification -->

Verification

All 385 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3992-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1459 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 387 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_319_ConductorRatioBound.md`

Checkpoint 319: A proved uniform upper bound on the chosen conductor-majorant ratio

A proved uniform upper bound on the chosen conductor-majorant ratio. These are proved arithmetic and annular estimates for the actual source components. Gauss decay is available at every exponent delta<1/2, and the selected conductor majorant divided by its level has one absolute bound. The annular high mask is zero on the narrow core and its nonzero onset forces the displayed decay. These are steps toward a genuine single-scale high operator estimate; no full high maximal or multifrequency conclusion is assumed.

Checked declarations

Module	Named supporting theorems
ConductorRatioBound.lean	2

- `conductorMajorant_linear_bound`
- `conductorMajorant_ratio_uniform`

<!-- checkpoint-verification -->

Verification

All 386 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3993-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1461 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 388 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_320_AnnularHighMask.md`

Checkpoint 320: The actual high mask and its precise frequency onset

The actual high mask and its precise frequency onset. These are proved arithmetic and annular estimates for the actual source components. Gauss decay is available at every exponent delta<1/2, and the selected conductor majorant divided by its level has one absolute bound. The annular high mask is zero on the narrow core and its nonzero onset forces the displayed decay. These are steps toward a genuine single-scale high operator estimate; no full high maximal or multifrequency conclusion is assumed.

Checked declarations

Module	Named supporting theorems
AnnularHighMask.lean	2

- `annularHighMask_range`
- `annularHighMask_onset`

<!-- checkpoint-verification -->

Verification

All 387 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3994-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1463 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 389 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

`Checkpoint_321_AnnularHighFactor.md`

Checkpoint 321: Uniform size and effective-cutoff decay of the actual annular high factor

Uniform size and effective-cutoff decay of the actual annular high factor. These are proved arithmetic and annular estimates for the actual source components. Gauss decay is available at every exponent delta<1/2, and the selected conductor majorant divided by its level has one absolute bound. The annular high mask is zero on the narrow core and its nonzero onset forces the displayed decay. These are steps toward a genuine single-scale high operator estimate; no full high maximal or multifrequency conclusion is assumed.

Checked declarations

Module	Named supporting theorems
AnnularHighFactor.lean	4

- `highMaskConstant_one_le`
- `annularHighFactor_le_annular`
- `annularHighFactor_half`
- `annularHighFactor_linear`

<!-- checkpoint-verification -->

Verification

All 388 project modules and the root were freshly rebuilt: 3995 jobs pass without warnings/errors. All 1467 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 390 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_322_HighFactorInterpolation.md`

Checkpoint 322: Fractional-power interpolation of the actual high size and tail bounds

Fractional-power interpolation of the actual high size and tail bounds. The original centered high symbol is retained. The effective exceptional cutoff is Q*q0/B0, and the source quotient condition and smoothing width are proved algebraically identical to that cutoff. Interpolation of the proved size and decay bounds gives a genuine fractional Q saving, with the complementary denominator power retained for the coming dyadic sum. No high estimate is assumed.

Checked declarations

Module	Named supporting theorems
HighFactorInterpolation.lean	3

- `min_one_le_fractional_power`
- `high_power_product`
- `two_bounds_rpow_interpolate`

<!-- checkpoint-verification -->

Verification

All 389 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3996-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1470 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 391 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_323_ExceptionalEffectiveCutoff.md`

Checkpoint 323: The exact effective exceptional cutoff and frequency width

The exact effective exceptional cutoff and frequency width. The original centered high symbol is retained. The effective exceptional cutoff is Q*q0/B0, and the source quotient condition and smoothing width are proved algebraically identical to that cutoff. Interpolation of the proved size and decay bounds gives a genuine fractional Q saving, with the complementary denominator power retained for the coming dyadic sum. No high estimate is assumed.

Checked declarations

Module	Named supporting theorems
ExceptionalEffectiveCutoff.lean	3

- `exceptional_effective_cutoff`
- `exceptional_effective_width`
- `exceptional_effective_ratio`

<!-- checkpoint-verification -->

Verification

All 390 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3997-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1473 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 392 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_324_CenteredHighMaskFormula.md`

Checkpoint 324: The exact original centered high symbol in ordinary and exceptional masks

The exact original centered high symbol in ordinary and exceptional masks. The original centered high symbol is retained. The effective exceptional cutoff is Q*q0/B0, and the source quotient condition and smoothing width are proved algebraically identical to that cutoff. Interpolation of the proved size and

decay bounds gives a genuine fractional Q saving, with the complementary denominator power retained for the coming dyadic sum. No high estimate is assumed.

Checked declarations

Module	Named supporting theorems
CenteredHighMaskFormula.lean	1

- `centeredHighSymbol_mask_formula`

<!-- checkpoint-verification -->

Verification

All 391 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3998-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1474 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 393 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_325_MaskedGaussPairBounds.md`

Checkpoint 325: Actual ordinary and exceptional high size and linear-decay bounds

Actual ordinary and exceptional high size and linear-decay bounds. Both Gauss coefficients are the actual proved sums, and the induced effective cutoff is retained. The proved size and frequency-decay estimates yield Q^(-eta)*den(r)^(-(delta-eta)) for 0<=eta<=1 and any available Gauss exponent delta. The global Gauss and conductor-ratio bounds are proved separately and will be discharged in the final uniform high theorem. This stage does not assume a high operator norm or countable maximal conclusion.

Checked declarations

Module	Named supporting theorems
MaskedGaussPairBounds.lean	2

- `norm_masked_gauss_pair`
- `maskedGaussPair_two_bounds`

<!-- checkpoint-verification -->

Verification

All 392 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 3999-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1476 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 394 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_326_CenteredHighPowerBound.md`

Checkpoint 326: A quantitative denominator-weighted power bound for the actual centered high symbol

A quantitative denominator-weighted power bound for the actual centered high symbol. Both Gauss coefficients are the actual proved sums, and the induced effective cutoff is retained. The proved size and frequency-decay estimates yield Q^(-eta)*den(r)^(-(delta-eta)) for 0<=eta<=1 and any available Gauss exponent delta. The global Gauss and conductor-ratio bounds are proved separately and will be discharged in the final uniform high theorem. This stage does not assume a high operator norm or countable maximal conclusion.

Checked declarations

Module	Named supporting theorems
CenteredHighPowerBound.lean	1

- `centeredHighSymbol_power_bound`

<!-- checkpoint-verification -->

Verification

All 393 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4000-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1477 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 395 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_327_HighBroadSupport.md`

Checkpoint 327: Exact broad-support vanishing of the actual centered high symbol

Exact broad-support vanishing of the actual centered high symbol. Both Gauss coefficients are the actual proved sums, and the induced effective cutoff is retained. The proved size and frequency-decay estimates yield Q^(-eta)*den(r)^(-(delta-eta)) for 0<=eta<=1 and any available Gauss exponent delta. The global Gauss and conductor-ratio bounds are proved separately and will be discharged in the final uniform high theorem. This stage does not assume a high operator norm or countable maximal conclusion.

Checked declarations

Module	Named supporting theorems
HighBroadSupport.lean	1

- `centeredHighSymbol_zero_of_broad`

<!-- checkpoint-verification -->

Verification

All 394 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4001-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1478 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 396 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_328_DyadicActiveBound.md`

Checkpoint 328: The exact active-center count in each denominator block

The exact active-center count in each denominator block. Rational separation gives at most one active center on every denominator block s>=2, while the two smallest scales retain all four possible small-denominator centers. The finite sums are partitioned exactly, including the denominator-one center and empty sets. The extra positive denominator exponent yields a genuinely summable geometric bound instead of a logarithm of the global cutoff.

Checked declarations

Module	Named supporting theorems
DyadicActiveBound.lean	1

- `dyadicActive_card`

<!-- checkpoint-verification -->

Verification

All 395 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4002-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1479 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 397 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_329_DyadicSparseSum.md`

Checkpoint 329: A proved sparse sum bound with exact zero support

A proved sparse sum bound with exact zero support. Rational separation gives at most one active center on every denominator block s>=2, while the two smallest scales retain all four possible small-denominator centers. The finite sums are partitioned exactly, including the denominator-one center and empty sets. The extra positive denominator exponent yields a genuinely summable geometric bound instead of a logarithm of the global cutoff.

Checked declarations

Module	Named supporting theorems
DyadicSparseSum.lean	2

- `rational_sum_eq_active`
- `dyadic_sparse_sum_bound`

<!-- checkpoint-verification -->

Verification

All 396 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4003-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1481 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 398 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_330_DyadicDenominatorPartition.md`

Checkpoint 330: Exact partition of all source rational centers by dyadic denominator

Exact partition of all source rational centers by dyadic denominator. Rational separation gives at most one active center on every denominator block s>=2, while the two smallest scales retain all four possible small-denominator centers. The finite sums are partitioned exactly, including the denominator-one center and empty sets. The extra positive denominator exponent yields a genuinely summable geometric bound instead of a logarithm of the global cutoff.

Checked declarations

Module	Named supporting theorems
DyadicDenominatorPartition.lean	1

- `rational_dyadic_partition`

<!-- checkpoint-verification -->

Verification

All 397 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4004-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1482 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 399 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

Checkpoint_331_DyadicPowerDecay.md

Checkpoint 331: Geometric decay of the retained denominator power

Geometric decay of the retained denominator power. Rational separation gives at most one active center on every denominator block s>=2, while the two smallest scales retain all four possible small-denominator centers. The finite sums are partitioned exactly, including the denominator-one center and empty sets. The extra positive denominator exponent yields a genuinely summable geometric bound instead of a logarithm of the global cutoff.

Checked declarations

Module	Named supporting theorems
DyadicPowerDecay.lean	1

- `dyadic_inverse_rpow`

<!-- checkpoint-verification -->

Verification

All 398 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4005-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1483 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 400 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_332_GeometricPowerBudget.md

Checkpoint 332: A cutoff-independent geometric budget for the retained denominator power

The geometric ratio 2^(-epsilon) is strictly between zero and one for epsilon>0. Every finite dyadic sum is bounded by 1/(1-2^(-epsilon)), proved through the convergent series in Mathlib.

Checked declarations

Module	Named supporting theorems
GeometricPowerBudget.lean	2

- `geometric_inverse_power_partial`
- `geometric_inverse_power_denominator_pos`

<!-- checkpoint-verification -->

Verification

All 399 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4006-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1485 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 401 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_333_RationalHighWeightedSum.md`

Checkpoint 333: The full denominator-uniform weighted rational high sum

For any finite canonical rational set and a function with the actual broad cutoff support, the dyadic separation and small-denominator count give a bound 4A/(1-2^(-epsilon)) from the pointwise bound A den^(-epsilon). No cardinality factor depending on the global denominator cap survives.

Checked declarations

Module	Named supporting theorems
RationalHighWeightedSum.lean	1

- `rational_high_weighted_sum`

<!-- checkpoint-verification -->

Verification

All 400 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4007-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1486 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 402 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_334_IntegerWindowSupport.md`

Checkpoint 334: The exact three-term periodization on the full unit interval

The integer period sum of a function vanishing for absolute value at least one equals the three-term window throughout [-1,1]. Boundary points are included exactly; this is the interval needed for a canonical rational center subtracted from [0,1].

Checked declarations

Module	Named supporting theorems
IntegerWindowSupport.lean	1

- `integerPeriodSum_unit_window`

<!-- checkpoint-verification -->

Verification

All 401 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4008-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1487 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 403 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_335_CircleHighWindow.md`

Checkpoint 335: The exact three-term real formula for every canonical source high term

The actual circle high term agrees exactly with the three-term compact period window when the real frequency lies in [0,1]. The proof uses the original centered high symbol support and the canonical rational range.

Checked declarations

Module	Named supporting theorems
CircleHighWindow.lean	1

- `circleHighTerm_unit_window`

<!-- checkpoint-verification -->

Verification

All 402 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4009-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1488 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 404 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_336_CircleHighFiniteWindow.md`

Checkpoint 336: Exact finite window assembly for the complete source high circle sum

The full circle high sum is the sum of three unperiodized rational high sums at xi-1, xi, xi+1. This exact identity lets the denominator-uniform separation estimate pass to the continuous circle.

Checked declarations

Module	Named supporting theorems
CircleHighFiniteWindow.lean	1

- `circleHighModel_finite_window`

<!-- checkpoint-verification -->

Verification

All 403 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4010-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1489 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 405 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_337_CircleHighPowerSymbolBound.md`

Checkpoint 337: The actual circle high symbol with a cutoff-independent power bound

The original continuous high model has norm at most H(C,T,delta,eta) Q^(-eta), provided eta<delta, 0<=eta<=1, the proved Gauss pair bound at delta, the conductor ratio budget and the displayed character coefficient bounds. H=12(2C K(1+T))/(1-2^(-(delta-eta))). The denominator cap and arithmetic length do not enter H. The next stages discharge the Gauss and conductor budgets; this is a single-scale bound, not a countable high maximal estimate.

Checked declarations

Module	Named supporting theorems
CircleHighPowerSymbolBound.lean	1

- `circleHighModel_power_bound`

<!-- checkpoint-verification -->

Verification

All 404 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4011-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1490 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 406 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_338_CircleHighL2PowerBound.md`

Checkpoint 338: The actual high l2 multiplier from its proved power symbol bound

The continuous-circle high estimate implies the genuine l2 operator norm bound with the same explicit H(C,T,delta,eta). Positivity of that constant is checked from the strict exponent gap. This is the actual source operator, not an abstract operator supplied with the desired norm as a hypothesis.

Checked declarations

Module	Named supporting theorems
CircleHighL2PowerBound.lean	2

- `highPowerConstant_nonneg`
- `circleHighOperator_power_bound`

<!-- checkpoint-verification -->

Verification

All 405 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4012-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1492 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 407 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_339_SourceHighPowerUniform.md`

Checkpoint 339: An unconditional single-scale high bound at every exponent below one half

For every 0<eta<1/2 there is H_eta>0, independent of N,Q,denominator cap,conductor and character family, such that the actual high circle symbol and its l2 operator are bounded by H_eta Q^(-eta), for N>=2,Q>0,beta in [0,1], canonical finite rational centers and coefficients of norm at most one. The Gauss and conductor bounds are discharged using the proved arithmetic theorems. This is weaker than the sharp printed half-power subexponential bound and does not prove a countable high maximum. The uniform estimate is for the actual defined circleHighModel complement. Identification with the printed high decomposition uses the proved low source bridge and source scale geometry.

Checked declarations

Module	Named supporting theorems
SourceHighPowerUniform.lean	1

- `sourceHigh_power_uniform`

<!-- checkpoint-verification -->

Verification

All 406 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4013-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1493 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 408 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_340_HighFiniteIndicatorPairing.md`

Checkpoint 340: The actual high finite-indicator pairing bound

Cauchy-Schwarz in the genuine l2 space gives the quantitative high pairing H_eta Q^(-eta) sqrt(|F|) sqrt(|G|) for every finite indicator input and test set. The constant is uniform across all actual source parameters in the preceding theorem.

Checked declarations

Module	Named supporting theorems
HighFiniteIndicatorPairing.lean	1

- `sourceHigh_finite_pairing`

<!-- checkpoint-verification -->

Verification

All 407 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4014-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1494 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 409 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_341_SourceRemainderOperator.md`

Checkpoint 341: The actual high-plus-error remainder and exact finite decomposition

The remainder is defined as the actual circle high operator plus the actual prime major-arc error operator. The annular von Mangoldt finite convolution equals the actual low output plus this remainder output, with l1 and l2 input realizations checked. A triangle inequality gives its norm from the separately proved components.

Checked declarations

Module	Named supporting theorems
SourceRemainderOperator.lean	2

- `annularKernel_low_remainder`
- `sourceRemainderOperator_norm`

<!-- checkpoint-verification -->

Verification

All 408 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4015-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1496 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 410 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint notes 342-416

Checkpoint_342_SourceRemainderPower.md

Checkpoint 342: The conditional source high-plus-error power estimate in the uniform paper range

The unconditional actual high norm is combined with the earlier conditional global circle approximation. The remainder has norm at most H_eta Q^(-eta)+2(2^A Q^(-1/2)(1+log N)^(-A)), uniformly for sufficiently large N and real Q in the displayed logarithmic range. The raw progression-prefix, Ren and Ghosh estimates are explicit CircleAnalyticInputs hypotheses. No source high norm or desired paper conclusion is assumed.

Checked declarations

Module	Named supporting theorems
SourceRemainderPower.lean	1

- `sourceRemainder_uniform`

<!-- checkpoint-verification -->

Verification

All 409 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4016-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1497 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 411 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_343_SourceErrorAbsorption.md

Checkpoint 343: Absorbing the conditional approximation error into the proved high power

For Q>=1, N>=1, A>=0 and eta<=1/2, the earlier error envelope is bounded by 2*2^A Q^(-eta). Thus the actual high-plus-error remainder has a uniform H Q^(-eta) l2 bound under the explicit raw CircleAnalyticInputs hypotheses. The raw deep analytic inputs remain unproved; the norm is derived rather than assumed.

Checked declarations

Module	Named supporting theorems
SourceErrorAbsorption.lean	2

- `sourceError_power_absorb`
- `sourceRemainder_uniform_absorbed`

<!-- checkpoint-verification -->

Verification

All 410 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4017-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1499 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 412 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_344_ActualSourceLowRemainderBounds.md`

Checkpoint 344: Uniform actual source low and remainder operator bounds

One common large-N threshold supplies the actual source low l1-to-linfinity norm 2 C_low Q log(2Q)/N^2 and the actual high-plus-error l2 norm H_eta Q^(-eta), uniformly in the full positive conductor range and optional quadratic exceptional data. The low norm is unconditional; the remainder uses the displayed raw progression-prefix, Ren and Ghosh hypotheses. The actual source global scale and logarithmic Q range are retained.

Checked declarations

Module	Named supporting theorems
ActualSourceLowRemainderBounds.lean	1

- `actualSource_low_remainder_uniform`

<!-- checkpoint-verification -->

Verification

All 411 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4018-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1500 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 413 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_345_SummableCutoffHighMaximum.md`

Checkpoint 345: A genuine countable high maximum under a proved summable cutoff budget

For actual high operators with arbitrary varying N, conductor, character and beta data, a summable sequence Q_j^(-eta) gives a genuine pointwise countable supremum realized in l2, of norm at most H_eta (sum Q_j^(-eta)) times the input norm. Geometric cutoffs Q_j=2^j Q satisfy the required summability, proved in Lean. This is a restricted varying-cutoff high maximum. It does not establish the fixed-Q high maximal or multifrequency estimate required by the paper.

Checked declarations

Module	Named supporting theorems
SummableCutoffHighMaximum.lean	2

- `dyadic_cutoff_inverse_power_summable`
- `actualHigh_summable_cutoff_maximum`

<!-- checkpoint-verification -->

Verification

All 412 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4019-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1502 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 414 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_346_ActualSourceRestrictedPairing.md`

Checkpoint 346: The conditional actual prime-square annular restricted pairing from proved low and high bounds

For every 0<eta<1/2, the actual annular von Mangoldt square kernel has finite restricted pairing bounded by 2 C_low Q log(2Q)/N^2 |F||G| + H_eta Q^(-eta) sqrt(|F|) sqrt(|G|), uniformly for all sufficiently large N and the displayed real-Q logarithmic range, every positive conductor and optional quadratic exceptional data. The exact finite decomposition, genuine l1 low norm and l2 high-plus-error norm are used; no scalar pairing estimate is assumed. The raw progression-prefix, Ren and Ghosh estimates remain explicit hypotheses. This is a single-scale weighted-annular consequence. It does not prove the sharp high maximal/multifrequency estimates, Proposition 4.19, Theorem 1.1, or the full paper. No new manuscript statement error or mandatory source replacement is established.

Checked declarations

Module	Named supporting theorems
ActualSourceRestrictedPairing.lean	1

- `actualSource_restricted_pairing`

<!-- checkpoint-verification -->

Verification

All 413 project modules and the root were freshly rebuilt: 4020 jobs pass without warnings/errors. All 1503 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 415 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_347_DyadicAnnularScales.md`

Checkpoint 347: Exact dyadic annular scale arithmetic

The natural dyadic annular scales remain at least two, compose exactly under index shifts, and increase with the index. These facts keep all later onset and tail arguments at the actual natural arithmetic scales.

Checked declarations

Module	Named supporting theorems
DyadicAnnularScales.lean	3

- `dyadic_annular_scale_lower`
- `dyadic_annular_scale_shift`
- `dyadic_annular_scale_mono`

<!-- checkpoint-verification -->

Verification

All 414 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4021-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1506 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 416 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_348_DyadicNarrowWidths.md`

Checkpoint 348: Exact dyadic growth and positivity of the narrow collar widths

The actual smoothing width q(2^j M)^2/Q equals 4^j q M^2/Q, with positivity proved from positive parameters. No asymptotic scaling identity is assumed.

Checked declarations

Module	Named supporting theorems
DyadicNarrowWidths.lean	2

- `dyadic_narrow_width`
- `dyadic_narrow_width_pos`

<!-- checkpoint-verification -->

Verification

All 415 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4022-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1508 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 417 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_349_PureCollarMask.md`

Checkpoint 349: The actual complementary narrow cutoff and its exact range

The pure collar mask 1-eta(q N^2 theta/Q) lies in [0,1], vanishes at theta=0, and agrees exactly with the earlier high mask whenever q<=Q. The pure mask is useful even for arbitrary positive effective cutoffs.

Checked declarations

Module	Named supporting theorems
PureCollarMask.lean	3

- `annularCollarMask_range`
- `annularCollarMask_at_zero`
- `annularHighMask_eq_collar`

<!-- checkpoint-verification -->

Verification

All 416 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4023-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1511 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 418 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_350_PureCollarOnset.md`

Checkpoint 350: The strict first possible frequency for a nonzero collar

A nonzero pure mask forces |theta|>Q/(2qN^2). Conversely the full closed core has zero mask. All boundary inequalities and positive denominators are checked.

Checked declarations

Module	Named supporting theorems
PureCollarOnset.lean	2

- `annularCollarMask_onset`
- `annularCollarMask_zero_of_core`

<!-- checkpoint-verification -->

Verification

All 417 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4024-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1513 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 419 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_351_PureAnnularCollar.md`

Checkpoint 351: The actual pure annular collar factor and its norm attenuation

The genuine oscillatory annular factor is multiplied by the pure narrow complement. Its norm is at most the unmasked annular norm, it vanishes whenever the mask vanishes, and it agrees with the previously defined high factor in the low-denominator range.

Checked declarations

Module	Named supporting theorems
PureAnnularCollar.lean	3

- `annularCollarFactor_le_annular`
- `annularCollarFactor_zero_of_mask`
- `annularHighFactor_eq_collar`

<!-- checkpoint-verification -->

Verification

All 418 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4025-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1516 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 420 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_352_FirstAnnularCollarScale.md`

Checkpoint 352: The exact first nonzero dyadic collar and the no-onset case

The mask sequence either vanishes at all natural dyadic indices or has a least nonzero index with every earlier mask exactly zero. This is the actual scale-dependent mask, not an assumed threshold.

Checked declarations

Module	Named supporting theorems
FirstAnnularCollarScale.lean	2

- `annularCollar_first_scale`
- `annularCollar_no_onset`

<!-- checkpoint-verification -->

Verification

All 419 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4026-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1518 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 421 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_353_FirstCollarFrequency.md

Checkpoint 353: Frequency onset and exact zero collar outputs before it

At the first active mask the actual frequency exceeds Q/(2qM_i^2). Every earlier masked annular factor is zero, even if beta varies with the index.

Checked declarations

Module	Named supporting theorems
FirstCollarFrequency.lean	2

- `firstCollar_frequency`
- `collarFactor_before_onset`

<!-- checkpoint-verification -->

Verification

All 420 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4027-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1520 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 422 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_354_FirstCollarTailDecay.md

Checkpoint 354: Uniform geometric decay after the actual first collar scale

After the first active mask, the genuine collar factor decays by 4^(-j) relative to 4q/(pi Q). The proof uses the already proved oscillatory integral bound and exact dyadic index composition. Beta may vary arbitrarily in [0,1].

Checked declarations

Module	Named supporting theorems
FirstCollarTailDecay.lean	1

- `collarFactor_after_onset`

<!-- checkpoint-verification -->

Verification

All 421 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4028-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1521 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 423 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_355_FirstCollarTailSquares.md`

Checkpoint 355: The actual collar square decay after onset

Squaring the genuine collar norm bound gives the exact geometric ratio 1/16, with the exponent and nonnegative norm hypotheses checked.

Checked declarations

Module	Named supporting theorems
FirstCollarTailSquares.lean	1

- `collarFactor_after_onset_square`

<!-- checkpoint-verification -->

Verification

All 422 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4029-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1522 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 424 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_356_CollarTailSummability.md`

Checkpoint 356: The summable actual collar tail and its quantitative total

The actual masked tail has summable squared norms and total at most (16/15)(4q/(pi Q))^2. This is derived from the geometric bound and retains arbitrary beta values in [0,1].

Checked declarations

Module	Named supporting theorems
CollarTailSummability.lean	2

- `collarFactor_tail_square_summable`
- `collarFactor_tail_square_total`

<!-- checkpoint-verification -->

Verification

All 423 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4030-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1524 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 425 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_357_FullCollarSquareSummability.md`

Checkpoint 357: Summability of the complete actual dyadic collar square sequence

The complete collar sequence has summable squared norms for every real frequency and beta values in [0,1]. The proof handles both no-onset and least-onset cases, using exact zero prefixes and the proved shifted tail.

Checked declarations

Module	Named supporting theorems
FullCollarSquareSummability.lean	1

- `collarFactor_square_summable`

<!-- checkpoint-verification -->

Verification

All 424 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4031-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1525 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 426 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_358_FullCollarSquareTotal.md`

Checkpoint 358: A fixed-cutoff bound on the complete actual collar square energy

For every frequency the actual masked annular square sum is at most (16/15)(4q/(pi Q))^2, independently of its first active scale. All beta values may vary in [0,1]. This discharges the scalar energy premise needed for an actual fixed-Q collar maximum.

Checked declarations

Module	Named supporting theorems
FullCollarSquareTotal.lean	1

- `collarFactor_square_total`

<!-- checkpoint-verification -->

Verification

All 425 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4032-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1526 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 427 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_359_FiniteCollarSquareEnergy.md`

Checkpoint 359: Uniform finite actual collar energies with a usable squared constant

Every finite selection of dyadic collar scales has the proved uniform energy bound. A convenient larger exact squared constant (8q/(pi Q))^2 is also established for the l2 square-function construction.

Checked declarations

Module	Named supporting theorems
FiniteCollarSquareEnergy.lean	2

- `collarFactor_finite_square_energy`
- `collarFactor_finite_square_simple`

<!-- checkpoint-verification -->

Verification

All 426 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4033-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1528 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 428 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_360_WeightedCollarSquareEnergy.md`

Checkpoint 360: Actual scalar collar energy with arbitrary bounded scale coefficients

Multiplying the genuine collar factors by arbitrary complex coefficients of norm at most b preserves the finite energy with the expected factor b^2. This permits frequency cutoffs, optional scale selectors and source truncations as actual attenuations rather than assumed operator estimates.

Checked declarations

Module	Named supporting theorems
WeightedCollarSquareEnergy.lean	1

- `collarFactor_weighted_energy`

<!-- checkpoint-verification -->

Verification

All 427 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4034-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1529 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 429 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_361_CollarFrequencyContinuity.md`

Checkpoint 361: Continuity of the actual pure collar masks and factors

The pure mask and the genuine oscillatory collar factor are continuous functions of the real frequency. All arithmetic scale and effective-cutoff parameters are retained.

Checked declarations

Module	Named supporting theorems
CollarFrequencyContinuity.lean	2

- `annularCollarMask_continuous`
- `annularCollarFactor_continuous`

<!-- checkpoint-verification -->

Verification

All 428 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4035-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1531 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 430 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_362_CollarBroadSymbol.md`

Checkpoint 362: The actual continuous centered collar symbol with broad cutoff

The centered continuous symbol multiplies the actual annular collar by the original broad cutoff eta_s. Its norm is bounded by the scalar collar norm through the proved broad cutoff range.

Checked declarations

Module	Named supporting theorems
CollarBroadSymbol.lean	1

- `annularCollarSymbol_le_factor`

<!-- checkpoint-verification -->

Verification

All 429 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4036-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1532 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 431 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_363_CollarBroadSupport.md`

Checkpoint 363: Exact broad-support and compact-support vanishing for collars

The centered collar vanishes whenever the original broad cutoff vanishes and, in particular, outside the closed unit support. These are exact zero identities needed for circle periodization and rational separation.

Checked declarations

Module	Named supporting theorems
CollarBroadSupport.lean	2

- `annularCollarSymbol_zero_of_broad`
- `annularCollarSymbol_support`

<!-- checkpoint-verification -->

Verification

All 430 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4037-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1534 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 432 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_364_CollarBroadEnergy.md

Checkpoint 364: The finite square energy of actual centered continuous collars

The complete finite dyadic energy of the actual broad-cutoff collar is at most (8q/(pi Q))^2 at every real frequency, even for varying beta in [0,1]. This discharges its scalar square-energy premise without any maximal inequality assumption.

Checked declarations

Module	Named supporting theorems
CollarBroadEnergy.lean	1

- `annularCollarSymbol_finite_energy`

<!-- checkpoint-verification -->

Verification

All 431 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4038-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1535 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 433 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_365_CircleAnnularCollar.md

Checkpoint 365: The genuine shifted continuous circle collar symbol

The actual centered collar is periodized and shifted to any real center. Its value at every real lift equals the complete integer period sum; continuity and compact support are proved rather than assumed.

Checked declarations

Module	Named supporting theorems
CircleAnnularCollar.lean	1

- `circleAnnularCollar_real`

<!-- checkpoint-verification -->

Verification

All 432 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4039-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1536 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 434 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_366_CircleCollarWindow.md`

Checkpoint 366: Exact circle collar windows and common frequency representatives

The actual circle collar equals its three-term real window on the full permitted interval. A shared representative of the circle point minus the center is independent of the dyadic index, which is essential for summing square energies.

Checked declarations

Module	Named supporting theorems
CircleCollarWindow.lean	2

- `circleAnnularCollar_window`
- `circleAnnularCollar_representative`

<!-- checkpoint-verification -->

Verification

All 433 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4040-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1538 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 435 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_367_ThreeTermSquareBound.md`

Checkpoint 367: The exact square-norm cost of the three-term period window

The squared complex norm of a three-term sum is at most three times the sum of squared norms. The proof retains all cross terms and derives the window inequality used by the circle energy calculation.

Checked declarations

Module	Named supporting theorems
ThreeTermSquareBound.lean	2

- `norm_three_sum_square`
- `compactPeriodWindow_square`

<!-- checkpoint-verification -->

Verification

All 434 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4041-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1540 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 436 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_368_CircleCollarSquareEnergy.md`

Checkpoint 368: A proved uniform finite square-energy bound on the actual circle collars

For every circle frequency, every finite dyadic index set, and arbitrary beta values in [0,1], the actual collar energy is at most (24q/(pi Q))^2. The three periodized pieces are handled explicitly; no circle energy conclusion is assumed.

Checked declarations

Module	Named supporting theorems
CircleCollarSquareEnergy.lean	1

- `circleAnnularCollar_finite_energy`

<!-- checkpoint-verification -->

Verification

All 435 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4042-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1541 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 437 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_369_CircleCollarCountableMaximum.md`

Checkpoint 369: The genuine fixed-cutoff countable l2 maximum for one collar frequency

The actual single-center circle collar has a genuine countable pointwise maximum in l2 of norm at most 24q/(pi Q) times the input norm. Q is fixed throughout all dyadic scales; this uses the proved finite energy and the earlier square-function theorem.

Checked declarations

Module	Named supporting theorems
CircleCollarCountableMaximum.lean	1

- `circleAnnularCollar_countable_maximum`

<!-- checkpoint-verification -->

Verification

All 436 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4043-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1542 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 438 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_370_WeightedCircleCollarEnergy.md`

Checkpoint 370: Actual circle collar energy with bounded complex scale coefficients

Arbitrary scale-dependent coefficients of norm at most b give the exact squared energy constant (24bq/(pi Q))^2. This includes genuine source selectors and fixed Gauss coefficients as special cases.

Checked declarations

Module	Named supporting theorems
WeightedCircleCollarEnergy.lean	1

- `circleAnnularCollar_weighted_energy`

<!-- checkpoint-verification -->

Verification

All 437 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4044-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1543 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 439 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_371_WeightedCircleCollarMaximum.md`

Checkpoint 371: The genuine weighted fixed-cutoff circle collar maximum

For bounded complex scale coefficients, the actual weighted circle collars admit a countable pointwise l2 maximum with norm at most 24bq/(pi Q) times the input norm. This is an actual fixed-Q maximum for the collar contribution. It does not establish the full high operator, whose large-denominator terms need separate estimates.

Checked declarations

Module	Named supporting theorems
WeightedCircleCollarMaximum.lean	1

- `circleAnnularCollar_weighted_maximum`

<!-- checkpoint-verification -->

Verification

All 438 project modules and the root were freshly rebuilt: 4045 jobs pass without warnings/errors. All 1544 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 440 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_372_FiniteSumSquareCardinality.md`

Checkpoint 372: The exact square-norm cost of a finite active frequency sum

The squared norm of a finite complex sum is bounded by its cardinality times its squared coefficient energy. This is proved from the actual finite Cauchy-Schwarz inequality and complex triangle inequality.

Checked declarations

Module	Named supporting theorems
FiniteSumSquareCardinality.lean	1

- `norm_finset_sum_square_card`

<!-- checkpoint-verification -->

Verification

All 439 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4046-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1545 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 441 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_373_DyadicSparseSquareEnergy.md`

Checkpoint 373: Uniform square-energy summation over the actual active rational centers

For one denominator block, actual broad support leaves at most four active canonical centers independent of the dyadic arithmetic index. Summing the individual finite time energies gives a full rational-sum energy at most (4A)^2, with every inactive term exactly zero. The bound is independent of the total number of rational centers.

Checked declarations

Module	Named supporting theorems
DyadicSparseSquareEnergy.lean	1

- `dyadic_sparse_square_energy`

<!-- checkpoint-verification -->

Verification

All 440 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4047-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1546 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 442 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_374_SourceBroadCutoffNorm.md`

Checkpoint 374: The actual broad cutoff is a complex norm contraction

The source broad cutoff has complex norm at most one and multiplying any complex coefficient by it decreases its norm. These proved pointwise facts allow the exact source broad support to remain in every energy argument.

Checked declarations

Module	Named supporting theorems
SourceBroadCutoffNorm.lean	2

- `sourceBroadCutoff_complex_norm`
- `sourceBroadCutoff_mul_norm`

<!-- checkpoint-verification -->

Verification

All 441 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4048-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1548 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 443 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_375_RationalCollarBlockEnergy.md

Checkpoint 375: The actual centered rational collar block square-energy estimate

At any real frequency, any fixed denominator block has total dyadic collar energy at most (32 b U/(pi Q))^2, where U bounds its denominators and b bounds its actual complex scale coefficients. Beta and coefficients may vary by scale and rational center. Rational separation and scalar collar energy discharge every analytic premise.

Checked declarations

Module	Named supporting theorems
RationalCollarBlockEnergy.lean	1

- `rationalCollarBlock_real_energy`

<!-- checkpoint-verification -->

Verification

All 442 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4049-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1549 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 444 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_376_CircleRationalCollarBlock.md

Checkpoint 376: The genuine finite rational collar circle block and its exact real window

The finite continuous circle symbol sums the actual shifted collar terms with their actual complex coefficients. On [0,1] its value equals three explicit centered rational sums; canonical rational endpoints and all modulation shifts are retained.

Checked declarations

Module	Named supporting theorems
CircleRationalCollarBlock.lean	1

- `rationalCollarBlock_real_window`

<!-- checkpoint-verification -->

Verification

All 443 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4050-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1550 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 445 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_377_CircleRationalCollarEnergy.md`

Checkpoint 377: The actual circle rational-block square-energy estimate

Every finite dyadic index set of one actual rational collar block has square energy at most (96 b U/(pi Q))^2 at every circle point. The proof combines the three real windows and the proved active rational-center energy without a rational-cardinality loss.

Checked declarations

Module	Named supporting theorems
CircleRationalCollarEnergy.lean	1

- `rationalCollarBlock_finite_energy`

<!-- checkpoint-verification -->

Verification

All 444 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4051-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1551 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 446 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_378_RationalCollarBlockMaximum.md`

Checkpoint 378: A genuine fixed-Q countable l2 maximum for a complete rational block

The actual finite rational collar block admits a countable pointwise l2 maximum with norm at most 96 b U/(pi Q) times the input norm. All scale-dependent coefficients and beta values remain explicit, and rational separation is proved.

Checked declarations

Module	Named supporting theorems
RationalCollarBlockMaximum.lean	1

- `rationalCollarBlock_countable_maximum`

<!-- checkpoint-verification -->

Verification

All 445 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4052-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1552 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 447 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_379_AttenuatedRationalCollars.md`

Checkpoint 379: The fixed-Q rational collar maximum under additional frequency attenuation

Any actual continuous circle family dominated pointwise in frequency modulus by the proved collar block inherits its genuine countable l2 maximum. This is a proved application of the symbol energy criterion; it assumes only pointwise attenuation, not an operator or maximal norm bound.

Checked declarations

Module	Named supporting theorems
AttenuatedRationalCollars.lean	1

- `rationalCollarBlock_attenuated_maximum`

<!-- checkpoint-verification -->

Verification

All 446 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4053-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1553 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 448 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_380_WeightedFiniteSquareSum.md`

Checkpoint 380: Weighted finite Cauchy-Schwarz without a block-count loss

A finite complex sum has square norm at most (sum B_i)(sum |F_i|^2/B_i) for positive weights. This is proved from the exact weighted finite Cauchy-Schwarz inequality and will combine different denominator blocks.

Checked declarations

Module	Named supporting theorems
WeightedFiniteSquareSum.lean	1

- `norm_finset_sum_square_weighted`

<!-- checkpoint-verification -->

Verification

All 447 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4054-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1554 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 449 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_381_FiniteSquareEnergyAssembly.md`

Checkpoint 381: Assembly of distinct finite square energies with their correct total constant

If each finite block has time energy at most B_i^2, the full sum has energy at most (sum B_i)^2. The proof interchanges the two finite sums and cancels the positive weights; no factor equal to the number of blocks is introduced.

Checked declarations

Module	Named supporting theorems
FiniteSquareEnergyAssembly.lean	1

- `finite_square_energy_assemble`

<!-- checkpoint-verification -->

Verification

All 448 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4055-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1555 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 450 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_382_FiniteCircleEnergyAssembly.md`

Checkpoint 382: Exact finite-block assembly of genuine circle square energies and countable maxima

Proved frequency energy bounds for finitely many continuous circle families combine with the sum of their constants. The resulting genuine pointwise countable maximum in l2 follows from the previously formalized Fourier square-energy theorem. These are abstract assembly implications; actual source block premises are supplied separately.

Checked declarations

Module	Named supporting theorems
FiniteCircleEnergyAssembly.lean	2

- `finite_circle_energy_assemble`
- `finite_circle_countable_maximum`

<!-- checkpoint-verification -->

Verification

All 449 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4056-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1557 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 451 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_383_DyadicCoefficientBudget.md`

Checkpoint 383: Actual denominator lower bounds and coefficient decay on dyadic blocks

For a canonical denominator block, 2^s<=den(r) is proved and a C den(r)^(-delta) coefficient bound gives the correct uniform block bound C(2^s)^(-delta) for delta>=0.

Checked declarations

Module	Named supporting theorems
DyadicCoefficientBudget.lean	2

- `dyadic_denominator_real_lower`
- `dyadic_coefficient_power_budget`

<!-- checkpoint-verification -->

Verification

All 450 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4057-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1559 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 452 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_384_DyadicCollarConstant.md

Checkpoint 384: Exact normalization of the denominator-block collar constant

The proved collar constant 96 C (2^s)^(-delta) 2^(s+1)/(pi Q) equals (192 C/(pi Q))(2^(1-delta))^s. All real power and natural power identities are retained exactly.

Checked declarations

Module	Named supporting theorems
DyadicCollarConstant.lean	1

- `dyadic_collar_constant_identity`

<!-- checkpoint-verification -->

Verification

All 451 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4058-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1560 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 453 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_385_DyadicGaussCollarEnergy.md

Checkpoint 385: The actual rational collar block energy under proved arithmetic power decay

A denominator coefficient envelope C den^(-delta) implies the exact finite dyadic energy constant ((192C/(pi Q))(2^(1-delta))^s)^2 for the actual continuous circle block. Rational ranges, small denominators and varying beta/coefficient data are retained. The following arithmetic stages sum these constants across denominators.

Checked declarations

Module	Named supporting theorems
DyadicGaussCollarEnergy.lean	1

- `rationalCollarBlock_power_energy`

<!-- checkpoint-verification -->

Verification

All 452 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4059-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1561 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 454 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_386_GrowingGeometricBudget.md`

Checkpoint 386: A quantitative finite budget for the growing dyadic collar series

For r>1, the exact finite geometric identity gives sum_{s<J} r^s<=r^J/(r-1). The collar ratio 2^(1-delta) is proved greater than one for delta<1.

Checked declarations

Module	Named supporting theorems
GrowingGeometricBudget.lean	2

- `growing_geometric_partial`
- `collar_dyadic_ratio_gt_one`

<!-- checkpoint-verification -->

Verification

All 453 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4060-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1563 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 455 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_387_RealCutoffDyadicBudget.md

Checkpoint 387: The exact real-cutoff dyadic endpoint budget

For real Q>=1, 2^(log2 floor Q)<=Q. Raising to the nonnegative exponent 1-delta gives the exact final-block budget without replacing the original real cutoff by a different smoothing width.

Checked declarations

Module	Named supporting theorems
RealCutoffDyadicBudget.lean	2

- `real_cutoff_log2_power`
- `real_cutoff_growing_power`

<!-- checkpoint-verification -->

Verification

All 454 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4061-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1565 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 456 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_388_CutoffGeometricSeries.md

Checkpoint 388: The full denominator-block geometric series in terms of the real cutoff

The sum of the growing dyadic factors through log2 floor Q is bounded by [2^(1-delta)/(2^(1-delta)-1)] Q^(1-delta), with every floor and endpoint checked.

Checked declarations

Module	Named supporting theorems
CutoffGeometricSeries.lean	1

- `cutoff_growing_series`

<!-- checkpoint-verification -->

Verification

All 455 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4062-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1566 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 457 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_389_CollarPowerSummation.md`

Checkpoint 389: The actual summed collar block constant decays as Q to minus delta

The positive explicit constant collarPowerConstant(C,delta) bounds the sum of the actual denominator-block collar constants by collarPowerConstant(C,delta) Q^(-delta), for C>=0, Q>=1 and delta<1. The exact cancellation Q^(1-delta)/Q=Q^(-delta) is proved.

Checked declarations

Module	Named supporting theorems
CollarPowerSummation.lean	3

- `cutoff_power_divide`
- `collarPowerConstant_pos`
- `collar_block_constant_sum`

<!-- checkpoint-verification -->

Verification

All 456 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4063-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1569 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 458 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_390_RationalCollarDyadicPartition.md`

Checkpoint 390: Exact dyadic partition of the complete continuous rational collar model

The full finite rational collar symbol is exactly the sum of its denominator blocks, in the continuous circle space. No frequency term, denominator-one center or boundary is dropped.

Checked declarations

Module	Named supporting theorems
RationalCollarDyadicPartition.lean	1

- `rationalCollarBlock_dyadic_partition`

<!-- checkpoint-verification -->

Verification

All 457 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4064-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1570 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 459 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_391_CompleteRationalCollarEnergy.md`

Checkpoint 391: The complete fixed-Q rational collar square-energy bound

For every finite canonical rational set with denominators at most floor Q, coefficients bounded by C den^(-delta), and beta values in [0,1], the actual full collar model has finite time energy at most (collarPowerConstant(C,delta) Q^(-delta))^2. Positive weighted finite energy assembly and the proved real-cutoff geometric budget discharge the complete frequency energy premise.

Checked declarations

Module	Named supporting theorems
CompleteRationalCollarEnergy.lean	1

- `completeRationalCollar_finite_energy`

<!-- checkpoint-verification -->

Verification

All 458 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4065-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1571 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 460 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_392_CompleteRationalCollarMaximum.md`

Checkpoint 392: The genuine full fixed-Q rational collar countable maximum

The actual full rational collar family has a genuine countable pointwise supremum in l2, bounded by collarPowerConstant(C,delta) Q^(-delta) times the input norm, for Q>=1, C>0 and 0<=delta<1. The coefficient power decay remains an explicit arithmetic premise here and is discharged for actual Gauss data in the next units.

Checked declarations

Module	Named supporting theorems
CompleteRationalCollarMaximum.lean	1

- `completeRationalCollar_countable_maximum`

<!-- checkpoint-verification -->

Verification

All 459 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4066-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1572 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 461 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_393_RationalGaussCoefficientBounds.md

Checkpoint 393: The actual rational principal and twisted Gauss coefficients with uniform decay

The actual reduced rational numerator is a residue unit. The proved uniform Gauss theorem gives one positive C_delta bounding every rational principal coefficient and every Dirichlet twisted coefficient by C_delta den(r)^(-delta), for delta<1/2. The principal twisted identity is also proved.

Checked declarations

Module	Named supporting theorems
RationalGaussCoefficientBounds.lean	2

- `rationalGauss_uniform_decay`
- `rationalTwistedGauss_principal`

<!-- checkpoint-verification -->

Verification

All 460 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4067-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1574 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 462 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_394_OrdinaryCollarUniformMaximum.md

Checkpoint 394: An unconditional complete ordinary fixed-Q collar maximum

For every 0<=delta<1/2, the full actual ordinary-Gauss rational collar family has a genuine countable l2 maximum at most H_delta Q^(-delta). The constant is uniform in Q>=1, the starting arithmetic scale M>=2, canonical rational subsets, beta values in [0,1], and bounded complex scale selectors. The arithmetic coefficient premise has been discharged by the proved Gauss theorem.

Checked declarations

Module	Named supporting theorems
OrdinaryCollarUniformMaximum.lean	1

- `ordinaryCollar_uniform_maximum`

<!-- checkpoint-verification -->

Verification

All 461 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4068-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1575 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 463 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_395_TwistedCollarUniformMaximum.md`

Checkpoint 395: An unconditional complete twisted fixed-Q collar maximum

The same genuine full fixed-Q collar maximum holds uniformly for actual twisted Gauss coefficients, even with Dirichlet characters and beta values varying by scale and rational center, and bounded scale selectors. No multifrequency theorem or desired maximal norm conclusion is assumed. The width Q is fixed; exceptional conductor-dependent effective widths are identified separately.

Checked declarations

Module	Named supporting theorems
TwistedCollarUniformMaximum.lean	1

- `twistedCollar_uniform_maximum`

<!-- checkpoint-verification -->

Verification

All 462 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4069-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1576 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 464 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_396_OrdinaryCenteredHighCollar.md

Checkpoint 396: The exact original ordinary centered high collar identity

In the low-denominator range den(r)<=Q, the original centered high symbol with no exceptional coefficient equals its actual principal Gauss coefficient times the proved centered collar symbol. This is an exact identity in the continuous real symbol space, retaining the original broad cutoff.

Checked declarations

Module	Named supporting theorems
OrdinaryCenteredHighCollar.lean	1

- `centeredHighSymbol_ordinary_collar`

<!-- checkpoint-verification -->

Verification

All 463 project modules and the root were freshly rebuilt: 4070 jobs pass without warnings/errors. All 1577 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 465 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_397_OrdinaryCircleHighCollar.md

Checkpoint 397: The exact original ordinary circle high collar identity

The original circle high term in the low-denominator range equals the principal Gauss coefficient times the actual shifted circle collar. The proof passes the centered identity through the genuine period window without replacing the circle operator by an abstract model.

Checked declarations

Module	Named supporting theorems
OrdinaryCircleHighCollar.lean	1

- `circleHighTerm_ordinary_collar`

<!-- checkpoint-verification -->

Verification

All 464 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4071-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1578 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 466 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_398_OrdinaryHighCollarSum.md`

Checkpoint 398: The full exact ordinary low-denominator high-symbol correspondence

For every finite rational set with den(r)<=Q, the original ordinary-only high circle sum is exactly the actual rational collar block with principal Gauss coefficients. No operator bound or paper conclusion is assumed in this source correspondence.

Checked declarations

Module	Named supporting theorems
OrdinaryHighCollarSum.lean	1

- `circleHighModel_ordinary_collar`

<!-- checkpoint-verification -->

Verification

All 465 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4072-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1579 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 467 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_399_OriginalOrdinaryCollarMaximum.md`

Checkpoint 399: The original ordinary low-denominator high family has a fixed-Q countable l2 maximum

For every 0<=delta<1/2, the original ordinary-only high family restricted to all canonical denominators <=Q has a genuine countable l2 maximum bounded by H_delta Q^(-delta). Q>=1 is fixed throughout the dyadic scales. The exact original symbol identities and the proved Gauss arithmetic discharge the generic collar premises. Large-denominator terms remain outside this theorem.

Checked declarations

Module	Named supporting theorems
OriginalOrdinaryCollarMaximum.lean	1

- `originalOrdinaryCollar_countable_maximum`

<!-- checkpoint-verification -->

Verification

All 466 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4073-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1580 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 468 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_400_CircleEnergyMaximalWitness.md`

Checkpoint 400: A genuine circle maximal witness with explicit pointwise domination

The circle finite-energy theorem supplies a real l2 representative of the actual countable supremum, proves that it is nonnegative, and proves that it dominates every scale output at every integer point. The norm bound is retained. No boundedness property of an unbounded supremum is silently assumed.

Checked declarations

Module	Named supporting theorems
CircleEnergyMaximalWitness.lean	1

- `circle_energy_maximal_witness`

<!-- checkpoint-verification -->

Verification

All 467 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4074-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1581 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 469 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_401_RealL2FinitePairing.md`

Checkpoint 401: The finite test-set bound for a nonnegative genuine real l2 majorant

Cauchy-Schwarz and the actual lp square-energy formula give sum_G g<=||g||_2 sqrt(|G|) for nonnegative real l2 functions. A complex function dominated by g inherits the same finite sum bound.

Checked declarations

Module	Named supporting theorems
RealL2FinitePairing.lean	2

- `real_l2_finite_sum_bound`
- `real_l2_majorant_finite_sum`

<!-- checkpoint-verification -->

Verification

All 468 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4075-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1583 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 470 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_402_CircleEnergyIndicatorMaximum.md`

Checkpoint 402: The actual finite-indicator maximal pairing supplied by proved circle energy

For a finite indicator input, a proved circle frequency energy bound yields a genuine dominating countable maximum and the finite-test-set bound B sqrt(|F|) sqrt(|G|). The exact indicator lp norm and real l2 majorant theorem are used, not an assumed maximal pairing estimate.

Checked declarations

Module	Named supporting theorems
CircleEnergyIndicatorMaximum.lean	1

- `circle_energy_indicator_maximal_pairing`

<!-- checkpoint-verification -->

Verification

All 469 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4076-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1584 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 471 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_403_OrdinaryGaussCollarEnergy.md`

Checkpoint 403: The actual ordinary Gauss collar family has uniform square energy

The actual ordinary Gauss collar family has a proved uniform finite dyadic square-energy bound (H_delta Q^(-delta))^2 at every circle point, for 0<=delta<1/2 and fixed Q>=1. Actual Gauss arithmetic, all reduced rational denominators, varying beta in [0,1], and bounded scale selectors are retained. This

strengthened energy statement supplies pointwise dominating maximal witnesses and finite-indicator maximal pairings.

Checked declarations

Module	Named supporting theorems
OrdinaryGaussCollarEnergy.lean	1

- `ordinaryGaussCollar_uniform_energy`

<!-- checkpoint-verification -->

Verification

All 470 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4077-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1585 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 472 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_404_TwistedGaussCollarEnergy.md`

Checkpoint 404: The actual twisted Gauss collar family has uniform square energy

The actual twisted Gauss collar family has a proved uniform finite dyadic square-energy bound (H_delta Q^(-delta))^2 at every circle point, for 0<=delta<1/2 and fixed Q>=1. Actual Gauss arithmetic, all reduced rational denominators, varying beta in [0,1], and bounded scale selectors are retained. This strengthened energy statement supplies pointwise dominating maximal witnesses and finite-indicator maximal pairings.

Checked declarations

Module	Named supporting theorems
TwistedGaussCollarEnergy.lean	1

- `twistedGaussCollar_uniform_energy`

<!-- checkpoint-verification -->

Verification

All 471 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4078-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1586 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 473 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_405_OrdinaryCollarMaximalPairing.md`

Checkpoint 405: The actual ordinary collar finite-indicator maximal pairing

The actual ordinary Gauss collar maximum has the finite-indicator pairing bound H_delta Q^(-delta) sqrt(|F|) sqrt(|G|), with a genuine real l2 supremum that explicitly dominates every scale output. The proof uses the newly proved frequency energy, not an assumed maximal norm.

Checked declarations

Module	Named supporting theorems
OrdinaryCollarMaximalPairing.lean	1

- `ordinaryCollar_uniform_maximal_pairing`

<!-- checkpoint-verification -->

Verification

All 472 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4079-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1587 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 474 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_406_OrdinarySourceFilter.md`

Checkpoint 406: The exact ordinary collar correspondence inside the globally truncated source

For a real global source radius R>=Q>0, filtering the original source rational set to den(r)<=Q gives exactly the canonical set through floor Q. The original ordinary high symbol on that set equals the actual full Gauss collar symbol. The real cutoff remains in every smoothing width.

Checked declarations

Module	Named supporting theorems
OrdinarySourceFilter.lean	1

- `ordinaryHigh_filtered_source`

<!-- checkpoint-verification -->

Verification

All 473 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4080-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1588 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 475 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_407_SourceOrdinaryCollarFamily.md`

Checkpoint 407: The original globally truncated ordinary collar family has the proved fixed-Q maximum

The original ordinary-only high family, restricted to den(r)<=Q within arbitrary real source radii R_j>=Q, has the uniform genuine countable l2 maximum H_delta Q^(-delta). The source radius, inert character and beta data may vary with scale; every exact source filtering identity is used.

Checked declarations

Module	Named supporting theorems
SourceOrdinaryCollarFamily.lean	1

- `sourceOrdinaryCollar_family_maximum`

<!-- checkpoint-verification -->

Verification

All 474 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4081-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1589 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 476 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_408_DyadicSourceLogBudget.md`

Checkpoint 408: The actual paper logarithmic cutoff budget persists on every dyadic tail

For c>0 and M>=2, the allowed real-Q logarithmic range at M implies the same range at every dyadic arithmetic scale 2^j M. The source radius at each of those scales also contains Q. All natural-to-real casts and positivity of the logarithmic coefficient are checked.

Checked declarations

Module	Named supporting theorems
DyadicSourceLogBudget.lean	2

- `source_log_budget_dyadic`
- `source_cutoff_le_dyadic_radius`

<!-- checkpoint-verification -->

Verification

All 475 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4082-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1591 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 477 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_409_DyadicSourceFourthPower.md`

Checkpoint 409: A uniform fourth-power source radius budget on the whole dyadic tail

One sufficiently large starting scale threshold ensures R(2^j M,Q)^4<=(2^j M)/2 simultaneously for every dyadic index j and every real Q in the fixed starting-scale logarithmic range.

Checked declarations

Module	Named supporting theorems
DyadicSourceFourthPower.lean	1

- `sourceRadius_uniform_dyadic_fourth`

<!-- checkpoint-verification -->

Verification

All 476 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4083-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1592 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 478 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_410_DyadicSourceEighthPower.md`

Checkpoint 410: The full source eighth-power geometry uniformly on every dyadic tail

The uniform fourth-power budget gives the exact geometry 2 R(2^j M,Q)^8<=(2^j M)^2 required by the original full-conductor low bridge, simultaneously for the entire dyadic tail. This is a proved numerical source condition, not an analytic hypothesis.

Checked declarations

Module	Named supporting theorems
DyadicSourceEighthPower.lean	1

- `sourceRadius_uniform_dyadic_eighth`

<!-- checkpoint-verification -->

Verification

All 477 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4084-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1593 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 479 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_411_DyadicActualLowData.md`

Checkpoint 411: The actual full-conductor low data on a source dyadic tail

The original source radius, optional exceptional selector, quadratic character and beta data form actual NaturalSourceLowData at every dyadic scale. Only numerical radius and character conditions enter the data, with no desired norm conclusion. The genuine low l1-to-linfinity norm is uniformly bounded by the starting-scale amplitude.

Checked declarations

Module	Named supporting theorems
DyadicActualLowData.lean	1

- `dyadicActualLowData_norm`

<!-- checkpoint-verification -->

Verification

All 478 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4085-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1594 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 480 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_412_DyadicActualLowMaximum.md`

Checkpoint 412: The genuine all-input countable maximum for actual dyadic source low data

The actual dyadic source low symbols admit a genuine countable l1-to-linfinity maximum. The outputs are the actual l2 Fourier multipliers applied to the l1 inclusion; the proof uses their proved l1 agreement and the actual numeric scale data.

Checked declarations

Module	Named supporting theorems
DyadicActualLowMaximum.lean	1

- `dyadicActualLow_countable_maximum`

<!-- checkpoint-verification -->

Verification

All 479 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4086-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1595 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 481 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_413_UniformSourceLowDyadicMaximum.md`

Checkpoint 413: The original source low countable maximum in the actual uniform paper range

One starting-scale threshold gives the original optional-exceptional source low maximum for all dyadic tails and all real Q>=2 in the starting-scale logarithmic range. The conductor, quadratic character, beta<=1 and exceptional activation may all vary by scale. The bound is 2 C_low Q log(2Q)/M^2 times the l1 input norm, with no new deep analytic input.

Checked declarations

Module	Named supporting theorems
UniformSourceLowDyadicMaximum.lean	1

- `actualSourceLow_uniform_dyadic_maximum`

<!-- checkpoint-verification -->

Verification

All 480 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4087-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1596 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 482 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_414_CircleHighDenominatorSplit.md`

Checkpoint 414: The exact denominator split of the original full high circle model

The original full high circle symbol is exactly the sum of its den(r)<=Q and den(r)>Q parts. Both cut-off boundary cases are retained for every real Q, with no convergence assumption because the source rational set is finite.

Checked declarations

Module	Named supporting theorems
CircleHighDenominatorSplit.lean	1

- `circleHighModel_denominator_split`

<!-- checkpoint-verification -->

Verification

All 481 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4088-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1597 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 483 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_415_LargeOrdinaryCenteredHigh.md`

Checkpoint 415: Exact vanishing of the ordinary low symbol beyond the cutoff

For den(r)>Q with no exceptional coefficient, the original centered low symbol is exactly zero and the original high symbol equals the full centered rational model. The statement needs no broad/narrow containment assumption.

Checked declarations

Module	Named supporting theorems
LargeOrdinaryCenteredHigh.lean	2

- `centeredLowSymbol_large_ordinary_zero`
- `centeredHighSymbol_large_ordinary`

<!-- checkpoint-verification -->

Verification

All 482 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4089-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1599 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 484 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_416_LargeOrdinaryCircleTerm.md`

Checkpoint 416: The exact original circle contribution beyond the ordinary cutoff

The original low circle term vanishes beyond Q; its high circle term therefore equals the actual full rational circle model term. The period window and exact low-plus-high circle identity are retained.

Checked declarations

Module	Named supporting theorems
LargeOrdinaryCircleTerm.lean	2

- `circleLowTerm_large_ordinary_zero`
- `circleHighTerm_large_ordinary`

<!-- checkpoint-verification -->

Verification

All 483 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4090-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1601 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 485 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint notes 417-491

Checkpoint_417_LargeOrdinaryCircleModel.md

Checkpoint 417: The actual complete ordinary large-denominator high circle sum

For any finite rational set entirely beyond Q, the original ordinary-only high circle model equals its full major-arc circle model. This is an exact identity, not a claimed maximal bound.

Checked declarations

Module	Named supporting theorems
LargeOrdinaryCircleModel.lean	1

- `circleHighModel_large_ordinary`

<!-- checkpoint-verification -->

Verification

All 484 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4091-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1602 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 486 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_418_FullOrdinaryHighSplit.md

Checkpoint 418: The exact full ordinary source high symbol is collar plus large denominators

The actual globally truncated source ordinary high symbol is exactly the proved complete collar symbol plus the full rational major-arc sum over Q<den(r)<=R. This identifies the contribution whose fixed-Q maximum remains to be controlled.

Checked declarations

Module	Named supporting theorems
FullOrdinaryHighSplit.lean	1

- `circleHighModel_ordinary_full_split`

<!-- checkpoint-verification -->

Verification

All 485 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4092-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1603 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 487 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_419_OrdinaryFourPieceSymbol.md`

Checkpoint 419: The actual annular prime-square symbol split into four exact ordinary pieces

The actual annular von Mangoldt square symbol equals the original low symbol, the fully identified ordinary collar, the actual large-denominator rational model, and the actual major-arc approximation error. No one of the remaining operator bounds is assumed.

Checked declarations

Module	Named supporting theorems
OrdinaryFourPieceSymbol.lean	1

- `annularSymbol_ordinary_four_piece`

<!-- checkpoint-verification -->

Verification

All 486 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4093-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1604 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 488 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_420_OrdinaryFourPieceOperator.md`

Checkpoint 420: The actual annular l2 operator split into the same four ordinary pieces

The exact four-piece continuous symbol identity is passed through the genuine integer Fourier multiplier. All terms are actual continuous linear maps on l2, including the original low model, proved collar, large-denominator model and approximation error.

Checked declarations

Module	Named supporting theorems
OrdinaryFourPieceOperator.lean	1

- `annularOperator_ordinary_four_piece`

<!-- checkpoint-verification -->

Verification

All 487 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4094-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1605 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 489 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_421_OrdinaryFourPieceConvolution.md`

Checkpoint 421: The exact finite-complex-input prime-square four-piece decomposition

For every finite complex input, the actual annular von Mangoldt square convolution equals its original l1 low output plus its l2 collar, large-denominator and approximation-error outputs. The l1/l2 finite input realizations and every original source cutoff are checked. Large-denominator maximal estimates and the headline conclusions remain open.

Checked declarations

Module	Named supporting theorems
OrdinaryFourPieceConvolution.lean	1

- `annularConvolution_ordinary_four_piece`

<!-- checkpoint-verification -->

Verification

All 488 project modules and the root were freshly rebuilt: 4095 jobs pass without warnings/errors. All 1606 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 490 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_422_EffectiveCollarCutoff.md`

Checkpoint 422: The positive effective exceptional collar cutoff

Defines the actual exceptional effective cutoff Q q_0/B(q_0), proves positivity and its exact equivalence to the source exceptional narrow activation condition. The ordinary cutoff Q is not substituted for it.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
EffectiveCollarCutoff.lean	2

- `effectiveCollarCutoff_pos`

- `effectiveCollarCutoff_condition`

<!-- checkpoint-verification -->

Verification

All 489 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4096-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1608 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 491 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_423_ConductorCollarCenters.md`

Checkpoint 423: The exact finite conductor-multiple collar frequency set

Defines a finite canonical frequency set using the effective exceptional cutoff and the actual conductor divisibility condition. Its membership theorem retains both the real-cutoff floor and canonical rational range.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ConductorCollarCenters.lean	1

- `mem_conductorCollarCenters`

<!-- checkpoint-verification -->

Verification

All 490 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4097-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1609 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 492 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_424_SubunitConductorCollars.md`

Checkpoint 424: Exact vanishing of the exceptional collar set below one

Discharges the small effective-cutoff case exactly: positive rational denominators leave no exceptional collar frequencies when the effective cutoff is below one.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SubunitConductorCollars.lean	1

- `conductorCollarCenters_eq_empty`

<!-- checkpoint-verification -->

Verification

All 491 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4098-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1610 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 493 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_425_ConductorPowerSeparation.md`

Checkpoint 425: Extracting a conductor power from denominator decay

Proves the conductor-sensitive algebraic gain q^{-(η+ε)} ≤ q₀^{-ε}q^{-η} whenever q₀≤q, and obtains that lower bound from actual conductor divisibility. This gain is independent of any zero-free-region input.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ConductorPowerSeparation.lean	2

- `conductor_power_separation`
- `conductor_divides_den_lower`

<!-- checkpoint-verification -->

Verification

All 492 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4099-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1612 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 494 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_426_ConductorGaussEnvelope.md

Checkpoint 426: A conductor-sensitive actual twisted Gauss envelope

Discharges the arithmetic coefficient premise with the actual uniformly bounded twisted square Gauss sums. The coefficient envelope gains $q_0^{-\varepsilon}$, provided $\eta+\varepsilon<1/2$. Characters need not be principal or fixed across the dyadic scales.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ConductorGaussEnvelope.lean	1

- `conductorGauss_envelope`

<!-- checkpoint-verification -->

Verification

All 493 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4100-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1613 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 495 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_427_CollarConstantLinearity.md

Checkpoint 427: Exact homogeneity of the complete collar constant

Proves the exact homogeneity needed to carry the conductor power through the full denominator-block energy constant, avoiding an unexplained change in the quantitative bound.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
CollarConstantLinearity.lean	1

- `collarPowerConstant_mul`

<!-- checkpoint-verification -->

Verification

All 494 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4101-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1614 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 496 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_428_EffectiveCutoffPower.md`

Checkpoint 428: Exact conversion of the exceptional cutoff power

Converts the effective-cutoff decay into the original Q decay and the exact conductor-majorant ratio. All real-power domain and positivity conditions are retained.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
EffectiveCutoffPower.lean	1

- `effectiveCollarCutoff_rpow`

<!-- checkpoint-verification -->

Verification

All 495 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4102-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1615 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 497 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_429_EffectiveCutoffUniformPower.md`

Checkpoint 429: Uniform control of the effective-cutoff decay

Shows that the actual exceptional effective-cutoff decay is uniformly controlled by Q^{-η}T^η once the already proved conductor-majorant ratio is bounded by T.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
EffectiveCutoffUniformPower.lean	1

- `effectiveCollarCutoff_power_bound`

<!-- checkpoint-verification -->

Verification

All 496 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4103-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1616 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 498 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_430_ConductorCollarEnvelopeEnergy.md`

Checkpoint 430: The refined finite exceptional collar square energy

Applies the proved complete rational collar energy to the actual conductor-multiple set with its refined coefficient envelope. The result retains the original effective-cutoff power and an explicit q₀^{-ε} factor.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ConductorCollarEnvelopeEnergy.lean	1

- `conductorCollar_envelope_energy`

<!-- checkpoint-verification -->

Verification

All 497 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4104-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1617 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 499 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_431_ConductorCollarConstantBudget.md`

Checkpoint 431: The complete refined collar constant in the original cutoff

Carries the effective exceptional cutoff through the quantitative estimate and isolates one universal factor T^η. The conductor gain is preserved without dropping a sign condition.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ConductorCollarConstantBudget.lean	1

- `conductorCollar_constant_budget`

<!-- checkpoint-verification -->

Verification

All 498 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4105-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1618 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 500 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_432_RefinedTwistedCollarEnergy.md`

Checkpoint 432: Uniform conductor-sensitive twisted collar energy above the cutoff threshold

Discharges the actual twisted Gauss envelope and the uniform conductor-ratio premise. For η,ε≥0 with η+ε<1/2, proves finite circle frequency energy bounded by (H Q^{-η}q₀^{-ε})² when the effective cutoff is at least one.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
RefinedTwistedCollarEnergy.lean	1

- `refinedTwistedCollar_energy_large`

<!-- checkpoint-verification -->

Verification

All 499 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4106-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1619 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 501 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_433_AllCutoffTwistedCollarEnergy.md`

Checkpoint 433: The conductor-sensitive energy for every positive cutoff

Removes the auxiliary effective-cutoff restriction using the proved empty-frequency case. This is an unconditional energy estimate for the actual twisted collar model for every Q>0 and positive conductor.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
AllCutoffTwistedCollarEnergy.lean	1

- `refinedTwistedCollar_energy`

<!-- checkpoint-verification -->

Verification

All 500 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4107-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1620 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 502 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_434_RefinedTwistedCollarMaximum.md`

Checkpoint 434: The genuine conductor-sensitive exceptional collar maximum

Constructs a genuine real ℓ^2 maximal witness for the conductor-sensitive exceptional collar family, with pointwise domination of every scale and norm H Q^{-η}q₀^{-ε}‖f‖₂. It is a consequence of proved frequency energy, not an assumed maximal inequality.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
RefinedTwistedCollarMaximum.lean	1

- `refinedTwistedCollar_maximum`

<!-- checkpoint-verification -->

Verification

All 501 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4108-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1621 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 503 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_435_RefinedTwistedCollarPairing.md`

Checkpoint 435: The actual exceptional collar finite-indicator maximal pairing

Supplies the finite test-set pairing for the true exceptional collar maximum with both square-root cardinality factors. It does not replace the missing large-denominator high estimate.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
RefinedTwistedCollarPairing.lean	1

- `refinedTwistedCollar_indicator_pairing`

<!-- checkpoint-verification -->

Verification

All 502 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4109-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1622 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 504 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_436_CenteredExceptionalHigh.md`

Checkpoint 436: The separate genuine centered exceptional high component

Separates the actual centered exceptional high component as the difference between the ordinary and complete original high symbols. It is continuous and has the exact compact support needed for circle periodization.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
CenteredExceptionalHigh.lean	1

- `centeredExceptionalHigh_support`

<!-- checkpoint-verification -->

Verification

All 503 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4110-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1623 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 505 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_437_CenteredExceptionalHighFormula.md

Checkpoint 437: The exact twisted Gauss and effective-mask formula

Derives the exact exceptional formula from the original high model. The narrow complement uses Qq_0/B, while the original Gauss coefficient, selector, β and broad cutoff are all retained.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
CenteredExceptionalHighFormula.lean	1

- `centeredExceptionalHigh_formula`

<!-- checkpoint-verification -->

Verification

All 504 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4111-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1624 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 506 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_438_CenteredExceptionalCollar.md

Checkpoint 438: The actual exceptional high collar correspondence

Identifies every actual exceptional high term below its own effective cutoff with the exact collar model already controlled by square energy. No ordinary-cutoff truncation is imposed on the exceptional term.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
CenteredExceptionalCollar.lean	1

- `centeredExceptionalHigh_collar`

<!-- checkpoint-verification -->

Verification

All 505 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4112-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1625 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 507 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_439_CircleExceptionalHigh.md

Checkpoint 439: The genuine exceptional high circle term and full high split

Periodizes the separate exceptional high component and proves the exact signed identity for the original circle high term. This preserves the exceptional contribution even when the two narrow cutoffs differ.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
CircleExceptionalHigh.lean	1

- `circleHighTerm_ordinary_sub_exceptional`

<!-- checkpoint-verification -->

Verification

All 506 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4113-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1626 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 508 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_440_CircleExceptionalCollar.md`

Checkpoint 440: The exact exceptional high circle collar identity

Transfers the exact centered correspondence to the genuine continuous circle term, with all period-window terms accounted for.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
CircleExceptionalCollar.lean	1

- `circleExceptionalHigh_collar`

<!-- checkpoint-verification -->

Verification

All 507 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4114-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1627 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 509 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_441_FiniteExceptionalHigh.md`

Checkpoint 441: Exact ordinary-minus-exceptional splitting of the finite high model

Proves the exact finite-family split before any cutoff filtering. It avoids silently identifying the ordinary and exceptional collar denominator sets.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
FiniteExceptionalHigh.lean	1

- `circleHighModel_ordinary_sub_exceptional`

<!-- checkpoint-verification -->

Verification

All 508 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4115-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1628 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 510 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_442_FiniteExceptionalCollar.md`

Checkpoint 442: The full exact exceptional collar sum

Identifies the entire finite exceptional collar sum with the exact rational collar block for arbitrary bounded source selectors. This is the source-model bridge needed for the conductor-sensitive maximum.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
FiniteExceptionalCollar.lean	1

- `circleExceptionalHighModel_collar`

<!-- checkpoint-verification -->

Verification

All 509 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4116-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1629 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 511 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_443_LargeCenteredExceptionalHigh.md`

Checkpoint 443: The exact exceptional high term above its effective cutoff

Identifies the remaining large-denominator exceptional high contribution with the full twisted model rather than a collar. No maximal estimate for this piece is claimed.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
LargeCenteredExceptionalHigh.lean	1

- `centeredExceptionalHigh_large`

<!-- checkpoint-verification -->

Verification

All 510 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4117-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1630 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 512 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_444_LargeCircleExceptionalHigh.md

Checkpoint 444: The exact large-denominator exceptional circle contribution

Proves the full circle identity for exceptional denominators above their effective cutoff. The remaining high-frequency estimate is precisely located on this full-model piece.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
LargeCircleExceptionalHigh.lean	1

- `circleExceptionalHigh_large`

<!-- checkpoint-verification -->

Verification

All 511 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4118-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1631 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 513 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_445_RationalCollarFilter.md

Checkpoint 445: Exact frequency filtering by scalar collar coefficients

Proves exact filtering of the genuine rational collar symbol by a scalar coefficient indicator. This permits a different source-radius cutoff at every dyadic scale while preserving one fixed finite frequency universe.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
RationalCollarFilter.lean	1

- `rationalCollarBlock_filter`

<!-- checkpoint-verification -->

Verification

All 512 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4119-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1632 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 514 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_446_CircleCollarSubsetExtension.md`

Checkpoint 446: Exact extension of a varying finite collar subset

Extends any source collar subset to a fixed containing frequency set using coefficients that vanish outside the subset. The identity handles arbitrary scale-dependent finite subsets, not only monotone cutoffs.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
CircleCollarSubsetExtension.lean	1

- `rationalCollarBlock_subset`

<!-- checkpoint-verification -->

Verification

All 513 project modules and the root were freshly rebuilt: 4120 jobs pass without warnings/errors. All 1633 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 515 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_447_SourceEffectiveConductorFilter.md`

Checkpoint 447: The exact source exceptional frequency-set identity

Connects the actual globally truncated source frequency set to the canonical conductor collar set when the global radius contains the effective cutoff. This uses the exceptional cutoff, which can differ from Q.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceEffectiveConductorFilter.lean	1

- `source_effective_conductor_filter`

<!-- checkpoint-verification -->

Verification

All 514 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4121-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1634 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 516 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_448_ExceptionalSelectorPruning.md`

Checkpoint 448: Exact removal of nonconductor frequencies from the exceptional collar

Proves that the source exceptional selector removes every nonmultiple of the conductor exactly, including inactive exceptional data. This justifies applying the conductor-sensitive envelope to the original source sum.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ExceptionalSelectorPruning.lean	1

- `rationalCollarBlock_exceptional_prune`

<!-- checkpoint-verification -->

Verification

All 515 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4122-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1635 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 517 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_449_SourceExceptionalCollarIdentity.md`

Checkpoint 449: The original exceptional high collar equals the refined model

Proves the exact equality between the paper source exceptional high collar and the conductor-sensitive circle model. It allows arbitrary active flags and characters and retains the original broad/narrow cutoff geometry.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceExceptionalCollarIdentity.lean	1

- `source_exceptional_collar_identity`

<!-- checkpoint-verification -->

Verification

All 516 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4123-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1636 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 518 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_450_SourceExceptionalCollarEnergy.md`

Checkpoint 450: Uniform energy of the original exceptional source collar

Transfers the proved conductor-sensitive energy to the literal original exceptional source collar, uniformly over changing global source radii, activity flags, character data and β values. No maximal input is assumed.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceExceptionalCollarEnergy.lean	1

- `sourceExceptionalCollar_energy`

<!-- checkpoint-verification -->

Verification

All 517 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4124-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1637 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 519 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_451_SourceExceptionalCollarMaximum.md`

Checkpoint 451: The original exceptional source collar countable maximum

Produces the actual countable ℓ^2 maximum for the original source exceptional collar, with the uniform conductor power gain. The finite-energy premise is fully discharged by the preceding source identity and Gauss bounds.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceExceptionalCollarMaximum.lean	1

- `sourceExceptionalCollar_maximum`

<!-- checkpoint-verification -->

Verification

All 518 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4125-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1638 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 520 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_452_ExceptionalHighDenominatorSplit.md`

Checkpoint 452: The exact exceptional high split at its own cutoff

Splits the entire exceptional high symbol at an independently specified cutoff, retaining the boundary in the collar set. This permits the actual exceptional cutoff without conflating it with the ordinary one.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ExceptionalHighDenominatorSplit.lean	1

- `circleExceptionalHighModel_denominator_split`

<!-- checkpoint-verification -->

Verification

All 519 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4126-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1639 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 521 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_453_LargeExceptionalHighModel.md`

Checkpoint 453: The remaining complete exceptional large-denominator model

Identifies the entire exceptional large-denominator contribution exactly with a difference of genuine major-arc models. The estimate still required for that piece is not hidden in a collar bound.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
LargeExceptionalHighModel.lean	1

- `circleExceptionalHighModel_large`

<!-- checkpoint-verification -->

Verification

All 520 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4127-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1640 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 522 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_454_FullSourceHighFourPieces.md`

Checkpoint 454: An exact four-piece decomposition of the complete source high model

Proves the full original source high symbol is the sum of ordinary collar and ordinary large denominators, minus exceptional collar and exceptional large denominators. Both actual collar correspondences are exact and the remaining two large pieces are explicit.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
FullSourceHighFourPieces.lean	1

- `circleHighModel_source_four_pieces`

<!-- checkpoint-verification -->

Verification

All 521 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4128-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1641 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 523 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_455_SourceSixPieceSymbol.md`

Checkpoint 455: The exact annular prime-square symbol decomposition with exceptional data

Gives an exact six-piece identity for the actual annular von Mangoldt prime-square symbol: low, two ordinary high parts, two exceptional high parts and the actual analytic error. This is an algebraic identity, not a bound on the remaining large pieces or error.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceSixPieceSymbol.lean	1

- `annularSymbol_source_six_pieces`

<!-- checkpoint-verification -->

Verification

All 522 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4129-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1642 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 524 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_456_SourceHighOperatorFourPieces.md`

Checkpoint 456: The actual high operator has the exact four-piece split

Transfers the exact source high decomposition to genuine bounded ℓ^2 convolution operators. No scalar formal multiplier is substituted for the actual operator.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceHighOperatorFourPieces.lean	1

- `circleHighOperator_source_four_pieces`

<!-- checkpoint-verification -->

Verification

All 523 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4130-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1643 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 525 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_457_ExceptionalCollarInactive.md`

Checkpoint 457: Exact disappearance of inactive exceptional collar data

Proves the exceptional collar is exactly zero when the source exceptional flag is inactive, including every frequency and every scale. This supports the subsequent changing-character energy assembly.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ExceptionalCollarInactive.lean	1

- `rationalCollarBlock_inactive`

<!-- checkpoint-verification -->

Verification

All 524 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4131-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1644 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 526 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_458_TimeClassFiniteSum.md`

Checkpoint 458: Exact finite partition of dyadic times by character class

Proves the exact finite time partition needed when the exceptional character changes between dyadic scales. Each scale is counted in exactly one class.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
TimeClassFiniteSum.lean	1

- `time_class_sum`

<!-- checkpoint-verification -->

Verification

All 525 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4132-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1645 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 527 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_459_DisjointTimeSquareEnergy.md`

Checkpoint 459: Square-energy assembly with one active class at each time

Proves a sum of squared class constants, rather than the square of their sum, when exactly one character class is selected at each time. No monotonicity of the selected class is required.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
DisjointTimeSquareEnergy.lean	1

- `disjoint_time_square_energy`

<!-- checkpoint-verification -->

Verification

All 526 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4133-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1646 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 528 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_460_ConductorClassPositivity.md`

Checkpoint 460: Positive conductors from dyadic class lower bounds

Records the precise numerical classification premise used in the changing-character argument: the conductor assigned to class i is at least 2^i. It proves positivity but does not assert a formalized Landau–Page theorem.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ConductorClassPositivity.lean	2

- `conductor_class_positive`
- `conductor_class_real_lower`

<!-- checkpoint-verification -->

Verification

All 527 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4134-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1648 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 529 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_461_ConductorClassPowerDecay.md`

Checkpoint 461: The geometric decay supplied by separated conductor classes

Turns the conductor-class lower bound into geometric decay, with all real/natural exponent conversions proved explicitly.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ConductorClassPowerDecay.lean	1

- `conductor_class_power_decay`

<!-- checkpoint-verification -->

Verification

All 528 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4135-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1649 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 530 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_462_ConductorClassSquareDecay.md`

Checkpoint 462: Summable squared conductor gains

Proves the exact square-decay exponent needed for energy summation over conductor classes. This avoids losing the factor of two in the summability parameter.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ConductorClassSquareDecay.lean	1

- `conductor_class_square_decay`

<!-- checkpoint-verification -->

Verification

All 529 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4136-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1650 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 531 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_463_FiniteConductorClassBudget.md`

Checkpoint 463: A class-count-independent finite conductor budget

Bounds the squared conductor gains over any finite set of classes by one geometric-series constant. The bound is independent of how many classes occur in the selected time set.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
FiniteConductorClassBudget.lean	1

- `finite_conductor_class_budget`

<!-- checkpoint-verification -->

Verification

All 530 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4137-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1651 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 532 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_464_ConductorClassEnergyConstant.md`

Checkpoint 464: The positive class energy constant and its exact square

Defines the explicit finite class energy constant and proves its positivity and exact square. This constant is the sole cost of allowing the character class to change arbitrarily with time.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ConductorClassEnergyConstant.lean	2

- `conductorClassEnergyConstant_pos`
- `conductorClassEnergyConstant_sq`

<!-- checkpoint-verification -->

Verification

All 531 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4138-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1653 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 533 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_465_WeightedConductorClassBudget.md`

Checkpoint 465: The complete uniform weighted class square budget

Scales the class budget by the actual Q-dependent energy amplitude. It is valid for every finite class image and therefore for arbitrary finite time tests.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
WeightedConductorClassBudget.lean	1

- `weighted_conductor_class_budget`

<!-- checkpoint-verification -->

Verification

All 532 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4139-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1654 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 534 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

Checkpoint_466_MixedCircleSquareEnergy.md

Checkpoint 466: Square energy for genuine circle symbols with changing conductor classes

Assembles actual continuous circle families with one selected conductor class per time into a uniform finite square-energy estimate. The class selection can change arbitrarily and need not be a time-tail.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
MixedCircleSquareEnergy.lean	1

- `mixed_circle_square_energy`

<!-- checkpoint-verification -->

Verification

All 533 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4140-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1655 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 535 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_467_MixedCircleMaximum.md

Checkpoint 467: A genuine countable maximum under changing conductor classes

Constructs the actual countable ℓ^2 maximum for the selected circle family from the proved mixed energy. The numerical conductor classification premise remains displayed and is not represented as an analytic theorem already formalized.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
MixedCircleMaximum.lean	1

- `mixed_circle_maximum`

<!-- checkpoint-verification -->

Verification

All 534 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4141-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1656 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 536 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_468_DyadicConductorCollar.md`

Checkpoint 468: The actual dyadic conductor collar family with a short interface

Packages the actual dyadic conductor collar family without changing any definition or estimate. Its proved energy interface is used to assemble different conductor classes without repeatedly rebuilding scale proof terms.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
DyadicConductorCollar.lean	1

- `dyadicConductorCollar_energy`

<!-- checkpoint-verification -->

Verification

All 535 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4142-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1657 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 537 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_469_ChangingConductorCollarEnergy.md`

Checkpoint 469: Uniform energy for collars with a changing conductor class

Proves the actual changing-conductor collar energy $K^2Q^{-2\eta}$ with a constant independent of the chosen time sequence. The only classification premise is $q_0(i)\geq 2^i$; the arithmetic and collar energy premises are discharged.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ChangingConductorCollarEnergy.lean	1

- `changingConductorCollar_energy`

<!-- checkpoint-verification -->

Verification

All 536 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4143-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1658 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 538 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_470_ChangingConductorCollarMaximum.md

Checkpoint 470: The actual fixed-Q maximum with changing conductor classes

Constructs the genuine fixed-Q countable maximum for arbitrary class selection at each dyadic time, with KQ^{-η}‖f‖₂. A formalized Landau–Page classification is not claimed.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ChangingConductorCollarMaximum.lean	1

- `changingConductorCollar_maximum`

<!-- checkpoint-verification -->

Verification

All 537 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4144-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1659 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 539 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_471_ClippedSourceConductorCenters.md

Checkpoint 471: The exact conductor collar set for arbitrary nonnegative source radius

Removes the assumption that the global source radius contains the whole exceptional collar. The source set is exactly the full conductor collar clipped at the actual global radius, including when the conductor lies beyond that radius.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ClippedSourceConductorCenters.lean	1

- `clipped_source_conductor_centers`

<!-- checkpoint-verification -->

Verification

All 538 project modules and the root were freshly rebuilt: 4145 jobs pass without warnings/errors. All 1660 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 540 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_472_ClippedSourceCollarIdentity.md`

Checkpoint 472: The exact exceptional source collar with arbitrary source clipping

Identifies the literal exceptional source collar exactly with the fixed conductor family and a bounded source-radius coefficient mask, for every nonnegative source radius. This removes an artificial containment condition.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ClippedSourceCollarIdentity.lean	1

- `clipped_source_collar_identity`

<!-- checkpoint-verification -->

Verification

All 539 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4146-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1661 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 541 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_473_ClippedSourceCollarEnergy.md`

Checkpoint 473: The original clipped exceptional collar has uniform refined energy

Proves the refined source collar energy for arbitrary changing nonnegative radii, even when the effective cutoff exceeds the source radius. The source clipping is handled by a proved coefficient contraction.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ClippedSourceCollarEnergy.lean	1

- `clippedSourceCollar_energy`

<!-- checkpoint-verification -->

Verification

All 540 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4147-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1662 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 542 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_474_ClippedSourceCollarMaximum.md`

Checkpoint 474: The actual source exceptional collar maximum without containment assumptions

Constructs the genuine original exceptional source collar maximum with no global-radius containment condition. The conductor power gain and arbitrary source-radius variation are both retained.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ClippedSourceCollarMaximum.lean	1

- `clippedSourceCollar_maximum`

<!-- checkpoint-verification -->

Verification

All 541 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4148-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1663 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 543 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_475_ChangingSourceCollarEnergy.md`

Checkpoint 475: Uniform energy of the actual source collar with changing conductor classes

Proves uniform energy of the literal exceptional source collar under arbitrary changing conductor-class selection, source clipping and exceptional activation. Only the displayed dyadic conductor classification premise remains.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ChangingSourceCollarEnergy.lean	1

- `changingSourceCollar_energy`

<!-- checkpoint-verification -->

Verification

All 542 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4149-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1664 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 544 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_476_ChangingSourceCollarMaximum.md`

Checkpoint 476: The actual source exceptional collar maximum with changing conductors

Constructs the genuine countable ℓ^2 maximum for the original clipped exceptional source collar even as its conductor class changes. The bound is KQ^{-η}‖f‖₂ in the displayed subhalf-power range.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ChangingSourceCollarMaximum.lean	1

- `changingSourceCollar_maximum`

<!-- checkpoint-verification -->

Verification

All 543 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4150-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1665 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 545 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_477_ChangingSourceCollarPairing.md`

Checkpoint 477: The original changing-character source collar finite-indicator pairing

Proves the finite-indicator restricted pairing for the actual changing-character source collar maximum. It supplies a checked component of the eventual high restricted estimate, with the large-denominator terms still separate.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ChangingSourceCollarPairing.lean	1

- `changingSourceCollar_indicator_pairing`

<!-- checkpoint-verification -->

Verification

All 544 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4151-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1666 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 546 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

Checkpoint_478_DenominatorActivation.md

Checkpoint 478: The literal source denominator activation and prefix identity

Defines the actual denominator activation and proves its exact finite prefix-sum identity. The cutoff acts separately on each denominator, even when all denominators lie in one dyadic block.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
DenominatorActivation.lean	2

- `denominator_activation_sum`
- `denominatorActivation_norm`

<!-- checkpoint-verification -->

Verification

All 545 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4152-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1668 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 547 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_479_IndividualDenominatorTail.md

Checkpoint 479: Individual denominator activation is a time tail under increasing radii

Proves the valid individual-tail statement: each fixed denominator stays active once the source radius reaches it. The statement does not identify an entire denominator block with one common time tail.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
IndividualDenominatorTail.lean	1

- `individual_denominator_tail`

<!-- checkpoint-verification -->

Verification

All 546 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4153-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1669 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 548 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_480_CommonTailMaskCounterexample.md`

Checkpoint 480: A checked obstruction to treating a denominator prefix as one common tail

Proves a finite exact counterexample to the common-tail mask claim: denominators 2 and 3, in the same dyadic block, have activation rows (1,0) and (1,1) at radii 2 and 3. This concerns a proof step in Proposition 6.21; it is not a counterexample to its conclusion. The appropriate repair is a denominator-prefix maximum, as in the paper's Corollary 6.19.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
CommonTailMaskCounterexample.lean	1

- `denominator_prefix_not_common_tail`

<!-- checkpoint-verification -->

Verification

All 547 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4154-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1670 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 549 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_481_IntervalComplexSum.md`

Checkpoint 481: Exact contiguous finite sums for denominator prefixes

Defines contiguous coefficient sums and proves exact concatenation and endpoint identities. These are the finite denominator-prefix objects used in the Rademacher–Menshov argument.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
IntervalComplexSum.lean	3

- `intervalComplexSum_add`
- `intervalComplexSum_zero`
- `intervalComplexSum_one`

<!-- checkpoint-verification -->

Verification

All 548 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4155-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1673 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 550 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_482_BinaryPrefixSplit.md`

Checkpoint 482: The exact binary split of a denominator prefix

Proves the exact prefix split across the midpoint of a binary interval and the correct bound for the residual prefix. No denominator-independent time-tail assumption is made.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryPrefixSplit.lean	2

- `intervalComplexSum_binary_prefix`
- `binary_prefix_remainder`

<!-- checkpoint-verification -->

Verification

All 549 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4156-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1675 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 551 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

Checkpoint_483_BinaryTreeSquareEnergy.md

Checkpoint 483: The actual finite binary interval square energy

Defines the exact sum of squared interval sums over the finite binary tree, including every level. Its nonnegativity is proved recursively.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryTreeSquareEnergy.lean	1

- `binaryTreeSquareEnergy_nonneg`

<!-- checkpoint-verification -->

Verification

All 550 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4157-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1676 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 552 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_484_BinaryTreeEnergyContainment.md

Checkpoint 484: The full interval and both child energies are controlled by the tree

Proves that the full interval sum and the total child energy are bounded by the exact tree energy. These are the nonnegative comparison steps used in the prefix induction.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryTreeEnergyContainment.lean	2

- `binaryTreeSquareEnergy_root`
- `binaryTreeSquareEnergy_children`

<!-- checkpoint-verification -->

Verification

All 551 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4158-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1678 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 553 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_485_TwoTermWeightedSquare.md`

Checkpoint 485: The weighted two-term square estimate for prefix induction

Proves the weighted two-term square inequality needed to retain a logarithmic, rather than exponential, prefix cost in the binary induction.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
TwoTermWeightedSquare.lean	1

- `norm_add_square_weighted`

<!-- checkpoint-verification -->

Verification

All 552 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4159-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1679 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 554 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_486_BinaryPrefixSquareBound.md`

Checkpoint 486: A proved logarithmic binary-prefix square bound

Proves by binary induction that every denominator prefix has squared norm at most (h+1) times the complete binary-tree square energy. This is the core finite Rademacher–Menshov prefix estimate.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryPrefixSquareBound.lean	1

- `binary_prefix_square_bound`

<!-- checkpoint-verification -->

Verification

All 553 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4160-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1680 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 555 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_487_BinaryPrefixPointwiseEnvelope.md`

Checkpoint 487: A simultaneous pointwise envelope for every finite prefix

Converts the exact prefix square bound into one nonnegative pointwise square-root envelope that dominates every truncation of the finite denominator sequence.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryPrefixPointwiseEnvelope.lean	1

- `binary_prefix_norm_envelope`

<!-- checkpoint-verification -->

Verification

All 554 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4161-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1681 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 556 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_488_L2PointwiseMagnitude.md`

Checkpoint 488: The actual real l2 magnitude of a complex l2 function

Constructs the genuine real ℓ^2 pointwise magnitude of a complex ℓ^2 function and proves exact norm preservation. This is used for measurable/normed prefix envelopes.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
L2PointwiseMagnitude.lean	1

- `l2Magnitude_norm`

<!-- checkpoint-verification -->

Verification

All 555 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4162-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1682 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 557 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_489_IntervalL2Sum.md`

Checkpoint 489: Actual finite interval sums in complex l2

Defines the actual ℓ^2 finite interval sum, proves its exact pointwise correspondence with the scalar sum and its concatenation identity.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
IntervalL2Sum.lean	2

- `intervalL2Sum_apply`
- `intervalL2Sum_add`

<!-- checkpoint-verification -->

Verification

All 556 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4163-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1684 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 558 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

Checkpoint_490_BinaryTreeL2Energy.md

Checkpoint 490: The finite tree of actual l2 interval norms

Defines the exact sum of squared norms of the genuine ℓ^2 interval sums over the binary tree. Its positivity is proved independently of any orthogonality assumption.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryTreeL2Energy.lean	1

- `binaryTreeL2Energy_nonneg`

<!-- checkpoint-verification -->

Verification

All 557 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4164-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1685 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 559 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_491_BinaryTreeCoordinateSummability.md

Checkpoint 491: Summability in space of the finite binary tree energy

Proves that the exact pointwise binary tree energy has a summable spatial total for arbitrary ℓ^2 inputs. This justifies the subsequent square-root envelope as an actual ℓ^2 object.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryTreeCoordinateSummability.lean	1

- `binaryTree_coordinate_summable`

<!-- checkpoint-verification -->

Verification

All 558 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4165-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1686 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 560 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint notes 492-566

Checkpoint_492_BinaryTreeSpatialTotal.md

Checkpoint 492: Exact Fubini identity for the finite binary tree

Proves the exact equality between the spatial sum of pointwise tree energies and the finite tree of actual ℓ^2 norms. All exchanges use established summability.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryTreeSpatialTotal.lean	1

- `binaryTree_spatial_total`

<!-- checkpoint-verification -->

Verification

All 559 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4166-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1687 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 561 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_493_BinaryTreeL2Envelope.md

Checkpoint 493: The genuine square-root binary tree envelope in real l2

Constructs the actual nonnegative real ℓ^2 binary-tree envelope from its pointwise square root, using the proved coordinate summability.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryTreeL2Envelope.lean	1

- `binaryTreeEnvelope_nonneg`

<!-- checkpoint-verification -->

Verification

All 560 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4167-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1688 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 562 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_494_BinaryTreeEnvelopeNorm.md`

Checkpoint 494: The exact norm of the binary tree envelope

Proves the exact norm-square identity and square-root norm formula for the genuine binary tree envelope, with no unspecified absolute constants.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryTreeEnvelopeNorm.lean	2

- `binaryTreeEnvelope_norm_square`
- `binaryTreeEnvelope_norm`

<!-- checkpoint-verification -->

Verification

All 561 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4168-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1690 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 563 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_495_PrefixL2Majorant.md`

Checkpoint 495: A genuine real l2 majorant of all denominator prefixes

Builds one actual nonnegative ℓ^2 function dominating every denominator prefix pointwise. Its membership is proved, rather than only its formal pointwise expression being specified.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
PrefixL2Majorant.lean	2

- `prefixL2Majorant_dominates`

- `prefixL2Majorant_nonneg`

<!-- checkpoint-verification -->

Verification

All 562 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4169-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1692 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 564 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_496_PrefixL2MajorantNorm.md`

Checkpoint 496: The quantitative finite-prefix l2 estimate from actual tree energy

Proves the exact ℓ^2 cost of the common prefix majorant in terms of the finite tree energy. The sharp logarithmic Rademacher–Menshov cost follows when the separate interval energy budget is supplied and proved.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
PrefixL2MajorantNorm.lean	1

- `prefixL2Majorant_norm`

<!-- checkpoint-verification -->

Verification

All 563 project modules and the root were freshly rebuilt: 4170 jobs pass without warnings/errors. All 1693 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 565 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_497_IntervalRealBudget.md`

Checkpoint 497: Exact nonnegative interval budgets for the prefix tree

Defines the exact leaf-energy budget and proves concatenation and nonnegativity. This keeps the interval norm assumptions separate from the prefix conclusion.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
IntervalRealBudget.lean	2

- `intervalRealBudget_add`
- `intervalRealBudget_nonneg`

<!-- checkpoint-verification -->

Verification

All 564 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4171-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1695 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 566 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_498_BinaryTreeUniformBudget.md`

Checkpoint 498: The logarithmic tree-energy budget from uniform interval bounds

Proves that a uniform dyadic interval norm bound gives total binary-tree norm energy at most (h+1) times the leaf budget. This is the second logarithmic factor in the squared prefix estimate.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryTreeUniformBudget.lean	1

- `binaryTree_uniform_budget`

<!-- checkpoint-verification -->

Verification

All 565 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4172-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1696 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 567 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

`Checkpoint_499_RademacherMenshovMajorant.md`

Checkpoint 499: A proved finite Rademacher–Menshov majorant with exact logarithmic cost

Proves a finite Rademacher–Menshov theorem for actual ℓ^2 interval sums: a genuine nonnegative majorant of every prefix has norm at most (h+1)B times the square root of leaf energy. The interval estimate is an explicit, independently meaningful premise.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
RademacherMenshovMajorant.lean	1

- `rademacherMenshov_majorant`

<!-- checkpoint-verification -->

Verification

All 566 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4173-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1697 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 568 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_500_OrthogonalFiniteSumEnergy.md`

Checkpoint 500: Exact Pythagoras for finite orthogonal sums

Proves exact finite Pythagoras for a pairwise orthogonal complex Hilbert-space family. This discharges the interval norm premise in the orthogonal specialization of the prefix theorem.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
OrthogonalFiniteSumEnergy.lean	1

- `orthogonal_finset_sum_energy`

<!-- checkpoint-verification -->

Verification

All 567 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4174-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1698 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 569 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_501_OrthogonalL2IntervalEnergy.md`

Checkpoint 501: The actual orthogonal l2 interval energy identity

Proves the exact interval energy identity for actual complex ℓ^2 functions, using injectivity of the interval index shift. It supplies the leaf budget with B=1.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
OrthogonalL2IntervalEnergy.lean	1

- `orthogonal_l2_interval_energy`

<!-- checkpoint-verification -->

Verification

All 568 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4175-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1699 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 570 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_502_OrthogonalRademacherMenshov.md`

Checkpoint 502: A fully proved orthogonal l2 prefix maximal estimate

Discharges every analytic premise in the orthogonal specialization and obtains the actual ℓ^2 finite-prefix maximal majorant with logarithmic cost h+1. The paper's nonorthogonal multifrequency family is a separate application still to be proved.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
OrthogonalRademacherMenshov.lean	1

- `orthogonal_rademacherMenshov`

<!-- checkpoint-verification -->

Verification

All 569 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4176-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1700 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 571 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_503_PrefixMajorantFinitePairing.md`

Checkpoint 503: Finite test-set control of every prefix by the proved majorant

Transfers the genuine prefix majorant to finite test-set pairings with the correct square-root cardinality factor. The exact tree-energy cost remains visible.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
PrefixMajorantFinitePairing.lean	1

- `prefix_majorant_finite_pairing`

<!-- checkpoint-verification -->

Verification

All 570 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4177-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1701 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 572 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_504_RealL2SquareEnergy.md`

Checkpoint 504: Exact summability and total of real l2 squares

Proves the exact spatial square summability and norm-square identity for genuine real ℓ^2 majorants. These facts allow interval maxima to be assembled without treating them as complex linear outputs.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
RealL2SquareEnergy.lean	2

- `real_l2_square_summable`
- `real_l2_square_total`

<!-- checkpoint-verification -->

Verification

All 571 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4178-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1703 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 573 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_505_BinaryNodeSquareEnergy.md`

Checkpoint 505: The actual binary tree energy of interval majorants

Defines the exact binary tree of squared interval majorants, independently of the particular time at which the underlying multiplier is evaluated.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryNodeSquareEnergy.lean	1

- `binaryNodeSquareEnergy_nonneg`

<!-- checkpoint-verification -->

Verification

All 572 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4179-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1704 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 574 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_506_BinaryNodeNormEnergy.md`

Checkpoint 506: The finite norm budget for interval maximal majorants

Defines the exact norm budget for all interval majorants in the finite tree and proves its nonnegativity. It is independent of the number of time parameters being maximized.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryNodeNormEnergy.lean	1

- `binaryNodeNormEnergy_nonneg`

<!-- checkpoint-verification -->

Verification

All 573 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4180-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1705 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 575 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_507_BinaryNodeCoordinateSummability.md`

Checkpoint 507: Spatial summability of the interval-maximal tree

Proves spatial summability of the actual finite tree of interval-maximal majorants, justifying its square-root ℓ^2 envelope.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryNodeCoordinateSummability.lean	1

- `binaryNode_coordinate_summable`

<!-- checkpoint-verification -->

Verification

All 574 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4181-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 1706 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 576 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_508_BinaryNodeSpatialTotal.md`

Checkpoint 508: Exact norm-energy total for the interval-maximal tree

Proves the exact spatial-total identity for the tree of genuine interval maximal majorants. The proof explicitly justifies each finite sum/tsum exchange.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryNodeSpatialTotal.lean	1

- `binaryNode_spatial_total`

<!-- checkpoint-verification -->

Verification

All 575 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4182-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1707 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 577 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_509_BinaryNodeL2Envelope.md`

Checkpoint 509: The genuine time-uniform binary interval envelope

Constructs the actual nonnegative real ℓ^2 tree envelope of interval maximal functions. This envelope is independent of the chosen time and denominator prefix.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryNodeL2Envelope.lean	1

- `binaryNodeEnvelope_nonneg`

<!-- checkpoint-verification -->

Verification

All 576 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4183-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1708 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 578 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_510_BinaryNodeEnvelopeNorm.md`

Checkpoint 510: The exact norm of the time-uniform interval envelope

Proves the exact ℓ^2 norm formula for the time-uniform binary interval envelope. The remaining logarithmic cost comes only from the finite binary prefix comparison and the interval norm budget.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryNodeEnvelopeNorm.lean	2

- `binaryNodeEnvelope_norm_square`
- `binaryNodeEnvelope_norm`

<!-- checkpoint-verification -->

Verification

All 577 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4184-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1710 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 579 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_511_BinaryTreeIntervalDomination.md`

Checkpoint 511: Domination of every scalar binary tree by interval majorants

Proves that interval-wise pointwise majorants dominate the entire scalar binary tree at every time. No positivity premise is needed beyond what follows from domination of norms.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryTreeIntervalDomination.lean	1

- `binaryTree_interval_domination`

<!-- checkpoint-verification -->

Verification

All 578 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4185-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1711 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 580 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_512_TimePrefixPointwiseEnvelope.md`

Checkpoint 512: One pointwise envelope for both time and denominator prefixes

Proves a single pointwise envelope for all times and all denominator prefixes. The required interval maximal control is displayed explicitly; no multifrequency theorem is silently assumed.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
TimePrefixPointwiseEnvelope.lean	1

- `time_prefix_pointwise_envelope`

<!-- checkpoint-verification -->

Verification

All 579 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4186-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1712 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 581 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_513_TimePrefixL2Majorant.md`

Checkpoint 513: The genuine l2 majorant of the double maximum

Constructs a genuine nonnegative real ℓ^2 majorant for the simultaneous time/prefix problem and proves its exact norm formula.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
TimePrefixL2Majorant.lean	2

- `timePrefixL2Majorant_nonneg`
- `timePrefixL2Majorant_norm`

<!-- checkpoint-verification -->

Verification

All 580 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4187-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1714 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 582 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_514_BinaryNodeUniformBudget.md`

Checkpoint 514: The exact logarithmic energy budget for interval maxima

Proves the logarithmic tree-energy budget directly for actual interval maximal majorants. This is the quantitative step needed for a second supremum over denominator prefixes.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
BinaryNodeUniformBudget.lean	1

- `binaryNode_uniform_budget`

<!-- checkpoint-verification -->

Verification

All 581 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4188-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1715 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 583 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_515_TimePrefixRademacherMenshov.md`

Checkpoint 515: A proved time-uniform Rademacher–Menshov prefix theorem

Proves the full abstract denominator-prefix step with a simultaneous time maximum: interval maximal majorants and their uniform energy budgets imply one ℓ^2 majorant of every time and prefix at cost h+1. The paper's multifrequency interval estimate remains the separate application premise.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
TimePrefixRademacherMenshov.lean	1

- `time_prefix_rademacherMenshov`

<!-- checkpoint-verification -->

Verification

All 582 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4189-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1716 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 584 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_516_TimePrefixFinitePairing.md`

Checkpoint 516: Finite pairings for the simultaneous time and prefix envelope

Supplies finite test-set control of the actual time/prefix majorant, with exact tree norm and cardinality factors. This supports a later restricted-pairing application without asserting the missing multifrequency bound.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
TimePrefixFinitePairing.lean	1

- `time_prefix_finite_pairing`

<!-- checkpoint-verification -->

Verification

All 583 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4190-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1717 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 585 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_517_FiniteNaturalPrefixSet.md

Checkpoint 517: The exact finite natural denominator-prefix set

Identifies a denominator-prefix filter exactly with the appropriate truncated natural range, including empty and endpoint cases.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
FiniteNaturalPrefixSet.lean	1

- `natural_prefix_filter`

<!-- checkpoint-verification -->

Verification

All 584 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4191-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1718 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 586 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_518_RealDenominatorPrefixSet.md

Checkpoint 518: The exact real-cutoff denominator-prefix set

Proves the exact floor conversion for real source cutoffs. The upper endpoint and the distinction between floor R and floor R plus one are explicit.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
RealDenominatorPrefixSet.lean	1

- `real_denominator_prefix_filter`

<!-- checkpoint-verification -->

Verification

All 585 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4192-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1719 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 587 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_519_RealCutoffScalarPrefix.md`

Checkpoint 519: The actual real source truncation is a contiguous scalar prefix

Identifies the literal real source activation sum with an actual contiguous prefix of a finite denominator sequence. This is the exact object controlled by the prefix maximal theorem.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
RealCutoffScalarPrefix.lean	1

- `real_cutoff_scalar_prefix`

<!-- checkpoint-verification -->

Verification

All 586 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4193-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1720 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 588 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_520_RationalDenominatorGrouping.md`

Checkpoint 520: Exact finite grouping of rational frequencies by denominator

Proves exact rational-frequency grouping by denominator for any additive commutative target, including actual continuous symbols and bounded operators.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
RationalDenominatorGrouping.lean	1

- `rational_denominator_grouping`

<!-- checkpoint-verification -->

Verification

All 587 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4194-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1721 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 589 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_521_RationalNaturalPrefixGrouping.md`

Checkpoint 521: Exact denominator-prefix grouping of a finite rational family

Proves the exact prefix grouping of a rational finite sum. Frequencies with the same denominator remain together, and no ordering of individual numerators is assumed.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
RationalNaturalPrefixGrouping.lean	1

- `rational_natural_prefix_grouping`

<!-- checkpoint-verification -->

Verification

All 588 project modules and the root were freshly rebuilt: 4195 jobs pass without warnings/errors. All 1722 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice

and Quot.sound. The source scan checks 590 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_522_RationalRealPrefixGrouping.md`

Checkpoint 522: Exact real-cutoff prefix grouping of rational families

Extends the exact grouping identity to the literal real source-radius cutoff. This connects the source finite rational sum to a natural denominator-prefix maximum without losing the boundary.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
RationalRealPrefixGrouping.lean	1

- `rational_real_prefix_grouping`

<!-- checkpoint-verification -->

Verification

All 589 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4196-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1723 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 591 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_523_CircleMajorArcPrefixIdentity.md`

Checkpoint 523: The actual circle major-arc truncation is a denominator prefix

Proves the exact prefix identity for the genuine continuous circle major-arc model, with the original characters, coefficients and β data. This is a model identity, not an assumed operator bound.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
CircleMajorArcPrefixIdentity.lean	1

- `circleMajorArcModel_real_prefix`

<!-- checkpoint-verification -->

Verification

All 590 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4197-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1724 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 592 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_524_SourceDenominatorBlockUniverse.md`

Checkpoint 524: A fixed canonical universe for a source denominator block

Proves that a globally truncated source family in a bounded denominator universe is exactly a real-radius prefix of one fixed canonical set. This isolates the changing source radius from the fixed block frequencies.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceDenominatorBlockUniverse.lean	1

- `source_denominator_block_universe`

<!-- checkpoint-verification -->

Verification

All 591 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4198-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1725 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 593 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_525_SourceCircleBlockPrefix.md`

Checkpoint 525: The actual globally truncated circle block has an exact prefix expansion

Expands the literal globally truncated source circle block into the denominator prefix to which the proved abstract prefix theorem applies. The remaining multifrequency interval norm premise stays separate.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceCircleBlockPrefix.lean	1

- `source_circle_block_prefix`

<!-- checkpoint-verification -->

Verification

All 592 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4199-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1726 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 594 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_526_ConductorMultiplePrefix.md`

Checkpoint 526: The exact conductor-multiple prefix cutoff

Proves the exact prefix cutoff after writing an exceptional denominator as $q_0 d$. It identifies the same source-radius issue in the exceptional large-denominator part of Proposition 6.21.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
ConductorMultiplePrefix.lean	1

- `conductor_multiple_prefix`

<!-- checkpoint-verification -->

Verification

All 593 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4200-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1727 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 595 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_527_SharpHighExponentWeight.md

Checkpoint 527: The actual sharp high exponent weight and its positive comparison

Defines the exact exponent scale √L/log L used in the sharp high bound and compares it with the previously formalized regularized Orlicz weight. This supports a precise polynomial-loss repair.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SharpHighExponentWeight.lean	2

- `sharpHighExponentWeight_pos`
- `orliczWeight_le_sharpHighExponentWeight`

<!-- checkpoint-verification -->

Verification

All 594 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4201-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1729 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 596 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_528_SharpHighPolynomialAbsorption.md

Checkpoint 528: Uniform absorption of fixed powers into the sharp high exponent

Proves uniform absorption of any fixed real polynomial power into an arbitrarily small addition to the sharp high exponent coefficient, including bounded L≥2.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SharpHighPolynomialAbsorption.lean	1

- `sharpHigh_polynomial_absorption`

<!-- checkpoint-verification -->

Verification

All 595 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4202-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1730 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 597 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_529_SharpHighPrefixLossAbsorption.md`

Checkpoint 529: The extra fixed logarithmic prefix loss is harmless for the leading exponent

Proves that any fixed power of 1+L is absorbed by an arbitrarily small sharp-exponent increment. This includes the additional logarithmic factor caused by the denominator-prefix repair.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SharpHighPrefixLossAbsorption.lean	1

- `sharpHigh_shifted_polynomial_absorption`

<!-- checkpoint-verification -->

Verification

All 596 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4203-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1731 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 598 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_530_SharpHighExponentCoefficient.md`

Checkpoint 530: Exact preservation of the leading coefficient after a prefix loss

Shows precisely how a fixed prefix polynomial changes c to c+ε in the exponent. It does not alter the limiting leading coefficient.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SharpHighExponentCoefficient.lean	1

- `sharpHigh_prefix_loss_coefficient`

<!-- checkpoint-verification -->

Verification

All 597 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4204-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1732 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 599 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_531_SharpHighSourceLogLoss.md

Checkpoint 531: The prefix-loss estimate in the original source cutoff Q

Specializes the proved polynomial absorption to the exact source parameter Q. This is the scale needed to justify retaining the sharp leading coefficient in Proposition 6.21 after using its prefix corollary.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SharpHighSourceLogLoss.lean	1

- `sharpHigh_source_log_loss`

<!-- checkpoint-verification -->

Verification

All 598 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4205-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1733 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 600 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_532_SharpHighLeadingTwoRepair.md

Checkpoint 532: The corrected logarithmic loss preserves the sharp leading two

Proves the exact Q^{-1/2} sharp-high expression still permits leading coefficient 2+ε after a fixed logarithmic prefix loss. This is an absorption lemma, not a proof of the paper's missing multifrequency estimate.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SharpHighLeadingTwoRepair.lean	1

- `sharpHigh_leading_two_prefix_repair`

<!-- checkpoint-verification -->

Verification

All 599 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4206-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1734 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 601 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_533_TwoTermSquareEnergyBound.md`

Checkpoint 533: The correct square-energy cost of subtracting two high components

Proves the exact two-term square-norm inequality used to combine the ordinary and exceptional collar energies. The subtraction sign does not require cancellation to be assumed.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
TwoTermSquareEnergyBound.lean	1

- `norm_sub_square_two_bound`

<!-- checkpoint-verification -->

Verification

All 600 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4207-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1735 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 602 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_534_DifferenceCircleEnergy.md`

Checkpoint 534: Uniform finite circle energy for the difference of two families

Assembles two genuine circle square-energy bounds into a uniform bound for their signed difference, with a fixed explicit constant and no time-count loss.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
DifferenceCircleEnergy.lean	1

- `difference_circle_energy`

<!-- checkpoint-verification -->

Verification

All 601 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4208-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1736 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 603 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_535_SourceCollarDifference.md`

Checkpoint 535: The exact combined ordinary and exceptional source collar

Defines the actual signed collar contribution of the original high model, retaining global clipping for the exceptional term and its own effective cutoff.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

The exact interpretation as the complete original source high collar retains Q<=R at every scale. The displayed continuous-symbol bound also covers the explicitly defined extension at smaller nonnegative radii where permitted by its signature. Identification with the printed cutoff-difference formula retains the earlier source geometry and low-symbol bridge.

Checked declarations

Module	Named supporting theorems
SourceCollarDifference.lean	1

- `sourceCollarDifference_apply`

<!-- checkpoint-verification -->

Verification

All 602 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4209-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1737 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 604 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_536_CombinedSourceCollarEnergy.md`

Checkpoint 536: Uniform energy of the complete signed source collar

Proves finite frequency square energy for the complete signed original source collar, uniformly over changing conductor classes, source radii and active flags. The actual arithmetic and model-identity premises are discharged; the numerical conductor-class condition remains explicit.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

The exact interpretation as the complete original source high collar retains Q<=R at every scale. The displayed continuous-symbol bound also covers the explicitly defined extension at smaller nonnegative radii where permitted by its signature. Identification with the printed cutoff-difference formula retains the earlier source geometry and low-symbol bridge.

Checked declarations

Module	Named supporting theorems
CombinedSourceCollarEnergy.lean	1

- `combinedSourceCollar_energy`

<!-- checkpoint-verification -->

Verification

All 603 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4210-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1738 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 605 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_537_CombinedSourceCollarMaximum.md`

Checkpoint 537: The true countable l2 maximum of the complete source collar

Constructs the genuine countable ℓ^2 maximum of the complete signed source collar, including pointwise domination of every dyadic scale. The source large-denominator remainder is not included in this claim.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

The exact interpretation as the complete original source high collar retains Q<=R at every scale. The displayed continuous-symbol bound also covers the explicitly defined extension at smaller nonnegative radii where permitted by its signature. Identification with the printed cutoff-difference formula retains the earlier source geometry and low-symbol bridge.

Checked declarations

Module	Named supporting theorems
CombinedSourceCollarMaximum.lean	1

- `combinedSourceCollar_maximum`

<!-- checkpoint-verification -->

Verification

All 604 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4211-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1739 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 606 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_538_CombinedSourceCollarPairing.md`

Checkpoint 538: The complete source collar finite-indicator maximal pairing

Proves the actual restricted finite-indicator pairing for the entire signed source collar maximum. It is a complete collar component, while the large-denominator maximal estimate and headline theorem remain open.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

The exact interpretation as the complete original source high collar retains Q<=R at every scale. The displayed continuous-symbol bound also covers the explicitly defined extension at smaller nonnegative radii where permitted by its signature. Identification with the printed cutoff-difference formula retains the earlier source geometry and low-symbol bridge.

Checked declarations

Module	Named supporting theorems
CombinedSourceCollarPairing.lean	1

- `combinedSourceCollar_indicator_pairing`

<!-- checkpoint-verification -->

Verification

All 605 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4212-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1740 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 607 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_539_SourceCollarInactiveIdentity.md`

Checkpoint 539: The signed source collar reduces exactly to the ordinary collar when inactive

Checks the exact inactive-data boundary case of the final source collar decomposition. The entire signed collar is precisely the ordinary collar, not merely bounded by its estimate.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceCollarInactiveIdentity.lean	1

- `sourceCollarDifference_inactive`

<!-- checkpoint-verification -->

Verification

All 606 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4213-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1741 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 608 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_540_SourceCollarOutsideConductor.md`

Checkpoint 540: Exact disappearance of the exceptional collar beyond the source radius

Proves the exact source-radius boundary case: if the conductor exceeds the global radius, the exceptional collar contribution vanishes, regardless of the active flag.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceCollarOutsideConductor.lean	1

- `sourceCollarDifference_outside_conductor`

<!-- checkpoint-verification -->

Verification

All 607 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4214-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1742 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 609 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_541_SourceLargeHighRemainder.md`

Checkpoint 541: The exact full source high split into proved collars and the remaining large terms

Locates the formalization frontier exactly: the complete original source high symbol is the proved signed collar plus the two remaining large-denominator high terms. The identity needs no exceptional-cutoff containment hypothesis.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceLargeHighRemainder.lean	1

- `circleHighModel_collar_remainder`

<!-- checkpoint-verification -->

Verification

All 608 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4215-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1743 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 610 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

Checkpoint_542_SourceHighOperatorFrontier.md

Checkpoint 542: The exact source high operator frontier

Transfers the exact proof frontier to the actual ℓ^2 operators. The remaining task is a quantitative maximal estimate for the displayed large-denominator operator.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceHighOperatorFrontier.lean	1

- `circleHighOperator_collar_remainder`

<!-- checkpoint-verification -->

Verification

All 609 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4216-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1744 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 611 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_543_SourceFiniteConvolutionFrontier.md

Checkpoint 543: The exact finite-input annular prime-square decomposition at the proof frontier

Proves the exact decomposition for actual finite complex input convolution: the original low operator on ℓ^1, the proved collar and remaining large high operator on ℓ^2, and the genuine analytic error on ℓ^2.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceFiniteConvolutionFrontier.lean	1

- `annularConvolution_source_frontier`

<!-- checkpoint-verification -->

Verification

All 610 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4217-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1745 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 612 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_544_SourceAnnularSymbolFrontier.md`

Checkpoint 544: The exact annular prime-square symbol at the final proof frontier

Provides the exact continuous-symbol form of the final source decomposition, with no analytic error or large-denominator estimate hidden in the statement.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourceAnnularSymbolFrontier.lean	1

- `annularSymbol_source_frontier`

<!-- checkpoint-verification -->

Verification

All 611 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4218-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1746 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 613 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_545_SourcePointwiseConvolutionBound.md`

Checkpoint 545: The unconditional pointwise triangle bound for the exact source decomposition

Proves an unconditional pointwise bound for the actual finite-input prime-square convolution by the four exact source pieces. The statement leaves every still-unbounded piece visible.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
SourcePointwiseConvolutionBound.lean	1

- `annularConvolution_source_triangle`

<!-- checkpoint-verification -->

Verification

All 612 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4219-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1747 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 614 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_546_SourceCollarSubhalfMaximum.md`

Checkpoint 546: The completed subhalf-power source collar maximum with no auxiliary epsilon parameter

Completes the actual signed original source collar maximum for every 0≤η<1/2, eliminating the auxiliary epsilon by an explicit admissible choice. It includes arbitrary source clipping, character activity and changing conductor-class selection subject to the displayed numerical class condition. The genuine real ℓ^2 witness dominates every scale and has norm Kη Q^{-η}‖f‖₂. Theorem 1.1, the sharp full high maximum and the large-denominator remainder remain open.

The claims are exactly the displayed Lean declarations. No new axiom or manuscript conclusion is assumed. The sharp full high maximal bound, Proposition 4.19 and Theorem 1.1 remain open.

The exact interpretation as the complete original source high collar retains Q<=R at every scale. The displayed continuous-symbol bound also covers the explicitly defined extension at smaller nonnegative radii where permitted by its signature. Identification with the printed cutoff-difference formula retains the earlier source geometry and low-symbol bridge.

Checked declarations

Module	Named supporting theorems
SourceCollarSubhalfMaximum.lean	1

- `sourceCollar_subhalf_maximum`

<!-- checkpoint-verification -->

Verification

All 613 project modules and the root were freshly rebuilt: 4220 jobs pass without warnings/errors. All 1748 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 615 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

Checkpoint_547_CircleMultiplyComposition.md

Checkpoint 547: Circle multiplication composes by the product of the actual symbols

Proves the composition identity for the existing continuous-symbol action on circle L2 classes. The proof works with almost-everywhere representatives, so no unjustified pointwise representative equality is used.

Checked declarations

Module	Named supporting theorems
CircleMultiplyComposition.lean	1

- `circleMultiply_comp_apply`

<!-- checkpoint-verification -->

Verification

All 614 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4221-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1749 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 616 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_548_IntegerMultiplierComposition.md

Checkpoint 548: The integer Fourier multiplier action respects symbol products

Transfers actual circle composition through the proved integer Fourier isometry. This supplies the localization identity used below, without postulating an abstract Plancherel operator.

Checked declarations

Module	Named supporting theorems
IntegerMultiplierComposition.lean	1

- `integerMultiplier_comp_apply`

<!-- checkpoint-verification -->

Verification

All 615 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4222-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1750 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 617 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_549_CircleDisjointInner.md`

Checkpoint 549: Disjoint circle symbols produce orthogonal L2 outputs

For arbitrary circle L2 inputs, pointwise disjoint continuous symbols give zero inner product of the multiplied outputs. Both inputs can differ; the support condition is explicitly displayed.

Checked declarations

Module	Named supporting theorems
CircleDisjointInner.lean	1

- `circleMultiply_disjoint_inner`

<!-- checkpoint-verification -->

Verification

All 616 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4223-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1751 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 618 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_550_DisjointIntegerMultipliers.md`

Checkpoint 550: Disjoint symbols give orthogonality on the integers

Proves the corresponding integer l2 inner-product identity using the actual Fourier isometric equivalence. It holds for different input functions and will discharge spectral orthogonality premises.

Checked declarations

Module	Named supporting theorems
DisjointIntegerMultipliers.lean	1

- `integerMultiplier_disjoint_inner`

<!-- checkpoint-verification -->

Verification

All 617 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4224-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1752 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 619 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_551_DisjointMultiplierParseval.md`

Checkpoint 551: Finite disjoint multiplier sums have exact square energy

Discharges the generic Hilbert-space Pythagoras premise for continuous circle symbols with pairwise disjoint pointwise supports. This is an exact identity, including empty finite sets.

Checked declarations

Module	Named supporting theorems
DisjointMultiplierParseval.lean	1

- `disjointMultiplier_finset_energy`

<!-- checkpoint-verification -->

Verification

All 618 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4225-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1753 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 620 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_552_DisjointMultiplierInterval.md`

Checkpoint 552: Denominator intervals inherit exact disjoint multiplier energy

Records the exact interval square norm for a disjoint symbol family, with arbitrary start and length. This supplies the real interval budget used by the prefix majorant.

Checked declarations

Module	Named supporting theorems
DisjointMultiplierInterval.lean	1

- `disjointMultiplier_interval_energy`

<!-- checkpoint-verification -->

Verification

All 619 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4226-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1754 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 621 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_553_DisjointMultiplierPrefix.md`

Checkpoint 553: Disjoint Fourier multipliers admit a genuine logarithmic prefix majorant

Constructs a nonnegative real l2 function dominating every prefix up to 2^h. Its norm is bounded by (h+1) times the square root of the exact sum of output square norms. Spectral disjointness is the displayed sufficient condition; it is not yet asserted for all source blocks.

Checked declarations

Module	Named supporting theorems
DisjointMultiplierPrefix.lean	1

- `disjointMultiplier_prefix_majorant`

<!-- checkpoint-verification -->

Verification

All 620 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4227-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1755 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 622 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_554_IntervalCircleSymbol.md`

Checkpoint 554: Actual continuous multiplier symbols can be summed over denominator intervals

Defines an interval sum in the continuous circle symbol space and proves its pointwise evaluation identity. This keeps the subsequent truncation identities tied to continuous multipliers.

Checked declarations

Module	Named supporting theorems
IntervalCircleSymbol.lean	1

- `intervalCircleSymbol_apply`

<!-- checkpoint-verification -->

Verification

All 621 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4228-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1756 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 623 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_555_IntervalIntegerMultiplier.md`

Checkpoint 555: Interval symbol sums act as the corresponding finite sums of operators

Proves finite linearity for the actual Fourier multiplier map and identifies every interval symbol action with intervalL2Sum, rather than leaving that bridge as an informal convention.

Checked declarations

Module	Named supporting theorems
IntervalIntegerMultiplier.lean	2

- `integerMultiplier_finset_sum`
- `integerMultiplier_interval`

<!-- checkpoint-verification -->

Verification

All 622 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4229-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1758 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 624 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_556_WeightedDisjointSymbols.md`

Checkpoint 556: Scalar weights preserve spectral disjointness

Proves pairwise disjointness after arbitrary complex coefficient weights. The arithmetic weights need no boundedness for this identity; quantitative size enters only in later norm budgets.

Checked declarations

Module	Named supporting theorems
WeightedDisjointSymbols.lean	1

- `weighted_symbols_disjoint`

<!-- checkpoint-verification -->

Verification

All 623 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4230-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1759 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 625 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_557_WeightedDisjointPrefix.md`

Checkpoint 557: Weighted disjoint multipliers retain the logarithmic prefix bound

Applies the discharged spectral orthogonality theorem after arbitrary complex weights. The majorant budget records the exact weighted output square norms, so no arithmetic coefficient estimate is silently inserted.

Checked declarations

Module	Named supporting theorems
WeightedDisjointPrefix.lean	1

- `weightedDisjoint_prefix_majorant`

<!-- checkpoint-verification -->

Verification

All 624 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4231-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1760 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 626 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_558_WeightedMultiplierBudget.md`

Checkpoint 558: Coefficient envelopes control the exact interval square budget

Proves a quantitative interval energy comparison for arbitrary multipliers after bounded complex scalar weights. It applies to each finite interval and keeps the unweighted localized input energy visible.

Checked declarations

Module	Named supporting theorems
WeightedMultiplierBudget.lean	1

- `weightedMultiplier_interval_budget`

<!-- checkpoint-verification -->

Verification

All 625 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4232-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1761 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 627 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_559_PaddedCircleSymbols.md`

Checkpoint 559: Finite symbol families extend by exact zero padding

Defines zero padding outside the original finite denominator list and proves its inside/outside values. Padding is used only to reach a binary tree capacity; it does not add frequencies.

Checked declarations

Module	Named supporting theorems
PaddedCircleSymbols.lean	2

- `paddedCircleSymbols_inside`
- `paddedCircleSymbols_outside`

<!-- checkpoint-verification -->

Verification

All 626 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4233-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1763 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 628 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_560_PaddedSymbolDisjointness.md`

Checkpoint 560: Zero padding preserves the exact finite spectral disjointness premise

Extends a disjointness condition imposed only on indices below L to the entire padded natural-indexed symbol family. This prevents an unnecessarily infinite source hypothesis.

Checked declarations

Module	Named supporting theorems
PaddedSymbolDisjointness.lean	1

- `paddedCircleSymbols_disjoint`

<!-- checkpoint-verification -->

Verification

All 627 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4234-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1764 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 629 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_561_PaddedPrefixIdentity.md`

Checkpoint 561: Every genuine finite prefix is unchanged by binary padding

Proves equality of continuous symbol prefixes before and after padding, including empty prefixes and K=L. Thus the binary envelope controls the original operator family exactly.

Checked declarations

Module	Named supporting theorems
PaddedPrefixIdentity.lean	1

- `paddedCircleSymbols_prefix`

<!-- checkpoint-verification -->

Verification

All 628 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4235-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1765 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 630 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_562_PaddedMultiplierEnergy.md`

Checkpoint 562: Binary padding does not enlarge the multiplier square budget

Proves exact equality between the padded budget at any capacity T>=L and the original finite square-energy sum. All extra padded operators are zero, not estimated by a cardinality loss.

Checked declarations

Module	Named supporting theorems
PaddedMultiplierEnergy.lean	1

- `paddedMultiplier_square_budget`

<!-- checkpoint-verification -->

Verification

All 629 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4236-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1766 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 631 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_563_FiniteDisjointPrefixMaximum.md`

Checkpoint 563: Finite disjoint symbol families have a true prefix majorant

Produces one nonnegative real l2 function for every prefix of a finite family. Its norm uses the exact original L-term energy and binary depth h whenever L<=2^h; the support hypothesis is finite.

Checked declarations

Module	Named supporting theorems
FiniteDisjointPrefixMaximum.lean	1

- `finiteDisjoint_prefix_maximum`

<!-- checkpoint-verification -->

Verification

All 630 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4237-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1767 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 632 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_564_RealTruncatedCirclePrefix.md`

Checkpoint 564: Real source cutoffs equal exact continuous-symbol prefixes

Defines the finite denominator activation symbol and proves the exact floor/min prefix formula for every nonnegative real cutoff. The boundary denominator is included, and the list length is clipped explicitly.

Checked declarations

Module	Named supporting theorems
RealTruncatedCirclePrefix.lean	1

- `realTruncatedCircleSymbol_prefix`

<!-- checkpoint-verification -->

Verification

All 631 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4238-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1768 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 633 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_565_RealCutoffDisjointMaximum.md`

Checkpoint 565: One l2 majorant controls every nonnegative real denominator cutoff

The finite disjoint-prefix witness simultaneously controls all real cutoffs R>=0, with no monotonicity assumption on a selected radius family and no common-tail claim.

Checked declarations

Module	Named supporting theorems
RealCutoffDisjointMaximum.lean	1

- `finiteDisjoint_real_cutoff_maximum`

<!-- checkpoint-verification -->

Verification

All 632 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4239-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1769 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 634 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_566_FiniteBinaryCapacity.md`

Checkpoint 566: Ceiling logarithms give a valid finite binary capacity

Defines the binary depth as Nat.clog 2 L and proves that its power-of-two capacity contains the original finite list, including L=0 and L=1. No manually chosen padding size remains.

Checked declarations

Module	Named supporting theorems
FiniteBinaryCapacity.lean	2

- `finiteBinaryDepth_capacity`
- `finiteBinaryDepth_zero`

<!-- checkpoint-verification -->

Verification

All 633 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4240-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1771 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 635 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint notes 567-641

Checkpoint_567_BinaryDepthLogarithm.md

Checkpoint 567: The binary depth has the expected logarithmic size

Proves finiteBinaryDepth L<=Nat.log2 L+1 and the corresponding real depth-factor comparison. This supplies an explicit logarithmic prefix cost without asymptotic shorthand.

Checked declarations

Module	Named supporting theorems
BinaryDepthLogarithm.lean	2

- `finiteBinaryDepth_le_log2`
- `finiteBinaryDepth_factor_le`

<!-- checkpoint-verification -->

Verification

All 634 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4241-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1773 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 636 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_568_LogarithmicDisjointMaximum.md

Checkpoint 568: The real-cutoff maximum has an explicit logarithmic energy bound

Eliminates the auxiliary binary depth: all nonnegative real denominator cutoffs are dominated by one real l2 function with norm at most (log2 L+2) times the square root of the exact finite multiplier energy.

Checked declarations

Module	Named supporting theorems
LogarithmicDisjointMaximum.lean	1

- `logarithmicDisjoint_real_cutoff_maximum`

<!-- checkpoint-verification -->

Verification

All 635 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4242-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1774 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 637 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_569_DisjointSymbolSumBound.md`

Checkpoint 569: A disjoint finite symbol sum has no cardinality loss

If at most one symbol is nonzero at each point, the norm of their finite sum is bounded by the common individual envelope. The proof handles both the all-zero point and the unique active term explicitly.

Checked declarations

Module	Named supporting theorems
DisjointSymbolSumBound.lean	1

- `disjointSymbol_sum_bound`

<!-- checkpoint-verification -->

Verification

All 636 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4243-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1775 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 638 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_570_DisjointMultiplierBessel.md`

Checkpoint 570: Bounded disjoint symbols give a finite input-energy estimate

Combines exact orthogonality with the actual multiplier operator bound to control the sum of output square norms by A^2 times the input square norm. This discharges the finite energy budget for bounded disjoint localizers.

Checked declarations

Module	Named supporting theorems
DisjointMultiplierBessel.lean	1

- `disjointMultiplier_bessel`

<!-- checkpoint-verification -->

Verification

All 637 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4244-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 1776 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 639 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_571_UniformDisjointRealMaximum.md`

Checkpoint 571: Bounded disjoint symbols have a logarithmic l2 cutoff maximum

Combines the finite real-cutoff prefix theorem with the discharged Bessel energy estimate. A genuine nonnegative l2 majorant has norm at most (log2 L+2) A times the input norm, independent of the number of active centers except for the logarithmic prefix cost.

Checked declarations

Module	Named supporting theorems
UniformDisjointRealMaximum.lean	1

- `uniformDisjoint_real_maximum`

<!-- checkpoint-verification -->

Verification

All 638 project modules and the root were freshly rebuilt: 4245 jobs pass without warnings/errors. All 1777 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 640 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_572_LocalizedMultiplierEstimate.md`

Checkpoint 572: A localized symbol acts only on its localized input

From the exact symbol identity m=m*p, proves equality T_m f=T_m(T_p f) and its norm bound. The localizer is not assumed to be an idempotent projection; its plateau identity is the precise required hypothesis.

Checked declarations

Module	Named supporting theorems
LocalizedMultiplierEstimate.lean	2

- `integerMultiplier_localized`
- `integerMultiplier_localized_norm`

<!-- checkpoint-verification -->

Verification

All 639 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4246-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1779 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 641 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_573_LocalizedCrossProducts.md`

Checkpoint 573: Disjoint localizers annihilate all off-diagonal symbol products

Proves m_i*p_j=0 for different indices from pairwise disjoint localizers and m_i=m_i*p_i. This is the exact algebra needed to localize interval sums.

Checked declarations

Module	Named supporting theorems
LocalizedCrossProducts.lean	1

- `localized_symbol_cross`

<!-- checkpoint-verification -->

Verification

All 640 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4247-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1780 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 642 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_574_FiniteSymbolLocalization.md`

Checkpoint 574: Finite symbol intervals localize to the sum of their own localizers

Sums the exact diagonal and off-diagonal identities to prove intervalSymbol(m)=intervalSymbol(m)*intervalSymbol(p). This checks the localization step for every interval, including empty intervals.

Checked declarations

Module	Named supporting theorems
FiniteSymbolLocalization.lean	1

- `intervalCircleSymbol_localized`

<!-- checkpoint-verification -->

Verification

All 641 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4248-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1781 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 643 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_575_LocalizedIntervalEnergy.md`

Checkpoint 575: Localized interval input norms equal their sum of individual energies

Identifies the square norm of the interval localizer applied to the actual input with the exact interval budget of individual localizer outputs. Disjointness is proved sufficient via actual Fourier orthogonality.

Checked declarations

Module	Named supporting theorems
LocalizedIntervalEnergy.lean	1

- `localized_interval_input_energy`

<!-- checkpoint-verification -->

Verification

All 642 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4249-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1782 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 644 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_576_LocalizedTimePrefixMajorant.md`

Checkpoint 576: Interval maximal budgets transfer to all times and prefixes

Applies the checked binary prefix principle to actual circle multiplier outputs. The hypothesis is an explicit family of interval majorants with norm budget in the corresponding localized input, rather than a presumed source prefix theorem.

Checked declarations

Module	Named supporting theorems
LocalizedTimePrefixMajorant.lean	1

- `localized_time_prefix_majorant`

<!-- checkpoint-verification -->

Verification

All 643 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4250-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1783 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 645 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_577_LocalizedIntervalSquareMaximum.md

Checkpoint 577: Frequency square energy produces actual interval maximal majorants

For one interval symbol family with a common localizer and a displayed finite square-energy bound across time, constructs the true countable l2 maximum. This is a sufficient frequency-energy criterion; the original multifrequency interval theorem is not declared proved.

Checked declarations

Module	Named supporting theorems
LocalizedIntervalSquareMaximum.lean	1

- `localized_circle_square_maximum`

<!-- checkpoint-verification -->

Verification

All 644 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4251-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1784 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 646 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_578_CircleIntervalMajorants.md

Checkpoint 578: Uniform interval energy data yield a whole family of interval majorants

Constructs all binary interval majorants for actual time-dependent multiplier symbols from exact interval localization and finite time square-energy assumptions. Choice only selects already proved l2 witnesses; it adds no mathematical axiom beyond the audited classical foundations.

Checked declarations

Module	Named supporting theorems
CircleIntervalMajorants.lean	1

- `circle_interval_majorants`

<!-- checkpoint-verification -->

Verification

All 645 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4252-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1785 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 647 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_579_CircleTimePrefixMaximum.md`

Checkpoint 579: A frequency-energy criterion controls all times and denominator prefixes

Proves a genuine time-and-prefix maximum for actual circle multipliers from explicit finite time square-energy bounds on each binary interval and disjoint localizers. The output norm has exactly one binary-depth factor and the square root of the localized input energy.

Checked declarations

Module	Named supporting theorems
CircleTimePrefixMaximum.lean	1

- `circle_time_prefix_maximum`

<!-- checkpoint-verification -->

Verification

All 646 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4253-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1786 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 648 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_580_CircleTimePrefixPairing.md`

Checkpoint 580: The actual time-prefix maximum controls finite test-set pairings

Derives the finite test-set pairing bound from the constructed time-prefix witness. The interval energy, localizer and square-energy hypotheses remain explicit, and no source interval estimate is inserted.

Checked declarations

Module	Named supporting theorems
CircleTimePrefixPairing.lean	1

- `circle_time_prefix_pairing`

<!-- checkpoint-verification -->

Verification

All 647 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4254-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1787 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 649 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_581_RationalCircleDenominatorBlocks.md`

Checkpoint 581: Actual rational circle symbols have exact denominator blocks

Defines denominator blocks for an arbitrary rational-indexed continuous circle symbol. The complete finite sum equals the sum of its blocks whenever all denominators are at most M, including the identically empty denominator-zero block.

Checked declarations

Module	Named supporting theorems
RationalCircleDenominatorBlocks.lean	1

- `rationalCircle_full_blocks`

<!-- checkpoint-verification -->

Verification

All 648 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4255-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1788 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 650 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_582_RationalCircleRealPrefix.md`

Checkpoint 582: Rational circle sums with real denominator cutoffs equal exact prefixes

Connects the generic real-cutoff multiplier symbol with the actual finite rational sum. The proof retains the nonnegative-radius premise and exact floor/min convention.

Checked declarations

Module	Named supporting theorems
RationalCircleRealPrefix.lean	1

- `rationalCircle_real_prefix`

<!-- checkpoint-verification -->

Verification

All 649 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4256-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1789 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 651 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_583_SourceCircleSymbolPrefix.md`

Checkpoint 583: Every finite source symbol admits the exact denominator-prefix action

For an arbitrary rational source symbol, identifies source radii clipped to a fixed denominator universe M with the exact real-cutoff multiplier prefix. This single bridge applies to ordinary, high and exceptional symbols without altering their formulas.

Checked declarations

Module	Named supporting theorems
SourceCircleSymbolPrefix.lean	1

- `sourceCircle_symbol_prefix`

<!-- checkpoint-verification -->

Verification

All 650 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4257-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 1790 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 652 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_584_SourceHighRealPrefix.md

Checkpoint 584: The actual high source symbol has the exact prefix representation

Instantiates the source bridge with circleHighTerm, including its actual conductor, narrow-cutoff and exceptional data. This is an identity for a fixed denominator cap M, not a maximal estimate on the unbounded union of caps.

Checked declarations

Module	Named supporting theorems
SourceHighRealPrefix.lean	1

- `sourceHigh_real_prefix`

<!-- checkpoint-verification -->

Verification

All 651 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4258-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1791 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 653 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_585_SourceExceptionalRealPrefix.md

Checkpoint 585: The actual exceptional high symbol has the exact prefix representation

Instantiates the same exact source bridge for circleExceptionalHigh, preserving all exceptional arithmetic and cutoff parameters. It avoids replacing the term by a fixed coefficient schematic model.

Checked declarations

Module	Named supporting theorems
SourceExceptionalRealPrefix.lean	1

- `sourceExceptional_real_prefix`

<!-- checkpoint-verification -->

Verification

All 652 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4259-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1792 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 654 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_586_RealDenominatorBandDifference.md`

Checkpoint 586: A real denominator band is an exact difference of prefixes

Proves the band formula Q<den<=R using the lower endpoint min(Q,R). It covers Q>R correctly as the empty band; no extra assumption ordering a conductor-dependent cutoff and the source radius is introduced.

Checked declarations

Module	Named supporting theorems
RealDenominatorBandDifference.lean	1

- `rationalCircle_real_band`

<!-- checkpoint-verification -->

Verification

All 653 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4260-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1793 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 655 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_587_PrefixDifferenceMajorant.md`

Checkpoint 587: A single prefix witness controls every difference of two prefixes

Proves the actual multiplier difference estimate and constructs the doubled nonnegative real l2 majorant. Its exact norm is at most twice the original norm, so band truncation adds only the displayed factor two.

Checked declarations

Module	Named supporting theorems
PrefixDifferenceMajorant.lean	1

- `realPrefix_difference_majorant`

<!-- checkpoint-verification -->

Verification

All 654 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4261-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1794 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 656 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_588_SourceOrdinaryLargeBand.md`

Checkpoint 588: The actual ordinary large-denominator block is a difference of prefixes

Identifies the ordinary large-denominator source contribution inside a fixed cap M as two prefixes of the unchanged circleRationalModelTerm with zero exceptional selector. It includes the empty-band case Q>R.

Checked declarations

Module	Named supporting theorems
SourceOrdinaryLargeBand.lean	1

- `sourceOrdinary_large_band`

<!-- checkpoint-verification -->

Verification

All 655 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4262-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1795 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 657 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_589_SourceExceptionalLargeBand.md`

Checkpoint 589: The exceptional large-denominator block has the same exact band structure

Proves the two-prefix identity for the actual exceptional high term with any nonnegative lower cutoff W. In the source application W=Q*q0/conductorMajorant(q0); min(W,R) handles conductors whose collar cutoff exceeds the radius.

Checked declarations

Module	Named supporting theorems
SourceExceptionalLargeBand.lean	1

- `sourceExceptional_large_band`

<!-- checkpoint-verification -->

Verification

All 656 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4263-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1796 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 658 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_590_SourceRadiusCapIdentity.md

Checkpoint 590: A source radius bounded by a fixed cap loses no rational centers

Proves that sourceRationalCenters R is unchanged by the denominator cap M whenever 0<=R<=M. This records exactly when the finite-block prefix identities apply to the whole original source sum.

Checked declarations

Module	Named supporting theorems
SourceRadiusCapIdentity.lean	1

- `sourceRadius_filter_cap`

<!-- checkpoint-verification -->

Verification

All 657 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4264-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1797 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 659 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_591_SourceTimePrefixFromIntervals.md

Checkpoint 591: Actual source-radius maxima follow from explicit interval majorants

Connects the binary prefix theorem directly to arbitrary time-dependent rational source symbols in a fixed denominator cap. The interval majorants and their norm budgets are explicit hypotheses, so this bridge does not certify the unproved source multifrequency interval estimate.

Checked declarations

Module	Named supporting theorems
SourceTimePrefixFromIntervals.lean	1

- `source_time_prefix_from_intervals`

<!-- checkpoint-verification -->

Verification

All 658 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4265-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1798 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 660 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_592_SourceTimeBandFromIntervals.md

Checkpoint 592: The same interval data control every source denominator band

Uses the one prefix witness twice to dominate every band Q<den<=R at every time, with factor two in the norm and the exact min(Q,R) boundary. No monotonicity of radii or common time tail is required.

Checked declarations

Module	Named supporting theorems
SourceTimeBandFromIntervals.lean	1

- `source_time_band_from_intervals`

<!-- checkpoint-verification -->

Verification

All 659 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4266-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1799 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 661 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_593_SourceHighIntervalBridge.md`

Checkpoint 593: Explicit interval data now reach the genuine capped source high family

Instantiates the interval-to-prefix bridge with the actual circleHighTerm at varying scales, conductors, exceptional selectors, characters and beta. The norm conclusion is conditional on the displayed interval witnesses and budget; no full source high theorem is claimed.

Checked declarations

Module	Named supporting theorems
SourceHighIntervalBridge.lean	1

- `sourceHigh_interval_bridge`

<!-- checkpoint-verification -->

Verification

All 660 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4267-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1800 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 662 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_594_SourceLargeRemainderPrefixBridge.md`

Checkpoint 594: Prefix witnesses control the actual signed large-denominator remainder

For a nonnegative source radius bounded by a fixed cap M, proves an explicit majorant for the original sourceLargeHighRemainder from ordinary and exceptional prefix witnesses. The displayed pointwise bound is 2*(go+ge); the doubled-majorant and l2 triangle estimates supply its norm budget. The interval/source estimates needed to obtain those witnesses remain open.

Checked declarations

Module	Named supporting theorems
SourceLargeRemainderPrefixBridge.lean	1

- `sourceLargeRemainder_prefix_bridge`

<!-- checkpoint-verification -->

Verification

All 661 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4268-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1801 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 663 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_595_SourceHighFromLargeMajorant.md`

Checkpoint 595: The proved source collar maximum reduces full high control to its large remainder

Reuses the fully proved signed collar maximum, including its changing conductor classes and beta range. Given an explicitly supplied majorant only for the remaining large-denominator source remainder, constructs a genuine majorant for the original full high operator, with the proved collar contribution added to its norm. The remainder premise is not discharged.

Checked declarations

Module	Named supporting theorems
SourceHighFromLargeMajorant.lean	1

- `sourceHigh_from_large_majorant`

<!-- checkpoint-verification -->

Verification

All 662 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4269-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1802 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 664 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_596_AnnularSourceFinitePairing.md`

Checkpoint 596: The exact annular decomposition yields a finite pairing assembly

Proves a concrete finite test-set bound from the original low operator, the signed collar, the genuine large remainder and the actual major-arc error. Every supplied bound is displayed separately. This is an assembly lemma; it does not assert that the remaining high/error inputs or Theorem 1.1 have been established.

Checked declarations

Module	Named supporting theorems
AnnularSourceFinitePairing.lean	1

- `annularSource_finite_pairing`

<!-- checkpoint-verification -->

Verification

All 663 project modules and the root were freshly rebuilt: 4270 jobs pass without warnings/errors. All 1803 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 665 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_597_CharacterMultiplierUnitMass.md

Checkpoint 597: The actual Fourier character multiplier shifts a unit mass exactly

Proves that the existing integer multiplier with symbol exp(-2*pi*i*j*theta) sends the unit mass at zero to the unit mass at j. The statement uses the proved Fourier convention and actual L2 representatives.

Checked declarations

Module	Named supporting theorems
CharacterMultiplierUnitMass.lean	1

- `characterMultiplier_unit_mass`

<!-- checkpoint-verification -->

Verification

All 664 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4271-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1804 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 666 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_598_UnitMassMajorantLowerBound.md

Checkpoint 598: A majorant for N distinct unit masses must have l2 norm at least sqrt N

Proves the exact square-norm lower bound using a finite set of distinct integer coordinates. This is a kernel-checked obstruction to confusing uniform individual operator norms with a countable maximal norm.

Checked declarations

Module	Named supporting theorems
UnitMassMajorantLowerBound.lean	1

- `unitMass_majorant_square_lower`

<!-- checkpoint-verification -->

Verification

All 665 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4272-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1805 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 667 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_599_NoL2UnitMassMajorant.md`

Checkpoint 599: Countably many distinct unit masses have no l2 pointwise majorant

Uses the finite lower bound and the Archimedean property to rule out any real l2 function that is at least one on every nonnegative integer. No numerical sampling or informal divergence argument substitutes for the proof.

Checked declarations

Module	Named supporting theorems
NoL2UnitMassMajorant.lean	1

- `no_l2_unit_mass_majorant`

<!-- checkpoint-verification -->

Verification

All 666 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4273-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1806 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 668 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_600_UniformMultiplierNormCounterexample.md`

Checkpoint 600: Uniform single-scale multiplier bounds do not imply a countable maximal bound

Closes this milestone with an explicit counterexample using the actual continuous circle symbols and existing integer Fourier multipliers: every character multiplier has operator norm at most one, yet their outputs on the unit mass at zero have no real l2 pointwise majorant. This proves why the remaining full

high maximum needs a separate argument. The counterexample concerns a general family, not a counterexample to the manuscript theorem. Theorem 1.1, Proposition 4.19 and the source multifrequency interval estimate remain open.

Checked declarations

Module	Named supporting theorems
UniformMultiplierNormCounterexample.lean	1

- `uniformMultiplier_norm_not_maximal`

<!-- checkpoint-verification -->

Verification

All 667 project modules and the root were freshly rebuilt: 4274 jobs pass without warnings/errors. All 1807 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 669 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_601_CanonicalLiftDenominatorBound.md

Checkpoint 601: Canonical rational lifts retain the denominator separation bound

Proves the exact 1/D^2 lower bound for the distance between different canonical rationals after every integer lift. This is the circle-frequency separation needed for actual source cutoff disjointness.

Checked declarations

Module	Named supporting theorems
CanonicalLiftDenominatorBound.lean	1

- `canonical_lift_denominator_bound`

<!-- checkpoint-verification -->

Verification

All 668 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4275-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1808 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 670 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_602_CanonicalLiftArcUniqueness.md

Checkpoint 602: Short arcs cannot meet two distinct canonical rational lifts

Proves circle uniqueness directly with two independent integer lifts, retaining arbitrary real center, radius and denominator bound. It prevents an invalid use of real-line separation at a circle boundary.

Checked declarations

Module	Named supporting theorems
CanonicalLiftArcUniqueness.lean	1

- `canonical_lift_small_arc_unique`

<!-- checkpoint-verification -->

Verification

All 669 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4276-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1809 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 671 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_603_SourceCutoffLiftUniqueness.md`

Checkpoint 603: The actual dyadic source cutoffs have unique canonical centers on the circle

Applies the exact source bump radius and dyadic width estimates to arbitrary integer lifts. For s>=2 and denominators at most 2^(s+1), two nonzero lifted cutoffs must have the same canonical rational center.

Checked declarations

Module	Named supporting theorems
SourceCutoffLiftUniqueness.lean	1

- `source_cutoff_lift_unique`

<!-- checkpoint-verification -->

Verification

All 670 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4277-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1810 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 672 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

Checkpoint_604_SourceCutoffLiftDisjointness.md

Checkpoint 604: Different canonical source cutoffs have disjoint lifted supports

Records the useful pointwise disjointness conclusion for the actual source cutoff, valid at every real representative and every pair of integer lifts. No abstract disjointness assumption is used.

Checked declarations

Module	Named supporting theorems
SourceCutoffLiftDisjointness.lean	1

- `source_cutoff_lift_disjoint`

<!-- checkpoint-verification -->

Verification

All 671 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4278-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1811 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 673 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_605_SourceCutoffLiftWidth.md

Checkpoint 605: The actual source cutoff is narrow enough for unique integer lifts

Proves 2/16^s<1 for every s>=1. This is the strict width estimate needed to rule out two different integer translates of the same center, including boundary equality cases.

Checked declarations

Module	Named supporting theorems
SourceCutoffLiftWidth.lean	1

- `source_cutoff_lift_width`

<!-- checkpoint-verification -->

Verification

All 672 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4279-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1812 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 674 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_606_SourceCutoffSingleLift.md`

Checkpoint 606: Each canonical source center has at most one nonzero integer lift

Combines integer separation with the actual source cutoff radius. At any fixed real point and scale s>=1, nonzero contributions from one rational center have exactly one possible integer translate.

Checked declarations

Module	Named supporting theorems
SourceCutoffSingleLift.lean	1

- `source_cutoff_single_lift`

<!-- checkpoint-verification -->

Verification

All 673 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4280-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1813 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 675 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_607_SourceCutoffContinuousMap.md`

Checkpoint 607: The actual scaled cutoff is a compactly supported continuous symbol

Constructs the complex continuous map of the actual source cutoff and proves its real-line support bound. This is the source bump, with no abstract support hypothesis.

Checked declarations

Module	Named supporting theorems
SourceCutoffContinuousMap.lean	1

- `sourceCutoffContinuousMap_support`

<!-- checkpoint-verification -->

Verification

All 674 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4281-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1814 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 676 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_608_CircleSourceCutoff.md`

Checkpoint 608: The actual source cutoff defines a continuous circle symbol

Defines the translated periodized circle cutoff for every rational center and proves its exact all-real integer-lift formula. It uses the already proved compact circle lift.

Checked declarations

Module	Named supporting theorems
CircleSourceCutoff.lean	1

- `circleSourceCutoff_real`

<!-- checkpoint-verification -->

Verification

All 675 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4282-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1815 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 677 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_609_CircleSourceCutoffWitness.md`

Checkpoint 609: A nonzero circle cutoff has a nonzero real lift

Proves the actual support witness needed to transfer separation estimates from the line to the circle. No choice of a common lift for different centers is imposed.

Checked declarations

Module	Named supporting theorems
CircleSourceCutoffWitness.lean	1

- `circleSourceCutoff_nonzero_lift`

<!-- checkpoint-verification -->

Verification

All 676 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4283-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1816 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 678 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_610_CircleSourceCutoffDisjoint.md`

Checkpoint 610: Distinct canonical source cutoffs are disjoint on the circle

Discharges actual circle support disjointness for canonical centers with denominators at most 2^(s+1), for s at least two. Independent integer lifts are allowed, including supports crossing the fundamental interval endpoints.

Checked declarations

Module	Named supporting theorems
CircleSourceCutoffDisjoint.lean	1

- `circleSourceCutoff_disjoint`

<!-- checkpoint-verification -->

Verification

All 677 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4284-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1817 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 679 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_611_CircleSourceCutoffProduct.md`

Checkpoint 611: Distinct source cutoffs have zero product as continuous symbols

Packages the pointwise source disjointness as equality of actual continuous circle products, suitable for the existing Fourier multiplier composition and orthogonality lemmas.

Checked declarations

Module	Named supporting theorems
CircleSourceCutoffProduct.lean	1

- `circleSourceCutoff_mul_eq_zero`

<!-- checkpoint-verification -->

Verification

All 678 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4285-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1818 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 680 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_612_CircleSourceCutoffLiftValue.md`

Checkpoint 612: A nonzero lift gives the exact circle cutoff value

Uses uniqueness of the nonzero integer lift for s at least one to evaluate the periodized cutoff at that lift. In particular periodization introduces no hidden multiplicity factor.

Checked declarations

Module	Named supporting theorems
CircleSourceCutoffLiftValue.lean	1

- `circleSourceCutoff_eq_lift`

<!-- checkpoint-verification -->

Verification

All 679 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4286-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1819 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 681 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_613_CircleSourceCutoffPointwiseBound.md`

Checkpoint 613: The actual periodized cutoff has pointwise norm at most one

Proves the sharp pointwise norm bound one, for every circle point and rational center when s is at least one. It derives the bound from the unique nonzero lift rather than the earlier crude three-window estimate.

Checked declarations

Module	Named supporting theorems
CircleSourceCutoffPointwiseBound.lean	1

- `circleSourceCutoff_pointwise_norm`

<!-- checkpoint-verification -->

Verification

All 680 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4287-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1820 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 682 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_614_CircleSourceCutoffNorm.md`

Checkpoint 614: The actual circle cutoff has continuous-map norm at most one

Turns the proved pointwise source bound into the continuous-symbol norm needed by Plancherel and multiplier estimates.

Checked declarations

Module	Named supporting theorems
CircleSourceCutoffNorm.lean	1

- `circleSourceCutoff_norm`

<!-- checkpoint-verification -->

Verification

All 681 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4288-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1821 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 683 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_615_CenteredRationalSupportWitness.md`

Checkpoint 615: The actual rational model vanishes wherever its source cutoff vanishes

Extracts the genuine source bump factor from the ordinary-minus-exceptional annular rational symbol. This support conclusion holds for arbitrary character, coefficient and beta data.

Checked declarations

Module	Named supporting theorems
CenteredRationalSupportWitness.lean	1

- `centeredRationalSymbol_cutoff_zero`

<!-- checkpoint-verification -->

Verification

All 682 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4289-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1822 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 684 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_616_CircleRationalModelWitness.md`

Checkpoint 616: A nonzero rational circle model has a nonzero source-cutoff lift

Proves an exact support witness for the actual periodized rational model, with arbitrary exceptional data. It does not infer nonzero summands from a zero sum.

Checked declarations

Module	Named supporting theorems
CircleRationalModelWitness.lean	1

- `circleRationalModel_nonzero_cutoff_lift`

<!-- checkpoint-verification -->

Verification

All 683 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4290-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1823 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 685 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_617_CircleSourceCutoffSupport.md`

Checkpoint 617: Nonzero real source lifts remain nonzero on the circle

Proves the reverse support implication, using the exact single-lift value. Combined with the witness lemma, this gives faithful real/circle support transfer for s at least one.

Checked declarations

Module	Named supporting theorems
CircleSourceCutoffSupport.lean	1

- `circleSourceCutoff_ne_zero_of_lift`

<!-- checkpoint-verification -->

Verification

All 684 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4291-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1824 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 686 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_618_CircleRationalModelSupport.md`

Checkpoint 618: Actual rational circle models are supported in their actual source cutoffs

Establishes support containment of the complete rational model, including exceptional terms, in the genuine circle source cutoff. The denominator scale is its actual floor binary logarithm.

Checked declarations

Module	Named supporting theorems
CircleRationalModelSupport.lean	1

- `circleRationalModel_cutoff_nonzero`

<!-- checkpoint-verification -->

Verification

All 685 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4292-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1825 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 687 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_619_CircleRationalModelDisjoint.md`

Checkpoint 619: Actual rational models at one source scale have disjoint circle supports

Discharges disjointness for different canonical rational centers in one dyadic source block, allowing different times, characters, coefficients and beta values for the two terms. This is an actual-symbol conclusion, not an abstract disjointness premise.

Checked declarations

Module	Named supporting theorems
CircleRationalModelDisjoint.lean	1

- `circleRationalModel_disjoint`

<!-- checkpoint-verification -->

Verification

All 686 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4293-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1826 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 688 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_620_FiniteOrthogonalEnergy.md`

Checkpoint 620: Finite orthogonal energy holds for the actual rational index set

Generalizes the earlier natural-number energy identity to any finite index set, with orthogonality required only for members of that set. This avoids an impossible globally injective bounded-denominator enumeration of infinitely many natural numbers.

Checked declarations

Module	Named supporting theorems
FiniteOrthogonalEnergy.lean	1

- `finite_orthogonal_energy`

<!-- checkpoint-verification -->

Verification

All 687 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4294-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1827 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 689 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_621_FiniteDisjointSymbolBound.md`

Checkpoint 621: Finite disjoint symbol sums have no cardinality loss

Proves the pointwise norm bound for sums over arbitrary finite rational index sets. All support and size assumptions are local to the actual finite set.

Checked declarations

Module	Named supporting theorems
FiniteDisjointSymbolBound.lean	1

- `finite_disjoint_symbol_bound`

<!-- checkpoint-verification -->

Verification

All 688 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4295-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1828 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 690 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_622_FiniteDisjointMultiplierBessel.md`

Checkpoint 622: Finite actual multipliers satisfy a cardinality-free Bessel estimate

Combines genuine integer Fourier orthogonality with the finite support bound to control the sum of squared output norms by A squared times the input energy. No abstract interval maximal estimate is used.

Checked declarations

Module	Named supporting theorems
FiniteDisjointMultiplierBessel.lean	1

- `finite_disjoint_multiplier_bessel`

<!-- checkpoint-verification -->

Verification

All 689 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4296-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1829 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 691 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_623_CircleSourceCutoffBessel.md`

Checkpoint 623: The actual source cutoffs obey a unit Bessel energy bound

Discharges both the disjointness and symbol-size hypotheses for the actual circle cutoffs on a canonical dyadic block. The total localized input energy is at most the original input energy.

Checked declarations

Module	Named supporting theorems
CircleSourceCutoffBessel.lean	1

- `circleSourceCutoff_bessel`

<!-- checkpoint-verification -->

Verification

All 690 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4297-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1830 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 692 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_624_CircleRationalModelOrthogonality.md`

Checkpoint 624: Actual rational annular multipliers are orthogonal at distinct centers

Proves actual integer-l2 orthogonality of rational annular outputs in one dyadic source block. The two inputs, arithmetic data and times can differ.

Checked declarations

Module	Named supporting theorems
CircleRationalModelOrthogonality.lean	1

- `circleRationalModel_orthogonal`

<!-- checkpoint-verification -->

Verification

All 691 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4298-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1831 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 693 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_625_CircleRationalModelFiniteEnergy.md`

Checkpoint 625: Actual rational model sums have exact finite orthogonal energy

Proves the exact norm-squared identity for any finite canonical source block, including arbitrary center-dependent input vectors. It instantiates the new local finite orthogonality theorem with the actual annular circle models.

Checked declarations

Module	Named supporting theorems
CircleRationalModelFiniteEnergy.lean	1

- `circleRationalModel_finite_energy`

<!-- checkpoint-verification -->

Verification

All 692 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4299-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1832 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 694 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_626_CircleRationalModelBessel.md`

Checkpoint 626: Actual rational model energy has no rational-count loss

Obtains the cardinality-free output energy bound for the actual finite source model, assuming only its displayed pointwise coefficient envelope A. The circle disjointness hypothesis is fully discharged.

Checked declarations

Module	Named supporting theorems
CircleRationalModelBessel.lean	1

- `circleRationalModel_bessel`

<!-- checkpoint-verification -->

Verification

All 693 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4300-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1833 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 695 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_627_CircleSourceCutoffMultiplierNorm.md`

Checkpoint 627: Actual source cutoff multipliers are contractions on integer l2

Proves operator norm at most one for each actual circle source cutoff, for s at least one. This single-scale statement is kept separate from the countable maximal problem.

Checked declarations

Module	Named supporting theorems
CircleSourceCutoffMultiplierNorm.lean	1

- `circleSourceCutoff_multiplier_norm`

<!-- checkpoint-verification -->

Verification

All 694 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4301-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1834 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 696 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_628_SourceDyadicRationalCenters.md`

Checkpoint 628: The actual dyadic rational block has an exact finite enumeration

Defines the source dyadic block by its actual denominator logarithm and proves exact canonical membership. The upper denominator box is a finite enumeration device only.

Checked declarations

Module	Named supporting theorems
SourceDyadicRationalCenters.lean	1

- `mem_sourceDyadicRationalCenters`

<!-- checkpoint-verification -->

Verification

All 695 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4302-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1835 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 697 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_629_SourceDyadicDenominatorBounds.md`

Checkpoint 629: Dyadic source membership gives the exact denominator range

Proves the two real denominator inequalities 2^s <= den(r) < 2^(s+1) for an actual source block member. Later source geometry no longer needs a separately supplied denominator estimate.

Checked declarations

Module	Named supporting theorems
SourceDyadicDenominatorBounds.lean	1

- `sourceDyadicRationalCenters_den_bounds`

<!-- checkpoint-verification -->

Verification

All 696 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4303-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1836 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 698 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_630_SourceDyadicCircleDisjoint.md`

Checkpoint 630: The actual dyadic block discharges circle cutoff disjointness

Combines exact dyadic membership with actual circle geometry. The only remaining restrictions are s at least two and distinct rational centers; canonical range and denominator bounds are now derived.

Checked declarations

Module	Named supporting theorems
SourceDyadicCircleDisjoint.lean	1

- `sourceDyadic_circleCutoff_disjoint`

<!-- checkpoint-verification -->

Verification

All 697 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4304-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1837 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 699 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_631_SourceDyadicCutoffBessel.md`

Checkpoint 631: The complete actual dyadic source block has unit localization energy

Proves the sum of squared localized input norms over the entire actual dyadic rational block is at most the input energy. All arithmetic support premises are discharged from block membership.

Checked declarations

Module	Named supporting theorems
SourceDyadicCutoffBessel.lean	1

- `sourceDyadic_cutoff_bessel`

<!-- checkpoint-verification -->

Verification

All 698 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4305-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1838 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 700 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_632_SourceDyadicModelDisjoint.md`

Checkpoint 632: Actual dyadic rational model terms are disjoint for arbitrary time data

Derives actual rational-model circle disjointness directly from membership in the source dyadic block. Character and beta data may change with time, while center separation remains fixed.

Checked declarations

Module	Named supporting theorems
SourceDyadicModelDisjoint.lean	1

- `sourceDyadic_model_disjoint`

<!-- checkpoint-verification -->

Verification

All 699 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4306-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1839 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 701 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_633_SourceDyadicModelEnergy.md

Checkpoint 633: The complete actual dyadic model has exact orthogonal output energy

Proves the finite energy identity for the complete actual rational source block and arbitrary center-dependent l2 inputs. This is exact Plancherel orthogonality for actual annular symbols.

Checked declarations

Module	Named supporting theorems
SourceDyadicModelEnergy.lean	1

- `sourceDyadic_model_energy`

<!-- checkpoint-verification -->

Verification

All 700 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4307-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1840 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 702 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_634_InitialTemporalMaximum.md

Checkpoint 634: Finite time maxima are defined by exact partial suprema

Defines the pointwise maximum of the first n+1 actual output magnitudes and proves nonnegativity and domination of every included time. It makes no claim about the l2 norm of a countable maximum.

Checked declarations

Module	Named supporting theorems
InitialTemporalMaximum.lean	2

- `initialTemporalMaximum_nonneg`
- `initialTemporalMaximum_dominates`

<!-- checkpoint-verification -->

Verification

All 701 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4308-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1842 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 703 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_635_FiniteTemporalMajorants.md`

Checkpoint 635: Uniform finite time majorants bound every coordinate

States the precise uniform finite-maximal hypothesis and proves pointwise boundedness of the entire time family. This hypothesis controls finite maxima with one common constant, which is stronger than a uniform individual operator norm.

Checked declarations

Module	Named supporting theorems
FiniteTemporalMajorants.lean	1

- `finiteTemporalMajorants_coordinate_bound`

<!-- checkpoint-verification -->

Verification

All 702 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4309-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1843 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 704 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_636_InitialTemporalMaximumL2.md`

Checkpoint 636: Every finite time maximum is an actual l2 vector

Under the displayed finite-maximal hypothesis, proves membership of the exact finite pointwise maximum in real l2 by domination. This provides genuine finite maximum vectors, rather than formal supremum notation alone.

Checked declarations

Module	Named supporting theorems
InitialTemporalMaximumL2.lean	1

- `initialTemporalMaximum_mem_l2`

<!-- checkpoint-verification -->

Verification

All 703 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4310-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1844 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 705 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_637_InitialTemporalMaximumNorm.md`

Checkpoint 637: The exact finite time maximum retains the uniform norm bound

Proves that the constructed finite maximum has l2 norm at most B, with no factor depending on the number of time scales.

Checked declarations

Module	Named supporting theorems
InitialTemporalMaximumNorm.lean	1

- `initialTemporalMaximumL2_norm`

<!-- checkpoint-verification -->

Verification

All 704 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4311-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1845 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 706 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_638_TemporalMaximumConvergence.md`

Checkpoint 638: Finite time maxima converge to the genuine countable supremum

Proves pointwise monotone convergence from exact finite maxima to the supremum of all time outputs. Boundedness is derived from the finite-maximal hypothesis.

Checked declarations

Module	Named supporting theorems
TemporalMaximumConvergence.lean	1

- `initialTemporalMaximum_tendsto`

<!-- checkpoint-verification -->

Verification

All 705 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4312-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1846 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 707 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_639_TemporalMaximumFiniteSpatialEnergy.md`

Checkpoint 639: The countable time supremum has bounded finite spatial energy

Passes the uniform finite-maximal l2 bound through a finite spatial sum by monotone pointwise convergence. This is the key finite-to-countable step, with the exact same constant B.

Checked declarations

Module	Named supporting theorems
TemporalMaximumFiniteSpatialEnergy.lean	1

- `temporalMaximum_finite_spatial_energy`

<!-- checkpoint-verification -->

Verification

All 706 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4313-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1847 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 708 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_640_TemporalMaximumMembership.md`

Checkpoint 640: A uniform finite-maximal bound gives a genuine countable l2 maximum

Proves real-l2 membership of the actual countable time supremum from its bounded finite spatial energy. No summability of individual time-output norms is assumed.

Checked declarations

Module	Named supporting theorems
TemporalMaximumMembership.lean	1

- `temporalMaximum_mem_l2`

<!-- checkpoint-verification -->

Verification

All 707 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4314-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1848 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 709 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_641_TemporalMaximumNorm.md`

Checkpoint 641: The genuine countable maximum preserves the finite-maximal constant

Proves that the constructed countable pointwise supremum has l2 norm at most B. It uses the full finite-maximal hypothesis and therefore does not conflict with the checkpoint 600 counterexample.

Checked declarations

Module	Named supporting theorems
TemporalMaximumNorm.lean	1

- `temporalMaximumL2_norm`

<!-- checkpoint-verification -->

Verification

All 708 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4315-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1849 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 710 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint notes 642-716

Checkpoint_642_FiniteToCountableMaximum.md

Checkpoint 642: Uniform finite time maxima produce the required countable majorant

Assembles the full finite-to-countable principle: a genuine nonnegative real-l2 majorant dominates every output pointwise and has norm at most the common finite-maximal bound. This is the form needed for deep finite-family multiplier inputs.

Checked declarations

Module	Named supporting theorems
FiniteToCountableMaximum.lean	1

- `finite_to_countable_maximum`

<!-- checkpoint-verification -->

Verification

All 709 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4316-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1850 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 711 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_643_FiniteTemporalCriterionEquivalence.md

Checkpoint 643: Uniform finite and countable maximal majorants are equivalent

Records the exact logical equivalence between uniformly bounded finite time maxima and a single countable pointwise l2 majorant. No individual-norm-only hypothesis appears.

Checked declarations

Module	Named supporting theorems
FiniteTemporalCriterionEquivalence.lean	1

- `finiteTemporalMajorants_iff_countable`

<!-- checkpoint-verification -->

Verification

All 710 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4317-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1851 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 712 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_644_FiniteFamilyCountableCriterion.md`

Checkpoint 644: Arbitrary finite time subfamilies suffice for the countable maximum

Converts a uniform majorant bound for every finite set of time indices into the genuine countable maximal bound. This matches finite-dimensional vector-valued transference inputs.

Checked declarations

Module	Named supporting theorems
FiniteFamilyCountableCriterion.lean	1

- `finite_family_countable_maximum`

<!-- checkpoint-verification -->

Verification

All 711 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4318-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1852 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 713 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_645_FiniteMultiplierMaximalCriterion.md`

Checkpoint 645: Finite actual multiplier estimates extend to a countable l2 maximum

Applies the proved finite-family limiting principle to the project actual integer Fourier multipliers. The finite-family estimate remains explicitly supplied; no Bourgain theorem is silently assumed.

Checked declarations

Module	Named supporting theorems
FiniteMultiplierMaximalCriterion.lean	1

- `finite_multiplier_countable_maximum`

<!-- checkpoint-verification -->

Verification

All 712 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4319-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1853 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 714 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_646_QuadraticPhaseSmallFrequency.md`

Checkpoint 646: The actual quadratic phase has a linear small-frequency bound

Proves the exact bound |exp(-2 pi i theta t squared)-1| <= 2 pi |theta| t squared for real t. It uses the standard complex exponential inequality, with the manuscript negative phase.

Checked declarations

Module	Named supporting theorems
QuadraticPhaseSmallFrequency.lean	1

- `quadraticOscillation_sub_one_norm`

<!-- checkpoint-verification -->

Verification

All 713 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4320-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1854 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 715 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_647_AnnularDifferenceIntegrand.md`

Checkpoint 647: The actual annular difference integrand has a uniform bound

Bounds the difference integrand on the real annulus N/2 <= t <= N, for N at least two and beta at most one. The arithmetic length has frequency scale N squared.

Checked declarations

Module	Named supporting theorems
AnnularDifferenceIntegrand.lean	1

- `annularDifference_integrand_bound`

<!-- checkpoint-verification -->

Verification

All 714 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4321-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1855 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 716 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_648_AnnularZeroFrequencyDifference.md`

Checkpoint 648: The actual annular model difference has its exact integral formula

Proves the normalized interval-integral identity for v(N,beta,theta)-v(N,beta,0), preserving real N/2 endpoints, the density t^(beta-1), and the factor 1/N.

Checked declarations

Module	Named supporting theorems
AnnularZeroFrequencyDifference.lean	1

- `annularOscillatoryModel_sub_zero`

<!-- checkpoint-verification -->

Verification

All 715 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4322-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1856 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 717 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_649_AnnularSmallFrequencyDifference.md`

Checkpoint 649: The actual annular model has uniform linear error at zero frequency

Proves |v(N,beta,theta)-v(N,beta,0)| <= pi N squared |theta|, uniformly in beta <= 1 and N >= 2. This supplies the small-frequency half of a summable projection-error argument.

Checked declarations

Module	Named supporting theorems
AnnularSmallFrequencyDifference.lean	1

- `annularOscillatoryModel_small_frequency`

<!-- checkpoint-verification -->

Verification

All 716 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4323-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1857 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 718 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_650_AnnularProjectionError.md`

Checkpoint 650: The actual annular model is compared to a smooth frequency projection

Defines the difference between the annular oscillatory model and its actual zero-frequency amplitude times the fixed source bump at scale N squared. Proves exact vanishing at zero frequency for every beta.

Checked declarations

Module	Named supporting theorems
AnnularProjectionError.lean	1

- `annularProjectionError_zero`

<!-- checkpoint-verification -->

Verification

All 717 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4324-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1858 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 719 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_651_AnnularProjectionErrorSize.md`

Checkpoint 651: The actual projection error has a uniform size bound

Proves norm at most one for the annular projection error, for N >= 2 and beta <= 1. The fixed bump and zero-frequency amplitude are bounded using already proved source estimates.

Checked declarations

Module	Named supporting theorems
AnnularProjectionErrorSize.lean	1

- `annularProjectionError_norm_one`

<!-- checkpoint-verification -->

Verification

All 718 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4325-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1859 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 720 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_652_AnnularProjectionErrorCore.md

Checkpoint 652: The projection error agrees with the exact annular difference in its core

Proves exact equality with v(theta)-v(0) whenever N squared times |theta| is at most one half. No Taylor approximation or numerical estimate is used.

Checked declarations

Module	Named supporting theorems
AnnularProjectionErrorCore.lean	1

- `annularProjectionError_core`

<!-- checkpoint-verification -->

Verification

All 719 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4326-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1860 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 721 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_653_AnnularProjectionErrorNear.md

Checkpoint 653: The full projection error is bounded linearly at small frequency

Proves the global bound |error| <= (pi+2) N squared |theta|. It combines the exact core difference with a uniform bound in the transition region of the bump.

Checked declarations

Module	Named supporting theorems
AnnularProjectionErrorNear.lean	1

- `annularProjectionError_near`

<!-- checkpoint-verification -->

Verification

All 720 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4327-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1861 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 722 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_654_AnnularProjectionErrorFar.md`

Checkpoint 654: The full projection error has reciprocal frequency decay

Proves (N squared |theta|) |error| <= 1+2/pi uniformly for beta in [0,1]. Outside the bump support, the error is exactly the original annular oscillatory model.

Checked declarations

Module	Named supporting theorems
AnnularProjectionErrorFar.lean	1

- `annularProjectionError_far`

<!-- checkpoint-verification -->

Verification

All 721 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4328-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1862 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 723 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

Checkpoint_655_AnnularProjectionErrorProfile.md

Checkpoint 655: One explicit constant controls both projection-error regimes

Packages the actual annular projection error with uniform linear and reciprocal bounds and a positive explicit constant. These are the hypotheses of the next telescoping square-energy argument.

Checked declarations

Module	Named supporting theorems
AnnularProjectionErrorProfile.lean	1

- `annularProjectionError_profile`

<!-- checkpoint-verification -->

Verification

All 722 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4329-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1863 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 724 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_656_AnnularProjectionErrorContinuity.md

Checkpoint 656: The actual projection error is continuous for every beta

Proves continuity of the exact annular projection error, with the existing real-annulus regularity. This permits a faithful continuous circle-symbol realization of its localized form.

Checked declarations

Module	Named supporting theorems
AnnularProjectionErrorContinuity.lean	1

- `annularProjectionError_continuous`

<!-- checkpoint-verification -->

Verification

All 723 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4330-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1864 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 725 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_657_GeometricEnergyPotential.md

Checkpoint 657: A bounded positive potential controls dyadic projection errors

Defines H(x)=1/(1+x squared) and proves its values lie in [0,1]. This potential will give a telescoping square-energy estimate without selecting a first crossing scale.

Checked declarations

Module	Named supporting theorems
GeometricEnergyPotential.lean	1

- `geometricEnergyPotential_range`

<!-- checkpoint-verification -->

Verification

All 724 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4331-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1865 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 726 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_658_GeometricPotentialDifference.md

Checkpoint 658: The dyadic potential difference has an exact positive formula

Proves H(x)-H(4x)=15 x squared / ((1+x squared)(1+16 x squared)). The identity is valid at zero and uses strictly positive denominators.

Checked declarations

Module	Named supporting theorems
GeometricPotentialDifference.lean	1

- `geometricEnergyPotential_difference`

<!-- checkpoint-verification -->

Verification

All 725 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4332-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1866 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 727 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_659_GeometricPotentialNear.md

Checkpoint 659: The potential difference dominates the small-frequency square

Proves x squared <= (34/15)(H(x)-H(4x)) on 0<=x<=1, with an explicit constant. This will sum the small-frequency errors geometrically.

Checked declarations

Module	Named supporting theorems
GeometricPotentialNear.lean	1

- `geometricEnergyPotential_near`

<!-- checkpoint-verification -->

Verification

All 726 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4333-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1867 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 728 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_660_GeometricPotentialFar.md

Checkpoint 660: The potential difference dominates the large-frequency square

Proves 1/x squared <= (34/15)(H(x)-H(4x)) for x>=1. Together with the near bound it handles both halves of the annular projection error.

Checked declarations

Module	Named supporting theorems
GeometricPotentialFar.lean	1

- `geometricEnergyPotential_far`

<!-- checkpoint-verification -->

Verification

All 727 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4334-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1868 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 729 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_661_TwoSidedErrorPotential.md`

Checkpoint 661: Linear and reciprocal bounds give one telescoping square estimate

Proves a general scalar error-square inequality from e<=Cx and xe<=C. It is a proved analytic inequality, not a hypothesis about the paper source maximal operator.

Checked declarations

Module	Named supporting theorems
TwoSidedErrorPotential.lean	1

- `twoSidedError_potential`

<!-- checkpoint-verification -->

Verification

All 728 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4335-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1869 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 730 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_662_GeometricPotentialTelescoping.md`

Checkpoint 662: Dyadic potential differences telescope exactly

Proves the exact finite telescoping identity over scales 4^j x. It avoids any ceiling, logarithm, or first-onset indexing convention.

Checked declarations

Module	Named supporting theorems
GeometricPotentialTelescoping.lean	1

- `geometricEnergyPotential_telescoping`

<!-- checkpoint-verification -->

Verification

All 729 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4336-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1870 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 731 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_663_GeometricErrorRangeEnergy.md`

Checkpoint 663: The scalar two-sided profile has a uniform finite scale energy bound

Shows that any nonnegative sequence satisfying e_j<=C 4^j x and (4^j x)e_j<=C has total squared error at most (34/15)C squared over every initial scale range, uniformly in x>=0.

Checked declarations

Module	Named supporting theorems
GeometricErrorRangeEnergy.lean	1

- `geometricError_range_energy`

<!-- checkpoint-verification -->

Verification

All 730 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4337-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1871 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 732 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_664_GeometricErrorFiniteEnergy.md`

Checkpoint 664: The scalar profile controls arbitrary finite sets of scales

Extends the uniform initial-range error bound to every finite scale subset by positivity. This is the exact finite-energy form consumed by the project Fourier square-function machinery.

Checked declarations

Module	Named supporting theorems
GeometricErrorFiniteEnergy.lean	1

- `geometricError_finite_energy`

<!-- checkpoint-verification -->

Verification

All 731 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4338-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1872 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 733 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_665_GeometricErrorSummability.md`

Checkpoint 665: The complete scalar projection-error square sum converges

Proves summability and the total bound (34/15)C squared for the entire scalar profile. It includes the zero-frequency case without division by zero.

Checked declarations

Module	Named supporting theorems
GeometricErrorSummability.lean	2

- `geometricError_square_summable`
- `geometricError_square_total`

<!-- checkpoint-verification -->

Verification

All 732 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4339-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1874 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 734 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_666_DyadicAnnularProjectionProfile.md`

Checkpoint 666: The actual dyadic annular error satisfies the two-sided scalar profile

Identifies the actual dyadic spatial scale (2^j M) squared with 4^j M squared and proves the uniform error bounds even when beta changes arbitrarily in [0,1] between times.

Checked declarations

Module	Named supporting theorems
DyadicAnnularProjectionProfile.lean	1

- `dyadicAnnularProjectionError_profile`

<!-- checkpoint-verification -->

Verification

All 733 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4340-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1875 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 735 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_667_DyadicAnnularProjectionEnergy.md

Checkpoint 667: The actual dyadic annular projection error has uniform finite energy

Proves a finite square-energy bound for the genuine annular model minus its smooth projection, independent of frequency, starting scale, number of times and beta choices in [0,1].

Checked declarations

Module	Named supporting theorems
DyadicAnnularProjectionEnergy.lean	1

- `dyadicAnnularProjectionError_finite_energy`

<!-- checkpoint-verification -->

Verification

All 734 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4341-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1876 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 736 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_668_DyadicAnnularProjectionTotal.md

Checkpoint 668: The actual dyadic annular projection-error square series converges

Proves actual full square summability and the uniform total bound for the annular projection error, with time-dependent beta in [0,1]. This controls the error contribution; the smooth main projection maximum is a separate stage.

Checked declarations

Module	Named supporting theorems
DyadicAnnularProjectionTotal.lean	2

- `dyadicAnnularProjectionError_summable`
- `dyadicAnnularProjectionError_total`

<!-- checkpoint-verification -->

Verification

All 735 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4342-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1878 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 737 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_669_LocalizedAnnularProjectionError.md`

Checkpoint 669: The projection error has a compact continuous source localization

Constructs the actual complex continuous error symbol times the original broad source bump and proves its compact support. No replacement cutoff or support assumption is used.

Checked declarations

Module	Named supporting theorems
LocalizedAnnularProjectionError.lean	1

- `localizedAnnularProjectionError_support`

<!-- checkpoint-verification -->

Verification

All 736 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4343-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1879 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 738 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_670_CircleAnnularProjectionError.md`

Checkpoint 670: The actual projection error defines a translated circle symbol

Defines its exact compact periodization and proves the all-real integer-lift formula, retaining the actual rational center and broad source cutoff.

Checked declarations

Module	Named supporting theorems
CircleAnnularProjectionError.lean	1

- `circleAnnularProjectionError_real`

<!-- checkpoint-verification -->

Verification

All 737 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4344-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1880 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 739 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_671_CircleAnnularErrorLiftValue.md`

Checkpoint 671: One fixed nonzero source lift evaluates every time-error symbol

Uses the time-independent broad cutoff to identify the periodized error with one real lift for every time and beta. This prevents the lift from varying silently between terms of the square sum.

Checked declarations

Module	Named supporting theorems
CircleAnnularErrorLiftValue.lean	1

- `circleAnnularProjectionError_eq_lift`

<!-- checkpoint-verification -->

Verification

All 738 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4345-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1881 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 740 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_672_CircleAnnularErrorSupport.md`

Checkpoint 672: The actual circle projection error is supported in the source cutoff

Proves exact vanishing of each time-error symbol wherever the actual circle source cutoff vanishes. The support is fixed across all time and beta parameters.

Checked declarations

Module	Named supporting theorems
CircleAnnularErrorSupport.lean	1

- `circleAnnularProjectionError_cutoff_zero`

<!-- checkpoint-verification -->

Verification

All 739 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4346-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1882 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 741 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_673_CircleAnnularErrorEnergy.md`

Checkpoint 673: The actual circle projection error has uniform finite time energy

Transfers the complete real annular error bound to the actual circle symbol with no periodization multiplicity loss. The same source lift works at every time, including for arcs crossing the fundamental interval endpoints.

Checked declarations

Module	Named supporting theorems
CircleAnnularErrorEnergy.lean	1

- `circleAnnularProjectionError_finite_energy`

<!-- checkpoint-verification -->

Verification

All 740 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4347-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 1883 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 742 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_674_CircleAnnularErrorMaximum.md`

Checkpoint 674: The actual single-center error has a genuine countable l2 majorant

Applies the proved circle square-energy principle to construct an actual nonnegative l2 majorant for every dyadic time-error output. Its norm is at most sqrt(34/15) times the explicit error constant times the input norm.

Checked declarations

Module	Named supporting theorems
CircleAnnularErrorMaximum.lean	3

- `annularProjectionErrorL2Constant_pos`
- `annularProjectionErrorL2Constant_sq`
- `circleAnnularProjectionError_maximum`

<!-- checkpoint-verification -->

Verification

All 741 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4348-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1886 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 743 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_675_CircleAnnularErrorDisjoint.md`

Checkpoint 675: Distinct canonical source errors have disjoint circle supports

Discharges actual support disjointness for the time-error symbols, allowing different times and beta values at the two centers. The bound depends on canonical denominator geometry, not coefficient size.

Checked declarations

Module	Named supporting theorems
CircleAnnularErrorDisjoint.lean	1

- `circleAnnularProjectionError_disjoint`

<!-- checkpoint-verification -->

Verification

All 742 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4349-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1887 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 744 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_676_RationalProjectionErrorBlock.md`

Checkpoint 676: A rational error block vanishes outside the actual source supports

Defines the finite weighted rational projection-error block and proves it vanishes when all its source cutoffs vanish. All coefficients and beta values remain explicit.

Checked declarations

Module	Named supporting theorems
RationalProjectionErrorBlock.lean	1

- `rationalProjectionErrorBlock_zero`

<!-- checkpoint-verification -->

Verification

All 743 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4350-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1888 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 745 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_677_RationalProjectionErrorActiveCenter.md`

Checkpoint 677: One actual source center controls every time in the rational error block

Proves exact collapse of the block to its one active canonical center. Because the broad supports are time independent, the chosen center can be held fixed throughout the time-energy sum.

Checked declarations

Module	Named supporting theorems
RationalProjectionErrorActiveCenter.lean	1

- `rationalProjectionErrorBlock_active`

<!-- checkpoint-verification -->

Verification

All 744 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4351-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1889 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 746 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_678_WeightedCircleAnnularErrorEnergy.md`

Checkpoint 678: Bounded varying coefficients preserve the actual single-center error energy

Proves the weighted finite time-energy estimate for the actual circle error, allowing coefficients and beta to change at every dyadic time. It retains the square of the coefficient envelope.

Checked declarations

Module	Named supporting theorems
WeightedCircleAnnularErrorEnergy.lean	1

- `circleAnnularProjectionError_weighted_energy`

<!-- checkpoint-verification -->

Verification

All 745 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4352-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1890 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 747 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_679_RationalProjectionErrorEnergy.md`

Checkpoint 679: The actual rational error block has cardinality-free time energy

Proves the full finite rational-block square-energy estimate with no number-of-centers loss. It combines one time-independent active source center with the proved annular projection-error bound.

Checked declarations

Module	Named supporting theorems
RationalProjectionErrorEnergy.lean	1

- `rationalProjectionErrorBlock_finite_energy`

<!-- checkpoint-verification -->

Verification

All 746 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4353-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1891 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 748 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_680_RationalProjectionErrorMaximum.md

Checkpoint 680: The actual weighted rational error block has a genuine countable l2 maximum

Constructs a countable maximal l2 majorant for the entire actual weighted rational error block with norm at most A sqrt(34/15) C_error times the input norm. Coefficients and beta may vary with time; the canonical centers and source scale are fixed.

Checked declarations

Module	Named supporting theorems
RationalProjectionErrorMaximum.lean	1

- `rationalProjectionErrorBlock_maximum`

<!-- checkpoint-verification -->

Verification

All 747 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4354-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1892 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 749 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_681_LocalizedAnnularSource.md

Checkpoint 681: The original broad annular source symbol is reused exactly

Names the existing annular cutoff symbol with its actual source scale 16^s and proves support for every natural source scale. This is an exact reuse of the already constructed real annular model.

Checked declarations

Module	Named supporting theorems
LocalizedAnnularSource.lean	1

- `localizedAnnularSource_support`

<!-- checkpoint-verification -->

Verification

All 748 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4355-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1893 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 750 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_682_CircleAnnularSource.md`

Checkpoint 682: The original broad annular source has an exact circle realization

Constructs the translated circle realization of the original source annular factor and proves its all-real lift formula. It retains the exact broad cutoff, annular normalization and phase.

Checked declarations

Module	Named supporting theorems
CircleAnnularSource.lean	1

- `circleAnnularSource_real`

<!-- checkpoint-verification -->

Verification

All 749 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4356-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1894 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 751 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

Checkpoint_683_LocalizedAnnularSmoothProjection.md

Checkpoint 683: The smooth comparison symbol retains the original broad source mask

Constructs the product of the fixed smooth projection at scale N squared and the original source cutoff. It proves continuity and compact support for every source scale.

Checked declarations

Module	Named supporting theorems
LocalizedAnnularSmoothProjection.lean	1

- `localizedAnnularSmoothProjection_support`

<!-- checkpoint-verification -->

Verification

All 750 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4357-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1895 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 752 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_684_CircleAnnularSmoothProjection.md

Checkpoint 684: The smooth comparison family has an exact continuous circle symbol

Defines the actual translated smooth comparison symbol on the circle and proves its full integer-periodization formula. No maximal estimate is asserted by this definition.

Checked declarations

Module	Named supporting theorems
CircleAnnularSmoothProjection.lean	1

- `circleAnnularSmoothProjection_real`

<!-- checkpoint-verification -->

Verification

All 751 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4358-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1896 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 753 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_685_LocalizedAnnularProjectionIdentity.md`

Checkpoint 685: The localized annular error is exactly source minus smooth main term

Proves equality of continuous real symbols between the localized error and the original annular source minus its actual zero-frequency amplitude times the smooth comparison. This is an exact decomposition.

Checked declarations

Module	Named supporting theorems
LocalizedAnnularProjectionIdentity.lean	1

- `localizedAnnularProjectionError_identity`

<!-- checkpoint-verification -->

Verification

All 752 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4359-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1897 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 754 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_686_CircleAnnularProjectionIdentity.md`

Checkpoint 686: The error decomposition holds for the actual circle symbols

Transfers the exact source-minus-smooth-main identity through the compact circle lift and rational translation. It preserves the actual circle symbol rather than assuming a line/circle transference estimate.

Checked declarations

Module	Named supporting theorems
CircleAnnularProjectionIdentity.lean	1

- `circleAnnularProjectionError_identity`

<!-- checkpoint-verification -->

Verification

All 753 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4360-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1898 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 755 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_687_CircleAnnularSourceDecomposition.md`

Checkpoint 687: The original circle annular source splits into a smooth main term and controlled error

Rearranges the proved circle identity into the form used in the maximal argument. The error already has a proved countable block maximum; the smooth main family remains a separate target.

Checked declarations

Module	Named supporting theorems
CircleAnnularSourceDecomposition.lean	1

- `circleAnnularSource_decomposition`

<!-- checkpoint-verification -->

Verification

All 754 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4361-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1899 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 756 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_688_CenteredGaussAnnularDecomposition.md`

Checkpoint 688: The actual centered Gauss model splits into its two annular source factors

Defines the genuine principal and exceptional rational coefficients and proves the exact real continuous-symbol decomposition. The exceptional sign and coefficient order are retained.

Checked declarations

Module	Named supporting theorems
CenteredGaussAnnularDecomposition.lean	1

- `centeredRationalSymbol_annular_decomposition`

<!-- checkpoint-verification -->

Verification

All 755 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4362-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 1900 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 757 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_689_CircleGaussAnnularDecomposition.md`

Checkpoint 689: The original rational circle model has its exact ordinary-minus-exceptional decomposition

Proves equality of the actual circle model with its principal coefficient times the ordinary annular source minus its exceptional coefficient times the beta annular source. It does not require any coefficient estimate.

Checked declarations

Module	Named supporting theorems
CircleGaussAnnularDecomposition.lean	1

- `circleRationalModel_annular_decomposition`

<!-- checkpoint-verification -->

Verification

All 756 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4363-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1901 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 758 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_690_CircleGaussSmoothErrorDecomposition.md`

Checkpoint 690: The original rational circle model splits exactly into smooth main and controlled errors

Connects the new error maximum to the original rational Gauss symbol. The smooth-main coefficient is the genuine principal zero-frequency amplitude minus the exceptional beta zero-frequency amplitude.

Checked declarations

Module	Named supporting theorems
CircleGaussSmoothErrorDecomposition.lean	1

- `circleRationalModel_smooth_error_decomposition`

<!-- checkpoint-verification -->

Verification

All 757 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4364-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1902 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 759 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_691_RationalGaussSmoothMainBlock.md`

Checkpoint 691: The original finite rational model has an exact smooth-main and error-block decomposition

Defines the actual smooth main block and proves its exact relation to the original circleMajorArcModel on one denominator scale. Both the ordinary and exceptional error blocks are the actual controlled symbols constructed earlier.

Checked declarations

Module	Named supporting theorems
RationalGaussSmoothMainBlock.lean	1

- `circleMajorArcModel_smooth_error_blocks`

<!-- checkpoint-verification -->

Verification

All 758 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4365-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1903 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 760 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_692_SignedProjectionErrorMaximum.md`

Checkpoint 692: The signed ordinary-minus-exceptional error has a countable l2 majorant

Combines the two actual weighted error maxima, preserving time-dependent coefficients and beta values. The resulting constant is the sum of the two coefficient envelopes times the proved error constant.

Checked declarations

Module	Named supporting theorems
SignedProjectionErrorMaximum.lean	1

- `signedProjectionErrorBlock_maximum`

<!-- checkpoint-verification -->

Verification

All 759 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4366-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1904 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 761 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_693_ActualGaussProjectionErrorMaximum.md

Checkpoint 693: The original rational model minus its smooth main block has a countable maximum

Connects the proved error maximum to the original circleMajorArcModel, including characters and exceptional coefficients that change between times. Only explicit coefficient envelopes and source block geometry remain as hypotheses.

Checked declarations

Module	Named supporting theorems
ActualGaussProjectionErrorMaximum.lean	1

- `actualGaussProjectionError_maximum`

<!-- checkpoint-verification -->

Verification

All 760 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4367-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1905 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 762 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_694_RationalGaussCoefficientUnitBounds.md

Checkpoint 694: The actual rational coefficients have unconditional unit envelopes

Discharges the two basic coefficient envelopes using the genuine principal and twisted normalized Gauss sums. The exceptional multiplier kappa is only required to have norm at most one.

Checked declarations

Module	Named supporting theorems
RationalGaussCoefficientUnitBounds.lean	1

- `rationalGaussCoefficient_unit_bounds`

<!-- checkpoint-verification -->

Verification

All 761 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4368-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1906 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 763 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_695_ActualGaussProjectionErrorUnitMaximum.md`

Checkpoint 695: The original model comparison error has a uniform countable block maximum

Discharges both coefficient envelopes for the actual ordinary-minus-exceptional source model. The genuine countable error maximum is bounded by twice the explicit projection-error constant, independently of the rational block cardinality and changing characters.

Checked declarations

Module	Named supporting theorems
ActualGaussProjectionErrorUnitMaximum.lean	1

- `actualGaussProjectionError_unit_maximum`

<!-- checkpoint-verification -->

Verification

All 762 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4369-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1907 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 764 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and

Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_696_RationalExceptionalCoefficientDecay.md

Checkpoint 696: The actual ordinary and exceptional coefficients have uniform subhalf decay

Reuses the fully proved character-uniform Gauss estimate to bound the exact ordinary coefficient and the exceptional coefficient including kappa. No PNT or extra arithmetic input is introduced.

Checked declarations

Module	Named supporting theorems
RationalExceptionalCoefficientDecay.lean	1

- `rationalGaussCoefficient_uniform_decay`

<!-- checkpoint-verification -->

Verification

All 763 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4370-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1908 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 765 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_697_SourceDyadicGaussCoefficientEnvelope.md

Checkpoint 697: Actual dyadic coefficients have a common geometric scale envelope

Derives C(2^s)^(-delta) for every actual source dyadic block coefficient, uniformly over all complex characters and kappa of norm at most one, for 0<=delta<1/2.

Checked declarations

Module	Named supporting theorems
SourceDyadicGaussCoefficientEnvelope.lean	1

- `sourceDyadicGaussCoefficient_envelope`

<!-- checkpoint-verification -->

Verification

All 764 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4371-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1909 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 766 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_698_SourceDyadicProjectionErrorPowerMaximum.md

Checkpoint 698: The actual Gauss-model comparison error has geometric dyadic decay

Proves a genuine countable maximal error bound H_delta (2^s)^(-delta) on any subset of the actual dyadic source block, for 0<=delta<1/2. All coefficient envelopes and support premises are discharged; changing character data and beta in [0,1] are allowed.

Checked declarations

Module	Named supporting theorems
SourceDyadicProjectionErrorPowerMaximum.lean	1

- `sourceDyadicProjectionError_power_maximum`

<!-- checkpoint-verification -->

Verification

All 765 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4372-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1910 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 767 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_699_SmoothMainCoefficientBound.md

Checkpoint 699: The actual smooth main coefficient retains the annular half-mass bound

Bounds the exact principal-minus-exceptional zero-frequency coefficient by half the sum of the coefficient envelopes. This is a single-symbol estimate and does not supply a time maximum.

Checked declarations

Module	Named supporting theorems
SmoothMainCoefficientBound.lean	1

- `rationalSmoothMain_coefficient_bound`

<!-- checkpoint-verification -->

Verification

All 766 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4373-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 1911 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 768 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_700_SourceBlockReductionToSmoothMaximum.md`

Checkpoint 700: The actual source block maximum reduces to the smooth finite-family estimate

Combines the proved uniform finite-family to countable maximal extension with the fully discharged annular comparison-error maximum. On an actual dyadic rational block, a supplied smooth-main finite-family maximal bound B yields a genuine countable source-block majorant with norm at most (B+2 C_error) times the input norm. The smooth-main premise remains explicit and unproved here. This is a reduction; it does not complete Lemma 6.16, Proposition 6.21, Proposition 4.19 or Theorem 1.1. This draft was reconstructed after recovering the confirmed checkpoint699 archive and is counted complete only after its new production build, full audits and save.

Checked declarations

Module	Named supporting theorems
SourceBlockReductionToSmoothMaximum.lean	1

- `sourceBlock_maximum_from_smooth_finite`

<!-- checkpoint-verification -->

Verification

All 767 project modules and the root were freshly rebuilt: 4374 jobs pass without warnings/errors. All 1912 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 769 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_701_ScaledSmoothCutoffMap.md`

Checkpoint 701: A general real scale of the fixed source bump gives a compact continuous map

A general real scale of the fixed source bump gives a compact continuous map. Reconstructed after recovering the confirmed checkpoint699 project. This is a supporting source/smooth-cutoff statement; it does not prove the smooth maximal theorem or Theorem1.1. Counted complete only after individual production compilation, full audits and separate save.

Checked declarations

Module	Named supporting theorems
ScaledSmoothCutoffMap.lean	2

- `scaledSmoothCutoff_radius`
- `scaledSmoothCutoffMap_support`

<!-- checkpoint-verification -->

Verification

All 768 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4375-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1914 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 770 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_702_CircleSmoothCutoff.md

Checkpoint 702: The general smooth cutoff has an exact circle periodization

The general smooth cutoff has an exact circle periodization. Reconstructed after recovering the confirmed checkpoint699 project. This is a supporting source/smooth-cutoff statement; it does not prove the smooth maximal theorem or Theorem1.1. Counted complete only after individual production compilation, full audits and separate save.

Checked declarations

Module	Named supporting theorems
CircleSmoothCutoff.lean	1

- `circleSmoothCutoff_real`

<!-- checkpoint-verification -->

Verification

All 769 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4376-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1915 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 771 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_703_ScaledSmoothSingleLift.md

Checkpoint 703: A sufficiently narrow smooth circle cutoff has a unique nonzero lift

A sufficiently narrow smooth circle cutoff has a unique nonzero lift. Reconstructed after recovering the confirmed checkpoint699 project. This is a supporting source/smooth-cutoff statement; it does not prove the smooth maximal theorem or Theorem1.1. Counted complete only after individual production compilation, full audits and separate save.

Checked declarations

Module	Named supporting theorems
ScaledSmoothSingleLift.lean	1

- `scaledSmoothCutoff_single_lift`

<!-- checkpoint-verification -->

Verification

All 770 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4377-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1916 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 772 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_704_CircleSmoothLiftValue.md`

Checkpoint 704: The narrow general smooth circle cutoff is evaluated at its nonzero lift

The narrow general smooth circle cutoff is evaluated at its nonzero lift. Reconstructed after recovering the confirmed checkpoint699 project. This is a supporting source/smooth-cutoff statement; it does not prove the smooth maximal theorem or Theorem1.1. Counted complete only after individual production compilation, full audits and separate save.

Checked declarations

Module	Named supporting theorems
CircleSmoothLiftValue.lean	1

- `circleSmoothCutoff_eq_lift`

<!-- checkpoint-verification -->

Verification

All 771 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4378-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1917 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 773 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_705_CircleSmoothPointwiseBound.md`

Checkpoint 705: The general narrow smooth cutoff has pointwise norm at most one

The general narrow smooth cutoff has pointwise norm at most one. Reconstructed after recovering the confirmed checkpoint699 project. This is a supporting source/smooth-cutoff statement; it does not prove the smooth maximal theorem or Theorem1.1. Counted complete only after individual production compilation, full audits and separate save.

Checked declarations

Module	Named supporting theorems
CircleSmoothPointwiseBound.lean	1

- `circleSmoothCutoff_pointwise_norm`

<!-- checkpoint-verification -->

Verification

All 772 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4379-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1918 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 774 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_706_CircleSmoothNorm.md`

Checkpoint 706: General smooth circle cutoff multipliers are l2 contractions

General smooth circle cutoff multipliers are l2 contractions. Reconstructed after recovering the confirmed checkpoint699 project. This is a supporting source/smooth-cutoff statement; it does not prove the smooth maximal theorem or Theorem1.1. Counted complete only after individual production compilation, full audits and separate save.

Checked declarations

Module	Named supporting theorems
CircleSmoothNorm.lean	2

- `circleSmoothCutoff_norm`
- `circleSmoothCutoff_multiplier_norm`

<!-- checkpoint-verification -->

Verification

All 773 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4380-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1920 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 775 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_707_SmoothProjectionSourceSupport.md`

Checkpoint 707: The actual masked smooth projection vanishes outside the fixed source cutoff

The actual masked smooth projection vanishes outside the fixed source cutoff. Reconstructed after recovering the confirmed checkpoint699 project. This is a supporting source/smooth-cutoff statement; it does not prove the smooth maximal theorem or Theorem1.1. Counted complete only after individual production compilation, full audits and separate save.

Checked declarations

Module	Named supporting theorems
SmoothProjectionSourceSupport.lean	1

- `circleAnnularSmoothProjection_cutoff_zero`

<!-- checkpoint-verification -->

Verification

All 774 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4381-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1921 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 776 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_708_SmoothProjectionLiftValue.md`

Checkpoint 708: The actual masked smooth projection has one fixed source lift

The actual masked smooth projection has one fixed source lift. Reconstructed after recovering the confirmed checkpoint699 project. This is a supporting source/smooth-cutoff statement; it does not prove the smooth maximal theorem or Theorem1.1. Counted complete only after individual production compilation, full audits and separate save.

Checked declarations

Module	Named supporting theorems
SmoothProjectionLiftValue.lean	1

- `circleAnnularSmoothProjection_eq_lift`

<!-- checkpoint-verification -->

Verification

All 775 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4382-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1922 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 777 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_709_SmoothCutoffSourceLiftCompatibility.md`

Checkpoint 709: The general narrow cutoff and broad source cutoff use the same integer lift

The general narrow cutoff and broad source cutoff use the same integer lift. Reconstructed after recovering the confirmed checkpoint699 project. This is a supporting source/smooth-cutoff statement; it does not prove the smooth maximal theorem or Theorem1.1. Counted complete only after individual production compilation, full audits and separate save.

Checked declarations

Module	Named supporting theorems
SmoothCutoffSourceLiftCompatibility.lean	2

- `smoothCutoff_source_lifts_unique`
- `circleSmoothCutoff_on_source_lift`

<!-- checkpoint-verification -->

Verification

All 776 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4383-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1924 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 778 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_710_SmoothProjectionCircleFactorization.md`

Checkpoint 710: The actual masked smooth projection factors as a circle product

The actual masked smooth projection factors as a circle product. Reconstructed after recovering the confirmed checkpoint699 project. This is a supporting source/smooth-cutoff statement; it does not prove

the smooth maximal theorem or Theorem1.1. Counted complete only after individual production compilation, full audits and separate save.

Checked declarations

Module	Named supporting theorems
SmoothProjectionCircleFactorization.lean	1

- `circleAnnularSmoothProjection_factorization`

<!-- checkpoint-verification -->

Verification

All 777 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4384-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1925 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 779 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_711_CircleSmoothSupportWitness.md`

Checkpoint 711: A nonzero smooth circle cutoff has a nonzero real lift

A nonzero smooth circle cutoff has a nonzero real lift. This is a supporting localization statement for the actual source model, with every separation and scale condition explicit. It does not prove the remaining multifrequency input or Theorem 1.1.

Checked declarations

Module	Named supporting theorems
CircleSmoothSupportWitness.lean	1

- `circleSmoothCutoff_nonzero_lift`

<!-- checkpoint-verification -->

Verification

All 778 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4385-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1926 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 780 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic

conclusion is assumed.

Checkpoint_712_SourceSmoothCrossLiftUniqueness.md

Checkpoint 712: Broad source and narrower smooth lifts at different canonical centers cannot overlap

Broad source and narrower smooth lifts at different canonical centers cannot overlap. This is a supporting localization statement for the actual source model, with every separation and scale condition explicit. It does not prove the remaining multifrequency input or Theorem 1.1.

Checked declarations

Module	Named supporting theorems
SourceSmoothCrossLiftUniqueness.lean	1

- `source_smooth_cross_lift_unique`

<!-- checkpoint-verification -->

Verification

All 779 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4386-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1927 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 781 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint_713_SourceSmoothCrossProduct.md

Checkpoint 713: Different canonical centers have zero broad-source times narrow-smooth product

Different canonical centers have zero broad-source times narrow-smooth product. This is a supporting localization statement for the actual source model, with every separation and scale condition explicit. It does not prove the remaining multifrequency input or Theorem 1.1.

Checked declarations

Module	Named supporting theorems
SourceSmoothCrossProduct.lean	1

- `source_smooth_cross_product`

<!-- checkpoint-verification -->

Verification

All 780 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4387-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1928 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 782 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_714_WeightedSourceCutoffInput.md`

Checkpoint 714: A weighted disjoint source cutoff gives an input bound without a cardinality loss

A weighted disjoint source cutoff gives an input bound without a cardinality loss. This is a supporting localization statement for the actual source model, with every separation and scale condition explicit. It does not prove the remaining multifrequency input or Theorem 1.1.

Checked declarations

Module	Named supporting theorems
WeightedSourceCutoffInput.lean	2

- `weightedSourceCutoff_norm`
- `weightedSourceCutoff_input_norm`

<!-- checkpoint-verification -->

Verification

All 781 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4388-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1930 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 783 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_715_WeightedSmoothSourceFactorization.md`

Checkpoint 715: The weighted masked smooth block factors through one time independent source input

The weighted masked smooth block factors through one time independent source input. This is a supporting localization statement for the actual source model, with every separation and scale condition explicit. It does not prove the remaining multifrequency input or Theorem 1.1.

Checked declarations

Module	Named supporting theorems
WeightedSmoothSourceFactorization.lean	1

- `weightedSmoothSource_factorization`

<!-- checkpoint-verification -->

Verification

All 782 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4389-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1931 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 784 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_716_WeightedSmoothSourceOperator.md`

Checkpoint 716: The smooth source factorization holds for the actual integer Fourier operators

The smooth source factorization holds for the actual integer Fourier operators. This is a supporting localization statement for the actual source model, with every separation and scale condition explicit. It does not prove the remaining multifrequency input or Theorem 1.1.

Checked declarations

Module	Named supporting theorems
WeightedSmoothSourceOperator.lean	1

- `weightedSmoothSource_operator`

<!-- checkpoint-verification -->

Verification

All 783 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4390-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1932 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 785 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

Checkpoint notes 717-791

`Checkpoint_717_GeometricSmoothCircleFamily.md`

Checkpoint 717: The actual smooth family uses geometric frequency scales

The actual smooth family uses geometric frequency scales. This is a supporting localization statement for the actual source model, with every separation and scale condition explicit. It does not prove the remaining multifrequency input or Theorem 1.1.

Checked declarations

Module	Named supporting theorems
GeometricSmoothCircleFamily.lean	3

- `smoothCircleScale_ge`
- `smoothCircleScale_four`
- `smoothCircleScale_one`

<!-- checkpoint-verification -->

Verification

All 784 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4391-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1935 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 786 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_718_SeparatedSmoothCircleSupports.md`

Checkpoint 718: Periodically separated smooth circle cutoffs have disjoint support

Periodically separated smooth circle cutoffs have disjoint support. This is a supporting localization statement for the actual source model, with every separation and scale condition explicit. It does not prove the remaining multifrequency input or Theorem 1.1.

Checked declarations

Module	Named supporting theorems
SeparatedSmoothCircleSupports.lean	1

- `separatedSmoothCircle_disjoint`

<!-- checkpoint-verification -->

Verification

All 785 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4392-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1936 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 787 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_719_SeparatedSmoothCircleNorm.md`

Checkpoint 719: Each separated smooth family member is an operator contraction

Each separated smooth family member is an operator contraction. This is a supporting localization statement for the actual source model, with every separation and scale condition explicit. It does not prove the remaining multifrequency input or Theorem 1.1.

Checked declarations

Module	Named supporting theorems
SeparatedSmoothCircleNorm.lean	1

- `separatedSmoothCircleFamily_norm`

<!-- checkpoint-verification -->

Verification

All 786 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4393-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1937 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 788 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. Published Ren and Ghosh quadratic exponential-sum estimates, when consumed, remain displayed raw theorem hypotheses; the progression-prefix estimate also remains an explicit unproved input. No own-paper analytic conclusion is assumed.

`Checkpoint_720_SmoothCircleBourgainInput.md`

Checkpoint 720: An explicit generic smooth multifrequency input yields the countable circle maximum

Defines the exact generic discrete smooth multifrequency estimate as an explicit unproved compound external input and proves its finite-family to countable extension. The input includes separation, normalization, sampling and the fixed bump; its existence has not been formalized. It is not a Lean axiom or an assumed own-paper annular conclusion. All subsequent uses remain conditional on this predicate. The logarithmic factor is squared. Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
SmoothCircleBourgainInput.lean	2

- `circleFrequencyLogFactor_nonneg`

- `smoothCircleBourgain_countable`

<!-- checkpoint-verification -->

Verification

All 787 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4394-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1939 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 789 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The raw progression-prefix and published Ren/Ghosh estimates remain explicit unproved hypotheses. SmoothCircleBourgain-Input is a newly explicit compound unproved external analytic boundary, comprising the general smooth multifrequency estimate, sampling, normalization and fixed bump. Its existence is not proved. The Lean theorems prove implications conditional on it; the axiom audit does not certify this hypothesis. No own-paper annular lemma is assumed.

`Checkpoint_721_BourgainWeightedSourceProjection.md`

Checkpoint 721: The generic smooth input applies to the actual weighted source projection

Assuming the explicitly unproved generic smooth multifrequency input, applies it to the actual weighted circle source projection. The fixed weighted cutoff is applied to the input once. Periodic rational separation and the two large-start inequalities are proved and retained explicitly, and the bound has no coefficient cardinality loss. This is a conditional application, not a proof of the external input, the full all-scale source estimate, or Theorem 1.1.

Checked declarations

Module	Named supporting theorems
BourgainWeightedSourceProjection.lean	1

- `bourgain_weightedSource_projection`

<!-- checkpoint-verification -->

Verification

All 788 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4395-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1940 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 790 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The raw progression-prefix and published Ren/Ghosh estimates remain explicit unproved hypotheses. SmoothCircleBourgain-Input is a newly explicit compound unproved external analytic boundary, comprising the general smooth multifrequency estimate, sampling, normalization and fixed bump. Its existence is not proved. The Lean theorems prove implications conditional on it; the axiom audit does not certify this hypothesis. No own-paper annular lemma is assumed.

`Checkpoint_722_SmoothMainCommonAmplitudes.md`

Checkpoint 722: The actual smooth main block splits into two common time amplitudes

The original smooth main block is exactly the ordinary fixed-coefficient projection times its common annular amplitude minus the exceptional fixed-coefficient projection times its common annular amplitude. This exact algebraic identity permits the previously proved half-mass amplitude bound to be used without introducing a time dependent coefficient assumption.

Checked declarations

Module	Named supporting theorems
SmoothMainCommonAmplitudes.lean	1

- `rationalGaussSmoothMain_common_amplitudes`

<!-- checkpoint-verification -->

Verification

All 789 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4396-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1941 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 791 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The raw progression-prefix and published Ren/Ghosh estimates remain explicit unproved hypotheses. SmoothCircleBourgain-Input is a newly explicit compound unproved external analytic boundary, comprising the general smooth multifrequency estimate, sampling, normalization and fixed bump. Its existence is not proved. The Lean theorems prove implications conditional on it; the axiom audit does not certify this hypothesis. No own-paper annular lemma is assumed.

Checkpoint_723_ScalarMultiplierMajorants.md

Checkpoint 723: A bounded scalar time amplitude preserves an actual multiplier majorant

Proves the actual Fourier-operator majorant is stable under arbitrary bounded scalar time amplitudes. The nonnegative real l2 majorant is scaled, and its precise norm loss is proved. This elementary step consumes no new external theorem.

Checked declarations

Module	Named supporting theorems
ScalarMultiplierMajorants.lean	1

- `scalar_multiplier_majorant`

<!-- checkpoint-verification -->

Verification

All 790 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4397-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1942 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 792 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. The raw progression-prefix and published Ren/Ghosh estimates remain explicit unproved hypotheses. SmoothCircleBourgain-Input is a newly explicit compound unproved external analytic boundary, comprising the general smooth multifrequency estimate, sampling, normalization and fixed bump. Its existence is not proved. The Lean theorems prove implications conditional on it; the axiom audit does not certify this hypothesis. No own-paper annular lemma is assumed.

`Checkpoint_724_LargeStartSmoothMainMaximum.md`

Checkpoint 724: The actual smooth main block has a conditional large-start countable maximum

Derives a genuine countable large-start smooth-main maximum for the actual principal-minus-exceptional Gauss block from the generic smooth multifrequency input. The two coefficient families are fixed in time; beta may vary. Both large-start scale inequalities and coefficient envelopes remain explicit. The common amplitude contributes exactly one half. No unconditional all-scale source maximum or Theorem 1.1 is claimed.

Checked declarations

Module	Named supporting theorems
LargeStartSmoothMainMaximum.lean	1

- `largeStart_smoothMain_maximum`

<!-- checkpoint-verification -->

Verification

All 791 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4398-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1943 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 793 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The raw progression-prefix and published Ren/Ghosh estimates remain explicit unproved hypotheses. SmoothCircleBourgain-Input is a newly explicit compound unproved external analytic boundary, comprising the general smooth multifrequency estimate, sampling, normalization and fixed bump. Its existence is not proved. The Lean theorems prove implications conditional on it; the axiom audit does not certify this hypothesis. No own-paper annular lemma is assumed.

`Checkpoint_725_LargeStartActualSourceMaximum.md`

Checkpoint 725: The original Gauss source block has a conditional large-start countable maximum

Combines the conditional smooth-main maximum with the fully proved annular projection-error maximum for the original source Gauss model. The resulting countable real-l2 majorant has bound (A0+A1)*(C*logFactor/2+E). Every support, scale, fixed-character, coefficient and beta hypothesis is displayed. This milestone is conditional on the generic external smooth multifrequency predicate; arbitrary starting scales, denominator/conductor summation, Proposition 6.21, Proposition 4.19 and Theorem 1.1 remain open.

Checked declarations

Module	Named supporting theorems
LargeStartActualSourceMaximum.lean	1

- `largeStart_actualSource_maximum`

<!-- checkpoint-verification -->

Verification

All 792 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4399-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1944 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 794 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The raw progression-prefix and published Ren/Ghosh estimates remain explicit unproved hypotheses. SmoothCircleBourgainInput is a newly explicit compound unproved external analytic boundary, comprising the general smooth multifrequency estimate, sampling, normalization and fixed bump. Its existence is not proved. The Lean theorems prove implications conditional on it; the axiom audit does not certify this hypothesis. No own-paper annular lemma is assumed.

`Checkpoint_726_WeightedAnnularLargeStartMaximum.md`

Checkpoint 726: A general weighted annular block has a conditional large-start countable maximum

Defines the actual weighted bare annular source block with fixed rational centers and coefficients, proves its exact smooth-plus-error decomposition, and derives a genuine countable real-l2 majorant conditional on SmoothCircleBourgainInput C. The bound is A*(C*circleFrequencyLogFactor/2+E) times the input norm. Beta may vary with time in [0,1]. Both large-start inequalities and s>=2 remain explicit. No new external hypothesis is introduced. Arbitrary starting scales, small source scales, denominator/conductor summation and the printed stronger multifrequency claims remain open. The paper-level checklist is saved and its 54-claim coverage is refreshed with every subsequent checkpoint; this supporting unit closes no new whole paper statement.

Checked declarations

Module	Named supporting theorems
WeightedAnnularLargeStartMaximum.lean	2

- `weightedAnnularSourceBlock_decomposition`
- `largeStart_weightedAnnular_maximum`

<!-- checkpoint-verification -->

Verification

All 793 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4400-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1946 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 795 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. The raw progression-prefix and published Ren/Ghosh estimates remain explicit unproved hypotheses. SmoothCircleBourgain-Input is a newly explicit compound unproved external analytic boundary, comprising the general smooth multifrequency estimate, sampling, normalization and fixed bump. Its existence is not proved. The Lean theorems prove implications conditional on it; the axiom audit does not certify this hypothesis. No own-paper annular lemma is assumed.

`Checkpoint_727_PrimeFactorThresholdBudget.md`

Checkpoint 727: A common prime-factor threshold controls every conductor product

Proves an arbitrary fixed-threshold reciprocal prime-factor budget and the corresponding bound for the logarithm of the actual conductorRho product, uniformly for every positive modulus including small moduli. The resulting exponential envelope is centered at a common threshold y and has the explicit slope gaussThresholdSlope y. These elementary statements use no PNT input; PNT will later identify the sharp leading constant of the envelope at y=log Q. This is the first arithmetic step on the fixed-scale route to Theorem1.1, which does not require the full countable high maximum. No whole source statement is newly closed.

Checked declarations

Module	Named supporting theorems
PrimeFactorThresholdBudget.lean	5

- `primeFactors_reciprocal_threshold`
- `gaussThresholdSlope_nonneg`
- `log_conductorRho_le`
- `conductorRho_log_threshold`
- `conductorRho_exp_threshold`

<!-- checkpoint-verification -->

Verification

All 794 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4401-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1951 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 796 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The raw progression-prefix and published Ren/Ghosh estimates remain explicit unproved hypotheses. SmoothCircleBourgain-Input is a newly explicit compound unproved external analytic boundary, comprising the general smooth multifrequency estimate, sampling, normalization and fixed bump. Its existence is not proved. The Lean theorems prove implications conditional on it; the axiom audit does not certify this hypothesis. No own-paper annular lemma is assumed.

`Checkpoint_728_ThresholdEnvelopeBounds.md`

Checkpoint 728: Sharp threshold envelopes on both sides of the denominator cutoff

Proves that the common-threshold slope tends to zero and is eventually at most one quarter. Under the existing two PNT inputs, the threshold exponent divided by the source regularized Orlicz weight tends to2, with its full epsilon consequence. The actual conductorRho product is bounded uniformly below the threshold and with a quarter-power relative loss above it. These are the arithmetic envelopes required for the fixed-scale high bound; no source high estimate or Theorem1.1 conclusion is assumed.

Checked declarations

Module	Named supporting theorems
ThresholdEnvelopeBounds.lean	6

- `gaussThresholdSlope_tendsto_zero`
- `gaussThresholdSlope_eventually_quarter`
- `gaussAnalyticMajorant_regularized_ratio_tendsto`
- `gaussAnalyticMajorant_regularized_eventually`
- `conductorRho_threshold_small`
- `conductorRho_threshold_large`

<!-- checkpoint-verification -->

Verification

All 795 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4402-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1957 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 797 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The raw progression-prefix and published Ren/Ghosh estimates remain explicit unproved hypotheses. SmoothCircleBourgainInput is a newly explicit compound unproved external analytic boundary, comprising the general smooth multifrequency estimate, sampling, normalization and fixed bump. Its existence is not proved. The Lean theorems prove implications conditional on it; the axiom audit does not certify this hypothesis. No own-paper annular lemma is assumed.

`Checkpoint_729_RationalRelativeCutoffSum.md`

Checkpoint 729: Uniform rational sums with two relative denominator cutoffs

Proves a denominator-relative cutoff profile and bounds a genuine finite rational-frequency sum by its dyadic profile, including all small dyadic scales. Reuses the already proved uniform interpolation-weight sum to obtain a two-cutoff bound independent of the ordinary and exceptional positive cutoff locations. This provides the geometric summation needed for the sharp fixed-scale high symbol; no multifrequency maximal input, PNT, or own-paper high estimate is assumed.

Checked declarations

Module	Named supporting theorems
RationalRelativeCutoffSum.lean	4

- `relativeCutoffProfile_nonneg`
- `relativeCutoffProfile_dyadic`
- `rational_dyadic_profile_sum`
- `rational_two_relative_cutoff_sum`

<!-- checkpoint-verification -->

Verification

All 796 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4403-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1961 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 798 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The raw progression-prefix and published Ren/Ghosh estimates remain explicit unproved hypotheses. SmoothCircleBourgain-Input is a newly explicit compound unproved external analytic boundary, comprising the general smooth multifrequency estimate, sampling, normalization and fixed bump. Its existence is not proved. The Lean theorems prove implications conditional on it; the axiom audit does not certify this hypothesis. No own-paper annular lemma is assumed.

`Checkpoint_730_ThresholdMaskedCoefficientBounds.md`

Checkpoint 730: Threshold bounds for masked coefficients

Proves the threshold estimate for an actual complex coefficient multiplied by the annular high mask. The bound uses the geometric relative-cutoff profile with exponent one quarter, retaining the sharp Gauss envelope and the square root of the conductor scale. Includes both sides of the cutoff and the exact inverse-square-root rescaling identity. This is a supporting estimate; no main paper theorem is declared completed. No deep external input is used in this checkpoint.

Checked declarations

Module	Named supporting theorems
ThresholdMaskedCoefficientBounds.lean	5

- `relativeCutoffProfile_small`
- `relativeCutoffProfile_large`
- `inverse_sqrt_relative`
- `conductorRho_scaled_threshold_large`
- `threshold_masked_product`

<!-- checkpoint-verification -->

Verification

All 797 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4404-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1966 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 799 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The raw progression-prefix and published Ren/Ghosh estimates remain explicit unproved hypotheses. SmoothCircleBourgain-Input is a newly explicit compound unproved external analytic boundary, comprising the general smooth multifrequency estimate, sampling, normalization and fixed bump. Its existence is not proved. The Lean theorems prove implications conditional on it; the axiom audit does not certify this hypothesis. No own-paper annular lemma is assumed.

`Checkpoint_731_InducedRationalCoefficientBudget.md`

Checkpoint 731: Conductor quotient coefficient bounds

Rewrites the proved induced-character Gauss bound in terms of the integer quotient of the denominator by the conductor, and specializes it to the actual source-induced rational character. Proves a uniform bound for conductorSigma times the square root of conductorMajorant using the already proved conductor decay. All are supporting unconditional coefficient estimates; Theorem 1.1 remains open.

Checked declarations

Module	Named supporting theorems
InducedRationalCoefficientBudget.lean	3

- `induced_twistedSquareSum_div_bound`
- `sourceInduced_rationalGauss_div_bound`
- `conductorSigma_majorant_uniform`

<!-- checkpoint-verification -->

Verification

All 798 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4405-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1969 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 800 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The raw progression-prefix and published Ren/Ghosh estimates remain explicit unproved hypotheses. SmoothCircleBourgainInput is a newly explicit compound unproved external analytic boundary, comprising the general smooth multifrequency estimate, sampling, normalization and fixed bump. Its existence is not proved. The Lean theorems prove implications conditional on it; the axiom audit does not certify this hypothesis. No own-paper annular lemma is assumed.

`Checkpoint_732_CenteredHighThresholdProfile.md`

Checkpoint 732: Actual centered high symbol at the two cutoffs

Proves the sharp threshold profile for the actual centered high symbol, with the source-induced quadratic character and the actual exceptional selector. The ordinary cutoff Q and exceptional cutoff Q*q0/conductorMajorant(q0) are retained separately. The induced estimate uses the quotient denominator and the checked conductorSigma square-root budget; the inactive and nondivisible cases are proved explicitly. No PNT or additional external theorem is used. This is a supporting high-symbol estimate, not yet a completed Theorem 1.1.

Checked declarations

Module	Named supporting theorems
CenteredHighThresholdProfile.lean	2

- `principal_rationalGauss_rho_bound`
- `centeredHighSymbol_threshold_profile`

<!-- checkpoint-verification -->

Verification

All 799 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4406-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1971 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 801 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains

two earlier explicit PNT limits; its external bridge is not integrated or used here. The raw progression-prefix and published Ren/Ghosh estimates remain explicit unproved hypotheses. SmoothCircleBourgain-Input is a newly explicit compound unproved external analytic boundary, comprising the general smooth multifrequency estimate, sampling, normalization and fixed bump. Its existence is not proved. The Lean theorems prove implications conditional on it; the axiom audit does not certify this hypothesis. No own-paper annular lemma is assumed.

`Checkpoint_733_SourceHighThresholdUniform.md`

Checkpoint 733: Uniform sharp-envelope fixed-scale high operator

Combines the checked ordinary and induced-character profiles with rational-frequency sparsity, uniform two-cutoff geometric summation, and the exact three-term periodization formula. Proves a uniform bound for the actual circle high model and its actual l2 operator, with Q^{-1/2} times the explicit Gauss majorant. The constant is uniform in the arithmetic scale, conductor, quadratic character, selector, cutoff and canonical finite rational set. The bound is unconditional when the explicit threshold-slope conditions hold. Theorem 1.1 remains the assembly target.

Checked declarations

Module	Named supporting theorems
SourceHighThresholdUniform.lean	1

- `sourceHigh_threshold_uniform`

<!-- checkpoint-verification -->

Verification

All 800 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4407-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1972 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 802 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The raw progression-prefix and published Ren/Ghosh estimates remain explicit unproved hypotheses. SmoothCircleBourgain-Input is a newly explicit compound unproved external analytic boundary, comprising the general smooth multifrequency estimate, sampling, normalization and fixed bump. Its existence is not proved. The Lean theorems prove implications conditional on it; the axiom audit does not certify this hypothesis. No own-paper annular lemma is assumed.

`Checkpoint_734_SourceHighOrliczUniform.md`

Checkpoint 734: Sharp fixed-scale high bound with every coefficient above two

Proves the actual fixed-scale high-model and high-operator bounds with Q^{-1/2} times exp(c1*sqrt(log Q)/log(log Q)) for every c1>2, under the existing precise PNT limits. The estimate is uniform at every dyadic cutoff Q>=2. The cutoff Q=2 is treated separately, since the unregularized log-log denominator prevents a uniform compact comparison across Q=e. Larger cutoffs use the checked common-threshold envelope, with the weaker unconditional high bound absorbing the finite range. No maximal estimate is claimed. This closes the sharp fixed-scale high bound needed for Theorem 1.1, while its final operator and interval assembly remains open.

Checked declarations

Module	Named supporting theorems
SourceHighOrliczUniform.lean	4

- `log_four_one_le`
- `orliczHighFactor_one_le`
- `gaussAnalyticMajorant_unregularized_eventually`
- `sourceHigh_orlicz_uniform`

<!-- checkpoint-verification -->

Verification

All 801 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4408-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1976 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 803 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. The raw progression-prefix and published Ren/Ghosh estimates remain explicit unproved hypotheses. SmoothCircleBourgain-Input is a newly explicit compound unproved external analytic boundary, comprising the general smooth multifrequency estimate, sampling, normalization and fixed bump. Its existence is not proved. The Lean theorems prove implications conditional on it; the axiom audit does not certify this hypothesis. No own-paper annular lemma is assumed.

`Checkpoint_735_IndependentSquareDecomposition.md`

Checkpoint 735: Independent square input and checked full-to-annular transfer

Introduces a precise, explicitly unproved external interface for the independent full unweighted-square decomposition. Its global norm form is a compound consequence of Han–Lacey–Yang's published Fourier construction and local estimate, including reflection; it is not claimed to have been imported from Mathlib. Proves the actual full-to-annular identity on finite indicators and deduces the annular high/low operator bounds at even cutoffs, with the spatial-scale and one-half factors checked in Lean. No prime-square conclusion is an assumption of this interface. Theorem 1.1 remains open pending assembly.

Checked declarations

Module	Named supporting theorems
IndependentSquareDecomposition.lean	2

- `annular_indicator_two_mul`
- `fullSquare_decomposition_annular_two_mul`

<!-- checkpoint-verification -->

Verification

All 802 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4409-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1978 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 804 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma

3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. The PNT limits, raw compatible progression/Ren/Ghosh estimates, and independent unweighted-square decomposition remain explicit hypotheses. The smooth multifrequency hypothesis is confined to the separate maximal development.

Checkpoint_736_PrimeSquareDistributionInputs.md

Checkpoint 736: Actual prime-square three-piece decomposition and raw input boundary

Collects exact compatible raw progression/Ren/Ghosh data in an explicit external input structure, with no own-paper conclusions assumed. Combines the checked actual circle low operator, sharp high operator and actual error operator to prove the three-piece OperatorOrliczBound for the genuine annular von-Mangoldt square kernel, for all sufficiently large arithmetic scales. The complete dyadic cutoff range and the change from arithmetic N to spatial N^2 are checked. The existing Lean range is (min(c/7,1))/8; this stronger, correctly parenthesized range is retained. The uniform low constant, high constant with every c1>2, and error constant eight are proved. Theorem 1.1's endpoint and reconstruction remain open.

Checked declarations

Module	Named supporting theorems
PrimeSquareDistributionInputs.lean	4

- `distribution_cutoff_constant_pos`
- `square_log_sqrt_le_double`
- `source_low_amplitude_orlicz`
- `primeSquare_orlicz_decomposition_large`

<!-- checkpoint-verification -->

Verification

All 803 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4410-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1982 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 805 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_737_PrimeSquareAnnularEndpoint.md

Checkpoint 737: Sharp endpoint for the actual annular prime-square operator

Checks the exact leading coefficient c1*(2*sqrt(3)/9) and chooses every high coefficient c1>2 below a requested endpoint coefficient cstar>4*sqrt(3)/9. Instantiates the completed abstract Theorem 4.1 with the genuine annular von-Mangoldt kernel and independent ordinary-square input. This proves the sharp restricted endpoint for all sufficiently large dyadic arithmetic scales, with all constants quantified uniformly before the scale and finite sets. The result is conditional only on the explicit PNT, raw distribution and independent unweighted-square inputs. Full averages and the second main conclusion are not yet declared completed.

Checked declarations

Module	Named supporting theorems
PrimeSquareAnnularEndpoint.lean	3

- `primeSquare_leading_coefficient`
- `exists_primeSquare_high_coefficient`
- `primeSquare_annular_endpoint_large`

<!-- checkpoint-verification -->

Verification

All 804 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4411-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1985 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 806 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_738_EndpointFiniteScaleBounds.md`

Checkpoint 738: Elementary completion of the finite arithmetic scales

Proves a uniform elementary point bound for the actual annular von-Mangoldt square kernel, using injectivity of natural-number squares and Lambda(n)<=log N<=N. Deduces a finite-scale restricted endpoint bound with an explicit T^2 constant and the exact paper Phi. This handles all arithmetic scales below a fixed threshold, including empty input sets, without placing a finite-scale decomposition among the external assumptions. The large-scale sharp endpoint was checked separately; the full-average reconstruction remains to be assembled.

Checked declarations

Module	Named supporting theorems
EndpointFiniteScaleBounds.lean	3

- `mangoldtSquareKernel_point_bound`
- `restricted_pairing_card_bound`
- `annular_endpoint_finite_scale`

<!-- checkpoint-verification -->

Verification

All 805 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4412-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1988 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 807 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data,

and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_739_PrimeSquareDyadicAnnularEndpoint.md`

Checkpoint 739: All dyadic annular scales and monotonicity of the exact Orlicz loss

Combines the checked large-scale annular endpoint with the elementary finite-scale bound to prove the sharp endpoint for every dyadic arithmetic cutoff, uniformly over finite input and test sets with density at most one. Also proves monotonicity of sqrt(L)/(2+log L) for L>=2 and the corresponding density monotonicity of the exact paper Phi. These facts allow summation over annuli without changing the endpoint coefficient. The two final Theorem 1.1 conclusions still require the full-average and arbitrary-cutoff transfers and strong interpolation.

Checked declarations

Module	Named supporting theorems
PrimeSquareDyadicAnnularEndpoint.lean	3

- `orliczWeight_half_mono`
- `paperPhi_half_antitone`
- `primeSquare_annular_endpoint_dyadic`

<!-- checkpoint-verification -->

Verification

All 806 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4413-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1991 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 808 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_740_OrliczScaleComparison.md`

Checkpoint 740: Exact Orlicz loss under bounded changes of density

Proves the bounded-additive-shift estimate for the exact square-root Orlicz weight and the resulting multiplicative comparison for Phi when two densities differ by at most a fixed factor. The endpoint coefficient is preserved exactly; the fixed factor only changes the outside constant. This is needed for the dyadic-to-arbitrary arithmetic cutoff and source interval normalization in Theorem 1.1. No new analytic input is used.

Checked declarations

Module	Named supporting theorems
OrliczScaleComparison.lean	2

- `orliczWeight_half_add_le`
- `paperPhi_half_fixed_rescale`

<!-- checkpoint-verification -->

Verification

All 807 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4414-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1993 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 809 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_741_DyadicEndpointReconstruction.md`

Checkpoint 741: Exact full-average pairing and endpoint rescaling

Proves exact reconstruction of the genuine full von-Mangoldt kernel pairing from descending dyadic annuli, using the already checked identity of actual continuous operators. Proves the exact change of finite-set density between two square spatial scales and the two-thirds endpoint scaling identity, including all normalization factors. These are equalities, not numerical checks or new analytic assumptions, and are used in summing the annular endpoint to the full average.

Checked declarations

Module	Named supporting theorems
DyadicEndpointReconstruction.lean	3

- `mangoldtPairing_dyadic_reconstruction`
- `square_density_ratio`
- `square_endpoint_rescale`

<!-- checkpoint-verification -->

Verification

All 808 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4415-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 1996 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 810 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_742_PrimeSquareFullDyadicEndpoint.md`

Checkpoint 742: Sharp full-average endpoint at every dyadic cutoff

Proves the sharp restricted endpoint for the genuine FULL von-Mangoldt square average at every dyadic arithmetic cutoff. The proof uses the exact descending-annulus pairing identity, extends each annular bound to density above one with the actual Chebyshev l2 estimate, preserves the paper Phi by density

monotonicity, and sums the resulting one-third-power geometric weights. Every rescaling factor and the uniform geometric constant are checked in Lean. This completes the full dyadic endpoint, conditional on the already recorded independent deep inputs; arbitrary arithmetic cutoffs, interval normalization and strong interpolation remain.

Checked declarations

Module	Named supporting theorems
PrimeSquareFullDyadicEndpoint.lean	4

- `finiteSetDensity_scale_antitone`
- `primeSquare_annular_comparison`
- `dyadic_descending_ratio`
- `primeSquare_full_endpoint_dyadic`

<!-- checkpoint-verification -->

Verification

All 809 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4416-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2000 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 811 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_743_PrimeSquareCutoffComparison.md`

Checkpoint 743: Positive full-average comparison and a dyadic upper cutoff

Proves positivity-based comparison of the genuine full von-Mangoldt kernel and restricted pairing at two arbitrary arithmetic cutoffs N<=M, retaining the exact M/N normalization. Also constructs a natural dyadic upper cutoff between N and 2N. These elementary deductions connect the checked dyadic full-average endpoint to arbitrary positive N without new analytic assumptions.

Checked declarations

Module	Named supporting theorems
PrimeSquareCutoffComparison.lean	3

- `mangoldtSquareKernel_cutoff_bound`
- `mangoldtSquarePairing_cutoff_bound`
- `exists_dyadic_upper_nat`

<!-- checkpoint-verification -->

Verification

All 810 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4417-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2003 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 812 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_744_PrimeSquareFullEndpoint.md`

Checkpoint 744: Sharp full-average endpoint for every positive arithmetic cutoff

Proves the sharp restricted endpoint for the actual full von-Mangoldt square average at EVERY positive integer cutoff N, uniformly over all finite sets with density at most one. The proof chooses an actual dyadic upper cutoff in [N,2N], uses the checked positivity comparison, bounds the density change by sixteen, and applies the checked fixed-factor comparison for the exact Phi. The outside constant changes, while every requested coefficient cstar>4*sqrt(3)/9 is preserved. The exact source interval normalization and strong conclusion are the remaining assembly steps.

Checked declarations

Module	Named supporting theorems
PrimeSquareFullEndpoint.lean	2

- `square_density_bounded_ratio`
- `primeSquare_full_endpoint`

<!-- checkpoint-verification -->

Verification

All 811 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4418-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2005 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 813 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_745_PrimeSquareSourceInterval.md`

Checkpoint 745: Theorem 1.1 restricted endpoint with the exact source interval

Proves the exact source interval I=[-N^2,2N^2] intersect Z has cardinality 3N^2+1, and checks its comparison with the spatial scale N^2. Exports theorem_1_1_restricted: the FIRST conclusion of Theorem 1.1 for the genuine full von-Mangoldt average, every positive N, all F,G contained in the exact interval, and every coefficient cstar>4*sqrt(3)/9. The proof also handles source densities whose N^2-normalized version exceeds one, using the actual l2 estimate. Constants and both normalizations are checked. The strong conclusion is still being assembled; Theorem 1.1 is not yet marked fully complete.

Checked declarations

Module	Named supporting theorems
PrimeSquareSourceInterval.lean	5

- `primeSquareSourceInterval_card`
- `primeSquareSourceInterval_card_bounds`
- `finiteSetDensity_scale_ratio`
- `primeSquare_full_comparison`
- `theorem_1_1_restricted`

<!-- checkpoint-verification -->

Verification

All 812 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4419-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2010 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 814 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_746_PrimeSquareStrongImproving.md`

Checkpoint 746: Strong prime-square improving by checked interpolation

Proves a uniform restricted power estimate for every 3/2<q<=2 by absorbing the checked endpoint Phi into an arbitrarily small density power. Uses the actual l2 bound to cover all finite sets and instantiates the checked strong interpolation theorem. The result is a genuine lp-to-lq continuous kernel realization with the sharp spatial scaling, for the entire range 3/2<p<=2; the p=2 case is handled by the actual Chebyshev l2 operator. No strong estimate is assumed as an external input. Identification with the literal finite weighted average and final source normalization follow next.

Checked declarations

Module	Named supporting theorems
PrimeSquareStrongImproving.lean	3

- `primeSquare_fixed_coefficient`
- `primeSquare_restricted_power_uniform`
- `primeSquare_strong_kernel_uniform`

<!-- checkpoint-verification -->

Verification

All 813 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4420-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2013 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 815 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency

boundary is only for later maximal applications.

`Checkpoint_747_PrimeSquareActualStrongOperator.md`

Checkpoint 747: Theorem 1.1 strong conclusion for the actual averaging operator

Identifies the interpolated kernel realization with the genuine finite von-Mangoldt square average by uniqueness on finite indicators. Proves the sharp scaling for this ACTUAL lp-to-lq operator, and exports theorem_1_1_strong with the exact source interval cardinality and normalization, including p=2. The inequality holds for every lp input, so it includes all source-supported inputs. Both mathematical conclusions now have checked declarations (theorem_1_1_restricted and theorem_1_1_strong); the final combined statement, literal real-exponent/sum form and paper-level completion review are being prepared before awarding the Theorem 1.1 milestone. The explicit independent deep hypotheses remain unchanged.

Checked declarations

Module	Named supporting theorems
PrimeSquareActualStrongOperator.lean	4

- `primeSquareImprovingOperator_apply`
- `primeSquare_actual_strong_uniform`
- `improving_scale_comparison`
- `theorem_1_1_strong`

<!-- checkpoint-verification -->

Verification

All 814 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4421-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2017 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 816 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. The abstract results retain their source hypotheses. Full Lemma 6.5, Proposition 4.19 and headline applications remain open. Lemma 3.21 retains two earlier explicit PNT limits; its external bridge is not integrated or used here. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_748_TheoremOneOne.md`

Checkpoint 748: Theorem 1.1: both conclusions and exact real-exponent statement

Completes the combined export PrimeSquares.theorem_1_1 and the literal real-exponent/sum-of-powers form of the second conclusion, including actual output membership in l^{p'}. Both statements are for the genuine full von-Mangoldt square average, all positive integer arithmetic cutoffs, exact source interval cardinality, every restricted coefficient cstar>4*sqrt(3)/9, and every real 3/2<p<=2. The strong estimate is proved for all lp inputs, hence includes the manuscript's supported inputs. The source-to-Lean quantifier review awards Theorem1.1 status CONDITIONAL: a complete implication under the explicit PNT limits, raw compatible progression/Ren/Ghosh data existence, and the independent ordinary-square decomposition plus its globalization/reflection. None is a new Lean axiom or an assumed prime-square conclusion. The separate maximal/sparse and later paper results remain unfinished. Canonical paper coverage and remaining-work reports are updated with this reviewed milestone.

Checked declarations

Module	Named supporting theorems
TheoremOneOne.lean	4

- `normalized_lp_norm_as_sum`
- `theorem_1_1_strong_sum`
- `theorem_1_1_strong_real`
- `theorem_1_1`

<!-- checkpoint-verification -->

Verification

All 815 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4422-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2021 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 817 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_749_InitialScaleMaximalCompletion.md`

Checkpoint 749: Complete a maximal estimate across finitely many initial scales

Proves the operator-level extension from a dominated infinite tail to all natural time indices, with a finite initial-scale cost and an actual nonnegative real-l2 majorant. This is the precise completion step needed to remove the starting-scale restrictions in the existing annular multifrequency bound. It introduces no deep hypothesis. Checkpoint748 and both Theorem1.1 conclusions are recovered and retained unchanged; the full project is freshly rebuilt after workspace maintenance. Proposition6.21 remains open until the actual annular family, truncations, sharp arithmetic and exceptional-character assembly are proved.

Checked declarations

Module	Named supporting theorems
InitialScaleMaximalCompletion.lean	1

- `initialScale_maximal_completion`

<!-- checkpoint-verification -->

Verification

All 816 project modules and the root were freshly rebuilt: 4423 jobs pass without warnings/errors. All 2022 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 818 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT

limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_750_WeightedAnnularBlockBounds.md`

Checkpoint 750: Bound each actual annular block uniformly at the initial scales

The actual annular source on the circle has a uniform pointwise bound. Its support lies inside the source cutoff, and distinct canonical centers in one sufficiently separated denominator block give disjoint supports. Consequently a fixed weighted annular block, with coefficients bounded by A, has multiplier operator norm at most (3/2)A, uniformly in the starting scale. These are bounds for the actual periodized oscillatory-integral symbols. They supply the finite initial-scale part of the maximal argument; they do not assert Proposition 6.21.

Checked declarations

Module	Named supporting theorems
WeightedAnnularBlockBounds.lean	4

- `circleAnnularSource_norm_bound`
- `circleAnnularSource_zero_of_cutoff`
- `weightedAnnularSourceBlock_norm_bound`
- `weightedAnnularSourceBlock_operator_bound`

<!-- checkpoint-verification -->

Verification

All 817 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4424-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2026 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 819 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_751_WeightedAnnularAllStartingScales.md`

Checkpoint 751: Complete the weighted annular maximum from every starting scale

The weighted annular-source maximum now holds from every starting scale M >= 2, under the existing general smooth multifrequency input. The proof chooses an explicit later starting scale 2^(2s+3) M at which both separation budgets hold, applies the previously checked infinite-tail result there, and adds the finitely many earlier operators using their actual uniform norm bound. The resulting loss is A times [(3/2)(2s+3)+(C/2)(1+log(card Gamma+1))^2+E]. No maximum, initial segment, or scale range is assumed as an extra input. The fixed polynomial logarithmic loss is compatible with the o(1) in Proposition 6.21; the stronger printed auxiliary logarithmic exponent is not claimed.

Checked declarations

Module	Named supporting theorems
WeightedAnnularAllStartingScales.lean	2

- `annularLargeStart_budgets`

- `weightedAnnular_allStartingScales`

<!-- checkpoint-verification -->

Verification

All 818 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4425-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2028 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 820 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_752_NestedSourceLocalization.md`

Checkpoint 752: Localize actual annular sources with disjoint wider cutoffs

The actual circle source is localized by a wider cutoff: the wider cutoff equals one everywhere that the original cutoff is nonzero. The proof passes through the unique real lift, with the actual smooth bump's core radius 1/2 and support radius 1. A general denominator-separation statement proves disjointness of the wider cutoffs. For denominator blocks s >= 4, the cutoff at s-1 is both sufficiently wide to localize the source at s and sufficiently narrow to remain disjoint between distinct canonical centers. These facts will let the time-and-denominator maximum use orthogonal filtered inputs rather than pay for the number of denominators.

Checked declarations

Module	Named supporting theorems
NestedSourceLocalization.lean	4

- `circleSourceCutoff_one_on_inner`
- `circleAnnularSource_localized`
- `circleSourceCutoff_general_disjoint`
- `predecessorSourceCutoff_geometry`

<!-- checkpoint-verification -->

Verification

All 819 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4426-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2032 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 821 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain

explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_753_FiniteFrequencyTruncationMaximum.md

Checkpoint 753: Control time and denominator truncations using localized input energy

A general finite-frequency theorem now controls the maximum jointly over time and every real denominator cutoff. Its hypotheses are localization by pairwise disjoint cutoff symbols of norm at most one and a hereditary time-maximal estimate on each frequency subset. The proof groups the frequencies by denominator, constructs each interval majorant on the corresponding filtered input, applies the already proved time-dependent Rademacher-Menshov inequality, and uses multiplier Bessel orthogonality to recover the original input norm. The final loss is h+1 when the maximum denominator plus one is at most 2^h. In particular, it does not introduce a square-root loss in the number of denominators. This is a proved abstract transfer theorem, with the actual source-family instantiation still to follow.

Checked declarations

Module	Named supporting theorems
FiniteFrequencyTruncationMaximum.lean	5

- `rationalCircle_interval_grouping`
- `finite_localized_symbol_sum`
- `rationalCircle_grouped_disjoint`
- `rationalCircle_grouped_norm`
- `finiteFrequency_denominator_maximum`

<!-- checkpoint-verification -->

Verification

All 820 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4427-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2037 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 822 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_754_AnnularBlockLogarithmicBudget.md

Checkpoint 754: Bound the actual frequency and truncation losses by a cubic logarithm

The frequency-cardinality loss is now bounded by the denominator-block index: (1+log(card Gamma+1))^2 <= 36(s+1)^2 for canonical centers of denominator at most 2^(s+1). The proof uses the actual finite enumeration bound, not an asymptotic cardinality assumption. Combining this with the finite initial-scale loss and the denominator-truncation factor gives an explicit constant K(C)=3(8+18C+E) multiplying A(s+1)^3. This fixed cubic logarithmic cost is harmless in the final sharp-exponent asymptotic, but no sharper auxiliary logarithmic exponent is asserted.

Checked declarations

Module	Named supporting theorems
AnnularBlockLogarithmicBudget.lean	4

- `circleFrequencyLogFactor_mono`
- `circleFrequencyLogFactor_denominator_budget`
- `annularTruncationConstant_nonneg`
- `annular_allStarting_budget`

<!-- checkpoint-verification -->

Verification

All 821 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4428-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2041 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 823 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_755_WeightedAnnularTruncationMaximum.md`

Checkpoint 755: Complete the actual annular maximum with changing denominator cutoffs

The actual weighted annular-source block now has a proved joint maximum over every dyadic scale from M >= 2 and every real upper denominator cutoff. For s >= 4 and coefficients bounded by A, the majorant has l2 norm at most K(C) A(s+1)^3 times the input norm. Its only analytic boundary is the previously explicit general smooth multifrequency input. The proof constructs disjoint wider source cutoffs, localizes the actual symbols, derives the hereditary time estimate from the all-starting-scale theorem, and applies the checked denominator-truncation theorem and cardinality budget. This supplies the formerly missing actual-source interval estimate without assuming a maximal statement from this paper.

Checked declarations

Module	Named supporting theorems
WeightedAnnularTruncationMaximum.lean	1

- `weightedAnnular_truncation_maximum`

<!-- checkpoint-verification -->

Verification

All 822 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4429-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2042 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 824 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT

limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_756_PolynomialDyadicProfileSum.md

Checkpoint 756: Sum polynomially weighted sharp cutoff profiles uniformly

For every positive decay exponent and every fixed polynomial degree, the two-sided dyadic cutoff profile is summable even after multiplication by (s+1)^m. Its total is bounded uniformly by an explicit constant times (1+log W)^m for W >= 1. The proof splits at a genuine dyadic scale adjacent to W, uses the previously verified uniform profile sum for the lower part, and sums a polynomial times a geometric sequence for the upper part. This provides the sharp arithmetic summation needed after the cubic logarithmic block loss, without weakening the Q^(-1/2) leading power.

Checked declarations

Module	Named supporting theorems
PolynomialDyadicProfileSum.lean	4

- `shifted_polynomial_geometric_summable`
- `polynomialDyadicProfileConstant_nonneg`
- `polynomial_dyadic_profile_summable`
- `polynomial_dyadic_profile_tsum`

<!-- checkpoint-verification -->

Verification

All 823 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4430-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2046 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 825 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_757_ThresholdCoefficientAmplitudes.md

Checkpoint 757: Extract sharp conductor-sensitive amplitudes on both sides of the cutoff

The common-threshold Gauss-product envelope now gives separate, exact coefficient bounds on both sides of an effective cutoff. After extracting A sqrt(B) Q^(-1/2) exp(G(log Q)), the small-denominator coefficient is bounded by t^(-1/2), and the large-denominator coefficient by t^(-1/4). These imply uniform amplitudes on each dyadic denominator block. Two elementary profile inequalities convert the actual collar gain and the large-block maximum into the same two-sided profile of exponent 1/4. The argument preserves the conductor factor rather than replacing it by a worse denominator estimate.

Checked declarations

Module	Named supporting theorems
ThresholdCoefficientAmplitudes.lean	7

- `sharpThresholdAmplitude_nonneg`

- `threshold_coefficient_small`
- `threshold_coefficient_large`
- `threshold_small_block_amplitude`
- `threshold_large_block_amplitude`
- `large_amplitude_profile`
- `small_amplitude_profile`

<!-- checkpoint-verification -->

Verification

All 824 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4431-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2053 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 826 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_758_ThresholdAnnularSource.md`

Checkpoint 758: Identify the actual high operator with the common threshold-annular family

The ordinary and exceptional pieces of the actual high symbol are identified exactly with a common threshold-annular family. Below the effective cutoff this family is the actual narrow-cutoff collar; above it the family is the actual broad annular source. This identity is proved for each source term and for the full finite high model, with the exceptional effective threshold Q q_chi/B_chi. A separate exact identity writes an arbitrary varying upper denominator cutoff in a collar block as a time-dependent coefficient bounded by the original coefficient. No maximal estimate is built into these definitions.

Checked declarations

Module	Named supporting theorems
ThresholdAnnularSource.lean	5

- `circleHighTerm_threshold_ordinary`
- `circleExceptionalHigh_threshold`
- `circleHighModel_threshold_decomposition`
- `thresholdAnnularBlock_split`
- `rationalCollarBlock_truncation`

<!-- checkpoint-verification -->

Verification

All 825 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4432-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2058 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 827 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition,

Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_759_SharpAnnularBlockMaximum.md

Checkpoint 759: Prove sharp maximal bounds for both actual high-block pieces

Both actual pieces of a dyadic high block now have sharp maximal majorants. The collar proof allows arbitrary changing upper denominator cutoffs by masking its coefficients and uses the already proved square-function estimate. The large-denominator proof uses the actual joint scale-and-cutoff maximum, not a scalar symbol norm. After extracting the common amplitude A sqrt(B) Q^(-1/2) exp(G(log Q)), both bounds have the same two-sided relative-cutoff profile; the large piece also has the verified cubic logarithmic factor. An effective cutoff at least 16 ensures that every nonempty large block satisfies the separation index required by the source theorem. Empty blocks and all smaller collar blocks are explicitly handled.

Checked declarations

Module	Named supporting theorems
SharpAnnularBlockMaximum.lean	2

- `sharpCollar_block_maximum`
- `sharpLarge_block_maximum`

<!-- checkpoint-verification -->

Verification

All 826 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4433-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2060 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 828 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_760_SharpThresholdFiniteMaximum.md

Checkpoint 760: Assemble and sum the full sharp high maximum over finite center sets

The collar and large-denominator majorants are now assembled for the exact threshold-annular high family. Summing over every dyadic denominator block gives a uniform maximal bound K(C) A sqrt(B) Q^(-1/2) exp(G(log Q)) (1+log W)^3. This theorem allows any finite canonical center set, has a constant independent of that set's maximum denominator, starts at every M >= 2, and allows an arbitrary nonnegative sequence of real denominator cutoffs. The proof uses actual continuous multipliers and lp majorants throughout. No fixed-scale symbol estimate is substituted for a maximal estimate.

Checked declarations

Module	Named supporting theorems
SharpThresholdFiniteMaximum.lean	4

- `sharpThresholdBlockConstant_nonneg`
- `sharpThresholdMaximalConstant_nonneg`
- `sharpThreshold_block_maximum`
- `sharpThreshold_finiteCenters_maximum`

<!-- checkpoint-verification -->

Verification

All 827 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4434-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2064 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 829 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_761_SharpThresholdCountableMaximum.md`

Checkpoint 761: Complete the sharp maximum for an unbounded changing denominator radius

The finite-center sharp estimate now extends to the actual countable family whose maximum denominator changes without bound as the dyadic scale increases. For each finite set of times, the proof forms a common finite center set from the largest actual denominator radius, applies the uniform finite-center theorem, and checks the exact set identity at every selected time. The established finite-to-countable maximal criterion then gives the full countable maximum. The coefficient/support hypotheses concern individual arithmetic coefficients, and no source maximal conclusion is assumed.

Checked declarations

Module	Named supporting theorems
SharpThresholdCountableMaximum.lean	1

- `sharpThreshold_countable_maximum`

<!-- checkpoint-verification -->

Verification

All 828 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4435-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2065 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 830 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the

later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_762_SharpOrdinaryExceptionalMaximum.md`

Checkpoint 762: Complete the sharp maximum for the actual ordinary and fixed-character families

The countable sharp theorem is instantiated for the two actual arithmetic families. The ordinary part uses the principal quadratic Gauss coefficient; the exceptional part uses the induced fixed real character, the exact quotient denominator q/q_chi, and its conductor-sensitive coefficient bound. The exceptional bound retains sigma_chi sqrt(B_chi), rather than discarding its decay. A proved mask identity handles both the active and inactive exceptional selectors and restricts the active sum exactly to conductor multiples. The upper denominator radius may change arbitrarily with scale.

Checked declarations

Module	Named supporting theorems
SharpOrdinaryExceptionalMaximum.lean	3

- `sharpOrdinary_source_maximum`
- `sharpExceptional_source_maximum`
- `thresholdAnnularBlock_mask`

<!-- checkpoint-verification -->

Verification

All 829 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4436-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2068 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 831 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_763_ConductorLogarithmicAbsorption.md`

Checkpoint 763: Retain summable conductor decay after absorbing all logarithmic losses

Every fixed logarithmic power is bounded uniformly by an arbitrarily small positive power of its argument. Applying this to the previously proved conductor decay absorbs (1+log q_chi)^3 while retaining q_chi^(-1/50). The effective-cutoff logarithm separates into a Q logarithm and a conductor logarithm. A separate unconditional arithmetic bound gives a fixed Q threshold above which every exceptional effective cutoff is at least 16. Finally, the retained conductor decay is summable even linearly over a family with at most one character in each dyadic conductor class. These results introduce no external input.

Checked declarations

Module	Named supporting theorems
ConductorLogarithmicAbsorption.lean	5

- `shifted_log_power_uniform`

- `conductor_logarithmic_decay`
- `effective_threshold_logarithmic_bound`
- `effective_threshold_large_uniform`
- `finite_conductor_class_linear_budget`

<!-- checkpoint-verification -->

Verification

All 830 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4437-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2073 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 832 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_764_SharpConductorMaximalRate.md`

Checkpoint 764: Complete uniform sharp piece estimates with summable conductor decay

The ordinary countable maximum is now expressed as a constant times Q^(-1/2) exp(G(log Q)) (1+log Q)^3. The exceptional countable maximum has the same Q rate, uniformly above one fixed arithmetic cutoff, and retains a factor q_chi^(-1/50) after every conductor logarithm is absorbed. The constants and the lower cutoff are independent of the conductor, starting scale, denominator-radius sequence, input, and the allowed beta sequence. This is the summable estimate needed to combine different exceptional characters over time.

Checked declarations

Module	Named supporting theorems
SharpConductorMaximalRate.lean	3

- `sharpHighMaximalRate_nonneg`
- `sharpOrdinary_rate_maximum`
- `sharpExceptional_conductor_rate`

<!-- checkpoint-verification -->

Verification

All 831 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4438-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2076 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 833 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT

limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_765_SharpChangingCharacterMaximum.md`

Checkpoint 765: Complete the actual high maximum for changing class-indexed exceptional characters

The full actual high operator now has a countable dyadic maximal majorant for a class-indexed family of fixed quadratic characters, allowing arbitrary character changes between times, arbitrary active/inactive selectors, and arbitrary beta and denominator-radius sequences in their stated ranges. The only class condition is 2^i <= q_i. The proof uses the exact ordinary-minus-exceptional symbol identity, the sharp ordinary majorant, each fixed-character majorant, and the retained summable conductor decay. Linear summation over occupied classes already converges, so no unproved class square-function estimate is needed. The Q rate remains Q^(-1/2) exp(G(log Q)) (1+log Q)^3, uniformly over starting scales. The next step proves the class parametrization from a precise independent Page input rather than assuming it for the source family.

Checked declarations

Module	Named supporting theorems
SharpChangingCharacterMaximum.lean	1

- `sharpChangingCharacter_maximum`

<!-- checkpoint-verification -->

Verification

All 832 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4439-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2077 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 834 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_766_RawPageUniqueness.md`

Checkpoint 766: Derive dyadic exceptional-character uniqueness from the raw Landau-Page input

The independent Landau-Page input is now stated precisely for primitive nonprincipal quadratic characters and their real zeros. A zero below 1 is represented by the limit of ordinary Dirichlet-series partial sums, not by an unconditional tsum; the analytic-continuation bridge is explicitly part of the unformalized input. The standard uniqueness region itself remains an explicit general hypothesis, with no operator conclusion assumed. From it, Lean proves uniqueness of the complete character-and-zero datum in each dyadic conductor class after halving the region constant, and proves that source-region eligibility implies conductor-region eligibility. The standard input is documented against Philippe Michel's 2006 lectures, section 1.2 (Landau/Page), https://www.epfl.ch/labs/tan/wp-content/uploads/2018/10/Parkcitylectures.pdf. This does not change Theorem 1.1's previously recorded inputs.

Checked declarations

Module	Named supporting theorems
RawPageUniqueness.lean	2

- `page_dyadic_uniqueness`
- `page_source_region_eligible`

<!-- checkpoint-verification -->

Verification

All 833 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4440-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2079 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 835 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_767_PageConductorClassSelection.md`

Checkpoint 767: Construct and verify the actual exceptional-character classes from Page uniqueness

The class parametrization is now proved from raw Page uniqueness. The construction chooses an occurring exceptional datum for each occupied dyadic conductor class and supplies a harmless principal-character fallback for empty classes. Lean proves that every active source datum equals its selected representative, that the representative conductor is at least 2^i, and that each selected character is quadratic. Source-region zero data imply the exact conductor-region eligibility used in the uniqueness proof. The selection includes the actual character and zero, not merely the conductor; no unverified identification is made when the exceptional character changes with scale.

Checked declarations

Module	Named supporting theorems
PageConductorClassSelection.lean	7

- `optionPageBeta_nonneg`
- `optionPageBeta_le_one`
- `pageClassRepresentative_bin`
- `pageClassRepresentative_eq`
- `pageClassConductor_lower`
- `pageClassCharacter_quadratic`
- `pageSourceRegion_family_eligible`

<!-- checkpoint-verification -->

Verification

All 834 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4441-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2086 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 836 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_768_ActualPageHighMaximum.md

Checkpoint 768: Complete the sharp maximum for the actual changing exceptional data

The sharp countable maximum is now proved for the actual scale-varying exceptional data. Each active datum contains the primitive real character, conductor, and zero; inactive data give the ordinary family exactly. Page uniqueness proves that the conductor-class map is injective on every finite set of occurring data. The proof sums majorants for those actual data and then passes to the countable maximum, avoiding any unverified change of character or dependent modulus. The resulting Q rate and constant are uniform in the starting scale and changing denominator radius. Page uniqueness, the general smooth multifrequency input, and the already recorded arithmetic inputs remain explicit; no own-paper operator estimate is assumed.

Checked declarations

Module	Named supporting theorems
ActualPageHighMaximum.lean	9

- `optionPageConductor_some`
- `optionPageConductor_none`
- `optionPageBeta_some`
- `optionPageBeta_none`
- `circleHighOperator_inactive`
- `pageHighOperator_none`
- `pageHighOperator_some`
- `pageFinite_conductor_budget`
- `actualPageHigh_maximum`

<!-- checkpoint-verification -->

Verification

All 835 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4442-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2095 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 837 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_769_SharpHighMaximalAsymptotic.md

Checkpoint 769: Recover the precise leading-two asymptotic after all maximal losses

The analytic Q rate is converted to the precise leading-two asymptotic. Under the two explicit prime asymptotic inputs, for every epsilon > 0 there are fixed positive constants and a lower Q cutoff such that Q^(-1/2) exp(G(log Q)) (1+log Q)^3 is bounded by a constant times Q^(-1/2) exp((2+epsilon) sqrt(log Q)/log log Q). The same cutoff ensures all threshold-slope conditions needed by the maximal proof. The constants are independent of the starting scale, source radius, and exceptional data. This step does not claim the separate bounded-Q completion.

Checked declarations

Module	Named supporting theorems
SharpHighMaximalAsymptotic.lean	1

- `sharpHighMaximalRate_asymptotic`

<!-- checkpoint-verification -->

Verification

All 836 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4443-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2096 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 838 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_770_Proposition621Asymptotic.md`

Checkpoint 770: Prove the actual source supremum with the sharp asymptotic uniformly in the starting threshold

The actual maximal conclusion of Proposition 6.21 is now proved in precise large-Q epsilon form, uniformly in the real starting threshold N0 >= 2. The multiplier uses the actual source radius R(M)=max(exp((log M)^(1/10)),Q^8), and the source zero-region hypotheses imply the exact Page eligibility used in the proof. Earlier dyadic times are excluded explicitly, and inactive exceptional data require no zero assumption. The resulting real lp function equals the pointwise countable supremum of the actual high outputs; it is not merely an unspecified majorant. Its norm has the sharp Q^(-1/2) exp((2+epsilon) sqrt(log Q)/log log Q) bound, and every individual output has the same bound. The constant and Q cutoff are independent of N0 and all compatible source data. The separate bounded-Q completion remains the next obligation before recording the full source proposition as complete.

Checked declarations

Module	Named supporting theorems
Proposition621Asymptotic.lean	3

- `exact_maximum_of_majorant`
- `norm_le_nonnegative_majorant`
- `proposition621_maximal_asymptotic`

<!-- checkpoint-verification -->

Verification

All 837 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4444-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2099 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 839 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_771_SmallSourceCutoffDifference.md`

Checkpoint 771: Control small denominator indices by a summable actual-cutoff difference

The small denominator indices are compared with the separated cutoff at index 4. The difference vanishes in the common central core and has actual multiplier norm at most an explicit constant divided by N^2, uniformly in beta in [0,1]. The proof uses the true source bump core, oscillatory-integral decay outside that core, and the finite circle-periodization norm bound. This summable error supplies a uniform treatment of the finitely many small denominator indices needed for the bounded-Q completion.

Checked declarations

Module	Named supporting theorems
SmallSourceCutoffDifference.lean	3

- `smallSourceDifferenceConstant_pos`
- `localizedAnnularSource_small_difference`
- `circleAnnularSource_small_difference`

<!-- checkpoint-verification -->

Verification

All 838 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4445-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2102 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 840 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_772_AllIndexAnnularTruncationMaximum.md`

Checkpoint 772: Complete the actual annular maximum at every denominator index

The actual annular joint scale-and-denominator-cutoff maximum is now proved for every denominator index, including 0, 1, 2 and 3. The small indices use the verified cutoff comparison, a geometric error

maximum, and the separated index-4 source estimate. A concrete finite cardinality bound controls the small center sets. Together with the previously proved large-index maximum, this yields one constant times A(s+1)^3 for every index, every starting M >= 2, every allowed beta sequence, and every nonnegative changing denominator radius. This closes the small-frequency issue required to prove a uniform bounded-Q high maximum.

Checked declarations

Module	Named supporting theorems
AllIndexAnnularTruncationMaximum.lean	6

- `dyadic_inverse_square_bound`
- `geometric_symbol_maximum`
- `weightedSmallSource_difference_bound`
- `allIndexAnnularConstant_pos`
- `smallIndex_annular_truncation_maximum`
- `allIndex_annular_truncation_maximum`

<!-- checkpoint-verification -->

Verification

All 839 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4446-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2108 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 841 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_773_PowerDecayingAnnularMaximum.md`

Checkpoint 773: Prove a uniform maximum for the full power-decaying annular family

The full bare annular family with denominator-dependent source indices now has a countable dyadic maximum whenever its fixed coefficients decay by any positive denominator power. The proof uses the all-index joint truncation estimate, the actual dyadic denominator partition, a convergent polynomial-geometric sum, and the verified finite-to-countable passage for the growing source radius. Any fixed lower-denominator restriction or conductor-multiple restriction may be included in the center set. The bound is uniform in the restriction, scale, and changing upper radius. This is the large-denominator input for the bounded-Q high estimate.

Checked declarations

Module	Named supporting theorems
PowerDecayingAnnularMaximum.lean	3

- `annularPowerConstant_pos`
- `powerAnnular_finiteCenters_maximum`
- `powerAnnular_countable_maximum`

<!-- checkpoint-verification -->

Verification

All 840 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4447-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2111 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 842 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_774_PowerThresholdUniformMaximum.md

Checkpoint 774: Prove the uniform threshold maximum needed for bounded Q

The full threshold-annular maximum now has a uniform bound whenever the effective cutoff has any fixed positive lower bound and the arithmetic coefficients have a fixed positive denominator-power decay. The collar part is proved directly from the complete collar square-function estimate, including the empty cutoff range below one; the large part uses the full annular maximum at every denominator index. Every upper source-radius cutoff may vary with time. The result is independent of the size of the effective cutoff above its lower bound. This supplies the separate bounded-Q estimate rather than assuming that it can be absorbed.

Checked declarations

Module	Named supporting theorems
PowerThresholdUniformMaximum.lean	2

- `thresholdPowerConstant_pos`
- `powerThreshold_countable_maximum`

<!-- checkpoint-verification -->

Verification

All 841 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4448-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2113 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 843 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_775_UniformPageHighMaximum.md

Checkpoint 775: Complete the actual high maximum throughout the bounded Q range

The actual high family now has a uniform countable dyadic maximum for every real Q >= 2, including the entire bounded-Q range. A fixed effective-cutoff lower bound follows from the unconditional conductor-majorant ratio bound. The general power-decay maximum supplies ordinary and exceptional majorants; the exceptional bound retains a summable conductor power. Raw Page uniqueness controls the actual finite set of occurring data, and the finite-to-countable argument gives the full source family. The source-radius and real starting-threshold restriction are included, and the output function equals the actual pointwise supremum. No PNT input is needed for this uniform bound. The final source review must also verify that the convenient broad-masked circle model agrees with the manuscript's complementary narrow cutoff in its allowed geometric range.

Checked declarations

Module	Named supporting theorems
UniformPageHighMaximum.lean	2

- `actualPageHigh_uniform_maximum`
- `sourcePageDyadicTail_uniform`

<!-- checkpoint-verification -->

Verification

All 842 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4449-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2115 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 844 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_776_Proposition621.md`

Checkpoint 776: Complete both high estimates including the full bounded Q range

Both maximal and individual-output estimates now hold for the broad-masked high model for every Q >= 2 and every real starting threshold N0 >= 2. The theorem combines the sharp large-Q estimate with the independently verified uniform bounded-Q maximum. Its precise epsilon formulation has the leading coefficient 2+epsilon and uses max(2,log Q) only in the harmless bounded-Q convention; it is exactly sqrt(log Q)/log log Q once log Q >= 2. The output is the actual pointwise countable supremum in real lp2. The last source-identification stage verifies that this convenient model equals the manuscript's unmasked complementary cutoff throughout the allowed source range before awarding the paper claim.

Checked declarations

Module	Named supporting theorems
Proposition621.lean	3

- `regularizedHighExponentWeight_pos`
- `regularizedHighExponentWeight_eq`
- `proposition621`

<!-- checkpoint-verification -->

Verification

All 843 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4450-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2118 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 845 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_777_SourceHighUniformGeometry.md

Checkpoint 777: Verify the original source range uniformly supplies the exact cutoff geometry

The manuscript's actual allowed range now implies the cutoff-containment geometry uniformly from every real N0 >= 2. One fixed positive c0 is chosen independently of epsilon. If Q >= 2 and log Q <= c0 sqrt(log N0), the constraint itself forces N0 past the existing uniform source threshold; hence every integer scale N >= N0 satisfies 2 R(N,Q)^8 <= N^2. This proof uses the established uniform fourth-power source budget and checks the exact parsing of its constant. The stronger squared working radius allows both ordinary and exceptional narrow cutoffs to be compared with the broad cutoff, without incorrectly assuming that the exceptional effective threshold is at most R.

Checked declarations

Module	Named supporting theorems
SourceHighUniformGeometry.lean	1

- `sourceHigh_uniform_geometry`

<!-- checkpoint-verification -->

Verification

All 844 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4451-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2119 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 846 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_778_Proposition621Source.md

Checkpoint 778: Complete the exact source Proposition 6.21, both maximal and individual clauses

Both clauses of the actual source Proposition 6.21 are complete checked implications. The final declaration chooses one positive c0 before epsilon, retains the full real Q >= 2 and N0 >= 2 allowed range, constructs the actual countable lp2 supremum and the individual-output bound, handles changing genuine exceptional data, preserves the sharp leading coefficient 2, and includes a separately proved bounded-Q range. The exact source identification is part of the conclusion: the circle multiplier equals the periodization of the manuscript's unmasked complementary narrow-cutoff formula at every valid scale. The uniform source-range geometry and the coverage of every dyadic scale are proved. PNT limits, the general smooth multifrequency/sampling input, raw Page uniqueness and source-compatible exceptional-data existence remain explicit independent boundaries; no own-paper maximal estimate is assumed. The source-level registry awards only Proposition 6.21, changing it from partial to complete conditional; Theorem 1.1 is preserved and the stronger auxiliary lemmas remain partial. The completion report, statement review and remaining-work plan are packaged. The manuscript and all accepted replacement hashes are unchanged.

Checked declarations

Module	Named supporting theorems
Proposition621Source.lean	4

- `pageHighOperator_symbol`
- `pageHighSymbol_paper_identity`
- `dyadic_tail_covers_source_scales`
- `proposition621_source`

<!-- checkpoint-verification -->

Verification

All 845 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4452-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2123 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 847 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. The whole paper, the full maximal clauses of Proposition 4.19/6.21 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_779_IntegerReflection.md`

Checkpoint 779: Integer reflection is an involutive lp isometry and reverses support intervals

Spatial reflection is defined by Rf(x)=f(-x). The new IntegerReflection module proves membership in every lp space is preserved, reflection is involutive, and its norm is unchanged for every p>=1, including p=infinity. It constructs an actual linear isometric equivalence over any normed scalar field, so the real and complex sequence spaces used by the paper are both covered. It also proves that finite input functions and finite indicators reflect exactly, that finite-set cardinalities are preserved, and that the integer interval [a,b] becomes [-b,-a]. These are unconditional algebraic and sequence-space proofs; the independent square-average theorem is not assumed or formalized here. This checkpoint is the foundation for the convolution and square-sign transfer. The completed conditional Theorem 1.1 and Proposition 6.21 are retained. A stale generated progress paragraph was corrected to agree with the already-verified Proposition 6.21 completion. The manuscript and all accepted replacements are unchanged.

Checked declarations

Module	Named supporting theorems
IntegerReflection.lean	13

- `reflectSequence_apply`
- `reflectSequence_involutive`
- `memLp_reflectSequence`
- `reflectLp_apply`
- `reflectLp_involutive`
- `reflectLp_norm`
- `integerReflection_apply`
- `mem_reflectedFinset`
- `reflectedFinset_involutive`
- `reflectedFinset_card`
- `reflectedFinset_Icc`
- `reflectLp_finiteIndicator`
- `reflectLp_finiteInput`

<!-- checkpoint-verification -->

Verification

All 846 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4453-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2136 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 848 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. Both source clauses of Proposition 6.21 are also complete checked implications under their recorded independent inputs. The whole paper, Proposition 4.19 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_780_ReflectedConvolution.md`

Checkpoint 780: Reflection preserves convolution pairings and all decomposition constants

For an arbitrary complex kernel K, reflection K^R(z)=K(-z) satisfies the exact finite convolution identity (K^R*f)(x)=(K*(Rf))(-x) when the finite input set is also reflected. The module proves the corresponding finite complex and indicator pairings, with both input and output sets reflected and their cardinalities preserved by checkpoint779. It constructs R T R for continuous linear operators between any lp and lq with p,q>=1, including infinity, and proves exact equality of their operator norms. The actual OperatorHighLowBound, OperatorOrliczBound and KernelL2Bound predicates are each equivalent for K and K^R with every numerical parameter and constant unchanged. The low/high/error maps need not be linear; their norm bounds transfer through the involutive isometry. No analytic input is used for these results. This closes the general convolution-estimate part of the requested sign transfer; the next checkpoint identifies the literal plus-square formulas and support windows.

Checked declarations

Module	Named supporting theorems
ReflectedConvolution.lean	19

- `reflectLpMap_apply`

- reflectLpMap_involutive
- reflectLpMap_norm
- reflectLpMap_bound_iff
- reflectContinuousLinearMap_apply
- reflectContinuousLinearMap_eq
- reflectContinuousLinearMap_involutive
- reflectContinuousLinearMap_norm_le
- reflectContinuousLinearMap_norm
- finiteConvolution_reflect
- indicatorConvolution_reflect
- finiteKernelPairing_reflect
- restrictedKernelPairing_reflect
- operatorHighLowBound_reflect
- operatorHighLowBound_reflect_iff
- operatorOrliczBound_reflect
- operatorOrliczBound_reflect_iff
- kernelL2Bound_reflect
- kernelL2Bound_reflect_iff

<!-- checkpoint-verification -->

Verification

All 847 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4454-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2155 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 849 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. Both source clauses of Proposition 6.21 are also complete checked implications under their recorded independent inputs. The whole paper, Proposition 4.19 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_781_SquareAverageReflection.md

Checkpoint 781: Full and annular plus-square formulas equal reflected minus-square averages

The plus-square operators and kernels are defined directly by finite sums, independently of reflection. Lean proves that their convolution kernel is the reflection of the existing minus-square kernel and their operator is R T_- R. The full weighted average is exactly N^-1 times the sum over 0<n<=N of w(n)f(x+n^2); the annular version retains the exact natural-index range N/2<n<=N. Both are equivalent by involutive conjugation to the existing x-n^2 operators, with equal operator norms. The proofs allow arbitrary real arithmetic weights; neither n nor w(n) is negated. They therefore apply unchanged to ordinary square weights and von-Mangoldt weights. The positive-translation kernel is supported in [-N^2,-1], and its local input window is [x,x+N^2]. Lean proves the interval reflection and the actual indicator-convolution localization on that window. This is an exact transfer after reflecting the geometry, not a claim that an asymmetric support interval or a particular dyadic grid stays fixed. No manuscript replacement is required. The external analytic square theorem is untouched.

Checked declarations

Module	Named supporting theorems
SquareAverageReflection.lean	18

- finiteSquareKernel_reflection

- finitePlusSquareOperator_apply
- finitePlusSquareOperator_reflection
- finitePlusSquareOperator_norm
- finitePlusSquareKernel_finiteInput
- weightedSquareAverageKernel_reflection
- annularWeightedSquareAverageKernel_reflection
- plusSquareAverage_apply
- annularPlusSquareAverage_apply
- plusSquareAverage_reflection
- annularPlusSquareAverage_reflection
- minusSquareAverage_from_plus
- annularMinusSquareAverage_from_plus
- plusSquareAverage_opNorm
- annularPlusSquareAverage_opNorm
- squareAverageWindow_reflection
- finitePlusSquareKernel_support_interval
- finitePlusSquareKernel_localization

<!-- checkpoint-verification -->

Verification

All 848 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4455-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2173 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 850 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. Both source clauses of Proposition 6.21 are also complete checked implications under their recorded independent inputs. The whole paper, Proposition 4.19 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_782_MaximalReflection.md

Checkpoint 782: Reflection transfers actual maximal bounds and temporal majorants

Spatial reflection transfers arbitrary families of lp-to-lq maps, including the l-infinity endpoint. The new LpFamilyMaximalBound predicate requires bounded coordinate norm ranges, an actual real-valued lp representative of their supremum, and the stated norm bound. The proof reflects that representative and proves equivalence of the maximal estimate for T_i and R T_i R with the same constant. No summability of operator norms or additional analytic maximal theorem is assumed. The already-used FiniteTemporalMajorants predicate is likewise equivalent for a family and its spatial reflection. Norms of operator-output differences are also preserved, recording the algebraic ingredient for variation and difference estimates without claiming to formalize a separate variational theorem. All these results are unconditional transfer lemmas. The independent ordinary-square analytic theorem and its globalization remain external until separately proved; this checkpoint supplies no new analytic assumption.

Checked declarations

Module	Named supporting theorems
MaximalReflection.lean	6

- familySupremum_reflect
- lpFamilyMaximalBound_reflect
- lpFamilyMaximalBound_reflect_iff

- `finiteTemporalMajorants_reflect`
- `finiteTemporalMajorants_reflect_iff`
- `reflectedMap_difference_norm`

<!-- checkpoint-verification -->

Verification

All 849 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4456-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2179 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 851 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. Both source clauses of Proposition 6.21 are also complete checked implications under their recorded independent inputs. The whole paper, Proposition 4.19 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the independent square decomposition remain explicit hypotheses. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_783_SquareDecompositionReflection.md`

Checkpoint 783: Complete the square-sign reflection and integrate it into Theorem 1.1

The requested square-sign reflection is complete. The new PositiveFullSquareDecompositionInput states the independent ordinary-square global high/low hypothesis using the literal x+n^2 kernel. Lean proves it equivalent to the existing FullSquareDecompositionInput at exactly the same constant C, then derives the existing minus-sign and plus-sign annular inputs with the already proved factor 3C. The final theorem_1_1_from_positive_square_input accepts the positive-sign global hypothesis and returns both existing conclusions of Theorem 1.1 for the paper's actual minus-sign prime-square average, without adding an analytic assumption. The literal von-Mangoldt sum and the reflected source interval [-2N^2,N^2] are checked. Together with checkpoints779-782 this proves the lp isometries through infinity, general convolution and pairing identities, high/low/error constant transfer, full/annular formulas, actual maximal-bound transfer and support/localization changes. Reflection is removed from the unformalized boundary; the independent ordinary-square analytic theorem and its local-to-global passage remain explicit global hypotheses. The manuscript is unchanged. Current external-input, statement-review, completion and work-plan documents are reconciled; paper claim statuses remain 10 proved,3 conditional,6 partial,1 external candidate,34 open. The blueprint-length report distinguishes topic/reference notes, individual checkpoint expositions and overlapping batch summaries; the old 19-page PDF is only the initial blueprint, and there is no current combined PDF. See docs/SQUARE_REFLECTION_COMPLETION.md and docs/BLUEPRINT_LENGTH.md.

Checked declarations

Module	Named supporting theorems
SquareDecompositionReflection.lean	8

- `fullSquareDecompositionInput_sign_iff`
- `fullSquare_decomposition_from_plus`
- `plusSquare_decomposition_from_minus`
- `positiveFullSquare_decomposition_annular`
- `positiveFullSquare_decomposition_annular_plus`
- `primeSquareSourceInterval_reflection`
- `primeSquareAverageValue_reflection`
- `theorem_1_1_from_positive_square_input`

<!-- checkpoint-verification -->

Verification

All 850 project modules and the root were freshly rebuilt: 4457 jobs pass without warnings/errors. All 2187 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 852 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. Both source clauses of Proposition 6.21 are also complete checked implications under their recorded independent inputs. The whole paper, Proposition 4.19 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the global independent square decomposition remain explicit hypotheses. Reflection between plus and minus square translations is proved without any extra analytic assumption. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_784_PageDistributionInputs.md`

Checkpoint 784: Compatible Page zero and raw distribution data for one common operator family

The refined PagePrimeSquareDistributionInputs interface supplies compatible optional genuine Page-zero data together with the existing raw progression-prefix and Ren/Ghosh estimates. Its existence remains an explicit independent number-theoretic hypothesis, not a new Lean axiom and not an own-paper operator conclusion. The module proves that this refined input supplies the already used PrimeSquareDistributionInputs, so it preserves the fixed-scale Theorem 1.1 route. It defines one chosen exceptional datum for each N,Q and proves its raw analytic and Page-region properties whenever the original range is admissible. The choice is independent of the high-bound epsilon and every error-decay exponent; this is needed to assemble all estimates for the same three operators. The optional trivial-character case and positive conductors are checked. The full Proposition 4.19 still requires the fixed operator-family construction, all-input local low bound, uniform all-scale error estimates and final quantifier review. Theorem 1.1, Proposition 6.21 and the completed reflection bridge are preserved, and the manuscript is unchanged.

Checked declarations

Module	Named supporting theorems
PageDistributionInputs.lean	4

- `optionPageConductor_pos`
- `optionPageCharacter_quadratic`
- `sourcePageDatum_spec`
- `pageDistribution_cutoff_pos`

<!-- checkpoint-verification -->

Verification

All 851 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4458-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2191 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 853 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. Both source clauses of Proposition 6.21 are also complete checked implications under their recorded independent inputs. The whole paper, Proposition 4.19 and the later

headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the global independent square decomposition remain explicit hypotheses. Reflection between plus and minus square translations is proved without any extra analytic assumption. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_785_SelectedPrimeSquareOperators.md

Checkpoint 785: Exact common low high and error operators on all inputs

Defines the actual low, high and error operators using one fixed optional Page datum and the manuscript radius sourceMajorArcScale N Q. None of these definitions depends on epsilon or an error exponent. Lean proves the exact continuous-linear-operator decomposition on all l2 inputs, not just indicators, using the exact annular Fourier identity. It also proves the identity on all l1 inputs with the low term realized as the genuine l1-to-l-infinity operator and the high/error inputs included into l2. The selected high tail is exactly the already verified dyadic source tail from Proposition 6.21, with the original starting-threshold mask and all dyadic scales at least two. This checkpoint establishes common operator identity and compatibility, without yet awarding any of the missing uniform bounds or closing Proposition 4.19. All existing completed results and the manuscript are preserved.

Checked declarations

Module	Named supporting theorems
SelectedPrimeSquareOperators.lean	4

- `selectedPrimeLow_agreement`
- `selectedPrime_decomposition_l2`
- `selectedPrime_decomposition_l1`
- `selectedPrimeHighTail_of_le`

<!-- checkpoint-verification -->

Verification

All 852 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4459-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2195 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 854 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. Both source clauses of Proposition 6.21 are also complete checked implications under their recorded independent inputs. The whole paper, Proposition 4.19 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the global independent square decomposition remain explicit hypotheses. Reflection between plus and minus square translations is proved without any extra analytic assumption. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_786_SourceLocalMass.md

Checkpoint 786: Literal real-radius local maximal average and extension to all l1 inputs

Defines the literal local average in Proposition 4.19 for every l1 input: the supremum over all real M>=N of M^-2 times the sum of |f(y)| over |x-y|<=M^2. Lean proves summability, positivity, boundedness of the supremum, and its l1 norm majorant. It proves that each finite dyadic local mass already used by the low-kernel analysis is dominated by this all-input source supremum, taking M=sqrt(2^j N^2). A general continuous convolution lemma then extends any existing finite-input dyadic local bound to the actual l1-to-l-infinity Fourier operator on all l1 inputs, using a proved convergent sum rather than a density assertion left informal. Thus the final local clause will use the printed source quantity with the absolute

radius constant C=1. This is an unconditional supporting proof and adds no analytic input. The selected family, independent-input boundaries, completed prior results and manuscript remain unchanged.

Checked declarations

Module	Named supporting theorems
SourceLocalMass.lean	9

- `sourceBall_summable`
- `sourceBallMass_nonneg`
- `sourceBallMass_uniform`
- `sourceLocalMass_bddAbove`
- `sourceBallMass_le_local`
- `sourceLocalMass_nonneg`
- `sourceLocalMass_uniform`
- `finiteDyadicLocalMass_le_source`
- `fourierL1Operator_local_of_finite`

<!-- checkpoint-verification -->

Verification

All 853 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4460-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2204 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 855 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. Both source clauses of Proposition 6.21 are also complete checked implications under their recorded independent inputs. The whole paper, Proposition 4.19 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the global independent square decomposition remain explicit hypotheses. Reflection between plus and minus square translations is proved without any extra analytic assumption. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_787_SelectedPrimeLowBounds.md`

Checkpoint 787: Both low clauses on every l1 input with an absolute source range

Both actual low-term clauses of Proposition 4.19 are now proved for the common selected operator family, on every l1 input. One positive absolute source-cutoff coefficient is chosen before all data, epsilons and error exponents. The exact l1-to-l-infinity norm is bounded by (2 lowSpatialConstant) N^-2 Q log(2Q), and the pointwise local bound by (8 lowSpatialConstant) Q log(2Q) times the literal all-real-radius sourceLocalMass. The latter is the printed source supremum with C=1. A proved general cutoff-forcing lemma shows that a sufficiently small fixed coefficient in log Q<=delta sqrt(log N), together with Q>=2, forces any fixed starting threshold. This handles the low-term threshold without weakening the source N-range. No own-paper low estimate or local-to-global assertion is assumed. The existing raw number-theoretic boundary remains explicit, and no manuscript replacement is needed.

Checked declarations

Module	Named supporting theorems
SelectedPrimeLowBounds.lean	2

- `cutoff_forcing_threshold`
- `selectedPrimeLow_source_bounds`

<!-- checkpoint-verification -->

Verification

All 854 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4461-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2206 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 856 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. Both source clauses of Proposition 6.21 are also complete checked implications under their recorded independent inputs. The whole paper, Proposition 4.19 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the global independent square decomposition remain explicit hypotheses. Reflection between plus and minus square translations is proved without any extra analytic assumption. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_788_FinitePageErrorBound.md`

Checkpoint 788: Uniform finite-scale error envelope for every real cutoff and Page datum

Proves an elementary error-operator envelope uniform over every real Q>=2 and every optional Page datum on each finite range 2<=N<=T, whenever the source cutoff geometry holds. The geometry implies sourceMajorArcScale N Q<=N, so all rational centers lie in one finite canonical set. The annular average is bounded by a finite sum of its actual operator norms, and the major-arc model by the already proved Gauss coefficient envelope. Their exact difference bounds the actual error. No asymptotic error estimate, distribution estimate or new analytic hypothesis is used. This finite-range result will close the small-scale part of every A-dependent error-decay estimate while preserving one A-independent source coefficient and one fixed selected operator family. It does not award the full Proposition 4.19 yet. The existing manuscript and completed results remain unchanged.

Checked declarations

Module	Named supporting theorems
FinitePageErrorBound.lean	3

- `sourceRadius_le_scale`
- `finitePageErrorEnvelope_pos`
- `finitePageError_bound`

<!-- checkpoint-verification -->

Verification

All 855 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4462-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2209 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 857 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. Both source clauses of Proposition 6.21 are also complete checked implications under their recorded independent inputs. The whole paper, Proposition 4.19 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the global independent square decomposition remain

explicit hypotheses. Reflection between plus and minus square translations is proved without any extra analytic assumption. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_789_SelectedPrimeErrorBounds.md

Checkpoint 789: Every error exponent with one fixed source range and operator family

The actual selected error operator now satisfies Q^(-1/2)(1+log N)^(-A) decay for every real A>=0, with one positive source-cutoff coefficient chosen before A. The same operator uses the same N,Q-dependent Page datum for every exponent. Large N uses the existing proved all-point circle approximation from raw progression/Ren/Ghosh data; the finitely many smaller arithmetic scales use the unconditional envelope from checkpoint788. Geometry bounds Q by the finite threshold, allowing the constant depending on A to absorb those scales. Thus no A-dependent narrowing of the final source range or A-dependent replacement of the error operator is used. The resulting bound has logarithmic Q power zero, which implies the source's allowed polynomial-logarithmic loss. Raw data existence remains the explicit refined independent input, not an assumed error estimate. This completes the error component needed by Proposition 4.19 without awarding the separate stronger source approximation lemma, whose full statement has not been reviewed here.

Checked declarations

Module	Named supporting theorems
SelectedPrimeErrorBounds.lean	1

- `selectedPrimeError_all_exponents`

<!-- checkpoint-verification -->

Verification

All 856 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4463-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2210 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 858 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. Both source clauses of Proposition 6.21 are also complete checked implications under their recorded independent inputs. The whole paper, Proposition 4.19 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the global independent square decomposition remain explicit hypotheses. Reflection between plus and minus square translations is proved without any extra analytic assumption. The smooth multifrequency boundary is only for later maximal applications.

Checkpoint_790_SelectedPrimeHighBounds.md

Checkpoint 790: Sharp fixed and countable maximal high bounds for the common selected family

The common selected high family now has both clauses required by Proposition 4.19: the exact fixed dyadic-scale l2 operator norm and the actual countable l2 supremum from every real starting threshold, for every real Q>=2. A single positive cutoff coefficient is selected before epsilon; one positive K for each epsilon works uniformly over Q, all scales and starting thresholds, and all inputs. Bounded Q and the sharp leading 2+epsilon coefficient are inherited from the fully checked Proposition 6.21. The proof constructs compatible genuine Page data at every valid scale from the refined raw distribution selector, using a proved range-forcing lemma. It retains the exact source periodized high-symbol identity and changing exceptional characters. No own-paper maximal theorem is an external assumption. The

only independent inputs used for this component are the two prime limits, the general smooth multifrequency/sampling input, raw Page uniqueness and compatible raw progression/Ren/Ghosh data existence. The fixed estimate is deduced from the very same maximal family. The final Proposition 4.19 export and source review remain to be packaged.

Checked declarations

Module	Named supporting theorems
SelectedPrimeHighBounds.lean	2

- `selectedPage_raw_source_range`
- `selectedPrimeHigh_source_bounds`

<!-- checkpoint-verification -->

Verification

All 857 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4464-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2212 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 859 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. In addition, Lemma 3.21 and BOTH conclusions of Theorem 1.1 are complete checked implications under their explicit independent deep inputs. The PNT limits are used in the sharp Gauss/high bound; an imported PNT bridge is not integrated. Both source clauses of Proposition 6.21 are also complete checked implications under their recorded independent inputs. The whole paper, Proposition 4.19 and the later headline applications remain unfinished. See docs/THEOREM_1_1_EXTERNAL_INPUTS.md. PNT limits, raw compatible progression/Ren/Ghosh data, and the global independent square decomposition remain explicit hypotheses. Reflection between plus and minus square translations is proved without any extra analytic assumption. The smooth multifrequency boundary is only for later maximal applications.

`Checkpoint_791_Proposition419.md`

Checkpoint 791: Complete every source clause of Proposition 4.19 with one common operator family

Completes every displayed clause of the actual source Proposition 4.19 as one checked implication. The final proposition419 export proves Proposition419Bounds for the same selected low, high and error family, with one positive cutoff coefficient chosen before epsilon and every error exponent. It includes the exact all-input decomposition, the genuine low l1-to-l-infinity norm, the literal real-radius local low bound for every l1 input, both sharp fixed and countable maximal high estimates for every real Q>=2 and every allowed starting threshold, every A>0 error rate without an extra Q logarithm, and the exact printed high-symbol identity. The dyadic coverage theorem excludes the impossible N=1 case and covers every admissible original dyadic scale. Genuine Page-data selection from the refined raw interface is integrated; raw data existence, the two prime limits, raw Page uniqueness and the general smooth multifrequency/sampling theorem remain explicit independent hypotheses. No own-paper operator conclusion is assumed and no manuscript correction is needed. The exact source review awards only Proposition 4.19, moving it from partial to complete conditional; the whole paper and stronger auxiliary statements remain unfinished. Theorem 1.1, Proposition 6.21 and reflection are preserved. The updated inventory has 54 claims:10 proved,4 conditional,5 partial,1 external candidate,34 open, with 4 definitions separate. The next source-level milestone is sparse domination and Theorem 1.3.

Checked declarations

Module	Named supporting theorems
Proposition419.lean	2

- `proposition419`

- `proposition419_dyadic_coverage`

<!-- checkpoint-verification -->

Verification

All 858 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4465-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2214 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 860 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

Checkpoint notes 792-866

Checkpoint_792_SparseIntegerIntervals.md

Checkpoint 792: Define genuine integer sparse families and the two source forms

Defines nonempty integer intervals, their exact cardinalities, fivefold dilation and support-ball containment. Defines the actual finite-set endpoint and power sparse forms with the printed densities and records sparsity through actual pairwise disjoint major subsets. The parameter is explicit: the preliminary definition uses 1/4 and Theorem 1.3 uses 1/10. Proves elementary positivity and monotonicity of sparsity. These are foundations only; no source sparse theorem is awarded from definitions. The previously completed conditional results remain unchanged.

Checked declarations

Module	Named supporting theorems
SparseIntegerIntervals.lean	11

- `card_points`
- `mem_points`
- `nonempty_points`
- `points_subset_five`
- `five_injective`
- `local_ball_subset_five`
- `integerSparseFamily_mono`
- `integerSparseFamily_empty`
- `integerSparseFamily_singleton`
- `integerEndpointSparseForm_nonneg`
- `integerPowerSparseForm_nonneg`

<!-- checkpoint-verification -->

Verification

All 859 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4466-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2225 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 861 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

Checkpoint_793_FiniteSparseMajorSets.md

Checkpoint 793: Construct actual disjoint major subsets and retain them under dilation

Proves that deleting the union of strict descendants in a finite laminar interval family gives actual pairwise disjoint major subsets, with the cardinality prescribed by the packing inequality. Proves fivefold dilation preserves those very subsets while dividing the sparsity constant exactly by five. Thus a 3/4-sparse parent family produces a genuine 1/10-sparse family of fivefold intervals, with no unproved grid replacement. The construction of the stopping intervals and the analytic local estimates remain separate unfinished deductions.

Checked declarations

Module	Named supporting theorems
FiniteSparseMajorSets.lean	7

- `integerMajorCore_subset`
- `integerMajorCore_disjoint_descendant`
- `integerMajorCore_pairwise`
- `integerMajorCore_card`
- `integerSparseFamily_of_laminar`
- `integerSparseFamily_five`
- `integerSparseFamily_five_source`

<!-- checkpoint-verification -->

Verification

All 860 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4467-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2232 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 862 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

Checkpoint_794_FiniteIntegerVitali.md

Checkpoint 794: Prove finite disjoint interval selection and threefold covering

Proves finite integer-interval disjoint selection by induction on the number of intervals, choosing a longest interval at each stage. Every original interval meets a selected interval at least as long and is contained in its threefold enlargement. Gives the resulting actual finite covering. This is a checked ingredient of the stopping packing proof, not an assumed Hardy-Littlewood weak theorem.

Checked declarations

Module	Named supporting theorems
FiniteIntegerVitali.lean	3

- `integerInterval_intersect_subset_three`
- `finiteIntegerVitali`
- `finiteIntegerVitali_cover`

<!-- checkpoint-verification -->

Verification

All 861 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4468-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2235 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 863 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

`Checkpoint_795_IntegerIntervalWeakCounting.md`

Checkpoint 795: Prove the finite weak density bound and quantitative stopping packing

Proves the counting-measure weak (1,1) inequality for the union of all intervals in a finite family whose F-density exceeds a given threshold. The constant is 3 and follows from the checked disjoint interval selection and exact finite intersection counts. Derives a concrete stopping packing bound: at threshold 64|F|/L the bad union has cardinality at most L/4. This supplies the quantitative packing ingredient without assuming a maximal-function theorem.

Checked declarations

Module	Named supporting theorems
IntegerIntervalWeakCounting.lean	4

- `disjoint_interval_intersection_count`
- `integerInterval_weak_counting`
- `integerBadIntervalUnion_counting`
- `integerBadIntervalUnion_packing`

<!-- checkpoint-verification -->

Verification

All 862 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4469-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2239 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 864 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

`Checkpoint_796_FiniteSparseGluing.md`

Checkpoint 796: Prove exact gluing of recursive interval costs and packing

Proves finite gluing below pairwise disjoint stopping children. The interval subfamilies are disjoint as finite sets, the strict-descendant union at every interior node stays in its original child construction, the

parent packing bound propagates, and the recursive cost sum is exactly the sum over the glued finite family. No sparse recursion is assumed in these algebraic and cardinality deductions.

Checked declarations

Module	Named supporting theorems
FiniteSparseGluing.lean	4

- `intervalSubfamilies_disjoint`
- `integerDescendantUnion_glue_subset`
- `integerSparse_glue_packing`
- `integerSparse_glue_cost`

<!-- checkpoint-verification -->

Verification

All 863 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4470-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2243 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 865 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

`Checkpoint_797_FiniteSparseRecursion.md`

Checkpoint 797: Prove termination and sparse cost summation of the finite stopping recursion

Proves termination and cost summation of finite interval stopping recursion by strong induction on integer length. A one-generation recursive inequality and quarter-packing produce a finite family with actual 3/4-sized pairwise disjoint major subsets and the required summed cost. The local one-generation inequality is explicitly a hypothesis of this abstract theorem and still has to be proved for the actual averages; it is not declared an external analytic input or the completed paper recursion.

Checked declarations

Module	Named supporting theorems
FiniteSparseRecursion.lean	3

- `integerLaminar_subset`
- `finiteSparseRecursion_certificate`
- `finiteSparseRecursion`

<!-- checkpoint-verification -->

Verification

All 864 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4471-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2246 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 866 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

`Checkpoint_798_StoppingPairingMajorants.md`

Checkpoint 798: Prove actual varying-scale high and low pairing bounds from Proposition 4.19

Proves finite variable-scale pairings from genuine real l2 pointwise majorants, then applies the exact countable high maximum and local low estimate of completed Proposition 4.19 at an arbitrary spatially varying dyadic arithmetic scale. The high pairing uses the actual supremum representative and its l2 norm. A fixed-scale norm is not substituted. These estimates are deductions from Proposition 4.19; no new external input is used. The local error maximum, ordinary-square stopping boundary and final sparse assembly are still separate work.

Checked declarations

Module	Named supporting theorems
StoppingPairingMajorants.lean	4

- `finite_pairing_real_l2_majorant`
- `finite_threePiece_majorant_bound`
- `selectedPrimeHigh_stopping_pairing`
- `selectedPrimeLow_stopping_pairing`

<!-- checkpoint-verification -->

Verification

All 865 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4472-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2250 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 867 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

`Checkpoint_799_LaminarSparseRecursion.md`

Checkpoint 799: Construct sparse families without an ambient dyadic-grid assumption

Strengthens the finite stopping recursion: pairwise disjoint children themselves generate a laminar output, even when the ambient finite collection contains every subinterval and is not a dyadic grid. Proves

laminarity under gluing, termination, actual major-subset packing and the exact final sparse cost bound. This permits the direct finite interval proof used for Theorem 1.3. Supplying the actual one-generation inequality remains unfinished.

Checked declarations

Module	Named supporting theorems
LaminarSparseRecursion.lean	3

- `integerLaminar_glue`
- `laminarSparseRecursion_certificate`
- `laminarSparseRecursion`

<!-- checkpoint-verification -->

Verification

All 866 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4473-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2253 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 868 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

`Checkpoint_800_IntegerIntervalGeometry.md`

Checkpoint 800: Prove interval inclusion, connected unions and dilation geometry

Proves exact inclusion criteria, endpoint membership, length comparison, connected union via interval hull, and preservation of inclusion under fivefold dilation for actual nonempty integer intervals. These facts support a direct stopping construction with maximal consecutive bad intervals; the output need not depend on a dyadic-grid replacement.

Checked declarations

Module	Named supporting theorems
IntegerIntervalGeometry.lean	10

- `integerInterval_left_mem`
- `integerInterval_last_mem`
- `integerInterval_subset_iff`
- `integerInterval_subset_length`
- `integerInterval_subset_equal_length`
- `integerInterval_hull_mem`
- `integerInterval_subset_hull_left`
- `integerInterval_subset_hull_right`
- `integerInterval_hull_eq_union`
- `integerInterval_five_subset`

<!-- checkpoint-verification -->

Verification

All 867 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4474-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2263 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 869 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

`Checkpoint_801_FiniteIntegerSubintervals.md`

Checkpoint 801: Construct maximal consecutive bad intervals and prove their exact disjoint covering

Constructs the finite collection of every nonempty integer subinterval of a parent, with an exact membership characterization. Constructs maximal consecutive component intervals of a finite bad set, proves that they cover that set exactly and are pairwise disjoint, and proves their inclusion in the parent. These are actual intervals and finite sets, not a chosen abstract partition.

Checked declarations

Module	Named supporting theorems
FiniteIntegerSubintervals.lean	7

- `mem_integerSubintervals`
- `integerSubintervals_self`
- `integerSubintervals_trans`
- `integerSingletonInterval_points`
- `integerMaximalComponents_subset`
- `integerMaximalComponents_cover`
- `integerMaximalComponents_disjoint`

<!-- checkpoint-verification -->

Verification

All 868 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4475-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2270 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 870 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator

conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

Checkpoint_802_SelectedPrimeErrorMaximum.md

Checkpoint 802: Prove the actual selected error maximum over all valid dyadic scales

Proves an actual countable l2 error maximum over every dyadic scale satisfying the fixed source cutoff. Invalid scales are masked to zero and valid scales are exactly the selected error operator of Proposition 4.19. Its A=2 clause gives summable operator norms and a genuine l2 supremum with norm at most K Q^(-1/2) times the input norm. This proves the variable-scale error control without an extra logarithmic loss or a new deep input.

Checked declarations

Module	Named supporting theorems
SelectedPrimeErrorMaximum.lean	2

- `selectedPrimeValidError_of_valid`
- `selectedPrimeError_countable_maximum`

<!-- checkpoint-verification -->

Verification

All 869 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4476-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2272 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 871 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

Checkpoint_803_IntegerBadComponentNeighbors.md

Checkpoint 803: Find genuine good neighboring points and prove strict stopping-length decrease

Proves that each maximal consecutive bad interval has good neighboring points whenever those neighbors lie in the parent. A quarter-packing bad set leaves a good point, each component is strictly shorter than the parent, and every point in a component has a good parent point at distance at most the component length. These facts supply both termination and the admissibility of the remaining stopping scales.

Checked declarations

Module	Named supporting theorems
IntegerBadComponentNeighbors.lean	7

- `integerInterval_extendLeft_points`
- `integerInterval_extendRight_points`
- `integerComponent_left_neighbor_good`

- integerComponent_right_neighbor_good
- integerComponent_near_good
- integerComponent_length_lt
- integer_good_point_of_quarter_packing

<!-- checkpoint-verification -->

Verification

All 870 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4477-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2279 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 872 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

Checkpoint_804_GoodPointLocalMass.md

Checkpoint 804: Control the full real-radius local average from a nearby good point

Proves that a good nearby point controls sourceLocalMass at every real radius M>=N, not merely at dyadic radii. Establishes exact finite-indicator ball counts, integer centered-ball geometry and the bound sourceLocalMass <= 5t from tested interval densities and the parent density. This supplies the analytic/geometric bridge required by the actual low operator of Proposition 4.19.

Checked declarations

Module	Named supporting theorems
GoodPointLocalMass.lean	5

- integerCenteredBall_mem
- integerCenteredBall_subset_five
- sourceBallMass_finiteIndicator
- sourceBallMass_le_interval_count
- goodPoint_sourceLocalMass

<!-- checkpoint-verification -->

Verification

All 871 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4478-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2284 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 873 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and

the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

`Checkpoint_805_PrimeStoppingGeometry.md`

Checkpoint 805: Construct quarter-packed stopping children and prove actual source admissibility

Constructs the actual parent bad set by testing every integer subinterval of the fivefold parent at density threshold 64|F|/|I|. Proves quarter packing, exact covering by disjoint maximal bad components, strict decrease of their lengths and the good-point density property. For every remaining arithmetic scale N, proves the literal sourceLocalMass bound <=320|F|/|I|, uniformly over every real radius M>=N. These deductions supply admissibility rather than assuming it.

Checked declarations

Module	Named supporting theorems
PrimeStoppingGeometry.lean	7

- `primeStoppingBadSet_subset`
- `primeStoppingBadSet_packing`
- `primeStoppingChildren_cover`
- `primeStoppingChildren_disjoint`
- `primeStoppingChildren_geometry`
- `primeStopping_good_density`
- `primeStopping_remaining_localMass`

<!-- checkpoint-verification -->

Verification

All 872 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4479-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2291 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 874 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

`Checkpoint_806_FiniteAnnularMaximum.md`

Checkpoint 806: Define and control actual truncated dyadic prime-square maxima

Defines the actual finite dyadic annular prime-square maximum with arithmetic cutoff 2^j*2 and spatial cutoff N^2<=L. Proves exact index coverage, monotonicity, attainment on nonempty finite scale sets, elementary maximum bounds and nonnegativity of the localized finite maximal pairing. The zero-scale convention is an actual nonnegative finite supremum.

Checked declarations

Module	Named supporting theorems
FiniteAnnularMaximum.lean	9

- `mem_dyadicSpatialIndices`
- `dyadicSpatialIndices_mono`
- `dyadicSpatialIndices_empty`
- `finiteAnnularMaximum_dominates`
- `finiteAnnularMaximum_mono`
- `finiteAnnularMaximum_attained`
- `finiteAnnularMaximum_le`
- `finiteAnnularMaximum_union`
- `annularLocalMaxPairing_nonneg`

<!-- checkpoint-verification -->

Verification

All 873 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4480-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2300 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 875 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

`Checkpoint_807_AnnularStoppingSupport.md`

Checkpoint 807: Prove exact compact-support localization for the actual stopping averages

Proves exact localization of the actual annular square kernel to the fivefold interval whenever its spatial scale N^2 is at most the interval length. Proves compatibility of parent and child restrictions, using actual compact kernel support. Also proves monotonicity of positive-kernel indicator outputs and specializes it to the actual von Mangoldt annular averages. Smoothed low kernels are not assigned the original support.

Checked declarations

Module	Named supporting theorems
AnnularStoppingSupport.lean	4

- `annularSquare_indicator_localize`
- `annularSquare_indicator_restrict_twice`
- `nonnegativeKernel_indicator_norm_mono`
- `annularMangoldt_indicator_norm_mono`

<!-- checkpoint-verification -->

Verification

All 874 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4481-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2304 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 876 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

`Checkpoint_808_PrimeRemainingScaleRecursion.md`

Checkpoint 808: Prove the exact one-generation recursion for the actual prime-square maximum

Proves the actual one-generation finite maximal recursion. Every dyadic arithmetic scale is assigned either to the admissible remaining-scale maximum or to the unique smaller stopped interval containing the point. The residual average is identified exactly with the child-localized original kernel using proved compact support, and the finite test pairing is bounded by the parent remaining contribution plus the exact sum of child maximal pairings. No final local endpoint estimate is assumed in this decomposition.

Checked declarations

Module	Named supporting theorems
PrimeRemainingScaleRecursion.lean	5

- `primeRemainingIndices_subset`
- `primeRemainingScale_admissible`
- `primeRemainingScale_pointwise`
- `finiteSum_interval_mask`
- `primeRemainingScale_pairing`

<!-- checkpoint-verification -->

Verification

All 875 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4482-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2309 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 877 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved.

`Checkpoint_809_IndependentStoppingInputs.md`

Checkpoint 809: Record the precise independent stopping and maximal analytic boundaries

Records two explicit independent analytic interfaces needed for sparse applications: the ordinary-square stopping low/high decomposition, and complementary-exponent prime-square maximal Lp pairings. Their existence is not proved or added as a Lean axiom. The ordinary input is stronger than the earlier fixed-scale square interface. The support, normalization, admissibility and sign adaptations remain within its unformalized scope, as do prime normalization and prime-power extension in the maximal input. Proves monotonicity of the square constant and restriction of the maximal test set. No endpoint improving or sparse conclusion of this paper is a field of these predicates.

Checked declarations

Module	Named supporting theorems
IndependentStoppingInputs.lean	2

- `ordinarySquareStoppingInput_mono`
- `independentPrimeMaximal_restrict_tests`

<!-- checkpoint-verification -->

Verification

All 876 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4483-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2311 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 878 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_810_AdmissibleAnnularLinearization.md`

Checkpoint 810: Prove exact admissible linearization and the sparse deduction from a local estimate

Proves exact finite linearization of the remaining-scale maximum on its actual active test set. The chosen dyadic scale attains that maximum, satisfies the parent spatial cutoff and the proved local-average admissibility bound. Combines the actual one-generation recursion with the terminating laminar construction to obtain a genuine finite 3/4-sparse bound from an explicit admissible local estimate. That local estimate remains a deduction to be supplied from the recorded analytic inputs, not a newly assumed independent result.

Checked declarations

Module	Named supporting theorems
AdmissibleAnnularLinearization.lean	5

- `finite_positive_pairing_norm`
- `annularDyadicValue_positive`

- `primeRemainingScale_linearized`
- `primeAnnularSparseStep`
- `primeAnnularSparse_from_local`

<!-- checkpoint-verification -->

Verification

All 877 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4484-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2316 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 879 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_811_LocalPrimeScalarBalance.md`

Checkpoint 811: Balance the local prime low high and maximal-error terms uniformly in density

Proves the uniform stopping scalar balance for the actual low, high and maximal-error terms, with every coefficient strictly above the sharp leading coefficient. The chosen dyadic Q obeys log Q <= B paper-Log(delta) with one fixed B, and the bound includes the full bounded-density range through an explicit compact estimate. No optimization or cutoff-choice conclusion is assumed.

Checked declarations

Module	Named supporting theorems
LocalPrimeScalarBalance.lean	2

- `localPrimeScalar_eq`
- `exists_localPrime_balance`

<!-- checkpoint-verification -->

Verification

All 878 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4485-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2318 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 880 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and

the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_812_UnrestrictedHighLowBalance.md`

Checkpoint 812: Prove the unrestricted two-term stopping high-low optimization

Proves the unconstrained two-term high/low scalar endpoint directly from the already checked exact balance and a dyadic splitting parameter. Covers both large and bounded preferred splitting scales and retains the exact density and logarithmic exponents. This is the optimization needed for the independent ordinary-square stopping low/high estimates.

Checked declarations

Module	Named supporting theorems
UnrestrictedHighLowBalance.lean	1

- `unrestricted_highLow_scalar`

<!-- checkpoint-verification -->

Verification

All 879 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4486-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2319 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 881 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_813_DyadicHighFactorComparison.md`

Checkpoint 813: Control all bounded dyadic cutoffs in the sharp high factor

Proves that the regularized sharp high factor of Proposition 4.19 is bounded by a fixed constant times the printed unregularized Orlicz high factor for every dyadic Q. The large-Q expressions agree exactly, and the remaining four dyadic indices are absorbed by an explicit positive finite sum of exponential ratios. No singular compact interval in the unregularized logarithm is incorrectly treated as uniformly harmless.

Checked declarations

Module	Named supporting theorems
DyadicHighFactorComparison.lean	1

- `regularizedHighFactor_dyadic`

<!-- checkpoint-verification -->

Verification

All 880 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4487-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2320 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 882 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_814_OrdinarySquareStoppingEndpoint.md`

Checkpoint 814: Deduce the ordinary-square stopping endpoint from the independent low-high input

Deduces the exact ordinary-square stopping endpoint with density exponent 2/3 and logarithmic exponent 4/3 from the explicitly independent ordinary-square low/high input. The finite test set may be restricted while the reference density retains a larger test set; this is proved before optimization and avoids assuming monotonicity of a logarithmic bump. Parent support, spatial cutoff and the 320 admissibility constant are retained explicitly.

Checked declarations

Module	Named supporting theorems
OrdinarySquareStoppingEndpoint.lean	2

- `ordinaryStopping_pairing_scalar`
- `ordinaryStopping_endpoint`

<!-- checkpoint-verification -->

Verification

All 881 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4488-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2322 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 883 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma

3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_815_NormalizedStoppingPairing.md`

Checkpoint 815: Normalize the genuine three-piece varying-scale pairing

Proves the normalized finite three-piece stopping pairing bound with a larger reference test set, a genuine l2 high maximum and a genuine l2 error maximum. Checks the density normalization exactly and identifies the result with the actual scalar low/high/error expression. No scale-dependent norm is substituted for a maximal estimate.

Checked declarations

Module	Named supporting theorems
NormalizedStoppingPairing.lean	1

- `normalizedStoppingPairing`

<!-- checkpoint-verification -->

Verification

All 882 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4489-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2323 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 884 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_816_PrimeLargeStoppingPairing.md`

Checkpoint 816: Prove the actual uniform large-stopping-scale prime pairing

Proves the actual uniform large-stopping-scale prime pairing from the common family in Proposition 4.19. All constants are chosen before the interval, input and reference sets, varying arithmetic scale, dyadic Q and starting threshold. The low local bound, actual countable high maximum and actual valid-error maximum are combined with the same selected Page data and normalized to the checked scalar expression. No own-paper pairing conclusion is an external input.

Checked declarations

Module	Named supporting theorems
PrimeLargeStoppingPairing.lean	3

- `annularDyadicValue_l2`
- `annularDyadicValue_selected_decomposition`
- `primeLargeStopping_pairing`

<!-- checkpoint-verification -->

Verification

All 883 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4490-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2326 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 885 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_817_PrimeStoppingThreshold.md`

Checkpoint 817: Construct the balanced source-range starting threshold

Constructs an explicit starting scale from Q and the positive source cutoff coefficient. Proves it is at least 2, satisfies the exact Proposition 4.19 range, and has logarithm at most a fixed multiple of paper-Log(delta)^2 whenever the balanced log Q bound holds. The coefficient is uniform in the interval, sets and density.

Checked declarations

Module	Named supporting theorems
PrimeStoppingThreshold.lean	1

- `primeStoppingStart_bounds`

<!-- checkpoint-verification -->

Verification

All 884 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4491-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2327 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 886 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma

3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_818_PrimeSmallStoppingDomination.md`

Checkpoint 818: Prove actual small-stopping-scale domination by ordinary squares

Proves actual pointwise prime-annular domination by the full ordinary-square average with the exact log N weight, using positivity and the proved full-to-annular identity. Proves its genuinely varying-scale finite pairing consequence on every set of small stopping scales. No square endpoint or maximal theorem is assumed in this elementary domination step.

Checked declarations

Module	Named supporting theorems
PrimeSmallStoppingDomination.lean	3

- `ordinaryDyadicValue_positive`
- `annularDyadicValue_square_domination`
- `primeSmallStopping_square_domination`

<!-- checkpoint-verification -->

Verification

All 885 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4492-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2330 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 887 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_819_PrimeStoppingLocalEndpoint.md`

Checkpoint 819: Prove the actual sharp admissible prime-square local endpoint

Proves the actual admissible prime-square local endpoint, uniformly in every parent interval, finite input/reference/test sets and varying dyadic arithmetic scale. The density retains the larger reference test set. Small scales use the proved prime-to-ordinary comparison and the independent ordinary low/high

input; large scales use the exact common Proposition 4.19 family and the proved scalar balance. The sharp coefficient is every cstar>4sqrt(3)/9, all bounded densities are included, and fixed logarithmic powers are absorbed by the actual paper Phi. No local endpoint conclusion is an assumed input.

Checked declarations

Module	Named supporting theorems
PrimeStoppingLocalEndpoint.lean	2

- `finiteSetDensity_five_bounds`
- `primeStopping_local_endpoint`

<!-- checkpoint-verification -->

Verification

All 886 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4493-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2332 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 888 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_820_PrimeSparseRoot.md`

Checkpoint 820: Construct a finite root containing every nonzero annular test scale

Constructs an explicit finite root interval for every pair of finite sets using a finite absolute-value radius. Proves that every actual annular dyadic scale outside its spatial cutoff vanishes on the actual test set, and every scale is dominated by the finite root maximum. Identifies the localized root pairing with the exact unrestricted-set finite maximum. This avoids any unproved limiting selection of sparse families.

Checked declarations

Module	Named supporting theorems
PrimeSparseRoot.lean	6

- `primeSparseRadius_abs`
- `primeSparseRoot_contains`
- `annularDyadicValue_zero_of_far`
- `primeSparseRoot_tail_zero`
- `primeSparseRoot_dominates`
- `primeSparseRoot_pairing`

<!-- checkpoint-verification -->

Verification

All 887 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4494-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2338 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 889 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_821_PrimeFullMaximalComparison.md`

Checkpoint 821: Control the literal all-cutoff full prime-square maximum by the finite root

Defines the literal countable full maximal indicator over every original integer cutoff N>=2 and proves its range is bounded and it dominates each cutoff. Proves positive-kernel comparison with a nearby dyadic full cutoff, then controls every full dyadic cutoff by the finite annular root through the exact full-to-annular identity. The resulting actual maximal pairing is at most four times the finite localized root pairing. No large-scale truncation, limiting sparse-family selection or maximal Lp hypothesis is used for this reduction.

Checked declarations

Module	Named supporting theorems
PrimeFullMaximalComparison.lean	10

- `fullPrimeKernel_cutoff_bound`
- `fullPrimeIndicator_cutoff_bound`
- `primeSparseRoot_full_dyadic`
- `primeSparseRoot_full_cutoff`
- `primeSquareAverageValue_indicator_kernel`
- `primeSquareMaximalIndicator_bdd`
- `primeSquareMaximalIndicator_dominates`
- `primeSquareMaximalIndicator_nonneg`
- `primeSquareMaximalPairing_nonneg`
- `primeSquareMaximalPairing_le_root`

<!-- checkpoint-verification -->

Verification

All 888 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4495-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2348 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 890 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma

3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

Checkpoint_822_PrimeEndpointSparse.md

Checkpoint 822: Complete the sharp endpoint sparse conclusion for the literal full maximum

Completes the endpoint sparse conclusion of Theorem 1.3 as an implication from the recorded independent analytic inputs. The literal full maximal indicator over every N>=2 is bounded by a finite family of actual integer intervals with actual pairwise disjoint major subsets of size at least one tenth. The sparse form has the printed 2/3 density exponent and every coefficient cstar>4sqrt(3)/9. No local endpoint or sparse conclusion is assumed. The open-region clause of Theorem 1.3 is still being assembled, so the full source theorem is not yet awarded.

Checked declarations

Module	Named supporting theorems
PrimeEndpointSparse.lean	2

- `primeSparse_from_admissible_local`
- `primeEndpointSparse`

<!-- checkpoint-verification -->

Verification

All 889 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4496-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2350 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 891 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

Checkpoint_823_SparseTriangleGeometry.md

Checkpoint 823: Cover the full open sparse triangle with a strict interpolation margin

Defines the exact open sparse triangle by its three strict edge inequalities and proves equivalence with strictly positive barycentric coordinates at the three printed vertices. Proves that every point in the full

open triangle admits interpolation between a diagonal exponent strictly below 2/3 and a strictly interior complementary-exponent point. This supplies the quantitative margin that absorbs the endpoint Orlicz factor.

Checked declarations

Module	Named supporting theorems
SparseTriangleGeometry.lean	2

- `sourceSparseTriangle_iff_barycentric`
- `sourceSparseTriangle_interpolation`

<!-- checkpoint-verification -->

Verification

All 890 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4497-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2352 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 892 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_824_RestrictedGeometricInterpolation.md`

Checkpoint 824: Prove exact restricted geometric interpolation and complementary normalization

Proves exact normalization of complementary count powers and geometric interpolation of two restricted scalar bounds. The interpolation is proved from logarithmic inequalities for positive values, with the zero value handled separately, and retains the exact target exponents. No abstract interpolation theorem or own-paper sparse estimate is assumed.

Checked declarations

Module	Named supporting theorems
RestrictedGeometricInterpolation.lean	2

- `complementary_count_normalization`
- `restricted_geometric_interpolation`

<!-- checkpoint-verification -->

Verification

All 891 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4498-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2354 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 893 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

Checkpoint_825_PrimeStoppingOpenEndpoint.md

Checkpoint 825: Prove the local estimate at every point of the full open sparse triangle

Completes the actual admissible local prime-square estimate at every point of the full open sparse triangle. Uses the proved strict interpolation geometry, the sharp local endpoint with its actual Phi loss, proved uniform absorption by an arbitrarily small density power, and the independent complementary-exponent maximal input. The geometric interpolation and exact interval normalization are checked, including the zero pairing, and no local power estimate or sparse conclusion is assumed.

Checked declarations

Module	Named supporting theorems
PrimeStoppingOpenEndpoint.lean	1

- `primeStopping_open_endpoint`

<!-- checkpoint-verification -->

Verification

All 892 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4499-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2355 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 894 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

Checkpoint_826_PrimeOpenSparse.md

Checkpoint 826: Complete the full open-region sparse conclusion

Completes the open-region sparse conclusion at every point of the full source triangle, including its reciprocal-exponent interpretation. The local estimate is supplied by the proved endpoint/density-loss interpolation and the independent complementary-exponent maximal input. The literal full all-cutoff maximum is bounded by a finite family of actual integer intervals with actual pairwise disjoint major subsets of size at least one tenth, and the power form has exactly the printed averages. The final combined source theorem and statement review are next.

Checked declarations

Module	Named supporting theorems
PrimeOpenSparse.lean	1

- `primeOpenSparse`

<!-- checkpoint-verification -->

Verification

All 893 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4500-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2356 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 895 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_827_SparseDyadicQualification.md`

Checkpoint 827: Check and document the required dyadic scope of the abstract recursion

Proves a concrete positive compactly supported convolution counterexample showing why the abstract recursion lemma must specify a dyadic maximum when its local hypothesis tests only dyadic stopping values. The kernel is zero at every dyadic index and equals the identity at N=3; all local dyadic pairings vanish but the unrestricted maximum is at least one on singleton test sets. Saves a proposed statement qualification for source Lemma 5.10 (l:sparserecursion) and a precise all-cutoff reduction paragraph for Theorem 1.3. These are proposed only, not applied to the current manuscript. The actual prime-square all-cutoff comparison and both sparse conclusions are already proved separately, so no new hypothesis is added to them.

Checked declarations

Module	Named supporting theorems
SparseDyadicQualification.lean	7

- `two_pow_ne_three`
- `sparseNonDyadicKernel_positive`
- `sparseNonDyadicKernel_support`

- `sparseNonDyadicKernel_dyadic_zero`
- `sparseNonDyadic_local_pairing_zero`
- `sparseNonDyadic_full_pairing_one`
- `sparseNonDyadic_maximum_positive`

<!-- checkpoint-verification -->

Verification

All 894 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4501-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2363 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 896 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed numbered results under their paper hypotheses remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, Theorem 4.1 and Corollary 4.15. Lemma 3.21, both conclusions of Theorem 1.1, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. The whole paper, the stronger printed auxiliary lemmas and the later sparse/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_828_TheoremOneThree.md`

Checkpoint 828: Complete both source conclusions of Theorem 1.3 with actual sparse families

Both source sparse conclusions are complete. The final theorem_1_3 proves the sharp endpoint for every cstar>4sqrt(3)/9 and the full open triangle, with actual finite interval families and pairwise disjoint major subsets of size at least one tenth. The literal full maximal indicator covers every integer N>=2. All local endpoint, stopping, support, packing, recursion, interpolation and all-cutoff deductions are proved. Independent raw distribution/Page, prime-limit, smooth multifrequency/sampling, ordinary-square stopping and complementary-exponent maximal inputs remain explicit hypotheses, not new Lean axioms. A checked counterexample motivates an unapplied dyadic scope clarification for abstract Lemma5.10; the actual prime-square theorem already has the required all-cutoff comparison. Only Theorem1.3 gains full source completion credit; three general Section5 auxiliary statements gain partial credit, and the rest of the paper remains unfinished.

Checked declarations

Module	Named supporting theorems
TheoremOneThree.lean	4

- `integerSparseFamily_iff_collection`
- `theorem_1_3`
- `theorem_1_3_endpoint`
- `theorem_1_3_open_pq`

<!-- checkpoint-verification -->

Verification

All 895 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4502-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2367 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 897 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_829_NearOneGauge.md`

Checkpoint 829: Literal Section7 Orlicz gauge and convexity

The literal Section7 integrand and its primitive Psi_C are defined. A continuous clamp extends the density to negative arguments and is proved identical to the printed formula for every nonnegative argument. Lean proves positivity, monotonicity and continuity of the density; the fundamental theorem of calculus for Psi_C; its continuity, monotonicity and convexity; Psi_C(0)=0; Psi_C(t)>=t for C>=0,t>=0; and the upper comparison Psi_C(t)<=t exp(C rate(t)). No analytic theorem is assumed. The matching lower comparison, doubling, quantitative optimization and all operator conclusions remain to be developed. This checkpoint does not close Theorem t:orlicz-maximal (archived number7.2) or either ergodic corollary.

Checked declarations

Module	Named supporting theorems
NearOneGauge.lean	17

- `nearOneLog_one_le`
- `nearOneLog_mono`
- `nearOneRate_nonneg`
- `nearOneRate_mono`
- `nearOneDensity_continuous`
- `nearOneDensity_pos`
- `nearOneDensity_one_le`
- `nearOneDensity_mono`
- `nearOnePsi_zero`
- `nearOnePsi_nonneg`
- `nearOnePsi_hasDerivAt`
- `nearOnePsi_continuous`
- `nearOnePsi_mono`
- `nearOnePsi_convex`
- `nearOnePsi_ge_self`
- `nearOnePsi_le_density`
- `nearOneGauge_literal`

<!-- checkpoint-verification -->

Verification

All 896 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4503-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2384 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 898 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_830_NearOneGaugeComparison.md`

Checkpoint 830: Orlicz gauge comparison and doubling with explicit constants

The matching lower comparison and doubling of the literal Section7 gauge are now proved. A direct algebraic logarithmic tangent estimate proves sqrt(y log(e+y)) <= sqrt(x log(e+x))+2(y-x) for 1<=x<=y. Consequently the actual density satisfies the fixed-rescaling bound exp(2 C log a) for every a>=1. Integration over the final half interval gives the lower gauge comparison, and together with the upper comparison proves the displayed asymptotic equivalence in e:orlicz-gauge-comparison with explicit constants and a doubling inequality. These are unconditional scalar and integration deductions. No IW, maximal, modular or ergodic bound is assumed or claimed here.

Checked declarations

Module	Named supporting theorems
NearOneGaugeComparison.lean	7

- `nearOne_innerLog_bounds`
- `nearOneCore_lipschitz`
- `nearOneRate_rescale`
- `nearOneDensity_rescale`
- `nearOnePsi_ge_half_density`
- `nearOnePsi_comparison`
- `nearOnePsi_doubling`

<!-- checkpoint-verification -->

Verification

All 897 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4504-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2391 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 899 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit

independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_831_NearOneQuantitativeGrowth.md`

Checkpoint 831: Quantitative Orlicz power absorption uniform near one

Lean proves the quantitative small-power absorption required near p=1. For every C>=1, 0<e<=1 and v>=1, C sqrt(v log(euler+v)) <= e v/2 + K(C)/e log(euler/e), where K(C)>0 is explicit and independent of e,v. Here euler means exp(1), while the variable e is p-1. The proof uses a logarithmic tangent inequality and a squared Young inequality, with no asymptotic implicit constants. Substituting the literal gauge logarithm gives density_C(t) <= exp(K(C)/e log(exp(1)/e)) (exp(1)+max(0,t))^(e/2) for every real t. This is the scalar dependence in c:orlicz-p; the operator distributional estimate and layer-cake integration are not assumed or claimed proved at this checkpoint.

Checked declarations

Module	Named supporting theorems
NearOneQuantitativeGrowth.lean	5

- `nearOne_log_affine`
- `root_product_absorption`
- `nearOneGrowthConstant_pos`
- `nearOneCore_quantitative`
- `nearOneDensity_small_power`

<!-- checkpoint-verification -->

Verification

All 898 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4505-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2396 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 900 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_832_FiniteDistributionBounds.md`

Checkpoint 832: Finite testing and actual distributional cardinal bounds

A finite-test distribution formalism is proved equivalent to finiteness and the actual level-set cardinal bound. Infinite level sets cannot silently receive cardinality zero: arbitrarily large finite subsets give a checked contradiction. Lean proves Chebyshev bounds for every positive real exponent, their actual lp-space specialization, monotonicity, the two-term distribution union estimate, and the finite-family sum estimate. These are the counting tools for Section7 high-low and layer-cake deductions. They contain no arithmetic, restricted endpoint or modular estimate as an assumed external theorem.

Checked declarations

Module	Named supporting theorems
FiniteDistributionBounds.lean	9

- `finiteDistribution_nonneg`
- `testedDistribution_finite`
- `testedDistribution_ncard`
- `testedDistribution_iff`
- `finiteDistribution_chebyshev`
- `finiteDistribution_lp`
- `finiteDistribution_mono`
- `finiteDistribution_two`
- `finiteDistribution_finite_sum`

<!-- checkpoint-verification -->

Verification

All 899 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4506-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2405 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 901 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_833_NearOneLayerAbsorption.md`

Checkpoint 833: Global absorption of dyadic layer losses by the literal gauge

The dyadic layer-cake loss is absorbed into the literal gauge by proved uniform rescaling estimates. Lean proves coefficient monotonicity, rate(t)>=sqrt(log(exp(1)+t)), log(v)<=2sqrt(v) for v>=1, and a general variable-rescaling inequality when log(a)<=L+d rate(t). In particular a factor bounded by D log(exp(1)+t)^2 changes C to 9C+4 with an explicit constant depending only on C,D. This replaces the informal little-o sentence in the source by a global bound, including all bounded layers. The maximal operator restricted estimate is not assumed here, and its arithmetic derivation remains to be connected.

Checked declarations

Module	Named supporting theorems
NearOneLayerAbsorption.lean	6

- `nearOneDensity_coefficient_mono`
- `nearOnePsi_coefficient_mono`
- `nearOneRate_ge_sqrt_log`
- `log_le_two_sqrt`
- `nearOnePsi_variable_rescale`

- `nearOnePsi_log_polynomial`

<!-- checkpoint-verification -->

Verification

All 900 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4507-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2411 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 902 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_834_PositiveAverageMaximum.md`

Checkpoint 834: Genuine positive averaging maxima and finite layer majorization

The genuine countable maximum of a general positive finite averaging family is constructed and controlled. The family records its actual integer shifts, finite index sets, nonnegative real weights and a uniform mass bound. For bounded inputs the supremum range is proved bounded, every original average is dominated, and the maximum is nonnegative. The complex finite averages satisfy the exact indicator identity and a finite layer-majorization theorem with the correct mass contribution of the bounded remainder. These are proved algebraic and order facts, ready for the actual square, von Mangoldt and prime normalizations. No endpoint or IW estimate is a field of the family.

Checked declarations

Module	Named supporting theorems
PositiveAverageMaximum.lean	10

- `finiteFunction_norm_bound`
- `positiveAverage_norm_bound`
- `positiveMaximal_bdd`
- `positiveMaximal_dominates`
- `positiveMaximal_nonneg`
- `positiveMaximal_norm_bound`
- `finiteIndicator_norm_le_one`
- `positiveMaximal_indicator_bound`
- `positiveAverage_indicator_real`
- `positiveMaximal_layer_majorization`

<!-- checkpoint-verification -->

Verification

All 901 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4508-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2421 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 903 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_835_DyadicFunctionLayers.md`

Checkpoint 835: Actual finite dyadic layers and exact cost partition

The finite dyadic layer decomposition is constructed for arbitrary complex finitely supported inputs. A chosen integer height satisfies the exact dyadic bracketing; layers are actual finite subsets, pairwise disjoint, and their indices form a finite set. Lean proves exact partition of layer costs and pointwise domination of the input by the bounded remainder plus the weighted indicators. The half-open convention at dyadic boundary values is explicit, and does not discard points. This step is algebraic and does not assume a restricted weak estimate.

Checked declarations

Module	Named supporting theorems
DyadicFunctionLayers.lean	7

- `functionLayerIndex_spec`
- `functionLayer_mem`
- `functionLayer_disjoint`
- `functionLayer_nonempty_index`
- `functionLayer_bounds`
- `functionLayer_partition_cost`
- `functionLayer_majorization`

<!-- checkpoint-verification -->

Verification

All 902 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4509-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2428 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 904 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit

independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_836_DyadicLayerGaugeCost.md`

Checkpoint 836: Uniform dyadic gauge costs and truncated level-set sum

The literal gauge pays for all dyadic layer costs globally. For K>=2 and every natural layer j, Lean proves Psi_C(4*2^j*(j+1)^2) <= L(C,K) Psi_(9C+4)(2^j/K) with an explicit positive constant. The proof includes the bounded layers and all dyadic endpoints. A checked finite inverse-square sum bound controls the threshold allocation, and exact layer partitioning gives the truncated modular cost summed over the actual set where |f|>lambda/K. No asymptotic sentence or unproved rearrangement of infinite sums is used.

Checked declarations

Module	Named supporting theorems
DyadicLayerGaugeCost.lean	4

- `layerGaugeConstant_pos`
- `dyadicLayer_gauge_cost`
- `finite_inverse_square_sum`
- `functionLayer_gauge_sum`

<!-- checkpoint-verification -->

Verification

All 903 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4510-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2432 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 905 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_837_RestrictedToModular.md`

Checkpoint 837: Complete finite layer-cake passage from restricted to truncated modular estimates

The general restricted-to-truncated-modular implication is proved for a genuine countable positive averaging maximum. Starting with a restricted indicator distribution bound (a reusable internal hypothesis, not an independent prime-square input), Lean constructs the finite layers, proves the threshold union inclusion, pays the threshold sum, applies the restricted estimate to every actual layer, and absorbs its cost by the literal Psi_(9C+4). The result has C0=2 times the actual mass bound and the exact input truncation |f|>lambda/C0. No circular claim that the actual arithmetic restricted estimate is already proved is made. The actual restricted estimate must still be obtained from the IW/arithmetic argument before source Theorem t:orlicz-maximal (archived number7.2) can be declared complete.

Checked declarations

Module	Named supporting theorems
RestrictedToModular.lean	2

- `finiteDistribution_positive_mul`
- `positiveRestricted_to_truncated_modular`

<!-- checkpoint-verification -->

Verification

All 904 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4511-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2434 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 906 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_838_ActualPositiveSquareFamilies.md`

Checkpoint 838: Actual square von Mangoldt and prime averaging families

All three averaging families in the printed Section7 theorem now instantiate the positive finite-family framework. They use the literal square shifts x-n^2. The von Mangoldt family uses Mathlib Lambda including prime powers and the proved Chebyshev mass bound. The ordinary-square family uses 1/N. The unweighted-prime family uses the actual primesLE finite set and Mathlib primeCounting pi(N), with exactly the normalization 1/pi(N). Their actual finite averages and countable maxima over every integer N>=2 are proved identical to the concrete formulas. The common uniform mass bound is proved without PNT. No modular, near-L1 or IW estimate is inferred from mass alone.

Checked declarations

Module	Named supporting theorems
ActualPositiveSquareFamilies.lean	7

- `mangoldtPositiveFamily_apply`
- `ordinaryPositiveFamily_apply`
- `primePositiveFamily_apply`
- `positiveSquareFamilies_uniform_mass`
- `mangoldtPositiveFamily_maximal`
- `ordinaryPositiveFamily_maximal`
- `primePositiveFamily_maximal`

<!-- checkpoint-verification -->

Verification

All 905 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4512-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2441 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 907 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_839_NearOneGaugeMoment.md`

Checkpoint 839: Actual improper Orlicz moment and quantitative power integral

The actual improper gauge moment in c:orlicz-p is now controlled. Lean proves a global power majorant for Psi_C(u) on every interval u>=a>0, with an explicit constant containing exp(K(C)/(p-1) log(exp(1)/(p-1))). It proves actual integrability of Psi_C(u) u^(-p-1) over (a,infinity), and bounds its integral by an explicitly integrated power. This uses the checked literal gauge, continuity and Bochner improper integrals, and never replaces a divergent totalized integral by a number. The operator layer-cake and actual restricted arithmetic estimate remain separate obligations; this scalar result alone does not close c:orlicz-p.

Checked declarations

Module	Named supporting theorems
NearOneGaugeMoment.lean	5

- `nearOnePowerConstant_pos`
- `nearOnePsi_power_majorant`
- `nearOneGaugeMoment_majorant`
- `nearOneGaugeMoment_integrable`
- `nearOneGaugeMoment_bound`

<!-- checkpoint-verification -->

Verification

All 906 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4513-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2446 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 908 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain

partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_840_QuantitativeIWParameters.md`

Checkpoint 840: Literal quantitative Ionescu-Wainger parameters and separation

The quantitative Ionescu-Wainger scalar parameters from e:IWparameters7 and e:orlicz-D are defined with their printed powers and logarithms. Lean proves 1<p_r<=4/3, the exact identities p_r-1=1/(2r-1) and 1/p_r=1-1/(2r), and the rho range, half-exponent and exact floor-rank relation for k>=4. It proves B>=1, its exact logarithm, epsilon>0 and the strict separation inequality 4 epsilon B^2<1. All are unconditional algebraic deductions; existence of the IW sets, multiplier and sampling estimates has not been supplied by an assumption or declared proved. Natural Q is enough for the later dyadic parameter choice; the full printed real-Q proposition is not awarded.

Checked declarations

Module	Named supporting theorems
QuantitativeIWParameters.lean	10

- `iwNearOneExponent_bounds`
- `iwNearOneExponent_sub_one`
- `iwNearOneExponent_reciprocal`
- `iwRho_bounds`
- `iwRho_half`
- `iwRho_rank`
- `iwDenominatorBound_one_le`
- `iwDenominatorBound_log`
- `iwWidth_pos`
- `iwWidth_denominator_separation`

<!-- checkpoint-verification -->

Verification

All 907 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4514-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2456 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 909 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations

remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

Checkpoint_841_NearOneMomentExponential.md

Checkpoint 841: Exact exponential dependence on p minus one in the gauge moment

The scalar moment has the exact near-L1 exponential dependence displayed in c:orlicz-p. For C,C0>=1 and every 1<p<=2, Lean proves actual integrability on (1/C0,infinity) and bounds the integral of Psi_C(u)u^(-p-1) by exp(K(C,C0)/(p-1) log(exp(1)/(p-1))). The constant is explicit and quantified before p. The proof includes the lower cutoff, reciprocal power, prefactor absorption, and all p in the stated range. This is an unconditional scalar theorem. It does not assume or declare proved the operator modular estimate, its layer-cake norm implication, or the final ergodic assertion.

Checked declarations

Module	Named supporting theorems
NearOneMomentExponential.lean	5

- `nearOne_log_reciprocal_bounds`
- `nearOneMoment_prefactor_one_le`
- `nearOneMomentConstant_pos`
- `nearOneGaugeMoment_exponential`
- `nearOneGaugeMoment_source_p`

<!-- checkpoint-verification -->

Verification

All 908 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4515-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2461 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 910 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

Checkpoint_842_InitialPositiveDistribution.md

Checkpoint 842: Actual initial-scale mass and Chebyshev contribution

The initial-scale contribution to the Section7 restricted estimate is proved for actual positive finite averages. Translation injectivity bounds the finite sum of a shifted indicator by its cardinality. The actual complex average therefore has l1 mass at most mass*|F|. The sum over the first J averages has l1 mass at most J*mass*|F|, giving exactly t J mass |F| in the distributional estimate. Generic real power budgets, their actual lp-space specialization, and Chebyshev are also proved. No deep theorem or source high-low existence statement is a hypothesis of these initial-scale results.

Checked declarations

Module	Named supporting theorems
InitialPositiveDistribution.lean	8

- `finite_shift_indicator_sum`
- `positiveAverage_indicator_l1`
- `initialPositiveSum_nonneg`
- `initialPositiveSum_dominates`
- `initialPositiveSum_l1`
- `finiteDistribution_real_power`
- `finitePowerBudget_lp`
- `initialPositiveSum_distribution`

<!-- checkpoint-verification -->

Verification

All 909 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4516-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2469 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 911 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_843_HighLowRestrictedDistribution.md`

Checkpoint 843: Exact high-low restricted distribution deduction

The distributional high-low deduction underlying e:orlicz-optimize is proved for the genuine all-index maximum. Given actual large-scale low and high envelopes with finite p and 2 power budgets, the level set is controlled by t J mass |F|+(2t)^p B_low+4t^2 B_high. The proof finds a violating original scale from the supremum, separates the first J averages, performs the exact threshold union, and applies the proved Chebyshev bounds. It also proves the actual level set finite. The large-scale envelope premises are internal hypotheses of this reusable deduction, not authorized replacements for the paper own high-low existence theorem. Their construction remains open for Section7.

Checked declarations

Module	Named supporting theorems
HighLowRestrictedDistribution.lean	2

- `highLow_restricted_distribution`
- `highLow_restricted_levelset_finite`

<!-- checkpoint-verification -->

Verification

All 910 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4517-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2471 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 912 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_844_QuantitativeIWBalance.md`

Checkpoint 844: Uniform integer optimization for the Ionescu-Wainger loss

The integer optimization behind the quasi-polynomial Orlicz loss is proved uniformly. For every v>=1, choose k=ceil(sqrt(v/log(exp(1)+v)))+4. Lean proves k>=4, its two-sided rounding bounds, and k log(exp(1)+v)+v/k <=12 sqrt(v log(exp(1)+v)); a weighted version has the same bound multiplied by any fixed A>=1. This admissible alternative to the source eventual floor choice handles every bounded v directly. It proves the scalar balancing step, without assuming existence of any IW model or claiming that the full r,k,Q,N0,D parameter comparison has been discharged.

Checked declarations

Module	Named supporting theorems
QuantitativeIWBalance.lean	3

- `iwBalancedRank_bounds`
- `iwBalancedRank_optimization`
- `iwBalancedRank_weighted`

<!-- checkpoint-verification -->

Verification

All 911 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4518-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2474 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 913 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit

independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion.

`Checkpoint_845_IWFiniteFrequencyGeometry.md`

Checkpoint 845: Actual finite IW frequency geometry from the independent raw construction

The actual finite IW frequency set is defined using canonical rational representatives and denominators dividing products of at most k positive coprime integers. Its exact membership and denominator bound are proved. A narrow independent construction interface records Tao arXiv:2008.05066v2 Lemma5.1 for arbitrary rho and natural Q: positive coprime components, rational coverage, a common period at most 3^Q and the raw component bound. Its existence remains unproved and it contains no multiplier, sampling or current-paper maximal statement. From that input Lean proves the quantitative frequency geometry, including every denominator<=B and 4 epsilon B^2<1. The full goodness, multiplier and sampling clauses of source p:IWinput7 remain unfinished, so no full source proposition is awarded.

Checked declarations

Module	Named supporting theorems
IWFiniteFrequencyGeometry.lean	4

- `iwFrequencySet_mem`
- `iwFrequencySet_denominator_bound`
- `taoArc_denominator_bound`
- `quantitative_iw_frequency_geometry`

<!-- checkpoint-verification -->

Verification

All 912 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4519-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2478 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 914 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_846_SectionSevenDeductions.md`

Checkpoint 846: Reviewed Section7 modular deductions and exact remaining analytic obligations

The checked Section7 modular deductions are exported with actual finite level sets and both truncated and full finite-input costs. Their restricted indicator estimate remains an internal premise, not a falsely

independent analytic assumption. The source-statement review records partial progress only for t:orlicz-maximal, c:orlicz-p and p:IWinput7; all other source claims and definitions are unchanged. The actual continuous/low/arithmetic high comparison and restricted bound, operator Lp deduction and ergodic consequences remain unfinished. Earlier checkpoint exposition numbering is corrected to archived Theorem7.2, without editing any Lean proof or the manuscript. See docs/SECTION_7_PROGRESS.md for completed deductions, exact independent boundary and next mathematical target.

Checked declarations

Module	Named supporting theorems
SectionSevenDeductions.lean	3

- `section7_truncated_modular_finite_card`
- `section7_modular_deduction`
- `section7_modular_finite_card`

<!-- checkpoint-verification -->

Verification

All 913 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4520-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2481 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 915 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_847_IWGoodness.md`

Checkpoint 847: Full quantitative Ionescu-Wainger goodness with an integer component maximum

The full Ionescu-Wainger goodness condition in Tao Definition1.3 (c=1/2) is now proved for the actual quantitative parameters. A natural integer maximum component is chosen without rounding the real component bound upwards; this avoids an erroneous factor exponential in r*k. The proof shows qmax^(2*r*k)<=B^(2*r) and the strict epsilon<1/(4*r*qmax^(2*r*k)) inequality, not just pairwise separation. Combined with the existing independent raw Tao construction, Lean now returns an actual finite frequency set with coverage, period, denominator cap and genuine goodness. The general multiplier/sampling and current-paper low/high estimates remain separate work and are not assumed here.

Checked declarations

Module	Named supporting theorems
IWGoodness.lean	3

- `finite_component_maximum`
- `quantitative_iw_goodness`
- `quantitative_iw_construction_good`

<!-- checkpoint-verification -->

Verification

All 914 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4521-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2484 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 916 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_848_RealLineFourierMultipliers.md`

Checkpoint 848: Actual real-line Fourier multipliers and Lp norm interfaces

The real-line Fourier multiplier is now an actual continuous linear operator on real-line L2, built from Mathlib's unitary Fourier transform, multiplication by a bounded continuous symbol, and inverse Fourier transform. Lean proves the Fourier identity and symbol-norm operator bounds. General-p real and integer multiplier bounds require both Lp membership and the explicit extended-norm inequality on the L2 intersection; neither bound is assumed merely by declaring its predicate. The newly needed pinned Mathlib Fourier Lp dependency was compiled from the unchanged pinned sources (dependency build log retained). This checkpoint provides the actual operator interface for a general independent IW theorem, but does not yet prove the current paper's quantitative low/high or restricted maximal estimate.

Checked declarations

Module	Named supporting theorems
RealLineFourierMultipliers.lean	14

- `realLineProduct_memLp`
- `realLineMultiplyValue_ae`
- `realLineMultiplyValue_norm_le`
- `realLineMultiply_ae`
- `realLineMultiply_norm_le`
- `realLineMultiply_opNorm_le`
- `realLineMultiply_opNorm_le_of_bound`
- `realLineMultiplier_apply`
- `realLineFourier_multiplier`
- `realLineMultiplier_norm_le`
- `realLineMultiplier_opNorm_le`
- `realLineMultiplier_opNorm_le_of_bound`

- realLpMultiplierBound_mono
- integerLpMultiplierBound_mono

<!-- checkpoint-verification -->

Verification

All 915 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4534-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2498 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 917 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

Checkpoint_849_CyclicSquareMeasures.md

Checkpoint 849: Concrete cyclic square measures and exact Fourier coefficients

The finite cyclic low-model measure is now concrete: it is the pushforward of the character-weighted uniform measure on units under u -> u^2, normalized by phi(P). Lean proves its action on every test function, total variation at most one for every complex Dirichlet character, and exact equality of its finite Fourier coefficients with the existing twisted quadratic sums, with the manuscript's minus sign and totient normalization. The principal character gives the existing principal square sums. These identities remove an arithmetic identification premise at the full period; passage to a smaller denominator and conductor cancellation are subsequent work. No quantitative low maximal or prime-square restricted estimate is assumed or claimed.

Checked declarations

Module	Named supporting theorems
CyclicSquareMeasures.lean	6

- cyclicSquareMeasure_test
- cyclicVariation_nonneg
- cyclicSquareMeasure_variation_le_one
- cyclicFourier_norm_le_variation
- cyclicSquareMeasure_fourier
- cyclicPrincipalSquareMeasure_fourier

<!-- checkpoint-verification -->

Verification

All 916 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4535-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2504 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 918 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_850_UniformFiniteGroupAverages.md`

Checkpoint 850: Uniform finite-group averages and kernel cancellation

Lean now proves two general finite-group identities needed by the arithmetic IW comparison. Pullback under any surjective finite-group homomorphism preserves the normalized complex average; a translation and double-counting argument gives the exact normalization without an unverified equal-fibre assertion. For any complex multiplicative character that is nontrivial on the kernel, every character-weighted pullback test sum is zero. These are proved supporting lemmas, with no external analytic assumptions. They will be applied to reduction of unit groups at divisors of the IW period.

Checked declarations

Module	Named supporting theorems
UniformFiniteGroupAverages.lean	2

- `finite_group_pullback_average`
- `finite_group_twisted_pullback_zero`

<!-- checkpoint-verification -->

Verification

All 917 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4536-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2506 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 919 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations

remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

Checkpoint_851_CyclicConductorCoefficients.md

Checkpoint 851: Conductor-sensitive cyclic Fourier coefficient identity

The coefficient identity used in the current-paper IW comparison is now proved at every q dividing the period P. The Fourier coefficient of the actual cyclic square measure descends exactly to the twisted quadratic sum for a character that factors through q, and is zero when its conductor does not divide q. Lean derives the cancellation witness from Mathlib's conductor and kernel criterion and applies the proved finite-group averaging identities. The rational circle Fourier coefficient is also identified with this reduced coefficient, retaining the negative phase; the principal measure reduces to the existing principal square sum. This removes the equal-fibre and character-cancellation assertions from the remaining comparison boundary. No low/high norm conclusion is a hypothesis or a claimed proved source result.

Checked declarations

Module	Named supporting theorems
CyclicConductorCoefficients.lean	7

- `cyclicReducedSquare_test`
- `cyclicReducedSquare_changeLevel`
- `character_kernel_witness`
- `cyclicReducedSquare_conductor_zero`
- `cyclicReducedSquare_conductor_formula`
- `cyclicRationalFourier_eq_reduced`
- `cyclicReducedPrincipalSquare`

<!-- checkpoint-verification -->

Verification

All 918 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4537-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2513 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 920 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

Checkpoint_852_IWMultiplierInterface.md

Checkpoint 852: Actual smooth IW multiplier interface and quantitative specialization

The independent IW multiplier boundary now acts on the actual Fourier operators, rather than unspecified normed objects. Its explicit hypothesis ranges over arbitrary good coprime finite sets, smooth compact real symbols, finite cyclic measures, and the full admissible p interval; the assertion includes actual output Lp membership and the extended-norm inequality. This is an unformalized independent theorem (Tao Theorem1.7, with finite cyclic specialization), not a new axiom and not a claim that Mathlib proves IW. Lean proves its specialization to the manuscript's checked geometry and constants. The actual finite periodized symbol has a proved real-frequency formula, and its cyclic coefficients have the proved variation bound. The smoothness and real Lp estimate for each concrete annular symbol must still be verified before using this interface; the paper's own low/high and restricted estimates are absent from the external hypothesis.

Checked declarations

Module	Named supporting theorems
IWMultiplierInterface.lean	3

- `cyclicRationalCoefficient_norm`
- `iwCyclicSymbol_real`
- `quantitative_iw_multiplier`

<!-- checkpoint-verification -->

Verification

All 919 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4538-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2516 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 921 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_853_IWAnnularDenominatorTails.md`

Checkpoint 853: Actual Q-to-minus-one-quarter annular denominator-tail maximum

The denominator tail with the original denominator-dependent cutoff now has an actual countable maximal L2 majorant of size K*Q^(-1/4). Lean proves the power split q^(-3/8)<=Q^(-1/4)*q^(-1/8), applies the previously checked power-decaying annular maximal theorem, and specializes the result to the actual principal and twisted quadratic coefficients using the project's proved Gauss decay. The result allows variable beta and variable upper denominator truncation, and its constant is uniform in Q, the input, and the character family. It uses only the already declared generic smooth multifrequency boundary. This proves the original-cutoff denominator-tail contribution; the common IW-cutoff tail, small-denominator difference, changing-character sum and low-term maximum remain necessary for the full comparison.

Checked declarations

Module	Named supporting theorems
IWAnnularDenominatorTails.lean	3

- `iw_denominator_tail_power`
- `iw_annular_denominator_tail`
- `iw_gauss_annular_denominator_tails`

<!-- checkpoint-verification -->

Verification

All 920 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4539-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2519 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 922 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_854_CommonCutoffGeometry.md`

Checkpoint 854: Common-cutoff factorization and contraction on each denominator band

Lean now proves the exact relation between a common narrow cutoff and the original denominator-band cutoff. When the common width lies in the band cutoff's core, the actual periodized annular symbol factors as the original annular symbol times the common cutoff. The proof treats integer periodization explicitly and rules out cross-lift contributions. For a separated denominator band, the sum of common cutoffs defines an actual integer Fourier multiplier of operator norm at most one. This is the projection needed for the IW-cutoff high tail; it is proved, rather than included in an external sampling or maximal premise.

Checked declarations

Module	Named supporting theorems
CommonCutoffGeometry.lean	9

- `circleWidthAnnular_real`
- `common_cutoff_source_core`
- `circleAnnularSource_eq_lift`
- `circleWidthAnnular_on_source_lift`
- `common_cutoff_zero_of_source_zero`
- `circleSmoothCutoff_supported_in_source`
- `circleWidthAnnular_factorization`

- `common_circle_cutoff_norm`
- `common_circle_projection_norm`

<!-- checkpoint-verification -->

Verification

All 921 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4540-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2528 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 923 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_855_CommonCutoffAnnularMaximum.md`

Checkpoint 855: Actual common-cutoff annular maximum on a denominator band

The common-cutoff annular block on a denominator band is now exactly the original annular block composed with a fixed Fourier projection. Lean proves that cross-frequency products vanish, the symbol factorization, and the resulting actual countable dyadic L2 maximum, uniformly in the common width and variable beta. The projection is applied separately on each band, avoiding unjustified cross terms from a single projection over all denominators. The only analytic hypothesis is the previously recorded general smooth multifrequency estimate.

Checked declarations

Module	Named supporting theorems
CommonCutoffAnnularMaximum.lean	3

- `common_annular_cross_zero`
- `weightedWidthAnnular_factorization`
- `common_cutoff_band_maximum`

<!-- checkpoint-verification -->

Verification

All 922 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4541-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2531 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 924 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_856_CommonCutoffPowerMaximum.md`

Checkpoint 856: Summed common-cutoff maximum and Q-to-minus-one-quarter tail

The separate denominator-band projections are now summed in Lean. The actual common-cutoff annular maximum is bounded uniformly by a convergent polynomial-geometric series for every positive coefficient decay exponent. Splitting q^(-3/8) then gives the required Q^(-1/4) bound for the common-cutoff denominator tail, for any finite canonical frequency set whose common width lies in the relevant band cores. The proof includes empty bands, the finite dyadic partition and the infinite scalar sum. The exact IW parameters and arithmetic cyclic coefficients will now be substituted into this checked estimate.

Checked declarations

Module	Named supporting theorems
CommonCutoffPowerMaximum.lean	2

- `common_cutoff_power_maximum`
- `common_cutoff_denominator_tail`

<!-- checkpoint-verification -->

Verification

All 923 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4542-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2533 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 925 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW

frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_857_CyclicRationalGaussDecay.md`

Checkpoint 857: Uniform Gauss decay of actual rational cyclic coefficients

The actual rational cyclic coefficients now inherit the project's proved uniform Gauss decay for every exponent below one half. Lean first identifies the rational circle coefficient with reduction at its exact reduced denominator, then proves the principal and induced-character identities and the nondividing-conductor zero case. The uniform bound is independent of the period and the character. This supplies the real coefficients needed to specialize the common-cutoff tail, without an assumed arithmetic coefficient bound.

Checked declarations

Module	Named supporting theorems
CyclicRationalGaussDecay.lean	5

- `cyclicRationalCoefficient_eq_reduced`
- `cyclicRationalCoefficient_principal`
- `cyclicRationalCoefficient_changeLevel`
- `cyclicRationalCoefficient_conductor_zero`
- `cyclicRationalSquare_uniform_decay`

<!-- checkpoint-verification -->

Verification

All 924 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4543-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2538 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 926 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_858_IWCommonCutoffTails.md`

Checkpoint 858: Actual quantitative IW common-cutoff denominator-tail maximum

The common-cutoff tail is now specialized to the exact quantitative IW parameters and the actual cyclic square measures. Lean proves that epsilon^(-1) lies inside every needed denominator-band core, then combines the actual frequency geometry and the proved uniform cyclic Gauss decay to obtain a countable dyadic L2 majorant of size K*Q^(-1/4). The constant is uniform in r,k,Q, period, complex character and variable beta. This proves the common-cutoff denominator-tail contribution for both the principal and any

fixed induced character. It does not require the external IW multiplier theorem or assume the low-model maximum; only the previously declared generic smooth multifrequency input is used.

Checked declarations

Module	Named supporting theorems
IWCommonCutoffTails.lean	4

- `iwCutoffScale_formula`
- `iwCutoffScale_four_le`
- `iwCutoffScale_band_core`
- `iw_cyclic_square_common_tail`

<!-- checkpoint-verification -->

Verification

All 925 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4544-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2542 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 927 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_859_IWSmallDenominatorDifference.md`

Checkpoint 859: Actual small-denominator cutoff-difference maximum

The remaining small-denominator cutoff difference now has a proved actual countable maximal L2 bound. The proof compares the two real cutoffs on their common core, uses the already proved annular Fourier decay outside that core, and passes through the actual circle periodization and integer Fourier operator. For a finite frequency set Gamma the bound is 24*card(Gamma)*A*epsilon^(-1)/(pi*M^2), uniformly in variable beta. This is a deliberately coarse finite-count bound, valid without separation, whose polynomial denominator cost can be absorbed by the manuscript's sufficiently large starting scale. The absorption into Q^(-1) and the final comparison assembly are still separate proof obligations.

Checked declarations

Module	Named supporting theorems
IWSmallDenominatorDifference.lean	4

- `annular_common_source_difference`
- `circle_annular_common_source_difference`
- `weighted_common_source_difference`
- `common_source_difference_maximum`

<!-- checkpoint-verification -->

Verification

All 926 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4545-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2546 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 928 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_860_IWLargeConductorMaximum.md`

Checkpoint 860: Actual Q-to-minus-one-quarter changing large-conductor maximum

The changing large-conductor exceptional contribution now has an actual countable maximal L2 majorant of size K*Q^(-1/4). Lean keeps the original rational annular symbols, permits the exceptional character to vary with scale, and proves uniformity by splitting the Gauss decay between Q, the conductor and the remaining denominator. The already checked Page uniqueness theorem bounds the sum over conductor classes. No fixed-character substitution and no current-paper exceptional maximal estimate is assumed. The only hypotheses are the independently recorded smooth multifrequency and raw Page uniqueness inputs.

Checked declarations

Module	Named supporting theorems
IWLargeConductorMaximum.lean	2

- `iw_conductor_denominator_power`
- `iw_large_conductor_maximum`

<!-- checkpoint-verification -->

Verification

All 927 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4546-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2548 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 929 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs.

The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_861_IWFixedSmallCharacter.md`

Checkpoint 861: Actual fixed small-character selection and IW period divisibility

Lean now constructs the fixed small-conductor Page datum used by the low model. It proves uniqueness across all scales in the fixed conductor region directly from the existing raw Page theorem, retaining the original scale-by-scale activity and allowing inactivity. It also proves that every small denominator q>=2 divides the actual IW period, using coverage of the rational point 1/q; hence the selected small character can actually be induced to that period. Neither the existence of a fixed character nor its period divisibility is left as a current-paper external premise.

Checked declarations

Module	Named supporting theorems
IWFixedSmallCharacter.lean	4

- `iw_period_divisible_small`
- `page_zero_in_fixed_region`
- `fixed_small_page_datum`
- `fixed_small_page_conductor_period`

<!-- checkpoint-verification -->

Verification

All 928 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4547-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2552 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 930 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are

not fields and remain unfinished.

Checkpoint_862_IWActualAnnularSymbols.md

Checkpoint 862: Actual annular IW symbols and primitive-character identification

The actual annular low symbol is now connected to the real-line/IW operator interface, with a proved bounded continuous real symbol and exact equality to the finite periodized circle symbol. Lean also identifies a primitive character induced to the IW period with its original masked quadratic coefficients at every IW rational frequency. The principal and induced-character whole-sum identities are proved, including the conductor-not-dividing zero case. This supplies a concrete low operator for the comparison; no general-p continuous or low maximal bound is claimed by this identification.

Checked declarations

Module	Named supporting theorems
IWActualAnnularSymbols.lean	7

- `annularCutoffSymbol_norm_half`
- `annularBoundedSymbol_apply`
- `annularBoundedSymbol_support`
- `iwCyclicSymbol_annular`
- `cyclicRationalCoefficient_primitive_induced`
- `iw_principal_width_symbol`
- `iw_induced_width_symbol`

<!-- checkpoint-verification -->

Verification

All 929 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4548-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2559 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 931 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

Checkpoint_863_IWFiniteModelComparison.md

Checkpoint 863: Exact original-versus-IW circle-symbol comparison

The circle-model comparison is now an exact identity of the actual continuous circle symbols. Lean decomposes the original principal/optional-character model into the corresponding annular blocks, proves that the IW and original frequency sets have exactly the same small-denominator part, and splits their difference into the original tail, the common-cutoff tail and the small-denominator cutoff difference. The

proof retains all periodized terms and the source's exceptional selector. No identification of operators or truncations is left as an informal premise in this finite comparison.

Checked declarations

Module	Named supporting theorems
IWFiniteModelComparison.lean	6

- `circleRationalModelTerm_annular_decomposition`
- `circleMajorArcModel_annular_decomposition`
- `pageCircleModel_annular_decomposition`
- `iw_small_frequency_set`
- `source_small_frequency_set`
- `iw_original_common_comparison`

<!-- checkpoint-verification -->

Verification

All 930 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4549-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2565 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 932 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_864_IWQuantitativeModelComparison.md`

Checkpoint 864: Quantitative actual original-versus-IW maximal L2 model comparison

The exact comparison and all three associated maximal estimates are now assembled in Lean into a genuine countable L2 model comparison with Q^(-1/4) rate. It compares the original denominator-dependent circle multiplier with the common-cutoff IW multiplier on the same dyadic scales and variable beta. The small-denominator contribution is absorbed under the explicit arithmetic starting budget (Q+1)^2*epsilon^(-1)*Q<=M^2. The coefficient decay and small coefficient bound are internal premises for this generic comparison, to be discharged by the actual principal/character coefficients; they are not added as independent analytic inputs. The source's large starting-scale formula must also be shown to imply that budget before the full source proposition is closed.

Checked declarations

Module	Named supporting theorems
IWQuantitativeModelComparison.lean	2

- `iw_small_difference_absorption`

- `iw_quantitative_model_comparison`

<!-- checkpoint-verification -->

Verification

All 931 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4550-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2567 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 933 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_865_IWCharacterModelComparison.md`

Checkpoint 865: Actual principal and primitive-character quantitative model comparisons

The quantitative model comparison is now discharged for the actual principal square measure and for every primitive small character induced to the actual IW period. Lean proves the required coefficient norm-one bounds and uniform Gauss decay, then obtains actual countable L2 comparison majorants with a common constant times Q^(-1/4). Both comparisons retain the actual cyclic measures in their low symbols, rather than replacing them by an unidentified coefficient family. The only independent analytic input is the recorded generic smooth multifrequency theorem; the explicit starting-scale arithmetic budget remains stated until it is checked from the manuscript's exponential scale.

Checked declarations

Module	Named supporting theorems
IWCharacterModelComparison.lean	4

- `rationalPrincipalGauss_norm_one`
- `rationalInducedSquareCoefficient_norm_one`
- `rationalInducedSquareCoefficient_uniform_decay`
- `iw_character_model_comparisons`

<!-- checkpoint-verification -->

Verification

All 932 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4551-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2571 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 934 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_866_IWStartingScaleBudget.md`

Checkpoint 866: Starting-scale budget proved from the manuscript exponential scale

The starting-scale budget in the quantitative model comparison is now proved from the manuscript's exponential scale, with an explicit constant 25. For C_*>=e, Lean proves log B>=1, log(epsilon^(-1))<=17*r*log B, and (Q+1)^2*epsilon^(-1)*Q<=exp(25*((log Q)^2+r*log B)). Consequently any integer M>=2 above this exponential threshold satisfies the squared-size budget needed to absorb the small-denominator cutoff difference. Thus the coarse finite-count estimate does fit the source's permitted sufficiently large absolute constant; no extra growth condition is being silently imposed.

Checked declarations

Module	Named supporting theorems
IWStartingScaleBudget.lean	4

- `iwDenominatorBound_log_one_le`
- `iwCutoffScale_log_bound`
- `iw_starting_budget_exp_bound`
- `iw_starting_budget_of_source_scale`

<!-- checkpoint-verification -->

Verification

All 933 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4552-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2575 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 935 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations

remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

Checkpoint notes 867-941

Checkpoint_867_IWOptionalModelIdentities.md

Checkpoint 867: Exact optional Page model and actual IW low-symbol identities

The fixed-small/large-character split and the complete optional circle-model comparison are now exact identities of actual symbols. The low operator retains the original activity at each scale, and its fixed character term is identified with the actual cyclic square measure induced to the IW period. Lean proves the masked-denominator reindexing, the small/large optional split, and the original-minus-low decomposition. This supplies the exact same operator family required to combine the already checked comparison and changing-character maximal bounds.

Checked declarations

Module	Named supporting theorems
IWOptionalModelIdentities.lean	6

- `largePageDatum_eq_some`
- `induced_annular_mask`
- `pageExceptionalAnnularSymbol_some`
- `page_exceptional_small_large_split`
- `page_model_IW_difference`
- `fixedPageIWSymbol_actual_measure`

<!-- checkpoint-verification -->

Verification

All 934 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4553-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2581 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 936 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

Checkpoint_868_IWPageModelMaximum.md

Checkpoint 868: Quantitative IW comparison for the actual varying-Page circle model

The principal, fixed small-character and varying large-character comparison bounds are now assembled for the actual circle model. Lean constructs the fixed small datum from the given genuine Page data and proves a countable L2 majorant of size K*Q^(-1/4) for V_N minus the actual optional IW low symbol, on the same dyadic scales and with the original activity. This is the full circle-model high comparison,

conditional only on the already recorded general smooth multifrequency and raw Page inputs and the explicit starting budget. The prime-average approximation error, quantitative low Lp maximum and final restricted optimization still need their assembly.

Checked declarations

Module	Named supporting theorems
IWPageModelMaximum.lean	1

- `iw_page_model_maximum`

<!-- checkpoint-verification -->

Verification

All 935 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4554-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2582 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 937 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_869_IWPrimeApproximationMaximum.md`

Checkpoint 869: Actual prime approximation-error maximum with compatible Page data

The actual prime-square approximation error now has a uniform countable L2 maximum of size K*Q^(-1/2) from every permitted integer dyadic starting scale M. Lean obtains it from the already proved raw-data circle approximation with exponent two, sums the operator-norm decay, and retains the same selected Page datum at all scales. One positive range coefficient also guarantees genuine raw admissibility at every later scale, so the Page data used by the IW comparison and by the approximation are compatible. No IW low/high or restricted conclusion is assumed.

Checked declarations

Module	Named supporting theorems
IWPrimeApproximationMaximum.lean	1

- `iw_prime_approximation_maximum`

<!-- checkpoint-verification -->

Verification

All 936 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4555-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2583 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 938 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_870_IWPrimeHighMaximum.md`

Checkpoint 870: Actual prime-square IW high maximal estimate with Q-to-minus-one-quarter rate

The actual von Mangoldt annular high operator is now constructed as the actual average minus the concrete IW low operator, with an exact operator decomposition. Lean combines the proved optional circle-model comparison with the compatible raw-data approximation error to obtain the full countable L2 high bound K*Q^(-1/4), including all prime powers and the varying Page character. The IW geometry is obtained from the independent raw Tao construction. The low measure identities are proved. The explicit residual source-range inequality is still stated here and will be absorbed into the sufficiently large exponential starting constant next. The quantitative low Lp maximum and actual restricted estimate remain unproved; this checkpoint does not assume either of them.

Checked declarations

Module	Named supporting theorems
IWPrimeHighMaximum.lean	3

- `iwPrime_decomposition`
- `iwPrimeHigh_error_model`
- `iw_prime_high_maximum`

<!-- checkpoint-verification -->

Verification

All 937 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4556-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2586 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 939 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes

PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are not fields and remain unfinished.

`Checkpoint_871_IWPrimeHighSourceScale.md`

Checkpoint 871: Actual prime high maximal estimate at the manuscript starting scale

The actual prime high-side estimate now holds under the manuscript's sufficiently large exponential starting scale, with no separate Q-range premise. Lean proves the uniform logarithmic cutoff implication and absorbs both the source-range and cutoff-difference requirements into one absolute starting constant. The conclusion constructs the genuine IW set and period, a permitted fixed small character, and the actual prime high operator with countable L2 majorant K*Q^(-1/4). Its independent assumptions are raw Tao construction, the existing generic smooth multifrequency theorem, raw Page uniqueness and compatible raw prime-distribution data. It does not use the new general IW multiplier boundary, assume the quantitative low maximum, or claim the actual restricted estimate. The ordinary-square companion and the low side are still open within the full source proposition.

Checked declarations

Module	Named supporting theorems
IWPrimeHighSourceScale.lean	2

- `iw_source_cutoff_of_exp`
- `iw_prime_high_source_scale`

<!-- checkpoint-verification -->

Verification

All 938 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4557-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2588 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 940 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Proposition 4.19 consumes PrimeAsymptoticInputs, the refined PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. These are independent raw analytic/data hypotheses, not own-paper operator conclusions or new Lean axioms. Theorem 1.1 retains its separately recorded older fixed-scale inputs and independent ordinary-square global decomposition; reflection is proved. Sparse applications additionally record OrdinarySquareStoppingInput CS and IndependentPrimeMaximalInput as explicit independent analytic boundaries. Their existence and the exact normalization/support/sign adaptations remain unproved; neither predicate contains this paper's endpoint or sparse conclusion. Section7 additionally records the independent RawTaoArcConstructionInput C, used only to obtain actual finite IW frequency geometry. Its existence is unproved; multiplier, sampling, low and high norm estimates are

not fields and remain unfinished.

Checkpoint_872_AnnularMassVariation.md

Checkpoint 872: Uniform annular mass variation and reviewed prime high-side milestone

Lean now proves the exact zero-frequency mass factor for the actual annular model, its uniform bound by one, positivity and monotonicity in the scale, and uniform total variation at most one along every finite dyadic range. These are beta-uniform proved ingredients for the continuous signed-increment argument; the complete continuous Lp lemma is not claimed. This checkpoint also reviews the exact source statements against the checked prime high theorem at871: p:orlicz-hilo moves from open to partial only. The actual restricted estimate, low maximum and ordinary-square companion remain unproved. Documentation and recovery metadata are reconciled with the new scope and 54-claim inventory.

Checked declarations

Module	Named supporting theorems
AnnularMassVariation.lean	7

- `annularMassFactor_nonneg`
- `annularMassFactor_le_one`
- `annularMass_formula`
- `annularMass_re_bounds`
- `annularMass_re_antitone`
- `annularMass_difference_norm`
- `annularMass_dyadic_variation`

<!-- checkpoint-verification -->

Verification

All 939 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4558-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2595 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 941 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint_873_DyadicAbsoluteProfile.md

Checkpoint 873: Uniform absolute dyadic profile sum for the continuous increment argument

Lean now proves a uniform absolute geometric profile sum: sum over j<J of min(4^j*x,(4^j*x)^(-1))<=10/3 for every nonnegative x, including zero. A positive rational potential gives exact telescoping and avoids any unverified cutoff at a logarithmic index. The companion two-sided profile lemma converts the already proved annular small- and large-frequency estimates into this summable form. These

are proved scalar ingredients for the continuous signed-increment estimate; no low maximal conclusion is assumed.

Checked declarations

Module	Named supporting theorems
DyadicAbsoluteProfile.lean	8

- `absoluteDyadicPotential_range`
- `absoluteDyadicPotential_difference`
- `absoluteDyadicPotential_near`
- `absoluteDyadicPotential_far`
- `absoluteDyadicProfile_bound`
- `two_sided_absolute_profile`
- `absoluteDyadicPotential_telescoping`
- `dyadic_absolute_profile_sum`

<!-- checkpoint-verification -->

Verification

All 940 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4559-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2603 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 942 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_874_AnnularAbsoluteError.md`

Checkpoint 874: Uniform absolute summation of the actual annular projection errors

The actual annular projection errors now have a proved uniform absolute dyadic sum, bounded by (10/3)*annularProjectionErrorConstant. The result permits beta to vary with the scale. Lean derives it from the existing actual small- and large-frequency bounds and the checked absolute geometric profile, including frequency zero. This strengthens the earlier square-energy bound in the form needed for the continuous signed-increment argument; no Lp low maximum is assumed or claimed.

Checked declarations

Module	Named supporting theorems
AnnularAbsoluteError.lean	2

- `annularProjectionError_absolute_profile`
- `dyadicAnnularProjectionError_absolute_sum`

<!-- checkpoint-verification -->

Verification

All 941 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4560-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2605 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 943 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_875_SourceCutoffVariation.md`

Checkpoint 875: Uniform dyadic variation of the actual source cutoff

The actual smooth cutoff now has a proved uniform dyadic total-variation bound, at most 80/3. The proof uses only its checked core, support and range properties; it does not assume monotonicity of the chosen bump. Together with the absolute annular projection-error sum and the mass variation, this gives the three quantitative ingredients for the uniform continuous signed-increment multiplier bound.

Checked declarations

Module	Named supporting theorems
SourceCutoffVariation.lean	5

- `sourceCutoff_difference_abs_le_one`
- `sourceCutoff_dilation_near`
- `sourceCutoff_dilation_far`
- `sourceCutoff_dilation_profile`
- `sourceCutoff_dyadic_variation`

<!-- checkpoint-verification -->

Verification

All 942 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4561-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2610 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 944 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and

Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_876_ContinuousAnnularIncrements.md`

Checkpoint 876: Uniform beta-independent continuous annular signed-increment multiplier bound

Lean now proves a uniform bound for the actual continuous dyadic increments, first as a sum of absolute multiplier differences and then for arbitrary complex signed sums with coefficient norms at most one. The proof combines the actual mass variation, actual cutoff variation and absolute projection-error sums. The bound is independent of beta, frequency, number of scales and every integer starting scale M>=2. The initial N=1 scale and actual real-line operator realization are handled next; no Calderon-Zygmund or low maximal bound is used here.

Checked declarations

Module	Named supporting theorems
ContinuousAnnularIncrements.lean	5

- `continuousAnnularIncrementConstant_pos`
- `annularMass_scaled_variation`
- `annularIncrement_norm_bound`
- `continuousAnnularIncrement_absolute_sum`
- `continuousAnnularSignedIncrement_bound`

<!-- checkpoint-verification -->

Verification

All 943 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4562-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2615 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 945 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_877_ContinuousAnnularInitialIncrement.md`

Checkpoint 877: Continuous signed increments including the initial N-equals-one scale

The continuous signed-increment multiplier estimate now includes the actual N=1 starting term in the manuscript. Lean proves the uniform annular symbol bound at every N>=1, including the beta-uniform edge case, and combines the first increment with the checked later-scale sum. The resulting constant is independent of beta, frequency, length and complex signs. This completes the scalar multiplier preparation for a genuine Plancherel L2 operator bound; no general-p continuous or maximal theorem is assumed.

Checked declarations

Module	Named supporting theorems
ContinuousAnnularInitialIncrement.lean	3

- `annularOscillatoryModel_norm_one`
- `continuousAnnularIncrement_one_start`
- `continuousAnnularSignedIncrement_one_start`

<!-- checkpoint-verification -->

Verification

All 944 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4563-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2618 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 946 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_878_ContinuousAnnularSignedL2.md`

Checkpoint 878: Actual beta-uniform continuous signed-increment L2 estimate

The first continuous signed-increment inequality is now proved at p=2 on the actual real-line Fourier L2 space, uniformly in beta, number of scales and complex signs, including the initial N=1 term. Lean proves the literal untruncated symbol formula and the smooth-cutoff variant, the operator-norm and all-input L2 bounds, and the actual extended Lp norm and membership formulation at exponent2. Mathlib Plancherel is used; no independent continuous inequality is assumed. The exact source review promotes only l:continuousestimates7 to partial. The full 1<p<2 and continuous maximal estimates, quantitative IW low maximum and actual restricted estimate remain unfinished.

Checked declarations

Module	Named supporting theorems
ContinuousAnnularSignedL2.lean	9

- `allAnnularCutoff_norm_one`

- allAnnularBoundedSymbol_apply
- continuousAnnularSignedSymbol_apply
- continuousAnnularSignedSymbol_norm
- continuousAnnularSignedSymbol_untruncated
- continuousAnnularSignedMultiplier_opNorm
- continuousAnnularSignedMultiplier_L2
- realLpMultiplierBound_two_of_bound
- continuousAnnularSignedMultiplier_Lp_two

<!-- checkpoint-verification -->

Verification

All 945 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4564-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2627 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 947 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint_879_ContinuousAnnularKernel.md

Checkpoint 879: Actual continuous annular kernel, integrability and Hardy-Littlewood box bound

The literal continuous annular kernel is now a concrete real density supported on [N^2/4,N^2]. Lean proves its real and complex integrability, positivity and the uniform box bound by N^(-2) on [0,N^2] for N>=2 and beta<=1. This is the own-paper kernel bound needed to apply a precise general Hardy-Littlewood theorem; no continuous maximal estimate is assumed. Its exact Fourier-transform identity and uniform signed-kernel regularity remain subsequent obligations. The current canonical progress, statement review, independent-boundary and recovery reports have also been saved with the checked878 scope.

Checked declarations

Module	Named supporting theorems
ContinuousAnnularKernel.lean	8

- continuousAnnularKernel_of_mem
- continuousAnnularKernel_of_not_mem
- annularKernelDensity_continuousOn
- continuousAnnularKernel_integrable
- continuousAnnularKernel_complex_integrable
- continuousAnnularKernel_nonneg
- annularKernelDensity_bound
- continuousAnnularKernel_box_bound

<!-- checkpoint-verification -->

Verification

All 946 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4565-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2635 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 948 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_880_ContinuousAnnularKernelFourier.md`

Checkpoint 880: Exact Fourier identity for the literal continuous annular kernel

The literal annular kernel now has a proved exact Fourier-transform identity with the actual annular oscillatory multiplier. Lean checks the substitution t -> t^2, its derivative, both endpoints, the real/complex scalar normalization and the negative Fourier phase. The proof treats the kernel as an actual integrable function and uses the interval change-of-variables theorem; it does not assume the paper's kernel formula or continuous operator estimate. This connects the checked continuous L2 multiplier to the physical kernel needed by the remaining general-p argument.

Checked declarations

Module	Named supporting theorems
ContinuousAnnularKernelFourier.lean	2

- `annularKernelDensity_substitution`
- `continuousAnnularKernel_fourier`

<!-- checkpoint-verification -->

Verification

All 947 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4566-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2637 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 949 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and

Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint_881_ContinuousAnnularKernelDerivative.md

Checkpoint 881: Uniform derivative bound for the actual continuous annular density

Lean now proves the derivative formula for the actual annular density and the uniform bound |K'_N(x)|<=4/N^4 on its annulus for 0<=beta<=1 and N>=2. The derivative is also identified exactly as (beta/2-1) times the density divided by x. Together with the checked support, box bound, integrability and exact Fourier identity, this supplies the own-paper local regularity data for the remaining uniform kernel argument. No full Calderon-Zygmund applicability, general-p increment estimate or low maximum is assumed or claimed.

Checked declarations

Module	Named supporting theorems
ContinuousAnnularKernelDerivative.lean	3

- `annularKernelDensity_hasDerivAt`
- `annularKernelDerivative_density`
- `annularKernelDerivative_bound`

<!-- checkpoint-verification -->

Verification

All 948 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4567-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2640 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 950 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint_882_ContinuousAnnularKernelRegularity.md

Checkpoint 882: Actual annular kernel local regularity and far-support vanishing

Lean proves the actual annular density is Lipschitz on its support, with constant 4/N^4, for N>=2 and 0<=beta<=1. It also proves that both a kernel and its translation vanish outside |x|>2|h| whenever N^2<=|h|. These are own-paper kernel estimates; the endpoint jumps are retained and will be treated by an integrated translation bound. No general-p estimate, low maximum or restricted maximal conclusion is assumed or claimed.

Checked declarations

Module	Named supporting theorems
ContinuousAnnularKernelRegularity.lean	2

- `annularKernelDensity_lipschitz`
- `continuousAnnularKernel_far_zero`

<!-- checkpoint-verification -->

Verification

All 949 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4568-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2642 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 951 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_883_IntervalKernelShiftGeometry.md`

Checkpoint 883: Translation envelope including both interval endpoint jumps

Lean now proves a general translation envelope for an interval-truncated kernel, explicitly including both endpoint jumps. A change in interval membership forces x into a collar of width 2|h| around one endpoint. For an inside bound B and inside Lipschitz constant L, the absolute translated difference is bounded pointwise by L|h| on the interval plus B on each endpoint collar. The result applies to arbitrary real densities; no annular maximal or Lp conclusion is assumed. The next step is to integrate this envelope and sum the dyadic scales.

Checked declarations

Module	Named supporting theorems
IntervalKernelShiftGeometry.lean	2

- `interval_shift_membership_boundary`
- `interval_kernel_shift_envelope`

<!-- checkpoint-verification -->

Verification

All 950 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4569-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2644 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 952 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_884_IntervalKernelShiftIntegral.md`

Checkpoint 884: Lebesgue integral bound for a translated interval-truncated kernel

Lean integrates the checked endpoint-jump envelope against actual Lebesgue measure. For any integrable interval-truncated real kernel with inside bound B and inside Lipschitz constant L, the full L1 translation difference is at most (L(b-a)+4B)|h|. Translation invariance, integrability of the translated difference, all three indicator integrals and their exact interval lengths are checked. This general calculus-and-measure argument adds no deep theorem assumption. Its annular specialization and dyadic summation remain next.

Checked declarations

Module	Named supporting theorems
IntervalKernelShiftIntegral.lean	3

- `intervalKernelShiftEnvelope_integrable`
- `intervalKernelShiftEnvelope_integral`
- `interval_kernel_shift_integral`

<!-- checkpoint-verification -->

Verification

All 951 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4570-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2647 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 953 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and

Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_885_ContinuousAnnularKernelTranslation.md`

Checkpoint 885: Actual annular kernel mass and scale-sensitive L1 translation bounds

Lean proves the actual annular kernel has precisely the mass given by its zero-frequency multiplier, that its translated difference is integrable, and that the full L1 translation difference is at most 2 for every N>=1, 0<beta<=1. For N>=2 the integrated endpoint-jump envelope yields the scale-sensitive bound 7|h|/N^2. All statements concern the literal kernel and actual Lebesgue measure. No Calderon-Zygmund conclusion, low maximum or restricted estimate is used as a premise. The next step restricts to |x|>2|h| and sums the dyadic scales uniformly.

Checked declarations

Module	Named supporting theorems
ContinuousAnnularKernelTranslation.lean	4

- `continuousAnnularKernel_integral`
- `continuousAnnularKernel_translation_integrable`
- `continuousAnnularKernel_translation_rough`
- `continuousAnnularKernel_translation_bound`

<!-- checkpoint-verification -->

Verification

All 952 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4571-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2651 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 954 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_886_ContinuousAnnularFarTranslation.md`

Checkpoint 886: Summable far-translation profile for the actual annular kernel

Lean now restricts the actual translated annular kernel difference to |x|>2|h|. The restricted function is nonnegative and integrable. Small scales N^2<=|h| vanish exactly; at larger scales the checked full translation estimate gives a bound by 7 min(N^2/|h|,(N^2/|h|)^(-1)). This is the summable dyadic profile needed for a uniform Hormander estimate. No general Lp, low maximum or restricted maximal conclusion has been assumed.

Checked declarations

Module	Named supporting theorems
ContinuousAnnularFarTranslation.lean	5

- `annularFarTranslation_nonneg`
- `annularFarTranslation_integrable`
- `annularFarTranslation_integral_le`
- `annularFarTranslation_small_zero`
- `annularFarTranslation_profile`

<!-- checkpoint-verification -->

Verification

All 953 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4572-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2656 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 955 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint_887_ContinuousAnnularDyadicTranslation.md

Checkpoint 887: Uniform dyadic sum of actual annular far-translation integrals

Lean sums the actual far-translation integrals over every finite set of consecutive dyadic scales. Scales starting at N=2 have total at most 70/3, using the checked min(y,1/y) dyadic profile. Including N=1 costs at most 2, with no dependence on the number of scales, beta or the translation. The zero-translation case is handled exactly. This is the uniform kernel regularity needed for arbitrary bounded signed increments; assembling that signed kernel and its precise Fourier symbol is next.

Checked declarations

Module	Named supporting theorems
ContinuousAnnularDyadicTranslation.lean	3

- `dyadic_annular_translation_ratio`

- `annularFarTranslation_dyadic_sum`
- `annularFarTranslation_all_dyadic_sum`

<!-- checkpoint-verification -->

Verification

All 954 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4573-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2659 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 956 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_888_ContinuousAnnularSignedKernel.md`

Checkpoint 888: Actual complex signed annular kernel and far-translation domination

Lean now defines the literal complex signed sum of consecutive dyadic annular kernels and proves its integrability. Its translated difference is bounded by the sum of the corresponding absolute single-kernel differences for arbitrary complex coefficients of modulus at most 1. The same domination is proved after restricting to the Hormander region |x|>2|h|. A reusable integrability lemma for that restricted norm is also checked. This prepares the actual uniform Hormander conclusion; no general Lp bound or low maximum is assumed.

Checked declarations

Module	Named supporting theorems
ContinuousAnnularSignedKernel.lean	4

- `continuousAnnularSignedKernel_integrable`
- `continuousAnnularSignedKernel_shift_norm`
- `farKernelTranslation_integrable`
- `continuousAnnularSignedKernel_far_bound`

<!-- checkpoint-verification -->

Verification

All 955 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4574-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2663 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 957 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_889_ContinuousAnnularHormander.md`

Checkpoint 889: Actual signed annular kernel satisfies a uniform integral Hormander condition

Lean proves the actual complex signed annular kernel satisfies the integral Hormander condition with the uniform explicit bound 50. This includes every translation, every finite number of dyadic increments, the initial N=1 kernel, all 0<beta<=1 and all complex coefficients of modulus at most 1. Integrability of each restricted translation norm is part of the checked condition. The proof uses the actual endpoint-jump estimates and summable dyadic profile, not an assumed statement from the paper. Matching the signed Fourier symbol and applying an independently stated general singular-integral theorem are next.

Checked declarations

Module	Named supporting theorems
ContinuousAnnularHormander.lean	1

- `continuousAnnularSignedKernel_hormander`

<!-- checkpoint-verification -->

Verification

All 956 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4575-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2664 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 958 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full

high/low conclusion are not external input fields.

`Checkpoint_890_RealFourierIntegralLinearity.md`

Checkpoint 890: Checked linearity of literal real Fourier integrals

Lean checks reusable literal Lebesgue Fourier-integral linearity for integrable complex functions: addition, subtraction, scalar multiplication, and arbitrary finite sums. The convergence hypotheses needed for addition and finite sums are supplied explicitly using Mathlib's Fourier-integral convergence theorem. This supports the exact Fourier identity for the already checked signed annular kernel. No analytic boundedness input is introduced.

Checked declarations

Module	Named supporting theorems
RealFourierIntegralLinearity.lean	4

- `realFourier_add`
- `realFourier_smul`
- `realFourier_sub`
- `realFourier_finsetSum`

<!-- checkpoint-verification -->

Verification

All 957 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4576-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2668 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 959 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_891_ContinuousAnnularSignedFourier.md`

Checkpoint 891: Exact Fourier identity for the actual signed annular kernel

Lean proves the exact Fourier transform of the literal complex signed annular kernel is the actual untruncated bounded signed-increment symbol used by the L2 realization. The proof supplies integrability of every summand, checks Fourier linearity, and invokes the already proved annular change-of-variables identity separately at both scales. Together with the checked uniform L2 bound and integral Hormander condition, this discharges all family-specific hypotheses for the independent scalar singular-integral theorem. No general-p bound is assumed or claimed at this checkpoint.

Checked declarations

Module	Named supporting theorems
ContinuousAnnularSignedFourier.lean	1

- `continuousAnnularSignedKernel_fourier`

<!-- checkpoint-verification -->

Verification

All 958 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4577-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2669 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 960 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_892_ScalarHormanderInterface.md`

Checkpoint 892: Full continuous signed-increment Lp implication from a general singular-integral theorem

Lean now proves the full actual continuous signed-increment Lp estimate for every 1<p<=2, uniformly in beta, the number of increments and arbitrary bounded complex signs, conditional only on an explicit general scalar Hormander singular-integral theorem. All family-specific integrability, exact Fourier, uniform L2 and endpoint-jump Hormander hypotheses have been proved. RawScalarHormanderInput C quantifies over arbitrary integrable kernels and their actual Fourier symbols; it contains no annular, IW, low/high, maximal or restricted conclusion. Its classical general existence and quantitative interpolation consequence are recorded as unproved independent analytic input, not Lean axioms. The second continuous maximal inequality, the quantitative IW low maximum and actual restricted maximal estimate remain unfinished.

Checked declarations

Module	Named supporting theorems
ScalarHormanderInterface.lean	2

- `scalar_hormander_multiplier`
- `continuousAnnularSignedMultiplier_Lp`

<!-- checkpoint-verification -->

Verification

All 959 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4578-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2671 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 961 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_893_ScaledCutoffL1.md`

Checkpoint 893: Actual cutoff kernels have invariant L1 norm and exact Fourier symbols

Lean checks the actual inverse-cutoff kernels at every positive dilation: integrability, the exact invariant L1 norm, and the exact forward Fourier transform eta(a theta). The invariant constant is the integral of the norm of the already constructed Schwartz inverse cutoff, so finiteness follows from the checked Schwartz integrability. A general positive real Fourier dilation identity is also proved from the existing inverse identity. This supplies concrete cutoff applicability data; it adds no analytic theorem input and does not assume a low or maximal bound.

Checked declarations

Module	Named supporting theorems
ScaledCutoffL1.lean	5

- `cutoffL1Constant_nonneg`
- `scaledCutoffKernel_integrable`
- `scaledCutoffKernel_integral_norm`
- `real_fourier_dilate`
- `scaledCutoffKernel_fourier`

<!-- checkpoint-verification -->

Verification

All 960 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4579-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2676 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 962 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization

are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint_894_ContinuousAnnularConvolution.md

Checkpoint 894: Actual continuous annular convolution is dominated by one-sided interval averages

Lean now defines the actual continuous annular convolution using Mathlib's Lebesgue convolution and proves its measurability. For every L2 input and every point x, the literal annular convolution norm is bounded by the one-sided interval average of |f| over [0,N^2], for N>=2 and beta<=1. Local integrability of each reflected translate, integrability of the actual convolution integrand and the kernel box domination are proved explicitly. This is the family-specific applicability step for the independent general Hardy-Littlewood maximal theorem; that theorem has not yet been invoked. No current-paper maximal conclusion is an assumption.

Checked declarations

Module	Named supporting theorems
ContinuousAnnularConvolution.lean	4

- `continuousAnnularConvolution_aestronglyMeasurable`
- `shiftedL2_integrableOn`
- `continuousAnnularKernel_product_box`
- `continuousAnnularConvolution_box_bound`

<!-- checkpoint-verification -->

Verification

All 961 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4580-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2680 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 963 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint_895_RealMultiplierComposition.md

Checkpoint 895: Composition and Lp product bounds for actual real Fourier multipliers

Lean proves composition of the actual real-line multiplication and Fourier-multiplier operators, including their almost-everywhere realization. It then proves the product rule for actual Lp multiplier bounds on

the L2 intersection, with output membership and the exact product of the two constants. These are algebraic/operator deductions from the checked Plancherel realization and extended-norm inequalities; no independent analytic theorem is added. This supplies the composition step for applying the invariant-L1 cutoff kernels to the actual continuous signed increments.

Checked declarations

Module	Named supporting theorems
RealMultiplierComposition.lean	3

- `realLineMultiply_comp_apply`
- `realLineMultiplier_comp_apply`
- `realLpMultiplierBound_mul`

<!-- checkpoint-verification -->

Verification

All 962 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4581-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2683 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 964 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_896_ContinuousAnnularMaximalInterface.md`

Checkpoint 896: Actual integer-scale continuous maximal Lp implication from general Hardy-Littlewood

Lean proves the actual countable continuous annular maximum over every integer N>=2 is in Lp with bound C/(p-1), for 1<p<=2, conditional only on the general one-sided Hardy-Littlewood maximal theorem. The input theorem ranges over arbitrary functions and all positive real interval lengths; it contains no annular or IW conclusion. Own-paper kernel domination, convolution integrability/measurability, simultaneous almost-everywhere domination of all integer scales, boundedness of the actual supremum and its Lp membership are checked. This includes all needed dyadic N>=2. The independent theorem existence is unproved, not a Lean axiom. The quantitative IW low maximum and actual restricted estimate remain unfinished.

Checked declarations

Module	Named supporting theorems
ContinuousAnnularMaximalInterface.lean	2

- `continuousAnnular_maximal_majorant`
- `continuousAnnular_maximal_Lp`

<!-- checkpoint-verification -->

Verification

All 963 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4582-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2685 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 965 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_897_AnnularFourierSmoothness.md`

Checkpoint 897: Actual annular C-infinity smoothness and corrected IW regularity interface

Lean proves integrability of every polynomial moment of the actual compact annular kernel, then proves the annular Fourier symbol, its source cutoff, and every bounded signed cutoff symbol are C-infinity. This checkpoint also corrects the IW interface's two regularity hypotheses from analytic top to C-infinity, as required by the independent theorem. The prior exact source and hashes are preserved; every other old proof source and all prior artifacts must remain authenticated. The changed module and all affected dependents are rebuilt before the full axiom audit. Seven negative/positive correction-validation cases pass. No prior nonzero use of the defective interface existed, and the actual prime high-side theorem does not use its predicate. No manuscript change or own-paper analytic assumption is introduced.

Checked declarations

Module	Named supporting theorems
AnnularFourierSmoothness.lean	4

- `continuousAnnularKernel_moments`
- `annularOscillatoryModel_contDiff`
- `annularCutoffSymbol_contDiff`
- `continuousAnnularSignedSymbol_contDiff`

<!-- checkpoint-verification -->

Verification

All 964 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4583-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2689 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 966 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Propo-

sition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_898_CutoffYoungInterface.md`

Checkpoint 898: Actual uniform cutoff Lp bound from general Young convolution

Lean derives the actual dilated source-cutoff Lp multiplier bound from a separately stated general Young L1 convolution theorem, with exact invariant constant cutoffL1Constant and no loss as p approaches one. The independent input is quantified over arbitrary integrable kernels and their exact actual Fourier symbols, including their standard L2/Fourier realization; it contains no cutoff, annular, IW or maximal conclusion. All concrete cutoff integrability, Fourier identity and invariant-L1 hypotheses are proved. Existence of the general theorem remains explicit and unproved, not a Lean axiom. The actual IW low maximum and restricted estimate remain unfinished.

Checked declarations

Module	Named supporting theorems
CutoffYoungInterface.lean	3

- `continuousCutoff_norm_one`
- `continuousCutoffSymbol_apply`
- `continuousCutoffMultiplier_Lp`

<!-- checkpoint-verification -->

Verification

All 965 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4584-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2692 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 967 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full

high/low conclusion are not external input fields.

Checkpoint_899_ContinuousAnnularCutoffLp.md

Checkpoint 899: Actual signed cutoff Lp bounds with a single near-one loss and initial-scale control

Lean proves the actual signed cutoff symbol factors exactly into the uncut signed symbol and source cutoff. Operator composition combines the checked continuous singular-integral implication with the invariant-L1 Young cutoff bound, retaining precisely one factor 1/(p-1), not an extra near-one loss. The actual annular kernel has integral norm at most 1 for every N>=1 and 0<beta<=1; Young therefore also supplies the actual individual annular and cutoff bounds, including N=1. No current-paper low maximum or restricted estimate is a premise. Independent general Hormander and Young theorem parameters remain explicit.

Checked declarations

Module	Named supporting theorems
ContinuousAnnularCutoffLp.lean	7

- `continuousAnnularSignedSymbol_cutoff`
- `continuousAnnularCutoffSignedMultiplier_Lp`
- `allAnnularBoundedSymbol_untruncated`
- `continuousAnnularKernel_integral_norm_le`
- `annularYoungMultiplier_Lp`
- `allAnnularBoundedSymbol_cutoff`
- `annularCutoffYoungMultiplier_Lp`

<!-- checkpoint-verification -->

Verification

All 966 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4585-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2699 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 968 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint_900_IWSymbolLinearity.md

Checkpoint 900: Exact subtraction and finite-combination identities for actual IW symbols

Lean proves exact linearity of the literal compact periodization and actual cyclic IW symbols under subtraction and arbitrary finite complex linear combinations. The support hypotheses are explicit for every source and target symbol. The proof works directly with the periodization window and the actual

rational-frequency sum, retaining all cyclic Fourier coefficients. No norm inequality or independent analytic input is introduced. This supplies the exact identification between the forthcoming transferred signed multiplier and signed differences of the actual IW annular low family.

Checked declarations

Module	Named supporting theorems
IWSymbolLinearity.lean	4

- `compactCircleLift_bounded_sub`
- `compactCircleLift_bounded_combination`
- `iwCyclicSymbol_sub`
- `iwCyclicSymbol_combination`

<!-- checkpoint-verification -->

Verification

All 967 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4586-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2703 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 969 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_901_IWSignedAnnularBounds.md`

Checkpoint 901: Actual IW signed annular multiplier bound with explicit r times C_(r,k) dependence

Lean transfers the actual bounded smooth signed annular cutoff symbol through the corrected general IW multiplier theorem at p_r=(2r)'. It proves the exact narrow-support applicability, C-infinity applicability, positivity and near-one arithmetic, yielding the quantitative actual integer Lp bound K*r*C_(r,k), with K depending only on the recorded general scalar Hormander constant and the fixed cutoff L1 norm. The estimate is uniform in Q, the finite cyclic measure of variation at most one, beta, the number of increments and arbitrary complex signs. All output membership is explicit. Only the already recorded general IW, Hormander and Young parameters are used; no current-paper low maximum or restricted result is a premise. The exact signed-family identity and finite maximal deduction are next.

Checked declarations

Module	Named supporting theorems
IWSignedAnnularBounds.lean	7

- `scaled_cutoff_boundary_zero`
- `allAnnularBoundedSymbol_boundary`

- `continuousAnnularSignedSymbol_boundary`
- `iwNearOneExponent_cost`
- `iwMultiplierConstant_nonneg`
- `iwAnnularIncrementCost_nonneg`
- `iw_signed_annular_multiplier`

<!-- checkpoint-verification -->

Verification

All 968 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4587-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2710 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 970 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_902_IWAnnularSignedIdentity.md`

Checkpoint 902: Exact identification with signed increments of the actual IW annular low family

Lean identifies the actual cyclic IW annular symbol for every N>=1 with the literal weighted-width Fourier block already used in the high comparison. It proves that the transferred signed continuous symbol is exactly the finite signed sum of differences of these concrete low symbols, and that the resulting integer Fourier operator is exactly the corresponding signed sum of the actual low operators. Canonical low and signed operator definitions are supplied. These are exact identities for arbitrary finite cyclic complex measures and arbitrary complex signs; no analytic theorem input or maximal conclusion is added.

Checked declarations

Module	Named supporting theorems
IWAnnularSignedIdentity.lean	3

- `iwCyclicSymbol_all_annular`
- `iwCyclicSymbol_signed_annular`
- `iw_signed_annular_operator_identity`

<!-- checkpoint-verification -->

Verification

All 969 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4588-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2713 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 971 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_903_IWAnnularLp.md`

Checkpoint 903: Actual IW annular single-scale and signed-increment Lp bounds

Lean proves the single-scale actual IW annular Lp bound C_(r,k) times the invariant cutoff L1 constant for every N>=1, including the initial scale. It also proves the actual signed sum of consecutive IW annular low operators has Lp bound K*r*C_(r,k), with membership and extended-norm control, by the checked exact signed-family identity. These statements are uniform in beta and arbitrary finite cyclic measures of variation at most one. Independent general IW, Hormander and Young theorem parameters remain explicit; no source low maximum or restricted estimate is a hypothesis. The finite short-scale maximum and the narrow large-scale sampling argument remain next.

Checked declarations

Module	Named supporting theorems
IWAnnularLp.lean	2

- `iw_annular_single_multiplier`
- `iw_annular_signed_increments`

<!-- checkpoint-verification -->

Verification

All 970 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4589-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2715 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 972 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities,

cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_904_FiniteSignedMaximalInterface.md`

Checkpoint 904: Precise general finite signed-sum logarithmic maximal interface

Lean records the precise general finite-sequence Rademacher-Menshov/Khintchine maximal consequence needed for short scales, including actual membership, exact partial sums, arbitrary complex signs and the logarithmic dependence on sequence length. The independent theorem quantifies over arbitrary scalar sequences and contains no Fourier, arithmetic, annular or IW object. Its existence and the general probabilistic/dyadic argument are explicitly unproved, not Lean axioms. Separate checked lemmas identify the literal partial-sum maximum, its nonnegativity and domination. Instantiation for the actual IW increment sequence and addition of the initial term remain next.

Checked declarations

Module	Named supporting theorems
FiniteSignedMaximalInterface.lean	3

- `finitePartialSumMaximum_nonneg`
- `finitePartialSumMaximum_dominates`
- `finite_signed_maximal`

<!-- checkpoint-verification -->

Verification

All 971 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4590-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2718 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 973 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_905_IWShortScaleIncrementMaximum.md`

Checkpoint 905: Actual finite IW increment maximum with logarithmic scale-count dependence

Lean applies the general finite signed-sum maximal theorem to the literal increments of the actual IW annular low operators. It proves the exact coordinate identity between the signed scalar sequence and the signed operator, supplies the already checked K*r*C_(r,k) bound for every complex sign choice, and derives the actual partial-sum maximum with logarithmic cost log(2+J), membership and extended-norm control. Only explicitly recorded general standard analytic inputs are used; no paper-specific maximum

is an input. Adding the separately checked initial annular term to obtain the actual finite low maximum is next.

Checked declarations

Module	Named supporting theorems
IWShortScaleIncrementMaximum.lean	2

- `iwAnnularIncrementSequence_signed`
- `iw_short_scale_increment_maximum`

<!-- checkpoint-verification -->

Verification

All 972 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4591-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2720 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 974 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_906_FiniteDifferenceMaximum.md`

Checkpoint 906: Checked telescoping and Lp assembly of a finite family maximum from its increments

Lean proves a general finite-family maximal assembly from consecutive differences: exact telescoping, pointwise domination by the initial norm plus the partial-sum maximum, and the actual Lp membership and extended-norm triangle bound. This applies to arbitrary L2-valued sequences on the integers and uses only already formalized algebra, finite suprema and Lp inequalities. It introduces no independent analytic input. Combining it with the checked actual IW initial and increment bounds will yield the finite low-family maximum.

Checked declarations

Module	Named supporting theorems
FiniteDifferenceMaximum.lean	3

- `finiteDifferenceSequence_telescope`
- `finiteDifferenceSequence_maximum_bound`
- `finiteDifferenceSequence_maximum_Lp`

<!-- checkpoint-verification -->

Verification

All 973 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4592-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2723 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 975 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_907_IWShortScaleMaximum.md`

Checkpoint 907: Actual finite IW annular low maximum including the initial scale

Lean proves the actual finite IW annular low maximum over every dyadic index 0<=j<=J, including the initial N=1 term. Output Lp membership and the explicit sum of the initial C_(r,k) cutoff cost and logarithmic K*r*C_(r,k) increment cost are checked. All exact symbol, signed-operator, telescoping and Lp triangle identities are supplied; only recorded general standard analytic inputs remain explicit. The source review moves only Proposition7.22 from open to partial: the full J_*, narrow large-scale sampling and cutoff-remainder argument, D(r,k,Q), and actual restricted estimate are still unfinished. No full Section7 statement is awarded.

Checked declarations

Module	Named supporting theorems
IWShortScaleMaximum.lean	1

- `iw_short_scale_maximum`

<!-- checkpoint-verification -->

Verification

All 974 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4593-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2724 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 976 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth

multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_908_IWNarrowWidth.md`

Checkpoint 908: Exact narrow-cutoff geometry and inverse-width logarithmic bound

Lean defines the exact narrower cutoff delta=min(epsilon/16,1/(100P)) from the source. It proves positivity, both defining upper bounds, the uniform smallness bound, exact containment in the core of the original cutoff, and the logarithmic inverse-width bound by the maximum of the original-width and period contributions. These are unconditional scalar applicability estimates and introduce no analytic input. The dyadic transition index and the quantitative logarithmic budget for the checked finite low maximum are next.

Checked declarations

Module	Named supporting theorems
IWNarrowWidth.lean	6

- `iwNarrowWidth_pos`
- `iwNarrowWidth_le_original`
- `iwNarrowWidth_le_period`
- `iwNarrowWidth_small`
- `iwNarrowWidth_cutoff_core`
- `iwNarrowWidth_log_inverse`

<!-- checkpoint-verification -->

Verification

All 975 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4594-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2730 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 977 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_909_IWTransitionScale.md`

Checkpoint 909: Exact natural-ceiling transition index and dyadic scale bounds

Lean constructs the source dyadic transition index by the natural ceiling of -log(delta)/(2 log 2). It proves the exact ceiling bounds and the actual geometric scale inequalities 1<=N_*^2 delta<4 and 1<=N_* sqrt(delta)<2 for every 0<delta<=1. The log-two lower bound is also checked. This supplies the integer transition scale rather than treating an approximate real cutoff as a natural one. No analytic input, low maximum or restricted conclusion is assumed.

Checked declarations

Module	Named supporting theorems
IWTransitionScale.lean	4

- `real_log_two_half_le`
- `iwTransitionIndex_bounds`
- `iwTransitionScale_squared_bounds`
- `iwTransitionScale_sqrt_bounds`

<!-- checkpoint-verification -->

Verification

All 976 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4595-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2734 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 978 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_910_IWFiniteMaximumCost.md`

Checkpoint 910: Finite low maximum has the required r times C_(r,k) logarithmic cost

Lean simplifies the exact finite IW maximum cost to a universal constant times r*C_(r,k)*log(2+J), without an extra near-one or parameter loss. It also proves C_(r,k)>=1 in the actual admissible range, so the finite maximum can be compared with the source r*C_(r,k)^2 cost. These are explicit scalar deductions from the actual parameter definitions and the fixed cutoff norm, not asymptotic assumptions. The exact transition-index logarithmic budget and its substitution into the finite maximum are next.

Checked declarations

Module	Named supporting theorems
IWFiniteMaximumCost.lean	2

- `iw_short_scale_cost_bound`
- `iwMultiplierConstant_one_le`

<!-- checkpoint-verification -->

Verification

All 977 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4596-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2736 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 979 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_911_IWTransitionLogBudget.md`

Checkpoint 911: Actual period width and integer transition-index logarithmic budgets

Lean proves the full source logarithmic budget for the actual transition index. The constructed period obeys log P<=2Q; the exact narrow width has -log(delta)<=100(Q+r log B); the natural ceiling index satisfies J_*+2<=101(Q+r log B); and log(2+J_*)<=101 log(e+Q+r log B). Every constant is absolute and every quantity is the actual source-defined one, with no extra parameter dependence. No analytic input or current-paper maximal estimate is assumed. Substitution into the actual finite low maximum is next.

Checked declarations

Module	Named supporting theorems
IWTransitionLogBudget.lean	4

- `iw_period_log_bound`
- `iw_narrow_log_inverse_budget`
- `iw_transition_index_linear_budget`
- `iw_transition_log_cost_budget`

<!-- checkpoint-verification -->

Verification

All 978 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4597-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2740 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 980 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain

partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_912_IWShortTransitionMaximum.md`

Checkpoint 912: Actual short-scale IW low maximum with the precise D(r,k,Q) cost

Lean proves the actual short-scale IW low maximum for every dyadic N<=N_* has the paper's precise K*D(r,k,Q) bound, uniformly in beta, the cyclic complex measure and the arithmetic parameters. This uses the actual natural-ceiling transition index, period P<=3^Q, exact original/narrow widths, the checked r*C_(r,k)*log(2+J) finite cost and C_(r,k)>=1. Membership and extended-norm control are explicit. K depends only on the fixed cutoff and the recorded general standard analytic constants. No own-paper low maximum is assumed. The narrow large-scale maximum, summed cutoff remainder and actual restricted estimate remain unfinished; source Proposition7.22 stays partial.

Checked declarations

Module	Named supporting theorems
IWShortTransitionMaximum.lean	3

- `iwFiniteLowConstant_nonneg`
- `iw_short_transition_cost_bound`
- `iw_short_transition_maximum`

<!-- checkpoint-verification -->

Verification

All 979 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4598-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2743 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 981 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_913_RealMultiplierSubtraction.md`

Checkpoint 913: Actual real multiplier subtraction and Lp triangle bounds

Lean proves subtraction for the actual real-line Fourier multipliers, using the checked Plancherel realization and almost-everywhere multiplication identities. It then proves the general Lp subtraction bound with exact K+L cost for every p>=1, including output membership. The proof retains the quotient representatives and uses the actual extended Lp triangle inequality. No external analytic input is introduced. This is the operator-algebra step for the original-minus-narrow cutoff remainder.

Checked declarations

Module	Named supporting theorems
RealMultiplierSubtraction.lean	2

- `realLineMultiplier_symbol_sub`
- `realLpMultiplierBound_sub`

<!-- checkpoint-verification -->

Verification

All 980 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4599-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2745 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 982 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_914_RieszThorinMultiplierInterface.md`

Checkpoint 914: General Riesz-Thorin interface and exact p_r interpolation parameter

Lean states the precise general Riesz-Thorin multiplier interpolation boundary and proves the specialization to p_r=(2r)' with interpolation parameter 1/r, including its admissible range and exact reciprocal-exponent identity. The independent theorem quantifies over arbitrary bounded continuous symbols and endpoint bounds; no annular, IW, cutoff, maximal or current-paper conclusion occurs in its fields. Its general existence remains unproved, not a Lean axiom. The actual remainder's endpoint hypotheses and subsequent transfer/summation remain own-paper proof obligations.

Checked declarations

Module	Named supporting theorems
RieszThorinMultiplierInterface.lean	1

- `real_interpolate_one_two_at_iw`

<!-- checkpoint-verification -->

Verification

All 981 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4600-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2746 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 983 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_915_AnnularCutoffRemainderBounds.md`

Checkpoint 915: Actual cutoff remainder identities smoothness and L1/L2 endpoint bounds

Lean proves the actual original-minus-narrow annular cutoff remainder: exact nested-cutoff core and product identities, literal remainder factorization, uniform domination by the actual annular Fourier symbol, C-infinity smoothness, and the sharper L2 decay 4b/(pi N^2). The general-p rough bound 2*cutoffL1Constant follows from the checked individual Young bounds and actual operator subtraction; no own remainder norm is assumed. All endpoint applicability is now checked. Interpolation through the separately recorded general Riesz-Thorin theorem and subsequent IW transfer/summation are next.

Checked declarations

Module	Named supporting theorems
AnnularCutoffRemainderBounds.lean	8

- `sourceArcCutoff_nested_core`
- `sourceArcCutoff_nested_product`
- `annularCutoffRemainderSymbol_factor`
- `annularCutoffRemainderSymbol_norm_le`
- `annularCutoffRemainderSymbol_decay`
- `annularCutoffRemainderSymbol_smooth`
- `annularCutoffRemainder_Lp_rough`
- `annularCutoffRemainder_L2`

<!-- checkpoint-verification -->

Verification

All 982 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4601-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2754 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 984 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_916_AnnularCutoffRemainderLp.md`

Checkpoint 916: Actual annular cutoff remainder interpolation at the IW exponent

Lean proves the actual original-minus-narrow annular cutoff remainder at the near-one exponent p_r: the general Young endpoint and the checked sharper L2 decay interpolate to A (b/N^2)^(1/r), with the fixed explicit constant A=1+2*cutoffL1Constant+4/pi. The reciprocal interpolation identity, positivity, exact scalar power simplification and actual operator endpoint applicability are checked. Only the separately recorded independent general Young and Riesz-Thorin theorem hypotheses are used. IW transfer, summation and large-scale sampling remain unfinished; the actual restricted maximal estimate is not yet proved.

Checked declarations

Module	Named supporting theorems
AnnularCutoffRemainderLp.lean	3

- `annularRemainderConstant_pos`
- `annularRemainderConstant_bounds`
- `annularCutoffRemainder_Lp`

<!-- checkpoint-verification -->

Verification

All 983 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4602-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2757 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 985 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete con-

tinuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_917_IWAnnularRemainder.md`

Checkpoint 917: Actual integer IW cutoff remainder with interpolated decay

The actual integer IW annular low operator minus its narrower-cutoff version satisfies the interpolated decay Crk*A*(b/N^2)^(1/r). Lean checks both support boundaries, the common IW width, C-infinity smoothness, exponent range, scalar positivity and the exact equality of the transferred cyclic symbol with the two actual width blocks. The only hypotheses are the previously recorded independent general IW, Young and Riesz-Thorin theorems; no own-paper remainder or maximal bound is assumed. Countable geometric summation and the separate narrow large-scale sampling step remain.

Checked declarations

Module	Named supporting theorems
IWAnnularRemainder.lean	1

- `iw_annular_remainder_multiplier`

<!-- checkpoint-verification -->

Verification

All 984 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4603-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2758 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 986 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_918_CountableLpMajorant.md`

Checkpoint 918: Countable Lp majorants from summable actual norms

Lean proves a general countable summation principle on the integers for every p>=1: if the actual eLp norms of arbitrary complex functions have finite total sum, their pointwise norm sum is a nonnegative real Lp function, dominates every term at every integer, and has eLp norm at most that total sum. The proof uses the pinned Mathlib Lp infinite-sum theorem, checks its hypotheses, and upgrades almost-everywhere identities to pointwise identities for counting measure. No external analytic theorem is added. This supplies the countable majorant required to sum the actual IW remainders, whose geometric scalar budget is being checked next.

Checked declarations

Module	Named supporting theorems
CountableLpMajorant.lean	1

- `countable_integer_Lp_majorant`

<!-- checkpoint-verification -->

Verification

All 985 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4605-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2759 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 987 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_919_IWRemainderGeometry.md`

Checkpoint 919: Exact IW transition geometry and summable remainder decay

Lean proves the full scalar geometry needed for countable cutoff-remainder summation. The actual narrow width has a positive inverse b, satisfies 2a<=b and b>=2, and its exact transition scale is at least two. At the literal scales 2^j*Nstar the interpolation factor (b/N^2)^(1/r) is bounded by exp(-1/r)^j; the geometric sum is at most 2r uniformly for integer r>=2. These statements use the already proved exact transition bounds, not an asymptotic replacement or a chosen unverified threshold. No external theorem is introduced. The next step combines this geometry with the checked actual integer IW remainder and countable Lp majorant.

Checked declarations

Module	Named supporting theorems
IWRemainderGeometry.lean	5

- `iwRemainderRatio_bounds`
- `iwRemainderRatio_sum_bound`
- `iwNarrow_inverse_geometry`
- `iwTransitionScale_two_le`
- `iw_transition_remainder_decay`

<!-- checkpoint-verification -->

Verification

All 986 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4606-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2764 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 988 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_920_IWRemainderMaximum.md`

Checkpoint 920: Actual countable IW cutoff remainder maximum

The actual large-scale IW original-minus-narrow remainder now has a countable nonnegative Lp majorant at every integer: all literal scales N=2^j*Nstar are dominated simultaneously, with eLp norm at most 2*A*r*Crk times the input norm. The proof combines the checked actual IW multiplier difference, exact narrow-width geometry, interpolation decay, the uniform geometric sum, and the general countable Lp theorem. It applies to every input in the L2/Lp intersection, not merely finite tests. No low maximal bound is assumed and no additional external analytic input is introduced. The finite short-scale K*D maximum is already checked; the narrow large-scale main term, full low estimate, and actual restricted estimate remain unfinished.

Checked declarations

Module	Named supporting theorems
IWRemainderMaximum.lean	2

- `geometric_integer_Lp_majorant`
- `iw_large_remainder_majorant`

<!-- checkpoint-verification -->

Verification

All 987 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4607-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2766 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 989 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization

are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint_921_IWSamplingSpaces.md

Checkpoint 921: Precise independent major-arc and finite-target Shannon sampling interfaces

Lean records precise independent sampling interfaces: scalar finite-cyclic Tao Theorem1.6 for arbitrary bandlimited Schwartz components and the finite-coordinate l-infinity specialization of Tao Proposition1.5 at width <=1/(100P). The normalized cyclic norm is explicit, as are actual diagonal values, exponent ranges, Fourier support and input membership. Lean proves the quantitative major-arc applicability wrapper from the checked IW goodness and periodic-denominator data, together with literal finite-coordinate maximum and synthesis identities. These general published sampling theorems are explicit hypotheses, not Lean axioms; neither contains an annular kernel or a current-paper low/high or maximal conclusion. The actual lift, its diagonal identity and the narrow large-scale application remain to be proved.

Checked declarations

Module	Named supporting theorems
IWSamplingSpaces.lean	5

- `cyclicSynthesisSchwartz_apply`
- `finiteCoordinateMaximum_nonneg`
- `finiteCoordinateMaximum_le`
- `finiteCoordinateMaximum_dominates`
- `quantitative_iw_sampling`

<!-- checkpoint-verification -->

Verification

All 988 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4608-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2771 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 990 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint_922_IWFiniteLiftIdentity.md

Checkpoint 922: Exact periodized Fourier kernels and finite-input sampling identities

Lean proves exact Fourier identities needed for the concrete sampling lift: translating a circle symbol multiplies its integer inverse kernel by the correctly signed character; finite-symbol sums commute with the inverse kernel; the actual cyclic IW symbol has the literal finite sum of modulated real inverse Fourier kernels; and the multiplier on a finite input equals the inverse kernel of the symbol times the actual finite Fourier polynomial. Periodization also commutes with multiplication by a circle function after the proper rational shift. These are equalities of the existing operators, with no additional analytic assumption. Constructing the compact smooth lift amplitudes and identifying their diagonal with the actual cutoff projection are next.

Checked declarations

Module	Named supporting theorems
IWFiniteLiftIdentity.lean	5

- `fourierKernel_translate`
- `fourierKernel_finsetSum`
- `iwCyclicSymbol_kernel`
- `integerMultiplier_finite_symbol_product`
- `compactCircleLift_mul_circle`

<!-- checkpoint-verification -->

Verification

All 989 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4609-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2776 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 991 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint_923_CyclicRealLpBounds.md

Checkpoint 923: Normalized cyclic Lp bounds and actual finite-cyclic Young inequality

Lean proves the normalized cyclic real-Lp norm inequalities used in the large-scale sampling argument: fibrewise norm comparison, invariance under cyclic translations, triangle inequality, scalar multiplication and finite sums. For the literal cyclic convolution of arbitrary Schwartz fibres, the resulting norm is at most the actual cyclic total variation times the original norm. The normalization is explicit, all relevant fibre memberships are checked, and no factor depending on the period is lost. These elementary analytic steps are proved rather than included in a new external input. The actual finite-input Fourier lift and narrow real-variable maximal application remain next.

Checked declarations

Module	Named supporting theorems
CyclicRealLpBounds.lean	9

- `cyclicRealLpSize_mono_bound`
- `eLpNorm_count_equiv`
- `cyclicRealLpSize_translate`
- `cyclicRealLpSize_add`
- `cyclicRealLpSize_smul`
- `cyclicRealLpSize_zero`
- `cyclicRealLpSize_finsetSum`
- `cyclicConvolutionSchwartz_apply`
- `cyclicConvolutionSchwartz_Lp`

<!-- checkpoint-verification -->

Verification

All 990 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4610-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2785 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 992 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_924_FiniteSmoothFourierLift.md`

Checkpoint 924: Concrete smooth finite-input Fourier lift and actual IW projection formula

Lean constructs the concrete compact smooth Fourier amplitudes for every finite complex integer input and arbitrary smooth compact real symbol, and their actual inverse Schwartz transforms. Their Fourier transforms, support boundary, smoothness, periodization and rational-shift identities are checked. The existing integer IW multiplier on the finite input is exactly the finite sum of these modulated real lift components with the actual cyclic Fourier coefficients. This establishes the Fourier algebra needed for sampling, rather than assuming an intertwining identity. The cyclic residue diagonal identification, lift norm bound and narrow main-term maximum remain next.

Checked declarations

Module	Named supporting theorems
FiniteSmoothFourierLift.lean	11

- `compact_support_of_unit_boundary`
- `finiteFourierPolynomial_shift_smooth`
- `finiteLiftFrequencyMap_boundary`
- `finiteLiftFrequencyMap_smooth`

- `finiteLiftFrequencySchwartz_apply`
- `finiteLiftComponent_fourier`
- `finiteLiftFrequency_periodization`
- `finiteLiftFrequency_shift_periodization`
- `finiteLiftComponent_inverse`
- `finite_arc_projection_formula`
- `iwCyclicSymbol_finite_lift`

<!-- checkpoint-verification -->

Verification

All 991 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4611-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2796 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 993 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_925_CyclicLiftCharacters.md`

Checkpoint 925: Actual cyclic diagonal identification and quantitative finite-input lift bound

Lean proves the exact rational-character descent from integers to the finite cyclic group, including subtraction and the actual residue-valued diagonal. Cyclic convolution of the synthesis is identified with multiplication by the literal cyclic Fourier coefficients. The point mass has total variation and Fourier coefficients exactly one. Consequently the actual diagonal of the constructed finite-input lift is precisely the existing IW cutoff projection, and its cyclic convolution has the actual weighted IW projection as diagonal. Combining these identities with the separately recorded independent major-arc sampling and IW multiplier hypotheses gives the lift norm bound Crk^2*K times the finite-input norm. No intertwining or own low maximum is assumed. The narrow annular real-variable maximum and Shannon application remain unfinished.

Checked declarations

Module	Named supporting theorems
CyclicLiftCharacters.lean	10

- `rationalCharacter_eq_of_cyclic_cast`
- `cyclicLiftCharacter_integer`
- `cyclicLiftCharacter_sub`
- `cyclicSynthesis_convolution`
- `cyclicSynthesis_diagonal`
- `cyclicPointMass_variation`
- `cyclicPointMass_coefficient`
- `iw_finite_lift_diagonal`

- `iw_cyclic_convolution_lift_diagonal`
- `iw_finite_lift_size`

<!-- checkpoint-verification -->

Verification

All 992 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4612-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2806 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 994 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_926_SchwartzMultiplierRealization.md`

Checkpoint 926: Actual Schwartz realization of real multipliers and cyclic norm transfer

Lean identifies the actual real L2 Fourier multiplier with the corresponding Schwartz function whenever their literal Fourier transforms agree. Its general Lp multiplier bound therefore controls the actual Schwartz representative, with both input and output almost-everywhere identities checked. Compact smooth multipliers are constructed on arbitrary Schwartz functions by their genuine inverse Fourier transform, their exact Fourier formula and Lp bound are proved, and the latter lifts to the normalized cyclic family norm. This removes a possible gap between abstract L2 operator norms and the pointwise functions later sampled. No additional external analytic input is introduced.

Checked declarations

Module	Named supporting theorems
SchwartzMultiplierRealization.lean	6

- `realLineFourier_schwartz`
- `realLineMultiplier_schwartz_of_fourier`
- `realLpMultiplierBound_on_schwartz`
- `compactSmoothMultiplier_fourier`
- `compactSmoothMultiplier_Lp`
- `cyclicRealLpSize_compactMultiplier`

<!-- checkpoint-verification -->

Verification

All 993 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4613-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2812 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 995 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_927_SchwartzConvolutionIdentity.md`

Checkpoint 927: Pointwise annular convolution identification for the sampled Schwartz functions

Lean proves that an integrable real convolution against a Schwartz function is exactly the Schwartz output whenever the literal Fourier transforms agree. The proof checks continuity, integrability, Fourier convolution and inversion, so it is pointwise rather than just an L2 identity. Applied to the actual annular kernel, this identifies its narrow-cutoff smooth multiplier with the actual annular convolution of the cutoff-filtered input. The cutoff and annular smoothness/support witnesses and finite-coordinate maximum continuity are proved explicitly. This supplies the final real-operator identification needed to apply the previously checked continuous maximal estimate to the narrow main term.

Checked declarations

Module	Named supporting theorems
SchwartzConvolutionIdentity.lean	6

- `integrable_convolution_eq_schwartz`
- `continuousCutoffSymbol_smooth`
- `continuousCutoffSymbol_boundary`
- `allAnnularBoundedSymbol_smooth`
- `compactSmoothMultiplier_annular_convolution`
- `finiteCoordinateMaximum_continuous`

<!-- checkpoint-verification -->

Verification

All 994 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4615-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2818 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 996 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth

multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint_928_SchwartzAnnularMaximum.md

Checkpoint 928: Actual narrow annular finite maximum on real and normalized cyclic Schwartz families

For arbitrary Schwartz inputs, Lean proves the literal finite maximum of the actual annular narrow-cutoff Schwartz outputs belongs to Lp and has norm at most (H/(p-1))*cutoffL1Constant times the input norm, uniformly over arbitrary finite collections of integer scales at least two and 0<=beta<=1. Its realization as the actual continuous annular convolution, finite maximum measurability, common Hardy-Littlewood majorant and Young cutoff cost are all checked. The estimate lifts to arbitrary normalized cyclic Schwartz families without any dependence on the number of scales or the period. Only the recorded independent general Hardy-Littlewood and Young inputs are used. The final narrow sampling and full low maximum assembly are next.

Checked declarations

Module	Named supporting theorems
SchwartzAnnularMaximum.lean	4

- `annularSchwartzOutput_convolution`
- `cutoffSchwartzOutput_Lp`
- `annularSchwartzOutput_maximum_Lp`
- `cyclic_annularSchwartzOutput_maximum`

<!-- checkpoint-verification -->

Verification

All 995 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4616-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2822 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 997 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint_929_SchwartzSynthesisMultipliers.md

Checkpoint 929: Exact multiplier synthesis and nested cutoff intertwining

Lean proves exact Fourier uniqueness on Schwartz space, the Fourier formula for the cyclic synthesis, its equality with actual cyclic convolution after inserting the cyclic Fourier coefficients, and the commutation of arbitrary compact smooth multipliers with a synthesis whose components satisfy the corresponding Fourier identity. The actual narrow annular symbol is proved to absorb the original wider cutoff using the checked nested-cutoff product identity. These exact algebraic steps supply the intertwining needed to turn the sampled narrow Schwartz outputs into the actual narrow IW operators; no sampling or maximal conclusion is assumed in their proofs.

Checked declarations

Module	Named supporting theorems
SchwartzSynthesisMultipliers.lean	6

- `schwartz_eq_of_fourier`
- `cyclicSynthesisSchwartz_fourier`
- `cyclicSynthesis_weighted_convolution`
- `compactSmoothMultiplier_synthesis`
- `annular_symbol_absorbs_wide_cutoff`
- `schwartz_smul_fourier`

<!-- checkpoint-verification -->

Verification

All 996 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4617-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2828 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 998 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_930_IWNarrowLiftIdentity.md`

Checkpoint 930: Actual narrow IW operator as the diagonal of its annular lift

Lean proves the concrete narrow-lift intertwining: apply the actual annular narrow multiplier to the cyclic convolution of the original cutoff lift, and the result is exactly the cyclic convolution of the finite-input annular lift. On the actual diagonal (n,n mod P), this is precisely the existing narrow IW low operator on the finite input. The proof checks absorption of the original cutoff, rational characters, actual Fourier transforms and the genuine finite-input operator formula. The narrow output Fourier band is proved explicitly. Thus the functions to which the finite-target Shannon hypothesis will be applied are identified with the paper's operators, rather than treated as a generic unnamed model.

Checked declarations

Module	Named supporting theorems
IWNarrowLiftIdentity.lean	3

- `annularSchwartzOutput_lift`
- `iw_narrow_lift_diagonal`
- `annularSchwartzOutput_band`

<!-- checkpoint-verification -->

Verification

All 997 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4618-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2831 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 999 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_931_IWNarrowFiniteMaximum.md`

Checkpoint 931: Actual narrow IW finite maximum with quantitative r Crk squared cost

The actual narrow IW finite maximum is proved, for every finite complex integer input and arbitrary finite collections of integer scales at least two. Its Lp norm is at most K*r*Crk^2 times the input norm, with K=2*S*H*cutoffL1Constant^2 independent of the input, beta, period and number of scales. The proof combines the checked actual cutoff lift bound, cyclic total-variation contraction, real annular maximum, exact narrow Fourier band, general finite-target Shannon theorem, and actual diagonal identity. Membership is explicit and no own-paper maximal bound is a premise. The countable extension and assembly with the already checked short-scale maximum and countable remainder are next; the actual restricted estimate is still unfinished.

Checked declarations

Module	Named supporting theorems
IWNarrowFiniteMaximum.lean	2

- `iwFiniteCutoffLift_size`
- `iw_narrow_finite_maximum`

<!-- checkpoint-verification -->

Verification

All 998 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4619-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2833 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1000 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_932_UniformFiniteLpMaximum.md`

Checkpoint 932: General uniform finite to countable actual Lp maximal extension

Lean proves a general uniform-finite-to-countable Lp maximal extension on the integers for every nonzero exponent, including infinity. A finite uniform extended-norm bound for every literal finite coordinate maximum yields one nonnegative actual Lp majorant dominating every member at every integer, with the same bound. Pointwise boundedness follows from counting-measure coordinate control; the convergence of finite maxima and the norm passage use the pinned Mathlib monotone-convergence and Fatou results with their hypotheses checked. No additional external theorem is introduced. This extends the newly checked narrow IW finite maxima to a genuine countable maximum.

Checked declarations

Module	Named supporting theorems
UniformFiniteLpMaximum.lean	1

- `uniform_finite_integer_Lp_majorant`

<!-- checkpoint-verification -->

Verification

All 999 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4621-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2834 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1001 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities,

cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_933_IWNarrowCountableMaximum.md`

Checkpoint 933: Actual narrow IW countable maximum over every integer scale

The actual narrow IW low operator now has one nonnegative Lp majorant simultaneously at every integer scale N>=2, for each finite complex input. Its norm retains the checked K*r*Crk^2 bound, with no dependence on the number of scales, the period or beta in [0,1]. The proof uses the actual finite maxima and the checked uniform finite-to-countable extension; no countable maximal conclusion is an external premise. In particular it covers all dyadic scales beyond the exact transition scale. The remaining low-side task is the assembly with the already checked short-scale maximum and countable cutoff remainder.

Checked declarations

Module	Named supporting theorems
IWNarrowCountableMaximum.lean	2

- `iwNarrowLowConstant_nonneg`
- `iw_narrow_integer_majorant`

<!-- checkpoint-verification -->

Verification

All 1000 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4622-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2836 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1002 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_934_IWFullLowMaximum.md`

Checkpoint 934: Full actual finite-input dyadic IW low maximum with D cost

The full actual dyadic IW low operator family on every finite complex input now has one nonnegative Lp majorant of norm at most K*D(r,k,Q) times the input norm. Lean assembles the exact short-scale maximum, the actual narrow countable maximum and the interpolated countable remainder, checks the exact dyadic reindexing and proves every cost fits D with K independent of the arithmetic parameters, beta, input and cyclic measure. All operator identities and applicability hypotheses are checked. Only the separately recorded independent general standard theorems are hypotheses; no own low/high/restricted estimate is assumed. The source ledger is reviewed without upgrading its partial claims: full printed all-input/real-Q clauses and the actual restricted maximal estimate remain unfinished.

Checked declarations

Module	Named supporting theorems
IWFullLowMaximum.lean	5

- `iwCompleteLowConstant_pos`
- `iwMaximalCost_lower_bounds`
- `integer_operator_difference_bound`
- `iwAnnularLowOperator_scale_congr`
- `iw_finite_input_low_majorant`

<!-- checkpoint-verification -->

Verification

All 1001 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4623-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2841 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1003 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_935_IntegerLpMeasureBridge.md`

Checkpoint 935: Exact integer Lp norm bridge and indicator power budget

Lean proves the exact bridge between counting-measure Lp functions and the existing integer lp sequences at every positive finite exponent, including equality of the extended measure norm with the integer sequence norm. Finite complex inputs are thereby proved to satisfy every Lp membership hypothesis used by the actual low theorem. The exact indicator norm power is checked, and an actual Lp majorant bounded by K times the indicator input norm is converted into the finite power budget K^p*|F| required by the restricted-distribution argument. These are proved norm and membership identities, not additional independent inputs.

Checked declarations

Module	Named supporting theorems
IntegerLpMeasureBridge.lean	6

- `mem$\ell$p_of_count_memLp`
- `eLpNorm_count_eq_lp_enorm`
- `count_memLp_of_mem$\ell$p`
- `finiteInputLp_count_memLp`
- `finiteIndicatorLp_norm_power`
- `count_Lp_indicator_power_budget`

<!-- checkpoint-verification -->

Verification

All 1002 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4624-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2847 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1004 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_936_IWPrimeLowMaximum.md`

Checkpoint 936: Actual prime low maximum with the selected Page character

The actual prime low operator used by the already checked high theorem now has its own full dyadic finite-input Lp majorant, with norm at most 2*K*D(r,k,Q) times the input norm. Lean identifies the operator exactly as the principal cyclic square term minus its selected fixed Page-character term, checks the actual induced character and its cyclic total variation, and uses the beta-uniform full IW low estimate for both. The varying activation mask is handled pointwise; it introduces no additional maximal hypothesis. Finite-input Lp membership is discharged by the norm bridge. No new external analytic input is introduced. The source review metadata now also records both exact active statement hashes. Actual low and high norms now refer to the same existing prime operators; their quantitative indicator distribution assembly and optimization are next.

Checked declarations

Module	Named supporting theorems
IWPrimeLowMaximum.lean	4

- `iwPrimeLowOperator_none`
- `iwPrimeLowOperator_some`
- `iwPrimeLowOperator_some_norm`
- `iw_prime_finite_low_majorant`

<!-- checkpoint-verification -->

Verification

All 1003 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4625-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2851 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1005 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs.

The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_937_FiniteInputMaximalExtension.md`

Checkpoint 937: All-input extension and literal countable integer Lp maxima

Lean proves a general finite-input-to-all-input extension for countable families of actual bounded integer L2 operators. Finite inputs converge to the original L2 input, each finite output maximum is continuous, truncating the input does not increase its counting-measure Lp norm, and the checked Fatou and finite-to-countable principles preserve the exact maximal constant. This extends a proved finite-input maximal bound, not an independently assumed own-paper conclusion. The resulting all-input property implies the literal countable maximum bound and the standard integer lp formulation for every positive finite p<=2, using the exact counting-norm bridge and canonical integer lp inclusion into l2. No additional external analytic theorem is introduced.

Checked declarations

Module	Named supporting theorems
FiniteInputMaximalExtension.lean	4

- `finiteInputLp_count_norm_le`
- `finite_input_maximal_extension`
- `allInputLpMaximalBound_integerLp`
- `allInputLpMaximalBound_maximum`

<!-- checkpoint-verification -->

Verification

All 1004 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4626-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2855 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1006 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full

high/low conclusion are not external input fields.

`Checkpoint_938_IWAllInputLowMaximum.md`

Checkpoint 938: Full actual natural-Q IW low maximum on all integer lp inputs

The natural-Q IW low estimate is now proved on all integer lp inputs at the near-one exponent, with a countable majorant of norm at most K*D(r,k,Q) times the actual lp input norm. The proved finite-input extension uses L2 continuity and Fatou; every integer lp input at p_r<=2 embeds canonically into l2, with exact counting-norm identification. The actual selected-Page prime low family also has its all-input maximal bound, with factor two. All own smoothness, Fourier, support, sampling, normalization and countable-limit obligations are discharged; the recorded general independent theorem hypotheses remain. The source ledger distinguishes the remaining printed real-Q/rough-symbol generality and the still unfinished actual restricted estimate.

Checked declarations

Module	Named supporting theorems
IWAllInputLowMaximum.lean	5

- `iwNearOneExponent_toReal_pos`
- `iwNearOneExponent_ennreal_le_two`
- `iw_full_low_maximal_bound`
- `iw_prime_low_maximal_bound`
- `iw_low_all_integer_lp`

<!-- checkpoint-verification -->

Verification

All 1005 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4627-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2860 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1007 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_939_MangoldtAnnularPositiveFamily.md`

Checkpoint 939: Actual positive dyadic annular von Mangoldt square averages

Lean constructs the actual positive dyadic annular von Mangoldt square-average family with its literal index interval, square shifts, normalization and all prime powers. The existing proved Chebyshev bound supplies its uniform mass; no prime number theorem is used. Every family member is identified exactly with the already formalized annular weighted-square L2 operator. This provides the concrete positive

maximal family to which the checked actual IW low/high bounds and the restricted-distribution lemma will be applied.

Checked declarations

Module	Named supporting theorems
MangoldtAnnularPositiveFamily.lean	2

- `mangoldtAnnularDyadicFamily_apply`
- `mangoldtAnnularDyadicFamily_mass`

<!-- checkpoint-verification -->

Verification

All 1006 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4628-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2862 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1008 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_940_IWPrimeRestrictedDistribution.md`

Checkpoint 940: Actual prime annular restricted-distribution inequality from checked IW low and high bounds

The quantitative restricted-distribution inequality is now proved for the actual positive dyadic annular von Mangoldt square maximum, including all prime powers. Lean uses the actual source-scale high theorem to choose the same IW geometry and fixed Page datum, applies the checked all-input prime low bound, checks both indicator power budgets and the exact dyadic high reindexing, and invokes the proved union/Chebyshev argument. The result contains precisely the initial-scale t*J cost, the near-one low cost and the t^2*Q^(-1/2) high cost. No current-paper low/high or restricted-distribution hypothesis remains in this theorem; only the recorded independent general and arithmetic inputs do. The uniform parameter optimization to the printed near-one gauge, all-integer cutoff transfer and unweighted prime normalization are still unfinished.

Checked declarations

Module	Named supporting theorems
IWPrimeRestrictedDistribution.lean	3

- `iwPrimeHighOperator_scale_congr`
- `primeHigh_indicator_power_budget`
- `iw_prime_annular_restricted_distribution`

<!-- checkpoint-verification -->

Verification

All 1007 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4629-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2865 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1009 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_941_IWRestrictedParameters.md`

Checkpoint 941: Explicit admissible integer parameters and endpoint losses

Lean verifies explicit integer parameter choices for every v>=1: r=ceil(v)+2, k=iwBalancedRank(v), and Q=2^(4r). Their rank and admissibility, r and k size bounds and the exact cutoff logarithm 2v<=log Q<=16v are checked. At v=nearOneLog(t), t>=1, the near-one exponent loss t^(p_r-1) is at most e and the high factor t*Q^(-1/2) is at most one; the full (2t)^p factor is at most 4e*t. These choices work uniformly including bounded t, so the scalar optimization does not rely on an unverified sufficiently-large range or rounding convention.

Checked declarations

Module	Named supporting theorems
IWRestrictedParameters.lean	7

- `iwRestrictedOrder_bounds`
- `iwRestrictedRank_bounds`
- `iwRestrictedCutoff_log`
- `iwRestrictedCutoff_bounds`
- `iwRestricted_near_one_power`
- `iwRestricted_high_cost`
- `iwRestricted_two_t_power`

<!-- checkpoint-verification -->

Verification

All 1008 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4630-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2872 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1010 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs.

The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint notes 942-1016

Checkpoint_942_IWOptimizedMultiplierCost.md

Checkpoint 942: Balanced-rank multiplier growth with explicit constants

Lean verifies that the balanced rank and rounded near-one order give log(C_{r,k}) <= C(log C+20) k log(e+v) uniformly for v>=1. It also checks 1<=sqrt(v log(e+v))<=2v and the corresponding exponential multiplier bound. These are scalar consequences of the previously proved explicit parameter choices; no new analytic theorem or optimization assumption is introduced.

Checked declarations

Module	Named supporting theorems
IWOptimizedMultiplierCost.lean	4

- `nearOneCore_basic_bounds`
- `iwMultiplierLogCoefficient_one_le`
- `iwRestrictedMultiplier_log_bound`
- `iwRestrictedMultiplier_exp_bound`

<!-- checkpoint-verification -->

Verification

All 1009 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4631-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2876 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1011 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint_943_IWOptimizedScaleCost.md

Checkpoint 943: Exact starting-scale expression fits the near-one rate

For the actual integer r(v), k(v), Q(v), Lean bounds the exact high-term starting expression S=(log Q)^2+r log B by (256+196 log C) exp(384 sqrt(v log(e+v))) for every v>=1 and C>=2. The proof checks the denominator exponent, the polynomial factors and the balanced-rank optimization. No starting-scale or asymptotic optimization statement is left as an independent input.

Checked declarations

Module	Named supporting theorems
IWOptimizedScaleCost.lean	5

- `iwRestrictedStartExpression_nonneg`
- `iwOptimizedScaleCoefficient_one_le`
- `iwRestrictedDenominator_log_bound`
- `iwRestrictedStartExpression_polynomial_bound`
- `iwRestrictedStartExpression_exp_bound`

<!-- checkpoint-verification -->

Verification

All 1010 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4632-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2881 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1012 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_944_IWOptimizedMaximalCost.md`

Checkpoint 944: The complete actual low maximal cost fits the near-one rate

Lean bounds the exact maximal cost D(r(v),k(v),Q(v)) by A_D(C) exp(B_D(C) sqrt(v log(e+v))) uniformly for v>=1. Both constants are explicit and positive. The logarithm log(e+Q+r log B), the squared multiplier constant and the order factor are all included; this verifies the complete scalar cost used by the actual all-input low maximal theorem.

Checked declarations

Module	Named supporting theorems
IWOptimizedMaximalCost.lean	3

- `iwOptimizedCostConstants_positive`
- `iwRestrictedMaximal_log_bound`
- `iwRestrictedMaximal_exp_bound`

<!-- checkpoint-verification -->

Verification

All 1011 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4633-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2884 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1013 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_945_MangoldtFullAnnularComparison.md`

Checkpoint 945: Actual full integer-cutoff maximum is controlled by the annular maximum

Lean checks the pointwise comparison of the actual full von Mangoldt square maximal operator over every integer cutoff N>=2 with four times the actual dyadic annular maximum. The proof uses the exact annular reconstruction, Lambda(1)=0 and positivity; it does not invoke the independent sparse or ordinary-square decomposition inputs. It also proves the corresponding restricted-gauge transfer with the checked doubling constant. This prepares the transfer of the optimized annular estimate to the literal full maximal operator.

Checked declarations

Module	Named supporting theorems
MangoldtFullAnnularComparison.lean	4

- `mangoldt_annular_indicator_kernel`
- `mangoldt_full_dyadic_annular_bound`
- `mangoldt_full_annular_maximal_bound`
- `positiveRestrictedGauge_mangoldt_full`

<!-- checkpoint-verification -->

Verification

All 1012 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4634-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2888 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1014 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities,

cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_946_IWOptimizedStartingIndex.md`

Checkpoint 946: A checked admissible dyadic starting index and its endpoint cost

Lean defines the starting index J=ceil(C_start S/log 2)+1 and proves the exact required exp(C_start S)<=2^J inequality, including all integer rounding. Its upper bound is (2 C_start (256+196 log C)+2) exp(384 sqrt(v log(e+v))). The admissibility inequality is checked without any asymptotic or sufficiently-large qualification.

Checked declarations

Module	Named supporting theorems
IWOptimizedStartingIndex.lean	2

- `iwRestrictedStartingIndex_admissible`
- `iwRestrictedStartingIndex_exp_bound`

<!-- checkpoint-verification -->

Verification

All 1013 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4635-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2890 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1015 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_947_IWOptimizedDistributionBudget.md`

Checkpoint 947: All three actual distribution costs fit one Orlicz density

Lean combines all three exact distribution costs at the rounded integer parameters: the initial-scale L1 union cost, the near-one low Lp cost and the Q^(-1/2) high L2 cost. For every t>=1 their sum is bounded by R0*t*exp(Cg*sqrt(v log(e+v))), v=log(e+t), with explicit positive R0 and nonnegative Cg. Both real powers and the high decay factor are included. The actual operator estimate is not an assumption of this scalar proof.

Checked declarations

Module	Named supporting theorems
IWOptimizedDistributionBudget.lean	3

- `iwRestrictedLow_power_bound`
- `iwRestrictedBudget_constants`
- `iwRestrictedDistribution_budget`

<!-- checkpoint-verification -->

Verification

All 1014 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4636-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2893 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1016 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_948_IWPrimeAnnularRestrictedGauge.md`

Checkpoint 948: Actual prime annular restricted Orlicz estimate from the quantitative IW comparison

The actual dyadic annular von Mangoldt square maximal operator now satisfies the claimed restricted near-one Orlicz estimate for every finite indicator and every t>=1. Lean proves existence of nonnegative Cg and positive Rg and proves PositiveRestrictedGauge for the actual arithmetic family. It instantiates the concrete checked low/high distribution inequality at the checked integer parameters and starting index, applies the complete scalar budget and compares t*exp(Cg sqrt(log(e+t) log(e+log(e+t)))) with the integral gauge Psi. The paper-specific restricted estimate is a conclusion, not an independent hypothesis. The general IW, sampling, continuous, interpolation, Page and distribution inputs retain their explicit boundaries. The all-integer cutoff transfer and unweighted normalization are separate assemblies.

Checked declarations

Module	Named supporting theorems
IWPrimeAnnularRestrictedGauge.lean	1

- `iw_prime_annular_restricted_gauge`

<!-- checkpoint-verification -->

Verification

All 1015 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4637-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2894 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1017 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_949_PrimeCountingNormalization.md`

Checkpoint 949: Prime-counting normalization from checked Mathlib Chebyshev bounds

Lean derives the uniform lower bound N<=512*pi(N)*log N and its inverse-normalization form for every integer N>=2 from Mathlib Chebyshev.pi_ge. The finite small range is handled uniformly using pi(N)>=1; the large range uses a checked logarithm-square-root bound. This supplies the needed prime-counting normalization without adding PNT or a new independent input. It does not replace the separate stronger PNT hypotheses used earlier for the sharp Gauss asymptotic.

Checked declarations

Module	Named supporting theorems
PrimeCountingNormalization.lean	4

- `real_log_le_two_sqrt`
- `primeCounting_positive_of_two_le`
- `primeCounting_log_lower_bound`
- `primeCounting_inverse_bound`

<!-- checkpoint-verification -->

Verification

All 1016 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4638-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2898 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1018 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities,

cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_950_IWPrimeFullRestrictedGauge.md`

Checkpoint 950: Actual full von Mangoldt restricted maximal and truncated modular estimates

The actual full von Mangoldt square maximal operator, taking the genuine supremum over every integer cutoff N>=2, now has the restricted Psi_C estimate. Lean transfers the proved annular estimate by the proved factor-four pointwise comparison and checked gauge doubling. It also instantiates the earlier restricted-to-modular proof: every finitely supported complex input has a finite actual superlevel set and the explicit truncated Orlicz cardinal bound. Thus the current-paper restricted bound is no longer an assumed premise for this arithmetic family. All independent general analytic and prime-distribution hypotheses remain explicit; this is a complete implication for the von Mangoldt family, not yet the full printed three-family Theorem7.2.

Checked declarations

Module	Named supporting theorems
IWPrimeFullRestrictedGauge.lean	2

- `iw_prime_full_restricted_gauge`
- `iw_prime_full_modular_estimates`

<!-- checkpoint-verification -->

Verification

All 1017 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4639-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2900 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1019 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_951_PrimeMangoldtSumComparison.md`

Checkpoint 951: Elementary weighted prime sums controlled by von Mangoldt partial sums

Lean proves an elementary comparison for every nonnegative weighted sequence: if all von Mangoldt partial sums up to m are at most m*B, then the prime sum up to N is at most 6*N*B/log N. The proof splits at floor(sqrt N), checks both prime logarithm bounds and the exact finite-sum inclusions. Combining it with the Mathlib Chebyshev normalization gives the normalized prime sum <=3072*B, for all N>=2. This is an elementary internal comparison, not a new analytic input.

Checked declarations

Module	Named supporting theorems
PrimeMangoldtSumComparison.lean	2

- `prime_sum_le_mangoldt_partial_bound`
- `normalized_prime_sum_le_mangoldt_bound`

<!-- checkpoint-verification -->

Verification

All 1018 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4640-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2902 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1020 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_952_PrimeMangoldtMaximalComparison.md`

Checkpoint 952: Actual unweighted prime maximal comparison without a PNT assumption

Lean identifies the actual von Mangoldt indicator partial sums with their full positive maximal bound and proves the actual pointwise comparison A^*{P^2} 1_F <=3072 A^*{Lambda} 1_F, uniformly over every integer cutoff. The earlier weighted prime-sum and Mathlib Chebyshev normalization lemmas discharge all arithmetic premises. A proved general gauge rescaling then transfers the restricted Orlicz estimate to the actual unweighted prime family. No PNT assumption and no independent prime-normalization hypothesis are used.

Checked declarations

Module	Named supporting theorems
PrimeMangoldtMaximalComparison.lean	4

- `mangoldt_indicator_partial_sum_bound`
- `prime_mangoldt_indicator_maximal_bound`
- `nearOnePsi_rescale_bound`
- `positiveRestrictedGauge_mangoldt_to_prime`

<!-- checkpoint-verification -->

Verification

All 1019 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4641-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 2906 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1021 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_953_IWUnweightedPrimeRestrictedGauge.md`

Checkpoint 953: Actual full weighted and unweighted prime restricted maximal estimates

The actual full unweighted prime-square maximal operator now satisfies the restricted Orlicz cardinal estimate, with a proved finite superlevel set for every finite F and t>=1. Its finitely supported complex-input truncated modular estimate is also instantiated. The new common-constant theorem proves the restricted gauge simultaneously for the actual full von Mangoldt and unweighted-prime families with common Cg,Rg>=1; coefficient and multiplicative-constant enlargement are proved. The quantitative low/high comparison, complete integer optimization, full-cutoff transfer and unweighted normalization are conclusions of this development under the recorded independent general analytic and prime-distribution inputs. No new PNT input is needed. The source review and roadmap are reconciled: the full printed three-family Theorem7.2 remains partial because the ordinary-square companion and final three-family assembly remain. See docs/SECTION_7_RESTRICTED_COMPLETION.md.

Checked declarations

Module	Named supporting theorems
IWUnweightedPrimeRestrictedGauge.lean	5

- `nearOnePsi_mono_coefficient`
- `positiveRestrictedGauge_enlarge`
- `iw_unweighted_prime_restricted_gauge`
- `iw_unweighted_prime_modular_estimates`
- `iw_prime_families_restricted_gauges`

<!-- checkpoint-verification -->

Verification

All 1020 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4642-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2911 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1022 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain

partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_954_CyclicFullSquareMeasures.md`

Checkpoint 954: The actual cyclic square measure for all integer residues

Lean defines the actual cyclic probability measure obtained by squaring every residue, including nonunits, with group-order normalization. Its test-function formula, total variation at most one, exact complete quadratic Gauss Fourier coefficient and norm-at-most-one bound are proved. This is the distinct measure needed for ordinary integer squares; the earlier unit-residue prime measure is unchanged. Workspace maintenance removed the compiler and dependency caches, so the exact pinned environment was restored and all 1020 recovered project modules plus the root were freshly compiled. All recovered Lean hashes agree with checkpoint953. The original compilation receipt is preserved, the process-path compatibility shim affects no Lean compiler/kernel code, and the new baseline and fresh complete audits are recorded. No ordinary restricted estimate or complete Theorem7.2 is claimed at this stage.

Checked declarations

Module	Named supporting theorems
CyclicFullSquareMeasures.lean	4

- `cyclicFullSquareMeasure_test`
- `cyclicFullSquareMeasure_variation_le_one`
- `cyclicFullSquareMeasure_fourier`
- `cyclicFullSquareMeasure_fourier_norm_le_one`

<!-- checkpoint-verification -->

Verification

All 1021 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4643-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2915 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1023 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full

high/low conclusion are not external input fields.

`Checkpoint_955_CompleteQuadraticGaussBound.md`

Checkpoint 955: Complete quadratic Gauss bounds for arbitrary positive moduli

Lean proves the complete quadratic Gauss bound for every positive modulus and unit coefficient, including even moduli. The proof changes to difference/second coordinates, applies exact additive-character orthogonality and proves that the kernel of multiplication by two has at most two residues. Consequently the squared norm is at most 2q and the norm is at most 2 sqrt(q). This is elementary finite arithmetic proved in the project; no independent Gauss bound is assumed. It supplies the ordinary-square coefficient decay needed for the quantitative IW comparison.

Checked declarations

Module	Named supporting theorems
CompleteQuadraticGaussBound.lean	4

- `zmod_two_kernel_card_le_two`
- `completeQuadraticGaussSum_autocorrelation`
- `completeQuadraticGaussSum_norm_sq_le`
- `completeQuadraticGaussSum_norm_le`

<!-- checkpoint-verification -->

Verification

All 1022 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4644-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2919 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1024 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_956_CyclicFullSquareCoefficients.md`

Checkpoint 956: Exact ordinary-square coefficients and quantitative decay

Lean proves normalized complete Gauss decay, the actual rational ordinary-square coefficient and its exact equality with the Fourier coefficient of the full-residue cyclic square measure at every denominator dividing the period. Change of level follows from a proved finite additive-group pullback identity and the surjective residue map. The coefficient norm is at most one and its uniform denominator decay is at most 2*q^(-3/8), using the elementary Gauss bound at955. Thus the precise coefficient agreement, normalization and decay needed by both quantitative IW sides are checked, without an independent arithmetic input.

Checked declarations

Module	Named supporting theorems
CyclicFullSquareCoefficients.lean	7

- `finite_add_group_pullback_average`
- `normalizedCompleteGauss_norm_one`
- `normalizedCompleteGauss_decay`
- `rationalFullSquareGauss_norm_one`
- `rationalFullSquareGauss_decay`
- `cyclicReducedFullSquare`
- `cyclicRationalCoefficient_fullSquare`

<!-- checkpoint-verification -->

Verification

All 1023 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4645-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2926 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1025 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_957_IWFullSquareLowComparison.md`

Checkpoint 957: Actual ordinary-square IW low maximum and model comparison

The actual ordinary-square IW low operator now has the full countable near-one lp maximal bound on every integer lp input, by applying the already checked general IW theorem to the proved all-residue cyclic measure. Its exact low/high decomposition and complete-Gauss coefficient identity are checked. The actual original-cutoff ordinary circle model and its actual common-cutoff IW low operator have a proved countable L2 difference bound K*Q^(-1/4), using the existing generic smooth multifrequency comparison and the proved coefficient decay. The comparison includes varying upper denominator limits and exact cutoff geometry. The remaining step on the high side is the approximation of the actual discrete ordinary-square average by that original circle model; no own restricted or high estimate is assumed.

Checked declarations

Module	Named supporting theorems
IWFullSquareLowComparison.lean	5

- `iwFullSquare_decomposition`
- `iw_full_square_width_symbol`
- `iw_full_square_low_maximal_bound`

- `iwFullSquareModelConstant_pos`
- `iw_full_square_model_comparison`

<!-- checkpoint-verification -->

Verification

All 1024 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4646-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2931 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1026 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_958_QuadraticIntegralTail.md`

Checkpoint 958: Quadratic oscillatory tails on arbitrary positive intervals

Lean proves quadratic integration by parts on every positive real interval [a,b], evaluates the inverse-square integral exactly, bounds the actual oscillatory remainder by 1/a and proves (4*pi*|theta|)*|integral_a^b exp(-2*pi*i*theta*t^2) dt|<=3/a. These are proved analytic lemmas, without an independent oscillatory-integral input. They prepare the full initial-interval bound required by the ordinary-square Weyl argument.

Checked declarations

Module	Named supporting theorems
QuadraticIntegralTail.lean	4

- `quadratic_integral_parts`
- `inverse_square_integral`
- `quadratic_integral_remainder_bound`
- `quadratic_integral_tail_scaled`

<!-- checkpoint-verification -->

Verification

All 1025 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4647-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2935 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1027 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs.

The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_959_QuadraticIntegralBound.md`

Checkpoint 959: The full quadratic oscillatory integral bound is proved

Lean proves the full initial-interval quadratic oscillatory integral bound by splitting at |theta|^(-1/2), using unit modulus on the initial segment and the proved integration-by-parts tail afterward. The exact constant is 1+3/(4*pi). The actual normalized initial continuous square model has norm at most one and the required 1/(N*sqrt(|theta|)) decay. These estimates are proved from Mathlib calculus, with no independent Fresnel or continuous-decay hypothesis.

Checked declarations

Module	Named supporting theorems
QuadraticIntegralBound.lean	5

- `quadraticIntegralConstant_one_le`
- `quadratic_integral_norm_le_length`
- `quadratic_integral_uniform_bound`
- `initialQuadraticModel_norm_one`
- `initialQuadraticModel_decay`

<!-- checkpoint-verification -->

Verification

All 1026 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4648-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2940 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1028 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full

high/low conclusion are not external input fields.

`Checkpoint_960_QuadraticWeylInput.md`

Checkpoint 960: Precise scalar Weyl boundary and proved rational-center uniqueness

Lean records a precise independent scalar quadratic Weyl approximation predicate and proves the rational-center uniqueness needed for its major-arc application. If the source center has denominator<=R, the selected denominator is<=8M, M lies between N/2 and N and 32R<=N, rational separation forces the selected center to equal the source center. The complete-Gauss half-power decay is also realized for the actual rational coefficient. The unproved scalar input, including its Dirichlet-selection/reduction/phase bridge to the published theorem, is described exactly in docs/SECTION_7_QUADRATIC_WEYL_BOUNDARY.md. It contains no own-paper circle sum, IW or maximal conclusion, and is not declared as a Lean axiom.

Checked declarations

Module	Named supporting theorems
QuadraticWeylInput.lean	2

- `rationalFullSquareGauss_half_decay`
- `quadratic_weyl_center_unique`

<!-- checkpoint-verification -->

Verification

All 1027 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4649-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2942 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1029 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_961_QuadraticWeylArcBounds.md`

Checkpoint 961: Actual initial square major- and minor-arc bounds from the scalar Weyl input

From the explicit independent scalar Weyl input, Lean proves the actual initial-square major-arc approximation by forcing the selected rational center to agree with the source center. It also proves the full initial-square minor-arc bound, using the already proved complete Gauss decay and full oscillatory integral estimate, with exact denominator, square-root and relative-scale inequalities. The only new independent ingredient used is the scalar Weyl predicate documented at960; no own circle model, high maximum or restricted conclusion is assumed. These bounds hold uniformly for the two initial scales needed to form a dyadic annulus.

Checked declarations

Module	Named supporting theorems
QuadraticWeylArcBounds.lean	4

- `quadratic_weyl_major_approximation`
- `quadratic_weyl_main_outside`
- `quadratic_weyl_error_scale`
- `quadratic_weyl_initial_outside`

<!-- checkpoint-verification -->

Verification

All 1028 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4650-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2946 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1030 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_962_FullSquareRealCircleModel.md`

Checkpoint 962: Actual ordinary-square circle model and outside-arc bounds

Lean realizes the actual ordinary-square finite circle model with complete-Gauss coefficients. Each rational term has the proved outside-arc 1/sqrt(R) bound. Exact active-denominator counting gives the logarithmic finite-model bound, and the actual periodic circle model is identified with its five real lifts and bounded outside all source arcs. The periodization, cutoffs, canonical rational centers and denominator normalizations are retained exactly. No discrete circle approximation or maximal conclusion is assumed here.

Checked declarations

Module	Named supporting theorems
FullSquareRealCircleModel.lean	6

- `fullSquareRationalModelTerm_norm`
- `fullSquareRationalModelTerm_outside`
- `fullSquareFiniteModel_active_norm`
- `fullSquareFiniteModel_outside`
- `fullSquareCircleModel_five`
- `fullSquareCircleModel_outside`

<!-- checkpoint-verification -->

Verification

All 1029 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4651-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2952 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1031 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_963_FullSquareAnnularWeylBounds.md`

Checkpoint 963: Actual ordinary annular Fourier and Weyl bounds

Lean proves the exact full-to-annular Fourier identity for every even square cutoff and the matching identity for the actual continuous quadratic integrals. It then applies the scalar Weyl input at the two initial scales and proves the actual annular major-arc approximation and minor-arc norm bound, with the same rational center and correct half normalization. These are complete implications of the recorded scalar input; no annular circle approximation or maximal theorem is assumed.

Checked declarations

Module	Named supporting theorems
FullSquareAnnularWeylBounds.lean	4

- `annularSquareSymbol_two_mul_value`
- `initialQuadraticModel_annular_two_mul`
- `quadratic_weyl_annular_major`
- `quadratic_weyl_annular_outside`

<!-- checkpoint-verification -->

Verification

All 1030 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4652-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2956 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1032 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and

Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint_964_FullSquareCircleError.md

Checkpoint 964: Actual ordinary-square circle approximation on every circle frequency

Lean now proves the actual ordinary annular discrete Fourier symbol minus its actual finite circle model is uniformly bounded by the explicit major/minor and logarithmic model-tail budget. The proof checks the major-arc main term, erased rational centers, all five periodic lifts, the outside-arc branch and every point of the circle. The scalar Weyl approximation is the only new independent arithmetic input; no own circle approximation is assumed. This closes the concrete pointwise approximation before source-scale logarithmic absorption and the L2 maximal deduction.

Checked declarations

Module	Named supporting theorems
FullSquareCircleError.lean	4

- `fullSquareRealTailBudget_nonneg`
- `fullSquareFiniteError_major`
- `fullSquareCircleError_major`
- `fullSquareCircleError_bound`

<!-- checkpoint-verification -->

Verification

All 1031 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4653-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2960 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1033 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

Checkpoint_965_FullSquareApproximationRate.md

Checkpoint 965: Uniform logarithmic rate for the actual ordinary-square approximation

Lean proves a uniform source-scale bound for the actual ordinary annular Fourier approximation error: at every circle point the error is at most 4 Q^(-1/2) (1+log N)^(-2), for sufficiently large even N under log Q ≤ (1/8) sqrt(log N). The explicit major/minor and periodic model-tail budget is absorbed using proved logarithmic estimates. The geometric hypotheses are verified at the actual source radius, with the source parameter chosen to give the required cutoff range. This implication consumes only the declared independent scalar quadratic Weyl approximation for its arithmetic step; the paper-specific Fourier approximation is proved.

Checked declarations

Module	Named supporting theorems
FullSquareApproximationRate.lean	5

- `fullSquareApproxBudget_log_bound`
- `eventually_fullSquareLogCost`
- `fullSquareLogCost_uniform`
- `sourceFullSquare_arc_conditions`
- `fullSquareCircleError_uniform`

<!-- checkpoint-verification -->

Verification

All 1032 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4654-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2965 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1034 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_966_FullSquareApproximationMaximum.md`

Checkpoint 966: Countable L2 maximum for the ordinary-square approximation error

Lean proves an actual countable L2 maximal majorant for the ordinary-square circle-approximation error operators at all scales 2^j M, starting from a sufficiently large even M. The uniform logarithmic decay is converted to a summable dyadic operator-norm sequence; the nonnegative majorant has norm at most 8 (log 2)^(-2) Q^(-1/2) times the input L2 norm. The conclusion concerns the concrete discrete average minus the original varying-radius circle model, rather than an assumed approximation operator. The independent scalar quadratic Weyl hypothesis remains explicit.

Checked declarations

Module	Named supporting theorems
FullSquareApproximationMaximum.lean	4

- `fullSquareCircleErrorOperator_eq`
- `fullSquareCircleErrorOperator_uniform`
- `fullSquareErrorMaximalConstant_pos`
- `full_square_approximation_maximum`

<!-- checkpoint-verification -->

Verification

All 1033 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4655-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2969 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1035 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_967_IWFullSquareHighSourceScale.md`

Checkpoint 967: Actual ordinary-square IW high-term maximal bound

Lean proves the actual ordinary-square IW high term has a countable L2 maximal majorant of size K Q^(-1/4) at all annular scales 2^j M, once the source exponential starting-scale condition holds and M is even. It derives this result by adding the checked discrete circle-approximation error and the checked circle-model to IW-low comparison. All period instances, denominator geometry, cutoff widths and starting-scale thresholds are verified. The only new arithmetic input is the explicitly scoped scalar quadratic Weyl approximation; the source-specific high-term estimate and its maximal conclusion are proved, not assumed. This ordinary-square route does not consume the earlier ordinary-square global stopping decomposition input.

Checked declarations

Module	Named supporting theorems
IWFullSquareHighSourceScale.lean	2

- `iwFullSquareHigh_error_model`
- `iw_full_square_high_source_scale`

<!-- checkpoint-verification -->

Verification

All 1034 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4656-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 2971 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1036 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_968_OrdinaryAnnularPositiveFamily.md`

Checkpoint 968: Actual ordinary-square positive annular family

Lean defines the actual ordinary-square positive annular dyadic family and proves its mass is bounded by one. It identifies every family average with the concrete unweighted annular square operator on L2. The family includes the scale N=1, which will supply the nonzero initial ordinary-square term in the full-to-annular reconstruction. No independent analytic input is used by this checkpoint.

Checked declarations

Module	Named supporting theorems
OrdinaryAnnularPositiveFamily.lean	2

- `ordinaryAnnularDyadicFamily_apply`
- `ordinaryAnnularDyadicFamily_mass`

<!-- checkpoint-verification -->

Verification

All 1035 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4657-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2973 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1037 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete con-

tinuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_969_IWFullSquareRestrictedDistribution.md`

Checkpoint 969: Restricted distribution from actual ordinary-square low and high estimates

Lean combines the actual ordinary-square IW low and high maximal bounds into a restricted distribution estimate for the full annular dyadic maximum. The proof checks indicator membership in the required near-one Lp space, Chebyshev power budgets, the scale reindexing above the chosen starting index and the exact decomposition of each concrete average. The finite initial range is estimated by positivity and mass one; its harmless enlargement to the previously optimized common mass constant permits reuse of the checked numerical parameter optimization. No ordinary-square restricted or maximal conclusion is assumed as an independent input.

Checked declarations

Module	Named supporting theorems
IWFullSquareRestrictedDistribution.lean	2

- `iwFullSquareHighOperator_scale_congr`
- `iw_full_square_annular_restricted_distribution`

<!-- checkpoint-verification -->

Verification

All 1036 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4658-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2975 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1038 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_970_IWFullSquareAnnularRestrictedGauge.md`

Checkpoint 970: Optimized ordinary-square annular restricted Orlicz estimate

Lean proves the actual ordinary-square annular restricted Orlicz estimate. It instantiates the previously checked integer choices for the exponent, rank, denominator cutoff and starting scale, applies the actual ordinary low-high distribution theorem, and absorbs all three distribution costs into the precise near-one gauge from the manuscript. The constants are uniform in the finite indicator set and in $t \geq 1$. The result is a proved implication from the declared general IW/multiplier inputs and the independent scalar quadratic Weyl approximation, without a source-specific restricted estimate hypothesis.

Checked declarations

Module	Named supporting theorems
IWFullSquareAnnularRestrictedGauge.lean	1

- `iw_full_square_annular_restricted_gauge`

<!-- checkpoint-verification -->

Verification

All 1037 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4659-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2976 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1039 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_971_OrdinaryFullAnnularComparison.md`

Checkpoint 971: Ordinary-square reconstruction for every integer cutoff

Lean proves the literal ordinary-square maximum over all integer cutoffs $N \geq 2$ is at most four times the actual ordinary annular dyadic maximum on indicators. The proof establishes the full-kernel cutoff comparison, the full-to-annular recurrence and its nonzero $N=1$ base case, and a genuine supremum bound. It then transfers the actual near-one restricted gauge through this comparison using the proved doubling estimate. This reconstruction is elementary and consumes no independent ordinary-square globalization or reflection assumption.

Checked declarations

Module	Named supporting theorems
OrdinaryFullAnnularComparison.lean	7

- `fullOrdinaryKernel_cutoff_bound`
- `fullOrdinaryIndicator_cutoff_bound`
- `ordinarySquareFullValue_indicator_kernel`
- `ordinary_annular_indicator_kernel`
- `ordinary_full_dyadic_annular_bound`
- `ordinary_full_annular_maximal_bound`
- `positiveRestrictedGauge_ordinary_full`

<!-- checkpoint-verification -->

Verification

All 1038 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4660-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2983 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1040 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_972_IWFullSquareRestrictedGauge.md`

Checkpoint 972: Ordinary-square full restricted and modular estimates

Lean proves the actual ordinary-square restricted Orlicz gauge for the maximum over every integer cutoff $N \geq 2$. It combines the actual optimized annular estimate with the checked full-to-annular reconstruction, and deduces finite-cardinality restricted and truncated modular estimates for complex finitely supported inputs. This completes the ordinary-square companion needed for the three-family headline theorem. Its hypotheses are independent general IW/multiplier inputs and the precisely declared scalar quadratic Weyl approximation; neither an ordinary-square maximal conclusion nor the old global stopping decomposition is assumed.

Checked declarations

Module	Named supporting theorems
IWFullSquareRestrictedGauge.lean	2

- `iw_full_square_restricted_gauge`
- `iw_full_square_modular_estimates`

<!-- checkpoint-verification -->

Verification

All 1039 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4661-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2985 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1041 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and

Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_973_IWThreeFamilyRestrictedGauge.md`

Checkpoint 973: Common restricted Orlicz constants for all three actual square families

Lean assembles actual restricted Orlicz bounds for all three manuscript families: von Mangoldt squares, ordinary squares and unweighted prime squares. It chooses common gauge and distribution constants at least one, proves the coefficient and multiplicative enlargements, and instantiates the actual family maxima. The ordinary branch is supplied by the newly proved scalar-Weyl-to-IW route; the prime branches use the previously checked prime-distribution route. This checkpoint proves the shared restricted estimate, while the final common truncated/unrestricted modular statement assembly is next.

Checked declarations

Module	Named supporting theorems
IWThreeFamilyRestrictedGauge.lean	1

- `iw_three_families_restricted_gauges`

<!-- checkpoint-verification -->

Verification

All 1040 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4662-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2986 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1042 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_974_SectionSevenThreeFamilyAssembly.md`

Checkpoint 974: One common set of constants for the three manuscript Orlicz conclusions

Lean proves the final source-shaped assembly from the actual shared three-family restricted gauges. It chooses one common enlarged gauge $C \geq 1$, one common truncation constant $C0 = 2$ times the checked uniform mass constant, and one common multiplicative constant $K \geq 1$. For each actual family it proves

finite level sets and all three manuscript conclusions: truncated modular, unrestricted modular and restricted. The proof checks the direction of the truncation inclusion and every nonnegative sum/constant enlargement. The internal gauge premise is already supplied by checkpoint 973; it is not a new independent theorem input. The raw-input wrapper and literal finite-support sums over the integers are completed next.

Checked declarations

Module	Named supporting theorems
SectionSevenThreeFamilyAssembly.lean	3

- `sectionSevenFamily_mass_cases`
- `sectionSevenFamily_mass_le`
- `section7_three_family_assembly`

<!-- checkpoint-verification -->

Verification

All 1041 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4663-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2989 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1043 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_975_TheoremSevenTwo.md`

Checkpoint 975: Raw-input three-family Theorem 7.2 assembly

Lean now derives the complete three-family finite-support formulation of Theorem 7.2 directly from the explicit independent raw inputs. The final theorem chooses common C, C0 and $K \geq 1$ and concludes all three manuscript estimates for the actual weighted-prime, ordinary-square and unweighted-prime maxima over all integer cutoffs. It does not take an actual restricted estimate, quantitative comparison or ordinary-square high/low conclusion as an input. The scalar Weyl, general multiplier/sampling and compatible prime-distribution boundaries remain explicit. The next checkpoint expresses finite-support sums literally over all integers and records the exact source-statement review.

Checked declarations

Module	Named supporting theorems
TheoremSevenTwo.lean	1

- `theorem_7_2`

<!-- checkpoint-verification -->

Verification

All 1042 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4664-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2990 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1044 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, both conclusions of Theorems 1.1 and 1.3, all displayed clauses of Proposition 4.19 and both clauses of Proposition 6.21 are complete checked implications under their separately recorded independent deep inputs. The square-sign reflection is proved. Theorem 1.3 has actual finite sparse families, disjoint major subsets and the literal full all-cutoff maximum. The three general Section5 auxiliary statements remain partial, and the later Orlicz/ergodic/sieve applications remain unfinished. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 preserve their separately recorded raw prime/distribution, Page, generic smooth multifrequency, ordinary-square and stopping hypotheses. Section7 additionally records RawTaoArcConstructionInput C and RawIWMultiplierInput C as explicit unproved independent general theorems, not Lean axioms. Actual geometry and full goodness and conditional smooth multiplier specialization are proved. The actual prime high-side comparison uses raw Tao construction, SmoothCircleBourgainInput, RawPageUniquenessInput and PagePrimeSquareDistributionInputs; its own coefficient identities, cutoffs, Page selection and approximation deductions are proved. General sampling and concrete continuous/low estimates remain unfinished. The current-paper low maximum, restricted estimate and full high/low conclusion are not external input fields.

`Checkpoint_976_TheoremSevenTwoFinitelySupported.md`

Checkpoint 976: Complete source Theorem 7.2 with literal finite-support sums and reviewed independent inputs

Lean completes the literal finite-support formulation of Theorem 7.2: all three actual families, every integer cutoff $N \geq 2$, arbitrary complex finite support and every positive level, common C, C0, $K \geq 1$, finite actual superlevel sets, the truncated modular estimate, unrestricted modular estimate and restricted estimate for all finite F and $t \geq 1$. The sums are explicitly over all integers; the conversion from finite support is proved rather than inferred. The exact source review upgrades only t:orlicz-maximal from partial to complete conditional implication. Its independent scalar Weyl input includes its unproved published-theorem selection/reduction/phase bridge; general IW/multiplier/sampling and compatible prime-distribution boundaries remain explicit. No own circle approximation, quantitative high/low or actual restricted conclusion is assumed. Corollaries 7.6 and 7.46 and stronger standalone auxiliary ranges remain separately unfinished. The manuscript is unchanged.

Checked declarations

Module	Named supporting theorems
TheoremSevenTwoFinitelySupported.lean	2

- `section7_source_sum_assembly`
- `theorem_7_2_finitely_supported`

<!-- checkpoint-verification -->

Verification

All 1043 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4665-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2992 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1045 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked

implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and strong lp/ergodic/sieve deductions remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route.

`Checkpoint_977_PositivePowerLayerCake.md`

Checkpoint 977: General power layer-cake identity for the strong maximal deduction

Lean proves the general positive-power layer-cake identity for an arbitrary measure space and an a.e. nonnegative measurable function, using the explicit primitive of p t^(p-1) for every p > 0. Both the primitive and the extended nonnegative integral identity are checked. This starts the actual operator deduction of Corollary 7.6 from the proved truncated modular estimate; no layer-cake or current-paper strong maximal conclusion is added as an independent input. The subsequent steps will evaluate the truncated power kernel, bound the actual maximum and extend finite inputs.

Checked declarations

Module	Named supporting theorems
PositivePowerLayerCake.lean	2

- `positivePower_primitive`
- `nonneg_power_layerCake`

<!-- checkpoint-verification -->

Verification

All 1044 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4666-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2994 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1046 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and strong lp/ergodic/sieve deductions remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route.

`Checkpoint_978_TruncatedPowerKernel.md`

Checkpoint 978: Exact truncated power kernel including the zero-input case

Lean evaluates the actual truncated positive power kernel in the extended nonnegative integral, including zero input and its singular endpoint at zero. For 0 < q < p it proves the exact identity with coefficient C0^(p-q)/(p-q) times a^p. The proof checks the indicator-domain change, integrability, power primitive and scaling algebra. This is the scalar integral needed for the strong maximal bound from the already proved truncated modular estimate; no new independent input is used.

Checked declarations

Module	Named supporting theorems
TruncatedPowerKernel.lean	2

- `truncated_rpow_kernel`
- `truncated_scaled_power_kernel`

<!-- checkpoint-verification -->

Verification

All 1045 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4667-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2996 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1047 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and strong lp/ergodic/sieve deductions remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route.

`Checkpoint_979_TruncatedDistributionStrongPower.md`

Checkpoint 979: Strong power estimate from the actual truncated distribution bound

Lean proves a general strong power bound from a truncated distribution inequality with exponent q strictly below p. It integrates the actual superlevel measure by the checked layer-cake identity, interchanges the finite sum and nonnegative integral, and applies the exact truncated kernel evaluation. All measurability, positivity and extended-integral steps are checked, so finiteness of the resulting Lp quantity is a conclusion rather than a premise. No new independent input is used.

Checked declarations

Module	Named supporting theorems
TruncatedDistributionStrongPower.lean	1

- `truncated_distribution_strong_power`

<!-- checkpoint-verification -->

Verification

All 1046 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4668-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2997 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1048 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and strong lp/ergodic/sieve deductions remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route.

`Checkpoint_980_TruncatedGaugeStrongPower.md`

Checkpoint 980: Actual finite-support strong power estimate from Theorem 7.2

Lean instantiates the strong-power argument on the actual positive averaging maximum, using the proved truncated Orlicz modular estimate. It checks that the actual maximum is nonnegative, converts finite level-set cardinalities to counting measure, and applies the precise gauge power majorant above the truncation threshold. The resulting finite-support Lp power bound is a proved operator consequence of Theorem 7.2, not an independent strong-bound premise. It retains the exact p-dependent constant for the next exponential-budget step.

Checked declarations

Module	Named supporting theorems
TruncatedGaugeStrongPower.lean	1

- `truncated_gauge_strong_power`

<!-- checkpoint-verification -->

Verification

All 1047 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4669-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 2998 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1049 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and strong lp/ergodic/sieve deductions remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction,

IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route.

`Checkpoint_981_NearOneStrongConstant.md`

Checkpoint 981: Uniform source exponential budget for the strong maximal constant

Lean bounds the complete strong-power constant by exp(D/(p-1) log(e/(p-1))) with D independent of p, the input function and the averaging family. The proof retains the truncation constant, distribution multiplier, endpoint factor 1/(p-1) and all power factors, and absorbs them using the checked reciprocal-logarithmic estimates. It also proves D is positive under the source constant ranges. This verifies the quantitative dependence needed for Corollary 7.6.

Checked declarations

Module	Named supporting theorems
NearOneStrongConstant.lean	3

- `nearOneMoment_prefactor_bound`
- `nearOneStrongConstant_pos`
- `nearOneStrong_power_budget`

<!-- checkpoint-verification -->

Verification

All 1048 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4670-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3001 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1050 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and strong lp/ergodic/sieve deductions remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route.

`Checkpoint_982_FiniteStrongMaximalNorm.md`

Checkpoint 982: Actual finite-support strong maximal norm with the source p dependence

Lean proves actual finite-support strong Lp membership and the displayed exponential norm bound for the genuine positive averaging maximum. It converts the checked power integral to the actual counting-measure Lp norm, proves finiteness, takes the p-th root with the correct inequality direction, and uses exact finite-support input norm identities. All constants remain uniform in the family, function and 1 <

p ≤ 2. The next step extends this proved estimate to every integer lp input through the already checked finite-input/Fatou machinery.

Checked declarations

Module	Named supporting theorems
FiniteStrongMaximalNorm.lean	3

- `finiteInput_enorm_power_sum`
- `count_strong_power_to_norm`
- `actual_finite_strong_maximal_norm`

<!-- checkpoint-verification -->

Verification

All 1049 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4671-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3004 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1051 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and strong lp/ergodic/sieve deductions remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route.

`Checkpoint_983_PositiveAverageStrongExtension.md`

Checkpoint 983: Actual strong maximal estimate on every integer lp input

Lean constructs the actual positive finite-average L2 operators and proves their exact pointwise agreement with the averaging family. It extends the proved finite-support strong estimate to every integer lp input for 1 < p ≤ 2 by the previously checked finite-input/Fatou extension. The conclusion supplies the actual full countable maximum as an lp function, with the same bound, not merely a finite-scale estimate or an unspecified dominating operator. No independent extension or current-paper strong theorem is assumed.

Checked declarations

Module	Named supporting theorems
PositiveAverageStrongExtension.lean	2

- `positiveAverageL2_apply`
- `positiveAverage_finite_strong_extension`

<!-- checkpoint-verification -->

Verification

All 1050 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4672-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3006 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1052 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and strong lp/ergodic/sieve deductions remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route.

`Checkpoint_984_CorollarySevenSix.md`

Checkpoint 984: Complete source Corollary 7.6 for all three actual full maximal operators

Lean completes the source Corollary 7.6 for all three actual averaging families, every $1 < p \le 2$ and every integer lp input. One absolute constant is chosen before the family, p and input; the actual full maximal function belongs to lp and satisfies the exact displayed exponential norm bound. The proof derives it from Theorem 7.2 using checked layer cake, singular integral evaluation, all constant absorption and finite-input/Fatou extension. The source review upgrades c:orlicz-p from partial to complete conditional implication. No new independent input is introduced; the Theorem 7.2 boundaries remain explicit.

Checked declarations

Module	Named supporting theorems
CorollarySevenSix.lean	2

- `section7_strong_assembly`
- `corollary_7_6`

<!-- checkpoint-verification -->

Verification

All 1051 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4673-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3008 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1053 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and later deductions retain their separately reviewed scopes. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation

(including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route.

`Checkpoint_985_ErgodicSquareAverages.md`

Checkpoint 985: Actual ergodic square averages and a proper extended countable maximum

Lean defines the actual ergodic positive averages for an invertible measurable transformation, using the exact integer action and all three source families. It proves the positive square-iterate specialization, measurability, linear difference identity and uniform bounded-function estimate. The countable ergodic maximum is defined in the extended nonnegative reals, with every average dominated unconditionally; finite real-valued norm claims will explicitly include an almost-everywhere finiteness certificate. This prevents an unbounded supremum from collapsing under a conditional real supremum convention.

Checked declarations

Module	Named supporting theorems
ErgodicSquareAverages.lean	10

- `integerSystemIterate_measurable`
- `integerSystemIterate_square`
- `sectionSevenFamily_shift`
- `ergodicPositiveAverage_square`
- `ergodicPositiveAverage_measurable`
- `ergodicMaximalENorm_measurable`
- `ergodicPositiveAverage_sub`
- `ergodicPositiveAverage_norm_bound`
- `ergodicMaximalENorm_dominates`
- `ergodicMaximalENorm_bounded`

<!-- checkpoint-verification -->

Verification

All 1052 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4674-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3018 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1054 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and later deductions retain their separately reviewed scopes. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW

route.

Checkpoint_986_ErgodicIndependentInputs.md

Checkpoint 986: Precise independent general transference and bounded-class convergence interfaces

Lean implements precise independent general strong and modular Calderon transference predicates and the independent bounded-class square convergence predicate. The strong interface retains the actual discrete all-input norm and requires almost-everywhere finiteness of the proper extended ergodic maximum. The modular interface ranges over arbitrary continuous nonnegative gauges and actual finite-input distribution bounds. The bounded-class input contains its explicitly unformalized published-theorem and normalization bridge and asserts no unbounded Orlicz result. A checked application lemma exposes the exact strong transfer. These are unproved independent hypotheses, not newly formalized published theorems or new Lean mathematical axioms.

Checked declarations

Module	Named supporting theorems
ErgodicIndependentInputs.lean	1

- `calderon_strong_apply`

<!-- checkpoint-verification -->

Verification

All 1053 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4675-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3019 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1055 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and later deductions retain their separately reviewed scopes. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route. The ergodic route now records precise general strong/modular Calderon interfaces and a bounded-class polynomial pointwise interface, including its unproved published-theorem and PNT/Abel/prime-power normalization bridge. No unbounded Orlicz or source prime-enumeration endpoint is assumed; these remain deductions to prove.

Checkpoint_987_ErgodicStrongMaximal.md

Checkpoint 987: Actual ergodic strong maximal bound with the source quantitative rate

Lean transfers the proved actual Corollary 7.6 to the actual ergodic maximal operators for all three families, with exactly the same exponential p dependence. One constant is chosen before the probability system, family, exponent and function. For every $1 < p \le 2$ and every Lp input, the conclusion includes almost-everywhere finiteness of the proper extended maximum, actual Lp membership of its real representative and the stated norm bound. Only the explicit independent general strong Calderon

theorem is newly consumed; no source-specific ergodic norm estimate is assumed. The modular and Orlicz convergence clauses remain subsequent deductions.

Checked declarations

Module	Named supporting theorems
ErgodicStrongMaximal.lean	1

- `ergodic_strong_maximal_source`

<!-- checkpoint-verification -->

Verification

All 1054 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4676-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3020 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1056 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and later deductions retain their separately reviewed scopes. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route. The ergodic route now records precise general strong/modular Calderon interfaces and a bounded-class polynomial pointwise interface, including its unproved published-theorem and PNT/Abel/prime-power normalization bridge. No unbounded Orlicz or source prime-enumeration endpoint is assumed; these remain deductions to prove.

`Checkpoint_988_ErgodicModularMaximal.md`

Checkpoint 988: Actual ergodic modular inequality from the general transference input

Lean proves the actual ergodic modular maximal inequality from the already proved Theorem 7.2 and the explicit independent general modular Calderon input. It converts finite counting cardinalities correctly, proves the discrete modular premise for the exact gauge (including any later coefficient enlargement), and applies transfer to the proper extended ergodic maximum. The conclusion uses the actual nonnegative gauge integral for every positive level and every a.e. strongly measurable complex function. The Orlicz truncation/convergence argument is not assumed and remains the next analytic deduction.

Checked declarations

Module	Named supporting theorems
ErgodicModularMaximal.lean	2

- `section7_positive_modular_bound`
- `section7_ergodic_modular_apply`

<!-- checkpoint-verification -->

Verification

All 1055 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4677-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3022 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1057 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and later deductions retain their separately reviewed scopes. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route. The ergodic route now records precise general strong/modular Calderon interfaces and a bounded-class polynomial pointwise interface, including its unproved published-theorem and PNT/Abel/prime-power normalization bridge. No unbounded Orlicz or source prime-enumeration endpoint is assumed; these remain deductions to prove.

`Checkpoint_989_PrimeCutoffEnumeration.md`

Checkpoint 989: Exact prime-counting and enumeration sum identities

Lean proves the exact arithmetic bridge between prime cutoffs and prime enumeration: pi(p_(k+1)) = k+1, pi(N) > 0 for N ≥ 2, the finite prime set is the injective image of the corresponding range under Nat.nth Prime, and sums over primes equal sums over that range. The result uses actual Mathlib prime counting and infinite prime enumeration and checks every index, with no PNT, asymptotic or enumeration hypothesis. This supplies the concrete normalization and sum identity required for Theorem1.5(ii).

Checked declarations

Module	Named supporting theorems
PrimeCutoffEnumeration.lean	4

- `primeCounting_nth_prime`
- `primeCutoff_count_pos`
- `primesLE_image_nth`
- `sum_primesLE_nth`

<!-- checkpoint-verification -->

Verification

All 1056 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4678-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3026 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1058 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual

square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and later deductions retain their separately reviewed scopes. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route. The ergodic route now records precise general strong/modular Calderon interfaces and a bounded-class polynomial pointwise interface, including its unproved published-theorem and PNT/Abel/prime-power normalization bridge. No unbounded Orlicz or source prime-enumeration endpoint is assumed; these remain deductions to prove.

`Checkpoint_990_PrimeErgodicEnumeration.md`

Checkpoint 990: Exact equality of prime-cutoff and prime-enumerated ergodic maxima

Lean proves the exact pointwise ergodic bridge from prime cutoffs to averages over the first m primes. It identifies the concrete finite sums and normalizations, treats the first prime, proves both supremum inequalities and concludes equality of the proper extended maxima and their real representatives. No arithmetic asymptotic, transference or enumeration premise is used for this identity. This is the complete actual reindexing step needed to apply the ergodic strong bound to Theorem1.5(ii).

Checked declarations

Module	Named supporting theorems
PrimeErgodicEnumeration.lean	4

- `ergodic_prime_cutoff_enumerated`
- `enumeratedPrimeSquareAverage_cutoff`
- `enumeratedPrimeMaximalENorm_eq`
- `enumeratedPrimeMaximalValue_eq`

<!-- checkpoint-verification -->

Verification

All 1057 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4679-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3030 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1059 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and later deductions retain their separately reviewed scopes. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW

route. The ergodic route now records precise general strong/modular Calderon interfaces and a bounded-class polynomial pointwise interface, including its unproved published-theorem and PNT/Abel/prime-power normalization bridge. No unbounded Orlicz or source prime-enumeration endpoint is assumed; these remain deductions to prove.

`Checkpoint_991_TheoremOneFivePartTwo.md`

Checkpoint 991: Complete source Theorem 1.5(ii) for the actual prime-enumerated maximal operator

Lean completes every clause requested in Theorem1.5(ii): one absolute positive constant, every invertible measure-preserving probability system, every $1 < p \le 2$, every Lp input and the actual maximum over averages along the first m prime squares for all $m \ge 1$. The proof uses the exact proved cut-off/enumeration identity and actual strong ergodic bound. Almost-everywhere finiteness, actual Lp membership and both extended and finite real norm bounds are explicit. The general strong Calderon input and earlier Theorem7.2 inputs remain unproved independent parameters; bounded-class convergence and modular transference are not used here. The whole Theorem1.5 is marked partial because part(i) remains open.

Checked declarations

Module	Named supporting theorems
TheoremOneFivePartTwo.lean	1

- `theorem_1_5_ii`

<!-- checkpoint-verification -->

Verification

All 1058 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4680-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3031 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1060 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and later deductions retain their separately reviewed scopes. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route. The ergodic route now records precise general strong/modular Calderon interfaces and a bounded-class polynomial pointwise interface, including its unproved published-theorem and PNT/Abel/prime-power normalization bridge. No unbounded Orlicz or source prime-enumeration endpoint is assumed; these remain deductions to prove.

`Checkpoint_992_OrliczTruncation.md`

Checkpoint 992: Orlicz truncation, integrability and vanishing tail at every positive scale

Lean defines bounded truncations and the exact measurable Orlicz class, and proves equivalence with finiteness of the nonnegative gauge integral at some positive scale. It proves pointwise vanishing of the

truncation tail, the gauge rescaling bound at every fixed positive level, L1 integrability from Psi(t) ≥ t, and dominated convergence of the actual tail modular integral to zero. Thus the density approximation required by Corollary7.46 is proved rather than assumed, including the scale change justified by the checked doubling/rescaling estimate.

Checked declarations

Module	Named supporting theorems
OrliczTruncation.lean	8

- `boundedTruncation_measurable`
- `boundedTruncation_norm_le`
- `boundedTruncation_tail_norm_le`
- `boundedTruncation_tail_tendsto`
- `nearOneOrliczMember_iff`
- `nearOnePsi_change_scale`
- `nearOneOrliczMember_integrable`
- `orlicz_truncation_modular_tail`

<!-- checkpoint-verification -->

Verification

All 1059 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4681-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3039 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1061 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and later deductions retain their separately reviewed scopes. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route. The ergodic route now records precise general strong/modular Calderon interfaces and a bounded-class polynomial pointwise interface, including its unproved published-theorem and PNT/Abel/prime-power normalization bridge. No unbounded Orlicz or source prime-enumeration endpoint is assumed; these remain deductions to prove.

`Checkpoint_993_OrliczMaximalCauchy.md`

Checkpoint 993: Own Orlicz density argument giving almost-everywhere Cauchy convergence

Lean proves the own Orlicz density/Cauchy deduction. It defines the actual tail-oscillation obstruction, proves that a Cauchy bounded approximant excludes that obstruction whenever the actual error maximum is small, bounds its measure by each truncation-tail modular integral and lets that integral tend to zero. A countable sequence of positive thresholds then gives almost-everywhere Cauchy convergence of the original averages. The maximal bound and bounded-class Cauchy facts are internal premises to be supplied by the proved modular transfer and the explicitly independent bounded-class theorem; no unbounded Orlicz convergence is an external input. L1 convergence is the remaining analytic deduction.

Checked declarations

Module	Named supporting theorems
OrliczMaximalCauchy.lean	3

- `tailOscillation_not_mem`
- `ae_cauchySeq_of_tailOscillation_null`
- `orlicz_ergodic_ae_cauchy`

<!-- checkpoint-verification -->

Verification

All 1060 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4682-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3042 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1062 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and later deductions retain their separately reviewed scopes. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route. The ergodic route now records precise general strong/modular Calderon interfaces and a bounded-class polynomial pointwise interface, including its unproved published-theorem and PNT/Abel/prime-power normalization bridge. No unbounded Orlicz or source prime-enumeration endpoint is assumed; these remain deductions to prove.

`Checkpoint_994_ErgodicL1Operators.md`

Checkpoint 994: Actual ergodic L1 operators, representative agreement and uniform norm

Lean constructs the genuine L1 continuous linear ergodic square-average operators from measure-preserving iterates, proves their almost-everywhere agreement with the actual source averages, checks invariance under almost-everywhere changes of the input, and handles arbitrary L1 representatives. It proves the uniform operator norm bound by the actual averaging mass. Thus the uniform L1 continuity needed for the density argument is proved from positivity and measure preservation, rather than added as an independent input.

Checked declarations

Module	Named supporting theorems
ErgodicL1Operators.lean	4

- `ergodicSquareL1Operator_apply`
- `ergodicPositiveAverage_congr_ae`
- `ergodicSquareL1Operator_apply_toLp`
- `ergodicSquareL1Operator_norm_le`

<!-- checkpoint-verification -->

Verification

All 1061 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4683-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3046 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1063 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and later deductions retain their separately reviewed scopes. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route. The ergodic route now records precise general strong/modular Calderon interfaces and a bounded-class polynomial pointwise interface, including its unproved published-theorem and PNT/Abel/prime-power normalization bridge. No unbounded Orlicz or source prime-enumeration endpoint is assumed; these remain deductions to prove.

`Checkpoint_995_L1TruncationApproximation.md`

Checkpoint 995: Checked bounded truncation density in the actual L1 space

Lean proves bounded truncations preserve L1 integrability, that their actual L1 tail norms tend to zero by dominated convergence, and that the corresponding actual L1 equivalence classes converge to the original input. The proof checks both pointwise representatives and quotient-space distances, and exports an approximation sequence with almost-everywhere agreement with every literal truncation. No density or L1 approximation theorem is assumed independently.

Checked declarations

Module	Named supporting theorems
L1TruncationApproximation.lean	4

- `boundedTruncation_norm_le_original`
- `boundedTruncation_integrable`
- `boundedTruncation_L1_tail`
- `L1_bounded_truncation_approximation`

<!-- checkpoint-verification -->

Verification

All 1062 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4684-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3050 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1064 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual

square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and later deductions retain their separately reviewed scopes. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route. The ergodic route now records precise general strong/modular Calderon interfaces and a bounded-class polynomial pointwise interface, including its unproved published-theorem and PNT/Abel/prime-power normalization bridge. No unbounded Orlicz or source prime-enumeration endpoint is assumed; these remain deductions to prove.

`Checkpoint_996_BoundedL1Cauchy.md`

Checkpoint 996: Bounded pointwise Cauchy convergence implies actual L1 Cauchy convergence

Lean proves the bounded L1 Cauchy criterion on a finite measure space: an almost-everywhere Cauchy sequence of actual L1 functions with a common almost-everywhere pointwise bound is Cauchy in L1. The proof applies dominated convergence to all pairs of indices, checks the product filter and actual L1 quotient distances, and concludes the full Cauchy property. This is the bounded-class L1 step required by the ergodic density argument; no bounded L1 convergence or density-extension theorem is assumed independently.

Checked declarations

Module	Named supporting theorems
BoundedL1Cauchy.lean	1

- `bounded_L1_cauchy_of_ae_cauchy`

<!-- checkpoint-verification -->

Verification

All 1063 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4685-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3051 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1065 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and later deductions retain their separately reviewed scopes. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route. The ergodic route now records precise general strong/modular Calderon interfaces and a bounded-class polynomial pointwise interface, including its unproved published-theorem and PNT/Abel/prime-

power normalization bridge. No unbounded Orlicz or source prime-enumeration endpoint is assumed; these remain deductions to prove.

`Checkpoint_997_UniformOperatorDensity.md`

Checkpoint 997: Uniform continuous-operator density argument for full L1 convergence

Lean proves the general uniform-operator density argument in normed spaces: a uniformly bounded sequence of actual continuous linear operators is Cauchy on an input whenever it is Cauchy on a sequence of approximants converging to that input. It checks the full two-index Cauchy condition and quantitative approximation errors. Applied to the proved L1 ergodic operators and bounded truncations, this supplies the unbounded L1 convergence extension without assuming that conclusion as an independent theorem.

Checked declarations

Module	Named supporting theorems
UniformOperatorDensity.lean	1

- `uniform_operator_dense_cauchy`

<!-- checkpoint-verification -->

Verification

All 1064 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4686-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3052 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1066 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and later deductions retain their separately reviewed scopes. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route. The ergodic route now records precise general strong/modular Calderon interfaces and a bounded-class polynomial pointwise interface, including its unproved published-theorem and PNT/Abel/prime-power normalization bridge. No unbounded Orlicz or source prime-enumeration endpoint is assumed; these remain deductions to prove.

`Checkpoint_998_OrliczErgodicConvergence.md`

Checkpoint 998: Actual Orlicz almost-everywhere and L1 convergence with the same limit

Lean completes the own Orlicz pointwise and L1 convergence deduction for the actual ergodic averages. It applies the exact bounded-class independent input to every checked truncation, proves L1 Cauchy convergence on the bounded class by dominated convergence, extends through the actual uniformly bounded L1 operators and truncation density, and obtains a genuine L1 limit. A convergent subsequence in measure and the proved almost-everywhere Cauchy property identify the full pointwise limit with that same L1 limit. Actual integrability of every average, vanishing L1 seminorm and vanishing integral norm of the difference are explicit. No unbounded Orlicz or L1 density-extension conclusion is an independent input. The remaining final step is common-constant/source assembly for Corollary7.46.

Checked declarations

Module	Named supporting theorems
OrliczErgodicConvergence.lean	2

- `bounded_truncation_cauchy_from_input`
- `orlicz_ergodic_convergence`

<!-- checkpoint-verification -->

Verification

All 1065 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4687-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3054 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1067 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The ten completed source results under their paper hypotheses remain unchanged. Lemma 3.21, Theorems 1.1 and 1.3, Propositions 4.19 and 6.21, and every clause of Theorem 7.2 are complete checked implications under their separately recorded independent inputs. Theorem 7.2 covers all three actual square families, every integer cutoff N >= 2, arbitrary complex finite support, common constants, actual finite level sets and truncated, unrestricted modular and restricted estimates. The square-sign reflection is proved. The stronger standalone auxiliary ranges and later deductions retain their separately reviewed scopes. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved published-theorem selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative high/low comparisons, restricted estimates, parameter optimization and final source assembly are proved. No such own-paper conclusion is an independent input. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs. The full-square global/local stopping hypotheses for the earlier theorems are not used by the new ordinary-square IW route. The ergodic route now records precise general strong/modular Calderon interfaces and a bounded-class polynomial pointwise interface, including its unproved published-theorem and PNT/Abel/prime-power normalization bridge. No unbounded Orlicz or source prime-enumeration endpoint is assumed; these remain deductions to prove.

`Checkpoint_999_CorollarySevenFortySix.md`

Checkpoint 999: Complete source Corollary 7.46: quantitative ergodic bounds and Orlicz convergence

Lean completes the full source Corollary7.46 for all three actual ergodic square families. One common positive constant governs the quantitative strong exponential rate and Orlicz gauge, with a common modular multiplier fixed before all systems, families, functions and levels. The proper extended maximal function has almost-everywhere finiteness, actual Lp membership and the source norm and modular bounds. Every member of the literal Orlicz class at some positive scale has actual averages converging almost everywhere and in L1 to the same genuine L1 limit; every average is integrable, and both the L1 seminorm and integral norm of the difference tend to zero. Own truncation, scale-change, tail-oscillation, L1 operator, density, completion and limit-identification deductions are proved. General strong/modular transference and bounded-class convergence, including their unformalized published-theorem/normalization bridges, remain precise independent hypotheses; none asserts unbounded convergence or an own-paper endpoint. Corollary7.6 was completed at984 and Theorem1.5(ii) at991. Theorem1.5(i) remains open, so the whole Theorem1.5 remains partial. The manuscript is unchanged, and each checkpoint retains separate build, audit, integrity and save evidence.

Checked declarations

Module	Named supporting theorems
CorollarySevenFortySix.lean	2

- `section7_ergodic_assembly`
- `corollary_7_46`

<!-- checkpoint-verification -->

Verification

All 1066 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4688-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3056 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1068 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1000_UnitSquareResidues.md`

Checkpoint 1000: Actual unit-square residues, CRT and prime representation

Lean defines the actual unit-square residue sets, proves their exact equivalence to unit squares, their CRT multiplicativity and nonemptiness, and proves that every such residue is represented by an actual enumerated prime square. The prime representation uses the pinned Mathlib Dirichlet theorem itself, with no new independent hypothesis. This begins Section 8; the correlation, spacing, auxiliary modulus and persistent-badness assemblies remain to be proved. All earlier checkpoint999 conclusions and their independent boundaries are preserved.

Checked declarations

Module	Named supporting theorems
UnitSquareResidues.lean	7

- `mem_unitSquareResidues`
- `isUnitSquare_iff`
- `isUnitSquare_ringEquiv`
- `isUnitSquare_prod`
- `unitSquareResidues_crt`
- `unitSquareResidues_nonempty`
- `unitSquareResidues_prime_represented`

<!-- checkpoint-verification -->

Verification

All 1067 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4857-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3063 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1069 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1001_QuadraticResidueIndicator.md`

Checkpoint 1001: Quadratic-character indicator and exact prime residue density

Lean proves the exact nonzero-square character indicator, the quadratic-character sum identity, the odd-prime cardinality 2*card(Lambda_p)=p-1 and the uniform density bounds 1/3 <= card(Lambda_p)/p <= 1/2. These are proved from pinned Mathlib finite-field facts without a Weil or other new independent hypothesis. The actual k-point correlation and remaining Section8 assembly are not yet claimed.

Checked declarations

Module	Named supporting theorems
QuadraticResidueIndicator.lean	7

- `unitSquareIndicator_bounds`
- `unitSquareIndicator_formula`
- `sum_unitSquareIndicator`
- `sum_quadraticCharReal`
- `unitSquareResidues_prime_card_real`
- `unitSquareResidues_prime_card`
- `unitSquareResidues_prime_density`

<!-- checkpoint-verification -->

Verification

All 1068 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4858-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3070 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1070 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and

reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1002_UnitSquareCRTCardinality.md`

Checkpoint 1002: CRT cardinality and exact prime-product square count

Lean proves an exact CRT image identity and multiplicativity of the actual unit-square cardinalities. It also proves the full prime-product cardinality formula by induction, including the modulus-one case, with coprimality and positivity checked. No independent CRT, finite-group cardinality or prime-cardinality input is introduced.

Checked declarations

Module	Named supporting theorems
UnitSquareCRTCardinality.lean	6

- `unitSquareResidues_crt_image`
- `unitSquareResidues_mul_card`
- `unitSquareResidues_one_card`
- `prime_coprime_product`
- `unitSquareCard_eq`
- `unitSquareResidues_primeProduct_card`

<!-- checkpoint-verification -->

Verification

All 1069 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4859-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3076 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1071 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-

power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1003_FiniteSpacingCounterexample.md`

Checkpoint 1003: Exact counterexample and recorded Section 8 spacing repair obligation

Lean certifies an exact rational counterexample to the general strict fraction inequality quoted from LaVictoire Lemma3.2 and used in the source Section8 proof. For the normalized circle configuration coming from {0,1,2,4} modulo6, theta=1/10 and J=12, the claimed left fraction is1/4 and its lower bound is1/2. The kernel checks the actual cyclic gaps and all four far-point proportions; no floating-point or native evaluation is used. This identifies a repair obligation, not a counterexample to persistent badness. SECTION_8_SPACING_CORRECTION.md records the details. The manuscript is unchanged, and the badness conclusion remains unclaimed until a correct replacement argument is proved.

Checked declarations

Module	Named supporting theorems
FiniteSpacingCounterexample.lean	4

- `spacing_counterexample_gap_tails`
- `spacing_counterexample_far_values`
- `spacing_counterexample_strict_fraction`
- `quoted_strict_spacing_bound_fails`

<!-- checkpoint-verification -->

Verification

All 1070 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4860-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3080 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1072 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1004_QuadraticCharacterExpansion.md`

Checkpoint 1004: Exact character subset expansion and precise general Weil boundary

Lean proves the exact soft quadratic-character indicator, its bounds, multiplicativity under arbitrary finite products, and the complete subset expansion with the constant term separated. It defines a precise independent general split-polynomial Weil character-sum input, without assuming any own quadratic-residue correlation or badness conclusion. The actual finite zero correction and correlation estimate remain deductions to prove.

Checked declarations

Module	Named supporting theorems
QuadraticCharacterExpansion.lean	5

- `quadraticSoftIndicator_eq`
- `quadraticSoftIndicator_bounds`
- `quadraticCharReal_prod`
- `finite_one_add_expansion`
- `quadratic_one_add_expansion`

<!-- checkpoint-verification -->

Verification

All 1071 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4861-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3085 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1073 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1005_QuadraticCorrelationZeroCorrection.md`

Checkpoint 1005: Exact finite correlation and zero-location correction

Lean proves that the literal finite quadratic-residue correlation equals the sum of exact indicator products. It proves the complete zero correction between that hard count and the soft character-product count: the soft count is larger, and its excess is at most the number of distinct shifts. Every zero location is counted explicitly by a finite double sum. No Weil input is needed for this correction.

Checked declarations

Module	Named supporting theorems
QuadraticCorrelationZeroCorrection.lean	5

- `quadraticCorrelation_as_product`
- `quadraticSoft_product_nonneg`
- `quadraticSoft_product_le_one`
- `quadraticSoft_product_correction`
- `quadraticSoftCorrelation_error`

<!-- checkpoint-verification -->

Verification

All 1072 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4862-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3090 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1074 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1006_QuadraticCorrelationWeilBound.md`

Checkpoint 1006: Actual k-point quadratic correlation from the general Weil estimate

Lean derives the actual k-point nonzero-quadratic-residue correlation from the independent general split-polynomial Weil estimate. For every finite set S of distinct shifts it proves the explicit bound abs(N(S)-p/2^card(S)) <= 2*card(S)*sqrt(p). The full subset expansion, all finite character sums, empty-set constant, powerset count and exact zero corrections are checked. No own residue-correlation estimate is an independent hypothesis. The relative density normalization and c=1/3 GK format remain the next step.

Checked declarations

Module	Named supporting theorems
QuadraticCorrelationWeilBound.lean	4

- `quadraticSoftCorrelation_expansion`
- `quadratic_character_expansion_error`
- `quadraticSoftCorrelation_Weil_error`
- `quadraticCorrelation_Weil_error`

<!-- checkpoint-verification -->

Verification

All 1073 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4863-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3094 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1075 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1007_BoundedPowerDifference.md`

Checkpoint 1007: Uniform finite power-difference estimate for correlation normalization

Lean proves a general explicit power-difference inequality on the unit interval: abs(x^k-y^k) <= k*abs(x-y). This supplies the exact finite normalization of the quadratic-residue correlation main term; it is a proved elementary inequality, not an extra asymptotic hypothesis.

Checked declarations

Module	Named supporting theorems
BoundedPowerDifference.lean	1

- `bounded_power_difference`

<!-- checkpoint-verification -->

Verification

All 1074 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4864-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3095 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1076 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth

multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1008_QuadraticCorrelationRelative.md`

Checkpoint 1008: Exact local-density normalization and relative quadratic correlation

Lean proves the exact local density formula and normalizes the absolute quadratic correlation. It checks the finite power main-term shift, proves abs(N(S)-r_p^card(S)*p) <= 3*card(S)*sqrt(p), and proves the explicit relative error abs(N(S)/(r_p^card(S)*p)-1) <= 3*card(S)*3^card(S)/sqrt(p). Positivity of all normalizing factors and every denominator cancellation are checked. The only independent input is the already recorded general Weil character-sum estimate.

Checked declarations

Module	Named supporting theorems
QuadraticCorrelationRelative.lean	6

- `primeResidueDensity_formula`
- `primeResidueDensity_bounds`
- `quadratic_correlation_mainterm_shift`
- `quadraticCorrelation_density_error`
- `primeResidueDensity_power_lower`
- `quadraticCorrelation_relative_error`

<!-- checkpoint-verification -->

Verification

All 1075 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4865-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3101 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1077 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate

recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1009_QuadraticGKCorrelation.md`

Checkpoint 1009: Full source quadratic correlation, GK exponent and subpower density

Lean completes the full source quadratic-residue correlation lemma: the literal finite count, both main terms, uniform injective tuples of shifts, explicit absolute and relative constants, the exact c=1/3 GK format and subpower reciprocal density. Only the independent general split-polynomial Weil character-sum input remains. The exact source review upgrades l:quadratic-GK from open to conditional; it does not award the separate spacing repair or Theorem1.5(i). The manuscript remains unchanged.

Checked declarations

Module	Named supporting theorems
QuadraticGKCorrelation.lean	5

- `quadratic_GK_absolute_error`
- `quadratic_GK_relative_error`
- `quadratic_GK_uniform_correlation`
- `quadratic_GK_tuple`
- `primeResidueDensity_subpower`

<!-- checkpoint-verification -->

Verification

All 1076 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4866-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3106 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1078 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1010_FiniteAdditiveResidueCounts.md`

Checkpoint 1010: Actual additive pair/triple counts and exact product identities

Lean defines literal finite additive pair and triple counts. It proves invariance under arbitrary additive equivalences, exact multiplication for product sets, the actual triple-fiber sum of pair counts, and the pair

cardinality and zero-shift identities. These are unconditional finite counting statements used to transport the local residue estimates through CRT.

Checked declarations

Module	Named supporting theorems
FiniteAdditiveResidueCounts.lean	8

- `mem_image_addEquiv`
- `additivePairCount_equiv`
- `additiveTripleCount_equiv`
- `additivePairCount_product`
- `additiveTripleCount_product`
- `additiveTripleCount_sum_pairs`
- `additivePairCount_le_card`
- `additivePairCount_zero`

<!-- checkpoint-verification -->

Verification

All 1077 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4867-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3114 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1079 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1011_QuadraticPairMoment.md`

Checkpoint 1011: Actual nonzero-shift quadratic pair moment

Lean identifies the actual additive pair count with the two-point quadratic correlation and proves its explicit normalized relative error and moment upper bound. Every nonzero shift modulo an odd prime has pair count at most (1+54/sqrt(p))*card(Lambda_p)^2/p, conditional only on the general Weil estimate. The zero-shift exception remains explicitly separate.

Checked declarations

Module	Named supporting theorems
QuadraticPairMoment.lean	3

- additivePairCount_quadraticCorrelation
- quadratic_pair_relative_error
- quadratic_pair_moment

<!-- checkpoint-verification -->

Verification

All 1078 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4868-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3117 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1080 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

Checkpoint_1012_QuadraticTripleMoment.md

Checkpoint 1012: Uniform actual quadratic triple moment with the zero-shift exception checked

Lean proves the actual uniform three-variable additive residue moment by summing the checked pair correlations over the third variable. It isolates and bounds the unique possible zero-shift exception, with all density and square-root comparisons checked. The triple count is at most (1+63/sqrt(p))*card(Lambda_p)^3/p for every target residue; both pair and triple counts are then bounded with the common coefficient64. These conclusions are derived from the general Weil input and are not new independent hypotheses.

Checked declarations

Module	Named supporting theorems
QuadraticTripleMoment.lean	5

- finite_one_exception_sum
- large_density_exception
- quadratic_triple_moment
- quadratic_pair_moment_common
- quadratic_triple_moment_common

<!-- checkpoint-verification -->

Verification

All 1079 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4869-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3122 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1081 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1013_UnitSquareMomentCRT.md`

Checkpoint 1013: Exact CRT multiplication of actual unit-square additive moments

Lean proves exact CRT multiplication for the actual unit-square additive pair and triple counts, including all signed target residues. It supplies total integer-modulus count functions with exact positive-modulus bridges, checks the modulus-one base cases and proves the prime-product factorizations by a general multiplicative induction. No approximate CRT or moment factorization is assumed.

Checked declarations

Module	Named supporting theorems
UnitSquareMomentCRT.lean	11

- `unitSquarePairCount_crt`
- `unitSquareTripleCount_crt`
- `unitSquarePairCount_eq`
- `unitSquareTripleCount_eq`
- `unitSquarePairCount_mul`
- `unitSquareTripleCount_mul`
- `unitSquarePairCount_one`
- `unitSquareTripleCount_one`
- `multiplicative_prime_product`
- `unitSquarePairCount_primeProduct`
- `unitSquareTripleCount_primeProduct`

<!-- checkpoint-verification -->

Verification

All 1080 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4870-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3133 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1082 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1014_FiniteUnionMomentBound.md`

Checkpoint 1014: Finite union lower bound from actual pair intersections

Lean proves the exact two-term finite inclusion-exclusion lower bound for arbitrary equal-size finite sets with a common bound on pairwise intersections, including the empty-family and one-set cases. It also proves the convenient quadratic lower bound and union upper bound. This is an unconditional finite combinatorial deduction for the Section8 thickening repair.

Checked declarations

Module	Named supporting theorems
FiniteUnionMomentBound.lean	3

- `finite_union_uniform_lower`
- `finite_union_uniform_quadratic_lower`
- `finite_union_uniform_upper`

<!-- checkpoint-verification -->

Verification

All 1081 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4871-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3136 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1083 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-

class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1015_UnitSquareDensityProducts.md`

Checkpoint 1015: Actual prime-product residue density and mean-spacing bounds

Lean proves exact multiplicative density and mean-spacing identities for the actual unit-square residue sets. For every product of n distinct odd primes it proves density between (1/3)^n and (1/2)^n and mean spacing between 2^n and3^n. It also checks positivity, the full probability bound and spacing at most the modulus. No density or spacing-growth input is assumed.

Checked declarations

Module	Named supporting theorems
UnitSquareDensityProducts.lean	12

- `unitSquareCard_pos`
- `unitSquareDensity_pos`
- `unitSquareSpacing_pos`
- `unitSquareDensity_inv`
- `unitSquareDensity_prime`
- `unitSquareCard_primeProduct`
- `unitSquareDensity_primeProduct`
- `unitSquareSpacing_primeProduct`
- `unitSquareDensity_primeProduct_bounds`
- `unitSquareSpacing_primeProduct_bounds`
- `unitSquareDensity_le_one`
- `unitSquareSpacing_le_modulus`

<!-- checkpoint-verification -->

Verification

All 1082 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4872-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3148 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1084 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate

recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1016_GlobalQuadraticMomentBound.md`

Checkpoint 1016: Actual global quadratic moments from the checked local error budget

Lean proves that a sum budget for the actual local Weil errors bounds the full prime-product pair and triple counts by three times their random-model expressions. The argument checks finite product/exponential bounds, exact CRT count factorizations and exact model normalization. It proves pair <= 3*card(Lambda_q)^2/q for shifts nonzero at every block prime, and triple <= 3*card(Lambda_q)^3/q for every target. The finite error budget is an explicit ordinary hypothesis to be supplied by the proved large-prime block construction, not an independent analytic input.

Checked declarations

Module	Named supporting theorems
GlobalQuadraticMomentBound.lean	5

- `finite_product_error_budget`
- `finite_product_perturbed_bound`
- `unitSquare_model_product`
- `global_quadratic_pair_moment`
- `global_quadratic_triple_moment`

<!-- checkpoint-verification -->

Verification

All 1083 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4873-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3153 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1085 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

Checkpoint notes 1017-1091

Checkpoint_1017_CyclicThickening.md

Checkpoint 1017: Actual cyclic thickening, translation counts and exact rounding

Lean defines the actual cyclic thickening as the finite union of positive integer translates, with the exact open-interval convention and no limiting model substituted. It proves exact translation/intersection cardinality identities, the exact natural-ceiling membership condition, the one-unit rounding bounds for the number of allowed shifts and the genuine probability bounds on the thickening density. These unconditional finite statements supply the discretization step of the Section8 repair.

Checked declarations

Module	Named supporting theorems
CyclicThickening.lean	7

- `mem_translatedResidues`
- `translatedResidues_card`
- `translatedResidues_inter_card`
- `mem_positiveIntegerShifts`
- `positiveIntegerShifts_card_bounds`
- `mem_cyclicThickening`
- `cyclicThickeningDensity_bounds`

<!-- checkpoint-verification -->

Verification

All 1084 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4874-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3160 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1086 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

Checkpoint_1018_CyclicThickeningMomentBound.md

Checkpoint 1018: Actual cyclic thickening deficit from finite pair moments

Lean proves the actual cyclic thickening estimate from its finite pair counts, without a spacing-distribution hypothesis. If the actual allowed translates have pair intersection cost at most

C*card(S)^2/q, their thickening density lies between gamma-card(S)/q-(C/2)*gamma^2 and gamma. The exact integer rounding, positive normalizations, finite union count and every quadratic inequality are checked. The actual prime-product application will supply the pair-count premise from the already proved Weil/CRT estimates.

Checked declarations

Module	Named supporting theorems
CyclicThickeningMomentBound.lean	1

- `cyclic_thickening_pair_bound`

<!-- checkpoint-verification -->

Verification

All 1085 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4875-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3161 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1087 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1019_PrimeProductThickening.md`

Checkpoint 1019: Actual prime-product thickening bound with every pair premise discharged

Lean applies the finite thickening theorem to the actual unit-square residue sets of prime products. The global pair bound is supplied by the proved Weil/CRT moment theorem; the exact small-shift hypothesis is checked by injectivity of natural residues below every block prime. Under the finite error budget and explicit radius bound, the actual density lies between gamma-rho_q-(3/2)*gamma^2 and gamma. No spacing distribution, Poisson limit or thickening conclusion is an independent input. The constructed blocks still need to supply the elementary budget and radius conditions.

Checked declarations

Module	Named supporting theorems
PrimeProductThickening.lean	2

- `small_integer_shift_difference_ne_zero`
- `prime_product_thickening_bound`

<!-- checkpoint-verification -->

Verification

All 1086 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4876-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3163 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1088 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1020_CyclicNearShifts.md`

Checkpoint 1020: Exact cyclic near-pair covering by signed integer shifts

Lean proves exact signed-integer shift cardinalities and a kernel-checked covering of the actual near-difference pairs by triple-count events. It proves that every short cyclic distance is represented by an integer shift in the precise floor-bounded interval, including zero and either direction around the circle. Far and near cardinalities partition the full ordered pair set. No general spacing inequality is assumed.

Checked declarations

Module	Named supporting theorems
CyclicNearShifts.lean	5

- `signedIntegerShifts_card`
- `signedIntegerShifts_card_bound`
- `cyclic_distance_short_shift`
- `cyclicFarPairs_add_near_card`
- `cyclicNearPairs_subset_shift_union`

<!-- checkpoint-verification -->

Verification

All 1087 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4877-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3168 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1089 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong

and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1021_CyclicDifferenceMomentBound.md`

Checkpoint 1021: Actual far-pair lower bound from the checked cyclic triple-count cover

Lean proves the genuine far-pair fraction is between zero and one and derives its quantitative lower bound from the actual triple moments. With moment coefficient C, the fraction is at least1-2*C*gamma-C*card(S)/q. The proof checks the exact near/far partition, covers every short cyclic distance by a signed integer shift, bounds the finite union and verifies the positive normalization. No finite-spacing fraction lemma is assumed.

Checked declarations

Module	Named supporting theorems
CyclicDifferenceMomentBound.lean	2

- `cyclicFarPairFraction_bounds`
- `cyclic_far_pair_triple_bound`

<!-- checkpoint-verification -->

Verification

All 1088 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4878-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3170 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1090 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both

limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1022_PrimeProductDifference.md`

Checkpoint 1022: Actual prime-product difference bound and same-modulus finite repair

Lean proves the actual prime-product far-pair lower bound 1-6*gamma-3*rho_q from the checked triple moment and exact signed-shift cover. It assembles this with the actual thickening bounds for the same residue set, modulus and finite error budget, and checks the source alpha=1/160 margin for gamma<1/16. The remaining arithmetic construction must supply the finite budget and radius assumptions; no Poisson spacing or false strict fraction inequality is used.

Checked declarations

Module	Named supporting theorems
PrimeProductDifference.lean	3

- `prime_product_far_pair_bound`
- `prime_product_finite_badness_conditions`
- `badness_difference_constant_margin`

<!-- checkpoint-verification -->

Verification

All 1089 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4879-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3173 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1091 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1023_LargePrimeBlocks.md`

Checkpoint 1023: Actual large-prime sequence and exact finite block construction

Lean constructs an actual strictly increasing sequence of primes exceeding every prescribed earlier threshold, using the proved infinitude and enumeration of primes. It defines the auxiliary prime sequence and the finite blocks through an explicit pairing injection, proves every block has exactly j+1 distinct odd

primes, verifies the precise growth threshold for every block member and proves each product modulus is positive. No large-prime existence or block-size hypothesis is left as an independent input; combined coprimality and the numerical error budget are the next deductions.

Checked declarations

Module	Named supporting theorems
LargePrimeBlocks.lean	11

- `badnessPrimeIndex_strictMono`
- `badnessPrimeIndex_dominates`
- `badnessAuxPrime_prime`
- `badnessAuxPrime_strictMono`
- `badnessAuxPrime_threshold`
- `badnessAuxPrime_gt_two`
- `badnessAuxPrime_odd`
- `badnessPrimeBlock_card`
- `badnessPrimeBlock_prime`
- `badnessPrimeBlock_threshold`
- `badnessBlockModulus_pos`

<!-- checkpoint-verification -->

Verification

All 1090 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4880-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3184 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1092 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1024_BadnessModuliCoprime.md`

Checkpoint 1024: Actual auxiliary modulus disjointness, oddness and combined coprimality

Lean proves that the actual auxiliary primes and finite prime blocks are mutually disjoint, and that the combined auxiliary-prime/product-modulus collection is pairwise coprime across all four index cases. It proves every auxiliary modulus is odd, every block product exceeds one and the needed positivity instances. These are exact properties of the constructed sequences, not assumptions about hypothetical moduli.

Checked declarations

Module	Named supporting theorems
BadnessModuliCoprime.lean	10

- badnessAuxU_prime
- badnessAuxU_odd
- badnessAuxU_not_mem_block
- badnessPrimeBlocks_disjoint
- badnessAuxU_pairwise_coprime
- badnessAuxU_block_coprime
- badnessBlockModuli_pairwise_coprime
- badness_combined_moduli_pairwise_coprime
- badnessBlockModulus_odd
- badnessBlockModulus_gt_one

<!-- checkpoint-verification -->

Verification

All 1091 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4881-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3194 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1093 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

Checkpoint_1025_BadnessBlockBudgets.md

Checkpoint 1025: Constructed prime blocks discharge every finite moment and radius premise

Lean proves that the actual constructed prime blocks satisfy every finite error-budget and small-radius premise of the repaired moment argument. It verifies the explicit growth inequalities, square-root error sum, exact density/spacing bounds and the common actual modulus. It then obtains the actual thickening deficit and far-pair bounds for every block and every0<gamma<=1, conditional only on the general Weil estimate. No large-prime, density, spacing, budget or own-paper finite endpoint premise remains unproved. The limiting conditions, prime-class frequency pushforward and general badness-principle assembly remain.

Checked declarations

Module	Named supporting theorems
BadnessBlockBudgets.lean	6

- `badnessPrimeBlock_large`
- `badnessPrimeBlock_error_budget`
- `badnessBlockSpacing_bounds`
- `badnessBlockDensity_bounds`
- `badnessBlock_small_radius`
- `badnessBlock_finite_conditions`

<!-- checkpoint-verification -->

Verification

All 1092 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4882-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3200 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1094 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1026_BadnessDensityLimits.md`

Checkpoint 1026: Actual density limits and fully justified real liminf comparisons

Lean proves the actual block densities tend to zero, the actual mean spacings tend to infinity, and every actual auxiliary-prime density is at least1/3. It also proves bounded-real liminf comparison lemmas with all boundedness and coboundedness premises explicit, so the next passage to the source limiting conditions cannot rely on a default real liminf outside its valid scope. These are unconditional consequences of the already proved exact finite cardinalities.

Checked declarations

Module	Named supporting theorems
BadnessDensityLimits.lean	5

- `badnessBlockDensity_tendsto_zero`
- `badnessBlockSpacing_tendsto_atTop`
- `badnessAuxU_density_lower`
- `liminf_lower_from_vanishing_error`
- `liminf_le_uniform_upper`

<!-- checkpoint-verification -->

Verification

All 1093 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4883-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3205 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1095 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1027_BadnessSpacingLimits.md`

Checkpoint 1027: Actual source liminf thickening deficit and difference condition

Lean passes the actual constructed-block moment estimates to the source real liminf conditions, using the proved density limit and explicit boundedness. The thickening deficit is nonnegative and at most (3/2)*gamma^2 for0<gamma<=1. The actual far-pair liminf is at least1-6*gamma and strictly exceeds5*(1/160) whenever0<gamma<1/16. Every statement concerns the literal unit-square sets for the constructed moduli. The little-o formulation is the next short analytic step.

Checked declarations

Module	Named supporting theorems
BadnessSpacingLimits.lean	3

- `badnessThickeningDeficit_bounds`
- `badness_far_pair_liminf`
- `badness_difference_condition`

<!-- checkpoint-verification -->

Verification

All 1094 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4884-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3208 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1096 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i)

remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1028_BadnessThickeningLittleO.md`

Checkpoint 1028: Full source little-o thickening condition for the actual constructed moduli

Lean proves the exact source little-o thickening condition for the actual constructed unit-square sets. It sends the deficit divided by gamma to zero on the positive-side neighborhood of zero using the checked quadratic deficit bound, then proves the genuine IsLittleO statement with the denominator's nonvanishing condition explicit. Together with the prior source difference condition, all density/thickening/difference limiting hypotheses are now proved under the general Weil input, without Poisson spacings or the refuted strict finite inequality.

Checked declarations

Module	Named supporting theorems
BadnessThickeningLittleO.lean	2

- `badnessThickeningDeficit_ratio_zero`
- `badness_thickening_little_o`

<!-- checkpoint-verification -->

Verification

All 1095 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4885-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3210 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1097 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both

limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1029_UniformUnitSquareFibers.md`

Checkpoint 1029: Exact uniform fibers of squaring on units

For every positive modulus, Lean proves that all fibers of squaring on the finite unit group over its image have equal size. The explicit multiplication bijection identifies each fiber with the kernel. Summing the fibers gives kernel size times the actual unit-square image size equals Euler totient. Consequently each unit-square class has exactly the fraction 1/|Lambda_q| of all unit residues. This includes composite moduli and supplies the exact normalization for prime-square frequencies without any exceptional-prime approximation.

Checked declarations

Module	Named supporting theorems
UniformUnitSquareFibers.lean	6

- `mem_unitSquareRoots`
- `unitSquareRoots_nonempty`
- `unitSquareRoots_card_eq_kernel`
- `unitSquareRoots_sum`
- `unitSquareRoots_kernel_factor`
- `unitSquareRoots_normalized`

<!-- checkpoint-verification -->

Verification

All 1096 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4886-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3216 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1098 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1030_PrimeSquareFrequency.md`

Checkpoint 1030: Exact prime-square residue frequencies from fixed-progression PNT

The actual frequency of the nth-prime squares in every unit-square residue class is proved to converge to 1/|Lambda_q|, for every positive modulus. The only new independent input is the standard fixed-progression PNT formulated for frequencies of enumerated primes in individual unit classes. Lean proves the exact disjoint root-class partition of the finite counts, sums the finite limits, and applies the checked uniform-fiber identity. It also proves the strict beta=1/2 liminf condition and strict increase of the actual prime-square sequence. No prime-square equidistribution conclusion is assumed as an input.

Checked declarations

Module	Named supporting theorems
PrimeSquareFrequency.lean	6

- `sequenceResidueFrequency_card`
- `square_indicator_unit_fibers`
- `primeSquareFrequency_partition`
- `primeSquareFrequency_limit`
- `primeSquareFrequency_strict_liminf`
- `primeSquareSequence_strictMono`

<!-- checkpoint-verification -->

Verification

All 1097 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4887-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3222 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1099 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

Checkpoint_1031_LaVictoireArithmetic.md

Checkpoint 1031: All source badness arithmetic hypotheses for the actual prime squares

The complete arithmetic hypotheses of the source LaVictoire criterion are now packaged as literal finite cyclic sets, CRT membership, represented residues, two odd pairwise-coprime sequences, density zero, a strictly positive density infimum, genuine right-side little-o thickening, positive liminf of the actual far-pair count, and strict prime-square residue frequencies for every product of distinct chosen moduli. Lean constructs this full arithmetic data for the actual enumerated prime squares. Only the general Weil and fixed-progression prime-frequency inputs are used. No arithmetic condition or prime-square badness conclusion is an input.

Checked declarations

Module	Named supporting theorems
LaVictoireArithmetic.lean	3

- `unitSquareFamily_density`
- `badness_aux_density_iInf_positive`
- `primeSquare_lavictoire_arithmetic`

<!-- checkpoint-verification -->

Verification

All 1098 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4888-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3225 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1100 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1032_SequenceBadnessMaximal.md`

Checkpoint 1032: Proper sequence maximal operator and pointwise divergence

Lean defines the literal complex ergodic averages along an arbitrary sequence and the proper extended-real maximum over any set of cutoffs. It proves measurability, the exact identity with the maximum along the increasing enumeration of an infinite cutoff set, and that an infinite maximum prevents convergence to any finite complex limit. Nonatomicity is defined by measurable splitting of every positive-measure set; the deprecated Mathlib NoAtoms alias only means null singletons and is deliberately not used. These deductions contain no external mathematical input.

Checked declarations

Module	Named supporting theorems
SequenceBadnessMaximal.lean	5

- `sequenceErgodicAverage_measurable`
- `restrictedSequenceMax_measurable`
- `restrictedSequenceMax_enumeration`
- `convergent_sequence_enorm_sup_finite`
- `maximal_infinite_implies_divergence`

<!-- checkpoint-verification -->

Verification

All 1099 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4890-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3230 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1101 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1033_PersistentBadnessCriterion.md`

Checkpoint 1033: Persistent badness from independent general weak maximal principles

Lean proves the full persistent-divergence adaptation of the original arithmetic badness theorem. The independent inputs are separately quantified general original LaVictoire weak-maximal failure, Sawyer maximal principle, and same-constant Conze transference. The proof uses contraposition to produce an integrable measurable function whose proper maximum is infinite on a measurable positive-measure set, then proves divergence along the increasing enumeration of every infinite set of cutoffs. It applies to truly nonatomic ergodic measure-preserving probability systems without assuming invertibility. The current paper's adapted criterion and prime-square badness are deductions, not input fields. The published general theorems and their precise Lean formulation/convention bridges remain unformalized.

Checked declarations

Module	Named supporting theorems
PersistentBadnessCriterion.lean	2

- `weak_failure_produces_infinite_maximal`
- `persistent_l1_badness_from_arithmetic`

<!-- checkpoint-verification -->

Verification

All 1100 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4891-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3232 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1102 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong

and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1034_TheoremOneFive.md`

Checkpoint 1034: Prime-square persistent badness and both parts of Theorem 1.5

Lean proves Theorem1.5(i) for the actual sequence of enumerated prime squares: every nonatomic ergodic measure-preserving probability system, every infinite set of cutoffs, a measurable integrable complex function and a measurable positive-measure set of pointwise divergence. Its independent inputs are the general split-polynomial Weil bound, fixed-progression PNT frequencies, original LaVictoire weak maximal theorem, Sawyer maximal principle and Conze transference; their existence and published-to-Lean convention bridges remain unproved. It also assembles both parts of Theorem1.5 with the already checked quantitative Lp conclusion of part(ii), retaining all of that part's existing independent inputs. The corrected spacing proof is used, and Granville–Kurlberg is not needed for this badness deduction. Source-level status review and exact replacement TeX remain before closing the Section8 milestone.

Checked declarations

Module	Named supporting theorems
TheoremOneFive.lean	2

- `theorem_1_5_i`
- `theorem_1_5`

<!-- checkpoint-verification -->

Verification

All 1101 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4892-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3234 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1103 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter

optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1035_CyclicPoissonSpacings.md`

Checkpoint 1035: Literal cyclic successor and joint Poisson spacing conclusion

The quoted Poisson-spacing conclusion is formalized using the actual finite cyclic successor, normalized gap and all joint tails of consecutive gaps. Lean proves that the successor stays in the original finite residue set and that joint Poisson convergence implies the source single-gap tail limit exp(-theta). These definitions distinguish full joint Poisson spacings from a single marginal limit. No Poisson convergence is assumed or used in the completed badness proof; the independent quoted general theorem interface follows separately.

Checked declarations

Module	Named supporting theorems
CyclicPoissonSpacings.lean	3

- `cyclicNextPoint_mem`
- `cyclicJointGapTail_one`
- `poisson_single_gap_tail`

<!-- checkpoint-verification -->

Verification

All 1102 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4893-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3237 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1104 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary

IW route.

Checkpoint_1036_GranvilleKurlbergInterface.md

Checkpoint 1036: General GK boundary and actual unit-square joint Poisson application

The quoted Granville–Kurlberg theorem now has a precise general interface: arbitrary CRT residue families, actual local correlations with a common positive exponent and constants chosen before primes and shifts, subpower reciprocal density, odd squarefree moduli and growing mean spacing imply full joint cyclic Poisson spacings. Its proof and published-to-Lean cyclic/joint formulation bridge remain unproved and explicitly external. Lean verifies every local and modulus hypothesis for the actual unit squares, proves the constructed block moduli squarefree, and derives joint Poisson convergence and the source single-gap tail exp(-theta) conditional on that general input and the Weil input. The completed prime-square badness theorem does not depend on this optional Poisson input.

Checked declarations

Module	Named supporting theorems
GranvilleKurlbergInterface.lean	6

- `unitSquareFamily_crt`
- `unitSquareFamily_prime_density`
- `unitSquareFamily_gk_local`
- `badnessBlockModulus_squarefree`
- `unitSquareBlock_poisson`
- `unitSquareBlock_gap_tail`

<!-- checkpoint-verification -->

Verification

All 1103 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4894-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3243 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1105 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

Checkpoint_1037_OddSquarefreeUnitSquareFormulas.md

Checkpoint 1037: All odd-squarefree image, root and mean-spacing formulas

For every odd squarefree modulus, Lean proves the exact image cardinality |Lambda_q|=phi(q)/2^omega(q), exactly 2^omega(q) unit roots of every unit-square class, and the mean-spacing product over prime divisors. It first proves the finite prime-product totient identity and then passes to the literal prime-factor set of any odd squarefree modulus, including q=1. These proofs complete the remaining elementary cardinality clauses of Section8 and require no independent analytic input. Together with the already saved Theorem1.5 assembly, only final source-statement review, replacement-proof documentation and report synchronization remain for this milestone.

Checked declarations

Module	Named supporting theorems
OddSquarefreeUnitSquareFormulas.lean	6

- `unitSquareCard_primeProduct_totient`
- `odd_primeFactors`
- `oddSquarefree_unitSquareCard_factor`
- `oddSquarefree_unitSquareCard_real`
- `oddSquarefree_unitSquareRoots_card`
- `oddSquarefree_unitSquareSpacing`

<!-- checkpoint-verification -->

Verification

All 1104 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4895-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3249 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1106 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Theorem7.2, Corollaries7.6/7.46 and Theorem1.5(ii) are complete checked implications under their explicitly recorded independent inputs. This includes the actual full maximal operators, quantitative strong and modular bounds, own Orlicz pointwise/L1 convergence and exact prime enumeration. Theorem1.5(i) remains open, so its whole source entry is partial. Earlier Theorems1.1/1.3, Propositions4.19/6.21 and reflection retain their reviewed scopes. Stronger standalone auxiliary statements and later arithmetic applications remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route.

`Checkpoint_1038_SectionEightMilestone.md`

Checkpoint 1038: Section 8 and Theorem 1.5 complete conditional milestone with checked proof repair

Checkpoint1038 closes Section8 and Theorem1.5(i) as complete checked conditional implications and retains the combined theorem for both parts of Theorem1.5. The final literal statement displays actual prime-square iterates and normalized finite sums, every truly nonatomic ergodic probability system, every infinite cutoff set, a measurable integrable function and a measurable positive-measure divergence set. The adapted criterion is proved from separately specified original LaVictoire, Sawyer and Conze

inputs; Weil and fixed-progression PNT supply general arithmetic inputs, while every own-paper arithmetic/moment/limit/frequency deduction is proved. The quoted general GK theorem remains unproved, with a precise joint Poisson interface and checked application that badness does not use. The exact source review updates only the two completed source claims. A Lean kernel counterexample identifies the strict finite-fraction error, and the checked moment replacement is supplied as both a proof passage and complete Section8 TeX with an insertion guide. The replacement compiles in isolated LaTeX; the hashed manuscript is unchanged. Final complete build, source audit, transitive theorem axiom audit, source-level registry validation and archive checks are required before this checkpoint is saved.

Checked declarations

Module	Named supporting theorems
SectionEightMilestone.lean	3

- `source_lavictoire_persistent_criterion`
- `theorem_1_5_i_literal`
- `section_eight_complete`

<!-- checkpoint-verification -->

Verification

All 1105 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4896-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3252 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1107 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

Both parts of Theorem1.5, the adapted LaVictoire criterion and the full Section8 own-paper deductions are complete checked implications under separately recorded general independent inputs. The actual prime-square badness proof uses Weil, fixed-progression PNT, original LaVictoire, Sawyer and Conze; their existence and published-to-Lean convention bridges remain unproved. All arithmetic, moment, limiting, frequency and persistent-divergence deductions are proved. The quoted general GK theorem remains unproved with a precise interface and checked joint Poisson application; it is not used by badness. The source strict finite-fraction proof requires the saved exact TeX replacement. The frozen manuscript is unchanged. Earlier headline, maximal, sparse and Orlicz scopes remain as reviewed. Stronger standalone auxiliary ranges and Sections9–11 remain separately unfinished. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

Checkpoint_1039_SectionSixSourceExports.md

Checkpoint 1039: Confirmed Section8 edits and full source approximation and low estimates

The user-confirmed three Section8 edits are recorded as applied in a new selected project manuscript reconstructed from the exact agreed passages; original files are preserved, all numbered statements and labels are unchanged, and no new full upload is claimed. The Sections3–8 source review finds twelve partial auxiliaries in Sections5–7. This checkpoint closes two of them: the full source approximation lemma and the full low-contribution proposition. Lean exports the literal selected error with exact Q^(-1/2)*(1+log N)^(-A) operator and all-l2 function norms for every A>0, with one source coefficient chosen before A. It also exports one uniform constant for the actual low operator's all-real-radius local bound and l1-to-linfinity norm, for every complex l1 input and real Q in the standing source range. Existing compatible raw distribution inputs remain explicit; no new general mathematical input is added. Ten other auxiliary claims, including a sparse-recursion statement requiring scale qualification, remain before Goldbach work.

Checked declarations

Module	Named supporting theorems
SectionSixSourceExports.lean	3

- `source_approximation_all_exponents`
- `source_approximation_function_bound`
- `source_low_local_and_norm`

<!-- checkpoint-verification -->

Verification

All 1106 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4897-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3255 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1108 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

Checkpoint_1040_RealAnnularKernelBounds.md

Checkpoint 1040: Literal annular kernels at every real scale

Lean extends the actual continuous annular density and kernel to every positive real scale, with exact agreement at natural scales. It proves integrability, nonnegativity and the uniform box-kernel bound for every real N>=2, uniformly in beta<=1. This removes the first integer-scale restriction in the standalone continuous auxiliary lemma. The full uncountable maximum and its measurability, common almost-everywhere bound and Lp estimate are separate next deductions; this checkpoint does not prematurely award the full source lemma. No independent input is used for these real-scale kernel proofs.

Checked declarations

Module	Named supporting theorems
RealAnnularKernelBounds.lean	7

- `realContinuousAnnularKernel_nat`
- `realAnnularKernelDensity_continuousOn`
- `realContinuousAnnularKernel_integrable`
- `realContinuousAnnularKernel_complex_integrable`
- `realContinuousAnnularKernel_nonneg`
- `realAnnularKernelDensity_bound`
- `realContinuousAnnularKernel_box_bound`

<!-- checkpoint-verification -->

Verification

All 1107 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4898-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3262 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1109 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

Checkpoint_1041_RealAnnularConvolution.md

Checkpoint 1041: All-input real annular convolution and continuity in scale

For arbitrary Lp inputs with p>=1, Lean proves the real-scale annular convolution is measurable and dominated pointwise by the one-sided interval average. It checks the literal kernel-to-interval integral identity and proves continuity in the real scale parameter at every spatial point, using a locally integrable clamped power weight and continuous interval primitives. These facts enable a fixed countable dense set of real scales to represent the proper uncountable maximum, rather than interchanging uncountably many almost-everywhere assertions. No external input is used in this checkpoint.

Checked declarations

Module	Named supporting theorems
RealAnnularConvolution.lean	7

- `shiftedLp_integrableOn`
- `realContinuousAnnularKernel_product_box`
- `realContinuousAnnularConvolution_box_bound`
- `realContinuousAnnularConvolution_aestronglyMeasurable`
- `realAnnularPrimitiveIntegrand_locallyIntegrable`
- `realContinuousAnnularConvolution_interval`
- `realContinuousAnnularConvolution_continuous_scale`

<!-- checkpoint-verification -->

Verification

All 1108 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4899-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3269 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1110 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

Checkpoint_1042_RealAnnularMaximal.md

Checkpoint 1042: Proper all-real-scale annular maximum on the full Lp domain

The proper maximum over every real N>=2 is now proved measurable by equality with one fixed countable dense set of scales, using the checked scale continuity. For every actual Lp input with 1<p<=2, Lean proves a common almost-everywhere bound for all real N, almost-everywhere finiteness of the extended-real maximum, Lp membership of its finite representative and the exact near-one norm bound. The standard one-sided Hardy–Littlewood input is stated on its full Lp domain; this general theorem remains unproved and contains no annular or current-paper conclusion. Its restriction to the earlier L2-intersection interface is proved. Integer box averages and an exact ceiling comparison avoid an uncountable union of exceptional sets. No paper-specific maximal conclusion is assumed. The full source continuous lemma will be reviewed only after its signed-multiplier clause and exact kernel/Fourier identification are assembled.

Checked declarations

Module	Named supporting theorems
RealAnnularMaximal.lean	8

- `full_one_sided_HL_implies_existing`
- `realAnnularDenseScales_countable`
- `realAnnularDenseScales_dense`
- `realContinuousAnnularMax_countable`
- `realContinuousAnnularMax_aemeasurable`
- `oneSidedBoxAverage_enlarge`
- `realContinuousAnnular_common_majorant`
- `realContinuousAnnular_maximal_Lp`

<!-- checkpoint-verification -->

Verification

All 1109 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4900-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3277 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1111 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and

persistence deductions are proved.

Checkpoint_1043_RealAnnularFourierIdentity.md

Checkpoint 1043: Exact Fourier identity for the source model at every real scale

Lean identifies the literal continuous annular kernel at every positive real scale with the exact source Fourier integral N^(-1) integral_(N/2)^N t^(beta-1)e(-theta*t^2) dt. The quadratic substitution, derivative, endpoints, kernel indicator and Fourier normalization are all checked. This closes the real-scale model-identification prerequisite for the standalone continuous source estimate; no external mathematical input is used.

Checked declarations

Module	Named supporting theorems
RealAnnularFourierIdentity.lean	2

- `realAnnularKernelDensity_substitution`
- `realContinuousAnnularKernel_fourier`

<!-- checkpoint-verification -->

Verification

All 1110 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4901-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3279 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1112 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

Checkpoint_1044_FullContinuousSource.md

Checkpoint 1044: Complete source continuous annular lemma on all Lp inputs and real scales

The entire printed continuous annular lemma is now a checked conditional implication. One positive constant precedes beta,p,J,signs and every Lp function. The actual signed-increment convolution has the

exact source multiplier and a uniform near-one bound on the full Lp domain. The proper maximum over every real N>=2 is measurable, finite almost everywhere, belongs to Lp and satisfies the same near-one rate. All annular Fourier, L2, Hormander, scale-continuity and dense-supremum deductions are proved. The general Hardy–Littlewood and scalar Hormander theorems are stated in precise full-domain forms and remain unproved independent inputs; no source-specific estimate is assumed. The source review changes only this auxiliary claim to conditional, leaving nine partial auxiliaries before Goldbach work.

Checked declarations

Module	Named supporting theorems
FullContinuousSource.lean	3

- `fullLpConvolutionBound_mono`
- `continuousAnnularSigned_full_Lp`
- `source_continuous_annular_estimates`

<!-- checkpoint-verification -->

Verification

All 1111 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4902-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3282 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1113 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1045_LinearLocalMass.md`

Checkpoint 1045: Literal linear-scale admissibility and exact finite window counts

The literal linear-scale local average and its supremum over every real radius are now formalized on actual l1 inputs, with convergence, positivity at positive scales and uniform bounds. For indicator inputs Lean proves the exact finite window-count formula and exports the source admissible dyadic stopping-map condition as an equivalence with its displayed all-real-radius finite-count inequality. This supplies the missing concrete admissibility objects for the general Section5 argument. No analytic or own-paper

conclusion is an assumed input. Completion of the definition's additional convention remarks is not claimed here.

Checked declarations

Module	Named supporting theorems
LinearLocalMass.lean	10

- `linearSourceBall_summable`
- `linearSourceBallMass_nonneg`
- `linearSourceBallMass_uniform`
- `linearSourceLocalMass_bddAbove`
- `linearSourceBallMass_le_local`
- `linearSourceLocalMass_nonneg`
- `linearSourceLocalMass_uniform`
- `linearSourceLocalMass_le_iff`
- `linearSourceBallMass_indicator`
- `sourceLinearAdmissible_iff_display`

<!-- checkpoint-verification -->

Verification

All 1112 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4903-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3292 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1114 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1046_LocalErrorSquareMaximum.md`

Checkpoint 1046: General local error maximum at the sharp square-summability threshold

Lean proves the countable l2 error maximum from the source's full assumption A>1/2. The proof uses square summability of the logarithmically decaying dyadic operator norms, not the stronger and unjustified requirement that their norms be summable. It constructs an actual nonnegative l2 majorant

dominating every scale pointwise and proves its norm bound with a finite constant depending only on A. This is the general error ingredient for the complete abstract localized stopping lemma. The current work plan is regenerated from exact pending source labels at each checkpoint, and the confirmed Section8 edits remain recorded.

Checked declarations

Module	Named supporting theorems
LocalErrorSquareMaximum.lean	3

- `dyadic_log_square_summable`
- `dyadicLogSquareConstant_nonneg`
- `localError_square_majorant`

<!-- checkpoint-verification -->

Verification

All 1113 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4904-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3295 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1115 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1047_HighLowCappedBalance.md`

Checkpoint 1047: Exact endpoint optimization on the full bounded density range

The general local endpoint optimization is extended from density at most one to every fixed bounded density range. Lean keeps the exact printed density and logarithmic exponents, handles densities above one by the splitting parameter J=2, and proves one constant uniform over all densities below the fixed cap. This is needed because support in a fixed dilation cI permits $|F||G|/|I|^2$ above one. It closes that normalization issue in the general Section5 argument without silently imposing a smaller density range or changing its exponents. No independent analytic input is used.

Checked declarations

Module	Named supporting theorems
HighLowCappedBalance.lean	2

- `highLowEndpoint_one_le`
- `unrestricted_highLow_scalar_capped`

<!-- checkpoint-verification -->

Verification

All 1114 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4905-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3297 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1116 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1048_LocalStoppingHypotheses.md`

Checkpoint 1048: Exact hypotheses and small scales for the abstract stopping lemma

The full abstract local hypotheses are encoded as bounds on the actual operator family, its low/high/error pieces, and the real-radius linear local average. The record contains neither a stopping pairing estimate nor an endpoint conclusion. The physical scale is 2^j, the splitting threshold is exactly (10J)^(s/(1-v)), and Lean proves the small-scale estimate with the printed powers of J and log J. This exposes the full mathematical assumptions needed for the general Section5 local argument rather than replacing it by the already-proved prime-square instance. No analytic input or new axiom is introduced.

Checked declarations

Module	Named supporting theorems
LocalStoppingHypotheses.lean	4

- `localSplittingThreshold_power`

- `small_local_scale_growth`
- `local_small_scale_log_absorption`
- `localStopping_small_pointwise`

<!-- checkpoint-verification -->

Verification

All 1115 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4906-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3301 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1117 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1049_LocalStoppingTradeoff.md`

Checkpoint 1049: The finite stopping tradeoff with actual square-summed errors

Lean proves the full low/high/error tradeoff for a family with uniform local hypotheses. The actual stopped output is split pointwise into the original small-scale contribution and the decomposed large-scale contribution. The high and error pieces are controlled by genuine l2 majorants; the error uses the square-summation theorem valid for every A>1/2. The resulting finite pairing bound is uniform over the stopping map and splitting parameter. The localized, single-indicator formulation in the printed lemma still needs its own export: the present record has stronger bounds uniform over indicators, so this supporting result alone is not counted as completing the source lemma.

Checked declarations

Module	Named supporting theorems
LocalStoppingTradeoff.lean	2

- `localStopping_low_pointwise`
- `localStopping_finite_tradeoff`

<!-- checkpoint-verification -->

Verification

All 1116 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4907-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3303 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1118 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1050_LocalizedStoppingTradeoff.md`

Checkpoint 1050: Localized hypotheses with uniform constants in the stopping argument

The tradeoff is now proved with the hypotheses localized exactly as needed in the source: local and low bounds are required only for the current indicator F and only on the testing set G. The common input constant is an explicit parameter, so subsequent endpoint constants are uniform over the sets and the chosen operators. No bound for unrelated indicators or points is assumed. The proof constructs and masks actual high and error outputs and uses genuine l2 majorants. This removes a possible quantifier gap between a convenient uniform-family supporting lemma and the printed local statement. No independent analytic input is introduced.

Checked declarations

Module	Named supporting theorems
LocalizedStoppingTradeoff.lean	2

- `localizedStopping_low_pointwise`
- `localizedStopping_finite_tradeoff`

<!-- checkpoint-verification -->

Verification

All 1117 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4908-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3305 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1119 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1051_AbstractLocalStoppingEndpoint.md`

Checkpoint 1051: The complete localized abstract stopping endpoint with exact exponents

The abstract local stopping endpoint is proved for every v<1, s>0, r>0, gamma>=beta>=0 and A>1/2, with exactly the source density and logarithmic exponents. One output constant is chosen before the interval length, sets, stopping map, and localized operator data; it depends only on the displayed fixed parameters, a common input bound and the fixed density cap. Empty sets, densities above one, every stopping map, and the localized low hypothesis are handled. No current-paper conclusion is an input. The remaining source export is the precise radius/scale/convolution convention bridge, not the endpoint inequality or its optimization.

Checked declarations

Module	Named supporting theorems
AbstractLocalStoppingEndpoint.lean	3

- `localizedTradeoffConstant_bounds`
- `localizedStopping_normalized_tradeoff`
- `abstract_local_stopping_endpoint`

<!-- checkpoint-verification -->

Verification

All 1118 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4909-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3308 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1120 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma

and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1052_LinearLocalRadiusComparison.md`

Checkpoint 1052: All-real-radius comparison for the admissibility conventions

The all-real-radius local average is now compared across arbitrary fixed positive radii. Lean proves that replacing C0 by C costs at most max(1,C/C0), by enlarging each real scale and checking both the window inclusion and the change in normalization. This supplies the missing admissibility-radius bridge in the abstract local stopping lemma. The proof holds for every l1 input, not merely finite indicators, and uses no external analytic theorem.

Checked declarations

Module	Named supporting theorems
LinearLocalRadiusComparison.lean	3

- `linearSourceBallMass_enlarge_radius`
- `linearSourceLocalMass_enlarge_radius`
- `linearSourceLocalMass_change_radius`

<!-- checkpoint-verification -->

Verification

All 1119 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4910-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3311 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1121 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted

estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1053_SourceAbstractLocalStopping.md`

Checkpoint 1053: The physical-scale and radius export of the abstract local lemma

The source conventions for the abstract stopping estimate are now exported: physical dyadic values are recovered exactly from their exponents, arbitrary positive admissibility radii are compared, support size in a fixed dilation is replaced by its explicit uniform density cap, and the parent interval length is the normalization. Lean proves the exact equivalence of the source splitting range J<=M^((1-v)/s)/10 with the threshold used by the proof and identifies M^(1-v) times the local average with the literal M^(-v) window count. The terminal source endpoint has the printed exponents and a constant chosen before all interval, set, stopping and operator data. A final Fourier-symbol-to-operator bridge is being added before awarding the printed source lemma completion.

Checked declarations

Module	Named supporting theorems
SourceAbstractLocalStopping.lean	4

- `localSplittingThreshold_le_iff`
- `linearSourceBallMass_indicator_power`
- `source_local_stopping_density_cap`
- `source_abstract_local_stopping_estimate`

<!-- checkpoint-verification -->

Verification

All 1120 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4911-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3315 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1122 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon

transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1054_SourceLocalStoppingFourier.md`

Checkpoint 1054: Completion of the full abstract local stopping lemma

The full abstract local stopping lemma l:localsparse is now matched to the source and proved under its printed hypotheses. The error symbol is an arbitrary circle function bounded almost everywhere, rather than a smooth or continuous restriction: Lean proves its operator norm from the actual integer Fourier isometry. The terminal theorem chooses one constant before all intervals, sets, physical dyadic stopping maps, operators and symbols, and has exactly the printed density and logarithmic exponents. It covers every A>1/2, localized low assumptions, arbitrary positive admissibility radii, all real radii, empty sets and densities above one. No own-paper conclusion or new independent analytic input is assumed. The reviewed registry now records 11 proved claims, 14 complete conditional implications, eight partial auxiliaries, one quoted external theorem and 20 open later claims. Section9 has not been started.

Checked declarations

Module	Named supporting theorems
SourceLocalStoppingFourier.lean	2

- `fourierSymbol_operator_bound`
- `source_local_stopping_from_fourier_hypotheses`

<!-- checkpoint-verification -->

Verification

All 1121 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4912-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3317 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1123 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propo-

sitions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1055_LinearOneSidedAdmissibility.md`

Checkpoint 1055: Exact one-sided admissibility comparison and a saved clarification

An exact Lean check resolves the one-sided convention attached to admissibility. Each one-sided window count is bounded by the symmetric count, and the symmetric count is at most the sum of the two directions. A singleton F={-1} at x=0 has zero forward local averages at every real radius M>=1, while its symmetric local average is at least one. Thus a single fixed direction cannot be interchanged pointwise merely by multiplying constants. A precise replacement using the maximum of both one-sided local averages is saved separately; it is not silently applied. The full abstract local stopping lemma uses the symmetric condition and remains proved. No new external input is used.

Checked declarations

Module	Named supporting theorems
LinearOneSidedAdmissibility.lean	5

- `linearWindowCount_oneSided_compare`
- `linearSourceLocalMass_of_both_oneSided`
- `rightLinearWindowCount_negative_singleton`
- `symmetricLocalMass_negative_singleton`
- `single_direction_does_not_control_symmetric`

<!-- checkpoint-verification -->

Verification

All 1122 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4913-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3322 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1124 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure,

Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1056_VariableAdmissibilityPairings.md`

Checkpoint 1056: Uniform stopping pairings for every fixed admissibility constant

The ordinary-square stopping pairing and endpoint, and the actual prime large-scale pairing, are extended from the convenient fixed admissibility constant 320 to every fixed B0>=1. Lean checks the dependence of all output constants before the interval, sets and stopping map. The original independent ordinary-square low/high input already allowed arbitrary B0, so no stronger external input is introduced. This removes a genuine source-range restriction in the prime-square local auxiliary rather than merely rescaling the final statement.

Checked declarations

Module	Named supporting theorems
VariableAdmissibilityPairings.lean	3

- `ordinaryStopping_pairing_scalar_with_constant`
- `ordinaryStopping_endpoint_with_constant`
- `primeLargeStopping_pairing_with_constant`

<!-- checkpoint-verification -->

Verification

All 1123 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4914-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3325 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1125 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and

persistence deductions are proved.

`Checkpoint_1057_VariableAdmissibilityPrimeEndpoint.md`

Checkpoint 1057: Prime-square local assembly with an arbitrary fixed admissibility constant

The full small/large-scale prime-square stopping assembly now works for every fixed admissibility constant B0>=1, retaining the sharp leading Orlicz coefficient cstar>4 sqrt(3)/9. All constants are chosen uniformly before the interval, finite sets and spatially varying dyadic scales. It uses the existing independent ordinary-square decomposition and the checked actual Proposition4.19, with no new external theorem input. The next export changes the parent normalization and covers arbitrary fixed support dilations and spatial radii.

Checked declarations

Module	Named supporting theorems
VariableAdmissibilityPrimeEndpoint.lean	1

- `primeStopping_local_endpoint_with_constant`

<!-- checkpoint-verification -->

Verification

All 1124 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4915-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3326 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1126 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1058_ParentEndpointRescaling.md`

Checkpoint 1058: Parent-interval geometry and full-density Orlicz rescaling

Lean proves the fixed-density rescaling needed to express the prime local endpoint using the parent interval length. The Orlicz comparison now covers target densities both below and above one, preserving

the sharp leading coefficient. It also constructs enlarged parent intervals, proves their support inclusions and proves the exact density relation. These are geometric and scalar deductions; no analytic estimate is assumed or modified.

Checked declarations

Module	Named supporting theorems
ParentEndpointRescaling.lean	6

- `enlargedStoppingParent_contains`
- `enlargedStoppingParent_support`
- `paperPhi_half_minimum`
- `paperPhi_half_fixed_rescale_all`
- `primeEndpoint_fixed_density_rescale`
- `enlargedStoppingParent_density`

<!-- checkpoint-verification -->

Verification

All 1125 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4916-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3332 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1127 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1059_QuadraticLocalRadiusBridge.md`

Checkpoint 1059: Exact quadratic-scale admissibility for arbitrary fixed spatial radii

The literal quadratic-scale admissibility condition is identified with the linear local average at the squared lower scale, using every real radius and the exact square-root inverse substitution. Lean then deduces the source radius-one bound from the printed condition with any fixed positive spatial radius C, with the explicit factor max(1,1/C). The finite window counts and normalization M^-2 are exact. This supplies the spatial-radius bridge for the general prime-square local endpoint without restricting the printed radius to a convenient value.

Checked declarations

Module	Named supporting theorems
QuadraticLocalRadiusBridge.lean	3

- `sourceBallMass_eq_linear`
- `sourceLocalMass_eq_linear_square`
- `sourceLocalMass_from_quadratic_radius`

<!-- checkpoint-verification -->

Verification

All 1126 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4917-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3335 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1128 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1060_GeneralParentPrimeEndpoint.md`

Checkpoint 1060: Arbitrary support dilations and the exact parent-normalized prime endpoint

The prime-square local endpoint now has the literal parent-length normalization and allows every fixed support dilation. The proof embeds the original parent in a fixed larger interval, converts its admissibility constant exactly, and compares the resulting densities and Orlicz factors without changing the leading coefficient. It retains a larger reference test set, so restricting to the nonzero stopping scales does not assume monotonicity of an Orlicz bump. The operator remains the actual annular von Mangoldt square average. No new independent input is introduced.

Checked declarations

Module	Named supporting theorems
GeneralParentPrimeEndpoint.lean	1

- `primeStopping_general_parent_endpoint`

<!-- checkpoint-verification -->

Verification

All 1127 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4918-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3336 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1129 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1061_PhysicalDyadicPrimeStopping.md`

Checkpoint 1061: The physical dyadic scales including the zero contribution at scale one

Lean verifies the exact reindexing from the physical dyadic stopping values used in the manuscript to the implementation's scales 2^j*2. Scale1 contributes exactly zero for the actual von Mangoldt kernel. The full physical pairing is therefore exactly the pairing over the test subset where the stopping value is at least2; no norm inequality or density monotonicity is substituted for this identity. Empty input sets are covered as well. This closes the final scale-convention issue for the source prime-square local endpoint.

Checked declarations

Module	Named supporting theorems
PhysicalDyadicPrimeStopping.lean	4

- `physicalAnnularPrimeValue_one`
- `physicalAnnularPrimeValue_empty`
- `physicalDyadic_above_one`
- `physicalDyadic_prime_pairing`

<!-- checkpoint-verification -->

Verification

All 1128 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4919-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 3340 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1130 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1062_SourcePrimeLocalEndpoint.md`

Checkpoint 1062: Completion of the full source prime-square local endpoint

The full source prime-square local endpoint l:primeendpointlocal is now complete as a conditional implication. The terminal theorem derives Proposition4.19 from the already-listed independent inputs and proves the actual annular von Mangoldt square pairing for arbitrary fixed real support dilation, positive spatial radius and admissibility constant, every real local radius, physical dyadic scales including1, empty sets, and the exact parent-length normalization. It preserves the sharp leading coefficient: every cstar>4sqrt(3)/9. No own-paper conclusion is a new independent input. The reviewed paper registry now records 11 proved claims, 15 complete conditional implications, seven partial auxiliaries, one quoted external theorem and20 open later claims. Section9 has not been started.

Checked declarations

Module	Named supporting theorems
SourcePrimeLocalEndpoint.lean	3

- `source_prime_local_endpoint`
- `source_prime_local_endpoint_real_dilation`
- `source_prime_local_endpoint_from_independent_inputs`

<!-- checkpoint-verification -->

Verification

All 1129 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4920-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3343 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1131 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1063_RealIWParameters.md`

Checkpoint 1063: Exact real-Q Ionescu-Wainger parameters and monotonicity

The exact quantitative Ionescu-Wainger denominator, width and maximal-cost formulas are now defined for real Q, with proved equality to the existing definitions at every integer Q. Lean proves denominator and cost monotonicity, width antitonicity and positivity, and preservation of the actual goodness condition when the width decreases. These checks support the real-Q extension without rounding the displayed width or changing the constants silently. No external analytic input is used in these parameter deductions.

Checked declarations

Module	Named supporting theorems
RealIWParameters.lean	9

- `realIWDenominatorBound_nat`
- `realIWWidth_nat`
- `realIWMaximalCost_nat`
- `realIWDenominatorBound_one_le`
- `realIWDenominatorBound_mono`
- `realIWWidth_pos`
- `realIWWidth_antitone`
- `iwGood_of_smaller_width`
- `realIWMaximalCost_mono`

<!-- checkpoint-verification -->

Verification

All 1130 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4921-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3352 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1132 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1064_RealIWConstruction.md`

Checkpoint 1064: The real-Q IW construction and exact-width smooth applications

The actual Ionescu-Wainger geometric construction is now proved for every real Q>=2^k. The proof uses floor(Q) only for the integer denominator range and retains the exact real-Q formulas for B and epsilon. Lean verifies P<=3^Q, coverage, denominator and component bounds, and actual goodness at the narrower width. It also proves the existing smooth multiplier and scalar Schwartz inverse-sampling applications at that exact width throughout p_r<=p<=2r. The broader rough-symbol/full-class/two-direction/vector-valued sampling clauses and the exact-real-width full low maximal assembly remain explicitly partial. No new deep input is introduced and no stronger source status is awarded without its proof.

Checked declarations

Module	Named supporting theorems
RealIWConstruction.lean	5

- `real_iw_goodness_from_floor`
- `real_iw_geometry_from_floor`
- `quantitative_iw_real_construction_good`
- `quantitative_real_iw_smooth_multiplier`
- `quantitative_real_iw_schwartz_sampling`

<!-- checkpoint-verification -->

Verification

All 1131 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4922-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3357 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1133 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sec-

tions3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Goldbach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1065_AuxiliaryScopeCertificates.md`

Checkpoint 1065: Consolidated pre-Goldbach source review with exact scope certificates

The pre-Goldbach source review is consolidated with exact Lean scope certificates. A nonnegative density profile disproves the blanket fixed-dilation comparison for arbitrary Psi in the sparse-recursion proof; the corrected dyadic recursion can retain the displayed dilated form, but still needs its proof. Two scalar certificates show why a logarithm-squared estimate cannot be relabeled as a first-power estimate by changing a constant. They do not assert a counterexample to the special sharper annular theorem. The source review records five newly completed numbered auxiliaries, seven precise remaining claims, the exact real-Q IW extension, and the separately proposed one-sided admissibility clarification. Section9 has not been started. No new independent analytic input is introduced.

Checked declarations

Module	Named supporting theorems
AuxiliaryScopeCertificates.lean	4

- `quadratic_log_loss_not_linear`
- `cubic_log_loss_not_quadratic`
- `densityProfileSpike_nonneg`
- `densityProfileSpike_no_dilation_comparison`

<!-- checkpoint-verification -->

Verification

All 1132 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4923-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3361 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1134 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The user confirmed the three Section8 edits; the selected project manuscript contains their exact agreed reconstruction, with all numbered statements and labels unchanged and original sources preserved. Sections3/4 and Section8 retain their complete reviewed conditional scopes. The full approximation lemma and low proposition of Section6 are now exported with exact real-Q, all-input, all-real-radius and exponent quantifiers. Remaining Sections5–7 auxiliary claims are being reviewed and completed before Gold-

bach work. Deep general inputs remain explicit and unproved. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1066_LinearGoodPointMass.md`

Checkpoint 1066: Accepted source conventions and general linear-scale stopping geometry

The accepted admissibility, dyadic sparse-scale, retained-dilation and factor-four passages are reconstructed in a new selected source. All labels and all other numbered statements are preserved; no new full manuscript upload is claimed. The corrected admissibility definition and both comparison directions are now fully formalized, including the exact factor two. Lean also proves the linear-scale good-point estimate and all-real-radius admissibility of the remaining stopping scales, for arbitrary fixed kernel radius C, directly from the checked elementary interval-packing construction. These are the geometric inputs to the general positive-kernel sparse recursion, not an assumed sparse conclusion. Seven numbered auxiliaries still await their complete source exports. No new analytic input or axiom is introduced.

Checked declarations

Module	Named supporting theorems
LinearGoodPointMass.lean	6

- `linearOneSided_bounds_of_symmetric`
- `linearSymmetric_bound_of_both_oneSided`
- `linearBallMass_le_interval_count`
- `goodPoint_linearSourceLocalMass`
- `linearStopping_remaining_localMass`
- `linearStopping_remaining_radius`

<!-- checkpoint-verification -->

Verification

All 1133 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4924-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3367 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1135 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic

Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1067_GenericDyadicKernels.md`

Checkpoint 1067: General positive dyadic kernels and arbitrary fixed support dilations

An arbitrary positive dyadic convolution family is now represented by its actual kernels with only positivity and compact spatial support as fields. Lean constructs every fixed odd interval dilation and proves its nesting and kernel localization, including repeated restriction from a parent to a stopping child. The dilation may be chosen for any fixed real kernel radius; no prime-square structure or analytic decomposition is assumed. These source-level support deductions are used next in the general sparse recursion. No new independent input is introduced.

Checked declarations

Module	Named supporting theorems
GenericDyadicKernels.lean	6

- `integerInterval_subset_oddDilate`
- `integerInterval_oddDilate_subset`
- `integerBall_subset_oddDilate`
- `positiveDyadicValue_positive`
- `positiveDyadicValue_localize`
- `positiveDyadicValue_restrict_twice`

<!-- checkpoint-verification -->

Verification

All 1134 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4925-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3373 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1136 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter

optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1068_GenericDyadicStopping.md`

Checkpoint 1068: The remaining-scale recursion for arbitrary positive convolution kernels

The remaining-scale recursion is now proved for arbitrary positive dyadic convolution kernels and every fixed odd support dilation covering the kernel radius. All physical dyadic scales, including1, are included. Lean proves the finite maximum, exact child localization, all-real-radius admissibility of remaining scales, and the full one-generation pointwise and pairing inequalities. The construction is independent of any density profile and assumes no monotonicity or dilation comparison for Psi. No current-paper maximal or sparse estimate is a field or new analytic input.

Checked declarations

Module	Named supporting theorems
GenericDyadicStopping.lean	6

- `mem_linearDyadicIndices`
- `genericDyadicMaximum_attained`
- `genericRemainingIndices_subset`
- `genericRemainingScale_admissible`
- `genericRemainingScale_pointwise`
- `genericRemainingScale_pairing`

<!-- checkpoint-verification -->

Verification

All 1135 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4926-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3379 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1137 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-

power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1069_GenericSparseLinearization.md`

Checkpoint 1069: Whole-set admissible linearization without a monotone density profile

The remaining maximum is linearized on the whole testing set, without discarding points or assuming that Psi is monotone. Where the remaining index set is empty, the stopping map uses the largest dyadic scale below the interval length; Lean proves that this default scale is admissible. Positivity then gives the required pairing inequality on the entire testing set. The local input is only the source stopping-time estimate, with its sufficiently large fixed admissibility constant explicit. No maximal or sparse conclusion is introduced as an input.

Checked declarations

Module	Named supporting theorems
GenericSparseLinearization.lean	3

- `genericDefaultIndex_bounds`
- `genericDefaultIndex_admissible`
- `genericRemainingScale_linearized`

<!-- checkpoint-verification -->

Verification

All 1136 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4927-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3382 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1138 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence

and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

Checkpoint_1070_GenericFiniteSparse.md

Checkpoint 1070: General finite sparse recursion for an arbitrary density profile

The general finite sparse recursion is proved from the local stopping-time hypothesis for arbitrary positive compact-support dyadic convolution families. The output has actual intervals and disjoint major subsets of at least three quarters their lengths. The exact density-profile cost uses F intersected with the fixed dilation and G intersected with the parent, divided by the parent length. Psi is arbitrary: no positivity, monotonicity or dilation-comparison property is added. Every recursive call shortens the interval, so termination and the cost sum are checked. Removing the outer scale truncation remains a separate proof step.

Checked declarations

Module	Named supporting theorems
GenericFiniteSparse.lean	3

- `genericPositiveSparseStep`
- `genericPositiveFiniteSparse`
- `genericFiniteSparse_from_local_profile`

<!-- checkpoint-verification -->

Verification

All 1137 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4928-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3385 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1139 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

Checkpoint_1071_GenericKernelTesting.md

Checkpoint 1071: Finiteness of the full maximum deduced from the local testing hypothesis

The local hypothesis itself now supplies a uniform bound for every finite-indicator output at every dyadic scale and spatial point. Lean proves translation invariance of the actual convolutions, compares each output with a dense interval test, and applies the local estimate to that dense test at its full parent scale. The resulting bound is K Psi(2m+1,1), independent of the indicator and scale. Thus finiteness of the full maximum is derived rather than assumed. No positivity or monotonicity condition on Psi is introduced.

Checked declarations

Module	Named supporting theorems
GenericKernelTesting.lean	4

- `positiveDyadicValue_translate`
- `positiveDyadicValue_le_interval_test`
- `genericConstantStop_admissible`
- `genericLocalHypothesis_uniform_indicator_bound`

<!-- checkpoint-verification -->

Verification

All 1138 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4929-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3389 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1140 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1072_GenericFullMaximum.md`

Checkpoint 1072: The untruncated dyadic maximum and a finite testing reduction

The full untruncated dyadic maximum is now a bounded real supremum, with boundedness derived from the local testing estimate. For every finite testing set, Lean selects finitely many dyadic scales whose maximum dominates at least half the full pairing. This handles suprema that need not be attained and removes the need for an unproved compactness or monotone-limit argument for sparse families. The finite selection is proved for the actual kernel outputs, and Psi remains arbitrary. No new independent input is introduced.

Checked declarations

Module	Named supporting theorems
GenericFullMaximum.lean	4

- `genericFullDyadicMaximum_bdd`
- `genericFullDyadicMaximum_nonneg`
- `genericFullDyadicMaximum_half_attained`
- `genericFullDyadicMaximum_finite_test`

<!-- checkpoint-verification -->

Verification

All 1139 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4930-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3393 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1141 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1073_GenericFullSparse.md`

Checkpoint 1073: The full general dyadic sparse implication with an arbitrary density profile

The general untruncated dyadic sparse implication is proved for every positive compact-support convolution family and arbitrary density profile. Lean constructs a finite root containing the input, testing points and the finitely selected scales, applies the terminating recursion, and obtains actual quarter-sparse major subsets. The constant is exactly twice the local testing constant and is uniform over all finite input and testing sets. No monotone Psi, scale decay, attained infinite maximum, limiting sparse family, or current-paper sparse conclusion is assumed. The final wrapper still checks the source convention that stopping maps are admissible on the entire parent interval.

Checked declarations

Module	Named supporting theorems
GenericFullSparse.lean	5

- genericFiniteTestRoot_contains
- genericFiniteTestRoot_scales
- genericFiniteTest_le_root
- genericPositiveFullSparse
- source_dyadic_sparse_recursion

<!-- checkpoint-verification -->

Verification

All 1140 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4931-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3398 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1142 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

Checkpoint_1074_GenericSourceSparse.md

Checkpoint 1074: Complete general sparse deduction with full-domain admissibility and explicit constants

The full generic dyadic sparse implication is proved from the normal local stopping hypothesis, with admissibility required on the entire parent interval. An explicit extension by a large default scale proves the needed local restriction without enlarging the hypothesis or assuming a monotone Psi. The full maximum is finite by a proved dense-testing argument, and an actual quarter-sparse finite family controls it with constant2K. One source-notation refinement remains: the window radius and sufficiently large RHS admissibility constant must be independent; A>=320max(1,C) is a sufficient explicit choice. Its exact LaTeX is saved separately, and the source claim stays conservatively partial pending that clarification. No new deep input is introduced.

Checked declarations

Module	Named supporting theorems
GenericSourceSparse.lean	2

- sourceFullLocal_implies_generic
- source_dyadic_sparse_recursion_full_domain

<!-- checkpoint-verification -->

Verification

All 1141 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4932-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3400 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1143 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1075_AnalyticIWLow.md`

Checkpoint 1075: Low maximal proof under its precise analytic data requirements

The full IW low maximal proof is now checked for an analytic data structure requiring only positive pairwise coprime components, a common period, its size bound, and the component bound. Rational coverage is not a hypothesis of this analytic theorem. Twenty-two existing deductions have been generalized and freshly compiled, from goodness and multiplier/sampling estimates through the narrow, transition, remainder, finite-input, and all-input maximal estimates. This introduces no external input. The original arithmetic construction and its coverage theorem remain available separately; this separation will allow the exact real-Q cutoff to be retained rather than rounded. The full paper proposition is not claimed complete at this intermediate step.

Checked declarations

Module	Named supporting theorems
AnalyticIWLow.lean	22

- `analytic_quantitative_iw_goodness`
- `analytic_quantitative_iw_multiplier`
- `analytic_quantitative_iw_sampling`
- `analytic_iw_finite_lift_size`
- `analytic_iwFiniteCutoffLift_size`
- `analytic_iw_narrow_finite_maximum`
- `analytic_iw_narrow_integer_majorant`
- `analytic_iw_annular_remainder_multiplier`

- analytic_iw_large_remainder_majorant
- analytic_iw_annular_single_multiplier
- analytic_iw_signed_annular_multiplier
- analytic_iw_annular_signed_increments
- analytic_iw_short_scale_increment_maximum
- analytic_iw_short_scale_maximum
- analytic_iw_period_log_bound
- analytic_iw_narrow_log_inverse_budget
- analytic_iw_transition_index_linear_budget
- analytic_iw_transition_log_cost_budget
- analytic_iw_short_transition_cost_bound
- analytic_iw_short_transition_maximum
- analytic_iw_finite_input_low_majorant
- analytic_iw_full_low_maximal_bound

<!-- checkpoint-verification -->

Verification

All 1142 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4933-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3422 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1144 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1076_RealIWBase.md`

Checkpoint 1076: Exact real-parameter width through a proved change of universal constant

Monotonicity in the universal IW constant is proved for the quantitative multiplier constant and for the two independent multiplier/sampling inputs. For real Q>=1 and k>=4, an adjusted constant between C and C^2 converts the natural parameter ceil(Q) to exactly the real-Q denominator bound with universal constant C^2. The denominator bound and the cutoff width agree identically, not merely up to an inequality. The exponent and ceiling estimates are checked, and no additional analytic assumption has been introduced.

Checked declarations

Module	Named supporting theorems
RealIWBase.lean	9

- `iwMultiplierConstant_mono_base`
- `rawIWMultiplierInput_mono_base`
- `rawMajorArcSamplingInput_mono_base`
- `realIWCeil_ratio_bounds`
- `realIWAdjusted_exponent_bounds`
- `realIWAdjustedBase_bounds`
- `realIWAdjustedBase_power`
- `realIWAdjustedBase_denominator`
- `realIWAdjustedBase_width`

<!-- checkpoint-verification -->

Verification

All 1143 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4934-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3431 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1145 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1077_RealIWAnalyticData.md`

Checkpoint 1077: Arithmetic data and cost at the exact real-Q width

The original floor-Q arithmetic construction is now converted to the weaker analytic data at ceil(Q), using the adjusted universal constant. The rational set and the period are unchanged. Its component bound, period bound, and exact real-Q width are checked. The complete natural-Q low cost at the adjusted base is bounded by twice the literal real-Q cost at base C^2. The proof keeps the exact logarithm and checks all signs, ceiling estimates, and the universal-constant comparison. Rational coverage through the original real Q is still supplied by the floor construction.

Checked declarations

Module	Named supporting theorems
RealIWAnalyticData.lean	2

- `realIWAdjusted_cost`
- `realIWAnalyticData_width`

<!-- checkpoint-verification -->

Verification

All 1144 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4935-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3433 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1146 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1078_RealIWLowMaximum.md`

Checkpoint 1078: Complete all-input low maximal bound at the literal real-Q width

The exact real-Q dyadic low family now has its full countable maximal bound on the L2/Lp intersection and on every integer lp input at p=p_r. The arithmetic set and period are the original floor-Q construction; the width is exactly the displayed real-Q width, with universal parameter constant C^2. The bound is twice the original universal low constant times the literal real-Q cost. A proved monotonicity step transports the independent standard multiplier and sampling inputs to the adjusted constant. The all-input extension and canonical inclusion into L2 are reused with all applicability checked. No new deep hypothesis or current-paper conclusion is assumed.

Checked declarations

Module	Named supporting theorems
RealIWLowMaximum.lean	5

- `realIWCutoffScale_four_le`
- `realIWAdjusted_cutoff`
- `allInputLpMaximalBound_mono_constant`

- `real_iw_full_low_maximal_bound`
- `real_iw_low_all_integer_lp`

<!-- checkpoint-verification -->

Verification

All 1145 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4936-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3438 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1147 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1079_RealIWSourceLow.md`

Checkpoint 1079: Complete real-Q source low proposition with its arithmetic construction

The real-Q geometric construction and the exact actual all-input low maximal estimate are now assembled in one theorem. It gives period<=3^Q, coverage of all rational denominators<=Q, the literal denominator bound and goodness, and the countable low estimate uniformly for all allowed measures and beta. The fixed universal parameter constant is enlarged from C to C^2; monotonicity of the independent Tao construction is proved as well. The paper registry now records p:orlicz-low as a complete conditional implication. No own low/maximal conclusion is assumed, and no new external theorem or Lean axiom has been introduced. The remaining broader IW and high-side statements are not awarded credit by this step.

Checked declarations

Module	Named supporting theorems
RealIWSourceLow.lean	2

- `rawTaoArcConstructionInput_mono_base`
- `real_iw_source_construction_and_low`

<!-- checkpoint-verification -->

Verification

All 1146 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4937-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3440 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1148 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1080_RealIWHighData.md`

Checkpoint 1080: Exact real-Q high-side data and decay-rate conversion

A second exact parameter conversion now transports the natural floor-Q high comparisons to the literal real-Q width. Its base constant is at least the original constant, and its natural denominator bound and cutoff equal the real ones exactly. The original arithmetic set, coverage and period are unchanged. The real floor bounds and the conversion of the floor-Q negative-quarter-power rate to at most twice the real-Q rate are checked. This is a geometric/algebraic conversion, not an additional high-side analytic assumption.

Checked declarations

Module	Named supporting theorems
RealIWHighData.lean	6

- `realIWFloor_bounds`
- `realIWUpperBase_lower`
- `realIWUpperBase_power`
- `realIWUpperBase_denominator`
- `realIWUpperBase_cutoff`
- `realIWFloor_quarter_power`

<!-- checkpoint-verification -->

Verification

All 1147 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4938-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 3446 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1149 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1081_RealIWHighLowAssembly.md`

Checkpoint 1081: Real-Q low and high estimates for one shared arithmetic construction

The complete high estimates have been strengthened and checked for every valid arithmetic construction, so the prime and ordinary-square comparisons can use exactly the same set and period. Their common real-Q assembly is now proved at the literal real-Q width and exponential starting scale, with the required negative-quarter-power rate for the sum of both high norms. The original general multifrequency, Page, distribution and quadratic-Weyl inputs remain explicit. Independently, the actual real-Q prime low maximum (including its selected exceptional character) and ordinary-square low maximum are now proved for all inputs, by a checked reduction to the already proved measure low theorem. Both use the same unchanged arithmetic construction and exact width. These deductions introduce no new deep input.

Checked declarations

Module	Named supporting theorems
RealIWHighLowAssembly.lean	7

- `iw_prime_high_maximum_for_data`
- `iw_full_square_high_source_scale_for_data`
- `real_iw_high_comparison`
- `realIWMaximalCost_nonneg`
- `iwPrime_low_from_measures`
- `real_iw_prime_low_maximal_bound`
- `real_iw_full_square_low_maximal_bound`

<!-- checkpoint-verification -->

Verification

All 1148 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4939-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 3453 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1150 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1082_RealIWSourceHighLow.md`

Checkpoint 1082: Complete real-Q source high-low proposition for both actual families

The complete printed real-Q high-low proposition is now assembled for actual prime and ordinary-square operators. One arithmetic construction retains the exact real-Q period bound, rational coverage, denominator bound and goodness. One selected Page datum is used for the prime low and high families. Both decomposition identities, both full all-input near-one low maxima, and the summed negative-quarter-power L2 high maximum above the exponential starting scale are checked. The universal parameter constant is enlarged once from C to C^2, and both low and high use the literal real-Q cutoff without rounding. The paper registry records p:orlicz-hilo as a complete conditional implication under the unchanged independent standard inputs. Five auxiliary source claims remain: four substantive incomplete claims and the completed sparse core awaiting its independently stated admissibility constant. Section9 remains untouched.

Checked declarations

Module	Named supporting theorems
RealIWSourceHighLow.lean	1

- `real_iw_source_high_low`

<!-- checkpoint-verification -->

Verification

All 1149 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4940-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3454 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1151 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1083_RealIWMaximumNorms.md`

Checkpoint 1083: Literal real-Q maximal norms and reconciled seven-claim review

The literal countable maximum is now realized as an actual integer lp element for every all-input maximal bound at p<=2. A checked two-family norm theorem gives the sum of the real-Q prime and ordinary-square low maximum norms, with constant6 times the fixed complete-low constant times the literal real-Q cost. The final seven-claim review distinguishes the two newly completed conditional propositions, the complete sparse core awaiting its constant notation, and the four substantive partial claims. Earlier accepted source edits remain hashed and preserved; the new constant clarification is proposed separately. The remaining-work narrative is corrected to reflect the now complete sparse proof. No new deep input, source edit, or Section9 work is introduced.

Checked declarations

Module	Named supporting theorems
RealIWMaximumNorms.lean	3

- `allInputLpMaximalBound_integerLp_maximum`
- `allInputLp_two_maximum_norms`
- `real_iw_two_low_maximum_norms`

<!-- checkpoint-verification -->

Verification

All 1150 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4941-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3457 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1152 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The accepted Section8, admissibility, dyadic sparse-scale and retained-dilation edits are reconstructed in the selected manuscript, with original source copies and every label preserved. Only the accepted definition and dyadic auxiliary statement change inside numbered environments; no new full user upload is claimed. The seven remaining auxiliaries are being proved from their normal source hypotheses and

the explicitly recorded independent standard inputs. No own-paper conclusion is a new deep assumption. Section9 has not been started. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1084_AcceptedSparseRecursion.md`

Checkpoint 1084: Accepted current manuscript and complete generic sparse recursion

The complete current TeX upload is preserved exactly, and its accepted independent admissibility coefficient is verified. The generic dyadic sparse theorem now matches the source definition and is exported with its uniformly chosen admissibility threshold. It is recorded proved under its normal local hypothesis; no external sparse estimate is assumed. The selected working copy restores the previously approved factor-four dyadic sparse application and the preferred upper-cutoff wording, with the unchanged upload and exact replacement preserved. Four substantive auxiliaries remain partial. The user has newly authorized Section9 mapping and reductions.

Checked declarations

Module	Named supporting theorems
AcceptedSparseRecursion.lean	2

- `accepted_sparse_admissibility_threshold`
- `accepted_general_dyadic_sparse_recursion`

<!-- checkpoint-verification -->

Verification

All 1151 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4942-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3459 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1153 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Four substantive auxiliaries remain partial. Section9 mapping and reductions are now authorized; later analytic and sieve inputs remain explicit. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumera-

tion proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1085_RationalFrequencyPeriod.md`

Checkpoint 1085: Rational common periods and the complete Section9 dependency map

Every numbered Section9 statement is now mapped: nine claims, their actual proof locations, forward dependencies on Sections10–11, the two CRT constructions and the final exceptional-set reduction. Mapping gives no proof credit and later own-paper conclusions are not new external inputs. Independently, an explicit common period for every denominator block is constructed by a factorial, and its rational-frequency logarithmic cost is proved at most 11(s+1). This is the arithmetic step needed for a sharper rational-frequency route; it does not assume the printed annular endpoint. The map flags the p2=z endpoint mismatch for a separate exact check.

Checked declarations

Module	Named supporting theorems
RationalFrequencyPeriod.lean	6

- `rationalFrequencyLoss_nonneg`
- `rationalFrequencyLoss_subset`
- `rationalBand_common_period`
- `rationalBand_log_card`
- `rationalBand_log_period`
- `rationalBand_loss_linear`

<!-- checkpoint-verification -->

Verification

All 1152 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4943-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3465 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1154 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Four substantive auxiliaries remain partial. Section9 mapping and reductions are now authorized; later analytic and sieve inputs remain explicit. Theorem7.2 consumes the precise independent scalar quadratic Weyl approximation (including its unproved selection/reduction/phase bridge), general Tao construction, IW multiplier, major-arc/Shannon sampling, continuous/interpolation, signed-sequence and smooth multifrequency inputs, Page uniqueness and compatible raw prime-distribution data. All own ordinary and prime circle approximations, quantitative low/high comparisons, restricted estimates, parameter optimization and final source assembly are proved. Corollary7.6 adds no independent input. Theorem1.5(ii) additionally consumes only general strong Calderon transference, with actual prime enumeration proved. Corollary7.46 additionally uses general strong/modular Calderon transference and bounded-class polynomial convergence, including its still-unformalized published-theorem/cutoff/PNT/Abel/prime-power normalization bridges. Own unbounded Orlicz/L1 density convergence and identification of both limits are proved. These Prop-valued independent inputs are unproved theorem parameters, not new

Lean mathematical axioms. Earlier Theorems1.1/1.3 and Propositions4.19/6.21 retain their separate recorded inputs; older ordinary global/local stopping hypotheses are not consumed by this new ordinary IW route. Theorem1.5(i) uses only the precise general split-polynomial Weil, fixed-progression prime-frequency PNT, original LaVictoire weak-maximal-failure, Sawyer and Conze inputs. Their existence and exact published-to-Lean convention bridges remain unproved. The optional quoted general GK input has a precise interface and checked joint Poisson application, but is not consumed by badness. All current-paper arithmetic, moment, limit, frequency and persistence deductions are proved.

`Checkpoint_1086_RationalSmoothWeighted.md`

Checkpoint 1086: Rational-frequency annular deduction under the precise independent smooth input

The independent rational smooth multifrequency and circle-transference boundary is now stated precisely for arbitrary center sets and common periods. It contains no own annular or prime-square endpoint. From that explicit general input, the weighted smooth projection and actual annular maximum are freshly proved at large and every earlier starting scale for s>=2. The actual comparison error remains internally proved. The quoted rational-frequency theory and its transfer remain unformalized standard inputs; no Lean axiom is introduced and no full printed source claim is awarded completion at this intermediate stage.

Checked declarations

Module	Named supporting theorems
RationalSmoothWeighted.lean	4

- `rationalSmoothCircle_countable`
- `rational_weightedSource_projection`
- `rational_largeStart_weightedAnnular_maximum`
- `rational_weightedAnnular_allStartingScales`

<!-- checkpoint-verification -->

Verification

All 1153 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4944-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3469 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1155 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Four substantive auxiliaries remain partial. Section9 mapping and reductions are now authorized; later analytic and sieve inputs remain explicit. The rational smooth multifrequency theorem together with circle transference is an explicit independent compound hypothesis; its existence is unproved. Its predicate contains no own-paper annular or truncated maximal conclusion. See docs/RATIONAL_MULTIFREQUENCY_BOUNDARY.md.

`Checkpoint_1087_RationalAnnularSourceMaximum.md`

Checkpoint 1087: Complete source annular multifrequency bound with first-power logarithmic loss

The full printed annular multifrequency lemma is now proved as a complete conditional implication with its first-power (1+s) bound, every dyadic source scale, all s>=0, scale1 and the small denominator cases. The precise independent rational smooth/transference input is explicit; the arithmetic period budget, own weighted projection, actual annular error and starting-scale arguments are checked. A matching rational-period proof passage is saved separately for the manuscript. The source lemma is not deduced by

dropping a logarithm from the older general smooth input. Its denominator-truncated corollary remains separately incomplete until all its own clauses are discharged.

Checked declarations

Module	Named supporting theorems
RationalAnnularSourceMaximum.lean	6

- `rationalAnnularConstant_pos`
- `rational_annular_allStarting`
- `circleAnnularSource_initial_norm`
- `weightedAnnularSource_initial_operator_norm`
- `sourceRationalAnnularConstant_pos`
- `source_rational_annular_maximum`

<!-- checkpoint-verification -->

Verification

All 1154 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4945-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3475 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1156 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Four substantive auxiliaries remain partial. Section9 mapping and reductions are now authorized; later analytic and sieve inputs remain explicit. The rational smooth multifrequency theorem together with circle transference is an explicit independent compound hypothesis; its existence is unproved. Its predicate contains no own-paper annular or truncated maximal conclusion. See docs/RATIONAL_MULTIFREQUENCY_BOUNDARY.md.

`Checkpoint_1088_RationalAnnularTruncationMaximum.md`

Checkpoint 1088: Simultaneous real-denominator and time maximum with square loss

A single nonnegative integer L2 majorant now controls every dyadic time and every nonnegative real denominator cutoff simultaneously, with the printed square (1+s)^2 loss. The proof uses actual localized multiplier blocks and disjoint Bessel energies for large s, all 33 possible bounded denominator cutoffs for the small indices, and the three finite cutoffs needed at scale1. It invokes exactly the explicit rational smooth/transference and general smooth inputs already recorded at checkpoints1086–1087. No fixed-multiplier clause is credited yet; the full corollary remains partial until that clause is proved.

Checked declarations

Module	Named supporting theorems
RationalAnnularTruncationMaximum.lean	7

- `rationalAnnularTruncationConstant_pos`
- `rational_weightedAnnular_truncation_maximum`
- `rationalAllTruncationConstant_pos`
- `rational_smallIndex_all_thresholds`
- `rational_annular_all_thresholds`
- `sourceRationalTruncationConstant_pos`
- `source_rational_annular_truncation_maximum`

<!-- checkpoint-verification -->

Verification

All 1155 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4946-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3482 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1157 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Four substantive auxiliaries remain partial. Section9 mapping and reductions are now authorized; later analytic and sieve inputs remain explicit. The rational smooth multifrequency theorem together with circle transference is an explicit independent compound hypothesis; its existence is unproved. Its predicate contains no own-paper annular or truncated maximal conclusion. See docs/RATIONAL_MULTIFREQUENCY_BOUNDARY.md.

`Checkpoint_1089_AEBoundedCircleMultipliers.md`

Checkpoint 1089: Measurable bounded circle multipliers and exact localized composition

Constructed actual bounded L2 multiplier operators for arbitrary almost everywhere strongly measurable, essentially bounded circle symbols. The checked Fourier representation, norm bounds and localization/composition identity work on all integer L2 inputs. This removes the continuity restriction before proving the fixed bounded multiplier clause of the denominator corollary. No analytic theorem is assumed for this construction, and no paper claim is newly credited at this supporting stage.

Checked declarations

Module	Named supporting theorems
AEBoundedCircleMultipliers.lean	11

- `aeCircleProduct_memLp`
- `aeCircleMultiplyValue_ae`
- `aeCircleMultiplyValue_norm_le`
- `aeCircleMultiply_ae`
- `aeCircleMultiply_norm_le`
- `aeCircleMultiply_opNorm_le`
- `aeIntegerMultiplier_apply`
- `integerFourier_aeMultiplier`
- `aeIntegerMultiplier_norm_le`
- `aeIntegerMultiplier_opNorm_le`
- `aeIntegerMultiplier_localized_comp`

<!-- checkpoint-verification -->

Verification

All 1156 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4947-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3493 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1158 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Four substantive auxiliaries remain partial. Section9 mapping and reductions are now authorized; later analytic and sieve inputs remain explicit. The rational smooth multifrequency theorem together with circle transference is an explicit independent compound hypothesis; its existence

is unproved. Its predicate contains no own-paper annular or truncated maximal conclusion. See docs/RATIONAL_MULTIFREQUENCY_BOUNDARY.md.

`Checkpoint_1090_DisjointAEMultiplierAssembly.md`

Checkpoint 1090: Disjoint measurable multiplier assembly independent of scale and cutoff

For a finite separated circle family, bounded measurable symbols supported in the respective disjoint cutoffs combine to one measurable symbol of essential supremum at most1. The exact localized operator composition is proved for arbitrary continuous outer symbols supported in those cutoffs. The merged symbol is fixed independently of the scale and of the denominator cutoff. This is an internal checked consequence of disjoint supports, with no further deep input.

Checked declarations

Module	Named supporting theorems
DisjointAEMultiplierAssembly.lean	3

- `finiteAEMultiplier_sum_measurable`
- `disjointAEMultiplier_sum_bound`
- `disjointAEMultiplierSymbol_local_comp`

<!-- checkpoint-verification -->

Verification

All 1157 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4948-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3496 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1159 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Four substantive auxiliaries remain partial. Section9 mapping and reductions are now authorized; later analytic and sieve inputs remain explicit. The rational smooth multifrequency theorem together with circle transference is an explicit independent compound hypothesis; its existence is unproved. Its predicate contains no own-paper annular or truncated maximal conclusion. See docs/RATIONAL_MULTIFREQUENCY_BOUNDARY.md.

`Checkpoint_1091_FixedAnnularTruncationMaximum.md`

Checkpoint 1091: Complete denominator corollary with fixed bounded measurable multipliers

Completed the entire printed denominator-truncated annular corollary, including the fixed arbitrary bounded measurable multiplier clause, under the precise independent smooth hypotheses. For separated cutoffs one merged symbol of essential supremum at most1 factors every time and real cutoff through the bare actual family. Small indices are controlled by at most25 individual-frequency majorants, retaining scale1. The statement ledger now has 12 proved claims,19 complete conditional implications,2 partial,1 quoted external and20 open; supporting theorem counts are not paper-claim counts. A complete proof passage is saved separately, not silently substituted into the selected source.

Checked declarations

Module	Named supporting theorems
FixedAnnularTruncationMaximum.lean	4

- `fixedAnnularMultiplier_disjoint_factor`
- `fixed_annular_finite_truncation_maximum`

- `fixedRationalTruncationConstant_pos`
- `source_fixed_annular_truncation_maximum`

<!-- checkpoint-verification -->

Verification

All 1158 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4949-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3500 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1160 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Four substantive auxiliaries remain partial. Section9 mapping and reductions are now authorized; later analytic and sieve inputs remain explicit. The rational smooth multifrequency theorem together with circle transference is an explicit independent compound hypothesis; its existence is unproved. Its predicate contains no own-paper annular or truncated maximal conclusion. See docs/RATIONAL_MULTIFREQUENCY_BOUNDARY.md.

Checkpoint notes 1092-1166

Checkpoint_1092_CorrectedMinorArcInput.md

Checkpoint 1092: Complete Ren bracket and sufficient full complementary minor estimate

Exported the complete normalized Ren bracket without dropping its first term and proved a sufficient full complementary R^(-1/4)*(log N)^d bound from the already explicit independent balanced Ren/Ghosh hypotheses. Matching minor statement, proof, downstream approximation paragraph and bibliography replacements are saved separately. The source stronger minor estimate is not claimed proved or disproved; its cited deduction does not establish the full stated range. The existing approximation theorem remains completely conditional through the checked weaker route. No paper claim is promoted and no new analytic hypothesis is added.

Checked declarations

Module	Named supporting theorems
CorrectedMinorArcInput.lean	3

- `annularMangoldt_complete_Ren_bound`
- `complementaryMinorBudget_quarter_bound`
- `annularMangoldt_outside_major_quarter_bound`

<!-- checkpoint-verification -->

Verification

All 1159 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4950-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3503 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1161 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Four substantive auxiliaries remain partial. Section9 mapping and reductions are now authorized; later analytic and sieve inputs remain explicit. The rational smooth multifrequency theorem together with circle transference is an explicit independent compound hypothesis; its existence is unproved. Its predicate contains no own-paper annular or truncated maximal conclusion. See docs/RATIONAL_MULTIFREQUENCY_BOUNDARY.md.

Checkpoint_1093_GoldbachUpperCutoffBoundary.md

Checkpoint 1093: Recovered full proof project and checked Section9 lower cutoff correction

Recovered checkpoint1092 from its exact saved ZIP, checked all3814 archived file hashes and restored the pinned Lean4.34.1 environment. A fresh full build and complete audits authenticate the recovered baseline. The new Section9 boundary check proves an exact admissible alpha and integer witness for the mismatch between z<=p2 in T_n and z<p2 in L. The two defining lower inequalities and all six later repeated bounds or sums have a separately saved replacement and exact eight-occurrence patch. This checks a canonical-inclusion gap, not falsity of the final asymptotic theorem. The nine-statement map is retained, and later circle-method/sieve statements are not new external facts.

Checked declarations

Module	Named supporting theorems
GoldbachUpperCutoffBoundary.lean	5

- `goldbach_upper_boundary_data`

- goldbach_seventy_no_strict_factor
- goldbachBoundaryAlpha_range
- goldbachBoundaryAlpha_cutoff
- goldbach_boundary_closed_geometry

<!-- checkpoint-verification -->

Verification

All 1160 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4951-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3508 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1162 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem together with circle transference is an explicit independent compound hypothesis; its existence is unproved. Its predicate contains no own-paper annular or truncated maximal conclusion. See docs/RATIONAL_MULTIFREQUENCY_BOUNDARY.md.

Checkpoint_1094_GoldbachFiniteRepresentations.md

Checkpoint 1094: Actual Goldbach representation sets and uniform prime-divisor exception bounds

Introduced the actual finite prime-plus-prime-square representation sets, primitive good set, residue-class admissible set, sifted set and explicit exception envelope indexed by prime divisors of n-1. Proved that a shared prime divisor of r-1 and n-1 must equal the represented square prime, bounds the entire exceptional set by omega(n-1), and gives the full uniform O_epsilon(N^epsilon) error for n-1<=N^2. The coprimality removal used in the upper reduction is proved as well. For n>12 in the source residue class, every representation has square prime>3 and r congruent11 modulo24. These arithmetic statements have no new external theorem inputs. The numbered sieve reductions are not promoted before their remaining set inclusions are proved.

Checked declarations

Module	Named supporting theorems
GoldbachFiniteRepresentations.lean	11

- mem_goldbachRepresentations
- goldbach_representation_unique_prime
- goldbach_shared_prime_eq
- goldbach_shared_divisor_exception
- goldbachDivisorExceptions_card
- goldbach_factor_coprime_of_not_exception
- goldbachSifted_antitone
- goldbach_prime_divisor_power_error
- goldbachDivisorExceptions_power_error
- goldbach_square_prime_large
- goldbach_representation_admissible

<!-- checkpoint-verification -->

Verification

All 1161 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4952-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3519 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1163 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem together with circle transference is an explicit independent compound hypothesis; its existence is unproved. Its predicate contains no own-paper annular or truncated maximal conclusion. See docs/RATIONAL_MULTIFREQUENCY_BOUNDARY.md.

`Checkpoint_1095_GoldbachFactorParity.md`

Checkpoint 1095: Multiplicity-sensitive modulo4 parity and the exactly-two factor count

Proved the exact modulo4 parity formula for lists whose factors are1 or3 modulo4, with bad factors retained with multiplicity. A product1 modulo4 has an even number of bad entries. The product lower bound shows that fewer than four entries can be at least z when the whole product is below z^4. If a bad factor exists, the bad list therefore has length exactly2, including repeated equal primes. These are unconditional arithmetic prerequisites for the actual Section9 truncation; no classical two-square criterion or Goldbach conclusion is assumed.

Checked declarations

Module	Named supporting theorems
GoldbachFactorParity.lean	7

- `mem_badModFourFactors`
- `mem_goodModFourFactors`
- `modFourFactors_product`
- `odd_list_product_mod_four`
- `modFourFactors_bad_count_even`
- `list_product_ge_length_power`
- `exactly_two_badModFourFactors`

<!-- checkpoint-verification -->

Verification

All 1162 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4953-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3526 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1164 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem together with circle transference is an explicit independent compound hypothesis; its existence is unproved. Its predicate contains no own-paper annular or truncated maximal

conclusion. See docs/RATIONAL_MULTIFREQUENCY_BOUNDARY.md.

`Checkpoint_1096_GoldbachFullSiftedReduction.md`

Checkpoint 1096: Complete first Section9 sifted-count reduction with its explicit classical criterion

Completed the first actual Section9 lower-sifted counting reduction with exact finite error omega(n-1) and the complete uniform O_epsilon(N^epsilon) statement. The r-to-r-1 reindexing is proved injective and no own Goldbach conclusion is an input. The classical primitive two-square criterion is explicitly isolated as an independent unproved arithmetic hypothesis; no existence or unconditional proof is claimed. The statement ledger has12 proved,20 complete conditional,2 partial,1 quoted external and19 open claims. Later circle-method and sieve conclusions remain internal obligations.

Checked declarations

Module	Named supporting theorems
GoldbachFullSiftedReduction.lean	4

- `goldbachSiftedValues_card`
- `goldbach_full_sifted_inclusion`
- `goldbach_full_sifted_count`
- `goldbach_from_full_sifted`

<!-- checkpoint-verification -->

Verification

All 1163 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4954-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3530 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1165 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1097_GoldbachTwoLargePrimeFactors.md`

Checkpoint 1097: Exact two-large-prime factorization with a primitive-compatible remaining factor

Proved the complete integer factorization underlying the truncated sieve. For m congruent1 modulo4 with m<z^4, if a prime3 modulo4 divides m and every such prime is at least z, then m=u*p1*p2 with positive u, ordered primes z<=p2<=p1, and every prime divisor of u congruent1 modulo4. Prime factor lists retain multiplicity, so p1=p2 is included. Positivity, exact products, ordering and all remaining prime divisors are proved. This is an unconditional arithmetic theorem with no primitive two-square, sieve or current-paper analytic hypothesis.

Checked declarations

Module	Named supporting theorems
GoldbachTwoLargePrimeFactors.lean	2

- `primeFactorsList_odd_mod_four`
- `primitive_obstruction_two_large_primes`

<!-- checkpoint-verification -->

Verification

All 1164 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4955-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3532 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1166 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1098_GoldbachTruncationSets.md`

Checkpoint 1098: Actual truncated bad set and both directions of the sieve classification

Defined the actual truncated bad representation set with ordered prime factors, the closed lower cutoff and all remaining factor conditions. Proved that every sifted nonprimitive representation outside the explicit prime-divisor envelope belongs to this set, using the exact multiplicity-sensitive factorization. Proved both converse inclusions: every truncated bad representation is sifted and is not primitive, while every good representation in the source n residue class is sifted. The only unformalized arithmetic input is the already explicit independent primitive two-square criterion. No circle-method or sieve asymptotic input is used.

Checked declarations

Module	Named supporting theorems
GoldbachTruncationSets.lean	5

- `prime_three_mod_four_dvd_half`
- `goldbach_sifted_bad_factorization`
- `goldbachTruncatedBad_subset_sifted`
- `goldbachTruncatedBad_not_good`
- `goldbach_good_subset_sifted`

<!-- checkpoint-verification -->

Verification

All 1165 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4956-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3537 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1167 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1099_GoldbachTruncationReduction.md

Checkpoint 1099: Complete truncated-sieve identity with an explicit uniform error witness

Completed the actual truncated-sieve equality with an explicit nonnegative integer error at most omega(n-1), and then the entire uniform O_epsilon(N^epsilon) statement for every real1<alpha<2. The exact scale z=N^(1/alpha), z^4>N^2, prime multiplicities, disjoint good/bad sets and r-1 image cardinalities are proved. Source n>12 is explicit, consistent with the large-n context. The classical primitive two-square criterion is the only independent unproved input. No Section10 or11 conclusion is assumed. The source ledger now has12 proved,21 complete conditional,2 partial,1 quoted external and18 open claims.

Checked declarations

Module	Named supporting theorems
GoldbachTruncationReduction.lean	4

- `goldbachTruncatedBadValues_card`
- `goldbach_truncation_exact_error`
- `goldbach_truncation_scale`
- `goldbach_truncationT`

<!-- checkpoint-verification -->

Verification

All 1166 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4957-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3541 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1168 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1100_GoldbachUpperFactorGeometry.md

Checkpoint 1100: Actual corrected upper-sieve sets and complete factor and cutoff geometry

Defined the corrected closed lower-cutoff L and L_n sets, the literal upper M_n(ell) representation set, the W sieve-prime predicate and actual quarter-power sifting level. Proved that every element of L is below N^2, so the finite indexing set loses no source element. Proved both factor geometry inequalities, the upper level bound by N^(1/2), the strict comparison with N^(1/alpha), and that a prime survives every smaller prime sieve. These are unconditional arithmetic/geometric facts. The eight-occurrence manuscript patch remains proposed and the selected source is unchanged; no full current-source upper-reduction credit is claimed.

Checked declarations

Module	Named supporting theorems
GoldbachUpperFactorGeometry.lean	6

- `goldbachClosedL_bound`
- `mem_goldbachClosedLSet`
- `goldbach_upper_factor_geometry`
- `goldbach_upper_level_le_sqrt`
- `goldbach_sqrt_lt_truncation`
- `prime_survives_smaller_sieve`

<!-- checkpoint-verification -->

Verification

All 1167 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4958-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3547 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1169 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1101_GoldbachUpperSieveInclusion.md

Checkpoint 1101: Full corrected upper-sieve inclusion on the actual arithmetic sets

Proved the full actual set inclusion for the upper reduction with the proposed closed L cutoff. Every truncated bad representation outside the explicit prime-divisor exception envelope produces an element ell in the actual coprime L_n and belongs to its actual upper sifted set. The proof discharges factor geometry, finite bounds, q>3, the shifted representation identity, coprimality and survival of every sieve prime. This implication is purely arithmetic; it does not assume the primitive criterion or any analytic/sieve theorem. The source remains unchanged, and promotion of the printed upper statement awaits the documented cutoff correction and the finite count assembly.

Checked declarations

Module	Named supporting theorems
GoldbachUpperSieveInclusion.lean	1

- `goldbach_upper_sifted_inclusion`

<!-- checkpoint-verification -->

Verification

All 1168 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4959-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3548 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1170 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1102_GoldbachMasterSieveReduction.md`

Checkpoint 1102: Full corrected upper counting estimate and actual master sieve assembly

Completed the full actual upper counting inequality and master sieve assembly for the proposed corrected closed L. The upper bound has error at most omega(n-1); the exact master has error at most twice that quantity. Both uniform O_epsilon(N^epsilon) formulations are proved. The upper deduction uses no independent theorem input; the master uses only the explicit classical primitive two-square criterion. All own lower, truncated and upper set/cardinality deductions are discharged. The current strict-L manuscript remains unchanged, so its upper and master labels remain source-matching partials pending the eight-occurrence correction. The source ledger is12 proved,21 complete conditional,4 partial,1 quoted external and16 open, including two earlier analytic partials and two corrected-source matching items.

Checked declarations

Module	Named supporting theorems
GoldbachMasterSieveReduction.lean	4

- `goldbach_upper_reduction_exact`
- `goldbach_upper_reduction`
- `goldbach_master_sieve_exact`
- `goldbach_master_sieve_reduction`

<!-- checkpoint-verification -->

Verification

All 1169 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4960-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3552 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1171 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two earlier source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including

measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1103_GoldbachExceptionalSetReduction.md`

Checkpoint 1103: Actual exceptional-set mean-square and Chebyshev reductions

Defined the actual finite exceptional set where the primitive prime-plus-prime-square representation count is zero. Proved its squared-threshold cardinal bound by the two actual mean-square errors, and the resulting Chebyshev cardinal inequality, using the already proved actual master reduction. The lower/upper sieve estimates, positive main-term separation and small divisor error are explicit internal hypotheses to be discharged later; they are not entered in the external-input registry or used to claim a completed headline Goldbach theorem. The arithmetic criterion remains explicit.

Checked declarations

Module	Named supporting theorems
GoldbachExceptionalSetReduction.lean	2

- `goldbach_exceptional_square_sum`
- `goldbach_exceptional_count`

<!-- checkpoint-verification -->

Verification

All 1170 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4961-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3554 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1172 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two earlier source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1104_GoldbachLogarithmicReduction.md`

Checkpoint 1104: Exact logarithmic saving from the actual exceptional-set square bound

Completed the exact logarithmic exponent bookkeeping in the exceptional-set argument. A mean-square bound of order N^4*(log N)^(-K) and a threshold at least kappa*N*(log N)^(-C) yield the actual exceptional cardinal bound (2B/kappa^2)*N^2*(log N)^(-A) whenever K>=A+2C. The algebra, divisions, positive-domain conditions and real exponents are proved. The own-paper lower/upper sieve and positive-gap statements remain explicit internal hypotheses; no headline completion or new external input is claimed.

Checked declarations

Module	Named supporting theorems
GoldbachLogarithmicReduction.lean	2

- `logarithmic_square_threshold_bound`
- `goldbach_logarithmic_exceptional_bound`

<!-- checkpoint-verification -->

Verification

All 1171 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4962-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3556 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1173 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two earlier source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1105_GoldbachSectionNineReduction.md`

Checkpoint 1105: Complete finite final-count reduction including the small-integer contribution

Completed the small-integer contribution and the actual all-admissible-n exceptional-set reduction, with explicit finite ceiling bounds. The source exponent choice K=A+2C+10 is used literally, producing the N^2*(log N)^(-A) bound from the unresolved internal lower/upper sieve, positive-gap and mean-square statements. Proved also that the positive-u closed L definition is exactly the source prime-divisor formulation: u=0 is excluded by the prime3 condition, so no additional restriction is hidden. This is a completed reduction, not a completed Goldbach headline theorem; internal analytic estimates remain unproved and are not new external hypotheses.

Checked declarations

Module	Named supporting theorems
GoldbachSectionNineReduction.lean	7

- `one_mod_four_prime_divisors_pos`
- `goldbachClosedL_source_form`
- `goldbachSmallNumbers_card_le_ceil`
- `goldbachLargeNumbers_range`
- `goldbach_exceptional_split`
- `goldbach_full_exceptional_count`
- `goldbach_section9_count_reduction`

<!-- checkpoint-verification -->

Verification

All 1172 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4963-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3563 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1174 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two earlier source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1106_GoldbachDivisorErrorAbsorption.md`

Checkpoint 1106: Complete elementary Section9 error absorption and final-count reduction

Completed uniform eventual absorption of the arithmetic prime-divisor error and the small-n cutoff, then exported the final actual N^2*(log N)^(-A) Section9 count reduction with that bookkeeping discharged. Only the internal analytic/sieve/main-term estimates remain in this reduction; they are not new external facts and the headline theorem is not claimed proved. The full nine-claim map, Sections3–8 review, pending manuscript guide and recovery review are reconciled. The current minor proof passage now uses the exact R/4 denominator lower bound already present in Lean; no Lean correction was needed. The current source remains unchanged by the proposed minor and closed-L edits.

Checked declarations

Module	Named supporting theorems
GoldbachDivisorErrorAbsorption.lean	2

- `goldbach_eventual_arithmetic_budgets`
- `goldbach_section9_eventual_count_reduction`

<!-- checkpoint-verification -->

Verification

All 1173 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4964-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3565 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1175 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two earlier source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1107_UniformPageProgressionRange.md`

Checkpoint 1107: Uniform fixed-level Page progression and extended major range

Preserved the exact current upload, verified all eight closed L cutoffs and the sufficient minor statement, and reconstructed a separate selected source with the already supplied approximation replacement and accepted sparse/wording restorations. The corrected upper counting lemma is now proved under its normal hypotheses, and the corrected master reduction is complete under the explicit classical primitive two-square criterion. Implemented the independent general UniformPageProgressionInputs interface with one genuine optional Page datum per N, fixed before Q, all positive moduli and every real endpoint in [N/2,N]. Proved the full extended q<=R^(11/10) major-arc application, including the actual induced character, selector, beta, phase width and inverse-fifth error. The progression input and fixed-level analytic bridge remain unproved; no paper endpoint is assumed. The corrected minor/major joint export is the next stage.

Checked declarations

Module	Named supporting theorems
UniformPageProgressionRange.lean	5

- `optionPageBeta_pos`
- `pageModulusCharacter_rational`
- `pageModulusSelector_rational`
- `extended_major_denominator_in_Page_range`
- `uniformPage_extended_major_arc`

<!-- checkpoint-verification -->

Verification

All 1174 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4965-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3570 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1176 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two earlier source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1108_CorrectedPrimeSquareInput.md`

Checkpoint 1108: Complete corrected prime-square major and minor input

Corrected prime-square input: complete conditional implication at checkpoint1108

The current user upload adopts the complementary minor estimate R^(-1/4)(log N)^C. The selected copy also restores the matching approximation proof. The complete source lemma is exported as source_corrected_prime_square_input.

The same genuine optional primitive Page zero is selected for each N before Q, every modulus, phase, residue and real endpoint. Its conductor and zero region at the fixed exponential level are retained. The general progression input is uniform through exp(eta sqrt(log N)); eta and the error-decay constant c are independent positive constants. Taking their minimum supplies a fixed positive Q-range coefficient. Lean proves that q<=R^(11/10) is inside that distribution range and that the exact prefix error is absorbed by the inverse-fifth major error. The annular partial summation, prime-power/nonunit error, induced character and missing-character selector are all actual proved formulas.

The independent Ren and Ghosh inputs retain their full raw brackets. For every real frequency outside all reduced rational major arcs, the checked balanced approximation gives the complementary R^(-1/4)(log N)^d bound. Lean proves R^4<=N/2 and log N>=1 uniformly beyond one fixed threshold, for all allowed Q. No original stronger minor claim is assumed or asserted.

The remaining independent hypotheses are UniformPageProgressionInputs and UniformRenGhoshInputs. The first is a general fixed-level uniform prime-progression theorem, including its analytic Page-zero bridge; the second packages the already recorded general Ren/Ghosh estimates before Q. Their existence is unproved. Neither contains a quadratic major-arc conclusion, an approximation theorem or a paper-specific maximal estimate.

All source clauses of the corrected l:prime-square-input are matched. This closes the paper-specific deduction within the agreed conditional boundary, not the independent deep theorems. The broad quantitative IW proposition is the sole remaining incomplete standalone claim in Sections3–8.

Checked declarations

Module	Named supporting theorems
CorrectedPrimeSquareInput.lean	3

- `pageModulusSelector_none`
- `pageModulusSelector_some`
- `source_corrected_prime_square_input`

<!-- checkpoint-verification -->

Verification

All 1175 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4966-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3573 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1177 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two earlier source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1109_CyclicProductLpOperators.md`

Checkpoint 1109: Full real Lp cyclic product operator bound

Implemented the normalized cyclic Lp norm on families of actual Banach-space elements and proved its pointwise domination, translation and finite-sum properties. Proved the product operator bound for arbitrary full real Lp fibres, every finite exponent p>=1, arbitrary bounded real operator and every finite cyclic complex measure: the exact norm is at most its total variation times the real operator norm. The TV<=1 consequence has no Schwartz or L2-intersection restriction and adds no independent theorem hypothesis. This is the full-domain product estimate used in the broad IW multiplier application. The remaining deep IW and sampling theorems are still explicit external inputs, and the full proposition is not yet complete.

Checked declarations

Module	Named supporting theorems
CyclicProductLpOperators.lean	5

- `cyclicFamilyLpSize_mono`
- `cyclicFamilyLpSize_translate`
- `cyclicFamilyLpSize_finsetSum`
- `cyclicRealOperator_Lp_bound`
- `cyclicRealOperator_TV_one_bound`

<!-- checkpoint-verification -->

Verification

All 1176 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4967-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3578 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1178 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two earlier source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1110_AEBoundedRealMultipliers.md`

Checkpoint 1110: Rough real Fourier symbols and full Lp realizations

Constructed arbitrary essentially bounded measurable real Fourier symbols, the actual bounded real-line L2 multipliers and their Fourier formulas without any continuity assumption. Proved exact scalar multiplication of symbols, operators and inverse Fourier kernels. Defined a full Lp Fourier realization as an actual bounded operator on the whole real Lp space agreeing with the given rough L2 multiplier on its genuine L2 intersection; this avoids mistaking an intersection norm for a full-domain operator. Its scalar-multiple realization is constructed with no deep input. These are foundations for the remaining broad IW clause, not an assertion of the deep theorem or a completion of the source proposition.

Checked declarations

Module	Named supporting theorems
AEBoundedRealMultipliers.lean	8

- `aeRealProduct_memLp`
- `aeRealMultiplyValue_ae`
- `aeRealMultiplyValue_norm_le`
- `aeRealMultiply_ae`
- `realLineFourier_aeMultiplier`
- `aeRealLineMultiplier_norm_le`
- `aeRealLineMultiplier_smul`
- `aeRealSymbol_smul_inverse_kernel`

<!-- checkpoint-verification -->

Verification

All 1177 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4968-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3586 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1179 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two earlier source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1111_FullSamplingFourierSupport.md`

Checkpoint 1111: Full Lp Fourier support and bandlimited sampling classes

Implemented Fourier support on the whole real Lp space using Mathlib's actual tempered-distribution Fourier transform, including p=infinity. Implemented the integer Fourier-support condition by its literal Schwartz test series and proved absolute convergence for every integer lp input, so no totalized divergent series can satisfy the condition vacuously. Implemented actual finite cyclic Fourier components and the full bandlimited real/cyclic Lp class, with continuous representatives retained for point sampling. Proved support enlargement and the equality between the full fibre-Lp norm and the previously normalized iterated real/cyclic norm. These classes do not impose Schwartz, integrability or L2 conditions. The general deep sampling theorems and source applications remain the next step.

Checked declarations

Module	Named supporting theorems
FullSamplingFourierSupport.lean	5

- `realLpSpectrumIn_mono`
- `realLpSpectrumIn_interval_mono`
- `schwartz_integer_test_summable`
- `integerLpSpectrumIn_mono`
- `cyclicFunctionLp_size`

<!-- checkpoint-verification -->

Verification

All 1178 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4969-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3591 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1180 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two earlier source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its

reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1112_FullAdelicIWApplication.md

Checkpoint 1112: Full rough-symbol adelic IW application and cyclic basis correction

Implemented the independent general full-domain adelic IW interface, with its rough-symbol limiting and finite-quotient/kernel bridges explicitly unproved. It quantifies over arbitrary good coprime data and arbitrary spatial cyclic families, not the source probability-measure specialization. Proved the exact rank-one kernel factorization and the full real-Q rough multiplier application. The output is an actual bounded operator on all integer lp inputs, with its literal finite-convolution Fourier kernel and operator norm <=C_rk times the actual full real Lp operator norm. The real/cyclic product estimate is discharged by the unconditional full-Lp module. Corrected the newly introduced sampling basis to denominators dividing P, with exact declared source hashes and required dependency rebuild; no source completion was awarded to the earlier incorrect class.

Checked declarations

Module	Named supporting theorems
FullAdelicIWApplication.lean	4

- `adelicProjectedKernel_rank_one`
- `iwRoughKernel_finite_formula`
- `quantitative_real_iw_full_rough_multiplier`
- `source_iw_rough_multiplier_norm`

<!-- checkpoint-verification -->

Verification

All 1179 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4970-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3595 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1181 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two earlier source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1113_FullTaoSamplingApplications.md

Checkpoint 1113: Full two-direction IW sampling and finite-dimensional Shannon application

Implemented separate full-domain general Tao major-arc and Shannon sampling interfaces, with their truth and general published-to-finite-quotient/full-class bridges explicitly unproved. Applied Theorem1.6 at the exact constructed real-Q width and full finite exponent range. Constructed the actual unique interpolation for every integer lp input with distributional Fourier support, proved its diagonal identity,

uniqueness and the literal two-sided source bounds including the inverse C_rk factor. Applied Proposition1.5 to every narrower band, every finite-dimensional normed complex target and every extended exponent 1<=p<=infinity, retaining both norm directions and actual diagonal lp membership. All applications use the corrected P-torsion cyclic Fourier basis. The final one-construction four-clause source assembly remains.

Checked declarations

Module	Named supporting theorems
FullTaoSamplingApplications.lean	5

- `quantitative_real_iw_full_sampling`
- `fullIWInterpolation_diagonal`
- `fullIWInterpolation_unique`
- `fullIWInterpolation_source_bounds`
- `quantitative_full_shannon_diagonal`

<!-- checkpoint-verification -->

Verification

All 1180 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4971-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3600 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1182 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

The current complete manuscript upload confirms the independent admissibility coefficient. The full generic sparse recursion is proved under its normal local hypothesis. The unchanged upload and a separate selected copy with the previously approved factor-four application restored are preserved. Two earlier source statements remain partial: the original strong minor-arc clause and the broad standalone quantitative IW function classes. The full multifrequency lemma and denominator corollary, including measurable fixed symbols, are complete conditional implications. Section9 mapping is saved and its reductions are authorized; later circle-method and sieve conclusions remain internal proof obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1114_QuantitativeIWSourceAssembly.md`

Checkpoint 1114: Complete all four quantitative IW clauses and close Sections 3–8 deductions

Full quantitative IW proposition: completed conditional deduction at checkpoint1114

The final theorem is proposition_quantitative_IW_full in `PrimeSquares/QuantitativeIWSourceAssembly.lean`. It chooses one arithmetic construction before the multiplier, p, complex measure, sampled input and target space. It is a complete implication under explicitly named independent general inputs, not an unconditional proof of Tao's theorems.

Source clause	Matched checked scope
(i), construction	Every real Q>=2^k, r>=2 and k>=4; exactly B=C^(k^2 Q^(1/(k-1/2))) and epsilon=(2^20 r B^(2r+8))^(-1); one positive integer P<=3^Q; every a/q modulo one for all integers a and all positive q<=Q, including unreduced fractions; all reduced denominators <=B; full (r,1/2)-goodness.

Source clause	Matched checked scope
(ii), multiplier	Arbitrary essentially bounded measurable real symbols supported almost everywhere in the closed narrow interval, arbitrary complex cyclic measures of total variation <=1, and every p in [p_r,2r]. The real norm is that of an operator on the whole real Lp space. The conclusion is an actual bounded operator on the whole integer lp space, with the literal projected inverse-Fourier convolution kernel and norm <=C_rk times the full real operator norm. Agreement on finite inputs determines this entire operator uniquely. No smoothness or L2-intersection restriction is substituted for the full operator.
(iii), major-arc sampling	The complete distributional Fourier-support lp class, both norm directions, the literal inverse-C_rk lower bound, unique interpolation and its exact sampled diagonal. The positive-exponent typeclass is proved from p>=p_r>1, not a new mathematical hypothesis. The normalized finite cyclic norm agrees with the earlier iterated real/cyclic norm.
(iv), Shannon sampling	Every finite-dimensional normed complex target, every extended exponent 1<=p<=infinity, every positive delta<=1/(100P), full real Lp fibres with distributional Fourier support, actual diagonal lp membership and both norm directions with one absolute sampling constant. No Hilbert, p<=2 or Schwartz restriction remains.

For point sampling, bandlimited functions are evaluated in their continuous representatives, as in the standard sampling convention. This does not impose Schwartz decay, integrability or L2 membership. An arbitrary modification of an Lp representative at individual points cannot define the sampling operator.

The real Fourier support is expressed using Mathlib's actual tempered-distribution Fourier transform. The integer support is expressed by its literal Schwartz-test Fourier series; absolute convergence is proved for every lp input. The finite cyclic Fourier basis is exactly the P-torsion rational set, not all denominators <=P. Its correction at1112 has exact source hashes and a complete dependency rebuild/audit; the final theorem explicitly proves every IW frequency belongs to the corrected basis.

The independent unproved inputs are RawTaoArcConstructionInput, RawFullAdelicIWInput, RawFullMajorArcSamplingInput and RawFullShannonInput. The last three general statements explicitly include their published-to-full-class/finite-quotient/kernel bridges; the general rough-symbol limiting argument is also not formalized. They quantify over arbitrary good coprime data or arbitrary narrow cyclic data, not the paper's chosen Q, probability measure, annular model, low/high estimate or maximal conclusion. The full real Lp product bound, coefficient specialization, exact construction applicability and source assembly are proved deductions. Checking their consuming implications does not prove these independent hypotheses true.

The earlier endpoint theorems retain their previously recorded, narrower assumptions. This broad standalone completion does not retroactively add the full-domain hypotheses to them. The Goldbach circle-method and sieve estimates remain open internal paper obligations.

Checked declarations

Module	Named supporting theorems
QuantitativeIWSourceAssembly.lean	9

- `iw_frequency_in_cyclic_basis`
- `rational_fract_circle`
- `iw_coverage_all_rational_lifts`
- `iw_coverage_all_fraction_points`
- `fullIntegerKernelMultiplier_single`
- `fullIntegerKernelMultiplier_unique`
- `iw_source_exponent_one_le`
- `full_iw_source_analytic_clauses`
- `proposition_quantitative_IW_full`

<!-- checkpoint-verification -->

Verification

All 1181 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4972-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3609 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1183 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1115_GoldbachFourierInversion.md`

Checkpoint 1115: Exact unweighted Goldbach Fourier inversion and full-cutoff removal

The Section10 Fourier identity is now proved for the literal unweighted prime sums. Prime cutoffs have arbitrary real endpoints, and membership is exactly primality together with m<=U. The progression cutoff retains ell*t<N^2 and t congruent to a modulo k. Neither a coprimality assumption nor a distribution theorem is needed for this identity.

The positive exponential convention is Mathlib's fourier j on the period-one circle with Haar mass one. Character orthogonality is proved from the existing Fourier basis; the product of the actual prime-square and progression polynomials is expanded before integration. The coefficient at n is exactly the complex embedding of the finite number of ordered prime pairs, including all real endpoint conventions. This is equation e:fourierinversion9, not an assumed Fourier identity.

For positive ell and n<N^2, goldbach_full_cutoffs_redundant proves that U=N, W=N^2 and ell*t<N^2 impose no additional restriction on the untruncated prime-pair count. All finiteness bounds are deduced from m^2+ell*t=n. The terminal goldbach_full_fourier_inversion therefore identifies the full representation count itself with the circle integral.

Eight new supporting theorems are unconditional. No analytic minor/major estimate, sieve bound, external theorem or paper endpoint is assumed or proved by this checkpoint. Lemma l:circmethod and the headline Goldbach theorem remain open; the exact Fourier bridge is a proved prerequisite. The user reports applying the complete approximation proof and the three earlier sparse/wording/strict-approximation edits. Work proceeds with the selected reconciled source; no new manuscript upload or verification of the user's editor is claimed.

Checked declarations

Module	Named supporting theorems
GoldbachFourierInversion.lean	8

- `mem_goldbachPrimeCutoff`
- `mem_goldbachProgressionCutoff`
- `goldbach_finite_character_integral`
- `goldbachPrimeSquarePolynomial_mul`
- `goldbach_cutoff_fourier_inversion`
- `mem_goldbachCutoffPrimePairs`
- `goldbach_full_cutoffs_redundant`
- `goldbach_full_fourier_inversion`

<!-- checkpoint-verification -->

Verification

All 1182 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4973-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3617 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1184 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1116_GoldbachMinorArcBessel.md`

Checkpoint 1116: Actual Goldbach minor-arc Bessel reduction and error assembly

The actual weighted progression sum is now defined with arbitrary complex coefficients and residues. The arithmetic-index set has precisely positive ell<L and positive k<V(ell), with strict arbitrary real cutoffs. Its membership equivalence is proved. The exact weighted Fourier identity and the full representation-count specialization are proved by finite linearity, not assumed.

The new goldbach_finite_fourier_bessel is a genuine finite Bessel inequality for arbitrary square-integrable circle functions, deduced from Mathlib's complete Fourier basis and the actual toLp representative. The set-restricted version applies to every measurable minor set and continuous prime/progression polynomials. It proves the literal bound sum_n |integral_minor P F e(-n theta)|^2 <= B^2 integral_circle |F|^2 from the pointwise bound |P|<=B on that set. Arbitrary finite integer output sets are allowed, so restriction to the paper's n<N^2 range costs nothing.

The theorem goldbach_actual_minor_bessel specializes this to the literal prime-square and weighted progression sums. A remaining bound on the actual progression energy is an INTERNAL proof obligation, as is the source's use of the independent raw prime exponential-sum estimate. Neither is renamed an external theorem. No logarithmic-saving hybrid estimate is claimed yet.

The exact major/minor decomposition of the weighted prime-pair count is proved for every measurable major set. The finite three-stage mean-square assembly proves the factor three in the source argument from the three successive INTERNAL error estimates. This assembles estimates once proved; it does not prove any of them by assumption. All eight supporting theorems use no new external deep input. The Bombieri–Vinogradov source lemma and the headline Goldbach theorem remain open.

Checked declarations

Module	Named supporting theorems
GoldbachMinorArcBessel.lean	8

- `mem_goldbachCoefficientIndices`
- `goldbach_weighted_fourier_inversion`
- `goldbach_weighted_full_fourier_inversion`
- `goldbach_finite_fourier_bessel`
- `goldbach_restricted_fourier_bessel`
- `goldbach_actual_minor_bessel`
- `goldbach_major_minor_partition`
- `goldbach_circle_three_stage_bound`

<!-- checkpoint-verification -->

Verification

All 1183 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4974-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3625 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1185 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1117_GoldbachProgressionEnergy.md`

Checkpoint 1117: Exact progression collision energy and off-diagonal prime geometry

The actual weighted progression energy has been opened by finite character orthogonality. The general goldbach_frequency_energy_identity proves the exact equality between the circle integral of the squared polynomial norm and the real part of the weighted sum over equal frequencies. Repeated frequencies are retained, so this is not an injectivity or disjointness assumption. Arbitrary complex coefficients are allowed.

For coefficients of norm at most one, goldbach_frequency_energy_le_collisions bounds the exact energy by the cardinality of the finite equal-frequency collision set. The actual progression events retain (ell,k,t), primality, the real W endpoint, ell*t<N^2 and the residue condition. The flat event polynomial is proved equal to the original nested progression polynomial. Thus goldbach_progression_energy_le_collisions is about the source's actual polynomial, not a replacement surrogate.

The composed goldbach_actual_minor_collision_bound has the literal natural-number output range n<N^2. It removes the free energy parameter from the earlier Bessel reduction and leaves only an explicit finite collision count and a pointwise prime-square minor bound. All coefficient and residue quantifiers remain arbitrary. Bounding that finite count by O(N^2 (log N)^6), and supplying the source uniform prime-square logarithmic bound, are still internal proof tasks; no such estimate is assumed to be a general external theorem.

The diagonal collision forces ell_1=ell_2. If t_1 and t_2 are distinct primes, the exact off-diagonal geometry ell_1=s*t_2 and ell_2=s*t_1 is proved, with a unique positive integer s whenever ell_1 is positive. This is the arithmetic starting point for the source dyadic Cauchy–Schwarz estimate.

All eight new supporting theorems are proved without a new independent deep input. Together, checkpoints1115–1117 prove the exact weighted Fourier bridge, measurable major/minor partition, actual Bessel inequality, explicit collision-energy reduction, diagonal/off-diagonal geometry and three-error assembly. The source hybrid logarithmic estimate, major approximation/completion, Bombieri–Vinogradov lemma, Section11 sieve/main-term/integral estimates and headline Goldbach theorem remain open. No manuscript replacement is newly required by this checkpoint.

Checked declarations

Module	Named supporting theorems
GoldbachProgressionEnergy.lean	8

- goldbach_frequency_energy_identity
- goldbach_frequency_energy_le_collisions
- mem_goldbachProgressionEvents
- goldbach_weighted_progression_as_frequency_polynomial
- goldbach_progression_energy_le_collisions
- goldbach_actual_minor_collision_bound
- goldbach_prime_collision_diagonal
- goldbach_prime_collision_off_diagonal

<!-- checkpoint-verification -->

Verification

All 1184 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4975-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3633 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1186 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1118_GoldbachReciprocalLCM.md

Checkpoint 1118: Explicit reciprocal-lcm logarithmic bound for the Goldbach energy

The reciprocal-lcm estimate used in the source diagonal and congruence-fibre square bounds is now proved with an explicit constant. For every natural V, the sum over 1<=a,b<=V of 1/lcm(a,b) is at most H_V^3, hence at most (1+log V)^3. The V=0 and V=1 endpoints are included. The strict real-cutoff version is also proved for every real V>=1 and all positive integers a,b<V; this matches the varying source modulus cutoffs without rounding them upward inside the logarithm.

The proof is elementary and uses no new deep input. Each denominator pair injects into the triple (gcd(a,b),a/gcd(a,b),b/gcd(a,b)). The lcm is proved equal to the product of these three quantities. For positive bounded denominators, all three entries lie in [1,V]. Nonnegative summation and the proved injection therefore bound the reciprocal-lcm sum by the full product of three harmonic sums. Mathlib's existing elementary harmonic/logarithm comparison supplies the explicit logarithmic bound.

The new five supporting theorems do not assume the reciprocal-lcm estimate, a sieve theorem, a mean-square result or the paper's hybrid lemma. They are direct Lean proofs. The next source step is the uniform count of simultaneous congruences in an interval, followed by the dyadic Cauchy–Schwarz collision bound O(N^2(log N)^6). The source prime-square minor bound must then be derived in its exact parameter range from independently recorded raw exponential-sum inputs. The full hybrid estimate and both major estimates remain open, so no source-level completion credit is awarded to l:hybridminorarc or l:circmethod. Section11 and the headline Goldbach theorem remain open.

Checkpoints1115,1116,1117 and1118 are saved separately. They respectively establish exact unweighted Fourier inversion, the actual weighted Bessel/partition/error assembly, exact collision energy and prime geometry, and the reciprocal-lcm logarithmic estimate. No new manuscript replacement is required by these four checkpoints. The user's reported manuscript edits are recorded without claiming a new upload or a remote edit.

Checked declarations

Module	Named supporting theorems
GoldbachReciprocalLCM.lean	5

- `goldbach_lcm_gcd_quotients`
- `goldbach_gcd_quotient_triple_injective`
- `goldbach_reciprocal_lcm_harmonic_cube`
- `goldbach_reciprocal_lcm_log_cube`
- `goldbach_reciprocal_lcm_strict_real_cutoff`

<!-- checkpoint-verification -->

Verification

All 1185 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4976-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3638 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1187 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1119_GoldbachCongruenceCounting.md`

Checkpoint 1119: Uniform simultaneous-congruence interval counting

The simultaneous-congruence interval count is proved with exact constants. For any finite subset F of [A,A+T) whose members are pairwise congruent modulo q, card(F)<=T/q+1, first with natural division and then with literal real division. The proof chooses the least member only in the nonempty case and injects t into (t-min(F))/q. It therefore does not presume a compatible residue class or a solution. The empty case is separate; zero lengths and even the formal zero modulus are included.

The actual simultaneous congruence set retains both residue conditions. For any two of its members, Mathlib's Nat.mod_lcm proves congruence modulo lcm(k1,k2). Specialization yields card <=T/lcm(k1,k2)+1 in natural and real forms. Neither CRT existence, coprimality of the moduli/residues nor a prime-distribution theorem is assumed. These four supporting theorems supply the exact interval-counting ingredient in the source diagonal estimate and equation e:wellsquare9.

The reciprocal-lcm logarithmic bound was already proved at1118. Their combination into the actual squared multiplicity and progression-energy estimate is the next internal task. No source hybrid estimate, major estimate, sieve bound or headline theorem is newly declared complete. No manuscript replacement is required.

Checked declarations

Module	Named supporting theorems
GoldbachCongruenceCounting.lean	4

- `goldbach_congruence_interval_card`

- goldbach_congruence_interval_card_real
- goldbach_simultaneous_congruence_count
- goldbach_simultaneous_congruence_count_real

<!-- checkpoint-verification -->

Verification

All 1186 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4977-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3642 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1188 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1120_GoldbachCongruenceMultiplicity.md

Checkpoint 1120: Exact squared congruence-multiplicity estimate

The actual congruence multiplicity w(t), defined as the cardinality of moduli k satisfying t congruent to a(k) modulo k, is formalized. Its finite square is expanded exactly into the sum over pairs of moduli of the cardinality of their simultaneous-congruence set. No mean-square or counting estimate is assumed in this identity.

The interval count proved at1119 yields sum_{A<=t<A+T} w(t)^2 <= T sum_{k,j in K} 1/lcm(k,j)+card(K)^2. A proved subset version of the reciprocal-lcm bound therefore gives T H_M^3+card(K)^2 whenever K is any subset of [1,M]. This retains the local modulus count rather than replacing it by the global bound M^2, which would be too costly after summing over ell.

For every real V>=1, the literal strict source cutoff 0<k<V gives the exact bound T(1+log V)^3+V^2. This is a complete direct proof of the elementary squared-multiplicity estimate used in e:wellsquare9, with arbitrary residues and all natural interval endpoints. There is no coprimality, compatibility, prime-distribution or own-paper analytic assumption. The five new supporting theorems are unconditional arithmetic/finite-sum deductions. The full source progression-energy and hybrid minor estimates remain open.

A possible efficient route to the global energy uses grouping at each prime-product frequency and its number of distinct prime divisors. It will be checked explicitly before replacing the source dyadic bookkeeping in the formal proof. No manuscript edit is currently required.

Checked declarations

Module	Named supporting theorems
GoldbachCongruenceMultiplicity.lean	5

- goldbach_congruence_square_identity
- goldbach_congruence_square_lcm_bound
- goldbach_reciprocal_lcm_subset
- goldbach_congruence_square_harmonic_bound
- goldbach_congruence_square_strict_real_cutoff

<!-- checkpoint-verification -->

Verification

All 1187 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4978-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3647 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1189 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1121_GoldbachPrimeFrequencyEnergy.md`

Checkpoint 1121: Prime-product frequency multiplicity and direct energy bound

A direct repeated-frequency energy principle is now proved. A finite exponential polynomial is regrouped exactly by its integer frequencies. Orthogonality and the already proved finite weighted square inequality give integral |sum c_i e(f_i theta)|^2 <= B sum |c_i|^2 whenever every frequency has at most B preimages. Repeated frequencies are counted, not silently treated as orthogonal.

For any finite set of pairs (ell,p) with positive ell and prime p, the fibre ell*p=m injects into the actual prime-divisor set of m by projection to p. Its cardinality is bounded by omega(m), independently of the coefficients. The elementary prime-product logarithmic budget already proved in the project yields omega(m)<=log X/log 2 whenever 0<m<=X. No PNT or distribution input is used.

Combining these statements gives a checked global energy bound for arbitrary complex coefficients indexed by positive ell and primes, with the factor log X/log 2 and the literal coefficient square sum. The seven new supporting theorems contain no new independent deep input. This alternative route can bypass the source dyadic off-diagonal bookkeeping once the arithmetic moduli are grouped correctly. That grouping and the final summation remain to be proved; no full progression-energy, hybrid or major-arc source claim is closed here. No manuscript replacement is required.

Checked declarations

Module	Named supporting theorems
GoldbachPrimeFrequencyEnergy.lean	7

- `goldbach_frequency_polynomial_regroup`
- `goldbach_distinct_frequency_energy`
- `goldbach_finite_sum_square`
- `goldbach_frequency_energy_multiplicity`
- `goldbach_prime_factor_count_log`
- `goldbach_prime_product_fibre_card`
- `goldbach_prime_product_energy`

<!-- checkpoint-verification -->

Verification

All 1188 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4979-job dependency-aware build with remote project artifact reuse

disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3654 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1190 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1122_GoldbachProgressionRows.md`

Checkpoint 1122: Exact prime-progression row grouping and energy reduction

The literal progression polynomial is now regrouped by its positive integer ell and its actual prime t. Each row retains exactly the moduli present in the original finite index set I. The membership equivalence for the row modulus set and the general row-sum identity are proved, so this regrouping does not duplicate or drop arithmetic indices.

The complex row coefficient is the sum of the original d(k,ell) over exactly those row moduli satisfying the source residue condition. Its norm is bounded by the actual congruence multiplicity when every coefficient has norm at most one. The full continuous circle polynomial is proved equal to the regrouped prime-product polynomial, with the original W cutoff and ell*t<N^2 restriction intact.

Applying the checked prime-frequency multiplicity bound yields an actual integral inequality: the energy is at most log(N^2)/log 2 times the sum over ell and 1<=t<=N^2/ell of the squared row congruence multiplicity. The extension from the original primes/W-cutoff to this finite interval is proved by an explicit injection and nonnegative summation. Only positivity of the source ell indices and the normal coefficient bound are needed. No prime-distribution, PNT or sieve theorem is assumed.

These five supporting theorems complete the exact bridge between the efficient frequency argument and the original progression polynomial. The remaining summation uses the row-square estimate proved at1120 and the printed local modulus-count bound. Its source-range verification is still to be supplied before claiming the full progression-energy estimate. No new manuscript replacement is required.

Checked declarations

Module	Named supporting theorems
GoldbachProgressionRows.lean	5

- `mem_goldbachModuliAt`
- `goldbach_index_row_sum`
- `goldbach_progression_row_coefficient_norm`
- `goldbach_progression_polynomial_by_rows`
- `goldbach_progression_energy_by_row_squares`

<!-- checkpoint-verification -->

Verification

All 1189 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4980-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3659 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1191 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1123_GoldbachLogarithmicEnergy.md`

Checkpoint 1123: Global logarithmic prime-progression energy bound

The global row-square summation is now proved. For arbitrary positive ell in [1,X], arbitrary row modulus sets contained in [1,M], and the local numerical constraint card(K_ell)^2<=X/ell, the total squared congruence multiplicity over 1<=t<=X/ell is at most X(H_M^3+1)H_X. The proof uses the exact interval square bound, natural-to-real division inequality and a nonnegative harmonic sum. It does not use a dyadic mean-square assertion.

For the actual weighted prime-progression polynomial, this gives the explicit integral bound (log(N^2)/log 2) N^2(H_N^3+1)H_(N^2), provided the arithmetic indices have positive ell<=N^2, row moduli at most N and the local row-cardinality constraint. These are elementary numerical conditions to be derived from the printed source cutoff, not new deep or analytic hypotheses.

The same theorem is converted to the explicit stronger bound (8/log 2) N^2(1+log N)^5 for every natural N>=1. Prime-frequency multiplicity was proved at1121, actual row grouping at1122 and every finite-sum step is checked. The four new supporting theorems use no PNT, progression-distribution or sieve input.

Before claiming the source estimate e:remainingL2, the printed varying V_ell cutoff and L0 range must still be shown to imply these numerical conditions. The next checkpoint supplies that source bridge and the required sixth-power logarithmic form. The full hybrid minor lemma, both major estimates and the headline remain open. No manuscript replacement is required; a formal proof may use a stronger route than the printed bookkeeping.

Checked declarations

Module	Named supporting theorems
GoldbachLogarithmicEnergy.lean	4

- `goldbach_harmonic_nonneg`
- `goldbach_total_row_square_bound`
- `goldbach_progression_energy_harmonic`
- `goldbach_progression_energy_log_five`

<!-- checkpoint-verification -->

Verification

All 1190 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4981-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3663 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1192 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1124_GoldbachSourceEnergy.md`

Checkpoint 1124: Complete the source logarithmic progression-energy estimate

The full source progression-energy bound e:remainingL2 is now proved from the printed numerical ranges. The exact terminal theorem goldbach_source_progression_energy_printed_range covers every natural N>=1 with log N>=1, every delta>0, every B>=0, every real L0<N^(2-delta), all varying real V_ell below sqrt(N^2/ell)(log N)^(-B), arbitrary real prime endpoint W, arbitrary residues and arbitrary complex coefficients of norm at most one.

Every elementary range condition of the efficient energy proof is discharged. The strict coefficient-index set gives positive ell and k; the L0 range implies ell<=N^2. For each nonempty row, V_ell>1. Monotonicity and the source cutoff give moduli at most N and card(K_ell)^2<=N^2/ell. Negative or too-small V_ell create empty rows and are not improperly squared. The logarithmic factor (log N)^(-B)<=1 and the real/Natural cutoff comparisons are proved explicitly.

The resulting literal circle integral is at most (256/log 2) N^2(log N)^6, with a constant independent of L0, V, residues, coefficients and W. The underlying general estimate at1123 is stronger, of order N^2(1+log N)^5. The proof groups arithmetic moduli before counting prime-product frequency multiplicities. It uses elementary prime-divisor logarithmic bounds and the checked congruence-square bound, and requires no PNT, Bombieri–Vinogradov, Siegel–Walfisz or new independent hypothesis.

The four new supporting theorems complete an unnumbered estimate inside the source hybrid-minor proof. The full l:hybridminorarc remains open until the actual prime-square logarithmic bound is supplied uniformly in its U range and combined with Bessel. The major-arc replacement/completion, combined l:circmethod, Section11 sieve/main-term/integral estimates and headline theorem remain open. No manuscript replacement is required by this alternative stronger formal proof.

Checked declarations

Module	Named supporting theorems
GoldbachSourceEnergy.lean	4

- `goldbach_source_row_cutoffs`
- `goldbach_source_energy_numerical_ranges`
- `goldbach_source_progression_energy`
- `goldbach_source_progression_energy_printed_range`

<!-- checkpoint-verification -->

Verification

All 1191 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4982-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3667 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1193 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1125_GoldbachHybridMinorReduction.md`

Checkpoint 1125: Hybrid minor-arc assembly with the prime-prefix obligation explicit

The full progression-energy estimate is now used in the literal minor-arc integral, with natural output range n<N^2 and the original weighted progression sums. For every measurable minor set, if the actual unweighted prime-square polynomial has pointwise norm at most C N(log N)^(-(A+6)/2), then its hybrid mean square is at most C^2(256/log 2) N^4(log N)^(-A). The exponent identity, all constants, source cutoff ranges and finite-output reindexing are proved explicitly. This reduction holds uniformly for every real prime endpoint U,W subject to that bound.

The displayed pointwise bound is still an INTERNAL analytic obligation. It is not a new independent deep input, not a Lean axiom and not a proof of the source l:hybridminorarc. The primary Ren Theorem1.1 was checked: it estimates annular von-Mangoldt sums, while the Goldbach polynomial uses unweighted prime prefixes. The actual partial-summation/prime-power/cutoff bridge remains to be proved rather than hidden in an external hypothesis. See docs/GOLDBACH_PRIME_INPUT_REVIEW.md.

These two supporting theorems discharge the entire mean-square assembly once that pointwise step is supplied. The general earlier prime exponential-sum input remains as previously recorded; no new independent hypothesis or manuscript replacement is introduced. Major-arc and sieve estimates remain open and can have their elementary arithmetic bridges prepared independently.

Checked declarations

Module	Named supporting theorems
GoldbachHybridMinorReduction.lean	2

- `goldbach_minor_logarithmic_exponents`
- `goldbach_hybrid_minor_from_pointwise`

<!-- checkpoint-verification -->

Verification

All 1192 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4983-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3669 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1194 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved.

The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1126_GoldbachMajorLCMReindexing.md

Checkpoint 1126: Major-arc lcm multiplicity and divisor-weighted error reindexing

The major-arc change from the modulus k to h=lcm(k,q) is now proved exactly. The fibre of k with lcm(k,q)=h is a subset of the actual divisors of h, hence contains at most tau(h) elements. No coprimality or guessed fibre multiplicity is imposed.

For every finite modulus set and nonnegative error function E, the sum of E(lcm(k,q)) is bounded by the sum of tau(h)E(h) over any interval covering all image moduli. The source tau(h)^2 version follows from an elementary natural-power inequality. The final theorem specializes this to complex progression errors bounded by R^2 E(lcm(k,q)), obtaining the exact weighted divisor-error sum used before the weighted Bombieri–Vinogradov input.

These five supporting theorems establish only the finite reindexing and multiplicity deductions. The actual progression error estimate, the weighted consequence of the general Bombieri–Vinogradov theorem and the full major approximation remain separate internal obligations. No new distribution theorem, own-paper estimate or mathematical axiom is assumed. The already completed progression energy and earlier source scopes are unchanged. No manuscript replacement is required.

Checked declarations

Module	Named supporting theorems
GoldbachMajorLCMReindexing.lean	5

- `goldbach_lcm_fibre_subset_divisors`
- `goldbach_lcm_fibre_card`
- `goldbach_lcm_weighted_sum`
- `goldbach_lcm_weighted_square_divisor_sum`
- `goldbach_major_progression_error_sum`

<!-- checkpoint-verification -->

Verification

All 1193 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4984-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3674 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1195 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1127_GoldbachSieveResidues.md

Checkpoint 1127: Exact lower and upper sieve CRT residues and common product class

The literal lower and upper sieve residue choices in e:lowerresidue9 and e:upperresidue9 are now constructed in Lean using CRT and actual modular units. The helper for solving a*r=b modulo n constructs a bounded natural representative from a unit of ZMod n; it includes n=1. Congruence preserves the required coprimality by the actual gcd identity.

For every positive d coprime to6, a lower residue a<24d is constructed with a=11 mod24, a=1 modd and gcd(a,24d)=1. The source condition that all prime divisors of d exceed3 is separately proved to imply positivity and coprimality to6, so the source lower application is included; squarefreeness is not needed for this stronger arithmetic statement.

For ell=2v with gcd(v,6)=1 and positive d coprime to12ell, an upper residue a<12d is constructed with v*a=5 mod12, ell*a+1=0 modd and gcd(a,12d)=1. The second common residue b<12d is also constructed using only d, with b=10 mod12, b+1=0 modd and gcd(b,12d)=2. Every permitted upper residue satisfies ell*a=b mod12d. Thus the printed independence from ell and the exact gcd-two conclusion are proved, rather than inferred from examples.

All six supporting theorems are unconditional elementary arithmetic deductions. They do not assert the lower/upper sieve estimates, a sieve main-term lower bound, normalized integral positivity or the headline Goldbach theorem. The full progression-energy estimate is already proved at1124. The source hybrid minor still needs the actual weighted-annular to unweighted-prime-prefix bridge; major approximation/completion and the later analytic sieve work remain open. No new independent deep hypothesis or manuscript replacement is introduced.

Checked declarations

Module	Named supporting theorems
GoldbachSieveResidues.lean	6

- `goldbach_coprime_of_congruence`
- `goldbach_modular_linear_residue`
- `goldbach_lower_sieve_residue`
- `goldbach_lower_sieve_residue_from_support`
- `goldbach_upper_sieve_residue`
- `goldbach_upper_sieve_common_product_residue`

<!-- checkpoint-verification -->

Verification

All 1194 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4985-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3680 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1196 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1128_GoldbachFiniteAbel.md

Checkpoint 1128: Finite Abel summation and logarithmic unweighting

A finite Abel-summation norm inequality is now proved directly from Mathlib's exact summation-by-parts identity. For nonnegative decreasing real weights a_i and complex coefficients z_i whose actual partial sums have norm at most B, the weighted sum over range(n) has norm at most B*a_0. The proof includes n=0, the boundary term, the variation terms and their exact telescoping total. It is a proved inequality, not an additional summation-by-parts hypothesis.

The logarithmic removal weight is the actual function 1/log(max(2,n)). Its nonnegativity, monotonicity, initial value1/log2 and equality to1/log n at every n>=2 are all proved. Clipping at2 is only a finite-index convention: von-Mangoldt weights at0 and1 vanish. It prevents an invalid monotonicity assertion across log1=0.

These five supporting theorems supply the discrete logarithmic unweighting step. The next step identifies the unweighted Mangoldt sum with the actual prime sum up to the endpoint, apart from the already bounded higher prime-power contribution. The full Goldbach minor, major and sieve estimates are not newly claimed proved. No new independent deep input or manuscript replacement is introduced.

Checked declarations

Module	Named supporting theorems
GoldbachFiniteAbel.lean	5

- `goldbach_finite_abel_bound`
- `goldbachLogRemovalWeight_nonneg`
- `goldbachLogRemovalWeight_antitone`
- `goldbachLogRemovalWeight_zero`
- `goldbachLogRemovalWeight_of_two_le`

<!-- checkpoint-verification -->

Verification

All 1195 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4986-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3685 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1197 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1129_GoldbachPrimeUnweighting.md`

Checkpoint 1129: Actual prime logarithmic unweighting and proper-prime-power error

The weighted-to-unweighted prime bridge is now proved for the literal Goldbach polynomial. Genuine Mangoldt partial sums and their logarithmically unweighted sums are defined with exact finite endpoints. Applying the finite Abel inequality at1128 gives norm at most B/log2 when all Mangoldt partial sums have norm at most B. The zero and one coefficients are handled explicitly.

For every prime p, logarithmic removal turns the actual von-Mangoldt coefficient into exactly one. The remainder is exactly the sum over nonprimes. Its norm is bounded using the already proved elementary

proper-prime-power mass estimate; no PNT or distribution input is added. For every M<=N with N>=1, the comparison error is at most (2/log2)sqrt(N)log N.

The real endpoint convention is proved through the actual natural floor, including endpoints at primes. The terminal theorem gives, for every 0<=U<=N and every circle frequency, |P_U| <= B/log2+(2/log2)sqrt(N)log N from a uniform bound B on the genuine Mangoldt partial sums up to N. The eight supporting theorems prove the complete logarithmic-unweighting and prime-power-removal step rather than assuming it.

It remains to construct the uniform Mangoldt partial-sum bound from the existing raw annular Ren/Ghosh input on the actual complementary arcs. The halving argument, arc rescaling and logarithmic parameter choice are the next internal steps. The full source hybrid minor estimate is not yet claimed complete. No new independent deep assumption or manuscript replacement is introduced.

Checked declarations

Module	Named supporting theorems
GoldbachPrimeUnweighting.lean	8

- `goldbach_sum_range_succ_eq_Ioc`
- `goldbachMangoldtPrefix_Ioc`
- `goldbach_log_removed_mangoldt_bound`
- `goldbach_log_removed_prime_term`
- `goldbach_log_removed_mangoldt_split`
- `goldbach_prime_power_log_removal_error`
- `goldbachPrimeSquarePolynomial_floor`
- `goldbach_prime_from_mangoldt_partials`

<!-- checkpoint-verification -->

Verification

All 1196 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4987-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3693 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1198 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1130_GoldbachHalvingPartials.md`

Checkpoint 1130: Exact Mangoldt halving reconstruction and uniform partial sums

The halving step is now proved by strong induction, with the exact integer floor at every scale. A bound on F(m)-F(floor(m/2)) above a real threshold H, together with a bound for the actual small partial sums, gives |F(m)|<=2 B m+E. The proof includes zero and odd scales and uses 2 floor(m/2)<=m; no unproved dyadic reconstruction is inserted.

For the genuine Mangoldt exponential sums, the half-scale identity is proved from the exact finite intervals. The positive Fourier phase is identified with the existing Ren convention. Its annular difference

has norm exactly m times the actual normalized annular symbol. A crude small-scale bound |S_m|<=m log N is proved directly from the elementary pointwise von-Mangoldt/logarithm inequality, without PNT or a prime-counting hypothesis.

The terminal theorem assembles the uniform bound |S_m|<=2 B N+H log N for all m<=N from a common annular symbol bound B above H. These seven supporting theorems discharge the discrete reconstruction and all finite cutoff conventions. The common annular estimate will next be supplied by the existing raw Ren/Ghosh inputs using explicit rescaled complementary arcs. No new independent deep hypothesis or manuscript replacement is introduced.

Checked declarations

Module	Named supporting theorems
GoldbachHalvingPartials.lean	7

- `goldbach_halving_partial_bound`
- `goldbachMangoldtPrefix_halving`
- `goldbach_positive_quadratic_character`
- `goldbachMangoldtPrefix_annular_Ren`
- `goldbachMangoldtPrefix_annular_norm`
- `goldbachMangoldtPrefix_crude`
- `goldbach_mangoldt_partials_from_annular`

<!-- checkpoint-verification -->

Verification

All 1197 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4988-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3700 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1199 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1131_GoldbachMinorArcGeometry.md`

Checkpoint 1131: Exact minor-arc region and rescaling between annular scales

The source minor-arc region is now an explicit measurable subset of the circle: the complement of the closed major arcs of uniform radius R/N^2 about reduced rational centres with denominator at most R. Its definition using all rational lifts is proved equivalent to the existing finite source-centre definition. Circle distance is bounded by the distance between any real lifts. These facts give a proved scale transfer: membership in the source minor arcs at scale N implies the annular Ren/Ghosh exclusion at every scale $1 \le m \le N$ with parameter $R(m/N)^2$. The denominator-dependent annular exclusion and all integer lifts are covered explicitly. No prime estimate is introduced as an independent hypothesis.

Checked declarations

Module	Named supporting theorems
GoldbachMinorArcGeometry.lean	5

- `goldbachMinorArcs_measurable`
- `goldbach_minor_arcs_source_centres`
- `goldbach_circle_distance_le_real`
- `goldbach_minor_arcs_smaller_annular`
- `goldbach_minor_arcs_rescale`

<!-- checkpoint-verification -->

Verification

All 1198 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4989-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3705 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1200 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1132_GoldbachUniformAnnularMinor.md`

Checkpoint 1132: Actual prime partial sums from raw annular Ren-Ghosh inputs

The common annular bound at all relevant smaller scales is derived from the already recorded raw Ren/Ghosh input. It uses the proved rescaling of the actual source minor arcs. The assumptions are explicit elementary inequalities on a common cutoff H and radius R; in particular $R^4 \le H/2$ and $1 \le R(H/N)^2$. The strong-induction halving identity reconstructs all von Mangoldt partial sums. Finite Abel summation then removes the logarithmic prime weight, and the proved proper-prime-power mass bound accounts for the nonprime terms. The result bounds the actual unweighted prime-square polynomial for every real $0 \le U \le N$. The only independent analytic assumption remains UniformRenGhoshInputs. The paper's own pointwise prime estimate is not assumed.

Checked declarations

Module	Named supporting theorems
GoldbachUniformAnnularMinor.lean	3

- `goldbachPrimeExponentialConstant_nonneg`
- `goldbach_common_annular_minor`
- `goldbach_prime_minor_from_numeric_parameters`

<!-- checkpoint-verification -->

Verification

All 1199 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4990-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 3708 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1201 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1133_GoldbachMinorLogParameters.md`

Checkpoint 1133: Explicit logarithmic parameters for the actual minor-arc prime saving

The logarithmic parameters are explicit. For target saving t > 0, set K=t+2, H=N(log N)^(-K), and take any C ≥ 6t+4d+8, where d is the logarithmic exponent in the existing raw Ren/Ghosh input. The rescaled annular radius is exactly (log N)^(C-2K). Its quarter-power saving absorbs both d and the target t. The small-scale term H log N equals N(log N)^(-t-1). Once the elementary eventual scale and square-root budgets hold, the actual prime polynomial has the uniform bound Z N(log N)^(-t), where Z=(2P+3)/log 2 and P is the sum of the proved Ren/Ghosh constants. Every C above the threshold is allowed. No own endpoint estimate is an assumption.

Checked declarations

Module	Named supporting theorems
GoldbachMinorLogParameters.lean	5

- `goldbachMinorPrimeConstant_pos`
- `goldbach_log_minor_radius_identity`
- `goldbach_log_minor_annular_budget`
- `goldbach_log_minor_small_cutoff`
- `goldbach_prime_minor_log_parameters`

<!-- checkpoint-verification -->

Verification

All 1200 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4991-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3713 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1202 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifre-

quency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1134_GoldbachMinorEventualCutoffs.md

Checkpoint 1134: Eventual cutoff and proper-prime-power budgets without extra inputs

All numeric assumptions needed by the prime minor-arc estimate are now proved eventually. The proof uses the formal logarithm-versus-positive-power limit with explicit constants. For every fixed target t and radius exponent C, a natural threshold $N_0 \geq 4$ works simultaneously for log N ≥ 1, H=N(log N)^(-(t+2)) ≥ max(4,startingScale,exp 1), (log N)^(4C) ≤ H/2, and sqrt(N) log N ≤ N(log N)^(-t). The threshold is independent of the real prime cutoff U, the circle frequency, the progression cutoff W, the coefficient family, and the residue classes. This discharges the finite-scale and proper-prime-power budgets without introducing a new analytic hypothesis.

Checked declarations

Module	Named supporting theorems
GoldbachMinorEventualCutoffs.lean	2

- `goldbach_minor_real_eventual_budgets`
- `goldbach_minor_eventual_numeric_parameters`

<!-- checkpoint-verification -->

Verification

All 1201 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4992-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3715 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1203 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1135_GoldbachPrimeMinorUniform.md

Checkpoint 1135: Uniform actual prime minor bound and the printed large-cutoff estimate

For each t > 0 and every C ≥ 6t+4d+8, one natural $N_0 \geq 4$ now gives the actual uniform circle estimate |P_U(θ)| ≤ Z N(log N)^(-t) on the literal source minor arcs, simultaneously for all real 0 ≤ U ≤ N. Quotient representatives are handled explicitly, so this is a theorem about the circle polynomial, not a chosen real lift. The threshold is independent of U, W, θ and all progression/coefficient data. Taking t=3s+1 also proves the stronger displayed source estimate |P_U| ≤ Z (U/log U)(log N)^(-s) whenever 2 ≤ U ≤ N and U ≥ N(log N)^(-2s). Thus the printed large-cutoff prime estimate is covered as well as the bound needed for Bessel. Only the existing raw UniformRenGhoshInputs is assumed.

Checked declarations

Module	Named supporting theorems
GoldbachPrimeMinorUniform.lean	4

- `goldbachMinorRadiusThreshold_pos`
- `goldbach_prime_minor_uniform`
- `goldbach_prime_minor_source_cutoff_algebra`
- `goldbach_prime_minor_source_high_range`

<!-- checkpoint-verification -->

Verification

All 1202 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4993-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3719 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1204 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1136_GoldbachHybridMinorComplete.md`

Checkpoint 1136: Complete source hybrid minor-arc estimate from raw Ren-Ghosh inputs

The full source hybrid minor-arc lemma l:hybridminorarc is now a complete implication under the existing independent raw UniformRenGhoshInputs. For every A>0, take C_0=3A+4d+26. Every real C≥C_0 is allowed; one N_0≥4 then works uniformly in all printed coefficient, residue and cutoff data. The actual mean square over n<N^2 is at most K N^4(log N)^(-A), with K=Z^2(256/log 2) and Z=(2P+3)/log 2. The theorem covers every B≥0, δ>0, the strict printed L_0 and V_ell ranges, all coefficients of norm at most1, all residue families, every real 0≤U≤N and arbitrary real W. Thus it includes the source ranges 2≤U≤N and 2≤W≤N^2. The minor set is the exact measurable complement of the closed source arcs of uniform radius (log N)^C/N^2. All Abel, proper-prime-power, halving, scale-transfer, logarithmic-parameter, progression-energy and Bessel steps are proved. No own-paper endpoint is assumed. This closes only the hybrid minor lemma; the full circle-method approximation and Section11 endpoints remain open.

Checked declarations

Module	Named supporting theorems
GoldbachHybridMinorComplete.lean	2

- `goldbachHybridMinorConstant_pos`
- `goldbach_hybrid_minor_complete`

<!-- checkpoint-verification -->

Verification

All 1203 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4994-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3721 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1205 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1137_GoldbachMajorStrictEndpoint.md`

Checkpoint 1137: Exact strict progression cutoff and the possible prime endpoint

The strict source cutoff ell*t<N^2 is retained. When W≤N^2/ell, the actual progression cutoff equals the ordinary prime progression cutoff with the possible equality ell*t=N^2 removed. There is at most one such prime, and the corresponding actual complex exponential-sum difference has norm at most1. Replacing W by min(W,N^2/ell) is proved to leave the strict source cutoff unchanged. This corrects the boundary bookkeeping required before partial summation: the manuscript sentence saying the strict condition is automatic under a non-strict bound is false at equality. No main theorem change is inferred; the finite endpoint term is explicit and still has to be included in the eventual major-arc bound. The source archive and selected manuscript are preserved unchanged.

Checked declarations

Module	Named supporting theorems
GoldbachMajorStrictEndpoint.lean	5

- `mem_goldbachPrimeProgressionCutoff`
- `goldbach_progression_strict_endpoint`
- `goldbach_progression_endpoint_card`
- `goldbach_progression_endpoint_error`
- `goldbach_progression_cutoff_min`

<!-- checkpoint-verification -->

Verification

All 1204 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4995-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3726 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1206 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough

full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1138_GoldbachPrimeCountingModel.md`

Checkpoint 1138: Actual prime progression counts and logarithmic-integral models

The unweighted prime progression coefficient and its actual counting function are defined from the existing real prime cutoff, with an exact finite-sum identity. The logarithmic integral is the actual integral from2 to x of1/log t. Its derivative, the progression model Li(x)/phi(k), the corresponding density, and the zero model value at2 are proved. Integrability and continuity are established on the correct nonsingular range x≥2. These are genuine functions; no weighted exponential estimate or distribution theorem is assumed.

Checked declarations

Module	Named supporting theorems
GoldbachPrimeCountingModel.lean	9

- `goldbachPrimeProgressionCoeff_zero`
- `goldbachPrimeProgressionCoeff_one`
- `goldbach_prime_progression_count_sum`
- `goldbach_log_density_continuousOn`
- `goldbachLi_hasDerivAt`
- `goldbachLi_two`
- `goldbachPrimeProgressionModel_two`
- `goldbachPrimeProgressionModel_hasDerivAt`
- `goldbachPrimeProgressionDensity_continuousOn`

<!-- checkpoint-verification -->

Verification

All 1205 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4996-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3735 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1207 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypoth-

esis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1139_GoldbachCountingPartialSummation.md`

Checkpoint 1139: Quantitative partial summation from actual prime-count discrepancies

Quantitative partial summation is proved for actual finite counting sums and a differentiable model with zero lower value at2. The prime at2 remains in the finite sum; no artificial lower-bound convention is assumed. The identity is derived from Mathlib Abel summation and integration by parts, with all integrability obligations proved. A unit-norm phase with derivative bounded by D transports a uniform counting discrepancy B to error at most B(1+D(U-2)). Specializing to the actual prime progression coefficient and actual Li/phi model gives the needed unweighted prime exponential-sum implication. This is an internal proved conversion, not an external weighted prime estimate.

Checked declarations

Module	Named supporting theorems
GoldbachCountingPartialSummation.lean	4

- `goldbach_counting_model_error_identity`
- `goldbach_counting_partial_error_bound`
- `goldbach_prime_weighted_count_sum`
- `goldbach_prime_progression_partial_error`

<!-- checkpoint-verification -->

Verification

All 1206 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4997-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3739 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1208 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1140_GoldbachPrimePhaseErrors.md`

Checkpoint 1140: Actual linear and quadratic major-arc errors from prime counts

The actual positive quadratic and linear logarithmic integrals are defined. The positive linear oscillation has its exact real derivative, continuity and unit norm; the existing quadratic derivative is reused with the explicit sign reversal. The actual progression-density integrals equal the logarithmic integrals divided by phi(k), with the factor pulled through the Bochner integral rigorously. The proved counting partial-summation identity now gives quadratic error B(1+4*pi*|gamma|*U^2) and linear error B(1+2*pi*|beta|*U). These are deductions from ordinary prime-count discrepancies for the literal finite sums. No SW/BV theorem, own weighted estimate or model equality is assumed.

Checked declarations

Module	Named supporting theorems
GoldbachPrimePhaseErrors.lean	7

- `goldbach_linear_hasDerivAt`
- `goldbachLinearDerivative_continuous`
- `goldbachLinearDerivative_norm`
- `goldbach_square_density_integral`
- `goldbach_linear_density_integral`
- `goldbach_prime_quadratic_count_error`
- `goldbach_prime_linear_count_error`

<!-- checkpoint-verification -->

Verification

All 1207 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4998-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3746 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1209 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1141_GoldbachSquareMajorCounts.md`

Checkpoint 1141: Exact square major-arc model and its prime-count discrepancy bound

The actual positive prime-square polynomial is identified with the previously proved finite square symbol under the exact phase reflection. Actual prime progression cutoffs agree with residue fibres modulo q. The exceptional nonunit primes are bounded unconditionally by q. The existing finite Gauss assembly then gives the literal source square major-arc model s(-a,q) I_U(gamma), with error at most q+phi(q) B(1+4*pi*|gamma|*U²) when the ordinary reduced prime-count discrepancies up to U are at most B. Both the sign and the totient factor are proved, and the nonunit remainder is retained explicitly. A raw Siegel–Walfisz theorem will supply the ordinary counting discrepancy; the own square approximation is not an external input.

Checked declarations

Module	Named supporting theorems
GoldbachSquareMajorCounts.lean	6

- `goldbach_prime_positive_symbol`
- `goldbach_prime_residue_cutoff`
- `goldbach_square_prime_residue_sum`
- `goldbach_prime_nonunit_card`
- `goldbach_prime_nonunit_symbol`

- goldbach_square_major_from_counts

<!-- checkpoint-verification -->

Verification

All 1208 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 4999-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3752 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1210 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1142_GoldbachMajorLCMQFibres.md

Checkpoint 1142: Fixed-denominator lcm fibres and reduction to ordinary BV

For every fixed major denominator q, a fibre of k↦lcm(k,q)=h has cardinality at most q. The injection is k↦h/k; its values lie in1..q and determine k exactly. Consequently the sum of unweighted distribution errors over k is at most q times the ordinary sum over moduli h. If q≤R and each progression error is at most R^2 times the error for its lcm modulus, the total is at most R^3 times the ordinary unweighted distribution-error sum. This proves an efficient route from classical unweighted BV to the actual progression approximation: all losses depending on q are powers of R=(log N)^C and can be absorbed in the requested logarithmic saving. No divisor-weighted BV theorem or divisor-moment input is needed for this route.

Checked declarations

Module	Named supporting theorems
GoldbachMajorLCMQFibres.lean	4

- goldbach_lcm_quotient_range
- goldbach_lcm_fibre_card_le_q
- goldbach_lcm_unweighted_sum
- goldbach_major_error_from_unweighted_distribution

<!-- checkpoint-verification -->

Verification

All 1209 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5000-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3756 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1211 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input

includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1143_GoldbachMajorCompatibleResidues.md

Checkpoint 1143: Compatible lcm unit residues and generalized CRT

The reduced residue classes modulo lcm(k,q) compatible with the prescribed residue modulo k are actual finite sets of natural representatives. Their reduction modulo q is injective, so their cardinality is at most q. The generalized CRT constructs every class compatible modulo gcd(k,q), with exact representative bounds and both unit conditions. The converse compatibility and unit properties are proved. These facts retain shared prime factors of k and q; no coprime-moduli simplification is imposed.

Checked declarations

Module	Named supporting theorems
GoldbachMajorCompatibleResidues.lean	5

- `mem_goldbachMajorResidues`
- `goldbach_major_residues_modq_injective`
- `goldbach_major_residues_card`
- `goldbach_major_compatible_lift`
- `goldbach_major_residue_compatible`

<!-- checkpoint-verification -->

Verification

All 1210 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5001-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3761 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1212 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1144_GoldbachMajorUpsilonIdentity.md

Checkpoint 1144: Exact Upsilon normalization from compatible lcm residues

The compatible lcm residues are proved to map bijectively onto the actual unit residues modulo q satisfying the gcd compatibility condition. The map preserves every additive character. The finite character sum divided by phi(lcm(k,q)) is therefore exactly Upsilon(q,k,a,-alpha)/phi(k), with the source negative argument and both totient factors. Shared-prime cases are included through generalized CRT, not excluded. No analytic distribution hypothesis is used in this arithmetic identity.

Checked declarations

Module	Named supporting theorems
GoldbachMajorUpsilonIdentity.lean	6

- `goldbach_reduction_val`
- `goldbach_major_residue_image`
- `goldbach_major_castq_injective`
- `goldbach_major_residue_sum`
- `goldbach_major_compatibility_character`
- `goldbach_major_upsilon_identity`

<!-- checkpoint-verification -->

Verification

All 1211 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5002-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3767 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1213 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1145_GoldbachRawPrimeDistribution.md`

Checkpoint 1145: Precise ordinary prime-count SW-BV inputs and elementary budgets

Precise independent general SW and BV interfaces are recorded only for actual ordinary unweighted prime counts relative to the actual logarithmic integral divided by phi(h). Their complex discrepancy is proved equal to the usual real absolute error. SW is uniform in the reduced residue, all endpoints up to N and every fixed logarithmic modulus range. BV has the classical endpoint maximum and an unweighted modulus sum up to sqrt(X)(log X)^(-L), with an honest error majorant for the actual counts. Neither interface mentions a quadratic phase, the paper's coefficient family or a circle-method conclusion. The external theorems remain unproved propositions, not Lean axioms. The source square approximation is reduced numerically to R+(1+4*pi)R² times its ordinary count discrepancy; the necessary logarithmic power identity and eventual radius-error absorption are proved. The full SW square application is the next internal deduction. The fixed-q lcm bound permits ordinary BV directly, with no extra divisor-weighted distribution input.

Checked declarations

Module	Named supporting theorems
GoldbachRawPrimeDistribution.lean	4

- `goldbach_prime_count_discrepancy_real`
- `goldbach_major_radius_eventual_budget`
- `goldbach_square_major_numeric`
- `goldbach_major_log_product`

<!-- checkpoint-verification -->

Verification

All 1212 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5003-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3771 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1214 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1146_GoldbachSquareMajorSW.md`

Checkpoint 1146: Full uniform SW square major-arc approximation from ordinary prime counts

The full actual square major approximation is now proved from the ordinary unweighted SW interface. For every t>0 and C≥0, a positive constant and one natural threshold work simultaneously for all positive q≤(log N)^C, all integer numerators, every real2≤U≤N and every |gamma|≤(log N)^C/N². The model is exactly s(-a,q) I_U(gamma), with the actual logarithmic integral. The error is at most K N(log N)^(-t). Every prime cutoff, rational sign, residue normalization, nonunit term and eventual power absorption is proved. No large-U-only restriction or weighted square approximation is an assumption. The classical ordinary SW theorem remains the explicit independent unproved input.

Checked declarations

Module	Named supporting theorems
GoldbachSquareMajorSW.lean	1

- `goldbach_square_major_siegel_walfisz`

<!-- checkpoint-verification -->

Verification

All 1213 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5004-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3772 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1215 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1147_GoldbachLinearResidueFibres.md`

Checkpoint 1147: Exact actual linear progression fibres and rational phases

The actual good prime progression cutoff is exactly regrouped into compatible residue fibres modulo lcm(k,q). Every fibre is proved equal to the ordinary prime progression cutoff for that larger modulus. The positive Fourier character is identified with its real exponential phase, and the rational phase splits with the precise additive character and ell*gamma phase. Consequently the actual good linear sum has the exact finite residue decomposition used by the source, with all original prime endpoints retained. No analytic hypothesis is used in these grouping identities.

Checked declarations

Module	Named supporting theorems
GoldbachLinearResidueFibres.lean	5

- `goldbach_good_linear_maps_to`
- `goldbach_good_linear_fibre`
- `goldbach_positive_linear_character`
- `goldbach_linear_rational_phase`
- `goldbach_good_linear_residue_grouping`

<!-- checkpoint-verification -->

Verification

All 1214 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5005-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3777 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1216 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The

classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1148_GoldbachLinearMajorCounts.md

Checkpoint 1148: Actual linear Upsilon model with every nonunit and endpoint error

The actual good linear prime sum is approximated by the exact source Upsilon coefficient times the logarithmic integral, with error at most q B(1+2*pi*|ell*gamma| W) from ordinary prime-count discrepancies for the lcm modulus. All compatible classes are present. Primes nonunit modulo lcm(k,q) are proved to divide q using the source coprime residue condition, and their total norm is at most q. The strict source endpoint contributes at most1. Thus the actual full progression polynomial has the source model with the explicit error1+q+q B(1+2*pi*|ell*gamma| W). No omitted exceptional prime or equality endpoint is hidden in a distribution assumption.

Checked declarations

Module	Named supporting theorems
GoldbachLinearMajorCounts.lean	5

- `goldbach_good_linear_model_error`
- `goldbach_linear_nonunit_card`
- `goldbach_linear_nonunit_error`
- `goldbach_closed_linear_fourier`
- `goldbach_linear_major_from_counts`

<!-- checkpoint-verification -->

Verification

All 1215 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5006-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3782 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1217 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1149_GoldbachLinearSourceIntegral.md

Checkpoint 1149: Exact strict Euclidean source integral and numeric major-arc error

The actual strict Euclidean source integral J_{ell,W} is defined using2≤v≤W and ell*v<N². Its domain is proved almost everywhere equal to (2,min(W,N²/ell)], so the integral equals the actual closed logarithmic integral at the clamped endpoint. Both the prime cutoff and the integral are therefore matched exactly under clamping, including equality cases. The actual full linear progression model is proved for every real W≥2 whenever N²/ell≥2. On a source major arc its error is at most1+q+(1+2*pi)R² B. Every domain and boundary correction is retained explicitly; the strict source condition is never replaced by an automatic non-strict condition.

Checked declarations

Module	Named supporting theorems
GoldbachLinearSourceIntegral.lean	4

- `goldbach_linear_source_domain_ae`
- `goldbach_linear_source_integral`
- `goldbach_linear_source_major_from_counts`
- `goldbach_linear_source_major_numeric`

<!-- checkpoint-verification -->

Verification

All 1216 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5007-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3786 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1218 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1150_GoldbachLinearBVRow.md`

Checkpoint 1150: Actual ordinary-BV row approximation with independent prime cutoffs

The actual source linear major model and error are explicit functions. The sum of error norms over an arbitrary finite row of positive moduli is bounded by (1+q)card(K)+(1+2*pi)R³ times the ordinary unweighted modulus-distribution error sum. Each progression may have its own real cutoff W_k≥2 and its own coprime residue. Applying RawPrimeBombieriVinogradovInput gives one level, constant and starting scale chosen before q, all residue families and cutoffs. The remaining terms1+q are exactly the retained nonunit/strict-endpoint contribution, not an unproved distribution assumption. This completes the genuine BV row conversion. Matching the paper's source level and summing over ell are the next internal steps.

Checked declarations

Module	Named supporting theorems
GoldbachLinearBVRow.lean	2

- `goldbach_linear_error_sum_unweighted`
- `goldbach_linear_major_bv_row`

<!-- checkpoint-verification -->

Verification

All 1217 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5008-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 3788 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1219 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1151_GoldbachBVSourceParameters.md`

Checkpoint 1151: Printed source ranges imply the ordinary BV level uniformly

The source arithmetic scale X_ell=N²/ell is proved to satisfy N^delta<X_ell≤N² under the printed strict ell range. The exact logarithmic comparisons delta log N≤log X_ell≤2 log N follow. Taking B≥C+L+1 and log N≥2^L puts V_ell R below the classical BV level sqrt(X_ell)(log X_ell)^(-L). The logarithmic saving at X_ell is converted to the saving at N with explicit factor delta^(-T). One natural threshold, independent of ell and the major denominator, is proved to ensure the required starting-scale and logarithmic inequalities. The lcm moduli then lie in the BV range. No source range is strengthened silently.

Checked declarations

Module	Named supporting theorems
GoldbachBVSourceParameters.lean	5

- `goldbach_progression_scale_logs`
- `goldbach_bv_log_level`
- `goldbach_bv_log_saving`
- `goldbach_bv_source_eventual_threshold`
- `goldbach_lcm_bv_level`

<!-- checkpoint-verification -->

Verification

All 1218 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5009-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3793 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1220 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The

classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1152_GoldbachMajorSourceCounts.md

Checkpoint 1152: Actual source row sums and coefficient-family cardinality bounds

The actual active row set lies in1..N^2, its reciprocal sum is bounded by the harmonic number, and the sum of X_ell=N^2/ell is at most3N^2 log N. The number of active rows is bounded by max(L0,0), including empty and negative-cutoff cases. Exact row regrouping and Cauchy–Schwarz give card(I)^2≤3N^2 max(L0,0) log N from the printed square-root row cutoffs. These are proved finite-count bounds for the actual coefficient family; they will absorb every retained prime endpoint/nonunit term with the source N^(-delta) margin.

Checked declarations

Module	Named supporting theorems
GoldbachMajorSourceCounts.lean	5

- `goldbach_source_row_subset`
- `goldbach_source_row_reciprocal`
- `goldbach_source_row_scale_sum`
- `goldbach_source_row_card`
- `goldbach_source_index_card_square`

<!-- checkpoint-verification -->

Verification

All 1219 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5010-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3798 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1221 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1153_GoldbachMajorFiniteErrorAbsorption.md

Checkpoint 1153: Uniform absorption of all strict-endpoint and nonunit prime errors

The actual coefficient-family cardinality satisfies card(I)^2≤3N^(4-delta) log N under the printed strict L0 and logarithmic V_ell ranges. An explicit logarithm-versus-power budget is proved eventually. Combining the square bounds proves2(log N)^C card(I)≤N^2(log N)^(-t) for every fixed desired saving, with one threshold independent of all cutoffs and residue data. Thus every retained1+q term from a strict prime endpoint or a prime dividing q is absorbed uniformly, since1+q≤2R. This closes the previously identified finite-boundary issue without changing the stated approximation theorem.

Checked declarations

Module	Named supporting theorems
GoldbachMajorFiniteErrorAbsorption.lean	4

- `goldbach_source_index_card_printed`
- `goldbach_major_finite_error_eventual`
- `goldbach_major_finite_error_numeric`
- `goldbach_source_finite_error_absorbed`

<!-- checkpoint-verification -->

Verification

All 1220 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5011-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3802 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1222 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1154_GoldbachLinearMajorBV.md`

Checkpoint 1154: Full actual source BV progression approximation uniformly in every cutoff

The full actual source progression approximation is proved from ordinary unweighted BV. For every desired saving t>0, every C≥0 and every delta>0, one B0≥0, positive constant and natural threshold work uniformly for every B≥B0, every printed strict L0 and V_ell range, every source major denominator/numerator/offset, all coprime residue families and all independently chosen real progression cutoffs W_i≥2. The sum of the actual source error norms is at most K N²(log N)^(-t). The models use the exact Upsilon factor and exact strict Euclidean J integral. The fixed-q lcm fibre bound, source BV level, logarithmic conversion, ell-scale sum and every endpoint/nonunit error are all discharged. Only RawPrimeBombieriVinogradovInput remains independent and unproved. This proves the source pointwise linear estimate, not yet its full circle mean-square assembly.

Checked declarations

Module	Named supporting theorems
GoldbachLinearMajorBV.lean	2

- `goldbach_bv_aggregate_log_product`
- `goldbach_linear_major_bombieri_vinogradov`

<!-- checkpoint-verification -->

Verification

All 1221 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5012-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3804 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1223 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1155_GoldbachMajorArcGeometry.md`

Checkpoint 1155: Exact source major-arc geometry measure and disjointness

The major set is exactly the complement of the already formalized source minor set and equals the finite union of the source closed balls of uniform radius R/N^2. Its measurability, rational-centre cardinality at most$4R^2$ and normalized Haar measure at most$8R^3/N^2$ are proved. Exact real lifts turn circle ball membership into the integer-shift inequalities used in the earlier rational separation theorem. Consequently the source major balls are pairwise disjoint once$2R^3<N^2$. A single eventual threshold proves this separation and radius<1/2 for R=(log N)^C and every fixed C≥0. No arc, wrap-around or measure convention is assumed.

Checked declarations

Module	Named supporting theorems
GoldbachMajorArcGeometry.lean	10

- `goldbachMajorArcs_measurable`
- `mem_goldbachMajorArcs`
- `goldbach_major_arcs_union`
- `goldbach_circle_volume_eq_haar`
- `goldbach_major_ball_measure`
- `goldbach_source_centres_card`
- `goldbach_major_arcs_measure`
- `goldbach_circle_ball_real_lift`
- `goldbach_major_balls_disjoint`
- `goldbach_major_geometry_eventual`

<!-- checkpoint-verification -->

Verification

All 1222 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5013-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3814 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1224 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other

previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1156_GoldbachMajorModelRegularity.md`

Checkpoint 1156: Measurable source coordinates and actual Euclidean model bounds

A canonical measurable local circle coordinate is constructed at every rational centre. Its real absolute value is exactly the circle distance, it lies in[-1/2,1/2), and it reconstructs the original circle point. The actual square and linear logarithmic integrals are strongly measurable in frequency, by the Bochner parameter-integral theorem. The strict source linear integral inherits this regularity through its proved clamping identity. Their uniform norm bounds are proved from the actual1/log weight: |I_U|≤U/log2 and |J_ell,W|≤W/log2 in the required range. No abstract Euclidean model, measurable representative, or size estimate is assumed.

Checked declarations

Module	Named supporting theorems
GoldbachMajorModelRegularity.lean	12

- `goldbachArcCoordinate_measurable`
- `goldbachArcCoordinate_coe`
- `goldbachArcCoordinate_bounds`
- `goldbachArcCoordinate_abs`
- `goldbachArcCoordinate_reconstruct`
- `goldbachSquareIntegral_stronglyMeasurable`
- `goldbachLinearIntegral_stronglyMeasurable`
- `goldbachLinearSourceIntegral_stronglyMeasurable`
- `goldbach_phase_log_weight_bound`
- `goldbachSquareIntegral_norm`
- `goldbachLinearIntegral_norm`
- `goldbachLinearSourceIntegral_norm`

<!-- checkpoint-verification -->

Verification

All 1223 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5014-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3826 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1225 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The

classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1157_GoldbachCircleMajorModels.md

Checkpoint 1157: Constructed measurable source major-arc models with all-Lp regularity

The source major-arc models are now actual measurable functions on the circle, constructed as finite indicators of the exact source major balls times the genuine local-coordinate integrals. A general localization theorem proves their precise source value on each arc using the checked pairwise disjointness. Uniform Euclidean bounds prove membership in every Lp, including infinity; this is not a measurable-model hypothesis. The square model has the exact Gauss coefficient and norm at most N/log2 on the major set. The progression model has the exact Upsilon/phi coefficient and strict source J integral. All quotient representatives, boundary values, and finite model sums are explicit.

Checked declarations

Module	Named supporting theorems
GoldbachCircleMajorModels.lean	8

- `goldbachArcModel_stronglyMeasurable`
- `goldbachArcModel_norm_bound`
- `goldbachArcModel_memLp`
- `goldbachArcModel_on_arc`
- `goldbach_square_model_norm_euclidean`
- `goldbachSquareMajorModel_memLp`
- `goldbachProgressionMajorModel_memLp`
- `goldbachSquareMajorModel_norm_on_major`

<!-- checkpoint-verification -->

Verification

All 1224 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5015-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3834 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1226 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1158_GoldbachCircleMajorEstimates.md

Checkpoint 1158: Full SW-BV bounds for the constructed exact circle models

The completed actual real-frequency SW and BV estimates are transported to the constructed source circle models. At every point of the exact source major set, the selected local coordinate reconstructs that point with its canonical reduced numerator and denominator. SW therefore bounds the actual square polynomial minus its circle model uniformly. BV bounds the sum of the actual progression-minus-model

norms over the entire source coefficient family, with independent real prime cutoffs and all source ranges. All constants and thresholds precede the arithmetic/cutoff data, and the active ell scales are proved at least2 for model integrability. No arc matching, choice of rational approximant, or global model comparison remains assumed.

Checked declarations

Module	Named supporting theorems
GoldbachCircleMajorEstimates.lean	2

- `goldbach_square_circle_major_sw`
- `goldbach_linear_circle_major_bv`

<!-- checkpoint-verification -->

Verification

All 1225 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5016-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3836 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1227 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1159_GoldbachMajorBesselAssembly.md`

Checkpoint 1159: Restricted Bessel assembly for the two major-arc replacements

The restricted Bessel inequality is proved for arbitrary measurable circle functions in L^2, by applying the finite Fourier Bessel inequality to their indicators. A uniform major-arc error costs the normalized measure of the major arcs. A bounded measurable multiplier costs its squared bound times the full progression energy.

For the actual two-factor replacement, the exact identity $PF-QG=Q(F-G)+(P-Q)F$ is integrated and the two squared errors are bounded by $2(BQ^2\ BE^2\ \mu(\text{major})+BP^2\ E)$. All integrability and finite Fourier summations are proved. No endpoint approximation is assumed. The independent ordinary SW and BV inputs are not used by this general assembly theorem.

Checked declarations

Module	Named supporting theorems
GoldbachMajorBesselAssembly.lean	4

- `goldbach_restricted_function_bessel`
- `goldbach_restricted_uniform_bessel`
- `goldbach_restricted_measurable_multiplier_bessel`
- `goldbach_two_model_major_bessel`

<!-- checkpoint-verification -->

Verification

All 1226 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5017-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3840 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1228 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1160_GoldbachMajorWeightedAssembly.md`

Checkpoint 1160: Weighted source coefficient identity and exact major-arc budget

The weighted progression major model is constructed from the actual source models and proved to lie in every Lp space on the circle. For coefficients of norm at most one, its difference from the actual weighted prime polynomial is bounded by the sum of the individual progression error norms.

The source sum of weighted major-arc Fourier integrals is proved exactly equal to the Fourier integral of the two weighted factors. All finite sum exchanges and integrability conditions are supplied. The final numerical identity combines the major-arc measure bound$8R^3/N^2$ with the SW loss N(log N)^(-(A+6)/2), the BV loss N^2(log N)^(-(A+3C)/2), and the proved progression-energy bound(256/log2)N^2(log N)^6. It yields an explicit constant times N^4(log N)^(-A). No own-paper endpoint is an input.

Checked declarations

Module	Named supporting theorems
GoldbachMajorWeightedAssembly.lean	5

- `goldbachWeightedProgressionMajorModel_memLp`
- `goldbach_weighted_major_error_le_sum`
- `goldbach_major_weighted_coefficient_identity`
- `goldbach_major_square_log_identity`
- `goldbach_major_mean_square_budget`

<!-- checkpoint-verification -->

Verification

All 1227 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5018-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3845 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1229 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough

full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1161_GoldbachMajorApproximationComplete.md`

Checkpoint 1161: Complete source major-arc approximation under ordinary SW and BV

The full source major-arc approximation, stable label l:majorapproximation9, is now a complete Lean implication under only RawPrimeSiegelWalfiszInput and RawPrimeBombieriVinogradovInput. These independent hypotheses concern ordinary unweighted prime progression counts; neither contains an oscillatory model, the paper coefficient family or its endpoint conclusion. Their analytic proofs remain external.

For every A>0, C≥0 and delta>0, the theorem selects B0, a positive error constant and a sufficiently large integer scale before all source data. It covers every B≥B0, every strict printed L0 and V range, every reduced progression residue and bounded complex coefficient, every real2≤U≤N and every realW≥2. In particular it covers the paper's W≤N^2 range. The sum over all natural n<N^2 is exactly the source weighted sum of model-replacement integrals on the actual circle major arcs, bounded by K N^4(log N)^(-A).

The actual Fourier signs, rational centres, normalized measure, strict prime cutoff, nonunit primes, Upsilon coefficient, genuine Euclidean integrals and measurable circle models are all supplied. SW uses exponent(A+6)/2; BV uses(A+3C)/2. Restricted Bessel, the measure bound8R^3/N^2 and the previously proved progression-energy bound give the explicit final constant2(8 KE^2/(log2)²+KP^2·256/log2).

A small proof correction is required: W≤N^2/ell does not make ell*t<N^2 automatic at equality. The endpoint and primes dividingq contribute at most1+q to each individual error. All these finite errors are proved absorbed uniformly. The separate replacement files retain the lemma statement and equation labels and give the checked unweighted fixed-q fibre route. The selected manuscript is unchanged.

This closes one paper-level claim after review of its exact source statement and quantifiers. The ledger is now13 proved,26 complete conditional implications,1 quoted external theorem and14 open. The full circle-method approximation l:circmethod still requires completion/arithmetic evaluation and the smooth Euclidean-cutoff stage. Section11 and the headline Goldbach theorem remain open.

Checked declarations

Module	Named supporting theorems
GoldbachMajorApproximationComplete.lean	1

- `goldbach_major_approximation_complete`

<!-- checkpoint-verification -->

Verification

All 1228 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5019-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3846 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1230 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional

Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1162_GoldbachArithmeticEuler.md

Checkpoint 1162: Absolute convergence and exact source arithmetic Euler factor

The actual source R-smooth coefficient series is defined with the inclusive prime cutoff p≤R. Its finite support is proved inside the explicit divisor set from the earlier arbitrary-multiplier Euler theorem. Absolute summability is proved for the norm series, with no analytic distribution input.

The complete sum is identified exactly with the source two-adic factor times the product of the actual unit-pair solution counts for2<p≤R. The floor conversion includes primes equal to R. An exact finite split at every real denominator cutoff Y is supplied. This closes the algebraic identity and convergence clauses of l:arithmeticcoefficients9; its quantitative bounds and the full completion/smoothing estimates remain open. The already completed SW/BV major replacement and hybrid minor scopes are unchanged.

Checked declarations

Module	Named supporting theorems
GoldbachArithmeticEuler.lean	6

- `goldbach_prime_inclusive_cutoff`
- `goldbach_smooth_coefficient_zero_off_bound`
- `goldbach_smooth_coefficient_abs_summable`
- `goldbach_smooth_series_divisor_sum`
- `goldbach_arithmetic_factor_euler`
- `goldbach_arithmetic_factor_finite_split`

<!-- checkpoint-verification -->

Verification

All 1229 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5020-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3852 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1231 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1163_GoldbachEuclideanDecay.md

Checkpoint 1163: Genuine reciprocal-log Euclidean integral decay

The actual reciprocal-log weight is proved antitone on every source interval. Mathlib's general vector-valued second mean value theorem, proved in the pinned dependency, transfers the existing unweighted quadratic oscillatory bound to the genuine source integral: norm I_U(gamma)≤(2 quadraticIntegralConstant/log2)/sqrt(abs gamma).

The actual linear exponential primitive is differentiated and integrated exactly. This gives norm of its interval integral≤1/(pi abs beta), then norm J_U(beta)≤1/(pi log2 abs beta). The strict-domain source integral is identified with its clamped endpoint and satisfies norm J_ell,W(gamma)≤1/(pi log2 ell abs gamma). No van der Corput or Fourier-decay assertion is assumed externally. The results are uniform for every real cutoff in the printed range. The quantitative arithmetic bounds, smoothing assembly and full l:circmethod remain open.

Checked declarations

Module	Named supporting theorems
GoldbachEuclideanDecay.lean	6

- `goldbach_inverse_log_antitone`
- `goldbach_quadratic_interval_decay`
- `goldbachSquareIntegral_decay`
- `goldbach_linear_interval_decay`
- `goldbachLinearIntegral_decay`
- `goldbachLinearSourceIntegral_decay`

<!-- checkpoint-verification -->

Verification

All 1230 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5023-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3858 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1232 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1164_GoldbachTotientHarmonic.md`

Checkpoint 1164: Unconditional reciprocal-totient harmonic bounds

The reciprocal totient is bounded by the earlier actual conductor-correlation product divided by n. Its previously proved weighted harmonic estimate gives sum_{n≤D}1/phi(n)≤exp(2) H_D without an external average-totient input.

For every positive divisor d, the sum restricted to d|n is bounded by exp(2) H_D/phi(d). The proof uses the injective quotient n↦n/d and the proved supermultiplicativity phi(d)phi(n/d)≤phi(n). The actual gcd is expanded exactly as the sum of phi(d) over common divisors. These are unconditional arithmetic prerequisites for the source family harmonic estimate; the quantitative coefficient bounds and full major completion remain open.

Checked declarations

Module	Named supporting theorems
GoldbachTotientHarmonic.lean	4

- `goldbach_reciprocal_totient_majorant`
- `goldbach_reciprocal_totient_sum`
- `goldbach_reciprocal_totient_multiples`
- `goldbach_gcd_totient_divisor_expansion`

<!-- checkpoint-verification -->

Verification

All 1231 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5024-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3862 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1233 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1165_GoldbachArithmeticHarmonic.md`

Checkpoint 1165: Complete source weighted arithmetic harmonic estimate

The complete source weighted gcd/totient harmonic estimate e:harmonicsum9 is proved with explicit constant9 exp(2) and divisor exponent2. It holds for the actual strict coefficient family in every printed $B\geq 0$ and $L\leq N^2$ range, including empty rows and negative cutoffs.

The proof expands gcd(q,n) exactly through common-divisor totients. Reciprocal multiples give sum $\gcd(q,n)/n\leq\tau(q)$ H_D. The previous totient-multiples theorem gives sum $\gcd(q,n)/\varphi(n)\leq\tau(q)$ exp(2) H_D. The proved divisor relation $\gcd(q,ab)\mid\gcd(q,a)\gcd(q,b)$ separates the double sum. Extending the actual family to the N^2 rectangle and using $H_{N^2}\leq 3\log N$ gives9 exp(2) $\tau(q)^2(\log N)^2$. No sieve, average-totient or other independent input is assumed. The arithmetic coefficient decay and full major completion remain internal open obligations.

Checked declarations

Module	Named supporting theorems
GoldbachArithmeticHarmonic.lean	6

- `goldbach_gcd_weighted_divisor_sum`
- `goldbach_gcd_reciprocal_sum`
- `goldbach_gcd_reciprocal_totient_sum`
- `goldbach_gcd_product_bound`
- `goldbach_arithmetic_harmonic_rectangle`
- `goldbach_source_arithmetic_harmonic`

<!-- checkpoint-verification -->

Verification

All 1232 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5025-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3868 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1234 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1166_GoldbachArithmeticPrime.md

Checkpoint 1166: Exact coefficient identities and odd-prime gcd decay

Three exact identities are proved for the genuine arbitrary-multiplier coefficient: when q divides k, it is C_q(n-ell*b)/phi(q); when q is coprime to k and ell is a unit modulo q, it is mu(q) C_q(n)/phi(q)²; when q is coprime to k and ell vanishes modulo q, it is C_q(n)/phi(q). The unit case uses an actual permutation of the residue units, not an assumed invariance.

The existing exact odd-prime Cohen evaluation gives norm C_p(n)≤p+1. Splitting p|k, p|ell and p∤k*ell yields the full bound norm G_p(ell,k,n)≤8 gcd(p,k*ell)/p for every odd prime and arbitrary source parameters. The coefficient retains the actual Gauss/Upsilon normalizations. Bounds for arbitrary moduli and for the individual Upsilon sums, together with smoothing and smooth-modulus tails, remain open; no full source arithmetic lemma or completion is claimed.

Checked declarations

Module	Named supporting theorems
GoldbachArithmeticPrime.lean	5

- `goldbach_general_coefficient_of_dvd`
- `goldbach_general_coefficient_of_coprime_unit`
- `goldbach_general_coefficient_of_coprime_zero`
- `goldbach_odd_prime_cohen_norm`
- `goldbach_general_coefficient_odd_prime_bound`

<!-- checkpoint-verification -->

Verification

All 1233 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5026-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3873 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1235 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input

includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint notes 1167-1241

Checkpoint_1167_GoldbachEuclideanProduct.md

Checkpoint 1167: Actual Euclidean product bounds and both improper tails

The source linear integral is bounded by $N^2/(\ell \log 2)$, uniformly in W, using the actual clamped strict-domain integral. Consequently the genuine product I_U(gamma) J_ell,W(gamma) has the flat bound $N^3/(\ell(\log 2)^2)$ and the decay bound(D/ell) abs(gamma)^(-3/2), where D=(2 quadraticIntegralConstant/log2)/(pi log2)>0.

For every c>0 and either frequency direction sigma with abs(sigma)=1, the actual complex product at sigma*gamma is proved integrable on(c,infinity). Its norm integral is bounded by2D/(ell sqrt c). This is obtained by domination with the actual integrable rpow tail and evaluation of that improper integral. No Fourier-decay, integrability or tail endpoint is an external assumption. These establish the Euclidean tails needed in smoothing; the actual cutoff replacement and full circle-method completion remain open.

Checked declarations

Module	Named supporting theorems
GoldbachEuclideanProduct.lean	6

- `goldbachEuclideanDecayConstant_pos`
- `goldbachLinearSourceIntegral_norm_scale`
- `goldbach_euclidean_product_flat`
- `goldbach_inverse_sqrt_power`
- `goldbach_euclidean_product_decay`
- `goldbach_euclidean_product_tail`

<!-- checkpoint-verification -->

Verification

All 1234 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5027-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3879 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1236 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1168_GoldbachPrimePowerBounds.md

Checkpoint 1168: All prime-power coefficients and explicit two-adic bounds

The exact Upsilon normalization phi(gcd(q,k))/phi(q) is proved from the totient lcm identity. Direct finite norm estimates yield norm Upsilon≤phi(gcd(q,k)) and norm $G_q \le q^2$. These coarse bounds are used only to control the finite two-adic cases, not claimed as the required general Upsilon decay.

For every prime p and positive exponent a, the actual source coefficient satisfies norm G_{p^a}(ell,k,n)≤512 gcd(p^a,k*ell)/p^a. At odd primes, only a=1 contributes, by the earlier proved higher-power vanishing. At2, exponents at most3 use the explicit q^2 estimate; exponents at least4 vanish. Thus the two-adic contribution is fully controlled with an absolute constant. General-modulus decay is the next multiplicative deduction; Upsilon decay and actual smoothing/completion remain open.

Checked declarations

Module	Named supporting theorems
GoldbachPrimePowerBounds.lean	6

- `goldbach_upsilon_gcd_normalization`
- `goldbach_progression_exponential_trivial`
- `goldbach_upsilon_trivial_gcd`
- `goldbach_general_coefficient_trivial`
- `goldbach_general_coefficient_small_two_power`
- `goldbach_general_coefficient_prime_power`

<!-- checkpoint-verification -->

Verification

All 1235 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5028-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3885 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1237 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1169_GoldbachArithmeticDecay.md`

Checkpoint 1169: General-modulus arithmetic and Gauss divisor decay

The genuine arithmetic coefficient is bounded at every positive modulus by tau(q)^9 gcd(q,k*ell)/q, uniformly in the source parameters. Its exact multiplicative factorization is combined with the proved prime-power estimates. The gcd ratio gcd(q,c)/q is itself factored by a proved coprime product identity. The elementary inequality2^omega(q)≤tau(q) turns the finite prime-factor weight512^omega(q) into tau(q)^9.

The actual principal square Gauss coefficient also satisfies norm s(a,q)≤tau(q)^3/sqrt(q) at every unit frequency, by the earlier unconditional twisted Gauss estimate and the same divisor comparison. No character bound, Gauss estimate or general-modulus coefficient decay is assumed as an external input. The remaining arithmetic clause in l:arithmeticcoefficients9 is the sharper individual Upsilon decay. The Euler identity, absolute convergence, full G_q decay, Gauss decay and source harmonic sum are now proved; full smoothing/completion and l:circmethod remain open.

Checked declarations

Module	Named supporting theorems
GoldbachArithmeticDecay.lean	7

- `goldbach_gcd_ratio_multiplicative`
- `goldbach_gcd_ratio_factorization`
- `goldbach_two_pow_primeFactors_le_divisors`
- `goldbach_prime_factor_weight_le_divisors`
- `goldbach_general_coefficient_factor_bound`
- `goldbach_general_coefficient_divisor_bound`
- `goldbach_principal_square_divisor_bound`

<!-- checkpoint-verification -->

Verification

All 1236 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5029-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3892 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1238 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1170_GoldbachArithmeticFamily.md`

Checkpoint 1170: Actual weighted arithmetic-family and finite-tail estimates

The full actual source family sum of norm G_q divided by ell phi(k) is bounded by9 exp(2) tau(q)^11(log N)²/q, uniformly in the residue family and n. The proof combines the actual general-modulus coefficient decay with the complete source harmonic estimate. The positive modulus and progression indices are justified from the exact strict finite family.

For any finite set S of positive denominators, bounded complex coefficients d, and kernel values H of norm at most a nonnegative M, the actual weighted arithmetic error is bounded by9 exp(2) M(log N)² times sum_{q in S}tau(q)^11/q. All finite sum exchanges and complex normalizations are proved. This applies to the explicit finite support obtained from the Euler series, but does not claim the missing quantitative smooth-denominator tail decay.

The separate progress map lists exactly what is complete and what remains: Upsilon decay, the actual cutoff replacement, addition of separated smooth denominators, the Rankin tail beyond Y, and final circle-method assembly. The full l:arithmeticcoefficients9, l:majorcompletion9 and l:circmethod remain open. The paper ledger is unchanged. No new external input or Lean axiom is introduced, and the selected manuscript is unchanged pending the user's updated source.

Checked declarations

Module	Named supporting theorems
GoldbachArithmeticFamily.lean	2

- `goldbach_source_arithmetic_coefficient_sum`

- `goldbach_source_finite_arithmetic_tail`

<!-- checkpoint-verification -->

Verification

All 1237 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5030-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3894 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1239 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1171_GoldbachFiniteUnitAverages.md`

Checkpoint 1171: Exact normalized averages under finite unit reduction

The normalized sum of an arbitrary complex function over a finite group is preserved under pullback by a surjective group homomorphism. The proof partitions into fibres, proves all fibre cardinalities equal and positive, and cancels the exact cardinality normalization. It is specialized to the surjective reduction maps on units of ZMod at every positive pair of dividing moduli.

This supplies the exact finite expectation identity needed for reduction of the actual Upsilon frequency to its primitive modulus. No asymptotic equidistribution or prime-distribution input is involved. The full arithmetic, smoothing and circle-method endpoints remain internal obligations. The newer uploaded manuscript was read completely and all58 active mathematical statements match the selected reference; it still contains the endpoint sentence pending the user's proof edit, so no automatic source substitution is made.

Checked declarations

Module	Named supporting theorems
GoldbachFiniteUnitAverages.lean	2

- `goldbach_normalized_group_sum_pullback`
- `goldbach_normalized_units_sum_pullback`

<!-- checkpoint-verification -->

Verification

All 1238 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5031-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3896 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1240 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input

includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1172_GoldbachPrimitiveProgression.md

Checkpoint 1172: Exact primitive progression cancellation and CRT bound

The actual progression exponential sum vanishes under a nontrivial square-zero translation preserving the progression. Such a translation is constructed whenever the divisor modulus and its complementary factor share a prime. The primitive additive frequency supplies the nontrivial phase, so the cancellation is exact.

In the coprime case, an explicit CRT unit permutation fixes the progression coordinate. The free coordinate gives a genuine Ramanujan sum at a unit, equal to the Mobius function. Consequently its norm is at most one. These primitive-frequency statements are proved directly from the source finite sums and the earlier elementary character identities, with no independent progression-sum estimate.

Checked declarations

Module	Named supporting theorems
GoldbachPrimitiveProgression.lean	3

- `goldbach_progression_exponential_zero_translation`
- `goldbach_progression_exponential_noncoprime`
- `goldbach_progression_exponential_coprime_norm`

<!-- checkpoint-verification -->

Verification

All 1239 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5032-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3899 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1241 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1173_GoldbachUpsilonReduction.md

Checkpoint 1173: Exact Upsilon unit expectation and frequency reduction

The actual unit sum is converted exactly to its finite canonical residue range. The lcm-compatible residue sum is identified with a unit sum on ZMod(lcm(k,q)), preserving the source residue restriction and all totient factors. Thus actual Upsilon is expressed as a normalized unit expectation with the source progression fixed.

The normalized pullback theorem proves the exact frequency reduction Upsilon(q*d,k,b,d*beta)=Upsilon(q,k,b,beta), for every positive denominator, multiplier and progression modulus and every coprime source residue. The proof checks both reduction maps and the exact additive-character scaling; no distribution or cancellation estimate is assumed. This is the precise nonprimitive-frequency bridge needed for the missing Upsilon decay.

Checked declarations

Module	Named supporting theorems
GoldbachUpsilonReduction.lean	5

- `goldbach_sum_units_range`
- `goldbach_unit_lcm_residue_sum`
- `goldbach_upsilon_lcm_unit_average`
- `goldbach_unit_reduction_comp`
- `goldbach_upsilon_frequency_reduction`

<!-- checkpoint-verification -->

Verification

All 1240 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5033-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3904 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1242 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1174_GoldbachUpsilonDecay.md`

Checkpoint 1174: Actual Upsilon gcd decay from primitive frequency reduction

The primitive progression norm bound is proved for every divisor/complement pair, including the exact zero branch. This gives norm Upsilon(q,k,b,alpha)≤gcd(q,k)/phi(q) at every primitive additive frequency and source coprime residue.

The nonprimitive frequency is reduced exactly by d=gcd(q,ell). The quotient frequency is proved primitive, the reciprocal totient is bounded by tau(q)/q, divisor-cardinality monotonicity is checked, and the gcd contribution is transported through the common factor. The actual source coefficient satisfies norm Upsilon(alpha*ell,q,b,k)≤tau(q) gcd(q,k*ell)/q, stronger than the unspecified divisor-power estimate in the paper. No independent Lacey progression estimate or other new input is assumed. Integer residue transport and the common-exponent source arithmetic assembly are next.

Checked declarations

Module	Named supporting theorems
GoldbachUpsilonDecay.lean	5

- `goldbach_upsilon_int_modulus_congr`
- `goldbach_progression_exponential_primitive_norm`
- `goldbach_upsilon_primitive_decay`
- `goldbach_reciprocal_totient_divisor_bound`
- `goldbach_upsilon_gcd_divisor_decay`

<!-- checkpoint-verification -->

Verification

All 1241 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5034-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3909 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1243 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1175_GoldbachArithmeticComplete.md`

Checkpoint 1175: Full arithmetic coefficient lemma and Euler product

Every clause of l:arithmeticcoefficients9 is now proved for the genuine source coefficients. The integer residue b is reduced to its canonical coprime representative; the Upsilon bound holds for arbitrary integer coprime b. The Gauss, Upsilon and G_q estimates share the absolute exponent C_1=9. Their constants can be chosen1, and the harmonic-family constant is9 exp(2).

The infinite local Euler factors are evaluated directly: all odd prime exponents beyond1 vanish, and the two-adic sum consists exactly of the exponents1,2,3. The full R-smooth series is absolutely summable and equals both the printed all-prime Euler product and the source arithmetic factor G_R. The statements hold for every R≥2, hence for the eventual logarithmic R in Section10. The harmonic theorem retains all strict real family cutoffs, every B≥0, and all positive arithmetic indices.

A clause-by-clause source review closes l:arithmeticcoefficients9 as proved. Positivity and smoothing of the actual singular integral, the full major completion and the final circle-method lemma remain internal obligations. No endpoint from this paper has become an external assumption. The selected manuscript is unchanged.

Checked declarations

Module	Named supporting theorems
GoldbachArithmeticComplete.lean	9

- `goldbach_upsilon_canonical_residue`
- `goldbach_upsilon_integer_gcd_decay`

- goldbach_divisor_card_one_le
- goldbach_arithmetic_uniform_bounds
- goldbach_arithmetic_harmonic_common_exponent
- goldbach_odd_prime_power_series
- goldbach_two_prime_power_series
- goldbach_all_primes_euler_series
- goldbach_arithmetic_euler_complete

<!-- checkpoint-verification -->

Verification

All 1242 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5035-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3918 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1244 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1176_GoldbachAutocorrelation.md

Checkpoint 1176: Actual autocorrelation cutoff and inverse Fourier positivity

The paper allows an arbitrary nonzero real smooth compactly supported seed; it does not assume that the seed itself is even or nonnegative. The formal construction now complexifies any real Schwartz seed, reflects it, and takes its genuine convolution. The convolution is proved equal to the actual real autocorrelation integral. Its value at zero is the integral of the seed squared, proved strictly positive from nonzeroness and continuity.

The Fourier transform of the reflection is the conjugate transform of the seed. Consequently the autocorrelation transform is the squared complex modulus. Normalization gives eta(0)=1, and its actual Euclidean inverse Fourier transform is real and nonnegative. These are proved facts, not additional assumptions about the paper kernel. Scaling, support and quantitative smoothing, the singular integral and major completion remain to be assembled.

Checked declarations

Module	Named supporting theorems
GoldbachAutocorrelation.lean	8

- goldbach_autocorrelation_integral
- goldbach_seed_mass_pos
- goldbach_autocorrelation_zero
- goldbach_real_seed_fourier_conj
- goldbach_reflected_seed_fourier
- goldbach_autocorrelation_fourier
- goldbach_eta_zero
- goldbach_eta_inverse_fourier_nonnegative

<!-- checkpoint-verification -->

Verification

All 1243 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5036-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3926 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1245 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1177_GoldbachScaledCutoff.md`

Checkpoint 1177: Scaled cutoff support and square positivity

Positive Euclidean dilation of a Fourier multiplier has its genuine inverse transform equal to the dilated inverse transform times the positive reciprocal scale. This proves inverse-transform positivity for the actual scaled eta_N, not a newly assumed kernel.

For every Schwartz multiplier with real nonnegative inverse transform, the inverse transform of its square is proved equal to the genuine convolution of its inverse transforms, and is real nonnegative. This applies to the exact eta_N squared in the manuscript.

The seed support inside (-1/8,1/8) gives eta=0 whenever |x|≥1/4. The scaled cutoff therefore vanishes whenever |gamma|≥R/(4N²). All conclusions use ordinary Mathlib Fourier inversion and actual integrals. The full singular-integral positivity and smoothing estimate and the major-completion lemma are not yet claimed.

Checked declarations

Module	Named supporting theorems
GoldbachScaledCutoff.lean	5

- `goldbach_inverse_fourier_scale`
- `goldbach_scaled_eta_inverse_nonnegative`
- `goldbach_cutoff_square_inverse_nonnegative`
- `goldbach_eta_support`
- `goldbach_scaled_eta_support`

<!-- checkpoint-verification -->

Verification

All 1244 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5037-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3931 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1246 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input

includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1178_GoldbachCutoffBounds.md`

Checkpoint 1178: All source cutoff bounds from the genuine seed

The normalized autocorrelation cutoff is proved real and even, and its modulus is at most1. The latter follows by integrating the elementary inequality2|uv|≤u²+v² and translation invariance, so it does not require nonnegativity of the original seed.

Evenness gives eta'(0)=0. Bounds on the second derivative of the actual Schwartz cutoff and two applications of the mean value theorem give the uniform quadratic estimate. In particular there exists a positive constant K depending only on the fixed seed such that |1-eta(x)²|≤K min(1,|x|²) for every real x. This is precisely the source cutoff bound before the singular-integral lemma, with its scale dependence retained through the proved dilation formula.

The cutoff properties are now proved from the source construction. They are not new independent hypotheses. The actual singular integral still needs its Fourier inversion, nonnegativity and quantitative comparison, followed by major completion and final assembly.

Checked declarations

Module	Named supporting theorems
GoldbachCutoffBounds.lean	8

- `goldbach_eta_real`
- `goldbach_eta_even`
- `goldbach_autocorrelation_norm`
- `goldbach_eta_norm_le_one`
- `goldbach_eta_deriv_zero`
- `goldbach_schwartz_derivative_difference`
- `goldbach_eta_quadratic`
- `goldbach_eta_square_cutoff_bound`

<!-- checkpoint-verification -->

Verification

All 1245 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5038-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3939 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1247 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved.

The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1179_GoldbachIntegrableDecay.md`

Checkpoint 1179: Two-sided Euclidean integrability and quantitative tail assembly

A strongly measurable complex function with a uniform bound M on |gamma|≤c and decay B|gamma|^(-3/2) outside that interval is proved integrable on the entire real line. Its actual L1 norm is at most2Mc+4B/sqrt(c). The proof evaluates both improper tails, controls the finite interval and proves the negative-half-line change of variables.

This general fact supplies the missing analytic justification for applying Fourier inversion to the genuine Euclidean product and for bounding smoothing errors. It has no unproved analytic input. The singular-integral and major-completion source statements remain open until their actual functions are substituted and the final comparisons are proved.

Checked declarations

Module	Named supporting theorems
GoldbachIntegrableDecay.lean	2

- `goldbach_halfline_integrable_decay`
- `goldbach_two_sided_integrable_decay`

<!-- checkpoint-verification -->

Verification

All 1246 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5039-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3941 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1248 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1180_GoldbachPositiveFourierAverage.md`

Checkpoint 1180: Exact Fubini identity for positive Fourier kernel averages

Fubini is proved for a genuine Schwartz cutoff and every integrable real weight on a product-space measure. The actual Euclidean frequency integral equals the weighted average of the cutoff inverse transform at the sum frequency. Absolute integrability of the joint integrand is proved from the product of the L1 norms.

If the weight is nonnegative and the cutoff inverse transform is real nonnegative, the resulting actual Fourier integral is real nonnegative. This is the analytic mechanism for positivity of H_(ell,R). It is proved from ordinary integration and Fourier inversion, not postulated. Substitution of the exact paper integrals and the quantitative comparison remain next.

Checked declarations

Module	Named supporting theorems
GoldbachPositiveFourierAverage.lean	2

- `goldbach_cutoff_fourier_fubini`
- `goldbach_positive_cutoff_fourier_average`

<!-- checkpoint-verification -->

Verification

All 1247 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5040-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3943 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1249 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1181_GoldbachSingularIntegralPositive.md`

Checkpoint 1181: Actual singular integral positivity for every arithmetic index

The genuine source H_(ell,R)(n;U,W) is now defined using the actual normalized seed cutoff, the exact quadratic integral and the strict progression-domain linear integral.

The product I_U J_(ell,W) is proved equal to its actual positive two-variable density Fourier integral. Substituting that identity into the proved Fubini theorem, and using the already-proved inverse positivity of eta_N squared, proves that H is real and nonnegative. This holds for every real n, all positive ell,N,R, and U,W≥2. If the strict progression domain is empty, the integral is exactly zero; this case is explicitly proved, so no upper bound on ell is hidden in the result.

Thus the full positivity clause of l:positiveH9 is proved. The O(N/sqrt(R)) comparison with the printed one-variable H_ell still needs its quantitative estimate and exact unsmoothed Fourier inversion. The lemma remains open in the ledger until both clauses are complete.

Checked declarations

Module	Named supporting theorems
GoldbachSingularIntegralPositive.lean	4

- `goldbach_euclidean_product_density`
- `goldbach_singular_integral_positive`
- `goldbach_singular_integral_empty`

- goldbach_singular_integral_nonnegative

<!-- checkpoint-verification -->

Verification

All 1248 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5041-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3947 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1250 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1182_GoldbachSingularIntegralNorm.md

Checkpoint 1182: Global Euclidean L1 and uniform singular-integral bounds

The genuine Euclidean product I_U J_(ell,W) is proved absolutely integrable over the full real frequency line. Splitting at$1/N^2$ and evaluating both improper tails gives an explicit uniform L1 bound C N/ell, where $C=2/(\log 2)^2+4D$ and D is the previously proved absolute decay constant.

The exact eta_N-squared singular-integral integrand is separately proved integrable and is dominated in modulus by the Euclidean product. Consequently $|H_{(ell,R)}(n;U,W)|\le C\ N$ for every real n and all positive ell, retaining U,W and the exact strict source domain. The empty-domain branch is included. This discharges the uniform kernel bound needed by the arithmetic tail in major completion.

The O(N/sqrt(R)) comparison with the printed H_ell and the full completed major-arc error still remain internal obligations.

Checked declarations

Module	Named supporting theorems
GoldbachSingularIntegralNorm.lean	5

- goldbachEuclideanL1Constant_pos
- goldbach_euclidean_product_integrable
- goldbach_singular_integrand_norm
- goldbach_singular_integrand_integrable
- goldbach_singular_integral_norm

<!-- checkpoint-verification -->

Verification

All 1249 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5042-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3952 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1251 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1183_GoldbachSmoothingDecay.md

Checkpoint 1183: Quantitative smoothing error from exact quadratic cancellation

A global quantitative smoothing lemma is now proved. If the actual error multiplier is bounded by K min(1,(|gamma|/c)²) and the other factor decays as D/(sqrt(|gamma|)|gamma|), their product is integrable and its L1 norm is at most6KD/sqrt(c).

The proof handles the zero frequency explicitly, proves the near-zero cancellation algebra, and evaluates both real-line tails. Substituting c=R/N² and the already-proved actual eta cutoff bound gives the scale N/sqrt(R) required by the source. No Fourier identity for the printed H_ell is assumed or claimed here; that identification remains to be proved.

Checked declarations

Module	Named supporting theorems
GoldbachSmoothingDecay.lean	2

- `goldbach_smoothing_power_identity`
- `goldbach_smoothing_decay_integrable`

<!-- checkpoint-verification -->

Verification

All 1250 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5043-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3954 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1252 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1184_GoldbachSingularSmoothing.md

Checkpoint 1184: Actual uniform singular-integral smoothing estimate

For the actual normalized seed and exact source I_U,J_(ell,W), the genuine smoothed integral differs from the genuine unsmoothed inverse integral by at most K N/sqrt(R), with K depending only on the fixed seed. The constant is selected before N,ell,R,U,W,n; all positive ell and all real n are included.

The proof substitutes the actual quadratic cutoff bound into the global smoothing lemma, proves every integral used in the subtraction is integrable, and handles the empty progression-domain case exactly. The scale R/N^2 is simplified without dropping ell or endpoint dependence.

The remaining clause for closing l:positiveH9 is the exact identification of the unsmoothed inverse integral with the printed positive one-variable H_ell. This identity has not become an external input. Major completion and final circle-method assembly also remain internal obligations.

Checked declarations

Module	Named supporting theorems
GoldbachSingularSmoothing.lean	1

- `goldbach_singular_smoothing_uniform`

<!-- checkpoint-verification -->

Verification

All 1251 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5044-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3955 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1253 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1185_GoldbachQuadraticDensity.md`

Checkpoint 1185: Exact quadratic density Fourier identity with all integrability conditions

The one-variable density associated with $u^2+a\,v$ is constructed as the actual affine average a^(-1) integral f(u)g((x-u²)/a) du. For every positive real a and arbitrary integrable measurable f,g, its joint integrand and the resulting density are proved globally integrable.

Its genuine Fourier transform is proved equal to the product of the quadratic transform of f and the scaled linear transform of g. The affine change of variables, the scale factor a, and the Fubini integrability are all proved explicitly. This provides the exact analytic route to the printed H_ell, rather than assuming the desired inversion identity.

The source logarithmic densities, pointwise continuity and the final inverse identity are next. No new external analytic input is introduced.

Checked declarations

Module	Named supporting theorems
GoldbachQuadraticDensity.lean	5

- `goldbach_affine_integral`
- `goldbach_forward_fourier_integral`
- `goldbach_quadratic_density_joint`
- `goldbach_quadratic_average_integrable`
- `goldbach_quadratic_average_fourier`

<!-- checkpoint-verification -->

Verification

All 1252 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5045-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3960 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1254 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1186_GoldbachSourceDensity.md`

Checkpoint 1186: Actual logarithmic density Fourier transform and endpoint continuity

The actual truncated logarithmic density is constructed, proved measurable, integrable and uniformly bounded. Its only possible discontinuities are its two endpoints, and continuity elsewhere is proved directly.

The exact quadratic average with the source cutoffs is proved globally integrable, and its Fourier transform is exactly I_U(-gamma)J_(ell,W)(-gamma). Pointwise continuity of the averaged density is also proved: the exceptional endpoint preimages are the explicitly excluded singleton square roots, which have zero measure. Dominated convergence uses the proved logarithmic bound, so no endpoint regularity is assumed.

These facts discharge every hypothesis of ordinary Fourier inversion for the actual source density. The next assembly identifies the inverse integral with the printed one-variable H_ell and closes the full positivity/comparison lemma.

Checked declarations

Module	Named supporting theorems
GoldbachSourceDensity.lean	8

- `goldbach_log_window_measurable`
- `goldbach_log_window_integrable`
- `goldbach_log_window_norm`
- `goldbach_log_window_continuousAt`
- `goldbach_log_window_oscillatory`
- `goldbach_source_density_integrable`

- goldbach_source_density_fourier
- goldbach_source_density_continuous

<!-- checkpoint-verification -->

Verification

All 1253 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5046-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3968 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1255 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1187_GoldbachPositiveSingularComplete.md

Checkpoint 1187: Complete source singular-integral positivity and comparison lemma

Every clause of l:positiveH9 is now proved for the genuine source functions. Fourier inversion applies to the actual continuous integrable logarithmic density, whose transform is exactly I_U(-gamma)J_(ell,W)(-gamma). The source n<N² support condition removes the strict progression restriction in the unsmoothed density, and the result is proved equal to the exact printed integral over2≤u≤U and2≤(n-u²)/ell≤W. Every endpoint and empty-domain case is included.

Combining that exact identity with the proved uniform smoothing comparison and actual-kernel positivity gives the complete lemma. One positive constant, depending only on the fixed seed, is chosen before N,ell,R,U,W,n. The theorem covers every real n<N², every positive ell,N,R and all U,W≥2 with U≤N; the source W≤N² restriction is unnecessary. Thus its range is at least the printed range.

The arbitrary real smooth compactly supported source seed is embedded into Schwartz space with nonzeroness proved. The normalized eta is proved equal to the actual autocorrelation integral divided by its actual squared-mass denominator, so the cutoff construction has not been substituted by an assumed surrogate.

A clause-by-clause source review closes l:positiveH9 as proved. All Section10 source statements are now complete except l:majorcompletion9. That major-completion lemma and the final l:circmethod assembly remain internal obligations. No new external theorem or own-paper endpoint was assumed. The manuscript is unchanged.

Checked declarations

Module	Named supporting theorems
GoldbachPositiveSingularComplete.lean	7

- goldbach_archimedean_integral_domain
- goldbach_source_density_archimedean
- goldbach_source_unsmoothed_inverse
- goldbach_unsmoothed_archimedean_complete
- goldbach_smooth_compact_seed_nonzero
- goldbach_eta_of_smooth_compact_seed

- goldbach_positive_singular_integral_complete

<!-- checkpoint-verification -->

Verification

All 1254 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5047-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3975 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1256 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1188_GoldbachRankinEuler.md

Checkpoint 1188: Exact Rankin tail reduction and positive divisor Euler product

The positive Rankin weight tau(q)^11 q^(-1+sigma) is proved multiplicative for every real sigma. Its local prime-power value is computed exactly, and the finite divisor sum is proved equal to the genuine prime-by-prime Euler product.

For every positive sigma,Y and every finite subset of the positive divisors of a nonzero integer, the portion with q>Y is bounded by Y^(-sigma) times that exact Euler product. The inequality is proved from real powers, with all positivity conditions retained. The factorial-cube support already proved for the actual smooth coefficients is a direct admissible specialization.

This proves the algebraic Rankin reduction required by major completion. The uniform logarithmic estimate for the Euler product and the actual major-arc completion still remain to be assembled. The full arithmetic and singular-integral source lemmas remain proved.

Checked declarations

Module	Named supporting theorems
GoldbachRankinEuler.lean	6

- goldbach_rankin_weight_nonnegative
- goldbach_rankin_weight_multiplicative
- goldbach_rankin_weight_prime_power
- goldbach_rankin_divisor_euler
- goldbach_rankin_pointwise
- goldbach_finite_rankin_euler_bound

<!-- checkpoint-verification -->

Verification

All 1255 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5048-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3981 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1257 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1189_GoldbachRankinLocal.md`

Checkpoint 1189: Uniform local Rankin Euler-factor bound

A uniform finite geometric estimate is proved, and the polynomial exponent factor is absorbed using the already-proved elementary exponential bound. This gives one positive absolute constant C such that every local Rankin factor satisfies sum_(j=0)^J (j+1)^11 p^(j(-1+sigma))≤1+C p^(-1+sigma), simultaneously for every prime p, sigma≤1/2 and every exponent cutoff J.

The proof uses an explicit auxiliary ratio2^(1/44), proves the resulting geometric ratio is below1, and retains the exact constant term1. That last point is essential for a logarithmic prime product rather than a factor growing once for each prime. No prime-distribution theorem or new external input is used.

Checked declarations

Module	Named supporting theorems
GoldbachRankinLocal.lean	3

- `goldbach_geometric_partial_bound`
- `goldbach_polynomial_local_bound`
- `goldbach_rankin_local_uniform`

<!-- checkpoint-verification -->

Verification

All 1256 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5049-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3984 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1258 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The

classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1190_GoldbachRankinLogProduct.md

Checkpoint 1190: Unconditional logarithmic Rankin Euler product and finite tail bound

The local Rankin bounds are assembled into a proved logarithmic Euler-product bound. The small real-power factor satisfies p^(-1+sigma)≤exp(1)/p whenever sigma≥0 and sigma log R≤1. Bernoulli's inequality then bounds each factor by a fixed integer power of1+1/p.

The already-proved elementary prime-product estimate gives a fixed natural exponent K such that the full divisor Rankin sum is at most(M log(2R))^K, uniformly in the supporting integer and its prime-power exponents. Here M=12+2/log3 is explicit. Combining this with the finite Rankin reduction gives the actual tail bound Y^(-sigma)(M log(2R))^K for every finite divisor subset with q>Y. No Mertens theorem, PNT or extra distribution input is assumed.

The eventual source-parameter saving and the major-arc Fourier assembly remain to be completed.

Checked declarations

Module	Named supporting theorems
GoldbachRankinLogProduct.lean	5

- `goldbachPrimeProductConstant_pos`
- `goldbach_rankin_prime_rpow`
- `goldbach_prime_subset_inclusive_log`
- `goldbach_rankin_euler_log`
- `goldbach_rankin_tail_log_bound`

<!-- checkpoint-verification -->

Verification

All 1257 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5050-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3989 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1259 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1191_GoldbachRankinDecay.md

Checkpoint 1191: Actual Rankin parameter decay with arbitrary logarithmic saving

The logarithmic-versus-power argument is proved with explicit sufficient inequalities. A general logarithmic quadratic budget gives the negative exponential Rankin saving while controlling the polynomial Euler-product factor.

For every fixed C≥1,M>0,K and t>0, the genuine parameter expression with R=x^C and Y=exp(x)/(100sqrt(R)) is proved at most x^(-t) for all sufficiently large real x. Every logarithm is evaluated with its positivity assumptions, and the numerical100 and square-root scale are retained. The threshold depends only on the fixed parameters, not on arithmetic moduli or coefficient families.

The final substitution x=log N and the actual smooth-divisor tail are next; the major-completion source statement remains open until the Fourier and arithmetic errors are assembled.

Checked declarations

Module	Named supporting theorems
GoldbachRankinDecay.lean	3

- `goldbach_eventually_log_square_budget`
- `goldbach_rankin_decay_log_budget`
- `goldbach_rankin_parameter_decay`

<!-- checkpoint-verification -->

Verification

All 1258 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5051-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3992 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1260 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1192_GoldbachRankinSource.md`

Checkpoint 1192: Actual source smooth-modulus tail with arbitrary logarithmic saving

The exact source parameters R=(log N)^C and Y=N/(100sqrt(R)) are now substituted into the complete Rankin estimate. For every fixed C≥1 and desired t>0, one natural threshold is chosen before every supporting integer D and every finite divisor subset S. If all prime factors of D are at most R and every q in S exceeds Y, then sum_(q in S) tau(q)^11/q≤(log N)^(-t).

The proof checks R≥2, the positive Rankin exponent, its upper bound1/2, the sigma log R budget, and the exact exp(log N)=N substitution. The threshold is uniform in D,S and all arithmetic coefficient data. This is the arbitrary logarithmic saving used in the source tail argument, proved without a new Mertens or distribution hypothesis.

The actual weighted arithmetic-family tail is next, followed by the remaining small-modulus and disjoint added-modulus major estimates and final assembly.

Checked declarations

Module	Named supporting theorems
GoldbachRankinSource.lean	1

- `goldbach_source_rankin_tail`

<!-- checkpoint-verification -->

Verification

All 1259 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5052-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3993 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1261 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1193_GoldbachRankinArithmeticTail.md`

Checkpoint 1193: Actual weighted arithmetic-family Rankin tail

The source Rankin saving is now applied to the genuine arithmetic-family tail weighted by the actual H_(ell,R). For every desired t>0 and C≥1, one threshold is chosen before all real L,V,B cutoffs, all coprime residues, all bounded complex coefficients, U,W, supporting smooth divisor sets and output integers n.

The proved O(N) bound for H and the full source harmonic/coefficient estimate give the actual tail norm at most(9 exp(2) C_H) N (log N)^(-t). The two logarithmic powers in the harmonic sum are explicitly absorbed by choosing the Rankin saving t+2. Positive indices and the exact strict real family bounds are retained; empty progression domains are already included in the H bound.

This closes the weighted arithmetic tail component of major completion. The full major-completion source statement still awaits the sharp small-modulus Fourier identity, its smoothing comparison and the disjoint added-modulus L2 estimate. No new external hypothesis is introduced.

Checked declarations

Module	Named supporting theorems
GoldbachRankinArithmeticTail.lean	2

- `goldbachArithmeticTailConstant_pos`
- `goldbach_source_rankin_arithmetic_tail`

<!-- checkpoint-verification -->

Verification

All 1260 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5053-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 3995 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1262 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input

includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1194_GoldbachArcLiftIntegral.md`

Checkpoint 1194: Exact arc-lift Fourier coefficients and Euclidean energy

A genuine real-frequency function supported in|gamma|<1/2 is lifted to the source circle using the actual centered coordinate. The circle integral is proved equal to the real-line integral by translation invariance and the fundamental-domain measure equivalence; no periodization identity is assumed.

The exact Fourier coefficient of a translated lift is proved equal to exp(-2pi i n r) times the actual Euclidean coefficient. The circle L2 energy is also proved exactly equal to the real-line energy. Bounded strongly measurable source functions are proved to give circle MemLp functions at every exponent.

These identities provide the exact normalization and phase needed for the remaining major-model Fourier estimates, including wrapped arcs.

Checked declarations

Module	Named supporting theorems
GoldbachArcLiftIntegral.lean	5

- `goldbach_arc_coordinate_real_shift`
- `goldbach_arc_lift_integral`
- `goldbach_arc_lift_fourier_integral`
- `goldbach_arc_lift_energy`
- `goldbach_arc_lift_memLp`

<!-- checkpoint-verification -->

Verification

All 1261 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5054-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4000 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1263 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypoth-

esis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1195_GoldbachDisjointArcBessel.md

Checkpoint 1195: Disjoint rational arc lifts and exact Euclidean Bessel bound

Rational arc lifts with denominator bound Y are proved disjoint whenever2epsilon<Y^(-2), using the exact source rational-center separation and the centered real lifts. The result includes wrapped arcs and arbitrary measurable functions supported within the stated radius.

For a finite disjoint family, the actual Fourier coefficients satisfy a Bessel bound by the sum of the exact Euclidean L2 energies. The proof checks circle MemLp, moves the finite sums through genuine integrals, proves the pointwise square-energy identity and applies the already-proved circle Bessel inequality. The Fourier phases and normalizations are the exact ones proved at1194.

This is the elementary analytic mechanism for adding R<q≤Y in major completion. No independent large-sieve or completion hypothesis is assumed.

Checked declarations

Module	Named supporting theorems
GoldbachDisjointArcBessel.lean	2

- `goldbach_arc_lifts_disjoint`
- `goldbach_disjoint_arc_bessel`

<!-- checkpoint-verification -->

Verification

All 1262 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5055-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4002 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1264 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1196_GoldbachMajorEnvelope.md

Checkpoint 1196: Uniform Euclidean major-envelope square energy

A single nonnegative measurable Euclidean envelope is constructed from the actual flat and inverse-three-halves decay bounds. It is proved integrable with mass O(N), and its squared integral is proved O(N^4). The L2 estimate follows from its actual uniform O(N^3) bound and the proved L1 mass.

For every positive ell and all source U,W, the genuine product ell|I_U J_(ell,W)| is bounded by this common envelope. The empty progression-domain branch is explicitly included. A general domination theorem then gives the exact L2 energy bound for any strongly measurable local integrand controlled by the envelope.

This supplies the quantitative analytic factor for the disjoint added-modulus estimate. The remaining steps are the actual local arithmetic coefficients, denominator summation and the final major-model identities.

Checked declarations

Module	Named supporting theorems
GoldbachMajorEnvelope.lean	9

- `goldbach_major_envelope_nonnegative`
- `goldbach_major_envelope_bound`
- `goldbach_major_envelope_decay`
- `goldbach_major_envelope_measurable`
- `goldbach_major_envelope_integrable`
- `goldbach_major_envelope_energy`
- `goldbach_major_envelope_dominated_energy`
- `goldbach_linear_source_integral_empty`
- `goldbach_euclidean_product_envelope`

<!-- checkpoint-verification -->

Verification

All 1263 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5056-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4011 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1265 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1197_GoldbachMajorFamilyIntegrand.md`

Checkpoint 1197: Actual local arithmetic-family major integrand and compact support

The genuine local major-family integrand is now defined from the actual principal Gauss coefficient, Upsilon progression coefficient, bounded family coefficients and Euclidean I_U,J_(ell,W). Its exact source-family norm is proved at most A_q E_N(gamma), where A_q=9 exp(2) tau(q)^6 (log N)^2/(q sqrt(q)). This uses the individual primitive Upsilon decay and the full harmonic-family estimate, with arbitrary integer coprime residues.

The integrand is proved strongly measurable. Multiplication by the actual eta_N squared preserves the norm bound and strong measurability. The smoothed integrand is proved exactly zero for$|\gamma| \geq R/(4N^2)$, directly from the original seed support. Every arithmetic index and strict source family cutoff is retained.

The already-proved envelope energy and disjoint Bessel theorem can now be applied to these actual local functions. The source major-completion and final circle-method endpoints remain open until the denominator sums and exact Fourier identities are assembled.

Checked declarations

Module	Named supporting theorems
GoldbachMajorFamilyIntegrand.lean	6

- `goldbach_major_arithmetic_prefactor_nonnegative`
- `goldbach_major_family_integrand_measurable`
- `goldbach_major_family_integrand_bound`
- `goldbach_smooth_major_family_measurable`
- `goldbach_smooth_major_family_norm`
- `goldbach_smooth_major_family_support`

<!-- checkpoint-verification -->

Verification

All 1264 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5057-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4017 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1266 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1198_GoldbachCenterArithmetic.md`

Checkpoint 1198: Exact signed rational-center and denominator arithmetic identity

The exact rational-center phase is proved equal to the actual ZMod additive character, including denominator1. The simultaneous negative numerator in the Gauss, progression and output-character factors is handled by a proved unit-group permutation.

Consequently the genuine arithmetic sum over source rational centers, with any denominator weight, is exactly the sum of the source G_q coefficients with that same weight. The proof uses the actual unit-to-rational bijection and retains all integer multipliers and residues. A zero-modulus extension only makes the nondependent function total; all contributing denominators are proved positive.

This is the exact arithmetic identity needed to turn the local Fourier model into the printed completed main term. No Euler factor or Fourier sign is postulated.

Checked declarations

Module	Named supporting theorems
GoldbachCenterArithmetic.lean	5

- `goldbach_center_arithmetic_eq`
- `goldbach_center_arithmetic_unit_sum`
- `goldbach_rational_negative_character`
- `goldbach_center_arithmetic_rational`
- `goldbach_centers_weighted_arithmetic`

<!-- checkpoint-verification -->

Verification

All 1265 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5058-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4022 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1267 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1199_GoldbachRadialMajorIdentity.md`

Checkpoint 1199: Exact actual smoothed major-family and arithmetic main-term identity

The actual local major-family Fourier expression is proved exactly equal to the arithmetic coefficient expression for every denominator weight and every integrable radial kernel. All finite sums are moved through genuine integrals with their integrability proved or explicitly supplied as ordinary function hypotheses.

For the paper's actual eta_N squared, those integrability hypotheses are discharged for every positive ell, including empty domains. The genuine smoothed rational-center expression therefore equals precisely the sum of d H_(ell,R)/(ell phi(k)) times the source denominator coefficients. Arbitrary denominator masks are retained, so the identity applies both to truncation and to the smooth added moduli.

No normalization, signed numerator, Euler coefficient or singular integral is substituted by an assumed quantity. The remaining completion estimates concern the denominator energy sum and sharp-cutoff comparison.

Checked declarations

Module	Named supporting theorems
GoldbachRadialMajorIdentity.lean	4

- `goldbach_radial_major_family_identity`
- `goldbach_smoothed_radial_kernel_integrable`
- `goldbach_smoothed_radial_kernel_integral`
- `goldbach_smoothed_major_family_identity`

<!-- checkpoint-verification -->

Verification

All 1266 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5059-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4026 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1268 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1200_GoldbachDivisorSquareTail.md`

Checkpoint 1200: Finite divisor-power denominator square tail

The finite inverse-three-halves tail is proved directly by integral comparison, including arbitrary finite subsets and real lower cutoffs R≥2. Its bound is4/sqrt(R), with the floor and endpoint conventions checked explicitly.

Combining that with the already-proved elementary divisor growth gives one positive absolute constant K such that sum_(q in S) tau(q)^12/q²≤K/sqrt(R) whenever every q in the finite set S exceeds R. The proof retains the full divisor power12 and has no smoothness restriction or upper denominator bound.

This is the denominator summation needed after counting the primitive numerators in the actual disjoint major-family energy estimate.

Checked declarations

Module	Named supporting theorems
GoldbachDivisorSquareTail.lean	3

- `goldbach_inverse_three_halves_interval`
- `goldbach_inverse_three_halves_tail`
- `goldbach_divisor_square_tail`

<!-- checkpoint-verification -->

Verification

All 1267 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5060-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4029 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1269 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypoth-

esis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1201_GoldbachDenominatorEnergy.md`

Checkpoint 1201: Actual rational-numerator count and added-denominator energy sum

The number of actual source rational centers of each denominator is proved exactly phi(q), including modulus1 and arbitrary denominator filters. No coarse center-count substitution is used in the energy sum.

After multiplying the actual squared local prefactor by phi(q), the denominator contribution is bounded by the fixed tau(q)^12/q^2 weight. The proved finite divisor tail therefore yields one absolute K such that the sum of squared local prefactors over every R<q≤Y is at most K(log N)^4/sqrt(R), uniformly in the real upper cutoff Y.

This is the complete arithmetic denominator-energy sum for the added source moduli; only the already-proved local analytic energy and disjoint Bessel inequality remain to be combined with it.

Checked declarations

Module	Named supporting theorems
GoldbachDenominatorEnergy.lean	4

- `goldbach_centers_denominator_sum`
- `goldbach_centers_filtered_denominator_sum`
- `goldbach_major_prefactor_count`
- `goldbach_added_denominator_energy_budget`

<!-- checkpoint-verification -->

Verification

All 1268 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5061-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4033 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1270 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1202_GoldbachAddedMajorBessel.md`

Checkpoint 1202: Actual quantitative added-modulus major-family Bessel estimate

The actual added-modulus Fourier family now satisfies the quantitative Bessel estimate: its squared coefficient sum over any finite output set is at most K N^4(log N)^4/sqrt(R). The constant is chosen before N, the arithmetic family, coefficients, cutoffs, upper denominator bound and output set.

The proof applies the exact actual integrands, their seed-derived support, the rational disjointness theorem, the proved envelope energies and the complete primitive-numerator denominator sum. It retains the elementary geometric conditions epsilon=R/(4N^2)<1/2 and2epsilon<Y^(-2); these will be discharged

for the printed Y=N/(100sqrt(R)). The finite rational set may be any subset of the added denominators, so the source R-smooth restriction is included.

This is the full analytic estimate for adding the intermediate smooth moduli, with no independent completion or large-sieve hypothesis. The remaining work is source parameter geometry and the sharp-to-smoothed small-modulus comparison, followed by final assembly.

Checked declarations

Module	Named supporting theorems
GoldbachAddedMajorBessel.lean	1

- `goldbach_smooth_added_major_bessel`

<!-- checkpoint-verification -->

Verification

All 1269 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5062-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4034 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1271 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1203_GoldbachCompletionParameters.md`

Checkpoint 1203: Printed completion geometry and logarithmic error rates

The elementary disjointness conditions are discharged for the exact printed Y=N/(100sqrt(R)), retaining the numerical100. If N≥10000,R≥1 and$R^3 \le N$, thenY≥R, the smoothing support is inside the centered half interval, and the translated supports have the required strict rational separation.

For every fixed C≥1, all these conditions are proved eventually forR=(log N)^C, with one natural threshold before all arithmetic families. A general rate lemma also proves(log N)^j/sqrt(R)≤(log N)^(-A) wheneverC≥2A+2j andlog N≥1. Thus the actual added-modulus bound has the required arbitrary logarithmic saving after choosing C large enough.

The remaining analytic estimate is the sharp-to-smoothed small-modulus comparison; exact source-model and final error assembly follow.

Checked declarations

Module	Named supporting theorems
GoldbachCompletionParameters.lean	3

- `goldbach_completion_geometry`
- `goldbach_completion_parameters_eventual`
- `goldbach_completion_log_power_rate`

<!-- checkpoint-verification -->

Verification

All 1270 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5063-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4037 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1272 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1204_GoldbachHardSmoothing.md`

Checkpoint 1204: Uniform sharp-to-smooth singular integral comparison

The actual sharp frequency cutoff is compared with the actual autocorrelation cutoff. The proof bounds both against the unsmoothed integral using the proved two-sided oscillatory tail and the proved quadratic cutoff estimate. It gives a constant depending only on the seed, chosen before every arithmetic family, with error at most K N / sqrt(R). The empty linear integration domain is included; no singular-integral comparison is postulated.

Checked declarations

Module	Named supporting theorems
GoldbachHardSmoothing.lean	7

- `goldbach_hard_cutoff_measurable`
- `goldbach_hard_cutoff_norm`
- `goldbach_euclidean_product_phase_integrable`
- `goldbach_frequency_tail_norm`
- `goldbach_euclidean_product_phase_tail`
- `goldbach_hard_unsmoothed_comparison`
- `goldbach_hard_singular_smoothing_uniform`

<!-- checkpoint-verification -->

Verification

All 1271 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5064-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4044 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1273 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional

Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1205_GoldbachSmallSmoothing.md`

Checkpoint 1205: Weighted small-modulus smoothing error

The divisor bound and the elementary harmonic estimate give the actual finite sum over q <= R of tau(q)^11/q at most K R^(1/4)(1+log R). Applying the proved arithmetic family inequality to the actual difference of the sharp and smooth singular integrals yields a uniform bound K N (log N)^2 (1+log R) / R^(1/4). The constants are fixed before all source arithmetic families. This slightly weaker power of R still yields every prescribed logarithmic saving by increasing the chosen C.

Checked declarations

Module	Named supporting theorems
GoldbachSmallSmoothing.lean	2

- `goldbach_divisor_harmonic_initial_sum`
- `goldbach_small_smoothing_uniform`

<!-- checkpoint-verification -->

Verification

All 1272 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5065-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4046 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1274 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1206_GoldbachHardMajorIdentity.md`

Checkpoint 1206: Exact actual sharp major model identity

The actual product of the square major model and weighted progression major model is identified pointwise with its disjoint rational-arc model. A proved coordinate change from the circle to the real line gives the exact sharp-cutoff Fourier coefficient, and finite Fubini plus the negative-unit permutation converts it to the actual generalCoefficient sum times the actual hard singular integral. Every coordinate, sign, boundary cutoff and finite sum is checked.

Checked declarations

Module	Named supporting theorems
GoldbachHardMajorIdentity.lean	6

- `goldbach_arc_model_zero_off_major`
- `goldbach_major_model_product`
- `goldbach_arc_model_fourier_identity`
- `goldbach_hard_radial_kernel_integral`
- `goldbach_hard_major_family_identity`
- `goldbach_actual_major_model_identity`

<!-- checkpoint-verification -->

Verification

All 1273 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5066-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4052 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1275 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1207_GoldbachCompletionArithmetic.md`

Checkpoint 1207: Exact arithmetic completion split

The actual initial coefficient sum is identified with its finite divisor support by proving vanishing for every omitted coefficient. The actual Euler factor is then split exactly into q <= R, R < q <= Y, and q > Y. The additional rational centers are identified with the middle divisor sum using the proved smoothed rational Fourier identity. No convergence or coefficient support assumption is added.

Checked declarations

Module	Named supporting theorems
GoldbachCompletionArithmetic.lean	7

- `goldbach_completion_bound_nezero`
- `goldbach_completion_bound_primes`
- `goldbach_initial_coefficients_divisor_sum`
- `goldbach_completion_arithmetic_split`
- `goldbach_added_moduli_filter`
- `goldbach_added_completion_centers_subset`
- `goldbach_added_completion_identity`

<!-- checkpoint-verification -->

Verification

All 1274 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5067-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4059 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1276 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1208_GoldbachCompletionErrorIdentity.md`

Checkpoint 1208: Actual major-completion error decomposition

The actual major-model coefficient minus the actual completed Euler-factor main term is proved equal to the small smoothing error minus the added-modulus Fourier error minus the far smooth arithmetic tail. This identity consumes the already checked actual integral and arithmetic identities, with all source families and signs retained. The three errors are defined by actual sums and integrals.

Checked declarations

Module	Named supporting theorems
GoldbachCompletionErrorIdentity.lean	1

- `goldbach_completion_error_identity`

<!-- checkpoint-verification -->

Verification

All 1275 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5068-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4060 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1277 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The

classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1209_GoldbachCompletionBudgets.md

Checkpoint 1209: Uniform completion budgets and geometry

The elementary logarithmic budgets are checked with explicit sufficient threshold C >= 2 A + 16. The squared pointwise small-modulus error summed over n < N^2 has the required N^4 log(N)^(-A) bound; the Rankin pointwise tail has the same mean-square scale. The actual hard and smooth arc separation inequalities are proved eventually for the chosen Y=N/(100 sqrt R), with all thresholds chosen before arithmetic families.

Checked declarations

Module	Named supporting theorems
GoldbachCompletionBudgets.lean	4

- `goldbach_uniform_finite_square_sum`
- `goldbach_small_completion_square_budget`
- `goldbach_far_completion_square_budget`
- `goldbach_completion_all_parameters_eventual`

<!-- checkpoint-verification -->

Verification

All 1276 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5069-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4064 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1278 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1210_GoldbachMajorCompletionBound.md

Checkpoint 1210: Full actual major-completion coefficient bound

The actual major coefficient minus the actual Euler-factor/singular-integral main term now satisfies the full mean-square estimate. The proof consumes the exact three-error identity, the small smoothing bound, the disjoint-arc Bessel estimate for added divisors, and the quantitative Rankin tail. Constants and thresholds precede N, all arithmetic families, coefficients and cutoffs. It proves a stronger coefficient formulation with B >= 0, L <= N^2 and W >= 2. No deep analytic input is assumed by this completion theorem.

Checked declarations

Module	Named supporting theorems
GoldbachMajorCompletionBound.lean	1

- `goldbach_major_completion_coefficient_complete`

<!-- checkpoint-verification -->

Verification

All 1277 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5070-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4065 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1279 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1211_GoldbachMajorCompletionComplete.md`

Checkpoint 1211: Complete source major-arc completion lemma

The literal source major-completion statement is now proved. The per-progression major integrals are identified with the checked weighted model coefficient by genuine integrability and finite integration. The source L<N^(2-delta) range discharges all elementary scale conditions uniformly. The consuming theorem uses the actual G_R and H_R, fixes constants before all arithmetic families, and needs no deep input. The source claim l:majorcompletion9 is awarded proved status only after this full clause review.

Checked declarations

Module	Named supporting theorems
GoldbachMajorCompletionComplete.lean	3

- `goldbach_model_weighted_coefficient`
- `goldbach_source_ranges_eventual`
- `goldbach_major_completion_complete`

<!-- checkpoint-verification -->

Verification

All 1278 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5071-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4068 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1280 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input

includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; the Section10/11 analytic and sieve estimates remain open internal obligations. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1212_GoldbachCircleMethodComplete.md`

Checkpoint 1212: Full circle-method theorem and Section10 completion

Completed the full actual circle-method approximation and its stronger uniform cutoff form. A single choice of constants and thresholds simultaneously satisfies the checked minor, major-replacement and major-completion estimates. Genuine Fourier inversion, complementary-arc integration, finite weighted identities and the three-error mean-square inequality are all proved. The final theorem uses the actual untruncated prime-pair count and actual G_R H_R/(ell phi(k)) main term. Its only deep hypotheses are the already recorded independent ordinary SW, ordinary BV and uniform raw Ren/Ghosh inputs. Every source statement in Section10 is now complete in the agreed conditional scope.

Checked declarations

Module	Named supporting theorems
GoldbachCircleMethodComplete.lean	2

- `goldbach_circle_method_cutoff_complete`
- `goldbach_circle_method_complete`

<!-- checkpoint-verification -->

Verification

All 1279 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5072-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4070 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1281 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1213_GoldbachCircleSourceSeed.md`

Checkpoint 1213: Final Section10 source interface and completion

The final source interface accepts the manuscript's arbitrary nonzero real C-infinity compact seed directly, with the literal normalized real autocorrelation integral in H_R. A proved identity identifies that integral with the previously checked Schwartz implementation. The full untruncated circle-method theorem and unconditional singular-integral positivity/comparison are exported for this source interface. The current paper ledger and continuation metadata are reconciled to 16 proved,27 conditional,1 external and10 open; Section10 has no remaining own-paper source conclusion.

Checked declarations

Module	Named supporting theorems
GoldbachCircleSourceSeed.lean	3

- `goldbach_source_singular_integral_eq`
- `goldbach_circle_method_source_complete`
- `goldbach_source_singular_integral_positive_comparison`

<!-- checkpoint-verification -->

Verification

All 1280 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5073-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4073 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1282 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1214_GoldbachFixedSieveComparison.md

Checkpoint 1214: Section11 fixed sieve sets and exact exceptional count

Started Section11 with the actual fixed lower and upper prime sets. The lower comparison and the exact upper exceptional-pair inclusion are proved. Every lost upper pair is indexed by a prime p dividing n, has r=n-p^2, and has multiplier ell dividing n-p^2-1. Finite pair counting gives the complete exact upper comparison with error sum over p|n of tau(n-p^2-1). The actual closed multiplier set and its n-dependent subset are used. No deep input is added; the uniform power error is the next proof obligation.

Checked declarations

Module	Named supporting theorems
GoldbachFixedSieveComparison.lean	11

- `mem_goldbachSievePairs`
- `goldbachSievePairs_card`
- `goldbach_fixed_lower_subset`

- goldbach_fixed_lower_values_subset
- goldbach_fixed_lower_count
- goldbach_fixed_upper_subset
- goldbach_fixed_upper_exception
- goldbach_fixed_sieve_exceptions_card
- goldbach_fixed_upper_count_finite
- goldbach_closed_multiplier_positive
- goldbach_fixed_upper_source_count_finite

<!-- checkpoint-verification -->

Verification

All 1281 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5074-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4084 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1283 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1215_GoldbachFixedSieveSets.md

Checkpoint 1215: Complete Section11 fixed-sieve comparison lemma

Completed the full fixed-sieve source lemma. The exact upper exceptional divisor sum is bounded by K_epsilon N^epsilon using the already proved elementary prime-factor and divisor growth estimates. Constants precede every source parameter and both comparisons use the actual fixed and n-dependent sieve sets. The theorem covers every positive n<=N^2 and all real truncation levels, including the full source retained range. No PNT, sieve theorem, SW or BV input is needed. The source label l:fixedsievesets10 is marked proved only after this full clause review.

Checked declarations

Module	Named supporting theorems
GoldbachFixedSieveSets.lean	3

- goldbach_uniform_divisor_card
- goldbach_fixed_sieve_divisor_sum
- goldbach_fixed_sieve_sets

<!-- checkpoint-verification -->

Verification

All 1282 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5075-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4087 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1284 Lean files pass

the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1216_GoldbachLowerSieveSequence.md`

Checkpoint 1216: Exact lower sieve progression sequence

The actual lower divisibility sequence is identified exactly with the untruncated prime-square/progression count at modulus24d. The CRT equivalence, injective removal of the r-1 shift, and finite count bijection are proved. A coprime progression residue exists and is chosen before every n, so no n-dependence is hidden in later sieve coefficients. The exact G_R H_R/phi(24d) plus circle-method remainder identity is exported. The local-density normalization by X_n and omega_n remains the next internal obligation.

Checked declarations

Module	Named supporting theorems
GoldbachLowerSieveSequence.lean	8

- `goldbach_lower_divisibility_values_card`
- `goldbach_sieve_coprime_twentyfour`
- `goldbach_lower_sieve_residue_iff`
- `mem_goldbach_progression_pairs`
- `goldbach_lower_sieve_sequence_count`
- `goldbach_sieve_support_coprime_six`
- `goldbach_lower_sequence_fixed_residue`
- `goldbach_lower_sequence_circle_identity`

<!-- checkpoint-verification -->

Verification

All 1283 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5076-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4095 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1285 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10

statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1217_GoldbachUpperSieveSequence.md

Checkpoint 1217: Exact upper sieve progression sequence

Proved the exact upper divisibility-sequence count as an untruncated prime-square/progression count at modulus12d and argument n-1. The congruence equivalence is proved in both directions. The progression conditions force the prime-square variable to exceed3 and the other prime to be3 modulo4. The actual closed multiplier set is proved to supply ell=2v with v=3 modulo4 and coprime to6 once z>3. A valid coprime residue is chosen before all n. The actual arithmetic/singular main term plus circle-method remainder identity is exported, with the natural n-1 argument retained exactly.

Checked declarations

Module	Named supporting theorems
GoldbachUpperSieveSequence.lean	7

- `goldbach_upper_sieve_residue_iff`
- `goldbach_upper_prime_conditions`
- `goldbach_upper_sieve_sequence_count`
- `goldbach_prime_support_one_mod_four`
- `goldbach_closed_multiplier_residue_data`
- `goldbach_upper_sequence_fixed_residue`
- `goldbach_upper_sequence_circle_identity`

<!-- checkpoint-verification -->

Verification

All 1284 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5077-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4102 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1286 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1218_GoldbachSieveRemainderTransport.md

Checkpoint 1218: Selected sieve remainder estimates and upper argument shift

The completed circle-method mean-square bound is transferred to every fixed finite selected subfamily of allowed arithmetic indices by explicit zero extension. Missing residues are extended by1 and verified

coprime, while all missing coefficients are zero. The actual n-1 upper-sieve shift is proved on any positive subset of the source range, and on the full natural range with a factor2 that handles the extra zero-boundary term. Constants remain fixed before the selected family and coefficients. Only the existing raw SW, BV and Ren/Ghosh inputs are used; no general sieve input or own-paper endpoint is added.

Checked declarations

Module	Named supporting theorems
GoldbachSieveRemainderTransport.lean	3

- `goldbach_pred_square_sum_bound`
- `goldbach_pred_full_square_sum_bound`
- `goldbach_sieve_remainders_selected`

<!-- checkpoint-verification -->

Verification

All 1285 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5078-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4105 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1287 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1219_GoldbachSieveRootCounts.md`

Checkpoint 1219: Actual root counts, density bounds and multiplicativity

The local root count is the actual finite number

$$\rho_p(a) = \#\{u \in (\mathbb{Z}/p\mathbb{Z})^\times : u^2 = a\}.$$

Opening the prime Ramanujan sum proves the exact identity

$$C_p(a) = p\rho_p(a) - (p-1).$$

The previously proved odd-prime Cohen evaluation then gives $\rho_p(a) = 2$ when its Legendre symbol is1 and0 otherwise. In particular $0 \le \rho_p(a) \le 2$, with zero roots at the zero residue. The progression-restricted solution count is proved equal to $\rho_p(n - \ell b)$; it is a cardinality identity for the actual pair set, not a replacement model.

The lower and upper candidate prime densities are defined with the source's inclusive cutoff $p \le R$. Their denominators are strictly positive for every prime $p > 3$. Both are nonnegative and bounded by $2/(p-3)$, which is strictly below1 for $p \ge 7$. At5, a kernel-checked exhaustive calculation proves that either $\rho_5(n) = 0$, or $\rho_5(n) = 2$ and $\rho_5(n-1) = 0$. This proves that the upper density is at most1/2, including the branch $p > R$, and hence lies in $[0, 1)$ at every allowed prime.

For any real prime density $g(p)$, the explicit function

$$\omega(d) = d \prod_{p|d} g(p)$$

satisfies $\omega(1) = 1$, is multiplicative on coprime natural arguments, and obeys $\omega(p)/p = g(p)$. It is defined at0 as well; the factor d makes that value zero. These statements require no unformalized number-theoretic input.

Scope: these files establish actual root counts, the candidate density bounds, and multiplicativity. The full paper local-density claims remain open until the candidates are connected to the actual normalized Euler products. The subsequent milestone supplies that connection and handles the possible zero upper base factor without dividing by it. The original manuscript is unchanged.

Checked declarations

Module	Named supporting theorems
GoldbachSieveRootCounts.lean	8
GoldbachSieveDensityBounds.lean	7
GoldbachSieveDensityMultiplicative.lean	4

- `goldbach_rho_eq`
- `goldbach_rho_congr`
- `goldbach_cohen_rho`
- `goldbach_rho_legendre`
- `goldbach_rho_le_two`
- `goldbach_rho_of_zero`
- `goldbach_rho_five_consecutive`
- `goldbach_prime_restricted_count`
- `goldbach_local_denominator_pos`
- `goldbach_local_density_nonneg`
- `goldbach_local_density_le`
- `goldbach_density_bound_lt_one`
- `goldbach_local_density_lt_one`
- `goldbach_upper_density_five`
- `goldbach_upper_density_range`
- `goldbach_sieve_omega_one`
- `goldbach_sieve_omega_prime`
- `goldbach_sieve_omega_mul`
- `goldbach_sieve_omega_div`

<!-- checkpoint-verification -->

Verification

All 1288 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5081-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4124 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1290 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional

explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1220_GoldbachSieveConductor.md

Checkpoint 1220: Actual local Euler factors and prime adjunction

The exact progression coefficient is evaluated whenever $q \mid \ell k$, rather than requiring $q \mid k$. Reduction of units to $\gcd(q,k)$, together with the normalized unit-fiber mass, gives

$$G_q(\ell,k,b,n) = \frac{C_q(n-\ell b)}{\phi(q)}.$$

The proof starts from the already checked character coefficient and proves the finite averaging identity internally. In particular this handles the upper sieve at modulus8 even though its base progression modulus is12 and its multiplier is even.

For every prime, the actual unrestricted Euler factor, with multiplier invertible modulo that prime, is

$$1 + G_p = \frac{p(p-1-\rho_p(n))}{(p-1)^2}.$$

For a fixed progression unit it is $p\rho_p(n-\ell b)/(p-1)$. When the multiplier vanishes modulo the prime, the common factor is $p\rho_p(n)/(p-1)$, which may be zero. The actual unrestricted pair count also satisfies $\#\text{pairs} + \rho_p(n) = p-1$.

Under $24 \mid \ell k$ and $n - \ell b \equiv 1 \pmod{24}$, the small-prime factors are exactly8 at2 and3 at3. Thus the original arithmetic factor is24 times the product over primes $p > 3$ with $p \le R$. Congruent progression residues give equal original coefficients and arithmetic factors.

Adjoining a prime $p > 3$, with $p \nmid k\ell$, and imposing $\ell b \equiv c \pmod{p}$, proves the exact normalized identity

$$\frac{G_R(\ell,kp,b,n)}{\phi(kp)} = \frac{G_R(\ell,k,b,n)}{\phi(k)} \begin{cases} \rho_p(n-c)/(p-1-\rho_p(n)), & p \le R, \\ 1/(p-1), & p > R. \end{cases}$$

Every other prime factor is proved unchanged. The proof multiplies local equalities and uses only the strictly positive unrestricted factor at the newly adjoined prime. It never divides by the complete base arithmetic factor, so common zero obstructions are retained correctly.

Scope: these are exact identities for the original finite character coefficients and Euler products, with no added deep inputs. The source local-density lemmas remain open at this saved milestone; the next milestone proves finite-prime assembly and the exact lower and upper CRT specializations.

Checked declarations

Module	Named supporting theorems
GoldbachSieveConductor.lean	2
GoldbachSievePrimeFactors.lean	4
GoldbachSieveSmallFactors.lean	4
GoldbachSieveResidueFactors.lean	4
GoldbachSievePrimeAdjunction.lean	3

- `goldbach_modular_multiplier_constant`
- `goldbach_general_coefficient_conductor`
- `goldbach_prime_unrestricted_factor`
- `goldbach_prime_fixed_factor`
- `goldbach_prime_zero_multiplier_factor`
- `goldbach_prime_unrestricted_count`
- `goldbach_two_factor_admissible`
- `goldbach_three_factor_admissible`
- `mem_goldbach_large_local_primes`
- `goldbach_arithmetic_factor_large_primes`
- `goldbach_general_coefficient_residue_congr`
- `goldbach_arithmetic_factor_coefficient_product`
- `goldbach_arithmetic_factor_residue_congr`

- goldbach_prime_factor_unchanged
- goldbach_prime_local_density_ratio
- goldbach_arithmetic_factor_adjoin_prime
- goldbach_normalized_factor_adjoin_prime

<!-- checkpoint-verification -->

Verification

All 1293 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5086-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4141 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1295 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1221_GoldbachSieveDensityFactorization.md

Checkpoint 1221: Complete lower and upper local-density lemmas

Both `l:lowerlocaldensity` and `l:upperlocaldensity` are proved for the retained source context. No new external input is used. The selected manuscript is unchanged. The clause-by-clause source review is in `verification/section11-local-density-source-review1221.json`.

The root count is the actual cardinality of the units satisfying the square equation. Consequently zero is excluded from the variable being squared, and rho is zero at the zero residue. Opening the prime Ramanujan sum proves

$$C_p(a) = p\rho_p(a) - (p-1), \qquad \rho_p(a) = \begin{cases} 2, & (a/p) = 1, \\ 0, & (a/p) \neq 1. \end{cases}$$

The normalized character coefficient agrees with the earlier original definition. Its unrestricted factor, when the multiplier is invertible, is

$$1 + G_p = \frac{p(p-1-\rho_p(n))}{(p-1)^2}.$$

If the progression fixes its unit variable to b, the factor becomes

$$1 + G_p = \frac{p\rho_p(n-\ell b)}{p-1}.$$

The proof also checks the multiplier-zero case, whose common factor can vanish. The actual unrestricted pair cardinality plus rho equals p-1.

The finite coefficient identity for q dividing ell*k is needed at the small primes. It is proved from reduction of units and their exact normalized fiber mass. Under the source congruence n-ell*b=1 modulo24, this gives the factor8 at2 and3 at3. In particular, the upper modulus12 with its even multiplier is treated correctly at8.

Adjoining one prime p>3 to a coprime progression modulus proves an identity between the original normalized arithmetic factors. The identity multiplies by

$$g(p) = \begin{cases} \rho_p(n-c)/(p-1-\rho_p(n)), & p \le R, \\ 1/(p-1), & p > R, \end{cases}$$

when ell*b=c modulo p. The cutoff is inclusive, including a prime equal to R. Every other prime factor is proved unchanged. Induction over a finite set of distinct primes, followed by the square-free prime-factor product identity, proves the full factorization for d.

For the lower sieve, the actual CRT conditions are b=11 modulo24 and b=1 modulo d, while n=12 modulo24. They give c=1. The base arithmetic factor at modulus24 is proved nonzero, so the paper's original quotient definition is identified with this density. The prime formula is proved for every p>3; the lower sieve's additional restriction p=3 modulo4 is therefore included.

For the upper sieve, the existing source multiplier result supplies ell=2v and v coprime6. The actual conditions vb=5 modulo12 and ell*b+1=0 modulo d imply the required congruence at24 and give c=-1 at every divisor prime. Unit cancellation shows that every representative reduces to the same base class modulo12. The resulting density is

$$g_{n,\ell}(p) = \begin{cases} \rho_p(n)/(p-1-\rho_p(n-1)), & p \le R, \\ 1/(p-1), & p > R. \end{cases}$$

One explicit function omega(d)=d times the product of these prime densities is constructed before all supported square-free d and all representatives of the source CRT classes. Its value at1 and its multiplicativity on coprime arguments are proved. The complete existence and factorization statement is `goldbach_upper_local_density`.

The denominator is positive for p>3. Both candidate densities are nonnegative and at most 2/(p-3), which is below1 for p>=7. At5, a kernel-checked finite calculation gives either rho_5(n)=0, or rho_5(n)=2 and rho_5(n-1)=0. The upper density is therefore at most1/2 at5. This proves its full required range [0,1).

The upper identity never divides by its base arithmetic factor. If a common obstructing local factor is zero, the multiplicative identity still proves equality with zero on both sides. A zero base is not an extra hypothesis or an omitted case.

The paper-level ledger is now19 proved claims,27 complete conditional implications,1 quoted external result and7 open claims, with4 definitions separate. Presieving and the later sieve estimates remain open. The later analytic use of these densities does not retroactively become an input to the local-density proofs.

Checked declarations

Module	Named supporting theorems
GoldbachSieveDensityFactorization.lean	2
GoldbachSieveLocalDensities.lean	7

- `goldbach_normalized_factor_prime_product`
- `goldbach_sieve_density_factorization`
- `goldbach_large_prime_coprime_twentyfour`
- `goldbach_lower_base_nonzero`
- `goldbach_lower_density_factorization`
- `goldbach_lower_local_density_formula`
- `goldbach_upper_phase_twentyfour`
- `goldbach_upper_density_factorization`
- `goldbach_upper_local_density`

<!-- checkpoint-verification -->

Verification

All 1295 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5088-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4150 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1297 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1222_GoldbachPresieveCoefficients.md`

Checkpoint 1222: Finite presieving coefficients, sequence bounds and quantitative error

The finite two-stage construction is proved in Lean. The original source lemma remains open: its dimension hypothesis must be corrected before the classical coefficient theorem can supply the stated sieve functions. Proposed replacement passages are in `Section11_Presieving_Replacements.tex`. They have not been applied to the selected manuscript.

The primary source checked is Yuta Suzuki's The Rosser–Iwaniec sieve, dated14 March2023. Definition8.2 requires the product bound with factor 1+K/log(w), rather than an unrestricted fixed factor K0. Lemma14.7 states the corresponding coefficient estimates with D>=z>=2. Proposition11.8 controls the sieve-function tails. The original official Iwaniec1980 PDF downloads returned403; this review does not claim they were read.

Three points need to be explicit in the manuscript. First, the refined dimension estimate is stronger than a coarse implied-constant estimate. A fixed K0 allows a concentration of local mass at a few large primes, while 1+K/log(w) approaches1 at large w. Consequently the coarse estimate alone is not the hypothesis of the cited coefficient theorem. The refined interval forms of Mertens' theorem are available for the actual outer densities in this application; they must be stated and used.

Second, the preliminary prime set includes p=R. A dimension condition restricted to p<y with y<=R omits that endpoint if R is prime. The proposed formulation bounds the truncated preliminary prime set over all relevant intervals and uses sifting limit R+1. This retains exactly the intended primes p<=R.

Third, D0<D by itself does not place the outer sieve in the range D1>=z, and log(D0)/log(R) tending to infinity by itself does not give the rate needed to absorb arbitrary outer logarithmic losses. The proposed passage states the actual scale R=(log N)^C, the bound R+1<=z<=D1<=N^A, and the intended uniformity. These conditions give preliminary relative errors bounded by a constant times exp(-c(log log N)^2), which dominate every fixed negative power of log N. Applying these scale conditions to the final lower and upper sieve parameters is a separate later obligation.

The checked finite construction does not assume the paper's two-stage conclusion. It starts from independent one-stage lower and upper weight inequalities. For an outer lower coefficient u(d), the preliminary choice is the lower weight when u(d)>=0 and the upper weight when u(d)<0. For an outer upper coefficient, the choices are reversed. Lean proves the resulting pointwise inequalities, then sums them over actual finite sequences. The sequence is represented by a finite set of indices and an arbitrary integer-valued map, so repeated values are counted correctly.

Disjoint prime supports imply coprimality of every preliminary divisor e and outer divisor d. This gives unique decomposition of m=e*d. The flattened coefficient is explicitly the sum over pairs with that product. Lean proves its absolute value is at most1 and that it vanishes for m>=D0*D1. It also proves exact reindexing into the sum over all natural m<D0*D1, using the natural ceiling of the real level. None of these coefficient definitions mentions the density function, the sequence, X or the remainders.

For the main term, let V0 be the preliminary Euler product, and suppose each preliminary coefficient sum differs from V0 by at most epsilon0*V0. The error from either signed selection is proved to be at

most

$$\varepsilon_0 V_0 \sum_d |u(d)|g(d).$$

Together with the outer coefficient estimates, this proves explicit finite lower and upper inequalities with the original signed remainders. No positivity assumption on an outer signed weight is introduced. The asymptotic coefficient theorem, the uniform Euler-product growth bound and the rate absorption must still be joined before calling the full corrected asymptotic lemma complete.

The source-level ledger remains19 proved claims,27 complete conditional implications,1 quoted external result and7 open claims. The presieving supporting theorems do not increase that count by themselves.

Checked declarations

Module	Named supporting theorems
GoldbachPresieveCoefficients.lean	5
GoldbachPresievePointwise.lean	9
GoldbachPresieveFiniteSequence.lean	7
GoldbachPresieveMainError.lean	3
GoldbachPresieveCoefficientAssembly.lean	5

- `goldbach_presieve_signed_sandwich`
- `goldbach_presieve_unique_product`
- `goldbach_presieve_pushforward_bound`
- `goldbach_presieve_pushforward_reindex`
- `goldbach_presieve_level`
- `goldbach_div_indicator_nonneg`
- `goldbach_sift_indicator_nonneg`
- `goldbach_div_indicator_product`
- `goldbach_sift_indicator_union`
- `goldbach_presieve_pair_bound`
- `goldbach_presieve_pair_sum`
- `goldbach_presieve_pair_divisor_sum`
- `goldbach_presieve_pointwise_lower`
- `goldbach_presieve_pointwise_upper`
- `goldbach_sequence_divisor_count_sum`
- `goldbach_sequence_sifted_count_sum`
- `goldbach_sequence_weighted_divisor_sum`
- `goldbach_presieve_sequence_lower`
- `goldbach_presieve_sequence_upper`
- `goldbach_presieve_sequence_expansion`
- `goldbach_presieve_disjoint_support_coprime`
- `goldbach_presieve_main_error`
- `goldbach_presieve_quantitative_lower`
- `goldbach_presieve_quantitative_upper`
- `goldbach_presieve_pushforward_off_image`
- `goldbach_presieve_pushforward_superset`
- `goldbach_presieve_coefficient_properties`
- `goldbach_presieve_product_range`
- `goldbach_presieve_remainder_level`

<!-- checkpoint-verification -->

Verification

All 1300 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5093-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4179 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1302 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1223_GoldbachPresieveDecay.md

Checkpoint 1223: Presieve Euler mass, exact scales and uniform error decay

The outer coefficient mass is bounded for the actual square-free divisor support. The proof first derives the prime-product expression from multiplicativity on supported square-free integers; no stronger global multiplicativity hypothesis is used. Injectivity of the prime-factor set on square-free integers then embeds the divisor sum into a sum over subsets of the outer prime set. Expanding the Euler product gives

$$M = \sum_d |u(d)|g(d) \le \prod_{p\in\mathfrak{P}_1}(1+g(p)), \qquad \frac{M}{V_1} \le V_1^{-2}.$$

Every factor of V1 is strictly positive. The estimate uses only0<=g(p)<1 and |u(d)|<=1; the outer weights may have either sign.

Given the dimension product bound at w=2 and log(z)<=A*log(N), the inverse outer Euler product is at most

$$K_2(\log N)^\kappa, \qquad K_2 = (1+K/\log 2)(A/\log 2)^\kappa.$$

The finite preliminary error therefore has a uniform logarithmic majorant independent of the density function.

Write t=log(log(N)), D0=exp(t^3), and use preliminary sifting limit exp(C*t)+1. Lean proves

$$s_0 \ge \frac{t^2}{C+1}$$

for t>=1, proves s0 tends to infinity, and proves D0 eventually exceeds that sifting limit. Gaussian decay against every fixed exponential-linear factor and power of t is proved from Mathlib's exponential asymptotics. Consequently the preliminary error K0*exp(-c*s0) absorbs every fixed power of log(N).

If D1 is at least exp(C*t)+1 eventually, it tends to infinity. For every delta>0, the outer error K1*(log(D1))^(-delta) tends to zero. Combining these deductions proves the exact common error rate used in the next assembly:

$$K_1(\log D_1)^{-\delta} + K_0 e^{-cs_0}\big(K_2(\log N)^\kappa\big)^2 \longrightarrow 0.$$

All these arithmetic and limit deductions are proved internally. The independent classical one-stage coefficient theorem is not used as an input to this milestone. The next milestone connects its two explicit interfaces to these bounds and proves the corrected full presieving implication. The original weaker manuscript statement remains unclaimed, with proposed replacements saved separately.

Checked declarations

Module	Named supporting theorems
GoldbachPresieveDecay.lean	11
GoldbachPresieveEulerMass.lean	5

- goldbach_presieve_gaussian_decay
- goldbach_presieve_polylog_decay
- goldbach_presieve_preliminary_ratio

- goldbach_presieve_preliminary_error_bound
- goldbach_presieve_preliminary_s_tendsto
- goldbach_presieve_preliminary_level_eventually
- goldbach_presieve_preliminary_absorption
- goldbach_presieve_outer_level_tendsto
- goldbach_presieve_outer_error_tendsto
- goldbach_presieve_total_error_tendsto
- goldbach_presieve_uniform_error_tendsto
- goldbach_presieve_density_prime_product
- goldbach_presieve_outer_mass
- goldbach_presieve_outer_product
- goldbach_presieve_outer_mass_ratio
- goldbach_presieve_inverse_polylog

<!-- checkpoint-verification -->

Verification

All 1302 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5095-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4195 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1304 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

`Checkpoint_1224_GoldbachPresieveUniformAssembly.md`

Checkpoint 1224: Corrected full presieving from independent classical inputs

The corrected full presieving implication is proved in `goldbach_corrected_presieving`, conditional on the independent general one-stage Rosser–Iwaniec fundamental and coefficient inputs. Their exact interfaces are documented in `docs/EXTERNAL_SIEVE_INPUTS.md`. The literal original source lemma remains open until its weaker hypotheses are corrected. Proposed replacement passages are in `Section11_Presieving_Replacements.tex`. They have not been applied to the selected manuscript.

The primary source checked is Yuta Suzuki's The Rosser–Iwaniec sieve, dated14 March2023. Definition8.2 requires the product bound with factor 1+K/log(w), rather than an unrestricted fixed factor K0. Lemma14.7 states the corresponding coefficient estimates with D>=z>=2. Proposition11.8 controls the sieve-function tails. The original official Iwaniec1980 PDF downloads returned403; this review does not claim they were read.

The corrected statement explicitly requires X>=0 and g(1)=1; X=0 is included. Three further points need to be explicit in the manuscript. First, the refined dimension estimate is stronger than a coarse implied-constant estimate. A fixed K0 allows a concentration of local mass at a few large primes, while 1+K/log(w) approaches1 at large w. Consequently the coarse estimate alone is not the hypothesis of the cited coefficient theorem. The refined interval forms of Mertens' theorem are available for the actual outer densities in this application; they must be stated and used.

Second, the preliminary prime set includes p=R. A dimension condition restricted to p<y with y<=R

omits that endpoint if R is prime. The proposed formulation bounds the truncated preliminary prime set over all relevant intervals and uses sifting limit R+1. This retains exactly the intended primes p<=R.

Third, D0<D by itself does not place the outer sieve in the range D1>=z, and log(D0)/log(R) tending to infinity by itself does not give the rate needed to absorb arbitrary outer logarithmic losses. The proposed passage states the actual scale R=(log N)^C, the bound R+1<=z<=D1<=N^A, and the intended uniformity. These conditions give preliminary relative errors bounded by a constant times exp(-c(log log N)^2), which dominate every fixed negative power of log N. Applying these scale conditions to the final lower and upper sieve parameters is a separate later obligation.

The checked finite construction does not assume the paper's two-stage conclusion. It starts from independent one-stage lower and upper weight inequalities. For an outer lower coefficient u(d), the preliminary choice is the lower weight when u(d)>=0 and the upper weight when u(d)<0. For an outer upper coefficient, the choices are reversed. Lean proves the resulting pointwise inequalities, then sums them over actual finite sequences. The sequence is represented by a finite set of indices and an arbitrary integer-valued map, so repeated values are counted correctly.

Disjoint prime supports imply coprimality of every preliminary divisor e and outer divisor d. This gives unique decomposition of m=e*d. The flattened coefficient is explicitly the sum over pairs with that product. Lean proves its absolute value is at most1 and that it vanishes for m>=D0*D1. It also proves exact reindexing into the sum over all natural m<D0*D1, using the natural ceiling of the real level. None of these coefficient definitions mentions the density function, the sequence, X or the remainders.

For the main term, let V0 be the preliminary Euler product, and suppose each preliminary coefficient sum differs from V0 by at most epsilon0*V0. The error from either signed selection is proved to be at most

$$\varepsilon_0 V_0 \sum_d |u(d)|g(d).$$

Together with the outer coefficient estimates, this proves explicit finite lower and upper inequalities with the original signed remainders. No positivity assumption on an outer signed weight is introduced. The uniform Euler-product bound and exact rate absorption are now proved and joined to the two independent classical one-stage interfaces in the complete corrected asymptotic deduction. No own-paper endpoint is assumed.

The source-level ledger remains19 proved claims,27 complete conditional implications,1 quoted external result and7 open claims. The presieving supporting theorems do not increase that count by themselves.

The signed outer mass M is proved to satisfy M/V1<=V1^(-2). The refined dimension condition at w=2 and log(z)<=A*log(N) give V1^(-1)<=K2*(log N)^kappa, with K2=(1+K/log 2)*(A/log 2)^kappa. Multiplicativity is needed only on supported square-free integers; its prime-product expression is proved internally.

For t=log(log N), D0=exp(t^3), and preliminary sifting limit exp(C*t)+1, the exact preliminary parameter satisfies s0>=t^2/(C+1) for t>=1. Lean proves this bound, eventual level admissibility and s0 tending to infinity. The explicit common relative error

$$\varepsilon(N) = K_1(\log D_1)^{-\delta} + K_0 e^{-cs_0}\big(K_2(\log N)^{\kappa}\big)^2 \longrightarrow 0$$

is proved uniformly before all admissible densities. For every sufficiently large N, the final coefficients are chosen before g, the finite sequence, X and the remainders. In the paper application they are therefore chosen before n. The main terms contain only the outer functions f_kappa and F_kappa. The natural-absolute-value bridge covers repeated and negative integer sequence values.

The original manuscript is unchanged. The corrected complete conditional variant is recorded separately from the literal-source inventory. Its independent one-stage inputs and classical function normalization remain unproved; the kernel audit verifies the consuming implication. The next mathematical milestone is the actual lower-sieve application.

Checked declarations

Module	Named supporting theorems
GoldbachPresieveUniformAssembly.lean	1
GoldbachPresieveClassicalInputs.lean	8

- `goldbach_presieve_uniform_assembly`

- goldbach_density_restrict
- goldbach_refined_dimension_restrict
- goldbach_refined_dimension_total
- goldbach_presieving_from_classical_coefficients
- goldbach_presieve_source_R
- goldbach_corrected_presieving
- goldbach_integer_sequence_divisor_count
- goldbach_integer_sequence_sifted_count

<!-- checkpoint-verification -->

Verification

All 1304 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5097-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4204 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1306 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The rational smooth multifrequency theorem with circle transference remains an explicit independent compound hypothesis. The classical primitive sum-of-two-squares criterion is an additional explicit independent arithmetic hypothesis for Section9 reductions; it contains no own-paper conclusion and is not declared as a Lean axiom.

Checkpoint_1225_GoldbachLowerSieveParameters.md

Checkpoint 1225: Actual lower-sieve levels and refined density conditions

The actual lower-sieve levels D=N/(24(log N)^B), D1=D/exp((log log N)^3), and z=N^(8/9) now have a proved normalized logarithmic limit log(D1)/log(N) -> 1 and sieve-parameter limit log(D1)/log(z) -> 9/8. The entire admissible range R+1<=z<=D1<=N^2 is proved eventually for every fixed B and C, and D0*D1=D is exact. The finite prime set and its disjoint preliminary/outer split are explicit. The actual unit-square local densities yield a supported multiplicative density with all prime factors in [0,1). Both refined dimension conditions are proved from independent classical interval Mertens products. The preliminary majorant is 2/(p-3), p>=7, of dimension2; the outer product is the ordinary p=3 mod4 comparison of dimension1/2. This input contains no n, Goldbach sequence, presieve conclusion, or own-paper endpoint. The real lower application and circle-error assembly remain to be joined. The manuscript is unchanged while the user edits the supplied replacements.

Checked declarations

Module	Named supporting theorems
GoldbachLowerSieveParameters.lean	6
GoldbachLowerSieveDensity.lean	7

- goldbach_lower_outer_level_pos
- goldbach_lower_outer_log
- goldbach_lower_outer_log_ratio
- goldbach_lower_sieve_parameter_tendsto
- goldbach_lower_sieve_levels_eventually
- goldbach_lower_product_level

- `mem_goldbach_lower_sieve_primes`
- `goldbach_lower_prime_ge_seven`
- `goldbach_lower_prime_split`
- `goldbach_lower_prime_density_range`
- `goldbach_lower_density_multiplicative`
- `goldbach_lower_preliminary_dimension`
- `goldbach_lower_outer_dimension`

<!-- checkpoint-verification -->

Verification

All 1306 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5099-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4217 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1308 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Existing independent classical inputs remain explicit. Classical interval Mertens products for the universal majorant and the progression3 mod4 are additional independent arithmetic inputs; no own-paper endpoint is assumed.

`Checkpoint_1226_GoldbachLowerSieveBridge.md`

Checkpoint 1226: Actual lower main term and circle remainder estimate

The actual lower sequence is now joined to the original arithmetic factor and smoothed singular integral. The base arithmetic factor is proved real and strictly positive at n=12 mod24; positivity of the singular integral proves X_n>=0 without a new assumption. CRT residue classes are fixed before n and are chosen to be units for every denominator. The exact real divisor-count identity and the sifted-count identity use the actual prime-square representations. The lower remainder sum is proved to satisfy the full N^4(log N)^(-A) mean-square bound, uniformly over every real coefficient family with absolute values at most1. This is a deduction from the existing ordinary SW/BV and Ren/Ghosh inputs, via the already checked circle-method theorem. The embedding d -> (1,24d), its injectivity, strict real modulus levels, reindexing and real-part norm comparison are all proved. A zero extension at denominator0 avoids an artificial zero-modulus term. The final presieve and limiting sieve-function assembly remain next. The selected manuscript is unchanged.

Checked declarations

Module	Named supporting theorems
GoldbachLowerSieveBridge.lean	6
GoldbachLowerSieveRemainder.lean	2

- `goldbach_lower_residue_properties`
- `goldbach_lower_residue_coprime`
- `goldbach_lower_base_real_positive`
- `goldbach_lower_main_term_nonneg`
- `goldbach_lower_divisor_real_identity`
- `goldbach_lower_sifted_sequence`
- `goldbach_lower_remainder_zero_extension`

- goldbach_lower_remainder_mean_square

<!-- checkpoint-verification -->

Verification

All 1308 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5101-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4225 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1310 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Existing independent classical inputs remain explicit. Classical interval Mertens products for the universal majorant and the progression3 mod4 are additional independent arithmetic inputs; no own-paper endpoint is assumed.

Checkpoint_1227_GoldbachLowerSieveApplication.md

Checkpoint 1227: Complete lower-sieve application and mean-square error

The full lower-sieve application, stable source label `p:sieveneedlower8`, is proved in `goldbach_lower_sieve_complete`. This is a complete implication under independent classical inputs, not an unconditional formalization of those inputs. Both clauses of the source proposition are covered.

For every positive prescribed logarithmic saving A, the theorem constructs fixed B>=0, C>0 and M>0, a common real error eta(N) tending to zero, and for every sufficiently large integer N a coefficient family c_d chosen before n, with |c_d|<=1. For every n=12 mod24 it proves

$$X_n\Theta_n(z)(f_{1/2}(9/8)-\eta(N))+\mathcal{E}_1(n)\le S(\mathcal{A}_n,\mathbb{P}_{3,n},z),\qquad z=N^{8/9},$$

and

$$\sum_{n<N^2}\mathcal{E}_1(n)^2\le MN^4(\log N)^{-A}.$$

The pointwise statement actually holds for every n in that residue class, so it includes the entire retained n range uniformly for every A0. The error is uniform because it is fixed before n; no density-dependent o(1) is introduced.

Actual levels and the common error

The definitions use D=N/(24(log N)^B), D0=exp((log log N)^3), D1=D/D0 and R=(log N)^C. Lean proves log(D1)/log(N)->1, log(D1)/log(z)->9/8, D0*D1=D, and the eventual complete range R+1<=z<=D1<=N^2. Thus the classical one-stage theorem is used in its admissible range.

Write epsilon(N) for the already proved explicit uniform presieving error. The new error is exactly

$$\eta(N)=\varepsilon(N)+\left|f_{1/2}(\log D_1/\log z)-f_{1/2}(9/8)\right|.$$

The first summand tends to zero by the quantitative presieve, and the second by the proved parameter limit and continuity of the classical sieve function at9/8. Positivity of the actual X_n and Theta_n permits the lower replacement. No numerical value or positivity of f_(1/2)(9/8) is assumed or needed here.

Actual sequence and main term

The sequence consists of r-1 for the actual admissible representations n=m^2+r with m and r prime and r=11 mod24. The finite-index representation counts each admissible r once, and the image r->r-1 is proved injective on this set. Its divisibility counts are exactly the progression counts with one CRT residue a_d modulo24d. Those residues are fixed independently of n. Their coprimality and congruences are proved, not supplied as new hypotheses.

The density is the product over prime factors of the actual root-count density rho_p(n-1)/(p-1-rho_p(n)) for p<=R, and 1/(p-1) for p>R. Multiplicativity, normalization at1, the interval[0,1), and both refined dimension conditions are proved. The preliminary condition is obtained from the universal majorant2/(p-3) on p>=7, with dimension2. The outer condition has dimension1/2 on the fixed primes3 mod4.

X_n is the real part of the original arithmetic factor G_R(1,24,n)/phi(24) times the original smoothed singular integral. Lean proves the base factor real and strictly positive for n=12 mod24, and uses the already proved singular-integral nonnegativity. The divisor-count expansion uses this exact X_n and the original circle-method remainder. Theta_n is exactly the Euler product of the same local densities. The two split Euler products are proved to equal it.

The corrected presieving construction chooses coefficients before all density functions, finite sequences and n. Its actual fixed-prime sifted count is bounded above by the original n-dependent sifted count using the previously proved fixed-sieve comparison.

Mean-square remainder

The remainder is the real signed sum over positive d<D of the original r_(24d,1)(n), with the fixed CRT residues. The embedding d->(1,24d) into the circle-method coefficient set is proved injective. Its moduli satisfy the strict bound24d<N(log N)^(-B). The coefficients have complex norm at most1. Thus the already proved circle-method theorem applies with the constant multiplier range L=2 and, for example, delta=1/2. Reindexing and |Re(z)|<=|z| give the stated real mean-square estimate. The zero denominator is assigned remainder0, so it introduces no artificial term.

Independent hypotheses and limits

The exact unproved inputs consumed are ordinary prime Siegel-Walfisz, ordinary Bombieri-Vinogradov, the existing raw Ren/Ghosh estimates, the general one-stage Rosser-Iwaniec fundamental and coefficient estimates, the two general interval Mertens products, and continuity of the classical lower sieve function at9/8. The last property is part of the classical sieve-function boundary. All are theorem parameters or proposition-valued interfaces; none is a new Lean axiom. The Mertens interface mentions only arbitrary prime intervals and universal comparison factors. It contains no n, Goldbach sequence, source presieve, lower-sieve proposition or own-paper endpoint.

The selected manuscript is unchanged. The user is editing the compact presieving replacements; their application is not claimed without a new source or confirmation. No further manuscript replacement was found necessary for this lower-sieve milestone.

Remaining work

Five substantive Goldbach claims remain: `l:maintermsize8`, `l:wumatomakisum`, `p:sieveneedupper8`, `l:numericalcomparison`, and `t:repmotthm`. The literal weaker presieving statement remains separately open pending manuscript synchronization, while its corrected conditional variant is already proved. The source ledger is now19 proved,28 complete conditional implications,1 quoted external theorem and6 open claims, plus4 definitions. Checkpoint numbers and supporting-theorem counts are not completion percentages.

Checked declarations

Module	Named supporting theorems
GoldbachLowerSieveApplication.lean	3

- `goldbach_lower_theta_positive`
- `goldbach_lower_presieve_application`
- `goldbach_lower_sieve_complete`

<!-- checkpoint-verification -->

Verification

All 1309 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5102-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4228 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1311 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The lower-sieve endpoint is proved from independent ordinary SW/BV, raw Ren/Ghosh, one-stage Rosser-Iwaniec, interval Mertens and classical sieve-function continuity inputs. No own-paper sieve or Goldbach endpoint is assumed. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The main-term size, weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1228_GoldbachArchimedeanLower.md`

Checkpoint 1228: Explicit archimedean and arithmetic product lower bounds

The lower-bound ingredients now include a kernel-checked explicit archimedean integral estimate and quantitative bounds for the actual local products. For64<=n<=N^2, the interval[sqrt(n)/4,sqrt(n)/2] lies inside the original integration domain. On it, the integrand is at least1/(2(log N)^2). Integrability, global nonnegativity, restriction monotonicity and the exact interval measure are proved, yielding H1(n)>=sqrt(n)/(8(log N)^2). The singular integral is the original one; no artificial substitute main term is introduced.

For every prime p>=7, the unrestricted arithmetic local factor is at least1-2/(p-3); the prime5 factor is at least1/2. The original arithmetic factor is exactly24 times the product of these real factors. From the already recorded general interval Mertens input, Lean proves its lower bound12/((1+K/log2)(log(R+1)/log2)^2). It also proves the uniform inverse bound Theta_n(z)^(-1)<=(1+K/log2)(log z/log2)^2 for the actual lower Euler product. These estimates hold uniformly in n. No new external analytic input is used. Smoothing absorption, a C-independent logarithmic exponent, and common-cutoff compatibility remain to be assembled before closing l:maintermsize8.

Checked declarations

Module	Named supporting theorems
GoldbachArchimedeanLower.lean	5
GoldbachLowerProductBounds.lean	6

- `goldbach_lower_arch_integrand_nonneg`
- `goldbach_lower_arch_integrand_integrable`
- `goldbach_lower_arch_real`
- `goldbach_lower_arch_interval`
- `goldbach_archimedean_lower`
- `goldbach_unrestricted_local_real_positive`
- `goldbach_unrestricted_local_majorant`
- `goldbach_unrestricted_local_five`
- `goldbach_lower_arithmetic_real_product`
- `goldbach_lower_theta_inverse`
- `goldbach_lower_arithmetic_polylog`

<!-- checkpoint-verification -->

Verification

All 1311 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5104-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4239 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1313 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The lower-sieve endpoint is proved from independent ordinary SW/BV, raw Ren/Ghosh, one-stage Rosser-Iwaniec, interval Mertens and classical sieve-function continuity inputs. No own-paper sieve or Goldbach endpoint is assumed. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The main-term size, weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1229_GoldbachLowerCutoffFreedom.md`

Checkpoint 1229: Common arbitrarily large cutoff for the full lower sieve

The circle-method and lower-sieve estimates now permit any prescribed minimum cutoff exponent Cmin. The proof selects C=max(max(CC,CM),Cmin) in the existing major/minor assembly and verifies every earlier threshold explicitly. The cutoff and full circle estimates, selected signed remainder estimates, lower mean-square estimate and full lower-sieve application are all exported with Cmin<=C. Thus enlarging C for the main-term smoothing argument is an actual theorem, rather than an inferred monotonicity assertion. No previous Lean proof is edited and no additional analytic hypothesis is used. Both lower-sieve clauses hold for one common selected R=(log N)^C, with all coefficients still chosen before n. The quantitative main-term size and its joint application are the next step.

Checked declarations

Module	Named supporting theorems
GoldbachLowerCutoffFreedom.lean	5

- `goldbach_circle_method_cutoff_above`
- `goldbach_circle_method_above`
- `goldbach_sieve_remainders_selected_above`
- `goldbach_lower_remainder_mean_square_above`
- `goldbach_lower_sieve_complete_above`

<!-- checkpoint-verification -->

Verification

All 1312 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5105-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4244 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1314 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The lower-sieve endpoint is proved from independent ordinary SW/BV, raw Ren/Ghosh, one-stage Rosser-Iwaniec, interval Mertens and classical sieve-function continuity inputs. No own-paper sieve or Goldbach endpoint is assumed. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The main-term size, weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1230_GoldbachLowerMainTermSize.md`

Checkpoint 1230: Complete lower-bound side with uniform main term and common cutoff

All mathematical source claims on the lower-bound side are now complete in the agreed conditional scope. The new main-term endpoint is `goldbach_lower_main_term_size`, matching `l:maintermsize8`. The existing full lower-sieve proposition is strengthened to allow any prescribed minimum cutoff exponent. Their joint application is `goldbach_lower_bounds_complete`, which uses the same chosen R=(log N)^C throughout.

Uniform actual main-term lower bound

For every A0>0, every fixed C>=A0+6, and all sufficiently large integer N, Lean proves uniformly for n=12 mod24 and N^2(log N)^(-A0)<=n<N^2 that

$$X_n\Theta_n(z) \geq k_C N(\log N)^{-(A_0/2+6)}, \qquad z = N^{8/9}, \qquad k_C > 0.$$

Thus C_(A0)=A0/2+6 is an explicit admissible exponent, independent of C. This is the exact uniformity required by the source lemma. The quantities are the actual original arithmetic factor, smoothed singular integral and local-density Euler product; no replacement main term is assumed.

For the dimension constant K, let L=(1+K/log2)/(log2)^2. The proof gives the explicit choice

$$k_C = \frac{3}{32L^2(C+1)^2}.$$

K and the smoothing seed are fixed classical/construction parameters, so the implied constant has the dependencies allowed in the paper.

Archimedean lower bound and smoothing

For64<=n<=N^2 the interval[sqrt(n)/4,sqrt(n)/2] lies inside the original domain2<=u<=N,2<=n-u^2<=N^2. On it the integrand is at least1/(2(log N)^2). Integrability, nonnegativity, interval restriction and the real part of the original complex integral are proved. Therefore

$$H_1(n) \geq \frac{\sqrt{n}}{8(\log N)^2} \geq \frac{N}{8(\log N)^{A_0/2+2}}.$$

The retained n range eventually implies n>=64; this growth claim is proved. The already proved smoothing estimate has error at most K_psi N/sqrt(R). If C>=A0+6, then for sufficiently large N this is at most N/(16(log N)^(A0/2+2)), uniformly over the retained n range. Hence the actual H_(1,R)(n) is at least N/(16(log N)^(A0/2+2)). No positivity or lower estimate for this actual function is a new hypothesis.

Actual local products

Every unrestricted factor p(p-1-rho_p(n))/(p-1)^2 with p>=7 is at least1-2/(p-3); the prime5 factor is at least1/2. The original arithmetic factor is proved to be exactly24 times their real product. The already recorded independent interval Mertens input gives

$$G_R(1,24,n) \geq \frac{12}{L(\log(R+1))^2} \geq \frac{12}{L(C+1)^2(\log N)^2}.$$

Separately, the same input and the actual local density bound give Theta_n(z)^(-1)<=L(log z)^2<=L(log N)^2. Together with phi(24)=8 and the smoothed archimedean lower bound, these yield the stated k_C and exponent. This direct product proof does not assume the paper's own main-term estimate or the more precise asymptotic normalization used in the later weighted upper calculation. That normalization remains part of the upper-side work; it is not needed to close this source lower-bound lemma.

One common sufficiently large cutoff

The earlier circle-method theorem returned an exponent C, but did not by its statement alone allow C to be enlarged. The new theorems explicitly select C=max(max(CC,CM),Cmin) in the original major/minor assembly. All applicability thresholds are checked again. The cutoff and full circle estimates, selected signed remainder estimate, lower mean-square estimate and complete lower-sieve application are exported with Cmin<=C.

Taking Cmin=A0+6 proves the lower-sieve inequality, its N^4(log N)^(-A) mean-square remainder estimate, and the actual main-term size simultaneously for one shared C and coefficient family chosen before n. This removes the possible mismatch between an existence statement for C and the smoothing requirement on C.

Positive coefficients and exceptions

Lean proves positivity of the explicit classical half-sieve value sqrt(8 exp(gamma)/(9 pi))*log2 for every real gamma. The classical sieve-function identification/normalization remains in the already recorded classical input boundary; this positivity lemma does not claim to reconstruct that deep theory.

The finite deduction from the lower inequality to positive counts is proved: if v>=T>0, f>0, eta<=f/2 and count>=v(f-eta)+E, then E>=-Tf/4 implies count>=Tf/4. The associated exceptional-set estimate is also proved:

$$\#\{n \in I : \mathrm{count}(n) < Tf/4\}(Tf/4)^2 \leq \sum_{n \in I} E(n)^2.$$

It is a general finite algebraic consequence, ready to apply to the actual common-cutoff lower theorem. No new own-paper endpoint is an external assumption. The numerical comparison with the upper sieve constant and the final Goldbach exceptional-set theorem are separate remaining results.

Conditional scope and source synchronization

No new deep hypothesis was introduced in this continuation. The main-term size consumes only the already recorded general interval Mertens products; its archimedean and smoothing arguments are internal proofs. The joint lower estimate retains the earlier ordinary SW/BV, raw Ren/Ghosh, general one-stage sieve, interval Mertens and classical function-continuity inputs. None is declared as a new Lean axiom.

The selected manuscript is unchanged. The user is editing the compact presieving replacements, and no completed edited source is claimed. Two minor notation corrections in the lower-bound exposition are saved in `blueprint/Section11_LowerBound_Notation_Fixes.tex`: use A0 in the final lower bound of e:H1elementary10 and in the smoothing-choice sentence of the main-term proof. These change notation, not the statements or mathematical argument.

The source ledger is now19 proved,29 complete conditional implications,1 quoted external theorem and5 open claims, plus4 definitions. Four open claims are substantive: the weighted upper main-term comparison, full upper sieve application, strict normalized numerical comparison and final Goldbach assembly. The fifth is the original weaker presieving statement awaiting manuscript synchronization. Supporting-theorem and checkpoint counts are not completion percentages.

Checked declarations

Module	Named supporting theorems
GoldbachLowerMainTermSize.lean	6
GoldbachLowerSmoothedSize.lean	5
GoldbachLowerBoundsComplete.lean	4

- `goldbach_mertens_product_constant_pos`

- goldbach_lower_log_preliminary
- goldbach_lower_arithmetic_logN
- goldbach_lower_theta_logN
- goldbach_lower_main_term_real
- goldbach_lower_main_term_size
- goldbach_retained_sqrt_lower
- goldbach_archimedean_retained_lower
- goldbach_smoothing_error_small
- goldbach_retained_height_eventually
- goldbach_lower_smoothed_size
- goldbach_lower_bounds_complete
- goldbach_half_sieve_closed_value_positive
- goldbach_lower_count_from_main_and_error
- goldbach_lower_bad_card

<!-- checkpoint-verification -->

Verification

All 1315 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5108-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4259 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1317 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. No new deep input was introduced. Main-term size consumes the existing interval Mertens input; joint lower sieve uses the previously recorded ordinary SW/BV, Ren/Ghosh and general classical one-stage sieve/function-continuity inputs. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

Checkpoint_1231_GoldbachWeightedLocalFactors.md

Checkpoint 1231: Weighted upper local cancellations and uniform prime weights

The first weighted upper-main-term component is checked. The local joint factor U_p, the exact prime weights h_R(p), their multiplicative extension, and B_R(n), C_R(n) are concrete definitions. Lean proves positivity of the joint denominator for every prime p>3, treating p=5 by the finite consecutive-residue calculation. It proves 0<=h_R(p)<=6 uniformly, both exact small-prime cancellation identities, the lower and upper local-density cancellations, multiplicativity and nonnegativity of h_R, and 0<B_R(n)<=C_R(n).

No new external input is used. These are internal arithmetic steps of l:wumatomakisum, not a proof of the full weighted comparison. Remaining: exact finite-product assembly with the original G and Theta; uniform multiplicative mean values and weighted summation; integration, smoothing and the final normalized constant. The source-level claim stays open. The large-m cutoff in Step6 must remain explicit; its displayed logarithmic-power integral cannot literally be extended to every m>0 without an indicator. The selected manuscript remains unchanged.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedLocalFactors.lean	13

- `goldbach_joint_denominator_positive`
- `goldbach_joint_local_positive`
- `goldbach_weighted_prime_nonneg`
- `goldbach_weighted_prime_le_six`
- `goldbach_weighted_small_prime_cancel`
- `goldbach_weighted_lower_local_cancel`
- `goldbach_weighted_upper_local_cancel`
- `goldbach_weighted_multiplicative_one`
- `goldbach_weighted_multiplicative_nonneg`
- `goldbach_weighted_multiplicative_mul`
- `goldbach_weighted_B_positive`
- `goldbach_weighted_B_le_C`
- `goldbach_weighted_C_positive`

<!-- checkpoint-verification -->

Verification

All 1316 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5109-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4272 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1318 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. No new external input in the local weighted upper arithmetic. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1232_GoldbachWeightedEulerProducts.md`

Checkpoint 1232: Original upper arithmetic products and multiplier geometry

The weighted upper quantities are now literal definitions: a canonical reduced base residue modulo12, the original arithmetic/smoothed-integral main term X_n(ell), its finite local-density product Theta, and the unrestricted or zero-multiplier arithmetic factors. Lean proves the base residue has the required congruence and is coprime to12; the original G_R(ell,12,n-1) is exactly24 times its real prime-factor product. Every factor is nonnegative and Theta is strictly positive. The lower small-prime normalization is exactly24*C_R(n), including both residue classes modulo4.

The multiplier geometry is also fully checked for every member of the actual closed L set, with z=N^(8/9): ell<N^(10/9), u<z_ell, N^(2/9)<z_ell<z<=p2, and in fact z_ell<p2. This includes z<=p2 rather than the unsupported strict endpoint in the printed intermediate argument. No new deep input is used. The full weighted source claim is still open; no uniform mean value or weighted-integral conclusion has been assumed.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedEulerProducts.lean	7
GoldbachWeightedGeometry.lean	3

- `goldbach_upper_base_residue`
- `mem_goldbach_upper_sieve_primes`
- `goldbach_upper_arithmetic_factor_real`
- `goldbach_upper_arithmetic_real_nonneg`
- `goldbach_upper_arithmetic_real_product`
- `goldbach_weighted_lower_small_product`
- `goldbach_upper_theta_positive`
- `goldbach_weighted_factor_geometry`
- `goldbach_weighted_level_geometry`
- `goldbach_weighted_multiplier_geometry`

<!-- checkpoint-verification -->

Verification

All 1318 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5111-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4282 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1320 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. No new external input in the weighted arithmetic and geometry. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1233_GoldbachWeightedMeanInputs.md`

Checkpoint 1233: Uniform prime balance and effective mean-value instantiation

The varying multiplicative weights now have a checked quantitative prime logarithmic balance. For every y>=R>=3, uniformly in n, the absolute value of sum_{p<=y}(h_R(p)-1/2)log(p)/p is at most6 log R+10K, from the independent ordinary/progression logarithmic Mertens input. The strong prime-power rule and the equality at prime arguments are checked.

The general nonnegative strongly multiplicative upper corollary of Tenenbaum's corrected effective Wirsing theorem is an explicit independent proposition-valued input. The exact instantiation to h_R is proved in goldbach_weighted_mean_from_tenenbaum, with Kmean and delta fixed before R,n,x,epsilon. All applicability requirements, including R<=x^epsilon and6 log R+10K<=epsilon^2 log x, are explicit and verified in the consuming proof. The input contains no Goldbach sequence, h_R, multiplier set, singular integral or paper main term. Its general corollary derivation and remaining external scope are recorded in docs/EXTERNAL_WEIGHTED_MEAN_INPUTS.md. No new Lean axiom or unchecked proof is introduced. Full weighted summation, integration and the final upper comparison remain internal obligations; l:wumatomakisum remains open.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedMeanInputs.lean	9

- `goldbach_half_mean_constant_positive`
- `goldbach_weighted_prime_power`
- `goldbach_weighted_at_prime`
- `goldbach_weighted_prime_above`
- `goldbach_baseline_prime_range`
- `goldbach_weighted_prime_difference`
- `goldbach_baseline_prime_log_balance`
- `goldbach_weighted_prime_log_balance`
- `goldbach_weighted_mean_from_tenenbaum`

<!-- checkpoint-verification -->

Verification

All 1319 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5112-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4291 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1321 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The effective Wirsing theorem and the independent prime logarithmic Mertens facts remain explicit inputs. Their exact application to the actual R-dependent weights is proved. See EXTERNAL_WEIGHTED_MEAN_INPUTS.md. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1234_GoldbachWeightedNormalization.md`

Checkpoint 1234: Exact lower and upper weighted Euler normalizations

Both arithmetic normalizations in the weighted comparison are now exact Lean identities for the original quantities. The lower identity is X_n*Theta_n(z)=3*C_R(n)*Re(H_(1,R)(n))*V_3(R,z), where V_3 is the literal product over R<p<z, p=3 mod4. For ell=2*u*p2, the upper identity is X_n(ell)*Theta_(n,ell)(z_ell)=6*B_R(n)*h_R(u)*Re(H_(ell,R)(n-1))/ell*V_all(R,z_ell). The explicit conditions are R>=3, R<z_ell, 0<u<z_ell<=p2, prime p2 and prime support of u in1 mod4. The actual closed multiplier geometry supplies these conditions once R< N^(2/9).

The proofs partition the actual finite prime sets at the inclusive small-prime cutoff, prove the small divisor equivalence p|ell iff p|u below p2, handle zero arithmetic factors without dividing by them, prove the prime-divisor product identity for h_R(u), and cancel the omitted outer factors exactly. The constants3 and6 are the exact24/phi(24) and24/phi(12), respectively. No asymptotic normalization is assumed. No new external input is used by these identities. The full weighted comparison remains open pending analytic summation, integration and smoothing.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedNormalization.lean	14

- `mem_goldbach_weighted_outer_primes`
- `goldbach_weighted_outer_product_positive`
- `goldbach_weighted_small_divisors`
- `goldbach_weighted_upper_small_product`
- `goldbach_weighted_lower_prime_partition`
- `goldbach_weighted_lower_theta_partition`
- `goldbach_weighted_lower_normalization`
- `goldbach_weighted_prime_coprime_twelve`
- `goldbach_weighted_upper_prime_partition`
- `goldbach_weighted_upper_theta_partition`
- `goldbach_weighted_divisor_product`
- `goldbach_weighted_upper_outer_product`
- `goldbach_weighted_upper_arithmetic_normalization`
- `goldbach_weighted_upper_normalization`

<!-- checkpoint-verification -->

Verification

All 1320 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5113-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4305 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1322 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. The effective Wirsing and prime Mertens inputs remain as recorded; this checkpoint proves exact arithmetic identities without consuming those inputs. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1235_GoldbachWeightedHarmonic.md`

Checkpoint 1235: Uniform harmonic control and normalized upper mean values

The actual varying weight has a checked uniform harmonic bound: sum_{u<=U}h_R(u)/u<=3*((1+K/log2)*(log(U+1)/log2)^2)^3 for U>=2, independently of n and R. The proof embeds the finite sum in the divisors of floor(U)!, expands the multiplicative divisor sum, bounds geometric prime-power sums, separates5, and uses the existing preliminary interval-product Mertens input. No Shiu conclusion is assumed.

The finite Euler product in the effective mean value is exactly(C_R/B_R) times the ordinary inverse prime product on R<p<=x, p=1 mod4. Consuming standard ordinary/progression Mertens ratios gives sum_{u<=x}h_R(u)<=(kappa+Kmean*epsilon^delta)*(1+Km/log R)*(C_R/B_R)*x/(sqrt(log R)*sqrt(log

x)), with kappa=exp(-gamma/2)/sqrt(pi). Kmean and delta are fixed before R,n,x andepsilon. Every condition of the effective mean-value input is retained and internally checked. The additional ratio input contains only fixed-progression prime products. No own-paper weighted summation or integration endpoint is assumed; l:wumatomakisum remains open.

A staging preflight issue was reconciled before compilation or any save attempt: identical draft copies were verified, and their temporary import/audit entries were removed and restaged after preparing the missing scope. The driver now reads the reviewed scope before staging. The reconciliation is recorded; no earlier proof source was changed.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedHarmonic.lean	5
GoldbachWeightedMeanValue.lean	3

- `goldbach_weighted_prime_harmonic_local`
- `goldbach_weighted_harmonic_divisors`
- `goldbach_weighted_harmonic_local_majorant`
- `goldbach_weighted_harmonic_product`
- `goldbach_weighted_harmonic_uniform`
- `goldbach_weighted_correction_product`
- `goldbach_weighted_euler_split`
- `goldbach_weighted_mean_normalized`

<!-- checkpoint-verification -->

Verification

All 1322 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5115-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4313 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1324 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Effective mean-value and ordinary/progression Mertens facts remain independent inputs; normalized applications and the harmonic estimate are Lean proofs. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1236_GoldbachWeightedAbel.md`

Checkpoint 1236: Quantitative Abel estimates and exact multiplier parametrization

Two quantitative real Abel estimates and their full model-error identity are kernel checked. One consumes an upper bound for a nonnegative summatory function against a decreasing positive weight, including the exact lower-boundary term. The other compares an increasing positive weighted sum to a differentiable model with bounded cumulative error, with the explicit error2B*f(b). All endpoints, finite floor cutoffs, integrability and integration by parts are proved.

The actual closed multiplier set L is proved to be the image of its finite(u,p2) pairs, and the map(u,p2)->2*u*p2 is injective on this set. Its proof uses the disjoint1 mod4 support of u and3 mod4 prime p2. Hence finite sums over the original L equal sums over the parametrizing pairs, with no multiplicity assumed away. The actual reciprocal weighted pair mass has a uniform logarithmic power7 bound derived from the preceding harmonic theorem and the elementary harmonic sum. No new deep input is used. The full weighted source inequality remains open; the new general Abel facts are proved tools for its internal analytic deduction.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedAbel.lean	3
GoldbachWeightedPairs.lean	6

- `goldbach_abel_model_identity`
- `goldbach_abel_decreasing_upper`
- `goldbach_abel_increasing_error`
- `goldbach_weighted_pair_multiplier`
- `mem_goldbach_weighted_pairs`
- `goldbach_weighted_pair_injective`
- `goldbach_weighted_pairs_image`
- `goldbach_weighted_pairs_sum`
- `goldbach_weighted_pairs_harmonic`

<!-- checkpoint-verification -->

Verification

All 1324 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5117-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4322 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1326 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Existing effective mean-value and Mertens inputs retained. This checkpoint introduces no new external input. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1237_GoldbachWeightedKernels.md`

Checkpoint 1237: Kernel calculus, exact W normalization and thin integration window

The rational logarithmic kernels used in both weighted summations have checked derivatives, signs and their precise quotient/log identity. The mean model A*x/sqrt(log x) has its checked derivative and upper density bound. These calculus facts apply on the explicit positive domains needed for Abel summation.

The constant integral is now the literal paper W(alpha), specialized to alpha=9/8. Lean evaluates the inner rational integral by a proved logarithmic primitive, proves its continuous dependence on r, its nonnegativity, and integrability of its r^(-1/2) weighted outer integral. An exact affine substitution proves

I_(9/8)=8*sqrt(2)*W(9/8), including the endpoint r=0, without a numerical approximation. The original displayed W integrand is interval integrable and W(9/8)>=0. The square-root endpoint is handled using the integrable power-1/2, not omitted.

The discarded integration window is also controlled: for n>=4 and0<=T<=n/4, the measure of{y>=2:0<n-1-y^2<T} is at most2T/sqrt(n), with the exact square-root interval inclusion and width proved. No new deep input is used. Weighted Abel applications to the actual coefficient families, two-variable summation, smoothing and final normalization remain before closing l:wumatomakisum.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedKernels.lean	7
GoldbachWeightedIntegralConstant.lean	10
GoldbachWeightedThinWindow.lean	3

- `goldbach_weighted_log_kernel_deriv`
- `goldbach_weighted_prime_kernel_deriv`
- `goldbach_weighted_counting_model_deriv`
- `goldbach_weighted_log_kernel_sign`
- `goldbach_weighted_prime_kernel_sign`
- `goldbach_weighted_counting_model_deriv_le`
- `goldbach_weighted_kernel_log_identity`
- `goldbach_weighted_inner_primitive_deriv`
- `goldbach_weighted_inner_integral`
- `goldbach_weighted_inner_closed`
- `goldbach_weighted_W_pointwise`
- `goldbach_weighted_W_identity`
- `goldbach_weighted_inner_continuous`
- `goldbach_weighted_inner_nonneg`
- `goldbach_weighted_outer_integrable`
- `goldbach_weighted_W_integrable`
- `goldbach_weighted_W_nonneg`
- `goldbach_weighted_thin_window_width`
- `goldbach_weighted_thin_window_subset`
- `goldbach_weighted_thin_window_measure`

<!-- checkpoint-verification -->

Verification

All 1327 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5120-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4342 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1329 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Earlier effective Wirsing and prime Mertens inputs retained. All calculus, W normalization and thin-window facts in this checkpoint are internal Lean proofs. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper

main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1238_GoldbachWeightedPrimeBalance.md`

Checkpoint 1238: Actual prime and varying-weight Abel applications

The actual primes3 mod4 logarithmic summatory function is now controlled from the independent ordinary and1 mod4 Mertens input. Lean proves the exact prime partition including the prime2 correction, and an absolute discrepancy bound2K+1. Its concrete increasing-kernel Abel application gives the prime weighted integral with the explicit endpoint error2*(2K+1)*g(b), with all derivatives, signs and integrability proved.

The decreasing-kernel application to the actual varying multiplicative weight is also complete. It gives an integral against A/(t*sqrt(log t)) plus the exact lower-boundary term A/(sqrt(log a)*L(a)^2). The theorem goldbach_weighted_mean_abel_effective instantiates this with the normalized effective mean-value theorem: Kmean and delta are chosen once before R,n and every interval. Conditions imposed at the lower interval endpoint propagate to every t in the interval by proved monotonicity of logs and powers. No interval-specific mean-value estimate is assumed. These two quantitative one-variable applications are the inputs to the remaining two-variable summation. The full source weighted comparison stays open; no new external input is introduced.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedPrimeBalance.lean	4
GoldbachWeightedMeanAbel.lean	2

- `goldbach_three_prime_log_partition`
- `goldbach_three_log_summatory`
- `goldbach_three_log_mertens`
- `goldbach_weighted_prime_abel`
- `goldbach_weighted_mean_abel`
- `goldbach_weighted_mean_abel_effective`

<!-- checkpoint-verification -->

Verification

All 1329 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5122-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4348 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1331 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Existing effective Wirsing and ordinary/progression Mertens inputs retained. No new external input in these concrete Abel applications. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper

main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1239_GoldbachWeightedFiniteIntegration.md`

Checkpoint 1239: Finite weighted integration and smoothing identities

Finite integration over variable upper intervals is checked with the original real endpoints and all integrability requirements: a finite sum of integrals over[a,b_i] equals the integral over[a,U] of the corresponding filtered finite sum. The ordinary outer prime products are proved to lie between0 and1. Two finite complex-to-real smoothing inequalities bound the weighted discrepancy by E times the reciprocal weight mass, with the extra outer factors absorbed using their proved[0,1] range.

The original archimedean integrals for every positive multiplier now have their literal real-integrand representation, uniform bounded-integrand integrability, nonnegativity, and the exact finite weighted sum/integral interchange. This includes the actual logarithmic window and the shift n-1; it does not substitute a power model for H_ell. These internally proved integration and smoothing tools will be applied to the quantitative weighted kernel bound. No new external input is used, and no own-paper endpoint has been assumed. The complete weighted comparison remains open until the two-variable estimate and final assembly are discharged.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedFiniteIntegration.lean	4
GoldbachWeightedArchimedean.lean	4

- `goldbach_finite_variable_interval`
- `goldbach_weighted_outer_product_le_one`
- `goldbach_finite_smoothing_upper`
- `goldbach_finite_smoothing_outer_upper`
- `goldbach_weighted_arch_integrand_integrable`
- `goldbach_weighted_arch_real`
- `goldbach_weighted_arch_sum_identity`
- `goldbach_weighted_arch_integrand_nonneg`

<!-- checkpoint-verification -->

Verification

All 1331 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5124-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4356 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1333 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Earlier effective Wirsing and prime Mertens inputs retained. No new external input in finite integration, smoothing or archimedean identities. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper

main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1240_GoldbachWeightedKernelIntegration.md`

Checkpoint 1240: Weighted upper main term: integration and logarithmic geometry

Weighted upper main term: integration and logarithmic geometry

The real Archimedean integrals now reduce to the actual finite weighted logarithmic kernel sum. The comparison uses H_1(n) directly. The discarded region is bounded by 2T/sqrt(n) times the reciprocal weight mass, divided by (log 2)^3. All support restrictions and integrability arguments are proved.

The two changes of variables p=exp(Dv), u=exp(Dr)/2 are checked. They identify the resulting iterated integral with the existing exact integral for 8 sqrt(2) W(9/8). This includes extension to the integrable r=0 endpoint. No numerical or asymptotic estimate is used in that identity.

This checkpoint does not assert the weighted upper main-term lemma. The double finite sum, endpoint errors, small-u contribution, and final uniform comparison remain to be assembled. The independent mean-value and Mertens inputs retain their previously documented scope; no new external input is introduced.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedKernelIntegration.lean	5
GoldbachWeightedMellin.lean	3

- `goldbach_weighted_arch_pointwise`
- `goldbach_weighted_arch_sum_pointwise`
- `goldbach_weighted_arch_retained_pointwise`
- `goldbach_weighted_arch_coarse`
- `goldbach_weighted_arch_kernel_integrated`
- `goldbach_weighted_inner_mellin`
- `goldbach_weighted_outer_mellin`
- `goldbach_weighted_outer_mellin_le_W`

<!-- checkpoint-verification -->

Verification

All 1333 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5126-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4364 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1335 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Previously documented independent mean-value/Mertens inputs only. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Gold-

bach conclusion remain open.

`Checkpoint_1241_GoldbachWeightedMeanCutoffs.md`

Checkpoint 1241: Weighted upper main term: small denominators and comparison losses

Weighted upper main term: small denominators and comparison losses

The strongly multiplicative weight now has a uniform coarse mean estimate for x at least R^D, with constants chosen before R and n. A finite Abel argument gives a harmonic tail estimate from a linear summatory bound. Combining this with the already checked finite Euler-product majorant controls all small denominators, including those outside the effective half-dimensional mean range.

Prime reciprocal sums are bounded from the independent logarithmic Mertens input. The factor introduced when replacing log u by log(2u), the retained-region logarithmic loss, and 3 sqrt(alpha) c_alpha = 48 exp(-gamma/2)/sqrt(pi) are checked exactly.

The weighted upper main-term source claim stays open until the finite double sum, integration, and uniform relative errors have all been assembled. No new external input is introduced and the selected manuscript is unchanged.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedMeanCutoffs.lean	4
GoldbachWeightedLogComparisons.lean	4

- `goldbach_weighted_mean_coarse`
- `goldbach_weighted_prime_reciprocal`
- `goldbach_harmonic_tail_linear`
- `goldbach_weighted_harmonic_refined`
- `goldbach_weighted_log_offset`
- `goldbach_weighted_log_loss`
- `goldbach_weighted_upper_constant_positive`
- `goldbach_weighted_upper_constant_normalization`

<!-- checkpoint-verification -->

Verification

All 1335 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5128-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4372 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1337 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Previously documented independent mean-value/Mertens inputs only. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

Checkpoint notes 1242-1265

Checkpoint_1242_GoldbachWeightedSmoothing.md

Checkpoint 1242: Weighted upper main term: source normalization and finite smoothing error

Weighted upper main term: the source sum and smoothing error

The actual source sum over multipliers is now exactly normalized as a finite sum over the unique pairs (u,p), with the concrete B_R, h_R, Euler factors, and smoothed singular integrals. The geometric bounds place every pair in the required sieve range. Applying the existing uniform smoothing theorem gives an explicit total error bounded by 6 B_R K N/sqrt(R) times the actual reciprocal weight mass.

A separate finite fiber argument bounds the weighted pair sum by the explicit prime-by-denominator rectangle. It preserves the p=z endpoint and the inclusive upper floor. The proof includes the coverage of the actual source pairs and all nonnegativity needed to enlarge the summation domain.

The weighted upper main-term source claim remains open pending the sharp double-sum bound and uniform error assembly. No new deep input is used, and no manuscript edits are applied.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedSmoothing.lean	3
GoldbachWeightedRectangles.lean	3

- `goldbach_weighted_pair_levels`
- `goldbach_weighted_main_term_sum_normalization`
- `goldbach_weighted_main_term_smoothing`
- `goldbach_finite_fiber_upper`
- `goldbach_weighted_pair_rectangle`
- `goldbach_weighted_kernel_rectangle`

<!-- checkpoint-verification -->

Verification

All 1337 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5130-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4378 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1339 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Previously documented independent mean-value/Mertens inputs only. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

Checkpoint_1243_GoldbachWeightedPrimeEndpoints.md

Checkpoint 1243: Weighted upper main term: two Abel estimates and prime endpoints

Weighted upper main term: two Abel estimates and prime endpoints

The prime-kernel estimate now applies to an arbitrary finite set of primes in the closed interval [z,b]. The possible prime at z contributes at most 1/(z (log z)^2); the logarithmic Mertens error contributes at most 2(2K+1)/(log z)^3. Thus the inclusive source convention is respected.

The actual strongly multiplicative denominator weight has a two-variable Abel reduction. Its finite sum is bounded by the integral of the remaining prime kernel plus an explicit summatory boundary term. The proof checks continuous integrability on the full rectangle, interchanges the finite variable intervals exactly, and retains the uniform mean hypothesis with its existing independent provenance.

The weighted upper main-term claim remains open until the integral estimate and all uniform relative errors are assembled. No new external input or manuscript modification is introduced.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedPrimeEndpoints.lean	2
GoldbachWeightedDoubleAbel.lean	1

- `goldbach_finite_real_endpoint`
- `goldbach_weighted_prime_finite_upper`
- `goldbach_weighted_double_abel`

<!-- checkpoint-verification -->

Verification

All 1339 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5132-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4381 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1341 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Previously documented independent mean-value/Mertens inputs only. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1244_GoldbachWeightedDoubleGeometry.md`

Checkpoint 1244: Weighted upper main term: logarithmic region and sharp integral

Weighted upper main term: the double-integral region

The logarithmic region for the two summation orders is checked: u <= exp(D)/(2p^2) is equivalent to p <= exp((D-log(2u))/2). Its bounds imply all derivative signs and nonvanishing denominators needed by the two Abel estimates.

The resulting integral is bounded sharply by sqrt(1+log(2)/log(a)) sqrt(D) 8 sqrt(2) W(9/8). The factor records the replacement of log(u) by log(2u) and tends to one at the intended cutoff. A separate explicit integral bound controls the prime endpoint and Mertens errors.

These are internal proved analytic steps. The independent mean-value and Mertens boundaries are unchanged. The source upper main-term comparison is still open pending finite assembly and uniform relative error estimates.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedDoubleGeometry.lean	4
GoldbachWeightedDoubleIntegral.lean	3

- `goldbach_weighted_u_bound_log`
- `goldbach_weighted_double_p_geometry`
- `goldbach_weighted_double_u_geometry`
- `goldbach_weighted_double_domain`
- `goldbach_weighted_log_harmonic_integral_le`
- `goldbach_weighted_double_integrands_continuous`
- `goldbach_weighted_double_integral_le_W`

<!-- checkpoint-verification -->

Verification

All 1341 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5134-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4388 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1343 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Previously documented independent mean-value/Mertens inputs only. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1245_GoldbachWeightedLargeRectangle.md`

Checkpoint 1245: Weighted upper main term: sharp finite double sum and cutoff split

Weighted upper main term: sharp finite rectangle estimate

The two checked Abel estimates and the checked logarithmic changes of variables now give the finite large-denominator estimate. Its leading coefficient is the exact paper integral W(9/8). Prime endpoints, logarithmic Mertens errors, and the summatory boundary are explicit additional terms.

The actual source pair kernel is split at a real cutoff a. The small-denominator part is bounded by its weighted harmonic sum and a prime reciprocal sum. The large part is placed in the active rectangle, with the actual floors and logarithmic denominators preserved. All enlargements use proved nonnegative weights.

No error term is silently discarded. The source comparison remains open until these explicit terms are shown to be uniformly negligible and inserted into the source main-term normalization. No new external theorem is assumed.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedLargeRectangle.lean	1
GoldbachWeightedKernelSplit.lean	3

- `goldbach_weighted_large_rectangle`
- `goldbach_weighted_pair_log_floor`
- `goldbach_weighted_kernel_small`
- `goldbach_weighted_kernel_large`

<!-- checkpoint-verification -->

Verification

All 1343 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5136-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4392 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1345 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Previously documented independent mean-value/Mertens inputs only. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1246_GoldbachWeightedFiniteKernel.md`

Checkpoint 1246: Weighted upper main term: actual finite kernel and scale limits

Weighted upper main term: quantitative source kernel and scale limits

The source pair kernel now has one checked quantitative upper bound. Its inputs are the internally proved summatory and harmonic estimates; its output retains the exact W(9/8) term and all prime, floor, and boundary errors. The estimate concerns the actual pair sum, with no abstract replacement of the paper's main term.

The logarithmic scale limits are proved separately: the leading coefficient tends to the half-dimensional constant times W, and the small-denominator and prime endpoint coefficients tend to zero. These results use the pinned Mathlib asymptotic lemmas and explicit inequalities.

The final weighted upper main-term lemma still requires the uniform cutoff budgets, moving-logarithm comparison, and insertion into the normalized source singular integrals. No new external input is introduced.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedFiniteKernel.lean	2
GoldbachWeightedScaleLimits.lean	5

- `goldbach_weighted_finite_kernel`
- `goldbach_weighted_prime_error_coarse`
- `goldbach_weighted_log_power_sqrt_limit`
- `goldbach_weighted_sqrt_log_inverse_limit`
- `goldbach_weighted_main_coefficient_limit`
- `goldbach_weighted_small_coefficient_limit`
- `goldbach_weighted_endpoint_coefficient_limit`

<!-- checkpoint-verification -->

Verification

All 1345 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5138-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4399 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1347 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Previously documented independent mean-value/Mertens inputs only. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1247_GoldbachWeightedScalarBounds.md`

Checkpoint 1247: Weighted upper main term: uniform cutoffs and scalar errors

Weighted upper main term: error absorption and valid cutoffs

The finite-kernel prime errors and Abel boundaries have explicit scalar majorants with the correct vanishing scale. The leading scalar identity preserves the exact W coefficient.

For R=(log N)^C and a=exp(sqrt(log N)), all hypotheses of the effective mean estimate are eventually valid: the lower scale, the prime-balance budget, the epsilon range, and the overlap with the coarse mean bound. These statements choose thresholds independently of n.

The small-denominator harmonic sum is bounded by (C_R/B_R) times a fixed constant times (log R)^6 plus the coarse mean tail. This retains all arithmetic dependence explicitly. The final weighted source comparison remains open until the uniform kernel and singular-integral assembly are complete.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedScalarBounds.lean	4
GoldbachWeightedCutoffBudgets.lean	1
GoldbachWeightedHarmonicCutoff.lean	1

- goldbach_weighted_prime_reciprocal_coarse
- goldbach_weighted_small_scalar
- goldbach_weighted_sharp_scalar_identity
- goldbach_weighted_large_scalar
- goldbach_weighted_cutoffs_eventually
- goldbach_weighted_harmonic_cutoff

<!-- checkpoint-verification -->

Verification

All 1348 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5141-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4405 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1350 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Previously documented independent mean-value/Mertens inputs only. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

Checkpoint_1248_GoldbachWeightedKernelCoefficient.md

Checkpoint 1248: Weighted upper main term: effective source kernel and coefficient

Weighted upper main term: effective arithmetic kernel

The actual finite kernel bound is now instantiated with the concrete strongly multiplicative mean and harmonic estimates. Its constants are chosen before the cutoff exponent, scale, and target integer. All remaining hypotheses are explicit scalar cutoff inequalities whose eventual validity has been proved.

The coefficient is normalized by (C_R/B_R)/(sqrt(log R) log(N) sqrt(log N)). Its exact factorization is checked, and its limit is (exp(-gamma/2)/sqrt(pi)+K epsilon^delta) W(9/8). The small-denominator and endpoint components tend to zero independently of n.

This is a complete effective kernel implication from the documented independent inputs. The original weighted main-term claim remains open pending the sharp eventual moving-kernel estimate and its singular-integral and relative-error assembly. No new external endpoint is assumed.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedKernelCoefficient.lean	3
GoldbachWeightedKernelEffective.lean	1

- goldbach_weighted_kernel_coefficient_limit
- goldbach_weighted_coefficient_factor
- goldbach_weighted_mean_interval
- goldbach_weighted_kernel_effective

<!-- checkpoint-verification -->

Verification

All 1350 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5143-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4409 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1352 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Previously documented independent mean-value/Mertens inputs only. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1249_GoldbachWeightedKernelUniform.md`

Checkpoint 1249: Weighted upper main term: sharp uniform moving kernel complete

Weighted upper main term: sharp uniform kernel estimate

The weighted kernel estimate is complete in epsilon form. For every fixed positive cutoff exponent C and every eta>0, the actual finite source pair kernel is eventually bounded by (kappa W(9/8)+eta)(C_R/B_R)/(sqrt(log R) log N sqrt(log N)), uniformly in n.

The same sharp bound holds for every m between N^2(log N)^(-A) and N^2. The proof controls the moving logarithmic denominator, respects its support filter, and chooses a positive tolerance using a checked continuity argument. All effective mean hypotheses are discharged by the proved cutoff budgets; no paper-specific estimate remains among the independent inputs.

The full source main-term claim still requires insertion into the singular integrals, total smoothing and discarded-region errors, and comparison with the lower main term. The existing independent mean-value and Mertens assumptions are unchanged.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedKernelUniform.lean	2
GoldbachWeightedMovingKernel.lean	3

- `goldbach_weighted_choose_mean_parameter`
- `goldbach_weighted_fixed_kernel_uniform`
- `goldbach_choose_positive_tolerance`
- `goldbach_weighted_moving_kernel_loss`
- `goldbach_weighted_moving_kernel_uniform`

<!-- checkpoint-verification -->

Verification

All 1352 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5145-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4414 named theorems pass a fresh complete

transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1354 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Previously documented independent mean-value/Mertens inputs only. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1250_GoldbachWeightedMass.md`

Checkpoint 1250: Weighted upper main term: source outer products and total mass

Weighted upper main term: source normalization and total error mass

The actual reciprocal mass over source pairs is bounded uniformly by a fixed constant times (log N)^7. The proof starts from the checked finite harmonic product estimate.

The original upper main-term sum is bounded by 24 B_R (1+K/log R) log R times the weighted Archimedean sum, plus its explicit total smoothing error. The lower main term is bounded below by 3 sqrt(9/8) (1-epsilon)^2 times the common reference C_R H_1(n) sqrt(log R/log N), whenever the two elementary relative error conditions hold.

Both results use the source's actual Euler products and singular integrals. The exact constant identity and arithmetic factors are preserved. The weighted upper comparison remains open pending the uniform relative errors and final assembly.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedMass.lean	3
GoldbachWeightedSourceOuter.lean	2

- `goldbach_weighted_mass_constant_positive`
- `goldbach_weighted_mass_log_bound`
- `goldbach_weighted_arch_real_nonneg`
- `goldbach_weighted_upper_outer`
- `goldbach_weighted_lower_outer`

<!-- checkpoint-verification -->

Verification

All 1354 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5147-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4419 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1356 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input

includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Previously documented independent mean-value/Mertens inputs only. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1251_GoldbachWeightedErrorBounds.md`

Checkpoint 1251: Weighted upper main term: all uniform relative errors

Weighted upper main term: uniform relative errors

Both errors are now bounded relative to the common positive reference C_R H_1(n) sqrt(log R/log N), using the actual retained range for n. With the truncation exponent A+14 and C >= A+24, the discarded-region error is bounded by a fixed multiple of (log N)^(-3), and the total smoothing error by a fixed multiple of (log N)^(-2), times that reference.

The lower singular integral has relative smoothing error at most 8K/log N. All scale restrictions, Mertens error tolerances, positive truncation heights, and the R-to-power-of-N comparisons are eventually valid with thresholds independent of n.

No paper-specific asymptotic error is assumed. These are proved inequalities and limits. The source weighted upper main-term lemma remains open only until the final checked assembly is staged.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedErrorBounds.lean	7
GoldbachWeightedSourceBudgets.lean	1

- `goldbach_weighted_smoothing_relative`
- `goldbach_weighted_reference_lower_scalar`
- `goldbach_weighted_thin_ratio_scalar`
- `goldbach_weighted_thin_error_relative`
- `goldbach_weighted_smooth_error_relative`
- `goldbach_weighted_error_coefficient_limit`
- `goldbach_weighted_thin_cutoff`
- `goldbach_weighted_source_budgets`

<!-- checkpoint-verification -->

Verification

All 1356 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5149-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4427 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1358 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional

Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 remains open. Previously documented independent mean-value/Mertens inputs only. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The weighted upper main-term sum, upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1252_GoldbachWeightedSourceBridge.md`

Checkpoint 1252: Weighted upper main-term comparison complete

Weighted upper main term: complete conditional formalization

The source lemma **l:wumatomakisum** is complete in uniform epsilon form. Its checked endpoint is `goldbach_weighted_upper_main_term_complete` in `PrimeSquares/GoldbachWeightedUpperMainTerm.lean`.

For every A>0, every fixed C>=A+24, and every eta>0, one threshold in N works for every n=12 modulo24 with N^2(log N)^(-A)<=n<N^2. With R=(log N)^C and alpha=9/8, the actual source sum satisfies

$$\sum_{\ell\in L} X_n(\ell)\Theta_{n,\ell}(z_\ell) \le (c_\alpha \mathcal{W}(\alpha)+\eta)X_n\Theta_n(z), \qquad c_\alpha=\frac{16e^{-\gamma/2}}{\sqrt{\pi\alpha}}.$$

This is the source uniform o(1) statement. The original main terms, multiplier set, local factors, Euler products and singular integrals are used. No paper conclusion is an external hypothesis.

The proved chain includes the arithmetic cancellation, strong multiplicativity and uniform prime-balance application, the small-denominator estimate, both finite Abel arguments, the prime endpoint p=z, the exact W integral, the sharp fixed and moving kernel, finite integration, total smoothing error, the discarded region, and the final positive lower comparison. A=the retained-range exponent, A+14 is a sufficient integration cutoff exponent, and C>=A+24 suffices for all error estimates.

The explicit independent hypotheses are the generic effective half-dimensional mean-value corollary of the corrected Tenenbaum theorem, ordinary/fixed-progression logarithmic Mertens estimates, the three ordinary relative Euler-product estimates, and the previously recorded interval-product Mertens bound. Their precise general interfaces and unformalized classical corollary reductions are in `docs/EXTERNAL_WEIGHTED_MEAN_INPUTS.md`. They are theorem parameters, not new Lean axioms. The entire weighted source application is proved. This milestone does not require a new Goldbach, weighted-prime-counting, or Siegel-Walfisz endpoint as an assumption.

The accompanying full build, source audit and exact axiom audit must pass before this checkpoint can be saved by the checkpoint driver. The selected manuscript and all accepted source hashes remain unchanged. The claim ledger moves from29 to30 completed conditional implications, leaving4 open entries: the upper sieve, numerical comparison, final Goldbach theorem, and the original weaker pre-sieving statement awaiting source synchronization. The latter is bookkeeping about the manuscript; its corrected stronger variant is already formalized.

Files and continuation

The complete project is in the checkpoint ZIP, including the pinned Lean/Mathlib configuration, all earlier proofs, verification records and recovery instructions. `blueprint/Weighted_Upper_Main_Term_Complete.md` presents the complete mathematical proof by topic. The next substantive milestone is the full upper-sieve application `p:sieveneedupper8`.

Suggested paper replacements

The lemma statement and constant need no change. The following two proof edits are saved in `docs/WEIGHTED_UPPER_MAIN_TERM_REPLACEMENTS.tex`; they have not been applied to the selected manuscript.

1. In Step1, replace the passage beginning “Finally, since $p_2 > z$ and $\ell = 2up_2 \ge p_2$, we have” through “all sufficiently large N” with:

```
Finally, since \(p_2\geq z\) and \(\ell=2up_2\geq p_2\), we have
\[
 z_\ell
 =\left(\frac{N^2}{\ell}\right)^{1/4}
 \leq\left(\frac{N^2}{p_2}\right)^{1/4}
 \leq\left(\frac{N^2}{z}\right)^{1/4}
 =N^{5/18}<N^{8/9}=z.
\]
Thus \(p_2\geq z>z_\ell\), and, since \(R=(\log N)^C\), also
\(p_2>R\) for all sufficiently large \(N\).
```

This preserves the inclusive endpoint in the definition of L.

2. In Step6, replace “Applying (`e:Wsum10`) on the remaining range gives” through “Therefore”, immediately before the existing display labelled `e:integratedW10`, with:

```
On the retained range \(m(y)\geq N^2(\log N)^{-A_1}\), we have
\(\log m(y)=(2+o(1))\log N\). Applying \eqref{e:Wsum10} therefore gives
\[
 \sum_{\substack{\ell=2up_2\in L\\2\ell\leq m(y)}}
 \frac{h_R(u)}{\ell(\log(m(y)/\ell))^2}
 \leq
 \left(\frac{e^{-\gamma/2}}{\sqrt\pi}\mathcal W(\alpha)+o(1)\right)
 \frac{C_R(n)}{B_R(n)}
 \frac{1}{\sqrt{\log R}(\log N)^{3/2}}.
\]
For these \(y\), we also have \(2\leq n-y^2\leq N^2\), so
\((\log y)^{-1}\leq
2\log N/(\log y\log(n-y^2))\).
The indicator in the definition of \(H_1(n)\) equals one on this range.
Integrating the preceding bound, and including the already estimated
 discarded region, gives the following inequality. For the latter error,
use \(C_R(n)/B_R(n)\geq1\) and \eqref{e:H1elementary10}, choosing
\(A_1\) sufficiently large in terms of \(A_0\).
```

Keep the existing `e:integratedW10` display unchanged. Equality at finitely many boundary points does not change the integrals. This edit retains the cutoff where the real logarithmic powers are valid and compares directly with H_1(n), without a separate shift asymptotic.

Checked declarations

Module	Named supporting theorems
GoldbachWeightedSourceBridge.lean	1
GoldbachWeightedUpperMainTerm.lean	1

- `goldbach_weighted_source_bridge`
- `goldbach_weighted_upper_main_term_complete`

<!-- checkpoint-verification -->

Verification

All 1358 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5151-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4429 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1360 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough

full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 is complete through the weighted main-term comparison; the upper sieve and final Goldbach deductions remain open. Independent general effective mean-value and classical Mertens hypotheses remain explicit and unproved; all own weighted upper main-term deductions are complete. Earlier independent theorem scopes are unchanged. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The full weighted upper main-term comparison l:wumatomakisum is a complete checked implication from the documented independent effective mean-value and Mertens inputs, with the exact printed constant and uniform retained-range quantifiers. The upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1253_GoldbachSieveNumericalComparison.md`

Checkpoint 1253: Exact normalized sieve comparison

The source numerical comparison l:numericalcomparison is complete conditional only on the classical normalization of the independent sieve functions. Lean proves 8 sqrt(2) W(9/8)<=9/32 by bounding the actual inner integrand, integrating the two real powers exactly, and using the previously checked W change of variables. It then proves c_(9/8) exp(gamma) W(9/8)<=3 exp(gamma/2)/(8 sqrt(pi)), whereas the lower coefficient is (2 sqrt(2)/3) log(2) exp(gamma/2)/sqrt(pi). The elementary exact inequality log(2)>=1/2 already makes the difference strictly positive. This proves more than a floating-point calculation: every integral comparison and algebraic inequality is kernel checked. The printed weaker W bound remains valid. No new manuscript correction is needed here. The existing decimal claim >0.17 is not used or credited by this checkpoint; the required strict positivity is proved.

The generic functions in the independent sieve interfaces retain explicit normalization identities at9/8 and2. These identities belong to the classical general sieve boundary (Iwaniec1980,(1.8)-(1.10)); they are not a paper conclusion or an added Lean axiom. A source retrieval attempt identified the publisher DOI10.4064/aa-36-2-171-202 but its PDF hosts denied access in this run; the formal theorem transparently retains both normalization identities as hypotheses.

Checked declarations

Module	Named supporting theorems
GoldbachSieveNumericalComparison.lean	4

- `goldbach_weighted_inner_linear_upper`
- `goldbach_weighted_W_rational_upper`
- `goldbach_sieve_numerical_comparison`
- `goldbach_normalized_sieve_comparison`

<!-- checkpoint-verification -->

Verification

All 1359 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5152-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4433 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1361 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional

Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 is complete through the weighted main-term comparison; the upper sieve and final Goldbach deductions remain open. Independent general effective mean-value and classical Mertens hypotheses remain explicit and unproved; all own weighted upper main-term deductions are complete. Earlier independent theorem scopes are unchanged. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The full weighted upper main-term comparison l:wumatomakisum is a complete checked implication from the documented independent effective mean-value and Mertens inputs, with the exact printed constant and uniform retained-range quantifiers. The upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1254_GoldbachUpperSieveDensity.md`

Checkpoint 1254: Upper-sieve densities and uniform level geometry

The upper sieve now has the exact density and scale prerequisites. Its preliminary dimension2 includes the exceptional prime5 with barrier1/2; its outer dimension1 follows from an explicitly independent all-prime interval Mertens input. The new interface contains only finite products over arbitrary prime subsets and no Goldbach sequence, local roots, main terms or paper conclusions. It is the all-prime counterpart of the already documented interval Mertens boundary.

For every fixed B>=0 and C, the actual levels z_ell=(N^2/ell)^(1/4) and D_ell=sqrt(N^2/ell)/(12(log N)^B) satisfy R+1<=z_ell<=D_ell/D0<=N^2 eventually, uniformly for1<=ell<=N^(10/9). The exact logarithmic identity proves log(D_ell/D0)/log z_ell tends to2 along every admissible varying multiplier path. The eventual uniformization and source application are next. No own-paper endpoint has been assumed.

Checked declarations

Module	Named supporting theorems
GoldbachUpperSieveDensity.lean	5
GoldbachUpperSieveParameters.lean	6

- `goldbach_upper_sieve_prime_ge_five`
- `goldbach_upper_prime_split`
- `goldbach_upper_density_multiplicative`
- `goldbach_upper_preliminary_dimension`
- `goldbach_upper_outer_dimension`
- `goldbach_upper_log_loss_ratio`
- `goldbach_upper_sieve_log_geometry`
- `goldbach_upper_outer_log`
- `goldbach_upper_sieve_levels_uniform`
- `goldbach_upper_parameter_along`
- `goldbach_upper_product_level`

<!-- checkpoint-verification -->

Verification

All 1361 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5154-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4444 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1363 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input

includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 is complete through the weighted main-term comparison; the upper sieve and final Goldbach deductions remain open. Independent general effective mean-value and classical Mertens hypotheses remain explicit and unproved; all own weighted upper main-term deductions are complete. Earlier independent theorem scopes are unchanged. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The full weighted upper main-term comparison l:wumatomakisum is a complete checked implication from the documented independent effective mean-value and Mertens inputs, with the exact printed constant and uniform retained-range quantifiers. The upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1255_GoldbachUpperPresieveUniform.md`

Checkpoint 1255: Uniform upper presieve and exact sequence bridge

The complete upper presieve prerequisite is now uniform over all1<=ell<=N^(10/9), with the actual levels and the coefficient quantifier preceding every density and sequence. A general filter selection proof derives uniformity from all admissible multiplier paths; the exact level geometry makes the classical two-stage input applicable along every such path. Continuity of the independent normalized F_1 at2 yields one common tolerance.

The actual source progression residue is selected independently of n, with a coprime fallback for irrelevant moduli. Lean proves the exact cardinality identity with the actual upper density, actual main term and shifted full circle remainder at n-1. Nonnegativity of the upper main term, including vanishing arithmetic factors, is established from the exact product and positive singular integral. No new independent input is added. The finite source sieve application, coefficient embedding and summed remainder are next.

Checked declarations

Module	Named supporting theorems
GoldbachUpperPresieveUniform.lean	2
GoldbachUpperSequenceBridge.lean	4

- `goldbach_eventually_uniform_of_paths`
- `goldbach_upper_presieving_uniform`
- `goldbach_upper_residue_properties`
- `goldbach_upper_residue_coprime`
- `goldbach_upper_main_term_nonneg`
- `goldbach_upper_divisor_real_identity`

<!-- checkpoint-verification -->

Verification

All 1363 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5156-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4450 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1365 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input

includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 is complete through the weighted main-term comparison; the upper sieve and final Goldbach deductions remain open. Independent general effective mean-value and classical Mertens hypotheses remain explicit and unproved; all own weighted upper main-term deductions are complete. Earlier independent theorem scopes are unchanged. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The full weighted upper main-term comparison l:wumatomakisum is a complete checked implication from the documented independent effective mean-value and Mertens inputs, with the exact printed constant and uniform retained-range quantifiers. The upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1256_GoldbachUpperPresieveApplication.md`

Checkpoint 1256: Actual upper sieve and summed error estimates

The actual source upper sieve is assembled at each multiplier, including the vanishing arithmetic-factor case, and summed over the corrected closed multiplier set. All coefficients have absolute value at most1 and are chosen before n. The original double sum of remainders is embedded injectively into the circle-method coefficient family with L0=N^(6/5) and delta=3/4. The shift n-1, the factor12, strict divisor levels and all coprime residue requirements are checked. The complete shifted mean square is bounded by2M N^4(log N)^(-A).

The fixed-sieve comparison uses exactly the previously proved K N^(1/4) correction. Adding it to the summed remainder yields a mean square at most(4M+2K^2)N^4(log N)^(-A), for large N. The proof conservatively bounds its squared total by O(N^3), which is sufficient for every logarithmic saving.

A new record collects only the previously explicit independent classical inputs and normalizations; it contains no manuscript endpoint. The exact numerical gap follows from these general normalized values. The final upper coefficient assembly and common-cutoff lower/upper closure are next. No new deep input is added.

Checked declarations

Module	Named supporting theorems
GoldbachUpperPresieveApplication.lean	3
GoldbachUpperFixedSieve.lean	1
GoldbachUpperSieveRemainder.lean	2
GoldbachUpperSieveErrors.lean	2
GoldbachSection11Inputs.lean	1

- `goldbach_upper_remainder_zero_extension`
- `goldbach_upper_sifted_sequence`
- `goldbach_upper_presieve_application`
- `goldbach_upper_fixed_sieve_analytic`
- `goldbach_sieve_circle_bound_exists_above`
- `goldbach_upper_remainder_from_circle`
- `goldbach_upper_error_square_sum`
- `goldbach_upper_arithmetic_error_eventually`
- `goldbach_section11_sieve_gap_positive`

<!-- checkpoint-verification -->

Verification

All 1368 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5161-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before

reuse; new/changed modules and the root are compiled. All 4459 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1370 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 is complete through the weighted main-term comparison; the upper sieve and final Goldbach deductions remain open. Independent general effective mean-value and classical Mertens hypotheses remain explicit and unproved; all own weighted upper main-term deductions are complete. Earlier independent theorem scopes are unchanged. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The full weighted upper main-term comparison l:wumatomakisum is a complete checked implication from the documented independent effective mean-value and Mertens inputs, with the exact printed constant and uniform retained-range quantifiers. The upper sieve, numerical comparison and final Goldbach conclusion remain open.

`Checkpoint_1257_GoldbachUpperSieveComplete.md`

Checkpoint 1257: Full upper-sieve proposition complete

The full source proposition p:sieveneedupper8 is complete. For every A0>0 and desired logarithmic mean-square saving A>0, a single B and C>=A0+24 is chosen before the tolerance. For every eta>0 and all large N, coefficients independent of n give the actual upper count at most(c_(9/8) F_1(2) W(9/8)+eta) X_n Theta_n(z)+E2(n), uniformly for retained admissible n. The actual E2 is the checked shifted double remainder plus the fixed-sieve K N^(1/4) correction; its full mean square is O(N^4(log N)^(-A)). No source main term, prime-counting weight or sieve endpoint is a hypothesis.

The checked theorem uses the explicit independent general input record described in docs/SECTION_11_UPPER_SIEVE_INPUTS.md. It includes the general sieve-function normalization and continuity, general one-stage sieve theorems, ordinary Mertens estimates and the effective mean-value theorem; their proofs remain external in the agreed scope.

The lower analytic theorem is now exported for every fixed B,C, and its exact mean-square remainder can be transported from the same circle bound as the upper remainder. This removes a possible mismatch between separately existential cutoff choices. The final common-cutoff Section11 theorem is next. The literal weaker presieving manuscript statement still requires its saved hypothesis replacements; no unsupported source claim is silently promoted.

Checked declarations

Module	Named supporting theorems
GoldbachUpperSieveComplete.lean	2
GoldbachLowerFixedSieve.lean	1
GoldbachLowerCommonCircle.lean	1

- `goldbach_upper_sieve_fixed_circle`
- `goldbach_upper_sieve_complete`
- `goldbach_lower_sieve_fixed_analytic`
- `goldbach_lower_remainder_from_circle`

<!-- checkpoint-verification -->

Verification

All 1371 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5164-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4463 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1373 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; Section11 lower, weighted-upper and full upper-sieve estimates and normalized numerical positivity are complete conditional implications; the final Goldbach assembly and manuscript presieving synchronization remain. Independent general effective mean-value and classical Mertens hypotheses remain explicit and unproved; all own weighted upper main-term deductions are complete. Earlier independent theorem scopes are unchanged. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The full weighted upper main-term comparison l:wumatomakisum is a complete checked implication from the documented independent effective mean-value and Mertens inputs, with the exact printed constant and uniform retained-range quantifiers. The complete upper-sieve proposition and strict numerical comparison are proved conditionally with exact constants and uniform quantifiers. The final Goldbach assembly and manuscript presieving synchronization remain.

`Checkpoint_1258_GoldbachSection11Complete.md`

Checkpoint 1258: Section11 complete at one shared cutoff

Section11 is complete for the reviewed corrected statements. The joint endpoint goldbach_section11_complete uses one B,C for the actual lower and upper sieve counts, actual common main term, the lower bound k N(log N)^(-(A0/2+6)), and both original remainder families. It gives the exact lower coefficient f_(1/2)(9/8), the exact upper coefficient c_(9/8) F_1(2) W(9/8), uniform tolerances, coefficients chosen before n and a combined mean-square saving of arbitrary logarithmic order. C can exceed any prescribed minimum and C>=A0+24 is sufficient. The exponent A0/2+6 is independent of C. Strict positivity of the normalized coefficient difference is already kernel checked.

The general one-stage sieve, SW/BV, Ren/Ghosh, effective multiplicative mean and classical Mertens inputs, plus classical sieve-function continuity and normalization, remain explicit independent hypotheses. No paper endpoint is an input or an added mathematical axiom.

A complete replacement copy is saved as sources/review/Section11_Reconciled.tex. It applies the earlier presieving hypotheses and endpoint correction, both lower-bound notation fixes, the weighted integration correction, the inclusive p2=z endpoint in both occurrences, and a shorter exact numerical-comparison proof matching Lean. Original manuscript files and their accepted hashes remain unchanged. The generated copy has unique labels and balanced environment counts; it has not been typeset as a standalone document. The reproduction script and exact replace-with manifest are included.

Crucial source distinction: the weaker original l:presieving hypothesis is not claimed proved. Its completed claim now points explicitly to the separately supplied corrected statement, whose bytes and original-statement bytes are guarded by hashes. The user manuscript still requires those edits. This review-copy completion does not assert that the user has applied them. The remaining open paper-level claim is the final quantitative Goldbach theorem; the original presieving version is retained as a superseded, unclaimed variant.

The weighted endpoint uses the checked effective upper-mean route. It does not add completion credit to

every stronger asymptotic displayed in the manuscript's alternative convolution argument. The exported lemma conclusions and the joint sieve estimates are what is certified.

Checked declarations

Module	Named supporting theorems
GoldbachSection11Complete.lean	1

- `goldbach_section11_complete`

<!-- checkpoint-verification -->

Verification

All 1372 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5165-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4464 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1374 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; All Section11 theorem statements, using the explicitly supplied corrected presieving replacement, are complete in the agreed conditional scope. The common-cutoff lower/upper assembly, exact constants and complete error estimates are proved; the original manuscript still needs the supplied replacements. Independent general effective mean-value and classical Mertens hypotheses remain explicit and unproved; all own weighted upper main-term deductions are complete. Earlier independent theorem scopes are unchanged. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The full weighted upper main-term comparison l:wumatomakisum is a complete checked implication from the documented independent effective mean-value and Mertens inputs, with the exact printed constant and uniform retained-range quantifiers. The complete upper-sieve proposition and strict numerical comparison are proved conditionally with exact constants and uniform quantifiers. The joint Section11 estimates are proved at one cutoff. The final Goldbach theorem is the next assembly; the supplied presieving replacement still needs to be copied into the user manuscript.

`Checkpoint_1259_GoldbachQuantitativeExceptionalSet.md`

Checkpoint 1259: Section11 and quantitative Goldbach theorem complete

Section11 is complete in the agreed conditional scope, and the final quantitative Goldbach theorem has also been assembled. The endpoint goldbach_headline_theorem proves, for every A>0 and all sufficiently large real X, the actual exceptional-set bound |E2(X)|<=D_A X(log X)^(-A).

The definition is matched exactly to positive n<X congruent12 mod24 with no representation a^2+b^2+p^2+1=n in which p and a^2+b^2+1 are prime and a,b are coprime. The finite prime representation, primitive sum-of-two-squares predicate and exceptional-set membership equivalence are proved. The common-cutoff Section11 theorem supplies both sieves, the lower main-term exponent A0/2+6 independent of C, exact normalized sieve constants and arbitrary combined logarithmic mean-square saving. The positive numerical gap, threshold choice, divisor-error absorption, Chebyshev step and small-n contribution are all discharged. Finally N=ceil(sqrt(X)), X<=N^2<=4X and log N>=log X/2 transfer the square-cutoff estimate to every real X with the explicit factor4*2^A.

The upper-sieve proposition is complete with the exact coefficient c_(9/8) F_1(2) W(9/8), coefficients chosen before n and the actual shifted double remainder. Its full mean square, including the fixed-sieve O(N^(1/4)) comparison, has arbitrary logarithmic saving. The rational integral proof gives8 sqrt(2) W(9/8)<=9/32 and proves strict positivity without floating point or an unchecked evaluator.

Independent hypotheses remain explicit: raw SW and BV, Ren/Ghosh, the general one-stage sieve theorems and their normalized values/continuity, the general effective multiplicative mean corollary, classical ordinary/progression Mertens estimates, and the primitive two-square criterion. Previously recorded independent inputs for earlier sections retain their scopes. These are theorem parameters, not added Lean axioms. No source sieve, weighted main term, numerical comparison or Goldbach endpoint is postulated.

The full replacement copy sources/review/Section11_Reconciled.tex is included, together with a reproducible reconciliation script and a manifest of exact replace-with passages. It includes the required strengthened presieving hypotheses and endpoint, the two lower-bound notation corrections, both inclusive p2=z endpoint occurrences, the retained-range integration correction and the exact numerical proof. Original manuscript files and all accepted hashes are unchanged. The corrected statement is the formal target; the literal weaker original presieving statement is not claimed proved and still needs replacement in the user manuscript. No user edit is assumed. The replacement copy has unique labels and balanced environments; no standalone LaTeX typesetting is claimed.

The claim ledger now has19 proved,34 complete conditional implications and1 quoted external theorem, with no remaining internal endpoint among the54 reviewed numbered targets. Presieving credit explicitly points to the corrected review copy with both old/new statement hashes retained. This does not claim to formalize every stronger unnumbered assertion in alternative proofs: the weighted main-term endpoint uses the checked effective-upper-mean route rather than the full convolution asymptotic printed in an alternate derivation. The independent external proofs remain future library work.

Checkpoint1258 already separately saved the full common-cutoff Section11 theorem. This checkpoint additionally closes the final Goldbach theorem and saves every earlier proof and validation record.

Checked declarations

Module	Named supporting theorems
GoldbachQuantitativeExceptionalSet.lean	1
GoldbachHeadlineTheorem.lean	5

- `goldbach_quantitative_exceptional_set_squares`
- `goldbach_good_representation_iff`
- `goldbach_source_exceptional_membership`
- `goldbach_ceiling_square_geometry`
- `goldbach_real_cutoff_from_square_bound`
- `goldbach_headline_theorem`

<!-- checkpoint-verification -->

Verification

All 1374 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5167-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4470 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1376 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; All Section11 theorem statements, using the explicitly supplied corrected presieving replacement, are com-

plete in the agreed conditional scope. The common-cutoff lower/upper assembly, exact constants and complete error estimates are proved; the original manuscript still needs the supplied replacements. Independent general effective mean-value and classical Mertens hypotheses remain explicit and unproved; all own weighted upper main-term deductions are complete. Earlier independent theorem scopes are unchanged. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The full weighted upper main-term comparison l:wumatomakisum is a complete checked implication from the documented independent effective mean-value and Mertens inputs, with the exact printed constant and uniform retained-range quantifiers. The complete upper-sieve proposition and strict numerical comparison are proved conditionally with exact constants and uniform quantifiers. The joint Section11 estimates are proved at one cutoff. The full quantitative Goldbach theorem is proved conditionally at every large real cutoff X; the supplied presieving replacement still needs to be copied into the user manuscript.

`Checkpoint_1260_GoldbachPrimitiveProducts.md`

Checkpoint 1260: Primitive two-square multiplication without an extra assumption

Recovered and authenticated checkpoint1259. The Section9 statement map was stale even though the curated current ledger already recorded the completed circle-method, both sieves, main-term size, numerical gap and final real-X exceptional-set theorem. This task audits the source section and seeks to remove the remaining elementary PrimitiveTwoSquaresCriterion hypothesis rather than manufacture another proof of already completed counting deductions.

Lean proves seven new arithmetic theorems. A prime-norm sum of squares has coprime coordinates. A common prime divisor of coordinates after multiplication can only be the determinant prime. At an odd split prime, the two conjugate orientations cannot both create that common divisor. Consequently primitive sums of two squares remain primitively represented after multiplication by every prime congruent1 mod4, including repeated prime factors. Integer coordinate signs are handled by natural absolute values, with exact norm preservation.

The only newly imported library result is the already proved Mathlib Fermat two-square theorem, Mathlib.NumberTheory.SumTwoSquares, at the unchanged pinned revision. Its1988-job dependency target was built successfully. No new unproved mathematical input is introduced. The general primitive criterion is not claimed complete until the next stage. Original manuscripts remain unchanged; the user says the edits are in progress, not finished.

Checked declarations

Module	Named supporting theorems
GoldbachPrimitiveProducts.lean	7

- `goldbach_coprime_zmod_not_both_zero`
- `goldbach_prime_norm_coprime`
- `goldbach_square_matrix_kernel`
- `goldbach_conjugate_kernel_separation`
- `goldbach_primitive_pair_from_integer_norm`
- `goldbach_primitive_prime_matrix`
- `goldbach_primitive_multiply_one_mod_four_prime`

<!-- checkpoint-verification -->

Verification

All 1375 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5172-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4477 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1377 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input

includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; All Section11 theorem statements, using the explicitly supplied corrected presieving replacement, are complete in the agreed conditional scope. The common-cutoff lower/upper assembly, exact constants and complete error estimates are proved; the original manuscript still needs the supplied replacements. Independent general effective mean-value and classical Mertens hypotheses remain explicit and unproved; all own weighted upper main-term deductions are complete. Earlier independent theorem scopes are unchanged. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The full weighted upper main-term comparison l:wumatomakisum is a complete checked implication from the documented independent effective mean-value and Mertens inputs, with the exact printed constant and uniform retained-range quantifiers. The complete upper-sieve proposition and strict numerical comparison are proved conditionally with exact constants and uniform quantifiers. The joint Section11 estimates are proved at one cutoff. The full quantitative Goldbach theorem is proved conditionally at every large real cutoff X; the supplied presieving replacement still needs to be copied into the user manuscript.

`Checkpoint_1261_GoldbachPrimitiveCriterion.md`

Checkpoint 1261: Primitive criterion and remaining Section9 prerequisites proved

The full primitive sum-of-two-squares criterion used in Section9 is now proved unconditionally: for n=2 mod4, n has a coprime sum-of-two-squares representation exactly when no prime3 mod4 divides n. The proof constructs a primitive representation of the odd cofactor using the checked prime-multiplication argument, then doubles it; parity excludes a common factor2. The converse uses Mathlib's already proved obstruction to -1 being a square modulo a3 mod4 prime. No external arithmetic criterion is assumed.

The source's unnumbered assertion that W(alpha) is finite for every alpha>1 is now proved across its full range, with interval integrability of the actual logarithmic kernel. The endpoint square-root singularity is handled by an integrable negative half power; nonnegativity is also proved for the complete range. This extends the earlier special-value formalization at alpha9/8.

A concrete smooth bump with inner radius1/32 and outer radius1/16 is converted to a Schwartz function. Lean verifies value1 at0, nonzero, compact support contained in(-1/8,1/8), and the source support cutoff. Thus the proof will no longer need an externally supplied test function at the final endpoint. No additional source replacement is required for these valid mathematical statements.

This checkpoint saves these prerequisite proofs separately. The next checked export specializes the legacy parameterized Section9 deductions with the proved criterion and fixed seed, and updates the source map. No source claim is promoted before that final statement review. The previously supplied manuscript edits remain in progress; no changed user source is assumed.

Checked declarations

Module	Named supporting theorems
GoldbachPrimitiveCriterion.lean	3
GoldbachGeneralWeightedIntegral.lean	2
GoldbachFixedSmoothSeed.lean	6

- `goldbach_primitive_one_mod_four_support`
- `goldbach_primitive_double_of_odd`
- `goldbach_primitive_two_squares_criterion`
- `goldbach_weighted_W_integrable_general`
- `goldbach_weighted_W_nonneg_general`
- `goldbach_fixed_smooth_seed_at_zero`

- goldbach_fixed_smooth_seed_nonzero
- goldbach_fixed_smooth_seed_support
- goldbach_fixed_smooth_seed_compact_support
- goldbach_fixed_smooth_seed_tsupport_subset
- goldbach_smooth_seed_exists

<!-- checkpoint-verification -->

Verification

All 1378 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5175-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4488 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1380 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. The Section9 arithmetic/final-count reductions are checked for the user-corrected closed L; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; All Section11 theorem statements, using the explicitly supplied corrected presieving replacement, are complete in the agreed conditional scope. The common-cutoff lower/upper assembly, exact constants and complete error estimates are proved; the original manuscript still needs the supplied replacements. Independent general effective mean-value and classical Mertens hypotheses remain explicit and unproved; all own weighted upper main-term deductions are complete. Earlier independent theorem scopes are unchanged. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The full weighted upper main-term comparison l:wumatomakisum is a complete checked implication from the documented independent effective mean-value and Mertens inputs, with the exact printed constant and uniform retained-range quantifiers. The complete upper-sieve proposition and strict numerical comparison are proved conditionally with exact constants and uniform quantifiers. The joint Section11 estimates are proved at one cutoff. The full quantitative Goldbach theorem is proved conditionally at every large real cutoff X; the supplied presieving replacement still needs to be copied into the user manuscript.

`Checkpoint_1262_GoldbachSectionNineComplete.md`

Checkpoint 1262: Section9 complete with the arithmetic criterion proved

Section9 has been reviewed against the actual source, not merely its stale planning map. Its nine numbered statements and the final quantitative exceptional-set theorem all have complete matched formal evidence. The final theorem already followed conditionally at1259; this task removes the remaining elementary arithmetic and seed hypotheses and closes additional source prerequisites.

The primitive sum-of-two-squares criterion is now proved from pinned Mathlib and project algebra. For n=2 mod4, a coprime sum-of-two-squares representation exists exactly when no3 mod4 prime divides n. The proof covers arbitrary prime powers by choosing a conjugate orientation at each split prime, and treats the single factor2 using parity. Consequently l:fromsiftedtop, l:truncationT and c:mastersievereduction are promoted from conditional to proved under their ordinary source hypotheses. The upper arithmetic reduction was already proved. Their actual finite sets, prime-factor multiplicities, closed cutoff and uniform error terms are retained.

The complete source arithmetic exports are goldbach_from_sifted_to_primes_complete, goldbach_truncation_complete, goldbach_section9_master_exact and goldbach_master_sieve_complete. No primitive criterion is a hypothesis. The source assertion that W(alpha) is finite for every alpha>1

is now proved over its full range, including the square-root endpoint singularity; nonnegativity is also proved.

A concrete compactly supported smooth function is constructed with value1 at0 and support inside(-1/8,1/8). The final export goldbach_section9_headline_complete therefore takes only GoldbachSection11Inputs and A>0, and yields the actual real-X exceptional-set estimate. It needs neither a supplied primitive criterion nor a supplied smoothing function. The almost-all statement is also exported as goldbach_section9_exceptional_density_zero.

The circle-method approximation, lower and upper sieve propositions, uniform main-term lower bound and strict numerical gap retain their complete checked conditional scopes. Remaining independent inputs are SW/BV, Ren/Ghosh, the general one-stage sieve and classical sieve normalizations/continuity, the effective multiplicative mean and classical Mertens estimates. None is a paper endpoint. The primitive criterion has been removed from every current claim's external-input list and is explicitly recorded as a proved internal arithmetic lemma.

The Section9 map is reconciled to the actual current nine source statements and their current theorem evidence, replacing the stale1085/1136 planning statuses. The old map is archived. Original manuscript and accepted source hashes remain unchanged. The user is applying the previously supplied Section11 edits; their application is not presumed. No additional Section9 LaTeX replacement was needed. The previously supplied presieving replacement remains the explicitly reviewed target.

The paper ledger now records22 proved under ordinary hypotheses,31 complete conditional implications and1 quoted external theorem among54 reviewed numbered targets, plus4 definitions. This is a statement inventory, not a percentage based on supporting theorem counts. Earlier stronger unnumbered assertions in alternative proof routes retain their separately documented scope; no new completion credit is awarded to them.

Checkpoint1260 separately saved the primitive multiplication proof, and1261 separately saved the complete criterion, general W integral and fixed smooth seed. This checkpoint saves their final Section9 consumption and source reconciliation.

Checked declarations

Module	Named supporting theorems
GoldbachSectionNineComplete.lean	6

- `goldbach_from_sifted_to_primes_complete`
- `goldbach_truncation_complete`
- `goldbach_section9_master_exact`
- `goldbach_master_sieve_complete`
- `goldbach_section9_headline_complete`
- `goldbach_section9_exceptional_density_zero`

<!-- checkpoint-verification -->

Verification

All 1379 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5176-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4494 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1381 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. All Section9 arithmetic reductions are proved for the adopted closed L without an extra primitive two-square hypothesis. The full real-X Goldbach estimate and exceptional-density limit follow from the documented independent analytic inputs, with the smooth seed constructed internally; all Section10 statements and

the full circle-method implication are complete in the agreed conditional scope; All Section11 theorem statements, using the explicitly supplied corrected presieving replacement, are complete in the agreed conditional scope. The common-cutoff lower/upper assembly, exact constants and complete error estimates are proved; the original manuscript still needs the supplied replacements. Independent general effective mean-value and classical Mertens hypotheses remain explicit and unproved; all own weighted upper main-term deductions are complete. Earlier independent theorem scopes are unchanged. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The full weighted upper main-term comparison l:wumatomakisum is a complete checked implication from the documented independent effective mean-value and Mertens inputs, with the exact printed constant and uniform retained-range quantifiers. The complete upper-sieve proposition and strict numerical comparison are proved conditionally with exact constants and uniform quantifiers. The joint Section11 estimates are proved at one cutoff. The full quantitative Goldbach theorem is proved conditionally at every large real cutoff X; the supplied presieving replacement still needs to be copied into the user manuscript. Section9 also includes a proved full primitive two-square criterion and convergence of W(alpha) for every alpha>1. The old parameterized theorems remain for compatibility; the current matched exports remove those discharged parameters. The user is applying manuscript edits and their application has not been verified against a new upload.

`Checkpoint_1263_FiniteSignedMaximalProof.md`

Checkpoint 1263: General finite signed maximal inequality proved

The general finite signed-sum maximal inequality is now proved without an independent analytic hypothesis. For any finite sequence of complex functions on the integers and every exponent at least one, a uniform Lp bound B for all sums with scalar coefficients of modulus at most one gives an Lp bound (ceil(log2 J)+1) B for the largest initial partial sum. The checked depth comparison gives the explicit bound 6 log(2+J) B. This proves the exact existing RawFiniteSignedMaximalInput interface at constant 6, including J=0, Lp membership, and extended-real B.

The proof uses exact finite Boolean-sign orthogonality and reconstruction. Each coordinate is dominated by the finite average of the norms of all signed sums. Minkowski then bounds the largest block at each level, and an aligned binary decomposition bounds every initial partial sum. There is no probabilistic oracle, Khintchine hypothesis, numerical testing dependency, new axiom, or unchecked evaluation in this argument.

This is a general theorem, not a new paper endpoint assumption. The next milestone consumes the proved constant in the public Section7 and near-L1 exports and completes the paper-wide source/assumption audit. All 54 numbered paper targets retain their existing source statuses; no extra paper result is counted merely because supporting lemmas were added. The deep inputs previously excluded from scope remain explicit. The user is applying manuscript edits, and no revised upload is presumed.

Checked declarations

Module	Named supporting theorems
FiniteSignedMaximalProof.lean	16

- `finiteBooleanSign_norm`
- `finiteBooleanSign_not`
- `finiteBooleanSign_self`
- `finiteBooleanSign_orthogonality`
- `finiteBooleanSign_reconstruction`
- `finiteSignAverage_nonneg`
- `finiteSignAverage_dominates`
- `finiteSignAverage_Lp`
- `signed_block_partition`
- `finiteBooleanSignLift_norm`
- `signedBlockFamily_sum`
- `signed_binary_strict_prefix_bound`
- `signed_binary_prefix_bound`

- `finite_signed_maximal_binary`
- `finite_signed_binary_log_bound`
- `rawFiniteSignedMaximalInput_proved`

<!-- checkpoint-verification -->

Verification

All 1380 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5177-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4510 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1382 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. All Section9 arithmetic reductions are proved for the adopted closed L without an extra primitive two-square hypothesis. The full real-X Goldbach estimate and exceptional-density limit follow from the documented independent analytic inputs, with the smooth seed constructed internally; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; All Section11 theorem statements, using the explicitly supplied corrected presieving replacement, are complete in the agreed conditional scope. The common-cutoff lower/upper assembly, exact constants and complete error estimates are proved; the original manuscript still needs the supplied replacements. Independent general effective mean-value and classical Mertens hypotheses remain explicit and unproved; all own weighted upper main-term deductions are complete. Earlier independent theorem scopes are unchanged. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The full weighted upper main-term comparison l:wumatomakisum is a complete checked implication from the documented independent effective mean-value and Mertens inputs, with the exact printed constant and uniform retained-range quantifiers. The complete upper-sieve proposition and strict numerical comparison are proved conditionally with exact constants and uniform quantifiers. The joint Section11 estimates are proved at one cutoff. The full quantitative Goldbach theorem is proved conditionally at every large real cutoff X; the supplied presieving replacement still needs to be copied into the user manuscript. Section9 also includes a proved full primitive two-square criterion and convergence of W(alpha) for every alpha>1. The old parameterized theorems remain for compatibility; the current matched exports remove those discharged parameters. The user is applying manuscript edits and their application has not been verified against a new upload.

`Checkpoint_1264_PaperClosedSignedMaximal.md`

Checkpoint 1264: Reviewed corrected paper complete in the agreed conditional scope

The paper-wide review is complete for the selected corrected target. Every declared theorem-like environment is checked against the source ledger: 54 numbered theorem/proposition/lemma/corollary targets and four definitions. The reviewed statuses are 22 proved under ordinary source hypotheses, 31 complete implications from explicit independent inputs, one quoted external Granville–Kurlberg theorem, and zero open internal numbered endpoints. Exact statements, quantifiers, source hashes and evidence are preserved; no supporting-theorem percentage is claimed.

The four current public Section7/near-L1 exports now consume the proved finite signed maximal theorem at constant6. theorem_7_2_closed_signed_maximal, corollary_7_6_closed_signed_maximal, corollary_7_46_closed_signed_maximal and theorem_1_5_ii_closed_signed_maximal have the full existing source conclusions without a supplied RawFiniteSignedMaximalInput or its parameter R. All other documented independent inputs retain their precise scopes. The older parameterized results remain for compatibility.

A reproducible paper completion audit checks all declared theorem-like environment types, label and hash coverage, every current evidence declaration type in Lean, and the actual discharge of the signed maximal and primitive-criterion parameters. The complete type output is saved. The full build, source scan and complete transitive axiom audit are run; no own-paper endpoint is added as an independent input.

The current plan, Section11 plans, earlier-sections summary and signed maximal boundary are reconciled. Prior planning versions are preserved in the archive directory. The exact independent analytic/ergodic/sieve boundaries are summarized in docs/INDEPENDENT_INPUTS_CURRENT.md. Proving those inputs is a separate optional development within the previously agreed scope.

The user is applying the previously supplied manuscript corrections. No new upload or application is presumed, the original sources remain unchanged, and the reviewed corrected presieving target is still explicit. No additional replacement passage is required by this continuation. The audit does not claim proofs of independent deep inputs or every stronger unnumbered assertion in alternative proof routes. See docs/PAPER_COMPLETION.md for the final result matrix and verification scope.

Checked declarations

Module	Named supporting theorems
PaperClosedSignedMaximal.lean	4

- `theorem_7_2_closed_signed_maximal`
- `corollary_7_6_closed_signed_maximal`
- `corollary_7_46_closed_signed_maximal`
- `theorem_1_5_ii_closed_signed_maximal`

<!-- checkpoint-verification -->

Verification

All 1381 project modules are covered by the authenticated compilation history, beginning with a fresh full build. This checkpoint passes a 5178-job dependency-aware build with remote project artifact reuse disabled. Unchanged earlier proof sources and compiled Lean artifacts are SHA256 checked before reuse; new/changed modules and the root are compiled. All 4514 named theorems pass a fresh complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound; all 1383 Lean files pass the source scan. No placeholders, added mathematical axioms or unchecked evaluation are used. Build receipts record exactly which modules were compiled at every step.

All reviewed current-paper statements in Sections3–8 are complete under their normal hypotheses or as complete implications from explicit independent general inputs. The corrected prime-square input includes one fixed-level Page datum per N, the full extended major range and the sufficient complementary minor estimate. The broad quantitative IW proposition includes the exact real-Q construction, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and finite-dimensional Shannon sampling for every1<=p<=infinity. The general Page/progression, Ren/Ghosh, Tao and other previously recorded independent theorems and their explicit bridges remain unproved. No own-paper endpoint is a new external assumption. The latest upload and selected reconciled copy are preserved. All Section9 arithmetic reductions are proved for the adopted closed L without an extra primitive two-square hypothesis. The full real-X Goldbach estimate and exceptional-density limit follow from the documented independent analytic inputs, with the smooth seed constructed internally; all Section10 statements and the full circle-method implication are complete in the agreed conditional scope; All Section11 theorem statements, using the explicitly supplied corrected presieving replacement, are complete in the agreed conditional scope. The common-cutoff lower/upper assembly, exact constants and complete error estimates are proved; the original manuscript still needs the supplied replacements. Independent general effective mean-value and classical Mertens hypotheses remain explicit and unproved; all own weighted upper main-term deductions are complete. Earlier independent theorem scopes are unchanged. The full Section11 lower-sieve proposition p:sieveneedlower8 is a complete conditional implication, including its actual main term, uniform o(1), n-independent coefficients and mean-square error. The uniform main-term lower bound l:maintermsize8 and its joint application with the same large cutoff C are also complete conditionally. The full weighted upper main-term comparison l:wumatomakisum is a complete checked implication from the documented independent effective mean-value and Mertens inputs, with the exact printed constant and uniform retained-range quantifiers. The complete upper-sieve proposition and strict numerical comparison are proved conditionally with exact constants and uniform quantifiers. The joint Section11 estimates are proved at one cutoff. The full quantitative Goldbach theorem is proved conditionally at every large real cutoff X; the supplied presieving replacement still needs to be copied into

the user manuscript. Section9 also includes a proved full primitive two-square criterion and convergence of W(alpha) for every alpha>1. The old parameterized theorems remain for compatibility; the current matched exports remove those discharged parameters. The user is applying manuscript edits and their application has not been verified against a new upload. The finite signed maximal theorem is proved internally at constant6 and consumed by current Section7, Corollaries7.6/7.46 and Theorem1.5(ii) exports. No supplied signed maximal hypothesis remains in those exports. Read docs/PAPER_COMPLETION.md and docs/INDEPENDENT_INPUTS_CURRENT.md for the complete current scope.

`Checkpoint_1265_ManuscriptReviewAndFullRebuild.md`

Checkpoint 1265: Current manuscript review and fresh full rebuild

The current complete manuscript has been reviewed directly against the checked source and Section11 replacement. It retains all54 numbered results and4 definitions with the same stable labels. Of the58 statements,57 are unchanged modulo whitespace. The only rewritten statement is the presieving lemma. Its corrected hypotheses and conclusion are retained, and the rewrite is covered by three new proved Lean equivalences: the old strict D0<D condition follows from the current level chain, and the inline preliminary/outer prime conditions define the same interval products.

The changed proof passages are reviewed separately: both approximation branches and periodized lifts, ordinary unweighted SW/BV major approximation with the strict endpoint and q+1 finite errors, refined interval dimensions, preliminary coefficient bounds, uniform lower/upper sieve levels, exact weighted integration constants and discarded-range absorption. They preserve the checked source conclusions and use the same documented independent inputs. No new independent mathematical hypothesis is introduced.

One cleanup is needed: two consecutive identical displays carry the label e:integratedW10. Keep the first and remove the preceding Therefore plus the second copy. No mathematical replacement is required. The original upload is preserved byte-for-byte; a complete cleaned TeX copy, exact deletion fragment and minimal patch are supplied. All numbered statement bodies are identical between the upload and cleaned copy; the cleanup removes the duplicate label without changing the formalization target.

The entire local project is freshly rebuilt from source, with project build traces removed and remote project artifact reuse disabled. All source scans, complete transitive theorem axiom audits and current paper evidence declaration-type checks are rerun. The project retains pinned Lean4.34.1 and Mathlib revisiond13f23b723b8a846827a245b89c10fc7d3f11612. The final verification below records the actual counts.

All54 numbered targets retain their agreed scopes:22 proved under ordinary source hypotheses,31 complete implications from explicit independent general inputs, one quoted external theorem, and no open internal numbered endpoint. All4 definitions are formalized. The independent deep analytic, sampling, ergodic, Weil and sieve theorem inputs and their formulation bridges remain unproved; the axiom audit checks the deductions and does not establish those input truths. No blanket claim is made for stronger unnumbered alternative-route assertions. The current upload's mathematical edits are verified; the earlier pending manuscript synchronization status is resolved.

Checked declarations

Module	Named supporting theorems
GoldbachPresieveSourceEquivalence.lean	3

- `goldbach_current_presieve_strict_level`
- `goldbach_current_presieve_preliminary_filter`
- `goldbach_current_presieve_outer_filter`

Bibliography check: the uploaded TeX cites Ghosh1981, which is absent from the bibliography saved with this project. The user's local bibliography was not supplied, so no absence is presumed there. A reviewed bibliography adds only that entry: A.Ghosh, The Distribution of alpha p squared Modulo1, Proceedings of the London Mathematical Society s3-42(2),252–269(1981), DOI10.1112/plms/s3-42.2.252. These details are checked against the primary publisher record at https://doi.org/10.1112/plms/s3-42.2.252. If the user's own .bib already contains that key, no bibliography edit is needed. The reviewed manuscript bundle supplies a matching BibPrimeSquares.bib; all earlier entries are preserved. This is a bibliography repair, not a new mathematical hypothesis or proof.

<!-- checkpoint-verification -->

Verification

All 1382 project modules and the root were freshly rebuilt: 5179 jobs pass without warnings/errors. All 4517 named theorems pass the complete transitive axiom audit, allowing only propext, Classical.choice and Quot.sound. The source scan checks 1384 Lean files. Exact unique import/audit coverage and unchanged frozen source hashes are checked before packaging. There are no placeholders or added mathematical axioms.

The complete28 October2026 user upload is reviewed directly: all54 numbered results and4 definitions are matched in the agreed conditional scope. The presieving rewrite retains the corrected hypotheses and is covered by three proved Lean equivalences; no mathematical replacement remains pending. One repeated e:integratedW10 display needs the supplied cosmetic cleanup. The entire project and root are freshly rebuilt from source with full source, theorem-axiom and declaration-type audits. Independent general inputs and their formulation bridges remain explicit and unproved. The original upload and earlier frozen sources are unchanged; a one-display cleanup copy is separately supplied.

Batch summaries

Batch_42_91_Report.md

Prime Squares: combined report for checkpoints 42–91

Fifty separately saved proof checkpoints continue checkpoint 41. This batch adds 95 named supporting theorems in 50 modules; the project now has 940 named supporting theorems in 158 modules. These counts describe supporting formal work, not newly completed numbered manuscript results.

What this batch establishes

1. The genuine finite real model's major-arc error has a logarithmic rate with a single natural-number threshold independent of Q, the chosen finite rational-center set and exceptional data, conditional on the explicitly displayed actual progression-prefix estimate.
2. Actual annular integrals and localized Gauss models are continuous in frequency. Their compact symbols have a genuine continuous unit-circle periodization, globally identified with the integer-translate sum. Boundary aliases are included.
3. Finite rational-center sums define genuine global continuous model, error, low and high candidate symbols. Their inverse Fourier kernels and compatible l1-to-linfinity / l2 operators are constructed. The actual annular von-Mangoldt square kernel and every finite complex input have exact low + high + error decompositions.
4. Infinite convolution by a bounded kernel is proved absolutely convergent on every l1 input. The resulting bounded linear operator agrees with the Fourier l2 operator on finite inputs. Kernel-sensitive bounds, denominator-weighted Gauss estimates and finite-family l2 envelopes are proved.
5. The narrow-to-broad cutoff containment is proved. The compact low and high real symbols equal the original paper formulas under the displayed scalar conditions, including Q*q0/B0<=R for the exceptional width. This condition is not silently inferred from unproved conductor data.
6. An explicit assembly theorem connects these actual operators to OperatorHighLowBound at the correct spatial scale N^2. Its sharp low-kernel and combined high/error-symbol bounds remain displayed inputs, not proved source estimates.

What remains open

Theorem 1.1 is NOT proved. Neither are full Lemma 6.5, global approximation 6.9, sharp low 6.14, multifrequency 6.16, high/maximal 6.21 or Proposition 4.19. The full-range minor-arc deduction from the verified Ren estimate remains unresolved. The exact manuscript center set, the full exceptional parameter range and the sharp analytic kernel/operator estimates must still be connected to the constructions here.

The raw progression-prefix estimate remains an explicit unproved deep-input hypothesis. No existence theorem for uniform exceptional data has been registered. Lemma 3.21 still uses its two earlier explicit PNT limits; the separate PNT development is not imported or claimed integrated.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, abstract Theorem 4.1 and abstract Corollary 4.15, with their previously recorded source hypotheses.

Correctness and verification

Lean 4.34.1 and Mathlib d13f23b723b8a846827a245b89c10fc7d3f11612 remain pinned. The recovered checkpoint-41 ZIP was tested and every archived file hash verified. The official Lean archive SHA256 and all nine exact dependency revisions were checked before continuing.

Every checkpoint has a successful dependency-aware build, complete source scan and fresh complete transitive-axiom audit for every named theorem. Reused old project proof sources and compiled artifacts are authenticated by SHA256 receipts. New modules and changed root imports are compiled with remote project artifact reuse disabled. Checkpoints 42, 74 and 91 use fresh full project rebuilds. The final build and audits must all pass before this report is saved.

One attempted checkpoint-74 root artifact failed the independent audit because it did not expose the new import, although direct import of the new module succeeded. The failed attempt was rejected and preserved in verification/rejected-attempt74. Build artifacts were isolated from workspace synchronization and the entire project freshly rebuilt and audited successfully before checkpoint 74 was saved. No rejected attempt counts as a completed checkpoint.

Only propext, Classical.choice and Quot.sound are allowed in the complete transitive axiom audit. There are no sorry/admit placeholders, added mathematical axioms, unsafe declarations or native_decide proofs. No new numerical experiment is substituted for an analytic proof. Earlier exact numerical reports and script hashes are preserved.

Every checkpoint ZIP contains the cumulative proof sources, expositions, source correspondence, current verification logs, assumption/scope notes, inventory and restart instructions. Its SHA256 manifest and every ZIP entry are verified. Reproducible dependency/build caches are excluded. The original manuscript and accepted revision fragments retain their frozen hashes.

Manuscript replacements

No additional mathematical error or justified replacement was established in this batch. The manuscript and earlier accepted corrections were left unchanged. An open proof obligation is not reported as a counterexample or converted into an unproved axiom. The corrected Lean sources and verification infrastructure are included in every subsequent cumulative archive.

Checkpoint ledger

Each archive below is cumulative. All 50 are saved separately; the final archive contains the full latest project. The checkpoint-specific exposition records the exact new declarations and limitations.

Checkpoint	Proof unit	New named supporting theorems	Archive
42	Uniform arithmetic thresholds for the major-arc model error	3	ZIP
43	Source parameter budgets imply both major-arc geometric conditions	2	ZIP
44	Uniform logarithmic major-arc approximation for the actual finite model	1	ZIP
45	Continuity in frequency of the genuine annular oscillatory integral	2	ZIP
46	Continuity of the genuine localized Gauss models	2	ZIP
47	The finite periodization window and its endpoint compatibility	3	ZIP
48	A continuous circle symbol from a compact real-frequency symbol	2	ZIP
49	Exact integer-sum interpretation of compact periodization	2	ZIP
50	Exact local agreement for narrow periodized symbols	2	ZIP
51	Linearity of compact periodization as a continuous circle map	2	ZIP
52	Global identification with genuine integer periodization	2	ZIP
53	The actual centered rational model and its compact support	2	ZIP
54	Genuine rational Gauss terms as continuous circle symbols	2	ZIP
55	Uniform Gauss decay survives actual circle periodization	2	ZIP
56	The genuine global finite circle model and its uniform Gauss bound	2	ZIP
57	Actual l2 model and error operators with exact annular decomposition	2	ZIP
58	Uniform inverse-kernel and finite-input convolution estimates	2	ZIP

Checkpoint	Proof unit	New named supporting theorems	Archive
59	Absolute convergence and bounds for all-input l1 convolution	2	ZIP
60	The bounded convolution sequence and its additive structure	2	ZIP
61	A genuine bounded linear l1-to-linfinity convolution operator	2	ZIP
62	Exact agreement of infinite convolution with finite and indicator inputs	2	ZIP
63	Compatible l1 and l2 realizations of continuous circle multipliers	2	ZIP
64	Operator estimates from sharper inverse-kernel bounds	2	ZIP
65	Pointwise linearity of actual inverse Fourier kernels	3	ZIP
66	Exact inverse-kernel assembly over finite symbol families	2	ZIP
67	Actual model and error kernels with exact annular decomposition	2	ZIP
68	Spatial model and error bounds from the genuine circle symbols	2	ZIP
69	Exact finite-input convolution formulas for model and error operators	2	ZIP
70	All-input l2 and l1-to-linfinity bounds for the finite circle model	2	ZIP
71	Exact support and core widths for arithmetic narrow cutoffs	2	ZIP
72	The narrow cutoff lies inside the broad cutoff core	2	ZIP
73	Continuity of the actual ordinary low-model ingredients	2	ZIP
74	Continuity of the actual exceptional low-model ingredients	2	ZIP
75	Continuity of the full arithmetic low term with both selectors	1	ZIP
76	A compact continuous realization of the centered low symbol	2	ZIP
77	The compact centered high symbol and exact total-model identity	2	ZIP
78	The actual periodized low term on the unit circle	1	ZIP
79	The periodized high term and exact low-plus-high identity	2	ZIP
80	Global finite low and high symbols with the actual annular error	2	ZIP
81	Actual inverse kernels for the finite low-high-error decomposition	2	ZIP
82	Genuine low l1 and high l2 operators with finite convolution formulas	2	ZIP
83	Exact actual averaging decomposition on every finite complex input	1	ZIP

Checkpoint	Proof unit	New named supporting theorems	Archive
84	Precise kernel and symbol inputs for the actual low and high operators	2	ZIP
85	The extra broad cutoff is redundant for the ordinary narrow term	1	ZIP
86	The extra broad cutoff is redundant under the exceptional width condition	1	ZIP
87	Exact identification of the compact low symbol with the source formula	1	ZIP
88	Exact identification of the compact high symbol with the source formula	1	ZIP
89	A genuine nonnegative l2 envelope for finite operator families	3	ZIP
90	Finite-family l2 maximal control through the explicit envelope	2	ZIP
91	Actual high-low assembly and completion of the fifty-checkpoint batch	1	ZIP

Resume

Extract the latest archive and read STATUS.md, docs/CONTINUE_HERE.md and docs/ASSUMPTIONS.md. Use the pinned toolchain and source revisions. A standard installation runs `lake build`, `python3 scripts/check_source.py` and `python3 scripts/check_axioms.py`. Local build artifacts are reproducible and are not included in the archive. The compatibility wrapper and narrow process-executable shim are documented in verification/ENVIRONMENT.md; no compiler or kernel modification is involved.

Next checkpoint: 92. Prioritize the full-range minor-arc/global approximation gap and the sharp source-specific low/high estimates; the circle/kernel/operator constructions in this batch should be reused.

`Checkpoints_92_96_Report.md`

Prime Squares: checkpoints 92–96

Five separately verified checkpoints add **28 named supporting theorems in five modules**. The final project has **163 modules and 968 named supporting theorems**. These are counts of supporting declarations, not counts of completed manuscript results.

The main advance is a proof of the actual periodized circle-model approximation on the major-arc and intermediate-denominator branches, conditional on the explicit progression-prefix estimate. The finite rational indexing and integer shifts are now handled exactly. The full minor-arc range and Theorem 1.1 remain open.

Saved checkpoints

Checkpoint	Mathematical unit	Added theorems	Project total	Archive and exposition
92	Exact reduced rational index set, including denominator one	7	947	ZIP / Proof notes
93	Separation and uniqueness for every integer lift	6	953	ZIP / Proof notes
94	Exact periodization and uniform logarithmic model tails	8	961	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Archive and exposition
95	Conditional major-arc error for the actual circle model	4	965	ZIP / Proof notes
96	Conditional intermediate-denominator circle approximation	3	968	ZIP / Proof notes

Each ZIP contains the complete project at that checkpoint, its proof notes, verification logs, frozen source materials, accepted revisions, dependency pins and continuation notes. Checkpoint 96 includes all five new modules and their expositions. Dependency caches are reproducible and excluded. Earlier checkpoints were saved before the subsequent production-tree changes.

What the proofs establish

`CanonicalRationalCenters.lean` constructs a finite set with membership exactly equivalent to $0 \le r < 1$ and $\mathrm{den}(r) \le R$. It accounts for all reduced canonical rationals, the empty zero cutoff, monotonicity and the denominator-one center zero. It does not assert the full character/coefficient summation correspondence with the manuscript.

`WrappedRationalArcGeometry.lean` proves separation of different rational-center/integer-lift pairs, uniqueness of sufficiently narrow arcs, and exclusion of all other lifted arcs. Integer shifts of a real representative give exactly the same point of the circle. Distinct lifts of the same rational are included in the proofs.

`PeriodizedModelTails.lean` writes the genuine periodization exactly as five shifts on the representative interval $[-1, 2]$. It proves an outside-arc norm bound of size $O((1+\log R)R^{-5/12})$ for the actual circle model. The fixed constants and budgets are defined in Lean. A common approximation budget is eventually at most $R^{-1/6}$ and has a threshold uniform over $Q \ge 1$ at the source scale.

`PeriodizedMajorArcError.lean` bounds the actual circle error on a major arc. The argument retains the four additional shifts of the real model and bounds them using the lifted separation theorem. It includes the shifted copies of the selected center itself. No periodicity is assumed for the continuous real-variable oscillatory integral.

`IntermediateCircleApproximation.lean` treats an approximating denominator $R < q \le R^{11/10}$ on the branch outside every small-denominator lifted major arc. It derives a bound for the actual von Mangoldt multiplier from the progression estimate and unconditional Gauss bounds, then subtracts the actual circle model. The geometric outside-arc condition and existence of the chosen approximation are explicit hypotheses of this branch.

Both conditional approximation branches give

$$\|E_{N,Q}(\xi)\| \le 2^A Q^{-1/2}(1+\log N)^{-A}$$

for all sufficiently large N, with one threshold independent of the permitted Q and exceptional data. Here $A \ge 0$, $Q \ge 1$, $R = \max(\exp((\log N)^{1/10}), Q^8)$, and $\log Q \le \min(c/7, 1)\sqrt{\log N}/8$. The relevant lifted arc, coefficient bounds, $0 < \beta \le 1$ and raw progression-prefix hypotheses remain explicit in the theorem statements.

Analytic assumptions and open work

The deep input is still a hypothesis bounding the actual `progressionMangoldtPrefix` minus `progressionPowerModel` by $NC\exp(-c\sqrt{\log N})$, for every unit residue and every real endpoint in $[N/2, N]$. The same character, coefficient and beta data are required throughout that interval. This input has not been proved in the project and is not introduced as an axiom. The source-scale inequalities are proved by existing threshold lemmas. The Gauss estimates used in these five units are already proved unconditionally in the project.

The missing range $q > R^{11/10}$ has not been filled by assuming the manuscript's minor-arc conclusion. Full Lemma 6.5, global Lemma 6.9, sharp low 6.14, multifrequency 6.16, high/maximal 6.21, Proposition 4.19 and **Theorem 1.1 remain open**. The full exceptional-parameter/source correspondence is also incomplete. Lemma 3.21 retains its two earlier explicit PNT limits; the separate PNT development has not been integrated by this batch.

The ten completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, abstract Theorem 4.1 and abstract Corollary 4.15 under their recorded source hypotheses. This batch does not establish a full-paper correctness or publication certification claim.

No new manuscript error has been established. No additional LaTeX replacement is prescribed. The frozen manuscript and the two accepted revisions remain unchanged.

Verification

The final checkpoint passes a fresh **3682-job full build of all 163 modules and the root**, with no warnings or errors. All **968 named theorems** pass a fresh complete transitive axiom audit allowing only `propext`, `Classical.choice` and `Quot.sound`. All **165 Lean files** pass the source scan: no placeholders, additional mathematical axioms, unchecked evaluation or unsafe code. Import and axiom-audit coverage are exact and unique. Complete output and source/artifact hashes are retained.

Checkpoint 92 also began with a fresh full rebuild after restoring the pinned environment. Checkpoints 93–95 used authenticated dependency-aware builds: earlier proof sources and compiled artifacts were SHA256 checked before reuse, and each new module and the root were compiled. Every checkpoint ran the complete source and transitive-axiom audits. The final full rebuild removes dependence on incremental history for the final verification result.

Lean 4.34.1 (commit `5045d0056413266e57c625dcd7c365b10e377c52`) and Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612` remain pinned. All nine dependency revisions were restored exactly. The official Lean archive hash was verified during recovery. The documented process-path compatibility shim does not modify Lean's compiler or kernel. GitHub CI is configured but has not run on GitHub.

Every checkpoint ZIP is CRC tested, and every archived project file is checked against its SHA256 manifest. The frozen manuscript hashes and both accepted revision hashes are checked. These new analytic results are Lean proofs; no new numerical experiment is used as a substitute. Earlier exact finite experiments and their script hashes are preserved.

Resume

Start from `PrimeSquares_Checkpoint_96.zip`, verify its manifest, and read `STATUS.md`, `docs/CONTINUE_HERE.md` and the five new expositions. The next checkpoint is **97**.

The next mathematical priority is the large-denominator/full-range minor-arc deduction. Check the exact uniform range and constants in a primary-source estimate before using it; do not vary a fixed epsilon with N or assume an uncontrolled own-paper conclusion. Then complete global approximation, connect the exceptional data and prove the sharp operator estimates needed for Proposition 4.19 and Theorem 1.1. The conditional major/intermediate circle branches and earlier exact operator assembly are already complete and should not be repeated.

`Checkpoints_97_146_Report.md`

Prime Squares: checkpoints 97–146

This batch contains **50 separately verified checkpoints**, adding **136 named supporting theorems in 50 modules**. The final project has **213 modules and 1104 named supporting theorems**. These are counts of supporting declarations, not counts of completed manuscript theorems.

The main mathematical advance is a conditional global circle-model approximation and a conditional quantitative countable dyadic maximal-error bound. A new complementary minor-arc argument resolves the range left uncovered by direct substitution into Ren's estimate. Theorem 1.1 and the separate sharp low/high inputs remain open.

Mathematical results

Write $x = N/2$ and $R = \max(\exp((\log N)^{1/10}), Q^8)$. The argument uses the actual annular von Mangoldt square multiplier, its actual continuous periodized circle model, and the corresponding genuine bounded operators on $\ell^2(\mathbb{Z})$.

1. **An unconditional divisor budget.** For every fixed positive exponent, the divisor count is bounded by a constant times that power of the denominator. The proof starts from the prime-power exponent factors, uses unique factorization, and retains all constants. In particular, $\tau(q)^{3/2} \le Dq^{1/12}$ with one fixed positive D.
2. **A compatible new rational selection.** Dirichlet approximation constructs a reduced rational a/q with $q \le x^{3/2}$ and $|\theta| \le 1/(qx^{3/2})$, where $\theta = \xi - a/q$. The strict error $|\theta| < 1/q^2$ is proved as well.

Normalization by the rational fractional part preserves the reduced denominator and identifies the exact integer lift of a canonical center.

3. **The complementary full-complement minor argument.** If $q \le \sqrt{x}$, the effective denominator $T = q(1+|\theta|x^2)$ is at most $2\sqrt{x}$. Exclusion from all small-denominator lifted source arcs forces $T \ge R/4$. Retaining the divisor loss, the three normalized Ren terms are bounded by

$$\sqrt{2}\,x^{-5/24}, \qquad x^{-19/120}, \qquad 4^{5/12}R^{-5/12}.$$

The source parameter budget implies $R^4 \le x$, so these give a bound of the form $K_{\mathrm{Ren}}(\log x)^d R^{-5/12}$. If $q > \sqrt{x}$, it lies in the balanced Ghosh range $[\sqrt{x}, x^{3/2}]$. The fixed choice $\varepsilon = 1/16$ gives a normalized saving $x^{-1/16} \le R^{-1/4}$. No fixed epsilon is changed with N. Combining the branches proves the alternative full-complement bound

$$|\widehat{\widetilde{A}}_{N,\Lambda}(\xi)| \le K_{\mathrm{Ren}}(\log(N/2))^d R^{-5/12} + K_{\mathrm{Gh}} R^{-1/4}.$$

The constants are defined in Lean, and the raw Ren and Ghosh inequalities remain explicit hypotheses. This is a weaker alternative to the stronger printed minor formula in Lemma 6.5; that printed formula is not claimed to be proved or disproved.

4. **Global circle approximation.** Logarithmic losses are absorbed uniformly in Q. A complete case split over the exact source rational set and every integer lift combines the major branch with the alternative minor branch. Every circle point is covered through a proved fundamental representative interval. Under the explicit analytic inputs, for every $A \ge 0$ and all sufficiently large N,

$$\|E_{Q,N}\|_{\ell^2 \to \ell^2} \le 2^{A+1} Q^{-1/2} (1+\log N)^{-A}.$$

The same bound holds for the actual circle error at every frequency. One threshold is chosen before all permitted Q and character/coefficient/beta data.

5. **A true countable maximal error.** The actual dyadic error operators have a summable norm tail. An absolutely convergent positive vector sum is constructed in real ℓ^2, its coordinate series is identified at every integer, and it is proved to dominate the genuine countable supremum. That supremum is itself realized in ℓ^2. At a starting dyadic scale $N_0 = 2^J$, the result is

$$\left\| \sup_{j \ge 0} |E_{Q,2^{J+j}} f| \right\|_2 \le C_A Q^{-1/2} (1+\log N_0)^{-A} \|f\|_2.$$

The Q condition is imposed at the starting scale and proved at every larger scale. The character, coefficient and beta data may vary between scales. The supremum covers the entire infinite tail, not only finite truncations.

6. **Continuation and pairing properties.** The actual maximal map is sublinear and Lipschitz. Norm-summable operator outputs vanish in ℓ^2 and at every integer coordinate, and their maximal tails vanish in ℓ^2. Exact operator identities identify the error with the actual average minus its circle model. The global error bound also yields finite-indicator pairings bounded by the error norm times $\sqrt{|F||G|}$, with the precise density normalization recorded.

Explicit inputs and scope

The hypothesis structure `CircleAnalyticInputs` contains beta/coefficient conditions and the following raw deep estimates. It contains no desired approximation or maximal-operator conclusion.

Input	What remains assumed
Progression prefix	The actual `progressionMangoldtPrefix` minus `progressionPowerModel` is bounded by $NP\exp(-c\sqrt{\log N})$ for every source denominator, unit residue and real endpoint in $[N/2, N]$. One set of scale data must work throughout that interval.
Ren	The displayed three-term estimate for the actual annular positive-phase prime-power sum, with its full $\tau(q)^{3/2}(\log x)^d$ prefactor, at compatible approximants.
Ghosh	The displayed fourth-root estimate for the actual quadratic sum at compatible approximants, with fixed $\varepsilon = 1/16$, the balanced denominator range and strict reciprocal-square error.

These published deep estimates are consumed as theorem hypotheses, as authorized. They are not formalized here, added as axioms, or replaced by numerical checks. The scalar constants are fixed before the varying scale, frequency, denominator and input function.

The common permitted range is $Q \ge 1$ and

$$\log Q \le \frac{\min(c/7,1)}{8}\sqrt{\log N}.$$

For a dyadic tail it suffices to impose this condition at its starting scale. Thresholds may depend on the fixed analytic constants and A, but not on Q or the later scale data.

The primary estimates were checked against Xiumin Ren's author preprint, On exponential sum over primes and application in Waring-Goldbach problem, Theorem 1.1 and equation (1.4). Ghosh's original article is The Distribution of alpha p squared Modulo 1, Theorem 2; its first-page transcription was also inspected at this copy. The original Ghosh publisher PDF was inaccessible; Ren's primary text supplies the directly displayed annular estimate. The exact raw inequalities and compatibility hypotheses used in Lean are visible in `RenBalancedInput`, `GhoshBalancedInput` and the consuming modules.

The stronger printed minor estimate of Lemma 6.5 remains open. The complete exceptional-parameter/source correspondence and exact double-sum bridge have not been silently certified by `CircleAnalyticInputs`. The full source version of Lemma 6.9 therefore still needs those bridges, even though the conditional global pointwise/operator approximation for the constructed circle model is proved. Sharp low 6.14, multifrequency 6.16, high/maximal 6.21, Proposition 4.19 and **Theorem 1.1 remain open**. The maximal error estimate above concerns the error component and does not bound the separate high operator.

The ten earlier completed numbered results remain 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, abstract Theorem 4.1 and abstract Corollary 4.15 under their recorded hypotheses. Lemma 3.21 retains its two earlier explicit PNT limits; the separate PNT project has not been integrated. There is no full-paper correctness or publication-certification claim.

No new manuscript statement error has been established, so no mandatory LaTeX replacement is prescribed. The alternative deduction is fully recorded in the new Lean modules and their expositions. The frozen manuscript and the two accepted revisions remain unchanged.

Verification

The final checkpoint passes a fresh **3738-job full build of all 213 modules and the root**, with no warnings or errors. All **1104 named theorems** pass a fresh complete transitive axiom audit permitting only `propext`, `Classical.choice` and `Quot.sound`. All **215 project Lean files** pass the source scan: no placeholders, extra mathematical axioms, unchecked evaluation or unsafe code. Import and audit coverage are exact and unique.

The baseline checkpoint 96 source/artifact hashes were verified before continuing. Every new checkpoint compiled its new module and the root and ran complete source and transitive-axiom audits. Unchanged earlier proof sources and compiled artifacts were SHA256 checked before incremental reuse. Checkpoints 121 and 146 performed fresh full project rebuilds. Verification logs and build receipts identify the exact compiled modules and complete dependency history.

Lean 4.34.1 (commit `5045d0056413266e57c625dcd7c365b10e377c52`) and Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612` remain pinned. No compiler/kernel modification or version upgrade was made. GitHub CI is configured but has not run on GitHub.

Each checkpoint ZIP is CRC tested, and every archived project file is checked against its SHA256 manifest. Frozen manuscript and accepted revision hashes are checked. These new analytic statements are Lean proofs; no new numerical experiment is used as a substitute. Earlier exact finite experiments and matching script hashes are preserved.

A final download check found 19 truncated archive transfers. Their complete entries were recovered with CRC checks and missing historical logs restored from the retained snapshots. The original file counts and every original Lean source hash were verified against the corresponding build receipts. The archives were replaced under their existing identities and checked again, including their SHA256 manifests. No mathematical proof was changed by this repair. The packaging script now publishes a complete ZIP by atomic rename so that a partially written archive cannot be selected during generation.

All fifty checkpoints

Each ZIP is a complete project snapshot with its proof notes, logs, source materials and restart instructions. The final ZIP includes all fifty new modules and expositions. Reproducible dependency caches are excluded. Each checkpoint was saved before the following production-tree mutation.

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
97	A fixed-epsilon Ghosh bound in the balanced denominator range	5	973	incremental	ZIP / Proof notes
98	Uniform prime-power exponent growth	3	976	incremental	ZIP / Proof notes
99	An unconditional uniform subpower bound for the divisor count	3	979	incremental	ZIP / Proof notes
100	A fixed subpower budget for the exact Ren divisor prefactor	3	982	incremental	ZIP / Proof notes
101	Effective-denominator bounds for the balanced approximation	3	985	incremental	ZIP / Proof notes
102	A proved rational selection at balanced precision	3	988	incremental	ZIP / Proof notes
103	The first two normalized Ren terms with the divisor loss retained	5	993	incremental	ZIP / Proof notes
104	The inverse effective-denominator term with divisor loss retained	2	995	incremental	ZIP / Proof notes
105	The complete weighted Ren bracket in the balanced regime	3	998	incremental	ZIP / Proof notes
106	The actual annular multiplier under the raw Ren estimate	2	1000	incremental	ZIP / Proof notes
107	A uniform fourth-power budget for the source scale	4	1004	incremental	ZIP / Proof notes
108	Canonical rational centers for an arbitrary selected approximation	3	1007	incremental	ZIP / Proof notes
109	Exclusion from all lifted source arcs and the effective lower bound	3	1010	incremental	ZIP / Proof notes
110	Converting the balanced Ren budget to a scale-only bound	3	1013	incremental	ZIP / Proof notes
111	Converting the balanced Ghosh saving to a scale-only bound	2	1015	incremental	ZIP / Proof notes
112	The small-denominator Ren branch at the actual source scale	2	1017	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
113	An alternative full-range minor-arc deduction under raw deep estimates	2	1019	incremental	ZIP / Proof notes
114	Absorbing fixed logarithmic and power budgets	3	1022	incremental	ZIP / Proof notes
115	Uniform logarithmic absorption at the actual source scale	3	1025	incremental	ZIP / Proof notes
116	A uniform source-scale complementary minor budget	2	1027	incremental	ZIP / Proof notes
117	A uniform error bound on the full circle minor-arc complement	3	1030	incremental	ZIP / Proof notes
118	The uniform error bound on all lifted major arcs	2	1032	incremental	ZIP / Proof notes
119	A conditional global pointwise approximation on the circle	1	1033	incremental	ZIP / Proof notes
120	From a fundamental representative interval to all circle points	3	1036	incremental	ZIP / Proof notes
121	A conditional global approximation for the actual l2 operator	2	1038	full	ZIP / Proof notes
122	Monotonicity of the allowed Q range and source cutoff	3	1041	incremental	ZIP / Proof notes
123	Exact dyadic logarithms and inverse-power comparison	4	1045	incremental	ZIP / Proof notes
124	Quantitative decay of the actual dyadic error operators	2	1047	incremental	ZIP / Proof notes
125	A finite reciprocal-square tail with exact telescoping	3	1050	incremental	ZIP / Proof notes
126	A quantitative infinite reciprocal-square tail	3	1053	incremental	ZIP / Proof notes
127	Weighted real-power finite tails	2	1055	incremental	ZIP / Proof notes
128	Summability and quantitative weighted infinite tails	2	1057	incremental	ZIP / Proof notes
129	Absolute summability and quantitative tails of operator norms	3	1060	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
130	A genuine countable positive l2 operator envelope	3	1063	incremental	ZIP / Proof notes
131	Exact coordinate identities for the countable l2 envelope	3	1066	incremental	ZIP / Proof notes
132	Pointwise domination by the countable envelope	3	1069	incremental	ZIP / Proof notes
133	A bounded true countable maximal function	4	1073	incremental	ZIP / Proof notes
134	A concrete l2 realization of the countable maximal function	2	1075	incremental	ZIP / Proof notes
135	An all-input countable maximal l2 bound	2	1077	incremental	ZIP / Proof notes
136	Sublinearity of the actual countable maximal function	3	1080	incremental	ZIP / Proof notes
137	A quantitative countable maximal tail from power decay	2	1082	incremental	ZIP / Proof notes
138	Monotonicity of the true countable maximal tails	3	1085	incremental	ZIP / Proof notes
139	Absolute norm summability of the actual dyadic error family	1	1086	incremental	ZIP / Proof notes
140	The genuine countable maximal error with a power tail	1	1087	incremental	ZIP / Proof notes
141	The conditional logarithmic dyadic maximal error estimate	3	1090	incremental	ZIP / Proof notes
142	Vanishing outputs of norm-summable operator families	3	1093	incremental	ZIP / Proof notes
143	Vanishing of the genuine countable maximal tails	2	1095	incremental	ZIP / Proof notes
144	Identifying the actual average-minus-model maximal error	3	1098	incremental	ZIP / Proof notes
145	Finite-indicator pairing bounds for the conditional error operator	3	1101	incremental	ZIP / Proof notes
146	Lipschitz continuity and final verification of the maximal error milestone	3	1104	full	ZIP / Proof notes

Continue from checkpoint 147

Recover `PrimeSquares_Checkpoint_146.zip`, verify its archive manifest, and read `STATUS.md`, `docs/CONTINUE_HERE.md`, this report and the relevant individual expositions. The next checkpoint is **147**.

The next mathematical priority is the genuine low-kernel estimate 6.14 and the sharp ordinary/exceptional high and multifrequency inputs 6.16/6.21. Complete the exceptional-data and exact source summation bridges before identifying the constructed model with every printed manuscript object. Then finish Proposition 4.19 and connect Theorem 1.1. The conditional global error, its alternative minor argument and its countable maximal-tail analysis are already complete and should not be repeated. Keep the stronger printed minor formula explicitly open unless a separate proof is supplied.

`Checkpoints_147_246_Report.md`

Prime Squares: checkpoints 147–246

This batch contains **100 separately verified checkpoints**, adding **217 named supporting theorems in 100 modules**. The final project has **313 modules and 1321 named supporting theorems**. These are supporting-declaration counts, not counts of completed manuscript results.

The main advance is a genuine denominator-indexed circle low multiplier, with its inverse kernel identified and its local and all-input operator estimates proved in the stated natural-cutoff range. The batch also develops countable square-function and maximal estimates, including an applicable criterion for actual circle multipliers. **Theorem 1.1 remains open.**

Mathematical results

1. **An actual smooth inverse cutoff.** The source smooth bump is proved to be Schwartz. Its inverse Fourier transform has rapid decay of every natural order. A proved real change of variables identifies the inverse transform of the frequency dilation with the physical dilation $a^{-1}\check{\eta}(x/a)$. Phase signs, complex scalars and normalizations are retained.
2. **The smoothed annular kernel.** For $N \geq 2$, $\beta \leq 1$ and $0 < a \leq N^2$, the actual kernel

$$K_{N,\beta,a}(x) = \frac{1}{N}\int_{N/2}^{N} t^{\beta-1}a^{-1}\check{\eta}((x-t^2)/a)\,dt$$

   is integrable in its defining variable and satisfies the proved rapid-decay envelope

$$|K_{N,\beta,a}(x)| \leq \frac{2^k C_k}{2a}\left(1+\frac{|x|}{N^2}\right)^{-k}.$$

   The constants C_k come from the actual inverse cutoff. The argument includes the scale comparison between x and $x-t^2$. It does not assume the desired spatial decay.
3. **Logarithmic arithmetic summation.** The ordinary truncated Cohen bound is assembled from the actual quadratic sums. For the exceptional arithmetic factor

$$\rho(n) = \prod_{p\mid n}\frac{p+1}{p-1},$$

   prime-subset expansion, divisibility and reciprocal-multiple sums prove

$$\sum_{n=1}^{D}\frac{\rho(n)}{n} \leq e^2 H_D \leq e^2\left(1+\frac{1}{\log 2}\right)\log(2D), \qquad D \geq 1.$$

   This estimate holds for all positive denominators; no squarefree restriction is needed for this upper bound. The correction Euler product is bounded by e^2. No PNT or Mertens hypothesis is used. The previously proved conductor-sensitive estimate for the actual induced twisted correlations supplies the exceptional coefficients.
4. **The ordinary minus exceptional physical low kernel.** The ordinary widths are qN^2/Q; the exceptional widths are $B_0 dN^2/Q$, with $B_0 = \texttt{conductorMajorant}(q_0)$. The arithmetic factor B_0 cancels against the exceptional smoothing width. For natural $N, Q \geq 2$, a positive conductor, a quadratic base character, $\beta \leq 1$, $S \subseteq \{1,\dots,Q\}$, and $B_0 d \leq Q$ for every $d \in S$, the actual complex kernel satisfies

$$|L_{N,Q}(x)| \leq C_{\text{low}}\frac{Q\log(2Q)}{N^2}\left(1+\frac{|x|}{N^2}\right)^{-10}.$$

The finite Gauss/unit sums and induced twisted coefficients are retained, with the exceptional term subtracted. The empty exceptional range is handled separately. The absolute constant is defined in Lean from the cutoff and arithmetic constants.

5. **Local convolution and genuine low operators.** A complete dyadic partition of integer distances includes every boundary and the inner shell. It gives, for every finite complex input, a convolution bound by $4C_{\mathrm{low}}Q\log(2Q)$ times the proved normalized dyadic local mass at scale N^2. The kernel also yields an all-input $\ell^1 \to \ell^\infty$ bound

$$\|L_{N,Q}\|_{\ell^1\to\ell^\infty} \le C_{\mathrm{low}}\frac{Q\log(2Q)}{N^2}.$$

Finite-indicator pairings and far-field decay follow. Uniformly bounded families admit a genuine countable $\ell^1 \to \ell^\infty$ maximum, including varying conductor, character and beta data between scales.

6. **The inverse-kernel identity for genuine circle symbols.** The compact frequency cutoff is passed to the circle, with the closed fundamental endpoints and every integer lift checked. Joint continuity and integrability justify the annular Fubini interchange. The resulting circle Fourier coefficient equals the actual smoothed physical kernel. Frequency shifts give the correct positive physical modulation. Summing over positive denominators and unit residues defines `denominatorCircleLow`, and its inverse kernel is proved equal to the physical low kernel above. The extra width budget $2Q \le N^2$ makes every contributing cutoff narrow enough for this circle construction. Thus the displayed local and ℓ^1 estimates hold for this actual Fourier multiplier, not just an abstract kernel majorant.
An all-input convolution series for every continuous circle symbol is also proved in ℓ^2 through density/continuous linear maps, and its ℓ^1 realization agrees with the actual ℓ^2 multiplier on the inclusion of ℓ^1. The `LowSymbolData` structure records only scale, character and width conditions; it contains no desired norm estimate. The exact all-real-lift formula for these circle symbols is recorded.

7. **Countable square functions and maxima.** For any sequence $F_j \in \ell^2(\mathbb{Z})$ with $\sum_j \|F_j\|_2^2 < \infty$, the double series of coordinate squares is summable. Its two orders of summation agree. The genuine pointwise square function is realized in real ℓ^2 with

$$\left\| \left(\sum_j |F_j|^2 \right)^{1/2} \right\|_2^2 = \sum_j \|F_j\|_2^2.$$

It dominates the actual countable pointwise supremum, which is itself realized in ℓ^2. Uniform finite energy bounds imply the countable maximal bound. This requires summability of squared output norms, a weaker condition than the earlier sum-of-output-norms construction. Operator versions, squared-operator-norm criteria, quantitative tail bounds, tail convergence to zero, sublinearity and Lipschitz control are proved.

8. **Actual annular collar decay and multiplier energy.** For $M \ge 2$, $\beta \in [0,1]$, $q, Q > 0$ and $|\theta| \ge Q/(2qM^2)$, the actual annular model at 2^jM satisfies

$$|\mathcal{A}_{2^jM,\beta}(\theta)| \le \frac{4q}{\pi Q}4^{-j}, \qquad \sum_{j\ge 0}|\mathcal{A}_{2^jM,\beta}(\theta)|^2 \le \frac{16}{15}\left(\frac{4q}{\pi Q}\right)^2.$$

First collar onset and exact finite energy decomposition under disjoint frequency support are proved as well. For continuous circle symbols m_j, the explicit finite-frequency energy condition

$$\sum_{j\in J}|m_j(\xi)|^2 \le B^2 \quad \text{for every finite } J \text{ and every circle point } \xi$$

implies the genuine countable operator bound

$$\left\| \sup_{j\ge 0}|T_{m_j}f| \right\|_2 \le B\|f\|_2.$$

The implication is proved by integration, the Fourier isometry and the countable square-function construction. Pointwise attenuation of the symbols preserves this conclusion. **The frequency energy condition has not yet been verified for the paper's complete high/multifrequency family.** This criterion is supporting infrastructure, not a completed proof of those source estimates.

Exact scope and remaining obligations

The new low estimates are unconditional within their displayed scale, character and finite-denominator hypotheses. The new square-function results are proved implications from their stated energy assumptions. No new deep analytic result, own-paper conclusion or mathematical axiom is introduced in this batch.

The low circle model uses natural $N, Q \geq 2$ with $2Q \leq N^2$ and the stated exceptional width budget. The general real-cutoff rounding and the exact identification with the earlier rational-indexed global approximation model remain open. The local result covers finite complex inputs; the global $\ell^1 \to \ell^\infty$ result covers all ℓ^1 inputs. These distinctions prevent a claim that every version and domain of the source low lemma 6.14 has been completed.

The earlier conditional global circle approximation and its countable dyadic maximal-error result remain valid under their explicit raw progression-prefix, Ren and Ghosh hypotheses. Those deep estimates have not been formalized here. The stronger printed minor formula in Lemma 6.5 and the complete exceptional-data/source summation correspondence remain open. The sharp high and multifrequency estimates 6.16/6.21, Proposition 4.19 and **Theorem 1.1 remain open**. The new low proof and generic square-energy criterion do not silently discharge these obligations.

The earlier ten completed numbered results remain recorded with their exact scopes: 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, abstract Theorem 4.1 and abstract Corollary 4.15. Lemma 3.21 retains its two earlier explicit PNT limits; no separate PNT formalization has been integrated. There is no claim of full-paper correctness or publication certification.

No new manuscript statement error has been established in this batch, so no mandatory LaTeX replacement is prescribed. The frozen manuscript, bibliography and two accepted source revisions remain unchanged. New deductions are explained in the Lean modules and individual proof notes.

Verification

The final checkpoint passes a fresh **3920-job full build of all 313 modules and the root**, with no warnings or errors. All **1321 named supporting theorems** pass a fresh complete transitive axiom audit allowing only `propext`, `Classical.choice` and `Quot.sound`. All **315 project Lean files** pass the source scan: no proof placeholders, extra mathematical axioms, unchecked evaluation or unsafe code. Import and audit coverage are exact and unique.

The baseline checkpoint 146 source/artifact hashes were checked before continuing. Every new checkpoint compiled its new module and the root and ran complete source and transitive-axiom audits. Before incremental reuse, SHA256 checks authenticated earlier proof sources and compiled artifacts. Checkpoints **171, 196, 221 and 246** performed fresh full project builds. Logs and build receipts record exact source/artifact hashes and build history.

Lean 4.34.1 (commit `5045d0056413266e57c625dcd7c365b10e377c52`) and Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612` remain pinned. Additional Fourier/Schwartz dependencies were built from that same Mathlib source. No compiler/kernel modification, toolchain upgrade or dependency version change was made. GitHub CI is configured but has not run on GitHub.

Every new ZIP is published atomically after CRC testing and checking every archived file against its SHA256 manifest. Frozen manuscript/revision hashes and source build receipts are checked. Every checkpoint is saved before the next production-tree mutation. Saved archive downloads are independently compared with the local archive bytes, CRC, internal manifests and Lean build-source receipts. This batch uses Lean proofs, not new numerical experiments, for its analytic conclusions. Earlier exact finite experiments and script hashes are preserved.

All one hundred checkpoints

Each ZIP is a complete project snapshot with proof notes, verification logs, source material and restart instructions. The final ZIP includes every new Lean module and exposition. Reproducible toolchain and dependency caches are excluded.

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
147	The truncated ordinary low arithmetic sum	3	1107	incremental	ZIP / Proof notes
148	Polynomial spatial decay at the square scale	4	1111	incremental	ZIP / Proof notes
149	Decay after a bounded spatial translation	2	1113	incremental	ZIP / Proof notes
150	Quantitative polynomial decay for Schwartz kernels	2	1115	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
151	The actual smooth cutoff as a Schwartz function	4	1119	incremental	ZIP / Proof notes
152	Rapid decay of the actual inverse cutoff transform	3	1122	incremental	ZIP / Proof notes
153	The continuous scaled inverse-cutoff kernel	3	1125	incremental	ZIP / Proof notes
154	Embedding narrower cutoff decay into the square scale	2	1127	incremental	ZIP / Proof notes
155	The actual annular convolution kernel	3	1130	incremental	ZIP / Proof notes
156	Uniform spatial decay after annular smoothing	1	1131	incremental	ZIP / Proof notes
157	Finite convolution against a decaying kernel	2	1133	incremental	ZIP / Proof notes
158	All-input l1 consequences of polynomial kernel decay	2	1135	incremental	ZIP / Proof notes
159	Finite near-far convolution decomposition	2	1137	incremental	ZIP / Proof notes
160	Polynomial decay on dyadic spatial shells	2	1139	incremental	ZIP / Proof notes
161	The exact summable tenth-order shell budget	2	1141	incremental	ZIP / Proof notes
162	The bounded dyadic local-mass envelope	3	1144	incremental	ZIP / Proof notes
163	Controlling every dyadic ball by the local envelope	4	1148	incremental	ZIP / Proof notes
164	A genuine dyadic cover of every distance	3	1151	incremental	ZIP / Proof notes
165	The core and annular cases of the least dyadic level	2	1153	incremental	ZIP / Proof notes
166	An exact finite partition into spatial shells	2	1155	incremental	ZIP / Proof notes
167	Exact volume-normalized shell summation	2	1157	incremental	ZIP / Proof notes
168	Local-mass control on each exact spatial shell	2	1159	incremental	ZIP / Proof notes
169	Summable local contributions from every shell	2	1161	incremental	ZIP / Proof notes
170	The local bound for the full polynomial profile	1	1162	incremental	ZIP / Proof notes
171	A full finite-input local convolution estimate	1	1163	full	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
172	Expanding the exceptional arithmetic factor	3	1166	incremental	ZIP / Proof notes
173	Prime subsets and exact divisibility	2	1168	incremental	ZIP / Proof notes
174	A common prime-set expansion for every denominator	2	1170	incremental	ZIP / Proof notes
175	A uniform harmonic bound for reciprocal multiples	1	1171	incremental	ZIP / Proof notes
176	Rearranging the actual weighted exceptional sum	1	1172	incremental	ZIP / Proof notes
177	The summable correction Euler product	3	1175	incremental	ZIP / Proof notes
178	An unconditional exceptional weighted harmonic bound	1	1176	incremental	ZIP / Proof notes
179	The exceptional logarithmic sum without Mertens	3	1179	incremental	ZIP / Proof notes
180	Logarithmic control for every contributing subset	2	1181	incremental	ZIP / Proof notes
181	The actual induced-character correlation at integer arguments	2	1183	incremental	ZIP / Proof notes
182	Weighted logarithmic control of induced correlations	1	1184	incremental	ZIP / Proof notes
183	The normalized ordinary arithmetic correlation	3	1187	incremental	ZIP / Proof notes
184	Summing arithmetic coefficients against spatial kernels	2	1189	incremental	ZIP / Proof notes
185	The exact ordinary low spatial width	3	1192	incremental	ZIP / Proof notes
186	Spatial decay of the ordinary physical low kernel	1	1193	incremental	ZIP / Proof notes
187	The exceptional spatial width and conductor cancellation	3	1196	incremental	ZIP / Proof notes
188	Uniform spatial decay of the induced exceptional kernel	1	1197	incremental	ZIP / Proof notes
189	The empty exceptional low range	2	1199	incremental	ZIP / Proof notes
190	The combined ordinary-minus-exceptional physical kernel	2	1201	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
191	Uniform low amplitude across arithmetic scales	2	1203	incremental	ZIP / Proof notes
192	The local low bound for finite complex inputs	1	1204	incremental	ZIP / Proof notes
193	A genuine all-input physical low operator	2	1206	incremental	ZIP / Proof notes
194	Finite-set bounds for the physical low operator	2	1208	incremental	ZIP / Proof notes
195	A quantitative far-field estimate for the physical low kernel	1	1209	incremental	ZIP / Proof notes
196	A genuine countable l1-to-linfinity maximal construction	4	1213	full	ZIP / Proof notes
197	The all-scale physical low maximal bound	2	1215	incremental	ZIP / Proof notes
198	The real Fourier phase and quadratic convention	4	1219	incremental	ZIP / Proof notes
199	The exact real inverse Fourier integral	3	1222	incremental	ZIP / Proof notes
200	Change of variables for the inverse Fourier transform	2	1224	incremental	ZIP / Proof notes
201	Identification of the scaled physical cutoff inverse	2	1226	incremental	ZIP / Proof notes
202	Reducing narrow inverse transforms to the fundamental interval	2	1228	incremental	ZIP / Proof notes
203	Exact periodization at both fundamental endpoints	1	1229	incremental	ZIP / Proof notes
204	The real-to-circle inverse-kernel identity	1	1230	incremental	ZIP / Proof notes
205	Joint regularity and product integrability for the annular inverse	2	1232	incremental	ZIP / Proof notes
206	A checked Fubini interchange on the annular rectangle	1	1233	incremental	ZIP / Proof notes
207	The actual narrow annular frequency symbol	2	1235	incremental	ZIP / Proof notes
208	Matching the annular and inverse Fourier phases	2	1237	incremental	ZIP / Proof notes
209	The exact annular-product inverse-transform formula	1	1238	incremental	ZIP / Proof notes
210	The genuine circle annular kernel equals the physical kernel	1	1239	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
211	Rational frequency shifts and physical modulation	3	1242	incremental	ZIP / Proof notes
212	Exact agreement with the rational additive character	2	1244	incremental	ZIP / Proof notes
213	The ordinary and twisted arithmetic phase sums	2	1246	incremental	ZIP / Proof notes
214	Lower spatial widths needed for exact circle lifting	2	1248	incremental	ZIP / Proof notes
215	The genuine ordinary denominator circle term	2	1250	incremental	ZIP / Proof notes
216	The exact kernel of the ordinary denominator model	1	1251	incremental	ZIP / Proof notes
217	The genuine induced exceptional circle term	2	1253	incremental	ZIP / Proof notes
218	The exact induced exceptional circle kernel	1	1254	incremental	ZIP / Proof notes
219	The genuine denominator-indexed low circle model	1	1255	incremental	ZIP / Proof notes
220	The low spatial envelope for the actual circle kernel	2	1257	incremental	ZIP / Proof notes
221	The local low estimate for the genuine Fourier multiplier	2	1259	full	ZIP / Proof notes
222	The all-input l1 bound for the genuine low multiplier	2	1261	incremental	ZIP / Proof notes
223	All-input Fourier convolution and l1-l2 agreement	3	1264	incremental	ZIP / Proof notes
224	All-input and finite-set low estimates	2	1266	incremental	ZIP / Proof notes
225	Uniform countable maximal bounds for actual low families	2	1268	incremental	ZIP / Proof notes
226	Source-range geometry and exact integer-lift formulas	3	1271	incremental	ZIP / Proof notes
227	Exact square energy of an integer l2 sequence	3	1274	incremental	ZIP / Proof notes
228	Joint square summability over scale and space	3	1277	incremental	ZIP / Proof notes
229	A rigorous interchange of scale and spatial energy	2	1279	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
230	The actual countable pointwise square function	3	1282	incremental	ZIP / Proof notes
231	Realizing the countable square function in l2	1	1283	incremental	ZIP / Proof notes
232	The exact l2 norm of the countable square function	2	1285	incremental	ZIP / Proof notes
233	The genuine countable maximum dominated by square energy	3	1288	incremental	ZIP / Proof notes
234	The countable l2 maximal bound from square summability	2	1290	incremental	ZIP / Proof notes
235	Finite energy budgets imply the full countable maximal bound	3	1293	incremental	ZIP / Proof notes
236	The countable operator maximal principle from square energy	2	1295	incremental	ZIP / Proof notes
237	A weaker squared-operator-norm summability criterion	3	1298	incremental	ZIP / Proof notes
238	Quantitative maximal tails from square energy	3	1301	incremental	ZIP / Proof notes
239	Convergence to zero of maximal square-energy tails	2	1303	incremental	ZIP / Proof notes
240	Sublinearity and Lipschitz control under square summability	3	1306	incremental	ZIP / Proof notes
241	The actual lower frequency at first collar onset	2	1308	incremental	ZIP / Proof notes
242	Geometric decay after an actual annular collar onset	2	1310	incremental	ZIP / Proof notes
243	The summable squared annular frequency tail	2	1312	incremental	ZIP / Proof notes
244	Exact finite energy decomposition under disjoint frequency support	2	1314	incremental	ZIP / Proof notes
245	A frequency square-energy criterion for actual l2 multipliers	5	1319	incremental	ZIP / Proof notes
246	The genuine countable maximal theorem for square-bounded symbols	2	1321	full	ZIP / Proof notes

Continue from checkpoint 247

Recover `PrimeSquares_Checkpoint_246.zip`, verify its manifest, and read `STATUS.md`, `docs/CONTINUE_HERE.md`, this report and the relevant individual proof notes. The next checkpoint is **247**.

The next mathematical priority is the exact correspondence between the new denominator/unit-indexed low symbol and the earlier rational-indexed approximation model, including exceptional-data and cutoff rounding. Then establish the source sharp high and multifrequency estimates, using the actual collar decay and square-energy machinery where applicable. Connect these proved estimates to Proposition 4.19 and ultimately Theorem 1.1. Preserve all displayed hypotheses and do not assume any of those target conclusions. The conditional global error and the new elementary low-kernel/arithmetic stages are already proved and should not be repeated.

`Checkpoints_247_346_Report.md`

Prime Squares: checkpoints 247–346

This batch contains **100 separately verified checkpoints**, adding **182 named supporting theorems in 100 modules**. The final project has **413 modules and 1503 named supporting theorems**. These are supporting declarations, not a count of completed manuscript statements.

The main advances are the exact bridge to the original rational-indexed low symbol, real cutoffs and the full conductor range; an unconditional single-scale high bound at every power below one half; and a conditional restricted-pairing bound for the actual annular von Mangoldt square kernel. **Theorem 1.1 remains open.**

The actual source low symbol

The canonical source rational set is identified exactly with the positive-denominator/unit-residue double sum. The map $u \mapsto u/q$ has its reduced numerator and denominator proved, including the denominator-one center zero. A concrete finite equivalence gives exact reindexing of complex sums and the totient cardinality identity. Induced characters, optional exceptional selectors, Gauss coordinates and real frequency shifts agree with the original source definitions.

For a real cutoff Q, integer floors enumerate the denominators only. The real Q itself stays in the smoothing widths. The ordinary width is qN^2/Q. For the exceptional conductor q_0, let $B_0 = \texttt{conductorMajorant}(q_0)$; the exceptional width is $B_0 dN^2/Q$. The contributing exceptional set is exactly

$$\{d \in \mathbb{N} : 1 \le d \le \lfloor Q \rfloor,\ B_0 d \le Q,\ q_0 d \le M\}.$$

All cutoff equality cases, natural/real conductor quotients, the exceptional minus sign and the inactive case are included. The actual induced correlation bound cancels B_0 against the exceptional width. The weighted arithmetic sum is bounded logarithmically using the previous elementary result; no PNT or Mertens assumption enters this low argument.

The inverse kernel is identified for the actual continuous circle symbol, first term by term and then for the complete sum. Exact all-real integer periodization and frequency-shift identities retain the normalizations and modulation signs. The combined ordinary-minus-exceptional model is proved equal to the original rational-indexed source low model. No desired kernel bound is assumed to obtain this identity.

The working radius may be enlarged to R^2 while the actual source denominator cap remains $\lfloor R \rfloor$. If $q_0 > R$, every exceptional selector on the source set is exactly zero and the exceptional denominator set is empty. These facts remove the intermediate exceptional-width range restriction and cover **every positive conductor**.

For natural $N \ge 2$, real $Q \ge 2$, $Q \le R$, $2R^8 \le N^2$, a positive conductor, a quadratic base character, an optional exceptional term and $\beta \le 1$, the actual source low kernel satisfies

$$|L_{N,Q}(x)| \le 2C_{\mathrm{low}} \frac{Q \log(2Q)}{N^2} \left(1 + \frac{|x|}{N^2}\right)^{-10}.$$

Its genuine all-input operator and finite-complex local estimates are

$$\|L_{N,Q}\|_{\ell^1 \to \ell^\infty} \le 2C_{\mathrm{low}} \frac{Q \log(2Q)}{N^2},$$

$$|L_{N,Q}f(x)| \le 8C_{\mathrm{low}}Q\log(2Q)\,\texttt{dyadicLocalMass}(N^2,F,|f|,x), \qquad \operatorname{supp} f \subseteq F \text{ finite}.$$

Here `dyadicLocalMass` is the exact normalized integer-shell quantity already defined in Lean. The absolute constant is constructed from the actual cutoff and arithmetic constants. Finite-indicator pairings and a genuine countable $\ell^1 \to \ell^\infty$ maximum follow, including families with varying exceptional data. The data structures record numerical and character conditions; they contain no desired norm conclusion.

The source paper budget $R^4 \le N/2$ implies the displayed geometry. For the actual source choice

$$R_{N,Q} = \max\{\exp((\log N)^{1/10}), Q^8\},$$

the low estimates therefore hold uniformly for all sufficiently large N and real $Q \ge 2$ satisfying

$$\log Q \le \frac{\min\{c/7,1\}}{8}\sqrt{\log N}, \qquad c > 0.$$

These low proofs use no new deep analytic input. The local conclusion is for finite complex inputs; the global norm conclusion is for all ℓ^1 inputs. Natural $N \ge 2$ and real $Q \ge 2$ are retained. We do not claim every domain or formulation of the printed low proposition (`p:lowcontribution`) beyond those proved hypotheses.

The actual single-scale high bound

For **every fixed** $0 < \eta < 1/2$, there is $H_\eta > 0$ such that the actual source high circle symbol and its genuine ℓ^2 multiplier satisfy

$$\sup_\theta |m^{\mathrm{high}}_{N,Q}(\theta)| \le H_\eta Q^{-\eta}, \qquad \|H_{N,Q}\|_{\ell^2\to\ell^2} \le H_\eta Q^{-\eta}.$$

This holds for natural $N \ge 2$, real $Q > 0$, $\beta \in [0,1]$, every positive conductor, every finite canonical rational set with bounded denominators, arbitrary Dirichlet-character data and source coefficients of norm at most one. The constant is independent of the denominator cap, conductor, character family and arithmetic length. The actual optional source selector satisfies the coefficient condition by a proved theorem. The uniform estimate is for the defined continuous complement `circleHighModel`, whose narrow low term retains the broad cutoff factor. Its identification with the printed low/high decomposition uses the proved source low identity and source geometry above; that identification is not asserted outside those hypotheses.

The proof is elementary relative to the already proved Gauss bounds:

1. For every $\delta < 1/2$, the principal and twisted unit Gauss sums have a uniform $C_\delta q^{-\delta}$ bound. A proved estimate also gives a universal upper bound on B_0/q_0.
2. The actual high mask is written exactly as $1 - \mathbf{1}_{q\le W}\eta(qN^2\theta/W)$, with ordinary $W = Q$ and exceptional $W = Qq_0/B_0$. Its nonzero onset and the actual annular oscillatory estimate give both size and q/W bounds.
3. Interpolation gives a centered high bound $DQ^{-\eta}q^{-(\delta-\eta)}$. The broad source cutoff is retained on both sides; no unjustified containment of broad and narrow cutoffs is used.
4. For each denominator block $s = \log_2 q$, the broad cutoffs have at most four active canonical centers. For $s \ge 2$ there is at most one; the small denominators are counted separately. Every inactive contribution is proved zero.
5. The retained factor satisfies $q^{-(\delta-\eta)} \le (2^{-(\delta-\eta)})^s$. Its geometric sum is bounded by $(1-2^{-(\delta-\eta)})^{-1}$ independently of the global denominator cap.
6. Exact three-term periodization on $[-1,1]$ passes the real sum to the circle. The resulting explicit constant is $12D/(1-2^{-(\delta-\eta)})$. Choosing $\eta < \delta < 1/2$ and discharging the Gauss and conductor budgets proves the claim.

Finite-indicator high pairings satisfy $H_\eta Q^{-\eta}\sqrt{|F|}\sqrt{|G|}$. For a sequence of actual high operators with varying cutoffs Q_j and arbitrary other allowed data, the proved condition $\sum_j Q_j^{-\eta} < \infty$ yields a genuine countable pointwise supremum in ℓ^2, with norm at most $H_\eta(\sum_j Q_j^{-\eta})\|f\|_2$. Geometric cutoffs $Q_j = 2^j Q$ satisfy this summability by a separate proved calculation.

These results do not supply the sharp printed high estimate or the fixed-cutoff countable high/multifrequency estimate. The printed factor is $Q^{-1/2}\exp((2+o(1))\sqrt{\log Q}/\log\log Q)$, which is stronger than the proved $Q^{-\eta}$ bounds. The varying-cutoff maximum has the explicit summability condition; it must not be substituted for the paper's remaining high maximal statement.

Conditional actual prime-square assembly

The remainder is defined as the actual circle high operator plus the actual global prime major-arc error operator. The annular von Mangoldt square kernel has an exact finite-input decomposition into the actual low output and this remainder output. The low input lies in ℓ^1 and the remainder input in ℓ^2; their finite coordinate agreement is checked.

The earlier conditional approximation bound gives an error norm at most

$$E_A(N,Q) = 2 \cdot 2^A Q^{-1/2}(1+\log N)^{-A}, \qquad A \geq 0.$$

The new high bound and a proved absorption $E_A(N,Q) \leq 2 \cdot 2^A Q^{-\eta}$ give an actual remainder norm $HQ^{-\eta}$ throughout the uniform source range. The **raw progression-prefix, Ren and Ghosh estimates remain explicit hypotheses** in `CircleAnalyticInputs`. That structure contains beta and coefficient conditions and those raw estimates, not the source high norm or the desired restricted pairing conclusion.

For the actual annular von Mangoldt square kernel K_N, every finite $F, G \subset \mathbb{Z}$ therefore satisfies the conditional result

$$|\langle K_N * \mathbf{1}_F, \mathbf{1}_G \rangle| \leq 2C_{\text{low}} \frac{Q \log(2Q)}{N^2} |F||G| + HQ^{-\eta}\sqrt{|F|}\sqrt{|G|}.$$

The quantifiers retain $0 < \eta < 1/2$, fixed nonnegative analytic/error parameters, a positive decay parameter c, sufficiently large natural N, the displayed real-$Q \geq 2$ logarithmic range, every positive conductor and optional quadratic exceptional data satisfying the raw analytic inputs. The bound is derived from the exact operators and their proved estimates; no scalar pairing estimate is assumed.

Remaining obligations and manuscript status

Theorem 1.1 and Proposition 4.19 remain open. This batch proves the displayed single-scale weighted-annular consequence. It does not establish the remaining sharp/fixed-cutoff high maximal or multifrequency estimates 6.16/6.21, nor silently complete the source prime-average/headline assembly. The stronger printed minor formula in Lemma 6.5 remains open; the earlier alternative minor estimate and conditional global approximation remain available.

The previously recorded ten numbered results retain their exact scopes: 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, abstract Theorem 4.1 and abstract Corollary 4.15. Lemma 3.21 retains its two explicit PNT limits. No separate PNT formalization has been integrated. The raw deep approximation inputs have not been formalized here. There is no claim of full-paper correctness or publication certification.

No new manuscript statement error has been established in this batch, so no mandatory LaTeX replacement is prescribed. The frozen manuscript, bibliography and both previously accepted source revisions are unchanged. All new work is recorded in Lean and the individual proof notes.

Verification and reproducibility

The final checkpoint passes a fresh **4020-job full build of all 413 modules and the root**, with no warnings or errors. All **1503 named supporting theorems** pass a fresh complete transitive axiom audit allowing only `propext`, `Classical.choice` and `Quot.sound`. All **415 project Lean files** pass the complete source scan: no proof placeholders, additional mathematical axioms, unchecked evaluation or unsafe code. Import and audit coverage are exact and unique.

The baseline checkpoint 246 source/artifact receipt was authenticated before any new production edits. Every new checkpoint compiled its actual new module and root and ran complete source and transitive-axiom audits. Before incremental reuse, SHA256 checks authenticated earlier Lean sources and compiled artifacts. Checkpoints **271, 296, 321 and 346** performed fresh full project builds. Logs and build receipts preserve the exact source/artifact hashes and compilation history. Scratch aggregate checks were used for proof development; they do not replace the individual production builds.

Lean 4.34.1, commit `5045d0056413266e57c625dcd7c365b10e377c52`, and Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612` remain pinned. Additional dependencies were built from that same Mathlib source. No compiler/kernel modification, dependency upgrade or mathematical axiom was introduced. The existing narrow process-executable path compatibility shim is retained solely for the container runtime. GitHub CI is configured but has not run on GitHub.

Each ZIP is written atomically after CRC testing and verification against its complete SHA256 manifest. Each checkpoint is saved before the next production-tree mutation, together with its proof notes and

the guarded stable progress-file update. Downloaded copies are independently checked for byte length, SHA256, every ZIP CRC, every manifest entry and all Lean source hashes in the checkpoint's build receipt. The final saved-files index records the resulting archive and final-document integrity checks. Lean proofs supply the new analytic conclusions; no new numerical experiment is substituted for a proof. Earlier exact numerical reports and their script hashes are preserved.

All one hundred checkpoints

Each ZIP is a complete project snapshot with the proof notes, verification logs, frozen source material and restart instructions. The final ZIP contains every new Lean module and exposition. Reproducible toolchain/dependency caches are excluded.

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
247	Canonical rational numerators as actual residue units	2	1323	incremental	ZIP / Proof notes
248	Exact reduced numerator and denominator of a unit fraction	2	1325	incremental	ZIP / Proof notes
249	Canonical rational centers from every residue unit	2	1327	incremental	ZIP / Proof notes
250	Recovery and injectivity of the unit-rational map	2	1329	incremental	ZIP / Proof notes
251	The exact finite denominator fibers	2	1331	incremental	ZIP / Proof notes
252	The actual bijection between residue units and a rational fiber	2	1333	incremental	ZIP / Proof notes
253	Exact finite-sum reindexing through the unit-fiber equivalence	1	1334	incremental	ZIP / Proof notes
254	Denominator fibers under the global rational truncation	2	1336	incremental	ZIP / Proof notes
255	Exact partition of the canonical sum by denominator	1	1337	incremental	ZIP / Proof notes
256	The complete rational to denominator-unit double-sum identity	2	1339	incremental	ZIP / Proof notes
257	Exact totient count of the canonical rational centers	2	1341	incremental	ZIP / Proof notes
258	Exact source frequency and numerator coordinates	2	1343	incremental	ZIP / Proof notes
259	The optional exceptional-conductor indicator with its actual norm	3	1346	incremental	ZIP / Proof notes
260	The induced character family at every rational denominator	2	1348	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
261	Exact induced-character and exceptional coefficient data on unit fractions	3	1351	incremental	ZIP / Proof notes
262	Exact rational to unit indexing for the actual low circle model	2	1353	incremental	ZIP / Proof notes
263	Removing the redundant broad cutoff in the actual source low symbol	2	1355	incremental	ZIP / Proof notes
264	Subtraction and zero in the compact circle lift	2	1357	incremental	ZIP / Proof notes
265	Equality of actual l2 and l1 operators from their inverse kernels	2	1359	incremental	ZIP / Proof notes
266	Exact linear algebra for shifted circle symbols	3	1362	incremental	ZIP / Proof notes
267	The actual centered ordinary low symbol with a real cutoff	2	1364	incremental	ZIP / Proof notes
268	The ordinary rational circle term from its actual centered source	1	1365	incremental	ZIP / Proof notes
269	Exact ordinary circle symbols under the rational-unit correspondence	2	1367	incremental	ZIP / Proof notes
270	The denominator-unit ordinary term for a real cutoff	1	1368	incremental	ZIP / Proof notes
271	The inverse kernel of the real-cutoff ordinary term	1	1369	full	ZIP / Proof notes
272	The actual ordinary circle low model at a real cutoff	1	1370	incremental	ZIP / Proof notes
273	Exact real-cutoff floors and logarithmic arithmetic budgets	3	1373	incremental	ZIP / Proof notes
274	The logarithmic ordinary kernel decay with the actual real cutoff	1	1374	incremental	ZIP / Proof notes
275	Uniform real-cutoff amplitudes and scale monotonicity	2	1376	incremental	ZIP / Proof notes
276	Rapid-decay and uniform ordinary low-kernel bounds at real Q	2	1378	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
277	The real-cutoff ordinary local convolution estimate	1	1379	incremental	ZIP / Proof notes
278	The real-cutoff ordinary all-input l1 operator bound	3	1382	incremental	ZIP / Proof notes
279	The real-cutoff ordinary local Fourier multiplier estimate	2	1384	incremental	ZIP / Proof notes
280	The genuine countable ordinary low maximum for real Q	1	1385	incremental	ZIP / Proof notes
281	The exact source low symbol in the absence of an exception	2	1387	incremental	ZIP / Proof notes
282	Extensionality of the actual compact circle lift	3	1390	incremental	ZIP / Proof notes
283	The exact ordinary-only source circle term	1	1391	incremental	ZIP / Proof notes
284	Exact denominator filtering for real source cutoffs	2	1393	incremental	ZIP / Proof notes
285	The full source ordinary-only low-symbol identity at real Q	1	1394	incremental	ZIP / Proof notes
286	The actual source ordinary-only low norm and local estimates	2	1396	incremental	ZIP / Proof notes
287	Exact real and natural conductor quotients and cutoffs	3	1399	incremental	ZIP / Proof notes
288	The exact finite set of contributing exceptional quotient denominators	2	1401	incremental	ZIP / Proof notes
289	Exact reindexing of conductor multiples and both cutoff boundaries	1	1402	incremental	ZIP / Proof notes
290	The actual induced exceptional centered cutoff symbol	1	1403	incremental	ZIP / Proof notes
291	The exact source exceptional sign and real smoothing width	1	1404	incremental	ZIP / Proof notes
292	Actual compact support of the induced exceptional centered term	1	1405	incremental	ZIP / Proof notes
293	The induced exceptional rational circle multiplier	2	1407	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
294	Exact exceptional unit and rational coordinates	2	1409	incremental	ZIP / Proof notes
295	The actual denominator-unit exceptional circle term	2	1411	incremental	ZIP / Proof notes
296	The exceptional inverse kernel with its exact conductor quotient	1	1412	full	ZIP / Proof notes
297	The real-cutoff induced exceptional low model and its inverse kernel	1	1413	incremental	ZIP / Proof notes
298	Uniform induced exceptional kernel decay for real cutoffs	1	1414	incremental	ZIP / Proof notes
299	The real-cutoff exceptional amplitude and l1 operator norm	3	1417	incremental	ZIP / Proof notes
300	The real-cutoff exceptional local convolution estimate	1	1418	incremental	ZIP / Proof notes
301	The full centered source low identity with an optional induced exception	1	1419	incremental	ZIP / Proof notes
302	The full source circle low term and its exact exceptional sign	2	1421	incremental	ZIP / Proof notes
303	Exact exceptional cutoff filtering in the unit sum	2	1423	incremental	ZIP / Proof notes
304	The complete source exceptional finite sum and conductor reindexing	2	1425	incremental	ZIP / Proof notes
305	The exact original source low symbol for real Q and optional exceptions	3	1428	incremental	ZIP / Proof notes
306	The actual combined real-cutoff low symbol and its inverse kernel	1	1429	incremental	ZIP / Proof notes
307	The complete real-cutoff low decay and l1 operator bounds	3	1432	incremental	ZIP / Proof notes
308	The complete real-cutoff low local Fourier estimate	2	1434	incremental	ZIP / Proof notes
309	Actual source low families and their genuine countable l1 maximum	3	1437	incremental	ZIP / Proof notes
310	Source scale budgets for the full conductor range	4	1441	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
311	Keeping the source denominator cap while using the squared working radius	1	1442	incremental	ZIP / Proof notes
312	Exact disappearance of exceptional terms beyond the source cutoff	5	1447	incremental	ZIP / Proof notes
313	The exact source low identity for every positive conductor	1	1448	incremental	ZIP / Proof notes
314	The source low kernel and operator bounds throughout the full conductor range	3	1451	incremental	ZIP / Proof notes
315	Actual finite-indicator pairings for the complete source low multiplier	3	1454	incremental	ZIP / Proof notes
316	Countable actual source low families throughout the paper scale budget	2	1456	incremental	ZIP / Proof notes
317	The complete source low estimates in the actual uniform paper parameter range	2	1458	incremental	ZIP / Proof notes
318	Uniform Gauss decay at every exponent below one half	1	1459	incremental	ZIP / Proof notes
319	A proved uniform upper bound on the chosen conductor-majorant ratio	2	1461	incremental	ZIP / Proof notes
320	The actual high mask and its precise frequency onset	2	1463	incremental	ZIP / Proof notes
321	Uniform size and effective-cutoff decay of the actual annular high factor	4	1467	full	ZIP / Proof notes
322	Fractional-power interpolation of the actual high size and tail bounds	3	1470	incremental	ZIP / Proof notes
323	The exact effective exceptional cutoff and frequency width	3	1473	incremental	ZIP / Proof notes
324	The exact original centered high symbol in ordinary and exceptional masks	1	1474	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
325	Actual ordinary and exceptional high size and linear-decay bounds	2	1476	incremental	ZIP / Proof notes
326	A quantitative denominator-weighted power bound for the actual centered high symbol	1	1477	incremental	ZIP / Proof notes
327	Exact broad-support vanishing of the actual centered high symbol	1	1478	incremental	ZIP / Proof notes
328	The exact active-center count in each denominator block	1	1479	incremental	ZIP / Proof notes
329	A proved sparse sum bound with exact zero support	2	1481	incremental	ZIP / Proof notes
330	Exact partition of all source rational centers by dyadic denominator	1	1482	incremental	ZIP / Proof notes
331	Geometric decay of the retained denominator power	1	1483	incremental	ZIP / Proof notes
332	A cutoff-independent geometric budget for the retained denominator power	2	1485	incremental	ZIP / Proof notes
333	The full denominator-uniform weighted rational high sum	1	1486	incremental	ZIP / Proof notes
334	The exact three-term periodization on the full unit interval	1	1487	incremental	ZIP / Proof notes
335	The exact three-term real formula for every canonical source high term	1	1488	incremental	ZIP / Proof notes
336	Exact finite window assembly for the complete source high circle sum	1	1489	incremental	ZIP / Proof notes
337	The actual circle high symbol with a cutoff-independent power bound	1	1490	incremental	ZIP / Proof notes
338	The actual high l2 multiplier from its proved power symbol bound	2	1492	incremental	ZIP / Proof notes
339	An unconditional single-scale high bound at every exponent below one half	1	1493	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
340	The actual high finite-indicator pairing bound	1	1494	incremental	ZIP / Proof notes
341	The actual high-plus-error remainder and exact finite decomposition	2	1496	incremental	ZIP / Proof notes
342	The conditional source high-plus-error power estimate in the uniform paper range	1	1497	incremental	ZIP / Proof notes
343	Absorbing the conditional approximation error into the proved high power	2	1499	incremental	ZIP / Proof notes
344	Uniform actual source low and remainder operator bounds	1	1500	incremental	ZIP / Proof notes
345	A genuine countable high maximum under a proved summable cutoff budget	2	1502	incremental	ZIP / Proof notes
346	The conditional actual prime-square annular restricted pairing from proved low and high bounds	1	1503	full	ZIP / Proof notes

Continue from checkpoint 347

Recover `PrimeSquares_Checkpoint_346.zip`, verify its manifest, and read `STATUS.md`, `docs/CONTINUE_HERE.md`, this report and the relevant proof notes. The next checkpoint is **347**.

The priorities are the sharp fixed-cutoff high maximal and multifrequency estimates and their exact role in Proposition 4.19 and Theorem 1.1. The actual original real-cutoff low source identity, full-conductor low bounds, single-scale sub-half-power high norm and conditional restricted pairing are now proved in their displayed ranges and should not be repeated. Reuse the countable square-energy machinery only after establishing its actual source energy premise; a uniform single-scale norm alone does not prove a fixed-cutoff countable maximum. Keep every raw deep analytic hypothesis explicit and preserve the exact manuscript/source comparison.

`Checkpoints_347_546_Report.md`

Prime Squares: checkpoints 347–546

This batch contains **200 separately verified and preserved proof checkpoints**, adding **245 named supporting theorems in 200 modules**. The final project has **613 modules, 615 Lean files and 1748 named supporting theorems**. Supporting theorem counts do not count completed manuscript statements. **Theorem 1.1 and Proposition 4.19 remain open.**

The main results are a genuine fixed-cutoff collar maximum, including the original exceptional source term with a conductor gain and changing conductor classes; exact source decompositions locating the remaining large-denominator high terms; and a proved time-and-denominator-prefix Rademacher–Menshov principle. The sharp full high maximal estimate remains open.

What is now proved

The actual complementary narrow cutoff is analyzed at fixed real Q. For dyadic scales $N_j = 2^j M$ with natural $M \geq 2$, the first nonzero collar scale is defined exactly. Every earlier term vanishes. After onset,

the actual annular oscillatory factor has geometric decay, uniformly for arbitrary $\beta_j \in [0,1]$. Consequently its complete scalar square energy is summable and quantitatively bounded. This is a fixed-Q argument; it does not require summable varying cutoffs.

The centered continuous symbol retains the actual broad source cutoff. Exact compact circle periodization accounts for every term of its three-term window. For a single center the finite circle square energy is at most $(24q/(\pi Q))^2$. Coefficients of norm at most one may change arbitrarily with scale. The countable maximum is constructed as a genuine nonnegative real ℓ^2 function, with pointwise domination and a proved norm bound.

On each denominator block the broad source cutoffs have at most four active canonical rational centers, including the small denominators. The rational-block circle energy constant is $96bU/(\pi Q)$ for denominators at most U and coefficients at most b. A weighted finite Cauchy–Schwarz argument sums denominator-block energies without an extra block-count factor. The complete collar family with coefficient envelope $Cq^{-\delta}$ has square energy bounded by $(K_\delta Q^{-\delta})^2$ and a genuine countable ℓ^2 maximum.

The actual principal and twisted unit square Gauss sums discharge that arithmetic envelope for every $0 \le \delta < 1/2$. Ordinary and twisted characters, scale-dependent selectors and scale-dependent $\beta \in [0,1]$ are included. Exact identities identify the original ordinary high terms with this collar below their actual narrow cutoff. The original globally truncated source low family also has its proved countable $\ell^1 \to \ell^\infty$ maximum throughout the displayed uniform source logarithmic range.

For a positive exceptional conductor q_0, put

$$B_0 = \text{conductorMajorant}(q_0), \qquad W = Qq_0/B_0.$$

The exceptional collar uses W, which need not equal Q. Its canonical frequencies are exactly $0 \le r < 1$, $\operatorname{den}(r) \le \lfloor W \rfloor$ and $q_0 \mid \operatorname{den}(r)$. If $W < 1$, the set is empty. For $\eta, \varepsilon \ge 0$ with $\eta + \varepsilon < 1/2$, actual Gauss decay and conductor divisibility give

$$q^{-(\eta+\varepsilon)} \le q_0^{-\varepsilon} q^{-\eta}.$$

The already proved uniform bound on B_0/q_0 then yields actual exceptional collar energy at most

$$(H_{\eta,\varepsilon} Q^{-\eta} q_0^{-\varepsilon})^2,$$

and the corresponding genuine countable maximum. This covers every $Q > 0$. The source global-radius restriction is identified exactly with a bounded coefficient mask, so no artificial requirement $W \le R$ remains. Inactive exceptional data and conductors beyond the source radius give exactly zero exceptional collar.

For changing characters, the displayed numerical classification premise is

$$q_0(i) \ge 2^i.$$

This is a **hypothesis on the family of conductors**, not a formalized Landau–Page zero theorem. With exactly one class selected at each dyadic time, finite times are partitioned by the selected class. The squared class constants sum geometrically. There is no assumption that the selected class is monotone or stays active on a common tail.

Combining the ordinary and exceptional contributions proves a genuine maximum for the **signed original source collar** for every $0 \le \eta < 1/2$:

$$\left\| \sup_j |C_{N_j,Q} f| \right\|_2 \le K_\eta Q^{-\eta} \|f\|_2.$$

Here $Q \ge 1$, $M \ge 2$, the source radii satisfy $Q \le R_j$, the active flags and character data can vary, $\beta \in [0,1]$, and the conductor-class condition above is retained. The Lean witness equals the pointwise countable supremum and dominates every individual output. Finite-indicator pairings have the corresponding $K_\eta Q^{-\eta}\sqrt{|F|}\sqrt{|G|}$ bound.

The continuous source model uses the already formalized broad-times-narrow low cutoff. Its identification with the printed cutoff-difference decomposition retains the earlier source geometry and exact low-symbol bridge. The extended symbol estimates at smaller nonnegative radii, where allowed by their Lean signatures, do not assert that printed-source identification.

This controls the collar contribution only. It does not establish the sharp full high bound $Q^{-1/2}\exp((2+o(1))\sqrt{\log Q}/\log\log Q)$, the remaining large-denominator maximum, or the headline theorem.

Prefix theorem and proposed proof repair

A checked two-denominator example shows that a source-radius cutoff in one denominator block need not be one common time tail: denominators 2 and 3 at radii 2 and 3 have activation rows $(1,0)$ and $(1,1)$. Each individual denominator does have a tail under increasing radii. The full block is a denominator prefix. Exact finite identities prove the natural/real cutoff floors and the rational grouping by denominator, including the genuine circle major-arc model and exceptional denominators $q_0 d$.

A finite binary tree argument now proves the Rademacher–Menshov prefix step. Every scalar prefix square is bounded by $(h+1)$ times the actual binary-tree energy. The associated square-root envelopes are constructed in real ℓ^2, with exact spatial-total and norm identities. Uniform interval norm budgets give a logarithmic $(h+1)$ prefix norm cost. The orthogonal ℓ^2 specialization is fully discharged. A second version dominates **every time and every denominator prefix simultaneously**, from explicitly displayed interval maximal majorants and their norm budgets.

The paper's underlying multifrequency interval estimate, its transference and its full source application are still open. They are not introduced as axioms or silently replaced by the prefix conclusion.

The proposed repair to the proof of Proposition 6.21 replaces the common-tail step by the manuscript's existing Corollary 6.19 (`c:orlicz-prefix`). The ordinary factor becomes $(1+s)^2$, and the exceptional large-denominator factor becomes $(1+\log(q_\chi d))^2$. These add only fixed logarithmic powers. The exact sharp-exponent absorption is proved in Lean: for every $\epsilon > 0$ and fixed natural n, uniformly for $Q > 0$ and $\log Q \geq 2$,

$$Q^{-1/2}(1+\log Q)^n \exp\left(2\frac{\sqrt{\log Q}}{\log\log Q}\right) \leq D_{n,\epsilon} Q^{-1/2}\exp\left((2+\epsilon)\frac{\sqrt{\log Q}}{\log\log Q}\right).$$

The replacement changes a proof justification, not the stated conclusion. It is saved separately as `proposed_revisions/Replacement_6_21_Denominator_Prefix.tex`; the frozen manuscript and both previously accepted revisions are unchanged. It remains a proposed local repair, not a certificate of the entire proposition.

Exact frontier and remaining work

The original source high symbol is identified exactly as

$$\text{High} = \text{signed collar} + \text{ordinary large denominators} - \text{exceptional large denominators}.$$

The final continuous-symbol, genuine ℓ^2 operator and finite-complex-input convolution identities retain this split. The annular von Mangoldt square convolution is exactly low plus signed collar plus the large-denominator remainder plus the genuine major-arc error. The low input lies in ℓ^1 and the other inputs in ℓ^2, with the finite coordinate agreement checked. No bound on a remaining term is inserted into an identity.

The next substantial task is the actual source multifrequency interval estimate and its large-denominator maximal application, followed by the sharp full high/source endpoint assembly in Proposition 4.19 and Theorem 1.1. The printed minor estimate in Lemma 6.5 also remains open in its stronger form; the earlier alternative estimate and conditional global approximation remain available.

The ten previously recorded numbered results retain their exact scopes: 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, abstract Theorem 4.1 and abstract Corollary 4.15. Lemma 3.21 still has its two explicit PNT limits. No separate PNT formalization was integrated. Raw progression-prefix, Ren and Ghosh approximation inputs remain explicit hypotheses where consumed. This batch introduces no new deep analytic axiom and no full-paper or publication-correctness claim.

Verification, saving and recovery

The final checkpoint passes a fresh **4220-job full project build of all 613 modules and the root**, with no warnings or errors. All **1748 named supporting theorems** pass a complete fresh transitive axiom audit allowing only `propext`, `Classical.choice` and `Quot.sound`. All **615 Lean files** pass the complete source scan: no placeholders, added mathematical axioms, unchecked evaluation or unsafe code. Root imports and theorem-audit coverage are exact and unique.

Baseline checkpoint 346 was authenticated before the first production edit. Every new checkpoint compiled its own actual new module and root and ran the full source and transitive-axiom scans. Incremental reuse authenticated all earlier source and compiled-artifact hashes. Fresh full project builds were performed at **371, 396, 421, 446, 471, 496, 521 and 546**. Aggregate scratch checks were used for development and do not replace those individual production builds. A read-only copy of authenticated old compilation objects was used solely for prototype checks during subsequent clean builds; it is excluded from the deliverable and does not affect production verification.

Lean 4.34.1, commit `5045d0056413266e57c625dcd7c365b10e377c52`, and Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612` remain pinned. No compiler/kernel modification or dependency upgrade was made. The existing narrow executable-path compatibility shim remains solely for this container runtime. GitHub CI is configured but has not run on GitHub.

Each archive is atomically packaged after CRC and complete SHA256-manifest checks. Every checkpoint is preserved before the next production mutation. Saved archive readbacks are checked for exact bytes, SHA256, every ZIP CRC, every manifest entry and the Lean source hashes in their authenticated build receipts. Readback completion is recorded separately after the final save; the extended checkpoints-347–600 saved-files index covers this entire batch and the subsequent continuation.

One save exception is explicit: checkpoint **352** passed every Lean/build/audit/package check, but its original standalone save response timed out and remained unconfirmed. That unknown write was not retried. Its exact archive and notes were instead preserved inside the distinct, confirmed checkpoint **353** archive, independently downloaded and fully integrity-verified there, and included again under `recovery/checkpoint352/` in final checkpoint **546**. Thus all 200 checkpoints are separately recoverable; the count must not be described as 200 confirmed standalone archive uploads. Checkpoint 372's later response timeout was reconciled by authoritative saved-file records and exact verified downloads of all three committed files, without repeating the write. Checkpoint 484's interrupted save was likewise reconciled by exact verified downloads of all three committed files, including progress version 482, with no repeated write. Checkpoint 416's discovery failure occurred before any upload-preparation request was issued; the guarded helper invocation then succeeded. The records distinguish these cases without fabricating success receipts.

No new numerical experiment is substituted for the new analytic proofs. Earlier exact numerical reports and their script hashes are retained; the new finite activation counterexample itself is proved in Lean.

All two hundred checkpoints

Each archive is a complete source project snapshot with proof notes, frozen source, current verification logs and continuation instructions. Starting at 485, redundant historical build receipts are hash-indexed instead of duplicated; their original archives and the working project retain them. Current build receipts, audits and every proof source remain included. The final ZIP contains all 200 new modules and expositions. Toolchain and dependency caches are reproducible and excluded.

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
347	Exact dyadic annular scale arithmetic	3	1506	incremental	ZIP / Proof notes
348	Exact dyadic growth and positivity of the narrow collar widths	2	1508	incremental	ZIP / Proof notes
349	The actual complementary narrow cutoff and its exact range	3	1511	incremental	ZIP / Proof notes
350	The strict first possible frequency for a nonzero collar	2	1513	incremental	ZIP / Proof notes
351	The actual pure annular collar factor and its norm attenuation	3	1516	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
352	The exact first nonzero dyadic collar and the no-onset case	2	1518	incremental	ZIP / Proof notes; exact archive recovered inside 353 and 546; original standalone write unconfirmed
353	Frequency onset and exact zero collar outputs before it	2	1520	incremental	ZIP / Proof notes
354	Uniform geometric decay after the actual first collar scale	1	1521	incremental	ZIP / Proof notes
355	The actual collar square decay after onset	1	1522	incremental	ZIP / Proof notes
356	The summable actual collar tail and its quantitative total	2	1524	incremental	ZIP / Proof notes
357	Summability of the complete actual dyadic collar square sequence	1	1525	incremental	ZIP / Proof notes
358	A fixed-cutoff bound on the complete actual collar square energy	1	1526	incremental	ZIP / Proof notes
359	Uniform finite actual collar energies with a usable squared constant	2	1528	incremental	ZIP / Proof notes
360	Actual scalar collar energy with arbitrary bounded scale coefficients	1	1529	incremental	ZIP / Proof notes
361	Continuity of the actual pure collar masks and factors	2	1531	incremental	ZIP / Proof notes
362	The actual continuous centered collar symbol with broad cutoff	1	1532	incremental	ZIP / Proof notes
363	Exact broad-support and compact-support vanishing for collars	2	1534	incremental	ZIP / Proof notes
364	The finite square energy of actual centered continuous collars	1	1535	incremental	ZIP / Proof notes
365	The genuine shifted continuous circle collar symbol	1	1536	incremental	ZIP / Proof notes
366	Exact circle collar windows and common frequency representatives	2	1538	incremental	ZIP / Proof notes
367	The exact square-norm cost of the three-term period window	2	1540	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
368	A proved uniform finite square-energy bound on the actual circle collars	1	1541	incremental	ZIP / Proof notes
369	The genuine fixed-cutoff countable l2 maximum for one collar frequency	1	1542	incremental	ZIP / Proof notes
370	Actual circle collar energy with bounded complex scale coefficients	1	1543	incremental	ZIP / Proof notes
371	The genuine weighted fixed-cutoff circle collar maximum	1	1544	full	ZIP / Proof notes
372	The exact square-norm cost of a finite active frequency sum	1	1545	incremental	ZIP / Proof notes
373	Uniform square-energy summation over the actual active rational centers	1	1546	incremental	ZIP / Proof notes
374	The actual broad cutoff is a complex norm contraction	2	1548	incremental	ZIP / Proof notes
375	The actual centered rational collar block square-energy estimate	1	1549	incremental	ZIP / Proof notes
376	The genuine finite rational collar circle block and its exact real window	1	1550	incremental	ZIP / Proof notes
377	The actual circle rational-block square-energy estimate	1	1551	incremental	ZIP / Proof notes
378	A genuine fixed-Q countable l2 maximum for a complete rational block	1	1552	incremental	ZIP / Proof notes
379	The fixed-Q rational collar maximum under additional frequency attenuation	1	1553	incremental	ZIP / Proof notes
380	Weighted finite Cauchy-Schwarz without a block-count loss	1	1554	incremental	ZIP / Proof notes
381	Assembly of distinct finite square energies with their correct total constant	1	1555	incremental	ZIP / Proof notes
382	Exact finite-block assembly of genuine circle square energies and countable maxima	2	1557	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
383	Actual denominator lower bounds and coefficient decay on dyadic blocks	2	1559	incremental	ZIP / Proof notes
384	Exact normalization of the denominator-block collar constant	1	1560	incremental	ZIP / Proof notes
385	The actual rational collar block energy under proved arithmetic power decay	1	1561	incremental	ZIP / Proof notes
386	A quantitative finite budget for the growing dyadic collar series	2	1563	incremental	ZIP / Proof notes
387	The exact real-cutoff dyadic endpoint budget	2	1565	incremental	ZIP / Proof notes
388	The full denominator-block geometric series in terms of the real cutoff	1	1566	incremental	ZIP / Proof notes
389	The actual summed collar block constant decays as Q to minus delta	3	1569	incremental	ZIP / Proof notes
390	Exact dyadic partition of the complete continuous rational collar model	1	1570	incremental	ZIP / Proof notes
391	The complete fixed-Q rational collar square-energy bound	1	1571	incremental	ZIP / Proof notes
392	The genuine full fixed-Q rational collar countable maximum	1	1572	incremental	ZIP / Proof notes
393	The actual rational principal and twisted Gauss coefficients with uniform decay	2	1574	incremental	ZIP / Proof notes
394	An unconditional complete ordinary fixed-Q collar maximum	1	1575	incremental	ZIP / Proof notes
395	An unconditional complete twisted fixed-Q collar maximum	1	1576	incremental	ZIP / Proof notes
396	The exact original ordinary centered high collar identity	1	1577	full	ZIP / Proof notes
397	The exact original ordinary circle high collar identity	1	1578	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
398	The full exact ordinary low-denominator high-symbol correspondence	1	1579	incremental	ZIP / Proof notes
399	The original ordinary low-denominator high family has a fixed-Q countable l2 maximum	1	1580	incremental	ZIP / Proof notes
400	A genuine circle maximal witness with explicit pointwise domination	1	1581	incremental	ZIP / Proof notes
401	The finite test-set bound for a nonnegative genuine real l2 majorant	2	1583	incremental	ZIP / Proof notes
402	The actual finite-indicator maximal pairing supplied by proved circle energy	1	1584	incremental	ZIP / Proof notes
403	The actual ordinary Gauss collar family has uniform square energy	1	1585	incremental	ZIP / Proof notes
404	The actual twisted Gauss collar family has uniform square energy	1	1586	incremental	ZIP / Proof notes
405	The actual ordinary collar finite-indicator maximal pairing	1	1587	incremental	ZIP / Proof notes
406	The exact ordinary collar correspondence inside the globally truncated source	1	1588	incremental	ZIP / Proof notes
407	The original globally truncated ordinary collar family has the proved fixed-Q maximum	1	1589	incremental	ZIP / Proof notes
408	The actual paper logarithmic cutoff budget persists on every dyadic tail	2	1591	incremental	ZIP / Proof notes
409	A uniform fourth-power source radius budget on the whole dyadic tail	1	1592	incremental	ZIP / Proof notes
410	The full source eighth-power geometry uniformly on every dyadic tail	1	1593	incremental	ZIP / Proof notes
411	The actual full-conductor low data on a source dyadic tail	1	1594	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
412	The genuine all-input countable maximum for actual dyadic source low data	1	1595	incremental	ZIP / Proof notes
413	The original source low countable maximum in the actual uniform paper range	1	1596	incremental	ZIP / Proof notes
414	The exact denominator split of the original full high circle model	1	1597	incremental	ZIP / Proof notes
415	Exact vanishing of the ordinary low symbol beyond the cutoff	2	1599	incremental	ZIP / Proof notes
416	The exact original circle contribution beyond the ordinary cutoff	2	1601	incremental	ZIP / Proof notes
417	The actual complete ordinary large-denominator high circle sum	1	1602	incremental	ZIP / Proof notes
418	The exact full ordinary source high symbol is collar plus large denominators	1	1603	incremental	ZIP / Proof notes
419	The actual annular prime-square symbol split into four exact ordinary pieces	1	1604	incremental	ZIP / Proof notes
420	The actual annular l2 operator split into the same four ordinary pieces	1	1605	incremental	ZIP / Proof notes
421	The exact finite-complex-input prime-square four-piece decomposition	1	1606	full	ZIP / Proof notes
422	The positive effective exceptional collar cutoff	2	1608	incremental	ZIP / Proof notes
423	The exact finite conductor-multiple collar frequency set	1	1609	incremental	ZIP / Proof notes
424	Exact vanishing of the exceptional collar set below one	1	1610	incremental	ZIP / Proof notes
425	Extracting a conductor power from denominator decay	2	1612	incremental	ZIP / Proof notes
426	A conductor-sensitive actual twisted Gauss envelope	1	1613	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
427	Exact homogeneity of the complete collar constant	1	1614	incremental	ZIP / Proof notes
428	Exact conversion of the exceptional cutoff power	1	1615	incremental	ZIP / Proof notes
429	Uniform control of the effective-cutoff decay	1	1616	incremental	ZIP / Proof notes
430	The refined finite exceptional collar square energy	1	1617	incremental	ZIP / Proof notes
431	The complete refined collar constant in the original cutoff	1	1618	incremental	ZIP / Proof notes
432	Uniform conductor-sensitive twisted collar energy above the cutoff threshold	1	1619	incremental	ZIP / Proof notes
433	The conductor-sensitive energy for every positive cutoff	1	1620	incremental	ZIP / Proof notes
434	The genuine conductor-sensitive exceptional collar maximum	1	1621	incremental	ZIP / Proof notes
435	The actual exceptional collar finite-indicator maximal pairing	1	1622	incremental	ZIP / Proof notes
436	The separate genuine centered exceptional high component	1	1623	incremental	ZIP / Proof notes
437	The exact twisted Gauss and effective-mask formula	1	1624	incremental	ZIP / Proof notes
438	The actual exceptional high collar correspondence	1	1625	incremental	ZIP / Proof notes
439	The genuine exceptional high circle term and full high split	1	1626	incremental	ZIP / Proof notes
440	The exact exceptional high circle collar identity	1	1627	incremental	ZIP / Proof notes
441	Exact ordinary-minus-exceptional splitting of the finite high model	1	1628	incremental	ZIP / Proof notes
442	The full exact exceptional collar sum	1	1629	incremental	ZIP / Proof notes
443	The exact exceptional high term above its effective cutoff	1	1630	incremental	ZIP / Proof notes
444	The exact large-denominator exceptional circle contribution	1	1631	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
445	Exact frequency filtering by scalar collar coefficients	1	1632	incremental	ZIP / Proof notes
446	Exact extension of a varying finite collar subset	1	1633	full	ZIP / Proof notes
447	The exact source exceptional frequency-set identity	1	1634	incremental	ZIP / Proof notes
448	Exact removal of nonconductor frequencies from the exceptional collar	1	1635	incremental	ZIP / Proof notes
449	The original exceptional high collar equals the refined model	1	1636	incremental	ZIP / Proof notes
450	Uniform energy of the original exceptional source collar	1	1637	incremental	ZIP / Proof notes
451	The original exceptional source collar countable maximum	1	1638	incremental	ZIP / Proof notes
452	The exact exceptional high split at its own cutoff	1	1639	incremental	ZIP / Proof notes
453	The remaining complete exceptional large-denominator model	1	1640	incremental	ZIP / Proof notes
454	An exact four-piece decomposition of the complete source high model	1	1641	incremental	ZIP / Proof notes
455	The exact annular prime-square symbol decomposition with exceptional data	1	1642	incremental	ZIP / Proof notes
456	The actual high operator has the exact four-piece split	1	1643	incremental	ZIP / Proof notes
457	Exact disappearance of inactive exceptional collar data	1	1644	incremental	ZIP / Proof notes
458	Exact finite partition of dyadic times by character class	1	1645	incremental	ZIP / Proof notes
459	Square-energy assembly with one active class at each time	1	1646	incremental	ZIP / Proof notes
460	Positive conductors from dyadic class lower bounds	2	1648	incremental	ZIP / Proof notes
461	The geometric decay supplied by separated conductor classes	1	1649	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
462	Summable squared conductor gains	1	1650	incremental	ZIP / Proof notes
463	A class-count-independent finite conductor budget	1	1651	incremental	ZIP / Proof notes
464	The positive class energy constant and its exact square	2	1653	incremental	ZIP / Proof notes
465	The complete uniform weighted class square budget	1	1654	incremental	ZIP / Proof notes
466	Square energy for genuine circle symbols with changing conductor classes	1	1655	incremental	ZIP / Proof notes
467	A genuine countable maximum under changing conductor classes	1	1656	incremental	ZIP / Proof notes
468	The actual dyadic conductor collar family with a short interface	1	1657	incremental	ZIP / Proof notes
469	Uniform energy for collars with a changing conductor class	1	1658	incremental	ZIP / Proof notes
470	The actual fixed-Q maximum with changing conductor classes	1	1659	incremental	ZIP / Proof notes
471	The exact conductor collar set for arbitrary nonnegative source radius	1	1660	full	ZIP / Proof notes
472	The exact exceptional source collar with arbitrary source clipping	1	1661	incremental	ZIP / Proof notes
473	The original clipped exceptional collar has uniform refined energy	1	1662	incremental	ZIP / Proof notes
474	The actual source exceptional collar maximum without containment assumptions	1	1663	incremental	ZIP / Proof notes
475	Uniform energy of the actual source collar with changing conductor classes	1	1664	incremental	ZIP / Proof notes
476	The actual source exceptional collar maximum with changing conductors	1	1665	incremental	ZIP / Proof notes
477	The original changing-character source collar finite-indicator pairing	1	1666	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
478	The literal source denominator activation and prefix identity	2	1668	incremental	ZIP / Proof notes
479	Individual denominator activation is a time tail under increasing radii	1	1669	incremental	ZIP / Proof notes
480	A checked obstruction to treating a denominator prefix as one common tail	1	1670	incremental	ZIP / Proof notes
481	Exact contiguous finite sums for denominator prefixes	3	1673	incremental	ZIP / Proof notes
482	The exact binary split of a denominator prefix	2	1675	incremental	ZIP / Proof notes
483	The actual finite binary interval square energy	1	1676	incremental	ZIP / Proof notes
484	The full interval and both child energies are controlled by the tree	2	1678	incremental	ZIP / Proof notes
485	The weighted two-term square estimate for prefix induction	1	1679	incremental	ZIP / Proof notes
486	A proved logarithmic binary-prefix square bound	1	1680	incremental	ZIP / Proof notes
487	A simultaneous pointwise envelope for every finite prefix	1	1681	incremental	ZIP / Proof notes
488	The actual real l2 magnitude of a complex l2 function	1	1682	incremental	ZIP / Proof notes
489	Actual finite interval sums in complex l2	2	1684	incremental	ZIP / Proof notes
490	The finite tree of actual l2 interval norms	1	1685	incremental	ZIP / Proof notes
491	Summability in space of the finite binary tree energy	1	1686	incremental	ZIP / Proof notes
492	Exact Fubini identity for the finite binary tree	1	1687	incremental	ZIP / Proof notes
493	The genuine square-root binary tree envelope in real l2	1	1688	incremental	ZIP / Proof notes
494	The exact norm of the binary tree envelope	2	1690	incremental	ZIP / Proof notes
495	A genuine real l2 majorant of all denominator prefixes	2	1692	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
496	The quantitative finite-prefix l2 estimate from actual tree energy	1	1693	full	ZIP / Proof notes
497	Exact nonnegative interval budgets for the prefix tree	2	1695	incremental	ZIP / Proof notes
498	The logarithmic tree-energy budget from uniform interval bounds	1	1696	incremental	ZIP / Proof notes
499	A proved finite Rademacher–Menshov majorant with exact logarithmic cost	1	1697	incremental	ZIP / Proof notes
500	Exact Pythagoras for finite orthogonal sums	1	1698	incremental	ZIP / Proof notes
501	The actual orthogonal l2 interval energy identity	1	1699	incremental	ZIP / Proof notes
502	A fully proved orthogonal l2 prefix maximal estimate	1	1700	incremental	ZIP / Proof notes
503	Finite test-set control of every prefix by the proved majorant	1	1701	incremental	ZIP / Proof notes
504	Exact summability and total of real l2 squares	2	1703	incremental	ZIP / Proof notes
505	The actual binary tree energy of interval majorants	1	1704	incremental	ZIP / Proof notes
506	The finite norm budget for interval maximal majorants	1	1705	incremental	ZIP / Proof notes
507	Spatial summability of the interval-maximal tree	1	1706	incremental	ZIP / Proof notes
508	Exact norm-energy total for the interval-maximal tree	1	1707	incremental	ZIP / Proof notes
509	The genuine time-uniform binary interval envelope	1	1708	incremental	ZIP / Proof notes
510	The exact norm of the time-uniform interval envelope	2	1710	incremental	ZIP / Proof notes
511	Domination of every scalar binary tree by interval majorants	1	1711	incremental	ZIP / Proof notes
512	One pointwise envelope for both time and denominator prefixes	1	1712	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
513	The genuine l2 majorant of the double maximum	2	1714	incremental	ZIP / Proof notes
514	The exact logarithmic energy budget for interval maxima	1	1715	incremental	ZIP / Proof notes
515	A proved time-uniform Rademacher–Menshov prefix theorem	1	1716	incremental	ZIP / Proof notes
516	Finite pairings for the simultaneous time and prefix envelope	1	1717	incremental	ZIP / Proof notes
517	The exact finite natural denominator-prefix set	1	1718	incremental	ZIP / Proof notes
518	The exact real-cutoff denominator-prefix set	1	1719	incremental	ZIP / Proof notes
519	The actual real source truncation is a contiguous scalar prefix	1	1720	incremental	ZIP / Proof notes
520	Exact finite grouping of rational frequencies by denominator	1	1721	incremental	ZIP / Proof notes
521	Exact denominator-prefix grouping of a finite rational family	1	1722	full	ZIP / Proof notes
522	Exact real-cutoff prefix grouping of rational families	1	1723	incremental	ZIP / Proof notes
523	The actual circle major-arc truncation is a denominator prefix	1	1724	incremental	ZIP / Proof notes
524	A fixed canonical universe for a source denominator block	1	1725	incremental	ZIP / Proof notes
525	The actual globally truncated circle block has an exact prefix expansion	1	1726	incremental	ZIP / Proof notes
526	The exact conductor-multiple prefix cutoff	1	1727	incremental	ZIP / Proof notes
527	The actual sharp high exponent weight and its positive comparison	2	1729	incremental	ZIP / Proof notes
528	Uniform absorption of fixed powers into the sharp high exponent	1	1730	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
529	The extra fixed logarithmic prefix loss is harmless for the leading exponent	1	1731	incremental	ZIP / Proof notes
530	Exact preservation of the leading coefficient after a prefix loss	1	1732	incremental	ZIP / Proof notes
531	The prefix-loss estimate in the original source cutoff Q	1	1733	incremental	ZIP / Proof notes
532	The corrected logarithmic loss preserves the sharp leading two	1	1734	incremental	ZIP / Proof notes
533	The correct square-energy cost of subtracting two high components	1	1735	incremental	ZIP / Proof notes
534	Uniform finite circle energy for the difference of two families	1	1736	incremental	ZIP / Proof notes
535	The exact combined ordinary and exceptional source collar	1	1737	incremental	ZIP / Proof notes
536	Uniform energy of the complete signed source collar	1	1738	incremental	ZIP / Proof notes
537	The true countable l2 maximum of the complete source collar	1	1739	incremental	ZIP / Proof notes
538	The complete source collar finite-indicator maximal pairing	1	1740	incremental	ZIP / Proof notes
539	The signed source collar reduces exactly to the ordinary collar when inactive	1	1741	incremental	ZIP / Proof notes
540	Exact disappearance of the exceptional collar beyond the source radius	1	1742	incremental	ZIP / Proof notes
541	The exact full source high split into proved collars and the remaining large terms	1	1743	incremental	ZIP / Proof notes
542	The exact source high operator frontier	1	1744	incremental	ZIP / Proof notes
543	The exact finite-input annular prime-square decomposition at the proof frontier	1	1745	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added theorems	Project total	Build	Archive and exposition
544	The exact annular prime-square symbol at the final proof frontier	1	1746	incremental	ZIP / Proof notes
545	The unconditional pointwise triangle bound for the exact source decomposition	1	1747	incremental	ZIP / Proof notes
546	The completed subhalf-power source collar maximum with no auxiliary epsilon parameter	1	1748	full	ZIP / Proof notes

Continue from checkpoint 547

Recover `PrimeSquares_Checkpoint_546.zip`, verify `verification/SHA256SUMS.txt`, and retain the separately saved files index alongside it. The final save receipt is issued after packaging. On a restart without the recorded compilation cache, perform a fresh full build to initialize new artifact receipts. Read `STATUS.md`, `docs/CONTINUE_HERE.md`, this report and the relevant proof notes. The next checkpoint is **547**. Reuse the completed fixed-Q ordinary and exceptional collar energy, actual changing-character collar maximum, exact source-radius prefix identities and proved abstract prefix theorem. Prove the actual multifrequency interval premise and large-denominator maximum next; do not repeat completed collar work or infer a full high maximum from a uniform single-scale norm.

`Checkpoints_347_600_Report.md`

Prime Squares: continuation through checkpoint 600

Checkpoint 484 is reconciled, and the project has continued through checkpoint 600. All three original checkpoint-484 saves were confirmed by authoritative records and exact verified downloads; its write was not repeated. Checkpoints 485–600 then each compiled their actual new proof module and root, passed the complete source and transitive-axiom audits, and were separately preserved before the next production mutation.

The current project has **667 project modules, 669 Lean files and 1807 named supporting theorems**. Across checkpoints 347–600 this is **254 proof units and 304 added named supporting theorems**; the new extension 547–600 adds **54 modules and 59 supporting theorems**. These counts are not counts of completed manuscript statements. **Theorem 1.1, Proposition 4.19 and the sharp full high maximal estimate remain open.**

What this continuation establishes

The earlier checkpoints complete the fixed-Q ordinary and exceptional collar estimates, including the signed original source collar, its conductor gain and changing-character conductor classes. They also prove the abstract time-and-prefix Rademacher–Menshov principle and exact source denominator-prefix identities. The separately saved checkpoints-347–546 report explains those results and the proposed local repair to Proposition 6.21.

The extension 547–600 connects that prefix principle to the actual circle and integer Fourier multipliers:

- Circle multiplication and the integer multiplier action respect symbol products. Almost-everywhere circle representatives are handled explicitly.
- Pointwise disjoint continuous symbols give exactly orthogonal integer l2 outputs, even for different inputs. Finite sums and intervals have exact Pythagorean square-energy identities.
- Arbitrary complex weights preserve spectral disjointness; bounded weights have the displayed square-budget comparison.
- Finite lists are padded by exact zeros to a binary capacity. Padding changes neither their genuine prefixes nor their square budget. The depth is Nat.clog 2 L, proved at most Nat.log2 L+1, including L=0 and L=1.

- For a finite disjoint family m_i, one nonnegative real l2 function dominates every nonnegative real denominator cutoff. Its norm is at most (Nat.log2 L+2) times the square root of the exact output energy. If the symbols are bounded by A, the energy is at most A^2 times the input square norm, with no cardinality loss.
- An exact localizer identity m_i=m_i*p_i and disjoint localizers imply all off-diagonal products vanish and every interval symbol localizes to the sum of its own localizers. No unjustified assertion that a smooth localizer is an idempotent projection is made.
- Explicit interval majorants and localized norm budgets yield a true majorant over every time and every denominator prefix. A separate sufficient criterion constructs those interval witnesses from finite time square-energy bounds on the actual circle symbols. **Those frequency-energy bounds are hypotheses of the criterion, not established properties of the paper's full high family.**
- Rational symbols are grouped exactly by denominator. The original high and exceptional high source symbols retain their actual formulas in these identities.
- A band Q<den<=R is exactly the difference of prefixes at R and min(Q,R). Thus the band is correctly empty when Q>R, including conductor-dependent exceptional cutoffs. A single prefix witness controls every such difference with norm cost two.
- The interval-to-prefix bridge reaches actual capped source high operators with varying scales, characters, conductors and beta. A fixed cap M applies to the entire source set only when 0<=R<=M; this cap restriction is proved and retained. No maximum over all unbounded caps is inferred from it.
- The already proved signed collar maximum is reused in an exact reduction of full original source high control to a supplied majorant for its remaining large-denominator remainder. This exposes the missing input instead of certifying it. A finite test-set pairing lemma also assembles the genuine low, collar, remainder and major-arc error contributions with each bound displayed separately.

A verified guard against an invalid maximal inference

The final four checkpoints prove a general obstruction using the existing actual Fourier convention. For j>=0, the continuous symbol exp(-2*pi*i*j*theta) defines an integer multiplier of operator norm at most one and sends the unit mass at zero exactly to the unit mass at j. Any real l2 majorant for the first N such outputs must have square norm at least N. Hence no real l2 function dominates the countable family on this one fixed input.

This is **a counterexample to the general inference from uniform individual operator bounds to a countable maximal bound**. It is not a counterexample to the manuscript theorem. It explains why the separate multifrequency/maximal argument is essential and cannot be replaced by the single-scale high estimate already proved.

Exact mathematical frontier

Proved: the signed source collar has a genuine fixed-Q countable l2 maximum K_eta Q^(-eta) for every 0<=eta<1/2, with Q>=1, dyadic starting scale M>=2, source radii R_j>=Q, beta in [0,1], and the explicit conductor-class condition q0(i)>=2^i. Character data and active flags may vary. That numerical conductor classification is a hypothesis; a Landau–Page zero classification is not claimed. Earlier exact source geometry and the broad-times-narrow low-symbol bridge remain required when identifying the continuous source model with the printed decomposition.

Proved with explicit hypotheses: the abstract and actual-multiplier interval-to-prefix principles, disjoint/localized spectral criteria, finite-cap source high bridge, and final assembly lemmas. Their source interval-energy or interval-majorant premises remain visible. These conditional results are not counted as completed numbered source conclusions.

Still required: the actual source multifrequency interval estimate, its needed transference and beta range, the remaining large-denominator high maximum, and the sharp full high/source endpoint assembly leading to Proposition 4.19 and Theorem 1.1. The stronger printed Lemma 6.5 estimate also remains open in the scope previously recorded. Single-scale high power decay and the completed collars do not close those gaps.

The ten previously recorded numbered results retain their scopes: 3.1, 3.11, 3.12, accepted revised 3.14 and Corollary 3.16, original 3.20, 3.24, 3.27, abstract Theorem 4.1 and abstract Corollary 4.15. Lemma 3.21 retains its two explicit PNT limits. A separate PNT formalization was not integrated. Published Ren/Ghosh estimates and the progression-prefix input remain explicit raw hypotheses where used. This continuation adds no deep theorem axiom.

The proposed proof-step repair in proposed_revisions/Replacement_6_21_Denominator_Prefix.tex remains separate: source-radius activation is a denominator prefix, so the prefix corollary supplies the extra fixed logarithmic factor. The checked polynomial absorption preserves the 2+epsilon sharp expo-

nent in its displayed range. This is a proposed local repair, not a full certificate of Proposition 6.21. The frozen manuscript and both accepted revisions are unchanged; no further mandatory source replacement is established here.

Verification and preserved files

The final checkpoint passes a fresh **4274-job full project build of all 667 modules and the root**, with no warnings or errors. All **1807 named supporting theorems** pass a fresh complete transitive axiom audit permitting only propext, Classical.choice and Quot.sound. All **669 Lean files** pass the source scan. There are no proof placeholders, added mathematical axioms, unsafe code or unchecked evaluation. Exact root-import coverage, unique named-theorem coverage, frozen manuscript hashes and the existing numerical-report script hashes are checked.

Every individual checkpoint compiles its new actual module and root, with remote project artifact-cache reuse disabled. Unchanged proof-source and compiled-artifact hashes are authenticated before incremental reuse. In this extended sequence full builds occur at 371, 396, 421, 446, 471, 496, 521, 546, 571, 596 and 600. Aggregate development checks are useful preparation and are not individual production completion certificates.

Lean 4.34.1, commit 5045d0056413266e57c625dcd7c365b10e377c52, and mathlib d13f23b723b8a846827a245b89c10fc7d3f11612 remain pinned. No compiler/kernel alteration or dependency upgrade was made. The narrow existing executable-path compatibility shim is only for this container runtime. Configured GitHub CI has not run on GitHub.

Each ZIP is atomically packaged and checked against every ZIP CRC and every included SHA256 manifest entry. Starting at checkpoint 485, only redundant historical build-receipt copies are omitted from new ZIPs and indexed by exact hash in verification/ARCHIVED_RECEIPTS.json; their original checkpoint archives and the working project retain them. Every proof source, the complete current build receipt and all current audits remain included. This reduces repeated archive size without changing proof verification. Reproducible toolchain and dependency caches are excluded.

Saved archive verification downloads the committed files and compares their byte lengths/SHA256, all ZIP CRCs, archived manifest entries and current build-receipt source hashes. The final readbacks occur after the final archive save. The separately saved index is created only after all 254 checkpoint recoveries have passed those checks; consult that index for completed readback coverage, per-file save identities, final progress version and the exception below. Retain that index alongside the final ZIP, because the final checkpoint's own save receipt is issued after packaging.

One earlier save exception remains explicit: checkpoint 352 passed Lean, audit and package checks, but its original standalone save outcome remained unconfirmed after a timeout. That write was never retried. Its exact standalone ZIP and notes were preserved inside confirmed checkpoint 353, independently recovered and fully checked there, and included again under recovery/checkpoint352 in checkpoints 546 and 600. Thus there are **253 confirmed standalone archive saves plus one exactly recovered checkpoint**, not 254 confirmed standalone uploads. Checkpoint 372 and the interrupted checkpoint 484 were each fully reconciled by exact verified downloads of all three committed files without repeated writes. Checkpoint 416's read-only discovery failure occurred before any upload request and was safely retried at the save step only. A later monitoring-session interruption left checkpoint 590 staged before any output package or upload request. Its new module and root were forced to rebuild, all audits reran, and its first save then succeeded; no source restaging or repeated write occurred.

The early working-draft bundles are restart aids with explicit uncompleted-draft labels. They are not substitutes for the later individually verified checkpoint ZIPs. No numerical sampling substitutes for these new Lean arguments; the finite activation obstruction and the final Fourier-multiplier obstruction themselves are proved in Lean.

Checkpoints 347–600

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
347	Exact dyadic annular scale arithmetic	3	1506	incremental	ZIP / Proof notes
348	Exact dyadic growth and positivity of the narrow collar widths	2	1508	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
349	The actual complementary narrow cutoff and its exact range	3	1511	incremental	ZIP / Proof notes
350	The strict first possible frequency for a nonzero collar	2	1513	incremental	ZIP / Proof notes
351	The actual pure annular collar factor and its norm attenuation	3	1516	incremental	ZIP / Proof notes
352	The exact first nonzero dyadic collar and the no-onset case	2	1518	incremental	ZIP / Proof notes; original standalone save unconfirmed, exact recovery in 353/546/600
353	Frequency onset and exact zero collar outputs before it	2	1520	incremental	ZIP / Proof notes
354	Uniform geometric decay after the actual first collar scale	1	1521	incremental	ZIP / Proof notes
355	The actual collar square decay after onset	1	1522	incremental	ZIP / Proof notes
356	The summable actual collar tail and its quantitative total	2	1524	incremental	ZIP / Proof notes
357	Summability of the complete actual dyadic collar square sequence	1	1525	incremental	ZIP / Proof notes
358	A fixed-cutoff bound on the complete actual collar square energy	1	1526	incremental	ZIP / Proof notes
359	Uniform finite actual collar energies with a usable squared constant	2	1528	incremental	ZIP / Proof notes
360	Actual scalar collar energy with arbitrary bounded scale coefficients	1	1529	incremental	ZIP / Proof notes
361	Continuity of the actual pure collar masks and factors	2	1531	incremental	ZIP / Proof notes
362	The actual continuous centered collar symbol with broad cutoff	1	1532	incremental	ZIP / Proof notes
363	Exact broad-support and compact-support vanishing for collars	2	1534	incremental	ZIP / Proof notes
364	The finite square energy of actual centered continuous collars	1	1535	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
365	The genuine shifted continuous circle collar symbol	1	1536	incremental	ZIP / Proof notes
366	Exact circle collar windows and common frequency representatives	2	1538	incremental	ZIP / Proof notes
367	The exact square-norm cost of the three-term period window	2	1540	incremental	ZIP / Proof notes
368	A proved uniform finite square-energy bound on the actual circle collars	1	1541	incremental	ZIP / Proof notes
369	The genuine fixed-cutoff countable l2 maximum for one collar frequency	1	1542	incremental	ZIP / Proof notes
370	Actual circle collar energy with bounded complex scale coefficients	1	1543	incremental	ZIP / Proof notes
371	The genuine weighted fixed-cutoff circle collar maximum	1	1544	full	ZIP / Proof notes
372	The exact square-norm cost of a finite active frequency sum	1	1545	incremental	ZIP / Proof notes
373	Uniform square-energy summation over the actual active rational centers	1	1546	incremental	ZIP / Proof notes
374	The actual broad cutoff is a complex norm contraction	2	1548	incremental	ZIP / Proof notes
375	The actual centered rational collar block square-energy estimate	1	1549	incremental	ZIP / Proof notes
376	The genuine finite rational collar circle block and its exact real window	1	1550	incremental	ZIP / Proof notes
377	The actual circle rational-block square-energy estimate	1	1551	incremental	ZIP / Proof notes
378	A genuine fixed-Q countable l2 maximum for a complete rational block	1	1552	incremental	ZIP / Proof notes
379	The fixed-Q rational collar maximum under additional frequency attenuation	1	1553	incremental	ZIP / Proof notes
380	Weighted finite Cauchy-Schwarz without a block-count loss	1	1554	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
381	Assembly of distinct finite square energies with their correct total constant	1	1555	incremental	ZIP / Proof notes
382	Exact finite-block assembly of genuine circle square energies and countable maxima	2	1557	incremental	ZIP / Proof notes
383	Actual denominator lower bounds and coefficient decay on dyadic blocks	2	1559	incremental	ZIP / Proof notes
384	Exact normalization of the denominator-block collar constant	1	1560	incremental	ZIP / Proof notes
385	The actual rational collar block energy under proved arithmetic power decay	1	1561	incremental	ZIP / Proof notes
386	A quantitative finite budget for the growing dyadic collar series	2	1563	incremental	ZIP / Proof notes
387	The exact real-cutoff dyadic endpoint budget	2	1565	incremental	ZIP / Proof notes
388	The full denominator-block geometric series in terms of the real cutoff	1	1566	incremental	ZIP / Proof notes
389	The actual summed collar block constant decays as Q to minus delta	3	1569	incremental	ZIP / Proof notes
390	Exact dyadic partition of the complete continuous rational collar model	1	1570	incremental	ZIP / Proof notes
391	The complete fixed-Q rational collar square-energy bound	1	1571	incremental	ZIP / Proof notes
392	The genuine full fixed-Q rational collar countable maximum	1	1572	incremental	ZIP / Proof notes
393	The actual rational principal and twisted Gauss coefficients with uniform decay	2	1574	incremental	ZIP / Proof notes
394	An unconditional complete ordinary fixed-Q collar maximum	1	1575	incremental	ZIP / Proof notes
395	An unconditional complete twisted fixed-Q collar maximum	1	1576	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
396	The exact original ordinary centered high collar identity	1	1577	full	ZIP / Proof notes
397	The exact original ordinary circle high collar identity	1	1578	incremental	ZIP / Proof notes
398	The full exact ordinary low-denominator high-symbol correspondence	1	1579	incremental	ZIP / Proof notes
399	The original ordinary low-denominator high family has a fixed-Q countable l2 maximum	1	1580	incremental	ZIP / Proof notes
400	A genuine circle maximal witness with explicit pointwise domination	1	1581	incremental	ZIP / Proof notes
401	The finite test-set bound for a nonnegative genuine real l2 majorant	2	1583	incremental	ZIP / Proof notes
402	The actual finite-indicator maximal pairing supplied by proved circle energy	1	1584	incremental	ZIP / Proof notes
403	The actual ordinary Gauss collar family has uniform square energy	1	1585	incremental	ZIP / Proof notes
404	The actual twisted Gauss collar family has uniform square energy	1	1586	incremental	ZIP / Proof notes
405	The actual ordinary collar finite-indicator maximal pairing	1	1587	incremental	ZIP / Proof notes
406	The exact ordinary collar correspondence inside the globally truncated source	1	1588	incremental	ZIP / Proof notes
407	The original globally truncated ordinary collar family has the proved fixed-Q maximum	1	1589	incremental	ZIP / Proof notes
408	The actual paper logarithmic cutoff budget persists on every dyadic tail	2	1591	incremental	ZIP / Proof notes
409	A uniform fourth-power source radius budget on the whole dyadic tail	1	1592	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
410	The full source eighth-power geometry uniformly on every dyadic tail	1	1593	incremental	ZIP / Proof notes
411	The actual full-conductor low data on a source dyadic tail	1	1594	incremental	ZIP / Proof notes
412	The genuine all-input countable maximum for actual dyadic source low data	1	1595	incremental	ZIP / Proof notes
413	The original source low countable maximum in the actual uniform paper range	1	1596	incremental	ZIP / Proof notes
414	The exact denominator split of the original full high circle model	1	1597	incremental	ZIP / Proof notes
415	Exact vanishing of the ordinary low symbol beyond the cutoff	2	1599	incremental	ZIP / Proof notes
416	The exact original circle contribution beyond the ordinary cutoff	2	1601	incremental	ZIP / Proof notes
417	The actual complete ordinary large-denominator high circle sum	1	1602	incremental	ZIP / Proof notes
418	The exact full ordinary source high symbol is collar plus large denominators	1	1603	incremental	ZIP / Proof notes
419	The actual annular prime-square symbol split into four exact ordinary pieces	1	1604	incremental	ZIP / Proof notes
420	The actual annular l2 operator split into the same four ordinary pieces	1	1605	incremental	ZIP / Proof notes
421	The exact finite-complex-input prime-square four-piece decomposition	1	1606	full	ZIP / Proof notes
422	The positive effective exceptional collar cutoff	2	1608	incremental	ZIP / Proof notes
423	The exact finite conductor-multiple collar frequency set	1	1609	incremental	ZIP / Proof notes
424	Exact vanishing of the exceptional collar set below one	1	1610	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
425	Extracting a conductor power from denominator decay	2	1612	incremental	ZIP / Proof notes
426	A conductor-sensitive actual twisted Gauss envelope	1	1613	incremental	ZIP / Proof notes
427	Exact homogeneity of the complete collar constant	1	1614	incremental	ZIP / Proof notes
428	Exact conversion of the exceptional cutoff power	1	1615	incremental	ZIP / Proof notes
429	Uniform control of the effective-cutoff decay	1	1616	incremental	ZIP / Proof notes
430	The refined finite exceptional collar square energy	1	1617	incremental	ZIP / Proof notes
431	The complete refined collar constant in the original cutoff	1	1618	incremental	ZIP / Proof notes
432	Uniform conductor-sensitive twisted collar energy above the cutoff threshold	1	1619	incremental	ZIP / Proof notes
433	The conductor-sensitive energy for every positive cutoff	1	1620	incremental	ZIP / Proof notes
434	The genuine conductor-sensitive exceptional collar maximum	1	1621	incremental	ZIP / Proof notes
435	The actual exceptional collar finite-indicator maximal pairing	1	1622	incremental	ZIP / Proof notes
436	The separate genuine centered exceptional high component	1	1623	incremental	ZIP / Proof notes
437	The exact twisted Gauss and effective-mask formula	1	1624	incremental	ZIP / Proof notes
438	The actual exceptional high collar correspondence	1	1625	incremental	ZIP / Proof notes
439	The genuine exceptional high circle term and full high split	1	1626	incremental	ZIP / Proof notes
440	The exact exceptional high circle collar identity	1	1627	incremental	ZIP / Proof notes
441	Exact ordinary-minus-exceptional splitting of the finite high model	1	1628	incremental	ZIP / Proof notes
442	The full exact exceptional collar sum	1	1629	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
443	The exact exceptional high term above its effective cutoff	1	1630	incremental	ZIP / Proof notes
444	The exact large-denominator exceptional circle contribution	1	1631	incremental	ZIP / Proof notes
445	Exact frequency filtering by scalar collar coefficients	1	1632	incremental	ZIP / Proof notes
446	Exact extension of a varying finite collar subset	1	1633	full	ZIP / Proof notes
447	The exact source exceptional frequency-set identity	1	1634	incremental	ZIP / Proof notes
448	Exact removal of nonconductor frequencies from the exceptional collar	1	1635	incremental	ZIP / Proof notes
449	The original exceptional high collar equals the refined model	1	1636	incremental	ZIP / Proof notes
450	Uniform energy of the original exceptional source collar	1	1637	incremental	ZIP / Proof notes
451	The original exceptional source collar countable maximum	1	1638	incremental	ZIP / Proof notes
452	The exact exceptional high split at its own cutoff	1	1639	incremental	ZIP / Proof notes
453	The remaining complete exceptional large-denominator model	1	1640	incremental	ZIP / Proof notes
454	An exact four-piece decomposition of the complete source high model	1	1641	incremental	ZIP / Proof notes
455	The exact annular prime-square symbol decomposition with exceptional data	1	1642	incremental	ZIP / Proof notes
456	The actual high operator has the exact four-piece split	1	1643	incremental	ZIP / Proof notes
457	Exact disappearance of inactive exceptional collar data	1	1644	incremental	ZIP / Proof notes
458	Exact finite partition of dyadic times by character class	1	1645	incremental	ZIP / Proof notes
459	Square-energy assembly with one active class at each time	1	1646	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
460	Positive conductors from dyadic class lower bounds	2	1648	incremental	ZIP / Proof notes
461	The geometric decay supplied by separated conductor classes	1	1649	incremental	ZIP / Proof notes
462	Summable squared conductor gains	1	1650	incremental	ZIP / Proof notes
463	A class-count-independent finite conductor budget	1	1651	incremental	ZIP / Proof notes
464	The positive class energy constant and its exact square	2	1653	incremental	ZIP / Proof notes
465	The complete uniform weighted class square budget	1	1654	incremental	ZIP / Proof notes
466	Square energy for genuine circle symbols with changing conductor classes	1	1655	incremental	ZIP / Proof notes
467	A genuine countable maximum under changing conductor classes	1	1656	incremental	ZIP / Proof notes
468	The actual dyadic conductor collar family with a short interface	1	1657	incremental	ZIP / Proof notes
469	Uniform energy for collars with a changing conductor class	1	1658	incremental	ZIP / Proof notes
470	The actual fixed-Q maximum with changing conductor classes	1	1659	incremental	ZIP / Proof notes
471	The exact conductor collar set for arbitrary nonnegative source radius	1	1660	full	ZIP / Proof notes
472	The exact exceptional source collar with arbitrary source clipping	1	1661	incremental	ZIP / Proof notes
473	The original clipped exceptional collar has uniform refined energy	1	1662	incremental	ZIP / Proof notes
474	The actual source exceptional collar maximum without containment assumptions	1	1663	incremental	ZIP / Proof notes
475	Uniform energy of the actual source collar with changing conductor classes	1	1664	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
476	The actual source exceptional collar maximum with changing conductors	1	1665	incremental	ZIP / Proof notes
477	The original changing-character source collar finite-indicator pairing	1	1666	incremental	ZIP / Proof notes
478	The literal source denominator activation and prefix identity	2	1668	incremental	ZIP / Proof notes
479	Individual denominator activation is a time tail under increasing radii	1	1669	incremental	ZIP / Proof notes
480	A checked obstruction to treating a denominator prefix as one common tail	1	1670	incremental	ZIP / Proof notes
481	Exact contiguous finite sums for denominator prefixes	3	1673	incremental	ZIP / Proof notes
482	The exact binary split of a denominator prefix	2	1675	incremental	ZIP / Proof notes
483	The actual finite binary interval square energy	1	1676	incremental	ZIP / Proof notes
484	The full interval and both child energies are controlled by the tree	2	1678	incremental	ZIP / Proof notes
485	The weighted two-term square estimate for prefix induction	1	1679	incremental	ZIP / Proof notes
486	A proved logarithmic binary-prefix square bound	1	1680	incremental	ZIP / Proof notes
487	A simultaneous pointwise envelope for every finite prefix	1	1681	incremental	ZIP / Proof notes
488	The actual real l2 magnitude of a complex l2 function	1	1682	incremental	ZIP / Proof notes
489	Actual finite interval sums in complex l2	2	1684	incremental	ZIP / Proof notes
490	The finite tree of actual l2 interval norms	1	1685	incremental	ZIP / Proof notes
491	Summability in space of the finite binary tree energy	1	1686	incremental	ZIP / Proof notes
492	Exact Fubini identity for the finite binary tree	1	1687	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
493	The genuine square-root binary tree envelope in real l2	1	1688	incremental	ZIP / Proof notes
494	The exact norm of the binary tree envelope	2	1690	incremental	ZIP / Proof notes
495	A genuine real l2 majorant of all denominator prefixes	2	1692	incremental	ZIP / Proof notes
496	The quantitative finite-prefix l2 estimate from actual tree energy	1	1693	full	ZIP / Proof notes
497	Exact nonnegative interval budgets for the prefix tree	2	1695	incremental	ZIP / Proof notes
498	The logarithmic tree-energy budget from uniform interval bounds	1	1696	incremental	ZIP / Proof notes
499	A proved finite Rademacher–Menshov majorant with exact logarithmic cost	1	1697	incremental	ZIP / Proof notes
500	Exact Pythagoras for finite orthogonal sums	1	1698	incremental	ZIP / Proof notes
501	The actual orthogonal l2 interval energy identity	1	1699	incremental	ZIP / Proof notes
502	A fully proved orthogonal l2 prefix maximal estimate	1	1700	incremental	ZIP / Proof notes
503	Finite test-set control of every prefix by the proved majorant	1	1701	incremental	ZIP / Proof notes
504	Exact summability and total of real l2 squares	2	1703	incremental	ZIP / Proof notes
505	The actual binary tree energy of interval majorants	1	1704	incremental	ZIP / Proof notes
506	The finite norm budget for interval maximal majorants	1	1705	incremental	ZIP / Proof notes
507	Spatial summability of the interval-maximal tree	1	1706	incremental	ZIP / Proof notes
508	Exact norm-energy total for the interval-maximal tree	1	1707	incremental	ZIP / Proof notes
509	The genuine time-uniform binary interval envelope	1	1708	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
510	The exact norm of the time-uniform interval envelope	2	1710	incremental	ZIP / Proof notes
511	Domination of every scalar binary tree by interval majorants	1	1711	incremental	ZIP / Proof notes
512	One pointwise envelope for both time and denominator prefixes	1	1712	incremental	ZIP / Proof notes
513	The genuine l2 majorant of the double maximum	2	1714	incremental	ZIP / Proof notes
514	The exact logarithmic energy budget for interval maxima	1	1715	incremental	ZIP / Proof notes
515	A proved time-uniform Rademacher–Menshov prefix theorem	1	1716	incremental	ZIP / Proof notes
516	Finite pairings for the simultaneous time and prefix envelope	1	1717	incremental	ZIP / Proof notes
517	The exact finite natural denominator-prefix set	1	1718	incremental	ZIP / Proof notes
518	The exact real-cutoff denominator-prefix set	1	1719	incremental	ZIP / Proof notes
519	The actual real source truncation is a contiguous scalar prefix	1	1720	incremental	ZIP / Proof notes
520	Exact finite grouping of rational frequencies by denominator	1	1721	incremental	ZIP / Proof notes
521	Exact denominator-prefix grouping of a finite rational family	1	1722	full	ZIP / Proof notes
522	Exact real-cutoff prefix grouping of rational families	1	1723	incremental	ZIP / Proof notes
523	The actual circle major-arc truncation is a denominator prefix	1	1724	incremental	ZIP / Proof notes
524	A fixed canonical universe for a source denominator block	1	1725	incremental	ZIP / Proof notes
525	The actual globally truncated circle block has an exact prefix expansion	1	1726	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
526	The exact conductor-multiple prefix cutoff	1	1727	incremental	ZIP / Proof notes
527	The actual sharp high exponent weight and its positive comparison	2	1729	incremental	ZIP / Proof notes
528	Uniform absorption of fixed powers into the sharp high exponent	1	1730	incremental	ZIP / Proof notes
529	The extra fixed logarithmic prefix loss is harmless for the leading exponent	1	1731	incremental	ZIP / Proof notes
530	Exact preservation of the leading coefficient after a prefix loss	1	1732	incremental	ZIP / Proof notes
531	The prefix-loss estimate in the original source cutoff Q	1	1733	incremental	ZIP / Proof notes
532	The corrected logarithmic loss preserves the sharp leading two	1	1734	incremental	ZIP / Proof notes
533	The correct square-energy cost of subtracting two high components	1	1735	incremental	ZIP / Proof notes
534	Uniform finite circle energy for the difference of two families	1	1736	incremental	ZIP / Proof notes
535	The exact combined ordinary and exceptional source collar	1	1737	incremental	ZIP / Proof notes
536	Uniform energy of the complete signed source collar	1	1738	incremental	ZIP / Proof notes
537	The true countable l2 maximum of the complete source collar	1	1739	incremental	ZIP / Proof notes
538	The complete source collar finite-indicator maximal pairing	1	1740	incremental	ZIP / Proof notes
539	The signed source collar reduces exactly to the ordinary collar when inactive	1	1741	incremental	ZIP / Proof notes
540	Exact disappearance of the exceptional collar beyond the source radius	1	1742	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
541	The exact full source high split into proved collars and the remaining large terms	1	1743	incremental	ZIP / Proof notes
542	The exact source high operator frontier	1	1744	incremental	ZIP / Proof notes
543	The exact finite-input annular prime-square decomposition at the proof frontier	1	1745	incremental	ZIP / Proof notes
544	The exact annular prime-square symbol at the final proof frontier	1	1746	incremental	ZIP / Proof notes
545	The unconditional pointwise triangle bound for the exact source decomposition	1	1747	incremental	ZIP / Proof notes
546	The completed subhalf-power source collar maximum with no auxiliary epsilon parameter	1	1748	full	ZIP / Proof notes
547	Circle multiplication composes by the product of the actual symbols	1	1749	incremental	ZIP / Proof notes
548	The integer Fourier multiplier action respects symbol products	1	1750	incremental	ZIP / Proof notes
549	Disjoint circle symbols produce orthogonal L2 outputs	1	1751	incremental	ZIP / Proof notes
550	Disjoint symbols give orthogonality on the integers	1	1752	incremental	ZIP / Proof notes
551	Finite disjoint multiplier sums have exact square energy	1	1753	incremental	ZIP / Proof notes
552	Denominator intervals inherit exact disjoint multiplier energy	1	1754	incremental	ZIP / Proof notes
553	Disjoint Fourier multipliers admit a genuine logarithmic prefix majorant	1	1755	incremental	ZIP / Proof notes
554	Actual continuous multiplier symbols can be summed over denominator intervals	1	1756	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
555	Interval symbol sums act as the corresponding finite sums of operators	2	1758	incremental	ZIP / Proof notes
556	Scalar weights preserve spectral disjointness	1	1759	incremental	ZIP / Proof notes
557	Weighted disjoint multipliers retain the logarithmic prefix bound	1	1760	incremental	ZIP / Proof notes
558	Coefficient envelopes control the exact interval square budget	1	1761	incremental	ZIP / Proof notes
559	Finite symbol families extend by exact zero padding	2	1763	incremental	ZIP / Proof notes
560	Zero padding preserves the exact finite spectral disjointness premise	1	1764	incremental	ZIP / Proof notes
561	Every genuine finite prefix is unchanged by binary padding	1	1765	incremental	ZIP / Proof notes
562	Binary padding does not enlarge the multiplier square budget	1	1766	incremental	ZIP / Proof notes
563	Finite disjoint symbol families have a true prefix majorant	1	1767	incremental	ZIP / Proof notes
564	Real source cutoffs equal exact continuous-symbol prefixes	1	1768	incremental	ZIP / Proof notes
565	One l2 majorant controls every nonnegative real denominator cutoff	1	1769	incremental	ZIP / Proof notes
566	Ceiling logarithms give a valid finite binary capacity	2	1771	incremental	ZIP / Proof notes
567	The binary depth has the expected logarithmic size	2	1773	incremental	ZIP / Proof notes
568	The real-cutoff maximum has an explicit logarithmic energy bound	1	1774	incremental	ZIP / Proof notes
569	A disjoint finite symbol sum has no cardinality loss	1	1775	incremental	ZIP / Proof notes
570	Bounded disjoint symbols give a finite input-energy estimate	1	1776	incremental	ZIP / Proof notes
571	Bounded disjoint symbols have a logarithmic l2 cutoff maximum	1	1777	full	ZIP / Proof notes

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
572	A localized symbol acts only on its localized input	2	1779	incremental	ZIP / Proof notes
573	Disjoint localizers annihilate all off-diagonal symbol products	1	1780	incremental	ZIP / Proof notes
574	Finite symbol intervals localize to the sum of their own localizers	1	1781	incremental	ZIP / Proof notes
575	Localized interval input norms equal their sum of individual energies	1	1782	incremental	ZIP / Proof notes
576	Interval maximal budgets transfer to all times and prefixes	1	1783	incremental	ZIP / Proof notes
577	Frequency square energy produces actual interval maximal majorants	1	1784	incremental	ZIP / Proof notes
578	Uniform interval energy data yield a whole family of interval majorants	1	1785	incremental	ZIP / Proof notes
579	A frequency-energy criterion controls all times and denominator prefixes	1	1786	incremental	ZIP / Proof notes
580	The actual time-prefix maximum controls finite test-set pairings	1	1787	incremental	ZIP / Proof notes
581	Actual rational circle symbols have exact denominator blocks	1	1788	incremental	ZIP / Proof notes
582	Rational circle sums with real denominator cutoffs equal exact prefixes	1	1789	incremental	ZIP / Proof notes
583	Every finite source symbol admits the exact denominator-prefix action	1	1790	incremental	ZIP / Proof notes
584	The actual high source symbol has the exact prefix representation	1	1791	incremental	ZIP / Proof notes
585	The actual exceptional high symbol has the exact prefix representation	1	1792	incremental	ZIP / Proof notes
586	A real denominator band is an exact difference of prefixes	1	1793	incremental	ZIP / Proof notes

Checkpoint	Mathematical unit	Added supporting theorems	Project total	Build	Archive and exposition
587	A single prefix witness controls every difference of two prefixes	1	1794	incremental	ZIP / Proof notes
588	The actual ordinary large-denominator block is a difference of prefixes	1	1795	incremental	ZIP / Proof notes
589	The exceptional large-denominator block has the same exact band structure	1	1796	incremental	ZIP / Proof notes
590	A source radius bounded by a fixed cap loses no rational centers	1	1797	incremental	ZIP / Proof notes
591	Actual source-radius maxima follow from explicit interval majorants	1	1798	incremental	ZIP / Proof notes
592	The same interval data control every source denominator band	1	1799	incremental	ZIP / Proof notes
593	Explicit interval data now reach the genuine capped source high family	1	1800	incremental	ZIP / Proof notes
594	Prefix witnesses control the actual signed large-denominator remainder	1	1801	incremental	ZIP / Proof notes
595	The proved source collar maximum reduces full high control to its large remainder	1	1802	incremental	ZIP / Proof notes
596	The exact annular decomposition yields a finite pairing assembly	1	1803	full	ZIP / Proof notes
597	The actual Fourier character multiplier shifts a unit mass exactly	1	1804	incremental	ZIP / Proof notes
598	A majorant for N distinct unit masses must have l2 norm at least sqrt N	1	1805	incremental	ZIP / Proof notes
599	Countably many distinct unit masses have no l2 pointwise majorant	1	1806	incremental	ZIP / Proof notes
600	Uniform single-scale multiplier bounds do not imply a countable maximal bound	1	1807	full	ZIP / Proof notes

Restart and next proof task

Use PrimeSquares_Checkpoint_600.zip and Saved_Files_Index_347_600.json. Verify verification/SHA256SUMS.txt, then read STATUS.md, docs/CONTINUE_HERE.md and this report. Restarting without the recorded compilation cache requires a fresh full build to create new artifact receipts; do not present an old cache receipt as evidence for newly generated binaries. The next checkpoint is **601**.

Focus the next mathematical milestone on the actual source multifrequency interval estimate and the large-denominator high maximal bound. Reuse the completed collars, prefix machinery and exact source identities. Keep raw deep external hypotheses distinct from own-paper targets and verify any published external theorem's precise primary-source statement before importing it as an allowed hypothesis. Theorem 1.1 remains an explicit eventual goal.

Earlier plans and completion notes

archive/PAPER_COMPLETION_Pre1265.md

Completion of the reviewed corrected paper

The internal formalization is complete within the agreed conditional scope. The complete inventory has 54 numbered theorem, proposition, lemma and corollary targets and four definitions. There are 22 claims proved under ordinary source hypotheses, 31 complete implications from explicit independent inputs, one quoted external theorem, and no open internal numbered endpoint. The four definitions are formalized. These counts measure statement coverage and are not proof-completion percentages.

Main result	Current checked evidence	Scope
Theorem 1.1	theorem_1_1, theorem_1_1_restricted, theorem_1_1_strong_real	Complete conditional improving estimates for the actual averages and source normalization
Theorem 1.3	theorem_1_3, theorem_1_3_endpoint, theorem_1_3_open_pq	Complete conditional sparse domination, endpoint and open triangle
Theorem 1.5(i)	theorem_1_5_i	Complete conditional persistent universal L1 badness
Theorem 1.5(ii)	theorem_1_5_ii_closed_signed_maximal	Complete conditional near-L1 ergodic maximal estimate; signed maximal input proved internally
Theorem 7.2	theorem_7_2_closed_signed_maximal	Complete conditional three-family quantitative Orlicz theorem
Corollaries 7.6 and 7.46	corollary_7_6_closed_signed_maximal, corollary_7_46_closed_signed_maximal	Complete conditional Lp maximal, transfer and convergence conclusions
Quantitative Goldbach theorem	goldbach_section9_headline_complete	Complete conditional estimate at all sufficiently large real cutoffs, with arithmetic criterion and smooth seed proved internally

The current continuation proves the general finite signed-sum maximal theorem at constant 6. Its binary bound holds for every exponent at least one, including all finite lengths and extended norm bounds. Exact Boolean-sign averaging and the Lp triangle inequality replace its former independent hypothesis. The public Section 7 and near-L1 exports consume this proof. The prior primitive two-square criterion, W(alpha) integrability for every alpha>1 and concrete smooth seed remain proved and integrated.

The paper-wide audit independently checks every declared theorem-like environment type, all source labels and statement hashes, all four definitions and all current evidence anchors. Lean prints the exact types of those declarations. The audit verifies removal of the signed maximal hypothesis from the four public exports and of the primitive criterion from the current Section 9 arithmetic/headline exports. Source scans, authenticated compilation receipts and complete transitive axiom audits cover the proof files. The only allowed foundational axioms are propext, Classical.choice and Quot.sound. Numerical experiments remain supplementary and are not Lean proofs; unchanged verified probes are preserved without redundant reruns.

Independent general inputs remain unproved, including the PNT/progression, Ren/Ghosh, multifrequency, IW/sampling, general real-variable and ergodic, Weil, and general sieve/Mertens/effective mean interfaces. Their exact scopes and still-unproved formulation bridges are listed in INDEPENDENT_INPUTS_CURRENT.md and the proof registry. This is a partial formalization of the mathematics used by the paper, with complete own-paper deductions in the agreed scope. An axiom audit does not establish the truth of explicit input hypotheses. The quoted Granville–Kurlberg theorem is external and optional for the badness conclusion.

The completeness target is the selected full source plus the explicitly supplied corrected Section 11 presieving statement. The weaker original presieving statement is not claimed. The user is applying manuscript edits; application is pending verification against a revised upload. No original source file is changed and no additional replacement passage is needed from this continuation. Stronger unnumbered assertions in alternative proof routes retain their separate documented scopes.

See blueprint/Paper_Proof_Status.json and blueprint/Paper_Proof_Status.csv for exact per-statement evidence. verification/paper-completion-current.json records the final coverage checks; verification/paper-declaration-types-current.log records the actual Lean types. Read docs/Remaining_Work_Plan.md for

source synchronization and the separately optional independent-input projects.

archive/Pre1264_SECTIONS_3_TO_8_REVIEW.md

Sections3–8 review at checkpoint1114

No reviewed current-paper statement in Sections3–8 remains partial or open within the agreed conditional scope. The explicitly quoted general external theorems remain unproved. This is not an unconditional formalization of all mathematics used by the paper.

The two previously incomplete standalone statements are now complete conditional implications. The corrected prime-square input is source_corrected_prime_square_input: one fixed-level genuine optional Page datum per N precedes Q, the full q<=R^(11/10) major range is proved, and the complementary R^(-1/4)*(log N)^d minor bound and required uniform thresholds are proved. The general fixed-level uniform progression theorem and raw Ren/Ghosh estimates remain unproved independent inputs.

The complete quantitative IW proposition is proposition_quantitative_IW_full: one common construction at every real Q>=2^k, the literal parameters, all rational fraction points, rough full-Lp multiplier norm, both full-space sampling directions, unique interpolation and arbitrary finite-dimensional normed complex Shannon targets for every 1<=p<=infinity. General Tao construction, full IW/sampling statements and their explicitly recorded general bridges remain unproved. Read FULL_QUANTITATIVE_IW_COMPLETION.md and FULL_IW_EXTERNAL_BOUNDARY.md for the exact boundary.

All earlier sparse, multifrequency, fixed-measurable-symbol denominator, maximal, Orlicz and Section8 scopes retain their reviewed statuses and their own hypotheses. The corrected cyclic Fourier model uses only denominators dividing P; its declared correction was rebuilt and audited before final completion credit.

The exact latest upload is preserved. It has all eight closed-L corrections and the corrected minor statement, but its approximation proof still uses the removed stronger minor estimate. The selected copy restores the already supplied approximation passage, the accepted dyadic factor-four application and preferred upper-cutoff wording, retaining all upload labels. The complete upload-to-selected patch is saved; the user's local manuscript is not edited remotely. See SOURCE_UPDATE_1107.md.

The next mathematical work is the five mapped Section10/11 circle-method, sieve, main-term and normalized integral estimates. Their own conclusions are internal proof obligations. The Section9 arithmetic/final-count deductions are already checked; their use of the classical primitive two-square criterion remains explicitly conditional. The headline Goldbach theorem remains open.

Source-only revision1 after checkpoint1114 supplies one complete approximation proof with explicit major and complementary minor cases, and removes a stray editing instruction in the earlier saved TeX. Every numbered statement, label, Lean source and compiled artifact is unchanged. See APPROXIMATION_PROOF_REVISION_1114_1.md.

archive/Pre1264_SECTION_11_PLAN.md

Remaining Section11 work after checkpoint1230

Completed in the agreed conditional scope: fixed-sieve comparison, local densities, corrected presieving, full lower-sieve application, uniform actual main-term size, arbitrarily large common cutoff and finite lower exception deductions. The main-term exponent A0/2+6 is independent of C. No new deep input was introduced in this continuation.

Four substantive source claims remain:

1. l:wumatomakisum: weighted upper main-term comparison, including smoothing/integration and the normalized constant.
2. p:sieveneedupper8: full upper application over the actual multiplier set.
3. l:numericalcomparison: strict normalized comparison with the upper sieve constant.
4. t:repmotthm: final Goldbach assembly.

The original weaker presieving statement remains separately open pending manuscript synchronization; its corrected conditional variant is proved. The user is editing the compact replacements. The manuscript is unchanged. Minor A/A0 notation fixes are saved in blueprint/Section11_LowerBound_Notation_Fixes.tex.

The ledger has19 proved,29 complete conditional implications,1 quoted external theorem and5 open claims, plus4 definitions. No checkpoint or supporting-theorem percentage is claimed. Read SECTION_11_LOWER_BOUND_COMPLETION.md and EXTERNAL_SIEVE_INPUTS.md.

archive/Pre1264_SECTION_11_REMAINING_WORK.md

Remaining Section11 work after checkpoint1230

Completed in the agreed conditional scope: fixed-sieve comparison, local densities, corrected presieving, full lower-sieve application, uniform actual main-term size, arbitrarily large common cutoff and finite lower exception deductions. The main-term exponent A0/2+6 is independent of C. No new deep input was introduced in this continuation.

Four substantive source claims remain:

1. l:wumatomakisum: weighted upper main-term comparison, including smoothing/integration and the normalized constant.
2. p:sieveneedupper8: full upper application over the actual multiplier set.
3. l:numericalcomparison: strict normalized comparison with the upper sieve constant.
4. t:repmotthm: final Goldbach assembly.

The original weaker presieving statement remains separately open pending manuscript synchronization; its corrected conditional variant is proved. The user is editing the compact replacements. The manuscript is unchanged. Minor A/A0 notation fixes are saved in blueprint/Section11_LowerBound_Notation_Fixes.tex.

The ledger has19 proved,29 complete conditional implications,1 quoted external theorem and5 open claims, plus4 definitions. No checkpoint or supporting-theorem percentage is claimed. Read SECTION_11_LOWER_BOUND_COMPLETION.md and EXTERNAL_SIEVE_INPUTS.md.

archive/Pre1264_SECTION_7_SIGNED_MAXIMAL_BOUNDARY.md

General finite signed-sum maximal boundary

Checkpoint 904 records `RawFiniteSignedMaximalInput C`. This is a general finite-sequence consequence of Rademacher-Menshov and scalar Khintchine, uniform for 1<=p<=2. It ranges over arbitrary complex sequences u_j(x) on integer counting measure, arbitrary finite length J and arbitrary extended norm bound B. If every complex signed sum with |sigma_j|<=1 has actual Lp membership and norm at most B, the maximum of all partial sums through J has actual Lp membership and norm at most C log(2+J) B. No Fourier symbol, arithmetic data, cutoff, annular kernel, IW construction or current-paper maximum occurs in its fields.

Its general existence is not proved in Lean and it is not a Lean axiom. The boundary includes the deterministic dyadic decomposition, scalar Khintchine, Fubini and the Lp triangle inequality. A primary source for the decomposition is Mirek, Stein and Zorin-Kranich, *A bootstrapping approach to jump inequalities and their applications*, Lemma 2.5, equation (2.7), https://arxiv.org/pdf/1808.09048 . Scalar Khintchine and its function-valued Fubini consequence are also recorded in Tao's *Lecture notes 4 for 247A*, Corollary 5.8, https://www.math.ucla.edu/~tao/247a.1.06f/notes4.pdf .

For clarity, the quantitative consequence is obtained by padding to a power of two and grouping increments in dyadic intervals at each level. Choosing one random sign per disjoint interval produces a permitted signed sum of the original u_j. Khintchine's lower bound at exponent 1, followed by monotonicity of probability Lp norms, bounds the square function uniformly for 1<=p<=2. The deterministic inequality and the Lp triangle inequality then give the number of levels, which is bounded by a universal multiple of log(2+J). This general argument remains in the explicit unformalized input, rather than being counted as a Lean proof.

The paper-specific signed sums are proved for the actual IW annular operators, with exact identities and K*r*C_(r,k) bounds. Their instantiation of this general theorem, and adding the checked initial term, are separate checked deductions. The entire low maximum, its narrow large-scale sampling argument, and the restricted maximal estimate are never external inputs.

archive/Remaining_Work_Plan_Pre1263.md

Remaining Section11 work after checkpoint1230

Completed in the agreed conditional scope: fixed-sieve comparison, local densities, corrected presieving, full lower-sieve application, uniform actual main-term size, arbitrarily large common cutoff and finite lower exception deductions. The main-term exponent A0/2+6 is independent of C. No new deep input was introduced in this continuation.

Four substantive source claims remain:

1. l:wumatomakisum: weighted upper main-term comparison, including smoothing/integration and the normalized constant.
2. p:sieveneedupper8: full upper application over the actual multiplier set.
3. l:numericalcomparison: strict normalized comparison with the upper sieve constant.
4. t:repmotthm: final Goldbach assembly.

The original weaker presieving statement remains separately open pending manuscript synchronization; its corrected conditional variant is proved. The user is editing the compact replacements. The manuscript is unchanged. Minor A/A0 notation fixes are saved in blueprint/Section11_LowerBound_Notation_Fixes.tex.

The ledger has19 proved,29 complete conditional implications,1 quoted external theorem and5 open claims, plus4 definitions. No checkpoint or supporting-theorem percentage is claimed. Read SECTION_11_LOWER_BOUND_COMPLETION.md and EXTERNAL_SIEVE_INPUTS.md.

archive/Remaining_Work_Plan_PreCompletion.md

Remaining Section11 work after checkpoint1230

Completed in the agreed conditional scope: fixed-sieve comparison, local densities, corrected presieving, full lower-sieve application, uniform actual main-term size, arbitrarily large common cutoff and finite lower exception deductions. The main-term exponent A0/2+6 is independent of C. No new deep input was introduced in this continuation.

Four substantive source claims remain:

1. l:wumatomakisum: weighted upper main-term comparison, including smoothing/integration and the normalized constant.
2. p:sieveneedupper8: full upper application over the actual multiplier set.
3. l:numericalcomparison: strict normalized comparison with the upper sieve constant.
4. t:repmotthm: final Goldbach assembly.

The original weaker presieving statement remains separately open pending manuscript synchronization; its corrected conditional variant is proved. The user is editing the compact replacements. The manuscript is unchanged. Minor A/A0 notation fixes are saved in blueprint/Section11_LowerBound_Notation_Fixes.tex.

The ledger has19 proved,29 complete conditional implications,1 quoted external theorem and5 open claims, plus4 definitions. No checkpoint or supporting-theorem percentage is claimed. Read SECTION_11_LOWER_BOUND_COMPLETION.md and EXTERNAL_SIEVE_INPUTS.md.

legacy/README.md

These files belong to the initial 96-page snapshot, not the corrected Draft5. Use ../Current_Statement_Inventory.csv, ../Cohen_Identity.md and ../Current_Roadmap.md. The historical PDF is not a status report for the current formalization.

Original LaTeX source fragments

Section11_LowerBound_Notation_Fixes.tex

```
% Minor notation fixes only; not applied to the selected manuscript.
% In e:H1elementary10, replace
%   \gg_A N(\log N)^{-A/2-2}
% with
\gg_{A_0} N(\log N)^{-A_0/2-2}

% In the proof of l:maintermsize8, replace
%   Taking \(C\) sufficiently large in terms of \(A\)
% with
Taking \(C\) sufficiently large in terms of \(A_0\)
```

Section11_Presieving_Replacements.tex

```
% Proposed replacements for l:presieving. The selected manuscript is unchanged.
% 1. Replace the passage beginning "Let \(2\leq R<z\)" and ending
% with the displayed condition \log D_0/\log R\longrightarrow\infty by:

Assume \(X\geq0\) and \(g(1)=1\).
Fix constants \(C,A>0\), put
\[
 R=(\log N)^C,\qquad
 D_0=\exp((\log\log N)^3),\qquad D_1=D/D_0,
\]
and work with sufficiently large \(N\). Assume
\[
 R+1\leq z\leq D_1\leq N^A,\qquad D_0<D.
\]
Write
\[
 \mathfrak P_0=\{p\in\mathfrak P:p\leq R\},\qquad
 \mathfrak P_1=\{p\in\mathfrak P:R<p<z\}.
\]
Suppose that there are fixed constants
\(\kappa_0,\kappa>0\) and \(K_0\geq2\), uniform in the sequence and
its density function, such that for every \(2\leq w<y\leq z\),
\begin{align}\label{e:tworangesievedimension}
 \prod_{\substack{w\leq p<y\\p\in\mathfrak P_0}}
 (1-g(p))^{-1}
 &\leq
 \left(1+\frac{K_0}{\log w}\right)
 \left(\frac{\log y}{\log w}\right)^{\kappa_0},\notag\\
 \prod_{\substack{w\leq p<y\\p\in\mathfrak P_1}}
 (1-g(p))^{-1}
 &\leq
 \left(1+\frac{K_0}{\log w}\right)
 \left(\frac{\log y}{\log w}\right)^\kappa.
\end{align}
All asymptotic assertions below are uniform when these constants are fixed.

% 2. In the first display of the proof, replace the definition of s_0 by:

\[
 s_0=\frac{\log D_0}{\log(R+1)}.
\]

% In the accompanying application of the preliminary coefficient theorem,
% use sifting limit R+1 on the prime set P_0. It includes exactly all primes
% p<=R, also when R itself is prime. The outer sifting limit remains z.
% Replace the subsequent large-s_0 justification by the following passage.

Apply the general Rosser--Iwaniec coefficient theorem to
\(\mathfrak P_0\), level \(D_0\), and sifting limit \(R+1\).
Its coefficient estimates and the decay of the sieve functions and error
majorant give constants \(c,C'>0\), depending only on the fixed sieve
parameters, such that
\[
 \left|\sum_e\lambda_{0,e}^{\pm}g(e)-V_0(R)\right|
 \leq C'V_0(R)e^{-c s_0}
\]
for sufficiently large \(N\). Since
\[
 s_0=\frac{(\log\log N)^3}{\log((\log N)^C+1)}
 \gg_C(\log\log N)^2,
\]
the relative preliminary error is smaller than every fixed negative
power of \(\log N\). The condition \(z\leq D_1\) places the outer
sieve in its stated range \(s\geq1\), and \(z\leq N^A\) permits this
preliminary error to absorb the logarithmic losses from the outer
coefficient sum. The signed coefficient construction in the remainder
```

```
of the proof is unchanged.

% 3. Strengthen the Mertens justification preceding l:presieving by adding
% the following passage. A coarse implied-constant estimate by itself
% does not supply the refined dimension hypothesis above.

We use the uniform interval forms of Mertens' theorem: for a suitable
constant \(K\) and every \(2\leq w<y\),
\begin{align*}
 \prod_{\substack{w\leq p<y\\p\equiv3\; (\mathrm{mod}\;4)}}
 \left(1-\frac1{p-1}\right)^{-1}
 &\leq
 \left(1+\frac K{\log w}\right)
 \left(\frac{\log y}{\log w}\right)^{1/2},\\
 \prod_{\substack{w\leq p<y\\p>2}}
 \left(1-\frac1{p-1}\right)^{-1}
 &\leq
 \left(1+\frac K{\log w}\right)
 \frac{\log y}{\log w}.
\end{align*}
Restricting these products to the relevant outer prime sets only
decreases them. For the preliminary prime sets, the local bounds
\(g(p)\leq2/(p-3)\) for \(p\geq7\), together with the bound
\(g(5)\leq1/2\) when that prime occurs, give the refined dimension
bound with \(\kappa_0=2\) and a uniform constant. Thus only the outer
dimensions \(1/2\) and \(1\) determine the final sieve functions.
```

legacy/Prime_Squares_Initial_Blueprint.tex

```
\documentclass[11pt]{article}
\usepackage[T1]{fontenc}
\usepackage{lmodern,amsmath,amssymb,amsthm,mathtools}
\usepackage[margin=0.86in]{geometry}
\usepackage{booktabs,longtable,array,enumitem,xcolor,fancyhdr,hyperref,xurl}
\hypersetup{colorlinks=true,linkcolor=blue!45!black,urlcolor=blue!45!black,
pdftitle={Prime Squares: Initial Formalization Blueprint and Checked Finite Core},
pdfauthor={Prepared for Hamed Mousavi}}
\definecolor{ink}{HTML}{16324F}
\definecolor{muted}{HTML}{526477}
\pagestyle{fancy}\fancyhf{}
\fancyhead[L]{\small\color{muted}Prime squares: initial formalization blueprint}
\fancyhead[R]{\small 20 October 2026}
\fancyfoot[C]{\small\thepage}
\setlength{\headheight}{14pt}
\setlength{\tabcolsep}{5pt}
\setlength{\parindent}{0pt}\setlength{\parskip}{6pt}
\setlist{nosep,leftmargin=1.5em}
\newcommand{\R}{\mathbb R}\newcommand{\Z}{\mathbb Z}\newcommand{\N}{\mathbb N}
\newcommand{\T}{\mathbb T}\newcommand{\PP}{\mathbb P}
\newcommand{\one}{\mathbf 1}\newcommand{\abs}[1]{\left|#1\right|}
\newcommand{\norm}[1]{\left\|#1\right\|}
\newcommand{\lean}[1]{\nolinkurl{#1}}
\newcommand{\status}[1]{\textbf{Status:} #1.}
\newtheorem{lemma}{Lemma}[section]
\newtheorem{proposition}[lemma]{Proposition}
\newtheorem{definition}[lemma]{Definition}
\newcommand{\page}{\newpage}
\begin{document}
\begin{center}
{\LARGE\bfseries\color{ink}Prime Squares}\\[6pt]
{\Large Initial Formalization Blueprint}\\[4pt]
{\large and a checked finite core}\\[18pt]
For Hamed Mousavi\\
\emph{On Equidistribution of Prime Squares and Its Applications}\\[10pt]
20 October 2026
\end{center}
\vspace{12pt}
\textbf{What has been done.} This package contains a first dependency blueprint for the
attached 96-page manuscript and a small Lean project. The project formalizes finite
exceptional-set inequalities, elementary arithmetic used in the paper, and positivity
of the elementary bracket in the last sieve comparison. These are useful components
of a larger development; none is a formalization of a headline theorem of the paper.

\textbf{What this document does not claim.} It is not an expanded proof of every lemma,
and it is not a certificate for the whole paper. In particular, the circle-method
approximation, high--low estimates, stopping-time proof, Ionescu--Wainger argument,
ergodic divergence argument and sieve estimates remain to be encoded and checked.
The word \emph{target} below always denotes unfinished work.

\textbf{Source lock.} Numbering and page references refer only to the uploaded file
\lean{Prime_Squares (32)(1).pdf}. Its SHA-256 is recorded in
\lean{PROVENANCE.json}. A revised manuscript must be compared with this snapshot
before the paper-to-Lean map is reused.

\textbf{Reproducibility.} Lean is pinned to 4.34.1. Mathlib is pinned to commit
\lean{d13f23b723b8a846827a245b89c10fc7d3f11612}. The source archive includes a lockfile,
the Lean modules, build instructions and scripts for checking proof dependencies.
The verification logs, rather than this prose, record what was actually run.
```

\textbf{Reading route.} Sections 1--3 define the scope and targets. Sections 4--5 give complete mathematical explanations of the checked components. Sections 6--11 map the unfinished arguments. Sections 12--14 record imports, audit points and the next inputs needed. The appendix inventories numbered statements in the attached PDF.

\page
\tableofcontents

\page
\section{The trust boundary}
There are three different objects in a formalization project.
\begin{enumerate}
\item A \textbf{blueprint} specifies mathematical objects, statements, dependencies and proof steps. It is written for humans and can itself contain mistakes.
\item A \textbf{Lean proof} is an actual term accepted by Lean's kernel. Elaborating a statement containing a placeholder is not the same as proving it.
\item A \textbf{conditional formalization} proves a result from explicitly declared external statements. Its mathematical scope depends on the accuracy and consistency of those statements as well as on the translation of the final theorem.
\end{enumerate}

For the present starter, no extra analytic-number-theory axioms are declared. Theorems in the starter have explicit hypotheses for the errors and main terms they use. Those hypotheses are not assertions that the required analytic estimates have already been established for prime squares.

For the completed project it is reasonable to import deep published results. Each import should have one stable declaration, a precise source, the full range of parameters, and a statement of constant dependence. The project should continue to prove the new arguments of this paper. Assuming its own high--low estimate or its final sieve comparison would leave that part of the paper unverified.

An axiom audit must examine the \emph{transitive} dependencies of each headline theorem, not just count explicit occurrences of \lean{axiom}. Standard Lean/Mathlib dependencies can include \lean{propext}, \lean{Classical.choice} and \lean{Quot.sound}. An unfinished proof introduces \lean{sorryAx}. The starter's scripts reject unapproved dependencies and scan the project sources for proof placeholders and new axioms.

\subsection*{What the Hua example establishes}
The supplied Hua blueprint contains both a current-paper development and substantial material from an earlier, broader draft. Its total length is therefore not a useful measure of how long the prime-squares blueprint should be.

At the repository snapshot inspected for this request, the trusted external declarations are \lean{Hua.siegel_walfisz} and \lean{Hua.hua_theorem10}. The second is an exponential-sum estimate, rather than the statement that sufficiently large admissible integers are sums of five prime squares. A static scan of its default library closure passed. That scan does not replace an independent rebuild or an audit of the meanings of its axioms and headline statements. This starter does not import that repository.

\page
\section{Objects and final statements to encode}
\status{Specifications only; the headline statements are not yet Lean declarations}

Use natural numbers for prime parameters and integer indices for discrete functions. Define, for \(N\ge2\),
\[
 A_{N,\Lambda}f(x)=\frac1N\sum_{1\le n\le N}\Lambda(n)f(x-n^2),\quad
 A_{N,\square}f(x)=\frac1N\sum_{1\le n\le N}f(x-n^2),
\]
\[
 A_{N,\PP^2}f(x)=\frac1{\pi(N)}\sum_{p\le N}f(x-p^2).
\]
The first and third operators differ in both weights and normalization. In the divergence theorem the parameter counts the first \(N\) primes instead of bounding their size. These conventions must not be silently identified.

Put \(L(t)=2+|\log t|\), \(L_2(t)=2+\log(2+|\log t|)\), and
\[
 \Phi_c(t)=\exp\!\left(c\frac{L(t)^{1/2}}{L_2(t)}\right)\quad(t>0).
\]
In bilinear forms define \(t^{2/3}\Phi_c(t)\) to be zero at \(t=0\).
For \(t\ge0\), define
\[
 \Psi_C(t)=\int_0^t
 \exp\!\left(C\sqrt{\log(e+u)\log(e+\log(e+u))}\right)\,du.
\]
The square root in this last expression covers the \emph{product} of the logarithms.

\textbf{Theorem 1.1.} For every \(\varepsilon>0\), there is \(C_\varepsilon\) such that the displayed restricted improving inequality holds for every allowed \(N,F,G\) with coefficient \(4\sqrt3/9+\varepsilon\). The \(o(1)\) in the paper is thus a quantifier over \(\varepsilon\), not an unspecified function added to the theorem. Strong improving bounds for \(3/2<p\le2\) are separate corollaries.

\textbf{Theorem 1.3.} Encode the endpoint sparse form and then the stated interior triangle of exponents. Sparsity uses subsets \(E_I\subset I\) of cardinality at least \(|I|/10\), pairwise disjoint over the family. Cardinalities, dilations and

normalizations must be explicit.

\textbf{Theorem 1.5.} Quantify over every nonatomic ergodic measure-preserving system and every infinite set of indices \(S\). The conclusion is existence of an integrable function whose averages along \(N\in S\) fail to converge on a set of positive measure. Do not weaken this to one system or to the full sequence only.

\textbf{Theorem 1.6.} Define \(\mathcal P\) as the primes represented by \(a^2+b^2+1\) with \((a,b)=1\). For every \(A>0\), specify constants \(C_A,X_A\) so that the nonrepresentable \(1\le n<X\), \(n\equiv12\pmod{24}\), have cardinality at most \(C_AX(\log X)^{-A}\) for \(X\ge X_A\).

\textbf{Section 7.} Theorem 7.2 and Corollaries 7.6 and 7.42 are additional targets: the modular estimate, quantitative \(L^p\) growth, and almost-everywhere and \(L^1\) convergence on the stated Orlicz class. They are not substitutes for Theorem 1.5.

\page
\section{The dependency architecture}
\status{Initial map, not a claim that the indicated modules exist already}

\begin{longtable}{p{.15\linewidth}p{.36\linewidth}p{.40\linewidth}}
\toprule
Branch & Principal nodes in the paper & Output and main dependencies\\\midrule
Arithmetic & Section 3, especially 3.1, 3.12, 3.14, 3.20, 3.21, 3.24, 3.27 & Cohen correlations, kernel bounds, Gauss sums and exceptional-conductor bounds.\\
Improving & 4.1, 4.15, 4.19 & Theorem 1.1; uses Section 3 and the decomposition in Section 6.\\
Sparse & 5.1, 5.10, 5.17 & Theorem 1.3; needs the local low estimate and maximal, rather than fixed-scale, high estimate.\\
High--low & 6.5, 6.9, 6.14, 6.16, 6.19, 6.21 & Proposition 4.19; uses primes in progressions with an exceptional term, minor-arc cancellation and multifrequency estimates.\\
Near \(L^1\) & 7.8, 7.15, 7.21, 7.28, 7.2 & Corollaries 7.6 and 7.42; uses quantitative Ionescu--Wainger/sampling inputs and the Section 6 model.\\
Divergence & 8.1, 8.6, 8.8 and the final assembly & Theorem 1.5; uses a character-sum bound, Poisson spacings, fixed-modulus prime equidistribution and the divergence criterion.\\
Goldbach & 9.2--9.10, Section 10, 11.13, 11.19, 9.17, 9.20, 9.22, 9.24 & Theorem 1.6; circle-method mean-square approximation, two-stage sieving, positive main term and exceptional-set assembly.\\\bottomrule
\end{longtable}

The high--low branch is shared by the improving, sparse and near-\(L^1\) results. The divergence branch shares arithmetic foundations but is logically separate from the maximal inequality. The Goldbach branch shares exponential sums and local correlations, but introduces its own mean-square approximation and sieve machinery.

\subsection*{Proposed eventual organization}
Keep declarations under \lean{PrimeSquares}. Suggested future directories are \lean{Arithmetic}, \lean{HighLow}, \lean{Improving}, \lean{Sparse}, \lean{Maximal}, \lean{Ergodic}, \lean{Goldbach} and \lean{External}. Maintain one table linking a manuscript label to each Lean declaration. Use stable LaTeX labels once the sources are available; printed theorem numbers alone will change during proofreading.

For each node record: the exact statement; normalization and parameter ranges; dependencies; an expanded proof; Lean name; external assumptions; build status. The appendix currently gives an inventory, not this complete node-by-node expansion.

\page
\section{A checked finite exceptional-set argument}
\status{Implemented in \lean{PrimeSquares/FiniteAssembly.lean}}

Let \(S\) be a finite set, let \(e_1,e_2:S\to\R\), and let \(t>0\). Define
\[
 B_t=\{n\in S:|e_1(n)|+|e_2(n)|>t\}.
\]
No number-theoretic interpretation is part of these definitions.

\begin{lemma}[Finite energy estimate]
If \(\sum_{n\in S}e_i(n)^2\le B_i\) for \(i=1,2\), then
\[
 |B_t|\le\frac{2(B_1+B_2)}{t^2}.
\]
\end{lemma}
\begin{proof}
For real \(a,b\), the nonnegativity of \((|a|-|b|)^2\) implies \((|a|+|b|)^2\le2(a^2+b^2)\). On \(B_t\), therefore, \(t^2\le2(e_1(n)^2+e_2(n)^2)\). Sum this inequality on \(B_t\), enlarge the nonnegative square sum to \(S\), and divide by \(t^2>0\).
\end{proof}

\begin{lemma}[Positivity outside the exceptional set]
Let \(R(n)\) be a nonnegative integer and suppose that
\[
 R(n)\ge M(n)+e_1(n)-e_2(n)-u(n),\quad
 M(n)\ge4t,\quad u(n)\le t.
\]
If \(n\notin B_t\), then \(R(n)\ge2t>0\).

\end{lemma}
\begin{proof}
Use \(e_1-e_2\ge-|e_1|-|e_2|\) and \(|e_1|+|e_2|\le t\). The lower
bound becomes \(4t-t-t=2t\).
\end{proof}

Combining the two lemmas gives the actual Lean result
\lean{PrimeSquares.zero_count_card_bound}:
\[
 \abs{\{n\in S:R(n)=0\}}\le\frac{2(B_1+B_2)}{t^2}.
\]
The hypotheses about \(M,R,u,e_i\) are part of its statement. In the paper's final
assembly take \(M=\Delta X_n\Theta_n(z)/2\),
\(t=cN(\log N)^{-C_{A_0}}\), and \(e_i=E_i\).
If each mean square is at most \(C_iN^4(\log N)^{-K}\), the bound becomes
\[
 \abs{\{R=0\}}\le
 \frac{2(C_1+C_2)}{c^2}N^2(\log N)^{-K+2C_{A_0}}.
\]
Choosing \(K\) large enough yields the desired logarithmic saving for this finite
range. The analytic estimates, eventual parameter choices, identification of \(R\)
with actual representations, and treatment of the initial short range remain
formalization work. Thus this is the finite last step of the Goldbach proof, not
Theorem 1.6 itself.

\page
\section{Other checked components}
\status{Implemented in \lean{LocalArithmetic.lean} and \lean{SieveConstant.lean}}

\subsection*{Congruences and the shared-prime observation}
For every natural number \(n\) with \(n\equiv1\pmod2\) and \(n\not\equiv0\pmod3\),
the module proves \(n^2\equiv1\pmod{24}\). The proof reduces to the 24 possible
residues; Lean checks the finite case split. For a prime \(p>3\), primality excludes
divisibility by 2 and 3, so the same conclusion follows.

Consequently, if \(r\equiv11\pmod{24}\) and \(p>3\) is prime, then
\(r+p^2\equiv12\pmod{24}\). The restriction \(p>3\) is included explicitly.
The starter also proves the shift equivalences
\[
 m=a^2+b^2+p^2\iff m+1=a^2+b^2+p^2+1,
 \qquad m\equiv11\pmod{24}\iff m+1\equiv12\pmod{24}.
\]
These clarify the relation between the abstract's formulation and Theorem 1.6;
they do not establish an exceptional-set estimate.

Suppose \(p,m\) are prime, \(n=m^2+r\), and \(p\mid n,r\). Then \(p\mid m^2\),
so \(p\mid m\); primality of \(m\) forces \(p=m\). This is the finite divisibility
observation used in Lemma 11.2. The divisor-counting estimate and the full sieve
comparison in that lemma are not yet formalized.

\subsection*{The elementary bracket in Lemma 9.24}
The manuscript reduces its final sieve comparison to
\[
 \Delta\ge\frac{e^{\gamma/2}}{\sqrt\pi}
 \left(\frac{2\sqrt2}{3}\log2-\frac{27}{64}\right).
\]
Lean proves positivity of the parenthesized expression. There is a particularly
short rigorous proof which does not need decimal approximations:
\[
 \sqrt2\ge\frac43,\qquad \log2\ge1-\frac12=\frac12,
\]
where the second inequality is the standard \(1-x^{-1}\le\log x\) for \(x>0\).
Both lower bounds are positive, and therefore
\[
 \frac{2\sqrt2}{3}\log2-\frac{27}{64}
 \ge\frac49-\frac{27}{64}=\frac{13}{576}>0.
\]
The declaration \lean{sieve_bracket_positive} proves this conclusion. The separate
declaration \lean{scaled_sieve_bracket_positive} allows multiplication by any
strictly positive real factor.

This does \emph{not} verify the integral evaluation for \(f_{1/2}(9/8)\), the upper
bound on \(\mathcal W(9/8)\), the factor \(e^{\gamma/2}/\sqrt\pi\), or the stronger
numerical inequality \(\Delta>0.17\). Those must be added before claiming Lemma 9.24.

\page
\section{Improving estimates: the scalar optimization and its inputs}
\status{Unfinished target; the following is the blueprint for Section 4}

Separate the prime parameter from the spatial scale. For prime squares, a sum up
to \(N\) has support at scale \(N^2\); the abstract theorem uses a spatial-scale
variable. A typed name such as \lean{primeScale} versus \lean{spaceScale} prevents
an incorrect substitution of \(N\) for \(N^2\).

Theorem 4.1 first supplies a comparison operator \(B_N\). Applied to indicators,
its decomposition gives, with \(\delta=|F||G|/N^2\),
\[
 N^{-1}|\langle B_N\one_F,\one_G\rangle|
 \lesssim\delta J^s(\log J)^\gamma+
 \delta^{1/2}J^{-r}(\log J)^\beta.
\]

The optimal J, up to bounded factors, is
$$
 J=\delta^{-1/(2(s+r))}(\operatorname{Log}\delta)^{(\beta-\gamma)/(s+r)}.
$$
Prove first the estimate when this J is in the permitted range. If it is too large, use the elementary local kernel estimate. This second branch is essential: choosing an inadmissible J is not an application of the decomposition.

For the prime operator the corresponding bound is
$$
 \delta Q^s(\log Q)^\alpha+
 \delta^{1/2}Q^{-r}
 \exp\!\left(c_1\frac{(\log Q)^\kappa}{\log\log Q}\right)
 +\text{error}.
$$
When $\operatorname{Log}\delta$ is small enough relative to $((\log N)^c$, take the dyadic Q specified by (4.14). Prove separately that it is admissible, that rounding changes the estimate by a bounded factor, and that the error is absorbed. In the complementary range, transfer the comparison estimate using the pointwise domination by B_N. This yields
$$
 \delta^{1-s/(2(s+r))}\Phi_{c_*,\kappa}(\delta),\qquad
 c_* = \frac{c_1s}{2^\kappa(s+r)^{\kappa+1}}+o(1).
$$
For $s=1,r=1/2,\kappa=1/2,c_1=2+\varepsilon$, the critical power is $2/3$ and the limiting coefficient is $4\sqrt3/9$.

\textbf{Separate Lean tasks.} Establish the finite bilinear norm inequalities; formalize the elementary optimization including the two parameter ranges; prove the dyadic-to-full annular summation; and then prove restricted-to-strong interpolation for $p>3/2$. None follows just from writing the numerical value of the coefficient into a declaration.

\page
\section{Sparse bounds: local control and recursion}
\status{Unfinished target; Sections 5 and the stopping-time conventions in Section 2}

The sparse proof needs more than Theorem 1.1. A stopping time $\tau(x)$ depends on the spatial point. Its high term must be controlled by a maximal ℓ^2 estimate over all permitted scales. The low estimate must be local enough that admissibility replaces its local density by the parent density.

The key tasks are these.
\begin{enumerate}
\item Define integer intervals, their dilations, a dyadic grid, and the admissible stopping times. Record the exact support radius of each kernel.
\item Prove Lemma 5.1 from the local kernel estimate, the local low estimate, the maximal high estimate and the square-summable errors. Include the small-scale branch, where the abstract high--low split is not available.
\item Prove Lemma 5.10. Select maximal bad dyadic intervals by the two density tests. Bound the sum of child lengths by $|I_0|/4$. Construct disjoint major subsets and iterate the local inequality.
\item In Lemma 5.17 split according to $\log\tau\le C_0L^2$ and its complement, where $L=\operatorname{Log}(\delta_0^{-1})$. The first range uses the square averages; the second uses the prime-square decomposition with an admissible denominator cutoff.
\item Convert the local sparse form with cI on the input side into the stated form. Prove the required grid/dilation reduction rather than identifying the two families by notation.
\end{enumerate}

For finite input sets, begin with a finite set of scales and a finite recursion. If using an infinite family later, specify its sum and prove the limiting step. This keeps the initial recursive construction elementary and exposes precisely which part needs an infinite argument.

There is a useful formalization trap here. An arbitrary real-valued supremum in Lean uses conditional completeness. Without a boundedness hypothesis, it must not be treated as a maximum with all the usual inequalities. One can first work with finite truncated maxima, or use extended nonnegative reals for a maximal function and prove finiteness afterward.

Finally, the strong sparse estimates for exponents in the interior of the triangle need their own interpolation/convexity argument. They should be separate declarations depending on the endpoint local bound and the trivial bounds, not an additional assumed theorem.

\page
\section{The high--low model and exceptional conductors}
\status{Unfinished target; the main shared arithmetic and analytic branch}

At prime scale N, the annular multiplier is approximated by the model V_N in (6.8). Its ordinary coefficients are normalized unit quadratic Gauss sums. If an exceptional character is present, its contribution has the minus sign displayed in that formula, an induced character modulo each multiple of its conductor, and a continuous model with parameter β.

Use explicit scale-indexed data. For example, define an optional datum at N containing (q_χ,χ,β), and define
$\mathcal M(\chi)$ to be the scales where that character is selected. Do not pick one character and use it at all scales. Page uniqueness must be strong

enough to justify the later regrouping by conductor.

\textbf{Arithmetic layer.} Prove the Cohen correlation by CRT, the odd prime-power evaluation and the explicit 2-adic cases. Build the restricted correlations and Euler products of Section 3 from these identities. Prove the ordinary Gauss bound with its subexponential factor, the real-character bound, and the stronger \(\sigma_\chi B_\chi^{1/2}\ll q_\chi^{-1/25}\) estimate. The last estimate is used to sum over changing exceptional characters.

\textbf{Approximation layer.} Encode Lemma 6.5 and then Lemma 6.9. The primes-in-progressions input must supply a single exceptional datum valid over the interval of partial summation \([N/2,N]\), with uniform errors. A fixed-modulus PNT alone does not provide this growing-modulus input.

\textbf{Exact decomposition.} The ordinary split is by \(q\le Q\). The exceptional split is by \(d=q/q_\chi\le Q/B_\chi\), with its own cutoff. Prove the multiplier identity (6.13) before norm estimates. Empty exceptional sums, including \(Q/B_\chi<1\), must be covered by the definitions.

\textbf{Low and high bounds.} Proposition 6.14 uses inverse transforms and the arithmetic kernel identities. Proposition 6.21 needs Lemma 6.16, the denominator-prefix extension 6.19, the cutoff-difference square functions, and a final sum over \(\chi\). At each scale there is at most one exceptional datum; the square sum over characters and Page uniqueness per dyadic conductor range are separate steps. Only after both bounds and the error estimate are checked does one obtain Proposition 4.19.

\page
\section{The Orlicz maximal estimate and convergence}
\status{Unfinished target; Section 7}

The quantitative Ionescu--Wainger input must retain its constants. Proposition 7.8 specifies a rational-frequency set containing every denominator at most \(Q\), a period \(P\), a denominator bound \(B\), a sufficiently small width, and norm comparisons for a range of exponents. It invokes Tao's Lemma 5.1, Theorems 1.6 and 1.7, and Proposition 1.5. These should have explicit external interfaces if they are not formalized here.

Next prove the continuous multiplier bounds uniformly in \(0<\beta\le1\) (Lemma 7.15), followed by the finite-group measure and sampling argument (Proposition 7.21). The measure is allowed to be complex but has total variation at most one. Dependence of constants on the period or on \(\beta\) would break the later optimization.

Proposition 7.28 combines this low model with an \(\ell^2\) high estimate. The exceptional characters of conductor above \(Q\) vary with scale and are retained in the high term. The low exceptional character, when it exists, is controlled using Page uniqueness. This assembly must use the same model and normalization as Section 6.

For the modular extension, do not stop at the restricted estimate on \(\one_F\). Let \(C_\infty\) bound the masses of the positive kernels. Remove \(|f|\le\lambda/(2C_\infty)\). Decompose the remaining function into level sets
\[
 E_j=\{2^j\lambda/(2C_\infty)<|f|\le2^{j+1}\lambda/(2C_\infty)\}.
\]
Allocate thresholds using a positive summable sequence such as \(a_j=6/(\pi^2(j+1)^2)\). Prove the union inclusion, apply the restricted estimate to each \(E_j\), and prove that the polynomial factor in \(j\) inside the threshold can be absorbed by enlarging the coefficient of \(\Psi_C\). This is a quantitative estimate for this particular Young function, not just sublinearity. It gives the truncated modular estimate (7.3), then (7.4).

Corollary 7.6 follows by layer cake and a one-variable integral estimate, with \(\varepsilon=p-1\). Corollary 7.42 first transfers finite truncated discrete estimates to an invertible probability-preserving system. Prove bounded-function convergence using the cited polynomial-prime result and control the higher prime powers there. Abel summation handles the unweighted prime averages. Finally use Orlicz truncation, doubling and the modular maximal estimate to pass to unbounded functions; use uniform \(L^1\) bounds for \(L^1\) convergence.

\page
\section{Persistent universal divergence}
\status{Unfinished target; Section 8}

For each modulus \(q\), define
\(\Lambda_q=\{b^2\bmod q:(b,q)=1\}\). The initial finite tasks are CRT compatibility, inclusion in the prime-square residue set, and, for odd squarefree \(q\),
\[
 |\Lambda_q|=\frac{\varphi(q)}{2^{\omega(q)}},\qquad
 s_q=\frac q{|\Lambda_q|}=\prod_{p\mid q}\frac{2p}{p-1}.
\]
The inclusion uses Dirichlet's theorem for a unit square root; it must not claim that every prime residue, including nonunits, is part of \(\Lambda_q\).

Lemma 8.8 expands each quadratic-residue indicator as \(\tfrac12(1+\chi_p-\one_{\{0\}})\). Terms containing a zero indicator are bounded by a constant depending on the number of shifts. For the others, apply the Weil bound to the squarefree product of the distinct shifted linear factors. This proves the correlation hypothesis for the imported Poisson-spacings theorem.

Choose pairwise disjoint blocks of odd primes with block sizes tending to
infinity, and a disjoint prime sequence \(u_j\). The block products \(q_j\)
satisfy \(\PP_{q_j}(\Lambda_{q_j})\le2^{-j}\), whereas
\(\PP_{u_j}(\Lambda_{u_j})\ge1/3\). The construction must ensure the combined
collection is pairwise coprime.

Theorem 8.6 supplies Poisson spacings. The thickening identity in the proof
expresses the covered proportion as
\[
 \PP_{q_j}(\Lambda_{q_j}^{\gamma})
 =\int_0^\gamma F_j(\theta)\,d\theta+O(s_{q_j}^{-1}).
\]
Dominated convergence gives \(1-e^{-\gamma}\), so the deficit is \(o(\gamma)\).
The pair-separation condition uses LaVictoire's Lemma 3.2 in addition to this
one-gap distribution; it is not an automatic restatement of Poisson convergence.

Finally, fixed-modulus prime equidistribution gives limiting mass
\(1/|\Lambda_Q|\) at every \(a\in\Lambda_Q\). The strict lower bound required
by (8.5) then holds with, for example, \(\beta=1/2\). Apply the full imported
divergence criterion, including its transference/Banach-principle conclusions,
to obtain exactly the persistence and system quantifiers of Theorem 1.5.

If the chosen formal ergodic infrastructure assumes invertibility, a separate
extension/transfer argument is needed to match the unqualified system statement
in Theorem 1.5. Simply adding invertibility would narrow the theorem.

\page
\section{The Goldbach branch}
\status{Only the finite final assembly and small supporting pieces are checked}

The reductions in Section 9 produce a lower sieve count minus an upper correction
count. Their errors must have mean-square bounds with an arbitrarily large
logarithmic saving. The common main term is
\(X_n\Theta_n(z)\); its positivity and uniform lower bound are separate lemmas.

\textbf{Circle method.} Lemma 9.10 is the principal target. Its coefficients are
independent of \(n\), and its conclusion is a mean square of a sum over auxiliary
moduli and \(\ell\). Section 10 supplies the minor-arc bound, the major-arc local
factor \(G_R\), the truncated continuous factor \(H_{\ell,R}\), the tail estimate
and the major-arc error. The same small-prime parameter \(R\) must be retained
in both main-term factors until its tails have been shown to be summable.

\textbf{Sieve.} Lemma 11.2 replaces variable sets of sieve primes by fixed sets,
which is needed before applying Lemma 9.10 to coefficients independent of \(n\).
The exact local densities are computed in Lemmas 11.6 and 11.8. For primes at
most \(R\), the uniform product dimensions are \(2/3\) and \(3/2\); one cannot
apply the desired half-dimensional and linear sieves over the whole range.

Lemma 11.13 removes \(p\le R\) first at level
\(D_0=\exp((\log\log N)^3)\). The remaining level is \(D/D_0\).
Because outer weights have both signs, the choice of preliminary upper or lower
weights must depend on that sign. Prove this finite algebra before estimating
the two stages. The coefficient bounds and independence of \(n\) must survive
the combination. The outer dimensions are then \(1/2\) and \(1\).

Lemma 11.19 performs the weighted sum over \(\ell\). It needs the cited mean-value
theorem for multiplicative functions and the relevant short-interval uniformity.
Track the cancellation of the factors at \(p\le R\) and the replacement of
\(H_{\ell,R}\) by \(H_\ell\); neither can be dropped at the definition stage.

The outputs are Propositions 9.17 and 9.20. Lemma 9.22 bounds the common main
term below. Lemma 9.24 compares the sieve constants. Their combination gives
(9.25), and the checked finite result in Section 4 then applies. The remaining
asymptotic task is to choose the constants in the correct order, show the
\(O(N^{\varepsilon})\) nuisance is smaller than the positive main term, discard
the initial range, and pass from scale \(N^2\) to general \(X\).

\page
\section{Proposed external-input register}
\status{Candidate interfaces; not declarations already installed in the starter}

The user's permission to leave deep theorems unformalized makes a conditional
development appropriate. The exact declarations should be approved mathematically
by comparison with their sources. The list below is an initial register, not a
claim that each result is absent from current Mathlib.

\small
\begin{longtable}{p{.11\linewidth}p{.39\linewidth}p{.41\linewidth}}
\toprule
ID & Published input & Required interface and use\\\midrule
NT01 & PNT in progressions with exceptional term and Landau--Page; paper (2.5)--(2.6), reference [22] &
Uniform growing-modulus error, one datum on an annulus, induced characters, and
uniqueness strong enough across scales. Used in Section 6.\\
NT02 & Siegel--Walfisz and fixed-modulus equidistribution &
Full quantifier order for modulus, residue, exponent and threshold. Used where
polylogarithmic moduli or fixed progressions suffice; not a replacement for NT01.\\
NT03 & Ren [40, Theorem 1] &
Quadratic prime exponential-sum cancellation with exactly the denominator and
arc-width ranges needed in (6.7) and (10.1).\\
NT04 & Bombieri--Vinogradov, Brun--Titchmarsh, Mertens-type estimates; Section 2.6 &

Maxima over residues and endpoints, divisor weights, and constants uniform in
the auxiliary moduli. Used in Section 10 and the sieve.\\
NT05 & Classical quadratic Gauss evaluations and Weil character-sum bounds &
Primitive and induced character conventions; primes and prime powers, including
2; squarefree-polynomial hypothesis for Lemma 8.8.\\
HA01 & Separated-frequency maximal inequality [25, Theorem 1.1] &
Precise separation and cardinality dependence. Prove its adaptation, transference
and uniformity in \(\beta\) in Lemma 6.16.\\
HA02 & Tao [45], Lemma 5.1, Theorems 1.6--1.7 and Proposition 1.5 &
Quantitative major-arc construction, multiplier theorem and sampling comparisons.
Keep \(r,k,B,P,\varepsilon,p\) and vector-valued constants explicit.\\
HA03 & Hardy--Littlewood, Calder\'on--Zygmund, Khintchine and Rademacher--Menshov &
Only use an external declaration where the pinned library does not already
provide the needed statement. Preserve the \(p\downarrow1\) dependence.\\
\bottomrule
\end{longtable}
\normalsize

Do not collapse these into a single assumption saying that all estimates needed
in the paper hold. An import should be independently recognizable as a published
theorem and should not assume the novel conclusion to be checked.

\page
\subsection*{External inputs continued}
\small
\begin{longtable}{p{.11\linewidth}p{.39\linewidth}p{.41\linewidth}}
\toprule
ID & Published input & Required interface and use\\\midrule
ER01 & LaVictoire [27, Theorem 3.1 and Lemma 3.2], with the principles used in the paper's Theorem 8.1 &
CRT families, two coprime sequences, thickening deficit, pair separation and
residue-frequency hypotheses; full persistent universal divergence conclusion.\\
ER02 & Granville--Kurlberg [14, Theorem 1], in the specialization 8.6 &
Uniform correlations for distinct shifts and the hypotheses on local densities;
convergence of normalized cyclic gaps.\\
ER03 & Polynomial-prime pointwise convergence [35, Corollary 2(ii)] &
The bounded-function dense class used in Corollary 7.42, with the correct weights
and systems. Prime powers and Abel summation remain internal tasks.\\
SV01 & Rosser--Iwaniec coefficient-form sieve and fundamental lemma [21] &
Weights, support, signs, coefficient bounds, product hypotheses, sieve functions
and quantitative remainder; prove the two-stage composition in Lemma 11.13.\\
SV02 & Granville--Koukoulopoulos [13] and Shiu [41] &
Mean values of the particular multiplicative weight and uniform estimates in the
retained ranges used by Lemma 11.19.\\\bottomrule
\end{longtable}
\normalsize

\subsection*{Represent imports as hypotheses first}
For a new branch it is often helpful to package the external inputs as a named
structure and prove a theorem with that structure as an explicit argument.
This makes the logical boundary visible without immediately adding global
axioms. Once the interfaces have been audited, a separate file can instantiate
the structure from selected quoted theorems.

In either design, dependency output for a headline theorem must agree with the
published import register. A proof conditional on an estimate proved nowhere in
the project or cited literature should be labeled incomplete, even if Lean
accepts it with that estimate as a hypothesis.

\subsection*{Possible reuse from the Hua project}
The Hua repository contains finite cyclic Fourier infrastructure, quadratic local
arithmetic and analytic assembly. Some may be reusable after comparing
definitions. Its prime-square weights are those of its own \(W\)-trick argument,
not automatically the operators \(A_{N,\Lambda}\) in this paper. Reuse requires
checking signs, normalizations, finite versus infinite groups, and transitive
axiom dependencies. No reuse has been counted as completed work here.

\page
\section{Points to settle during manuscript proofreading}
These are formalization requirements or visible editorial issues, not a verdict
that the main mathematical arguments are wrong.

\begin{enumerate}
\item \textbf{Freeze the version.} The attached paper has 96 pages and Section 7.
Its current numbering is the one used in this document. Match the final LaTeX
labels rather than importing numbering from older discussions.
\item \textbf{Repair the Section 7 references.} The opening of Section 7 and the
last sentence before the proof of Corollary 7.6 refer to Theorem 1.5. In this
snapshot Theorem 1.5 is the divergence theorem, so these references cannot name
the maximal-inequality consequence intended there.
\item \textbf{Preserve the two endpoint functions.} The improving loss is
\(\exp(c\sqrt{L}/L_2)\); the modular function uses
\(\exp(C\sqrt{\log(e+t)\log(e+\log(e+t))})\). These are different scales.
\item \textbf{Separate bounded and unbounded input.} A negligible prime-power
contribution for bounded functions is sufficient for the dense-class convergence
step; it is not itself a uniform error bound for every \(L^1\) function.
\item \textbf{Make exceptional data consistent.} Record the annular data used in
partial summation and prove the regrouping over \(\mathcal M(\chi)\). Do not
replace this by a single exceptional character fixed for all \(N\).
\item \textbf{Expose uniformity in the Goldbach range.} All \(o(1)\), tail errors
and positivity bounds must be uniform in admissible \(n,\ell\) and auxiliary
moduli. Write the order of choosing \(A_0,K,C\) and all thresholds explicitly.

```
\item \textbf{Retain fixed sieve-prime sets.} The independence from \(n\) after
Lemma 11.2 is required by the mean-square estimate. The coefficient combination
in the pre-sieve must preserve this independence.
\item \textbf{Check strictness and small cases.} Empty sets, \(q=1\), powers of 2,
\(N\ge2\), \(p=2,3\), and equality at density/threshold boundaries should be
covered by exact definitions, not silently discarded.
\end{enumerate}

The present work has not audited every cited theorem or every proof in the PDF.
In particular, successful checks of the starter do not validate any unfinished
node elsewhere in the map.

\page
\section{What is needed to continue}
The PDFs were enough to start. The next useful input is the final LaTeX project:
all source files, bibliography, custom macros and any code used for numerical
estimates, together with the PDF generated from those files. This makes the
statements and labels unambiguous and lets the map track proofreading changes.

GitHub is useful for versioning and automated builds, but an account is not
required to write or check Lean locally. This archive is a normal Lean project;
it can be uploaded later. The toolchain and lockfile are already selected.
There is no need for the author to learn Lean before mathematical work can begin.

The next mathematical milestone should be one coherent branch. A practical
sequence is:
\begin{enumerate}
\item Encode the final target statements and audit their quantifiers against the
LaTeX. Build a stable statement map.
\item Complete a substantial finite arithmetic module, such as the quadratic
Cohen identity with all prime-power cases, or complete the scalar optimization
in Theorem 4.1.
\item Specify and audit the deep-input interfaces. Prove one full branch from
those interfaces, without assuming its novel intermediate estimates.
\item Add the remaining branches and report the exact dependencies of each
headline theorem separately.
\end{enumerate}

For the Goldbach branch, the existing checked finite assembly supplies an endpoint
to which the circle-method and sieve components can connect. For the harmonic-
analysis branch, the shared high--low decomposition is the principal reusable
milestone. The divergence branch can be developed separately once its external
criterion and Poisson-spacings interfaces are audited.

\textbf{Completion criterion.} A full conditional formalization requires a fresh
build of every target theorem, a transitive axiom audit with no unfinished proof
axiom, and a human comparison of the final statements and imported inputs with
the manuscript and sources. A long blueprint, a clean source scan, or a passing
build of the present starter alone does not meet that criterion.

\textbf{Current deliverable.} This is an initial, reproducible starting point with
small checked components and an explicit account of the much larger unfinished
work. It should not yet be described in an arXiv or journal submission as a
formalization of the paper.

\page
\appendix
\section{Inventory of numbered statements}
The following inventory was extracted from headings in the attached PDF. Planned
names are suggestions only. A status of ``target'' means no proof of that numbered
statement is present in the starter. Checked supporting subclaims are described
separately in Sections 4--5; they do not change the status of the enclosing paper
theorem or lemma.
\small
\input{Statement_Inventory.tex}
\normalsize
\end{document}
```

legacy/Statement_Inventory.tex

```
\begin{longtable}{p{.23\linewidth}r p{.53\linewidth}p{.1\linewidth}}
\toprule Statement & Page & Planned name (after \texttt{PrimeSquares.}) & Status\\\midrule
\endhead
Theorem 1.1 & 3 & \lean{Main.theorem_1_1} & target\\
Theorem 1.3 & 3 & \lean{Main.theorem_1_3} & target\\
Theorem 1.5 & 4 & \lean{Main.theorem_1_5} & target\\
Theorem 1.6 & 5 & \lean{Main.theorem_1_6} & target\\
Definition 2.10 & 12 & \lean{Foundations.definition_2_10} & target\\
Definition 2.11 & 13 & \lean{Foundations.definition_2_11} & target\\
Lemma 3.1 & 15 & \lean{Arithmetic.lemma_3_1} & target\\
Definition 3.7 & 16 & \lean{Arithmetic.definition_3_7} & target\\
Definition 3.9 & 17 & \lean{Arithmetic.definition_3_9} & target\\
Lemma 3.11 & 17 & \lean{Arithmetic.lemma_3_11} & target\\
Lemma 3.12 & 17 & \lean{Arithmetic.lemma_3_12} & target\\
Lemma 3.14 & 17 & \lean{Arithmetic.lemma_3_14} & target\\
Corollary 3.16 & 18 & \lean{Arithmetic.corollary_3_16} & target\\
Lemma 3.20 & 19 & \lean{Arithmetic.lemma_3_20} & target\\
Lemma 3.21 & 19 & \lean{Arithmetic.lemma_3_21} & target\\
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Lemma 3.24 & 20 & \lean{Arithmetic.lemma_3_24} & target\\
Lemma 3.27 & 21 & \lean{Arithmetic.lemma_3_27} & target\\
Theorem 4.1 & 24 & \lean{Improving.theorem_4_1} & target\\
Corollary 4.15 & 25 & \lean{Improving.corollary_4_15} & target\\
Proposition 4.19 & 26 & \lean{Improving.proposition_4_19} & target\\
Lemma 5.1 & 28 & \lean{Sparse.lemma_5_1} & target\\
Lemma 5.10 & 30 & \lean{Sparse.lemma_5_10} & target\\
Lemma 5.17 & 31 & \lean{Sparse.lemma_5_17} & target\\
Lemma 6.5 & 34 & \lean{HighLow.lemma_6_5} & target\\
Lemma 6.9 & 35 & \lean{HighLow.lemma_6_9} & target\\
Proposition 6.14 & 37 & \lean{HighLow.proposition_6_14} & target\\
Lemma 6.16 & 39 & \lean{HighLow.lemma_6_16} & target\\
Corollary 6.19 & 40 & \lean{HighLow.corollary_6_19} & target\\
Proposition 6.21 & 41 & \lean{HighLow.proposition_6_21} & target\\
Theorem 7.2 & 44 & \lean{Maximal.theorem_7_2} & target\\
Corollary 7.6 & 44 & \lean{Maximal.corollary_7_6} & target\\
Proposition 7.8 & 45 & \lean{Maximal.proposition_7_8} & target\\
Lemma 7.15 & 47 & \lean{Maximal.lemma_7_15} & target\\
Proposition 7.21 & 48 & \lean{Maximal.proposition_7_21} & target\\
Proposition 7.28 & 51 & \lean{Maximal.proposition_7_28} & target\\
Corollary 7.42 & 56 & \lean{Maximal.corollary_7_42} & target\\
Theorem 8.1 & 57 & \lean{Divergence.theorem_8_1} & target\\
Theorem 8.6 & 58 & \lean{Divergence.theorem_8_6} & target\\
Lemma 8.8 & 58 & \lean{Divergence.lemma_8_8} & target\\
Lemma 9.2 & 61 & \lean{Goldbach.lemma_9_2} & target\\
Lemma 9.3 & 61 & \lean{Goldbach.lemma_9_3} & target\\
Lemma 9.7 & 62 & \lean{Goldbach.lemma_9_7} & target\\
Corollary 9.9 & 63 & \lean{Goldbach.corollary_9_9} & target\\
Lemma 9.10 & 63 & \lean{Goldbach.lemma_9_10} & target\\
Proposition 9.17 & 64 & \lean{Goldbach.proposition_9_17} & target\\
Proposition 9.20 & 64 & \lean{Goldbach.proposition_9_20} & target\\
Lemma 9.22 & 65 & \lean{Goldbach.lemma_9_22} & target\\
Lemma 9.24 & 65 & \lean{Goldbach.lemma_9_24} & target\\
Lemma 10.2 & 66 & \lean{GoldbachCircle.lemma_10_2} & target\\
Lemma 10.12 & 68 & \lean{GoldbachCircle.lemma_10_12} & target\\
Lemma 10.15 & 68 & \lean{GoldbachCircle.lemma_10_15} & target\\
Lemma 10.20 & 72 & \lean{GoldbachCircle.lemma_10_20} & target\\
Lemma 10.24 & 73 & \lean{GoldbachCircle.lemma_10_24} & target\\
Lemma 11.2 & 76 & \lean{GoldbachSieve.lemma_11_2} & target\\
Lemma 11.6 & 76 & \lean{GoldbachSieve.lemma_11_6} & target\\
Lemma 11.8 & 77 & \lean{GoldbachSieve.lemma_11_8} & target\\
Lemma 11.13 & 78 & \lean{GoldbachSieve.lemma_11_13} & target\\
Lemma 11.19 & 82 & \lean{GoldbachSieve.lemma_11_19} & target\\
\bottomrule
\end{longtable}
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Archive record: Current_Statement_Inventory.csv

Current_Statement_Inventory.csv

key,kind,source_number,source_pdf_page,section,section_title,source_line,status,preview_latex
t:improvingfixed,theorem,1.1,3,1,Introduction,212,"conditional: Both conclusions for the genuine full von-Mangoldt square average, every positive integer N, the exact source interval and its cardinality, all finite F,G subsets, every coefficient cstar>4 sqrt(3)/9, and every real 3/2<p<=2. The strong result proves actual output lp membership and the literal normalized sum form, for all lp inputs, hence also the supported inputs in the source. The implication is complete under the explicit independent deep inputs. PNT limits, compatible progression/Ren/Ghosh data existence, and the independent ordinary-square global decomposition remain unformalized hypotheses. Its analytic globalization remains external; the plus/minus spatial reflection is proved with the same constants at checkpoint783. Theorem1.1 and Propositions4.19/6.21 are complete conditional implications under their separate inputs; the current corrected numbered paper targets are all matched, with independent inputs and stronger unnumbered alternative-route assertions retaining their documented scopes.","Let \(I=[-N^2,2N^2]\cap\mathbb Z\). For all \(F,G\subseteq I\), \begin{align} \frac1{|I|} \bigl|\langle A_{N,\Lambda}\mathbf1_F,\mathbf1_G\rangle\bigr| &\ll \left(\frac{|F|}{|I|}\frac{|G|}{|I|}\right)^{2/3} \Phi_{4\sqrt3/9+o(1)} \left(\frac{|F|}{|I|}\frac{|G|}{|I|}\right). \end{align} Consequently, "
t:sparsebound,theorem,1.3,3,1,Introduction,287,"conditional: Both displayed conclusions are complete conditional implications: the literal full maximal indicator for every integer N>=2, arbitrary finite F,G including empty sets, actual finite interval collections with actual pairwise disjoint major subsets of size at least one tenth, the exact endpoint density exponent and paper Phi for every cstar>4sqrt(3)/9, and every point of the full open triangle in exact reciprocal/barycentric coordinates. Constants precede F,G. All stopping, packing, support, local endpoint, interpolation and sparse deductions are proved. No paper-specific deduction in either conclusion remains. The recorded prime limits, compatible raw Page/distribution data, Page uniqueness, general smooth multifrequency/sampling, independent ordinary-square stopping interface and complementary-exponent prime maximal input remain unproved explicit hypotheses, including their stated compatibility boundaries. The general auxiliary Section5 statements are separately partial. The proposed dyadic qualification of l:sparserecursion is unapplied; the actual prime-square all-cutoff reduction is proved.","Let \(F,G\subset\mathbb Z\) be finite. There is a sparse collection of intervals \(\mathcal S\) such that \begin{align} \bigl|\langle A_\Lambda^*\mathbf1_F,\mathbf1_G\rangle\bigr| &\ll \sum_{I\in\mathcal S}|I| \left(\frac{|F\cap I||G\cap I|}{|I|^2}\right)^{2/3} \Phi_{4\sqrt3/9+o(1)} \left(\frac{|F\c"
t:pet,theorem,1.5,4,1,Introduction,376,"conditional: Both parts are complete conditional implications. Part(i): the actual enumerated prime squares are persistently universally L1-bad on every truly nonatomic ergodic measure-preserving probability system, with every infinite cutoff set and an actual measurable positive-measure divergence set. CRT, Dirichlet representation, correlation counts, large-prime block construction, all moment/thickening/difference limits, uniform square-root fibers and the fixed-progression PNT pushforward are proved. The original badness theorem, Sawyer, Conze, general Weil and fixed-progression PNT remain separate unproved general inputs. Granville–Kurlberg is not used by part(i). A checked moment proof replaces the source strict finite-fraction argument, which has an exact kernel counterexample. Part(ii) is a complete conditional implication: an absolute positive constant before every invertible measure-preserving probability system, every 1 < p <= 2 and every Lp input; the proper actual maximum over the first m prime squares for all m >= 1 is finite almost everywhere, belongs to Lp and has the exact displayed exponential norm bound. The prime-counting/enumeration normalization, finite sums, first-prime case and equality of both full maxima are proved with no PNT or enumeration input. The proof uses the already proved Corollary7.6 and the precise independent general strong Calderon input, plus the earlier Theorem7.2 independent inputs. It does not use the bounded-class convergence input. At checkpoint1264 the current public export consumes the proved finite signed maximal theorem at constant6; a supplied RawFiniteSignedMaximalInput and its constant R are no longer hypotheses. No remaining source clause within the agreed conditional scope. General inputs and their exact convention bridges remain unproved. The user confirmed the three Section8 edits; the selected project manuscript now contains the agreed moment proof.","Let \(p_k\) denote the \(k\)-th prime, and put \(n_k=p_k^2\). \begin{enumerate}[label=\textup{(\roman*)}] \item The sequence \((n_k)\) is persistently universally \(L^1\)-bad. More precisely, for every nonatomic ergodic measure-preserving probability system \((X,\mu,T)\) and every infinite set \(\ma"
t:repmotthm,theorem,1.6,5,1,Introduction,453,"conditional: The full quantitative exceptional-set theorem is complete: for every A>0, the actual exceptional set of positive n<X congruent12 mod24 without a prime plus a square-of-prime representation whose first prime is a primitive sum of two squares plus1 has size O_A(X(log X)^(-A)), eventually at every real X. The representation and exceptional-set membership are exact. The common-cutoff Section11 estimates, positive numerical gap, arbitrary logarithmic mean-square saving, all arithmetic losses, small n and the passage from N^2 to real X are discharged. Explicit independent general SW, BV, Ren/Ghosh, one-stage sieve, effective multiplicative mean, Mertens, normalized sieve values/continuity and primitive two-square criterion remain unproved theorem parameters. No own-paper endpoint is a hypothesis. At1259: Section11 and the final quantitative Goldbach theorem are complete in the agreed conditional scope, using the supplied corrected presieving statement. Every paper endpoint is discharged; only explicit independent general classical inputs and their normalizations remain unproved. Original manuscripts are unchanged and the supplied Section11 replacements still require user application. No additional completion credit is claimed for stronger unnumbered assertions in alternate proof routes. At1261: Proved the full PrimitiveTwoSquaresCriterion from pinned Mathlib and elementary algebra, and proved W(alpha) convergence/nonnegativity for every alpha>1. Constructed a concrete nonzero compactly supported smooth seed. The next stage exports the Section9 arithmetic and headline statements without the former criterion or seed hypotheses, then reconciles the stale Section9 map. User manuscript edits remain in progress. At1262 the final endpoint is exported from GoldbachSection11Inputs alone: the primitive two-square criterion is proved internally and a fixed nonzero smooth compact seed is constructed. The real-X bound and zero exceptional density need no separate arithmetic or seed assumption. At1262: Every Section9 numbered source claim and its final quantitative theorem is complete in the agreed scope. All elementary arithmetic reductions are now proved without an extra primitive two-square assumption. W(alpha) convergence is proved for every alpha>1 and the smooth seed is constructed. Only the explicit independent analytic inputs remain for the circle-method/sieve/headline estimates. The user is applying previous manuscript edits; no application is assumed and original sources are unchanged.","[Quantitative exceptional set] Almost all integers congruent to 12 modulo 24 can be written as \[a^2+b^2+p^2+1, \qquad a^2+b^2+1\in\mathcal P,\quad p\in\mathbb P. \] More precisely, if \[E_2(X)=\{1\leq n<X:n\equiv12\pmod{24},\ n\notin\mathcal P+\mathbb P^2\}, \] then, for every $A>0$, \[|E_2("
d:sparse,definition,2.9,14,2,Notation and preliminaries,1235,"formalized: General interval collections

with actual disjoint finite major subsets; eta=1/4 gives the preliminary definition, and eta=1/10 gives the introductory theorem convention. The finite-family specialization is proved equivalent.","A collection $\mathcal S$ of intervals is sparse if, for every $I\in\mathcal S$, there is $E_I\subset I$ such that $|E_I|\geq|I|/4$ and the sets E_I, $I\in\mathcal S$, are pairwise disjoint."
d:admiss,definition,2.10,14,2,Notation and preliminaries,1251,"formalized: The literal standalone linear-scale admissible dyadic stopping-map predicate is exported on actual finite indicator inputs. Lean proves equality with the displayed all-real-radius finite-count inequality and all convergence/boundedness prerequisites. The earlier actual prime-square spatial N² admissibility is also proved. The accepted fixed-radius and two-direction comparison remarks are also proved, including the exact factor two. The current full upload explicitly separates the fixed radius and the sufficiently large admissibility coefficient, matching the checked generic sparse theorem.","Let $f=\mathbf1_F$ be supported in $3I_0$. A map $\tau:I_0\longrightarrow\{2^j:1\leq2^j\leq|I_0|\}$ is admissible if, for every $x\in I_0$, \begin{align} \sup_{M\geq\tau(x)}\frac1M \sum_{|y-x|\leq CM}f(y) \leq C_{\rm adm}\frac{|F|}{|I_0|}. \end{align} where $C>1$ is a fixed radius constant and $C_{\"
l:cohenquadratic,lemma,3.1,17,3,Elementary number-theoretic estimates,1533,"proved: Complete positive-modulus identity, all local factors, CRT and bound.","[Cohen's identity with square shifts] Let $2^a\parallel q$. Then \begin{align} \sum_{r\in\mathbb A_q}c_q(n-r^2) =|\mu(q/2^a)| T_a(n) \prod_{\substack{p\mid q\\p>2}} \begin{cases} p+1,&\left(\frac np\right)=1,\\ 1-p,&\left(\frac np\right)\ne1, \end{cases} \end{align} where \begin{align} T_a(n)= \begi"
anonymous-s3-line1636,definition,,,3,Elementary number-theoretic estimates,1636,open,"[Exponential sums along squares] For a Dirichlet character χ modulo q, put \begin{align} \mathfrak s_\chi(a,q) =\frac1{\phi(q)}\sum_{r\in\mathbb A_q} \chi(r)e\left(-\frac{ar^2}{q}\right). \end{align} For the principal character we write \begin{align} \mathfrak s(a,q) =\frac1{\phi(q)}\sum_{r\i"
anonymous-s3-line1651,definition,,,3,Elementary number-theoretic estimates,1651,open,"Let $g=(q,y)$ and $(b,g)=1$. Define \begin{align} \Upsilon(a,q,b,y)=\frac{\phi(y)}{\phi([y,q])} \sum_{\substack{r\in\mathbb A_q\\r\equiv b\pmod g}} e\left(-\frac{ar}{q}\right). \end{align}"
l:progressioncohen,lemma,3.11,19,3,Elementary number-theoretic estimates,1660,"proved: Complete identity, including the shared-prime vanishing case.","Let $g\mid q$ and $(b,g)=1$. Then \begin{align} \sum_{\substack{r\in\mathbb A_q\\r\equiv b\pmod g}}c_q(n-r) = \begin{cases} 0,&(g,q/g)>1,\\ \mu(q/g)c_{q/g}(n)c_g(n-b),&(g,q/g)=1. \end{cases} \end{align}"
l:sumoverramanujan,lemma,3.12,19,3,Elementary number-theoretic estimates,1698,"proved: Actual normalized sums, squarefree vanishing and prime-power factorization.","Let $g=(q,y)$ and $(b,g)=1$. Then \begin{align} &\frac{\phi(q)\phi([y,q])}{\phi(y)} \sum_{a\in\mathbb A_q}\Upsilon(a,q,b,y) \mathfrak s(a,q)e\left(\frac{ax}{q}\right)\notag\\ &\qquad=\mu(q/g)\mathbf1_{(g,q/g)=1} \sum_{u\in\mathbb A_q}c_{q/g}(x-u^2)c_g(x-b-u^2). \end{align} In particular, the right-h"
l:motamakemajorprimesquareprog,lemma,3.14,19,3,Elementary number-theoretic estimates,1737,"proved: Accepted revision: Q>2, positive k, and p<Q for the obstruction. The submitted revision is preserved.","Let $Q>2$, let $k\geq1$ be an integer, and let $x,b\in\mathbb Z$ satisfy $(b,k)=1$. Then \begin{align} &\sum_{P^+(q)<Q}\sum_{a\in\mathbb A_q} \Upsilon(a,q,b,k)\mathfrak s(a,q)e(ax/q)\notag\\ &=\mathcal T_2 \prod_{\substack{2<p<Q\\p\mid k\\(\frac{x-b}{p})=1}} \left(1+\frac{p+1}{p-1}\right) \prod_{\su"
c:generalmajorprimesquareprog,corollary,3.16,20,3,Elementary number-theoretic estimates,1817,"proved: Accepted Q>2 and positive-k ranges, arbitrary integer multiplier, actual local counts and dyadic factor.","Let $Q>2$, let $k\geq1$ be an integer, and let $\ell,n,b\in\mathbb Z$ satisfy $(b,k)=1$. Then \begin{align} &\sum_{P^+(q)<Q}\sum_{a\in\mathbb A_q} \Upsilon(a\ell,q,b,k)\mathfrak s(a,q)e(an/q)\notag\\ &\qquad= \mathcal T_2(\ell,k,n;b) \prod_{2<p<Q} \frac{p\phi((p,k))}{(p-1)^2} \left|\left\{(u,v)\in\m"
l:lowramanujan,lemma,3.20,21,3,Elementary number-theoretic estimates,1916,proved: All real Q>=3 and integer n; an elementary prime-product argument supplies the bound.,"For $Q\geq3$ and every integer n, \begin{align} \sum_{P^+(q)<Q}\frac1{q\phi(q)} \left|\sum_{r\in\mathbb A_q}c_q(n-r^2)\right| \ll\log Q. \end{align}"
l:inftyboundsaq,lemma,3.21,21,3,Elementary number-theoretic estimates,1943,"conditional: Complete implication for 2+o(1), uniform epsilon and eventual coefficient-three bounds. External PNT limits and the bridge from an imported PNT are not proved here. Under the agreed external-input boundary, the paper-specific deduction is complete.","If $(a,q)=1$, then \begin{align} |\mathfrak s(a,q)| \leq q^{-1/2} \exp\left((2+o(1))\frac{(\log q)^{1/2}}{\log\log q}\right). \end{align} In particular, the right-hand side may be replaced by $q^{-1/2}\exp(3(\log q)^{1/2}/\log\log q)$ for all sufficiently large q."
l:Gcharbound,lemma,3.24,22,3,Elementary number-theoretic estimates,1993,"proved: Complete statement, strengthened to all complex Dirichlet characters.","Let \(\chi\) be a real Dirichlet character modulo \(q\), and let \((a,q)=1\). Then, for every \(\varepsilon>0\), \begin{align} |\mathfrak s_\chi(a,q)|\ll_\varepsilon q^{-1/2+\varepsilon}. \end{align} Equivalently, \(|\mathfrak s_\chi(a,q)|\ll q^{-1/2+o(1)}\)."
l:exceptionalarithmetic,lemma,3.27,23,3,Elementary number-theoretic estimates,2075,"proved: All three bounds, logarithmic consequence and relative support; strengthened to all real base characters.","Let \(x\in\mathbb Z\). Let χ_0 be a primitive real character of conductor q_0, and for each multiple q of q_0 let χ_q denote the character modulo q induced by χ_0. For \(0\leq j\leq5\), let \(\chi_{2^jq_0}\) have the same meaning, and set \[\sigma_0:= \max_{0\leq j\leq5} \sup"
t:improvinggeneral,theorem,4.1,25,4,Proof of the improving inequalities,2330,proved: Complete abstract implication from the hypotheses printed in the source. Verifying those hypotheses for prime squares is a separate obligation.,"Let A_N and B_N be positive convolution operators whose kernels are supported in an interval of length $O(N)$. Suppose that \begin{align} |A_Nf(x)|&\leq(\log N)^aB_N|f|(x),\\ |B_Nf(x)|&\ll N^{-v}\|f\mathbf1_{[x-CN,x+CN]}\|_1, \qquad 0\leq v<1, \end{align} and that $s\geq1-v$, $r>0$, and, wheneve"
c:improvinggeneral,corollary,4.15,27,4,Proof of the improving inequalities,2463,"proved: Complete strong lp-to-lp statement, including p=2 and the source operator hypotheses.","Assume in addition that A_N and B_N are uniformly bounded on ℓ^2. If \begin{align*} \frac{2(r+s)}{2r+s}<p\leq2, \end{align*} then, for $f\in \ell^p(\mathbb{Z})$, \begin{align} \|B_Nf\|_{p'}+\|A_Nf\|_{p'} \ll_p N^{1/p'-1/p}\|f\|_p. \end{align}"
p:decompositionofan,proposition,4.19,28,4,Proof of the improving inequalities,2541,"conditional: All displayed clauses for the actual common selected source low/high/error family: exact all-input annular decomposition, actual l1-to-l-infinity low norm, local bound for all l1 inputs with the supremum over every real M>=N, sharp fixed dyadic and actual countable maximal high bounds for every real Q>=2, every allowed starting threshold, every error exponent A>0, bounded Q and the exact printed high-symbol identity. One cutoff coefficient is chosen before epsilon and A; the operator family never changes with these parameters. The implication is complete under PrimeAsymptoticInputs, PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. Their independent analytic truths and compatible raw-data existence remain external. No own-paper decomposition, low bound, approximation error or maximal conclusion is an input. The current corrected numbered paper targets are all matched; independent inputs and stronger unnumbered alternative-route assertions retain their documented scopes.","Let \(N\) be dyadic, let \(Q\geq2\), and suppose that \["

\log Q\leq c_0(\log N)^{1/2}, \] where \(c_0>0\) is a sufficiently small absolute constant. Then \begin{align} \widetilde A_{N,\Lambda} = \operatorname{Low}_{Q,N} +\operatorname{High}_{Q,N}+E_{Q,N}, \end{align} where \begin{align} \|\operatorn"

l:localsparse,lemma,5.1,30,5,Proof of the sparse bounds,2721,"proved: Complete abstract implication with the exact source exponents, all 0<=v<1, s>=1-v, r>0, gamma>=beta>=0 and A>1/2, arbitrary physical dyadic stopping times, actual localized low/high/error outputs, all-real-radius admissibility, arbitrary fixed positive radii, parent-length normalization, empty sets and the full bounded density range. The error symbol may be an arbitrary almost-everywhere bounded function; its operator norm follows from the proved Fourier isometry. The normal decomposition and norm estimates are the printed hypotheses, not extra deep inputs.","Fix constants \(C\geq1\) and \(c>2C+1\). Let \((B_M)\) be a family of positive convolution operators satisfying \begin{align} |B_M f(x)| \ll M^{-v}\|f\mathbf1_{[x-CM,x+CM]}\|_1, \qquad 0\leq v<1. \end{align} Let \(s\geq1-v\), \(r>0\), and \(\gamma\geq\beta\geq0\), and suppose that, whenever \(M\) is"

l:sparserecursion,lemma,5.10,31,5,Proof of the sparse bounds,2905,"proved: The actual prime-square finite stopping construction, disjoint components, real-radius admissibility, exact kernel localization, terminating cost recursion and actual sparse major subsets are proved, with a proved all-cutoff reduction for the original averages. A general finite recursive-cost implication is also proved. The complete general dyadic implication is now proved for arbitrary positive compact-support kernels, every fixed odd support dilation, arbitrary Psi, the full untruncated maximum and genuine quarter-sparse major sets. The uniform bound and finite approximation of the full maximum are deduced from the local hypothesis. The local hypothesis is imposed only on maps admissible on the whole interval; an extension argument proves the needed restricted version. No own-paper deduction remains under the accepted dyadic and independent-admissibility-constant conventions. The proof constructs actual sparse sets and the full maximum from the stated local hypothesis; no additional analytic estimate is assumed.","Suppose that every admissible τ satisfies \begin{align} |\langle T_\tau\mathbf1_F,\mathbf1_G\rangle| \ll |I_0|\,\Psi\left(\frac{|F|}{|I_0|}, \frac{|G|}{|I_0|}\right) \end{align} for $F\subset cI_0$ and $G\subset I_0$, where $(T_N)_{N\in2^{\mathbb N}}$ is a family of positive convolution operato"

l:primeendpointlocal,lemma,5.17,33,5,Proof of the sparse bounds,3047,"conditional: Complete source implication for the actual annular von Mangoldt square average: every fixed real support dilation >=1, every positive spatial radius, every fixed admissibility constant, physical dyadic stopping values including scale1, literal all-real-radius quadratic averages, empty sets and the exact parent-length normalization. The bound holds for every cstar>4sqrt(3)/9. Proposition4.19 is derived from the already-listed independent inputs in the terminal theorem, rather than assumed as an extra own-paper input.","Let $f=\mathbf1_F$ and $g=\mathbf1_G$, where $F\subset cI_0$ and $G\subset I_0$, and put $\delta_0=\frac{|F||G|}{|I_0|^2}.$ Let $\tau:I_0\to\{2^k:(2^k)^2\leq|I_0|\}$ be admissible at the spatial scale $\tau(x)^2$, in the sense that \begin{align} \sup_{M\geq\tau(x)}M^{-2} \sum_{|y-x|\leq CM^2}f(y) \l"

l:prime-square-input,lemma,6.5,36,6,The high--low estimates,3322,"conditional: Complete corrected source lemma, with one genuine optional Page datum per N selected before Q, all q<=R^(11/10), the exact induced character/selector/exceptional model and inverse-fifth major error, and the full complementary R^(-1/4)*(log N)^d minor bound. The source Q-range, uniform sufficiently-large-N threshold, R^4<=N/2 and log N>=1 are proved. The general fixed-level uniform Page-prefix theorem and raw Ren/Ghosh estimates remain explicit independent unproved hypotheses.","If \((a,q)=1\), \(q\leq R^{11/10}\), and \(|\theta|\leq R/(qN^2)\), then \begin{align} \widehat{\widetilde A}_{N,\Lambda}(a/q+\theta) &=\mathfrak s(a,q)\widehat{\widetilde M}_N(\theta) -\mathbf1_{q_0\mid q}\mathfrak s_{\chi_q}(a,q) \widehat{\widetilde M}_{N,\beta}(\theta) +O(R^{-5}). \end{align} Her"

l:aaplemma,lemma,6.9,37,6,The high--low estimates,3466,"conditional: Complete printed approximation lemma for the literal selected source error family, every real Q>=2 and every integer N in the Section6 source range. One positive source cutoff coefficient is chosen before all A>0; the exact Q^(-1/2)*(1+log N)^(-A) operator and all-l2 function bounds are proved. The optional Page data and radius are the same actual selected family as Proposition4.19. The previously explicit compatible raw distribution/Ren/Ghosh data remain unproved general inputs; no new own-paper estimate is assumed. No remaining source clause within the agreed conditional scope.","[Approximation] For every \(A>0\), \begin{align} \|E_{Q,N}f\|_2 \ll_A Q^{-1/2}(1+\log N)^{-A}\|f\|_2. \end{align}"

p:lowcontribution,proposition,6.14,39,6,The high--low estimates,3789,"conditional: Complete printed low proposition for the actual selected source low operator: every real Q>=2, every integer N in the common source range, one absolute positive constant for the literal all-real-radius local bound and the l1-to-linfinity norm bound. The local result holds for every complex l1 input using |f| and hence the stated nonnegative inputs, with fixed spatial constant1. Ordinary and induced exceptional terms use the complete conductor range and exact source cutoffs. Only the previously recorded compatible source-data boundary remains; no new mathematical input is added. No remaining source clause within the agreed conditional scope.","For nonnegative \(f\), \begin{align} |\operatorname{Low}_{Q,N}f(x)| \ll Q\log(2Q) \sup_{M\geq N}M^{-2} \sum_{|y-x|\leq CM^2}f(y). \end{align} Consequently, \[\|\operatorname{Low}_{Q,N}f\|_\infty \ll N^{-2}Q\log(2Q)\|f\|_1. \]"

l:notBourgain,lemma,6.16,41,6,The high--low estimates,4009,"conditional: The full actual annular source maximum with the printed first power of (1+s) is proved under the precise independent rational smooth multifrequency/circle-transference input and the previously recorded general smooth input. Every source dyadic scale N>=2^s, every s>=0, scale1 and the small denominator indices are included. The rational common period, linear cost, weight localization, actual annular square-function error and all earlier scale costs are proved. Beta may vary with the scale, a stronger conclusion than the source fixed-beta statement. No own annular deduction remains. The independent smooth multifrequency theorems and sampling remain explicit unproved inputs. The proof uses the rational-frequency refinement rather than replacing the general squared-log loss by a first power; a matching expository replacement proof is saved separately.","[Multifrequency estimate] Let \(0<\beta\leq1\), and let \(\mathcal R_s\) be any subset of the reduced fractions with denominators \(q\sim2^s\). Then \begin{align} \left\| \sup_{\substack{N\in2^{\mathbb N}\\N\gtrsim2^s}} \left| \left(\sum_{a/q\in\mathcal R_s} c_{a,q}\widehat{\widetilde M}_{N,\beta}("

c:orlicz-prefix,corollary,6.19,42,6,The high--low estimates,4081,"conditional: The full source bound with square (1+s)^2 loss is proved for all s>=0, all source dyadic times including scale1, every real denominator cutoff simultaneously and every integer L2 input. The additional fixed bounded measurable multiplier clause is included, with actual essentially bounded Fourier operators and no continuity restriction. Beta may also vary with the scale. No own corollary deduction remains. The independent rational smooth/circle-transference theorem and the prior general smooth input remain explicit unproved hypotheses; the full additional fixed measurable multiplier clause is proved, not added as an external input.","Let \(0<\beta\leq1\), \(s\geq0\), and \(|c_{a,q}|\leq b\). Then \begin{align} &\left\| \sup_{\substack{N\in2^{\mathbb N}\\N\geq2^s}} \sup_K \left| \left(\sum_{\substack{q\sim2^s\\q\leq K}} \sum_{a\in\mathbb A_q} c_{a,q} \widehat{\widetilde M}_{N,\beta}(\xi-a/q) \eta_s(\xi-a/q)\widehat f(\xi) \right"

p:highupperbound,proposition,6.21,43,6,The high--low estimates,4143,"conditional: Both source clauses are complete checked implications: the actual countable dyadic supremum and every individual output, all real Q>=2 and N0>=2 in the allowed logQ<=c0 sqrt(logN0) range, one positive c0 independent of epsilon,

the exact source radius and complementary narrow-cutoff multiplier, changing genuine exceptional data, and the leading coefficient 2+o(1) in precise epsilon form. A separate uniform estimate covers bounded Q; its harmless regularization agrees with sqrt(logQ)/loglogQ once logQ>=2. No paper-specific deduction in this proposition remains. The two prime asymptotic limits, the general smooth multifrequency/sampling input, raw Landau-Page uniqueness (including the L-function bridge), and the existence/selection of source-compatible exceptional zero data remain explicit external boundaries. The stronger auxiliary statements 6.16 and 6.19 are not awarded completion.","Let $N_0\geq2$ and suppose that $\log Q\leq c_0(\log N_0)^{1/2}.$ Then, for every dyadic $N\geq N_0$, \begin{align} \|\operatorname{High}_{Q,N}f\|_2 &\ll Q^{-1/2} \exp\left((2+o(1))\frac{\sqrt{\log Q}}{\log\log Q} \right)\|f\|_2, \\ \left\| \sup_{\substack{M\geq N_0\\M\in2^{\mathbb N}}} |\operatorn"

t:orlicz-maximal,theorem,7.2,46,7,Quantitative maximal estimates near \(L^1\),4494,"conditional: Complete printed Theorem 7.2 as an implication of explicit independent standard inputs: all three actual families (von Mangoldt squares, ordinary squares, unweighted prime squares), every integer cutoff N >= 2, every complex finitely supported function and positive level, with common absolute C, C0 and K >= 1. All three conclusions are proved, including finite level sets: the truncated modular bound, unrestricted modular bound and restricted bound for every finite F and t >= 1. The final theorem is stated with literal sums over all integers and finite support, using the actual integral gauge. The ordinary branch derives the actual discrete circle-model error, maximal comparison, low/high restricted bound and full-cutoff reconstruction from the independent scalar Weyl input and general IW inputs; no own restricted or high/low conclusion is assumed. The unweighted-prime transfer uses proved Mathlib Chebyshev bounds without a new PNT input. At checkpoint1264 the current public export consumes the proved finite signed maximal theorem at constant6; a supplied RawFiniteSignedMaximalInput and its constant R are no longer hypotheses. No remaining clause of Theorem7.2 or Corollaries7.6/7.46 within the agreed conditional scope. The explicitly recorded independent inputs and their published-theorem bridges remain unformalized. Stronger standalone auxiliary parameter/full-generality ranges and later badness/sieve results remain separately tracked.","[Orlicz maximal estimate] There are absolute constants \(C,C_0\geq1\) such that the following holds. Choose \(A_N\) from \(A_{N,\Lambda}\), \(A_{N,\mathbb N^2}\), and \(A_{N,\mathbb P^2}\), and put \(A^*f=\sup_{N\geq2}|A_Nf|\). Then, for every finitely supported \(f:\mathbb Z\to\mathbb C\) and every"

c:orlicz-p,corollary,7.6,46,7,Quantitative maximal estimates near \(L^1\),4528,"conditional: Complete printed Corollary 7.6 as an implication of the already recorded Theorem 7.2 independent inputs. One absolute D > 0 works for all three actual families, every 1 < p <= 2 and every integer lp input. The literal all-integer-cutoff maximal function is proved to belong to lp with norm at most exp(D/(p-1)*log(exp(1)/(p-1))) times the input norm. The actual truncated modular estimate, general layer cake, singular power integral, quantitative constant absorption, p-th-root conversion and all-input finite/Fatou extension are proved. At checkpoint1264 the current public export consumes the proved finite signed maximal theorem at constant6; a supplied RawFiniteSignedMaximalInput and its constant R are no longer hypotheses. No remaining source clause within the agreed conditional scope; no new strong maximal or layer-cake input is introduced. Theorem7.2's independent hypotheses remain explicit. The ergodic Corollary7.46 and Theorem1.5(ii) are now separately completed under their reviewed independent inputs.","For \(1<p\leq2\) and \(f\in\ell^p(\mathbb Z)\), \begin{align} \|A^*f\|_{\ell^p} \leq \exp\left(\frac{C}{p-1}\log\frac{e}{p-1} \right)\|f\|_{\ell^p}, \end{align} after enlarging the absolute constant \(C\)."

p:IWinput7,proposition,7.8,47,7,Quantitative maximal estimates near \(L^1\),4598,"conditional: All four source clauses are complete conditional implications for one common construction at every real Q>=2^k, r>=2 and k>=4. The literal B, epsilon and C_rk are retained; P<=3^Q, every a/q modulo one with q<=Q, all denominator bounds and full goodness are proved. The rough multiplier conclusion is an actual bounded operator on the entire integer lp space, with the exact finite-convolution Fourier kernel, uniqueness and norm <=C_rk times the actual full real Lp operator norm. The general full-Lp real/cyclic product bound and rank-one coefficient factorization are proved. Both major-arc sampling directions, actual unique interpolation on the full distributional integer lp Fourier-support class, the inverse-C_rk source inequality, and all finite-dimensional normed complex Shannon targets throughout 1<=p<=infinity are retained. Point sampling uses the standard continuous representative convention. General Tao construction, full adelic IW and full sampling theorems, including the explicitly described rough/full-class/finite-quotient realization bridges, remain independent unproved inputs. The earlier narrower endpoint results retain their own hypotheses.","Let \(r,k\in\mathbb N\), with \(r\geq2\) and \(k\geq4\), and let \(Q\geq2^k\). Put \begin{equation} \begin{gathered} p_r=\frac{2r}{2r-1}, \qquad B=C_*^{\,k^2Q^{1/(k-1/2)}},\\ \varepsilon=(2^{20}rB^{2r+8})^{-1}, \qquad C_{r,k}=(C_*r\log(2+k))^{C_*k}, \end{gathered} \end{equation} where \(C_*>1\) is a"

l:continuousestimates7,lemma,7.16,49,7,Quantitative maximal estimates near \(L^1\),4838,"conditional: Complete printed continuous lemma on the full Lp domain. One absolute positive constant is chosen before beta, p, J, signs and f. The actual signed-increment kernel has exactly the source multiplier and a full-Lp bound for every1<p<=2, arbitrary complex signs of modulus<=1 and every0<beta<=1. The maximum is over every real N>=2, represented as a proper extended-real supremum; a fixed countable dense set and proved scale continuity establish measurability, with a common almost-everywhere bound, finiteness and actual Lp membership. The literal real-scale kernel/Fourier identity and all annular L2/Hormander hypotheses are proved. The general one-sided Hardy–Littlewood and scalar Hormander theorems in their full-Lp forms remain unproved explicit independent inputs. No remaining source-specific clause within the agreed conditional scope. The full-domain general theorem interfaces and their exact published-to-Lean formulations remain external; they are not claimed to have been derived from the earlier restricted-domain inputs.","Uniformly in \(0<\beta\leq1\), \(J\geq1\), and \(|\sigma_j|\leq1\), one has, for \(1<p\leq2\), \begin{align} \left\| T_{\sum_{j=1}^J\sigma_j (\widehat{\widetilde M}_{2^j,\beta} -\widehat{\widetilde M}_{2^{j-1},\beta})}f \right\|_{L^p(\mathbb R)} &\ll \frac1{p-1}\|f\|_{L^p(\mathbb R)}, \\ \left\| \su"

p:orlicz-low,proposition,7.22,50,7,Quantitative maximal estimates near \(L^1\),5006,"conditional: The full actual dyadic IW low maximum is proved at every real Q>=2^k, uniformly in the cyclic measure of variation at most one and 0<beta<=1, for the real-Q geometric construction and for all integer lp inputs at p_r. The width, denominator bound and logarithmic cost are the literal real-Q quantities. A single harmless enlargement of the universal parameter constant from C to C^2 is made once. The arithmetic set, period, rational coverage through Q, denominator bound and goodness are retained; the maximal estimate uses exactly that set, period and real-Q cutoff. The adjusted-base proof and cost comparison are checked, as are all own symbol, smoothness, sampling, short-scale, narrow, remainder and all-input extension steps. The independent general standard inputs remain explicit; no current-paper conclusion is an external assumption. No own-paper deduction remains for this proposition. The recorded independent general construction, IW multiplier, sampling, continuous and signed-maximal theorems are explicit unformalized inputs, not proved Lean axioms.","Let \(r,k,Q,P,\Sigma_{\leq k},B,\varepsilon,p_r\), and \(C_{r,k}\) be as in Proposition~\ref{p:IWinput7}. Let \(\nu\) be a complex measure on \(\mathbb Z_P\) with \(\|\nu\|_{\mathrm{TV}}\leq1\), and let \(0<\beta\leq1\). Define \(L_N^{\nu,\beta}\) by \begin{align} \widehat{L_N^{\nu,\beta}}(\xi) = \s"

p:orlicz-hilo,proposition,7.30,53,7,Quantitative maximal estimates near \(L^1\),5322,"conditional: The complete real-Q high-low proposition is proved as an implication from the recorded independent standard

inputs. For every real Q>=2^k, one unchanged floor-Q arithmetic construction supplies both prime and ordinary-square decompositions at the literal real-Q cutoff. It retains period<=3^Q, rational coverage through Q, the literal denominator bound and goodness. Both actual countable low bounds hold on all integer lp inputs at p_r, with the exact logarithmic real-Q cost. Both actual high maxima share the same arithmetic set, period and selected Page datum and have a proved summed Q^(-1/4) L2 bound above the stated exponential starting scale. All operator identities and applicability are checked. The fixed absolute parameter constant is enlarged once from C to C^2; no width is rounded and no own-paper endpoint is an external premise. No own-paper deduction remains for this proposition. The recorded independent general construction, multiplier/sampling, continuous/signed-maximal, smooth multifrequency, Page/prime-distribution and scalar quadratic-Weyl inputs remain explicit and unproved. Their hypotheses are not Lean axioms or assertions of truth.","Using the assumptions and notation of Propositions~\ref{p:IWinput7} and \ref{p:orlicz-low}, put \begin{equation} D(r,k,Q)=rC_{r,k}^2\log(e+Q+r\log B). \end{equation} For dyadic \(N\geq N_0\), there are decompositions \[\widetilde A_{N,\Lambda}=L_N+H_N, \qquad \widetilde A_{N,\mathbb N^2}=L_N^\squar"

c:orlicz-ergodic,corollary,7.46,59,7,Quantitative maximal estimates near \(L^1\),6002,"conditional: Complete printed Corollary7.46 as an implication of the explicit independent inputs, for all three actual families and every invertible measure-preserving probability system. One common C >= 1 serves the strong exponential rate and source Orlicz gauge; one common modular multiplier is fixed before all systems, families, functions and levels. Every 1 < p <= 2 has the actual strong maximal norm with almost-everywhere finiteness and actual Lp membership. The proper extended maximum satisfies the source modular inequality. Every measurable Orlicz member at some positive scale has actual averages converging almost everywhere and in L1 to the same genuine L1 limit, with integrability of every average and both L1 seminorm/integral-norm convergence checked. Own truncation, rescaling, Cauchy/density, L1 operator and limit-identification deductions are proved. At checkpoint1264 the current public export consumes the proved finite signed maximal theorem at constant6; a supplied RawFiniteSignedMaximalInput and its constant R are no longer hypotheses. No remaining printed clause within the agreed conditional-formalization scope. The independent general Calderon strong/modular and bounded-class polynomial convergence inputs, including their unproved published-theorem/normalization bridges, remain explicit, as do the earlier Theorem7.2 independent inputs. Stronger standalone Section7 auxiliary ranges remain separately partial.","For \(1<p\leq2\), \begin{align} \|A_T^*f\|_{L^p(X)} \leq \exp\left(\frac{C}{p-1}\log\frac{e}{p-1} \right)\|f\|_{L^p(X)}. \end{align} Moreover, \begin{align} \mu\{A_T^*f>\lambda\} \ll \int_X\Psi_C\left(\frac{|f|}{\lambda}\right)\,d\mu, \qquad \lambda>0. \end{align} If $ \int_X\Psi_C(|f|/a)\,d\mu<\in"

t:l1badblackbox,theorem,8.1,60,8,Persistent universal \(L^1\)-badness of prime squares,6128,"conditional: Complete persistent-divergence adaptation from original general LaVictoire weak maximal failure and separately specified Sawyer and Conze principles. The arithmetic record contains actual cyclic residue sets, CRT, represented classes, odd coprime moduli, exact density infimum, little-o thickening, far-pair liminf and strict finite-product residue frequencies. For every nonatomic ergodic probability system and every infinite cutoff set, Lean produces a measurable integrable function and a measurable positive-measure set with no finite pointwise limit. Nonatomicity means measurable splitting, not merely null singletons. Positive moduli include the usual modulus-one convention; prime and cutoff indices are zero-based enumerations of the source positive-index sequences. No remaining persistent-divergence deduction within the agreed conditional scope. The original LaVictoire, Sawyer and Conze inputs and their precise published-to-Lean convention bridges are unproved.","Let (n_k) be an increasing sequence of natural numbers. Suppose that, for every $q>1$, there is a set $\Lambda_q\subset\{n_k\bmod q:k\geq1\}$ which is multiplicative under the Chinese remainder theorem: if $(q,r)=1$, then Λ_{qr} corresponds to $\Lambda_q\times\Lambda_r$. Suppose also that "

t:grankul,theorem,8.6,60,8,Persistent universal \(L^1\)-badness of prime squares,6170,"external_candidate: Quoted general deep theorem remains unproved. A precise interface now specifies arbitrary CRT residue families, uniform actual local correlations, one positive common exponent, constants chosen before primes/shifts, subpower reciprocal local density and growing spacing along odd squarefree moduli. The full joint cyclic gap conclusion is implemented. All its unit-square hypotheses and the actual joint and single-gap applications are proved conditional on that independent general input and Weil. The quoted general GK theorem and its published-to-Lean cyclic joint formulation bridge remain external and unproved. It is not needed for the repaired Theorem1.5(i) proof.","[Granville--Kurlberg \cite{GK08}] For each odd prime p, let $\Omega_p\subset\mathbb Z/p\mathbb Z$, put $r_p=|\Omega_p|/p$ and $s_p=r_p^{-1}$, and define Ω_q from the Ω_p, for squarefree q, by the Chinese remainder theorem. Suppose that, for every fixed $k\geq1$ and all distinct $0,"

l:quadratic-GK,lemma,8.8,61,8,Persistent universal \(L^1\)-badness of prime squares,6190,"conditional: Complete printed quadratic-residue correlation lemma as an implication of RawSplitQuadraticWeilInput, the general distinct-linear-factor character-sum bound. The literal finite residue correlation has explicit absolute error at most2*k*sqrt(p), actual density-normalized relative error at most3*k*3^k/sqrt(p), and the source GK error with c=1/3 and constant6*k*3^k. Quantification is uniform over every odd prime and every injective tuple of k shifts, including the source h0=0 case. All zero corrections, density factors, positivity, finite subset sums and reciprocal local-density subpower bounds are proved. No remaining printed correlation clause in the agreed conditional scope. The general Weil input remains unproved. The separate moment repair and badness deductions are now also complete; their inputs are separately recorded.","Let p be an odd prime and let \[\Lambda_p=\{x^2\pmod p:x\not\equiv0\pmod p\}. \] For every fixed $k\geq1$ and distinct $0,h_1,\ldots,h_{k-1}\pmod p$, \[N_k(\mathbf h,\Lambda_p)=\frac{p}{2^k}+O_k(p^{1/2}) =\left(\frac{p-1}{2p}\right)^kp\big(1+O_k(p^{-1/2})\big). \] Consequently \eqref{e:GK-hyp} h"

l:fromsiftedtop,lemma,9.2,63,9,Quantitative Goldbach-type estimate: reductions,6577,"proved: The complete first Section9 reduction is proved from its ordinary range hypotheses. The primitive sum-of-two-squares criterion is now a project theorem, not an additional input. Actual finite sifted and good representation sets and the uniform O(N^epsilon) exception are retained. At1262: Every Section9 numbered source claim and its final quantitative theorem is complete in the agreed scope. All elementary arithmetic reductions are now proved without an extra primitive two-square assumption. W(alpha) convergence is proved for every alpha>1 and the smooth seed is constructed. Only the explicit independent analytic inputs remain for the circle-method/sieve/headline estimates. The user is applying previous manuscript edits; no application is assumed and original sources are unchanged.","For every $\varepsilon>0$, \begin{align} |\{r\in\mathcal P:n-r\in\mathbb P^2\}| \geq S(\mathcal A_n,\mathbb P_{3,n},N^2)-O_\varepsilon(N^\varepsilon). \end{align}"

l:truncationT,lemma,9.3,64,9,Quantitative Goldbach-type estimate: reductions,6605,"proved: The full source truncation identity is proved with an explicit nonnegative error bounded by O(N^epsilon), including repeated3 mod4 prime factors and the actual source factorization set. The primitive criterion is internally proved; only the normal source range hypotheses remain. At1262: Every Section9 numbered source claim and its final quantitative theorem is complete in the agreed scope. All elementary arithmetic reductions are now proved without an extra primitive two-square assumption. W(alpha) convergence is proved for every alpha>1 and the smooth seed is constructed. Only the explicit

independent analytic inputs remain for the circle-method/sieve/headline estimates. The user is applying previous manuscript edits; no application is assumed and original sources are unchanged.","For every \(\varepsilon>0\), \begin{align} S(\mathcal A_n,\mathbb P_{3,n},z) - |\{r\in\mathcal P:n-r\in\mathbb P^2\}| = |T_n|+O_\varepsilon(N^\varepsilon), \end{align} where \begin{align} T_n= \left\{ r-1=2up_1p_2: \begin{array}{l} r\in\mathbb P,\quad n-r\in\mathbb P^2,\\ z\leq p_2\leq p_1,\quad p_i"
l:upperreduction,lemma,9.7,65,9,Quantitative Goldbach-type estimate: reductions,6706,"proved: For the user-corrected closed-cutoff L, the entire actual upper-sifted inclusion, exact finite cardinal upper bound with error omega(n-1), and uniform O_epsilon(N^epsilon) bound are proved. No analytic, sieve, primitive two-square or own-paper endpoint is assumed. All eight closed cutoff occurrences are verified in the latest upload.","For every $\varepsilon>0$, \begin{align} |T_n| \leq\sum_{\ell\in L_n} S(M_n(\ell),W_{\ell,n},z_\ell)+O_\varepsilon(N^\varepsilon). \end{align}"
c:mastersievereduction,corollary,9.9,65,9,Quantitative Goldbach-type estimate: reductions,6758,"proved: The full source master sieve reduction is proved from ordinary source hypotheses and the adopted closed multiplier cutoff. Its exact finite error is at most2 omega(n-1), and its uniform O(N^epsilon) consequence is exported. No primitive-criterion or analytic assumption is needed for this arithmetic reduction. At1262: Every Section9 numbered source claim and its final quantitative theorem is complete in the agreed scope. All elementary arithmetic reductions are now proved without an extra primitive two-square assumption. W(alpha) convergence is proved for every alpha>1 and the smooth seed is constructed. Only the explicit independent analytic inputs remain for the circle-method/sieve/headline estimates. The user is applying previous manuscript edits; no application is assumed and original sources are unchanged.","For every $\varepsilon>0$, \begin{align} |\{r\in\mathcal P:n-r\in\mathbb P^2\}| &\geq S(\mathcal A_n,\mathbb P_{3,n},z)-\sum_{\ell\in L_n} S(M_n(\ell),W_{\ell,n},z_\ell) -O_\varepsilon(N^\varepsilon). \end{align}"
l:circmethod,lemma,9.10,65,9,Quantitative Goldbach-type estimate: reductions,6783,"conditional: Complete source circle-method mean-square approximation for every A,delta>0, with B0,C,K,N0 chosen before N and every arithmetic family. Covers all B>=B0, exact strict real L<N^(2-delta), positive ell,k with k<V(ell), every coprime residue class, arbitrary complex coefficients of norm<=1, and all n<N^2. The stronger uniform cutoff statement holds for every 2<=U<=N,W>=2. The final source theorem removes cutoffs using the proved equation-imposed bounds, and uses the actual finite prime-pair count, actual local Euler factor G_R and actual autocorrelation singular integral H_R. Fourier inversion, all three error identifications, shared constants, minor estimate, major replacement, arithmetic completion and smoothing are consumed internally. Its only unformalized hypotheses are independent ordinary SW prime counts, ordinary unweighted BV prime-count distribution, and uniform raw Ren/Ghosh quadratic estimates. No own-paper conclusion is assumed. At1212: Completed all Section10 source statements and the full circle-method implication; the three independent raw deep inputs remain explicit and unproved. At1213: Completed every Section10 source statement and the full source circle-method implication, including the original smooth compact seed and exact normalized integral. Only the three agreed independent raw deep inputs remain for this circle-method implication. None within the full source circle-method implication. The three independent raw distribution/exponential-sum inputs remain unproved; Section11 and the headline Goldbach result remain separate work.","[Bombieri--Vinogradov approximation] Let \(A,\delta>0\) and \(L_0<N^{2-\delta}\). There are sufficiently large constants \(B=B(A,\delta)\) and \(C=C(A,\delta)\) such that the following holds. Put \[R=(\log N)^C,\qquad \eta_N(\theta)=\eta\left(\frac{N^2\theta}{R}\right), \] and suppose that $V_\ell\"
p:sieveneedlower8,proposition,9.17,67,9,Quantitative Goldbach-type estimate: reductions,6898,"conditional: Full source lower-sieve proposition, including the actual representation sequence, original X_n and Euler product, coefficients before n, a common error tending to zero, and the N^4(log N)^(-A) mean-square remainder bound. The pointwise theorem holds throughout the source retained range (indeed all n=12 mod24). General classical sieve/Mertens/function-continuity and ordinary SW/BV/Ren-Ghosh inputs remain explicit and unproved. At1229: Common cutoff compatibility for the circle-method and full lower-sieve estimates is proved for every requested lower bound on C; main-term size remains to be joined. At1230: The full source main-term size and the entire lower-bound side are complete conditionally, including an exponent independent of C, uniform smoothing, one common large cutoff and finite error-to-exception deductions. At1230 the source main-term size and the full lower sieve are proved jointly for the same C>=A0+6; finite positive-count and exceptional-cardinality deductions are checked. At1258: All Section11 numbered claims are complete in the agreed conditional scope for the explicitly supplied corrected presieving statement. The original weaker statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction.","[Lower sieve] For $N^2(\log N)^{-A_0}\leq n<N^2$, $n\equiv12\pmod{24}$, \begin{align} S(\mathcal A_n,\mathbb P_{3,n},z) \geq X_n\Theta_n(z)\bigl(f_{1/2}(9/8)+o(1)\bigr) +\mathcal E_1(n), \end{align} where \begin{align} \sum_{n<N^2}|\mathcal E_1(n)|^2 \ll_{A_0,K} N^4(\log N)^{-K}. \end{align}"
p:sieveneedupper8,proposition,9.20,67,9,Quantitative Goldbach-type estimate: reductions,6922,"conditional: The full source upper sieve is complete in uniform epsilon form, with the exact c_(9/8) F_1(2) W(9/8) coefficient, original actual main term and closed multiplier set. All coefficients precede n; source divisor counts, refined preliminary/outer dimensions, uniform varying levels, the shifted double remainder sum, its circle-method embedding and the fixed-sieve O(N^(1/4)) correction are proved. The total error has arbitrarily strong logarithmic mean-square saving. Only independent general SW, BV, Ren/Ghosh, classical one-stage sieve, effective multiplicative mean and Mertens inputs and general sieve normalizations remain unproved; no own-paper endpoint is assumed. At1257: Full upper-sieve proposition p:sieveneedupper8 complete in uniform epsilon form from explicit independent classical inputs, with the exact coefficient and arbitrary logarithmic mean-square saving. The lower analytic estimate and both remainder transports are available at a shared prescribed cutoff. The final joint Section11 assembly and manuscript synchronization are next. At1258: All Section11 numbered claims are complete in the agreed conditional scope for the explicitly supplied corrected presieving statement. The original weaker statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction.","[Upper sieve] For $N^2(\log N)^{-A_0}\leq n<N^2$, $n\equiv12\pmod{24}$, \begin{align} \sum_{\ell\in L_n}S(M_n(\ell),W_{\ell,n},z_\ell) &\leq \bigl(c_{9/8}F_1(2)\mathcal W(9/8)+o(1)\bigr) X_n\Theta_n(z)+\mathcal E_2(n), \end{align} where \begin{align} c_\alpha=\frac{16e^{-\gamma/2}}{\sqrt{\pi\alpha}}"
l:maintermsize8,lemma,9.22,67,9,Quantitative Goldbach-type estimate: reductions,6951,"conditional: Full source lemma proved for every A0>0 and every fixed C>=A0+6, uniformly over the exact retained n range. The actual X_n Theta_n(z) is at least a positive constant times N(log N)^(-(A0/2+6)); the exponent is independent of C. Actual archimedean, smoothing and local-product arguments are proved; only the already recorded independent interval Mertens products remain external. The same large C is joined to the complete lower sieve. At1258: All Section11 numbered claims are complete in the agreed conditional scope for the explicitly supplied corrected presieving statement. The original weaker statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary combined logarithmic

mean-square saving. The final Goldbach theorem remains the next deduction.","Let \(A_0>0\). There is \(C_{A_0}>0\) such that, for every fixed sufficiently large \(C\), with \(R=(\log N)^C\), one has \begin{align} X_n\Theta_n(z)\gg_{A_0,C} N(\log N)^{-C_{A_0}} \end{align} whenever $n\equiv12\pmod{24}$ and $N^2(\log N)^{-A_0}\leq n<N^2$."
l:numericalcomparison,lemma,9.24,67,9,Quantitative Goldbach-type estimate: reductions,6963,"conditional: The actual W integral has the unconditional exact bound 8 sqrt(2) W(9/8)<=9/32, proved by positive interval integration and exact rpow primitives. The strict comparison is proved for the classical normalized sieve values f_(1/2)(9/8)=sqrt(8 exp(gamma)/(9 pi)) log2 and F_1(2)=exp(gamma). Identifying the generic independent sieve functions with those classical normalized values remains explicit, within GENERAL_SIEVE; the paper comparison is not assumed. No floating point evaluation or unchecked evaluator is used. At1253: Proved the exact actual W integral bound, the normalized half/linear sieve comparison and strict positivity. Classical sieve-function normalizations remain explicit general input identities. The upper-sieve application and final Goldbach theorem remain open; no paper endpoint is assumed. At1258: All Section11 numbered claims are complete in the agreed conditional scope for the explicitly supplied corrected presieving statement. The original weaker statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction.","With the normalization of \cite[(1.8)--(1.10)]{I80}, \begin{align} \Delta>0. \end{align}"
l:hybridminorarc,lemma,10.2,69,10,Quantitative Goldbach-type estimate: the circle method,7094,"conditional: Complete source hybrid minor estimate at1136: the actual circle integral and finite n<N² mean square, exact complementary source arcs, every sufficiently large radius exponent C, and a single natural threshold uniform in all coefficients, residues, real U and W and printed strict L₀/V_ell ranges. The theorem allows B≥0 and 0≤U≤N, with arbitrary W and residues, so it is stronger than the stated ranges. Explicit bound K N⁴(log N)^(-A), K=Z²(256/log 2), Z=(2P+3)/log 2, C₀=3A+4d+26. Finite Abel, prime-power removal, halving reconstruction, scale transfer, eventual cutoffs, source progression energy and Bessel are proved. Only the already recorded independent raw Ren/Ghosh input is assumed. The independent raw UniformRenGhoshInputs remains explicit and unproved. The source hybrid minor conclusion and all conversion steps are discharged; no internal own-paper hypothesis remains.","Under the hypotheses of Lemma~\ref{l:circmethod}, uniformly for \(2\leq U\leq N\) and \(2\leq W\leq N^2\), \begin{align} \sum_{n<N^2}\left| \int_{\mathfrak m}P_U(\theta) \sum_{\ell<L_0}\sum_{k<V_\ell} d_{k,\ell}F_{k,\ell,W}(\theta)e(-n\theta)\,d\theta \right|^2 \ll_{A,\delta}N^4(\log N)^{-A}. \end{a"
l:positiveH9,lemma,10.12,70,10,Quantitative Goldbach-type estimate: the circle method,7266,"proved: At1176: The actual normalized autocorrelation cutoff has a proved positive denominator, exact convolution integral, eta(0)=1 and real nonnegative inverse Fourier transform. At1177: The scaled actual eta_N and its square have proved real nonnegative inverse Fourier transforms and exact compact support inherited from the seed. At1178: The actual eta is real, even, bounded by1, has zero derivative at0 and satisfies the full global quadratic source cutoff bound. At1179: Two-sided absolute integrability and an explicit L1 estimate are proved from the exact finite-scale bound and inverse-three-halves tail. At1180: The exact cutoff-weighted Fourier integral is a proved positive average of the inverse kernel, with Fubini integrability discharged. At1181: The actual source singular integral is proved real nonnegative, uniformly in all positive ell and every real n, including the empty-domain branch. At1182: The actual Euclidean product and smoothed integrand are globally integrable, and the exact source singular integral satisfies a proved uniform O(N) bound for all positive ell. At1183: The full quantitative real-line smoothing error bound is proved with absolute constant6 and every integrability condition discharged. At1184: The actual smoothed and unsmoothed source integrals satisfy the full uniform O(N/sqrt(R)) comparison, with all integrability and empty-domain branches proved. At1185: The actual quadratic affine density is globally integrable and has the proved exact product Fourier transform, including the scale factor. At1186: The genuine logarithmic quadratic density is continuous and integrable, with its exact source product Fourier transform proved and endpoint exceptional sets checked. At1187: The full source l:positiveH9 is proved: the exact autocorrelation cutoff, real nonnegative smoothed singular integral, its exact printed one-variable H_ell domain, and the uniform O(N/sqrt(R)) comparison, including all positive ell and empty-domain cases. There is no independent analytic input for this lemma. At1213: Completed every Section10 source statement and the full source circle-method implication, including the original smooth compact seed and exact normalized integral. Only the three agreed independent raw deep inputs remain for this circle-method implication. None. Positivity and the comparison with the exact printed one-variable domain are proved for the actual source cutoff and integrals, without extra external analytic input.","Uniformly in \(\ell,U,W\) and \(n<N^2\), \[H_{\ell,R}(n;U,W)\geq0, \qquad H_{\ell,R}(n;U,W) = H_\ell(n;U,W)+O(NR^{-1/2}), \] where \[H_\ell(n;U,W) = \int_{\substack{2\leq u\leq U\\ 2\leq(n-u^2)/\ell\leq W}} \frac{du}{\log u\log((n-u^2)/\ell)}. \]"
l:majorapproximation9,lemma,10.15,71,10,Quantitative Goldbach-type estimate: the circle method,7351,"conditional: At1126: The exact lcm-fibre divisor bound and weighted progression-error reindexing for the major arcs are proved. At1137: Exact strict-cutoff reduction and a uniform one-term endpoint error are proved for the actual progression sums. At1138: The exact prime counting and logarithmic-integral progression models, derivatives and lower-endpoint identities are proved. At1139: The actual prime-count discrepancy is transported through every smooth unit phase with an explicit quantitative error, including the lower prime endpoint. At1140: Exact positive-phase linear and quadratic partial-summation errors are proved from ordinary prime-counting discrepancies, with signs and totient normalizations checked. At1141: The actual prime-square major-arc model and explicit nonunit/count-discrepancy error are proved with the exact source sign and normalization. At1142: The fixed-q lcm fibre bound and complete reduction to ordinary unweighted modulus-distribution errors are proved. At1143: The compatible lcm residue sets, q-cardinality bound and generalized CRT existence/converse are proved for arbitrary positive moduli. At1144: The exact Upsilon coefficient for the actual compatible lcm residue classes is proved, including both signs and totient factors. At1145: The exact independent ordinary prime-count SW/BV boundaries are implemented; real-error equivalence, square-model numeric bound and logarithmic absorption are proved. At1146: The full uniform source SW prime-square major approximation is a complete deduction from the independent ordinary prime-count SW input. At1147: Exact prime progression residue fibres and positive rational-phase grouping are proved for the actual finite sums. At1148: The actual source linear model and all nonunit/strict-endpoint errors are proved from ordinary lcm-modulus prime-count discrepancies. At1149: The actual strict source J integral, clamped prime cutoff, and uniform numeric linear model error are matched exactly. At1150: Actual BV row approximation is proved under the independent ordinary unweighted BV input, with individual progression cutoffs and all finite exceptional terms retained. At1151: The printed source ranges imply the ordinary BV level and comparable logarithmic saving, with a threshold uniform in every active ell. At1152: The actual source row-scale sum and coefficient-family cardinality square bound are proved, including all empty-cutoff cases. At1153: Every retained prime endpoint/nonunit term is absorbed uniformly from the exact printed source ranges, using a proved power saving. At1154: The full uniform source BV progression-error norm sum is complete under the independent ordinary unweighted BV input, with all individual cutoffs and finite errors included. At1155: Exact source major-set geometry,

normalized measure, wrap-around lifts, pairwise disjointness and eventual separation are proved. At1156: Measurable local coordinates and genuine Euclidean model regularity and norm bounds are proved for the circle Bessel assembly. At1157: The exact source major models are constructed on the circle, with proved localization and all-Lp regularity and the required square size bound. At1158: Both full SW/BV major estimates now hold for the constructed exact source circle models, uniformly in every printed parameter. At1159: The restricted measurable Bessel inequalities and two-model major-arc mean-square assembly are proved. At1160: The actual weighted source coefficient identity, model Lp bounds and complete logarithmic mean-square budget are proved. At1161: The full source major-arc mean-square approximation l:majorapproximation9 is complete under ordinary SW and BV, uniformly in all printed data, with strict endpoints and finite errors discharged. No unmatched clause remains in this lemma. The independent ordinary SW and BV proofs remain external; apply docs/MAJOR_ARC_PROOF_REPLACEMENT.tex to correct the endpoint bookkeeping in its source proof.","Uniformly for \(2\leq U\leq N\) and \(2\leq W\leq N^2\), \begin{align} \sum_{n<N^2}\left| \sum_{\ell<L_0}\sum_{k<V_\ell}d_{k,\ell} \int_{\mathfrak M} \bigl(P_UF_{k,\ell,W} -P_U^{(0)}F_{k,\ell,W}^{(0)}\bigr)(\theta) e(-n\theta)\,d\theta \right|^2 \ll_{A,\delta}N^4(\log N)^{-A}. \end{align}" l:arithmeticcoefficients9,lemma,10.20,74,10,Quantitative Goldbach-type estimate: the circle method,7612,"proved: Completed Section 3 identities are reusable; this full statement is not counted as partial merely because it cites them. At1162: The actual inclusive R-smooth arithmetic series is absolutely summable, equals the source finite Euler product and admits an exact finite cutoff/tail split. At1163: The genuine Euclidean source integrals have proved quadratic and linear frequency decay, with explicit constants and all cutoffs retained. At1164: Unconditional reciprocal-totient harmonic estimates, including arbitrary divisor progressions, and the exact gcd divisor expansion are proved. At1165: The full source arithmetic-family harmonic estimate is proved unconditionally with explicit divisor exponent2 and constant9 exp(2). At1166: Exact arbitrary-multiplier coefficient identities and the uniform odd-prime bound8 gcd(p,k*ell)/p are proved. At1168: The genuine coefficient has a uniform prime-power gcd bound, with every two-adic exponent handled explicitly. At1169: General-modulus source coefficient decay tau(q)^9 gcd(q,k*ell)/q and principal Gauss decay tau(q)^3/sqrt(q) are proved unconditionally. At1170: The actual weighted source arithmetic-family and finite denominator-tail bounds are proved; the full source arithmetic and completion lemmas remain open with explicit remaining clauses. At1171: Normalized finite unit averages are proved invariant under reduction to every divisor modulus. At1172: Primitive progression exponential cancellation and the coprime norm bound are proved from actual finite unit sums. At1173: The actual Upsilon coefficient is an exact lcm-unit expectation and satisfies exact simultaneous frequency/denominator reduction. At1174: The actual Upsilon coefficient has the proved sharp gcd/divisor decay tau(q) gcd(q,k*ell)/q. At1175: The full source arithmetic-coefficient lemma is proved: absolute convergence, the all-prime infinite local Euler series and its exact source G_R value, all three individual bounds with the common divisor exponent9, and the complete real-cutoff harmonic family estimate. There are no external inputs for this lemma. None. All clauses are proved under the source hypotheses; no extra external mathematical input.","The series \[\sum_{\substack{q\geq1\\P^+(q)\leq R}} \mathfrak G_q(\ell,k,n) \] converges absolutely and satisfies \begin{align} \sum_{\substack{q\geq1\\P^+(q)\leq R}} \mathfrak G_q(\ell,k,n) &= \prod_{p\leq R} \left(1+\sum_{\nu\geq1} \mathfrak G_{p^\nu}(\ell,k,n)\right) =G_R(\ell,k,n), \end{align} " l:majorcompletion9,lemma,10.24,75,10,Quantitative Goldbach-type estimate: the circle method,7713,"proved: Complete actual source major-completion mean-square estimate for every A,delta>0, all sufficiently large C, all B>=0 and sufficiently large N, exact strict real arithmetic cutoffs, arbitrary coprime residue representatives and coefficients of norm at most one, and every 2<=U<=N,W>=2. Constants and thresholds precede N and all families. The actual major models, Euler factor G_R and singular integral H_R are used. The sharp cutoff comparison, separated additional denominators, finite arithmetic completion and quantitative Rankin tail are proved internally. No independent deep input is used. The source W<=N^2 restriction is unnecessary. The proof uses a weaker intermediate power of R that yields the same prescribed logarithmic saving after increasing C. At1211: Completed the full source major-completion lemma without external deep inputs. None within the source major-completion statement. The independent SW/BV replacement and Ren/Ghosh estimates belong to separate lemmas.","Under the hypotheses of Lemma~\ref{l:circmethod}, uniformly for \(2\leq U\leq N\) and \(2\leq W\leq N^2\), \begin{align} & \sum_{n<N^2}\left| \sum_{\ell<L_0}\sum_{k<V_\ell}d_{k,\ell} \left(\int_{\mathfrak M} P_U^{(0)}(\theta)F_{k,\ell,W}^{(0)}(\theta)e(-n\theta)\,d\theta - \frac{G_R(\ell,k,n)}{\ell" l:fixedsievesets10,lemma,11.2,78,11,Quantitative Goldbach-type estimate: sieve arguments,8013,"proved: Complete lower and upper fixed-sieve comparison, with actual sieve prime sets and actual closed multiplier set. The lower inequality is proved by inclusion, including the prime3 boundary. The upper comparison uses an exact finite exceptional-pair inclusion and the bound sum over p|n of tau(n-p^2-1) <= K_epsilon N^epsilon. One positive constant is chosen before N,n,z. Holds for every N>=1,0<n<=N^2 and every real z, hence the full retained source range. No deep input is used. At1215: The full source fixed-sieve comparison is proved without deep inputs. The local density, presieving and final sieve estimates remain open. None within the source fixed-sieve comparison lemma.","For every \(\varepsilon>0\), \begin{align} S(\mathcal A_n,\mathbb P_{3,n},z) &\geq S(\mathcal A_n,\mathbb P_3,z), \\ \sum_{\ell\in L_n}S(M_n(\ell),W_{\ell,n},z_\ell) &\leq \sum_{\ell\in L}S(M_n(\ell),W_\ell,z_\ell) +O_\varepsilon(N^\varepsilon). \end{align}" l:lowerlocaldensity,lemma,11.6,79,11,Quantitative Goldbach-type estimate: sieve arguments,8098,"proved: At1216: Proved the exact lower sieve sequence/progression count and its actual circle-method remainder decomposition. The full fixed-sieve comparison remains proved; local densities and later sieve conclusions remain open. At1219: Actual unit-square root counts, both candidate prime-density bounds including p=5, and the explicit multiplicative sieve function are proved without deep inputs. Full local-density normalization is the next internal obligation. At1220: Proved the conductor-sensitive coefficient identity, all actual prime-factor cases, the exact factors at2 and3, residue invariance, and normalized prime adjunction without division by the global base factor. Full local-density source closure is reserved for the final product and CRT assembly. At1221: complete actual normalized Euler-product factorization, exact source CRT residues, inclusive prime cutoff, multiplicativity and all density bounds are proved without external inputs. The upper base factor may vanish; no division by it is used. At1221: Both full source local-density lemmas are proved without new external inputs: exact original Euler normalization, inclusive cutoff, lower quotient identification, source CRT specializations, upper multiplicativity and p=5, with common zero upper base factors retained. Presieving and the later sieve estimates remain open. None within the source statement and retained admissible context.","For \(p>3\), \(p\equiv3\pmod4\), \begin{align} \frac{\omega_n(p)}p = \begin{cases} \displaystyle \frac{\rho_p(n-1)}{p-1-\rho_p(n)},&p\leq R,\\[7pt] \displaystyle\frac1{p-1},&p>R. \end{cases} \end{align} In particular, for \(p\geq7\), \(0\leq\omega_n(p)/p\leq2/(p-3)<1\)." l:upperlocaldensity,lemma,11.8,80,11,Quantitative Goldbach-type estimate: sieve arguments,8159,"proved: At1217: Proved the exact upper sieve sequence/progression count, including all prime exclusions, source multiplier arithmetic and residue independence from n. Local-density factorization and the later analytic sieve estimates remain open. At1219: Actual unit-square root counts, both candidate prime-density bounds including p=5, and the explicit multiplicative sieve function are proved without deep inputs. Full local-density normalization is the next internal obligation. At1220: Proved the conductor-sensitive coefficient identity, all actual prime-factor cases, the exact factors at2 and3,

residue invariance, and normalized prime adjunction without division by the global base factor. Full local-density source closure is reserved for the final product and CRT assembly. At1221: complete actual normalized Euler-product factorization, exact source CRT residues, inclusive prime cutoff, multiplicativity and all density bounds are proved without external inputs. The upper base factor may vanish; no division by it is used. At1221: Both full source local-density lemmas are proved without new external inputs: exact original Euler normalization, inclusive cutoff, lower quotient identification, source CRT specializations, upper multiplicativity and p=5, with common zero upper base factors retained. Presieving and the later sieve estimates remain open. None within the source statement and retained admissible context.","There is a multiplicative function \(\omega_n(\ell,\cdot)\), defined on square-free integers supported on \(W_\ell\), such that \begin{align} \frac{G_R(\ell,12d,n-1)}{\phi(12d)} = \frac{G_R(\ell,12,n-1)}{\phi(12)} \frac{\omega_n(\ell,d)}d . \end{align} For \(p\nmid12\ell\), its factors are \begin{al" l:presieving,lemma,11.13,81,11,Quantitative Goldbach-type estimate: sieve arguments,8273,"conditional: The corrected two-range presieving statement is a complete implication from the explicit independent general one-stage sieve inputs. The exact source matched is the separately supplied Section11_Reconciled.tex: X>=0, g(1)=1, fixed R=(log N)^C, R+1<=z<=D1<=N^A, refined1+K/logw dimension bounds, and the preliminary endpointR+1. Coefficients precede the density/sequence and n, have absolute value<=1 and exact level support; the common relative error tends to zero. The literal original weaker manuscript statement is not claimed proved; replacement is required in the user manuscript. No original source file was changed. At1258: All Section11 numbered claims are complete in the agreed conditional scope for the explicitly supplied corrected presieving statement. The original weaker statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction. At1265 the uploaded human-readable statement is reviewed directly: D0<D follows from its level chain, and the inline interval prime conditions are proved equivalent in Lean. The corrected mathematical hypotheses are verified as present in the actual upload; no presieving replacement remains outstanding.","Suppose that \begin{align} |\mathcal F_d|=Xg(d)+r_d, \end{align} where \(X\geq0\), and \(g\) is multiplicative on the square-free integers supported on a fixed set of primes \(\mathfrak P\), with \(g(1)=1\) and \(0\leq g(p)<1\). Fix \(C,A>0\), set \(R=(\log N)^C\), \(D_0=\exp((\log\log N)^3)\), and " l:wumatomakisum,lemma,11.19,85,11,Quantitative Goldbach-type estimate: sieve arguments,8636,"conditional: Complete source weighted upper main-term comparison, in uniform epsilon form, with the actual multiplier set, local factors, finite Euler products and singular integrals. For every A>0, every fixed C>=A+24 and eta>0, one threshold in N works for all n=12 mod24 with N^2(log N)^(-A)<=n<N^2. The coefficient is exactly c_(9/8) W(9/8)+eta. Strong multiplicativity, the uniform mean instantiation, both finite Abel arguments, W identity, all small-denominator and endpoint terms, moving logarithms, integration, smoothing and relative errors are proved. Only the independent generic effective mean-value corollary and classical fixed-progression Mertens bounds remain explicit unproved hypotheses. The source lemma statement is unchanged; two proof replacements are supplied but not applied. At1252: The full source weighted upper main-term comparison l:wumatomakisum is complete as a checked implication from independent generic effective mean-value and classical Mertens inputs, with the exact printed constant, actual source definitions, uniform retained-range quantifiers, and every summation, integration and relative-error obligation discharged.","Let \(\alpha=9/8\). Uniformly in the range \eqref{e:nrange10}, \begin{align} \sum_{\ell\in L}X_n(\ell)\Theta_{n,\ell}(z_\ell) \leq \bigl(c_\alpha\mathcal W(\alpha)+o(1)\bigr) X_n\Theta_n(z), \qquad c_\alpha=\frac{16e^{-\gamma/2}}{\sqrt{\pi\alpha}}. \end{align} Here \(\mathcal W(\alpha)\) is the inte"

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s,expositions,source_line,active_statement_sha256
1.1,theorem,t:improvingfixed,Main prime-square improving theorem,Prime-square high-low,conditional,"Both
 conclusions for the genuine full von-Mangoldt square average, every positive integer N, the exact
source interval and its cardinality, all finite F,G subsets, every coefficient cstar>4 sqrt(3)/9, and
every real 3/2<p<=2. The strong result proves actual output lp membership and the literal normalized sum
 form, for all lp inputs, hence also the supported inputs in the source.","The implication is complete
under the explicit independent deep inputs. PNT limits, compatible progression/Ren/Ghosh data existence,
 and the independent ordinary-square global decomposition remain unformalized hypotheses. Its analytic
globalization remains external; the plus/minus spatial reflection is proved with the same constants at
checkpoint783. Theorem1.1 and Propositions4.19/6.21 are complete conditional implications under their
separate inputs; the current corrected numbered paper targets are all matched, with independent inputs
and stronger unnumbered alternative-route assertions retaining their documented scopes.",PNT_LIMITS;
PROGRESSION_ESTIMATE; REN_GHOSH; FULL_SQUARE_DECOMPOSITION,theorem_1_1; theorem_1_1_restricted;
theorem_1_1_strong; theorem_1_1_strong_real; theorem_1_1_strong_sum; primeSquareImprovingOperator_apply;
 primeSquareSourceInterval_card; theorem_1_1_from_positive_square_input;
fullSquareDecompositionInput_sign_iff,docs/THEOREM_1_1_COMPLETION.md;
docs/THEOREM_1_1_STATEMENT_REVIEW.md; docs/THEOREM_1_1_EXTERNAL_INPUTS.md;
docs/SQUARE_REFLECTION_COMPLETION.md; blueprint/Checkpoint_783_SquareDecompositionReflection.md,212,cb25
c450dab863eeab6d1c5ec968919bcc842ef221a0fe2ea0218f469288c70e
1.3,theorem,t:sparsebound,Main sparse domination theorem,Sparse estimates,conditional,"Both displayed
conclusions are complete conditional implications: the literal full maximal indicator for every integer
N>=2, arbitrary finite F,G including empty sets, actual finite interval collections with actual pairwise
 disjoint major subsets of size at least one tenth, the exact endpoint density exponent and paper Phi
for every cstar>4sqrt(3)/9, and every point of the full open triangle in exact reciprocal/barycentric
coordinates. Constants precede F,G. All stopping, packing, support, local endpoint, interpolation and
sparse deductions are proved.","No paper-specific deduction in either conclusion remains. The recorded
prime limits, compatible raw Page/distribution data, Page uniqueness, general smooth
multifrequency/sampling, independent ordinary-square stopping interface and complementary-exponent prime
 maximal input remain unproved explicit hypotheses, including their stated compatibility boundaries. The
 general auxiliary Section5 statements are separately partial. The proposed dyadic qualification of
l:sparserecursion is unapplied; the actual prime-square all-cutoff reduction is proved.",PNT_LIMITS;
PROGRESSION_ESTIMATE; REN_GHOSH; SMOOTH_MULTIFREQUENCY; PAGE_CLASSES; ORDINARY_SQUARE_STOPPING;
PRIME_MAXIMAL_LP,theorem_1_3; theorem_1_3_endpoint; theorem_1_3_open_pq; primeEndpointSparse;
primeOpenSparse; primeSquareMaximalIndicator_dominates;
sourceSparseTriangle_iff_barycentric,docs/THEOREM_1_3_COMPLETION.md;
docs/THEOREM_1_3_STATEMENT_REVIEW.md; docs/INDEPENDENT_STOPPING_INPUTS.md;
docs/SPARSE_DYADIC_QUALIFICATION.md,287,19cf5a5a689f1a159f79417390972e69b01247af360a496b77fcee5210fb5017
1.5,theorem,t:pet,Persistent L1 badness and near-L1 maximal bound,Badness and
convergence,conditional,"Both parts are complete conditional implications. Part(i): the actual
enumerated prime squares are persistently universally L1-bad on every truly nonatomic ergodic
measure-preserving probability system, with every infinite cutoff set and an actual measurable
positive-measure divergence set. CRT, Dirichlet representation, correlation counts, large-prime block
construction, all moment/thickening/difference limits, uniform square-root fibers and the
fixed-progression PNT pushforward are proved. The original badness theorem, Sawyer, Conze, general Weil
and fixed-progression PNT remain separate unproved general inputs. Granville–Kurlberg is not used by
part(i). A checked moment proof replaces the source strict finite-fraction argument, which has an exact
kernel counterexample. Part(ii) is a complete conditional implication: an absolute positive constant
before every invertible measure-preserving probability system, every 1 < p <= 2 and every Lp input; the
proper actual maximum over the first m prime squares for all m >= 1 is finite almost everywhere, belongs
 to Lp and has the exact displayed exponential norm bound. The prime-counting/enumeration normalization,
 finite sums, first-prime case and equality of both full maxima are proved with no PNT or enumeration
input. The proof uses the already proved Corollary7.6 and the precise independent general strong
Calderon input, plus the earlier Theorem7.2 independent inputs. It does not use the bounded-class
convergence input. At checkpoint1264 the current public export consumes the proved finite signed maximal
 theorem at constant6; a supplied RawFiniteSignedMaximalInput and its constant R are no longer
hypotheses.",No remaining source clause within the agreed conditional scope. General inputs and their
exact convention bridges remain unproved. The user confirmed the three Section8 edits; the selected
project manuscript now contains the agreed moment proof.,LAVICTOIRE; WEIL; PROGRESSION_ESTIMATE;
ERGODIC_TRANSFER; QUANTITATIVE_IW; GENERAL_VARIATION_SAMPLING; REN_GHOSH; SMOOTH_MULTIFREQUENCY;
PAGE_CLASSES; QUADRATIC_WEYL_FJK; SAWYER; CONZE; FIXED_PROGRESSION_PNT,primeCounting_nth_prime;
primesLE_image_nth; sum_primesLE_nth; ergodic_prime_cutoff_enumerated;
enumeratedPrimeSquareAverage_cutoff; enumeratedPrimeMaximalENorm_eq; ergodic_strong_maximal_source;
theorem_1_5_ii; primeSquare_lavictoire_arithmetic; theorem_1_5_i; theorem_1_5_i_literal; theorem_1_5;
theorem_1_5_ii_closed_signed_maximal,docs/THEOREM_1_5_II_COMPLETION.md;
docs/ERGODIC_INDEPENDENT_BOUNDARY.md; blueprint/Checkpoint_991_TheoremOneFivePartTwo.md;
docs/THEOREM_1_5_I_COMPLETION.md; docs/SECTION_8_COMPLETION.md; docs/SECTION_8_SPACING_CORRECTION.md;
verification/section8-source-review-checkpoint1038.json;
docs/PAPER_COMPLETION.md,376,837e103c9047baed31fc086e1ed6343e09673a03c232e5a07365eb23ff53c000
1.6,theorem,t:repmotthm,Quantitative Goldbach-type exceptional set,Goldbach and sieve,conditional,"The
full quantitative exceptional-set theorem is complete: for every A>0, the actual exceptional set of
positive n<X congruent12 mod24 without a prime plus a square-of-prime representation whose first prime
is a primitive sum of two squares plus1 has size O_A(X(log X)^(-A)), eventually at every real X. The
representation and exceptional-set membership are exact. The common-cutoff Section11 estimates, positive
 numerical gap, arbitrary logarithmic mean-square saving, all arithmetic losses, small n and the passage
 from N^2 to real X are discharged. Explicit independent general SW, BV, Ren/Ghosh, one-stage sieve,
effective multiplicative mean, Mertens, normalized sieve values/continuity and primitive two-square
criterion remain unproved theorem parameters. No own-paper endpoint is a hypothesis. At1259: Section11
and the final quantitative Goldbach theorem are complete in the agreed conditional scope, using the
supplied corrected presieving statement. Every paper endpoint is discharged; only explicit independent
general classical inputs and their normalizations remain unproved. Original manuscripts are unchanged
and the supplied Section11 replacements still require user application. No additional completion credit
is claimed for stronger unnumbered assertions in alternate proof routes. At1261: Proved the full

PrimitiveTwoSquaresCriterion from pinned Mathlib and elementary algebra, and proved W(alpha) convergence/nonnegativity for every alpha>1. Constructed a concrete nonzero compactly supported smooth seed. The next stage exports the Section9 arithmetic and headline statements without the former criterion or seed hypotheses, then reconciles the stale Section9 map. User manuscript edits remain in progress. At1262 the final endpoint is exported from GoldbachSection11Inputs alone: the primitive two-square criterion is proved internally and a fixed nonzero smooth compact seed is constructed. The real-X bound and zero exceptional density need no separate arithmetic or seed assumption. At1262: Every Section9 numbered source claim and its final quantitative theorem is complete in the agreed scope. All elementary arithmetic reductions are now proved without an extra primitive two-square assumption. W(alpha) convergence is proved for every alpha>1 and the smooth seed is constructed. Only the explicit independent analytic inputs remain for the circle-method/sieve/headline estimates. The user is applying previous manuscript edits; no application is assumed and original sources are unchanged.",,SIEGEL_WALFISZ_PRIME_COUNTS; BOMBIERI_VINOGRADOV; REN_GHOSH; ROSSER_IWANIEC_SINGLE_STAGE; EFFECTIVE_HALF_DIMENSIONAL_MEAN; CLASSICAL_INTERVAL_MERTENS; CLASSICAL_FULL_INTERVAL_MERTENS; CLASSICAL_PRIME_LOG_MERTENS; CLASSICAL_PRIME_MERTENS_RATIOS,goldbach_exceptional_square_sum; goldbach_exceptional_count; logarithmic_square_threshold_bound; goldbach_logarithmic_exceptional_bound; goldbachSmallNumbers_card_le_ceil; goldbach_full_exceptional_count; goldbach_section9_count_reduction; goldbach_eventual_arithmetic_budgets; goldbach_section9_eventual_count_reduction; goldbach_quantitative_exceptional_set_squares; goldbach_good_representation_iff; goldbach_source_exceptional_membership; goldbach_ceiling_square_geometry; goldbach_real_cutoff_from_square_bound; goldbach_headline_theorem; goldbach_primitive_one_mod_four_support; goldbach_primitive_double_of_odd; goldbach_primitive_two_squares_criterion; goldbach_weighted_W_integrable_general; goldbach_weighted_W_nonneg_general; goldbach_fixed_smooth_seed_at_zero; goldbach_fixed_smooth_seed_nonzero; goldbach_fixed_smooth_seed_support; goldbach_fixed_smooth_seed_compact_support; goldbach_fixed_smooth_seed_tsupport_subset; goldbach_smooth_seed_exists; goldbach_from_sifted_to_primes_complete; goldbach_truncation_complete; goldbach_section9_master_exact; goldbach_master_sieve_complete; goldbach_section9_headline_complete; goldbach_section9_exceptional_density_zero,docs/SECTION_9_REDUCTION_COMPLETION.md; docs/RECOVERY_AND_SECTION9_REVIEW_1106.md; blueprint/Checkpoint_1259_GoldbachQuantitativeExceptionalSet.md; blueprint/Checkpoint_1261_GoldbachPrimitiveCriterion.md; blueprint/Checkpoint_1262_GoldbachSectionNineComplete.md,453,1684f0a759031ccbad229d24e321cb48ebbcfff43a873a8e978415beb575c68c

3.1,lemma,l:cohenquadratic,Quadratic Cohen identity and uniform bound,Arithmetic,proved,"Complete positive-modulus identity, all local factors, CRT and bound.",,,quadratic_cohen_identity; quadratic_cohen_bound,blueprint/Cohen_Identity.md,1533,8d0f0ea77770384b04410123cbf79aeb74f5f2348ffd66f657481d50e77767c4

3.11,lemma,l:progressioncohen,Progression-restricted Cohen identity,Arithmetic,proved,"Complete identity, including the shared-prime vanishing case.",,,progression_cohen_identity_int,blueprint/Progression_Cohen.md,1660,7a90bef55c95aae8fc7a1f23a024420f56812c743cc59a9f8957cbf523b9f36e

3.12,lemma,l:sumoverramanujan,Normalized progression correlation and factorization,Arithmetic,proved,"Actual normalized sums, squarefree vanishing and prime-power factorization.",,,sum_upsilon_square_identity; sum_upsilon_square_eq_zero_of_not_squarefree; sum_upsilon_square_prime_factorization,blueprint/Progression_Cohen.md,1698,4d3879e51609946440c3b96a84eae12c86cc0283428ea8ad3acf135190f73006

3.14,lemma,l:motamakemajorprimesquareprog,Corrected progression Euler product,Arithmetic,proved,"Accepted revision: Q>2, positive k, and p<Q for the obstruction. The submitted revision is preserved.",,,progression_euler_product_real; progression_euler_product_eq_zero_of_obstruction,blueprint/Euler_Product.md; docs/SOURCE_UPDATE_601.md,1737,2a806383575e1cd3406dfdfd8b8ac91591845ee70a46a817735310acf26c4988

3.16,corollary,c:generalmajorprimesquareprog,Corrected multiplier Euler product,Arithmetic,proved,"Accepted Q>2 and positive-k ranges, arbitrary integer multiplier, actual local counts and dyadic factor.",,,general_progression_euler_product_real,blueprint/General_Euler_Product.md; docs/SOURCE_UPDATE_601.md,1817,9c521c5fe85d1bfc41a6990df7ea95c8361e39076c9c24ae0cb68a9f0c17c081

3.20,lemma,l:lowramanujan,Uniform logarithmic Cohen bound,Arithmetic,proved,All real Q>=3 and integer n; an elementary prime-product argument supplies the bound.,,,low_ramanujan_uniform_bound,blueprint/Low_Ramanujan_Bound.md,1916,e3063d0290305807a7da8993f7d038a218c710168d9f768220554059195798f

3.21,lemma,l:inftyboundsaq,Sharp principal quadratic Gauss bound,Arithmetic,conditional,"Complete implication for 2+o(1), uniform epsilon and eventual coefficient-three bounds.","External PNT limits and the bridge from an imported PNT are not proved here. Under the agreed external-input boundary, the paper-specific deduction is complete.",PNT_LIMITS,principalSquareSum_two_add_o_one; principalSquareSum_sharp_bound_int; principalSquareSum_three_bound_int,blueprint/Sharp_Gauss_Asymptotic.md,1943,cfec3690d08892a4e2c9f95e1a2e57871ed5e7b80494153484844b0d52d2d753

3.24,lemma,l:Gcharbound,Uniform character-twisted quadratic Gauss bound,Arithmetic,proved,"Complete statement, strengthened to all complex Dirichlet characters.",,,twistedSquareSum_uniform_bound_int,blueprint/Twisted_Gauss_Bound.md,1993,c4979c02982ce1e0f18c0b1cc94b3cc24d286290089f10503502f3b2cd42e42f

3.27,lemma,l:exceptionalarithmetic,Conductor-sensitive exceptional arithmetic,Arithmetic,proved,"All three bounds, logarithmic consequence and relative support; strengthened to all real base characters.",,,conductor_sensitive_refinement,blueprint/Conductor_Refinement.md,2075,e3c22d4a79d8d475946d17f493351d4aedbf64fcfb6643248be2946ee47a9a33

4.1,theorem,t:improvinggeneral,Abstract endpoint improving theorem,Abstract improving,proved,Complete abstract implication from the hypotheses printed in the source. Verifying those hypotheses for prime squares is a separate obligation.,,,highLow_AB_endpoint,blueprint/Restricted_Improving.md; blueprint/Strong_Improving_Complete.md,2330,d796041cc73506aaea2e41d3bccd6173da332f8307518e6023db921b95dd773b

4.15,corollary,c:improvinggeneral,Abstract strong improving corollary,Abstract improving,proved,"Complete strong lp-to-lp statement, including p=2 and the source operator hypotheses.",,,highLow_AB_strong_improving,blueprint/Strong_Improving_Complete.md,2463,51e92b00a15429d558c739f763fb82036d19af599fbbe27d80b94950cc25671e

4.19,proposition,p:decompositionofan,Prime-square high-low decomposition,Prime-square high-low,conditional,"All displayed clauses for the actual common selected source low/high/error family: exact all-input annular decomposition, actual l1-to-l-infinity low norm, local bound for all l1 inputs with the supremum over every real M>=N, sharp fixed dyadic and actual countable maximal high bounds for every real Q>=2, every allowed starting threshold, every error exponent A>0, bounded Q and the exact printed high-symbol identity. One cutoff coefficient is chosen before epsilon and A; the operator family never changes with these parameters.","The implication is complete under PrimeAsymptoticInputs, PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. Their independent analytic truths and compatible raw-data existence remain external. No own-paper decomposition, low bound, approximation error or maximal conclusion is an input. The current corrected numbered paper targets are all matched; independent inputs and stronger unnumbered alternative-route assertions retain their documented scopes.",PNT_LIMITS; PROGRESSION_ESTIMATE; REN_GHOSH;

SMOOTH_MULTIFREQUENCY; PAGE_CLASSES,proposition419; proposition419_dyadic_coverage;
selectedPrime_decomposition_l2; selectedPrime_decomposition_l1; selectedPrimeLow_source_bounds;
selectedPrimeError_all_exponents; selectedPrimeHigh_source_bounds;
selectedPage_raw_source_range,docs/PROPOSITION_4_19_COMPLETION.md;
docs/PROPOSITION_4_19_STATEMENT_REVIEW.md; blueprint/Checkpoint_791_Proposition419.md,2541,6ae0920c69c38
7ee9bd07abba2fc239d47a8ff9aa1453e62c538bef43f88611e
5.1,lemma,l:localsparse,Abstract localized stopping-time estimate,Sparse estimates,proved,"Complete
abstract implication with the exact source exponents, all 0<=v<1, s>=1-v, r>0, gamma>=beta>=0 and A>1/2,
 arbitrary physical dyadic stopping times, actual localized low/high/error outputs, all-real-radius
admissibility, arbitrary fixed positive radii, parent-length normalization, empty sets and the full
bounded density range. The error symbol may be an arbitrary almost-everywhere bounded function; its
operator norm follows from the proved Fourier isometry. The normal decomposition and norm estimates are
the printed hypotheses, not extra deep inputs.",,,source_local_stopping_from_fourier_hypotheses;
source_abstract_local_stopping_estimate; abstract_local_stopping_endpoint; fourierSymbol_operator_bound;
 localError_square_majorant;
linearSourceLocalMass_change_radius,docs/ABSTRACT_LOCAL_STOPPING_COMPLETION.md; blueprint/Checkpoint_105
4_SourceLocalStoppingFourier.md,2721,1d84a26b30352c605f57d17de17ab8f0d32a3eb13c5f3fd8a63d85aa5543b988
5.10,lemma,l:sparserecursion,Sparse recursion from local estimates,Sparse estimates,proved,"The actual
prime-square finite stopping construction, disjoint components, real-radius admissibility, exact kernel
localization, terminating cost recursion and actual sparse major subsets are proved, with a proved
all-cutoff reduction for the original averages. A general finite recursive-cost implication is also
proved. The complete general dyadic implication is now proved for arbitrary positive compact-support
kernels, every fixed odd support dilation, arbitrary Psi, the full untruncated maximum and genuine
quarter-sparse major sets. The uniform bound and finite approximation of the full maximum are deduced
from the local hypothesis. The local hypothesis is imposed only on maps admissible on the whole
interval; an extension argument proves the needed restricted version.",No own-paper deduction remains
under the accepted dyadic and independent-admissibility-constant conventions. The proof constructs
actual sparse sets and the full maximum from the stated local hypothesis; no additional analytic
estimate is assumed.,,finiteSparseRecursion; laminarSparseRecursion; primeAnnularSparse_from_local;
primeSparse_from_admissible_local; sparseNonDyadic_local_pairing_zero; sparseNonDyadic_maximum_positive;
 densityProfileSpike_nonneg; densityProfileSpike_no_dilation_comparison;
source_dyadic_sparse_recursion_full_domain; genericPositiveFullSparse;
genericLocalHypothesis_uniform_indicator_bound; genericFullDyadicMaximum_finite_test;
accepted_sparse_admissibility_threshold;
accepted_general_dyadic_sparse_recursion,docs/THEOREM_1_3_COMPLETION.md;
docs/SPARSE_DYADIC_QUALIFICATION.md; docs/AUXILIARY_SCOPE_REVIEW_1065.md; docs/SOURCE_UPDATE_1066.md;
docs/GENERAL_DYADIC_SPARSE_COMPLETION.md;
docs/SOURCE_UPDATE_1084.md,2905,7c63ffcdc8aca70d79b7f2c65f11d7c511f7b13fd713f418144bd6c5b2d4167a
5.17,lemma,l:primeendpointlocal,Prime-square local endpoint estimate,Sparse
estimates,conditional,"Complete source implication for the actual annular von Mangoldt square average:
every fixed real support dilation >=1, every positive spatial radius, every fixed admissibility
constant, physical dyadic stopping values including scale1, literal all-real-radius quadratic averages,
empty sets and the exact parent-length normalization. The bound holds for every cstar>4sqrt(3)/9.
Proposition4.19 is derived from the already-listed independent inputs in the terminal theorem, rather
than assumed as an extra own-paper input.",,PNT_LIMITS; PROGRESSION_ESTIMATE; REN_GHOSH;
SMOOTH_MULTIFREQUENCY; PAGE_CLASSES;
ORDINARY_SQUARE_STOPPING,source_prime_local_endpoint_from_independent_inputs;
source_prime_local_endpoint_real_dilation; source_prime_local_endpoint;
primeStopping_general_parent_endpoint; primeStopping_local_endpoint_with_constant;
sourceLocalMass_from_quadratic_radius;
physicalDyadic_prime_pairing,docs/PRIME_LOCAL_SOURCE_COMPLETION.md; blueprint/Checkpoint_1062_SourcePrim
eLocalEndpoint.md,3047,5ce4517b5c21cced305979b404562cbb6647886daf38c36a4772fccf9b81ee54
6.5,lemma,l:prime-square-input,Major- and minor-arc prime-square inputs,Prime-square
high-low,conditional,"Complete corrected source lemma, with one genuine optional Page datum per N
selected before Q, all q<=R^(11/10), the exact induced character/selector/exceptional model and
inverse-fifth major error, and the full complementary R^(-1/4)*(log N)^d minor bound. The source
Q-range, uniform sufficiently-large-N threshold, R^4<=N/2 and log N>=1 are proved. The general
fixed-level uniform Page-prefix theorem and raw Ren/Ghosh estimates remain explicit independent unproved
 hypotheses.",,UNIFORM_PAGE_PREFIX; REN_GHOSH,annularMangoldt_majorArc_inverse_fifth;
annularMangoldt_complete_Ren_bound; complementaryMinorBudget_quarter_bound;
annularMangoldt_outside_major_quarter_bound; uniformPage_extended_major_arc;
source_corrected_prime_square_input; pageModulusSelector_none;
pageModulusSelector_some,docs/CONTINUE_HERE.md; docs/MINOR_ARC_REPLACEMENT_REVIEW_1092.md;
docs/SOURCE_UPDATE_1107.md; docs/CORRECTED_PRIME_SQUARE_INPUT_COMPLETION.md,3322,e7e3e86ffe5c4e366414fbb
3bf9a177dc64d7aba38cb000e1c0d8d5404fca48f
6.9,lemma,l:aaplemma,Full approximation error with arbitrary logarithmic decay,Prime-square
high-low,conditional,"Complete printed approximation lemma for the literal selected source error family,
 every real Q>=2 and every integer N in the Section6 source range. One positive source cutoff
coefficient is chosen before all A>0; the exact Q^(-1/2)*(1+log N)^(-A) operator and all-l2 function
bounds are proved. The optional Page data and radius are the same actual selected family as
Proposition4.19. The previously explicit compatible raw distribution/Ren/Ghosh data remain unproved
general inputs; no new own-paper estimate is assumed.",No remaining source clause within the agreed
conditional scope.,PROGRESSION_ESTIMATE; REN_GHOSH,source_approximation_all_exponents;
source_approximation_function_bound; selectedPrimeError_all_exponents;
selectedPage_raw_source_range,docs/SECTIONS_3_TO_8_REVIEW.md; blueprint/Checkpoint_1039_SectionSixSource
Exports.md,3466,4c61455f65090530120ecc6ae2f15baa109361f4d2cba44f0d516db5cf62045d
6.14,proposition,p:lowcontribution,Pointwise low contribution and its L1-to-Linfinity bound,Prime-square
 high-low,conditional,"Complete printed low proposition for the actual selected source low operator:
every real Q>=2, every integer N in the common source range, one absolute positive constant for the
literal all-real-radius local bound and the l1-to-linfinity norm bound. The local result holds for every
 complex l1 input using |f| and hence the stated nonnegative inputs, with fixed spatial constant1.
Ordinary and induced exceptional terms use the complete conductor range and exact source cutoffs. Only
the previously recorded compatible source-data boundary remains; no new mathematical input is added.",No
 remaining source clause within the agreed conditional scope.,PROGRESSION_ESTIMATE;
REN_GHOSH,source_low_local_and_norm; selectedPrimeLow_source_bounds; sourceLocalMass;
sourceFullLow_squared,docs/SECTIONS_3_TO_8_REVIEW.md; blueprint/Checkpoint_1039_SectionSixSourceExports.
md,3789,f5e94073753a34f43ecab5a354703cd3e88b921dce497a0ae733e13aa70b6fbb
6.16,lemma,l:notBourgain,Printed annular multifrequency maximum,Prime-square high-low,conditional,"The
full actual annular source maximum with the printed first power of (1+s) is proved under the precise
independent rational smooth multifrequency/circle-transference input and the previously recorded general
 smooth input. Every source dyadic scale N>=2^s, every s>=0, scale1 and the small denominator indices

are included. The rational common period, linear cost, weight localization, actual annular
square-function error and all earlier scale costs are proved. Beta may vary with the scale, a stronger
conclusion than the source fixed-beta statement.",No own annular deduction remains. The independent
smooth multifrequency theorems and sampling remain explicit unproved inputs. The proof uses the
rational-frequency refinement rather than replacing the general squared-log loss by a first power; a
matching expository replacement proof is saved separately.,SMOOTH_MULTIFREQUENCY;
RATIONAL_SMOOTH_MULTIFREQUENCY,bourgain_weightedSource_projection; rationalProjectionErrorBlock_maximum;
 largeStart_weightedAnnular_maximum; weightedAnnular_allStartingScales; quadratic_log_loss_not_linear;
rationalBand_common_period; rationalBand_loss_linear; rational_annular_allStarting;
circleAnnularSource_initial_norm; source_rational_annular_maximum,docs/NEXT_ANALYTIC_INPUT_700.md;
blueprint/Checkpoint_725_LargeStartActualSourceMaximum.md;
blueprint/Checkpoint_726_WeightedAnnularLargeStartMaximum.md; docs/AUXILIARY_SCOPE_REVIEW_1065.md;
docs/RATIONAL_MULTIFREQUENCY_BOUNDARY.md; docs/RATIONAL_ANNULAR_COMPLETION.md,4009,e751e8dfc0084cf20685b
968c76bb5e10f01eada79b7ea5cf055d50cea35d5dc
6.19,corollary,c:orlicz-prefix,Denominator-truncated annular multifrequency maximum,Prime-square
high-low,conditional,"The full source bound with square (1+s)^2 loss is proved for all s>=0, all source
dyadic times including scale1, every real denominator cutoff simultaneously and every integer L2 input.
The additional fixed bounded measurable multiplier clause is included, with actual essentially bounded
Fourier operators and no continuity restriction. Beta may also vary with the scale.","No own corollary
deduction remains. The independent rational smooth/circle-transference theorem and the prior general
smooth input remain explicit unproved hypotheses; the full additional fixed measurable multiplier clause
 is proved, not added as an external input.",SMOOTH_MULTIFREQUENCY;
RATIONAL_SMOOTH_MULTIFREQUENCY,source_time_prefix_from_intervals;
source_rational_annular_truncation_maximum; aeIntegerMultiplier_norm_le;
aeIntegerMultiplier_localized_comp; disjointAEMultiplierSymbol_local_comp;
fixedAnnularMultiplier_disjoint_factor; fixed_annular_finite_truncation_maximum;
source_fixed_annular_truncation_maximum,docs/NEXT_ANALYTIC_INPUT_700.md; docs/SOURCE_UPDATE_601.md; docs
/RATIONAL_TRUNCATION_COMPLETION.md,4081,1c1cc225e998129e05d985b54837ab7434794776734e647d8faaf44142492bbc
6.21,proposition,p:highupperbound,Sharp high contribution and countable maximum,Prime-square
high-low,conditional,"Both source clauses are complete checked implications: the actual countable dyadic
 supremum and every individual output, all real Q>=2 and N0>=2 in the allowed logQ<=c0 sqrt(logN0)
range, one positive c0 independent of epsilon, the exact source radius and complementary narrow-cutoff
multiplier, changing genuine exceptional data, and the leading coefficient 2+o(1) in precise epsilon
form. A separate uniform estimate covers bounded Q; its harmless regularization agrees with
sqrt(logQ)/loglogQ once logQ>=2.","No paper-specific deduction in this proposition remains. The two
prime asymptotic limits, the general smooth multifrequency/sampling input, raw Landau-Page uniqueness
(including the L-function bridge), and the existence/selection of source-compatible exceptional zero
data remain explicit external boundaries. The stronger auxiliary statements 6.16 and 6.19 are not
awarded completion.",PNT_LIMITS; SMOOTH_MULTIFREQUENCY; PAGE_CLASSES,proposition621_source;
proposition621; proposition621_maximal_asymptotic; sourcePageDyadicTail_uniform;
pageHighSymbol_paper_identity; sourceHigh_uniform_geometry; dyadic_tail_covers_source_scales;
regularizedHighExponentWeight_eq,docs/PROPOSITION_6_21_COMPLETION.md;
docs/PROPOSITION_6_21_STATEMENT_REVIEW.md; blueprint/Checkpoint_778_Proposition621Source.md,4143,726a641
9f6d781c946109a3fd51f147cd9ec560a4357d4434d207b00af1ae450
7.2,theorem,t:orlicz-maximal,Quantitative Orlicz maximal theorem,Orlicz and ergodic
estimates,conditional,"Complete printed Theorem 7.2 as an implication of explicit independent standard
inputs: all three actual families (von Mangoldt squares, ordinary squares, unweighted prime squares),
every integer cutoff N >= 2, every complex finitely supported function and positive level, with common
absolute C, C0 and K >= 1. All three conclusions are proved, including finite level sets: the truncated
modular bound, unrestricted modular bound and restricted bound for every finite F and t >= 1. The final
theorem is stated with literal sums over all integers and finite support, using the actual integral
gauge. The ordinary branch derives the actual discrete circle-model error, maximal comparison, low/high
restricted bound and full-cutoff reconstruction from the independent scalar Weyl input and general IW
inputs; no own restricted or high/low conclusion is assumed. The unweighted-prime transfer uses proved
Mathlib Chebyshev bounds without a new PNT input. At checkpoint1264 the current public export consumes
the proved finite signed maximal theorem at constant6; a supplied RawFiniteSignedMaximalInput and its
constant R are no longer hypotheses.",No remaining clause of Theorem7.2 or Corollaries7.6/7.46 within
the agreed conditional scope. The explicitly recorded independent inputs and their published-theorem
bridges remain unformalized. Stronger standalone auxiliary parameter/full-generality ranges and later
badness/sieve results remain separately tracked.,QUANTITATIVE_IW; GENERAL_VARIATION_SAMPLING;
PROGRESSION_ESTIMATE; REN_GHOSH; SMOOTH_MULTIFREQUENCY; PAGE_CLASSES;
QUADRATIC_WEYL_FJK,nearOnePsi_convex; nearOnePsi_comparison; nearOnePsi_doubling;
highLow_restricted_distribution; positiveRestricted_to_truncated_modular;
section7_truncated_modular_finite_card; section7_modular_finite_card; mangoldtPositiveFamily_apply;
ordinaryPositiveFamily_apply; primePositiveFamily_apply; iw_prime_annular_restricted_distribution;
iw_prime_annular_restricted_gauge; iw_prime_full_restricted_gauge; iw_prime_full_modular_estimates;
iw_unweighted_prime_restricted_gauge; iw_unweighted_prime_modular_estimates;
iw_prime_families_restricted_gauges; mangoldt_full_annular_maximal_bound;
prime_mangoldt_indicator_maximal_bound; primeCounting_log_lower_bound; iwRestrictedDistribution_budget;
iw_full_square_low_maximal_bound; iw_full_square_model_comparison; fullSquareCircleError_bound;
fullSquareCircleError_uniform; full_square_approximation_maximum; iw_full_square_high_source_scale;
iw_full_square_annular_restricted_distribution; iw_full_square_annular_restricted_gauge;
ordinary_full_annular_maximal_bound; iw_full_square_restricted_gauge; iw_full_square_modular_estimates;
iw_three_families_restricted_gauges; section7_three_family_assembly; theorem_7_2;
section7_source_sum_assembly; theorem_7_2_finitely_supported;
theorem_7_2_closed_signed_maximal,docs/SECTION_7_PROGRESS.md; docs/SECTION_7_STATEMENT_REVIEW.md;
blueprint/Checkpoint_837_RestrictedToModular.md;
blueprint/Checkpoint_843_HighLowRestrictedDistribution.md;
blueprint/Checkpoint_846_SectionSevenDeductions.md; docs/SECTION_7_RESTRICTED_COMPLETION.md;
blueprint/Checkpoint_948_IWPrimeAnnularRestrictedGauge.md;
blueprint/Checkpoint_950_IWPrimeFullRestrictedGauge.md;
blueprint/Checkpoint_953_IWUnweightedPrimeRestrictedGauge.md; docs/THEOREM_7_2_COMPLETION.md;
docs/SECTION_7_QUADRATIC_WEYL_BOUNDARY.md; blueprint/Checkpoint_967_IWFullSquareHighSourceScale.md;
blueprint/Checkpoint_970_IWFullSquareAnnularRestrictedGauge.md;
blueprint/Checkpoint_971_OrdinaryFullAnnularComparison.md;
blueprint/Checkpoint_972_IWFullSquareRestrictedGauge.md;
blueprint/Checkpoint_973_IWThreeFamilyRestrictedGauge.md;
blueprint/Checkpoint_974_SectionSevenThreeFamilyAssembly.md;
blueprint/Checkpoint_975_TheoremSevenTwo.md;
blueprint/Checkpoint_976_TheoremSevenTwoFinitelySupported.md;
docs/PAPER_COMPLETION.md,4494,f6934e115cdc14c9fc15ac263345e6240446462e41925fb091f0cc4cd530a890

7.6,corollary,c:orlicz-p,Quantitative lp maximal bounds near p=1,Orlicz and ergodic
estimates,conditional,"Complete printed Corollary 7.6 as an implication of the already recorded Theorem
7.2 independent inputs. One absolute D > 0 works for all three actual families, every 1 < p <= 2 and
every integer lp input. The literal all-integer-cutoff maximal function is proved to belong to lp with
norm at most exp(D/(p-1)*log(exp(1)/(p-1))) times the input norm. The actual truncated modular estimate,
 general layer cake, singular power integral, quantitative constant absorption, p-th-root conversion and
 all-input finite/Fatou extension are proved. At checkpoint1264 the current public export consumes the
proved finite signed maximal theorem at constant6; a supplied RawFiniteSignedMaximalInput and its
constant R are no longer hypotheses.",No remaining source clause within the agreed conditional scope; no
 new strong maximal or layer-cake input is introduced. Theorem7.2's independent hypotheses remain
explicit. The ergodic Corollary7.46 and Theorem1.5(ii) are now separately completed under their reviewed
 independent inputs.,QUANTITATIVE_IW; GENERAL_VARIATION_SAMPLING; PROGRESSION_ESTIMATE; REN_GHOSH;
SMOOTH_MULTIFREQUENCY; PAGE_CLASSES; QUADRATIC_WEYL_FJK,nearOneCore_quantitative;
nearOneGaugeMoment_integrable; nearOneGaugeMoment_source_p; positivePower_primitive;
nonneg_power_layerCake; truncated_rpow_kernel; truncated_scaled_power_kernel;
truncated_distribution_strong_power; truncated_gauge_strong_power; nearOneStrong_power_budget;
actual_finite_strong_maximal_norm; positiveAverage_finite_strong_extension; section7_strong_assembly;
corollary_7_6; corollary_7_6_closed_signed_maximal,docs/SECTION_7_PROGRESS.md;
docs/SECTION_7_STATEMENT_REVIEW.md; blueprint/Checkpoint_839_NearOneGaugeMoment.md;
blueprint/Checkpoint_841_NearOneMomentExponential.md; docs/COROLLARY_7_6_COMPLETION.md;
blueprint/Checkpoint_984_CorollarySevenSix.md;
docs/PAPER_COMPLETION.md,4528,e20c53dc1d2b86da97674402674f5c1be6309892a88e8dab399230ec2d5e59cb
7.8,proposition,p:IWinput7,Quantitative Ionescu-Wainger application,Orlicz and ergodic
estimates,conditional,"All four source clauses are complete conditional implications for one common
construction at every real Q>=2^k, r>=2 and k>=4. The literal B, epsilon and C_rk are retained; P<=3^Q,
every a/q modulo one with q<=Q, all denominator bounds and full goodness are proved. The rough
multiplier conclusion is an actual bounded operator on the entire integer lp space, with the exact
finite-convolution Fourier kernel, uniqueness and norm <=C_rk times the actual full real Lp operator
norm. The general full-Lp real/cyclic product bound and rank-one coefficient factorization are proved.
Both major-arc sampling directions, actual unique interpolation on the full distributional integer lp
Fourier-support class, the inverse-C_rk source inequality, and all finite-dimensional normed complex
Shannon targets throughout 1<=p<=infinity are retained. Point sampling uses the standard continuous
representative convention. General Tao construction, full adelic IW and full sampling theorems,
including the explicitly described rough/full-class/finite-quotient realization bridges, remain
independent unproved inputs. The earlier narrower endpoint results retain their own
hypotheses.",,QUANTITATIVE_IW; FULL_DOMAIN_ADELIC_IW;
FULL_DOMAIN_TAO_SAMPLING,iwNearOneExponent_sub_one; iwNearOneExponent_reciprocal; iwRho_rank;
iwWidth_denominator_separation; iwFrequencySet_mem; taoArc_denominator_bound;
quantitative_iw_frequency_geometry; quantitative_iw_goodness; quantitative_iw_construction_good;
quantitative_iw_multiplier; iwCyclicSymbol_real; quantitative_iw_sampling; iw_finite_lift_diagonal;
iw_finite_lift_size; iw_narrow_lift_diagonal; quantitative_iw_real_construction_good;
quantitative_real_iw_smooth_multiplier; quantitative_real_iw_schwartz_sampling;
cyclicRealOperator_Lp_bound; cyclicRealOperator_TV_one_bound; schwartz_integer_test_summable;
cyclicFunctionLp_size; source_iw_rough_multiplier_norm; quantitative_real_iw_full_sampling;
fullIWInterpolation_diagonal; fullIWInterpolation_unique; fullIWInterpolation_source_bounds;
quantitative_full_shannon_diagonal; iw_frequency_in_cyclic_basis; iw_coverage_all_fraction_points;
fullIntegerKernelMultiplier_unique; iw_source_exponent_one_le; full_iw_source_analytic_clauses;
proposition_quantitative_IW_full,docs/SECTION_7_PROGRESS.md; docs/SECTION_7_STATEMENT_REVIEW.md;
docs/SECTION_7_IW_BOUNDARY.md; blueprint/Checkpoint_840_QuantitativeIWParameters.md;
blueprint/Checkpoint_845_IWFiniteFrequencyGeometry.md; blueprint/Checkpoint_847_IWGoodness.md;
blueprint/Checkpoint_852_IWMultiplierInterface.md; docs/SECTION_7_SAMPLING_BOUNDARY.md;
blueprint/Checkpoint_925_CyclicLiftCharacters.md; blueprint/Checkpoint_930_IWNarrowLiftIdentity.md;
docs/REAL_IW_EXTENSION_REVIEW.md; docs/FULL_IW_EXTERNAL_BOUNDARY.md; docs/FULL_QUANTITATIVE_IW_COMPLETIO
N.md,4598,6dcd3eff30886df0f9ab8ba39abb4dc0be168065d1270cb3560476bb59d65dd9
7.16,lemma,l:continuousestimates7,Uniform continuous annular increment and maximal estimates,Orlicz and
ergodic estimates,conditional,"Complete printed continuous lemma on the full Lp domain. One absolute
positive constant is chosen before beta, p, J, signs and f. The actual signed-increment kernel has
exactly the source multiplier and a full-Lp bound for every1<p<=2, arbitrary complex signs of modulus<=1
 and every0<beta<=1. The maximum is over every real N>=2, represented as a proper extended-real
supremum; a fixed countable dense set and proved scale continuity establish measurability, with a common
 almost-everywhere bound, finiteness and actual Lp membership. The literal real-scale kernel/Fourier
identity and all annular L2/Hormander hypotheses are proved. The general one-sided Hardy–Littlewood and
scalar Hormander theorems in their full-Lp forms remain unproved explicit independent inputs.",No
remaining source-specific clause within the agreed conditional scope. The full-domain general theorem
interfaces and their exact published-to-Lean formulations remain external; they are not claimed to have
been derived from the earlier restricted-domain
inputs.,FULL_DOMAIN_CONTINUOUS_STANDARDS,continuousAnnularSignedMultiplier_L2;
continuousAnnularSignedKernel_fourier; continuousAnnularSignedKernel_hormander;
continuousAnnularSignedMultiplier_Lp; continuousAnnular_maximal_Lp;
continuousAnnularCutoffSignedMultiplier_Lp; annularCutoffYoungMultiplier_Lp;
annularOscillatoryModel_contDiff; realContinuousAnnularKernel_fourier;
realContinuousAnnularConvolution_continuous_scale; realContinuousAnnularMax_countable;
realContinuousAnnular_maximal_Lp; continuousAnnularSigned_full_Lp;
source_continuous_annular_estimates,docs/SECTIONS_3_TO_8_REVIEW.md;
docs/FULL_CONTINUOUS_SOURCE_COMPLETION.md; blueprint/Checkpoint_1044_FullContinuousSource.md,4838,d36373
8205aead2d0ac7c8fd176bd35c6d38d201045193a436f7a23ee26767c4
7.22,proposition,p:orlicz-low,Quantitative Orlicz low estimate,Orlicz and ergodic
estimates,conditional,"The full actual dyadic IW low maximum is proved at every real Q>=2^k, uniformly
in the cyclic measure of variation at most one and 0<beta<=1, for the real-Q geometric construction and
for all integer lp inputs at p_r. The width, denominator bound and logarithmic cost are the literal
real-Q quantities. A single harmless enlargement of the universal parameter constant from C to C^2 is
made once. The arithmetic set, period, rational coverage through Q, denominator bound and goodness are
retained; the maximal estimate uses exactly that set, period and real-Q cutoff. The adjusted-base proof
and cost comparison are checked, as are all own symbol, smoothness, sampling, short-scale, narrow,
remainder and all-input extension steps. The independent general standard inputs remain explicit; no
current-paper conclusion is an external assumption.","No own-paper deduction remains for this
proposition. The recorded independent general construction, IW multiplier, sampling, continuous and
signed-maximal theorems are explicit unformalized inputs, not proved Lean axioms.",QUANTITATIVE_IW;
GENERAL_VARIATION_SAMPLING,iw_annular_single_multiplier; iw_annular_signed_increments;
iw_short_scale_maximum; iw_signed_annular_operator_identity; iw_signed_annular_multiplier;
iw_short_transition_maximum; iw_short_transition_cost_bound; iw_transition_log_cost_budget;

iw_finite_input_low_majorant; iw_narrow_integer_majorant; iw_large_remainder_majorant;
iw_narrow_lift_diagonal; iw_finite_lift_size; iw_full_low_maximal_bound; iw_low_all_integer_lp;
finite_input_maximal_extension; allInputLpMaximalBound_maximum; iw_full_square_low_maximal_bound;
analytic_iw_full_low_maximal_bound; realIWAdjustedBase_width; realIWAdjusted_cost;
real_iw_full_low_maximal_bound; real_iw_low_all_integer_lp; real_iw_source_construction_and_low;
allInputLpMaximalBound_integerLp_maximum,docs/SECTION_7_PROGRESS.md; docs/SECTION_7_STATEMENT_REVIEW.md;
docs/SECTION_7_IW_BOUNDARY.md; docs/SECTION_7_CONTINUOUS_BOUNDARY.md;
docs/SECTION_7_SIGNED_MAXIMAL_BOUNDARY.md; blueprint/Checkpoint_903_IWAnnularLp.md;
blueprint/Checkpoint_907_IWShortScaleMaximum.md; blueprint/Checkpoint_912_IWShortTransitionMaximum.md;
docs/SECTION_7_SAMPLING_BOUNDARY.md; blueprint/Checkpoint_934_IWFullLowMaximum.md;
blueprint/Checkpoint_938_IWAllInputLowMaximum.md;
docs/REAL_IW_LOW_COMPLETION.md,5006,898c4f8f77d3f6133cf7f4cfcf9343a1acec4478fe447b1ca7ff49063ff2fd7d
7.30,proposition,p:orlicz-hilo,Quantitative Orlicz high-low estimate,Orlicz and ergodic
estimates,conditional,"The complete real-Q high-low proposition is proved as an implication from the
recorded independent standard inputs. For every real Q>=2^k, one unchanged floor-Q arithmetic
construction supplies both prime and ordinary-square decompositions at the literal real-Q cutoff. It
retains period<=3^Q, rational coverage through Q, the literal denominator bound and goodness. Both
actual countable low bounds hold on all integer lp inputs at p_r, with the exact logarithmic real-Q
cost. Both actual high maxima share the same arithmetic set, period and selected Page datum and have a
proved summed Q^(-1/4) L2 bound above the stated exponential starting scale. All operator identities and
applicability are checked. The fixed absolute parameter constant is enlarged once from C to C^2; no
width is rounded and no own-paper endpoint is an external premise.","No own-paper deduction remains for
this proposition. The recorded independent general construction, multiplier/sampling,
continuous/signed-maximal, smooth multifrequency, Page/prime-distribution and scalar quadratic-Weyl
inputs remain explicit and unproved. Their hypotheses are not Lean axioms or assertions of
truth.",QUANTITATIVE_IW; GENERAL_VARIATION_SAMPLING; PROGRESSION_ESTIMATE; REN_GHOSH;
SMOOTH_MULTIFREQUENCY; PAGE_CLASSES; QUADRATIC_WEYL_FJK,iwPrime_decomposition; iwPrimeHigh_error_model;
iw_prime_high_maximum; iw_prime_high_source_scale; cyclicReducedSquare_conductor_zero;
cyclicRationalCoefficient_primitive_induced; fixed_small_page_datum; iw_original_common_comparison;
iw_prime_low_maximal_bound; iw_prime_annular_restricted_distribution; iwFullSquare_decomposition;
iw_full_square_low_maximal_bound; iw_full_square_high_source_scale;
iw_full_square_annular_restricted_distribution; iw_prime_high_maximum_for_data;
iw_full_square_high_source_scale_for_data; realIWUpperBase_cutoff; real_iw_high_comparison;
real_iw_prime_low_maximal_bound; real_iw_full_square_low_maximal_bound; real_iw_source_high_low;
real_iw_two_low_maximum_norms,docs/SECTION_7_PROGRESS.md; docs/SECTION_7_STATEMENT_REVIEW.md;
docs/SECTION_7_IW_BOUNDARY.md; blueprint/Checkpoint_865_IWCharacterModelComparison.md;
blueprint/Checkpoint_868_IWPageModelMaximum.md; blueprint/Checkpoint_871_IWPrimeHighSourceScale.md;
blueprint/Checkpoint_938_IWAllInputLowMaximum.md;
blueprint/Checkpoint_940_IWPrimeRestrictedDistribution.md; docs/SECTION_7_RESTRICTED_COMPLETION.md;
docs/THEOREM_7_2_COMPLETION.md; docs/SECTION_7_QUADRATIC_WEYL_BOUNDARY.md;
blueprint/Checkpoint_967_IWFullSquareHighSourceScale.md;
blueprint/Checkpoint_969_IWFullSquareRestrictedDistribution.md; docs/REAL_IW_HIGH_LOW_COMPLETION.md,5322
,1a92c006599a6cbf381912a2d148a819ee489cf772e3362b08908c826b653bea
7.46,corollary,c:orlicz-ergodic,Ergodic transfer and quantitative endpoint conclusions,Orlicz and
ergodic estimates,conditional,"Complete printed Corollary7.46 as an implication of the explicit
independent inputs, for all three actual families and every invertible measure-preserving probability
system. One common C >= 1 serves the strong exponential rate and source Orlicz gauge; one common modular
multiplier is fixed before all systems, families, functions and levels. Every 1 < p <= 2 has the actual
strong maximal norm with almost-everywhere finiteness and actual Lp membership. The proper extended
maximum satisfies the source modular inequality. Every measurable Orlicz member at some positive scale
has actual averages converging almost everywhere and in L1 to the same genuine L1 limit, with
integrability of every average and both L1 seminorm/integral-norm convergence checked. Own truncation,
rescaling, Cauchy/density, L1 operator and limit-identification deductions are proved. At checkpoint1264
the current public export consumes the proved finite signed maximal theorem at constant6; a supplied
RawFiniteSignedMaximalInput and its constant R are no longer hypotheses.","No remaining printed clause
within the agreed conditional-formalization scope. The independent general Calderon strong/modular and
bounded-class polynomial convergence inputs, including their unproved published-theorem/normalization
bridges, remain explicit, as do the earlier Theorem7.2 independent inputs. Stronger standalone Section7
auxiliary ranges remain separately partial.",QUANTITATIVE_IW; GENERAL_VARIATION_SAMPLING;
PROGRESSION_ESTIMATE; REN_GHOSH; SMOOTH_MULTIFREQUENCY; PAGE_CLASSES; QUADRATIC_WEYL_FJK;
ERGODIC_TRANSFER; BOUNDED_SQUARE_POINTWISE,ergodic_strong_maximal_source;
section7_positive_modular_bound; section7_ergodic_modular_apply; nearOneOrliczMember_iff;
orlicz_truncation_modular_tail; orlicz_ergodic_ae_cauchy; ergodicSquareL1Operator_apply_toLp;
ergodicSquareL1Operator_norm_le; L1_bounded_truncation_approximation; bounded_L1_cauchy_of_ae_cauchy;
uniform_operator_dense_cauchy; bounded_truncation_cauchy_from_input; orlicz_ergodic_convergence;
section7_ergodic_assembly; corollary_7_46;
corollary_7_46_closed_signed_maximal,docs/COROLLARY_7_46_COMPLETION.md;
docs/ERGODIC_INDEPENDENT_BOUNDARY.md; docs/SECTION_7_PROGRESS.md; docs/SECTION_7_STATEMENT_REVIEW.md;
blueprint/Checkpoint_998_OrliczErgodicConvergence.md;
blueprint/Checkpoint_999_CorollarySevenFortySix.md;
docs/PAPER_COMPLETION.md,6002,1f6aa0255fe3599c207ff79c72a65dcc46b673c53150103116977e682a830123
8.1,theorem,t:l1badblackbox,The paper-specific persistent badness criterion,Badness and
convergence,conditional,"Complete persistent-divergence adaptation from original general LaVictoire weak
maximal failure and separately specified Sawyer and Conze principles. The arithmetic record contains
actual cyclic residue sets, CRT, represented classes, odd coprime moduli, exact density infimum,
little-o thickening, far-pair liminf and strict finite-product residue frequencies. For every nonatomic
ergodic probability system and every infinite cutoff set, Lean produces a measurable integrable function
and a measurable positive-measure set with no finite pointwise limit. Nonatomicity means measurable
splitting, not merely null singletons. Positive moduli include the usual modulus-one convention; prime
and cutoff indices are zero-based enumerations of the source positive-index sequences.","No remaining
persistent-divergence deduction within the agreed conditional scope. The original LaVictoire, Sawyer and
Conze inputs and their precise published-to-Lean convention bridges are unproved.",LAVICTOIRE; SAWYER;
CONZE,weak_failure_produces_infinite_maximal; persistent_l1_badness_from_arithmetic;
source_lavictoire_persistent_criterion,docs/SECTION_8_COMPLETION.md; docs/SECTION_8_INDEPENDENT_INPUTS.m
d,6128,81c703d95864ff9ec5b263ccc034ef86477105d8147039a5017e8562d0e9a818
8.6,theorem,t:grankul,Quoted Granville-Kurlberg spacing theorem,Badness and
convergence,external_candidate,"Quoted general deep theorem remains unproved. A precise interface now
specifies arbitrary CRT residue families, uniform actual local correlations, one positive common
exponent, constants chosen before primes/shifts, subpower reciprocal local density and growing spacing
along odd squarefree moduli. The full joint cyclic gap conclusion is implemented. All its unit-square
hypotheses and the actual joint and single-gap applications are proved conditional on that independent

general input and Weil.",The quoted general GK theorem and its published-to-Lean cyclic joint formulation bridge remain external and unproved. It is not needed for the repaired Theorem1.5(i) proof.,GRANVILLE_KURLBERG,RawGranvilleKurlbergInput; GKLocalHypotheses; CyclicPoissonSpacings; unitSquareFamily_gk_local; badnessBlockModulus_squarefree; unitSquareBlock_poisson; unitSquareBlock_gap_tail,docs/SECTION_8_COMPLETION.md; docs/SECTION_8_INDEPENDENT_INPUTS.md; blueprint/Checkpoint_1036_GranvilleKurlbergInterface.md,6170,e6d691f13c7f2544c4e052291276ac15619cfbaeaa4293a23fc481794f63c0f7
8.8,lemma,l:quadratic-GK,Quadratic-residue correlation input for spacing,Badness and convergence,conditional,"Complete printed quadratic-residue correlation lemma as an implication of RawSplitQuadraticWeilInput, the general distinct-linear-factor character-sum bound. The literal finite residue correlation has explicit absolute error at most2*k*sqrt(p), actual density-normalized relative error at most3*k*3^k/sqrt(p), and the source GK error with c=1/3 and constant6*k*3^k. Quantification is uniform over every odd prime and every injective tuple of k shifts, including the source h0=0 case. All zero corrections, density factors, positivity, finite subset sums and reciprocal local-density subpower bounds are proved.",No remaining printed correlation clause in the agreed conditional scope. The general Weil input remains unproved. The separate moment repair and badness deductions are now also complete; their inputs are separately recorded.,WEIL,unitSquareIndicator_formula; unitSquareResidues_prime_card_real; quadraticCorrelation_as_product; quadraticSoftCorrelation_error; quadraticCorrelation_Weil_error; quadraticCorrelation_relative_error; quadratic_GK_absolute_error; quadratic_GK_relative_error; quadratic_GK_uniform_correlation; quadratic_GK_tuple; primeResidueDensity_subpower,docs/SECTION_8_QUADRATIC_CORRELATION_COMPLETION.md; docs/SECTION_8_INDEPENDENT_INPUTS.md; blueprint/Checkpoint_1009_QuadraticGKCorrelation.md,6190,51ede37ed48e112cfefdc6523debfbb7b45c73df651fefac5e63db4b4e3420ed
9.2,lemma,l:fromsiftedtop,Reduction from a sifted set to prime-square representations,Goldbach and sieve,proved,"The complete first Section9 reduction is proved from its ordinary range hypotheses. The primitive sum-of-two-squares criterion is now a project theorem, not an additional input. Actual finite sifted and good representation sets and the uniform O(N^epsilon) exception are retained. At1262: Every Section9 numbered source claim and its final quantitative theorem is complete in the agreed scope. All elementary arithmetic reductions are now proved without an extra primitive two-square assumption. W(alpha) convergence is proved for every alpha>1 and the smooth seed is constructed. Only the explicit independent analytic inputs remain for the circle-method/sieve/headline estimates. The user is applying previous manuscript edits; no application is assumed and original sources are unchanged.",,,goldbach_shared_prime_eq; goldbach_shared_divisor_exception; goldbachDivisorExceptions_card; goldbach_prime_divisor_power_error; goldbach_representation_admissible; goldbach_factor_coprime_of_not_exception; goldbachSiftedValues_card; goldbach_full_sifted_inclusion; goldbach_full_sifted_count; goldbach_from_full_sifted; goldbach_coprime_zmod_not_both_zero; goldbach_prime_norm_coprime; goldbach_square_matrix_kernel; goldbach_conjugate_kernel_separation; goldbach_primitive_pair_from_integer_norm; goldbach_primitive_prime_matrix; goldbach_primitive_multiply_one_mod_four_prime; goldbach_primitive_one_mod_four_support; goldbach_primitive_double_of_odd; goldbach_primitive_two_squares_criterion; goldbach_weighted_W_integrable_general; goldbach_weighted_W_nonneg_general; goldbach_fixed_smooth_seed_at_zero; goldbach_fixed_smooth_seed_nonzero; goldbach_fixed_smooth_seed_support; goldbach_fixed_smooth_seed_compact_support; goldbach_fixed_smooth_seed_tsupport_subset; goldbach_smooth_seed_exists; goldbach_from_sifted_to_primes_complete; goldbach_truncation_complete; goldbach_section9_master_exact; goldbach_master_sieve_complete; goldbach_section9_headline_complete; goldbach_section9_exceptional_density_zero,docs/SECTION_9_FIRST_REDUCTION.md; blueprint/Checkpoint_1260_GoldbachPrimitiveProducts.md; blueprint/Checkpoint_1261_GoldbachPrimitiveCriterion.md; blueprint/Checkpoint_1262_GoldbachSectionNineComplete.md,6577,80b38572490f78eff5de2025b191aa7c569a91cc545911c5e463681866aaf2a3
9.3,lemma,l:truncationT,Truncation identity and the residual set T,Goldbach and sieve,proved,"The full source truncation identity is proved with an explicit nonnegative error bounded by O(N^epsilon), including repeated3 mod4 prime factors and the actual source factorization set. The primitive criterion is internally proved; only the normal source range hypotheses remain. At1262: Every Section9 numbered source claim and its final quantitative theorem is complete in the agreed scope. All elementary arithmetic reductions are now proved without an extra primitive two-square assumption. W(alpha) convergence is proved for every alpha>1 and the smooth seed is constructed. Only the explicit independent analytic inputs remain for the circle-method/sieve/headline estimates. The user is applying previous manuscript edits; no application is assumed and original sources are unchanged.",,,goldbach_shared_prime_eq; goldbach_shared_divisor_exception; goldbachDivisorExceptions_card; goldbach_prime_divisor_power_error; goldbach_representation_admissible; goldbach_factor_coprime_of_not_exception; primitive_obstruction_two_large_primes; goldbach_sifted_bad_factorization; goldbachTruncatedBad_subset_sifted; goldbachTruncatedBad_not_good; goldbach_good_subset_sifted; goldbachTruncatedBadValues_card; goldbach_truncation_exact_error; goldbach_truncation_scale; goldbach_truncationT; goldbach_coprime_zmod_not_both_zero; goldbach_prime_norm_coprime; goldbach_square_matrix_kernel; goldbach_conjugate_kernel_separation; goldbach_primitive_pair_from_integer_norm; goldbach_primitive_prime_matrix; goldbach_primitive_multiply_one_mod_four_prime; goldbach_primitive_one_mod_four_support; goldbach_primitive_double_of_odd; goldbach_primitive_two_squares_criterion; goldbach_weighted_W_integrable_general; goldbach_weighted_W_nonneg_general; goldbach_fixed_smooth_seed_at_zero; goldbach_fixed_smooth_seed_nonzero; goldbach_fixed_smooth_seed_support; goldbach_fixed_smooth_seed_compact_support; goldbach_fixed_smooth_seed_tsupport_subset; goldbach_smooth_seed_exists; goldbach_truncation_complete; goldbach_from_sifted_to_primes_complete; goldbach_section9_master_exact; goldbach_master_sieve_complete; goldbach_section9_headline_complete; goldbach_section9_exceptional_density_zero,docs/SECTION_9_TRUNCATION.md; blueprint/Checkpoint_1260_GoldbachPrimitiveProducts.md; blueprint/Checkpoint_1261_GoldbachPrimitiveCriterion.md; blueprint/Checkpoint_1262_GoldbachSectionNineComplete.md,6605,f87d8421b6f7544d3297232dd34e4500a62e187b1b958021e5b8ff9130a916d3
9.7,lemma,l:upperreduction,Upper reduction of T to the family of sifted sets,Goldbach and sieve,proved,"For the user-corrected closed-cutoff L, the entire actual upper-sifted inclusion, exact finite cardinal upper bound with error omega(n-1), and uniform O_epsilon(N^epsilon) bound are proved. No analytic, sieve, primitive two-square or own-paper endpoint is assumed. All eight closed cutoff occurrences are verified in the latest upload.",,,goldbach_shared_prime_eq; goldbach_shared_divisor_exception; goldbachDivisorExceptions_card; goldbach_prime_divisor_power_error; goldbach_representation_admissible; goldbach_factor_coprime_of_not_exception; goldbachClosedL_bound; goldbach_upper_sifted_inclusion; goldbach_upper_reduction_exact; goldbach_upper_reduction; goldbachClosedL_source_form,docs/SECTION_9_MASTER_REDUCTION.md; docs/SECTION_9_CUTOFF_CHECK_1093.md; docs/SOURCE_UPDATE_1107.md,6706,4ffc06f1022ed58eb24be09ce63cb187ce9e013f42f81bd04b0b256551e6a945
9.9,corollary,c:mastersievereduction,Combined lower and upper sieve reduction,Goldbach and sieve,proved,"The full source master sieve reduction is proved from ordinary source hypotheses and the

adopted closed multiplier cutoff. Its exact finite error is at most2 omega(n-1), and its uniform
O(N^epsilon) consequence is exported. No primitive-criterion or analytic assumption is needed for this
arithmetic reduction. At1262: Every Section9 numbered source claim and its final quantitative theorem is
 complete in the agreed scope. All elementary arithmetic reductions are now proved without an extra
primitive two-square assumption. W(alpha) convergence is proved for every alpha>1 and the smooth seed is
 constructed. Only the explicit independent analytic inputs remain for the circle-method/sieve/headline
estimates. The user is applying previous manuscript edits; no application is assumed and original
sources are unchanged.",,,goldbach_master_sieve_exact; goldbach_master_sieve_reduction;
goldbach_coprime_zmod_not_both_zero; goldbach_prime_norm_coprime; goldbach_square_matrix_kernel;
goldbach_conjugate_kernel_separation; goldbach_primitive_pair_from_integer_norm;
goldbach_primitive_prime_matrix; goldbach_primitive_multiply_one_mod_four_prime;
goldbach_primitive_one_mod_four_support; goldbach_primitive_double_of_odd;
goldbach_primitive_two_squares_criterion; goldbach_weighted_W_integrable_general;
goldbach_weighted_W_nonneg_general; goldbach_fixed_smooth_seed_at_zero;
goldbach_fixed_smooth_seed_nonzero; goldbach_fixed_smooth_seed_support;
goldbach_fixed_smooth_seed_compact_support; goldbach_fixed_smooth_seed_tsupport_subset;
goldbach_smooth_seed_exists; goldbach_master_sieve_complete; goldbach_from_sifted_to_primes_complete;
goldbach_truncation_complete; goldbach_section9_master_exact; goldbach_section9_headline_complete;
goldbach_section9_exceptional_density_zero,docs/SECTION_9_MASTER_REDUCTION.md;
docs/SOURCE_UPDATE_1107.md; blueprint/Checkpoint_1260_GoldbachPrimitiveProducts.md;
blueprint/Checkpoint_1261_GoldbachPrimitiveCriterion.md; blueprint/Checkpoint_1262_GoldbachSectionNineCo
mplete.md,6758,6602557c3014d261f87af428332e38894bd5c2acbf86b21022dea54e12fb01f4
9.10,lemma,l:circmethod,Weighted Bombieri-Vinogradov circle-method approximation,Goldbach and
sieve,conditional,"Complete source circle-method mean-square approximation for every A,delta>0, with
B0,C,K,N0 chosen before N and every arithmetic family. Covers all B>=B0, exact strict real
L<N^(2-delta), positive ell,k with k<V(ell), every coprime residue class, arbitrary complex coefficients
 of norm<=1, and all n<N^2. The stronger uniform cutoff statement holds for every 2<=U<=N,W>=2. The
final source theorem removes cutoffs using the proved equation-imposed bounds, and uses the actual
finite prime-pair count, actual local Euler factor G_R and actual autocorrelation singular integral H_R.
 Fourier inversion, all three error identifications, shared constants, minor estimate, major
replacement, arithmetic completion and smoothing are consumed internally. Its only unformalized
hypotheses are independent ordinary SW prime counts, ordinary unweighted BV prime-count distribution,
and uniform raw Ren/Ghosh quadratic estimates. No own-paper conclusion is assumed. At1212: Completed all
 Section10 source statements and the full circle-method implication; the three independent raw deep
inputs remain explicit and unproved. At1213: Completed every Section10 source statement and the full
source circle-method implication, including the original smooth compact seed and exact normalized
integral. Only the three agreed independent raw deep inputs remain for this circle-method
implication.",None within the full source circle-method implication. The three independent raw
distribution/exponential-sum inputs remain unproved; Section11 and the headline Goldbach result remain
separate work.,REN_GHOSH; SIEGEL_WALFISZ_PRIME_COUNTS;
BOMBIERI_VINOGRADOV,goldbach_cutoff_fourier_inversion; goldbach_full_cutoffs_redundant;
goldbach_full_fourier_inversion; goldbach_weighted_full_fourier_inversion; goldbach_actual_minor_bessel;
 goldbach_major_minor_partition; goldbach_circle_three_stage_bound; goldbach_frequency_energy_identity;
goldbach_progression_energy_le_collisions; goldbach_actual_minor_collision_bound;
goldbach_prime_collision_off_diagonal; goldbach_reciprocal_lcm_harmonic_cube;
goldbach_reciprocal_lcm_log_cube; goldbach_reciprocal_lcm_strict_real_cutoff;
goldbach_congruence_interval_card; goldbach_simultaneous_congruence_count_real;
goldbach_congruence_square_identity; goldbach_congruence_square_harmonic_bound;
goldbach_congruence_square_strict_real_cutoff; goldbach_frequency_energy_multiplicity;
goldbach_prime_product_fibre_card; goldbach_prime_product_energy; goldbach_index_row_sum;
goldbach_progression_polynomial_by_rows; goldbach_progression_energy_by_row_squares;
goldbach_total_row_square_bound; goldbach_progression_energy_harmonic;
goldbach_progression_energy_log_five; goldbach_source_energy_numerical_ranges;
goldbach_source_progression_energy; goldbach_source_progression_energy_printed_range;
goldbach_minor_logarithmic_exponents; goldbach_hybrid_minor_from_pointwise; goldbach_lcm_fibre_card;
goldbach_lcm_weighted_square_divisor_sum; goldbach_major_progression_error_sum;
goldbach_lower_sieve_residue_from_support; goldbach_upper_sieve_residue;
goldbach_upper_sieve_common_product_residue; goldbach_finite_abel_bound;
goldbachLogRemovalWeight_antitone; goldbachLogRemovalWeight_of_two_le;
goldbach_log_removed_mangoldt_bound; goldbach_prime_power_log_removal_error;
goldbach_prime_from_mangoldt_partials; goldbach_halving_partial_bound;
goldbachMangoldtPrefix_annular_norm; goldbach_mangoldt_partials_from_annular;
goldbachMinorArcs_measurable; goldbach_minor_arcs_source_centres; goldbach_minor_arcs_rescale;
goldbach_common_annular_minor; goldbach_prime_minor_from_numeric_parameters;
goldbach_log_minor_radius_identity; goldbach_log_minor_annular_budget;
goldbach_prime_minor_log_parameters; goldbach_minor_real_eventual_budgets;
goldbach_minor_eventual_numeric_parameters; goldbach_prime_minor_uniform;
goldbach_prime_minor_source_cutoff_algebra; goldbach_prime_minor_source_high_range;
goldbachHybridMinorConstant_pos; goldbach_hybrid_minor_complete; goldbach_progression_strict_endpoint;
goldbach_progression_endpoint_card; goldbach_progression_endpoint_error;
goldbach_progression_cutoff_min; goldbach_prime_progression_count_sum; goldbachLi_hasDerivAt;
goldbachPrimeProgressionModel_hasDerivAt; goldbachPrimeProgressionDensity_continuousOn;
goldbach_counting_model_error_identity; goldbach_counting_partial_error_bound;
goldbach_prime_weighted_count_sum; goldbach_prime_progression_partial_error;
goldbach_square_density_integral; goldbach_linear_density_integral;
goldbach_prime_quadratic_count_error; goldbach_prime_linear_count_error; goldbach_prime_positive_symbol;
 goldbach_square_prime_residue_sum; goldbach_prime_nonunit_symbol; goldbach_square_major_from_counts;
goldbach_lcm_quotient_range; goldbach_lcm_fibre_card_le_q; goldbach_lcm_unweighted_sum;
goldbach_major_error_from_unweighted_distribution; goldbach_major_residues_modq_injective;
goldbach_major_residues_card; goldbach_major_compatible_lift; goldbach_major_residue_compatible;
goldbach_major_residue_image; goldbach_major_residue_sum; goldbach_major_compatibility_character;
goldbach_major_upsilon_identity; goldbach_prime_count_discrepancy_real;
goldbach_major_radius_eventual_budget; goldbach_square_major_numeric; goldbach_major_log_product;
goldbach_square_major_siegel_walfisz; goldbach_good_linear_maps_to; goldbach_good_linear_fibre;
goldbach_linear_rational_phase; goldbach_good_linear_residue_grouping; goldbach_good_linear_model_error;
 goldbach_linear_nonunit_card; goldbach_linear_nonunit_error; goldbach_linear_major_from_counts;
goldbach_linear_source_domain_ae; goldbach_linear_source_integral;
goldbach_linear_source_major_from_counts; goldbach_linear_source_major_numeric;
goldbach_linear_error_sum_unweighted; goldbach_linear_major_bv_row; goldbach_progression_scale_logs;
goldbach_bv_log_level; goldbach_bv_log_saving; goldbach_bv_source_eventual_threshold;
goldbach_lcm_bv_level; goldbach_source_row_subset; goldbach_source_row_scale_sum;

goldbach_source_row_card; goldbach_source_index_card_square; goldbach_source_index_card_printed;
goldbach_major_finite_error_eventual; goldbach_major_finite_error_numeric;
goldbach_source_finite_error_absorbed; goldbach_bv_aggregate_log_product;
goldbach_linear_major_bombieri_vinogradov; mem_goldbachMajorArcs; goldbach_major_arcs_measure;
goldbach_circle_ball_real_lift; goldbach_major_balls_disjoint; goldbach_major_geometry_eventual;
goldbachArcCoordinate_measurable; goldbachArcCoordinate_abs; goldbachArcCoordinate_reconstruct;
goldbachSquareIntegral_stronglyMeasurable; goldbachLinearSourceIntegral_stronglyMeasurable;
goldbachSquareIntegral_norm; goldbachLinearSourceIntegral_norm; goldbachArcModel_stronglyMeasurable;
goldbachArcModel_memLp; goldbachArcModel_on_arc; goldbachSquareMajorModel_memLp;
goldbachProgressionMajorModel_memLp; goldbachSquareMajorModel_norm_on_major;
goldbach_square_circle_major_sw; goldbach_linear_circle_major_bv; goldbach_restricted_function_bessel;
goldbach_restricted_uniform_bessel; goldbach_restricted_measurable_multiplier_bessel;
goldbach_two_model_major_bessel; goldbachWeightedProgressionMajorModel_memLp;
goldbach_weighted_major_error_le_sum; goldbach_major_weighted_coefficient_identity;
goldbach_major_square_log_identity; goldbach_major_mean_square_budget;
goldbach_major_approximation_complete; goldbach_smooth_coefficient_abs_summable;
goldbach_smooth_series_divisor_sum; goldbach_arithmetic_factor_euler;
goldbach_arithmetic_factor_finite_split; goldbach_quadratic_interval_decay;
goldbachSquareIntegral_decay; goldbach_linear_interval_decay; goldbachLinearIntegral_decay;
goldbachLinearSourceIntegral_decay; goldbach_reciprocal_totient_majorant;
goldbach_reciprocal_totient_sum; goldbach_reciprocal_totient_multiples;
goldbach_gcd_totient_divisor_expansion; goldbach_gcd_weighted_divisor_sum; goldbach_gcd_reciprocal_sum;
goldbach_gcd_reciprocal_totient_sum; goldbach_gcd_product_bound; goldbach_arithmetic_harmonic_rectangle;
goldbach_source_arithmetic_harmonic; goldbach_general_coefficient_of_dvd;
goldbach_general_coefficient_of_coprime_unit; goldbach_general_coefficient_of_coprime_zero;
goldbach_odd_prime_cohen_norm; goldbach_general_coefficient_odd_prime_bound;
goldbachLinearSourceIntegral_norm_scale; goldbach_euclidean_product_flat;
goldbach_euclidean_product_decay; goldbach_euclidean_product_tail; goldbach_upsilon_gcd_normalization;
goldbach_progression_exponential_trivial; goldbach_upsilon_trivial_gcd;
goldbach_general_coefficient_trivial; goldbach_general_coefficient_small_two_power;
goldbach_general_coefficient_prime_power; goldbach_gcd_ratio_multiplicative;
goldbach_gcd_ratio_factorization; goldbach_two_pow_primeFactors_le_divisors;
goldbach_prime_factor_weight_le_divisors; goldbach_general_coefficient_factor_bound;
goldbach_general_coefficient_divisor_bound; goldbach_principal_square_divisor_bound;
goldbach_source_arithmetic_coefficient_sum; goldbach_source_finite_arithmetic_tail;
goldbach_normalized_group_sum_pullback; goldbach_normalized_units_sum_pullback;
goldbach_progression_exponential_zero_translation; goldbach_progression_exponential_noncoprime;
goldbach_progression_exponential_coprime_norm; goldbach_sum_units_range; goldbach_unit_lcm_residue_sum;
goldbach_upsilon_lcm_unit_average; goldbach_unit_reduction_comp; goldbach_upsilon_frequency_reduction;
goldbach_progression_exponential_primitive_norm; goldbach_upsilon_primitive_decay;
goldbach_reciprocal_totient_divisor_bound; goldbach_upsilon_gcd_divisor_decay;
goldbach_upsilon_integer_gcd_decay; goldbach_arithmetic_uniform_bounds;
goldbach_arithmetic_harmonic_common_exponent; goldbach_odd_prime_power_series;
goldbach_two_prime_power_series; goldbach_all_primes_euler_series; goldbach_arithmetic_euler_complete;
goldbach_autocorrelation_integral; goldbach_seed_mass_pos; goldbach_eta_zero;
goldbach_autocorrelation_fourier; goldbach_eta_inverse_fourier_nonnegative;
goldbach_inverse_fourier_scale; goldbach_scaled_eta_inverse_nonnegative;
goldbach_cutoff_square_inverse_nonnegative; goldbach_eta_support; goldbach_scaled_eta_support;
goldbach_eta_real; goldbach_eta_even; goldbach_autocorrelation_norm; goldbach_eta_norm_le_one;
goldbach_eta_deriv_zero; goldbach_eta_quadratic; goldbach_eta_square_cutoff_bound;
goldbach_halfline_integrable_decay; goldbach_two_sided_integrable_decay; goldbach_cutoff_fourier_fubini;
goldbach_positive_cutoff_fourier_average; goldbach_euclidean_product_density;
goldbach_singular_integral_positive; goldbach_singular_integral_empty;
goldbach_singular_integral_nonnegative; goldbach_euclidean_product_integrable;
goldbach_singular_integrand_integrable; goldbach_singular_integral_norm;
goldbach_smoothing_power_identity; goldbach_smoothing_decay_integrable;
goldbach_singular_smoothing_uniform; goldbach_affine_integral; goldbach_quadratic_density_joint;
goldbach_quadratic_average_integrable; goldbach_quadratic_average_fourier;
goldbach_log_window_integrable; goldbach_log_window_continuousAt; goldbach_source_density_integrable;
goldbach_source_density_fourier; goldbach_source_density_continuous;
goldbach_archimedean_integral_domain; goldbach_source_density_archimedean;
goldbach_source_unsmoothed_inverse; goldbach_unsmoothed_archimedean_complete;
goldbach_smooth_compact_seed_nonzero; goldbach_eta_of_smooth_compact_seed;
goldbach_positive_singular_integral_complete; goldbach_rankin_weight_prime_power;
goldbach_rankin_divisor_euler; goldbach_rankin_pointwise; goldbach_finite_rankin_euler_bound;
goldbach_geometric_partial_bound; goldbach_polynomial_local_bound; goldbach_rankin_local_uniform;
goldbach_rankin_prime_rpow; goldbach_prime_subset_inclusive_log; goldbach_rankin_euler_log;
goldbach_rankin_tail_log_bound; goldbach_eventually_log_square_budget; goldbach_rankin_decay_log_budget;
goldbach_rankin_parameter_decay; goldbach_source_rankin_tail; goldbach_source_rankin_arithmetic_tail;
goldbach_arc_coordinate_real_shift; goldbach_arc_lift_integral; goldbach_arc_lift_fourier_integral;
goldbach_arc_lift_energy; goldbach_arc_lift_memLp; goldbach_arc_lifts_disjoint;
goldbach_disjoint_arc_bessel; goldbach_major_envelope_integrable; goldbach_major_envelope_energy;
goldbach_major_envelope_dominated_energy; goldbach_euclidean_product_envelope;
goldbach_major_family_integrand_measurable; goldbach_major_family_integrand_bound;
goldbach_smooth_major_family_measurable; goldbach_smooth_major_family_norm;
goldbach_smooth_major_family_support; goldbach_center_arithmetic_unit_sum;
goldbach_rational_negative_character; goldbach_center_arithmetic_rational;
goldbach_centers_weighted_arithmetic; goldbach_radial_major_family_identity;
goldbach_smoothed_radial_kernel_integrable; goldbach_smoothed_radial_kernel_integral;
goldbach_smoothed_major_family_identity; goldbach_inverse_three_halves_interval;
goldbach_inverse_three_halves_tail; goldbach_divisor_square_tail; goldbach_centers_denominator_sum;
goldbach_centers_filtered_denominator_sum; goldbach_major_prefactor_count;
goldbach_added_denominator_energy_budget; goldbach_smooth_added_major_bessel;
goldbach_completion_geometry; goldbach_completion_parameters_eventual;
goldbach_completion_log_power_rate; goldbach_hard_singular_smoothing_uniform;
goldbach_hard_unsmoothed_comparison; goldbach_divisor_harmonic_initial_sum;
goldbach_small_smoothing_uniform; goldbach_major_model_product; goldbach_arc_model_fourier_identity;
goldbach_actual_major_model_identity; goldbach_initial_coefficients_divisor_sum;
goldbach_completion_arithmetic_split; goldbach_added_completion_identity;
goldbach_completion_error_identity; goldbach_small_completion_square_budget;
goldbach_far_completion_square_budget; goldbach_completion_all_parameters_eventual;

goldbach_major_completion_coefficient_complete; goldbach_major_completion_complete;
goldbach_model_weighted_coefficient; goldbach_source_ranges_eventual;
goldbach_circle_method_cutoff_complete; goldbach_circle_method_complete;
goldbach_source_singular_integral_eq; goldbach_circle_method_source_complete; goldbach_source_singular_i
ntegral_positive_comparison,blueprint/Checkpoint_1115_GoldbachFourierInversion.md;
blueprint/Checkpoint_1116_GoldbachMinorArcBessel.md;
blueprint/Checkpoint_1117_GoldbachProgressionEnergy.md;
blueprint/Checkpoint_1118_GoldbachReciprocalLCM.md;
blueprint/Checkpoint_1119_GoldbachCongruenceCounting.md;
blueprint/Checkpoint_1120_GoldbachCongruenceMultiplicity.md;
blueprint/Checkpoint_1121_GoldbachPrimeFrequencyEnergy.md;
blueprint/Checkpoint_1122_GoldbachProgressionRows.md;
blueprint/Checkpoint_1123_GoldbachLogarithmicEnergy.md;
blueprint/Checkpoint_1124_GoldbachSourceEnergy.md;
blueprint/Checkpoint_1125_GoldbachHybridMinorReduction.md;
blueprint/Checkpoint_1126_GoldbachMajorLCMReindexing.md;
blueprint/Checkpoint_1127_GoldbachSieveResidues.md; blueprint/Checkpoint_1128_GoldbachFiniteAbel.md;
blueprint/Checkpoint_1129_GoldbachPrimeUnweighting.md;
blueprint/Checkpoint_1130_GoldbachHalvingPartials.md;
blueprint/Checkpoint_1131_GoldbachMinorArcGeometry.md;
blueprint/Checkpoint_1132_GoldbachUniformAnnularMinor.md;
blueprint/Checkpoint_1133_GoldbachMinorLogParameters.md;
blueprint/Checkpoint_1134_GoldbachMinorEventualCutoffs.md;
blueprint/Checkpoint_1135_GoldbachPrimeMinorUniform.md;
blueprint/Checkpoint_1136_GoldbachHybridMinorComplete.md;
blueprint/Checkpoint_1137_GoldbachMajorStrictEndpoint.md;
blueprint/Checkpoint_1138_GoldbachPrimeCountingModel.md;
blueprint/Checkpoint_1139_GoldbachCountingPartialSummation.md;
blueprint/Checkpoint_1140_GoldbachPrimePhaseErrors.md;
blueprint/Checkpoint_1141_GoldbachSquareMajorCounts.md;
blueprint/Checkpoint_1142_GoldbachMajorLCMQFibres.md;
blueprint/Checkpoint_1143_GoldbachMajorCompatibleResidues.md;
blueprint/Checkpoint_1144_GoldbachMajorUpsilonIdentity.md;
blueprint/Checkpoint_1145_GoldbachRawPrimeDistribution.md;
blueprint/Checkpoint_1146_GoldbachSquareMajorSW.md;
blueprint/Checkpoint_1147_GoldbachLinearResidueFibres.md;
blueprint/Checkpoint_1148_GoldbachLinearMajorCounts.md;
blueprint/Checkpoint_1149_GoldbachLinearSourceIntegral.md;
blueprint/Checkpoint_1150_GoldbachLinearBVRow.md;
blueprint/Checkpoint_1151_GoldbachBVSourceParameters.md;
blueprint/Checkpoint_1152_GoldbachMajorSourceCounts.md;
blueprint/Checkpoint_1153_GoldbachMajorFiniteErrorAbsorption.md;
blueprint/Checkpoint_1154_GoldbachLinearMajorBV.md;
blueprint/Checkpoint_1155_GoldbachMajorArcGeometry.md;
blueprint/Checkpoint_1156_GoldbachMajorModelRegularity.md;
blueprint/Checkpoint_1157_GoldbachCircleMajorModels.md;
blueprint/Checkpoint_1158_GoldbachCircleMajorEstimates.md;
blueprint/Checkpoint_1159_GoldbachMajorBesselAssembly.md;
blueprint/Checkpoint_1160_GoldbachMajorWeightedAssembly.md;
blueprint/Checkpoint_1161_GoldbachMajorApproximationComplete.md;
blueprint/Checkpoint_1162_GoldbachArithmeticEuler.md;
blueprint/Checkpoint_1163_GoldbachEuclideanDecay.md;
blueprint/Checkpoint_1164_GoldbachTotientHarmonic.md;
blueprint/Checkpoint_1165_GoldbachArithmeticHarmonic.md;
blueprint/Checkpoint_1166_GoldbachArithmeticPrime.md;
blueprint/Checkpoint_1167_GoldbachEuclideanProduct.md;
blueprint/Checkpoint_1168_GoldbachPrimePowerBounds.md;
blueprint/Checkpoint_1169_GoldbachArithmeticDecay.md;
blueprint/Checkpoint_1170_GoldbachArithmeticFamily.md; docs/GOLDBACH_COMPLETION_PROGRESS_1170.md;
blueprint/Checkpoint_1171_GoldbachFiniteUnitAverages.md;
blueprint/Checkpoint_1172_GoldbachPrimitiveProgression.md;
blueprint/Checkpoint_1173_GoldbachUpsilonReduction.md;
blueprint/Checkpoint_1174_GoldbachUpsilonDecay.md;
blueprint/Checkpoint_1175_GoldbachArithmeticComplete.md;
blueprint/Checkpoint_1176_GoldbachAutocorrelation.md; blueprint/Checkpoint_1177_GoldbachScaledCutoff.md;
blueprint/Checkpoint_1178_GoldbachCutoffBounds.md;
blueprint/Checkpoint_1179_GoldbachIntegrableDecay.md;
blueprint/Checkpoint_1180_GoldbachPositiveFourierAverage.md;
blueprint/Checkpoint_1181_GoldbachSingularIntegralPositive.md;
blueprint/Checkpoint_1182_GoldbachSingularIntegralNorm.md;
blueprint/Checkpoint_1183_GoldbachSmoothingDecay.md;
blueprint/Checkpoint_1184_GoldbachSingularSmoothing.md;
blueprint/Checkpoint_1185_GoldbachQuadraticDensity.md;
blueprint/Checkpoint_1186_GoldbachSourceDensity.md;
blueprint/Checkpoint_1187_GoldbachPositiveSingularComplete.md;
blueprint/Checkpoint_1188_GoldbachRankinEuler.md; blueprint/Checkpoint_1189_GoldbachRankinLocal.md;
blueprint/Checkpoint_1190_GoldbachRankinLogProduct.md; blueprint/Checkpoint_1191_GoldbachRankinDecay.md;
blueprint/Checkpoint_1192_GoldbachRankinSource.md;
blueprint/Checkpoint_1193_GoldbachRankinArithmeticTail.md;
blueprint/Checkpoint_1194_GoldbachArcLiftIntegral.md;
blueprint/Checkpoint_1195_GoldbachDisjointArcBessel.md;
blueprint/Checkpoint_1196_GoldbachMajorEnvelope.md;
blueprint/Checkpoint_1197_GoldbachMajorFamilyIntegrand.md;
blueprint/Checkpoint_1198_GoldbachCenterArithmetic.md;
blueprint/Checkpoint_1199_GoldbachRadialMajorIdentity.md;
blueprint/Checkpoint_1200_GoldbachDivisorSquareTail.md;
blueprint/Checkpoint_1201_GoldbachDenominatorEnergy.md;
blueprint/Checkpoint_1202_GoldbachAddedMajorBessel.md;
blueprint/Checkpoint_1203_GoldbachCompletionParameters.md;
blueprint/Checkpoint_1204_GoldbachHardSmoothing.md; blueprint/Checkpoint_1205_GoldbachSmallSmoothing.md;
blueprint/Checkpoint_1206_GoldbachHardMajorIdentity.md;

blueprint/Checkpoint_1207_GoldbachCompletionArithmetic.md;
blueprint/Checkpoint_1208_GoldbachCompletionErrorIdentity.md;
blueprint/Checkpoint_1209_GoldbachCompletionBudgets.md;
blueprint/Checkpoint_1210_GoldbachMajorCompletionBound.md;
blueprint/Checkpoint_1211_GoldbachMajorCompletionComplete.md;
blueprint/Checkpoint_1212_GoldbachCircleMethodComplete.md; blueprint/Checkpoint_1213_GoldbachCircleSourc
eSeed.md,6783,3f5dd83379b996836000219a93279463c1628676e32d55d664fbfeaba63a25b8
9.17,proposition,p:sieveneedlower8,Lower sieve application,Goldbach and sieve,conditional,"Full source
lower-sieve proposition, including the actual representation sequence, original X_n and Euler product,
coefficients before n, a common error tending to zero, and the N^4(log N)^(-A) mean-square remainder
bound. The pointwise theorem holds throughout the source retained range (indeed all n=12 mod24). General
 classical sieve/Mertens/function-continuity and ordinary SW/BV/Ren-Ghosh inputs remain explicit and
unproved. At1229: Common cutoff compatibility for the circle-method and full lower-sieve estimates is
proved for every requested lower bound on C; main-term size remains to be joined. At1230: The full
source main-term size and the entire lower-bound side are complete conditionally, including an exponent
independent of C, uniform smoothing, one common large cutoff and finite error-to-exception deductions.
At1230 the source main-term size and the full lower sieve are proved jointly for the same C>=A0+6;
finite positive-count and exceptional-cardinality deductions are checked. At1258: All Section11 numbered
 claims are complete in the agreed conditional scope for the explicitly supplied corrected presieving
statement. The original weaker statement is not proved and must be replaced in the user manuscript. The
complete joint lower/upper estimates use one cutoff and n-independent coefficients, exact constants,
C_A0=A0/2+6 and arbitrary combined logarithmic mean-square saving. The final Goldbach theorem remains
the next deduction.",,ROSSER_IWANIEC_SINGLE_STAGE; CLASSICAL_INTERVAL_MERTENS;
SIEGEL_WALFISZ_PRIME_COUNTS; BOMBIERI_VINOGRADOV; REN_GHOSH,goldbach_lower_sieve_residue_from_support;
goldbach_upper_sieve_residue; goldbach_upper_sieve_common_product_residue;
goldbach_lower_sieve_sequence_count; goldbach_lower_sequence_fixed_residue;
goldbach_lower_sequence_circle_identity; goldbach_sieve_remainders_selected;
goldbach_pred_square_sum_bound; goldbach_pred_full_square_sum_bound;
goldbach_lower_sieve_parameter_tendsto; goldbach_lower_sieve_levels_eventually;
goldbach_lower_density_multiplicative; goldbach_lower_preliminary_dimension;
goldbach_lower_outer_dimension; goldbach_lower_base_real_positive; goldbach_lower_main_term_nonneg;
goldbach_lower_divisor_real_identity; goldbach_lower_sifted_sequence;
goldbach_lower_remainder_mean_square; goldbach_lower_theta_positive;
goldbach_lower_presieve_application; goldbach_lower_sieve_complete; goldbach_circle_method_above;
goldbach_lower_remainder_mean_square_above; goldbach_lower_sieve_complete_above;
goldbach_lower_main_term_size; goldbach_lower_smoothed_size; goldbach_lower_bounds_complete;
goldbach_lower_bad_card; goldbach_section11_complete,blueprint/Checkpoint_1127_GoldbachSieveResidues.md;
 blueprint/Checkpoint_1216_GoldbachLowerSieveSequence.md;
blueprint/Checkpoint_1218_GoldbachSieveRemainderTransport.md;
blueprint/Checkpoint_1225_GoldbachLowerSieveParameters.md;
blueprint/Checkpoint_1226_GoldbachLowerSieveBridge.md;
blueprint/Checkpoint_1227_GoldbachLowerSieveApplication.md;
blueprint/Checkpoint_1229_GoldbachLowerCutoffFreedom.md;
blueprint/Checkpoint_1230_GoldbachLowerMainTermSize.md; blueprint/Checkpoint_1258_GoldbachSection11Compl
ete.md,6898,a275396d64d20d1ebf9cc847f544d24dc59b6d4139b78144d8369b10e26895ff
9.20,proposition,p:sieveneedupper8,Upper sieve application and summed weights,Goldbach and
sieve,conditional,"The full source upper sieve is complete in uniform epsilon form, with the exact
c_(9/8) F_1(2) W(9/8) coefficient, original actual main term and closed multiplier set. All coefficients
 precede n; source divisor counts, refined preliminary/outer dimensions, uniform varying levels, the
shifted double remainder sum, its circle-method embedding and the fixed-sieve O(N^(1/4)) correction are
proved. The total error has arbitrarily strong logarithmic mean-square saving. Only independent general
SW, BV, Ren/Ghosh, classical one-stage sieve, effective multiplicative mean and Mertens inputs and
general sieve normalizations remain unproved; no own-paper endpoint is assumed. At1257: Full upper-sieve
 proposition p:sieveneedupper8 complete in uniform epsilon form from explicit independent classical
inputs, with the exact coefficient and arbitrary logarithmic mean-square saving. The lower analytic
estimate and both remainder transports are available at a shared prescribed cutoff. The final joint
Section11 assembly and manuscript synchronization are next. At1258: All Section11 numbered claims are
complete in the agreed conditional scope for the explicitly supplied corrected presieving statement. The
 original weaker statement is not proved and must be replaced in the user manuscript. The complete joint
 lower/upper estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and
arbitrary combined logarithmic mean-square saving. The final Goldbach theorem remains the next
deduction.",,BOMBIERI_VINOGRADOV; PROGRESSION_ESTIMATE; REN_GHOSH; ROSSER_IWANIEC_SINGLE_STAGE;
EFFECTIVE_HALF_DIMENSIONAL_MEAN; CLASSICAL_INTERVAL_MERTENS; CLASSICAL_FULL_INTERVAL_MERTENS;
CLASSICAL_PRIME_LOG_MERTENS; CLASSICAL_PRIME_MERTENS_RATIOS,goldbach_lower_sieve_residue_from_support;
goldbach_upper_sieve_residue; goldbach_upper_sieve_common_product_residue;
goldbach_upper_sieve_sequence_count; goldbach_closed_multiplier_residue_data;
goldbach_upper_sequence_fixed_residue; goldbach_upper_sequence_circle_identity;
goldbach_sieve_remainders_selected; goldbach_pred_square_sum_bound; goldbach_pred_full_square_sum_bound;
 goldbach_upper_sieve_prime_ge_five; goldbach_upper_prime_split; goldbach_upper_density_multiplicative;
goldbach_upper_preliminary_dimension; goldbach_upper_outer_dimension; goldbach_upper_log_loss_ratio;
goldbach_upper_sieve_log_geometry; goldbach_upper_outer_log; goldbach_upper_sieve_levels_uniform;
goldbach_upper_parameter_along; goldbach_upper_product_level; goldbach_eventually_uniform_of_paths;
goldbach_upper_presieving_uniform; goldbach_upper_residue_properties; goldbach_upper_residue_coprime;
goldbach_upper_main_term_nonneg; goldbach_upper_divisor_real_identity;
goldbach_upper_remainder_zero_extension; goldbach_upper_sifted_sequence;
goldbach_upper_presieve_application; goldbach_upper_fixed_sieve_analytic;
goldbach_sieve_circle_bound_exists_above; goldbach_upper_remainder_from_circle;
goldbach_upper_error_square_sum; goldbach_upper_arithmetic_error_eventually;
goldbach_section11_sieve_gap_positive; goldbach_upper_sieve_fixed_circle; goldbach_upper_sieve_complete;
 goldbach_lower_sieve_fixed_analytic; goldbach_lower_remainder_from_circle;
goldbach_section11_complete,blueprint/Checkpoint_1127_GoldbachSieveResidues.md;
blueprint/Checkpoint_1217_GoldbachUpperSieveSequence.md;
blueprint/Checkpoint_1218_GoldbachSieveRemainderTransport.md;
blueprint/Checkpoint_1254_GoldbachUpperSieveDensity.md;
blueprint/Checkpoint_1255_GoldbachUpperPresieveUniform.md;
blueprint/Checkpoint_1256_GoldbachUpperPresieveApplication.md;
blueprint/Checkpoint_1257_GoldbachUpperSieveComplete.md; blueprint/Checkpoint_1258_GoldbachSection11Comp
lete.md,6922,247bf33a5924247fe7c22732d456b5b4d74dc276ec5d5a2c45706ca0f3c7c5ca
9.22,lemma,l:maintermsize8,Uniform lower bound for the sieve main term,Goldbach and
sieve,conditional,"Full source lemma proved for every A0>0 and every fixed C>=A0+6, uniformly over the
exact retained n range. The actual X_n Theta_n(z) is at least a positive constant times N(log

N)^(-(A0/2+6)); the exponent is independent of C. Actual archimedean, smoothing and local-product arguments are proved; only the already recorded independent interval Mertens products remain external. The same large C is joined to the complete lower sieve. At1258: All Section11 numbered claims are complete in the agreed conditional scope for the explicitly supplied corrected presieving statement. The original weaker statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction.",,CLASSICAL_INTERVAL_MERTENS,goldbach_archimedean_lower; goldbach_lower_arch_real; goldbach_lower_arithmetic_polylog; goldbach_lower_theta_inverse; goldbach_lower_main_term_size; goldbach_lower_smoothed_size; goldbach_lower_bounds_complete; goldbach_lower_bad_card; goldbach_section11_complete,blueprint/Checkpoint_1228_GoldbachArchimedeanLower.md; blueprint/Checkpoint_1230_GoldbachLowerMainTermSize.md; blueprint/Checkpoint_1258_GoldbachSection11Complete.md,6951,51bd2bb6d6448ded5e4c3ccd31a0fb3fb215c10682f374d1fb2c4802c4275158

9.24,lemma,l:numericalcomparison,Strict positivity of the sieve comparison,Goldbach and sieve,conditional,"The actual W integral has the unconditional exact bound 8 sqrt(2) W(9/8)<=9/32, proved by positive interval integration and exact rpow primitives. The strict comparison is proved for the classical normalized sieve values f_(1/2)(9/8)=sqrt(8 exp(gamma)/(9 pi)) log2 and F_1(2)=exp(gamma). Identifying the generic independent sieve functions with those classical normalized values remains explicit, within GENERAL_SIEVE; the paper comparison is not assumed. No floating point evaluation or unchecked evaluator is used. At1253: Proved the exact actual W integral bound, the normalized half/linear sieve comparison and strict positivity. Classical sieve-function normalizations remain explicit general input identities. The upper-sieve application and final Goldbach theorem remain open; no paper endpoint is assumed. At1258: All Section11 numbered claims are complete in the agreed conditional scope for the explicitly supplied corrected presieving statement. The original weaker statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction.",,GENERAL_SIEVE,goldbach_weighted_inner_linear_upper; goldbach_weighted_W_rational_upper; goldbach_sieve_numerical_comparison; goldbach_normalized_sieve_comparison; goldbach_section11_complete,blueprint/Checkpoint_1253_GoldbachSieveNumericalComparison.md; blueprint/Checkpoint_1258_GoldbachSection11Complete.md,6963,f00144cc10f16f6c3e1fb11f0dea31de17044c8759302b963fe4ca9c308eaeb7

10.2,lemma,l:hybridminorarc,Hybrid minor-arc mean-square estimate,Goldbach and sieve,conditional,"Complete source hybrid minor estimate at1136: the actual circle integral and finite $n<N^2$ mean square, exact complementary source arcs, every sufficiently large radius exponent C, and a single natural threshold uniform in all coefficients, residues, real U and W and printed strict L_0/V_{ell} ranges. The theorem allows $B\ge 0$ and $0\le U\le N$, with arbitrary W and residues, so it is stronger than the stated ranges. Explicit bound $K\,N^4(\log N)^{-A}$, $K=Z^2(256/\log 2)$, $Z=(2P+3)/\log 2$, $C_0=3A+4d+26$. Finite Abel, prime-power removal, halving reconstruction, scale transfer, eventual cutoffs, source progression energy and Bessel are proved. Only the already recorded independent raw Ren/Ghosh input is assumed.",The independent raw UniformRenGhoshInputs remains explicit and unproved. The source hybrid minor conclusion and all conversion steps are discharged; no internal own-paper hypothesis remains.,REN_GHOSH,goldbach_weighted_full_fourier_inversion; goldbach_actual_minor_bessel; goldbach_major_minor_partition; goldbach_circle_three_stage_bound; goldbach_frequency_energy_identity; goldbach_progression_energy_le_collisions; goldbach_actual_minor_collision_bound; goldbach_prime_collision_off_diagonal; goldbach_reciprocal_lcm_harmonic_cube; goldbach_reciprocal_lcm_log_cube; goldbach_reciprocal_lcm_strict_real_cutoff; goldbach_congruence_interval_card; goldbach_simultaneous_congruence_count_real; goldbach_congruence_square_identity; goldbach_congruence_square_harmonic_bound; goldbach_congruence_square_strict_real_cutoff; goldbach_frequency_energy_multiplicity; goldbach_prime_product_fibre_card; goldbach_prime_product_energy; goldbach_index_row_sum; goldbach_progression_polynomial_by_rows; goldbach_progression_energy_by_row_squares; goldbach_total_row_square_bound; goldbach_progression_energy_harmonic; goldbach_progression_energy_log_five; goldbach_source_energy_numerical_ranges; goldbach_source_progression_energy; goldbach_source_progression_energy_printed_range; goldbach_minor_logarithmic_exponents; goldbach_hybrid_minor_from_pointwise; goldbach_finite_abel_bound; goldbachLogRemovalWeight_antitone; goldbachLogRemovalWeight_of_two_le; goldbach_log_removed_mangoldt_bound; goldbach_prime_power_log_removal_error; goldbach_prime_from_mangoldt_partials; goldbach_halving_partial_bound; goldbachMangoldtPrefix_annular_norm; goldbach_mangoldt_partials_from_annular; goldbachMinorArcs_measurable; goldbach_minor_arcs_source_centres; goldbach_minor_arcs_rescale; goldbach_common_annular_minor; goldbach_prime_minor_from_numeric_parameters; goldbach_log_minor_radius_identity; goldbach_log_minor_annular_budget; goldbach_prime_minor_log_parameters; goldbach_minor_real_eventual_budgets; goldbach_minor_eventual_numeric_parameters; goldbach_prime_minor_uniform; goldbach_prime_minor_source_cutoff_algebra; goldbach_prime_minor_source_high_range; goldbachHybridMinorConstant_pos; goldbach_hybrid_minor_complete,blueprint/Checkpoint_1116_GoldbachMinorArcBessel.md; blueprint/Checkpoint_1117_GoldbachProgressionEnergy.md; blueprint/Checkpoint_1118_GoldbachReciprocalLCM.md; blueprint/Checkpoint_1119_GoldbachCongruenceCounting.md; blueprint/Checkpoint_1120_GoldbachCongruenceMultiplicity.md; blueprint/Checkpoint_1121_GoldbachPrimeFrequencyEnergy.md; blueprint/Checkpoint_1122_GoldbachProgressionRows.md; blueprint/Checkpoint_1123_GoldbachLogarithmicEnergy.md; blueprint/Checkpoint_1124_GoldbachSourceEnergy.md; blueprint/Checkpoint_1125_GoldbachHybridMinorReduction.md; blueprint/Checkpoint_1128_GoldbachFiniteAbel.md; blueprint/Checkpoint_1129_GoldbachPrimeUnweighting.md; blueprint/Checkpoint_1130_GoldbachHalvingPartials.md; blueprint/Checkpoint_1131_GoldbachMinorArcGeometry.md; blueprint/Checkpoint_1132_GoldbachUniformAnnularMinor.md; blueprint/Checkpoint_1133_GoldbachMinorLogParameters.md; blueprint/Checkpoint_1134_GoldbachMinorEventualCutoffs.md; blueprint/Checkpoint_1135_GoldbachPrimeMinorUniform.md; blueprint/Checkpoint_1136_GoldbachHybridMinorComplete.md,7094,dbf2e41e83c506da3fab97ff32314503684b1f887470d2ffc7cfe10d73d8e6fb

10.12,lemma,l:positiveH9,Positivity and truncation of the singular integral,Goldbach and sieve,proved," At1176: The actual normalized autocorrelation cutoff has a proved positive denominator, exact convolution integral, eta(0)=1 and real nonnegative inverse Fourier transform. At1177: The scaled actual eta_N and its square have proved real nonnegative inverse Fourier transforms and exact compact support inherited from the seed. At1178: The actual eta is real, even, bounded by1, has zero derivative at0 and satisfies the full global quadratic source cutoff bound. At1179: Two-sided absolute integrability and an

explicit L1 estimate are proved from the exact finite-scale bound and inverse-three-halves tail. At1180: The exact cutoff-weighted Fourier integral is a proved positive average of the inverse kernel, with Fubini integrability discharged. At1181: The actual source singular integral is proved real nonnegative, uniformly in all positive ell and every real n, including the empty-domain branch. At1182: The actual Euclidean product and smoothed integrand are globally integrable, and the exact source singular integral satisfies a proved uniform O(N) bound for all positive ell. At1183: The full quantitative real-line smoothing error bound is proved with absolute constant6 and every integrability condition discharged. At1184: The actual smoothed and unsmoothed source integrals satisfy the full uniform O(N/sqrt(R)) comparison, with all integrability and empty-domain branches proved. At1185: The actual quadratic affine density is globally integrable and has the proved exact product Fourier transform, including the scale factor. At1186: The genuine logarithmic quadratic density is continuous and integrable, with its exact source product Fourier transform proved and endpoint exceptional sets checked. At1187: The full source l:positiveH9 is proved: the exact autocorrelation cutoff, real nonnegative smoothed singular integral, its exact printed one-variable H_ell domain, and the uniform O(N/sqrt(R)) comparison, including all positive ell and empty-domain cases. There is no independent analytic input for this lemma. At1213: Completed every Section10 source statement and the full source circle-method implication, including the original smooth compact seed and exact normalized integral. Only the three agreed independent raw deep inputs remain for this circle-method implication.","None. Positivity and the comparison with the exact printed one-variable domain are proved for the actual source cutoff and integrals, without extra external analytic input.",,goldbach_autocorrelation_integral; goldbach_seed_mass_pos; goldbach_eta_zero; goldbach_autocorrelation_fourier; goldbach_eta_inverse_fourier_nonnegative; goldbach_inverse_fourier_scale; goldbach_scaled_eta_inverse_nonnegative; goldbach_cutoff_square_inverse_nonnegative; goldbach_eta_support; goldbach_scaled_eta_support; goldbach_eta_real; goldbach_eta_even; goldbach_autocorrelation_norm; goldbach_eta_norm_le_one; goldbach_eta_deriv_zero; goldbach_eta_quadratic; goldbach_eta_square_cutoff_bound; goldbach_halfline_integrable_decay; goldbach_two_sided_integrable_decay; goldbach_cutoff_fourier_fubini; goldbach_positive_cutoff_fourier_average; goldbach_euclidean_product_density; goldbach_singular_integral_positive; goldbach_singular_integral_empty; goldbach_singular_integral_nonnegative; goldbach_euclidean_product_integrable; goldbach_singular_integrand_integrable; goldbach_singular_integral_norm; goldbach_smoothing_power_identity; goldbach_smoothing_decay_integrable; goldbach_singular_smoothing_uniform; goldbach_affine_integral; goldbach_quadratic_density_joint; goldbach_quadratic_average_integrable; goldbach_quadratic_average_fourier; goldbach_log_window_integrable; goldbach_log_window_continuousAt; goldbach_source_density_integrable; goldbach_source_density_fourier; goldbach_source_density_continuous; goldbach_archimedean_integral_domain; goldbach_source_density_archimedean; goldbach_source_unsmoothed_inverse; goldbach_unsmoothed_archimedean_complete; goldbach_smooth_compact_seed_nonzero; goldbach_eta_of_smooth_compact_seed; goldbach_positive_singular_integral_complete; goldbach_source_singular_integral_eq; goldbach_circle_method_source_complete; goldbach_source_singular_integral_positive_comparison,blueprint/Checkpoint_1176_GoldbachAutocorrelation.md; blueprint/Checkpoint_1177_GoldbachScaledCutoff.md; blueprint/Checkpoint_1178_GoldbachCutoffBounds.md; blueprint/Checkpoint_1179_GoldbachIntegrableDecay.md; blueprint/Checkpoint_1180_GoldbachPositiveFourierAverage.md; blueprint/Checkpoint_1181_GoldbachSingularIntegralPositive.md; blueprint/Checkpoint_1182_GoldbachSingularIntegralNorm.md; blueprint/Checkpoint_1183_GoldbachSmoothingDecay.md; blueprint/Checkpoint_1184_GoldbachSingularSmoothing.md; blueprint/Checkpoint_1185_GoldbachQuadraticDensity.md; blueprint/Checkpoint_1186_GoldbachSourceDensity.md; blueprint/Checkpoint_1187_GoldbachPositiveSingularComplete.md; blueprint/Checkpoint_1213_GoldbachCircleSourceSeed.md,7266,1419a1d8a16e5afefcc947e401c77b09483f681945a947c342ce052c3e3002e2
10.15,lemma,l:majorapproximation9,Replacement by the major-arc models,Goldbach and sieve,conditional,"At1126: The exact lcm-fibre divisor bound and weighted progression-error reindexing for the major arcs are proved. At1137: Exact strict-cutoff reduction and a uniform one-term endpoint error are proved for the actual progression sums. At1138: The exact prime counting and logarithmic-integral progression models, derivatives and lower-endpoint identities are proved. At1139: The actual prime-count discrepancy is transported through every smooth unit phase with an explicit quantitative error, including the lower prime endpoint. At1140: Exact positive-phase linear and quadratic partial-summation errors are proved from ordinary prime-counting discrepancies, with signs and totient normalizations checked. At1141: The actual prime-square major-arc model and explicit nonunit/count-discrepancy error are proved with the exact source sign and normalization. At1142: The fixed-q lcm fibre bound and complete reduction to ordinary unweighted modulus-distribution errors are proved. At1143: The compatible lcm residue sets, q-cardinality bound and generalized CRT existence/converse are proved for arbitrary positive moduli. At1144: The exact Upsilon coefficient for the actual compatible lcm residue classes is proved, including both signs and totient factors. At1145: The exact independent ordinary prime-count SW/BV boundaries are implemented; real-error equivalence, square-model numeric bound and logarithmic absorption are proved. At1146: The full uniform source SW prime-square major approximation is a complete deduction from the independent ordinary prime-count SW input. At1147: Exact prime progression residue fibres and positive rational-phase grouping are proved for the actual finite sums. At1148: The actual source linear model and all nonunit/strict-endpoint errors are proved from ordinary lcm-modulus prime-count discrepancies. At1149: The actual strict source J integral, clamped prime cutoff, and uniform numeric linear model error are matched exactly. At1150: Actual BV row approximation is proved under the independent ordinary unweighted BV input, with individual progression cutoffs and all finite exceptional terms retained. At1151: The printed source ranges imply the ordinary BV level and comparable logarithmic saving, with a threshold uniform in every active ell. At1152: The actual source row-scale sum and coefficient-family cardinality square bound are proved, including all empty-cutoff cases. At1153: Every retained prime endpoint/nonunit term is absorbed uniformly from the exact printed source ranges, using a proved power saving. At1154: The full uniform source BV progression-error norm sum is complete under the independent ordinary unweighted BV input, with all individual cutoffs and finite errors included. At1155: Exact source major-set geometry, normalized measure, wrap-around lifts, pairwise disjointness and eventual separation are proved. At1156: Measurable local coordinates and genuine Euclidean model regularity and norm bounds are proved for the circle Bessel assembly. At1157: The exact source major models are constructed on the circle, with proved localization and all-Lp regularity and the required square size bound. At1158: Both full SW/BV major estimates now hold for the constructed exact source circle models, uniformly in every printed parameter. At1159: The restricted measurable Bessel inequalities and two-model major-arc mean-square assembly are proved. At1160: The actual weighted source coefficient identity, model Lp bounds and complete logarithmic mean-square budget are proved. At1161: The full source major-arc mean-square approximation l:majorapproximation9 is complete under ordinary SW and BV, uniformly in all printed data, with strict endpoints and finite errors discharged.",No unmatched clause remains in this lemma. The independent ordinary SW and BV proofs remain external; apply

docs/MAJOR_ARC_PROOF_REPLACEMENT.tex to correct the endpoint bookkeeping in its source
proof.,SIEGEL_WALFISZ_PRIME_COUNTS; BOMBIERI_VINOGRADOV,goldbach_lcm_fibre_card;
goldbach_lcm_weighted_square_divisor_sum; goldbach_major_progression_error_sum;
goldbach_progression_strict_endpoint; goldbach_progression_endpoint_card;
goldbach_progression_endpoint_error; goldbach_progression_cutoff_min;
goldbach_prime_progression_count_sum; goldbachLi_hasDerivAt; goldbachPrimeProgressionModel_hasDerivAt;
goldbachPrimeProgressionDensity_continuousOn; goldbach_counting_model_error_identity;
goldbach_counting_partial_error_bound; goldbach_prime_weighted_count_sum;
goldbach_prime_progression_partial_error; goldbach_square_density_integral;
goldbach_linear_density_integral; goldbach_prime_quadratic_count_error;
goldbach_prime_linear_count_error; goldbach_prime_positive_symbol; goldbach_square_prime_residue_sum;
goldbach_prime_nonunit_symbol; goldbach_square_major_from_counts; goldbach_lcm_quotient_range;
goldbach_lcm_fibre_card_le_q; goldbach_lcm_unweighted_sum;
goldbach_major_error_from_unweighted_distribution; goldbach_major_residues_modq_injective;
goldbach_major_residues_card; goldbach_major_compatible_lift; goldbach_major_residue_compatible;
goldbach_major_residue_image; goldbach_major_residue_sum; goldbach_major_compatibility_character;
goldbach_major_upsilon_identity; goldbach_prime_count_discrepancy_real;
goldbach_major_radius_eventual_budget; goldbach_square_major_numeric; goldbach_major_log_product;
goldbach_square_major_siegel_walfisz; goldbach_good_linear_maps_to; goldbach_good_linear_fibre;
goldbach_linear_rational_phase; goldbach_good_linear_residue_grouping; goldbach_good_linear_model_error;
 goldbach_linear_nonunit_card; goldbach_linear_nonunit_error; goldbach_linear_major_from_counts;
goldbach_linear_source_domain_ae; goldbach_linear_source_integral;
goldbach_linear_source_major_from_counts; goldbach_linear_source_major_numeric;
goldbach_linear_error_sum_unweighted; goldbach_linear_major_bv_row; goldbach_progression_scale_logs;
goldbach_bv_log_level; goldbach_bv_log_saving; goldbach_bv_source_eventual_threshold;
goldbach_lcm_bv_level; goldbach_source_row_subset; goldbach_source_row_scale_sum;
goldbach_source_row_card; goldbach_source_index_card_square; goldbach_source_index_card_printed;
goldbach_major_finite_error_eventual; goldbach_major_finite_error_numeric;
goldbach_source_finite_error_absorbed; goldbach_bv_aggregate_log_product;
goldbach_linear_major_bombieri_vinogradov; mem_goldbachMajorArcs; goldbach_major_arcs_measure;
goldbach_circle_ball_real_lift; goldbach_major_balls_disjoint; goldbach_major_geometry_eventual;
goldbachArcCoordinate_measurable; goldbachArcCoordinate_abs; goldbachArcCoordinate_reconstruct;
goldbachSquareIntegral_stronglyMeasurable; goldbachLinearSourceIntegral_stronglyMeasurable;
goldbachSquareIntegral_norm; goldbachLinearSourceIntegral_norm; goldbachArcModel_stronglyMeasurable;
goldbachArcModel_memLp; goldbachArcModel_on_arc; goldbachSquareMajorModel_memLp;
goldbachProgressionMajorModel_memLp; goldbachSquareMajorModel_norm_on_major;
goldbach_square_circle_major_sw; goldbach_linear_circle_major_bv; goldbach_restricted_function_bessel;
goldbach_restricted_uniform_bessel; goldbach_restricted_measurable_multiplier_bessel;
goldbach_two_model_major_bessel; goldbachWeightedProgressionMajorModel_memLp;
goldbach_weighted_major_error_le_sum; goldbach_major_weighted_coefficient_identity;
goldbach_major_square_log_identity; goldbach_major_mean_square_budget;
goldbach_major_approximation_complete,blueprint/Checkpoint_1126_GoldbachMajorLCMReindexing.md;
blueprint/Checkpoint_1137_GoldbachMajorStrictEndpoint.md;
blueprint/Checkpoint_1138_GoldbachPrimeCountingModel.md;
blueprint/Checkpoint_1139_GoldbachCountingPartialSummation.md;
blueprint/Checkpoint_1140_GoldbachPrimePhaseErrors.md;
blueprint/Checkpoint_1141_GoldbachSquareMajorCounts.md;
blueprint/Checkpoint_1142_GoldbachMajorLCMQFibres.md;
blueprint/Checkpoint_1143_GoldbachMajorCompatibleResidues.md;
blueprint/Checkpoint_1144_GoldbachMajorUpsilonIdentity.md;
blueprint/Checkpoint_1145_GoldbachRawPrimeDistribution.md;
blueprint/Checkpoint_1146_GoldbachSquareMajorSW.md;
blueprint/Checkpoint_1147_GoldbachLinearResidueFibres.md;
blueprint/Checkpoint_1148_GoldbachLinearMajorCounts.md;
blueprint/Checkpoint_1149_GoldbachLinearSourceIntegral.md;
blueprint/Checkpoint_1150_GoldbachLinearBVRow.md;
blueprint/Checkpoint_1151_GoldbachBVSourceParameters.md;
blueprint/Checkpoint_1152_GoldbachMajorSourceCounts.md;
blueprint/Checkpoint_1153_GoldbachMajorFiniteErrorAbsorption.md;
blueprint/Checkpoint_1154_GoldbachLinearMajorBV.md;
blueprint/Checkpoint_1155_GoldbachMajorArcGeometry.md;
blueprint/Checkpoint_1156_GoldbachMajorModelRegularity.md;
blueprint/Checkpoint_1157_GoldbachCircleMajorModels.md;
blueprint/Checkpoint_1158_GoldbachCircleMajorEstimates.md;
blueprint/Checkpoint_1159_GoldbachMajorBesselAssembly.md;
blueprint/Checkpoint_1160_GoldbachMajorWeightedAssembly.md;
blueprint/Checkpoint_1161_GoldbachMajorApproximationComplete.md; docs/GOLDBACH_MAJOR_ARC_COMPLETION.md;
docs/MAJOR_ARC_PROOF_CORRECTION.md,7351,9bc962b0b6e7d6b519fea3a8fa5c5421bbb9ece669b322d6ead9997c98efef20
10.20,lemma,l:arithmeticcoefficients9,Goldbach arithmetic coefficients and Euler series,Goldbach and
sieve,proved,"Completed Section 3 identities are reusable; this full statement is not counted as partial
 merely because it cites them. At1162: The actual inclusive R-smooth arithmetic series is absolutely
summable, equals the source finite Euler product and admits an exact finite cutoff/tail split. At1163:
The genuine Euclidean source integrals have proved quadratic and linear frequency decay, with explicit
constants and all cutoffs retained. At1164: Unconditional reciprocal-totient harmonic estimates,
including arbitrary divisor progressions, and the exact gcd divisor expansion are proved. At1165: The
full source arithmetic-family harmonic estimate is proved unconditionally with explicit divisor
exponent2 and constant9 exp(2). At1166: Exact arbitrary-multiplier coefficient identities and the
uniform odd-prime bound8 gcd(p,k*ell)/p are proved. At1168: The genuine coefficient has a uniform
prime-power gcd bound, with every two-adic exponent handled explicitly. At1169: General-modulus source
coefficient decay tau(q)^9 gcd(q,k*ell)/q and principal Gauss decay tau(q)^3/sqrt(q) are proved
unconditionally. At1170: The actual weighted source arithmetic-family and finite denominator-tail bounds
 are proved; the full source arithmetic and completion lemmas remain open with explicit remaining
clauses. At1171: Normalized finite unit averages are proved invariant under reduction to every divisor
modulus. At1172: Primitive progression exponential cancellation and the coprime norm bound are proved
from actual finite unit sums. At1173: The actual Upsilon coefficient is an exact lcm-unit expectation
and satisfies exact simultaneous frequency/denominator reduction. At1174: The actual Upsilon coefficient
 has the proved sharp gcd/divisor decay tau(q) gcd(q,k*ell)/q. At1175: The full source
arithmetic-coefficient lemma is proved: absolute convergence, the all-prime infinite local Euler series
and its exact source G_R value, all three individual bounds with the common divisor exponent9, and the
complete real-cutoff harmonic family estimate. There are no external inputs for this lemma.",None. All
clauses are proved under the source hypotheses; no extra external mathematical

input.,,goldbach_smooth_coefficient_abs_summable; goldbach_smooth_series_divisor_sum;
goldbach_arithmetic_factor_euler; goldbach_arithmetic_factor_finite_split;
goldbach_quadratic_interval_decay; goldbachSquareIntegral_decay; goldbach_linear_interval_decay;
goldbachLinearIntegral_decay; goldbachLinearSourceIntegral_decay; goldbach_reciprocal_totient_majorant;
goldbach_reciprocal_totient_sum; goldbach_reciprocal_totient_multiples;
goldbach_gcd_totient_divisor_expansion; goldbach_gcd_weighted_divisor_sum; goldbach_gcd_reciprocal_sum;
goldbach_gcd_reciprocal_totient_sum; goldbach_gcd_product_bound; goldbach_arithmetic_harmonic_rectangle;
 goldbach_source_arithmetic_harmonic; goldbach_general_coefficient_of_dvd;
goldbach_general_coefficient_of_coprime_unit; goldbach_general_coefficient_of_coprime_zero;
goldbach_odd_prime_cohen_norm; goldbach_general_coefficient_odd_prime_bound;
goldbach_upsilon_gcd_normalization; goldbach_progression_exponential_trivial;
goldbach_upsilon_trivial_gcd; goldbach_general_coefficient_trivial;
goldbach_general_coefficient_small_two_power; goldbach_general_coefficient_prime_power;
goldbach_gcd_ratio_multiplicative; goldbach_gcd_ratio_factorization;
goldbach_two_pow_primeFactors_le_divisors; goldbach_prime_factor_weight_le_divisors;
goldbach_general_coefficient_factor_bound; goldbach_general_coefficient_divisor_bound;
goldbach_principal_square_divisor_bound; goldbach_source_arithmetic_coefficient_sum;
goldbach_source_finite_arithmetic_tail; goldbach_normalized_group_sum_pullback;
goldbach_normalized_units_sum_pullback; goldbach_progression_exponential_zero_translation;
goldbach_progression_exponential_noncoprime; goldbach_progression_exponential_coprime_norm;
goldbach_sum_units_range; goldbach_unit_lcm_residue_sum; goldbach_upsilon_lcm_unit_average;
goldbach_unit_reduction_comp; goldbach_upsilon_frequency_reduction;
goldbach_progression_exponential_primitive_norm; goldbach_upsilon_primitive_decay;
goldbach_reciprocal_totient_divisor_bound; goldbach_upsilon_gcd_divisor_decay;
goldbach_upsilon_integer_gcd_decay; goldbach_arithmetic_uniform_bounds;
goldbach_arithmetic_harmonic_common_exponent; goldbach_odd_prime_power_series;
goldbach_two_prime_power_series; goldbach_all_primes_euler_series;
goldbach_arithmetic_euler_complete,blueprint/Checkpoint_1162_GoldbachArithmeticEuler.md;
blueprint/Checkpoint_1163_GoldbachEuclideanDecay.md;
blueprint/Checkpoint_1164_GoldbachTotientHarmonic.md;
blueprint/Checkpoint_1165_GoldbachArithmeticHarmonic.md;
blueprint/Checkpoint_1166_GoldbachArithmeticPrime.md;
blueprint/Checkpoint_1168_GoldbachPrimePowerBounds.md;
blueprint/Checkpoint_1169_GoldbachArithmeticDecay.md;
blueprint/Checkpoint_1170_GoldbachArithmeticFamily.md; docs/GOLDBACH_COMPLETION_PROGRESS_1170.md;
blueprint/Checkpoint_1171_GoldbachFiniteUnitAverages.md;
blueprint/Checkpoint_1172_GoldbachPrimitiveProgression.md;
blueprint/Checkpoint_1173_GoldbachUpsilonReduction.md;
blueprint/Checkpoint_1174_GoldbachUpsilonDecay.md; blueprint/Checkpoint_1175_GoldbachArithmeticComplete.
md,7612,bd38614c74c31e405def8942353ad0568798b5051e2daedc4787450fffea9e00
10.24,lemma,l:majorcompletion9,Completion of the major-arc contribution,Goldbach and
sieve,proved,"Complete actual source major-completion mean-square estimate for every A,delta>0, all
sufficiently large C, all B>=0 and sufficiently large N, exact strict real arithmetic cutoffs, arbitrary
 coprime residue representatives and coefficients of norm at most one, and every 2<=U<=N,W>=2. Constants
 and thresholds precede N and all families. The actual major models, Euler factor G_R and singular
integral H_R are used. The sharp cutoff comparison, separated additional denominators, finite arithmetic
 completion and quantitative Rankin tail are proved internally. No independent deep input is used. The
source W<=N^2 restriction is unnecessary. The proof uses a weaker intermediate power of R that yields
the same prescribed logarithmic saving after increasing C. At1211: Completed the full source
major-completion lemma without external deep inputs.",None within the source major-completion statement.
 The independent SW/BV replacement and Ren/Ghosh estimates belong to separate
lemmas.,,goldbach_smooth_coefficient_abs_summable; goldbach_smooth_series_divisor_sum;
goldbach_arithmetic_factor_euler; goldbach_arithmetic_factor_finite_split;
goldbach_quadratic_interval_decay; goldbachSquareIntegral_decay; goldbach_linear_interval_decay;
goldbachLinearIntegral_decay; goldbachLinearSourceIntegral_decay; goldbach_reciprocal_totient_majorant;
goldbach_reciprocal_totient_sum; goldbach_reciprocal_totient_multiples;
goldbach_gcd_totient_divisor_expansion; goldbach_gcd_weighted_divisor_sum; goldbach_gcd_reciprocal_sum;
goldbach_gcd_reciprocal_totient_sum; goldbach_gcd_product_bound; goldbach_arithmetic_harmonic_rectangle;
 goldbach_source_arithmetic_harmonic; goldbach_general_coefficient_of_dvd;
goldbach_general_coefficient_of_coprime_unit; goldbach_general_coefficient_of_coprime_zero;
goldbach_odd_prime_cohen_norm; goldbach_general_coefficient_odd_prime_bound;
goldbachLinearSourceIntegral_norm_scale; goldbach_euclidean_product_flat;
goldbach_euclidean_product_decay; goldbach_euclidean_product_tail; goldbach_upsilon_gcd_normalization;
goldbach_progression_exponential_trivial; goldbach_upsilon_trivial_gcd;
goldbach_general_coefficient_trivial; goldbach_general_coefficient_small_two_power;
goldbach_general_coefficient_prime_power; goldbach_gcd_ratio_multiplicative;
goldbach_gcd_ratio_factorization; goldbach_two_pow_primeFactors_le_divisors;
goldbach_prime_factor_weight_le_divisors; goldbach_general_coefficient_factor_bound;
goldbach_general_coefficient_divisor_bound; goldbach_principal_square_divisor_bound;
goldbach_source_arithmetic_coefficient_sum; goldbach_source_finite_arithmetic_tail;
goldbach_normalized_group_sum_pullback; goldbach_normalized_units_sum_pullback;
goldbach_progression_exponential_zero_translation; goldbach_progression_exponential_noncoprime;
goldbach_progression_exponential_coprime_norm; goldbach_sum_units_range; goldbach_unit_lcm_residue_sum;
goldbach_upsilon_lcm_unit_average; goldbach_unit_reduction_comp; goldbach_upsilon_frequency_reduction;
goldbach_progression_exponential_primitive_norm; goldbach_upsilon_primitive_decay;
goldbach_reciprocal_totient_divisor_bound; goldbach_upsilon_gcd_divisor_decay;
goldbach_upsilon_integer_gcd_decay; goldbach_arithmetic_uniform_bounds;
goldbach_arithmetic_harmonic_common_exponent; goldbach_odd_prime_power_series;
goldbach_two_prime_power_series; goldbach_all_primes_euler_series; goldbach_arithmetic_euler_complete;
goldbach_autocorrelation_integral; goldbach_seed_mass_pos; goldbach_eta_zero;
goldbach_autocorrelation_fourier; goldbach_eta_inverse_fourier_nonnegative;
goldbach_inverse_fourier_scale; goldbach_scaled_eta_inverse_nonnegative;
goldbach_cutoff_square_inverse_nonnegative; goldbach_eta_support; goldbach_scaled_eta_support;
goldbach_eta_real; goldbach_eta_even; goldbach_autocorrelation_norm; goldbach_eta_norm_le_one;
goldbach_eta_deriv_zero; goldbach_eta_quadratic; goldbach_eta_square_cutoff_bound;
goldbach_halfline_integrable_decay; goldbach_two_sided_integrable_decay; goldbach_cutoff_fourier_fubini;
 goldbach_positive_cutoff_fourier_average; goldbach_euclidean_product_density;
goldbach_singular_integral_positive; goldbach_singular_integral_empty;
goldbach_singular_integral_nonnegative; goldbach_euclidean_product_integrable;
goldbach_singular_integrand_integrable; goldbach_singular_integral_norm;
goldbach_smoothing_power_identity; goldbach_smoothing_decay_integrable;

goldbach_singular_smoothing_uniform; goldbach_affine_integral; goldbach_quadratic_density_joint;
goldbach_quadratic_average_integrable; goldbach_quadratic_average_fourier;
goldbach_log_window_integrable; goldbach_log_window_continuousAt; goldbach_source_density_integrable;
goldbach_source_density_fourier; goldbach_source_density_continuous;
goldbach_archimedean_integral_domain; goldbach_source_density_archimedean;
goldbach_source_unsmoothed_inverse; goldbach_unsmoothed_archimedean_complete;
goldbach_smooth_compact_seed_nonzero; goldbach_eta_of_smooth_compact_seed;
goldbach_positive_singular_integral_complete; goldbach_rankin_weight_prime_power;
goldbach_rankin_divisor_euler; goldbach_rankin_pointwise; goldbach_finite_rankin_euler_bound;
goldbach_geometric_partial_bound; goldbach_polynomial_local_bound; goldbach_rankin_local_uniform;
goldbach_rankin_prime_rpow; goldbach_prime_subset_inclusive_log; goldbach_rankin_euler_log;
goldbach_rankin_tail_log_bound; goldbach_eventually_log_square_budget; goldbach_rankin_decay_log_budget;
 goldbach_rankin_parameter_decay; goldbach_source_rankin_tail; goldbach_source_rankin_arithmetic_tail;
goldbach_arc_coordinate_real_shift; goldbach_arc_lift_integral; goldbach_arc_lift_fourier_integral;
goldbach_arc_lift_energy; goldbach_arc_lift_memLp; goldbach_arc_lifts_disjoint;
goldbach_disjoint_arc_bessel; goldbach_major_envelope_integrable; goldbach_major_envelope_energy;
goldbach_major_envelope_dominated_energy; goldbach_euclidean_product_envelope;
goldbach_major_family_integrand_measurable; goldbach_major_family_integrand_bound;
goldbach_smooth_major_family_measurable; goldbach_smooth_major_family_norm;
goldbach_smooth_major_family_support; goldbach_center_arithmetic_unit_sum;
goldbach_rational_negative_character; goldbach_center_arithmetic_rational;
goldbach_centers_weighted_arithmetic; goldbach_radial_major_family_identity;
goldbach_smoothed_radial_kernel_integrable; goldbach_smoothed_radial_kernel_integral;
goldbach_smoothed_major_family_identity; goldbach_inverse_three_halves_interval;
goldbach_inverse_three_halves_tail; goldbach_divisor_square_tail; goldbach_centers_denominator_sum;
goldbach_centers_filtered_denominator_sum; goldbach_major_prefactor_count;
goldbach_added_denominator_energy_budget; goldbach_smooth_added_major_bessel;
goldbach_completion_geometry; goldbach_completion_parameters_eventual;
goldbach_completion_log_power_rate; goldbach_hard_singular_smoothing_uniform;
goldbach_hard_unsmoothed_comparison; goldbach_divisor_harmonic_initial_sum;
goldbach_small_smoothing_uniform; goldbach_major_model_product; goldbach_arc_model_fourier_identity;
goldbach_actual_major_model_identity; goldbach_initial_coefficients_divisor_sum;
goldbach_completion_arithmetic_split; goldbach_added_completion_identity;
goldbach_completion_error_identity; goldbach_small_completion_square_budget;
goldbach_far_completion_square_budget; goldbach_completion_all_parameters_eventual;
goldbach_major_completion_coefficient_complete; goldbach_major_completion_complete;
goldbach_model_weighted_coefficient;
goldbach_source_ranges_eventual,blueprint/Checkpoint_1162_GoldbachArithmeticEuler.md;
blueprint/Checkpoint_1163_GoldbachEuclideanDecay.md;
blueprint/Checkpoint_1164_GoldbachTotientHarmonic.md;
blueprint/Checkpoint_1165_GoldbachArithmeticHarmonic.md;
blueprint/Checkpoint_1166_GoldbachArithmeticPrime.md;
blueprint/Checkpoint_1167_GoldbachEuclideanProduct.md;
blueprint/Checkpoint_1168_GoldbachPrimePowerBounds.md;
blueprint/Checkpoint_1169_GoldbachArithmeticDecay.md;
blueprint/Checkpoint_1170_GoldbachArithmeticFamily.md; docs/GOLDBACH_COMPLETION_PROGRESS_1170.md;
blueprint/Checkpoint_1171_GoldbachFiniteUnitAverages.md;
blueprint/Checkpoint_1172_GoldbachPrimitiveProgression.md;
blueprint/Checkpoint_1173_GoldbachUpsilonReduction.md;
blueprint/Checkpoint_1174_GoldbachUpsilonDecay.md;
blueprint/Checkpoint_1175_GoldbachArithmeticComplete.md;
blueprint/Checkpoint_1176_GoldbachAutocorrelation.md; blueprint/Checkpoint_1177_GoldbachScaledCutoff.md;
 blueprint/Checkpoint_1178_GoldbachCutoffBounds.md;
blueprint/Checkpoint_1179_GoldbachIntegrableDecay.md;
blueprint/Checkpoint_1180_GoldbachPositiveFourierAverage.md;
blueprint/Checkpoint_1181_GoldbachSingularIntegralPositive.md;
blueprint/Checkpoint_1182_GoldbachSingularIntegralNorm.md;
blueprint/Checkpoint_1183_GoldbachSmoothingDecay.md;
blueprint/Checkpoint_1184_GoldbachSingularSmoothing.md;
blueprint/Checkpoint_1185_GoldbachQuadraticDensity.md;
blueprint/Checkpoint_1186_GoldbachSourceDensity.md;
blueprint/Checkpoint_1187_GoldbachPositiveSingularComplete.md;
blueprint/Checkpoint_1188_GoldbachRankinEuler.md; blueprint/Checkpoint_1189_GoldbachRankinLocal.md;
blueprint/Checkpoint_1190_GoldbachRankinLogProduct.md; blueprint/Checkpoint_1191_GoldbachRankinDecay.md;
 blueprint/Checkpoint_1192_GoldbachRankinSource.md;
blueprint/Checkpoint_1193_GoldbachRankinArithmeticTail.md;
blueprint/Checkpoint_1194_GoldbachArcLiftIntegral.md;
blueprint/Checkpoint_1195_GoldbachDisjointArcBessel.md;
blueprint/Checkpoint_1196_GoldbachMajorEnvelope.md;
blueprint/Checkpoint_1197_GoldbachMajorFamilyIntegrand.md;
blueprint/Checkpoint_1198_GoldbachCenterArithmetic.md;
blueprint/Checkpoint_1199_GoldbachRadialMajorIdentity.md;
blueprint/Checkpoint_1200_GoldbachDivisorSquareTail.md;
blueprint/Checkpoint_1201_GoldbachDenominatorEnergy.md;
blueprint/Checkpoint_1202_GoldbachAddedMajorBessel.md;
blueprint/Checkpoint_1203_GoldbachCompletionParameters.md;
blueprint/Checkpoint_1204_GoldbachHardSmoothing.md; blueprint/Checkpoint_1205_GoldbachSmallSmoothing.md;
 blueprint/Checkpoint_1206_GoldbachHardMajorIdentity.md;
blueprint/Checkpoint_1207_GoldbachCompletionArithmetic.md;
blueprint/Checkpoint_1208_GoldbachCompletionErrorIdentity.md;
blueprint/Checkpoint_1209_GoldbachCompletionBudgets.md;
blueprint/Checkpoint_1210_GoldbachMajorCompletionBound.md; blueprint/Checkpoint_1211_GoldbachMajorComple
tionComplete.md,7713,f6b649ad0680308d113b3aeb890f97ab223297f258315b01886a6036e0e2025d
11.2,lemma,l:fixedsievesets10,Comparison with fixed sieve prime sets,Goldbach and sieve,proved,"Complete
 lower and upper fixed-sieve comparison, with actual sieve prime sets and actual closed multiplier set.
The lower inequality is proved by inclusion, including the prime3 boundary. The upper comparison uses an
 exact finite exceptional-pair inclusion and the bound sum over p|n of tau(n-p^2-1) <= K_epsilon
N^epsilon. One positive constant is chosen before N,n,z. Holds for every N>=1,0<n<=N^2 and every real z,
 hence the full retained source range. No deep input is used. At1215: The full source fixed-sieve
comparison is proved without deep inputs. The local density, presieving and final sieve estimates remain
 open.",None within the source fixed-sieve comparison lemma.,,goldbach_fixed_lower_count;

goldbach_fixed_upper_exception; goldbach_fixed_upper_source_count_finite; goldbach_uniform_divisor_card;
 goldbach_fixed_sieve_divisor_sum;
goldbach_fixed_sieve_sets,blueprint/Checkpoint_1214_GoldbachFixedSieveComparison.md; blueprint/Checkpoin
t_1215_GoldbachFixedSieveSets.md,8013,633956d5fc07ad6076a32d15e26d853228ac3e45b734017a7580c93d53785206
11.6,lemma,l:lowerlocaldensity,Lower-sieve local density formula,Goldbach and sieve,proved," At1216:
Proved the exact lower sieve sequence/progression count and its actual circle-method remainder
decomposition. The full fixed-sieve comparison remains proved; local densities and later sieve
conclusions remain open. At1219: Actual unit-square root counts, both candidate prime-density bounds
including p=5, and the explicit multiplicative sieve function are proved without deep inputs. Full
local-density normalization is the next internal obligation. At1220: Proved the conductor-sensitive
coefficient identity, all actual prime-factor cases, the exact factors at2 and3, residue invariance, and
 normalized prime adjunction without division by the global base factor. Full local-density source
closure is reserved for the final product and CRT assembly. At1221: complete actual normalized
Euler-product factorization, exact source CRT residues, inclusive prime cutoff, multiplicativity and all
 density bounds are proved without external inputs. The upper base factor may vanish; no division by it
is used. At1221: Both full source local-density lemmas are proved without new external inputs: exact
original Euler normalization, inclusive cutoff, lower quotient identification, source CRT
specializations, upper multiplicativity and p=5, with common zero upper base factors retained.
Presieving and the later sieve estimates remain open.",None within the source statement and retained
admissible context.,,goldbach_lower_sieve_sequence_count; goldbach_lower_sequence_fixed_residue;
goldbach_lower_sequence_circle_identity; goldbach_rho_eq; goldbach_rho_congr; goldbach_cohen_rho;
goldbach_rho_legendre; goldbach_rho_le_two; goldbach_rho_of_zero; goldbach_rho_five_consecutive;
goldbach_prime_restricted_count; goldbach_local_denominator_pos; goldbach_local_density_nonneg;
goldbach_local_density_le; goldbach_density_bound_lt_one; goldbach_local_density_lt_one;
goldbach_upper_density_five; goldbach_upper_density_range; goldbach_sieve_omega_one;
goldbach_sieve_omega_prime; goldbach_sieve_omega_mul; goldbach_sieve_omega_div;
goldbach_modular_multiplier_constant; goldbach_general_coefficient_conductor;
goldbach_prime_unrestricted_factor; goldbach_prime_fixed_factor; goldbach_prime_zero_multiplier_factor;
goldbach_prime_unrestricted_count; goldbach_two_factor_admissible; goldbach_three_factor_admissible;
mem_goldbach_large_local_primes; goldbach_arithmetic_factor_large_primes;
goldbach_general_coefficient_residue_congr; goldbach_arithmetic_factor_coefficient_product;
goldbach_arithmetic_factor_residue_congr; goldbach_prime_factor_unchanged;
goldbach_prime_local_density_ratio; goldbach_arithmetic_factor_adjoin_prime;
goldbach_normalized_factor_adjoin_prime; goldbach_lower_base_nonzero;
goldbach_lower_density_factorization; goldbach_lower_local_density_formula;
goldbach_normalized_factor_prime_product; goldbach_sieve_density_factorization;
goldbach_large_prime_coprime_twentyfour; goldbach_upper_phase_twentyfour;
goldbach_upper_density_factorization;
goldbach_upper_local_density,blueprint/Checkpoint_1216_GoldbachLowerSieveSequence.md;
blueprint/Checkpoint_1219_GoldbachSieveRootCounts.md;
blueprint/Checkpoint_1220_GoldbachSieveConductor.md; blueprint/Checkpoint_1221_GoldbachSieveDensityFacto
rization.md,8098,a704a00d7508baee78bb0bd80230ee87d355237bedc219653ce6d7c52b73f94e
11.8,lemma,l:upperlocaldensity,Upper-sieve local density and multiplicativity,Goldbach and
sieve,proved," At1217: Proved the exact upper sieve sequence/progression count, including all prime
exclusions, source multiplier arithmetic and residue independence from n. Local-density factorization
and the later analytic sieve estimates remain open. At1219: Actual unit-square root counts, both
candidate prime-density bounds including p=5, and the explicit multiplicative sieve function are proved
without deep inputs. Full local-density normalization is the next internal obligation. At1220: Proved
the conductor-sensitive coefficient identity, all actual prime-factor cases, the exact factors at2 and3,
 residue invariance, and normalized prime adjunction without division by the global base factor. Full
local-density source closure is reserved for the final product and CRT assembly. At1221: complete actual
 normalized Euler-product factorization, exact source CRT residues, inclusive prime cutoff,
multiplicativity and all density bounds are proved without external inputs. The upper base factor may
vanish; no division by it is used. At1221: Both full source local-density lemmas are proved without new
external inputs: exact original Euler normalization, inclusive cutoff, lower quotient identification,
source CRT specializations, upper multiplicativity and p=5, with common zero upper base factors
retained. Presieving and the later sieve estimates remain open.",None within the source statement and
retained admissible context.,,goldbach_upper_sieve_sequence_count;
goldbach_closed_multiplier_residue_data; goldbach_upper_sequence_fixed_residue;
goldbach_upper_sequence_circle_identity; goldbach_rho_eq; goldbach_rho_congr; goldbach_cohen_rho;
goldbach_rho_legendre; goldbach_rho_le_two; goldbach_rho_of_zero; goldbach_rho_five_consecutive;
goldbach_prime_restricted_count; goldbach_local_denominator_pos; goldbach_local_density_nonneg;
goldbach_local_density_le; goldbach_density_bound_lt_one; goldbach_local_density_lt_one;
goldbach_upper_density_five; goldbach_upper_density_range; goldbach_sieve_omega_one;
goldbach_sieve_omega_prime; goldbach_sieve_omega_mul; goldbach_sieve_omega_div;
goldbach_modular_multiplier_constant; goldbach_general_coefficient_conductor;
goldbach_prime_unrestricted_factor; goldbach_prime_fixed_factor; goldbach_prime_zero_multiplier_factor;
goldbach_prime_unrestricted_count; goldbach_two_factor_admissible; goldbach_three_factor_admissible;
mem_goldbach_large_local_primes; goldbach_arithmetic_factor_large_primes;
goldbach_general_coefficient_residue_congr; goldbach_arithmetic_factor_coefficient_product;
goldbach_arithmetic_factor_residue_congr; goldbach_prime_factor_unchanged;
goldbach_prime_local_density_ratio; goldbach_arithmetic_factor_adjoin_prime;
goldbach_normalized_factor_adjoin_prime; goldbach_upper_local_density;
goldbach_upper_density_factorization; goldbach_sieve_density_factorization;
goldbach_normalized_factor_prime_product; goldbach_large_prime_coprime_twentyfour;
goldbach_lower_base_nonzero; goldbach_lower_density_factorization; goldbach_lower_local_density_formula;
 goldbach_upper_phase_twentyfour,blueprint/Checkpoint_1217_GoldbachUpperSieveSequence.md;
blueprint/Checkpoint_1219_GoldbachSieveRootCounts.md;
blueprint/Checkpoint_1220_GoldbachSieveConductor.md; blueprint/Checkpoint_1221_GoldbachSieveDensityFacto
rization.md,8159,6f29ddcaacdd521c4caa880908ea1298db4570e472a698f4245afaab8898693f
11.13,lemma,l:presieving,Two-range presieving deduction,Goldbach and sieve,conditional,"The corrected
two-range presieving statement is a complete implication from the explicit independent general one-stage
 sieve inputs. The exact source matched is the separately supplied Section11_Reconciled.tex: X>=0,
g(1)=1, fixed R=(log N)^C, R+1<=z<=D1<=N^A, refined1+K/logw dimension bounds, and the preliminary
endpointR+1. Coefficients precede the density/sequence and n, have absolute value<=1 and exact level
support; the common relative error tends to zero. The literal original weaker manuscript statement is
not claimed proved; replacement is required in the user manuscript. No original source file was changed.
 At1258: All Section11 numbered claims are complete in the agreed conditional scope for the explicitly
supplied corrected presieving statement. The original weaker statement is not proved and must be
replaced in the user manuscript. The complete joint lower/upper estimates use one cutoff and
n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary combined logarithmic mean-square

saving. The final Goldbach theorem remains the next deduction. At1265 the uploaded human-readable
statement is reviewed directly: D0<D follows from its level chain, and the inline interval prime
conditions are proved equivalent in Lean. The corrected mathematical hypotheses are verified as present
in the actual upload; no presieving replacement remains outstanding.",,GENERAL_SIEVE;
ROSSER_IWANIEC_SINGLE_STAGE,goldbach_presieve_signed_sandwich; goldbach_presieve_unique_product;
goldbach_presieve_pushforward_bound; goldbach_presieve_pushforward_reindex; goldbach_presieve_level;
goldbach_div_indicator_nonneg; goldbach_sift_indicator_nonneg; goldbach_div_indicator_product;
goldbach_sift_indicator_union; goldbach_presieve_pair_bound; goldbach_presieve_pair_sum;
goldbach_presieve_pair_divisor_sum; goldbach_presieve_pointwise_lower;
goldbach_presieve_pointwise_upper; goldbach_sequence_divisor_count_sum;
goldbach_sequence_sifted_count_sum; goldbach_sequence_weighted_divisor_sum;
goldbach_presieve_sequence_lower; goldbach_presieve_sequence_upper;
goldbach_presieve_sequence_expansion; goldbach_presieve_disjoint_support_coprime;
goldbach_presieve_main_error; goldbach_presieve_quantitative_lower;
goldbach_presieve_quantitative_upper; goldbach_presieve_pushforward_off_image;
goldbach_presieve_pushforward_superset; goldbach_presieve_coefficient_properties;
goldbach_presieve_product_range; goldbach_presieve_remainder_level; goldbach_presieve_gaussian_decay;
goldbach_presieve_polylog_decay; goldbach_presieve_preliminary_ratio;
goldbach_presieve_preliminary_error_bound; goldbach_presieve_preliminary_s_tendsto;
goldbach_presieve_preliminary_level_eventually; goldbach_presieve_preliminary_absorption;
goldbach_presieve_outer_level_tendsto; goldbach_presieve_outer_error_tendsto;
goldbach_presieve_total_error_tendsto; goldbach_presieve_uniform_error_tendsto;
goldbach_presieve_density_prime_product; goldbach_presieve_outer_mass; goldbach_presieve_outer_product;
goldbach_presieve_outer_mass_ratio; goldbach_presieve_inverse_polylog; goldbach_corrected_presieving;
goldbach_presieving_from_classical_coefficients; goldbach_presieve_uniform_assembly;
goldbach_density_restrict; goldbach_refined_dimension_restrict; goldbach_refined_dimension_total;
goldbach_presieve_source_R; goldbach_integer_sequence_divisor_count;
goldbach_integer_sequence_sifted_count; goldbach_section11_complete;
goldbach_current_presieve_strict_level; goldbach_current_presieve_preliminary_filter;
goldbach_current_presieve_outer_filter,blueprint/Checkpoint_1222_GoldbachPresieveCoefficients.md;
blueprint/Checkpoint_1223_GoldbachPresieveDecay.md;
blueprint/Checkpoint_1224_GoldbachPresieveUniformAssembly.md;
blueprint/Checkpoint_1258_GoldbachSection11Complete.md; docs/MANUSCRIPT_REVIEW_2026_10_28.md,8273,1fe187
e8904291eb9dc516fd677f555338966f01d534f77d4b4249325cc43f0e
11.19,lemma,l:wumatomakisum,Weighted upper main-term comparison,Goldbach and sieve,conditional,"Complete
 source weighted upper main-term comparison, in uniform epsilon form, with the actual multiplier set,
local factors, finite Euler products and singular integrals. For every A>0, every fixed C>=A+24 and
eta>0, one threshold in N works for all n=12 mod24 with N^2(log N)^(-A)<=n<N^2. The coefficient is
exactly c_(9/8) W(9/8)+eta. Strong multiplicativity, the uniform mean instantiation, both finite Abel
arguments, W identity, all small-denominator and endpoint terms, moving logarithms, integration,
smoothing and relative errors are proved. Only the independent generic effective mean-value corollary
and classical fixed-progression Mertens bounds remain explicit unproved hypotheses. The source lemma
statement is unchanged; two proof replacements are supplied but not applied. At1252: The full source
weighted upper main-term comparison l:wumatomakisum is complete as a checked implication from
independent generic effective mean-value and classical Mertens inputs, with the exact printed constant,
actual source definitions, uniform retained-range quantifiers, and every summation, integration and
relative-error obligation discharged.",,EFFECTIVE_HALF_DIMENSIONAL_MEAN; CLASSICAL_PRIME_LOG_MERTENS;
CLASSICAL_PRIME_MERTENS_RATIOS; CLASSICAL_INTERVAL_MERTENS,goldbach_weighted_small_prime_cancel;
goldbach_weighted_lower_local_cancel; goldbach_weighted_upper_local_cancel;
goldbach_weighted_prime_le_six; goldbach_upper_arithmetic_real_product;
goldbach_weighted_lower_small_product; goldbach_weighted_multiplier_geometry;
goldbach_weighted_prime_log_balance; goldbach_weighted_mean_from_tenenbaum;
goldbach_weighted_lower_normalization; goldbach_weighted_upper_normalization;
goldbach_weighted_upper_arithmetic_normalization; goldbach_weighted_harmonic_uniform;
goldbach_weighted_mean_normalized; goldbach_weighted_euler_split; goldbach_abel_model_identity;
goldbach_abel_decreasing_upper; goldbach_abel_increasing_error; goldbach_weighted_pairs_sum;
goldbach_weighted_pairs_harmonic; goldbach_weighted_W_identity; goldbach_weighted_W_integrable;
goldbach_weighted_outer_integrable; goldbach_weighted_thin_window_measure;
goldbach_weighted_log_kernel_deriv; goldbach_weighted_prime_kernel_deriv; goldbach_three_log_mertens;
goldbach_weighted_prime_abel; goldbach_weighted_mean_abel_effective; goldbach_finite_variable_interval;
goldbach_finite_smoothing_outer_upper; goldbach_weighted_arch_sum_identity; goldbach_weighted_arch_real;
 goldbach_weighted_arch_kernel_integrated; goldbach_weighted_inner_mellin;
goldbach_weighted_outer_mellin_le_W; goldbach_weighted_harmonic_refined;
goldbach_weighted_prime_reciprocal; goldbach_weighted_log_loss;
goldbach_weighted_upper_constant_normalization; goldbach_weighted_main_term_smoothing;
goldbach_weighted_pair_rectangle; goldbach_weighted_kernel_rectangle;
goldbach_weighted_prime_finite_upper; goldbach_weighted_double_abel; goldbach_weighted_double_domain;
goldbach_weighted_double_integral_le_W; goldbach_weighted_log_harmonic_integral_le;
goldbach_weighted_large_rectangle; goldbach_weighted_kernel_small; goldbach_weighted_kernel_large;
goldbach_weighted_finite_kernel; goldbach_weighted_prime_error_coarse;
goldbach_weighted_main_coefficient_limit; goldbach_weighted_small_coefficient_limit;
goldbach_weighted_endpoint_coefficient_limit; goldbach_weighted_large_scalar;
goldbach_weighted_cutoffs_eventually; goldbach_weighted_harmonic_cutoff;
goldbach_weighted_kernel_effective; goldbach_weighted_kernel_coefficient_limit;
goldbach_weighted_mean_interval; goldbach_weighted_fixed_kernel_uniform;
goldbach_weighted_moving_kernel_uniform; goldbach_weighted_mass_log_bound;
goldbach_weighted_upper_outer; goldbach_weighted_lower_outer; goldbach_weighted_thin_error_relative;
goldbach_weighted_smooth_error_relative; goldbach_weighted_source_budgets;
goldbach_weighted_source_bridge;
goldbach_weighted_upper_main_term_complete,blueprint/Checkpoint_1231_GoldbachWeightedLocalFactors.md;
blueprint/Checkpoint_1232_GoldbachWeightedEulerProducts.md;
blueprint/Checkpoint_1233_GoldbachWeightedMeanInputs.md;
blueprint/Checkpoint_1234_GoldbachWeightedNormalization.md;
blueprint/Checkpoint_1235_GoldbachWeightedHarmonic.md;
blueprint/Checkpoint_1236_GoldbachWeightedAbel.md; blueprint/Checkpoint_1237_GoldbachWeightedKernels.md;
 blueprint/Checkpoint_1238_GoldbachWeightedPrimeBalance.md;
blueprint/Checkpoint_1239_GoldbachWeightedFiniteIntegration.md;
blueprint/Checkpoint_1240_GoldbachWeightedKernelIntegration.md;
blueprint/Checkpoint_1241_GoldbachWeightedMeanCutoffs.md;
blueprint/Checkpoint_1242_GoldbachWeightedSmoothing.md;
blueprint/Checkpoint_1243_GoldbachWeightedPrimeEndpoints.md;

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blueprint/Checkpoint_1244_GoldbachWeightedDoubleGeometry.md;
blueprint/Checkpoint_1245_GoldbachWeightedLargeRectangle.md;
blueprint/Checkpoint_1246_GoldbachWeightedFiniteKernel.md;
blueprint/Checkpoint_1247_GoldbachWeightedScalarBounds.md;
blueprint/Checkpoint_1248_GoldbachWeightedKernelCoefficient.md;
blueprint/Checkpoint_1249_GoldbachWeightedKernelUniform.md;
blueprint/Checkpoint_1250_GoldbachWeightedMass.md;
blueprint/Checkpoint_1251_GoldbachWeightedErrorBounds.md;
blueprint/Weighted_Upper_Main_Term_Complete.md; blueprint/Checkpoint_1252_GoldbachWeightedSourceBridge.m
d,8636,efb02e188d9c5ce5c0855e7a43156faf7239d644a4d0ab86973b381bc6539945
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Archive record: Paper_Proof_Status.json

Paper_Proof_Status.json


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labels and the hashed current statement are authoritative. A fresh typeset attempt was blocked by
missing amsrefs.sty, an environment dependency; no manuscript error or number change is inferred.",
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 unproved hypotheses.",
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      "scope": "PrimeAsymptoticInputs: theta(y)/y tends to 1 and the weighted reciprocal-square-root
prime sum has leading constant 2. The external PNT bridge is not integrated.",
      "declared_as_Lean_axiom": false
    },
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      "id": "PROGRESSION_ESTIMATE",
      "status": "consumed_explicit_hypothesis",
      "scope": "The older PrimeSquareDistributionInputs supplies raw compatible progression/Ren/Ghosh
data for Theorem1.1. The refined PagePrimeSquareDistributionInputs c additionally returns genuine
optional primitive-character Page-zero data in the source region, and is proved to supply the older
interface. One selected family is integrated into all clauses of Proposition4.19 and the Proposition6.21
 high maximum. Existence of the refined raw data remains an independent analytic hypothesis; it is not
derived from the older interface alone. No own-paper operator conclusion is assumed.",
      "declared_as_Lean_axiom": false
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      "status": "consumed_explicit_hypotheses",
      "scope": "Published raw quadratic exponential-sum bounds are explicit inputs. An own-paper final
minor-arc estimate is not assumed.",
      "declared_as_Lean_axiom": false
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      "id": "SMOOTH_MULTIFREQUENCY",
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      "scope": "SmoothCircleBourgainInput C: a general discrete smooth multifrequency maximum. Existence
```

```
 is unproved. The boundary includes smooth real theory, sampling, normalization and the fixed bump; it
is not a theorem imported from Mathlib. Used in Proposition 6.21, whose actual annular,
all-starting-scale and denominator-truncation deductions are now proved. The same independent input is
consumed by the full Proposition4.19 assembly. This hypothesis is not used by Theorem 1.1.",
      "declared_as_Lean_axiom": false
    },
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      "id": "PAGE_CLASSES",
      "status": "consumed_explicit_general_hypothesis_source_consistency_proved",
      "scope": "RawPageUniquenessInput c: standard Landau-Page uniqueness for primitive nonprincipal
real Dirichlet characters and their real zeros, expressed using ordinary partial sums below 1; its
analytic-continuation bridge remains external. The source-region to conductor-region deduction,
uniqueness in each dyadic conductor class, actual-data finite-class injection and the changing-character
 maximum are proved. Source-compatible PageZeroDatum existence remains the refined raw distribution
hypothesis. Its scale-by-scale selection and integration into the common Proposition4.19 operator family
 are proved; existence is not proved from the older fixed-scale interface. No operator conclusion is
assumed and no Lean axiom is declared. See docs/PROPOSITION_6_21_COMPLETION.md.",
      "declared_as_Lean_axiom": false
    },
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RawNarrowShannonInput are precise independent Tao Lemma5.1, Theorems1.7/1.6 and Proposition1.5
specializations. Their existence is unproved; they are explicit parameters, not axioms. Actual
finite-input Fourier lifts, normalization, diagonal, smooth applicability and the full finite-input
dyadic low bound are proved. The analytic-top typo was corrected with an exact hash-declared rebuild
at897. The prime high theorem does not use the multiplier predicate. See docs/SECTION_7_IW_BOUNDARY.md
and docs/SECTION_7_SAMPLING_BOUNDARY.md.",
      "declared_as_Lean_axiom": false
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      "status": "explicit_independent_general_inputs_consumed_actual_low_applications_proved",
      "scope": "RawScalarHormanderInput, RawOneSidedHLInput, RawYoungMultiplierInput,
RawRieszThorinMultiplierInput are precise independent general standard theorem hypotheses for arbitrary
kernels, functions, multipliers or scalar sequences. All annular applicability, actual continuous and
narrow cyclic maxima, exact Fourier identifications, countable remainder and finite-input full IW low
assembly are proved. No own-paper low or restricted bound is a field. See the continuous, signed-maximal
 and sampling boundary documents. The finite signed maximal interface is proved internally at constant6
by rawFiniteSignedMaximalInput_proved (checkpoint1263), and is no longer an independent assumption.",
      "declared_as_Lean_axiom": false
    },
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      "status": "precise_unproved_independent_interfaces_implemented",
      "scope": "RawCalderonStrongInput and RawCalderonModularInput: general transference for arbitrary
positive finite convolution families, exponents/norm constants and continuous nonnegative gauges. Their
independent truth and exact published-theorem/action/modular bridge remain unproved. No own-paper
endpoint is an input. See docs/ERGODIC_INDEPENDENT_BOUNDARY.md.",
      "declared_as_Lean_axiom": false
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      "status": "precise_unproved_general_input_consumed",
      "scope": "RawOriginalLaVictoireInput: original Theorem3.1 for arbitrary increasing sequences with
literal cyclic arithmetic hypotheses, concluding failure of any universal weak L1 maximal constant along
 every infinite cutoff set. Its existence and exact zero-based/probability/weak-norm formulation bridge
remain unproved. The adapted own-paper persistent criterion is proved using separate Sawyer and Conze
inputs.",
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      "id": "GRANVILLE_KURLBERG",
      "status": "precise_unproved_quoted_general_interface_consumed",
      "scope": "RawGranvilleKurlbergInput: general CRT-family local-correlation/subpower hypotheses
imply full joint cyclic Poisson gap limits. Published-to-Lean cyclic joint formulation bridge and
theorem proof remain unformalized. Actual unit-square hypotheses/application are proved. This optional
input is not used by Theorem1.5(i).",
      "declared_as_Lean_axiom": false
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      "status": "consumed_explicit_general_hypothesis",
      "scope": "RawSplitQuadraticWeilInput is the precise weakened classical quadratic character-sum
bound for arbitrary nonempty sets of distinct linear factors over every odd prime field. Its existence
remains unproved. It supplies no own quadratic-residue correlation, spacing, moment or badness
conclusion; the source-specific applications remain deductions.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "BOMBIERI_VINOGRADOV",
      "status": "consumed_explicit_hypothesis",
      "scope": "RawPrimeBombieriVinogradovInput: classical general ordinary prime-count distribution
with endpoint maximum, unweighted modulus sum, square-root level and logarithmic saving. The fixed-q lcm
 fibre theorem reduces the paper reindexing to this ordinary input; the own-paper circle-method
conclusion remains internal. The actual linear major-error row sum is deduced at1150, with all
exceptional terms explicit. The complete full source progression major estimate is deduced at1154 and
the source major-arc mean-square lemma at1161.",
      "declared_as_Lean_axiom": false
    },
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two-stage presieve, full lower and upper applications, numerical comparison, common-cutoff assembly and
final Goldbach theorem are complete checked conditional implications.",
      "declared_as_Lean_axiom": false
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      "id": "WEIGHTED_PRIME_COUNTS",
      "status": "retired_unconsumed_planning_placeholder",
      "scope": "No weighted-prime or paper main-term conclusion is an independent input. The source
l:wumatomakisum is proved from the independently stated general effective mean and classical Mertens
interfaces. This earlier planning placeholder is not consumed.",
      "declared_as_Lean_axiom": false
    },
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      "status": "consumed_explicit_compound_hypothesis",
      "scope": "FullSquareDecompositionInput C: the independent ordinary-square high/low estimate in
global operator norm form. The analytic theorem and its globalization remain external. The plus-sign
PositiveFullSquareDecompositionInput is proved equivalent with exactly the same C by
fullSquareDecompositionInput_sign_iff at checkpoint783. The literal weighted formulas, general
convolution/norm/maximal transfer and reflected support are proved, so reflection is no longer
unformalized. The full-to-annular normalization is also proved. No prime-square conclusion is assumed.
See docs/THEOREM_1_1_EXTERNAL_INPUTS.md and docs/SQUARE_REFLECTION_COMPLETION.md.",
      "declared_as_Lean_axiom": false
    },
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      "status": "consumed_explicit_compound_hypothesis",
      "scope": "OrdinarySquareStoppingInput CS: independent local ordinary-square low/high stopping
estimates. Its existence and exact support, normalization, admissibility and published-convention
compatibility remain unproved. It is stronger than the earlier fixed-scale FullSquareDecompositionInput.
 See docs/INDEPENDENT_STOPPING_INPUTS.md. No prime local endpoint or sparse conclusion is an input
field.",
      "declared_as_Lean_axiom": false
    },
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      "id": "PRIME_MAXIMAL_LP",
      "status": "consumed_explicit_compound_hypothesis",
      "scope": "IndependentPrimeMaximalInput: complementary-exponent restricted consequence of the
independent prime polynomial maximal Lp theorem, in the actual von Mangoldt annular normalization.
Published prime normalization and prime-power extension remain within its unproved scope. Used only for
the open triangle, not for the endpoint. No sparse conclusion is a field.",
      "declared_as_Lean_axiom": false
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approximation in rational-center form, a consequence of Fiedler-Jurkat-Korner as restated in
Han-Lacey-Yang arXiv1907.05734v2 Theorem3.1. Its Dirichlet selection/reduction/phase bridge and
existence remain unproved parts of this explicit independent input. It has no own circle-model, IW or
maximal conclusion. Complete Gauss arithmetic and full quadratic oscillatory integral bounds are proved
separately. See docs/SECTION_7_QUADRATIC_WEYL_BOUNDARY.md.",
      "declared_as_Lean_axiom": false
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convergence, including the still-unformalized Mehlhop-Slomian one-dimensional/cutoff and
PNT/Abel/prime-power normalization bridge. It supplies no unbounded Orlicz convergence, strong norm,
modular inequality or density-extension conclusion. See docs/ERGODIC_INDEPENDENT_BOUNDARY.md.",
      "declared_as_Lean_axiom": false
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probability system, almost-everywhere finite proper maxima for all measurable integrable functions imply
 some positive weak L1 maximal constant. The general theorem and its exact complex/probability/cutoff
convention bridge remain unproved.",
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    },
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      "scope": "RawConzeMaximalInput: for arbitrary sequence averages, transfer the same weak maximal
constant from a nonatomic ergodic probability system to every probability system. General theorem and
exact convention bridge remain unproved.",
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      "scope": "RawFixedPrimeProgressionFrequencyInput: enumerated primes have limiting frequency
1/phi(q) in each unit residue class of every fixed positive modulus. This standard
PNT-in-fixed-progressions input remains unproved. Exact prime-square fiber partition and pushforward to
1/|Lambda_q| are proved.",
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      "declared_as_Lean_axiom": false
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      "id": "FULL_DOMAIN_CONTINUOUS_STANDARDS",
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      "scope": "RawFullOneSidedHLInput and RawFullScalarHormanderInput: general one-sided interval
maximal and scalar Hormander convolution theorems on arbitrary Lp inputs for1<p<=2. These broaden the
earlier L2-intersection formulations; their proofs and exact publication/convention bridges remain
unformalized. The full-HL restriction to the earlier interface is proved. The scalar full-domain version
 is not claimed to follow from the earlier interface. All own annular kernels, Fourier formulas, uniform
 L2/Hormander conditions, real-scale continuity, measurable proper supremum and source estimates are
deductions.",
      "declared_as_Lean_axiom": false
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    {
      "id": "RATIONAL_SMOOTH_MULTIFREQUENCY",
      "status": "consumed_explicit_compound_hypothesis",
      "scope": "RationalSmoothCircleInput C: the independent rational smooth multifrequency inequality
with a common denominator P, its rational logarithmic cost, fixed smooth bump normalization and circle
transference. The interface is quantified over arbitrary rational center sets, periods, separated
widths, finite scale sets and all integer L2 inputs. It contains no annular kernel, beta, arithmetic
coefficients, denominator truncation or current-paper endpoint. The independent rational-frequency
theorem and transference remain unproved here; existence is not asserted. It is the rational refinement
associated with Lacey1997 Theorem1.2 and its sampling specialization, rather than the general
separated-frequency squared-log bound.",
      "declared_as_Lean_axiom": false
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      "status": "proved_in_project_not_an_external_assumption",
      "scope": "Complete primitive two-square criterion for n=2 mod4, proved from pinned Mathlib
Fermat/negative-one obstruction and checked primitive multiplication/parity algebra. No additional
mathematical hypothesis.",
      "declared_as_Lean_axiom": false,
      "proof": "PrimeSquares.goldbach_primitive_two_squares_criterion",
      "proof_file": "PrimeSquares/GoldbachPrimitiveCriterion.lean",
      "proof_checkpoint": 1261,
      "classification": "proved_arithmetic_lemma",
      "consumed_by_completed_exports": true,
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      "source": "blueprint/Section9_Primitive_Two_Squares.md"
    },
    {
      "id": "UNIFORM_PAGE_PREFIX",
      "status": "consumed_explicit_general_hypothesis",
      "scope": "UniformPageProgressionInputs: a general prime progression prefix estimate through
exp(eta sqrt(log N)), uniform over every positive modulus, unit residue and real t in [N/2,N], with one
genuine optional Page datum per N selected before Q. The decay, modulus and prefix constants are
independent of N. Its existence and fixed-level analytic bridge remain unproved; no quadratic phase,
major-arc estimate or own-paper endpoint is assumed.",
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      "status": "consumed_explicit_compound_general_hypothesis",
      "scope": "RawFullAdelicIWInput C: independent full-domain Tao adelic IW theorem for arbitrary good
 coprime data and arbitrary finite-arithmetic spatial operator kernels and rough compact real symbols.
Its rough limiting and published-to-actual-kernel/quotient realization bridges remain unproved parts of
this explicit hypothesis. No Q, selected construction, probability measure, annular, low/high or own
endpoint is a field. The full real Lp product bound and source coefficient/geometry/norm application are
 proved.",
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    {
      "id": "FULL_DOMAIN_TAO_SAMPLING",
      "status": "consumed_explicit_compound_general_hypotheses",
      "scope": "RawFullMajorArcSamplingInput C and RawFullShannonInput S: general Tao Theorem1.6 on
arbitrary good coprime data, with full distributional lp Fourier support, both norm directions and
unique interpolation; and Proposition1.5 at fixed c=1/100 for all finite-dimensional normed complex
targets and every 1<=p<=infinity. Their truth and published/full-class/finite-quotient realization
bridges remain unproved. No source Q, B, annular operator, maximal estimate or own endpoint is assumed.
The actual real-Q applicability, full inverse, two-sided source constants and narrower-band
finite-target application are proved.",
      "declared_as_Lean_axiom": false
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      "status": "consumed_explicit_hypothesis",
      "scope": "RawPrimeSiegelWalfiszInput: general ordinary pi/Li prime counts, every endpoint y\u2264N
 and fixed logarithmic modulus range. No oscillatory weighted, coefficient-family or own-paper major-arc
 endpoint is assumed. The complete actual source square major approximation is deduced at1146. Together
with ordinary BV, it supplies the complete source major-arc mean-square replacement at1161.",
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      "id": "ROSSER_IWANIEC_SINGLE_STAGE",
      "status": "consumed_explicit_general_hypotheses_for_corrected_variant",
      "interfaces": [
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        "GoldbachClassicalCoefficientInput"
      ],
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      "scope": "General one-stage Rosser\u2013Iwaniec weights chosen before all supported densities
satisfying the refined dimension hypothesis, with pointwise sieve bounds, level/square-free support,
coefficient bound1, exponential fundamental-lemma error for large s, and the standard outer
f_kappa/F_kappa coefficient bounds with a logarithmic error. The full signed two-stage deduction, exact
scale, common error limit and integer-sequence bridges are proved. No own-paper presieving,
weighted-prime, lower/upper sieve or Goldbach endpoint is an input. The lower application also consumes
continuity of the classical lower sieve function at9/8; the actual parameter and uniform-error limits
are proved.",
      "source": "Yuta Suzuki, The Rosser\u2013Iwaniec sieve, 2023, Definition8.2, Proposition11.8 and
Lemma14.7; exact independent interfaces and classical corollary scope in docs/EXTERNAL_SIEVE_INPUTS.md",
      "declared_as_Lean_axiom": false
    },
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      "id": "CLASSICAL_INTERVAL_MERTENS",
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      "scope": "GoldbachClassicalIntervalMertens K: uniform interval Euler-product bounds for the
universal factor(1-2/(p-3))^(-1) on primes>=7 with dimension2, and(1-1/(p-1))^(-1) on primes3 mod4 with
dimension1/2. Quantified over all finite prime subsets of intervals. Contains no sequence, n, Goldbach
estimate or own-paper endpoint. Both actual density applications are proved.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "EFFECTIVE_HALF_DIMENSIONAL_MEAN",
      "status": "consumed_explicit_independent_general_hypothesis",
      "interfaces": [
        "ClassicalHalfMeanValueInput"
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      "scope": "Uniform effective nonnegative strongly multiplicative upper mean theorem for arbitrary
functions with prime values at most6 and the explicit prime logarithmic balance condition. It is the
general rho=1/2 upper corollary of Tenenbaum Theorem1.2 (corrected author version arXiv:1709.01315v13),
including the documented general corollary reductions. Those external reductions and the source theorem
are not formalized. No paper weight, root count, multiplier, singular integral, Goldbach or own-paper
endpoint is a field. All such applications and the complete weighted main-term comparison are
deductions.",
      "source": "docs/EXTERNAL_WEIGHTED_MEAN_INPUTS.md",
      "declared_as_Lean_axiom": false
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    {
      "id": "CLASSICAL_PRIME_LOG_MERTENS",
      "status": "consumed_explicit_independent_general_hypothesis",
      "interfaces": [
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      ],
      "scope": "Ordinary and1 mod4 sums of log(p)/p with fixed bounded errors, and the bounded
logarithmic higher-prime-power tail. Quantified over all real x>=2; no paper weight or endpoint. The3
mod4 partition and varying-weight prime balance are proved internally.",
      "source": "docs/EXTERNAL_WEIGHTED_MEAN_INPUTS.md",
      "declared_as_Lean_axiom": false
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      "status": "consumed_explicit_independent_general_hypothesis",
      "interfaces": [
        "ClassicalPrimeMertensRatios"
      ],
      "scope": "Ordinary all-prime and fixed1/3 mod4 Euler-product ratios with relative error K/log R
and explicit endpoint conventions. No varying weight or paper main-term conclusion. Exact source
local-factor and Euler-product normalizations, the effective mean application and the final weighted
comparison are proved.",
      "source": "docs/EXTERNAL_WEIGHTED_MEAN_INPUTS.md",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "CLASSICAL_FULL_INTERVAL_MERTENS",
      "status": "consumed_explicit_independent_general_hypothesis",
      "interfaces": [
        "GoldbachClassicalFullIntervalMertens"
      ],
      "scope": "Arbitrary finite prime-subset products for the p5/large-prime preliminary barrier and
outer1/(p-1) factors, with refined1+K/logw losses. Independent unproved general classical Mertens
corollary. No Goldbach data or endpoints.",
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      "archived_pdf_page": "3",
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      "source_line": 212,
      "statement_latex": "\\label{t:improvingfixed}\nLet \\(I=[-N^2,2N^2]\\cap\\mathbb Z\\).  For all
\\(F,G\\subseteq I\\),\n\\begin{align}\\label{e:introimproving}\n \\frac1{|I|}\n \\bigl|\\langle
A_{N,\\Lambda}\\mathbf1_F,\\mathbf1_G\\rangle\\bigr|\n &\\ll\n
\\left(\\frac{|F|}{|I|}\\frac{|G|}{|I|}\\right)^{2/3}\n \\Phi_{4\\sqrt3/9+o(1)}\n
\\left(\\frac{|F|}{|I|}\\frac{|G|}{|I|}\\right).\n\\end{align}\nConsequently, for every
\\(3/2<p\\leq2\\) and every\n\\(f\\in\\ell^p(\\mathbb Z)\\) supported in $I$,\n\\[\n
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\\left(\\frac1{|I|}\\sum_{x\\in\\mathbb Z}\n |A_{N,\\Lambda}f(x)|^{p'}\\right)^{1/p'}\n \\lesssim_p\n
\\left(\\frac1{|I|}\\sum_{x\\in\\mathbb Z}|f(x)|^p\\right)^{1/p}.\n\\]",
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        "theorem_1_1_strong",
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        "theorem_1_1_strong_sum",
        "primeSquareImprovingOperator_apply",
        "primeSquareSourceInterval_card",
        "theorem_1_1_from_positive_square_input",
        "fullSquareDecompositionInput_sign_iff"
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      "expositions": [
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        "docs/THEOREM_1_1_EXTERNAL_INPUTS.md",
        "docs/SQUARE_REFLECTION_COMPLETION.md",
        "blueprint/Checkpoint_783_SquareDecompositionReflection.md"
      ],
      "scope": "Both conclusions for the genuine full von-Mangoldt square average, every positive
integer N, the exact source interval and its cardinality, all finite F,G subsets, every coefficient
cstar>4 sqrt(3)/9, and every real 3/2<p<=2. The strong result proves actual output lp membership and the
 literal normalized sum form, for all lp inputs, hence also the supported inputs in the source.",
      "remaining": "The implication is complete under the explicit independent deep inputs. PNT limits,
compatible progression/Ren/Ghosh data existence, and the independent ordinary-square global
decomposition remain unformalized hypotheses. Its analytic globalization remains external; the
plus/minus spatial reflection is proved with the same constants at checkpoint783. Theorem1.1 and
Propositions4.19/6.21 are complete conditional implications under their separate inputs; the current
corrected numbered paper targets are all matched, with independent inputs and stronger unnumbered
alternative-route assertions retaining their documented scopes.",
      "external_inputs": [
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        "PROGRESSION_ESTIMATE",
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        "FULL_SQUARE_DECOMPOSITION"
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      "status_display": "Complete implication; external inputs explicit",
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          "file": "PrimeSquares/PrimeSquareSourceInterval.lean",
          "kind": "theorem",
          "in_complete_theorem_axiom_audit": true,
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        {
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          "file": "PrimeSquares/PrimeSquareActualStrongOperator.lean",
          "kind": "theorem",
          "in_complete_theorem_axiom_audit": true,
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          "in_complete_theorem_axiom_audit": true,
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          "file": "PrimeSquares/TheoremOneOne.lean",
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          "file": "PrimeSquares/PrimeSquareSourceInterval.lean",
          "kind": "theorem",
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      "source_line": 287,
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 sparse collection of\nintervals \\(\\mathcal S\\) such that\n\\begin{align}\\label{e:introsparse}\n
\\bigl|\\langle A_\\Lambda^*\\mathbf1_F,\\mathbf1_G\\rangle\\bigr|\n &\\ll \\sum_{I\\in\\mathcal S}|I|\n
 \\left(\\frac{|F\\cap I||G\\cap I|}{|I|^2}\\right)^{2/3}\n \\Phi_{4\\sqrt3/9+o(1)}\n
\\left(\\frac{|F\\cap I||G\\cap I|}{|I|^2}\\right).\n\\end{align}\nMoreover, if \\((1/p,1/q)\\) lies in
the interior of the triangle with\nvertices $(2/3,2/3),$ $(1,0)$, and $(0,1)$, \nthen there is a sparse
collection \\(\\mathcal S_{p,q}\\) such that\n\\begin{align}\\label{e:introsparseopen}\n \\bigl|\\langle
 A_\\Lambda^*\\mathbf1_F,\\mathbf1_G\\rangle\\bigr|\n &\\ll_{p,q}\n \\sum_{I\\in\\mathcal S_{p,q}}|I|\n
\\left(\\frac{|F\\cap I|}{|I|}\\right)^{1/p}\n \\left(\\frac{|G\\cap
I|}{|I|}\\right)^{1/q}.\n\\end{align}",
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        "primeOpenSparse",
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        "sourceSparseTriangle_iff_barycentric"
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        "docs/INDEPENDENT_STOPPING_INPUTS.md",
        "docs/SPARSE_DYADIC_QUALIFICATION.md"
      ],
      "scope": "Both displayed conclusions are complete conditional implications: the literal full
maximal indicator for every integer N>=2, arbitrary finite F,G including empty sets, actual finite
interval collections with actual pairwise disjoint major subsets of size at least one tenth, the exact
endpoint density exponent and paper Phi for every cstar>4sqrt(3)/9, and every point of the full open
triangle in exact reciprocal/barycentric coordinates. Constants precede F,G. All stopping, packing,
support, local endpoint, interpolation and sparse deductions are proved.",
      "remaining": "No paper-specific deduction in either conclusion remains. The recorded prime limits,
 compatible raw Page/distribution data, Page uniqueness, general smooth multifrequency/sampling,
independent ordinary-square stopping interface and complementary-exponent prime maximal input remain
unproved explicit hypotheses, including their stated compatibility boundaries. The general auxiliary
Section5 statements are separately partial. The proposed dyadic qualification of l:sparserecursion is
unapplied; the actual prime-square all-cutoff reduction is proved.",
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        "PAGE_CLASSES",
        "ORDINARY_SQUARE_STOPPING",
        "PRIME_MAXIMAL_LP"
      ],
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          "file": "PrimeSquares/PrimeOpenSparse.lean",
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        },
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          "kind": "theorem",
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\\(n_k=p_k^2\\).\n\\begin{enumerate}[label=\\textup{(\\roman*)}]\n\\item\nThe sequence \\((n_k)\\) is
persistently universally \\(L^1\\)-bad.\nMore precisely, for every nonatomic ergodic
measure-preserving\nprobability system \\((X,\\mu,T)\\) and every infinite set\n\\(\\mathcal
S\\subset\\mathbb N\\), there is an \\(f\\in L^1(X)\\)\nfor which\n\\[\n \\frac1N\\sum_{k\\leq
N}f(T^{n_k}x),\n \\qquad N\\in\\mathcal S,\n\\]\ndiverges on a set of positive measure.\n\n\\item\nThere
 is an absolute constant \\(C>0\\) such that, for every\ninvertible measure-preserving probability
system \\((X,\\mu,T)\\),\nevery \\(1<p\\leq2\\), and every \\(f\\in L^p(X)\\),\n\\[\n \\left\\|\n
\\sup_{N\\geq1}\n \\left|\\frac1N\\sum_{k\\leq N}f(T^{n_k}x)\\right|\n \\right\\|_{L^p(X)}\n \\leq\n
\\exp\\left(\n \\frac{C}{p-1}\\log\\frac{e}{p-1}\n \\right)\n \\|f\\|_{L^p(X)}.\n\\]\n\\end{enumerate}",
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        "docs/SECTION_8_COMPLETION.md",
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      ],
      "scope": "Both parts are complete conditional implications. Part(i): the actual enumerated prime
squares are persistently universally L1-bad on every truly nonatomic ergodic measure-preserving
probability system, with every infinite cutoff set and an actual measurable positive-measure divergence
set. CRT, Dirichlet representation, correlation counts, large-prime block construction, all
moment/thickening/difference limits, uniform square-root fibers and the fixed-progression PNT
pushforward are proved. The original badness theorem, Sawyer, Conze, general Weil and fixed-progression
PNT remain separate unproved general inputs. Granville\u2013Kurlberg is not used by part(i). A checked
moment proof replaces the source strict finite-fraction argument, which has an exact kernel
counterexample. Part(ii) is a complete conditional implication: an absolute positive constant before
every invertible measure-preserving probability system, every 1 < p <= 2 and every Lp input; the proper
actual maximum over the first m prime squares for all m >= 1 is finite almost everywhere, belongs to Lp
and has the exact displayed exponential norm bound. The prime-counting/enumeration normalization, finite
 sums, first-prime case and equality of both full maxima are proved with no PNT or enumeration input.
The proof uses the already proved Corollary7.6 and the precise independent general strong Calderon
input, plus the earlier Theorem7.2 independent inputs. It does not use the bounded-class convergence
input. At checkpoint1264 the current public export consumes the proved finite signed maximal theorem at
constant6; a supplied RawFiniteSignedMaximalInput and its constant R are no longer hypotheses.",
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congruent to $12$ modulo $24$ can be written as\n\\[\n a^2+b^2+p^2+1,\n \\qquad a^2+b^2+1\\in\\mathcal
P,\\quad p\\in\\mathbb P.\n\\]\nMore precisely, if\n\\[\n E_2(X)=\\{1\\leq n<X:n\\equiv12\\pmod{24},\\\n
 n\\notin\\mathcal P+\\mathbb P^2\\},\n\\]\nthen, for every $A>0$,\n\\[\n |E_2(X)|\\ll_A X(\\log
X)^{-A}.\n\\]",
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exceptional set of positive n<X congruent12 mod24 without a prime plus a square-of-prime representation
whose first prime is a primitive sum of two squares plus1 has size O_A(X(log X)^(-A)), eventually at
every real X. The representation and exceptional-set membership are exact. The common-cutoff Section11
estimates, positive numerical gap, arbitrary logarithmic mean-square saving, all arithmetic losses,
small n and the passage from N^2 to real X are discharged. Explicit independent general SW, BV,
Ren/Ghosh, one-stage sieve, effective multiplicative mean, Mertens, normalized sieve values/continuity
and primitive two-square criterion remain unproved theorem parameters. No own-paper endpoint is a
hypothesis. At1259: Section11 and the final quantitative Goldbach theorem are complete in the agreed
conditional scope, using the supplied corrected presieving statement. Every paper endpoint is
discharged; only explicit independent general classical inputs and their normalizations remain unproved.
 Original manuscripts are unchanged and the supplied Section11 replacements still require user
application. No additional completion credit is claimed for stronger unnumbered assertions in alternate
proof routes. At1261: Proved the full PrimitiveTwoSquaresCriterion from pinned Mathlib and elementary
algebra, and proved W(alpha) convergence/nonnegativity for every alpha>1. Constructed a concrete nonzero
 compactly supported smooth seed. The next stage exports the Section9 arithmetic and headline statements
 without the former criterion or seed hypotheses, then reconciles the stale Section9 map. User
manuscript edits remain in progress. At1262 the final endpoint is exported from GoldbachSection11Inputs
alone: the primitive two-square criterion is proved internally and a fixed nonzero smooth compact seed
is constructed. The real-X bound and zero exceptional density need no separate arithmetic or seed
assumption. At1262: Every Section9 numbered source claim and its final quantitative theorem is complete
in the agreed scope. All elementary arithmetic reductions are now proved without an extra primitive
two-square assumption. W(alpha) convergence is proved for every alpha>1 and the smooth seed is
constructed. Only the explicit independent analytic inputs remain for the circle-method/sieve/headline
estimates. The user is applying previous manuscript edits; no application is assumed and original
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$2^a\\parallel q$.  Then\n\\begin{align}\\label{e:cohenquadratic}\n \\sum_{r\\in\\mathbb
A_q}c_q(n-r^2)\n =|\\mu(q/2^a)| T_a(n)\n \\prod_{\\substack{p\\mid q\\\\p>2}}\n \\begin{cases}\n
p+1,&\\left(\\frac np\\right)=1,\\\\\n  1-p,&\\left(\\frac np\\right)\\ne1,\n
\\end{cases}\n\\end{align}\nwhere\n\\begin{align}\\label{e:Ta}\n T_a(n)=\n \\begin{cases}\n
1,&a=0,\\\\\n  (-1)^{n+1},&a=1,\\\\\n  0,&a=2,3\\text{ and }n\\not\\equiv1\\pmod{2^{a-1}},\\\\\n
\\phi(2^a)c_{2^a}(n-1)=\\pm\\phi(2^a)^2,\n    &a=2,3\\text{ and }n\\equiv1\\pmod{2^{a-1}},\\\\\n
0,&a\\geq4.\n \\end{cases}\n\\end{align}\nConsequently,\n\\begin{align}\\label{e:CohenQuadratic}\n
\\left|\\sum_{r\\in\\mathbb A_q}c_q(n-r^2)\\right|\n \\ll |\\mu(q/2^a)|\\prod_{p\\mid
q}(p+1).\n\\end{align}\nHere $(\\frac np)$ is the Legendre symbol, with value zero when $p\\mid n$.",
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Then\n\\begin{align}\\label{e:progcohenid}\n \\sum_{\\substack{r\\in\\mathbb A_q\\\\r\\equiv b\\pmod
g}}c_q(n-r)\n =\n \\begin{cases}\n  0,&(g,q/g)>1,\\\\\n  \\mu(q/g)c_{q/g}(n)c_g(n-b),&(g,q/g)=1.\n
\\end{cases}\n\\end{align}",
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Then\n\\begin{align}\\label{e:eqjustforLHS}\n &\\frac{\\phi(q)\\phi([y,q])}{\\phi(y)}\n
\\sum_{a\\in\\mathbb A_q}\\Upsilon(a,q,b,y)\n \\mathfrak
s(a,q)e\\left(\\frac{ax}{q}\\right)\\notag\\\\\n &\\qquad=\\mu(q/g)\\mathbf1_{(g,q/g)=1}\n
\\sum_{u\\in\\mathbb A_q}c_{q/g}(x-u^2)c_g(x-b-u^2).\n\\end{align}\nIn particular, the right-hand side
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integer, and let\n$x,b\\in\\mathbb Z$ satisfy $(b,k)=1$.\nThen\n\\begin{align}\\label{e:corupss}\n
&\\sum_{P^+(q)<Q}\\sum_{a\\in\\mathbb A_q}\n \\Upsilon(a,q,b,k)\\mathfrak s(a,q)e(ax/q)\\notag\\\\\n
&=\\mathcal T_2\n \\prod_{\\substack{2<p<Q\\\\p\\mid k\\\\(\\frac{x-b}{p})=1}}\n
\\left(1+\\frac{p+1}{p-1}\\right)\n \\prod_{\\substack{2<p<Q\\\\p\\mid k\\\\(\\frac{x-b}{p})\\ne1}}0 \n
\\prod_{\\substack{2<p<Q\\\\p\\nmid k\\\\(\\frac{x}{p})=1}}\n \\left(1-\\frac{p+1}{(p-1)^2}\\right)\n
\\prod_{\\substack{2<p<Q\\\\p\\nmid k\\\\(\\frac{x}{p})\\ne1}}\n
\\left(1+\\frac1{p-1}\\right),\n\\end{align}\nwhere, writing
\\(h=v_2(k)\\),\n\\begin{align}\\label{e:T2factor}\n \\mathcal T_2=\n \\begin{cases}\n
1+(-1)^x,&h=0,\\\\\n  2^{\\min\\{h,3\\}}\n
\\mathbf1_{\\{x-b\\equiv1\\;(\\mathrm{mod}\\;2^{\\min\\{h,3\\}})\\}},\n  &h\\geq1.\n
\\end{cases}\n\\end{align}\nThus the product is zero if some odd prime $p<Q$ dividing $k$\nsatisfies
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Then\n\\begin{align}\\label{e:generalcorupss}\n &\\sum_{P^+(q)<Q}\\sum_{a\\in\\mathbb A_q}\n
\\Upsilon(a\\ell,q,b,k)\\mathfrak s(a,q)e(an/q)\\notag\\\\\n &\\qquad=\n \\mathcal T_2(\\ell,k,n;b)\n
\\prod_{2<p<Q}\n \\frac{p\\phi((p,k))}{(p-1)^2}\n \\left|\\left\\{(u,v)\\in\\mathbb A_p^2:\n
\\begin{array}{l}\n  u^2+\\ell v\\equiv n\\pmod p,\\\\\n  v\\equiv b\\pmod{(p,k)}\n
\\end{array}\\right\\}\\right|,\n\\end{align}\nwhere \n\\begin{align}\\label{e:generalT2}\n \\mathcal
T_2(\\ell,k,n;b)\n :=\n 1+\\sum_{\\nu=1}^{3}\n \\frac{\\phi(k)}\n {\\phi(2^\\nu)\\phi([k,2^\\nu])}\n
\\sum_{\\substack{u,v\\in\\mathbb A_{2^\\nu}\\\\\n v\\equiv b\\;(\\mathrm{mod}\\,(2^\\nu,k))}}\n
c_{2^\\nu}(n-u^2-\\ell v).\n\\end{align}",
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$n$,\n\\begin{align}\n \\sum_{P^+(q)<Q}\\frac1{q\\phi(q)}\n \\left|\\sum_{r\\in\\mathbb
A_q}c_q(n-r^2)\\right|\n \\ll\\log Q.\n\\end{align}",
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\\(\\varepsilon>0\\),\n\\begin{align}\\label{e:Gcharbound}\n |\\mathfrak s_\\chi(a,q)|\\ll_\\varepsilon
q^{-1/2+\\varepsilon}.\n\\end{align}\nEquivalently, \\( |\\mathfrak s_\\chi(a,q)|\\ll
q^{-1/2+o(1)}\\).",
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the character modulo $q$\ninduced by $\\chi_0$.  For \\(0\\leq j\\leq5\\), let\n\\(\\chi_{2^jq_0}\\)
have the same meaning, and set\n\\[\n \\sigma_0:=\n \\max_{0\\leq j\\leq5}\n \\sup_{a\\in\\mathbb
A_{2^jq_0}}\n |\\mathfrak s_{\\chi_{2^jq_0}}(a,2^jq_0)|,\n \\qquad\n \\rho(d):=\\prod_{p\\mid
d}(1-p^{-1/2})^{-1}.\n\\]\nThere exists $\\mathcal B_0\\geq1$, depending only on $q_0$, such that,\nfor
every \\(q=q_0d\\),\n\\begin{align}\\label{e:exceptionalarithmetic}\n \\left|\n \\sum_{a\\in\\mathbb
A_q}\n \\mathfrak s_{\\chi_q}(a,q)e(ax/q)\n \\right|\n &\\ll\n \\mathcal B_0\\prod_{p\\mid
d}\\frac{p+1}{p-1},\\notag\\\\n |\\mathfrak s_{\\chi_q}(a,q)|\n &\\ll \\sigma_0d^{-1/2}\\rho(d),\n
\\qquad \\textup{ for } a\\in \\mathbb{A}_q\\\\n  \\sigma_0\\mathcal B_0^{1/2}&\\ll
q_0^{-1/25}.\n\\end{align}\nIn particular,\n\\[\n (1+\\log q_0)\\sigma_0\\mathcal
B_0^{1/2}\\ll1.\n\\]\nMoreover, if\n\\(a\\mapsto\\mathfrak s_{\\chi_q}(a,q)\\) is not identically zero,
then\n\\begin{align}\\label{e:sqfreeconds}\n    d=2^\\nu d_1,\\qquad\n 0\\leq\\nu\\leq5,\\qquad\n
\\mu^2(d_1)=1,\\qquad\n (d_1,2q_0)=1.\n\\end{align}",
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operators whose kernels are\nsupported in an interval of length $O(N)$.  Suppose that\n\\begin{align}\n
|A_Nf(x)|&\\leq(\\log N)^aB_N|f|(x),\\label{e:anbnsupbound}\\\\\n |B_Nf(x)|&\\ll
N^{-v}\\|f\\mathbf1_{[x-CN,x+CN]}\\|_1,\n \\qquad 0\\leq v<1,\\label{e:bnv}\n\\end{align}\nand that
$s\\geq1-v$, $r>0$, and, whenever\n$2\\leq J\\leq N^{(1-v)/s}/10$,\n\\begin{align}\n
B_N=Lo_{J,N}+Hi_{J,N},\\qquad\n \\|Lo_{J,N}f\\|_\\infty&\\ll N^{-1}J^s(\\log J)^\\gamma\\|f\\|_1,\n
\\label{e:lobn1}\\\\\n \\|Hi_{J,N}f\\|_2&\\ll J^{-r}(\\log J)^\\beta\\|f\\|_2,\n
\\label{e:hibn1}\n\\end{align}\nwhere $\\gamma\\geq\\beta\\geq0$.  Suppose also that, for\n$2\\leq Q$
and $\\log Q\\leq c_0(\\log N)^c$,\n\\begin{align}\n
A_N&=\\operatorname{Low}_{Q,N}+\\operatorname{High}_{Q,N}+E_{Q,N},\n \\label{e:decompan}\\\\\n
\\|\\operatorname{Low}_{Q,N}f\\|_\\infty\n &\\ll N^{-1}Q^s(\\log
Q)^\\alpha\\|f\\|_1,\\label{e:lowgeneral}\\\\\n \\|\\operatorname{High}_{Q,N}f\\|_2\n &\\ll
Q^{-r}\\exp\\left(c_1\\frac{(\\log Q)^\\kappa}{\\log\\log Q}\\right)\\|f\\|_2,\n
\\label{e:highgeneral}\\\\\n \\|E_{Q,N}f\\|_2&\\ll_A Q^{-r}(\\log Q)^{C_A}\n (1+\\log N)^{-A}\\|f\\|_2\n
 \\qquad(A>0),\\label{e:errorgeneral}\n\\end{align}\nwhere \\(0<\\kappa<1\\) and
\\(a,\\alpha,c_0,c_1,c>0\\).  In\n\\eqref{e:errorgeneral}, the constant \\(C_A\\) may depend on \\(A\\).
  Then, for\n$F,G$ contained in an interval of length $N$,\n\\begin{align}\n \\frac{|\\langle
B_N\\mathbf1_F,\\mathbf1_G\\rangle|}{N}\n &\\ll (|F||G|/N^2)^{1-\\frac{s}{2(s+r)}}\n (\\Log
(|F||G|/N^2))^{\\gamma+\\frac{s(\\beta-\\gamma)}{s+r}},\\label{e:endimprovbn}\\\\\n \\frac{|\\langle
A_N\\mathbf1_F,\\mathbf1_G\\rangle|}{N}\n &\\ll
(|F||G|/N^2)^{1-\\frac{s}{2(s+r)}}\\Phi_{c^*,\\kappa}(|F||G|/N^2),\n
\\label{e:endimprovan}\n\\end{align}\nwhere \n\\begin{align}\\label{e:cstar}\n
c^*=\\frac{c_1s}{2^\\kappa(s+r)^{\\kappa+1}}+o(1).\n\\end{align}\nHere and below the $o(1)$ in a
subscript means that the estimate holds\nwith $c^*$ replaced by any fixed larger constant.",
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\\frac{2(r+s)}{2r+s}<p\\leq2,\n\\end{align*}\nthen, for $f\\in \\ell^p(\\mathbb{Z})$,\n\\begin{align}\n
\\|B_Nf\\|_{p'}+\\|A_Nf\\|_{p'}\n \\ll_p N^{1/p'-1/p}\\|f\\|_p.\\label{e:lpimproving}\n\\end{align}",
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suppose that\n\\[\n \\log Q\\leq c_0(\\log N)^{1/2},\n\\]\nwhere \\(c_0>0\\) is a sufficiently small
absolute constant.  Then\n\\begin{align}\\label{e:primedecomposition}\n \\widetilde A_{N,\\Lambda}\n =\n
 \\operatorname{Low}_{Q,N}\n +\\operatorname{High}_{Q,N}+E_{Q,N},\n\\end{align}\nwhere\n\\begin{align}\n
 \\|\\operatorname{Low}_{Q,N}f\\|_\\infty\n &\\ll N^{-2}Q\\log(2Q)\\,\\|f\\|_1,\n
```

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\\label{e:primelow}\\\\\n \\|\\operatorname{High}_{Q,N}f\\|_2\n &\\ll Q^{-1/2}\n \\exp\\left(\n
(2+o(1))\\frac{(\\log Q)^{1/2}}{\\log\\log Q}\n \\right)\\|f\\|_2,\n \\label{e:primehigh}\\\\\n
\\|E_{Q,N}f\\|_2\n &\\ll_A Q^{-1/2}(\\log(2Q))^{O_A(1)}\n (1+\\log N)^{-A}\\|f\\|_2\n
\\label{e:primeerror}\n\\end{align}\nfor every \\(A>0\\).  More precisely, for some absolute
\\(C>0\\),\nthe low term satisfies\n\\begin{align}\\label{e:primelowlocal}\n
|\\operatorname{Low}_{Q,N}f(x)|\n \\ll Q\\log(2Q)\\sup_{M\\geq N}M^{-2}\n \\sum_{|y-x|\\leq
CM^2}|f(y)|.\n\\end{align}\nThe maximal estimate needed in Section~\\ref{s:sparse}
is\n\\begin{align}\\label{e:primehighmax}\n \\left\\|\n \\sup_{\\substack{M\\geq N_0\\\\M\\in2^{\\mathbb
 N}}}\n |\\operatorname{High}_{Q,M}f|\n \\right\\|_2\n &\\ll Q^{-1/2}\n \\exp\\left(\n
(2+o(1))\\frac{(\\log Q)^{1/2}}{\\log\\log Q}\n \\right)\\|f\\|_2,\n\\end{align}\nprovided\n\\[\n \\log
Q\\leq c_0(\\log N_0)^{1/2}.\n\\]\n Here \\(o(1)\\) is taken as \\(Q\\to\\infty\\), uniformly in the
displayed\nranges of \\(N\\) and \\(N_0\\); bounded values of \\(Q\\) are absorbed into\nthe implicit
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satisfying\n\\begin{align}\\label{e:bnv-local}\n |B_M f(x)|\n \\ll
M^{-v}\\|f\\mathbf1_{[x-CM,x+CM]}\\|_1,\n \\qquad 0\\leq v<1.\n\\end{align}\nLet \\(s\\geq1-v\\),
\\(r>0\\), and \\(\\gamma\\geq\\beta\\geq0\\), and suppose\nthat, whenever \\(M\\) is dyadic and
\\(2\\leq J\\leq M^{(1-v)/s}/10\\),\n\\begin{align}\\label{e:localdecomposition}\n
B_M=Lo_{J,M}+Hi_{J,M}+Er_{J,M}.\n\\end{align}\nLet \\(I_0\\) be an interval, and
let\n\\(f=\\mathbf1_F\\) and \\(g=\\mathbf1_G\\), where\n\\(F\\subset cI_0\\) and \\(G\\subset I_0\\).
Suppose that\n\\[\n \\tau:I_0\\longrightarrow\n
\\{2^k:2^k\\leq|I_0|\\}\n\\]\nsatisfies\n\\begin{align}\\label{e:admisslocal}\n
\\sup_{M\\geq\\tau(x)}M^{-1}\n \\sum_{|y-x|\\leq C_0M}f(y)\n \\ll \\langle f\\rangle_{cI_0,1},\n \\qquad
 x\\in I_0.\n\\end{align}\nWrite \\(B_\\tau f(x)=B_{\\tau(x)}f(x)\\).\n\nAssume that, for every
\\(J\\geq2\\) and every \\(x\\in I_0\\) such that\n\\(\\tau(x)\\geq(10J)^{s/(1-v)}\\),\n\\begin{align}\n
 |Lo_{J,\\tau(x)}f(x)|\n &\\ll J^s(\\log J)^\\gamma\n \\sup_{M\\geq\\tau(x)}M^{-1}\n \\sum_{|y-x|\\leq
CM}f(y),\n \\label{e:localLow}\n\\end{align}\nand that, for every \\(h\\in\\ell^2(\\mathbb
Z)\\),\n\\begin{align}\n \\left\\|\n \\sup_{\\substack{M\\geq(10J)^{s/(1-v)}\\\\M\\ {\\rm dyadic}}}\n
|Hi_{J,M}h|\n \\right\\|_2\n &\\ll J^{-r}(\\log J)^\\beta\\|h\\|_2.\n
\\label{e:localHigh}\n\\end{align}\nFinally, suppose that, for some fixed \\(A>1/2\\), uniformly in
dyadic \\(M\\),\n\\begin{align}\n \\|\\widehat{Er}_{J,M}\\|_{L^\\infty(\\mathbb T)}\n \\ll_A\n
J^{-r}(\\log J)^\\beta(1+\\log M)^{-A}.\n
\\label{e:localError}\n\\end{align}\nThen\n\\begin{align}\\label{e:sparsesquares1}\n |I_0|^{-1}|\\langle
 B_\\tau f,g\\rangle|\n \\ll_c\n \\left(\\frac{|F||G|}{|I_0|^2}\\right)^{\n 1-\\frac{s}{2(s+r)}}\n
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      "source_line": 2905,
      "statement_latex": "\\label{l:sparserecursion}\nSuppose that every admissible $\\tau$
satisfies\n\\begin{align}\\label{e:localform}\n |\\langle T_\\tau\\mathbf1_F,\\mathbf1_G\\rangle|\n \\ll
 |I_0|\\,\\Psi\\left(\\frac{|F|}{|I_0|},\n
\\frac{|G|}{|I_0|}\\right)\n\\end{align}\nfor $F\\subset cI_0$ and $G\\subset I_0$, where
$(T_N)_{N\\in2^{\\mathbb N}}$ is a family of positive convolution operators whose kernel is supported
in\n$[-CN,CN]$ for some fixed $C>0$.  Then there is a sparse family $\\mathcal S$ such
that\n\\begin{align}\\label{e:recursionconclusion}\n \\langle\\sup_{N\\in2^{\\mathbb
N}}|T_N\\mathbf1_F|,\\mathbf1_G\\rangle\n \\ll\\sum_{I\\in\\mathcal S}|I|\\,\n
\\Psi\\left(\\frac{|F\\cap cI|}{|I|},\n            \\frac{|G\\cap I|}{|I|}\\right).\n\\end{align}",
      "active_statement_sha256": "7c63ffcdc8aca70d79b7f2c65f11d7c511f7b13fd713f418144bd6c5b2d4167a",
      "title": "Sparse recursion from local estimates",
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        "primeAnnularSparse_from_local",
        "primeSparse_from_admissible_local",
        "sparseNonDyadic_local_pairing_zero",
        "sparseNonDyadic_maximum_positive",
        "densityProfileSpike_nonneg",
        "densityProfileSpike_no_dilation_comparison",
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        "genericPositiveFullSparse",
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        "genericFullDyadicMaximum_finite_test",
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        "docs/SPARSE_DYADIC_QUALIFICATION.md",
        "docs/AUXILIARY_SCOPE_REVIEW_1065.md",
        "docs/SOURCE_UPDATE_1066.md",
        "docs/GENERAL_DYADIC_SPARSE_COMPLETION.md",
        "docs/SOURCE_UPDATE_1084.md"
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      "scope": "The actual prime-square finite stopping construction, disjoint components, real-radius
admissibility, exact kernel localization, terminating cost recursion and actual sparse major subsets are
 proved, with a proved all-cutoff reduction for the original averages. A general finite recursive-cost
implication is also proved. The complete general dyadic implication is now proved for arbitrary positive
 compact-support kernels, every fixed odd support dilation, arbitrary Psi, the full untruncated maximum
and genuine quarter-sparse major sets. The uniform bound and finite approximation of the full maximum
are deduced from the local hypothesis. The local hypothesis is imposed only on maps admissible on the
whole interval; an extension argument proves the needed restricted version.",
      "remaining": "No own-paper deduction remains under the accepted dyadic and
independent-admissibility-constant conventions. The proof constructs actual sparse sets and the full
maximum from the stated local hypothesis; no additional analytic estimate is assumed.",
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          "kind": "theorem",
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where\n$F\\subset cI_0$ and $G\\subset I_0$, and put $\\delta_0=\\frac{|F||G|}{|I_0|^2}.$\n Let
$\\tau:I_0\\to\\{2^k:(2^k)^2\\leq|I_0|\\}$ be admissible at the\nspatial scale $\\tau(x)^2$, in the
sense that\n\\begin{align}\\label{e:admissprime}\n \\sup_{M\\geq\\tau(x)}M^{-2}\n \\sum_{|y-x|\\leq
CM^2}f(y)\n \\ll\\langle f\\rangle_{cI_0,1}.\n\\end{align}\nUnder the hypotheses of
Proposition~\\ref{p:decompositionofan},\n\\begin{align}\\label{e:primeendpointlocal}\n |I_0|^{-1}\n
\\big|\\langle\\widetilde A_{\\tau,\\Lambda}\\mathbf1_F,\n \\mathbf1_G\\rangle\\big|\n
\\ll\\delta_0^{2/3}\n \\Phi_{4\\sqrt3/9+o(1)}(\\delta_0).\n\\end{align}",
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        "primeStopping_local_endpoint_with_constant",
        "sourceLocalMass_from_quadratic_radius",
        "physicalDyadic_prime_pairing"
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        "blueprint/Checkpoint_1062_SourcePrimeLocalEndpoint.md"
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      "scope": "Complete source implication for the actual annular von Mangoldt square average: every
fixed real support dilation >=1, every positive spatial radius, every fixed admissibility constant,
physical dyadic stopping values including scale1, literal all-real-radius quadratic averages, empty sets
 and the exact parent-length normalization. The bound holds for every cstar>4sqrt(3)/9. Proposition4.19
is derived from the already-listed independent inputs in the terminal theorem, rather than assumed as an
 extra own-paper input.",
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        "REN_GHOSH",
        "SMOOTH_MULTIFREQUENCY",
        "PAGE_CLASSES",
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      ],
      "status_display": "Complete implication; external inputs explicit",
      "lean_evidence": [
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        "e:primelowlocal",
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        "l:localsparse",
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      "source_file": "sources/updates/PrimeSquares_User_2026_10_28.tex",
      "source_line": 3322,
      "statement_latex": "\\label{l:prime-square-input}\nIf \\((a,q)=1\\), \\(q\\leq R^{11/10}\\),
and\n\\(|\\theta|\\leq R/(qN^2)\\), then\n\\begin{align}\\label{e:usualnearzerosiegel}\n
\\widehat{\\widetilde A}_{N,\\Lambda}(a/q+\\theta)\n &=\\mathfrak s(a,q)\\widehat{\\widetilde
M}_N(\\theta)\n -\\mathbf1_{q_0\\mid q}\\mathfrak s_{\\chi_q}(a,q)\n   \\widehat{\\widetilde
M}_{N,\\beta}(\\theta)\n +O(R^{-5}).\n\\end{align}\nHere \\(q_0=q_0(N)\\), \\(\\beta=\\beta(N)\\),
and\n\\(\\chi_0=\\chi_0(N)\\) are the exceptional conductor, zero, and primitive\ncharacter supplied by
the Landau--Page theorem at scale \\(N\\), and\n\\(\\chi_q\\) is induced by \\(\\chi_0\\).  These
quantities may change when\n\\(N\\) changes. The second term is omitted if no exceptional
character\nexists. If $\\xi$ lies outside all major arcs\n\\[\n
\\left\\|\\xi-a/q\\right\\|\\leq\\frac{R}{qN^2},\n \\qquad q\\leq R,\\quad(a,q)=1,\n\\]\nwhere distance
is taken modulo one, then\n\\begin{align}\\label{e:prime-square-minor-input}\n |\\widehat{\\widetilde
A}_{N,\\Lambda}(\\xi)|\n \\ll R^{-1/4}(\\log N)^{C_0}.\n\\end{align}\nHere $R\\geq1$, $N$ is
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sufficiently large, and $R^4\\leq N/2$,\nas is the case for the choice in \\eqref{e:defR}.",
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        "annularMangoldt_outside_major_quarter_bound",
        "uniformPage_extended_major_arc",
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        "pageModulusSelector_none",
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        "docs/MINOR_ARC_REPLACEMENT_REVIEW_1092.md",
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        "docs/CORRECTED_PRIME_SQUARE_INPUT_COMPLETION.md"
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before Q, all q<=R^(11/10), the exact induced character/selector/exceptional model and inverse-fifth
major error, and the full complementary R^(-1/4)*(log N)^d minor bound. The source Q-range, uniform
sufficiently-large-N threshold, R^4<=N/2 and log N>=1 are proved. The general fixed-level uniform
Page-prefix theorem and raw Ren/Ghosh estimates remain explicit independent unproved hypotheses.",
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\\(A>0\\),\n\\begin{align}\\label{e:aaplemma}\n \\|E_{Q,N}f\\|_2\n \\ll_A Q^{-1/2}(1+\\log
N)^{-A}\\|f\\|_2.\n\\end{align}",
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      "title": "Full approximation error with arbitrary logarithmic decay",
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        "blueprint/Checkpoint_1039_SectionSixSourceExports.md"
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 real Q>=2 and every integer N in the Section6 source range. One positive source cutoff coefficient is
chosen before all A>0; the exact Q^(-1/2)*(1+log N)^(-A) operator and all-l2 function bounds are proved.
 The optional Page data and radius are the same actual selected family as Proposition4.19. The
previously explicit compatible raw distribution/Ren/Ghosh data remain unproved general inputs; no new
own-paper estimate is assumed.",
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        "e:defR",
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        "l:inftyboundsaq",
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\\(f\\),\n\\begin{align}\\label{e:lowlocal}\n |\\operatorname{Low}_{Q,N}f(x)|\n \\ll Q\\log(2Q)\n
\\sup_{M\\geq N}M^{-2}\n \\sum_{|y-x|\\leq CM^2}f(y).\n\\end{align}\nConsequently,\n\\[\n
\\|\\operatorname{Low}_{Q,N}f\\|_\\infty\n \\ll N^{-2}Q\\log(2Q)\\|f\\|_1.\n\\]",
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        "blueprint/Checkpoint_1039_SectionSixSourceExports.md"
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 Q>=2, every integer N in the common source range, one absolute positive constant for the literal
all-real-radius local bound and the l1-to-linfinity norm bound. The local result holds for every complex
 l1 input using |f| and hence the stated nonnegative inputs, with fixed spatial constant1. Ordinary and
induced exceptional terms use the complete conductor range and exact source cutoffs. Only the previously
 recorded compatible source-data boundary remains; no new mathematical input is added.",
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 let \\(\\mathcal R_s\\) be any subset of the\nreduced fractions with denominators \\(q\\sim2^s\\).
Then\n\\begin{align}\\label{e:notBourgain}\n\\left\\|\n \\sup_{\\substack{N\\in2^{\\mathbb
N}\\\\N\\gtrsim2^s}}\n \\left|\n \\left(\n \\sum_{a/q\\in\\mathcal R_s}\n c_{a,q}\\widehat{\\widetilde
M}_{N,\\beta}(\\xi-a/q)\n \\eta_s(\\xi-a/q)\\widehat f(\\xi)\n \\right)^\\vee\n \\right|\n
\\right\\|_2\n \\ll\n (1+s)\\sup_{a/q\\in\\mathcal R_s}|c_{a,q}|\\,\\|f\\|_2.\n\\end{align}",
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        "source_rational_annular_maximum"
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        "blueprint/Checkpoint_725_LargeStartActualSourceMaximum.md",
        "blueprint/Checkpoint_726_WeightedAnnularLargeStartMaximum.md",
        "docs/AUXILIARY_SCOPE_REVIEW_1065.md",
        "docs/RATIONAL_MULTIFREQUENCY_BOUNDARY.md",
        "docs/RATIONAL_ANNULAR_COMPLETION.md"
      ],
      "scope": "The full actual annular source maximum with the printed first power of (1+s) is proved
under the precise independent rational smooth multifrequency/circle-transference input and the
previously recorded general smooth input. Every source dyadic scale N>=2^s, every s>=0, scale1 and the
small denominator indices are included. The rational common period, linear cost, weight localization,
actual annular square-function error and all earlier scale costs are proved. Beta may vary with the
scale, a stronger conclusion than the source fixed-beta statement.",
      "remaining": "No own annular deduction remains. The independent smooth multifrequency theorems and
 sampling remain explicit unproved inputs. The proof uses the rational-frequency refinement rather than
replacing the general squared-log loss by a first power; a matching expository replacement proof is
saved separately.",
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          "file": "PrimeSquares/RationalProjectionErrorMaximum.lean",
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and\n\\(|c_{a,q}|\\leq b\\).  Then\n\\begin{align}\\label{e:orlicz-prefix}\n &\\left\\|\n
\\sup_{\\substack{N\\in2^{\\mathbb N}\\\\N\\geq2^s}}\n \\sup_K\n \\left|\n \\left(\n
\\sum_{\\substack{q\\sim2^s\\\\q\\leq K}}\n \\sum_{a\\in\\mathbb A_q}\n c_{a,q}\n \\widehat{\\widetilde
M}_{N,\\beta}(\\xi-a/q)\n \\eta_s(\\xi-a/q)\\widehat f(\\xi)\n \\right)^\\vee\n \\right|\n
\\right\\|_2\n \\ll (1+s)^2b\\|f\\|_2.\n\\end{align}\nThe same estimate holds if each summand is
multiplied by a fixed\nmultiplier of modulus at most one supported
where\n\\(\\eta_s(\\xi-a/q)\\neq0\\).",
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        "source_fixed_annular_truncation_maximum"
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        "docs/RATIONAL_TRUNCATION_COMPLETION.md"
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 times including scale1, every real denominator cutoff simultaneously and every integer L2 input. The
additional fixed bounded measurable multiplier clause is included, with actual essentially bounded
Fourier operators and no continuity restriction. Beta may also vary with the scale.",
      "remaining": "No own corollary deduction remains. The independent rational
smooth/circle-transference theorem and the prior general smooth input remain explicit unproved
hypotheses; the full additional fixed measurable multiplier clause is proved, not added as an external
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c_0(\\log N_0)^{1/2}.$\nThen, for every dyadic $N\\geq N_0$,\n\\begin{align}\n
\\|\\operatorname{High}_{Q,N}f\\|_2\n &\\ll Q^{-1/2}\n \\exp\\left(\n (2+o(1))\\frac{\\sqrt{\\log
Q}}{\\log\\log Q}\n \\right)\\|f\\|_2,\n \\label{e:highcontribution}\\\\\n \\left\\|\n
\\sup_{\\substack{M\\geq N_0\\\\M\\in2^{\\mathbb N}}}\n |\\operatorname{High}_{Q,M}f|\n \\right\\|_2\n
&\\ll Q^{-1/2}\n \\exp\\left(\n (2+o(1))\\frac{\\sqrt{\\log Q}}{\\log\\log Q}\n \\right)\\|f\\|_2.\n
\\label{e:highcontributionmaximal}\n\\end{align}",
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supremum and every individual output, all real Q>=2 and N0>=2 in the allowed logQ<=c0 sqrt(logN0) range,
 one positive c0 independent of epsilon, the exact source radius and complementary narrow-cutoff
multiplier, changing genuine exceptional data, and the leading coefficient 2+o(1) in precise epsilon
form. A separate uniform estimate covers bounded Q; its harmless regularization agrees with
sqrt(logQ)/loglogQ once logQ>=2.",
      "remaining": "No paper-specific deduction in this proposition remains. The two prime asymptotic
limits, the general smooth multifrequency/sampling input, raw Landau-Page uniqueness (including the
L-function bridge), and the existence/selection of source-compatible exceptional zero data remain
explicit external boundaries. The stronger auxiliary statements 6.16 and 6.19 are not awarded
completion.",
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constants \\(C,C_0\\geq1\\) such that the\nfollowing holds. Choose \\(A_N\\) from
\\(A_{N,\\Lambda}\\),\n\\(A_{N,\\mathbb N^2}\\), and \\(A_{N,\\mathbb P^2}\\), and
put\n\\(A^*f=\\sup_{N\\geq2}|A_Nf|\\).\nThen, for every finitely supported \\(f:\\mathbb Z\\to\\mathbb
C\\) and\nevery \\(\\lambda>0\\),\n\\begin{align}\\label{e:orlicz-truncated}\n \\bigl|\\{x\\in\\mathbb
Z:A^*f(x)>\\lambda\\}\\bigr|\n \\ll\n \\sum_{\\substack{x\\in\\mathbb Z\\\\\n |f(x)|>\\lambda/C_0}}\n \\
Psi_C\\left(\\frac{|f(x)|}{\\lambda}\\right).\n\\end{align}\nConsequently,\n\\begin{align}\\label{e:orli
cz-modular}\n \\bigl|\\{x\\in\\mathbb Z:A^*f(x)>\\lambda\\}\\bigr|\n \\ll\n \\sum_{x\\in\\mathbb Z}\n
\\Psi_C\\left(\\frac{|f(x)|}{\\lambda}\\right).\n\\end{align}\nIn particular, for every finite
\\(F\\subset\\mathbb Z\\) and \\(t\\geq1\\),\n\\begin{align}\\label{e:orlicz-restricted}\n
\\bigl|\\{x\\in\\mathbb Z:A^*\\mathbf1_F(x)>t^{-1}\\}\\bigr|\n \\ll |F|\\Psi_C(t).\n\\end{align}",
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all three actual families (von Mangoldt squares, ordinary squares, unweighted prime squares), every
integer cutoff N >= 2, every complex finitely supported function and positive level, with common
absolute C, C0 and K >= 1. All three conclusions are proved, including finite level sets: the truncated
modular bound, unrestricted modular bound and restricted bound for every finite F and t >= 1. The final
theorem is stated with literal sums over all integers and finite support, using the actual integral
gauge. The ordinary branch derives the actual discrete circle-model error, maximal comparison, low/high
restricted bound and full-cutoff reconstruction from the independent scalar Weyl input and general IW
inputs; no own restricted or high/low conclusion is assumed. The unweighted-prime transfer uses proved
Mathlib Chebyshev bounds without a new PNT input. At checkpoint1264 the current public export consumes
the proved finite signed maximal theorem at constant6; a supplied RawFiniteSignedMaximalInput and its
constant R are no longer hypotheses.",
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Z)\\),\n\\begin{align}\\label{e:orlicz-p}\n \\|A^*f\\|_{\\ell^p}\n \\leq\n \\exp\\left(\n
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independent inputs. One absolute D > 0 works for all three actual families, every 1 < p <= 2 and every
integer lp input. The literal all-integer-cutoff maximal function is proved to belong to lp with norm at
 most exp(D/(p-1)*log(exp(1)/(p-1))) times the input norm. The actual truncated modular estimate,
general layer cake, singular power integral, quantitative constant absorption, p-th-root conversion and
all-input finite/Fatou extension are proved. At checkpoint1264 the current public export consumes the
proved finite signed maximal theorem at constant6; a supplied RawFiniteSignedMaximalInput and its
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\\(k\\geq4\\),\nand let \\(Q\\geq2^k\\). Put\n\\begin{equation}\\label{e:IWparameters7}\n
\\begin{gathered}\n p_r=\\frac{2r}{2r-1},\n \\qquad B=C_*^{\\,k^2Q^{1/(k-1/2)}},\\\\\n
\\varepsilon=(2^{20}rB^{2r+8})^{-1},\n \\qquad C_{r,k}=(C_*r\\log(2+k))^{C_*k},\n
\\end{gathered}\n\\end{equation}\nwhere \\(C_*>1\\) is a sufficiently large absolute constant.\nThen
there exist an integer \\(P\\leq3^Q\\) and a\nIonescu--Wainger set\n\\(\\Sigma=\\Sigma_{\\leq k}\\subset
 P^{-1}\\mathbb Z/\\mathbb Z\\)\nwith the following
properties.\n\n\\begin{enumerate}[label=\\textup{(\\roman*)}]\n\\item\nEvery rational point of
denominator at most \\(Q\\) belongs to\n\\(\\Sigma\\):\n\\begin{align}\\label{e:IWcontainment7}\n
\\bigcup_{1\\leq q\\leq Q}\n \\left\\{\\frac aq\\pmod1:a\\in\\mathbb Z\\right\\}\n
\\subseteq\\Sigma.\n\\end{align}\nEvery element of \\(\\Sigma\\), when written in reduced form,
has\ndenominator at most \\(B\\). With the choice of \\(\\varepsilon\\) in \\eqref{e:IWparameters7},
the\ncorresponding major-arc parameter set is\n\\((r,\\frac12)\\)-good.\n\n\\item\nLet \\(\\nu\\) be a
complex measure on \\(\\mathbb Z_P\\) with\n\\(\\|\\nu\\|_{\\mathrm{TV}}\\leq1\\), and let \\(m\\) be a
multiplier on\n\\(\\mathbb R\\) supported in \\([-\\varepsilon,\\varepsilon]\\).  Then, for\n\\(p_r\\leq
 p\\leq2r\\),\n\\begin{align}\\label{e:IWmultiplier7}\n \\left\\|\n T_{\\displaystyle\n
\\sum_{\\alpha\\in\\Sigma}\n   \\widehat\\nu(\\alpha)m(\\,\\cdot-\\alpha)}\n
\\right\\|_{\\ell^p(\\mathbb Z)\\to\\ell^p(\\mathbb Z)}\n \\leq\n  C_{r,k}\n \\|T_m\\|_{L^p(\\mathbb
R)\\to L^p(\\mathbb R)},\n\\end{align}\nwhere the translates of \\(m\\) are periodized on \\(\\mathbb
T\\).\n\n\\item\nPut $\\Omega=[-\\varepsilon,\\varepsilon]\\times\\Sigma\n \\subset\\mathbb
R\\times\\mathbb Q/\\mathbb Z.$\nIf \\(f\\in\\ell^p(\\mathbb Z)\\) has Fourier support
in\n\\(\\pi(\\Omega)\\), then\n\\begin{align}\\label{e:IWsampling7}\n
C_{r,k}^{-1}\\|f\\|_{\\ell^p(\\mathbb Z)}\n \\leq\n \\|\\mathcal S_\\Omega^{-1}f\\|_{L^p(\\mathbb
A_{\\mathbb Z})}\n \\leq\n  C_{r,k}\\|f\\|_{\\ell^p(\\mathbb Z)}\n\\end{align}\nfor \\(p_r\\leq
p\\leq2r\\).\n\n\\item\nLet \\(V\\) be a finite-dimensional normed space
and\n\\(0<\\delta\\leq(100P)^{-1}\\).  If\n\\(G:\\mathbb R\\times\\mathbb Z_P\\to V\\) has real Fourier
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p\\leq\\infty\\),\n\\begin{align}\\label{e:IWdiagonal7}\n \\left\\|\n \\bigl(G(n,n\\bmod
P)\\bigr)_{n\\in\\mathbb Z}\n \\right\\|_{\\ell^p(\\mathbb Z;V)}\n \\asymp\n \\|G\\|_{L^p(\\mathbb
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construction at every real Q>=2^k, r>=2 and k>=4. The literal B, epsilon and C_rk are retained; P<=3^Q,
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multiplier conclusion is an actual bounded operator on the entire integer lp space, with the exact
finite-convolution Fourier kernel, uniqueness and norm <=C_rk times the actual full real Lp operator
norm. The general full-Lp real/cyclic product bound and rank-one coefficient factorization are proved.
Both major-arc sampling directions, actual unique interpolation on the full distributional integer lp
Fourier-support class, the inverse-C_rk source inequality, and all finite-dimensional normed complex
Shannon targets throughout 1<=p<=infinity are retained. Point sampling uses the standard continuous
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\\(J\\geq1\\), and\n\\(|\\sigma_j|\\leq1\\), one has, for \\(1<p\\leq2\\),\n\\begin{align}\n \\left\\|\n
 T_{\\sum_{j=1}^J\\sigma_j\n   (\\widehat{\\widetilde M}_{2^j,\\beta}\n    -\\widehat{\\widetilde
M}_{2^{j-1},\\beta})}f\n \\right\\|_{L^p(\\mathbb R)}\n &\\ll\n \\frac1{p-1}\\|f\\|_{L^p(\\mathbb R)},\n
 \\label{e:continuousincrements7}\\\\\n \\left\\|\n \\sup_{N\\geq2}\n |\\widetilde M_{N,\\beta}*f|\n
\\right\\|_{L^p(\\mathbb R)}\n &\\ll\n \\frac1{p-1}\\|f\\|_{L^p(\\mathbb R)}.\n
\\label{e:continuousmaximal7}\n\\end{align}\nHere \\(\\widetilde M_{N,\\beta}\\) is the continuous
annular measure\ndefined in \\eqref{e:M}.  Since \\eqref{e:continuousmaximal7} holds with\nthe supremum
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is chosen before beta, p, J, signs and f. The actual signed-increment kernel has exactly the source
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every0<beta<=1. The maximum is over every real N>=2, represented as a proper extended-real supremum; a
fixed countable dense set and proved scale continuity establish measurability, with a common
almost-everywhere bound, finiteness and actual Lp membership. The literal real-scale kernel/Fourier
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 and\n\\(C_{r,k}\\) be as in Proposition~\\ref{p:IWinput7}.\nLet \\(\\nu\\) be a complex measure on
\\(\\mathbb Z_P\\) with\n\\(\\|\\nu\\|_{\\mathrm{TV}}\\leq1\\), and let \\(0<\\beta\\leq1\\).
Define\n\\(L_N^{\\nu,\\beta}\\) by\n\\begin{align}\\label{e:deforliczlow7}\n
\\widehat{L_N^{\\nu,\\beta}}(\\xi)\n =\n \\sum_{\\alpha\\in\\Sigma_{\\leq k}}\n \\widehat\\nu(\\alpha)\n
 \\widehat{\\widetilde M}_{N,\\beta}(\\xi-\\alpha)\n \\eta\\left(\\frac{\\xi-\\alpha}{\\varepsilon}\\rig
ht).\n\\end{align}\nThen\n\\begin{align}\\label{e:orlicz-low}\n \\left\\|\n
\\sup_{\\substack{N\\geq1\\\\N\\in2^{\\mathbb N}}}\n |L_N^{\\nu,\\beta}f|\n
\\right\\|_{\\ell^{p_r}(\\mathbb Z)}\n \\ll\n rC_{r,k}^2\n \\log\\bigl(e+Q+r\\log B\\bigr)\n
\\|f\\|_{\\ell^{p_r}(\\mathbb Z)}.\n\\end{align}\nThe implied constant is independent of \\(\\nu\\) and
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quantities. A single harmless enlargement of the universal parameter constant from C to C^2 is made
once. The arithmetic set, period, rational coverage through Q, denominator bound and goodness are
retained; the maximal estimate uses exactly that set, period and real-Q cutoff. The adjusted-base proof
and cost comparison are checked, as are all own symbol, smoothness, sampling, short-scale, narrow,
remainder and all-input extension steps. The independent general standard inputs remain explicit; no
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of\nPropositions~\\ref{p:IWinput7} and \\ref{p:orlicz-low}, put\n\\begin{equation}\\label{e:orlicz-D}\n
D(r,k,Q)=rC_{r,k}^2\\log(e+Q+r\\log B).\n\\end{equation}\nFor dyadic \\(N\\geq N_0\\), there are
decompositions\n\\[\n \\widetilde A_{N,\\Lambda}=L_N+H_N,\n \\qquad\n \\widetilde A_{N,\\mathbb
N^2}=L_N^\\square+H_N^\\square,\n\\]\nsuch that\n\\begin{align}\n \\left\\|\\sup_{N\\geq
N_0}|L_Nf|\\right\\|_{\\ell^{p_r}(\\mathbb Z)}\n +\n \\left\\|\\sup_{N\\geq N_0}|L_N^\\square
f|\\right\\|_{\\ell^{p_r}(\\mathbb Z)}\n &\\ll D(r,k,Q)\\|f\\|_{\\ell^{p_r}(\\mathbb Z)},\n
\\label{e:orlicz-low-comparison}\\\\\n \\left\\|\\sup_{N\\geq N_0}|H_Nf|\\right\\|_{\\ell^2(\\mathbb
Z)}\n +\n \\left\\|\\sup_{N\\geq N_0}|H_N^\\square f|\\right\\|_{\\ell^2(\\mathbb Z)}\n &\\ll
Q^{-1/4}\\|f\\|_{\\ell^2(\\mathbb Z)}.\n \\label{e:orlicz-high-comparison}\n\\end{align}",
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independent standard inputs. For every real Q>=2^k, one unchanged floor-Q arithmetic construction
supplies both prime and ordinary-square decompositions at the literal real-Q cutoff. It retains
period<=3^Q, rational coverage through Q, the literal denominator bound and goodness. Both actual
countable low bounds hold on all integer lp inputs at p_r, with the exact logarithmic real-Q cost. Both
actual high maxima share the same arithmetic set, period and selected Page datum and have a proved
summed Q^(-1/4) L2 bound above the stated exponential starting scale. All operator identities and
applicability are checked. The fixed absolute parameter constant is enlarged once from C to C^2; no
width is rounded and no own-paper endpoint is an external premise.",
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general construction, multiplier/sampling, continuous/signed-maximal, smooth multifrequency,
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\\(1<p\\leq2\\),\n\\begin{align}\\label{e:orlicz-ergodic-lp}\n \\|A_T^*f\\|_{L^p(X)}\n \\leq\n
\\exp\\left(\n \\frac{C}{p-1}\\log\\frac{e}{p-1}\n
\\right)\\|f\\|_{L^p(X)}.\n\\end{align}\nMoreover,\n\\begin{align}\\label{e:orlicz-ergodic}\n
\\mu\\{A_T^*f>\\lambda\\}\n \\ll\n \\int_X\\Psi_C\\left(\\frac{|f|}{\\lambda}\\right)\\,d\\mu,\n \\qquad
 \\lambda>0.\n\\end{align}\nIf $ \\int_X\\Psi_C(|f|/a)\\,d\\mu<\\infty$\nfor some \\(a>0\\), then the
corresponding averages converge almost\neverywhere and in \\(L^1(X)\\).",
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 all three actual families and every invertible measure-preserving probability system. One common C >= 1
 serves the strong exponential rate and source Orlicz gauge; one common modular multiplier is fixed
before all systems, families, functions and levels. Every 1 < p <= 2 has the actual strong maximal norm
with almost-everywhere finiteness and actual Lp membership. The proper extended maximum satisfies the
source modular inequality. Every measurable Orlicz member at some positive scale has actual averages
converging almost everywhere and in L1 to the same genuine L1 limit, with integrability of every average
 and both L1 seminorm/integral-norm convergence checked. Own truncation, rescaling, Cauchy/density, L1
operator and limit-identification deductions are proved. At checkpoint1264 the current public export
consumes the proved finite signed maximal theorem at constant6; a supplied RawFiniteSignedMaximalInput
and its constant R are no longer hypotheses.",
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independent general Calderon strong/modular and bounded-class polynomial convergence inputs, including
their unproved published-theorem/normalization bridges, remain explicit, as do the earlier Theorem7.2
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numbers.  Suppose that, for\nevery $q>1$, there is a set $\\Lambda_q\\subset\\{n_k\\bmod
q:k\\geq1\\}$\nwhich is multiplicative under the Chinese remainder theorem: if $(q,r)=1$,\nthen
$\\Lambda_{qr}$ corresponds to $\\Lambda_q\\times\\Lambda_r$. Suppose also that there are two sequences
\\((u_j)\\) and \\((q_j)\\) of\nodd integers whose combined collection is pairwise coprime and such
that\n\\begin{align}\n &\\mathbb P_{q_j}(\\Lambda_{q_j})\\longrightarrow0,\n \\qquad\n \\inf_j\\mathbb
P_{u_j}(\\Lambda_{u_j})>0,\n \\label{e:l1-density}\\\\\n &\\gamma-\\liminf_{j\\to\\infty}\n
\\mathbb P_{q_j}(\\Lambda_{q_j}^{\\gamma})=o(\\gamma)\n
\\quad(\\gamma\\downarrow0),\\label{e:l1-thickening}\n\\end{align}\nand that there is an $\\alpha>0$
such that, for every sufficiently small\n$\\gamma>0$,\n\\begin{align}\n
\\liminf_{j\\to\\infty}\\frac1{|\\Lambda_{q_j}|^2}\n \\big|\\{(u,v)\\in\\Lambda_{q_j}^2:\n
\\operatorname{dist}_{q_j}(u-v,\\Lambda_{q_j})>\n       \\gamma s_{q_j}\\}\\big|>5\\alpha.
\\label{e:l1-difference}\n\\end{align}\nFinally, suppose that some $\\beta>0$ has the following
property. If \\(Q\\) is any product of distinct terms from the two sequences\n\\((u_j)\\) and
\\((q_j)\\), then for every \\(a\\in\\Lambda_Q\\),\n\\begin{align}\n \\liminf_{N\\to\\infty}\\frac1N\n
\\big|\\{1\\leq k\\leq N:n_k\\equiv a\\pmod Q\\}\\big|\n       >\\frac{\\beta}{|\\Lambda_Q|}.
                \\label{e:l1-equidistribution}\n\\end{align}\nThen $(n_k)$ is persistently universally
$L^1$-bad.",
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failure and separately specified Sawyer and Conze principles. The arithmetic record contains actual
cyclic residue sets, CRT, represented classes, odd coprime moduli, exact density infimum, little-o
thickening, far-pair liminf and strict finite-product residue frequencies. For every nonatomic ergodic
probability system and every infinite cutoff set, Lean produces a measurable integrable function and a
measurable positive-measure set with no finite pointwise limit. Nonatomicity means measurable splitting,
 not merely null singletons. Positive moduli include the usual modulus-one convention; prime and cutoff
indices are zero-based enumerations of the source positive-index sequences.",
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The original LaVictoire, Sawyer and Conze inputs and their precise published-to-Lean convention bridges
are unproved.",
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let $\\Omega_p\\subset\\mathbb Z/p\\mathbb Z$, put\n$r_p=|\\Omega_p|/p$ and $s_p=r_p^{-1}$, and define
$\\Omega_q$ from the\n$\\Omega_p$, for squarefree $q$, by the Chinese remainder theorem.  Suppose\nthat,
 for every fixed $k\\geq1$ and all distinct\n$0,h_1,\\ldots,h_{k-1}\\pmod p$,\n\\begin{align}\n
N_k(\\mathbf h,\\Omega_p)\n &:=\\big|\\{t\\in \\mathbb{Z}/p\\mathbb{Z}:t,t+h_1,\\ldots,t+h_{k-1}\\in\\Om
ega_p\\}\\big|=r_p^kp\\big(1+O_k((1-r_p)p^{-c})\\big)             \\label{e:GK-hyp}\n\\end{align}\nfor
some $c>0$, and suppose that $s_p=p^{o(1)}$.  Then the normalized spacings\nof $\\Omega_q$ are Poisson
distributed as $s_q=q/|\\Omega_q|\\to\\infty$.\nIn particular, if the elements of $\\Omega_q$ are listed
 cyclically and\n$F_q(\\theta)$ is the proportion of normalized gaps exceeding $\\theta$,
then\n$F_q(\\theta)\\to e^{-\\theta}$ for every $\\theta>0$.",
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arbitrary CRT residue families, uniform actual local correlations, one positive common exponent,
constants chosen before primes/shifts, subpower reciprocal local density and growing spacing along odd
squarefree moduli. The full joint cyclic gap conclusion is implemented. All its unit-square hypotheses
and the actual joint and single-gap applications are proved conditional on that independent general
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\\Lambda_p=\\{x^2\\pmod p:x\\not\\equiv0\\pmod p\\}.\n\\]\nFor every fixed $k\\geq1$ and
distinct\n$0,h_1,\\ldots,h_{k-1}\\pmod p$,\n\\[\n N_k(\\mathbf
h,\\Lambda_p)=\\frac{p}{2^k}+O_k(p^{1/2})\n
=\\left(\\frac{p-1}{2p}\\right)^kp\\big(1+O_k(p^{-1/2})\\big).\n\\]\nConsequently \\eqref{e:GK-hyp}
holds, for example with $c=1/3$.",
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RawSplitQuadraticWeilInput, the general distinct-linear-factor character-sum bound. The literal finite
residue correlation has explicit absolute error at most2*k*sqrt(p), actual density-normalized relative
error at most3*k*3^k/sqrt(p), and the source GK error with c=1/3 and constant6*k*3^k. Quantification is
uniform over every odd prime and every injective tuple of k shifts, including the source h0=0 case. All
zero corrections, density factors, positivity, finite subset sums and reciprocal local-density subpower
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bounds are proved.",
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$\\varepsilon>0$,\n\\begin{align}\\label{e:fromsiftedtop}\n |\\{r\\in\\mathcal P:n-r\\in\\mathbb
P^2\\}|\n \\geq S(\\mathcal A_n,\\mathbb P_{3,n},N^2)-O_\\varepsilon(N^\\varepsilon).\n\\end{align}",
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primitive sum-of-two-squares criterion is now a project theorem, not an additional input. Actual finite
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Section9 numbered source claim and its final quantitative theorem is complete in the agreed scope. All
elementary arithmetic reductions are now proved without an extra primitive two-square assumption.
W(alpha) convergence is proved for every alpha>1 and the smooth seed is constructed. Only the explicit
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rather than r is matched by an injective reindexing. The only extra input is the classical primitive
sum-of-two-squares criterion for m congruent2 modulo4. At1260: The Section9 final quantitative counting
endpoint was already complete conditionally at1259. This checkpoint proves the elementary
primitive-representation multiplication step from Mathlib Fermat and explicit finite-field algebra. The
full primitive two-square criterion and its consumption are the next obligations; no source claim is
newly promoted yet. The Section11 edits remain in progress, not assumed applied. At1261: Proved the full
 PrimitiveTwoSquaresCriterion from pinned Mathlib and elementary algebra, and proved W(alpha)
convergence/nonnegativity for every alpha>1. Constructed a concrete nonzero compactly supported smooth
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|T_n|+O_\\varepsilon(N^\\varepsilon),\n\\end{align}\nwhere\n\\begin{align}\\label{e:Tfactorization}\n
    T_n=\n \\left\\{\n r-1=2up_1p_2:\n \\begin{array}{l}\n r\\in\\mathbb P,\\quad n-r\\in\\mathbb
P^2,\\\\\n z\\leq p_2\\leq p_1,\\quad p_i\\in\\mathbb P_{3,n},\\\\\n p\\mid u\\Longrightarrow
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 adopted closed multiplier cutoff. Its exact finite error is at most2 omega(n-1), and its uniform
O(N^epsilon) consequence is exported. No primitive-criterion or analytic assumption is needed for this
arithmetic reduction. At1262: Every Section9 numbered source claim and its final quantitative theorem is
 complete in the agreed scope. All elementary arithmetic reductions are now proved without an extra
primitive two-square assumption. W(alpha) convergence is proved for every alpha>1 and the smooth seed is
 constructed. Only the explicit independent analytic inputs remain for the circle-method/sieve/headline
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uniform O_epsilon(N^epsilon) consequence. Its own truncated equality and upper bound are discharged by
the earlier proved modules. The independent classical primitive two-square criterion remains explicit.
At1260: The Section9 final quantitative counting endpoint was already complete conditionally at1259.
This checkpoint proves the elementary primitive-representation multiplication step from Mathlib Fermat
and explicit finite-field algebra. The full primitive two-square criterion and its consumption are the
next obligations; no source claim is newly promoted yet. The Section11 edits remain in progress, not
assumed applied. At1261: Proved the full PrimitiveTwoSquaresCriterion from pinned Mathlib and elementary
 algebra, and proved W(alpha) convergence/nonnegativity for every alpha>1. Constructed a concrete
nonzero compactly supported smooth seed. The next stage exports the Section9 arithmetic and headline
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\\(A,\\delta>0\\) and \\(L_0<N^{2-\\delta}\\).  There are sufficiently\nlarge constants
\\(B=B(A,\\delta)\\) and \\(C=C(A,\\delta)\\) such that the\nfollowing holds.  Put\n\\[\n R=(\\log
N)^C,\\qquad\n \\eta_N(\\theta)=\\eta\\left(\\frac{N^2\\theta}{R}\\right),\n\\]\nand suppose that
$V_\\ell\\leq (N^2/\\ell)^{1/2}(\\log N)^{-B}$ for $\\ell<L_0.$\nLet \\(|d_{k,\\ell}|\\leq1\\)
and\n\\((a_{k,\\ell},k)=1\\).\nDefine \n\\begin{align}\\quad \\quad\\label{e:defGR9}\n G_R(\\ell,k,n)\n
&:=\n \\mathcal T_2(\\ell,k,n;a_{k,\\ell})\n \\prod_{2<p\\leq R}\n \\frac{p\\phi((p,k))}{(p-1)^2}\n
\\left|\\left\\{(u,v)\\in\\mathbb A_p^2:\n \\begin{array}{l}\n  u^2+\\ell v\\equiv n\\pmod p,\\\\\n
v\\equiv a_{k,\\ell}\\pmod{(p,k)}\n \\end{array}\\right\\}\\right|,\n\\end{align}\nwhere $ \\mathcal
T_2(\\ell,k,n;a_{k,\\ell}) $ is defined in \\eqref{e:generalT2} and the second congruence is understood
to be trivial when \\(p\\nmid k\\).\nAlso define\n\\begin{align}\\label{e:defHR9}\n H_{\\ell,R}(n)\n
:=\n \\ell\\int_{\\mathbb R}\\eta_N(\\theta)^2\n \\left(\\int_2^N\\frac{e(\\theta u^2)}{\\log
u}\\,du\\right)\n \\left(\\int_{\\substack{2\\leq v\\leq N^2\\\\\\ell v<N^2}}\n \\frac{e(\\ell\\theta
v)}{\\log v}\\,dv\\right)\n
e(-n\\theta)\\,d\\theta.\n\\end{align}\nThen\n\\begin{align}\\label{e:circmethod}\n
\\sum_{n<N^2}\\left|\n \\sum_{\\ell<L_0}\\sum_{k<V_\\ell}\n d_{k,\\ell}r_{k,\\ell}(n)\\right|^2\n
\\ll_{A,\\delta}N^4(\\log N)^{-A},\n\\end{align}\nwhere\n\\begin{align}\\label{e:unweightedremainder}\n
r_{k,\\ell}(n)\n &=\n \\sum_{\\substack{m^2+\\ell t=n\\\\\n t\\equiv a_{k,\\ell}\\pmod k}}\n
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    "blueprint/Checkpoint_1128_GoldbachFiniteAbel.md",
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B0,C,K,N0 chosen before N and every arithmetic family. Covers all B>=B0, exact strict real
L<N^(2-delta), positive ell,k with k<V(ell), every coprime residue class, arbitrary complex coefficients
 of norm<=1, and all n<N^2. The stronger uniform cutoff statement holds for every 2<=U<=N,W>=2. The
final source theorem removes cutoffs using the proved equation-imposed bounds, and uses the actual
finite prime-pair count, actual local Euler factor G_R and actual autocorrelation singular integral H_R.
 Fourier inversion, all three error identifications, shared constants, minor estimate, major
replacement, arithmetic completion and smoothing are consumed internally. Its only unformalized
hypotheses are independent ordinary SW prime counts, ordinary unweighted BV prime-count distribution,
and uniform raw Ren/Ghosh quadratic estimates. No own-paper conclusion is assumed. At1212: Completed all
 Section10 source statements and the full circle-method implication; the three independent raw deep
inputs remain explicit and unproved. At1213: Completed every Section10 source statement and the full
source circle-method implication, including the original smooth compact seed and exact normalized
integral. Only the three agreed independent raw deep inputs remain for this circle-method implication.",
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distribution/exponential-sum inputs remain unproved; Section11 and the headline Goldbach result remain
separate work.",
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P_{3,n},z)\n \\geq X_n\\Theta_n(z)\\bigl(f_{1/2}(9/8)+o(1)\\bigr)\n +\\mathcal
E_1(n),\n\\end{align}\nwhere\n\\begin{align}\\label{e:E1section8}\n\\sum_{n<N^2}|\\mathcal
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(indeed all n=12 mod24). General classical sieve/Mertens/function-continuity and ordinary
SW/BV/Ren-Ghosh inputs remain explicit and unproved. At1229: Common cutoff compatibility for the
circle-method and full lower-sieve estimates is proved for every requested lower bound on C; main-term
size remains to be joined. At1230: The full source main-term size and the entire lower-bound side are
complete conditionally, including an exponent independent of C, uniform smoothing, one common large
cutoff and finite error-to-exception deductions. At1230 the source main-term size and the full lower
sieve are proved jointly for the same C>=A0+6; finite positive-count and exceptional-cardinality
deductions are checked. At1258: All Section11 numbered claims are complete in the agreed conditional
scope for the explicitly supplied corrected presieving statement. The original weaker statement is not
proved and must be replaced in the user manuscript. The complete joint lower/upper estimates use one
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L_n}$(M_n(\\ell),W_{\\ell,n},z_\\ell)\n &\\leq\n \\bigl(c_{9/8}F_1(2)\\mathcal W(9/8)+o(1)\\bigr)\n
X_n\\Theta_n(z)+\\mathcal E_2(n),\n\\end{align}\nwhere\n\\begin{align}\\label{e:calphadef8}\n c_\\alpha=
\\frac{16e^{-\\gamma/2}}{\\sqrt{\\pi\\alpha}}\n\\end{align}\nand\n\\begin{align}\\label{e:E2section8}\n\
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F_1(2) W(9/8) coefficient, original actual main term and closed multiplier set. All coefficients precede
 n; source divisor counts, refined preliminary/outer dimensions, uniform varying levels, the shifted
double remainder sum, its circle-method embedding and the fixed-sieve O(N^(1/4)) correction are proved.
The total error has arbitrarily strong logarithmic mean-square saving. Only independent general SW, BV,
Ren/Ghosh, classical one-stage sieve, effective multiplicative mean and Mertens inputs and general sieve
 normalizations remain unproved; no own-paper endpoint is assumed. At1257: Full upper-sieve proposition
p:sieveneedupper8 complete in uniform epsilon form from explicit independent classical inputs, with the
exact coefficient and arbitrary logarithmic mean-square saving. The lower analytic estimate and both
remainder transports are available at a shared prescribed cutoff. The final joint Section11 assembly and
 manuscript synchronization are next. At1258: All Section11 numbered claims are complete in the agreed
conditional scope for the explicitly supplied corrected presieving statement. The original weaker
statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper
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that, for every fixed\nsufficiently large \\(C\\), with \\(R=(\\log N)^C\\), one
has\n\\begin{align}\\label{e:maintermlower8}\n X_n\\Theta_n(z)\\gg_{A_0,C}\n N(\\log
N)^{-C_{A_0}}\n\\end{align}\nwhenever $n\\equiv12\\pmod{24}$ and $N^2(\\log N)^{-A_0}\\leq n<N^2$.",
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exact retained n range. The actual X_n Theta_n(z) is at least a positive constant times N(log
N)^(-(A0/2+6)); the exponent is independent of C. Actual archimedean, smoothing and local-product
arguments are proved; only the already recorded independent interval Mertens products remain external.
The same large C is joined to the complete lower sieve. At1258: All Section11 numbered claims are
complete in the agreed conditional scope for the explicitly supplied corrected presieving statement. The
 original weaker statement is not proved and must be replaced in the user manuscript. The complete joint
 lower/upper estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and
arbitrary combined logarithmic mean-square saving. The final Goldbach theorem remains the next
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\\cite[(1.8)--(1.10)]{I80},\n\\begin{align}\\label{e:Deltanumerical}\n \\Delta>0.\n\\end{align}",
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by positive interval integration and exact rpow primitives. The strict comparison is proved for the
classical normalized sieve values f_(1/2)(9/8)=sqrt(8 exp(gamma)/(9 pi)) log2 and F_1(2)=exp(gamma).
Identifying the generic independent sieve functions with those classical normalized values remains
explicit, within GENERAL_SIEVE; the paper comparison is not assumed. No floating point evaluation or
unchecked evaluator is used. At1253: Proved the exact actual W integral bound, the normalized
half/linear sieve comparison and strict positivity. Classical sieve-function normalizations remain
explicit general input identities. The upper-sieve application and final Goldbach theorem remain open;
no paper endpoint is assumed. At1258: All Section11 numbered claims are complete in the agreed
conditional scope for the explicitly supplied corrected presieving statement. The original weaker
statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper
estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary
combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction.",
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uniformly for\n\\(2\\leq U\\leq N\\) and \\(2\\leq W\\leq N^2\\),\n\\begin{align}\\label{e:BVinput9}\n
\\sum_{n<N^2}\\left|\n \\int_{\\mathfrak m}P_U(\\theta)\n \\sum_{\\ell<L_0}\\sum_{k<V_\\ell}\n
d_{k,\\ell}F_{k,\\ell,W}(\\theta)e(-n\\theta)\\,d\\theta\n \\right|^2\n \\ll_{A,\\delta}N^4(\\log
N)^{-A}.\n\\end{align}",
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n<N\u00b2 mean square, exact complementary source arcs, every sufficiently large radius exponent C, and
a single natural threshold uniform in all coefficients, residues, real U and W and printed strict
L\u2080/V_ell ranges. The theorem allows B\u22650 and 0\u2264U\u2264N, with arbitrary W and residues, so
 it is stronger than the stated ranges. Explicit bound K N\u2074(log N)^(-A), K=Z\u00b2(256/log 2),
Z=(2P+3)/log 2, C\u2080=3A+4d+26. Finite Abel, prime-power removal, halving reconstruction, scale
transfer, eventual cutoffs, source progression energy and Bessel are proved. Only the already recorded
independent raw Ren/Ghosh input is assumed.",
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hybrid minor conclusion and all conversion steps are discharged; no internal own-paper hypothesis
remains.",
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H_{\\ell,R}(n;U,W)\\geq0,\n \\qquad\n H_{\\ell,R}(n;U,W)\n =\n
H_\\ell(n;U,W)+O(NR^{-1/2}),\n\\]\nwhere\n\\[\n H_\\ell(n;U,W)\n =\n \\int_{\\substack{2\\leq u\\leq
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        "blueprint/Checkpoint_1178_GoldbachCutoffBounds.md",
        "blueprint/Checkpoint_1179_GoldbachIntegrableDecay.md",
        "blueprint/Checkpoint_1180_GoldbachPositiveFourierAverage.md",
        "blueprint/Checkpoint_1181_GoldbachSingularIntegralPositive.md",
        "blueprint/Checkpoint_1182_GoldbachSingularIntegralNorm.md",
        "blueprint/Checkpoint_1183_GoldbachSmoothingDecay.md",
        "blueprint/Checkpoint_1184_GoldbachSingularSmoothing.md",
        "blueprint/Checkpoint_1185_GoldbachQuadraticDensity.md",
        "blueprint/Checkpoint_1186_GoldbachSourceDensity.md",
        "blueprint/Checkpoint_1187_GoldbachPositiveSingularComplete.md",
        "blueprint/Checkpoint_1213_GoldbachCircleSourceSeed.md"
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      "scope": " At1176: The actual normalized autocorrelation cutoff has a proved positive denominator,
 exact convolution integral, eta(0)=1 and real nonnegative inverse Fourier transform. At1177: The scaled
 actual eta_N and its square have proved real nonnegative inverse Fourier transforms and exact compact
support inherited from the seed. At1178: The actual eta is real, even, bounded by1, has zero derivative
at0 and satisfies the full global quadratic source cutoff bound. At1179: Two-sided absolute
integrability and an explicit L1 estimate are proved from the exact finite-scale bound and
inverse-three-halves tail. At1180: The exact cutoff-weighted Fourier integral is a proved positive
average of the inverse kernel, with Fubini integrability discharged. At1181: The actual source singular
integral is proved real nonnegative, uniformly in all positive ell and every real n, including the
empty-domain branch. At1182: The actual Euclidean product and smoothed integrand are globally
integrable, and the exact source singular integral satisfies a proved uniform O(N) bound for all
positive ell. At1183: The full quantitative real-line smoothing error bound is proved with absolute
constant6 and every integrability condition discharged. At1184: The actual smoothed and unsmoothed
source integrals satisfy the full uniform O(N/sqrt(R)) comparison, with all integrability and
empty-domain branches proved. At1185: The actual quadratic affine density is globally integrable and has
 the proved exact product Fourier transform, including the scale factor. At1186: The genuine logarithmic
 quadratic density is continuous and integrable, with its exact source product Fourier transform proved
and endpoint exceptional sets checked. At1187: The full source l:positiveH9 is proved: the exact
autocorrelation cutoff, real nonnegative smoothed singular integral, its exact printed one-variable
H_ell domain, and the uniform O(N/sqrt(R)) comparison, including all positive ell and empty-domain
cases. There is no independent analytic input for this lemma. At1213: Completed every Section10 source
statement and the full source circle-method implication, including the original smooth compact seed and
exact normalized integral. Only the three agreed independent raw deep inputs remain for this
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-P_U^{(0)}F_{k,\\ell,W}^{(0)}\\bigr)(\\theta)\n e(-n\\theta)\\,d\\theta\n \\right|^2\n
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 the major arcs are proved. At1137: Exact strict-cutoff reduction and a uniform one-term endpoint error
are proved for the actual progression sums. At1138: The exact prime counting and logarithmic-integral
progression models, derivatives and lower-endpoint identities are proved. At1139: The actual prime-count
 discrepancy is transported through every smooth unit phase with an explicit quantitative error,
including the lower prime endpoint. At1140: Exact positive-phase linear and quadratic partial-summation
errors are proved from ordinary prime-counting discrepancies, with signs and totient normalizations
checked. At1141: The actual prime-square major-arc model and explicit nonunit/count-discrepancy error
are proved with the exact source sign and normalization. At1142: The fixed-q lcm fibre bound and
complete reduction to ordinary unweighted modulus-distribution errors are proved. At1143: The compatible
 lcm residue sets, q-cardinality bound and generalized CRT existence/converse are proved for arbitrary
positive moduli. At1144: The exact Upsilon coefficient for the actual compatible lcm residue classes is
proved, including both signs and totient factors. At1145: The exact independent ordinary prime-count
SW/BV boundaries are implemented; real-error equivalence, square-model numeric bound and logarithmic
absorption are proved. At1146: The full uniform source SW prime-square major approximation is a complete
 deduction from the independent ordinary prime-count SW input. At1147: Exact prime progression residue
fibres and positive rational-phase grouping are proved for the actual finite sums. At1148: The actual
source linear model and all nonunit/strict-endpoint errors are proved from ordinary lcm-modulus
prime-count discrepancies. At1149: The actual strict source J integral, clamped prime cutoff, and
uniform numeric linear model error are matched exactly. At1150: Actual BV row approximation is proved
under the independent ordinary unweighted BV input, with individual progression cutoffs and all finite
exceptional terms retained. At1151: The printed source ranges imply the ordinary BV level and comparable
 logarithmic saving, with a threshold uniform in every active ell. At1152: The actual source row-scale
sum and coefficient-family cardinality square bound are proved, including all empty-cutoff cases.
At1153: Every retained prime endpoint/nonunit term is absorbed uniformly from the exact printed source
ranges, using a proved power saving. At1154: The full uniform source BV progression-error norm sum is
complete under the independent ordinary unweighted BV input, with all individual cutoffs and finite
errors included. At1155: Exact source major-set geometry, normalized measure, wrap-around lifts,
pairwise disjointness and eventual separation are proved. At1156: Measurable local coordinates and
genuine Euclidean model regularity and norm bounds are proved for the circle Bessel assembly. At1157:
The exact source major models are constructed on the circle, with proved localization and all-Lp
regularity and the required square size bound. At1158: Both full SW/BV major estimates now hold for the
constructed exact source circle models, uniformly in every printed parameter. At1159: The restricted
measurable Bessel inequalities and two-model major-arc mean-square assembly are proved. At1160: The
actual weighted source coefficient identity, model Lp bounds and complete logarithmic mean-square budget
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\\sum_{\\substack{q\\geq1\\\\P^+(q)\\leq R}}\n \\mathfrak G_q(\\ell,k,n)\n\\]\nconverges absolutely and
satisfies\n\\begin{align}\\label{e:localGR92}\n \\sum_{\\substack{q\\geq1\\\\P^+(q)\\leq R}}\n
\\mathfrak G_q(\\ell,k,n)\n &=\n \\prod_{p\\leq R}\n \\left(1+\\sum_{\\nu\\geq1}\n \\mathfrak
G_{p^\\nu}(\\ell,k,n)\\right)\n =G_R(\\ell,k,n),\n\\end{align}\nwhere \\(G_R\\) is defined in
\\eqref{e:defGR9}.\nMoreover, for some absolute
\\(C_1>0\\),\n\\begin{align}\\label{e:arithmeticcoefficients9}\n |\\mathfrak s(a,q)|\n &\\ll
q^{-1/2}\\tau(q)^{C_1},\\notag\\\\n |\\Upsilon(a\\ell,q,a_{k,\\ell},k)|\n &\\ll
q^{-1}(q,k\\ell)\\tau(q)^{C_1},\\\\n |\\mathfrak G_q(\\ell,k,n)|\n &\\ll
q^{-1}(q,k\\ell)\\tau(q)^{C_1},\\notag\n\\end{align}\nand\n\\begin{align}\\label{e:harmonicsum9}\n
\\sum_{\\ell<L_0}\\sum_{k<V_\\ell}\n \\frac{(q,k\\ell)}{\\ell\\phi(k)}\n \\ll \\tau(q)^{C_1}(\\log
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partial merely because it cites them. At1162: The actual inclusive R-smooth arithmetic series is
absolutely summable, equals the source finite Euler product and admits an exact finite cutoff/tail
split. At1163: The genuine Euclidean source integrals have proved quadratic and linear frequency decay,
with explicit constants and all cutoffs retained. At1164: Unconditional reciprocal-totient harmonic
estimates, including arbitrary divisor progressions, and the exact gcd divisor expansion are proved.
At1165: The full source arithmetic-family harmonic estimate is proved unconditionally with explicit
divisor exponent2 and constant9 exp(2). At1166: Exact arbitrary-multiplier coefficient identities and
the uniform odd-prime bound8 gcd(p,k*ell)/p are proved. At1168: The genuine coefficient has a uniform
prime-power gcd bound, with every two-adic exponent handled explicitly. At1169: General-modulus source
coefficient decay tau(q)^9 gcd(q,k*ell)/q and principal Gauss decay tau(q)^3/sqrt(q) are proved
unconditionally. At1170: The actual weighted source arithmetic-family and finite denominator-tail bounds
 are proved; the full source arithmetic and completion lemmas remain open with explicit remaining
clauses. At1171: Normalized finite unit averages are proved invariant under reduction to every divisor
modulus. At1172: Primitive progression exponential cancellation and the coprime norm bound are proved
from actual finite unit sums. At1173: The actual Upsilon coefficient is an exact lcm-unit expectation
and satisfies exact simultaneous frequency/denominator reduction. At1174: The actual Upsilon coefficient
 has the proved sharp gcd/divisor decay tau(q) gcd(q,k*ell)/q. At1175: The full source
arithmetic-coefficient lemma is proved: absolute convergence, the all-prime infinite local Euler series
and its exact source G_R value, all three individual bounds with the common divisor exponent9, and the
complete real-cutoff harmonic family estimate. There are no external inputs for this lemma.",
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N^2\\),\n\\begin{align}\\label{e:majorcompletion9}\n& \\sum_{n<N^2}\\left|\n
\\sum_{\\ell<L_0}\\sum_{k<V_\\ell}d_{k,\\ell}\n \\left(\n \\int_{\\mathfrak M}\n
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P_U^{(0)}(\\theta)F_{k,\\ell,W}^{(0)}(\\theta)e(-n\\theta)\\,d\\theta\n -\n
\\frac{G_R(\\ell,k,n)}{\\ell\\phi(k)}\n H_{\\ell,R}(n;U,W)\n \\right)\\right|^2\\\\\n&\\hspace{7cm}
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 and thresholds precede N and all families. The actual major models, Euler factor G_R and singular
integral H_R are used. The sharp cutoff comparison, separated additional denominators, finite arithmetic
 completion and quantitative Rankin tail are proved internally. No independent deep input is used. The
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\\label{e:fixedlower10}\\\\\n \\sum_{\\ell\\in L_n}S(M_n(\\ell),W_{\\ell,n},z_\\ell)\n &\\leq\n
\\sum_{\\ell\\in L}S(M_n(\\ell),W_\\ell,z_\\ell)\n +O_\\varepsilon(N^\\varepsilon).\n
\\label{e:fixedupper10}\n\\end{align}",
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 closed multiplier set. The lower inequality is proved by inclusion, including the prime3 boundary. The
upper comparison uses an exact finite exceptional-pair inclusion and the bound sum over p|n of
tau(n-p^2-1) <= K_epsilon N^epsilon. One positive constant is chosen before N,n,z. Holds for every
N>=1,0<n<=N^2 and every real z, hence the full retained source range. No deep input is used. At1215: The
 full source fixed-sieve comparison is proved without deep inputs. The local density, presieving and
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\\(p\\equiv3\\pmod4\\),\n\\begin{align}\\label{e:lowerlocaldensity10}\n \\frac{\\omega_n(p)}p\n =\n
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\\displaystyle\\frac1{p-1},&p>R.\n \\end{cases}\n\\end{align}\nIn particular, for
\\(p\\geq7\\),\n\\(0\\leq\\omega_n(p)/p\\leq2/(p-3)<1\\).",
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prime-density bounds including p=5, and the explicit multiplicative sieve function are proved without
deep inputs. Full local-density normalization is the next internal obligation. At1220: Proved the
conductor-sensitive coefficient identity, all actual prime-factor cases, the exact factors at2 and3,
residue invariance, and normalized prime adjunction without division by the global base factor. Full
local-density source closure is reserved for the final product and CRT assembly. At1221: complete actual
 normalized Euler-product factorization, exact source CRT residues, inclusive prime cutoff,
multiplicativity and all density bounds are proved without external inputs. The upper base factor may
vanish; no division by it is used. At1221: Both full source local-density lemmas are proved without new
external inputs: exact original Euler normalization, inclusive cutoff, lower quotient identification,
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such that\n\\begin{align}\\label{e:upperfactorization10}\n \\frac{G_R(\\ell,12d,n-1)}{\\phi(12d)}\n =\n
\\frac{G_R(\\ell,12,n-1)}{\\phi(12)}\n \\frac{\\omega_n(\\ell,d)}d .\n\\end{align}\nFor
\\(p\\nmid12\\ell\\), its  factors are\n\\begin{align}\\label{e:upperlocaldensity10}\n
\\frac{\\omega_n(\\ell,p)}p\n =\n \\begin{cases}\n \\displaystyle\n
\\frac{\\rho_p(n)}{p-1-\\rho_p(n-1)},&p\\leq R,\\\\[7pt]\n \\displaystyle\\frac1{p-1},&p>R.\n
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candidate prime-density bounds including p=5, and the explicit multiplicative sieve function are proved
without deep inputs. Full local-density normalization is the next internal obligation. At1220: Proved
the conductor-sensitive coefficient identity, all actual prime-factor cases, the exact factors at2 and3,
 residue invariance, and normalized prime adjunction without division by the global base factor. Full
local-density source closure is reserved for the final product and CRT assembly. At1221: complete actual
 normalized Euler-product factorization, exact source CRT residues, inclusive prime cutoff,
multiplicativity and all density bounds are proved without external inputs. The upper base factor may
vanish; no division by it is used. At1221: Both full source local-density lemmas are proved without new
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square-free\nintegers supported on a fixed set of primes \\(\\mathfrak P\\),\nwith \\(g(1)=1\\) and
\\(0\\leq g(p)<1\\).\nFix \\(C,A>0\\), set \\(R=(\\log N)^C\\),\n\\(D_0=\\exp((\\log\\log N)^3)\\), and
\\(D_1=D/D_0\\), and assume\n\\(R+1\\leq z\\leq D_1\\leq N^A\\) for sufficiently large \\(N\\).\nSuppose
 that fixed constants \\(\\kappa_0,\\kappa>0\\) and\n\\(K_0\\geq2\\) satisfy, for every \\(2\\leq
w<y\\leq z\\),\n\\begin{align}\\label{e:tworangesievedimension}\n \\prod_{\\substack{w\\leq
p<y\\\\p\\in\\mathfrak P,\\ p\\leq R}}\n (1-g(p))^{-1}\n &\\leq\n \\left(1+\\frac{K_0}{\\log
w}\\right)\n \\left(\\frac{\\log y}{\\log w}\\right)^{\\kappa_0},\\notag\\\\\n \\prod_{\\substack{w\\leq
 p<y\\\\p\\in\\mathfrak P,\\ p>R}}\n (1-g(p))^{-1}\n &\\leq\n \\left(1+\\frac{K_0}{\\log w}\\right)\n
\\left(\\frac{\\log y}{\\log w}\\right)^\\kappa.\n\\end{align}\nAll \\(o(1)\\) terms below are uniform
for fixed\n\\(C,A,\\kappa_0,\\kappa,K_0\\).\nThen\n\\begin{align}\\label{e:presievedbounds10}\n
S(\\mathcal F,\\mathfrak P,z)\n &\\geq XV(z)\\bigl(f_\\kappa(s)+o(1)\\bigr)\n
+\\sum_{m<D}c_m^-r_m,\\notag\\\\\n S(\\mathcal F,\\mathfrak P,z)\n &\\leq
XV(z)\\bigl(F_\\kappa(s)+o(1)\\bigr)\n +\\sum_{m<D}c_m^+r_m,\n\\end{align}\nwhere\n\\[\n
V(z)=\\prod_{\\substack{p<z\\\\p\\in\\mathfrak P}}(1-g(p)),\n \\qquad\n s=\\frac{\\log D_1}{\\log z},\n
\\qquad\n |c_m^\\pm|\\leq1.\n\\]\nThe coefficients \\(c_m^\\pm\\) depend only on the fixed prime sets
and\nthe sieve levels, and not on the values of \\(g\\).  \nThe dimension\n\\(\\kappa_0\\) is used only
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L}X_n(\\ell)\\Theta_{n,\\ell}(z_\\ell)\n \\leq\n \\bigl(c_\\alpha\\mathcal W(\\alpha)+o(1)\\bigr)\n
X_n\\Theta_n(z),\n \\qquad\n
c_\\alpha=\\frac{16e^{-\\gamma/2}}{\\sqrt{\\pi\\alpha}}.\n\\end{align}\nHere \\(\\mathcal W(\\alpha)\\)
is the integral defined in\n\\eqref{e:Wdef8}.",
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coefficient is exactly c_(9/8) W(9/8)+eta. Strong multiplicativity, the uniform mean instantiation, both
 finite Abel arguments, W identity, all small-denominator and endpoint terms, moving logarithms,
integration, smoothing and relative errors are proved. Only the independent generic effective mean-value
 corollary and classical fixed-progression Mertens bounds remain explicit unproved hypotheses. The
source lemma statement is unchanged; two proof replacements are supplied but not applied. At1252: The
full source weighted upper main-term comparison l:wumatomakisum is complete as a checked implication
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$\\tau:I_0\\longrightarrow\\{2^j:1\\leq2^j\\leq|I_0|\\}$\nis admissible if, for every $x\\in
I_0$,\n\\begin{align}\\label{d:admissible}\n \\sup_{M\\geq\\tau(x)}\\frac1M\n \\sum_{|y-x|\\leq
CM}f(y)\n \\leq C_{\\rm adm}\\frac{|F|}{|I_0|}.\n\\end{align}\nwhere $C>1$ is a fixed radius constant
and $C_{\\rm adm}>1$ is a\nsufficiently large fixed constant depending on $C$.\nChanging the fixed
radius changes only the required admissibility\nconstant. One may also replace the symmetric local
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 on actual finite indicator inputs. Lean proves equality with the displayed all-real-radius finite-count
 inequality and all convergence/boundedness prerequisites. The earlier actual prime-square spatial
N\u00b2 admissibility is also proved. The accepted fixed-radius and two-direction comparison remarks are
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=\\frac1{\\phi(q)}\\sum_{r\\in\\mathbb A_q}\n
\\chi(r)e\\left(-\\frac{ar^2}{q}\\right).\n\\end{align}\nFor the principal character we
write\n\\begin{align}\\label{e:mathfraks}\n \\mathfrak s(a,q)\n =\\frac1{\\phi(q)}\\sum_{r\\in\\mathbb
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\\Upsilon(a,q,b,y)=\\frac{\\phi(y)}{\\phi([y,q])}\n \\sum_{\\substack{r\\in\\mathbb A_q\\\\r\\equiv
b\\pmod g}}\n e\\left(-\\frac{ar}{q}\\right).\n\\end{align}",
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Archive record: Proof_Coverage_Registry.json

Proof_Coverage_Registry.json


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is preserved.",
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bounds.",
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the agreed external-input boundary, the paper-specific deduction is complete.",
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 base characters.",
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        "blueprint/Strong_Improving_Complete.md"
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 hypotheses for prime squares is a separate obligation.",
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        "primeSquareImprovingOperator_apply",
        "primeSquareSourceInterval_card",
        "theorem_1_1_from_positive_square_input",
        "fullSquareDecompositionInput_sign_iff"
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      "expositions": [
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        "docs/THEOREM_1_1_STATEMENT_REVIEW.md",
        "docs/THEOREM_1_1_EXTERNAL_INPUTS.md",
        "docs/SQUARE_REFLECTION_COMPLETION.md",
        "blueprint/Checkpoint_783_SquareDecompositionReflection.md"
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      "scope": "Both conclusions for the genuine full von-Mangoldt square average, every positive
integer N, the exact source interval and its cardinality, all finite F,G subsets, every coefficient
cstar>4 sqrt(3)/9, and every real 3/2<p<=2. The strong result proves actual output lp membership and the
 literal normalized sum form, for all lp inputs, hence also the supported inputs in the source.",
      "remaining": "The implication is complete under the explicit independent deep inputs. PNT limits,
compatible progression/Ren/Ghosh data existence, and the independent ordinary-square global
decomposition remain unformalized hypotheses. Its analytic globalization remains external; the
plus/minus spatial reflection is proved with the same constants at checkpoint783. Theorem1.1 and
Propositions4.19/6.21 are complete conditional implications under their separate inputs; the current
corrected numbered paper targets are all matched, with independent inputs and stronger unnumbered
alternative-route assertions retaining their documented scopes.",
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        "selectedPrimeLow_source_bounds",
        "selectedPrimeError_all_exponents",
        "selectedPrimeHigh_source_bounds",
        "selectedPage_raw_source_range"
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        "docs/PROPOSITION_4_19_STATEMENT_REVIEW.md",
        "blueprint/Checkpoint_791_Proposition419.md"
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      "scope": "All displayed clauses for the actual common selected source low/high/error family: exact
 all-input annular decomposition, actual l1-to-l-infinity low norm, local bound for all l1 inputs with
the supremum over every real M>=N, sharp fixed dyadic and actual countable maximal high bounds for every
 real Q>=2, every allowed starting threshold, every error exponent A>0, bounded Q and the exact printed
high-symbol identity. One cutoff coefficient is chosen before epsilon and A; the operator family never
changes with these parameters.",
      "remaining": "The implication is complete under PrimeAsymptoticInputs,
PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. Their
independent analytic truths and compatible raw-data existence remain external. No own-paper
decomposition, low bound, approximation error or maximal conclusion is an input. The current corrected
numbered paper targets are all matched; independent inputs and stronger unnumbered alternative-route
assertions retain their documented scopes.",
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        "annularMangoldt_majorArc_inverse_fifth",
        "annularMangoldt_complete_Ren_bound",
        "complementaryMinorBudget_quarter_bound",
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        "annularMangoldt_outside_major_quarter_bound",
        "uniformPage_extended_major_arc",
        "source_corrected_prime_square_input",
        "pageModulusSelector_none",
        "pageModulusSelector_some"
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      "expositions": [
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        "docs/MINOR_ARC_REPLACEMENT_REVIEW_1092.md",
        "docs/SOURCE_UPDATE_1107.md",
        "docs/CORRECTED_PRIME_SQUARE_INPUT_COMPLETION.md"
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      "scope": "Complete corrected source lemma, with one genuine optional Page datum per N selected
before Q, all q<=R^(11/10), the exact induced character/selector/exceptional model and inverse-fifth
major error, and the full complementary R^(-1/4)*(log N)^d minor bound. The source Q-range, uniform
sufficiently-large-N threshold, R^4<=N/2 and log N>=1 are proved. The general fixed-level uniform
Page-prefix theorem and raw Ren/Ghosh estimates remain explicit independent unproved hypotheses.",
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        "source_approximation_function_bound",
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        "selectedPage_raw_source_range"
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      "expositions": [
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        "blueprint/Checkpoint_1039_SectionSixSourceExports.md"
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      "scope": "Complete printed approximation lemma for the literal selected source error family, every
 real Q>=2 and every integer N in the Section6 source range. One positive source cutoff coefficient is
chosen before all A>0; the exact Q^(-1/2)*(1+log N)^(-A) operator and all-l2 function bounds are proved.
 The optional Page data and radius are the same actual selected family as Proposition4.19. The
previously explicit compatible raw distribution/Ren/Ghosh data remain unproved general inputs; no new
own-paper estimate is assumed.",
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        "selectedPrimeLow_source_bounds",
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        "sourceFullLow_squared"
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        "blueprint/Checkpoint_1039_SectionSixSourceExports.md"
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 Q>=2, every integer N in the common source range, one absolute positive constant for the literal
all-real-radius local bound and the l1-to-linfinity norm bound. The local result holds for every complex
 l1 input using |f| and hence the stated nonnegative inputs, with fixed spatial constant1. Ordinary and
induced exceptional terms use the complete conductor range and exact source cutoffs. Only the previously
 recorded compatible source-data boundary remains; no new mathematical input is added.",
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        "largeStart_weightedAnnular_maximum",
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        "quadratic_log_loss_not_linear",
        "rationalBand_common_period",
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        "docs/AUXILIARY_SCOPE_REVIEW_1065.md",
        "docs/RATIONAL_MULTIFREQUENCY_BOUNDARY.md",
        "docs/RATIONAL_ANNULAR_COMPLETION.md"
      ],
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under the precise independent rational smooth multifrequency/circle-transference input and the
previously recorded general smooth input. Every source dyadic scale N>=2^s, every s>=0, scale1 and the
small denominator indices are included. The rational common period, linear cost, weight localization,
actual annular square-function error and all earlier scale costs are proved. Beta may vary with the
scale, a stronger conclusion than the source fixed-beta statement.",
      "remaining": "No own annular deduction remains. The independent smooth multifrequency theorems and
 sampling remain explicit unproved inputs. The proof uses the rational-frequency refinement rather than
replacing the general squared-log loss by a first power; a matching expository replacement proof is
saved separately.",
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        "RATIONAL_SMOOTH_MULTIFREQUENCY"
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        "aeIntegerMultiplier_localized_comp",
        "disjointAEMultiplierSymbol_local_comp",
        "fixedAnnularMultiplier_disjoint_factor",
        "fixed_annular_finite_truncation_maximum",
        "source_fixed_annular_truncation_maximum"
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        "docs/SOURCE_UPDATE_601.md",
        "docs/RATIONAL_TRUNCATION_COMPLETION.md"
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 times including scale1, every real denominator cutoff simultaneously and every integer L2 input. The
additional fixed bounded measurable multiplier clause is included, with actual essentially bounded
Fourier operators and no continuity restriction. Beta may also vary with the scale.",
      "remaining": "No own corollary deduction remains. The independent rational
smooth/circle-transference theorem and the prior general smooth input remain explicit unproved
hypotheses; the full additional fixed measurable multiplier clause is proved, not added as an external
input.",
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        "proposition621_maximal_asymptotic",
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        "pageHighSymbol_paper_identity",
        "sourceHigh_uniform_geometry",
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        "regularizedHighExponentWeight_eq"
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        "blueprint/Checkpoint_778_Proposition621Source.md"
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supremum and every individual output, all real Q>=2 and N0>=2 in the allowed logQ<=c0 sqrt(logN0) range,
 one positive c0 independent of epsilon, the exact source radius and complementary narrow-cutoff
multiplier, changing genuine exceptional data, and the leading coefficient 2+o(1) in precise epsilon
form. A separate uniform estimate covers bounded Q; its harmless regularization agrees with
sqrt(logQ)/loglogQ once logQ>=2.",
      "remaining": "No paper-specific deduction in this proposition remains. The two prime asymptotic
limits, the general smooth multifrequency/sampling input, raw Landau-Page uniqueness (including the
L-function bridge), and the existence/selection of source-compatible exceptional zero data remain
explicit external boundaries. The stronger auxiliary statements 6.16 and 6.19 are not awarded
completion.",
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        "sourceSparseTriangle_iff_barycentric"
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        "docs/INDEPENDENT_STOPPING_INPUTS.md",
        "docs/SPARSE_DYADIC_QUALIFICATION.md"
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maximal indicator for every integer N>=2, arbitrary finite F,G including empty sets, actual finite
interval collections with actual pairwise disjoint major subsets of size at least one tenth, the exact
endpoint density exponent and paper Phi for every cstar>4sqrt(3)/9, and every point of the full open
triangle in exact reciprocal/barycentric coordinates. Constants precede F,G. All stopping, packing,
support, local endpoint, interpolation and sparse deductions are proved.",
      "remaining": "No paper-specific deduction in either conclusion remains. The recorded prime limits,
 compatible raw Page/distribution data, Page uniqueness, general smooth multifrequency/sampling,
independent ordinary-square stopping interface and complementary-exponent prime maximal input remain
unproved explicit hypotheses, including their stated compatibility boundaries. The general auxiliary
Section5 statements are separately partial. The proposed dyadic qualification of l:sparserecursion is
unapplied; the actual prime-square all-cutoff reduction is proved.",
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        "source_abstract_local_stopping_estimate",
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        "localError_square_majorant",
        "linearSourceLocalMass_change_radius"
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      "expositions": [
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        "blueprint/Checkpoint_1054_SourceLocalStoppingFourier.md"
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      "scope": "Complete abstract implication with the exact source exponents, all 0<=v<1, s>=1-v, r>0,
gamma>=beta>=0 and A>1/2, arbitrary physical dyadic stopping times, actual localized low/high/error
outputs, all-real-radius admissibility, arbitrary fixed positive radii, parent-length normalization,
empty sets and the full bounded density range. The error symbol may be an arbitrary almost-everywhere
bounded function; its operator norm follows from the proved Fourier isometry. The normal decomposition
and norm estimates are the printed hypotheses, not extra deep inputs.",
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      "external_inputs": []
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      "work_group": "Sparse estimates",
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        "laminarSparseRecursion",
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        "primeSparse_from_admissible_local",
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        "sparseNonDyadic_maximum_positive",
        "densityProfileSpike_nonneg",
        "densityProfileSpike_no_dilation_comparison",
        "source_dyadic_sparse_recursion_full_domain",
        "genericPositiveFullSparse",
        "genericLocalHypothesis_uniform_indicator_bound",
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        "docs/AUXILIARY_SCOPE_REVIEW_1065.md",
        "docs/SOURCE_UPDATE_1066.md",
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        "docs/GENERAL_DYADIC_SPARSE_COMPLETION.md",
        "docs/SOURCE_UPDATE_1084.md"
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      "scope": "The actual prime-square finite stopping construction, disjoint components, real-radius
admissibility, exact kernel localization, terminating cost recursion and actual sparse major subsets are
 proved, with a proved all-cutoff reduction for the original averages. A general finite recursive-cost
implication is also proved. The complete general dyadic implication is now proved for arbitrary positive
 compact-support kernels, every fixed odd support dilation, arbitrary Psi, the full untruncated maximum
and genuine quarter-sparse major sets. The uniform bound and finite approximation of the full maximum
are deduced from the local hypothesis. The local hypothesis is imposed only on maps admissible on the
whole interval; an extension argument proves the needed restricted version.",
      "remaining": "No own-paper deduction remains under the accepted dyadic and
independent-admissibility-constant conventions. The proof constructs actual sparse sets and the full
maximum from the stated local hypothesis; no additional analytic estimate is assumed.",
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        "primeStopping_local_endpoint_with_constant",
        "sourceLocalMass_from_quadratic_radius",
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        "docs/PRIME_LOCAL_SOURCE_COMPLETION.md",
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      "scope": "Complete source implication for the actual annular von Mangoldt square average: every
fixed real support dilation >=1, every positive spatial radius, every fixed admissibility constant,
physical dyadic stopping values including scale1, literal all-real-radius quadratic averages, empty sets
 and the exact parent-length normalization. The bound holds for every cstar>4sqrt(3)/9. Proposition4.19
is derived from the already-listed independent inputs in the terminal theorem, rather than assumed as an
 extra own-paper input.",
      "remaining": "",
      "external_inputs": [
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        "ORDINARY_SQUARE_STOPPING"
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        "nearOnePsi_doubling",
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        "section7_modular_finite_card",
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        "primePositiveFamily_apply",
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        "iw_unweighted_prime_restricted_gauge",
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        "iw_prime_families_restricted_gauges",
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        "iw_full_square_model_comparison",
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        "fullSquareCircleError_uniform",
        "full_square_approximation_maximum",
        "iw_full_square_high_source_scale",
        "iw_full_square_annular_restricted_distribution",
        "iw_full_square_annular_restricted_gauge",
        "ordinary_full_annular_maximal_bound",
        "iw_full_square_restricted_gauge",
        "iw_full_square_modular_estimates",
        "iw_three_families_restricted_gauges",
        "section7_three_family_assembly",
        "theorem_7_2",
        "section7_source_sum_assembly",
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        "theorem_7_2_finitely_supported",
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        "blueprint/Checkpoint_846_SectionSevenDeductions.md",
        "docs/SECTION_7_RESTRICTED_COMPLETION.md",
        "blueprint/Checkpoint_948_IWPrimeAnnularRestrictedGauge.md",
        "blueprint/Checkpoint_950_IWPrimeFullRestrictedGauge.md",
        "blueprint/Checkpoint_953_IWUnweightedPrimeRestrictedGauge.md",
        "docs/THEOREM_7_2_COMPLETION.md",
        "docs/SECTION_7_QUADRATIC_WEYL_BOUNDARY.md",
        "blueprint/Checkpoint_967_IWFullSquareHighSourceScale.md",
        "blueprint/Checkpoint_970_IWFullSquareAnnularRestrictedGauge.md",
        "blueprint/Checkpoint_971_OrdinaryFullAnnularComparison.md",
        "blueprint/Checkpoint_972_IWFullSquareRestrictedGauge.md",
        "blueprint/Checkpoint_973_IWThreeFamilyRestrictedGauge.md",
        "blueprint/Checkpoint_974_SectionSevenThreeFamilyAssembly.md",
        "blueprint/Checkpoint_975_TheoremSevenTwo.md",
        "blueprint/Checkpoint_976_TheoremSevenTwoFinitelySupported.md",
        "docs/PAPER_COMPLETION.md"
      ],
      "scope": "Complete printed Theorem 7.2 as an implication of explicit independent standard inputs:
all three actual families (von Mangoldt squares, ordinary squares, unweighted prime squares), every
integer cutoff N >= 2, every complex finitely supported function and positive level, with common
absolute C, C0 and K >= 1. All three conclusions are proved, including finite level sets: the truncated
modular bound, unrestricted modular bound and restricted bound for every finite F and t >= 1. The final
theorem is stated with literal sums over all integers and finite support, using the actual integral
gauge. The ordinary branch derives the actual discrete circle-model error, maximal comparison, low/high
restricted bound and full-cutoff reconstruction from the independent scalar Weyl input and general IW
inputs; no own restricted or high/low conclusion is assumed. The unweighted-prime transfer uses proved
Mathlib Chebyshev bounds without a new PNT input. At checkpoint1264 the current public export consumes
the proved finite signed maximal theorem at constant6; a supplied RawFiniteSignedMaximalInput and its
constant R are no longer hypotheses.",
      "remaining": "No remaining clause of Theorem7.2 or Corollaries7.6/7.46 within the agreed
conditional scope. The explicitly recorded independent inputs and their published-theorem bridges remain
 unformalized. Stronger standalone auxiliary parameter/full-generality ranges and later badness/sieve
results remain separately tracked.",
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        "truncated_rpow_kernel",
        "truncated_scaled_power_kernel",
        "truncated_distribution_strong_power",
        "truncated_gauge_strong_power",
        "nearOneStrong_power_budget",
        "actual_finite_strong_maximal_norm",
        "positiveAverage_finite_strong_extension",
        "section7_strong_assembly",
        "corollary_7_6",
        "corollary_7_6_closed_signed_maximal"
      ],
      "expositions": [
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        "docs/SECTION_7_STATEMENT_REVIEW.md",
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        "blueprint/Checkpoint_841_NearOneMomentExponential.md",
        "docs/COROLLARY_7_6_COMPLETION.md",
        "blueprint/Checkpoint_984_CorollarySevenSix.md",
        "docs/PAPER_COMPLETION.md"
      ],
      "scope": "Complete printed Corollary 7.6 as an implication of the already recorded Theorem 7.2
independent inputs. One absolute D > 0 works for all three actual families, every 1 < p <= 2 and every
integer lp input. The literal all-integer-cutoff maximal function is proved to belong to lp with norm at
 most exp(D/(p-1)*log(exp(1)/(p-1))) times the input norm. The actual truncated modular estimate,
general layer cake, singular power integral, quantitative constant absorption, p-th-root conversion and
all-input finite/Fatou extension are proved. At checkpoint1264 the current public export consumes the
proved finite signed maximal theorem at constant6; a supplied RawFiniteSignedMaximalInput and its
constant R are no longer hypotheses.",
      "remaining": "No remaining source clause within the agreed conditional scope; no new strong
maximal or layer-cake input is introduced. Theorem7.2's independent hypotheses remain explicit. The
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ergodic Corollary7.46 and Theorem1.5(ii) are now separately completed under their reviewed independent
inputs.",
      "external_inputs": [
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        "PROGRESSION_ESTIMATE",
        "REN_GHOSH",
        "SMOOTH_MULTIFREQUENCY",
        "PAGE_CLASSES",
        "QUADRATIC_WEYL_FJK"
      ]
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      "work_group": "Orlicz and ergodic estimates",
      "status": "conditional",
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        "iwNearOneExponent_reciprocal",
        "iwRho_rank",
        "iwWidth_denominator_separation",
        "iwFrequencySet_mem",
        "taoArc_denominator_bound",
        "quantitative_iw_frequency_geometry",
        "quantitative_iw_goodness",
        "quantitative_iw_construction_good",
        "quantitative_iw_multiplier",
        "iwCyclicSymbol_real",
        "quantitative_iw_sampling",
        "iw_finite_lift_diagonal",
        "iw_finite_lift_size",
        "iw_narrow_lift_diagonal",
        "quantitative_iw_real_construction_good",
        "quantitative_real_iw_smooth_multiplier",
        "quantitative_real_iw_schwartz_sampling",
        "cyclicRealOperator_Lp_bound",
        "cyclicRealOperator_TV_one_bound",
        "schwartz_integer_test_summable",
        "cyclicFunctionLp_size",
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        "fullIWInterpolation_unique",
        "fullIWInterpolation_source_bounds",
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        "iw_frequency_in_cyclic_basis",
        "iw_coverage_all_fraction_points",
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        "docs/SECTION_7_IW_BOUNDARY.md",
        "blueprint/Checkpoint_840_QuantitativeIWParameters.md",
        "blueprint/Checkpoint_845_IWFiniteFrequencyGeometry.md",
        "blueprint/Checkpoint_847_IWGoodness.md",
        "blueprint/Checkpoint_852_IWMultiplierInterface.md",
        "docs/SECTION_7_SAMPLING_BOUNDARY.md",
        "blueprint/Checkpoint_925_CyclicLiftCharacters.md",
        "blueprint/Checkpoint_930_IWNarrowLiftIdentity.md",
        "docs/REAL_IW_EXTENSION_REVIEW.md",
        "docs/FULL_IW_EXTERNAL_BOUNDARY.md",
        "docs/FULL_QUANTITATIVE_IW_COMPLETION.md"
      ],
      "scope": "All four source clauses are complete conditional implications for one common
construction at every real Q>=2^k, r>=2 and k>=4. The literal B, epsilon and C_rk are retained; P<=3^Q,
every a/q modulo one with q<=Q, all denominator bounds and full goodness are proved. The rough
multiplier conclusion is an actual bounded operator on the entire integer lp space, with the exact
finite-convolution Fourier kernel, uniqueness and norm <=C_rk times the actual full real Lp operator
norm. The general full-Lp real/cyclic product bound and rank-one coefficient factorization are proved.
Both major-arc sampling directions, actual unique interpolation on the full distributional integer lp
Fourier-support class, the inverse-C_rk source inequality, and all finite-dimensional normed complex
Shannon targets throughout 1<=p<=infinity are retained. Point sampling uses the standard continuous
representative convention. General Tao construction, full adelic IW and full sampling theorems,
including the explicitly described rough/full-class/finite-quotient realization bridges, remain
independent unproved inputs. The earlier narrower endpoint results retain their own hypotheses.",
      "remaining": "",
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        "FULL_DOMAIN_TAO_SAMPLING"
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      "title": "Uniform continuous annular increment and maximal estimates",
      "work_group": "Orlicz and ergodic estimates",
      "status": "conditional",
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        "continuousAnnularSignedMultiplier_L2",
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        "continuousAnnularSignedMultiplier_Lp",
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        "continuousAnnularCutoffSignedMultiplier_Lp",
        "annularCutoffYoungMultiplier_Lp",
        "annularOscillatoryModel_contDiff",
        "realContinuousAnnularKernel_fourier",
        "realContinuousAnnularConvolution_continuous_scale",
        "realContinuousAnnularMax_countable",
        "realContinuousAnnular_maximal_Lp",
        "continuousAnnularSigned_full_Lp",
        "source_continuous_annular_estimates"
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      "expositions": [
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        "docs/FULL_CONTINUOUS_SOURCE_COMPLETION.md",
        "blueprint/Checkpoint_1044_FullContinuousSource.md"
      ],
      "scope": "Complete printed continuous lemma on the full Lp domain. One absolute positive constant
is chosen before beta, p, J, signs and f. The actual signed-increment kernel has exactly the source
multiplier and a full-Lp bound for every1<p<=2, arbitrary complex signs of modulus<=1 and
every0<beta<=1. The maximum is over every real N>=2, represented as a proper extended-real supremum; a
fixed countable dense set and proved scale continuity establish measurability, with a common
almost-everywhere bound, finiteness and actual Lp membership. The literal real-scale kernel/Fourier
identity and all annular L2/Hormander hypotheses are proved. The general one-sided Hardy\u2013Littlewood
 and scalar Hormander theorems in their full-Lp forms remain unproved explicit independent inputs.",
      "remaining": "No remaining source-specific clause within the agreed conditional scope. The
full-domain general theorem interfaces and their exact published-to-Lean formulations remain external;
they are not claimed to have been derived from the earlier restricted-domain inputs.",
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      "work_group": "Orlicz and ergodic estimates",
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        "iw_signed_annular_operator_identity",
        "iw_signed_annular_multiplier",
        "iw_short_transition_maximum",
        "iw_short_transition_cost_bound",
        "iw_transition_log_cost_budget",
        "iw_finite_input_low_majorant",
        "iw_narrow_integer_majorant",
        "iw_large_remainder_majorant",
        "iw_narrow_lift_diagonal",
        "iw_finite_lift_size",
        "iw_full_low_maximal_bound",
        "iw_low_all_integer_lp",
        "finite_input_maximal_extension",
        "allInputLpMaximalBound_maximum",
        "iw_full_square_low_maximal_bound",
        "analytic_iw_full_low_maximal_bound",
        "realIWAdjustedBase_width",
        "realIWAdjusted_cost",
        "real_iw_full_low_maximal_bound",
        "real_iw_low_all_integer_lp",
        "real_iw_source_construction_and_low",
        "allInputLpMaximalBound_integerLp_maximum"
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        "docs/SECTION_7_CONTINUOUS_BOUNDARY.md",
        "docs/SECTION_7_SIGNED_MAXIMAL_BOUNDARY.md",
        "blueprint/Checkpoint_903_IWAnnularLp.md",
        "blueprint/Checkpoint_907_IWShortScaleMaximum.md",
        "blueprint/Checkpoint_912_IWShortTransitionMaximum.md",
        "docs/SECTION_7_SAMPLING_BOUNDARY.md",
        "blueprint/Checkpoint_934_IWFullLowMaximum.md",
        "blueprint/Checkpoint_938_IWAllInputLowMaximum.md",
        "docs/REAL_IW_LOW_COMPLETION.md"
      ],
      "scope": "The full actual dyadic IW low maximum is proved at every real Q>=2^k, uniformly in the
cyclic measure of variation at most one and 0<beta<=1, for the real-Q geometric construction and for all
 integer lp inputs at p_r. The width, denominator bound and logarithmic cost are the literal real-Q
quantities. A single harmless enlargement of the universal parameter constant from C to C^2 is made
once. The arithmetic set, period, rational coverage through Q, denominator bound and goodness are
retained; the maximal estimate uses exactly that set, period and real-Q cutoff. The adjusted-base proof
and cost comparison are checked, as are all own symbol, smoothness, sampling, short-scale, narrow,
remainder and all-input extension steps. The independent general standard inputs remain explicit; no
current-paper conclusion is an external assumption.",
      "remaining": "No own-paper deduction remains for this proposition. The recorded independent
general construction, IW multiplier, sampling, continuous and signed-maximal theorems are explicit
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unformalized inputs, not proved Lean axioms.",
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        "iw_prime_high_maximum",
        "iw_prime_high_source_scale",
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        "cyclicRationalCoefficient_primitive_induced",
        "fixed_small_page_datum",
        "iw_original_common_comparison",
        "iw_prime_low_maximal_bound",
        "iw_prime_annular_restricted_distribution",
        "iwFullSquare_decomposition",
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        "iw_full_square_high_source_scale",
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        "iw_full_square_high_source_scale_for_data",
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        "real_iw_high_comparison",
        "real_iw_prime_low_maximal_bound",
        "real_iw_full_square_low_maximal_bound",
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        "blueprint/Checkpoint_868_IWPageModelMaximum.md",
        "blueprint/Checkpoint_871_IWPrimeHighSourceScale.md",
        "blueprint/Checkpoint_938_IWAllInputLowMaximum.md",
        "blueprint/Checkpoint_940_IWPrimeRestrictedDistribution.md",
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        "docs/THEOREM_7_2_COMPLETION.md",
        "docs/SECTION_7_QUADRATIC_WEYL_BOUNDARY.md",
        "blueprint/Checkpoint_967_IWFullSquareHighSourceScale.md",
        "blueprint/Checkpoint_969_IWFullSquareRestrictedDistribution.md",
        "docs/REAL_IW_HIGH_LOW_COMPLETION.md"
      ],
      "scope": "The complete real-Q high-low proposition is proved as an implication from the recorded
independent standard inputs. For every real Q>=2^k, one unchanged floor-Q arithmetic construction
supplies both prime and ordinary-square decompositions at the literal real-Q cutoff. It retains
period<=3^Q, rational coverage through Q, the literal denominator bound and goodness. Both actual
countable low bounds hold on all integer lp inputs at p_r, with the exact logarithmic real-Q cost. Both
actual high maxima share the same arithmetic set, period and selected Page datum and have a proved
summed Q^(-1/4) L2 bound above the stated exponential starting scale. All operator identities and
applicability are checked. The fixed absolute parameter constant is enlarged once from C to C^2; no
width is rounded and no own-paper endpoint is an external premise.",
      "remaining": "No own-paper deduction remains for this proposition. The recorded independent
general construction, multiplier/sampling, continuous/signed-maximal, smooth multifrequency,
Page/prime-distribution and scalar quadratic-Weyl inputs remain explicit and unproved. Their hypotheses
are not Lean axioms or assertions of truth.",
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        "PROGRESSION_ESTIMATE",
        "REN_GHOSH",
        "SMOOTH_MULTIFREQUENCY",
        "PAGE_CLASSES",
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      "work_group": "Orlicz and ergodic estimates",
      "status": "conditional",
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        "ergodic_strong_maximal_source",
        "section7_positive_modular_bound",
        "section7_ergodic_modular_apply",
        "nearOneOrliczMember_iff",
        "orlicz_truncation_modular_tail",
        "orlicz_ergodic_ae_cauchy",
        "ergodicSquareL1Operator_apply_toLp",
        "ergodicSquareL1Operator_norm_le",
        "L1_bounded_truncation_approximation",
        "bounded_L1_cauchy_of_ae_cauchy",
        "uniform_operator_dense_cauchy",
        "bounded_truncation_cauchy_from_input",
        "orlicz_ergodic_convergence",
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        "section7_ergodic_assembly",
        "corollary_7_46",
        "corollary_7_46_closed_signed_maximal"
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      "expositions": [
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        "docs/ERGODIC_INDEPENDENT_BOUNDARY.md",
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        "docs/SECTION_7_STATEMENT_REVIEW.md",
        "blueprint/Checkpoint_998_OrliczErgodicConvergence.md",
        "blueprint/Checkpoint_999_CorollarySevenFortySix.md",
        "docs/PAPER_COMPLETION.md"
      ],
      "scope": "Complete printed Corollary7.46 as an implication of the explicit independent inputs, for
 all three actual families and every invertible measure-preserving probability system. One common C >= 1
 serves the strong exponential rate and source Orlicz gauge; one common modular multiplier is fixed
before all systems, families, functions and levels. Every 1 < p <= 2 has the actual strong maximal norm
with almost-everywhere finiteness and actual Lp membership. The proper extended maximum satisfies the
source modular inequality. Every measurable Orlicz member at some positive scale has actual averages
converging almost everywhere and in L1 to the same genuine L1 limit, with integrability of every average
 and both L1 seminorm/integral-norm convergence checked. Own truncation, rescaling, Cauchy/density, L1
operator and limit-identification deductions are proved. At checkpoint1264 the current public export
consumes the proved finite signed maximal theorem at constant6; a supplied RawFiniteSignedMaximalInput
and its constant R are no longer hypotheses.",
      "remaining": "No remaining printed clause within the agreed conditional-formalization scope. The
independent general Calderon strong/modular and bounded-class polynomial convergence inputs, including
their unproved published-theorem/normalization bridges, remain explicit, as do the earlier Theorem7.2
independent inputs. Stronger standalone Section7 auxiliary ranges remain separately partial.",
      "external_inputs": [
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        "REN_GHOSH",
        "SMOOTH_MULTIFREQUENCY",
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        "BOUNDED_SQUARE_POINTWISE"
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        "sum_primesLE_nth",
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        "enumeratedPrimeMaximalENorm_eq",
        "ergodic_strong_maximal_source",
        "theorem_1_5_ii",
        "primeSquare_lavictoire_arithmetic",
        "theorem_1_5_i",
        "theorem_1_5_i_literal",
        "theorem_1_5",
        "theorem_1_5_ii_closed_signed_maximal"
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      "expositions": [
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        "docs/ERGODIC_INDEPENDENT_BOUNDARY.md",
        "blueprint/Checkpoint_991_TheoremOneFivePartTwo.md",
        "docs/THEOREM_1_5_I_COMPLETION.md",
        "docs/SECTION_8_COMPLETION.md",
        "docs/SECTION_8_SPACING_CORRECTION.md",
        "verification/section8-source-review-checkpoint1038.json",
        "docs/PAPER_COMPLETION.md"
      ],
      "scope": "Both parts are complete conditional implications. Part(i): the actual enumerated prime
squares are persistently universally L1-bad on every truly nonatomic ergodic measure-preserving
probability system, with every infinite cutoff set and an actual measurable positive-measure divergence
set. CRT, Dirichlet representation, correlation counts, large-prime block construction, all
moment/thickening/difference limits, uniform square-root fibers and the fixed-progression PNT
pushforward are proved. The original badness theorem, Sawyer, Conze, general Weil and fixed-progression
PNT remain separate unproved general inputs. Granville\u2013Kurlberg is not used by part(i). A checked
moment proof replaces the source strict finite-fraction argument, which has an exact kernel
counterexample. Part(ii) is a complete conditional implication: an absolute positive constant before
every invertible measure-preserving probability system, every 1 < p <= 2 and every Lp input; the proper
actual maximum over the first m prime squares for all m >= 1 is finite almost everywhere, belongs to Lp
and has the exact displayed exponential norm bound. The prime-counting/enumeration normalization, finite
 sums, first-prime case and equality of both full maxima are proved with no PNT or enumeration input.
The proof uses the already proved Corollary7.6 and the precise independent general strong Calderon
input, plus the earlier Theorem7.2 independent inputs. It does not use the bounded-class convergence
input. At checkpoint1264 the current public export consumes the proved finite signed maximal theorem at
constant6; a supplied RawFiniteSignedMaximalInput and its constant R are no longer hypotheses.",
      "remaining": "No remaining source clause within the agreed conditional scope. General inputs and
their exact convention bridges remain unproved. The user confirmed the three Section8 edits; the
selected project manuscript now contains the agreed moment proof.",
      "external_inputs": [
        "LAVICTOIRE",
```

```
        "WEIL",
        "PROGRESSION_ESTIMATE",
        "ERGODIC_TRANSFER",
        "QUANTITATIVE_IW",
        "GENERAL_VARIATION_SAMPLING",
        "REN_GHOSH",
        "SMOOTH_MULTIFREQUENCY",
        "PAGE_CLASSES",
        "QUADRATIC_WEYL_FJK",
        "SAWYER",
        "CONZE",
        "FIXED_PROGRESSION_PNT"
      ]
    },
    "t:l1badblackbox": {
      "title": "The paper-specific persistent badness criterion",
      "work_group": "Badness and convergence",
      "status": "conditional",
      "lean_anchors": [
        "weak_failure_produces_infinite_maximal",
        "persistent_l1_badness_from_arithmetic",
        "source_lavictoire_persistent_criterion"
      ],
      "expositions": [
        "docs/SECTION_8_COMPLETION.md",
        "docs/SECTION_8_INDEPENDENT_INPUTS.md"
      ],
      "scope": "Complete persistent-divergence adaptation from original general LaVictoire weak maximal
failure and separately specified Sawyer and Conze principles. The arithmetic record contains actual
cyclic residue sets, CRT, represented classes, odd coprime moduli, exact density infimum, little-o
thickening, far-pair liminf and strict finite-product residue frequencies. For every nonatomic ergodic
probability system and every infinite cutoff set, Lean produces a measurable integrable function and a
measurable positive-measure set with no finite pointwise limit. Nonatomicity means measurable splitting,
 not merely null singletons. Positive moduli include the usual modulus-one convention; prime and cutoff
indices are zero-based enumerations of the source positive-index sequences.",
      "remaining": "No remaining persistent-divergence deduction within the agreed conditional scope.
The original LaVictoire, Sawyer and Conze inputs and their precise published-to-Lean convention bridges
are unproved.",
      "external_inputs": [
        "LAVICTOIRE",
        "SAWYER",
        "CONZE"
      ]
    },
    "t:grankul": {
      "title": "Quoted Granville-Kurlberg spacing theorem",
      "work_group": "Badness and convergence",
      "status": "external_candidate",
      "lean_anchors": [
        "RawGranvilleKurlbergInput",
        "GKLocalHypotheses",
        "CyclicPoissonSpacings",
        "unitSquareFamily_gk_local",
        "badnessBlockModulus_squarefree",
        "unitSquareBlock_poisson",
        "unitSquareBlock_gap_tail"
      ],
      "expositions": [
        "docs/SECTION_8_COMPLETION.md",
        "docs/SECTION_8_INDEPENDENT_INPUTS.md",
        "blueprint/Checkpoint_1036_GranvilleKurlbergInterface.md"
      ],
      "scope": "Quoted general deep theorem remains unproved. A precise interface now specifies
arbitrary CRT residue families, uniform actual local correlations, one positive common exponent,
constants chosen before primes/shifts, subpower reciprocal local density and growing spacing along odd
squarefree moduli. The full joint cyclic gap conclusion is implemented. All its unit-square hypotheses
and the actual joint and single-gap applications are proved conditional on that independent general
input and Weil.",
      "remaining": "The quoted general GK theorem and its published-to-Lean cyclic joint formulation
bridge remain external and unproved. It is not needed for the repaired Theorem1.5(i) proof.",
      "external_inputs": [
        "GRANVILLE_KURLBERG"
      ]
    },
    "l:quadratic-GK": {
      "title": "Quadratic-residue correlation input for spacing",
      "work_group": "Badness and convergence",
      "status": "conditional",
      "lean_anchors": [
        "unitSquareIndicator_formula",
        "unitSquareResidues_prime_card_real",
        "quadraticCorrelation_as_product",
        "quadraticSoftCorrelation_error",
        "quadraticCorrelation_Weil_error",
        "quadraticCorrelation_relative_error",
        "quadratic_GK_absolute_error",
        "quadratic_GK_relative_error",
        "quadratic_GK_uniform_correlation",
        "quadratic_GK_tuple",
        "primeResidueDensity_subpower"
      ],
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      "expositions": [
        "docs/SECTION_8_QUADRATIC_CORRELATION_COMPLETION.md",
        "docs/SECTION_8_INDEPENDENT_INPUTS.md",
        "blueprint/Checkpoint_1009_QuadraticGKCorrelation.md"
      ],
      "scope": "Complete printed quadratic-residue correlation lemma as an implication of
RawSplitQuadraticWeilInput, the general distinct-linear-factor character-sum bound. The literal finite
residue correlation has explicit absolute error at most2*k*sqrt(p), actual density-normalized relative
error at most3*k*3^k/sqrt(p), and the source GK error with c=1/3 and constant6*k*3^k. Quantification is
uniform over every odd prime and every injective tuple of k shifts, including the source h0=0 case. All
zero corrections, density factors, positivity, finite subset sums and reciprocal local-density subpower
bounds are proved.",
      "remaining": "No remaining printed correlation clause in the agreed conditional scope. The general
 Weil input remains unproved. The separate moment repair and badness deductions are now also complete;
their inputs are separately recorded.",
      "external_inputs": [
        "WEIL"
      ]
    },
    "t:repmotthm": {
      "title": "Quantitative Goldbach-type exceptional set",
      "work_group": "Goldbach and sieve",
      "status": "conditional",
      "lean_anchors": [
        "goldbach_exceptional_square_sum",
        "goldbach_exceptional_count",
        "logarithmic_square_threshold_bound",
        "goldbach_logarithmic_exceptional_bound",
        "goldbachSmallNumbers_card_le_ceil",
        "goldbach_full_exceptional_count",
        "goldbach_section9_count_reduction",
        "goldbach_eventual_arithmetic_budgets",
        "goldbach_section9_eventual_count_reduction",
        "goldbach_quantitative_exceptional_set_squares",
        "goldbach_good_representation_iff",
        "goldbach_source_exceptional_membership",
        "goldbach_ceiling_square_geometry",
        "goldbach_real_cutoff_from_square_bound",
        "goldbach_headline_theorem",
        "goldbach_primitive_one_mod_four_support",
        "goldbach_primitive_double_of_odd",
        "goldbach_primitive_two_squares_criterion",
        "goldbach_weighted_W_integrable_general",
        "goldbach_weighted_W_nonneg_general",
        "goldbach_fixed_smooth_seed_at_zero",
        "goldbach_fixed_smooth_seed_nonzero",
        "goldbach_fixed_smooth_seed_support",
        "goldbach_fixed_smooth_seed_compact_support",
        "goldbach_fixed_smooth_seed_tsupport_subset",
        "goldbach_smooth_seed_exists",
        "goldbach_from_sifted_to_primes_complete",
        "goldbach_truncation_complete",
        "goldbach_section9_master_exact",
        "goldbach_master_sieve_complete",
        "goldbach_section9_headline_complete",
        "goldbach_section9_exceptional_density_zero"
      ],
      "expositions": [
        "docs/SECTION_9_REDUCTION_COMPLETION.md",
        "docs/RECOVERY_AND_SECTION9_REVIEW_1106.md",
        "blueprint/Checkpoint_1259_GoldbachQuantitativeExceptionalSet.md",
        "blueprint/Checkpoint_1261_GoldbachPrimitiveCriterion.md",
        "blueprint/Checkpoint_1262_GoldbachSectionNineComplete.md"
      ],
      "scope": "The full quantitative exceptional-set theorem is complete: for every A>0, the actual
exceptional set of positive n<X congruent12 mod24 without a prime plus a square-of-prime representation
whose first prime is a primitive sum of two squares plus1 has size O_A(X(log X)^(-A)), eventually at
every real X. The representation and exceptional-set membership are exact. The common-cutoff Section11
estimates, positive numerical gap, arbitrary logarithmic mean-square saving, all arithmetic losses,
small n and the passage from N^2 to real X are discharged. Explicit independent general SW, BV,
Ren/Ghosh, one-stage sieve, effective multiplicative mean, Mertens, normalized sieve values/continuity
and primitive two-square criterion remain unproved theorem parameters. No own-paper endpoint is a
hypothesis. At1259: Section11 and the final quantitative Goldbach theorem are complete in the agreed
conditional scope, using the supplied corrected presieving statement. Every paper endpoint is
discharged; only explicit independent general classical inputs and their normalizations remain unproved.
 Original manuscripts are unchanged and the supplied Section11 replacements still require user
application. No additional completion credit is claimed for stronger unnumbered assertions in alternate
proof routes. At1261: Proved the full PrimitiveTwoSquaresCriterion from pinned Mathlib and elementary
algebra, and proved W(alpha) convergence/nonnegativity for every alpha>1. Constructed a concrete nonzero
 compactly supported smooth seed. The next stage exports the Section9 arithmetic and headline statements
 without the former criterion or seed hypotheses, then reconciles the stale Section9 map. User
manuscript edits remain in progress. At1262 the final endpoint is exported from GoldbachSection11Inputs
alone: the primitive two-square criterion is proved internally and a fixed nonzero smooth compact seed
is constructed. The real-X bound and zero exceptional density need no separate arithmetic or seed
assumption. At1262: Every Section9 numbered source claim and its final quantitative theorem is complete
in the agreed scope. All elementary arithmetic reductions are now proved without an extra primitive
two-square assumption. W(alpha) convergence is proved for every alpha>1 and the smooth seed is
constructed. Only the explicit independent analytic inputs remain for the circle-method/sieve/headline
estimates. The user is applying previous manuscript edits; no application is assumed and original
sources are unchanged.",
      "remaining": "",
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      "external_inputs": [
        "SIEGEL_WALFISZ_PRIME_COUNTS",
        "BOMBIERI_VINOGRADOV",
        "REN_GHOSH",
        "ROSSER_IWANIEC_SINGLE_STAGE",
        "EFFECTIVE_HALF_DIMENSIONAL_MEAN",
        "CLASSICAL_INTERVAL_MERTENS",
        "CLASSICAL_FULL_INTERVAL_MERTENS",
        "CLASSICAL_PRIME_LOG_MERTENS",
        "CLASSICAL_PRIME_MERTENS_RATIOS"
      ],
      "internal_obligations_not_external_inputs": [],
      "corrected_source_arithmetic_reduction_complete": true,
      "previous_internal_obligations_discharged": [
        "l:circmethod",
        "p:sieveneedlower8",
        "p:sieveneedupper8",
        "l:maintermsize8",
        "l:numericalcomparison"
      ]
    },
    "l:fromsiftedtop": {
      "title": "Reduction from a sifted set to prime-square representations",
      "work_group": "Goldbach and sieve",
      "status": "proved",
      "lean_anchors": [
        "goldbach_shared_prime_eq",
        "goldbach_shared_divisor_exception",
        "goldbachDivisorExceptions_card",
        "goldbach_prime_divisor_power_error",
        "goldbach_representation_admissible",
        "goldbach_factor_coprime_of_not_exception",
        "goldbachSiftedValues_card",
        "goldbach_full_sifted_inclusion",
        "goldbach_full_sifted_count",
        "goldbach_from_full_sifted",
        "goldbach_coprime_zmod_not_both_zero",
        "goldbach_prime_norm_coprime",
        "goldbach_square_matrix_kernel",
        "goldbach_conjugate_kernel_separation",
        "goldbach_primitive_pair_from_integer_norm",
        "goldbach_primitive_prime_matrix",
        "goldbach_primitive_multiply_one_mod_four_prime",
        "goldbach_primitive_one_mod_four_support",
        "goldbach_primitive_double_of_odd",
        "goldbach_primitive_two_squares_criterion",
        "goldbach_weighted_W_integrable_general",
        "goldbach_weighted_W_nonneg_general",
        "goldbach_fixed_smooth_seed_at_zero",
        "goldbach_fixed_smooth_seed_nonzero",
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        "goldbach_fixed_smooth_seed_compact_support",
        "goldbach_fixed_smooth_seed_tsupport_subset",
        "goldbach_smooth_seed_exists",
        "goldbach_from_sifted_to_primes_complete",
        "goldbach_truncation_complete",
        "goldbach_section9_master_exact",
        "goldbach_master_sieve_complete",
        "goldbach_section9_headline_complete",
        "goldbach_section9_exceptional_density_zero"
      ],
      "expositions": [
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        "blueprint/Checkpoint_1260_GoldbachPrimitiveProducts.md",
        "blueprint/Checkpoint_1261_GoldbachPrimitiveCriterion.md",
        "blueprint/Checkpoint_1262_GoldbachSectionNineComplete.md"
      ],
      "scope": "The complete first Section9 reduction is proved from its ordinary range hypotheses. The
primitive sum-of-two-squares criterion is now a project theorem, not an additional input. Actual finite
sifted and good representation sets and the uniform O(N^epsilon) exception are retained. At1262: Every
Section9 numbered source claim and its final quantitative theorem is complete in the agreed scope. All
elementary arithmetic reductions are now proved without an extra primitive two-square assumption.
W(alpha) convergence is proved for every alpha>1 and the smooth seed is constructed. Only the explicit
independent analytic inputs remain for the circle-method/sieve/headline estimates. The user is applying
previous manuscript edits; no application is assumed and original sources are unchanged.",
      "remaining": "",
      "external_inputs": [],
      "prior_scope_before_arithmetic_discharge": "The actual finite lower sifted set and primitive
prime-plus-prime-square representation set satisfy the exact cardinal inequality with error at most
omega(n-1). A uniform O_epsilon(N^epsilon) bound is proved with the source quantifiers. The use of r-1
rather than r is matched by an injective reindexing. The only extra input is the classical primitive
sum-of-two-squares criterion for m congruent2 modulo4. At1260: The Section9 final quantitative counting
endpoint was already complete conditionally at1259. This checkpoint proves the elementary
primitive-representation multiplication step from Mathlib Fermat and explicit finite-field algebra. The
full primitive two-square criterion and its consumption are the next obligations; no source claim is
newly promoted yet. The Section11 edits remain in progress, not assumed applied. At1261: Proved the full
 PrimitiveTwoSquaresCriterion from pinned Mathlib and elementary algebra, and proved W(alpha)
convergence/nonnegativity for every alpha>1. Constructed a concrete nonzero compactly supported smooth
seed. The next stage exports the Section9 arithmetic and headline statements without the former
criterion or seed hypotheses, then reconciles the stale Section9 map. User manuscript edits remain in
progress."
```

```
    },
    "l:truncationT": {
      "title": "Truncation identity and the residual set T",
      "work_group": "Goldbach and sieve",
      "status": "proved",
      "lean_anchors": [
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        "goldbach_shared_divisor_exception",
        "goldbachDivisorExceptions_card",
        "goldbach_prime_divisor_power_error",
        "goldbach_representation_admissible",
        "goldbach_factor_coprime_of_not_exception",
        "primitive_obstruction_two_large_primes",
        "goldbach_sifted_bad_factorization",
        "goldbachTruncatedBad_subset_sifted",
        "goldbachTruncatedBad_not_good",
        "goldbach_good_subset_sifted",
        "goldbachTruncatedBadValues_card",
        "goldbach_truncation_exact_error",
        "goldbach_truncation_scale",
        "goldbach_truncationT",
        "goldbach_coprime_zmod_not_both_zero",
        "goldbach_prime_norm_coprime",
        "goldbach_square_matrix_kernel",
        "goldbach_conjugate_kernel_separation",
        "goldbach_primitive_pair_from_integer_norm",
        "goldbach_primitive_prime_matrix",
        "goldbach_primitive_multiply_one_mod_four_prime",
        "goldbach_primitive_one_mod_four_support",
        "goldbach_primitive_double_of_odd",
        "goldbach_primitive_two_squares_criterion",
        "goldbach_weighted_W_integrable_general",
        "goldbach_weighted_W_nonneg_general",
        "goldbach_fixed_smooth_seed_at_zero",
        "goldbach_fixed_smooth_seed_nonzero",
        "goldbach_fixed_smooth_seed_support",
        "goldbach_fixed_smooth_seed_compact_support",
        "goldbach_fixed_smooth_seed_tsupport_subset",
        "goldbach_smooth_seed_exists",
        "goldbach_truncation_complete",
        "goldbach_from_sifted_to_primes_complete",
        "goldbach_section9_master_exact",
        "goldbach_master_sieve_complete",
        "goldbach_section9_headline_complete",
        "goldbach_section9_exceptional_density_zero"
      ],
      "expositions": [
        "docs/SECTION_9_TRUNCATION.md",
        "blueprint/Checkpoint_1260_GoldbachPrimitiveProducts.md",
        "blueprint/Checkpoint_1261_GoldbachPrimitiveCriterion.md",
        "blueprint/Checkpoint_1262_GoldbachSectionNineComplete.md"
      ],
      "scope": "The full source truncation identity is proved with an explicit nonnegative error bounded
 by O(N^epsilon), including repeated3 mod4 prime factors and the actual source factorization set. The
primitive criterion is internally proved; only the normal source range hypotheses remain. At1262: Every
Section9 numbered source claim and its final quantitative theorem is complete in the agreed scope. All
elementary arithmetic reductions are now proved without an extra primitive two-square assumption.
W(alpha) convergence is proved for every alpha>1 and the smooth seed is constructed. Only the explicit
independent analytic inputs remain for the circle-method/sieve/headline estimates. The user is applying
previous manuscript edits; no application is assumed and original sources are unchanged.",
      "remaining": "",
      "external_inputs": [],
      "prior_scope_before_arithmetic_discharge": "The full actual truncated-sieve equality is proved for
 source n>12 in its residue class and N>=2, every real1<alpha<2, z=N^(1/alpha), with an explicit
nonnegative integer error at most omega(n-1), hence O_epsilon(N^epsilon) uniformly. The exact two-prime
factorization retains multiplicity, positive remaining factor, ordering, both directions of the
sifted-set classification and r-1 cardinal reindexing. At1260: The Section9 final quantitative counting
endpoint was already complete conditionally at1259. This checkpoint proves the elementary
primitive-representation multiplication step from Mathlib Fermat and explicit finite-field algebra. The
full primitive two-square criterion and its consumption are the next obligations; no source claim is
newly promoted yet. The Section11 edits remain in progress, not assumed applied. At1261: Proved the full
 PrimitiveTwoSquaresCriterion from pinned Mathlib and elementary algebra, and proved W(alpha)
convergence/nonnegativity for every alpha>1. Constructed a concrete nonzero compactly supported smooth
seed. The next stage exports the Section9 arithmetic and headline statements without the former
criterion or seed hypotheses, then reconciles the stale Section9 map. User manuscript edits remain in
progress."
    },
    "l:upperreduction": {
      "title": "Upper reduction of T to the family of sifted sets",
      "work_group": "Goldbach and sieve",
      "status": "proved",
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        "goldbachDivisorExceptions_card",
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        "goldbach_representation_admissible",
        "goldbach_factor_coprime_of_not_exception",
        "goldbachClosedL_bound",
        "goldbach_upper_sifted_inclusion",
        "goldbach_upper_reduction_exact",
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        "goldbach_upper_reduction",
        "goldbachClosedL_source_form"
      ],
      "expositions": [
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        "docs/SECTION_9_CUTOFF_CHECK_1093.md",
        "docs/SOURCE_UPDATE_1107.md"
      ],
      "scope": "For the user-corrected closed-cutoff L, the entire actual upper-sifted inclusion, exact
finite cardinal upper bound with error omega(n-1), and uniform O_epsilon(N^epsilon) bound are proved. No
 analytic, sieve, primitive two-square or own-paper endpoint is assumed. All eight closed cutoff
occurrences are verified in the latest upload.",
      "remaining": "",
      "external_inputs": [],
      "corrected_source_deduction_complete": true
    },
    "c:mastersievereduction": {
      "title": "Combined lower and upper sieve reduction",
      "work_group": "Goldbach and sieve",
      "status": "proved",
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        "goldbach_square_matrix_kernel",
        "goldbach_conjugate_kernel_separation",
        "goldbach_primitive_pair_from_integer_norm",
        "goldbach_primitive_prime_matrix",
        "goldbach_primitive_multiply_one_mod_four_prime",
        "goldbach_primitive_one_mod_four_support",
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        "goldbach_primitive_two_squares_criterion",
        "goldbach_weighted_W_integrable_general",
        "goldbach_weighted_W_nonneg_general",
        "goldbach_fixed_smooth_seed_at_zero",
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        "goldbach_fixed_smooth_seed_support",
        "goldbach_fixed_smooth_seed_compact_support",
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        "goldbach_smooth_seed_exists",
        "goldbach_master_sieve_complete",
        "goldbach_from_sifted_to_primes_complete",
        "goldbach_truncation_complete",
        "goldbach_section9_master_exact",
        "goldbach_section9_headline_complete",
        "goldbach_section9_exceptional_density_zero"
      ],
      "expositions": [
        "docs/SECTION_9_MASTER_REDUCTION.md",
        "docs/SOURCE_UPDATE_1107.md",
        "blueprint/Checkpoint_1260_GoldbachPrimitiveProducts.md",
        "blueprint/Checkpoint_1261_GoldbachPrimitiveCriterion.md",
        "blueprint/Checkpoint_1262_GoldbachSectionNineComplete.md"
      ],
      "scope": "The full source master sieve reduction is proved from ordinary source hypotheses and the
 adopted closed multiplier cutoff. Its exact finite error is at most2 omega(n-1), and its uniform
O(N^epsilon) consequence is exported. No primitive-criterion or analytic assumption is needed for this
arithmetic reduction. At1262: Every Section9 numbered source claim and its final quantitative theorem is
 complete in the agreed scope. All elementary arithmetic reductions are now proved without an extra
primitive two-square assumption. W(alpha) convergence is proved for every alpha>1 and the smooth seed is
 constructed. Only the explicit independent analytic inputs remain for the circle-method/sieve/headline
estimates. The user is applying previous manuscript edits; no application is assumed and original
sources are unchanged.",
      "remaining": "",
      "external_inputs": [],
      "corrected_source_deduction_complete": true,
      "prior_scope_before_arithmetic_discharge": "The master actual finite-set reduction is completely
assembled for the user-corrected closed-cutoff L, with error at most twice omega(n-1) and the full
uniform O_epsilon(N^epsilon) consequence. Its own truncated equality and upper bound are discharged by
the earlier proved modules. The independent classical primitive two-square criterion remains explicit.
At1260: The Section9 final quantitative counting endpoint was already complete conditionally at1259.
This checkpoint proves the elementary primitive-representation multiplication step from Mathlib Fermat
and explicit finite-field algebra. The full primitive two-square criterion and its consumption are the
next obligations; no source claim is newly promoted yet. The Section11 edits remain in progress, not
assumed applied. At1261: Proved the full PrimitiveTwoSquaresCriterion from pinned Mathlib and elementary
 algebra, and proved W(alpha) convergence/nonnegativity for every alpha>1. Constructed a concrete
nonzero compactly supported smooth seed. The next stage exports the Section9 arithmetic and headline
statements without the former criterion or seed hypotheses, then reconciles the stale Section9 map. User
 manuscript edits remain in progress."
    },
    "l:circmethod": {
      "title": "Weighted Bombieri-Vinogradov circle-method approximation",
      "work_group": "Goldbach and sieve",
      "status": "conditional",
      "lean_anchors": [
        "goldbach_cutoff_fourier_inversion",
        "goldbach_full_cutoffs_redundant",
        "goldbach_full_fourier_inversion",
        "goldbach_weighted_full_fourier_inversion",
        "goldbach_actual_minor_bessel",
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"goldbach_major_minor_partition",
"goldbach_circle_three_stage_bound",
"goldbach_frequency_energy_identity",
"goldbach_progression_energy_le_collisions",
"goldbach_actual_minor_collision_bound",
"goldbach_prime_collision_off_diagonal",
"goldbach_reciprocal_lcm_harmonic_cube",
"goldbach_reciprocal_lcm_log_cube",
"goldbach_reciprocal_lcm_strict_real_cutoff",
"goldbach_congruence_interval_card",
"goldbach_simultaneous_congruence_count_real",
"goldbach_congruence_square_identity",
"goldbach_congruence_square_harmonic_bound",
"goldbach_congruence_square_strict_real_cutoff",
"goldbach_frequency_energy_multiplicity",
"goldbach_prime_product_fibre_card",
"goldbach_prime_product_energy",
"goldbach_index_row_sum",
"goldbach_progression_polynomial_by_rows",
"goldbach_progression_energy_by_row_squares",
"goldbach_total_row_square_bound",
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"goldbach_minor_logarithmic_exponents",
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"goldbach_major_progression_error_sum",
"goldbach_lower_sieve_residue_from_support",
"goldbach_upper_sieve_residue",
"goldbach_upper_sieve_common_product_residue",
"goldbach_finite_abel_bound",
"goldbachLogRemovalWeight_antitone",
"goldbachLogRemovalWeight_of_two_le",
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"goldbach_minor_arcs_rescale",
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"goldbach_log_minor_radius_identity",
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B0,C,K,N0 chosen before N and every arithmetic family. Covers all B>=B0, exact strict real
L<N^(2-delta), positive ell,k with k<V(ell), every coprime residue class, arbitrary complex coefficients
 of norm<=1, and all n<N^2. The stronger uniform cutoff statement holds for every 2<=U<=N,W>=2. The
final source theorem removes cutoffs using the proved equation-imposed bounds, and uses the actual
finite prime-pair count, actual local Euler factor G_R and actual autocorrelation singular integral H_R.
 Fourier inversion, all three error identifications, shared constants, minor estimate, major
replacement, arithmetic completion and smoothing are consumed internally. Its only unformalized
hypotheses are independent ordinary SW prime counts, ordinary unweighted BV prime-count distribution,
and uniform raw Ren/Ghosh quadratic estimates. No own-paper conclusion is assumed. At1212: Completed all
 Section10 source statements and the full circle-method implication; the three independent raw deep
inputs remain explicit and unproved. At1213: Completed every Section10 source statement and the full
source circle-method implication, including the original smooth compact seed and exact normalized
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integral. Only the three agreed independent raw deep inputs remain for this circle-method implication.",
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distribution/exponential-sum inputs remain unproved; Section11 and the headline Goldbach result remain
separate work.",
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original X_n and Euler product, coefficients before n, a common error tending to zero, and the N^4(log
N)^(-A) mean-square remainder bound. The pointwise theorem holds throughout the source retained range
(indeed all n=12 mod24). General classical sieve/Mertens/function-continuity and ordinary
SW/BV/Ren-Ghosh inputs remain explicit and unproved. At1229: Common cutoff compatibility for the
circle-method and full lower-sieve estimates is proved for every requested lower bound on C; main-term
size remains to be joined. At1230: The full source main-term size and the entire lower-bound side are
complete conditionally, including an exponent independent of C, uniform smoothing, one common large
cutoff and finite error-to-exception deductions. At1230 the source main-term size and the full lower
sieve are proved jointly for the same C>=A0+6; finite positive-count and exceptional-cardinality
deductions are checked. At1258: All Section11 numbered claims are complete in the agreed conditional
scope for the explicitly supplied corrected presieving statement. The original weaker statement is not
proved and must be replaced in the user manuscript. The complete joint lower/upper estimates use one
cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary combined logarithmic
mean-square saving. The final Goldbach theorem remains the next deduction.",
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        "blueprint/Checkpoint_1218_GoldbachSieveRemainderTransport.md",
        "blueprint/Checkpoint_1254_GoldbachUpperSieveDensity.md",
        "blueprint/Checkpoint_1255_GoldbachUpperPresieveUniform.md",
        "blueprint/Checkpoint_1256_GoldbachUpperPresieveApplication.md",
        "blueprint/Checkpoint_1257_GoldbachUpperSieveComplete.md",
        "blueprint/Checkpoint_1258_GoldbachSection11Complete.md"
      ],
      "scope": "The full source upper sieve is complete in uniform epsilon form, with the exact c_(9/8)
F_1(2) W(9/8) coefficient, original actual main term and closed multiplier set. All coefficients precede
 n; source divisor counts, refined preliminary/outer dimensions, uniform varying levels, the shifted
double remainder sum, its circle-method embedding and the fixed-sieve O(N^(1/4)) correction are proved.
The total error has arbitrarily strong logarithmic mean-square saving. Only independent general SW, BV,
Ren/Ghosh, classical one-stage sieve, effective multiplicative mean and Mertens inputs and general sieve
 normalizations remain unproved; no own-paper endpoint is assumed. At1257: Full upper-sieve proposition
p:sieveneedupper8 complete in uniform epsilon form from explicit independent classical inputs, with the
exact coefficient and arbitrary logarithmic mean-square saving. The lower analytic estimate and both
remainder transports are available at a shared prescribed cutoff. The final joint Section11 assembly and
 manuscript synchronization are next. At1258: All Section11 numbered claims are complete in the agreed
conditional scope for the explicitly supplied corrected presieving statement. The original weaker
statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper
estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary
combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction.",
      "remaining": "",
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        "PROGRESSION_ESTIMATE",
        "REN_GHOSH",
        "ROSSER_IWANIEC_SINGLE_STAGE",
        "EFFECTIVE_HALF_DIMENSIONAL_MEAN",
        "CLASSICAL_INTERVAL_MERTENS",
        "CLASSICAL_FULL_INTERVAL_MERTENS",
        "CLASSICAL_PRIME_LOG_MERTENS",
        "CLASSICAL_PRIME_MERTENS_RATIOS"
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    "l:maintermsize8": {
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        "goldbach_lower_main_term_size",
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        "goldbach_lower_bad_card",
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      "expositions": [
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        "blueprint/Checkpoint_1230_GoldbachLowerMainTermSize.md",
        "blueprint/Checkpoint_1258_GoldbachSection11Complete.md"
      ],
      "scope": "Full source lemma proved for every A0>0 and every fixed C>=A0+6, uniformly over the
exact retained n range. The actual X_n Theta_n(z) is at least a positive constant times N(log
N)^(-(A0/2+6)); the exponent is independent of C. Actual archimedean, smoothing and local-product
arguments are proved; only the already recorded independent interval Mertens products remain external.
The same large C is joined to the complete lower sieve. At1258: All Section11 numbered claims are
complete in the agreed conditional scope for the explicitly supplied corrected presieving statement. The
 original weaker statement is not proved and must be replaced in the user manuscript. The complete joint
 lower/upper estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and
arbitrary combined logarithmic mean-square saving. The final Goldbach theorem remains the next
deduction.",
      "remaining": "",
      "external_inputs": [
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      "title": "Strict positivity of the sieve comparison",
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      "status": "conditional",
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      "expositions": [
        "blueprint/Checkpoint_1253_GoldbachSieveNumericalComparison.md",
        "blueprint/Checkpoint_1258_GoldbachSection11Complete.md"
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      "scope": "The actual W integral has the unconditional exact bound 8 sqrt(2) W(9/8)<=9/32, proved
by positive interval integration and exact rpow primitives. The strict comparison is proved for the
classical normalized sieve values f_(1/2)(9/8)=sqrt(8 exp(gamma)/(9 pi)) log2 and F_1(2)=exp(gamma).
Identifying the generic independent sieve functions with those classical normalized values remains
explicit, within GENERAL_SIEVE; the paper comparison is not assumed. No floating point evaluation or
unchecked evaluator is used. At1253: Proved the exact actual W integral bound, the normalized
half/linear sieve comparison and strict positivity. Classical sieve-function normalizations remain
explicit general input identities. The upper-sieve application and final Goldbach theorem remain open;
no paper endpoint is assumed. At1258: All Section11 numbered claims are complete in the agreed
conditional scope for the explicitly supplied corrected presieving statement. The original weaker
statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper
estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary
combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction.",
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        "goldbach_prime_minor_log_parameters",
        "goldbach_minor_real_eventual_budgets",
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        "blueprint/Checkpoint_1122_GoldbachProgressionRows.md",
        "blueprint/Checkpoint_1123_GoldbachLogarithmicEnergy.md",
        "blueprint/Checkpoint_1124_GoldbachSourceEnergy.md",
        "blueprint/Checkpoint_1125_GoldbachHybridMinorReduction.md",
        "blueprint/Checkpoint_1128_GoldbachFiniteAbel.md",
        "blueprint/Checkpoint_1129_GoldbachPrimeUnweighting.md",
        "blueprint/Checkpoint_1130_GoldbachHalvingPartials.md",
        "blueprint/Checkpoint_1131_GoldbachMinorArcGeometry.md",
        "blueprint/Checkpoint_1132_GoldbachUniformAnnularMinor.md",
        "blueprint/Checkpoint_1133_GoldbachMinorLogParameters.md",
        "blueprint/Checkpoint_1134_GoldbachMinorEventualCutoffs.md",
        "blueprint/Checkpoint_1135_GoldbachPrimeMinorUniform.md",
        "blueprint/Checkpoint_1136_GoldbachHybridMinorComplete.md"
      ],
      "scope": "Complete source hybrid minor estimate at1136: the actual circle integral and finite
n<N\u00b2 mean square, exact complementary source arcs, every sufficiently large radius exponent C, and
a single natural threshold uniform in all coefficients, residues, real U and W and printed strict
L\u2080/V_ell ranges. The theorem allows B\u22650 and 0\u2264U\u2264N, with arbitrary W and residues, so
 it is stronger than the stated ranges. Explicit bound K N\u2074(log N)^(-A), K=Z\u00b2(256/log 2),
Z=(2P+3)/log 2, C\u2080=3A+4d+26. Finite Abel, prime-power removal, halving reconstruction, scale
transfer, eventual cutoffs, source progression energy and Bessel are proved. Only the already recorded
independent raw Ren/Ghosh input is assumed.",
      "remaining": "The independent raw UniformRenGhoshInputs remains explicit and unproved. The source
hybrid minor conclusion and all conversion steps are discharged; no internal own-paper hypothesis
remains.",
      "external_inputs": [
        "REN_GHOSH"
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    "l:positiveH9": {
      "title": "Positivity and truncation of the singular integral",
      "work_group": "Goldbach and sieve",
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        "goldbach_cutoff_square_inverse_nonnegative",
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        "goldbach_scaled_eta_support",
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        "blueprint/Checkpoint_1179_GoldbachIntegrableDecay.md",
        "blueprint/Checkpoint_1180_GoldbachPositiveFourierAverage.md",
        "blueprint/Checkpoint_1181_GoldbachSingularIntegralPositive.md",
        "blueprint/Checkpoint_1182_GoldbachSingularIntegralNorm.md",
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        "blueprint/Checkpoint_1187_GoldbachPositiveSingularComplete.md",
        "blueprint/Checkpoint_1213_GoldbachCircleSourceSeed.md"
      ],
      "scope": " At1176: The actual normalized autocorrelation cutoff has a proved positive denominator,
 exact convolution integral, eta(0)=1 and real nonnegative inverse Fourier transform. At1177: The scaled
 actual eta_N and its square have proved real nonnegative inverse Fourier transforms and exact compact
support inherited from the seed. At1178: The actual eta is real, even, bounded by1, has zero derivative
at0 and satisfies the full global quadratic source cutoff bound. At1179: Two-sided absolute
integrability and an explicit L1 estimate are proved from the exact finite-scale bound and
inverse-three-halves tail. At1180: The exact cutoff-weighted Fourier integral is a proved positive
average of the inverse kernel, with Fubini integrability discharged. At1181: The actual source singular
integral is proved real nonnegative, uniformly in all positive ell and every real n, including the
empty-domain branch. At1182: The actual Euclidean product and smoothed integrand are globally
integrable, and the exact source singular integral satisfies a proved uniform O(N) bound for all
positive ell. At1183: The full quantitative real-line smoothing error bound is proved with absolute
constant6 and every integrability condition discharged. At1184: The actual smoothed and unsmoothed
source integrals satisfy the full uniform O(N/sqrt(R)) comparison, with all integrability and
empty-domain branches proved. At1185: The actual quadratic affine density is globally integrable and has
 the proved exact product Fourier transform, including the scale factor. At1186: The genuine logarithmic
 quadratic density is continuous and integrable, with its exact source product Fourier transform proved
and endpoint exceptional sets checked. At1187: The full source l:positiveH9 is proved: the exact
autocorrelation cutoff, real nonnegative smoothed singular integral, its exact printed one-variable
H_ell domain, and the uniform O(N/sqrt(R)) comparison, including all positive ell and empty-domain
cases. There is no independent analytic input for this lemma. At1213: Completed every Section10 source
statement and the full source circle-method implication, including the original smooth compact seed and
exact normalized integral. Only the three agreed independent raw deep inputs remain for this
circle-method implication.",
      "remaining": "None. Positivity and the comparison with the exact printed one-variable domain are
proved for the actual source cutoff and integrals, without extra external analytic input.",
      "external_inputs": []
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    "goldbachWeightedProgressionMajorModel_memLp",
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        "blueprint/Checkpoint_1160_GoldbachMajorWeightedAssembly.md",
        "blueprint/Checkpoint_1161_GoldbachMajorApproximationComplete.md",
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        "docs/MAJOR_ARC_PROOF_CORRECTION.md"
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      "scope": " At1126: The exact lcm-fibre divisor bound and weighted progression-error reindexing for
 the major arcs are proved. At1137: Exact strict-cutoff reduction and a uniform one-term endpoint error
are proved for the actual progression sums. At1138: The exact prime counting and logarithmic-integral
progression models, derivatives and lower-endpoint identities are proved. At1139: The actual prime-count
 discrepancy is transported through every smooth unit phase with an explicit quantitative error,
including the lower prime endpoint. At1140: Exact positive-phase linear and quadratic partial-summation
errors are proved from ordinary prime-counting discrepancies, with signs and totient normalizations
checked. At1141: The actual prime-square major-arc model and explicit nonunit/count-discrepancy error
are proved with the exact source sign and normalization. At1142: The fixed-q lcm fibre bound and
complete reduction to ordinary unweighted modulus-distribution errors are proved. At1143: The compatible
 lcm residue sets, q-cardinality bound and generalized CRT existence/converse are proved for arbitrary
positive moduli. At1144: The exact Upsilon coefficient for the actual compatible lcm residue classes is
proved, including both signs and totient factors. At1145: The exact independent ordinary prime-count
SW/BV boundaries are implemented; real-error equivalence, square-model numeric bound and logarithmic
absorption are proved. At1146: The full uniform source SW prime-square major approximation is a complete
 deduction from the independent ordinary prime-count SW input. At1147: Exact prime progression residue
fibres and positive rational-phase grouping are proved for the actual finite sums. At1148: The actual
source linear model and all nonunit/strict-endpoint errors are proved from ordinary lcm-modulus
prime-count discrepancies. At1149: The actual strict source J integral, clamped prime cutoff, and
uniform numeric linear model error are matched exactly. At1150: Actual BV row approximation is proved
under the independent ordinary unweighted BV input, with individual progression cutoffs and all finite
exceptional terms retained. At1151: The printed source ranges imply the ordinary BV level and comparable
 logarithmic saving, with a threshold uniform in every active ell. At1152: The actual source row-scale
sum and coefficient-family cardinality square bound are proved, including all empty-cutoff cases.
At1153: Every retained prime endpoint/nonunit term is absorbed uniformly from the exact printed source
ranges, using a proved power saving. At1154: The full uniform source BV progression-error norm sum is
complete under the independent ordinary unweighted BV input, with all individual cutoffs and finite
errors included. At1155: Exact source major-set geometry, normalized measure, wrap-around lifts,
pairwise disjointness and eventual separation are proved. At1156: Measurable local coordinates and
genuine Euclidean model regularity and norm bounds are proved for the circle Bessel assembly. At1157:
The exact source major models are constructed on the circle, with proved localization and all-Lp
regularity and the required square size bound. At1158: Both full SW/BV major estimates now hold for the
constructed exact source circle models, uniformly in every printed parameter. At1159: The restricted
measurable Bessel inequalities and two-model major-arc mean-square assembly are proved. At1160: The
actual weighted source coefficient identity, model Lp bounds and complete logarithmic mean-square budget
 are proved. At1161: The full source major-arc mean-square approximation l:majorapproximation9 is
complete under ordinary SW and BV, uniformly in all printed data, with strict endpoints and finite
errors discharged.",
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partial merely because it cites them. At1162: The actual inclusive R-smooth arithmetic series is
absolutely summable, equals the source finite Euler product and admits an exact finite cutoff/tail
split. At1163: The genuine Euclidean source integrals have proved quadratic and linear frequency decay,
with explicit constants and all cutoffs retained. At1164: Unconditional reciprocal-totient harmonic
estimates, including arbitrary divisor progressions, and the exact gcd divisor expansion are proved.
At1165: The full source arithmetic-family harmonic estimate is proved unconditionally with explicit
divisor exponent2 and constant9 exp(2). At1166: Exact arbitrary-multiplier coefficient identities and
the uniform odd-prime bound8 gcd(p,k*ell)/p are proved. At1168: The genuine coefficient has a uniform
prime-power gcd bound, with every two-adic exponent handled explicitly. At1169: General-modulus source
coefficient decay tau(q)^9 gcd(q,k*ell)/q and principal Gauss decay tau(q)^3/sqrt(q) are proved
unconditionally. At1170: The actual weighted source arithmetic-family and finite denominator-tail bounds
 are proved; the full source arithmetic and completion lemmas remain open with explicit remaining
clauses. At1171: Normalized finite unit averages are proved invariant under reduction to every divisor
modulus. At1172: Primitive progression exponential cancellation and the coprime norm bound are proved
from actual finite unit sums. At1173: The actual Upsilon coefficient is an exact lcm-unit expectation
and satisfies exact simultaneous frequency/denominator reduction. At1174: The actual Upsilon coefficient
 has the proved sharp gcd/divisor decay tau(q) gcd(q,k*ell)/q. At1175: The full source
arithmetic-coefficient lemma is proved: absolute convergence, the all-prime infinite local Euler series
and its exact source G_R value, all three individual bounds with the common divisor exponent9, and the
complete real-cutoff harmonic family estimate. There are no external inputs for this lemma.",
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sufficiently large C, all B>=0 and sufficiently large N, exact strict real arithmetic cutoffs, arbitrary
 coprime residue representatives and coefficients of norm at most one, and every 2<=U<=N,W>=2. Constants
 and thresholds precede N and all families. The actual major models, Euler factor G_R and singular
integral H_R are used. The sharp cutoff comparison, separated additional denominators, finite arithmetic
 completion and quantitative Rankin tail are proved internally. No independent deep input is used. The
source W<=N^2 restriction is unnecessary. The proof uses a weaker intermediate power of R that yields
the same prescribed logarithmic saving after increasing C. At1211: Completed the full source
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upper comparison uses an exact finite exceptional-pair inclusion and the bound sum over p|n of
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N>=1,0<n<=N^2 and every real z, hence the full retained source range. No deep input is used. At1215: The
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        "blueprint/Checkpoint_1221_GoldbachSieveDensityFactorization.md"
      ],
      "scope": " At1216: Proved the exact lower sieve sequence/progression count and its actual
circle-method remainder decomposition. The full fixed-sieve comparison remains proved; local densities
and later sieve conclusions remain open. At1219: Actual unit-square root counts, both candidate
prime-density bounds including p=5, and the explicit multiplicative sieve function are proved without
deep inputs. Full local-density normalization is the next internal obligation. At1220: Proved the
conductor-sensitive coefficient identity, all actual prime-factor cases, the exact factors at2 and3,
residue invariance, and normalized prime adjunction without division by the global base factor. Full
local-density source closure is reserved for the final product and CRT assembly. At1221: complete actual
 normalized Euler-product factorization, exact source CRT residues, inclusive prime cutoff,
multiplicativity and all density bounds are proved without external inputs. The upper base factor may
vanish; no division by it is used. At1221: Both full source local-density lemmas are proved without new
external inputs: exact original Euler normalization, inclusive cutoff, lower quotient identification,
source CRT specializations, upper multiplicativity and p=5, with common zero upper base factors
retained. Presieving and the later sieve estimates remain open.",
      "remaining": "None within the source statement and retained admissible context.",
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      "status": "proved",
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        "goldbach_rho_congr",
        "goldbach_cohen_rho",
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        "goldbach_rho_le_two",
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        "goldbach_sieve_density_factorization",
        "goldbach_normalized_factor_prime_product",
        "goldbach_large_prime_coprime_twentyfour",
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        "blueprint/Checkpoint_1220_GoldbachSieveConductor.md",
        "blueprint/Checkpoint_1221_GoldbachSieveDensityFactorization.md"
      ],
      "scope": " At1217: Proved the exact upper sieve sequence/progression count, including all prime
exclusions, source multiplier arithmetic and residue independence from n. Local-density factorization
and the later analytic sieve estimates remain open. At1219: Actual unit-square root counts, both
candidate prime-density bounds including p=5, and the explicit multiplicative sieve function are proved
without deep inputs. Full local-density normalization is the next internal obligation. At1220: Proved
the conductor-sensitive coefficient identity, all actual prime-factor cases, the exact factors at2 and3,
 residue invariance, and normalized prime adjunction without division by the global base factor. Full
local-density source closure is reserved for the final product and CRT assembly. At1221: complete actual
 normalized Euler-product factorization, exact source CRT residues, inclusive prime cutoff,
multiplicativity and all density bounds are proved without external inputs. The upper base factor may
vanish; no division by it is used. At1221: Both full source local-density lemmas are proved without new
external inputs: exact original Euler normalization, inclusive cutoff, lower quotient identification,
source CRT specializations, upper multiplicativity and p=5, with common zero upper base factors
retained. Presieving and the later sieve estimates remain open.",
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        "goldbach_div_indicator_product",
        "goldbach_sift_indicator_union",
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        "goldbach_presieve_pair_sum",
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        "goldbach_presieve_sequence_lower",
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        "goldbach_presieve_disjoint_support_coprime",
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        "goldbach_presieve_preliminary_error_bound",
        "goldbach_presieve_preliminary_s_tendsto",
        "goldbach_presieve_preliminary_level_eventually",
        "goldbach_presieve_preliminary_absorption",
        "goldbach_presieve_outer_level_tendsto",
        "goldbach_presieve_outer_error_tendsto",
        "goldbach_presieve_total_error_tendsto",
        "goldbach_presieve_uniform_error_tendsto",
        "goldbach_presieve_density_prime_product",
        "goldbach_presieve_outer_mass",
        "goldbach_presieve_outer_product",
        "goldbach_presieve_outer_mass_ratio",
        "goldbach_presieve_inverse_polylog",
        "goldbach_corrected_presieving",
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        "goldbach_integer_sequence_sifted_count",
        "goldbach_section11_complete",
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        "goldbach_current_presieve_preliminary_filter",
        "goldbach_current_presieve_outer_filter"
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        "blueprint/Checkpoint_1222_GoldbachPresieveCoefficients.md",
        "blueprint/Checkpoint_1223_GoldbachPresieveDecay.md",
        "blueprint/Checkpoint_1224_GoldbachPresieveUniformAssembly.md",
        "blueprint/Checkpoint_1258_GoldbachSection11Complete.md",
        "docs/MANUSCRIPT_REVIEW_2026_10_28.md"
      ],
      "scope": "The corrected two-range presieving statement is a complete implication from the explicit
 independent general one-stage sieve inputs. The exact source matched is the separately supplied
Section11_Reconciled.tex: X>=0, g(1)=1, fixed R=(log N)^C, R+1<=z<=D1<=N^A, refined1+K/logw dimension
bounds, and the preliminary endpointR+1. Coefficients precede the density/sequence and n, have absolute
value<=1 and exact level support; the common relative error tends to zero. The literal original weaker
manuscript statement is not claimed proved; replacement is required in the user manuscript. No original
source file was changed. At1258: All Section11 numbered claims are complete in the agreed conditional
scope for the explicitly supplied corrected presieving statement. The original weaker statement is not
proved and must be replaced in the user manuscript. The complete joint lower/upper estimates use one
cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary combined logarithmic
mean-square saving. The final Goldbach theorem remains the next deduction. At1265 the uploaded
human-readable statement is reviewed directly: D0<D follows from its level chain, and the inline
interval prime conditions are proved equivalent in Lean. The corrected mathematical hypotheses are
verified as present in the actual upload; no presieving replacement remains outstanding.",
      "remaining": "",
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        "external_inputs": [
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        "label": "l:presieving",
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        "applied_to_user_manuscript": false,
        "original_weaker_statement_claimed": false
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        "source_file": "sources/updates/PrimeSquares_User_2026_10_28.tex",
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        "equivalent_to_previous_corrected_target": true,
        "equivalence_evidence": "PrimeSquares/GoldbachPresieveSourceEquivalence.lean",
        "no_remaining_mathematical_replacement": true
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      "work_group": "Goldbach and sieve",
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    "goldbach_weighted_lower_small_product",
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    "goldbach_weighted_lower_normalization",
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    "goldbach_weighted_double_integral_le_W",
    "goldbach_weighted_log_harmonic_integral_le",
    "goldbach_weighted_large_rectangle",
    "goldbach_weighted_kernel_small",
    "goldbach_weighted_kernel_large",
    "goldbach_weighted_finite_kernel",
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    "goldbach_weighted_large_scalar",
    "goldbach_weighted_cutoffs_eventually",
    "goldbach_weighted_harmonic_cutoff",
    "goldbach_weighted_kernel_effective",
    "goldbach_weighted_kernel_coefficient_limit",
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    "goldbach_weighted_thin_error_relative",
    "goldbach_weighted_smooth_error_relative",
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    "blueprint/Checkpoint_1234_GoldbachWeightedNormalization.md",
    "blueprint/Checkpoint_1235_GoldbachWeightedHarmonic.md",
    "blueprint/Checkpoint_1236_GoldbachWeightedAbel.md",
    "blueprint/Checkpoint_1237_GoldbachWeightedKernels.md",
    "blueprint/Checkpoint_1238_GoldbachWeightedPrimeBalance.md",
    "blueprint/Checkpoint_1239_GoldbachWeightedFiniteIntegration.md",
    "blueprint/Checkpoint_1240_GoldbachWeightedKernelIntegration.md",
    "blueprint/Checkpoint_1241_GoldbachWeightedMeanCutoffs.md",
    "blueprint/Checkpoint_1242_GoldbachWeightedSmoothing.md",
    "blueprint/Checkpoint_1243_GoldbachWeightedPrimeEndpoints.md",
    "blueprint/Checkpoint_1244_GoldbachWeightedDoubleGeometry.md",
    "blueprint/Checkpoint_1245_GoldbachWeightedLargeRectangle.md",
    "blueprint/Checkpoint_1246_GoldbachWeightedFiniteKernel.md",
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        "blueprint/Checkpoint_1247_GoldbachWeightedScalarBounds.md",
        "blueprint/Checkpoint_1248_GoldbachWeightedKernelCoefficient.md",
        "blueprint/Checkpoint_1249_GoldbachWeightedKernelUniform.md",
        "blueprint/Checkpoint_1250_GoldbachWeightedMass.md",
        "blueprint/Checkpoint_1251_GoldbachWeightedErrorBounds.md",
        "blueprint/Weighted_Upper_Main_Term_Complete.md",
        "blueprint/Checkpoint_1252_GoldbachWeightedSourceBridge.md"
      ],
      "scope": "Complete source weighted upper main-term comparison, in uniform epsilon form, with the
actual multiplier set, local factors, finite Euler products and singular integrals. For every A>0, every
 fixed C>=A+24 and eta>0, one threshold in N works for all n=12 mod24 with N^2(log N)^(-A)<=n<N^2. The
coefficient is exactly c_(9/8) W(9/8)+eta. Strong multiplicativity, the uniform mean instantiation, both
 finite Abel arguments, W identity, all small-denominator and endpoint terms, moving logarithms,
integration, smoothing and relative errors are proved. Only the independent generic effective mean-value
 corollary and classical fixed-progression Mertens bounds remain explicit unproved hypotheses. The
source lemma statement is unchanged; two proof replacements are supplied but not applied. At1252: The
full source weighted upper main-term comparison l:wumatomakisum is complete as a checked implication
from independent generic effective mean-value and classical Mertens inputs, with the exact printed
constant, actual source definitions, uniform retained-range quantifiers, and every summation,
integration and relative-error obligation discharged.",
      "remaining": "",
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        "EFFECTIVE_HALF_DIMENSIONAL_MEAN",
        "CLASSICAL_PRIME_LOG_MERTENS",
        "CLASSICAL_PRIME_MERTENS_RATIOS",
        "CLASSICAL_INTERVAL_MERTENS"
      ]
    }
  },
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      "id": "PNT_LIMITS",
      "status": "consumed_explicit_hypothesis",
      "scope": "PrimeAsymptoticInputs: theta(y)/y tends to 1 and the weighted reciprocal-square-root
prime sum has leading constant 2. The external PNT bridge is not integrated.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "PROGRESSION_ESTIMATE",
      "status": "consumed_explicit_hypothesis",
      "scope": "The older PrimeSquareDistributionInputs supplies raw compatible progression/Ren/Ghosh
data for Theorem1.1. The refined PagePrimeSquareDistributionInputs c additionally returns genuine
optional primitive-character Page-zero data in the source region, and is proved to supply the older
interface. One selected family is integrated into all clauses of Proposition4.19 and the Proposition6.21
 high maximum. Existence of the refined raw data remains an independent analytic hypothesis; it is not
derived from the older interface alone. No own-paper operator conclusion is assumed.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "REN_GHOSH",
      "status": "consumed_explicit_hypotheses",
      "scope": "Published raw quadratic exponential-sum bounds are explicit inputs. An own-paper final
minor-arc estimate is not assumed.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "SMOOTH_MULTIFREQUENCY",
      "status": "consumed_explicit_compound_hypothesis",
      "scope": "SmoothCircleBourgainInput C: a general discrete smooth multifrequency maximum. Existence
 is unproved. The boundary includes smooth real theory, sampling, normalization and the fixed bump; it
is not a theorem imported from Mathlib. Used in Proposition 6.21, whose actual annular,
all-starting-scale and denominator-truncation deductions are now proved. The same independent input is
consumed by the full Proposition4.19 assembly. This hypothesis is not used by Theorem 1.1.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "PAGE_CLASSES",
      "status": "consumed_explicit_general_hypothesis_source_consistency_proved",
      "scope": "RawPageUniquenessInput c: standard Landau-Page uniqueness for primitive nonprincipal
real Dirichlet characters and their real zeros, expressed using ordinary partial sums below 1; its
analytic-continuation bridge remains external. The source-region to conductor-region deduction,
uniqueness in each dyadic conductor class, actual-data finite-class injection and the changing-character
 maximum are proved. Source-compatible PageZeroDatum existence remains the refined raw distribution
hypothesis. Its scale-by-scale selection and integration into the common Proposition4.19 operator family
 are proved; existence is not proved from the older fixed-scale interface. No operator conclusion is
assumed and no Lean axiom is declared. See docs/PROPOSITION_6_21_COMPLETION.md.",
      "declared_as_Lean_axiom": false
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      "status": "consumed_explicit_general_construction_multiplier_and_sampling_hypotheses",
      "scope": "RawTaoArcConstructionInput, RawIWMultiplierInput, RawMajorArcSamplingInput and
RawNarrowShannonInput are precise independent Tao Lemma5.1, Theorems1.7/1.6 and Proposition1.5
specializations. Their existence is unproved; they are explicit parameters, not axioms. Actual
finite-input Fourier lifts, normalization, diagonal, smooth applicability and the full finite-input
dyadic low bound are proved. The analytic-top typo was corrected with an exact hash-declared rebuild
at897. The prime high theorem does not use the multiplier predicate. See docs/SECTION_7_IW_BOUNDARY.md
and docs/SECTION_7_SAMPLING_BOUNDARY.md.",
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      "id": "GENERAL_VARIATION_SAMPLING",
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RawRieszThorinMultiplierInput are precise independent general standard theorem hypotheses for arbitrary
kernels, functions, multipliers or scalar sequences. All annular applicability, actual continuous and
narrow cyclic maxima, exact Fourier identifications, countable remainder and finite-input full IW low
assembly are proved. No own-paper low or restricted bound is a field. See the continuous, signed-maximal
 and sampling boundary documents. The finite signed maximal interface is proved internally at constant6
by rawFiniteSignedMaximalInput_proved (checkpoint1263), and is no longer an independent assumption.",
      "declared_as_Lean_axiom": false
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      "status": "precise_unproved_independent_interfaces_implemented",
      "scope": "RawCalderonStrongInput and RawCalderonModularInput: general transference for arbitrary
positive finite convolution families, exponents/norm constants and continuous nonnegative gauges. Their
independent truth and exact published-theorem/action/modular bridge remain unproved. No own-paper
endpoint is an input. See docs/ERGODIC_INDEPENDENT_BOUNDARY.md.",
      "declared_as_Lean_axiom": false
    },
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      "id": "LAVICTOIRE",
      "status": "precise_unproved_general_input_consumed",
      "scope": "RawOriginalLaVictoireInput: original Theorem3.1 for arbitrary increasing sequences with
literal cyclic arithmetic hypotheses, concluding failure of any universal weak L1 maximal constant along
 every infinite cutoff set. Its existence and exact zero-based/probability/weak-norm formulation bridge
remain unproved. The adapted own-paper persistent criterion is proved using separate Sawyer and Conze
inputs.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "GRANVILLE_KURLBERG",
      "status": "precise_unproved_quoted_general_interface_consumed",
      "scope": "RawGranvilleKurlbergInput: general CRT-family local-correlation/subpower hypotheses
imply full joint cyclic Poisson gap limits. Published-to-Lean cyclic joint formulation bridge and
theorem proof remain unformalized. Actual unit-square hypotheses/application are proved. This optional
input is not used by Theorem1.5(i).",
      "declared_as_Lean_axiom": false
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      "id": "WEIL",
      "status": "consumed_explicit_general_hypothesis",
      "scope": "RawSplitQuadraticWeilInput is the precise weakened classical quadratic character-sum
bound for arbitrary nonempty sets of distinct linear factors over every odd prime field. Its existence
remains unproved. It supplies no own quadratic-residue correlation, spacing, moment or badness
conclusion; the source-specific applications remain deductions.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "BOMBIERI_VINOGRADOV",
      "status": "consumed_explicit_hypothesis",
      "scope": "RawPrimeBombieriVinogradovInput: classical general ordinary prime-count distribution
with endpoint maximum, unweighted modulus sum, square-root level and logarithmic saving. The fixed-q lcm
 fibre theorem reduces the paper reindexing to this ordinary input; the own-paper circle-method
conclusion remains internal. The actual linear major-error row sum is deduced at1150, with all
exceptional terms explicit. The complete full source progression major estimate is deduced at1154 and
the source major-arc mean-square lemma at1161.",
      "declared_as_Lean_axiom": false
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      "status": "independent_single_stage_interfaces_implemented",
      "scope": "See ROSSER_IWANIEC_SINGLE_STAGE for the precise independent general inputs. The signed
two-stage presieve, full lower and upper applications, numerical comparison, common-cutoff assembly and
final Goldbach theorem are complete checked conditional implications.",
      "declared_as_Lean_axiom": false
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    {
      "id": "WEIGHTED_PRIME_COUNTS",
      "status": "retired_unconsumed_planning_placeholder",
      "scope": "No weighted-prime or paper main-term conclusion is an independent input. The source
l:wumatomakisum is proved from the independently stated general effective mean and classical Mertens
interfaces. This earlier planning placeholder is not consumed.",
      "declared_as_Lean_axiom": false
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      "status": "consumed_explicit_compound_hypothesis",
      "scope": "FullSquareDecompositionInput C: the independent ordinary-square high/low estimate in
global operator norm form. The analytic theorem and its globalization remain external. The plus-sign
PositiveFullSquareDecompositionInput is proved equivalent with exactly the same C by
fullSquareDecompositionInput_sign_iff at checkpoint783. The literal weighted formulas, general
convolution/norm/maximal transfer and reflected support are proved, so reflection is no longer
unformalized. The full-to-annular normalization is also proved. No prime-square conclusion is assumed.
See docs/THEOREM_1_1_EXTERNAL_INPUTS.md and docs/SQUARE_REFLECTION_COMPLETION.md.",
      "declared_as_Lean_axiom": false
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      "status": "consumed_explicit_compound_hypothesis",
      "scope": "OrdinarySquareStoppingInput CS: independent local ordinary-square low/high stopping
```

```
estimates. Its existence and exact support, normalization, admissibility and published-convention
compatibility remain unproved. It is stronger than the earlier fixed-scale FullSquareDecompositionInput.
 See docs/INDEPENDENT_STOPPING_INPUTS.md. No prime local endpoint or sparse conclusion is an input
field.",
      "declared_as_Lean_axiom": false
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    {
      "id": "PRIME_MAXIMAL_LP",
      "status": "consumed_explicit_compound_hypothesis",
      "scope": "IndependentPrimeMaximalInput: complementary-exponent restricted consequence of the
independent prime polynomial maximal Lp theorem, in the actual von Mangoldt annular normalization.
Published prime normalization and prime-power extension remain within its unproved scope. Used only for
the open triangle, not for the endpoint. No sparse conclusion is a field.",
      "declared_as_Lean_axiom": false
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      "id": "QUADRATIC_WEYL_FJK",
      "status": "precise_unproved_independent_scalar_interface_implemented",
      "scope": "RawQuadraticWeylApproximationInput E records the general initial quadratic Weyl
approximation in rational-center form, a consequence of Fiedler-Jurkat-Korner as restated in
Han-Lacey-Yang arXiv1907.05734v2 Theorem3.1. Its Dirichlet selection/reduction/phase bridge and
existence remain unproved parts of this explicit independent input. It has no own circle-model, IW or
maximal conclusion. Complete Gauss arithmetic and full quadratic oscillatory integral bounds are proved
separately. See docs/SECTION_7_QUADRATIC_WEYL_BOUNDARY.md.",
      "declared_as_Lean_axiom": false
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    {
      "id": "BOUNDED_SQUARE_POINTWISE",
      "status": "precise_unproved_independent_bounded_class_interface_consumed",
      "scope": "RawBoundedSquarePointwiseInput: independent bounded-class square/prime-square pointwise
convergence, including the still-unformalized Mehlhop-Slomian one-dimensional/cutoff and
PNT/Abel/prime-power normalization bridge. It supplies no unbounded Orlicz convergence, strong norm,
modular inequality or density-extension conclusion. See docs/ERGODIC_INDEPENDENT_BOUNDARY.md.",
      "declared_as_Lean_axiom": false
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      "id": "SAWYER",
      "status": "precise_unproved_general_input_consumed",
      "scope": "RawSawyerMaximalInput: for arbitrary sequence averages on a nonatomic ergodic
probability system, almost-everywhere finite proper maxima for all measurable integrable functions imply
 some positive weak L1 maximal constant. The general theorem and its exact complex/probability/cutoff
convention bridge remain unproved.",
      "declared_as_Lean_axiom": false
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      "scope": "RawConzeMaximalInput: for arbitrary sequence averages, transfer the same weak maximal
constant from a nonatomic ergodic probability system to every probability system. General theorem and
exact convention bridge remain unproved.",
      "declared_as_Lean_axiom": false
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      "id": "FIXED_PROGRESSION_PNT",
      "status": "precise_unproved_general_input_consumed",
      "scope": "RawFixedPrimeProgressionFrequencyInput: enumerated primes have limiting frequency
1/phi(q) in each unit residue class of every fixed positive modulus. This standard
PNT-in-fixed-progressions input remains unproved. Exact prime-square fiber partition and pushforward to
1/|Lambda_q| are proved.",
      "declared_as_Lean_axiom": false
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      "id": "FULL_DOMAIN_CONTINUOUS_STANDARDS",
      "status": "precise_unproved_general_full_lp_inputs_consumed",
      "scope": "RawFullOneSidedHLInput and RawFullScalarHormanderInput: general one-sided interval
maximal and scalar Hormander convolution theorems on arbitrary Lp inputs for1<p<=2. These broaden the
earlier L2-intersection formulations; their proofs and exact publication/convention bridges remain
unformalized. The full-HL restriction to the earlier interface is proved. The scalar full-domain version
 is not claimed to follow from the earlier interface. All own annular kernels, Fourier formulas, uniform
 L2/Hormander conditions, real-scale continuity, measurable proper supremum and source estimates are
deductions.",
      "declared_as_Lean_axiom": false
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      "status": "consumed_explicit_compound_hypothesis",
      "scope": "RationalSmoothCircleInput C: the independent rational smooth multifrequency inequality
with a common denominator P, its rational logarithmic cost, fixed smooth bump normalization and circle
transference. The interface is quantified over arbitrary rational center sets, periods, separated
widths, finite scale sets and all integer L2 inputs. It contains no annular kernel, beta, arithmetic
coefficients, denominator truncation or current-paper endpoint. The independent rational-frequency
theorem and transference remain unproved here; existence is not asserted. It is the rational refinement
associated with Lacey1997 Theorem1.2 and its sampling specialization, rather than the general
separated-frequency squared-log bound.",
      "declared_as_Lean_axiom": false
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      "id": "PRIMITIVE_TWO_SQUARE_CRITERION",
      "status": "proved_in_project_not_an_external_assumption",
      "scope": "Complete primitive two-square criterion for n=2 mod4, proved from pinned Mathlib
Fermat/negative-one obstruction and checked primitive multiplication/parity algebra. No additional
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mathematical hypothesis.",
      "declared_as_Lean_axiom": false,
      "proof": "PrimeSquares.goldbach_primitive_two_squares_criterion",
      "proof_file": "PrimeSquares/GoldbachPrimitiveCriterion.lean",
      "proof_checkpoint": 1261,
      "classification": "proved_arithmetic_lemma",
      "consumed_by_completed_exports": true,
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      "source": "blueprint/Section9_Primitive_Two_Squares.md"
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      "id": "UNIFORM_PAGE_PREFIX",
      "status": "consumed_explicit_general_hypothesis",
      "scope": "UniformPageProgressionInputs: a general prime progression prefix estimate through
exp(eta sqrt(log N)), uniform over every positive modulus, unit residue and real t in [N/2,N], with one
genuine optional Page datum per N selected before Q. The decay, modulus and prefix constants are
independent of N. Its existence and fixed-level analytic bridge remain unproved; no quadratic phase,
major-arc estimate or own-paper endpoint is assumed.",
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      "status": "consumed_explicit_compound_general_hypothesis",
      "scope": "RawFullAdelicIWInput C: independent full-domain Tao adelic IW theorem for arbitrary good
 coprime data and arbitrary finite-arithmetic spatial operator kernels and rough compact real symbols.
Its rough limiting and published-to-actual-kernel/quotient realization bridges remain unproved parts of
this explicit hypothesis. No Q, selected construction, probability measure, annular, low/high or own
endpoint is a field. The full real Lp product bound and source coefficient/geometry/norm application are
 proved.",
      "declared_as_Lean_axiom": false
    },
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      "id": "FULL_DOMAIN_TAO_SAMPLING",
      "status": "consumed_explicit_compound_general_hypotheses",
      "scope": "RawFullMajorArcSamplingInput C and RawFullShannonInput S: general Tao Theorem1.6 on
arbitrary good coprime data, with full distributional lp Fourier support, both norm directions and
unique interpolation; and Proposition1.5 at fixed c=1/100 for all finite-dimensional normed complex
targets and every 1<=p<=infinity. Their truth and published/full-class/finite-quotient realization
bridges remain unproved. No source Q, B, annular operator, maximal estimate or own endpoint is assumed.
The actual real-Q applicability, full inverse, two-sided source constants and narrower-band
finite-target application are proved.",
      "declared_as_Lean_axiom": false
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      "id": "SIEGEL_WALFISZ_PRIME_COUNTS",
      "status": "consumed_explicit_hypothesis",
      "scope": "RawPrimeSiegelWalfiszInput: general ordinary pi/Li prime counts, every endpoint y\u2264N
 and fixed logarithmic modulus range. No oscillatory weighted, coefficient-family or own-paper major-arc
 endpoint is assumed. The complete actual source square major approximation is deduced at1146. Together
with ordinary BV, it supplies the complete source major-arc mean-square replacement at1161.",
      "declared_as_Lean_axiom": false
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      "id": "ROSSER_IWANIEC_SINGLE_STAGE",
      "status": "consumed_explicit_general_hypotheses_for_corrected_variant",
      "interfaces": [
        "GoldbachClassicalFundamentalInput",
        "GoldbachClassicalCoefficientInput"
      ],
      "scope": "General one-stage Rosser\u2013Iwaniec weights chosen before all supported densities
satisfying the refined dimension hypothesis, with pointwise sieve bounds, level/square-free support,
coefficient bound1, exponential fundamental-lemma error for large s, and the standard outer
f_kappa/F_kappa coefficient bounds with a logarithmic error. The full signed two-stage deduction, exact
scale, common error limit and integer-sequence bridges are proved. No own-paper presieving,
weighted-prime, lower/upper sieve or Goldbach endpoint is an input. The lower application also consumes
continuity of the classical lower sieve function at9/8; the actual parameter and uniform-error limits
are proved.",
      "source": "Yuta Suzuki, The Rosser\u2013Iwaniec sieve, 2023, Definition8.2, Proposition11.8 and
Lemma14.7; exact independent interfaces and classical corollary scope in docs/EXTERNAL_SIEVE_INPUTS.md",
      "declared_as_Lean_axiom": false
    },
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      "id": "CLASSICAL_INTERVAL_MERTENS",
      "status": "consumed_explicit_independent_general_hypothesis",
      "scope": "GoldbachClassicalIntervalMertens K: uniform interval Euler-product bounds for the
universal factor(1-2/(p-3))^(-1) on primes>=7 with dimension2, and(1-1/(p-1))^(-1) on primes3 mod4 with
dimension1/2. Quantified over all finite prime subsets of intervals. Contains no sequence, n, Goldbach
estimate or own-paper endpoint. Both actual density applications are proved.",
      "declared_as_Lean_axiom": false
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      "id": "EFFECTIVE_HALF_DIMENSIONAL_MEAN",
      "status": "consumed_explicit_independent_general_hypothesis",
      "interfaces": [
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      "scope": "Uniform effective nonnegative strongly multiplicative upper mean theorem for arbitrary
functions with prime values at most6 and the explicit prime logarithmic balance condition. It is the
general rho=1/2 upper corollary of Tenenbaum Theorem1.2 (corrected author version arXiv:1709.01315v13),
including the documented general corollary reductions. Those external reductions and the source theorem
are not formalized. No paper weight, root count, multiplier, singular integral, Goldbach or own-paper
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endpoint is a field. All such applications and the complete weighted main-term comparison are
deductions.",
      "source": "docs/EXTERNAL_WEIGHTED_MEAN_INPUTS.md",
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      "scope": "Ordinary and1 mod4 sums of log(p)/p with fixed bounded errors, and the bounded
logarithmic higher-prime-power tail. Quantified over all real x>=2; no paper weight or endpoint. The3
mod4 partition and varying-weight prime balance are proved internally.",
      "source": "docs/EXTERNAL_WEIGHTED_MEAN_INPUTS.md",
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      "scope": "Ordinary all-prime and fixed1/3 mod4 Euler-product ratios with relative error K/log R
and explicit endpoint conventions. No varying weight or paper main-term conclusion. Exact source
local-factor and Euler-product normalizations, the effective mean application and the final weighted
comparison are proved.",
      "source": "docs/EXTERNAL_WEIGHTED_MEAN_INPUTS.md",
      "declared_as_Lean_axiom": false
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    {
      "id": "CLASSICAL_FULL_INTERVAL_MERTENS",
      "status": "consumed_explicit_independent_general_hypothesis",
      "interfaces": [
        "GoldbachClassicalFullIntervalMertens"
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      "scope": "Arbitrary finite prime-subset products for the p5/large-prime preliminary barrier and
outer1/(p-1) factors, with refined1+K/logw losses. Independent unproved general classical Mertens
corollary. No Goldbach data or endpoints.",
      "source": "docs/SECTION_11_UPPER_SIEVE_INPUTS.md",
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        "IntegerSparseFamily",
        "integerSparseFamily_iff_collection",
        "integerInterval_card_points"
      ],
      "scope": "General interval collections with actual disjoint finite major subsets; eta=1/4 gives
the preliminary definition, and eta=1/10 gives the introductory theorem convention. The finite-family
specialization is proved equivalent.",
      "remaining": ""
    },
    "d:admiss": {
      "title": "Admissible stopping maps",
      "status": "formalized",
      "lean_anchors": [
        "sourceLocalMass",
        "AdmissibleAnnularLocalBound",
        "primeRemainingScale_admissible",
        "SourceLinearAdmissible",
        "sourceLinearAdmissible_iff_display",
        "linearSourceBallMass_indicator",
        "linearSourceLocalMass_le_iff",
        "linearWindowCount_oneSided_compare",
        "linearSourceLocalMass_of_both_oneSided",
        "single_direction_does_not_control_symmetric",
        "linearOneSided_bounds_of_symmetric",
        "linearSymmetric_bound_of_both_oneSided",
        "linearSourceLocalMass_change_radius"
      ],
      "scope": "The literal standalone linear-scale admissible dyadic stopping-map predicate is exported
 on actual finite indicator inputs. Lean proves equality with the displayed all-real-radius finite-count
 inequality and all convergence/boundedness prerequisites. The earlier actual prime-square spatial
N\u00b2 admissibility is also proved. The accepted fixed-radius and two-direction comparison remarks are
 also proved, including the exact factor two. The current full upload explicitly separates the fixed
radius and the sufficiently large admissibility coefficient, matching the checked generic sparse
theorem.",
      "remaining": "",
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        "docs/SOURCE_UPDATE_1066.md"
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      "status": "formalized",
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        "upsilon"
      ]
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      "file": "PrimeSquares/FiniteSignedMaximalProof.lean",
      "proof_checkpoint": 1263,
      "public_export_integration_checkpoint": 1264,
      "range": "The proved binary bound holds for every p>=1; the former interface only
required1<=p<=2.",
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Archive record: Section11_Statement_Map.json

Section11_Statement_Map.json


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 S(\\mathcal A_n,\\mathbb P_{3,n},z)\n &\\geq S(\\mathcal A_n,\\mathbb P_3,z),\n
\\label{e:fixedlower10}\\\\\n \\sum_{\\ell\\in L_n}S(M_n(\\ell),W_{\\ell,n},z_\\ell)\n &\\leq\n
\\sum_{\\ell\\in L}S(M_n(\\ell),W_\\ell,z_\\ell)\n +O_\\varepsilon(N^\\varepsilon).\n
\\label{e:fixedupper10}\n\\end{align}",
      "status": "proved",
      "remaining": "None within the source fixed-sieve comparison lemma.",
      "lean_anchor": "goldbach_fixed_sieve_sets",
      "scope": "Complete lower and upper fixed-sieve comparison, with actual sieve prime sets and actual
 closed multiplier set. The lower inequality is proved by inclusion, including the prime3 boundary. The
upper comparison uses an exact finite exceptional-pair inclusion and the bound sum over p|n of
tau(n-p^2-1) <= K_epsilon N^epsilon. One positive constant is chosen before N,n,z. Holds for every
N>=1,0<n<=N^2 and every real z, hence the full retained source range. No deep input is used. At1215: The
 full source fixed-sieve comparison is proved without deep inputs. The local density, presieving and
final sieve estimates remain open.",
      "lean_anchors": [
        "goldbach_fixed_lower_count",
        "goldbach_fixed_upper_exception",
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        "goldbach_uniform_divisor_card",
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        "goldbach_fixed_sieve_sets"
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      "statement_latex": "\\label{l:lowerlocaldensity}\nFor \\(p>3\\),
\\(p\\equiv3\\pmod4\\),\n\\begin{align}\\label{e:lowerlocaldensity10}\n \\frac{\\omega_n(p)}p\n =\n
\\begin{cases}\n \\displaystyle\n \\frac{\\rho_p(n-1)}{p-1-\\rho_p(n)},&p\\leq R,\\\\[7pt]\n
\\displaystyle\\frac1{p-1},&p>R.\n \\end{cases}\n\\end{align}\nIn particular, for
\\(p\\geq7\\),\n\\(0\\leq\\omega_n(p)/p\\leq2/(p-3)<1\\).",
      "status": "proved",
      "remaining": "None within the source statement and retained admissible context.",
      "lean_anchor": "goldbach_lower_local_density_formula",
      "completed_checkpoint": 1221,
      "scope": " At1216: Proved the exact lower sieve sequence/progression count and its actual
circle-method remainder decomposition. The full fixed-sieve comparison remains proved; local densities
and later sieve conclusions remain open. At1219: Actual unit-square root counts, both candidate
prime-density bounds including p=5, and the explicit multiplicative sieve function are proved without
deep inputs. Full local-density normalization is the next internal obligation. At1220: Proved the
conductor-sensitive coefficient identity, all actual prime-factor cases, the exact factors at2 and3,
residue invariance, and normalized prime adjunction without division by the global base factor. Full
local-density source closure is reserved for the final product and CRT assembly. At1221: complete actual
 normalized Euler-product factorization, exact source CRT residues, inclusive prime cutoff,
multiplicativity and all density bounds are proved without external inputs. The upper base factor may
vanish; no division by it is used. At1221: Both full source local-density lemmas are proved without new
external inputs: exact original Euler normalization, inclusive cutoff, lower quotient identification,
source CRT specializations, upper multiplicativity and p=5, with common zero upper base factors
retained. Presieving and the later sieve estimates remain open.",
      "lean_anchors": [
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        "goldbach_cohen_rho",
        "goldbach_rho_legendre",
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        "goldbach_prime_restricted_count",
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        "goldbach_local_denominator_pos",
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        "goldbach_general_coefficient_conductor",
        "goldbach_prime_unrestricted_factor",
        "goldbach_prime_fixed_factor",
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        "goldbach_prime_unrestricted_count",
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        "goldbach_lower_local_density_formula",
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      "section": 11,
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      "active_statement_sha256": "6f29ddcaacdd521c4caa880908ea1298db4570e472a698f4245afaab8898693f",
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function\n\\(\\omega_n(\\ell,\\cdot)\\), defined on square-free integers supported on\n\\(W_\\ell\\),
such that\n\\begin{align}\\label{e:upperfactorization10}\n \\frac{G_R(\\ell,12d,n-1)}{\\phi(12d)}\n =\n
\\frac{G_R(\\ell,12,n-1)}{\\phi(12)}\n \\frac{\\omega_n(\\ell,d)}d .\n\\end{align}\nFor
\\(p\\nmid12\\ell\\), its  factors are\n\\begin{align}\\label{e:upperlocaldensity10}\n
\\frac{\\omega_n(\\ell,p)}p\n =\n \\begin{cases}\n \\displaystyle\n
\\frac{\\rho_p(n)}{p-1-\\rho_p(n-1)},&p\\leq R,\\\\[7pt]\n \\displaystyle\\frac1{p-1},&p>R.\n
\\end{cases}\n\\end{align}\nIn particular, \\(0\\leq\\omega_n(\\ell,p)/p<1\\).",
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exclusions, source multiplier arithmetic and residue independence from n. Local-density factorization
and the later analytic sieve estimates remain open. At1219: Actual unit-square root counts, both
candidate prime-density bounds including p=5, and the explicit multiplicative sieve function are proved
without deep inputs. Full local-density normalization is the next internal obligation. At1220: Proved
the conductor-sensitive coefficient identity, all actual prime-factor cases, the exact factors at2 and3,
 residue invariance, and normalized prime adjunction without division by the global base factor. Full
local-density source closure is reserved for the final product and CRT assembly. At1221: complete actual
 normalized Euler-product factorization, exact source CRT residues, inclusive prime cutoff,
multiplicativity and all density bounds are proved without external inputs. The upper base factor may
vanish; no division by it is used. At1221: Both full source local-density lemmas are proved without new
external inputs: exact original Euler normalization, inclusive cutoff, lower quotient identification,
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|\\mathcal F_d|=Xg(d)+r_d,\n\\end{align}\nwhere \\(X\\geq0\\), and \\(g\\) is multiplicative on the
square-free\nintegers supported on a fixed set of primes \\(\\mathfrak P\\),\nwith \\(g(1)=1\\) and
\\(0\\leq g(p)<1\\).\nFix \\(C,A>0\\), set \\(R=(\\log N)^C\\),\n\\(D_0=\\exp((\\log\\log N)^3)\\), and
\\(D_1=D/D_0\\), and assume\n\\(R+1\\leq z\\leq D_1\\leq N^A\\) for sufficiently large \\(N\\).\nSuppose
 that fixed constants \\(\\kappa_0,\\kappa>0\\) and\n\\(K_0\\geq2\\) satisfy, for every \\(2\\leq
w<y\\leq z\\),\n\\begin{align}\\label{e:tworangesievedimension}\n \\prod_{\\substack{w\\leq
p<y\\\\p\\in\\mathfrak P,\\ p\\leq R}}\n (1-g(p))^{-1}\n &\\leq\n \\left(1+\\frac{K_0}{\\log
w}\\right)\n \\left(\\frac{\\log y}{\\log w}\\right)^{\\kappa_0},\\notag\\\\n \\prod_{\\substack{w\\leq
 p<y\\\\p\\in\\mathfrak P,\\ p>R}}\n (1-g(p))^{-1}\n &\\leq\n \\left(1+\\frac{K_0}{\\log w}\\right)\n
\\left(\\frac{\\log y}{\\log w}\\right)^\\kappa.\n\\end{align}\nAll \\(o(1)\\) terms below are uniform
for fixed\n\\(C,A,\\kappa_0,\\kappa,K_0\\).\nThen\n\\begin{align}\\label{e:presievedbounds10}\n
S(\\mathcal F,\\mathfrak P,z)\n &\\geq XV(z)\\bigl(f_\\kappa(s)+o(1)\\bigr)\n
+\\sum_{m<D}c_m^-r_m,\\notag\\\\n S(\\mathcal F,\\mathfrak P,z)\n &\\leq
XV(z)\\bigl(F_\\kappa(s)+o(1)\\bigr)\n +\\sum_{m<D}c_m^+r_m,\n\\end{align}\nwhere\n\\[\n
V(z)=\\prod_{\\substack{p<z\\\\p\\in\\mathfrak P}}(1-g(p)),\n \\qquad\n s=\\frac{\\log D_1}{\\log z},\n
\\qquad\n |c_m^\\pm|\\leq1.\n\\]\nThe coefficients \\(c_m^\\pm\\) depend only on the fixed prime sets
and\nthe sieve levels, and not on the values of \\(g\\).  \nThe dimension\n\\(\\kappa_0\\) is used only
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bounds, and the preliminary endpointR+1. Coefficients precede the density/sequence and n, have absolute
value<=1 and exact level support; the common relative error tends to zero. The literal original weaker
manuscript statement is not claimed proved; replacement is required in the user manuscript. No original
source file was changed. At1258: All Section11 numbered claims are complete in the agreed conditional
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scope for the explicitly supplied corrected presieving statement. The original weaker statement is not
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\\(0\\leq g(p)<1\\).  Let\n\\(2\\leq R<z\\), and suppose that there are fixed
constants\n\\(\\kappa_0,\\kappa,K_0>0\\) such that\n\\begin{align}\\label{e:tworangesievedimension}\n
\\prod_{\\substack{w\\leq p<y\\\\p\\in\\mathfrak P}}\n \\left(1-g(p)\\right)^{-1}\n &\\leq\n
K_0\\left(\\frac{\\log y}{\\log w}\\right)^{\\kappa_0},\n &&2\\leq w<y\\leq R,\\notag\\\\n
\\prod_{\\substack{w\\leq p<y\\\\p\\in\\mathfrak P}}\n \\left(1-g(p)\\right)^{-1}\n &\\leq\n
K_0\\left(\\frac{\\log y}{\\log w}\\right)^\\kappa,\n &&R<w<y\\leq z.\n\\end{align}\nDefine \n\\[\n
D_0=\\exp((\\log\\log N)^3),\\qquad D_1=D/D_0, \\qquad D_0<D , \\qquad \\frac{\\log D_0}{\\log
R}\\longrightarrow\\infty.\n\\]\nThen\n\\begin{align}\\label{e:presievedbounds10}\n S(\\mathcal
F,\\mathfrak P,z)\n &\\geq XV(z)\\bigl(f_\\kappa(s)+o(1)\\bigr)\n +\\sum_{m<D}c_m^-r_m,\\notag\\\\n
S(\\mathcal F,\\mathfrak P,z)\n &\\leq XV(z)\\bigl(F_\\kappa(s)+o(1)\\bigr)\n
+\\sum_{m<D}c_m^+r_m,\n\\end{align}\nwhere\n\\[\n V(z)=\\prod_{\\substack{p<z\\\\p\\in\\mathfrak
P}}(1-g(p)),\n \\qquad\n s=\\frac{\\log D_1}{\\log z},\n \\qquad\n |c_m^\\pm|\\leq1.\n\\]\nThe
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coefficients \\(c_m^\\pm\\) depend only on the fixed prime sets and\nthe sieve levels, and not on the
values of \\(g\\).  \nThe dimension\n\\(\\kappa_0\\) is used only in the preliminary sieve over
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L}X_n(\\ell)\\Theta_{n,\\ell}(z_\\ell)\n \\leq\n \\bigl(c_\\alpha\\mathcal W(\\alpha)+o(1)\\bigr)\n
X_n\\Theta_n(z),\n \\qquad\n
c_\\alpha=\\frac{16e^{-\\gamma/2}}{\\sqrt{\\pi\\alpha}}.\n\\end{align}\nHere \\(\\mathcal W(\\alpha)\\)
is the integral defined in\n\\eqref{e:Wdef8}.",
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coefficient is exactly c_(9/8) W(9/8)+eta. Strong multiplicativity, the uniform mean instantiation, both
 finite Abel arguments, W identity, all small-denominator and endpoint terms, moving logarithms,
integration, smoothing and relative errors are proved. Only the independent generic effective mean-value
 corollary and classical fixed-progression Mertens bounds remain explicit unproved hypotheses. The
source lemma statement is unchanged; two proof replacements are supplied but not applied. At1252: The
full source weighted upper main-term comparison l:wumatomakisum is complete as a checked implication
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P_{3,n},z)\n \\geq X_n\\Theta_n(z)\\bigl(f_{1/2}(9/8)+o(1)\\bigr)\n +\\mathcal
E_1(n),\n\\end{align}\nwhere\n\\begin{align}\\label{e:E1section8}\n\\sum_{n<N^2}|\\mathcal
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(indeed all n=12 mod24). General classical sieve/Mertens/function-continuity and ordinary
SW/BV/Ren-Ghosh inputs remain explicit and unproved. At1229: Common cutoff compatibility for the
circle-method and full lower-sieve estimates is proved for every requested lower bound on C; main-term
size remains to be joined. At1230: The full source main-term size and the entire lower-bound side are
complete conditionally, including an exponent independent of C, uniform smoothing, one common large
cutoff and finite error-to-exception deductions. At1230 the source main-term size and the full lower
sieve are proved jointly for the same C>=A0+6; finite positive-count and exceptional-cardinality
deductions are checked. At1258: All Section11 numbered claims are complete in the agreed conditional
scope for the explicitly supplied corrected presieving statement. The original weaker statement is not
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L_n}S(M_n(\\ell),W_{\\ell,n},z_\\ell)\n &\\leq\n \\bigl(c_{9/8}F_1(2)\\mathcal W(9/8)+o(1)\\bigr)\n
X_n\\Theta_n(z)+\\mathcal E_2(n),\n\\end{align}\nwhere\n\\begin{align}\\label{e:calphadef8}\n c_\\alpha=
\\frac{16e^{-\\gamma/2}}{\\sqrt{\\pi\\alpha}}\n\\end{align}\nand\n\\begin{align}\\label{e:E2section8}\n\
\sum_{n<N^2}|\\mathcal E_2(n)|^2\n\\ll_{A_0,K} N^4(\\log N)^{-K}.\n\\end{align}",
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F_1(2) W(9/8) coefficient, original actual main term and closed multiplier set. All coefficients precede
 n; source divisor counts, refined preliminary/outer dimensions, uniform varying levels, the shifted
double remainder sum, its circle-method embedding and the fixed-sieve O(N^(1/4)) correction are proved.
The total error has arbitrarily strong logarithmic mean-square saving. Only independent general SW, BV,
Ren/Ghosh, classical one-stage sieve, effective multiplicative mean and Mertens inputs and general sieve
 normalizations remain unproved; no own-paper endpoint is assumed. At1257: Full upper-sieve proposition
p:sieveneedupper8 complete in uniform epsilon form from explicit independent classical inputs, with the
exact coefficient and arbitrary logarithmic mean-square saving. The lower analytic estimate and both
remainder transports are available at a shared prescribed cutoff. The final joint Section11 assembly and
 manuscript synchronization are next. At1258: All Section11 numbered claims are complete in the agreed
conditional scope for the explicitly supplied corrected presieving statement. The original weaker
statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper
estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary
combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction.",
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that, for every fixed\nsufficiently large \\(C\\), with \\(R=(\\log N)^C\\), one
has\n\\begin{align}\\label{e:maintermlower8}\n X_n\\Theta_n(z)\\gg_{A_0,C}\n N(\\log
N)^{-C_{A_0}}\n\\end{align}\nwhenever $n\\equiv12\\pmod{24}$ and $N^2(\\log N)^{-A_0}\\leq n<N^2$.",
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N)^(-(A0/2+6)); the exponent is independent of C. Actual archimedean, smoothing and local-product
arguments are proved; only the already recorded independent interval Mertens products remain external.
The same large C is joined to the complete lower sieve. At1258: All Section11 numbered claims are
complete in the agreed conditional scope for the explicitly supplied corrected presieving statement. The
 original weaker statement is not proved and must be replaced in the user manuscript. The complete joint
 lower/upper estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and
arbitrary combined logarithmic mean-square saving. The final Goldbach theorem remains the next
deduction.",
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\\cite[(1.8)--(1.10)]{I80},\n\\begin{align}\\label{e:Deltanumerical}\n \\Delta>0.\n\\end{align}",
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by positive interval integration and exact rpow primitives. The strict comparison is proved for the
classical normalized sieve values f_(1/2)(9/8)=sqrt(8 exp(gamma)/(9 pi)) log2 and F_1(2)=exp(gamma).
Identifying the generic independent sieve functions with those classical normalized values remains
explicit, within GENERAL_SIEVE; the paper comparison is not assumed. No floating point evaluation or
unchecked evaluator is used. At1253: Proved the exact actual W integral bound, the normalized
half/linear sieve comparison and strict positivity. Classical sieve-function normalizations remain
explicit general input identities. The upper-sieve application and final Goldbach theorem remain open;
no paper endpoint is assumed. At1258: All Section11 numbered claims are complete in the agreed
conditional scope for the explicitly supplied corrected presieving statement. The original weaker
statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper
estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary
combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction.",
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Archive record: Section9_Statement_Map.json

Section9_Statement_Map.json


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P^2\\}|\n \\geq S(\\mathcal A_n,\\mathbb P_{3,n},N^2)-O_\\varepsilon(N^\\varepsilon).\n\\end{align}",
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primitive sum-of-two-squares criterion is now a project theorem, not an additional input. Actual finite
sifted and good representation sets and the uniform O(N^epsilon) exception are retained. At1262: Every
Section9 numbered source claim and its final quantitative theorem is complete in the agreed scope. All
elementary arithmetic reductions are now proved without an extra primitive two-square assumption.
W(alpha) convergence is proved for every alpha>1 and the smooth seed is constructed. Only the explicit
independent analytic inputs remain for the circle-method/sieve/headline estimates. The user is applying
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|\\{r\\in\\mathcal P:n-r\\in\\mathbb P^2\\}|\n =\n
|T_n|+O_\\varepsilon(N^\\varepsilon),\n\\end{align}\nwhere\n\\begin{align}\\label{e:Tfactorization}\n
    T_n=\n \\left\\{\n r-1=2up_1p_2:\n \\begin{array}{l}\n r\\in\\mathbb P,\\quad n-r\\in\\mathbb
P^2,\\\\\n z\\leq p_2\\leq p_1,\\quad p_i\\in\\mathbb P_{3,n},\\\\\n p\\mid u\\Longrightarrow
p\\equiv1\\pmod4\n \\end{array}\n \\right\\}.\n\\end{align}",
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 by O(N^epsilon), including repeated3 mod4 prime factors and the actual source factorization set. The
primitive criterion is internally proved; only the normal source range hypotheses remain. At1262: Every
Section9 numbered source claim and its final quantitative theorem is complete in the agreed scope. All
elementary arithmetic reductions are now proved without an extra primitive two-square assumption.
W(alpha) convergence is proved for every alpha>1 and the smooth seed is constructed. Only the explicit
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S(M_n(\\ell),W_{\\ell,n},z_\\ell)+O_\\varepsilon(N^\\varepsilon).\n\\end{align}",
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P^2\\}|\n &\\geq S(\\mathcal A_n,\\mathbb P_{3,n},z)-\\sum_{\\ell\\in L_n}\n
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 adopted closed multiplier cutoff. Its exact finite error is at most2 omega(n-1), and its uniform
O(N^epsilon) consequence is exported. No primitive-criterion or analytic assumption is needed for this
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arithmetic reduction. At1262: Every Section9 numbered source claim and its final quantitative theorem is
 complete in the agreed scope. All elementary arithmetic reductions are now proved without an extra
primitive two-square assumption. W(alpha) convergence is proved for every alpha>1 and the smooth seed is
 constructed. Only the explicit independent analytic inputs remain for the circle-method/sieve/headline
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\\(A,\\delta>0\\) and \\(L_0<N^{2-\\delta}\\).  There are sufficiently\nlarge constants
\\(B=B(A,\\delta)\\) and \\(C=C(A,\\delta)\\) such that the\nfollowing holds.  Put\n\\[\n R=(\\log
N)^C,\\qquad\n \\eta_N(\\theta)=\\eta\\left(\\frac{N^2\\theta}{R}\\right),\n\\]\nand suppose that
$V_\\ell\\leq (N^2/\\ell)^{1/2}(\\log N)^{-B}$ for $\\ell<L_0.$\nLet \\(|d_{k,\\ell}|\\leq1\\)
and\n\\((a_{k,\\ell},k)=1\\).\nDefine \n\\begin{align}\\quad \\quad\\label{e:defGR9}\n G_R(\\ell,k,n)\n
&:=\n \\mathcal T_2(\\ell,k,n;a_{k,\\ell})\n \\prod_{2<p\\leq R}\n \\frac{p\\phi((p,k))}{(p-1)^2}\n
\\left|\\left\\{(u,v)\\in\\mathbb A_p^2:\n \\begin{array}{l}\n  u^2+\\ell v\\equiv n\\pmod p,\\\\\n
v\\equiv a_{k,\\ell}\\pmod{(p,k)}\n \\end{array}\\right\\}\\right|,\n\\end{align}\nwhere $ \\mathcal
T_2(\\ell,k,n;a_{k,\\ell}) $ is defined in \\eqref{e:generalT2} and the second congruence is understood
to be trivial when \\(p\\nmid k\\).\nAlso define\n\\begin{align}\\label{e:defHR9}\n H_{\\ell,R}(n)\n
:=\n \\ell\\int_{\\mathbb R}\\eta_N(\\theta)^2\n \\left(\\int_2^N\\frac{e(\\theta u^2)}{\\log
u}\\,du\\right)\n \\left(\\int_{\\substack{2\\leq v\\leq N^2\\\\\\ell v<N^2}}\n \\frac{e(\\ell\\theta
v)}{\\log v}\\,dv\\right)\n
e(-n\\theta)\\,d\\theta.\n\\end{align}\nThen\n\\begin{align}\\label{e:circmethod}\n
\\sum_{n<N^2}\\left|\n \\sum_{\\ell<L_0}\\sum_{k<V_\\ell}\n d_{k,\\ell}r_{k,\\ell}(n)\\right|^2\n
\\ll_{A,\\delta}N^4(\\log N)^{-A},\n\\end{align}\nwhere\n\\begin{align}\\label{e:unweightedremainder}\n
r_{k,\\ell}(n)\n &=\n \\sum_{\\substack{m^2+\\ell t=n\\\\\n t\\equiv a_{k,\\ell}\\pmod k}}\n
\\one_{\\mathbb P}(m)\\one_{\\mathbb P}(t)\n -\n
\\frac{G_R(\\ell,k,n)}{\\ell\\phi(k)}H_{\\ell,R}(n).\n\\end{align}",
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B0,C,K,N0 chosen before N and every arithmetic family. Covers all B>=B0, exact strict real
L<N^(2-delta), positive ell,k with k<V(ell), every coprime residue class, arbitrary complex coefficients
 of norm<=1, and all n<N^2. The stronger uniform cutoff statement holds for every 2<=U<=N,W>=2. The
final source theorem removes cutoffs using the proved equation-imposed bounds, and uses the actual
finite prime-pair count, actual local Euler factor G_R and actual autocorrelation singular integral H_R.
 Fourier inversion, all three error identifications, shared constants, minor estimate, major
replacement, arithmetic completion and smoothing are consumed internally. Its only unformalized
hypotheses are independent ordinary SW prime counts, ordinary unweighted BV prime-count distribution,
and uniform raw Ren/Ghosh quadratic estimates. No own-paper conclusion is assumed. At1212: Completed all
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P_{3,n},z)\n \\geq X_n\\Theta_n(z)\\bigl(f_{1/2}(9/8)+o(1)\\bigr)\n +\\mathcal
E_1(n),\n\\end{align}\nwhere\n\\begin{align}\\label{e:E1section8}\n\\sum_{n<N^2}|\\mathcal
E_1(n)|^2\n\\ll_{A_0,K} N^4(\\log N)^{-K}.\n\\end{align}",
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original X_n and Euler product, coefficients before n, a common error tending to zero, and the N^4(log
N)^(-A) mean-square remainder bound. The pointwise theorem holds throughout the source retained range
(indeed all n=12 mod24). General classical sieve/Mertens/function-continuity and ordinary
SW/BV/Ren-Ghosh inputs remain explicit and unproved. At1229: Common cutoff compatibility for the
circle-method and full lower-sieve estimates is proved for every requested lower bound on C; main-term
size remains to be joined. At1230: The full source main-term size and the entire lower-bound side are
complete conditionally, including an exponent independent of C, uniform smoothing, one common large
cutoff and finite error-to-exception deductions. At1230 the source main-term size and the full lower
sieve are proved jointly for the same C>=A0+6; finite positive-count and exceptional-cardinality
deductions are checked. At1258: All Section11 numbered claims are complete in the agreed conditional
scope for the explicitly supplied corrected presieving statement. The original weaker statement is not
proved and must be replaced in the user manuscript. The complete joint lower/upper estimates use one
cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary combined logarithmic
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 $n\\equiv12\\pmod{24}$,\n\\begin{align}\\label{e:sieveneedupper8}\n \\sum_{\\ell\\in
L_n}S(M_n(\\ell),W_{\\ell,n},z_\\ell)\n &\\leq\n \\bigl(c_{9/8}F_1(2)\\mathcal W(9/8)+o(1)\\bigr)\n
X_n\\Theta_n(z)+\\mathcal E_2(n),\n\\end{align}\nwhere\n\\begin{align}\\label{e:calphadef8}\n c_\\alpha=
\\frac{16e^{-\\gamma/2}}{\\sqrt{\\pi\\alpha}}\n\\end{align}\nand\n\\begin{align}\\label{e:E2section8}\n\
\sum_{n<N^2}|\\mathcal E_2(n)|^2\n\\ll_{A_0,K} N^4(\\log N)^{-K}.\n\\end{align}",
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F_1(2) W(9/8) coefficient, original actual main term and closed multiplier set. All coefficients precede
 n; source divisor counts, refined preliminary/outer dimensions, uniform varying levels, the shifted
double remainder sum, its circle-method embedding and the fixed-sieve O(N^(1/4)) correction are proved.
The total error has arbitrarily strong logarithmic mean-square saving. Only independent general SW, BV,
Ren/Ghosh, classical one-stage sieve, effective multiplicative mean and Mertens inputs and general sieve
 normalizations remain unproved; no own-paper endpoint is assumed. At1257: Full upper-sieve proposition
p:sieveneedupper8 complete in uniform epsilon form from explicit independent classical inputs, with the
exact coefficient and arbitrary logarithmic mean-square saving. The lower analytic estimate and both
remainder transports are available at a shared prescribed cutoff. The final joint Section11 assembly and
 manuscript synchronization are next. At1258: All Section11 numbered claims are complete in the agreed
conditional scope for the explicitly supplied corrected presieving statement. The original weaker
statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper
estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary
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that, for every fixed\nsufficiently large \\(C\\), with \\(R=(\\log N)^C\\), one
has\n\\begin{align}\\label{e:maintermlower8}\n X_n\\Theta_n(z)\\gg_{A_0,C}\n N(\\log
N)^{-C_{A_0}}\n\\end{align}\nwhenever $n\\equiv12\\pmod{24}$ and $N^2(\\log N)^{-A_0}\\leq n<N^2$.",
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exact retained n range. The actual X_n Theta_n(z) is at least a positive constant times N(log
N)^(-(A0/2+6)); the exponent is independent of C. Actual archimedean, smoothing and local-product
arguments are proved; only the already recorded independent interval Mertens products remain external.
The same large C is joined to the complete lower sieve. At1258: All Section11 numbered claims are
complete in the agreed conditional scope for the explicitly supplied corrected presieving statement. The
 original weaker statement is not proved and must be replaced in the user manuscript. The complete joint
 lower/upper estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and
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by positive interval integration and exact rpow primitives. The strict comparison is proved for the
classical normalized sieve values f_(1/2)(9/8)=sqrt(8 exp(gamma)/(9 pi)) log2 and F_1(2)=exp(gamma).
Identifying the generic independent sieve functions with those classical normalized values remains
explicit, within GENERAL_SIEVE; the paper comparison is not assumed. No floating point evaluation or
unchecked evaluator is used. At1253: Proved the exact actual W integral bound, the normalized
half/linear sieve comparison and strict positivity. Classical sieve-function normalizations remain
explicit general input identities. The upper-sieve application and final Goldbach theorem remain open;
no paper endpoint is assumed. At1258: All Section11 numbered claims are complete in the agreed
conditional scope for the explicitly supplied corrected presieving statement. The original weaker
statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper
estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary
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archive/Pre1265_Section11_Statement_Map.json


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 S(\\mathcal A_n,\\mathbb P_{3,n},z)\n &\\geq S(\\mathcal A_n,\\mathbb P_3,z),\n
\\label{e:fixedlower10}\\\\\n \\sum_{\\ell\\in L_n}S(M_n(\\ell),W_{\\ell,n},z_\\ell)\n &\\leq\n
\\sum_{\\ell\\in L}S(M_n(\\ell),W_\\ell,z_\\ell)\n +O_\\varepsilon(N^\\varepsilon).\n
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upper comparison uses an exact finite exceptional-pair inclusion and the bound sum over p|n of
tau(n-p^2-1) <= K_epsilon N^epsilon. One positive constant is chosen before N,n,z. Holds for every
N>=1,0<n<=N^2 and every real z, hence the full retained source range. No deep input is used. At1215: The
 full source fixed-sieve comparison is proved without deep inputs. The local density, presieving and
final sieve estimates remain open.",
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\\(p\\equiv3\\pmod4\\),\n\\begin{align}\\label{e:lowerlocaldensity10}\n \\frac{\\omega_n(p)}p\n =\n
\\begin{cases}\n \\displaystyle\n \\frac{\\rho_p(n-1)}{p-1-\\rho_p(n)},&p\\leq R,\\\\[7pt]\n
\\displaystyle\\frac1{p-1},&p>R.\n \\end{cases}\n\\end{align}\nIn particular, for
\\(p\\geq7\\),\n\\(0\\leq\\omega_n(p)/p\\leq2/(p-3)<1\\).",
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circle-method remainder decomposition. The full fixed-sieve comparison remains proved; local densities
and later sieve conclusions remain open. At1219: Actual unit-square root counts, both candidate
prime-density bounds including p=5, and the explicit multiplicative sieve function are proved without
deep inputs. Full local-density normalization is the next internal obligation. At1220: Proved the
conductor-sensitive coefficient identity, all actual prime-factor cases, the exact factors at2 and3,
residue invariance, and normalized prime adjunction without division by the global base factor. Full
local-density source closure is reserved for the final product and CRT assembly. At1221: complete actual
 normalized Euler-product factorization, exact source CRT residues, inclusive prime cutoff,
multiplicativity and all density bounds are proved without external inputs. The upper base factor may
vanish; no division by it is used. At1221: Both full source local-density lemmas are proved without new
external inputs: exact original Euler normalization, inclusive cutoff, lower quotient identification,
source CRT specializations, upper multiplicativity and p=5, with common zero upper base factors
retained. Presieving and the later sieve estimates remain open.",
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function\n\\(\\omega_n(\\ell,\\cdot)\\), defined on square-free integers supported on\n\\(W_\\ell\\),
such that\n\\begin{align}\\label{e:upperfactorization10}\n \\frac{G_R(\\ell,12d,n-1)}{\\phi(12d)}\n =\n
\\frac{G_R(\\ell,12,n-1)}{\\phi(12)}\n \\frac{\\omega_n(\\ell,d)}d .\n\\end{align}\nFor
\\(p\\nmid12\\ell\\), its  factors are\n\\begin{align}\\label{e:upperlocaldensity10}\n
\\frac{\\omega_n(\\ell,p)}p\n =\n \\begin{cases}\n \\displaystyle\n
\\frac{\\rho_p(n)}{p-1-\\rho_p(n-1)},&p\\leq R,\\\\[7pt]\n \\displaystyle\\frac1{p-1},&p>R.\n
\\end{cases}\n\\end{align}\nIn particular, \\(0\\leq\\omega_n(\\ell,p)/p<1\\).",
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exclusions, source multiplier arithmetic and residue independence from n. Local-density factorization
and the later analytic sieve estimates remain open. At1219: Actual unit-square root counts, both
candidate prime-density bounds including p=5, and the explicit multiplicative sieve function are proved
without deep inputs. Full local-density normalization is the next internal obligation. At1220: Proved
the conductor-sensitive coefficient identity, all actual prime-factor cases, the exact factors at2 and3,
 residue invariance, and normalized prime adjunction without division by the global base factor. Full
local-density source closure is reserved for the final product and CRT assembly. At1221: complete actual
 normalized Euler-product factorization, exact source CRT residues, inclusive prime cutoff,
multiplicativity and all density bounds are proved without external inputs. The upper base factor may
vanish; no division by it is used. At1221: Both full source local-density lemmas are proved without new
external inputs: exact original Euler normalization, inclusive cutoff, lower quotient identification,
source CRT specializations, upper multiplicativity and p=5, with common zero upper base factors
retained. Presieving and the later sieve estimates remain open.",
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|\\mathcal F_d|=Xg(d)+r_d,\n\\end{align}\nwhere \\(g\\) is multiplicative on the square-free integers
supported on\na fixed set of primes \\(\\mathfrak P\\), and \\(0\\leq g(p)<1\\).  Assume \\(X\\geq0\\)
and \\(g(1)=1\\).\nFix constants \\(C,A>0\\), put\n\\[\n R=(\\log N)^C,\\qquad\n D_0=\\exp((\\log\\log
N)^3),\\qquad D_1=D/D_0,\n\\]\nand work with sufficiently large \\(N\\). Assume\n\\[\n R+1\\leq z\\leq
D_1\\leq N^A,\\qquad D_0<D.\n\\]\nWrite\n\\[\n \\mathfrak P_0=\\{p\\in\\mathfrak P:p\\leq R\\},\\qquad\n
 \\mathfrak P_1=\\{p\\in\\mathfrak P:R<p<z\\}.\n\\]\nSuppose that there are fixed
constants\n\\(\\kappa_0,\\kappa>0\\) and \\(K_0\\geq2\\), uniform in the sequence and\nits density
function, such that for every \\(2\\leq w<y\\leq
z\\),\n\\begin{align}\\label{e:tworangesievedimension}\n \\prod_{\\substack{w\\leq
p<y\\\\p\\in\\mathfrak P_0}}\n (1-g(p))^{-1}\n &\\leq\n \\left(1+\\frac{K_0}{\\log w}\\right)\n
\\left(\\frac{\\log y}{\\log w}\\right)^{\\kappa_0},\\notag\\\\\n \\prod_{\\substack{w\\leq
p<y\\\\p\\in\\mathfrak P_1}}\n (1-g(p))^{-1}\n &\\leq\n \\left(1+\\frac{K_0}{\\log w}\\right)\n
\\left(\\frac{\\log y}{\\log w}\\right)^\\kappa.\n\\end{align}\nAll asymptotic assertions below are
uniform when these constants are fixed.\n\nThen\n\\begin{align}\\label{e:presievedbounds10}\n
S(\\mathcal F,\\mathfrak P,z)\n &\\geq XV(z)\\bigl(f_\\kappa(s)+o(1)\\bigr)\n
+\\sum_{m<D}c_m^-r_m,\\notag\\\\\n S(\\mathcal F,\\mathfrak P,z)\n &\\leq
XV(z)\\bigl(F_\\kappa(s)+o(1)\\bigr)\n +\\sum_{m<D}c_m^+r_m,\n\\end{align}\nwhere\n\\[\n
V(z)=\\prod_{\\substack{p<z\\\\p\\in\\mathfrak P}}(1-g(p)),\n \\qquad\n s=\\frac{\\log D_1}{\\log z},\n
\\qquad\n |c_m^\\pm|\\leq1.\n\\]\nThe coefficients \\(c_m^\\pm\\) depend only on the fixed prime sets
and\nthe sieve levels, and not on the values of \\(g\\).  \nThe dimension\n\\(\\kappa_0\\) is used only
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Section11_Reconciled.tex: X>=0, g(1)=1, fixed R=(log N)^C, R+1<=z<=D1<=N^A, refined1+K/logw dimension
bounds, and the preliminary endpointR+1. Coefficients precede the density/sequence and n, have absolute
value<=1 and exact level support; the common relative error tends to zero. The literal original weaker
manuscript statement is not claimed proved; replacement is required in the user manuscript. No original
source file was changed. At1258: All Section11 numbered claims are complete in the agreed conditional
scope for the explicitly supplied corrected presieving statement. The original weaker statement is not
proved and must be replaced in the user manuscript. The complete joint lower/upper estimates use one
cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary combined logarithmic
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that\n\\begin{align}\\label{e:circestsieve}\n |\\mathcal F_d|=Xg(d)+r_d,\n\\end{align}\nwhere \\(g\\) is
 multiplicative on the square-free integers supported on\na fixed set of primes \\(\\mathfrak P\\), and
\\(0\\leq g(p)<1\\).  Let\n\\(2\\leq R<z\\), and suppose that there are fixed
constants\n\\(\\kappa_0,\\kappa,K_0>0\\) such that\n\\begin{align}\\label{e:tworangesievedimension}\n
\\prod_{\\substack{w\\leq p<y\\\\p\\in\\mathfrak P}}\n \\left(1-g(p)\\right)^{-1}\n &\\leq\n
K_0\\left(\\frac{\\log y}{\\log w}\\right)^{\\kappa_0},\n &&2\\leq w<y\\leq R,\\notag\\\\\n
\\prod_{\\substack{w\\leq p<y\\\\p\\in\\mathfrak P}}\n \\left(1-g(p)\\right)^{-1}\n &\\leq\n
K_0\\left(\\frac{\\log y}{\\log w}\\right)^\\kappa,\n &&R<w<y\\leq z.\n\\end{align}\nDefine \n\\[\n
D_0=\\exp((\\log\\log N)^3),\\qquad D_1=D/D_0, \\qquad D_0<D , \\qquad \\frac{\\log D_0}{\\log
R}\\longrightarrow\\infty.\n\\]\nThen\n\\begin{align}\\label{e:presievedbounds10}\n S(\\mathcal
F,\\mathfrak P,z)\n &\\geq XV(z)\\bigl(f_\\kappa(s)+o(1)\\bigr)\n +\\sum_{m<D}c_m^-r_m,\\notag\\\\\n
S(\\mathcal F,\\mathfrak P,z)\n &\\leq XV(z)\\bigl(F_\\kappa(s)+o(1)\\bigr)\n
+\\sum_{m<D}c_m^+r_m,\n\\end{align}\nwhere\n\\[\n V(z)=\\prod_{\\substack{p<z\\\\p\\in\\mathfrak
P}}(1-g(p)),\n \\qquad\n s=\\frac{\\log D_1}{\\log z},\n \\qquad\n |c_m^\\pm|\\leq1.\n\\]\nThe
coefficients \\(c_m^\\pm\\) depend only on the fixed prime sets and\nthe sieve levels, and not on the
values of \\(g\\).  \nThe dimension\n\\(\\kappa_0\\) is used only in the preliminary sieve over
\\(p\\leq R\\);\nthe sieve functions in \\eqref{e:presievedbounds10} are determined by\nthe outer
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\\eqref{e:nrange10},\n\\begin{align}\\label{e:wumatomakisum10}\n \\sum_{\\ell\\in
L}X_n(\\ell)\\Theta_{n,\\ell}(z_\\ell)\n \\leq\n \\bigl(c_\\alpha\\mathcal W(\\alpha)+o(1)\\bigr)\n
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X_n\\Theta_n(z),\n \\qquad\n
c_\\alpha=\\frac{16e^{-\\gamma/2}}{\\sqrt{\\pi\\alpha}}.\n\\end{align}\nHere \\(\\mathcal W(\\alpha)\\)
is the integral defined in\n\\eqref{e:Wdef8}.",
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actual multiplier set, local factors, finite Euler products and singular integrals. For every A>0, every
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coefficient is exactly c_(9/8) W(9/8)+eta. Strong multiplicativity, the uniform mean instantiation, both
 finite Abel arguments, W identity, all small-denominator and endpoint terms, moving logarithms,
integration, smoothing and relative errors are proved. Only the independent generic effective mean-value
 corollary and classical fixed-progression Mertens bounds remain explicit unproved hypotheses. The
source lemma statement is unchanged; two proof replacements are supplied but not applied. At1252: The
full source weighted upper main-term comparison l:wumatomakisum is complete as a checked implication
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P_{3,n},z)\n \\geq X_n\\Theta_n(z)\\bigl(f_{1/2}(9/8)+o(1)\\bigr)\n +\\mathcal
E_1(n),\n\\end{align}\nwhere\n\\begin{align}\\label{e:E1section8}\n\\sum_{n<N^2}|\\mathcal
E_1(n)|^2\n\\ll_{A_0,K} N^4(\\log N)^{-K}.\n\\end{align}",
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(indeed all n=12 mod24). General classical sieve/Mertens/function-continuity and ordinary
SW/BV/Ren-Ghosh inputs remain explicit and unproved. At1229: Common cutoff compatibility for the
circle-method and full lower-sieve estimates is proved for every requested lower bound on C; main-term
size remains to be joined. At1230: The full source main-term size and the entire lower-bound side are
complete conditionally, including an exponent independent of C, uniform smoothing, one common large
cutoff and finite error-to-exception deductions. At1230 the source main-term size and the full lower
sieve are proved jointly for the same C>=A0+6; finite positive-count and exceptional-cardinality
deductions are checked. At1258: All Section11 numbered claims are complete in the agreed conditional
scope for the explicitly supplied corrected presieving statement. The original weaker statement is not
proved and must be replaced in the user manuscript. The complete joint lower/upper estimates use one
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 $n\\equiv12\\pmod{24}$,\n\\begin{align}\\label{e:sieveneedupper8}\n \\sum_{\\ell\\in
L_n}S(M_n(\\ell),W_{\\ell,n},z_\\ell)\n &\\leq\n \\bigl(c_{9/8}F_1(2)\\mathcal W(9/8)+o(1)\\bigr)\n
X_n\\Theta_n(z)+\\mathcal E_2(n),\n\\end{align}\nwhere\n\\begin{align}\\label{e:calphadef8}\n c_\\alpha=
\\frac{16e^{-\\gamma/2}}{\\sqrt{\\pi\\alpha}}\n\\end{align}\nand\n\\begin{align}\\label{e:E2section8}\n\
\sum_{n<N^2}|\\mathcal E_2(n)|^2\n\\ll_{A_0,K} N^4(\\log N)^{-K}.\n\\end{align}",
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 n; source divisor counts, refined preliminary/outer dimensions, uniform varying levels, the shifted
double remainder sum, its circle-method embedding and the fixed-sieve O(N^(1/4)) correction are proved.
The total error has arbitrarily strong logarithmic mean-square saving. Only independent general SW, BV,
Ren/Ghosh, classical one-stage sieve, effective multiplicative mean and Mertens inputs and general sieve
 normalizations remain unproved; no own-paper endpoint is assumed. At1257: Full upper-sieve proposition
p:sieveneedupper8 complete in uniform epsilon form from explicit independent classical inputs, with the
exact coefficient and arbitrary logarithmic mean-square saving. The lower analytic estimate and both
remainder transports are available at a shared prescribed cutoff. The final joint Section11 assembly and
 manuscript synchronization are next. At1258: All Section11 numbered claims are complete in the agreed
conditional scope for the explicitly supplied corrected presieving statement. The original weaker
statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper
estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary
combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction.",
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that, for every fixed\nsufficiently large \\(C\\), with \\(R=(\\log N)^C\\), one
has\n\\begin{align}\\label{e:maintermlower8}\n X_n\\Theta_n(z)\\gg_{A_0,C}\n N(\\log
N)^{-C_{A_0}}\n\\end{align}\nwhenever $n\\equiv12\\pmod{24}$ and $N^2(\\log N)^{-A_0}\\leq n<N^2$.",
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exact retained n range. The actual X_n Theta_n(z) is at least a positive constant times N(log
N)^(-(A0/2+6)); the exponent is independent of C. Actual archimedean, smoothing and local-product
arguments are proved; only the already recorded independent interval Mertens products remain external.
The same large C is joined to the complete lower sieve. At1258: All Section11 numbered claims are
complete in the agreed conditional scope for the explicitly supplied corrected presieving statement. The
 original weaker statement is not proved and must be replaced in the user manuscript. The complete joint
 lower/upper estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and
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arbitrary combined logarithmic mean-square saving. The final Goldbach theorem remains the next
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by positive interval integration and exact rpow primitives. The strict comparison is proved for the
classical normalized sieve values f_(1/2)(9/8)=sqrt(8 exp(gamma)/(9 pi)) log2 and F_1(2)=exp(gamma).
Identifying the generic independent sieve functions with those classical normalized values remains
explicit, within GENERAL_SIEVE; the paper comparison is not assumed. No floating point evaluation or
unchecked evaluator is used. At1253: Proved the exact actual W integral bound, the normalized
half/linear sieve comparison and strict positivity. Classical sieve-function normalizations remain
explicit general input identities. The upper-sieve application and final Goldbach theorem remain open;
no paper endpoint is assumed. At1258: All Section11 numbered claims are complete in the agreed
conditional scope for the explicitly supplied corrected presieving statement. The original weaker
statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper
estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary
combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction.",
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archive/Pre1265_Section9_Statement_Map.json


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sifted and good representation sets and the uniform O(N^epsilon) exception are retained. At1262: Every
Section9 numbered source claim and its final quantitative theorem is complete in the agreed scope. All
elementary arithmetic reductions are now proved without an extra primitive two-square assumption.
W(alpha) convergence is proved for every alpha>1 and the smooth seed is constructed. Only the explicit
independent analytic inputs remain for the circle-method/sieve/headline estimates. The user is applying
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|\\{r\\in\\mathcal P:n-r\\in\\mathbb P^2\\}|\n =\n
|T_n|+O_\\varepsilon(N^\\varepsilon),\n\\end{align}\nwhere\n\\begin{align}\\label{e:Tfactorization}\n
    T_n=\n \\left\\{\n r-1=2up_1p_2:\n \\begin{array}{l}\n r\\in\\mathbb P,\\quad n-r\\in\\mathbb
P^2,\\\\\n z\\leq p_2\\leq p_1,\\quad p_i\\in\\mathbb P_{3,n},\\\\\n p\\mid u\\Longrightarrow
p\\equiv1\\pmod4\n \\end{array}\n \\right\\}.\n\\end{align}",
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Section9 numbered source claim and its final quantitative theorem is complete in the agreed scope. All
elementary arithmetic reductions are now proved without an extra primitive two-square assumption.
W(alpha) convergence is proved for every alpha>1 and the smooth seed is constructed. Only the explicit
independent analytic inputs remain for the circle-method/sieve/headline estimates. The user is applying
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P^2\\}|\n &\\geq S(\\mathcal A_n,\\mathbb P_{3,n},z)-\\sum_{\\ell\\in L_n}\n
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arithmetic reduction. At1262: Every Section9 numbered source claim and its final quantitative theorem is
 complete in the agreed scope. All elementary arithmetic reductions are now proved without an extra
primitive two-square assumption. W(alpha) convergence is proved for every alpha>1 and the smooth seed is
 constructed. Only the explicit independent analytic inputs remain for the circle-method/sieve/headline
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estimates. The user is applying previous manuscript edits; no application is assumed and original
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\\(A,\\delta>0\\) and \\(L_0<N^{2-\\delta}\\).  There are sufficiently\nlarge constants
\\(B=B(A,\\delta)\\) and \\(C=C(A,\\delta)\\) such that the\nfollowing holds.  Put\n\\[\n R=(\\log
N)^C,\\qquad\n \\eta_N(\\theta)=\\eta\\left(\\frac{N^2\\theta}{R}\\right),\n\\]\nand suppose that
$V_\\ell\\leq (N^2/\\ell)^{1/2}(\\log N)^{-B}$ for $\\ell<L_0.$\nLet \\(|d_{k,\\ell}|\\leq1\\)
and\n\\((a_{k,\\ell},k)=1\\).\nDefine \n\\begin{align}\\quad \\quad\\label{e:defGR9}\n G_R(\\ell,k,n)\n
&:=\n \\mathcal T_2(\\ell,k,n;a_{k,\\ell})\n \\prod_{2<p\\leq R}\n \\frac{p\\phi((p,k))}{(p-1)^2}\n
\\left|\\left\\{(u,v)\\in\\mathbb A_p^2:\n \\begin{array}{l}\n  u^2+\\ell v\\equiv n\\pmod p,\\\\\n
v\\equiv a_{k,\\ell}\\pmod{(p,k)}\n \\end{array}\\right\\}\\right|,\n\\end{align}\nwhere $ \\mathcal
T_2(\\ell,k,n;a_{k,\\ell}) $ is defined in \\eqref{e:generalT2} and the second congruence is understood
to be trivial when \\(p\\nmid k\\).\nAlso define\n\\begin{align}\\label{e:defHR9}\n H_{\\ell,R}(n)\n
:=\n \\ell\\int_{\\mathbb R}\\eta_N(\\theta)^2\n \\left(\\int_2^N\\frac{e(\\theta u^2)}{\\log
u}\\,du\\right)\n \\left(\\int_{\\substack{2\\leq v\\leq N^2\\\\\\ell v<N^2}}\n \\frac{e(\\ell\\theta
v)}{\\log v}\\,dv\\right)\n
e(-n\\theta)\\,d\\theta.\n\\end{align}\nThen\n\\begin{align}\\label{e:circmethod}\n
\\sum_{n<N^2}\\left|\n \\sum_{\\ell<L_0}\\sum_{k<V_\\ell}\n d_{k,\\ell}r_{k,\\ell}(n)\\right|^2\n
\\ll_{A,\\delta}N^4(\\log N)^{-A},\n\\end{align}\nwhere\n\\begin{align}\\label{e:unweightedremainder}\n
r_{k,\\ell}(n)\n &=\n \\sum_{\\substack{m^2+\\ell t=n\\\\\n t\\equiv a_{k,\\ell}\\pmod k}}\n
\\one_{\\mathbb P}(m)\\one_{\\mathbb P}(t)\n -\n
\\frac{G_R(\\ell,k,n)}{\\ell\\phi(k)}H_{\\ell,R}(n).\n\\end{align}",
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B0,C,K,N0 chosen before N and every arithmetic family. Covers all B>=B0, exact strict real
L<N^(2-delta), positive ell,k with k<V(ell), every coprime residue class, arbitrary complex coefficients
 of norm<=1, and all n<N^2. The stronger uniform cutoff statement holds for every 2<=U<=N,W>=2. The
final source theorem removes cutoffs using the proved equation-imposed bounds, and uses the actual
finite prime-pair count, actual local Euler factor G_R and actual autocorrelation singular integral H_R.
 Fourier inversion, all three error identifications, shared constants, minor estimate, major
replacement, arithmetic completion and smoothing are consumed internally. Its only unformalized
hypotheses are independent ordinary SW prime counts, ordinary unweighted BV prime-count distribution,
and uniform raw Ren/Ghosh quadratic estimates. No own-paper conclusion is assumed. At1212: Completed all
 Section10 source statements and the full circle-method implication; the three independent raw deep
inputs remain explicit and unproved. At1213: Completed every Section10 source statement and the full
source circle-method implication, including the original smooth compact seed and exact normalized
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P_{3,n},z)\n \\geq X_n\\Theta_n(z)\\bigl(f_{1/2}(9/8)+o(1)\\bigr)\n +\\mathcal
E_1(n),\n\\end{align}\nwhere\n\\begin{align}\\label{e:E1section8}\n\\sum_{n<N^2}|\\mathcal
E_1(n)|^2\n\\ll_{A_0,K} N^4(\\log N)^{-K}.\n\\end{align}",
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N)^(-A) mean-square remainder bound. The pointwise theorem holds throughout the source retained range
(indeed all n=12 mod24). General classical sieve/Mertens/function-continuity and ordinary
SW/BV/Ren-Ghosh inputs remain explicit and unproved. At1229: Common cutoff compatibility for the
circle-method and full lower-sieve estimates is proved for every requested lower bound on C; main-term
size remains to be joined. At1230: The full source main-term size and the entire lower-bound side are
complete conditionally, including an exponent independent of C, uniform smoothing, one common large
cutoff and finite error-to-exception deductions. At1230 the source main-term size and the full lower
sieve are proved jointly for the same C>=A0+6; finite positive-count and exceptional-cardinality
deductions are checked. At1258: All Section11 numbered claims are complete in the agreed conditional
scope for the explicitly supplied corrected presieving statement. The original weaker statement is not
proved and must be replaced in the user manuscript. The complete joint lower/upper estimates use one
cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary combined logarithmic
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L_n}S(M_n(\\ell),W_{\\ell,n},z_\\ell)\n &\\leq\n \\bigl(c_{9/8}F_1(2)\\mathcal W(9/8)+o(1)\\bigr)\n
X_n\\Theta_n(z)+\\mathcal E_2(n),\n\\end{align}\nwhere\n\\begin{align}\\label{e:calphadef8}\n c_\\alpha=
\\frac{16e^{-\\gamma/2}}{\\sqrt{\\pi\\alpha}}\n\\end{align}\nand\n\\begin{align}\\label{e:E2section8}\n\
\sum_{n<N^2}|\\mathcal E_2(n)|^2\n\\ll_{A_0,K} N^4(\\log N)^{-K}.\n\\end{align}",
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F_1(2) W(9/8) coefficient, original actual main term and closed multiplier set. All coefficients precede
 n; source divisor counts, refined preliminary/outer dimensions, uniform varying levels, the shifted
double remainder sum, its circle-method embedding and the fixed-sieve O(N^(1/4)) correction are proved.
The total error has arbitrarily strong logarithmic mean-square saving. Only independent general SW, BV,
Ren/Ghosh, classical one-stage sieve, effective multiplicative mean and Mertens inputs and general sieve
 normalizations remain unproved; no own-paper endpoint is assumed. At1257: Full upper-sieve proposition
p:sieveneedupper8 complete in uniform epsilon form from explicit independent classical inputs, with the
exact coefficient and arbitrary logarithmic mean-square saving. The lower analytic estimate and both
remainder transports are available at a shared prescribed cutoff. The final joint Section11 assembly and
 manuscript synchronization are next. At1258: All Section11 numbered claims are complete in the agreed
conditional scope for the explicitly supplied corrected presieving statement. The original weaker
statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper
estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary
combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction.",
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that, for every fixed\nsufficiently large \\(C\\), with \\(R=(\\log N)^C\\), one
has\n\\begin{align}\\label{e:maintermlower8}\n X_n\\Theta_n(z)\\gg_{A_0,C}\n N(\\log
N)^{-C_{A_0}}\n\\end{align}\nwhenever $n\\equiv12\\pmod{24}$ and $N^2(\\log N)^{-A_0}\\leq n<N^2$.",
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exact retained n range. The actual X_n Theta_n(z) is at least a positive constant times N(log
N)^(-(A0/2+6)); the exponent is independent of C. Actual archimedean, smoothing and local-product
arguments are proved; only the already recorded independent interval Mertens products remain external.
The same large C is joined to the complete lower sieve. At1258: All Section11 numbered claims are
complete in the agreed conditional scope for the explicitly supplied corrected presieving statement. The
 original weaker statement is not proved and must be replaced in the user manuscript. The complete joint
 lower/upper estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and
arbitrary combined logarithmic mean-square saving. The final Goldbach theorem remains the next
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\\cite[(1.8)--(1.10)]{I80},\n\\begin{align}\\label{e:Deltanumerical}\n \\Delta>0.\n\\end{align}",
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by positive interval integration and exact rpow primitives. The strict comparison is proved for the
classical normalized sieve values f_(1/2)(9/8)=sqrt(8 exp(gamma)/(9 pi)) log2 and F_1(2)=exp(gamma).
Identifying the generic independent sieve functions with those classical normalized values remains
explicit, within GENERAL_SIEVE; the paper comparison is not assumed. No floating point evaluation or
unchecked evaluator is used. At1253: Proved the exact actual W integral bound, the normalized
half/linear sieve comparison and strict positivity. Classical sieve-function normalizations remain
explicit general input identities. The upper-sieve application and final Goldbach theorem remain open;
no paper endpoint is assumed. At1258: All Section11 numbered claims are complete in the agreed
conditional scope for the explicitly supplied corrected presieving statement. The original weaker
statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper
estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary
combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction.",
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estimates",
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archive/Proof_Coverage_Registry_Pre1265.json


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        "sum_upsilon_square_prime_factorization"
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is preserved.",
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bounds.",
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the agreed external-input boundary, the paper-specific deduction is complete.",
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 hypotheses for prime squares is a separate obligation.",
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        "docs/THEOREM_1_1_EXTERNAL_INPUTS.md",
        "docs/SQUARE_REFLECTION_COMPLETION.md",
        "blueprint/Checkpoint_783_SquareDecompositionReflection.md"
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      "scope": "Both conclusions for the genuine full von-Mangoldt square average, every positive
integer N, the exact source interval and its cardinality, all finite F,G subsets, every coefficient
cstar>4 sqrt(3)/9, and every real 3/2<p<=2. The strong result proves actual output lp membership and the
 literal normalized sum form, for all lp inputs, hence also the supported inputs in the source.",
      "remaining": "The implication is complete under the explicit independent deep inputs. PNT limits,
compatible progression/Ren/Ghosh data existence, and the independent ordinary-square global
decomposition remain unformalized hypotheses. Its analytic globalization remains external; the
plus/minus spatial reflection is proved with the same constants at checkpoint783. Theorem1.1 and
Propositions4.19/6.21 are complete conditional implications under their separate inputs; the current
corrected numbered paper targets are all matched, with independent inputs and stronger unnumbered
alternative-route assertions retaining their documented scopes.",
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      "scope": "All displayed clauses for the actual common selected source low/high/error family: exact
 all-input annular decomposition, actual l1-to-l-infinity low norm, local bound for all l1 inputs with
the supremum over every real M>=N, sharp fixed dyadic and actual countable maximal high bounds for every
 real Q>=2, every allowed starting threshold, every error exponent A>0, bounded Q and the exact printed
high-symbol identity. One cutoff coefficient is chosen before epsilon and A; the operator family never
changes with these parameters.",
      "remaining": "The implication is complete under PrimeAsymptoticInputs,
PagePrimeSquareDistributionInputs c, RawPageUniquenessInput c and SmoothCircleBourgainInput C. Their
independent analytic truths and compatible raw-data existence remain external. No own-paper
decomposition, low bound, approximation error or maximal conclusion is an input. The current corrected
numbered paper targets are all matched; independent inputs and stronger unnumbered alternative-route
assertions retain their documented scopes.",
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        "docs/SOURCE_UPDATE_1107.md",
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before Q, all q<=R^(11/10), the exact induced character/selector/exceptional model and inverse-fifth
major error, and the full complementary R^(-1/4)*(log N)^d minor bound. The source Q-range, uniform
sufficiently-large-N threshold, R^4<=N/2 and log N>=1 are proved. The general fixed-level uniform
Page-prefix theorem and raw Ren/Ghosh estimates remain explicit independent unproved hypotheses.",
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 real Q>=2 and every integer N in the Section6 source range. One positive source cutoff coefficient is
chosen before all A>0; the exact Q^(-1/2)*(1+log N)^(-A) operator and all-l2 function bounds are proved.
 The optional Page data and radius are the same actual selected family as Proposition4.19. The
previously explicit compatible raw distribution/Ren/Ghosh data remain unproved general inputs; no new
own-paper estimate is assumed.",
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        "blueprint/Checkpoint_1039_SectionSixSourceExports.md"
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      "scope": "Complete printed low proposition for the actual selected source low operator: every real
 Q>=2, every integer N in the common source range, one absolute positive constant for the literal
all-real-radius local bound and the l1-to-linfinity norm bound. The local result holds for every complex
 l1 input using |f| and hence the stated nonnegative inputs, with fixed spatial constant1. Ordinary and
induced exceptional terms use the complete conductor range and exact source cutoffs. Only the previously
 recorded compatible source-data boundary remains; no new mathematical input is added.",
      "remaining": "No remaining source clause within the agreed conditional scope.",
      "external_inputs": [
        "PROGRESSION_ESTIMATE",
        "REN_GHOSH"
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    "l:notBourgain": {
      "title": "Printed annular multifrequency maximum",
      "work_group": "Prime-square high-low",
      "status": "conditional",
      "lean_anchors": [
        "bourgain_weightedSource_projection",
        "rationalProjectionErrorBlock_maximum",
        "largeStart_weightedAnnular_maximum",
        "weightedAnnular_allStartingScales",
        "quadratic_log_loss_not_linear",
        "rationalBand_common_period",
        "rationalBand_loss_linear",
        "rational_annular_allStarting",
        "circleAnnularSource_initial_norm",
        "source_rational_annular_maximum"
      ],
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      "expositions": [
        "docs/NEXT_ANALYTIC_INPUT_700.md",
        "blueprint/Checkpoint_725_LargeStartActualSourceMaximum.md",
        "blueprint/Checkpoint_726_WeightedAnnularLargeStartMaximum.md",
        "docs/AUXILIARY_SCOPE_REVIEW_1065.md",
        "docs/RATIONAL_MULTIFREQUENCY_BOUNDARY.md",
        "docs/RATIONAL_ANNULAR_COMPLETION.md"
      ],
      "scope": "The full actual annular source maximum with the printed first power of (1+s) is proved
under the precise independent rational smooth multifrequency/circle-transference input and the
previously recorded general smooth input. Every source dyadic scale N>=2^s, every s>=0, scale1 and the
small denominator indices are included. The rational common period, linear cost, weight localization,
actual annular square-function error and all earlier scale costs are proved. Beta may vary with the
scale, a stronger conclusion than the source fixed-beta statement.",
      "remaining": "No own annular deduction remains. The independent smooth multifrequency theorems and
 sampling remain explicit unproved inputs. The proof uses the rational-frequency refinement rather than
replacing the general squared-log loss by a first power; a matching expository replacement proof is
saved separately.",
      "external_inputs": [
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        "RATIONAL_SMOOTH_MULTIFREQUENCY"
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    },
    "c:orlicz-prefix": {
      "title": "Denominator-truncated annular multifrequency maximum",
      "work_group": "Prime-square high-low",
      "status": "conditional",
      "lean_anchors": [
        "source_time_prefix_from_intervals",
        "source_rational_annular_truncation_maximum",
        "aeIntegerMultiplier_norm_le",
        "aeIntegerMultiplier_localized_comp",
        "disjointAEMultiplierSymbol_local_comp",
        "fixedAnnularMultiplier_disjoint_factor",
        "fixed_annular_finite_truncation_maximum",
        "source_fixed_annular_truncation_maximum"
      ],
      "expositions": [
        "docs/NEXT_ANALYTIC_INPUT_700.md",
        "docs/SOURCE_UPDATE_601.md",
        "docs/RATIONAL_TRUNCATION_COMPLETION.md"
      ],
      "scope": "The full source bound with square (1+s)^2 loss is proved for all s>=0, all source dyadic
 times including scale1, every real denominator cutoff simultaneously and every integer L2 input. The
additional fixed bounded measurable multiplier clause is included, with actual essentially bounded
Fourier operators and no continuity restriction. Beta may also vary with the scale.",
      "remaining": "No own corollary deduction remains. The independent rational
smooth/circle-transference theorem and the prior general smooth input remain explicit unproved
hypotheses; the full additional fixed measurable multiplier clause is proved, not added as an external
input.",
      "external_inputs": [
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        "RATIONAL_SMOOTH_MULTIFREQUENCY"
      ]
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      "title": "Sharp high contribution and countable maximum",
      "work_group": "Prime-square high-low",
      "status": "conditional",
      "lean_anchors": [
        "proposition621_source",
        "proposition621",
        "proposition621_maximal_asymptotic",
        "sourcePageDyadicTail_uniform",
        "pageHighSymbol_paper_identity",
        "sourceHigh_uniform_geometry",
        "dyadic_tail_covers_source_scales",
        "regularizedHighExponentWeight_eq"
      ],
      "expositions": [
        "docs/PROPOSITION_6_21_COMPLETION.md",
        "docs/PROPOSITION_6_21_STATEMENT_REVIEW.md",
        "blueprint/Checkpoint_778_Proposition621Source.md"
      ],
      "scope": "Both source clauses are complete checked implications: the actual countable dyadic
supremum and every individual output, all real Q>=2 and N0>=2 in the allowed logQ<=c0 sqrt(logN0) range,
 one positive c0 independent of epsilon, the exact source radius and complementary narrow-cutoff
multiplier, changing genuine exceptional data, and the leading coefficient 2+o(1) in precise epsilon
form. A separate uniform estimate covers bounded Q; its harmless regularization agrees with
sqrt(logQ)/loglogQ once logQ>=2.",
      "remaining": "No paper-specific deduction in this proposition remains. The two prime asymptotic
limits, the general smooth multifrequency/sampling input, raw Landau-Page uniqueness (including the
L-function bridge), and the existence/selection of source-compatible exceptional zero data remain
explicit external boundaries. The stronger auxiliary statements 6.16 and 6.19 are not awarded
completion.",
      "external_inputs": [
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        "SMOOTH_MULTIFREQUENCY",
        "PAGE_CLASSES"
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      "title": "Main sparse domination theorem",
      "work_group": "Sparse estimates",
      "status": "conditional",
      "lean_anchors": [
        "theorem_1_3",
        "theorem_1_3_endpoint",
        "theorem_1_3_open_pq",
        "primeEndpointSparse",
        "primeOpenSparse",
        "primeSquareMaximalIndicator_dominates",
        "sourceSparseTriangle_iff_barycentric"
      ],
      "expositions": [
        "docs/THEOREM_1_3_COMPLETION.md",
        "docs/THEOREM_1_3_STATEMENT_REVIEW.md",
        "docs/INDEPENDENT_STOPPING_INPUTS.md",
        "docs/SPARSE_DYADIC_QUALIFICATION.md"
      ],
      "scope": "Both displayed conclusions are complete conditional implications: the literal full
maximal indicator for every integer N>=2, arbitrary finite F,G including empty sets, actual finite
interval collections with actual pairwise disjoint major subsets of size at least one tenth, the exact
endpoint density exponent and paper Phi for every cstar>4sqrt(3)/9, and every point of the full open
triangle in exact reciprocal/barycentric coordinates. Constants precede F,G. All stopping, packing,
support, local endpoint, interpolation and sparse deductions are proved.",
      "remaining": "No paper-specific deduction in either conclusion remains. The recorded prime limits,
 compatible raw Page/distribution data, Page uniqueness, general smooth multifrequency/sampling,
independent ordinary-square stopping interface and complementary-exponent prime maximal input remain
unproved explicit hypotheses, including their stated compatibility boundaries. The general auxiliary
Section5 statements are separately partial. The proposed dyadic qualification of l:sparserecursion is
unapplied; the actual prime-square all-cutoff reduction is proved.",
      "external_inputs": [
        "PNT_LIMITS",
        "PROGRESSION_ESTIMATE",
        "REN_GHOSH",
        "SMOOTH_MULTIFREQUENCY",
        "PAGE_CLASSES",
        "ORDINARY_SQUARE_STOPPING",
        "PRIME_MAXIMAL_LP"
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    "l:localsparse": {
      "title": "Abstract localized stopping-time estimate",
      "work_group": "Sparse estimates",
      "status": "proved",
      "lean_anchors": [
        "source_local_stopping_from_fourier_hypotheses",
        "source_abstract_local_stopping_estimate",
        "abstract_local_stopping_endpoint",
        "fourierSymbol_operator_bound",
        "localError_square_majorant",
        "linearSourceLocalMass_change_radius"
      ],
      "expositions": [
        "docs/ABSTRACT_LOCAL_STOPPING_COMPLETION.md",
        "blueprint/Checkpoint_1054_SourceLocalStoppingFourier.md"
      ],
      "scope": "Complete abstract implication with the exact source exponents, all 0<=v<1, s>=1-v, r>0,
gamma>=beta>=0 and A>1/2, arbitrary physical dyadic stopping times, actual localized low/high/error
outputs, all-real-radius admissibility, arbitrary fixed positive radii, parent-length normalization,
empty sets and the full bounded density range. The error symbol may be an arbitrary almost-everywhere
bounded function; its operator norm follows from the proved Fourier isometry. The normal decomposition
and norm estimates are the printed hypotheses, not extra deep inputs.",
      "remaining": "",
      "external_inputs": []
    },
    "l:sparserecursion": {
      "title": "Sparse recursion from local estimates",
      "work_group": "Sparse estimates",
      "status": "proved",
      "lean_anchors": [
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        "laminarSparseRecursion",
        "primeAnnularSparse_from_local",
        "primeSparse_from_admissible_local",
        "sparseNonDyadic_local_pairing_zero",
        "sparseNonDyadic_maximum_positive",
        "densityProfileSpike_nonneg",
        "densityProfileSpike_no_dilation_comparison",
        "source_dyadic_sparse_recursion_full_domain",
        "genericPositiveFullSparse",
        "genericLocalHypothesis_uniform_indicator_bound",
        "genericFullDyadicMaximum_finite_test",
        "accepted_sparse_admissibility_threshold",
        "accepted_general_dyadic_sparse_recursion"
      ],
      "expositions": [
        "docs/THEOREM_1_3_COMPLETION.md",
        "docs/SPARSE_DYADIC_QUALIFICATION.md",
        "docs/AUXILIARY_SCOPE_REVIEW_1065.md",
        "docs/SOURCE_UPDATE_1066.md",
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        "docs/GENERAL_DYADIC_SPARSE_COMPLETION.md",
        "docs/SOURCE_UPDATE_1084.md"
      ],
      "scope": "The actual prime-square finite stopping construction, disjoint components, real-radius
admissibility, exact kernel localization, terminating cost recursion and actual sparse major subsets are
 proved, with a proved all-cutoff reduction for the original averages. A general finite recursive-cost
implication is also proved. The complete general dyadic implication is now proved for arbitrary positive
 compact-support kernels, every fixed odd support dilation, arbitrary Psi, the full untruncated maximum
and genuine quarter-sparse major sets. The uniform bound and finite approximation of the full maximum
are deduced from the local hypothesis. The local hypothesis is imposed only on maps admissible on the
whole interval; an extension argument proves the needed restricted version.",
      "remaining": "No own-paper deduction remains under the accepted dyadic and
independent-admissibility-constant conventions. The proof constructs actual sparse sets and the full
maximum from the stated local hypothesis; no additional analytic estimate is assumed.",
      "external_inputs": [],
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    "l:primeendpointlocal": {
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      "work_group": "Sparse estimates",
      "status": "conditional",
      "lean_anchors": [
        "source_prime_local_endpoint_from_independent_inputs",
        "source_prime_local_endpoint_real_dilation",
        "source_prime_local_endpoint",
        "primeStopping_general_parent_endpoint",
        "primeStopping_local_endpoint_with_constant",
        "sourceLocalMass_from_quadratic_radius",
        "physicalDyadic_prime_pairing"
      ],
      "expositions": [
        "docs/PRIME_LOCAL_SOURCE_COMPLETION.md",
        "blueprint/Checkpoint_1062_SourcePrimeLocalEndpoint.md"
      ],
      "scope": "Complete source implication for the actual annular von Mangoldt square average: every
fixed real support dilation >=1, every positive spatial radius, every fixed admissibility constant,
physical dyadic stopping values including scale1, literal all-real-radius quadratic averages, empty sets
 and the exact parent-length normalization. The bound holds for every cstar>4sqrt(3)/9. Proposition4.19
is derived from the already-listed independent inputs in the terminal theorem, rather than assumed as an
 extra own-paper input.",
      "remaining": "",
      "external_inputs": [
        "PNT_LIMITS",
        "PROGRESSION_ESTIMATE",
        "REN_GHOSH",
        "SMOOTH_MULTIFREQUENCY",
        "PAGE_CLASSES",
        "ORDINARY_SQUARE_STOPPING"
      ]
    },
    "t:orlicz-maximal": {
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      "work_group": "Orlicz and ergodic estimates",
      "status": "conditional",
      "lean_anchors": [
        "nearOnePsi_convex",
        "nearOnePsi_comparison",
        "nearOnePsi_doubling",
        "highLow_restricted_distribution",
        "positiveRestricted_to_truncated_modular",
        "section7_truncated_modular_finite_card",
        "section7_modular_finite_card",
        "mangoldtPositiveFamily_apply",
        "ordinaryPositiveFamily_apply",
        "primePositiveFamily_apply",
        "iw_prime_annular_restricted_distribution",
        "iw_prime_annular_restricted_gauge",
        "iw_prime_full_restricted_gauge",
        "iw_prime_full_modular_estimates",
        "iw_unweighted_prime_restricted_gauge",
        "iw_unweighted_prime_modular_estimates",
        "iw_prime_families_restricted_gauges",
        "mangoldt_full_annular_maximal_bound",
        "prime_mangoldt_indicator_maximal_bound",
        "primeCounting_log_lower_bound",
        "iwRestrictedDistribution_budget",
        "iw_full_square_low_maximal_bound",
        "iw_full_square_model_comparison",
        "fullSquareCircleError_bound",
        "fullSquareCircleError_uniform",
        "full_square_approximation_maximum",
        "iw_full_square_high_source_scale",
        "iw_full_square_annular_restricted_distribution",
        "iw_full_square_annular_restricted_gauge",
        "ordinary_full_annular_maximal_bound",
        "iw_full_square_restricted_gauge",
        "iw_full_square_modular_estimates",
        "iw_three_families_restricted_gauges",
        "section7_three_family_assembly",
        "theorem_7_2",
        "section7_source_sum_assembly",
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        "theorem_7_2_finitely_supported",
        "theorem_7_2_closed_signed_maximal"
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      "expositions": [
        "docs/SECTION_7_PROGRESS.md",
        "docs/SECTION_7_STATEMENT_REVIEW.md",
        "blueprint/Checkpoint_837_RestrictedToModular.md",
        "blueprint/Checkpoint_843_HighLowRestrictedDistribution.md",
        "blueprint/Checkpoint_846_SectionSevenDeductions.md",
        "docs/SECTION_7_RESTRICTED_COMPLETION.md",
        "blueprint/Checkpoint_948_IWPrimeAnnularRestrictedGauge.md",
        "blueprint/Checkpoint_950_IWPrimeFullRestrictedGauge.md",
        "blueprint/Checkpoint_953_IWUnweightedPrimeRestrictedGauge.md",
        "docs/THEOREM_7_2_COMPLETION.md",
        "docs/SECTION_7_QUADRATIC_WEYL_BOUNDARY.md",
        "blueprint/Checkpoint_967_IWFullSquareHighSourceScale.md",
        "blueprint/Checkpoint_970_IWFullSquareAnnularRestrictedGauge.md",
        "blueprint/Checkpoint_971_OrdinaryFullAnnularComparison.md",
        "blueprint/Checkpoint_972_IWFullSquareRestrictedGauge.md",
        "blueprint/Checkpoint_973_IWThreeFamilyRestrictedGauge.md",
        "blueprint/Checkpoint_974_SectionSevenThreeFamilyAssembly.md",
        "blueprint/Checkpoint_975_TheoremSevenTwo.md",
        "blueprint/Checkpoint_976_TheoremSevenTwoFinitelySupported.md",
        "docs/PAPER_COMPLETION.md"
      ],
      "scope": "Complete printed Theorem 7.2 as an implication of explicit independent standard inputs:
all three actual families (von Mangoldt squares, ordinary squares, unweighted prime squares), every
integer cutoff N >= 2, every complex finitely supported function and positive level, with common
absolute C, C0 and K >= 1. All three conclusions are proved, including finite level sets: the truncated
modular bound, unrestricted modular bound and restricted bound for every finite F and t >= 1. The final
theorem is stated with literal sums over all integers and finite support, using the actual integral
gauge. The ordinary branch derives the actual discrete circle-model error, maximal comparison, low/high
restricted bound and full-cutoff reconstruction from the independent scalar Weyl input and general IW
inputs; no own restricted or high/low conclusion is assumed. The unweighted-prime transfer uses proved
Mathlib Chebyshev bounds without a new PNT input. At checkpoint1264 the current public export consumes
the proved finite signed maximal theorem at constant6; a supplied RawFiniteSignedMaximalInput and its
constant R are no longer hypotheses.",
      "remaining": "No remaining clause of Theorem7.2 or Corollaries7.6/7.46 within the agreed
conditional scope. The explicitly recorded independent inputs and their published-theorem bridges remain
 unformalized. Stronger standalone auxiliary parameter/full-generality ranges and later badness/sieve
results remain separately tracked.",
      "external_inputs": [
        "QUANTITATIVE_IW",
        "GENERAL_VARIATION_SAMPLING",
        "PROGRESSION_ESTIMATE",
        "REN_GHOSH",
        "SMOOTH_MULTIFREQUENCY",
        "PAGE_CLASSES",
        "QUADRATIC_WEYL_FJK"
      ]
    },
    "c:orlicz-p": {
      "title": "Quantitative lp maximal bounds near p=1",
      "work_group": "Orlicz and ergodic estimates",
      "status": "conditional",
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        "nearOneCore_quantitative",
        "nearOneGaugeMoment_integrable",
        "nearOneGaugeMoment_source_p",
        "positivePower_primitive",
        "nonneg_power_layerCake",
        "truncated_rpow_kernel",
        "truncated_scaled_power_kernel",
        "truncated_distribution_strong_power",
        "truncated_gauge_strong_power",
        "nearOneStrong_power_budget",
        "actual_finite_strong_maximal_norm",
        "positiveAverage_finite_strong_extension",
        "section7_strong_assembly",
        "corollary_7_6",
        "corollary_7_6_closed_signed_maximal"
      ],
      "expositions": [
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        "docs/SECTION_7_STATEMENT_REVIEW.md",
        "blueprint/Checkpoint_839_NearOneGaugeMoment.md",
        "blueprint/Checkpoint_841_NearOneMomentExponential.md",
        "docs/COROLLARY_7_6_COMPLETION.md",
        "blueprint/Checkpoint_984_CorollarySevenSix.md",
        "docs/PAPER_COMPLETION.md"
      ],
      "scope": "Complete printed Corollary 7.6 as an implication of the already recorded Theorem 7.2
independent inputs. One absolute D > 0 works for all three actual families, every 1 < p <= 2 and every
integer lp input. The literal all-integer-cutoff maximal function is proved to belong to lp with norm at
 most exp(D/(p-1)*log(exp(1)/(p-1))) times the input norm. The actual truncated modular estimate,
general layer cake, singular power integral, quantitative constant absorption, p-th-root conversion and
all-input finite/Fatou extension are proved. At checkpoint1264 the current public export consumes the
proved finite signed maximal theorem at constant6; a supplied RawFiniteSignedMaximalInput and its
constant R are no longer hypotheses.",
      "remaining": "No remaining source clause within the agreed conditional scope; no new strong
maximal or layer-cake input is introduced. Theorem7.2's independent hypotheses remain explicit. The
```

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ergodic Corollary7.46 and Theorem1.5(ii) are now separately completed under their reviewed independent
inputs.",
      "external_inputs": [
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        "PROGRESSION_ESTIMATE",
        "REN_GHOSH",
        "SMOOTH_MULTIFREQUENCY",
        "PAGE_CLASSES",
        "QUADRATIC_WEYL_FJK"
      ]
    },
    "p:IWinput7": {
      "title": "Quantitative Ionescu-Wainger application",
      "work_group": "Orlicz and ergodic estimates",
      "status": "conditional",
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        "iwNearOneExponent_reciprocal",
        "iwRho_rank",
        "iwWidth_denominator_separation",
        "iwFrequencySet_mem",
        "taoArc_denominator_bound",
        "quantitative_iw_frequency_geometry",
        "quantitative_iw_goodness",
        "quantitative_iw_construction_good",
        "quantitative_iw_multiplier",
        "iwCyclicSymbol_real",
        "quantitative_iw_sampling",
        "iw_finite_lift_diagonal",
        "iw_finite_lift_size",
        "iw_narrow_lift_diagonal",
        "quantitative_iw_real_construction_good",
        "quantitative_real_iw_smooth_multiplier",
        "quantitative_real_iw_schwartz_sampling",
        "cyclicRealOperator_Lp_bound",
        "cyclicRealOperator_TV_one_bound",
        "schwartz_integer_test_summable",
        "cyclicFunctionLp_size",
        "source_iw_rough_multiplier_norm",
        "quantitative_real_iw_full_sampling",
        "fullIWInterpolation_diagonal",
        "fullIWInterpolation_unique",
        "fullIWInterpolation_source_bounds",
        "quantitative_full_shannon_diagonal",
        "iw_frequency_in_cyclic_basis",
        "iw_coverage_all_fraction_points",
        "fullIntegerKernelMultiplier_unique",
        "iw_source_exponent_one_le",
        "full_iw_source_analytic_clauses",
        "proposition_quantitative_IW_full"
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      "expositions": [
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        "docs/SECTION_7_STATEMENT_REVIEW.md",
        "docs/SECTION_7_IW_BOUNDARY.md",
        "blueprint/Checkpoint_840_QuantitativeIWParameters.md",
        "blueprint/Checkpoint_845_IWFiniteFrequencyGeometry.md",
        "blueprint/Checkpoint_847_IWGoodness.md",
        "blueprint/Checkpoint_852_IWMultiplierInterface.md",
        "docs/SECTION_7_SAMPLING_BOUNDARY.md",
        "blueprint/Checkpoint_925_CyclicLiftCharacters.md",
        "blueprint/Checkpoint_930_IWNarrowLiftIdentity.md",
        "docs/REAL_IW_EXTENSION_REVIEW.md",
        "docs/FULL_IW_EXTERNAL_BOUNDARY.md",
        "docs/FULL_QUANTITATIVE_IW_COMPLETION.md"
      ],
      "scope": "All four source clauses are complete conditional implications for one common
construction at every real Q>=2^k, r>=2 and k>=4. The literal B, epsilon and C_rk are retained; P<=3^Q,
every a/q modulo one with q<=Q, all denominator bounds and full goodness are proved. The rough
multiplier conclusion is an actual bounded operator on the entire integer lp space, with the exact
finite-convolution Fourier kernel, uniqueness and norm <=C_rk times the actual full real Lp operator
norm. The general full-Lp real/cyclic product bound and rank-one coefficient factorization are proved.
Both major-arc sampling directions, actual unique interpolation on the full distributional integer lp
Fourier-support class, the inverse-C_rk source inequality, and all finite-dimensional normed complex
Shannon targets throughout 1<=p<=infinity are retained. Point sampling uses the standard continuous
representative convention. General Tao construction, full adelic IW and full sampling theorems,
including the explicitly described rough/full-class/finite-quotient realization bridges, remain
independent unproved inputs. The earlier narrower endpoint results retain their own hypotheses.",
      "remaining": "",
      "external_inputs": [
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        "FULL_DOMAIN_ADELIC_IW",
        "FULL_DOMAIN_TAO_SAMPLING"
      ]
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    "l:continuousestimates7": {
      "title": "Uniform continuous annular increment and maximal estimates",
      "work_group": "Orlicz and ergodic estimates",
      "status": "conditional",
      "lean_anchors": [
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        "continuousAnnularSignedMultiplier_L2",
        "continuousAnnularSignedKernel_fourier",
        "continuousAnnularSignedKernel_hormander",
        "continuousAnnularSignedMultiplier_Lp",
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        "continuousAnnularCutoffSignedMultiplier_Lp",
        "annularCutoffYoungMultiplier_Lp",
        "annularOscillatoryModel_contDiff",
        "realContinuousAnnularKernel_fourier",
        "realContinuousAnnularConvolution_continuous_scale",
        "realContinuousAnnularMax_countable",
        "realContinuousAnnular_maximal_Lp",
        "continuousAnnularSigned_full_Lp",
        "source_continuous_annular_estimates"
      ],
      "expositions": [
        "docs/SECTIONS_3_TO_8_REVIEW.md",
        "docs/FULL_CONTINUOUS_SOURCE_COMPLETION.md",
        "blueprint/Checkpoint_1044_FullContinuousSource.md"
      ],
      "scope": "Complete printed continuous lemma on the full Lp domain. One absolute positive constant
is chosen before beta, p, J, signs and f. The actual signed-increment kernel has exactly the source
multiplier and a full-Lp bound for every1<p<=2, arbitrary complex signs of modulus<=1 and
every0<beta<=1. The maximum is over every real N>=2, represented as a proper extended-real supremum; a
fixed countable dense set and proved scale continuity establish measurability, with a common
almost-everywhere bound, finiteness and actual Lp membership. The literal real-scale kernel/Fourier
identity and all annular L2/Hormander hypotheses are proved. The general one-sided Hardy\u2013Littlewood
 and scalar Hormander theorems in their full-Lp forms remain unproved explicit independent inputs.",
      "remaining": "No remaining source-specific clause within the agreed conditional scope. The
full-domain general theorem interfaces and their exact published-to-Lean formulations remain external;
they are not claimed to have been derived from the earlier restricted-domain inputs.",
      "external_inputs": [
        "FULL_DOMAIN_CONTINUOUS_STANDARDS"
      ]
    },
    "p:orlicz-low": {
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      "work_group": "Orlicz and ergodic estimates",
      "status": "conditional",
      "lean_anchors": [
        "iw_annular_single_multiplier",
        "iw_annular_signed_increments",
        "iw_short_scale_maximum",
        "iw_signed_annular_operator_identity",
        "iw_signed_annular_multiplier",
        "iw_short_transition_maximum",
        "iw_short_transition_cost_bound",
        "iw_transition_log_cost_budget",
        "iw_finite_input_low_majorant",
        "iw_narrow_integer_majorant",
        "iw_large_remainder_majorant",
        "iw_narrow_lift_diagonal",
        "iw_finite_lift_size",
        "iw_full_low_maximal_bound",
        "iw_low_all_integer_lp",
        "finite_input_maximal_extension",
        "allInputLpMaximalBound_maximum",
        "iw_full_square_low_maximal_bound",
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        "realIWAdjustedBase_width",
        "realIWAdjusted_cost",
        "real_iw_full_low_maximal_bound",
        "real_iw_low_all_integer_lp",
        "real_iw_source_construction_and_low",
        "allInputLpMaximalBound_integerLp_maximum"
      ],
      "expositions": [
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        "docs/SECTION_7_STATEMENT_REVIEW.md",
        "docs/SECTION_7_IW_BOUNDARY.md",
        "docs/SECTION_7_CONTINUOUS_BOUNDARY.md",
        "docs/SECTION_7_SIGNED_MAXIMAL_BOUNDARY.md",
        "blueprint/Checkpoint_903_IWAnnularLp.md",
        "blueprint/Checkpoint_907_IWShortScaleMaximum.md",
        "blueprint/Checkpoint_912_IWShortTransitionMaximum.md",
        "docs/SECTION_7_SAMPLING_BOUNDARY.md",
        "blueprint/Checkpoint_934_IWFullLowMaximum.md",
        "blueprint/Checkpoint_938_IWAllInputLowMaximum.md",
        "docs/REAL_IW_LOW_COMPLETION.md"
      ],
      "scope": "The full actual dyadic IW low maximum is proved at every real Q>=2^k, uniformly in the
cyclic measure of variation at most one and 0<beta<=1, for the real-Q geometric construction and for all
 integer lp inputs at p_r. The width, denominator bound and logarithmic cost are the literal real-Q
quantities. A single harmless enlargement of the universal parameter constant from C to C^2 is made
once. The arithmetic set, period, rational coverage through Q, denominator bound and goodness are
retained; the maximal estimate uses exactly that set, period and real-Q cutoff. The adjusted-base proof
and cost comparison are checked, as are all own symbol, smoothness, sampling, short-scale, narrow,
remainder and all-input extension steps. The independent general standard inputs remain explicit; no
current-paper conclusion is an external assumption.",
      "remaining": "No own-paper deduction remains for this proposition. The recorded independent
general construction, IW multiplier, sampling, continuous and signed-maximal theorems are explicit
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```
unformalized inputs, not proved Lean axioms.",
      "external_inputs": [
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      "status": "conditional",
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        "iw_prime_high_source_scale",
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        "cyclicRationalCoefficient_primitive_induced",
        "fixed_small_page_datum",
        "iw_original_common_comparison",
        "iw_prime_low_maximal_bound",
        "iw_prime_annular_restricted_distribution",
        "iwFullSquare_decomposition",
        "iw_full_square_low_maximal_bound",
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        "iw_full_square_high_source_scale_for_data",
        "realIWUpperBase_cutoff",
        "real_iw_high_comparison",
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        "real_iw_two_low_maximum_norms"
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      "expositions": [
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        "docs/SECTION_7_STATEMENT_REVIEW.md",
        "docs/SECTION_7_IW_BOUNDARY.md",
        "blueprint/Checkpoint_865_IWCharacterModelComparison.md",
        "blueprint/Checkpoint_868_IWPageModelMaximum.md",
        "blueprint/Checkpoint_871_IWPrimeHighSourceScale.md",
        "blueprint/Checkpoint_938_IWAllInputLowMaximum.md",
        "blueprint/Checkpoint_940_IWPrimeRestrictedDistribution.md",
        "docs/SECTION_7_RESTRICTED_COMPLETION.md",
        "docs/THEOREM_7_2_COMPLETION.md",
        "docs/SECTION_7_QUADRATIC_WEYL_BOUNDARY.md",
        "blueprint/Checkpoint_967_IWFullSquareHighSourceScale.md",
        "blueprint/Checkpoint_969_IWFullSquareRestrictedDistribution.md",
        "docs/REAL_IW_HIGH_LOW_COMPLETION.md"
      ],
      "scope": "The complete real-Q high-low proposition is proved as an implication from the recorded
independent standard inputs. For every real Q>=2^k, one unchanged floor-Q arithmetic construction
supplies both prime and ordinary-square decompositions at the literal real-Q cutoff. It retains
period<=3^Q, rational coverage through Q, the literal denominator bound and goodness. Both actual
countable low bounds hold on all integer lp inputs at p_r, with the exact logarithmic real-Q cost. Both
actual high maxima share the same arithmetic set, period and selected Page datum and have a proved
summed Q^(-1/4) L2 bound above the stated exponential starting scale. All operator identities and
applicability are checked. The fixed absolute parameter constant is enlarged once from C to C^2; no
width is rounded and no own-paper endpoint is an external premise.",
      "remaining": "No own-paper deduction remains for this proposition. The recorded independent
general construction, multiplier/sampling, continuous/signed-maximal, smooth multifrequency,
Page/prime-distribution and scalar quadratic-Weyl inputs remain explicit and unproved. Their hypotheses
are not Lean axioms or assertions of truth.",
      "external_inputs": [
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        "GENERAL_VARIATION_SAMPLING",
        "PROGRESSION_ESTIMATE",
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        "PAGE_CLASSES",
        "QUADRATIC_WEYL_FJK"
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      "title": "Ergodic transfer and quantitative endpoint conclusions",
      "work_group": "Orlicz and ergodic estimates",
      "status": "conditional",
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        "ergodic_strong_maximal_source",
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        "section7_ergodic_modular_apply",
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        "orlicz_ergodic_ae_cauchy",
        "ergodicSquareL1Operator_apply_toLp",
        "ergodicSquareL1Operator_norm_le",
        "L1_bounded_truncation_approximation",
        "bounded_L1_cauchy_of_ae_cauchy",
        "uniform_operator_dense_cauchy",
        "bounded_truncation_cauchy_from_input",
        "orlicz_ergodic_convergence",
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        "section7_ergodic_assembly",
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        "corollary_7_46_closed_signed_maximal"
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      "expositions": [
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        "docs/ERGODIC_INDEPENDENT_BOUNDARY.md",
        "docs/SECTION_7_PROGRESS.md",
        "docs/SECTION_7_STATEMENT_REVIEW.md",
        "blueprint/Checkpoint_998_OrliczErgodicConvergence.md",
        "blueprint/Checkpoint_999_CorollarySevenFortySix.md",
        "docs/PAPER_COMPLETION.md"
      ],
      "scope": "Complete printed Corollary7.46 as an implication of the explicit independent inputs, for
 all three actual families and every invertible measure-preserving probability system. One common C >= 1
 serves the strong exponential rate and source Orlicz gauge; one common modular multiplier is fixed
before all systems, families, functions and levels. Every 1 < p <= 2 has the actual strong maximal norm
with almost-everywhere finiteness and actual Lp membership. The proper extended maximum satisfies the
source modular inequality. Every measurable Orlicz member at some positive scale has actual averages
converging almost everywhere and in L1 to the same genuine L1 limit, with integrability of every average
 and both L1 seminorm/integral-norm convergence checked. Own truncation, rescaling, Cauchy/density, L1
operator and limit-identification deductions are proved. At checkpoint1264 the current public export
consumes the proved finite signed maximal theorem at constant6; a supplied RawFiniteSignedMaximalInput
and its constant R are no longer hypotheses.",
      "remaining": "No remaining printed clause within the agreed conditional-formalization scope. The
independent general Calderon strong/modular and bounded-class polynomial convergence inputs, including
their unproved published-theorem/normalization bridges, remain explicit, as do the earlier Theorem7.2
independent inputs. Stronger standalone Section7 auxiliary ranges remain separately partial.",
      "external_inputs": [
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        "PAGE_CLASSES",
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        "ERGODIC_TRANSFER",
        "BOUNDED_SQUARE_POINTWISE"
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      "status": "conditional",
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        "primesLE_image_nth",
        "sum_primesLE_nth",
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        "enumeratedPrimeSquareAverage_cutoff",
        "enumeratedPrimeMaximalENorm_eq",
        "ergodic_strong_maximal_source",
        "theorem_1_5_ii",
        "primeSquare_lavictoire_arithmetic",
        "theorem_1_5_i",
        "theorem_1_5_i_literal",
        "theorem_1_5",
        "theorem_1_5_ii_closed_signed_maximal"
      ],
      "expositions": [
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        "docs/ERGODIC_INDEPENDENT_BOUNDARY.md",
        "blueprint/Checkpoint_991_TheoremOneFivePartTwo.md",
        "docs/THEOREM_1_5_I_COMPLETION.md",
        "docs/SECTION_8_COMPLETION.md",
        "docs/SECTION_8_SPACING_CORRECTION.md",
        "verification/section8-source-review-checkpoint1038.json",
        "docs/PAPER_COMPLETION.md"
      ],
      "scope": "Both parts are complete conditional implications. Part(i): the actual enumerated prime
squares are persistently universally L1-bad on every truly nonatomic ergodic measure-preserving
probability system, with every infinite cutoff set and an actual measurable positive-measure divergence
set. CRT, Dirichlet representation, correlation counts, large-prime block construction, all
moment/thickening/difference limits, uniform square-root fibers and the fixed-progression PNT
pushforward are proved. The original badness theorem, Sawyer, Conze, general Weil and fixed-progression
PNT remain separate unproved general inputs. Granville\u2013Kurlberg is not used by part(i). A checked
moment proof replaces the source strict finite-fraction argument, which has an exact kernel
counterexample. Part(ii) is a complete conditional implication: an absolute positive constant before
every invertible measure-preserving probability system, every 1 < p <= 2 and every Lp input; the proper
actual maximum over the first m prime squares for all m >= 1 is finite almost everywhere, belongs to Lp
and has the exact displayed exponential norm bound. The prime-counting/enumeration normalization, finite
 sums, first-prime case and equality of both full maxima are proved with no PNT or enumeration input.
The proof uses the already proved Corollary7.6 and the precise independent general strong Calderon
input, plus the earlier Theorem7.2 independent inputs. It does not use the bounded-class convergence
input. At checkpoint1264 the current public export consumes the proved finite signed maximal theorem at
constant6; a supplied RawFiniteSignedMaximalInput and its constant R are no longer hypotheses.",
      "remaining": "No remaining source clause within the agreed conditional scope. General inputs and
their exact convention bridges remain unproved. The user confirmed the three Section8 edits; the
selected project manuscript now contains the agreed moment proof.",
      "external_inputs": [
        "LAVICTOIRE",
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        "WEIL",
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        "ERGODIC_TRANSFER",
        "QUANTITATIVE_IW",
        "GENERAL_VARIATION_SAMPLING",
        "REN_GHOSH",
        "SMOOTH_MULTIFREQUENCY",
        "PAGE_CLASSES",
        "QUADRATIC_WEYL_FJK",
        "SAWYER",
        "CONZE",
        "FIXED_PROGRESSION_PNT"
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      "work_group": "Badness and convergence",
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        "source_lavictoire_persistent_criterion"
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      "expositions": [
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        "docs/SECTION_8_INDEPENDENT_INPUTS.md"
      ],
      "scope": "Complete persistent-divergence adaptation from original general LaVictoire weak maximal
failure and separately specified Sawyer and Conze principles. The arithmetic record contains actual
cyclic residue sets, CRT, represented classes, odd coprime moduli, exact density infimum, little-o
thickening, far-pair liminf and strict finite-product residue frequencies. For every nonatomic ergodic
probability system and every infinite cutoff set, Lean produces a measurable integrable function and a
measurable positive-measure set with no finite pointwise limit. Nonatomicity means measurable splitting,
 not merely null singletons. Positive moduli include the usual modulus-one convention; prime and cutoff
indices are zero-based enumerations of the source positive-index sequences.",
      "remaining": "No remaining persistent-divergence deduction within the agreed conditional scope.
The original LaVictoire, Sawyer and Conze inputs and their precise published-to-Lean convention bridges
are unproved.",
      "external_inputs": [
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        "SAWYER",
        "CONZE"
      ]
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    "t:grankul": {
      "title": "Quoted Granville-Kurlberg spacing theorem",
      "work_group": "Badness and convergence",
      "status": "external_candidate",
      "lean_anchors": [
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        "GKLocalHypotheses",
        "CyclicPoissonSpacings",
        "unitSquareFamily_gk_local",
        "badnessBlockModulus_squarefree",
        "unitSquareBlock_poisson",
        "unitSquareBlock_gap_tail"
      ],
      "expositions": [
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        "docs/SECTION_8_INDEPENDENT_INPUTS.md",
        "blueprint/Checkpoint_1036_GranvilleKurlbergInterface.md"
      ],
      "scope": "Quoted general deep theorem remains unproved. A precise interface now specifies
arbitrary CRT residue families, uniform actual local correlations, one positive common exponent,
constants chosen before primes/shifts, subpower reciprocal local density and growing spacing along odd
squarefree moduli. The full joint cyclic gap conclusion is implemented. All its unit-square hypotheses
and the actual joint and single-gap applications are proved conditional on that independent general
input and Weil.",
      "remaining": "The quoted general GK theorem and its published-to-Lean cyclic joint formulation
bridge remain external and unproved. It is not needed for the repaired Theorem1.5(i) proof.",
      "external_inputs": [
        "GRANVILLE_KURLBERG"
      ]
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    "l:quadratic-GK": {
      "title": "Quadratic-residue correlation input for spacing",
      "work_group": "Badness and convergence",
      "status": "conditional",
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        "unitSquareIndicator_formula",
        "unitSquareResidues_prime_card_real",
        "quadraticCorrelation_as_product",
        "quadraticSoftCorrelation_error",
        "quadraticCorrelation_Weil_error",
        "quadraticCorrelation_relative_error",
        "quadratic_GK_absolute_error",
        "quadratic_GK_relative_error",
        "quadratic_GK_uniform_correlation",
        "quadratic_GK_tuple",
        "primeResidueDensity_subpower"
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        "docs/SECTION_8_INDEPENDENT_INPUTS.md",
        "blueprint/Checkpoint_1009_QuadraticGKCorrelation.md"
      ],
      "scope": "Complete printed quadratic-residue correlation lemma as an implication of
RawSplitQuadraticWeilInput, the general distinct-linear-factor character-sum bound. The literal finite
residue correlation has explicit absolute error at most2*k*sqrt(p), actual density-normalized relative
error at most3*k*3^k/sqrt(p), and the source GK error with c=1/3 and constant6*k*3^k. Quantification is
uniform over every odd prime and every injective tuple of k shifts, including the source h0=0 case. All
zero corrections, density factors, positivity, finite subset sums and reciprocal local-density subpower
bounds are proved.",
      "remaining": "No remaining printed correlation clause in the agreed conditional scope. The general
 Weil input remains unproved. The separate moment repair and badness deductions are now also complete;
their inputs are separately recorded.",
      "external_inputs": [
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      "status": "conditional",
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        "goldbach_full_exceptional_count",
        "goldbach_section9_count_reduction",
        "goldbach_eventual_arithmetic_budgets",
        "goldbach_section9_eventual_count_reduction",
        "goldbach_quantitative_exceptional_set_squares",
        "goldbach_good_representation_iff",
        "goldbach_source_exceptional_membership",
        "goldbach_ceiling_square_geometry",
        "goldbach_real_cutoff_from_square_bound",
        "goldbach_headline_theorem",
        "goldbach_primitive_one_mod_four_support",
        "goldbach_primitive_double_of_odd",
        "goldbach_primitive_two_squares_criterion",
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        "goldbach_weighted_W_nonneg_general",
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        "goldbach_fixed_smooth_seed_nonzero",
        "goldbach_fixed_smooth_seed_support",
        "goldbach_fixed_smooth_seed_compact_support",
        "goldbach_fixed_smooth_seed_tsupport_subset",
        "goldbach_smooth_seed_exists",
        "goldbach_from_sifted_to_primes_complete",
        "goldbach_truncation_complete",
        "goldbach_section9_master_exact",
        "goldbach_master_sieve_complete",
        "goldbach_section9_headline_complete",
        "goldbach_section9_exceptional_density_zero"
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      "expositions": [
        "docs/SECTION_9_REDUCTION_COMPLETION.md",
        "docs/RECOVERY_AND_SECTION9_REVIEW_1106.md",
        "blueprint/Checkpoint_1259_GoldbachQuantitativeExceptionalSet.md",
        "blueprint/Checkpoint_1261_GoldbachPrimitiveCriterion.md",
        "blueprint/Checkpoint_1262_GoldbachSectionNineComplete.md"
      ],
      "scope": "The full quantitative exceptional-set theorem is complete: for every A>0, the actual
exceptional set of positive n<X congruent12 mod24 without a prime plus a square-of-prime representation
whose first prime is a primitive sum of two squares plus1 has size O_A(X(log X)^(-A)), eventually at
every real X. The representation and exceptional-set membership are exact. The common-cutoff Section11
estimates, positive numerical gap, arbitrary logarithmic mean-square saving, all arithmetic losses,
small n and the passage from N^2 to real X are discharged. Explicit independent general SW, BV,
Ren/Ghosh, one-stage sieve, effective multiplicative mean, Mertens, normalized sieve values/continuity
and primitive two-square criterion remain unproved theorem parameters. No own-paper endpoint is a
hypothesis. At1259: Section11 and the final quantitative Goldbach theorem are complete in the agreed
conditional scope, using the supplied corrected presieving statement. Every paper endpoint is
discharged; only explicit independent general classical inputs and their normalizations remain unproved.
 Original manuscripts are unchanged and the supplied Section11 replacements still require user
application. No additional completion credit is claimed for stronger unnumbered assertions in alternate
proof routes. At1261: Proved the full PrimitiveTwoSquaresCriterion from pinned Mathlib and elementary
algebra, and proved W(alpha) convergence/nonnegativity for every alpha>1. Constructed a concrete nonzero
 compactly supported smooth seed. The next stage exports the Section9 arithmetic and headline statements
 without the former criterion or seed hypotheses, then reconciles the stale Section9 map. User
manuscript edits remain in progress. At1262 the final endpoint is exported from GoldbachSection11Inputs
alone: the primitive two-square criterion is proved internally and a fixed nonzero smooth compact seed
is constructed. The real-X bound and zero exceptional density need no separate arithmetic or seed
assumption. At1262: Every Section9 numbered source claim and its final quantitative theorem is complete
in the agreed scope. All elementary arithmetic reductions are now proved without an extra primitive
two-square assumption. W(alpha) convergence is proved for every alpha>1 and the smooth seed is
constructed. Only the explicit independent analytic inputs remain for the circle-method/sieve/headline
estimates. The user is applying previous manuscript edits; no application is assumed and original
sources are unchanged.",
      "remaining": "",
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        "REN_GHOSH",
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        "EFFECTIVE_HALF_DIMENSIONAL_MEAN",
        "CLASSICAL_INTERVAL_MERTENS",
        "CLASSICAL_FULL_INTERVAL_MERTENS",
        "CLASSICAL_PRIME_LOG_MERTENS",
        "CLASSICAL_PRIME_MERTENS_RATIOS"
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        "p:sieveneedlower8",
        "p:sieveneedupper8",
        "l:maintermsize8",
        "l:numericalcomparison"
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      "title": "Reduction from a sifted set to prime-square representations",
      "work_group": "Goldbach and sieve",
      "status": "proved",
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        "goldbach_shared_prime_eq",
        "goldbach_shared_divisor_exception",
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        "goldbach_prime_divisor_power_error",
        "goldbach_representation_admissible",
        "goldbach_factor_coprime_of_not_exception",
        "goldbachSiftedValues_card",
        "goldbach_full_sifted_inclusion",
        "goldbach_full_sifted_count",
        "goldbach_from_full_sifted",
        "goldbach_coprime_zmod_not_both_zero",
        "goldbach_prime_norm_coprime",
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        "goldbach_conjugate_kernel_separation",
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        "goldbach_truncation_complete",
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        "goldbach_master_sieve_complete",
        "goldbach_section9_headline_complete",
        "goldbach_section9_exceptional_density_zero"
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      "expositions": [
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        "blueprint/Checkpoint_1260_GoldbachPrimitiveProducts.md",
        "blueprint/Checkpoint_1261_GoldbachPrimitiveCriterion.md",
        "blueprint/Checkpoint_1262_GoldbachSectionNineComplete.md"
      ],
      "scope": "The complete first Section9 reduction is proved from its ordinary range hypotheses. The
primitive sum-of-two-squares criterion is now a project theorem, not an additional input. Actual finite
sifted and good representation sets and the uniform O(N^epsilon) exception are retained. At1262: Every
Section9 numbered source claim and its final quantitative theorem is complete in the agreed scope. All
elementary arithmetic reductions are now proved without an extra primitive two-square assumption.
W(alpha) convergence is proved for every alpha>1 and the smooth seed is constructed. Only the explicit
independent analytic inputs remain for the circle-method/sieve/headline estimates. The user is applying
previous manuscript edits; no application is assumed and original sources are unchanged.",
      "remaining": "",
      "external_inputs": [],
      "prior_scope_before_arithmetic_discharge": "The actual finite lower sifted set and primitive
prime-plus-prime-square representation set satisfy the exact cardinal inequality with error at most
omega(n-1). A uniform O_epsilon(N^epsilon) bound is proved with the source quantifiers. The use of r-1
rather than r is matched by an injective reindexing. The only extra input is the classical primitive
sum-of-two-squares criterion for m congruent2 modulo4. At1260: The Section9 final quantitative counting
endpoint was already complete conditionally at1259. This checkpoint proves the elementary
primitive-representation multiplication step from Mathlib Fermat and explicit finite-field algebra. The
full primitive two-square criterion and its consumption are the next obligations; no source claim is
newly promoted yet. The Section11 edits remain in progress, not assumed applied. At1261: Proved the full
 PrimitiveTwoSquaresCriterion from pinned Mathlib and elementary algebra, and proved W(alpha)
convergence/nonnegativity for every alpha>1. Constructed a concrete nonzero compactly supported smooth
seed. The next stage exports the Section9 arithmetic and headline statements without the former
criterion or seed hypotheses, then reconciles the stale Section9 map. User manuscript edits remain in
progress."
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    },
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        "goldbach_factor_coprime_of_not_exception",
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 by O(N^epsilon), including repeated3 mod4 prime factors and the actual source factorization set. The
primitive criterion is internally proved; only the normal source range hypotheses remain. At1262: Every
Section9 numbered source claim and its final quantitative theorem is complete in the agreed scope. All
elementary arithmetic reductions are now proved without an extra primitive two-square assumption.
W(alpha) convergence is proved for every alpha>1 and the smooth seed is constructed. Only the explicit
independent analytic inputs remain for the circle-method/sieve/headline estimates. The user is applying
previous manuscript edits; no application is assumed and original sources are unchanged.",
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 source n>12 in its residue class and N>=2, every real1<alpha<2, z=N^(1/alpha), with an explicit
nonnegative integer error at most omega(n-1), hence O_epsilon(N^epsilon) uniformly. The exact two-prime
factorization retains multiplicity, positive remaining factor, ordering, both directions of the
sifted-set classification and r-1 cardinal reindexing. At1260: The Section9 final quantitative counting
endpoint was already complete conditionally at1259. This checkpoint proves the elementary
primitive-representation multiplication step from Mathlib Fermat and explicit finite-field algebra. The
full primitive two-square criterion and its consumption are the next obligations; no source claim is
newly promoted yet. The Section11 edits remain in progress, not assumed applied. At1261: Proved the full
 PrimitiveTwoSquaresCriterion from pinned Mathlib and elementary algebra, and proved W(alpha)
convergence/nonnegativity for every alpha>1. Constructed a concrete nonzero compactly supported smooth
seed. The next stage exports the Section9 arithmetic and headline statements without the former
criterion or seed hypotheses, then reconciles the stale Section9 map. User manuscript edits remain in
progress."
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finite cardinal upper bound with error omega(n-1), and uniform O_epsilon(N^epsilon) bound are proved. No
 analytic, sieve, primitive two-square or own-paper endpoint is assumed. All eight closed cutoff
occurrences are verified in the latest upload.",
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 adopted closed multiplier cutoff. Its exact finite error is at most2 omega(n-1), and its uniform
O(N^epsilon) consequence is exported. No primitive-criterion or analytic assumption is needed for this
arithmetic reduction. At1262: Every Section9 numbered source claim and its final quantitative theorem is
 complete in the agreed scope. All elementary arithmetic reductions are now proved without an extra
primitive two-square assumption. W(alpha) convergence is proved for every alpha>1 and the smooth seed is
 constructed. Only the explicit independent analytic inputs remain for the circle-method/sieve/headline
estimates. The user is applying previous manuscript edits; no application is assumed and original
sources are unchanged.",
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assembled for the user-corrected closed-cutoff L, with error at most twice omega(n-1) and the full
uniform O_epsilon(N^epsilon) consequence. Its own truncated equality and upper bound are discharged by
the earlier proved modules. The independent classical primitive two-square criterion remains explicit.
At1260: The Section9 final quantitative counting endpoint was already complete conditionally at1259.
This checkpoint proves the elementary primitive-representation multiplication step from Mathlib Fermat
and explicit finite-field algebra. The full primitive two-square criterion and its consumption are the
next obligations; no source claim is newly promoted yet. The Section11 edits remain in progress, not
assumed applied. At1261: Proved the full PrimitiveTwoSquaresCriterion from pinned Mathlib and elementary
 algebra, and proved W(alpha) convergence/nonnegativity for every alpha>1. Constructed a concrete
nonzero compactly supported smooth seed. The next stage exports the Section9 arithmetic and headline
statements without the former criterion or seed hypotheses, then reconciles the stale Section9 map. User
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        "blueprint/Checkpoint_1167_GoldbachEuclideanProduct.md",
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        "docs/GOLDBACH_COMPLETION_PROGRESS_1170.md",
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        "blueprint/Checkpoint_1211_GoldbachMajorCompletionComplete.md",
        "blueprint/Checkpoint_1212_GoldbachCircleMethodComplete.md",
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      "scope": "Complete source circle-method mean-square approximation for every A,delta>0, with
B0,C,K,N0 chosen before N and every arithmetic family. Covers all B>=B0, exact strict real
L<N^(2-delta), positive ell,k with k<V(ell), every coprime residue class, arbitrary complex coefficients
 of norm<=1, and all n<N^2. The stronger uniform cutoff statement holds for every 2<=U<=N,W>=2. The
final source theorem removes cutoffs using the proved equation-imposed bounds, and uses the actual
finite prime-pair count, actual local Euler factor G_R and actual autocorrelation singular integral H_R.
 Fourier inversion, all three error identifications, shared constants, minor estimate, major
replacement, arithmetic completion and smoothing are consumed internally. Its only unformalized
hypotheses are independent ordinary SW prime counts, ordinary unweighted BV prime-count distribution,
and uniform raw Ren/Ghosh quadratic estimates. No own-paper conclusion is assumed. At1212: Completed all
 Section10 source statements and the full circle-method implication; the three independent raw deep
inputs remain explicit and unproved. At1213: Completed every Section10 source statement and the full
source circle-method implication, including the original smooth compact seed and exact normalized
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integral. Only the three agreed independent raw deep inputs remain for this circle-method implication.",
      "remaining": "None within the full source circle-method implication. The three independent raw
distribution/exponential-sum inputs remain unproved; Section11 and the headline Goldbach result remain
separate work.",
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        "blueprint/Checkpoint_1226_GoldbachLowerSieveBridge.md",
        "blueprint/Checkpoint_1227_GoldbachLowerSieveApplication.md",
        "blueprint/Checkpoint_1229_GoldbachLowerCutoffFreedom.md",
        "blueprint/Checkpoint_1230_GoldbachLowerMainTermSize.md",
        "blueprint/Checkpoint_1258_GoldbachSection11Complete.md"
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      "scope": "Full source lower-sieve proposition, including the actual representation sequence,
original X_n and Euler product, coefficients before n, a common error tending to zero, and the N^4(log
N)^(-A) mean-square remainder bound. The pointwise theorem holds throughout the source retained range
(indeed all n=12 mod24). General classical sieve/Mertens/function-continuity and ordinary
SW/BV/Ren-Ghosh inputs remain explicit and unproved. At1229: Common cutoff compatibility for the
circle-method and full lower-sieve estimates is proved for every requested lower bound on C; main-term
size remains to be joined. At1230: The full source main-term size and the entire lower-bound side are
complete conditionally, including an exponent independent of C, uniform smoothing, one common large
cutoff and finite error-to-exception deductions. At1230 the source main-term size and the full lower
sieve are proved jointly for the same C>=A0+6; finite positive-count and exceptional-cardinality
deductions are checked. At1258: All Section11 numbered claims are complete in the agreed conditional
scope for the explicitly supplied corrected presieving statement. The original weaker statement is not
proved and must be replaced in the user manuscript. The complete joint lower/upper estimates use one
cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary combined logarithmic
mean-square saving. The final Goldbach theorem remains the next deduction.",
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        "blueprint/Checkpoint_1254_GoldbachUpperSieveDensity.md",
        "blueprint/Checkpoint_1255_GoldbachUpperPresieveUniform.md",
        "blueprint/Checkpoint_1256_GoldbachUpperPresieveApplication.md",
        "blueprint/Checkpoint_1257_GoldbachUpperSieveComplete.md",
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      "scope": "The full source upper sieve is complete in uniform epsilon form, with the exact c_(9/8)
F_1(2) W(9/8) coefficient, original actual main term and closed multiplier set. All coefficients precede
 n; source divisor counts, refined preliminary/outer dimensions, uniform varying levels, the shifted
double remainder sum, its circle-method embedding and the fixed-sieve O(N^(1/4)) correction are proved.
The total error has arbitrarily strong logarithmic mean-square saving. Only independent general SW, BV,
Ren/Ghosh, classical one-stage sieve, effective multiplicative mean and Mertens inputs and general sieve
 normalizations remain unproved; no own-paper endpoint is assumed. At1257: Full upper-sieve proposition
p:sieveneedupper8 complete in uniform epsilon form from explicit independent classical inputs, with the
exact coefficient and arbitrary logarithmic mean-square saving. The lower analytic estimate and both
remainder transports are available at a shared prescribed cutoff. The final joint Section11 assembly and
 manuscript synchronization are next. At1258: All Section11 numbered claims are complete in the agreed
conditional scope for the explicitly supplied corrected presieving statement. The original weaker
statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper
estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary
combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction.",
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        "EFFECTIVE_HALF_DIMENSIONAL_MEAN",
        "CLASSICAL_INTERVAL_MERTENS",
        "CLASSICAL_FULL_INTERVAL_MERTENS",
        "CLASSICAL_PRIME_LOG_MERTENS",
        "CLASSICAL_PRIME_MERTENS_RATIOS"
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      "scope": "Full source lemma proved for every A0>0 and every fixed C>=A0+6, uniformly over the
exact retained n range. The actual X_n Theta_n(z) is at least a positive constant times N(log
N)^(-(A0/2+6)); the exponent is independent of C. Actual archimedean, smoothing and local-product
arguments are proved; only the already recorded independent interval Mertens products remain external.
The same large C is joined to the complete lower sieve. At1258: All Section11 numbered claims are
complete in the agreed conditional scope for the explicitly supplied corrected presieving statement. The
 original weaker statement is not proved and must be replaced in the user manuscript. The complete joint
 lower/upper estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and
arbitrary combined logarithmic mean-square saving. The final Goldbach theorem remains the next
deduction.",
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      "expositions": [
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      "scope": "The actual W integral has the unconditional exact bound 8 sqrt(2) W(9/8)<=9/32, proved
by positive interval integration and exact rpow primitives. The strict comparison is proved for the
classical normalized sieve values f_(1/2)(9/8)=sqrt(8 exp(gamma)/(9 pi)) log2 and F_1(2)=exp(gamma).
Identifying the generic independent sieve functions with those classical normalized values remains
explicit, within GENERAL_SIEVE; the paper comparison is not assumed. No floating point evaluation or
unchecked evaluator is used. At1253: Proved the exact actual W integral bound, the normalized
half/linear sieve comparison and strict positivity. Classical sieve-function normalizations remain
explicit general input identities. The upper-sieve application and final Goldbach theorem remain open;
no paper endpoint is assumed. At1258: All Section11 numbered claims are complete in the agreed
conditional scope for the explicitly supplied corrected presieving statement. The original weaker
statement is not proved and must be replaced in the user manuscript. The complete joint lower/upper
estimates use one cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary
combined logarithmic mean-square saving. The final Goldbach theorem remains the next deduction.",
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      "scope": "Complete source hybrid minor estimate at1136: the actual circle integral and finite
n<N\u00b2 mean square, exact complementary source arcs, every sufficiently large radius exponent C, and
a single natural threshold uniform in all coefficients, residues, real U and W and printed strict
L\u2080/V_ell ranges. The theorem allows B\u22650 and 0\u2264U\u2264N, with arbitrary W and residues, so
 it is stronger than the stated ranges. Explicit bound K N\u2074(log N)^(-A), K=Z\u00b2(256/log 2),
Z=(2P+3)/log 2, C\u2080=3A+4d+26. Finite Abel, prime-power removal, halving reconstruction, scale
transfer, eventual cutoffs, source progression energy and Bessel are proved. Only the already recorded
independent raw Ren/Ghosh input is assumed.",
      "remaining": "The independent raw UniformRenGhoshInputs remains explicit and unproved. The source
hybrid minor conclusion and all conversion steps are discharged; no internal own-paper hypothesis
remains.",
      "external_inputs": [
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    "l:positiveH9": {
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 exact convolution integral, eta(0)=1 and real nonnegative inverse Fourier transform. At1177: The scaled
 actual eta_N and its square have proved real nonnegative inverse Fourier transforms and exact compact
support inherited from the seed. At1178: The actual eta is real, even, bounded by1, has zero derivative
at0 and satisfies the full global quadratic source cutoff bound. At1179: Two-sided absolute
integrability and an explicit L1 estimate are proved from the exact finite-scale bound and
inverse-three-halves tail. At1180: The exact cutoff-weighted Fourier integral is a proved positive
average of the inverse kernel, with Fubini integrability discharged. At1181: The actual source singular
integral is proved real nonnegative, uniformly in all positive ell and every real n, including the
empty-domain branch. At1182: The actual Euclidean product and smoothed integrand are globally
integrable, and the exact source singular integral satisfies a proved uniform O(N) bound for all
positive ell. At1183: The full quantitative real-line smoothing error bound is proved with absolute
constant6 and every integrability condition discharged. At1184: The actual smoothed and unsmoothed
source integrals satisfy the full uniform O(N/sqrt(R)) comparison, with all integrability and
empty-domain branches proved. At1185: The actual quadratic affine density is globally integrable and has
 the proved exact product Fourier transform, including the scale factor. At1186: The genuine logarithmic
 quadratic density is continuous and integrable, with its exact source product Fourier transform proved
and endpoint exceptional sets checked. At1187: The full source l:positiveH9 is proved: the exact
autocorrelation cutoff, real nonnegative smoothed singular integral, its exact printed one-variable
H_ell domain, and the uniform O(N/sqrt(R)) comparison, including all positive ell and empty-domain
cases. There is no independent analytic input for this lemma. At1213: Completed every Section10 source
statement and the full source circle-method implication, including the original smooth compact seed and
exact normalized integral. Only the three agreed independent raw deep inputs remain for this
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 the major arcs are proved. At1137: Exact strict-cutoff reduction and a uniform one-term endpoint error
are proved for the actual progression sums. At1138: The exact prime counting and logarithmic-integral
progression models, derivatives and lower-endpoint identities are proved. At1139: The actual prime-count
 discrepancy is transported through every smooth unit phase with an explicit quantitative error,
including the lower prime endpoint. At1140: Exact positive-phase linear and quadratic partial-summation
errors are proved from ordinary prime-counting discrepancies, with signs and totient normalizations
checked. At1141: The actual prime-square major-arc model and explicit nonunit/count-discrepancy error
are proved with the exact source sign and normalization. At1142: The fixed-q lcm fibre bound and
complete reduction to ordinary unweighted modulus-distribution errors are proved. At1143: The compatible
 lcm residue sets, q-cardinality bound and generalized CRT existence/converse are proved for arbitrary
positive moduli. At1144: The exact Upsilon coefficient for the actual compatible lcm residue classes is
proved, including both signs and totient factors. At1145: The exact independent ordinary prime-count
SW/BV boundaries are implemented; real-error equivalence, square-model numeric bound and logarithmic
absorption are proved. At1146: The full uniform source SW prime-square major approximation is a complete
 deduction from the independent ordinary prime-count SW input. At1147: Exact prime progression residue
fibres and positive rational-phase grouping are proved for the actual finite sums. At1148: The actual
source linear model and all nonunit/strict-endpoint errors are proved from ordinary lcm-modulus
prime-count discrepancies. At1149: The actual strict source J integral, clamped prime cutoff, and
uniform numeric linear model error are matched exactly. At1150: Actual BV row approximation is proved
under the independent ordinary unweighted BV input, with individual progression cutoffs and all finite
exceptional terms retained. At1151: The printed source ranges imply the ordinary BV level and comparable
 logarithmic saving, with a threshold uniform in every active ell. At1152: The actual source row-scale
sum and coefficient-family cardinality square bound are proved, including all empty-cutoff cases.
At1153: Every retained prime endpoint/nonunit term is absorbed uniformly from the exact printed source
ranges, using a proved power saving. At1154: The full uniform source BV progression-error norm sum is
complete under the independent ordinary unweighted BV input, with all individual cutoffs and finite
errors included. At1155: Exact source major-set geometry, normalized measure, wrap-around lifts,
pairwise disjointness and eventual separation are proved. At1156: Measurable local coordinates and
genuine Euclidean model regularity and norm bounds are proved for the circle Bessel assembly. At1157:
The exact source major models are constructed on the circle, with proved localization and all-Lp
regularity and the required square size bound. At1158: Both full SW/BV major estimates now hold for the
constructed exact source circle models, uniformly in every printed parameter. At1159: The restricted
measurable Bessel inequalities and two-model major-arc mean-square assembly are proved. At1160: The
actual weighted source coefficient identity, model Lp bounds and complete logarithmic mean-square budget
 are proved. At1161: The full source major-arc mean-square approximation l:majorapproximation9 is
complete under ordinary SW and BV, uniformly in all printed data, with strict endpoints and finite
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partial merely because it cites them. At1162: The actual inclusive R-smooth arithmetic series is
absolutely summable, equals the source finite Euler product and admits an exact finite cutoff/tail
split. At1163: The genuine Euclidean source integrals have proved quadratic and linear frequency decay,
with explicit constants and all cutoffs retained. At1164: Unconditional reciprocal-totient harmonic
estimates, including arbitrary divisor progressions, and the exact gcd divisor expansion are proved.
At1165: The full source arithmetic-family harmonic estimate is proved unconditionally with explicit
divisor exponent2 and constant9 exp(2). At1166: Exact arbitrary-multiplier coefficient identities and
the uniform odd-prime bound8 gcd(p,k*ell)/p are proved. At1168: The genuine coefficient has a uniform
prime-power gcd bound, with every two-adic exponent handled explicitly. At1169: General-modulus source
coefficient decay tau(q)^9 gcd(q,k*ell)/q and principal Gauss decay tau(q)^3/sqrt(q) are proved
unconditionally. At1170: The actual weighted source arithmetic-family and finite denominator-tail bounds
 are proved; the full source arithmetic and completion lemmas remain open with explicit remaining
clauses. At1171: Normalized finite unit averages are proved invariant under reduction to every divisor
modulus. At1172: Primitive progression exponential cancellation and the coprime norm bound are proved
from actual finite unit sums. At1173: The actual Upsilon coefficient is an exact lcm-unit expectation
and satisfies exact simultaneous frequency/denominator reduction. At1174: The actual Upsilon coefficient
 has the proved sharp gcd/divisor decay tau(q) gcd(q,k*ell)/q. At1175: The full source
arithmetic-coefficient lemma is proved: absolute convergence, the all-prime infinite local Euler series
and its exact source G_R value, all three individual bounds with the common divisor exponent9, and the
complete real-cutoff harmonic family estimate. There are no external inputs for this lemma.",
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      "scope": "Complete actual source major-completion mean-square estimate for every A,delta>0, all
sufficiently large C, all B>=0 and sufficiently large N, exact strict real arithmetic cutoffs, arbitrary
 coprime residue representatives and coefficients of norm at most one, and every 2<=U<=N,W>=2. Constants
 and thresholds precede N and all families. The actual major models, Euler factor G_R and singular
integral H_R are used. The sharp cutoff comparison, separated additional denominators, finite arithmetic
 completion and quantitative Rankin tail are proved internally. No independent deep input is used. The
source W<=N^2 restriction is unnecessary. The proof uses a weaker intermediate power of R that yields
the same prescribed logarithmic saving after increasing C. At1211: Completed the full source
major-completion lemma without external deep inputs.",
      "remaining": "None within the source major-completion statement. The independent SW/BV replacement
 and Ren/Ghosh estimates belong to separate lemmas.",
      "external_inputs": []
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      "expositions": [
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      "scope": "Complete lower and upper fixed-sieve comparison, with actual sieve prime sets and actual
 closed multiplier set. The lower inequality is proved by inclusion, including the prime3 boundary. The
upper comparison uses an exact finite exceptional-pair inclusion and the bound sum over p|n of
tau(n-p^2-1) <= K_epsilon N^epsilon. One positive constant is chosen before N,n,z. Holds for every
N>=1,0<n<=N^2 and every real z, hence the full retained source range. No deep input is used. At1215: The
 full source fixed-sieve comparison is proved without deep inputs. The local density, presieving and
final sieve estimates remain open.",
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      "scope": " At1216: Proved the exact lower sieve sequence/progression count and its actual
circle-method remainder decomposition. The full fixed-sieve comparison remains proved; local densities
and later sieve conclusions remain open. At1219: Actual unit-square root counts, both candidate
prime-density bounds including p=5, and the explicit multiplicative sieve function are proved without
deep inputs. Full local-density normalization is the next internal obligation. At1220: Proved the
conductor-sensitive coefficient identity, all actual prime-factor cases, the exact factors at2 and3,
residue invariance, and normalized prime adjunction without division by the global base factor. Full
local-density source closure is reserved for the final product and CRT assembly. At1221: complete actual
 normalized Euler-product factorization, exact source CRT residues, inclusive prime cutoff,
multiplicativity and all density bounds are proved without external inputs. The upper base factor may
vanish; no division by it is used. At1221: Both full source local-density lemmas are proved without new
external inputs: exact original Euler normalization, inclusive cutoff, lower quotient identification,
source CRT specializations, upper multiplicativity and p=5, with common zero upper base factors
retained. Presieving and the later sieve estimates remain open.",
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        "blueprint/Checkpoint_1220_GoldbachSieveConductor.md",
        "blueprint/Checkpoint_1221_GoldbachSieveDensityFactorization.md"
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      "scope": " At1217: Proved the exact upper sieve sequence/progression count, including all prime
exclusions, source multiplier arithmetic and residue independence from n. Local-density factorization
and the later analytic sieve estimates remain open. At1219: Actual unit-square root counts, both
candidate prime-density bounds including p=5, and the explicit multiplicative sieve function are proved
without deep inputs. Full local-density normalization is the next internal obligation. At1220: Proved
the conductor-sensitive coefficient identity, all actual prime-factor cases, the exact factors at2 and3,
 residue invariance, and normalized prime adjunction without division by the global base factor. Full
local-density source closure is reserved for the final product and CRT assembly. At1221: complete actual
 normalized Euler-product factorization, exact source CRT residues, inclusive prime cutoff,
multiplicativity and all density bounds are proved without external inputs. The upper base factor may
vanish; no division by it is used. At1221: Both full source local-density lemmas are proved without new
external inputs: exact original Euler normalization, inclusive cutoff, lower quotient identification,
source CRT specializations, upper multiplicativity and p=5, with common zero upper base factors
retained. Presieving and the later sieve estimates remain open.",
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 independent general one-stage sieve inputs. The exact source matched is the separately supplied
Section11_Reconciled.tex: X>=0, g(1)=1, fixed R=(log N)^C, R+1<=z<=D1<=N^A, refined1+K/logw dimension
bounds, and the preliminary endpointR+1. Coefficients precede the density/sequence and n, have absolute
value<=1 and exact level support; the common relative error tends to zero. The literal original weaker
manuscript statement is not claimed proved; replacement is required in the user manuscript. No original
source file was changed. At1258: All Section11 numbered claims are complete in the agreed conditional
scope for the explicitly supplied corrected presieving statement. The original weaker statement is not
proved and must be replaced in the user manuscript. The complete joint lower/upper estimates use one
cutoff and n-independent coefficients, exact constants, C_A0=A0/2+6 and arbitrary combined logarithmic
mean-square saving. The final Goldbach theorem remains the next deduction.",
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        "blueprint/Checkpoint_1241_GoldbachWeightedMeanCutoffs.md",
        "blueprint/Checkpoint_1242_GoldbachWeightedSmoothing.md",
        "blueprint/Checkpoint_1243_GoldbachWeightedPrimeEndpoints.md",
        "blueprint/Checkpoint_1244_GoldbachWeightedDoubleGeometry.md",
        "blueprint/Checkpoint_1245_GoldbachWeightedLargeRectangle.md",
        "blueprint/Checkpoint_1246_GoldbachWeightedFiniteKernel.md",
        "blueprint/Checkpoint_1247_GoldbachWeightedScalarBounds.md",
        "blueprint/Checkpoint_1248_GoldbachWeightedKernelCoefficient.md",
        "blueprint/Checkpoint_1249_GoldbachWeightedKernelUniform.md",
        "blueprint/Checkpoint_1250_GoldbachWeightedMass.md",
        "blueprint/Checkpoint_1251_GoldbachWeightedErrorBounds.md",
        "blueprint/Weighted_Upper_Main_Term_Complete.md",
        "blueprint/Checkpoint_1252_GoldbachWeightedSourceBridge.md"
      ],
      "scope": "Complete source weighted upper main-term comparison, in uniform epsilon form, with the
actual multiplier set, local factors, finite Euler products and singular integrals. For every A>0, every
 fixed C>=A+24 and eta>0, one threshold in N works for all n=12 mod24 with N^2(log N)^(-A)<=n<N^2. The
coefficient is exactly c_(9/8) W(9/8)+eta. Strong multiplicativity, the uniform mean instantiation, both
 finite Abel arguments, W identity, all small-denominator and endpoint terms, moving logarithms,
integration, smoothing and relative errors are proved. Only the independent generic effective mean-value
 corollary and classical fixed-progression Mertens bounds remain explicit unproved hypotheses. The
source lemma statement is unchanged; two proof replacements are supplied but not applied. At1252: The
```

```
full source weighted upper main-term comparison l:wumatomakisum is complete as a checked implication
from independent generic effective mean-value and classical Mertens inputs, with the exact printed
constant, actual source definitions, uniform retained-range quantifiers, and every summation,
integration and relative-error obligation discharged.",
      "remaining": "",
      "external_inputs": [
        "EFFECTIVE_HALF_DIMENSIONAL_MEAN",
        "CLASSICAL_PRIME_LOG_MERTENS",
        "CLASSICAL_PRIME_MERTENS_RATIOS",
        "CLASSICAL_INTERVAL_MERTENS"
      ]
    }
  },
  "external_inputs": [
    {
      "id": "PNT_LIMITS",
      "status": "consumed_explicit_hypothesis",
      "scope": "PrimeAsymptoticInputs: theta(y)/y tends to 1 and the weighted reciprocal-square-root
prime sum has leading constant 2. The external PNT bridge is not integrated.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "PROGRESSION_ESTIMATE",
      "status": "consumed_explicit_hypothesis",
      "scope": "The older PrimeSquareDistributionInputs supplies raw compatible progression/Ren/Ghosh
data for Theorem1.1. The refined PagePrimeSquareDistributionInputs c additionally returns genuine
optional primitive-character Page-zero data in the source region, and is proved to supply the older
interface. One selected family is integrated into all clauses of Proposition4.19 and the Proposition6.21
 high maximum. Existence of the refined raw data remains an independent analytic hypothesis; it is not
derived from the older interface alone. No own-paper operator conclusion is assumed.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "REN_GHOSH",
      "status": "consumed_explicit_hypotheses",
      "scope": "Published raw quadratic exponential-sum bounds are explicit inputs. An own-paper final
minor-arc estimate is not assumed.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "SMOOTH_MULTIFREQUENCY",
      "status": "consumed_explicit_compound_hypothesis",
      "scope": "SmoothCircleBourgainInput C: a general discrete smooth multifrequency maximum. Existence
 is unproved. The boundary includes smooth real theory, sampling, normalization and the fixed bump; it
is not a theorem imported from Mathlib. Used in Proposition 6.21, whose actual annular,
all-starting-scale and denominator-truncation deductions are now proved. The same independent input is
consumed by the full Proposition4.19 assembly. This hypothesis is not used by Theorem 1.1.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "PAGE_CLASSES",
      "status": "consumed_explicit_general_hypothesis_source_consistency_proved",
      "scope": "RawPageUniquenessInput c: standard Landau-Page uniqueness for primitive nonprincipal
real Dirichlet characters and their real zeros, expressed using ordinary partial sums below 1; its
analytic-continuation bridge remains external. The source-region to conductor-region deduction,
uniqueness in each dyadic conductor class, actual-data finite-class injection and the changing-character
 maximum are proved. Source-compatible PageZeroDatum existence remains the refined raw distribution
hypothesis. Its scale-by-scale selection and integration into the common Proposition4.19 operator family
 are proved; existence is not proved from the older fixed-scale interface. No operator conclusion is
assumed and no Lean axiom is declared. See docs/PROPOSITION_6_21_COMPLETION.md.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "QUANTITATIVE_IW",
      "status": "consumed_explicit_general_construction_multiplier_and_sampling_hypotheses",
      "scope": "RawTaoArcConstructionInput, RawIWMultiplierInput, RawMajorArcSamplingInput and
RawNarrowShannonInput are precise independent Tao Lemma5.1, Theorems1.7/1.6 and Proposition1.5
specializations. Their existence is unproved; they are explicit parameters, not axioms. Actual
finite-input Fourier lifts, normalization, diagonal, smooth applicability and the full finite-input
dyadic low bound are proved. The analytic-top typo was corrected with an exact hash-declared rebuild
at897. The prime high theorem does not use the multiplier predicate. See docs/SECTION_7_IW_BOUNDARY.md
and docs/SECTION_7_SAMPLING_BOUNDARY.md.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "GENERAL_VARIATION_SAMPLING",
      "status": "explicit_independent_general_inputs_consumed_actual_low_applications_proved",
      "scope": "RawScalarHormanderInput, RawOneSidedHLInput, RawYoungMultiplierInput,
RawRieszThorinMultiplierInput are precise independent general standard theorem hypotheses for arbitrary
kernels, functions, multipliers or scalar sequences. All annular applicability, actual continuous and
narrow cyclic maxima, exact Fourier identifications, countable remainder and finite-input full IW low
assembly are proved. No own-paper low or restricted bound is a field. See the continuous, signed-maximal
 and sampling boundary documents. The finite signed maximal interface is proved internally at constant6
by rawFiniteSignedMaximalInput_proved (checkpoint1263), and is no longer an independent assumption.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "ERGODIC_TRANSFER",
      "status": "precise_unproved_independent_interfaces_implemented",
      "scope": "RawCalderonStrongInput and RawCalderonModularInput: general transference for arbitrary
positive finite convolution families, exponents/norm constants and continuous nonnegative gauges. Their
```

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independent truth and exact published-theorem/action/modular bridge remain unproved. No own-paper
endpoint is an input. See docs/ERGODIC_INDEPENDENT_BOUNDARY.md.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "LAVICTOIRE",
      "status": "precise_unproved_general_input_consumed",
      "scope": "RawOriginalLaVictoireInput: original Theorem3.1 for arbitrary increasing sequences with
literal cyclic arithmetic hypotheses, concluding failure of any universal weak L1 maximal constant along
 every infinite cutoff set. Its existence and exact zero-based/probability/weak-norm formulation bridge
remain unproved. The adapted own-paper persistent criterion is proved using separate Sawyer and Conze
inputs.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "GRANVILLE_KURLBERG",
      "status": "precise_unproved_quoted_general_interface_consumed",
      "scope": "RawGranvilleKurlbergInput: general CRT-family local-correlation/subpower hypotheses
imply full joint cyclic Poisson gap limits. Published-to-Lean cyclic joint formulation bridge and
theorem proof remain unformalized. Actual unit-square hypotheses/application are proved. This optional
input is not used by Theorem1.5(i).",
      "declared_as_Lean_axiom": false
    },
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      "id": "WEIL",
      "status": "consumed_explicit_general_hypothesis",
      "scope": "RawSplitQuadraticWeilInput is the precise weakened classical quadratic character-sum
bound for arbitrary nonempty sets of distinct linear factors over every odd prime field. Its existence
remains unproved. It supplies no own quadratic-residue correlation, spacing, moment or badness
conclusion; the source-specific applications remain deductions.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "BOMBIERI_VINOGRADOV",
      "status": "consumed_explicit_hypothesis",
      "scope": "RawPrimeBombieriVinogradovInput: classical general ordinary prime-count distribution
with endpoint maximum, unweighted modulus sum, square-root level and logarithmic saving. The fixed-q lcm
 fibre theorem reduces the paper reindexing to this ordinary input; the own-paper circle-method
conclusion remains internal. The actual linear major-error row sum is deduced at1150, with all
exceptional terms explicit. The complete full source progression major estimate is deduced at1154 and
the source major-arc mean-square lemma at1161.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "GENERAL_SIEVE",
      "status": "independent_single_stage_interfaces_implemented",
      "scope": "See ROSSER_IWANIEC_SINGLE_STAGE for the precise independent general inputs. The signed
two-stage presieve, full lower and upper applications, numerical comparison, common-cutoff assembly and
final Goldbach theorem are complete checked conditional implications.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "WEIGHTED_PRIME_COUNTS",
      "status": "retired_unconsumed_planning_placeholder",
      "scope": "No weighted-prime or paper main-term conclusion is an independent input. The source
l:wumatomakisum is proved from the independently stated general effective mean and classical Mertens
interfaces. This earlier planning placeholder is not consumed.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "FULL_SQUARE_DECOMPOSITION",
      "status": "consumed_explicit_compound_hypothesis",
      "scope": "FullSquareDecompositionInput C: the independent ordinary-square high/low estimate in
global operator norm form. The analytic theorem and its globalization remain external. The plus-sign
PositiveFullSquareDecompositionInput is proved equivalent with exactly the same C by
fullSquareDecompositionInput_sign_iff at checkpoint783. The literal weighted formulas, general
convolution/norm/maximal transfer and reflected support are proved, so reflection is no longer
unformalized. The full-to-annular normalization is also proved. No prime-square conclusion is assumed.
See docs/THEOREM_1_1_EXTERNAL_INPUTS.md and docs/SQUARE_REFLECTION_COMPLETION.md.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "ORDINARY_SQUARE_STOPPING",
      "status": "consumed_explicit_compound_hypothesis",
      "scope": "OrdinarySquareStoppingInput CS: independent local ordinary-square low/high stopping
estimates. Its existence and exact support, normalization, admissibility and published-convention
compatibility remain unproved. It is stronger than the earlier fixed-scale FullSquareDecompositionInput.
 See docs/INDEPENDENT_STOPPING_INPUTS.md. No prime local endpoint or sparse conclusion is an input
field.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "PRIME_MAXIMAL_LP",
      "status": "consumed_explicit_compound_hypothesis",
      "scope": "IndependentPrimeMaximalInput: complementary-exponent restricted consequence of the
independent prime polynomial maximal Lp theorem, in the actual von Mangoldt annular normalization.
Published prime normalization and prime-power extension remain within its unproved scope. Used only for
the open triangle, not for the endpoint. No sparse conclusion is a field.",
      "declared_as_Lean_axiom": false
    },
    {
```

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      "id": "QUADRATIC_WEYL_FJK",
      "status": "precise_unproved_independent_scalar_interface_implemented",
      "scope": "RawQuadraticWeylApproximationInput E records the general initial quadratic Weyl
approximation in rational-center form, a consequence of Fiedler-Jurkat-Korner as restated in
Han-Lacey-Yang arXiv1907.05734v2 Theorem3.1. Its Dirichlet selection/reduction/phase bridge and
existence remain unproved parts of this explicit independent input. It has no own circle-model, IW or
maximal conclusion. Complete Gauss arithmetic and full quadratic oscillatory integral bounds are proved
separately. See docs/SECTION_7_QUADRATIC_WEYL_BOUNDARY.md.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "BOUNDED_SQUARE_POINTWISE",
      "status": "precise_unproved_independent_bounded_class_interface_consumed",
      "scope": "RawBoundedSquarePointwiseInput: independent bounded-class square/prime-square pointwise
convergence, including the still-unformalized Mehlhop-Slomian one-dimensional/cutoff and
PNT/Abel/prime-power normalization bridge. It supplies no unbounded Orlicz convergence, strong norm,
modular inequality or density-extension conclusion. See docs/ERGODIC_INDEPENDENT_BOUNDARY.md.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "SAWYER",
      "status": "precise_unproved_general_input_consumed",
      "scope": "RawSawyerMaximalInput: for arbitrary sequence averages on a nonatomic ergodic
probability system, almost-everywhere finite proper maxima for all measurable integrable functions imply
 some positive weak L1 maximal constant. The general theorem and its exact complex/probability/cutoff
convention bridge remain unproved.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "CONZE",
      "status": "precise_unproved_general_input_consumed",
      "scope": "RawConzeMaximalInput: for arbitrary sequence averages, transfer the same weak maximal
constant from a nonatomic ergodic probability system to every probability system. General theorem and
exact convention bridge remain unproved.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "FIXED_PROGRESSION_PNT",
      "status": "precise_unproved_general_input_consumed",
      "scope": "RawFixedPrimeProgressionFrequencyInput: enumerated primes have limiting frequency
1/phi(q) in each unit residue class of every fixed positive modulus. This standard
PNT-in-fixed-progressions input remains unproved. Exact prime-square fiber partition and pushforward to
1/|Lambda_q| are proved.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "FULL_DOMAIN_CONTINUOUS_STANDARDS",
      "status": "precise_unproved_general_full_lp_inputs_consumed",
      "scope": "RawFullOneSidedHLInput and RawFullScalarHormanderInput: general one-sided interval
maximal and scalar Hormander convolution theorems on arbitrary Lp inputs for1<p<=2. These broaden the
earlier L2-intersection formulations; their proofs and exact publication/convention bridges remain
unformalized. The full-HL restriction to the earlier interface is proved. The scalar full-domain version
 is not claimed to follow from the earlier interface. All own annular kernels, Fourier formulas, uniform
 L2/Hormander conditions, real-scale continuity, measurable proper supremum and source estimates are
deductions.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "RATIONAL_SMOOTH_MULTIFREQUENCY",
      "status": "consumed_explicit_compound_hypothesis",
      "scope": "RationalSmoothCircleInput C: the independent rational smooth multifrequency inequality
with a common denominator P, its rational logarithmic cost, fixed smooth bump normalization and circle
transference. The interface is quantified over arbitrary rational center sets, periods, separated
widths, finite scale sets and all integer L2 inputs. It contains no annular kernel, beta, arithmetic
coefficients, denominator truncation or current-paper endpoint. The independent rational-frequency
theorem and transference remain unproved here; existence is not asserted. It is the rational refinement
associated with Lacey1997 Theorem1.2 and its sampling specialization, rather than the general
separated-frequency squared-log bound.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "PRIMITIVE_TWO_SQUARE_CRITERION",
      "status": "proved_in_project_not_an_external_assumption",
      "scope": "Complete primitive two-square criterion for n=2 mod4, proved from pinned Mathlib
Fermat/negative-one obstruction and checked primitive multiplication/parity algebra. No additional
mathematical hypothesis.",
      "declared_as_Lean_axiom": false,
      "proof": "PrimeSquares.goldbach_primitive_two_squares_criterion",
      "proof_file": "PrimeSquares/GoldbachPrimitiveCriterion.lean",
      "proof_checkpoint": 1261,
      "classification": "proved_arithmetic_lemma",
      "consumed_by_completed_exports": true,
      "no_longer_an_external_assumption_at": 1262,
      "source": "blueprint/Section9_Primitive_Two_Squares.md"
    },
    {
      "id": "UNIFORM_PAGE_PREFIX",
      "status": "consumed_explicit_general_hypothesis",
      "scope": "UniformPageProgressionInputs: a general prime progression prefix estimate through
exp(eta sqrt(log N)), uniform over every positive modulus, unit residue and real t in [N/2,N], with one
genuine optional Page datum per N selected before Q. The decay, modulus and prefix constants are
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independent of N. Its existence and fixed-level analytic bridge remain unproved; no quadratic phase,
major-arc estimate or own-paper endpoint is assumed.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "FULL_DOMAIN_ADELIC_IW",
      "status": "consumed_explicit_compound_general_hypothesis",
      "scope": "RawFullAdelicIWInput C: independent full-domain Tao adelic IW theorem for arbitrary good
 coprime data and arbitrary finite-arithmetic spatial operator kernels and rough compact real symbols.
Its rough limiting and published-to-actual-kernel/quotient realization bridges remain unproved parts of
this explicit hypothesis. No Q, selected construction, probability measure, annular, low/high or own
endpoint is a field. The full real Lp product bound and source coefficient/geometry/norm application are
 proved.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "FULL_DOMAIN_TAO_SAMPLING",
      "status": "consumed_explicit_compound_general_hypotheses",
      "scope": "RawFullMajorArcSamplingInput C and RawFullShannonInput S: general Tao Theorem1.6 on
arbitrary good coprime data, with full distributional lp Fourier support, both norm directions and
unique interpolation; and Proposition1.5 at fixed c=1/100 for all finite-dimensional normed complex
targets and every 1<=p<=infinity. Their truth and published/full-class/finite-quotient realization
bridges remain unproved. No source Q, B, annular operator, maximal estimate or own endpoint is assumed.
The actual real-Q applicability, full inverse, two-sided source constants and narrower-band
finite-target application are proved.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "SIEGEL_WALFISZ_PRIME_COUNTS",
      "status": "consumed_explicit_hypothesis",
      "scope": "RawPrimeSiegelWalfiszInput: general ordinary pi/Li prime counts, every endpoint y\u2264N
 and fixed logarithmic modulus range. No oscillatory weighted, coefficient-family or own-paper major-arc
 endpoint is assumed. The complete actual source square major approximation is deduced at1146. Together
with ordinary BV, it supplies the complete source major-arc mean-square replacement at1161.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "ROSSER_IWANIEC_SINGLE_STAGE",
      "status": "consumed_explicit_general_hypotheses_for_corrected_variant",
      "interfaces": [
        "GoldbachClassicalFundamentalInput",
        "GoldbachClassicalCoefficientInput"
      ],
      "scope": "General one-stage Rosser\u2013Iwaniec weights chosen before all supported densities
satisfying the refined dimension hypothesis, with pointwise sieve bounds, level/square-free support,
coefficient bound1, exponential fundamental-lemma error for large s, and the standard outer
f_kappa/F_kappa coefficient bounds with a logarithmic error. The full signed two-stage deduction, exact
scale, common error limit and integer-sequence bridges are proved. No own-paper presieving,
weighted-prime, lower/upper sieve or Goldbach endpoint is an input. The lower application also consumes
continuity of the classical lower sieve function at9/8; the actual parameter and uniform-error limits
are proved.",
      "source": "Yuta Suzuki, The Rosser\u2013Iwaniec sieve, 2023, Definition8.2, Proposition11.8 and
Lemma14.7; exact independent interfaces and classical corollary scope in docs/EXTERNAL_SIEVE_INPUTS.md",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "CLASSICAL_INTERVAL_MERTENS",
      "status": "consumed_explicit_independent_general_hypothesis",
      "scope": "GoldbachClassicalIntervalMertens K: uniform interval Euler-product bounds for the
universal factor(1-2/(p-3))^(-1) on primes>=7 with dimension2, and(1-1/(p-1))^(-1) on primes3 mod4 with
dimension1/2. Quantified over all finite prime subsets of intervals. Contains no sequence, n, Goldbach
estimate or own-paper endpoint. Both actual density applications are proved.",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "EFFECTIVE_HALF_DIMENSIONAL_MEAN",
      "status": "consumed_explicit_independent_general_hypothesis",
      "interfaces": [
        "ClassicalHalfMeanValueInput"
      ],
      "scope": "Uniform effective nonnegative strongly multiplicative upper mean theorem for arbitrary
functions with prime values at most6 and the explicit prime logarithmic balance condition. It is the
general rho=1/2 upper corollary of Tenenbaum Theorem1.2 (corrected author version arXiv:1709.01315v13),
including the documented general corollary reductions. Those external reductions and the source theorem
are not formalized. No paper weight, root count, multiplier, singular integral, Goldbach or own-paper
endpoint is a field. All such applications and the complete weighted main-term comparison are
deductions.",
      "source": "docs/EXTERNAL_WEIGHTED_MEAN_INPUTS.md",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "CLASSICAL_PRIME_LOG_MERTENS",
      "status": "consumed_explicit_independent_general_hypothesis",
      "interfaces": [
        "ClassicalPrimeLogMertens"
      ],
      "scope": "Ordinary and1 mod4 sums of log(p)/p with fixed bounded errors, and the bounded
logarithmic higher-prime-power tail. Quantified over all real x>=2; no paper weight or endpoint. The3
mod4 partition and varying-weight prime balance are proved internally.",
      "source": "docs/EXTERNAL_WEIGHTED_MEAN_INPUTS.md",
      "declared_as_Lean_axiom": false
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    },
    {
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      "status": "consumed_explicit_independent_general_hypothesis",
      "interfaces": [
        "ClassicalPrimeMertensRatios"
      ],
      "scope": "Ordinary all-prime and fixed1/3 mod4 Euler-product ratios with relative error K/log R
and explicit endpoint conventions. No varying weight or paper main-term conclusion. Exact source
local-factor and Euler-product normalizations, the effective mean application and the final weighted
comparison are proved.",
      "source": "docs/EXTERNAL_WEIGHTED_MEAN_INPUTS.md",
      "declared_as_Lean_axiom": false
    },
    {
      "id": "CLASSICAL_FULL_INTERVAL_MERTENS",
      "status": "consumed_explicit_independent_general_hypothesis",
      "interfaces": [
        "GoldbachClassicalFullIntervalMertens"
      ],
      "scope": "Arbitrary finite prime-subset products for the p5/large-prime preliminary barrier and
outer1/(p-1) factors, with refined1+K/logw losses. Independent unproved general classical Mertens
corollary. No Goldbach data or endpoints.",
      "source": "docs/SECTION_11_UPPER_SIEVE_INPUTS.md",
      "declared_as_Lean_axiom": false
    }
  ],
  "definitions": {
    "d:sparse": {
      "title": "Sparse families and their major subsets",
      "status": "formalized",
      "lean_anchors": [
        "IntegerSparseCollection",
        "IntegerSparseFamily",
        "integerSparseFamily_iff_collection",
        "integerInterval_card_points"
      ],
      "scope": "General interval collections with actual disjoint finite major subsets; eta=1/4 gives
the preliminary definition, and eta=1/10 gives the introductory theorem convention. The finite-family
specialization is proved equivalent.",
      "remaining": ""
    },
    "d:admiss": {
      "title": "Admissible stopping maps",
      "status": "formalized",
      "lean_anchors": [
        "sourceLocalMass",
        "AdmissibleAnnularLocalBound",
        "primeRemainingScale_admissible",
        "SourceLinearAdmissible",
        "sourceLinearAdmissible_iff_display",
        "linearSourceBallMass_indicator",
        "linearSourceLocalMass_le_iff",
        "linearWindowCount_oneSided_compare",
        "linearSourceLocalMass_of_both_oneSided",
        "single_direction_does_not_control_symmetric",
        "linearOneSided_bounds_of_symmetric",
        "linearSymmetric_bound_of_both_oneSided",
        "linearSourceLocalMass_change_radius"
      ],
      "scope": "The literal standalone linear-scale admissible dyadic stopping-map predicate is exported
 on actual finite indicator inputs. Lean proves equality with the displayed all-real-radius finite-count
 inequality and all convergence/boundedness prerequisites. The earlier actual prime-square spatial
N\u00b2 admissibility is also proved. The accepted fixed-radius and two-direction comparison remarks are
 also proved, including the exact factor two. The current full upload explicitly separates the fixed
radius and the sufficiently large admissibility coefficient, matching the checked generic sparse
theorem.",
      "remaining": "",
      "expositions": [
        "docs/ADMISSIBILITY_ONE_SIDED_REVIEW.md",
        "docs/SOURCE_UPDATE_1066.md"
      ]
    },
    "anonymous-s3-line1634": {
      "title": "Actual quadratic exponential sums",
      "status": "formalized",
      "lean_anchors": [
        "twistedSquareSum",
        "principalSquareSum"
      ]
    },
    "anonymous-s3-line1649": {
      "title": "Actual normalized progression sum Upsilon",
      "status": "formalized",
      "lean_anchors": [
        "upsilon"
      ]
    }
  },
  "anonymous_statement_ids": {
    "5dbf61f06a1b4be9a52880b88084e8cc90b7556d0f2f0b3f3da750291f4a3958": "anonymous-s3-line1634",
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    "d9aa2de21484852b2269bc2f661bd9f2b5d4ffb7e883cdeb60f303aa4c8612ea": "anonymous-s3-line1649"
  },
  "source_only_revision_after_checkpoint1114": "docs/APPROXIMATION_PROOF_REVISION_1114_1.md",
  "proved_general_inputs": [
    {
      "interface": "RawFiniteSignedMaximalInput",
      "constant": 6,
      "theorem": "PrimeSquares.rawFiniteSignedMaximalInput_proved",
      "file": "PrimeSquares/FiniteSignedMaximalProof.lean",
      "proof_checkpoint": 1263,
      "public_export_integration_checkpoint": 1264,
      "range": "The proved binary bound holds for every p>=1; the former interface only
required1<=p<=2.",
      "independent_hypothesis": false
    }
  ]
}
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Archive record: archive/Section11_Statement_Map_1218.json

archive/Section11_Statement_Map_1218.json


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 S(\\mathcal A_n,\\mathbb P_{3,n},z)\n &\\geq S(\\mathcal A_n,\\mathbb P_3,z),\n
\\label{e:fixedlower10}\\\\\n \\sum_{\\ell\\in L_n}S(M_n(\\ell),W_{\\ell,n},z_\\ell)\n &\\leq\n
\\sum_{\\ell\\in L}S(M_n(\\ell),W_\\ell,z_\\ell)\n +O_\\varepsilon(N^\\varepsilon).\n
\\label{e:fixedupper10}\n\\end{align}",
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\\(p\\equiv3\\pmod4\\),\n\\begin{align}\\label{e:lowerlocaldensity10}\n \\frac{\\omega_n(p)}p\n =\n
\\begin{cases}\n \\displaystyle\n \\frac{\\rho_p(n-1)}{p-1-\\rho_p(n)},&p\\leq R,\\\\[7pt]\n
\\displaystyle\\frac1{p-1},&p>R.\n \\end{cases}\n\\end{align}\nIn particular, for
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function\n\\(\\omega_n(\\ell,\\cdot)\\), defined on square-free integers supported on\n\\(W_\\ell\\),
such that\n\\begin{align}\\label{e:upperfactorization10}\n \\frac{G_R(\\ell,12d,n-1)}{\\phi(12d)}\n =\n
\\frac{G_R(\\ell,12,n-1)}{\\phi(12)}\n \\frac{\\omega_n(\\ell,d)}d .\n\\end{align}\nFor
\\(p\\nmid12\\ell\\), its  factors are\n\\begin{align}\\label{e:upperlocaldensity10}\n
\\frac{\\omega_n(\\ell,p)}p\n =\n \\begin{cases}\n \\displaystyle\n
\\frac{\\rho_p(n)}{p-1-\\rho_p(n-1)},&p\\leq R,\\\\[7pt]\n \\displaystyle\\frac1{p-1},&p>R.\n
\\end{cases}\n\\end{align}\nIn particular, \\(0\\leq\\omega_n(\\ell,p)/p<1\\).",
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|\\mathcal F_d|=Xg(d)+r_d,\n\\end{align}\nwhere \\(g\\) is multiplicative on the square-free integers
supported on\na fixed set of primes \\(\\mathfrak P\\), and \\(0\\leq g(p)<1\\).  Let\n\\(2\\leq R<z\\),
 and suppose that there are fixed constants\n\\(\\kappa_0,\\kappa,K_0>0\\) such
that\n\\begin{align}\\label{e:tworangesievedimension}\n \\prod_{\\substack{w\\leq p<y\\\\p\\in\\mathfrak
 P}}\n \\left(1-g(p)\\right)^{-1}\n &\\leq\n K_0\\left(\\frac{\\log y}{\\log w}\\right)^{\\kappa_0},\n
&&2\\leq w<y\\leq R,\\notag\\\\\n \\prod_{\\substack{w\\leq p<y\\\\p\\in\\mathfrak P}}\n
\\left(1-g(p)\\right)^{-1}\n &\\leq\n K_0\\left(\\frac{\\log y}{\\log w}\\right)^\\kappa,\n &&R<w<y\\leq
 z.\n\\end{align}\nDefine \n\\[\n D_0=\\exp((\\log\\log N)^3),\\qquad D_1=D/D_0, \\qquad D_0<D , \\qquad
 \\frac{\\log D_0}{\\log
R}\\longrightarrow\\infty.\n\\]\nThen\n\\begin{align}\\label{e:presievedbounds10}\n S(\\mathcal
F,\\mathfrak P,z)\n &\\geq XV(z)\\bigl(f_\\kappa(s)+o(1)\\bigr)\n +\\sum_{m<D}c_m^-r_m,\\notag\\\\\n
S(\\mathcal F,\\mathfrak P,z)\n &\\leq XV(z)\\bigl(F_\\kappa(s)+o(1)\\bigr)\n
+\\sum_{m<D}c_m^+r_m,\n\\end{align}\nwhere\n\\[\n V(z)=\\prod_{\\substack{p<z\\\\p\\in\\mathfrak
P}}(1-g(p)),\n \\qquad\n s=\\frac{\\log D_1}{\\log z},\n \\qquad\n |c_m^\\pm|\\leq1.\n\\]\nThe
coefficients \\(c_m^\\pm\\) depend only on the fixed prime sets and\nthe sieve levels, and not on the
values of \\(g\\).  \nThe dimension\n\\(\\kappa_0\\) is used only in the preliminary sieve over
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from independent general one-stage Rosser\u2013Iwaniec inputs; the literal weaker printed statement is
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L}X_n(\\ell)\\Theta_{n,\\ell}(z_\\ell)\n \\leq\n \\bigl(c_\\alpha\\mathcal W(\\alpha)+o(1)\\bigr)\n
X_n\\Theta_n(z),\n \\qquad\n
c_\\alpha=\\frac{16e^{-\\gamma/2}}{\\sqrt{\\pi\\alpha}}.\n\\end{align}\nHere \\(\\mathcal W(\\alpha)\\)
is the integral defined in\n\\eqref{e:Wdef8}.",
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P_{3,n},z)\n \\geq X_n\\Theta_n(z)\\bigl(f_{1/2}(9/8)+o(1)\\bigr)\n +\\mathcal
E_1(n),\n\\end{align}\nwhere\n\\begin{align}\\label{e:E1section8}\n\\sum_{n<N^2}|\\mathcal
E_1(n)|^2\n\\ll_{A_0,K} N^4(\\log N)^{-K}.\n\\end{align}",
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L_n}S(M_n(\\ell),W_{\\ell,n},z_\\ell)\n &\\leq\n \\bigl(c_{9/8}F_1(2)\\mathcal W(9/8)+o(1)\\bigr)\n
X_n\\Theta_n(z)+\\mathcal E_2(n),\n\\end{align}\nwhere\n\\begin{align}\\label{e:calphadef8}\n c_\\alpha=
\\frac{16e^{-\\gamma/2}}{\\sqrt{\\pi\\alpha}}\n\\end{align}\nand\n\\begin{align}\\label{e:E2section8}\n\
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that, for every fixed\nsufficiently large \\(C\\), with \\(R=(\\log N)^C\\), one
has\n\\begin{align}\\label{e:maintermlower8}\n X_n\\Theta_n(z)\\gg_{A_0,C}\n N(\\log
N)^{-C_{A_0}}\n\\end{align}\nwhenever $n\\equiv12\\pmod{24}$ and $N^2(\\log N)^{-A_0}\\leq n<N^2$.",
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\\cite[(1.8)--(1.10)]{I80},\n\\begin{align}\\label{e:Deltanumerical}\n \\Delta>0.\n\\end{align}",
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archive/Section9_Statement_Map_Pre1262.json


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P^2\\}|\n \\geq S(\\mathcal A_n,\\mathbb P_{3,n},N^2)-O_\\varepsilon(N^\\varepsilon).\n\\end{align}",
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\\(\\varepsilon>0\\),\n\\begin{align}\\label{e:truncationT}\n S(\\mathcal A_n,\\mathbb P_{3,n},z)\n -\n
|\\{r\\in\\mathcal P:n-r\\in\\mathbb P^2\\}|\n =\n
|T_n|+O_\\varepsilon(N^\\varepsilon),\n\\end{align}\nwhere\n\\begin{align}\\label{e:Tfactorization}\n
    T_n=\n \\left\\{\n r-1=2up_1p_2:\n \\begin{array}{l}\n r\\in\\mathbb P,\\quad n-r\\in\\mathbb
P^2,\\\\\n z\\leq p_2\\leq p_1,\\quad p_i\\in\\mathbb P_{3,n},\\\\\n p\\mid u\\Longrightarrow
p\\equiv1\\pmod4\n \\end{array}\n \\right\\}.\n\\end{align}",
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$\\varepsilon>0$,\n\\begin{align}\\label{e:upperreduction}\n |T_n|\n \\leq\\sum_{\\ell\\in L_n}\n
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P^2\\}|\n &\\geq S(\\mathcal A_n,\\mathbb P_{3,n},z)-\\sum_{\\ell\\in L_n}\n
S(M_n(\\ell),W_{\\ell,n},z_\\ell)\n -O_\\varepsilon(N^\\varepsilon).\n\\end{align}",
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\\(A,\\delta>0\\) and \\(L_0<N^{2-\\delta}\\).  There are sufficiently\nlarge constants
\\(B=B(A,\\delta)\\) and \\(C=C(A,\\delta)\\) such that the\nfollowing holds.  Put\n\\[\n R=(\\log
N)^C,\\qquad\n \\eta_N(\\theta)=\\eta\\left(\\frac{N^2\\theta}{R}\\right),\n\\]\nand suppose that
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$V_\\ell\\leq (N^2/\\ell)^{1/2}(\\log N)^{-B}$ for $\\ell<L_0.$\nLet \\(|d_{k,\\ell}|\\leq1\\)
and\n\\((a_{k,\\ell},k)=1\\).\nDefine \n\\begin{align}\\quad \\quad\\label{e:defGR9}\n G_R(\\ell,k,n)\n
&:=\n \\mathcal T_2(\\ell,k,n;a_{k,\\ell})\n \\prod_{2<p\\leq R}\n \\frac{p\\phi((p,k))}{(p-1)^2}\n
\\left|\\left\\{(u,v)\\in\\mathbb A_p^2:\n \\begin{array}{l}\n  u^2+\\ell v\\equiv n\\pmod p,\\\\\n
v\\equiv a_{k,\\ell}\\pmod{(p,k)}\n \\end{array}\\right\\}\\right|,\n\\end{align}\nwhere $ \\mathcal
T_2(\\ell,k,n;a_{k,\\ell}) $ is defined in \\eqref{e:generalT2} and the second congruence is understood
to be trivial when \\(p\\nmid k\\).\nAlso define\n\\begin{align}\\label{e:defHR9}\n H_{\\ell,R}(n)\n
:=\n \\ell\\int_{\\mathbb R}\\eta_N(\\theta)^2\n \\left(\\int_2^N\\frac{e(\\theta u^2)}{\\log
u}\\,du\\right)\n \\left(\\int_{\\substack{2\\leq v\\leq N^2\\\\\\ell v<N^2}}\n \\frac{e(\\ell\\theta
v)}{\\log v}\\,dv\\right)\n
e(-n\\theta)\\,d\\theta.\n\\end{align}\nThen\n\\begin{align}\\label{e:circmethod}\n
\\sum_{n<N^2}\\left|\n \\sum_{\\ell<L_0}\\sum_{k<V_\\ell}\n d_{k,\\ell}r_{k,\\ell}(n)\\right|^2\n
\\ll_{A,\\delta}N^4(\\log N)^{-A},\n\\end{align}\nwhere\n\\begin{align}\\label{e:unweightedremainder}\n
r_{k,\\ell}(n)\n &=\n \\sum_{\\substack{m^2+\\ell t=n\\\\\n t\\equiv a_{k,\\ell}\\pmod k}}\n
\\one_{\\mathbb P}(m)\\one_{\\mathbb P}(t)\n -\n
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P_{3,n},z)\n \\geq X_n\\Theta_n(z)\\bigl(f_{1/2}(9/8)+o(1)\\bigr)\n +\\mathcal
E_1(n),\n\\end{align}\nwhere\n\\begin{align}\\label{e:E1section8}\n\\sum_{n<N^2}|\\mathcal
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L_n}S(M_n(\\ell),W_{\\ell,n},z_\\ell)\n &\\leq\n \\bigl(c_{9/8}F_1(2)\\mathcal W(9/8)+o(1)\\bigr)\n
X_n\\Theta_n(z)+\\mathcal E_2(n),\n\\end{align}\nwhere\n\\begin{align}\\label{e:calphadef8}\n c_\\alpha=
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has\n\\begin{align}\\label{e:maintermlower8}\n X_n\\Theta_n(z)\\gg_{A_0,C}\n N(\\log
N)^{-C_{A_0}}\n\\end{align}\nwhenever $n\\equiv12\\pmod{24}$ and $N^2(\\log N)^{-A_0}\\leq n<N^2$.",
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Archive record: legacy/Statement_Inventory.csv

legacy/Statement_Inventory.csv

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Theorem 1.1,3,PrimeSquares.Main.theorem_1_1,target - not implemented,"Let I = [−N 2 , 2N 2] ∩ Z. For all F, G ⊆ I,"
Theorem 1.3,3,PrimeSquares.Main.theorem_1_3,target - not implemented,"Let F, G ⊂ Z be finite. There is a sparse collection of intervals S such that"
Theorem 1.5,4,PrimeSquares.Main.theorem_1_5,target - not implemented,Let pk denote the k-th prime. The sequence (p2k) is persistently universally
Theorem 1.6,5,PrimeSquares.Main.theorem_1_6,target - not implemented,Quantitative exceptional set: Almost all integers congruent to 12 modulo 24
Definition 2.10,12,PrimeSquares.Foundations.definition_2_10,target - not implemented,"A collection S of intervals is sparse if, for every I ∈ S, there is EI ⊂ I such"
Definition 2.11,13,PrimeSquares.Foundations.definition_2_11,target - not implemented,Let f = 1F be supported in 3I0 . A map τ : I0 −→ {2j : 1 ≤ 2j ≤ |I0 |} is
Lemma 3.1,15,PrimeSquares.Arithmetic.lemma_3_1,target - not implemented,Cohen’s identity with square shifts: Let 2a ∥ q. Then
Definition 3.7,16,PrimeSquares.Arithmetic.definition_3_7,target - not implemented,"Exponential sums along squares: For a Dirichlet character χ modulo q, put"
Definition 3.9,17,PrimeSquares.Arithmetic.definition_3_9,target - not implemented,"Let g = (q, y) and (b, g) = 1. Define"
Lemma 3.11,17,PrimeSquares.Arithmetic.lemma_3_11,target - not implemented,"Let g | q and (b, g) = 1. Then"
Lemma 3.12,17,PrimeSquares.Arithmetic.lemma_3_12,target - not implemented,"Let g = (q, y) and (b, g) = 1. Then"
Lemma 3.14,17,PrimeSquares.Arithmetic.lemma_3_14,target - not implemented,"Let k, x, b ∈ Z, (b, k) = 1. Then"
Corollary 3.16,18,PrimeSquares.Arithmetic.corollary_3_16,target - not implemented,"Let ℓ, k, n, b ∈ Z, with (b, k) = 1. Then"
Lemma 3.20,19,PrimeSquares.Arithmetic.lemma_3_20,target - not implemented,"For Q ≥ 3 and every integer n,"
Lemma 3.21,19,PrimeSquares.Arithmetic.lemma_3_21,target - not implemented,"If (a, q) = 1, then"
Lemma 3.24,20,PrimeSquares.Arithmetic.lemma_3_24,target - not implemented,"Let χ be a real Dirichlet character modulo q, and let (a, q) = 1. Then, for every"
Lemma 3.27,21,PrimeSquares.Arithmetic.lemma_3_27,target - not implemented,"Let x ∈ Z. Let χ0 be a primitive real character of conductor q0 , and for each"
Theorem 4.1,24,PrimeSquares.Improving.theorem_4_1,target - not implemented,Let AN and BN be positive convolution operators whose kernels are supported in
Corollary 4.15,25,PrimeSquares.Improving.corollary_4_15,target - not implemented,Assume in addition that AN and BN are uniformly bounded on ℓ2 . If
Proposition 4.19,26,PrimeSquares.Improving.proposition_4_19,target - not implemented,"Let N be dyadic, let Q ≥ 2, and suppose that"
Lemma 5.1,28,PrimeSquares.Sparse.lemma_5_1,target - not implemented,Fix constants C ≥ 1 and c > 2C + 1. Let (BM) be a family of positive convolution
Lemma 5.10,30,PrimeSquares.Sparse.lemma_5_10,target - not implemented,Suppose that every admissible τ satisfies
Lemma 5.17,31,PrimeSquares.Sparse.lemma_5_17,target - not implemented,"Let f = 1F and g = 1G , where F ⊂ cI0 and G ⊂ I0 , and put δ0 = |F||I||G| 0|"
Lemma 6.5,34,PrimeSquares.HighLow.lemma_6_5,target - not implemented,"If (a, q) = 1, q ≤ R11/10 , and |θ| ≤ R/(qN 2), then"
Lemma 6.9,35,PrimeSquares.HighLow.lemma_6_9,target - not implemented,"Approximation: For every A > 0,"
Proposition 6.14,37,PrimeSquares.HighLow.proposition_6_14,target - not implemented,"For nonnegative f ,"
Lemma 6.16,39,PrimeSquares.HighLow.lemma_6_16,target - not implemented,"Multifrequency estimate: Let 0 < β ≤ 1, and let Rs be any subset of the reduced"
Corollary 6.19,40,PrimeSquares.HighLow.corollary_6_19,target - not implemented,"Denominator-prefix maximal estimate: Let 0 < β ≤ 1, s ≥ 0, and |ca,q | ≤ b."
Proposition 6.21,41,PrimeSquares.HighLow.proposition_6_21,target - not implemented,"Let N0 ≥ 2 and suppose that log Q ≤ c0 (log N0)1/2 . Then, for every dyadic"
Theorem 7.2,44,PrimeSquares.Maximal.theorem_7_2,target - not implemented,"Orlicz maximal estimate: There are absolute constants C, C0 ≥ 1 such that the"
Corollary 7.6,44,PrimeSquares.Maximal.corollary_7_6,target - not implemented,"For 1 < p ≤ 2,"
Proposition 7.8,45,PrimeSquares.Maximal.proposition_7_8,target - not implemented,"Let r ≥ 2, k ≥ 4, and Q ≥ 2k , and put"
Lemma 7.15,47,PrimeSquares.Maximal.lemma_7_15,target - not implemented,"Uniformly in 0 < β ≤ 1, J ≥ 1, and |σj | ≤ 1, one has, for 1 < p ≤ 2,"
Proposition 7.21,48,PrimeSquares.Maximal.proposition_7_21,target - not implemented,"Let r, k, Q, P, Σ≤k , B, ε, and Cr,k be as in Proposition 7.8, and put pr = 2r−1 ."
Proposition 7.28,51,PrimeSquares.Maximal.proposition_7_28,target - not implemented,"Using the assumptions and notation of Propositions 7.8 and 7.21, put"
Corollary 7.42,56,PrimeSquares.Maximal.corollary_7_42,target - not implemented,"For 1 < p ≤ 2,"
Theorem 8.1,57,PrimeSquares.Divergence.theorem_8_1,target - not implemented,"Let (nk) be an increasing sequence of natural numbers. Suppose that, for every"
Theorem 8.6,58,PrimeSquares.Divergence.theorem_8_6,target - not implemented,"Granville–Kurlberg [14]: For each odd prime p, let Ωp ⊂ Z/pZ, put rp = |Ωp |/p"
Lemma 8.8,58,PrimeSquares.Divergence.lemma_8_8,target - not implemented,Let p be an odd prime and let
Lemma 9.2,61,PrimeSquares.Goldbach.lemma_9_2,target - not implemented,"For every ε > 0,"
Lemma 9.3,61,PrimeSquares.Goldbach.lemma_9_3,target - not implemented,"For every ε > 0,"
Lemma 9.7,62,PrimeSquares.Goldbach.lemma_9_7,target - not implemented,"For every ε > 0,"
Corollary 9.9,63,PrimeSquares.Goldbach.corollary_9_9,target - not implemented,"For every ε > 0,"
Lemma 9.10,63,PrimeSquares.Goldbach.lemma_9_10,target - not implemented,"Bombieri–Vinogradov approximation: Let A, δ > 0 and L0 < N 2−δ . There are"
Proposition 9.17,64,PrimeSquares.Goldbach.proposition_9_17,target - not implemented,"Lower sieve: For N 2 (log N)−A0 ≤ n < N 2 , n ≡ 12 (mod 24),"
Proposition 9.20,64,PrimeSquares.Goldbach.proposition_9_20,target - not implemented,"Upper sieve: For N 2 (log N)−A0 ≤ n < N 2 , n ≡ 12 (mod 24),"

Lemma 9.22,65,PrimeSquares.Goldbach.lemma_9_22,target - not implemented,"Let A0 > 0. There is CA0 > 0 such that, for every fixed sufficiently large C, with"
Lemma 9.24,65,PrimeSquares.Goldbach.lemma_9_24,target - not implemented,"With the normalization of [21, (1.8)–(1.10)],"
Lemma 10.2,66,PrimeSquares.GoldbachCircle.lemma_10_2,target - not implemented,"Under the hypotheses of Lemma 9.10, uniformly for 2 ≤ U ≤ N and 2 ≤ W ≤ N 2 ,"
Lemma 10.12,68,PrimeSquares.GoldbachCircle.lemma_10_12,target - not implemented,"Uniformly in ℓ, U, W and n < N 2 ,"
Lemma 10.15,68,PrimeSquares.GoldbachCircle.lemma_10_15,target - not implemented,"Uniformly for 2 ≤ U ≤ N and 2 ≤ W ≤ N 2 ,"
Lemma 10.20,72,PrimeSquares.GoldbachCircle.lemma_10_20,target - not implemented,The series
X
Lemma 10.24,73,PrimeSquares.GoldbachCircle.lemma_10_24,target - not implemented,"Under the hypotheses of Lemma 9.10, uniformly for 2 ≤ U ≤ N and 2 ≤ W ≤"
Lemma 11.2,76,PrimeSquares.GoldbachSieve.lemma_11_2,target - not implemented,"For every ε > 0,"
Lemma 11.6,76,PrimeSquares.GoldbachSieve.lemma_11_6,target - not implemented,"For p > 3, p ≡ 3 (mod 4),"
Lemma 11.8,77,PrimeSquares.GoldbachSieve.lemma_11_8,target - not implemented,"There is a multiplicative function ωn (ℓ, ·), defined on square-free integers supported"
Lemma 11.13,78,PrimeSquares.GoldbachSieve.lemma_11_13,target - not implemented,Suppose that
Lemma 11.19,82,PrimeSquares.GoldbachSieve.lemma_11_19,target - not implemented,"Let α = 9/8. Uniformly in the range (11.1),"

Prime Squares

Initial Formalization Blueprint

and a checked finite core

For Hamed Mousavi
On Equidistribution of Prime Squares and Its Applications

20 October 2026

What has been done. This package contains a first dependency blueprint for the attached 96-page manuscript and a small Lean project. The project formalizes finite exceptional-set inequalities, elementary arithmetic used in the paper, and positivity of the elementary bracket in the last sieve comparison. These are useful components of a larger development; none is a formalization of a headline theorem of the paper.

What this document does not claim. It is not an expanded proof of every lemma, and it is not a certificate for the whole paper. In particular, the circle-method approximation, high–low estimates, stopping-time proof, Ionescu–Wainger argument, ergodic divergence argument and sieve estimates remain to be encoded and checked. The word *target* below always denotes unfinished work.

Source lock. Numbering and page references refer only to the uploaded file `Prime_Squares(32)(1).pdf`. Its SHA-256 is recorded in `PROVENANCE.json`. A revised manuscript must be compared with this snapshot before the paper-to-Lean map is reused.

Reproducibility. Lean is pinned to 4.34.1. Mathlib is pinned to commit `d13f23b723b8a846827a245b89c10fc7d3f11612`. The source archive includes a lockfile, the Lean modules, build instructions and scripts for checking proof dependencies. The verification logs, rather than this prose, record what was actually run.

Reading route. Sections 1–3 define the scope and targets. Sections 4–5 give complete mathematical explanations of the checked components. Sections 6–11 map the unfinished arguments. Sections 12–14 record imports, audit points and the next inputs needed. The appendix inventories numbered statements in the attached PDF.

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1 The trust boundary

There are three different objects in a formalization project.

1. A **blueprint** specifies mathematical objects, statements, dependencies and proof steps. It is written for humans and can itself contain mistakes.
2. A **Lean proof** is an actual term accepted by Lean's kernel. Elaborating a statement containing a placeholder is not the same as proving it.
3. A **conditional formalization** proves a result from explicitly declared external statements. Its mathematical scope depends on the accuracy and consistency of those statements as well as on the translation of the final theorem.

For the present starter, no extra analytic-number-theory axioms are declared. Theorems in the starter have explicit hypotheses for the errors and main terms they use. Those hypotheses are not assertions that the required analytic estimates have already been established for prime squares.

For the completed project it is reasonable to import deep published results. Each import should have one stable declaration, a precise source, the full range of parameters, and a statement of constant dependence. The project should continue to prove the new arguments of this paper. Assuming its own high–low estimate or its final sieve comparison would leave that part of the paper unverified.

An axiom audit must examine the *transitive* dependencies of each headline theorem, not just count explicit occurrences of `axiom`. Standard Lean/Mathlib dependencies can include `propext`, `Classical.choice` and `Quot.sound`. An unfinished proof introduces `sorryAx`. The starter's scripts reject unapproved dependencies and scan the project sources for proof placeholders and new axioms.

What the Hua example establishes

The supplied Hua blueprint contains both a current-paper development and substantial material from an earlier, broader draft. Its total length is therefore not a useful measure of how long the prime-squares blueprint should be.

At the repository snapshot inspected for this request, the trusted external declarations are `Hua.siegel_walfisz` and `Hua.hua_theorem10`. The second is an exponential-sum estimate, rather than the statement that sufficiently large admissible integers are sums of five prime squares. A static scan of its default library closure passed. That scan does not replace an independent rebuild or an audit of the meanings of its axioms and headline statements. This starter does not import that repository.

2 Objects and final statements to encode

Status: Specifications only; the headline statements are not yet Lean declarations.

Use natural numbers for prime parameters and integer indices for discrete functions. Define, for $N \geq 2$,

$$A_{N,\Lambda} f(x) = \frac{1}{N} \sum_{1 \leq n \leq N} \Lambda(n) f(x - n^2), \quad A_{N,\square} f(x) = \frac{1}{N} \sum_{1 \leq n \leq N} f(x - n^2),$$

$$A_{N,\mathbb{P}^2} f(x) = \frac{1}{\pi(N)} \sum_{p \leq N} f(x - p^2).$$

The first and third operators differ in both weights and normalization. In the divergence theorem the parameter counts the first N primes instead of bounding their size. These conventions must not be silently identified.

Put $L(t) = 2 + |\log t|$, $L_2(t) = 2 + \log(2 + |\log t|)$, and

$$\Phi_c(t) = \exp\left(c \frac{L(t)^{1/2}}{L_2(t)}\right) \quad (t > 0).$$

In bilinear forms define $t^{2/3}\Phi_c(t)$ to be zero at $t = 0$. For $t \geq 0$, define

$$\Psi_C(t) = \int_0^t \exp\left(C \sqrt{\log(e + u) \log(e + \log(e + u))}\right) du.$$

The square root in this last expression covers the *product* of the logarithms.

Theorem 1.1. For every $\varepsilon > 0$, there is C_ε such that the displayed restricted improving inequality holds for every allowed N, F, G with coefficient $4\sqrt{3}/9 + \varepsilon$. The $o(1)$ in the paper is thus a quantifier over ε, not an unspecified function added to the theorem. Strong improving bounds for $3/2 < p \leq 2$ are separate corollaries.

Theorem 1.3. Encode the endpoint sparse form and then the stated interior triangle of exponents. Sparsity uses subsets $E_I \subset I$ of cardinality at least $|I|/10$, pairwise disjoint over the family. Cardinalities, dilations and normalizations must be explicit.

Theorem 1.5. Quantify over every nonatomic ergodic measure-preserving system and every infinite set of indices S. The conclusion is existence of an integrable function whose averages along $N \in S$ fail to converge on a set of positive measure. Do not weaken this to one system or to the full sequence only.

Theorem 1.6. Define $\mathcal{P}$ as the primes represented by $a^2 + b^2 + 1$ with $(a, b) = 1$. For every $A > 0$, specify constants C_A, X_A so that the nonrepresentable $1 \leq n < X$, $n \equiv 12 \pmod{24}$, have cardinality at most $C_A X (\log X)^{-A}$ for $X \geq X_A$.

Section 7. Theorem 7.2 and Corollaries 7.6 and 7.42 are additional targets: the modular estimate, quantitative L^p growth, and almost-everywhere and L^1 convergence on the stated Orlicz class. They are not substitutes for Theorem 1.5.

3 The dependency architecture

Status: Initial map, not a claim that the indicated modules exist already.

Branch	Principal nodes in the paper	Output and main dependencies
Arithmetic	Section 3, especially 3.1, 3.12, 3.14, 3.20, 3.21, 3.24, 3.27	Cohen correlations, kernel bounds, Gauss sums and exceptional-conductor bounds.
Improving	4.1, 4.15, 4.19	Theorem 1.1; uses Section 3 and the decomposition in Section 6.
Sparse	5.1, 5.10, 5.17	Theorem 1.3; needs the local low estimate and maximal, rather than fixed-scale, high estimate.
High–low	6.5, 6.9, 6.14, 6.16, 6.19, 6.21	Proposition 4.19; uses primes in progressions with an exceptional term, minor-arc cancellation and multifrequency estimates.
Near L^1	7.8, 7.15, 7.21, 7.28, 7.2	Corollaries 7.6 and 7.42; uses quantitative Ionescu–Wainger/sampling inputs and the Section 6 model.
Divergence	8.1, 8.6, 8.8 and the final assembly	Theorem 1.5; uses a character-sum bound, Poisson spacings, fixed-modulus prime equidistribution and the divergence criterion.
Goldbach	9.2–9.10, Section 10, 11.13, 11.19, 9.17, 9.20, 9.22, 9.24	Theorem 1.6; circle-method mean-square approximation, two-stage sieving, positive main term and exceptional-set assembly.

The high–low branch is shared by the improving, sparse and near-L^1 results. The divergence branch shares arithmetic foundations but is logically separate from the maximal inequality. The Goldbach branch shares exponential sums and local correlations, but introduces its own mean-square approximation and sieve machinery.

Proposed eventual organization

Keep declarations under `PrimeSquares`. Suggested future directories are `Arithmetic`, `HighLow`, `Improving`, `Sparse`, `Maximal`, `Ergodic`, `Goldbach` and `External`. Maintain one table linking a manuscript label to each Lean declaration. Use stable LaTeX labels once the sources are available; printed theorem numbers alone will change during proofreading.

For each node record: the exact statement; normalization and parameter ranges; dependencies; an expanded proof; Lean name; external assumptions; build status. The appendix currently gives an inventory, not this complete node-by-node expansion.

4 A checked finite exceptional-set argument

Status: Implemented in `PrimeSquares/FiniteAssembly.lean`.

Let S be a finite set, let $e_1, e_2 : S \to \mathbb{R}$, and let $t > 0$. Define

$$B_t = \{n \in S : |e_1(n)| + |e_2(n)| > t\}.$$

No number-theoretic interpretation is part of these definitions.

Lemma 4.1 (Finite energy estimate)**.** *If $\sum_{n\in S} e_i(n)^2 \le B_i$ for $i = 1, 2$, then*

$$|B_t| \le \frac{2(B_1 + B_2)}{t^2}.$$

Proof. For real a, b, the nonnegativity of $(|a| - |b|)^2$ implies $(|a| + |b|)^2 \le 2(a^2 + b^2)$. On B_t, therefore, $t^2 \le 2(e_1(n)^2 + e_2(n)^2)$. Sum this inequality on B_t, enlarge the nonnegative square sum to S, and divide by $t^2 > 0$. □

Lemma 4.2 (Positivity outside the exceptional set)**.** *Let $R(n)$ be a nonnegative integer and suppose that*

$$R(n) \ge M(n) + e_1(n) - e_2(n) - u(n), \quad M(n) \ge 4t, \quad u(n) \le t.$$

If $n \notin B_t$, then $R(n) \ge 2t > 0$.

Proof. Use $e_1 - e_2 \ge -|e_1| - |e_2|$ and $|e_1| + |e_2| \le t$. The lower bound becomes $4t - t - t = 2t$. □

Combining the two lemmas gives the actual Lean result `PrimeSquares.zero_count_card_bound`:

$$|\{n \in S : R(n) = 0\}| \le \frac{2(B_1 + B_2)}{t^2}.$$

The hypotheses about M, R, u, e_i are part of its statement. In the paper's final assembly take $M = \Delta X_n \Theta_n(z)/2$, $t = cN(\log N)^{-C_{A_0}}$, and $e_i = E_i$. If each mean square is at most $C_i N^4 (\log N)^{-K}$, the bound becomes

$$|\{R = 0\}| \le \frac{2(C_1 + C_2)}{c^2} N^2 (\log N)^{-K+2C_{A_0}}.$$

Choosing K large enough yields the desired logarithmic saving for this finite range. The analytic estimates, eventual parameter choices, identification of R with actual representations, and treatment of the initial short range remain formalization work. Thus this is the finite last step of the Goldbach proof, not Theorem 1.6 itself.

5 Other checked components

Status: Implemented in `LocalArithmetic.lean` and `SieveConstant.lean`.

Congruences and the shared-prime observation

For every natural number n with $n \equiv 1 \pmod 2$ and $n \not\equiv 0 \pmod 3$, the module proves $n^2 \equiv 1 \pmod{24}$. The proof reduces to the 24 possible residues; Lean checks the finite case split. For a prime $p > 3$, primality excludes divisibility by 2 and 3, so the same conclusion follows.

Consequently, if $r \equiv 11 \pmod{24}$ and $p > 3$ is prime, then $r + p^2 \equiv 12 \pmod{24}$. The restriction $p > 3$ is included explicitly. The starter also proves the shift equivalences

$$m = a^2 + b^2 + p^2 \iff m + 1 = a^2 + b^2 + p^2 + 1, \qquad m \equiv 11 \pmod{24} \iff m + 1 \equiv 12 \pmod{24}.$$

These clarify the relation between the abstract's formulation and Theorem 1.6; they do not establish an exceptional-set estimate.

Suppose p, m are prime, $n = m^2 + r$, and $p \mid n, r$. Then $p \mid m^2$, so $p \mid m$; primality of m forces $p = m$. This is the finite divisibility observation used in Lemma 11.2. The divisor-counting estimate and the full sieve comparison in that lemma are not yet formalized.

The elementary bracket in Lemma 9.24

The manuscript reduces its final sieve comparison to

$$\Delta \geq \frac{e^{\gamma/2}}{\sqrt{\pi}} \left(\frac{2\sqrt{2}}{3} \log 2 - \frac{27}{64} \right).$$

Lean proves positivity of the parenthesized expression. There is a particularly short rigorous proof which does not need decimal approximations:

$$\sqrt{2} \geq \frac{4}{3}, \qquad \log 2 \geq 1 - \frac{1}{2} = \frac{1}{2},$$

where the second inequality is the standard $1 - x^{-1} \leq \log x$ for $x > 0$. Both lower bounds are positive, and therefore

$$\frac{2\sqrt{2}}{3} \log 2 - \frac{27}{64} \geq \frac{4}{9} - \frac{27}{64} = \frac{13}{576} > 0.$$

The declaration `sieve_bracket_positive` proves this conclusion. The separate declaration `scaled_sieve_bracket_positive` allows multiplication by any strictly positive real factor.

This does *not* verify the integral evaluation for $f_{1/2}(9/8)$, the upper bound on $\mathcal{W}(9/8)$, the factor $e^{\gamma/2}/\sqrt{\pi}$, or the stronger numerical inequality $\Delta > 0.17$. Those must be added before claiming Lemma 9.24.

6 Improving estimates: the scalar optimization and its inputs

Status: Unfinished target; the following is the blueprint for Section 4.

Separate the prime parameter from the spatial scale. For prime squares, a sum up to N has support at scale N^2; the abstract theorem uses a spatial-scale variable. A typed name such as `primeScale` versus `spaceScale` prevents an incorrect substitution of N for N^2.

Theorem 4.1 first supplies a comparison operator B_N. Applied to indicators, its decomposition gives, with $\delta = |F||G|/N^2$,

$$N^{-1}|\langle B_N \mathbf{1}_F, \mathbf{1}_G\rangle| \lesssim \delta J^s (\log J)^\gamma + \delta^{1/2} J^{-r} (\log J)^\beta.$$

The optimal J, up to bounded factors, is

$$J = \delta^{-1/(2(s+r))} (\operatorname{Log} \delta)^{(\beta-\gamma)/(s+r)}.$$

Prove first the estimate when this J is in the permitted range. If it is too large, use the elementary local kernel estimate. This second branch is essential: choosing an inadmissible J is not an application of the decomposition.

For the prime operator the corresponding bound is

$$\delta Q^s (\log Q)^\alpha + \delta^{1/2} Q^{-r} \exp\left(c_1 \frac{(\log Q)^\kappa}{\log\log Q} \right) + \text{error}.$$

When $\operatorname{Log}\delta$ is small enough relative to $(\log N)^c$, take the dyadic Q specified by (4.14). Prove separately that it is admissible, that rounding changes the estimate by a bounded factor, and that the error is absorbed. In the complementary range, transfer the comparison estimate using the pointwise domination by B_N. This yields

$$\delta^{1-s/(2(s+r))} \Phi_{c_*,\kappa}(\delta), \qquad c_* = \frac{c_1 s}{2^\kappa (s+r)^{\kappa+1}} + o(1).$$

For $s = 1, r = 1/2, \kappa = 1/2, c_1 = 2 + \varepsilon$, the critical power is $2/3$ and the limiting coefficient is $4\sqrt{3}/9$.

Separate Lean tasks. Establish the finite bilinear norm inequalities; formalize the elementary optimization including the two parameter ranges; prove the dyadic-to-full annular summation; and then prove restricted-to-strong interpolation for $p > 3/2$. None follows just from writing the numerical value of the coefficient into a declaration.

7 Sparse bounds: local control and recursion

Status: Unfinished target; Sections 5 and the stopping-time conventions in Section 2.

The sparse proof needs more than Theorem 1.1. A stopping time $\tau(x)$ depends on the spatial point. Its high term must be controlled by a maximal ℓ^2 estimate over all permitted scales. The low estimate must be local enough that admissibility replaces its local density by the parent density.

The key tasks are these.

1. Define integer intervals, their dilations, a dyadic grid, and the admissible stopping times. Record the exact support radius of each kernel.
2. Prove Lemma 5.1 from the local kernel estimate, the local low estimate, the maximal high estimate and the square-summable errors. Include the small-scale branch, where the abstract high–low split is not available.
3. Prove Lemma 5.10. Select maximal bad dyadic intervals by the two density tests. Bound the sum of child lengths by $|I_0|/4$. Construct disjoint major subsets and iterate the local inequality.
4. In Lemma 5.17 split according to $\log \tau \le C_0 L^2$ and its complement, where $L = \mathrm{Log}(\delta_0^{-1})$. The first range uses the square averages; the second uses the prime-square decomposition with an admissible denominator cutoff.
5. Convert the local sparse form with cI on the input side into the stated form. Prove the required grid/dilation reduction rather than identifying the two families by notation.

For finite input sets, begin with a finite set of scales and a finite recursion. If using an infinite family later, specify its sum and prove the limiting step. This keeps the initial recursive construction elementary and exposes precisely which part needs an infinite argument.

There is a useful formalization trap here. An arbitrary real-valued supremum in Lean uses conditional completeness. Without a boundedness hypothesis, it must not be treated as a maximum with all the usual inequalities. One can first work with finite truncated maxima, or use extended nonnegative reals for a maximal function and prove finiteness afterward.

Finally, the strong sparse estimates for exponents in the interior of the triangle need their own interpolation/convexity argument. They should be separate declarations depending on the endpoint local bound and the trivial bounds, not an additional assumed theorem.

8 The high–low model and exceptional conductors

Status: Unfinished target; the main shared arithmetic and analytic branch.

At prime scale N, the annular multiplier is approximated by the model V_N in (6.8). Its ordinary coefficients are normalized unit quadratic Gauss sums. If an exceptional character is present, its contribution has the minus sign displayed in that formula, an induced character modulo each multiple of its conductor, and a continuous model with parameter β.

Use explicit scale-indexed data. For example, define an optional datum at N containing (q_χ, χ, β), and define $\mathcal{M}(\chi)$ to be the scales where that character is selected. Do not pick one character and use it at all scales. Page uniqueness must be strong enough to justify the later regrouping by conductor.

Arithmetic layer. Prove the Cohen correlation by CRT, the odd prime-power evaluation and the explicit 2-adic cases. Build the restricted correlations and Euler products of Section 3 from these identities. Prove the ordinary Gauss bound with its subexponential factor, the real-character bound, and the stronger $\sigma_\chi B_\chi^{1/2} \ll q_\chi^{-1/25}$ estimate. The last estimate is used to sum over changing exceptional characters.

Approximation layer. Encode Lemma 6.5 and then Lemma 6.9. The primes-in- progressions input must supply a single exceptional datum valid over the interval of partial summation $[N/2, N]$, with uniform errors. A fixed-modulus PNT alone does not provide this growing-modulus input.

Exact decomposition. The ordinary split is by $q \le Q$. The exceptional split is by $d = q/q_\chi \le Q/B_\chi$, with its own cutoff. Prove the multiplier identity (6.13) before norm estimates. Empty exceptional sums, including $Q/B_\chi < 1$, must be covered by the definitions.

Low and high bounds. Proposition 6.14 uses inverse transforms and the arithmetic kernel identities. Proposition 6.21 needs Lemma 6.16, the denominator- prefix extension 6.19, the cutoff-difference square functions, and a final sum over χ. At each scale there is at most one exceptional datum; the square sum over characters and Page uniqueness per dyadic conductor range are separate steps. Only after both bounds and the error estimate are checked does one obtain Proposition 4.19.

9 The Orlicz maximal estimate and convergence

Status: Unfinished target; Section 7.

The quantitative Ionescu–Wainger input must retain its constants. Proposition 7.8 specifies a rational-frequency set containing every denominator at most Q, a period P, a denominator bound B, a sufficiently small width, and norm comparisons for a range of exponents. It invokes Tao's Lemma 5.1, Theorems 1.6 and 1.7, and Proposition 1.5. These should have explicit external interfaces if they are not formalized here.

Next prove the continuous multiplier bounds uniformly in $0 < \beta \leq 1$ (Lemma 7.15), followed by the finite-group measure and sampling argument (Proposition 7.21). The measure is allowed to be complex but has total variation at most one. Dependence of constants on the period or on β would break the later optimization.

Proposition 7.28 combines this low model with an ℓ^2 high estimate. The exceptional characters of conductor above Q vary with scale and are retained in the high term. The low exceptional character, when it exists, is controlled using Page uniqueness. This assembly must use the same model and normalization as Section 6.

For the modular extension, do not stop at the restricted estimate on $\mathbf{1}_F$. Let C_∞ bound the masses of the positive kernels. Remove $|f| \leq \lambda/(2C_\infty)$. Decompose the remaining function into level sets

$$E_j = \{2^j\lambda/(2C_\infty) < |f| \leq 2^{j+1}\lambda/(2C_\infty)\}.$$

Allocate thresholds using a positive summable sequence such as $a_j = 6/(\pi^2(j+1)^2)$. Prove the union inclusion, apply the restricted estimate to each E_j, and prove that the polynomial factor in j inside the threshold can be absorbed by enlarging the coefficient of Ψ_C. This is a quantitative estimate for this particular Young function, not just sublinearity. It gives the truncated modular estimate (7.3), then (7.4).

Corollary 7.6 follows by layer cake and a one-variable integral estimate, with $\varepsilon = p - 1$. Corollary 7.42 first transfers finite truncated discrete estimates to an invertible probability-preserving system. Prove bounded-function convergence using the cited polynomial-prime result and control the higher prime powers there. Abel summation handles the unweighted prime averages. Finally use Orlicz truncation, doubling and the modular maximal estimate to pass to unbounded functions; use uniform L^1 bounds for L^1 convergence.

10 Persistent universal divergence

Status: Unfinished target; Section 8.

For each modulus q, define $\Lambda_q = \{b^2 \bmod q : (b,q) = 1\}$. The initial finite tasks are CRT compatibility, inclusion in the prime-square residue set, and, for odd squarefree q,

$$|\Lambda_q| = \frac{\varphi(q)}{2^{\omega(q)}}, \qquad s_q = \frac{q}{|\Lambda_q|} = \prod_{p \mid q} \frac{2p}{p-1}.$$

The inclusion uses Dirichlet's theorem for a unit square root; it must not claim that every prime residue, including nonunits, is part of Λ_q.

Lemma 8.8 expands each quadratic-residue indicator as $\frac{1}{2}(1 + \chi_p - \mathbf{1}_{\{0\}})$. Terms containing a zero indicator are bounded by a constant depending on the number of shifts. For the others, apply the Weil bound to the squarefree product of the distinct shifted linear factors. This proves the correlation hypothesis for the imported Poisson-spacings theorem.

Choose pairwise disjoint blocks of odd primes with block sizes tending to infinity, and a disjoint prime sequence u_j. The block products q_j satisfy $\mathbb{P}_{q_j}(\Lambda_{q_j}) \le 2^{-j}$, whereas $\mathbb{P}_{u_j}(\Lambda_{u_j}) \ge 1/3$. The construction must ensure the combined collection is pairwise coprime.

Theorem 8.6 supplies Poisson spacings. The thickening identity in the proof expresses the covered proportion as

$$\mathbb{P}_{q_j}(\Lambda_{q_j}^{\gamma}) = \int_0^{\gamma} F_j(\theta)\, d\theta + O(s_{q_j}^{-1}).$$

Dominated convergence gives $1 - e^{-\gamma}$, so the deficit is $o(\gamma)$. The pair-separation condition uses LaVictoire's Lemma 3.2 in addition to this one-gap distribution; it is not an automatic restatement of Poisson convergence.

Finally, fixed-modulus prime equidistribution gives limiting mass $1/|\Lambda_Q|$ at every $a \in \Lambda_Q$. The strict lower bound required by (8.5) then holds with, for example, $\beta = 1/2$. Apply the full imported divergence criterion, including its transference/Banach-principle conclusions, to obtain exactly the persistence and system quantifiers of Theorem 1.5.

If the chosen formal ergodic infrastructure assumes invertibility, a separate extension/transfer argument is needed to match the unqualified system statement in Theorem 1.5. Simply adding invertibility would narrow the theorem.

11 The Goldbach branch

Status: Only the finite final assembly and small supporting pieces are checked.

The reductions in Section 9 produce a lower sieve count minus an upper correction count. Their errors must have mean-square bounds with an arbitrarily large logarithmic saving. The common main term is $X_n\Theta_n(z)$; its positivity and uniform lower bound are separate lemmas.

Circle method. Lemma 9.10 is the principal target. Its coefficients are independent of n, and its conclusion is a mean square of a sum over auxiliary moduli and ℓ. Section 10 supplies the minor-arc bound, the major-arc local factor G_R, the truncated continuous factor $H_{\ell,R}$, the tail estimate and the major-arc error. The same small-prime parameter R must be retained in both main-term factors until its tails have been shown to be summable.

Sieve. Lemma 11.2 replaces variable sets of sieve primes by fixed sets, which is needed before applying Lemma 9.10 to coefficients independent of n. The exact local densities are computed in Lemmas 11.6 and 11.8. For primes at most R, the uniform product dimensions are $2/3$ and $3/2$; one cannot apply the desired half-dimensional and linear sieves over the whole range.

Lemma 11.13 removes $p \le R$ first at level $D_0 = \exp((\log\log N)^3)$. The remaining level is D/D_0. Because outer weights have both signs, the choice of preliminary upper or lower weights must depend on that sign. Prove this finite algebra before estimating the two stages. The coefficient bounds and independence of n must survive the combination. The outer dimensions are then $1/2$ and 1.

Lemma 11.19 performs the weighted sum over ℓ. It needs the cited mean-value theorem for multiplicative functions and the relevant short-interval uniformity. Track the cancellation of the factors at $p \le R$ and the replacement of $H_{\ell,R}$ by H_ℓ; neither can be dropped at the definition stage.

The outputs are Propositions 9.17 and 9.20. Lemma 9.22 bounds the common main term below. Lemma 9.24 compares the sieve constants. Their combination gives (9.25), and the checked finite result in Section 4 then applies. The remaining asymptotic task is to choose the constants in the correct order, show the $O(N^\varepsilon)$ nuisance is smaller than the positive main term, discard the initial range, and pass from scale N^2 to general X.

12 Proposed external-input register

Status: Candidate interfaces; not declarations already installed in the starter.

The user's permission to leave deep theorems unformalized makes a conditional development appropriate. The exact declarations should be approved mathematically by comparison with their sources. The list below is an initial register, not a claim that each result is absent from current Mathlib.

ID	Published input	Required interface and use
NT01	PNT in progressions with exceptional term and Landau–Page; paper (2.5)–(2.6), reference [22]	Uniform growing-modulus error, one datum on an annulus, induced characters, and uniqueness strong enough across scales. Used in Section 6.
NT02	Siegel–Walfisz and fixed-modulus equidistribution	Full quantifier order for modulus, residue, exponent and threshold. Used where polylogarithmic moduli or fixed progressions suffice; not a replacement for NT01.
NT03	Ren [40, Theorem 1]	Quadratic prime exponential-sum cancellation with exactly the denominator and arc-width ranges needed in (6.7) and (10.1).
NT04	Bombieri–Vinogradov, Brun–Titchmarsh, Mertens-type estimates; Section 2.6	Maxima over residues and endpoints, divisor weights, and constants uniform in the auxiliary moduli. Used in Section 10 and the sieve.
NT05	Classical quadratic Gauss evaluations and Weil character-sum bounds	Primitive and induced character conventions; primes and prime powers, including 2; squarefree-polynomial hypothesis for Lemma 8.8.
HA01	Separated-frequency maximal inequality [25, Theorem 1.1]	Precise separation and cardinality dependence. Prove its adaptation, transference and uniformity in β in Lemma 6.16.
HA02	Tao [45], Lemma 5.1, Theorems 1.6–1.7 and Proposition 1.5	Quantitative major-arc construction, multiplier theorem and sampling comparisons. Keep $r, k, B, P, \varepsilon, p$ and vector-valued constants explicit.
HA03	Hardy–Littlewood, Calderón–Zygmund, Khintchine and Rademacher–Menshov	Only use an external declaration where the pinned library does not already provide the needed statement. Preserve the $p \downarrow 1$ dependence.

Do not collapse these into a single assumption saying that all estimates needed in the paper hold. An import should be independently recognizable as a published theorem and should not assume the novel conclusion to be checked.

External inputs continued

ID	Published input	Required interface and use
ER01	LaVictoire [27, Theorem 3.1 and Lemma 3.2], with the principles used in the paper's Theorem 8.1	CRT families, two coprime sequences, thickening deficit, pair separation and residue-frequency hypotheses; full persistent universal divergence conclusion.
ER02	Granville–Kurlberg [14, Theorem 1], in the specialization 8.6	Uniform correlations for distinct shifts and the hypotheses on local densities; convergence of normalized cyclic gaps.
ER03	Polynomial-prime pointwise convergence [35, Corollary 2(ii)]	The bounded-function dense class used in Corollary 7.42, with the correct weights and systems. Prime powers and Abel summation remain internal tasks.
SV01	Rosser–Iwaniec coefficient-form sieve and fundamental lemma [21]	Weights, support, signs, coefficient bounds, product hypotheses, sieve functions and quantitative remainder; prove the two-stage composition in Lemma 11.13.
SV02	Granville–Koukoulopoulos [13] and Shiu [41]	Mean values of the particular multiplicative weight and uniform estimates in the retained ranges used by Lemma 11.19.

Represent imports as hypotheses first

For a new branch it is often helpful to package the external inputs as a named structure and prove a theorem with that structure as an explicit argument. This makes the logical boundary visible without immediately adding global axioms. Once the interfaces have been audited, a separate file can instantiate the structure from selected quoted theorems.

In either design, dependency output for a headline theorem must agree with the published import register. A proof conditional on an estimate proved nowhere in the project or cited literature should be labeled incomplete, even if Lean accepts it with that estimate as a hypothesis.

Possible reuse from the Hua project

The Hua repository contains finite cyclic Fourier infrastructure, quadratic local arithmetic and analytic assembly. Some may be reusable after comparing definitions. Its prime-square weights are those of its own W-trick argument, not automatically the operators $A_{N,\Lambda}$ in this paper. Reuse requires checking signs, normalizations, finite versus infinite groups, and transitive axiom dependencies. No reuse has been counted as completed work here.

13 Points to settle during manuscript proofreading

These are formalization requirements or visible editorial issues, not a verdict that the main mathematical arguments are wrong.

1. **Freeze the version.** The attached paper has 96 pages and Section 7. Its current numbering is the one used in this document. Match the final LaTeX labels rather than importing numbering from older discussions.
2. **Repair the Section 7 references.** The opening of Section 7 and the last sentence before the proof of Corollary 7.6 refer to Theorem 1.5. In this snapshot Theorem 1.5 is the divergence theorem, so these references cannot name the maximal-inequality consequence intended there.
3. **Preserve the two endpoint functions.** The improving loss is $\exp(c\sqrt{L}/L_2)$; the modular function uses $\exp(C\sqrt{\log(e+t)\log(e+\log(e+t))})$. These are different scales.
4. **Separate bounded and unbounded input.** A negligible prime-power contribution for bounded functions is sufficient for the dense-class convergence step; it is not itself a uniform error bound for every L^1 function.
5. **Make exceptional data consistent.** Record the annular data used in partial summation and prove the regrouping over $\mathcal{M}(\chi)$. Do not replace this by a single exceptional character fixed for all N.
6. **Expose uniformity in the Goldbach range.** All $o(1)$, tail errors and positivity bounds must be uniform in admissible n, ℓ and auxiliary moduli. Write the order of choosing A_0, K, C and all thresholds explicitly.
7. **Retain fixed sieve-prime sets.** The independence from n after Lemma 11.2 is required by the mean-square estimate. The coefficient combination in the pre-sieve must preserve this independence.
8. **Check strictness and small cases.** Empty sets, $q = 1$, powers of 2, $N \geq 2$, $p = 2, 3$, and equality at density/threshold boundaries should be covered by exact definitions, not silently discarded.

The present work has not audited every cited theorem or every proof in the PDF. In particular, successful checks of the starter do not validate any unfinished node elsewhere in the map.

14 What is needed to continue

The PDFs were enough to start. The next useful input is the final LaTeX project: all source files, bibliography, custom macros and any code used for numerical estimates, together with the PDF generated from those files. This makes the statements and labels unambiguous and lets the map track proofreading changes.

GitHub is useful for versioning and automated builds, but an account is not required to write or check Lean locally. This archive is a normal Lean project; it can be uploaded later. The toolchain and lockfile are already selected. There is no need for the author to learn Lean before mathematical work can begin.

The next mathematical milestone should be one coherent branch. A practical sequence is:

1. Encode the final target statements and audit their quantifiers against the LaTeX. Build a stable statement map.
2. Complete a substantial finite arithmetic module, such as the quadratic Cohen identity with all prime-power cases, or complete the scalar optimization in Theorem 4.1.
3. Specify and audit the deep-input interfaces. Prove one full branch from those interfaces, without assuming its novel intermediate estimates.
4. Add the remaining branches and report the exact dependencies of each headline theorem separately.

For the Goldbach branch, the existing checked finite assembly supplies an endpoint to which the circle-method and sieve components can connect. For the harmonic- analysis branch, the shared high–low decomposition is the principal reusable milestone. The divergence branch can be developed separately once its external criterion and Poisson-spacings interfaces are audited.

Completion criterion. A full conditional formalization requires a fresh build of every target theorem, a transitive axiom audit with no unfinished proof axiom, and a human comparison of the final statements and imported inputs with the manuscript and sources. A long blueprint, a clean source scan, or a passing build of the present starter alone does not meet that criterion.

Current deliverable. This is an initial, reproducible starting point with small checked components and an explicit account of the much larger unfinished work. It should not yet be described in an arXiv or journal submission as a formalization of the paper.

A Inventory of numbered statements

The following inventory was extracted from headings in the attached PDF. Planned names are suggestions only. A status of "target" means no proof of that numbered statement is present in the starter. Checked supporting subclaims are described separately in Sections 4–5; they do not change the status of the enclosing paper theorem or lemma.

Statement	Page	Planned name (after `PrimeSquares.`)	Status
Theorem 1.1	3	`Main.theorem_1_1`	target
Theorem 1.3	3	`Main.theorem_1_3`	target
Theorem 1.5	4	`Main.theorem_1_5`	target
Theorem 1.6	5	`Main.theorem_1_6`	target
Definition 2.10	12	`Foundations.definition_2_10`	target
Definition 2.11	13	`Foundations.definition_2_11`	target
Lemma 3.1	15	`Arithmetic.lemma_3_1`	target
Definition 3.7	16	`Arithmetic.definition_3_7`	target
Definition 3.9	17	`Arithmetic.definition_3_9`	target
Lemma 3.11	17	`Arithmetic.lemma_3_11`	target
Lemma 3.12	17	`Arithmetic.lemma_3_12`	target
Lemma 3.14	17	`Arithmetic.lemma_3_14`	target
Corollary 3.16	18	`Arithmetic.corollary_3_16`	target
Lemma 3.20	19	`Arithmetic.lemma_3_20`	target
Lemma 3.21	19	`Arithmetic.lemma_3_21`	target
Lemma 3.24	20	`Arithmetic.lemma_3_24`	target
Lemma 3.27	21	`Arithmetic.lemma_3_27`	target
Theorem 4.1	24	`Improving.theorem_4_1`	target
Corollary 4.15	25	`Improving.corollary_4_15`	target
Proposition 4.19	26	`Improving.proposition_4_19`	target
Lemma 5.1	28	`Sparse.lemma_5_1`	target
Lemma 5.10	30	`Sparse.lemma_5_10`	target
Lemma 5.17	31	`Sparse.lemma_5_17`	target
Lemma 6.5	34	`HighLow.lemma_6_5`	target
Lemma 6.9	35	`HighLow.lemma_6_9`	target
Proposition 6.14	37	`HighLow.proposition_6_14`	target
Lemma 6.16	39	`HighLow.lemma_6_16`	target
Corollary 6.19	40	`HighLow.corollary_6_19`	target
Proposition 6.21	41	`HighLow.proposition_6_21`	target
Theorem 7.2	44	`Maximal.theorem_7_2`	target
Corollary 7.6	44	`Maximal.corollary_7_6`	target
Proposition 7.8	45	`Maximal.proposition_7_8`	target
Lemma 7.15	47	`Maximal.lemma_7_15`	target
Proposition 7.21	48	`Maximal.proposition_7_21`	target
Proposition 7.28	51	`Maximal.proposition_7_28`	target
Corollary 7.42	56	`Maximal.corollary_7_42`	target
Theorem 8.1	57	`Divergence.theorem_8_1`	target
Theorem 8.6	58	`Divergence.theorem_8_6`	target
Lemma 8.8	58	`Divergence.lemma_8_8`	target
Lemma 9.2	61	`Goldbach.lemma_9_2`	target
Lemma 9.3	61	`Goldbach.lemma_9_3`	target
Lemma 9.7	62	`Goldbach.lemma_9_7`	target
Corollary 9.9	63	`Goldbach.corollary_9_9`	target
Lemma 9.10	63	`Goldbach.lemma_9_10`	target
Proposition 9.17	64	`Goldbach.proposition_9_17`	target
Proposition 9.20	64	`Goldbach.proposition_9_20`	target

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Lemma 9.22	65	`Goldbach.lemma_9_22`	target
Lemma 9.24	65	`Goldbach.lemma_9_24`	target
Lemma 10.2	66	`GoldbachCircle.lemma_10_2`	target
Lemma 10.12	68	`GoldbachCircle.lemma_10_12`	target
Lemma 10.15	68	`GoldbachCircle.lemma_10_15`	target
Lemma 10.20	72	`GoldbachCircle.lemma_10_20`	target
Lemma 10.24	73	`GoldbachCircle.lemma_10_24`	target
Lemma 11.2	76	`GoldbachSieve.lemma_11_2`	target
Lemma 11.6	76	`GoldbachSieve.lemma_11_6`	target
Lemma 11.8	77	`GoldbachSieve.lemma_11_8`	target
Lemma 11.13	78	`GoldbachSieve.lemma_11_13`	target
Lemma 11.19	82	`GoldbachSieve.lemma_11_19`	target